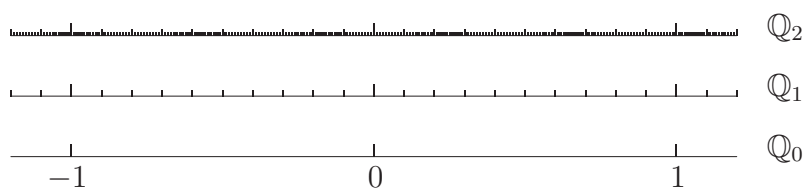
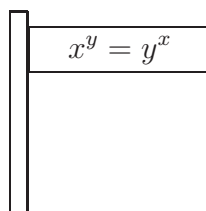


CALCULUS FENNICUS

Advanced first-year mathematics at Helsinki
University of Technology (2000–2013)

Juhani Pitkäranta



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Preface

This book grew out of the mathematics lectures that the author gave at Helsinki University of Technology (TKK, later Aalto University) between 2000 and 2013. It previously appeared in two volumes covering the advanced first-year mathematics core courses I–II at TKK.

Anyone who sets out to reform the basic teaching of university-level mathematics (and probably mathematics at any level) inevitably faces many difficult questions and choices. On the one hand, mathematics seems rather like a church, in which the fundamental truths are permanent and even liturgical changes are fiercely resisted. On the other hand, the surrounding world is changing all the time, and sooner or later the pressure to change reaches mathematics as well. The author has had particular cause to ponder the possible long-term effect of computers and powerful calculators on the teaching of mathematics at university level. — When numerical and symbolic manipulations can be carried out conveniently by machine, what sort of mathematical skill ought one to teach people? Should we perhaps place more emphasis than before on the practical side of mathematics and on applications? Or, on the contrary, on mathematics as the pure art of abstract thought?

Let the reader judge whether these musings have had any effect on the outcome (and if so, in which direction). One feature peculiar to this textbook, departing from the established teaching tradition of mathematics, is nonetheless worth mentioning: the book introduces sequences and series of numbers right at the outset, so that the real numbers can then be defined just as they ‘appear’ in the world of calculators and computers, namely as infinite decimals.

As an international point of comparison for this book one might mention the American Calculus books. Compared with those, this book is more ambitious in its aims. The goal is to give a unified, logically progressing introduction — starting from the very foundations — to the basic ideas of mathematics and to modern computational technique. With an eye to logical completeness, the level of abstraction has here and there been raised above the typical ‘street level’ characteristic of, for example, the Calculus books just mentioned. The book is

therefore intended for a fairly demanding audience, for whom mathematics is not only computational technique and ‘formulas’ but also an intellectual challenge and a genuine object of interest.

On the home ground of Otaniemi, the predecessors of this book are the Finnish-language textbooks and lecture notes written by associate professor Harri Rikkinen around the turn of the 1960s and 1970s, and by lecturer Simo K. Kivelä in the 1980s and 1990s. The influence of that tradition is visible in this book as well — it is truly difficult to break away from tradition. Besides this inherited lore, the many lively discussions the author has had with members of the TKK mathematics department have been very useful. Among the many active interlocutors during the book’s writing, special mention is due to lecturer (now emeritus) Simo Kivelä, who as a ‘ $\text{\LaTeX}$ guru’ deserves a separate word of thanks for his numerous pieces of advice and technical help on mathematical typesetting. Thanks are also due to lecturer Pekka Alestalo for the more than a hundred critical comments he recorded in 2006, when he served as substitute lecturer for the advanced core courses I–II.

The first, handwritten version of the book was produced in the academic year 2000–2001. Antti H. Niemi (then an engineering student, now D.Sc. (Tech.)), a member of that year’s audience, worked the text into its first computerized version in the summer of 2002. The text first took the form of a book in the academic year 2005–2006. Revised editions appeared in 2007 and 2009. In the ‘final’ version now appearing, the content of the book has been further corrected and streamlined, and the exercises too have been substantially reworked.

The author’s first contact with the advanced mathematics of TKK took place as a student in the autumn of 1968. The Long Basic Course in Mathematics of that time was lectured by Harri Rikkinen in Hall D of the main building — the same hall in which the author gave his own first lectures on the advanced core course 32 years later. Since those days many things have changed by the time of this writing: TKK no longer exists, nor (after the teaching reform of autumn 2013) does the advanced mathematics course. This book therefore no longer has any immediate use as a course text. Let it remain, however, at least as an online publication, a reminder of the tradition of advanced mathematics teaching that arose in its own distinctive way at TKK and continued for half a century.

I thank the Department of Mathematics and Systems Analysis of Aalto University for the financial support that encouraged the completion of this textbook even in retirement. I remember with warmth my entire audience from the autumn of 2000 to the spring of 2013, and with this book I thank them once more for their critical feedback.

Helsinki, June 15, 2015

Juhani Pitkäranta

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Chapter I

Numbers and sequences

"God created the natural numbers — all else is the work of man" was the aphorism of the German mathematician **Leopold Kronecker** (1823-1891) and his stance on the origin of mathematics. — Undoubtedly the differentiation in natural language of the words 'one', 'two', etc. describing quantity or order has been one starting point and a basic prerequisite for the birth of mathematics. Signs of a sense of numbers and ratios have survived from as far back as 30 000 years ago, so mathematics can justifiably be regarded, within human culture, as *sui generis*, one of a kind. As a cultural phenomenon, mathematics can to some extent be understood by comparing it to other kinds of culture. For instance, if one examines the history and cumulative structure of mathematics, or its modes of use, one can see many points of contact with other grand manifestations of the human spirit as well, especially natural languages and technology.

Mathematics, however, can be learned and understood more deeply only by studying mathematics. In this chapter, study begins 'from the beginning', that is, from the concept of (mathematical) *number*. The guiding thread of the chapter is the construction of the *real numbers* starting from the 'reasonable', i.e. *rational*, numbers. In the history of mathematics the concept of the real number was made finally precise only in the 19th century. The historical problem — and still a problem in teaching — is the idea of 'endlessness' inevitably contained (in one form or another) in the concept. In this text the real numbers are approached through the *sequences* formed by rational numbers, and their special cases, the *infinite decimals*. In this way the concept of the real number becomes tied to 'genuine computation' and also to ordinary numeral notation. Nonterminating sequences will come into use later in many other contexts as well. On them rest ultimately both many computational methods (algorithms) and the concepts of the mathematical analysis based on the real numbers, i.e. *real analysis*.

I.1 A survey of the rational numbers

We denote the set of *natural numbers* by

$$\mathbb{N} = \{1, 2, 3, 4, \dots\}.$$

More generally, a *set* consists of *elements*, which in the case of the set $\mathbb{N}$ are the numbers 1, 2, 3, etc. The notations

$$8 \in \mathbb{N}, \quad \infty \notin \mathbb{N}$$

are read 'eight belongs to $\mathbb{N}$ ', 'infinity does not belong to $\mathbb{N}$ ', meaning simply that $\mathbb{N}$ contains an element called 'eight' but no element called 'infinity'.

The set $A = \{1, 2, 3\}$ is a *subset* of $\mathbb{N}$. We write

$$A \subset \mathbb{N} \quad \text{or} \quad \mathbb{N} \supset A$$

and read ' A belongs to / is contained in $\mathbb{N}$ ' or ' $\mathbb{N}$ contains A '. In general the notation $A \subset B$ means that every element of A is also an element of B ; in other words,

$$x \in A \quad \Rightarrow \quad x \in B.$$

Here ' $\Rightarrow$ ' is the so-called *implication arrow*, one of the symbols of logic: the notation $P \Rightarrow Q$ is read ' Q follows from P ', or 'if P , then Q '. The symbol ' $\subset$ ' (containment as a subset) is thus defined by means of the symbol ' $\in$ ' (membership as an element). The latter symbol is a so-called *primitive*, which can no longer be defined by means of other symbols. When speaking of sets, one may thus regard the notations $x \in A$ or $x \notin A$ as the starting point of all logical thinking.

EXAMPLE 1 If $A = \{2, 1, 1, 2, 1\}$ and $B = \{1, 2\}$, then $A \subset B$, for if $x \in A$, then either $x = 1$, in which case $x \in B$, or $x = 2$, in which case also $x \in B$. By the same reasoning one sees that $B \subset A$ holds as well. $\square$

If both $A \subset B$ and $B \subset A$ hold for sets A and B , as in the example, then A and B are said to be (as sets) *equal*, and we write $A = B$.^{*} Notations of logic and set theory are treated somewhat more broadly in Section I.3. Let us note in this connection one more notation common in defining a subset,

$$B = \{x \in A \mid E\},$$

where E is some *condition* that, for each element $x \in A$, either holds, in which case $x \in B$, or does not hold, in which case $x \notin B$.

^{*}If $A \subset B$ and A is not the same as B , then A is said to be a *proper* (proper) subset of B . In some mathematical texts the symbol ' $\subset$ ' has this more restricted meaning, in which case the possibility $A = B$ is reserved for the notation $A \subseteq B$. In this text, therefore, $A \subset A$ holds.

EXAMPLE 2 The *even* and *odd* natural numbers are defined as subsets of $\mathbb{N}$

$$\begin{aligned} A &= \{ x \in \mathbb{N} \mid x = 2 \cdot y \text{ for some } y \in \mathbb{N} \} = \{ 2, 4, 6, \dots \}, \\ B &= \{ x \in \mathbb{N} \mid x \notin A \} = \{ 1, 3, 5, \dots \} \quad \square \end{aligned}$$

The example above already referred to the *arithmetic operations* defined (in the familiar way) on the set of natural numbers, of which there are two:

$$\begin{aligned} \text{addition: } x \in \mathbb{N} \ \& \ y \in \mathbb{N} \quad \mapsto \quad x + y \in \mathbb{N} \quad \text{'}x \text{ plus } y\text{'}, \\ \text{multiplication: } x \in \mathbb{N} \ \& \ y \in \mathbb{N} \quad \mapsto \quad x \cdot y \in \mathbb{N} \quad \text{'}x \text{ times } y\text{'}, \end{aligned}$$

Here ' $\&$ ' is a notation of logic, read 'and' (the more formal notation is ' $\wedge$ '), and ' $\mapsto$ ' denotes *assignment*: to the numbers x and y , i.e. to a *number pair* of $\mathbb{N}$, one assigns a (unique) number $x + y \in \mathbb{N}$, called the *sum* of x and y , and $x \cdot y \in \mathbb{N}$, called the *product* of x and y . An arithmetic operation thus means, in the case of individual numbers x, y , a 'rule of assignment', more generally a 'system of assignment rules'. The rules can be thought of either as derived from some more general principles (computational laws), or they can be understood simply as a 'list' covering all pairs of numbers. Once the content of addition and multiplication has been commonly agreed upon (and printed in the textbooks of comprehensive school), the result is an *algebra** built on the foundation of the natural numbers, which we denote $(\mathbb{N}, +, \cdot)$.

The following well-known basic laws hold for the arithmetic operations on the natural numbers:

- (L1) $x + y = y + x$
- (L2) $x \cdot y = y \cdot x$
- (L3) $x + (y + z) = (x + y) + z$
- (L4) $x \cdot (y \cdot z) = (x \cdot y) \cdot z$
- (L5) $x \cdot (y + z) = x \cdot y + x \cdot z$
- (L6) $x \cdot 1 = x \quad \forall x \in \mathbb{N}$

Here (L1), (L2) are the *commutative laws*, (L3), (L4) are the *associative laws*, and (L5) is the *distributive law*. The last-mentioned also has a parallel form

*Algebra, i.e. 'letter-reckoning', is the more abstract form of computing with numbers, i.e. of *arithmetic*. The development of algebra began within Arab culture in the 1st millennium AD.

$$(L5') \quad (x + y) \cdot z = x \cdot z + y \cdot z,$$

which follows by combining (L2) and (L5). In law (L6), ' $\forall$ ' is a symbol of logic, read 'for all'. This law expresses that the number 1 is the so-called *unit element* of multiplication. As will be seen later, the computational laws (L1)–(L6) are common to many algebras. In more general considerations these laws are indeed often taken as assumed basic laws, i.e. as *axioms*. *

In the computational laws (L1)–(L6) the normal ordering rules for combining operations are assumed, that is, expressions inside parentheses are evaluated first, and multiplications always before additions if there are no parentheses. Note that this is not a matter of any extra assumptions; rather, for example, the rule

$$x + y \cdot z = x + (y \cdot z)$$

is a parenthesis-omission rule, i.e. a matter of notational convention. When combining the same operation, by the associative laws (L3), (L4) the result is always the same regardless of the order of operations, so the parentheses can be omitted (notational convention!) and one writes

$$x + y + x + \dots, \quad x \cdot y \cdot z \dots$$

Among the notational conventions of ordinary number algebra is also the freedom to leave the multiplication sign unwritten insofar as there is no danger of confusion:

$$x \cdot y = xy, \quad x \cdot 2 = 2x, \quad 2 \cdot (1 + 2) = 2(1 + 2) = 2 \cdot 3 \neq 23.$$

*In the case of the natural numbers, the computational laws (L1)–(L6) are derivable from the **Peano axioms** underlying the natural numbers. The axioms were put forward by the Italian mathematician **Giuseppe Peano** (1858–1932) in 1889. Peano's axioms for the natural numbers are: (P1) $1 \in \mathbb{N}$. (P2) Every $x \in \mathbb{N}$ has a unique *successor* $x' \in \mathbb{N}$. (P3) The number 1 is not the successor of any $x \in \mathbb{N}$. (P4) Every $x \in \mathbb{N}$ is the successor of at most one number, i.e. if $x, y \in \mathbb{N}$ and $x' = y'$, then $x = y$. (P5) If $S \subset \mathbb{N}$ and (i) $1 \in S$ and (ii) $x' \in S$ whenever $x \in S$, then $S = \mathbb{N}$. The successor of a number appearing in axiom (P2) is, in ordinary usage, the 'next number'. By axiom (P3) the number 1 is the 'first' of the natural numbers; on the other hand there is no 'last' number, since every number has a successor (axiom (P2)) and no number can appear twice in the chain of successors (axioms (P3) and (P4)). Thus there are infinitely many natural numbers. Axiom (P5) states the so-called *induction principle*, see Section I.4 below.

The addition and multiplication of natural numbers are defined on the basis of axioms (P1)–(P5) by imposing two axioms for each operation. On this basis the result of any desired operation can be determined by means of the numbers 1, $1' = 2$, $2' = 3$, etc. The axioms of addition: (Y1) $x + 1 = x'$, (Y2) $x + y' = (x + y)'$. The axioms of multiplication: (K1) = (L6), (K2) $x \cdot y' = x \cdot y + x$.

When computing successive products, the n -fold ($n \in \mathbb{N}$) multiplication of the same number is customarily called *raising to a power*, and the notation used is

$$\underbrace{x \cdot x \cdots x}_{(n \text{ factors})} = x^n \quad \text{'}x \text{ to the power } n\text{' } (n \in \mathbb{N}).$$

The integers $\mathbb{Z}$

The set $\mathbb{Z}$ of *integers* is the first step in the 'human work' by which the number system is extended, starting from the natural numbers. The extension consists of two sub-steps, the first of which is the introduction of the number *zero* ('nothing') and the second of the *negative* ('how many are missing'). The set of numbers arising as a result of these ideas is denoted

$$\mathbb{Z} = \{ \dots, -2, -1, 0, 1, 2, \dots \} = \{ 0, \pm 1, \pm 2, \dots \}.$$

The extension of the number domain is done axiomatically by assuming the following additional axioms, which thus hold in $\mathbb{Z}$ but not in $\mathbb{N}$.

(L7) There exists a *number* 0, for which $x + 0 = x \quad \forall x \in \mathbb{Z}$.

(L8) Every $x \in \mathbb{Z}$ has a *negative* $-x$, for which $x + (-x) = 0$.

The abbreviated notation

$$x + (-y) = x - y \quad \text{'}x \text{ minus } y\text{'}$$

introduces into $\mathbb{Z}$ a new operation, *subtraction*, whose result is called the *difference* of the numbers x and y . This is thus a matter of combining addition and the 'sign change' $x \mapsto -x$ (which one can also interpret as an operation).

When computing with integers it is taken for granted that the number 0, as well as the negatives $-1, -2, \dots$, are unique. Also familiar are the computational rules $-(-x) = x$, $0 \cdot x = 0$, and $(-1) \cdot x = -x$. These assumptions/rules need not, however, be included among the axioms of the integers, for they are consequences of the axioms:

THEOREM I.1.1 * For the integers, by axioms (L1)–(L8), the following hold:

*When in a mathematical text one wishes to bring out some clearly formulated, true statement — e.g. because one wishes to refer to it later, or when one wishes to emphasize the novelty of the result — the terms *Theorem*, *Proposition*, *Lemma*, and *Corollary* are used. In this text these designations (Theorem, Proposition, Lemma, and Corollary) are used somewhat inconsistently. A Proposition is a statement of lesser weight or more technical content than a theorem — the term is used mainly when one wishes to classify theorems according to their weight. A Lemma is usually an intermediate stage in proving some theorem/proposition, and a corollary a straightforward consequence of an already proved theorem or proposition.

- (a) The number 0 is unique, as is the negative $-x$ for each $x \in \mathbb{Z}$.
- (b) $-(-x) = x \quad \forall x \in \mathbb{Z}$.
- (c) $0 \cdot x = 0 \quad \forall x \in \mathbb{Z}$.
- (d) $(-1) \cdot x = -x \quad \forall x \in \mathbb{Z}$.

Theorem I.1.1 will be proved in the next section as part of a more general theorem, so here we content ourselves with proving only the easiest part (b): since by axioms (L1) and (L8) we have

$$-x + x = x + (-x) = 0,$$

it follows, by the definition of the negative, that $-(-x) = x$. $\square$

EXAMPLE 3 By combining the computational rules (b) and (d) of Theorem I.1.1, one sees that $(-1) \cdot (-1) = -(-1) = 1$. $\square$

Number systems

In practice the integers must be expressed in some *number system*. The most common one, established as the 'mother tongue' of mathematics, is the *decimal system*, in which one writes

$$\begin{aligned} 1 + 1 &= 2 && \text{'two'}, \\ 2 + 1 &= 3 && \text{'three'}, \\ &\vdots \\ 8 + 1 &= 9 && \text{'nine'}, \\ 9 + 1 &= 10 && \text{'ten'}. \end{aligned}$$

Here the symbols $0 \dots 9$ are the *digits of the decimal system* (Latin *digitus* = finger, toe). The number 'ten' is the first of the natural numbers that has no symbol of its own in this system; instead it is written as the combination 'one zero'. This is the *base* of the system. Thereafter the numbers of the decimal system are expressed symbolically in the form

$$x = eabcd \dots \in \mathbb{Z},$$

where e is the *sign* ($e = +$ or $e = -$; 'empty' = $+$), and $a, b, c, d, \dots$ are digits of the decimal system. Depending on the number of digits, the interpretation is

$$ea = \pm a, \quad eab = \pm(a \cdot 10 + b), \quad eabc = \pm(a \cdot 10^2 + b \cdot 10 + c), \quad \text{etc.}$$

Here the sign is that given by e .^{*}

More generally, if the base of a number system is a number $k \in \mathbb{N}$, $k \neq 1$, then each of the numbers $0, 1, \dots, k-1$ of this k -system must have its own numeral symbol.[†] If $a, b, c, \dots$ are such symbols, then the number $x = eabcd\dots$ is interpreted by the same rules as above. One need only note that the number '10' is now not 'ten', but denotes the number $1 \cdot k + 0 = k$. The symbol of the base is thus always '10'; the manner of pronunciation, however (e.g. 'ten', 'dozen', 'score'), must be agreed separately in each number system (and in each language). If nothing has been agreed, one must read 'one zero'. The number system with the simplest set of numeral symbols is the *binary system*, in which the base is 'the successor of one', i.e. 2. The digits, or *bits*, of the binary system are only 0 and 1.

EXAMPLE 4 The decimal number 1253 can be read 'one two five three', in which case one merely pronounces the 'name' of the number. In the reading 'one thousand two hundred fifty-three', the number is already given an interpretation: $1253 =$ 'thousand plus two times hundred plus five times ten plus three'. Here 'hundred' and 'thousand' are pronunciations agreed in the English language (and taught in schools) for the numbers $100 = k^2$ and $1000 = k^3$, where $k =$ 'ten'. $\square$

EXAMPLE 5 Express the decimal number 11 in the binary system.

Solution In the decimal system one can write $11 = 1 \cdot 2^3 + 0 \cdot 2^2 + 1 \cdot 2 + 1$. The binary representation is obtained from this by writing '10' in place of the symbol '2':

$$11 = 1 \cdot 10^3 + 0 \cdot 10^2 + 1 \cdot 10 + 1 = 1011 \quad (\text{binary system}). \quad \square$$

EXAMPLE 6 Compute the sum of the numbers 145 and 66 of the base-7 system and convert the result to the decimal system.

Solution Writing $k = 7$, then by applying the computational laws (L1)–(L6) we

^{*}By the interpretation of the numeral notation, $\pm 0abc.. = \pm abc..$, i.e. 'leading zeros' do not affect the interpretation of the number.

[†]The idea of number systems was invented in Mesopotamia about 4000 years ago. In the earliest systems the base was 60. The numeral symbols were combinations of cuneiform wedge-marks.

obtain

$$\begin{aligned}
 145 + 66 &= (1 \cdot k^2 + 4 \cdot k + 5) + (6 \cdot k + 6) \\
 &= 1 \cdot k^2 + (4 + 6) \cdot k + (5 + 6) \\
 &= 1 \cdot k^2 + (1 \cdot k + 3) \cdot k + (1 \cdot k + 4) \\
 &= (1 + 1) \cdot k^2 + (3 + 1) \cdot k + 4 \\
 &= 2 \cdot k^2 + 4 \cdot k + 4 \\
 &= 244.
 \end{aligned}$$

Expressed in the decimal system, the final result is $2 \cdot 7^2 + 4 \cdot 7 + 4 = 130$. $\square$

The rational numbers $\mathbb{Q}$

In passing from $\mathbb{Z}$ to the set $\mathbb{Q}$ of *rational numbers*, only one more additional axiom is needed:

(L9) Every $x \neq 0$ has a *reciprocal* x^{-1} such that $x \cdot x^{-1} = 1$.

The abbreviated notation

$$x \cdot y^{-1} = x/y = \frac{x}{y} \quad \text{'}x \text{ over } y\text{'}$$

defines the fourth operation, *division*, whose result is called the *quotient* of the numbers x and y . Division is thus a matter of combining the 'inversion operation' $y \mapsto y^{-1}$ and the multiplication $x, y^{-1} \mapsto x \cdot y^{-1}$. The number x is said, in this operation, to be the *numerator* and the number y the *denominator*. By the definition of division and axioms (L2), (L6),

$$x^{-1} = x^{-1} \cdot 1 = 1 \cdot x^{-1} = 1/x.$$

When the number set $\mathbb{Z}$ is extended so that in the extended set $\mathbb{Q}$ the axioms (L1)-(L9) hold, the result is a new algebra, denoted $(\mathbb{Q}, +, \cdot)$. By axiom (L9) and the definition of division, the elements of $\mathbb{Q}$ include at least the numbers of the form $x = p/q$, where $p \in \mathbb{Z}$ and $q \in \mathbb{N}$. It turns out (see Example 7 below and Exercise 7) that, starting from numbers of the said form, the results of the operations $x \mapsto -x$, $x \mapsto x^{-1}$, $x, y \mapsto x + y$, and $x, y \mapsto x \cdot y$ are always of the same form, i.e. there are no numbers of other types in $\mathbb{Q}$. The set $\mathbb{Q}$ of rational numbers can therefore be defined as

$$\mathbb{Q} = \{ x = p/q \mid p \in \mathbb{Z} \ \& \ q \in \mathbb{N} \}.$$

From axioms (L1)-(L9) all the usual computational rules of the rational numbers can be derived. Below are two examples (see also Exercise 7).

EXAMPLE 7 Appealing to axioms (L1)–(L9) and Theorem I.1.1, justify the computational rules of the rational numbers

$$\text{a) } \frac{0}{q} = 0 \quad (q \in \mathbb{N}), \quad \text{b) } \frac{1}{q_1} \cdot \frac{1}{q_2} = \frac{1}{q_1 q_2} \quad (q_1, q_2 \in \mathbb{N}).$$

Solution a) Starting from the definition of a rational number and appealing to statement (c) of Theorem I.1.1 as well as to axioms (L4), (L9), and (L6), we conclude:

$$\begin{aligned} 0/q &= 0 \cdot q^{-1} = (0 \cdot q) \cdot q^{-1} \\ &= 0 \cdot (q \cdot q^{-1}) \\ &= 0 \cdot 1 \\ &= 0. \end{aligned}$$

b) We claim that $q_1^{-1} q_2^{-1} = (q_1 q_2)^{-1}$, i.e. that $y = q_1^{-1} q_2^{-1}$ is the reciprocal of the number $x = q_1 q_2$. Check:

$$\begin{aligned} x \cdot y &= (q_1 q_2) \cdot (q_1^{-1} q_2^{-1}) = (q_1 q_2) \cdot (q_2^{-1} q_1^{-1}) \\ &= q_1 (q_2 q_2^{-1}) q_1^{-1} \\ &= q_1 \cdot 1 \cdot q_1^{-1} \\ &= q_1 q_1^{-1} = 1. \end{aligned}$$

Here we appealed to axioms (L2), (L4), (L9), and (L6). $\square$

When, in raising a reciprocal to a power, one agrees on the notation

$$(x^{-1})^n = x^{-n}, \quad n \in \mathbb{N},$$

x^{-1} can also be read ' x to the power minus 1'. Taking axioms (L2), (L4), (L6), and (L9) into account, it also follows that

$$x \neq 0 \ \& \ m, n \in \mathbb{Z} \ \& \ m, n \neq 0 \quad \Rightarrow \quad x^m \cdot x^n = \begin{cases} x^{m+n}, & \text{when } m+n \neq 0, \\ 1, & \text{when } m+n = 0. \end{cases}$$

To simplify this rule, it is natural to define:

$$x^0 = 1, \quad \text{when } x \neq 0 \quad (x \in \mathbb{Q}).$$

Then one obtains the general computational rules

$$x^m x^n = x^{m+n}, \quad x^n y^n = (xy)^n, \quad (x^m)^n = x^{mn}, \quad x, y \neq 0, \ m, n \in \mathbb{Z}.$$

Also $0^n = 0 \ \forall n \in \mathbb{N}$. On the other hand, 0^0 and 0^{-n} , $n \in \mathbb{N}$, remain undefined, for the reason that 0 has no reciprocal.

The order relation of $\mathbb{Q}$

The rational numbers can also, in the familiar way, be *ordered*, i.e. one can define an *order relation*

$$< : \quad \text{'less than'}.$$

The resulting algebra can be denoted $(\mathbb{Q}, +, \cdot, <)$. The following general axioms (i.e. assumed basic laws) are associated with the order relation:

(J1) Of the alternatives $\{x < y, x = y, x > y\}$ exactly one holds.

(J2) $x < y \ \& \ y < z \Rightarrow x < z$.

(J3) $x < y \Rightarrow x + z < y + z \ \forall z$.

(J4) $x > 0 \ \& \ y > 0 \Rightarrow x \cdot y > 0$.

Here the notation $x > y$ (' x greater than y ') is an alternative form of the notation $y < x$. In practice, comparison is most quickly carried out using the computational laws (L7), (L8), and (L1), for from these and axiom (J3) it can be deduced that

$$\begin{cases} x < y & \Leftrightarrow & x - y < 0, \\ x = y & \Leftrightarrow & x - y = 0, \\ x > y & \Leftrightarrow & x - y > 0. \end{cases}$$

Here ' $\Leftrightarrow$ ' is the *equivalence arrow*, one of the symbols of logic, which in this context can be read

$$P \Leftrightarrow Q : \quad \text{'P exactly when Q'}.$$

By the above criterion, the comparison question $x ? y$ can thus be settled by examining whether the number $x - y$ is *positive* (> 0), *negative* (< 0), or $= 0$. For a rational number in standard form $x = p/q$, ($p \in \mathbb{Z}$, $q \in \mathbb{N}$), in turn, one agrees that the alternatives $x > 0$ and $x = 0$ correspond to the alternatives $p \in \mathbb{N}$ and $p = 0$; otherwise $x < 0$. This agreement is possible because, by rule (a) of Example 7, the numbers $0/q$ are identified with the number 0:

$$0 = \frac{0}{1} = \frac{0}{2} = \dots$$

On the other hand, there are no other forms in which 0 appears in $\mathbb{Q}$, for if $x = p/q = 0$, then it follows that $(p/q) \cdot q = p = 0 \cdot q = 0$ (axioms (L4), (L2), (L9), (L6) and statement (c) of Theorem I.1.1). Axiom (J1) therefore holds. The validity of axioms (J2)–(J4) is also easily deduced (Exercise 8).

Order and identification are examples of (set-theoretic) *relations*, which concern the 'relationship' between two elements of a set. Relation symbols associated with order/identification, besides those mentioned, are also ' $\neq$ ', ' $\leq$ ', and ' $\geq$ ', which are defined by

$$\begin{aligned} x \neq y &\Leftrightarrow \text{it does not hold that } x = y, \\ x \leq y &\Leftrightarrow x < y \text{ or } x = y, \\ x \geq y &\Leftrightarrow x > y \text{ or } x = y. \end{aligned}$$

The notation $P \Leftrightarrow Q$ can here be read ' P means the same as Q '.

Sum and product notation

When adding or multiplying together several different numbers in succession, it is convenient to number, i.e. *index*, the numbers, so that abbreviated notations can be used. These are

$$\text{Sum notation:} \quad x_1 + x_2 + \dots + x_n = \sum_{i=1}^n x_i$$

$$\text{Product notation:} \quad x_1 \cdot x_2 \cdot \dots \cdot x_n = \prod_{i=1}^n x_i$$

In the sum notation, the symbol i is called the *summation index*. This usually (not always) takes integer values, in which case the symbol is most commonly chosen from the set $\{i, j, k, l, m, n\}$. Since it is only a matter of abbreviated notation, the choice of symbol naturally does not affect the computation itself:

$$\sum_{i=1}^n a_i = \sum_{j=1}^n a_j = \sum_{k=1}^n a_k = \dots$$

In successive products, the numbers being multiplied are called the *factors* of the product. Associated with successive products of integers is the term *n-factorial*, which is denoted $n!$ and defined by

$$0! = 1, \quad 1! = 1, \quad n! = 1 \cdot 2 \cdot \dots \cdot n = \prod_{k=1}^n k, \quad n = 2, 3, \dots$$

EXERCISES

1. Show that $A = B$ for the following sets, i.e. that A and B are equal as sets:

$$\text{a) } A = \{3n + 2 \mid n \in \mathbb{Z}\}, \quad B = \{3n - 7 \mid n \in \mathbb{Z}\}$$

$$\text{b) } A = \{7n + 3 \mid n \in \mathbb{Z}\}, \quad B = \{7n - 32 \mid n \in \mathbb{Z}\}$$

2. Appealing to axioms (L1)–(L8) and the statements of Theorem I.1.1, justify the computational rules for integers a) $-x + (-y) = -(x + y)$, b) $(-x) \cdot y = -x \cdot y$, c) $(-x) \cdot (-y) = x \cdot y$.
3. Construct the multiplication tables of the number systems corresponding to the bases $k = 2$, $k = 3$, and $k = 4$.
4. In the dozenal system, in addition to the symbols of the decimal system, the following numeral symbols are used (on the left, the decimal notation)

$$10 = \diamond \quad \text{'diamond'}$$

$$11 = \heartsuit \quad \text{'heart'}$$

$$12 = 10 \quad \text{'dozen' (base)}$$

Convert the decimal number 155 to the dozenal system and the dozenal number $9\diamond 07\heartsuit$ to the decimal system.

5. Without converting to the decimal system, carry out the following operations in the dozenal system (see the previous exercise):

$$\text{a) } \heartsuit\heartsuit\diamond + \diamond\diamond\heartsuit \quad \text{b) } 247 - 19\diamond \quad \text{c) } 23 \cdot 34$$

6. In the raisin system, the decimal number 7 is pronounced 'raisin', the numbers $1 \dots 6$ are written and pronounced as in the decimal system, and the raisin-system numbers 100 and 1000 are pronounced 'bun' and 'cake'. How is the decimal number 2331 written and pronounced in the raisin system?
7. In the following computational rules for rational numbers, the result is expressed in standard form for a rational number, assuming that $p, p_1, p_2 \in \mathbb{Z}$ and $q, q_1, q_2 \in \mathbb{N}$. Justify the rules by appealing to axioms (L1)–(L9),

Theorem I.1.1, and the rules of Example 7.

- a) Reduction (expansion): $(p \cdot m)/(q \cdot m) = p/q$, $m \in \mathbb{Z}$, $m \neq 0$.
- b) Negative: $-(p/q) = (-p)/q$.
- c) Reciprocal: $(p/q)^{-1} = \begin{cases} q/p, & \text{if } p \in \mathbb{N}, \\ (-q)/(-p), & \text{if } p \notin \mathbb{N}, p \neq 0. \end{cases}$
- d) Addition: $p_1/q_1 + p_2/q_2 = (p_1q_2 + p_2q_1)/(q_1q_2)$.
- e) Multiplication: $(p_1/q_1)(p_2/q_2) = (p_1p_2)/(q_1q_2)$.

8. Appealing to the computational rules of Exercise 7 and the definition of the order relation of the rational numbers, show that axioms (J2), (J3), and (J4) hold for the order relation.
 9. Solve the following equations/inequalities in $\mathbb{Q}$, reducing to the simplest possible form. Justify the steps of the computation using axioms (L1)–(L9), (J1)–(J4)!
- a) $2x + 7 = 9x - 4$
 - b) $\frac{x}{5} + \frac{2}{3} < \frac{x}{3} + \frac{1}{2}$
 - c) $(x + 1)/(x - 2) = 4$
 - d) $(2x - 1)/(3x + 2) > 3$
10. a) Show that the following inference rules are valid both for integers ($x_k \in \mathbb{Z}$) and for rational numbers ($x_k \in \mathbb{Q}$). You may appeal both to the axioms and to the computational rules derived from them (Theorem I.1.1, Exercise 7).

$$\prod_{k=1}^n x_k = 0 \quad \Leftrightarrow \quad x_k = 0 \text{ for some } k \in \{1, \dots, n\}.$$

$$\prod_{k=1}^n x_k < 0 \quad \Leftrightarrow \quad \text{an odd number of the } x_k \text{ are negative, the rest } > 0.$$

Applying the conclusions of part a), solve in $\mathbb{Q}$:

- b) $2x^2 - 3x - 2 \geq 0$
- c) $63x^2 - 32x \leq 63$
- d) $9x^3 \geq 169x$
- e) $\frac{x+1}{x-1} \geq \frac{x}{x+1}$
- f) $\frac{5}{6x-7} \leq x$
- g) $6x + \frac{2x+12}{x^2+x} \geq 13$

11. Simplify (assuming $n \in \mathbb{N}$)

$$\text{a) } \sum_{k=1}^n n \quad \text{b) } \sum_{i=1}^{n^2} \frac{2}{n} - \sum_{j=1}^n 1 - \sum_{k=1}^n 1 \quad \text{c) } \prod_{k=1}^n \left(1 + \frac{1}{k}\right)$$

12. The double sum $\sum_{i=1}^n \sum_{j=1}^m a_{ij}$ ($n, m \in \mathbb{N}$, $a_{ij} \in \mathbb{Q}$) is defined by

$$\sum_{i=1}^n \sum_{j=1}^m a_{ij} = \sum_{i=1}^n \left(\sum_{j=1}^m a_{ij} \right) = \sum_{j=1}^m \left(\sum_{i=1}^n a_{ij} \right) = \sum_{j=1}^m \sum_{i=1}^n a_{ij}.$$

a) Show that here the removal of parentheses and the change of order are indeed possible, i.e. that the two middle expressions are equal.

- b) Simplify: $\sum_{i=1}^n \sum_{j=1}^n (a_i + a_j)$. c) Show: $\left(\sum_{i=1}^n a_i \right)^2 = \sum_{i=1}^n \sum_{j=1}^n a_i a_j$.

13. (*) The sum $\sum_{k=0}^n a_k$ is called a *telescoping sum* if $a_k = b_{k+1} - b_k \forall k$, where the numbers b_k , $k = 0 \dots n+1$, are known.

- a) Determine the value of the telescoping sum in terms of the numbers b_k .
 b) Where does the name 'telescoping sum' come from?
 c) Using the telescoping idea, compute the sums

$$\sum_{k=0}^n (2k+1) \quad \text{and} \quad \sum_{k=1}^n \frac{1}{k(k+1)}, \quad n \in \mathbb{N}.$$

- d) Using telescoping sums, show: $\sum_{k=1}^n k^2 = \frac{1}{6}n(2n^2 + 3n + 1)$, $n \in \mathbb{N}$.

I.2 Fields

The rational numbers, together with their operations, are an example (so far the only one) of an algebraic structure called a *field* (Swedish *kropp*, German *Körper*). If $\mathbb{K}$ is some set of numbers (it could also be a more general set), then $(\mathbb{K}, +, \cdot)$ is said to be a field if, first,

- The operations $x, y \mapsto x+y$ and $x, y \mapsto x \cdot y$ are uniquely defined $\forall x, y \in \mathbb{K}$, and $x + y \in \mathbb{K}$ and $x \cdot y \in \mathbb{K}$ hold,

and, in addition, the following **field axioms** hold:

- Commutative, associative, and distributive laws

$$(K1) \quad x + y = y + x$$

$$(K2) \quad x \cdot y = y \cdot x$$

$$(K3) \quad x + (y + z) = (x + y) + z$$

$$(K4) \quad x \cdot (y \cdot z) = (x \cdot y) \cdot z$$

$$(K5) \quad x \cdot (y + z) = x \cdot y + x \cdot z$$

- Existence of a zero element and a negative element

$$(K6) \quad \text{There exists a } \textit{zero element} \ 0 \in \mathbb{K}, \text{ for which } x + 0 = x \ \forall x \in \mathbb{K}.$$

$$(K7) \quad \text{Every } x \in \mathbb{K} \text{ has a } \textit{negative element} \ -x \in \mathbb{K}, \text{ for which } x + (-x) = 0.$$

- Existence of a unit element and an inverse element

$$(K8) \quad \text{There exists a } \textit{unit element} \ 1 \in \mathbb{K}, \text{ for which } x \cdot 1 = x \ \forall x \in \mathbb{K}.$$

$$(K9) \quad \text{Every } x \in \mathbb{K}, \ x \neq 0 \text{ has an } \textit{inverse element} \ x^{-1} \in \mathbb{K}, \text{ for which } x \cdot x^{-1} = 1.$$

- Distinctness axiom

$$(K10) \quad 0 \neq 1$$

The initial assumption, separate from the axioms, is usually considered to be part of the definition of the operations of $\mathbb{K}$. As a separate matter this assumption needs to be taken into account mainly when the operations were originally defined on some larger set $\mathbb{A} \supset \mathbb{K}$, or when one wishes to extend the set $\mathbb{K}$ under consideration. In that case one must verify that the result of the operations stays

in the set $\mathbb{K}$ under consideration, see the examples below. In addition to the basic operations $(+, \cdot)$, one can always define in a field the composite operations $x, y \mapsto x - y = x + (-y)$ (subtraction) and $x, y \mapsto x/y = x \cdot y^{-1}$ (division, $y \neq 0$), where $-x = 0 - x$ and $x^{-1} = 1/x$. Since axiom (K10) means '0 = 1 does not hold', in a field one generally needs only the identification relation (the principle of distinguishing the elements of $\mathbb{K}$), not an order relation. *

EXAMPLE 1 The simplest possible field is obtained by choosing $\mathbb{K} = \{0, 1\}$ (where $0 \neq 1$) and agreeing on the computational rule $1 + 1 = 0$ (!). Then $-1 = 1$ and the other computational rules can be deduced from axioms (K1), (K2), (K5), (K6), and (K8):

$$\begin{aligned} 0 + 0 &= 0, & 0 + 1 &= 1 + 0 = 1, & 1 \cdot 1 &= 1, & 0 \cdot 1 &= 1 \cdot 0 = 0, \\ 0 \cdot 0 &= 0 \cdot (1 + 1) = 0 \cdot 1 + 0 \cdot 1 = 0. \end{aligned}$$

With these rules axioms (K1)-(K9) all hold (as does the assumption (K0)), so this is a field. $\square$

THEOREM I.2.1 In every field $(\mathbb{K}, +, \cdot)$ the following hold:

- (a) The zero element, negative element, unit element, and inverse element are unique.
- (b) $-(-x) = x \quad \forall x \in \mathbb{K}, \quad (x^{-1})^{-1} = x \quad \forall x \in \mathbb{K}, x \neq 0.$
- (c) $0 \cdot x = 0 \quad \forall x \in \mathbb{K}.$
- (d) $-x = (-1) \cdot x \quad \forall x \in \mathbb{K}.$

Proof (a) Suppose that $0 \in \mathbb{K}$ and $\theta \in \mathbb{K}$ are two zero elements. Then by axiom (K6),

$$x = x + 0 \quad \forall x \in \mathbb{K}, \quad y + \theta = y \quad \forall y \in \mathbb{K}.$$

Choosing $x = \theta$ and $y = 0$ and using axiom (K1), it follows that

$$\theta = \theta + 0 = 0 + \theta = 0,$$

*Of the identification relation '=' one always requires the axioms (S1) $x = x \quad \forall x$, (S2) $x = y \Rightarrow y = x$, and (S3) $x = y \ \& \ y = z \Rightarrow x = z$. Of a field's identification relation one further requires compatibility with the operations, so that the result of an operation is always unique. By the commutative laws (K1)-(K2) and axiom (S3), this requirement is met by assuming:

(SK1) $x = y \Rightarrow x + z = y + z \quad \forall z$ and (SK2) $x = y \Rightarrow x \cdot z = y \cdot z \quad \forall z$.

i.e. $\theta = 0$. Thus the zero element is unique. To establish the uniqueness of the negative element, suppose that $a \in \mathbb{K}$ and $b \in \mathbb{K}$ are negative elements of the same element $x \in \mathbb{K}$. Then by axioms (K6), (K3), and (K1),

$$a = a + 0 = a + (x + b) = (a + x) + b = (x + a) + b = 0 + b = b + 0 = b,$$

i.e. $a = b$. Thus the negative element too is unique. The other asserted uniqueness statements follow by similar reasoning.

(b) The first part of the statement was proved in the previous section (Theorem I.1.1(b)). To prove the second part, we apply axioms (K2) and (K9):

$$x^{-1} \cdot x = x \cdot x^{-1} = 1 \Rightarrow (x^{-1})^{-1} = x.$$

(c) Writing $0 \cdot x = y \in \mathbb{K}$, then by axioms (K6), (K2), and (K5)

$$y = 0 \cdot x = (0 + 0) \cdot x = 0 \cdot x + 0 \cdot x = y + y. \quad (\star)$$

Using this result and the field axioms, we conclude:

$$\begin{aligned} 0 &= y + (-y) && \text{(K7)} \\ &= (y + y) + (-y) && (\star) \\ &= y + [y + (-y)] && \text{(K3)} \\ &= y + 0 && \text{(K7)} \\ &= y. && \text{(K6)} \end{aligned}$$

(d) By result (c) and the field axioms,

$$\begin{aligned} x + (-1) \cdot x &= x \cdot 1 + x \cdot (-1) && \text{(K8), (K2)} \\ &= x \cdot [1 + (-1)] && \text{(K5)} \\ &= x \cdot 0 && \text{(K7)} \\ &= 0. \quad \square && \text{(K2), (c)} \end{aligned}$$

In a general field, raising to a power can be defined in the same way as in the field of rational numbers, cf. the previous section.

EXAMPLE 2 Derive the binomial formulas for the expressions $(x + y)^2$ and $(x + y)^3$ in a field $(\mathbb{K}, +, \cdot)$ in which $1 + 1 = \spadesuit$, $\spadesuit + 1 = \clubsuit$.

Solution By the definition of raising to a power and the field axioms,

$$\begin{aligned} (x + y)^2 &= (x + y) \cdot (x + y) = x \cdot x + x \cdot y + y \cdot x + y \cdot y \\ &= x^2 + (x \cdot y + x \cdot y) + y^2. \end{aligned}$$

Here, further, by the field axioms and the assumption,

$$x \cdot y + x \cdot y = (x \cdot y) \cdot (1 + 1) = (x \cdot y) \cdot \spadesuit = \spadesuit \cdot x \cdot y,$$

so the first requested binomial formula is

$$(x + y)^2 = x^2 + \spadesuit \cdot x \cdot y + y^2.$$

By similar reasoning the second formula is obtained as

$$(x + y)^3 = x^3 + \clubsuit \cdot x^2 \cdot y + \clubsuit \cdot x \cdot y^2 + y^3. \quad \square$$

The following result of field algebra turns out to be useful later (Exercise 3a).

PROPOSITION I.2.2 In every field,

$$\begin{aligned} x^n - y^n &= (x - y) (x^{n-1} + x^{n-2}y + \dots + y^{n-1}) \\ &= (x - y) \sum_{k=0}^{n-1} x^{n-1-k} y^k, \quad n \in \mathbb{N}. \end{aligned}$$

Ordered fields

If an order relation has been defined on a field in accordance with axioms (J1)-(J4) of the previous section, then it is called an *ordered field*. In an ordered field every element is, by axiom (J1), either positive ($x > 0$), negative ($x < 0$), or $= 0$. So far the only example of an ordered field is the field of rational numbers.

THEOREM I.2.1 (continued) In every ordered field $(\mathbb{K}, +, \cdot, <)$ the following hold:

- (e) $x > 0 \Rightarrow -x < 0, \quad x > 0 \Rightarrow x^{-1} > 0.$
- (f) $x > 0 \ \& \ y > 0 \Rightarrow x + y > 0.$
- (g) $0 < 1.$

Proof (e) Suppose that $0 < x$. Then by axiom (J3) also

$$0 + (-x) < x + (-x),$$

which by the field axioms reduces to the first statement $-x < 0$. To prove the second part of the statement, we rule out the other alternatives left by axiom (J1). If $x^{-1} = 0$, then by axioms (K8), (K2) and result (c), $1 = x \cdot x^{-1} = x \cdot 0 = 0$,

i.e. $0 = 1$, which contradicts axiom (K10). Thus the possibility $x^{-1} = 0$ is ruled out. If $x^{-1} < 0$, then by result (d) and the already proved first part of statement (e), $(-1) \cdot x^{-1} > 0$. Then by axiom (J4) and the assumption $x > 0$, also $(-1) \cdot x^{-1} \cdot x > 0$, which by the field axioms and the mentioned results simplifies to the form $1 < 0$. This too is impossible by axiom (J1) and the still-unproved statement (g), so insofar as (g) is true, the only remaining alternative is $x^{-1} > 0$.

(f) We first conclude, by axioms (J3), (K1), and (K6):

$$0 < y \Rightarrow x < x + y.$$

Continuing from this by applying axiom (J3):

$$0 < x \ \& \ x < x + y \Rightarrow 0 < x + y.$$

(g) By axioms (J1) and (K10), either $0 < 1$ or $1 < 0$ must hold. If we assume the latter alternative, then by axiom (J3) in that case

$$1 + (-1) < 0 + (-1),$$

which by axioms (K7), (K1), and (K6) simplifies to the form

$$0 < -1.$$

Then by axiom (J4) it must also hold that

$$0 < (-1) \cdot (-1).$$

But by statements (b), (d), $1 = -(-1) = (-1) \cdot (-1)$, so we have deduced that in the assumed alternative $1 < 0$, $0 < 1$ also holds. By axiom (J1), however, this is impossible, so the only remaining alternative is $0 < 1$. The reasoning did not require the still-open statement (e), so the proof of that statement too was thereby completed. $\square$

In an ordered field, to each element of the field one can associate an 'absolute element', i.e. the *absolute value* of that element, by the definition

$$|x| = \max\{x, -x\} = \begin{cases} x, & \text{if } x \geq 0, \\ -x, & \text{if } x < 0. \end{cases}$$

By means of the absolute value one can further define the *distance* between two numbers

$$d(x, y) = |x - y|,$$

whereby such expressions as ' y is closer to x than z ' acquire an (algebraic) meaning. The following result concerning the absolute value is needed very often in mathematical analysis.

THEOREM I.2.3 (**Triangle inequality**) In an ordered field $(\mathbb{K}, +, \cdot, <)$,

$$||x| - |y|| \leq |x + y| \leq |x| + |y|, \quad x, y \in \mathbb{K}.$$

Proof The main result is the latter of the inequalities, for the former follows from the field axioms, the latter inequality, and the definition of the absolute value, by the reasoning

$$\begin{aligned} \begin{cases} |x| &= |(x + y) + (-y)| \leq |x + y| + |-y| = |x + y| + |y| \\ |y| &= |(x + y) + (-x)| \leq |x + y| + |-x| = |x + y| + |x| \end{cases} \\ \Rightarrow \quad \pm(|x| - |y|) \leq |x + y| \quad \Rightarrow \quad ||x| - |y|| \leq |x + y|. \end{aligned}$$

In the latter part of the triangle inequality, by the definition of the absolute value, it is claimed that

$$\pm(x + y) \leq |x| + |y|.$$

Since, by the same definition,

$$\pm x \leq |x| = a, \quad \pm y \leq |y| = b,$$

one sees that the claimed inequality follows from the statement

LEMMA I.2.4 In an ordered field,

$$x \leq a \quad \& \quad y \leq b \quad \Rightarrow \quad x + y \leq a + b.$$

Proof The statement has four alternatives: a) $x < a$, $y < b$, b) $x < a$, $y = b$, c) $x = a$, $y < b$, d) $x = a$, $y = b$. In alternative a), $x < a \Leftrightarrow a - x > 0$ and $y < b \Leftrightarrow b - y > 0$, so that from statement (f) of Theorem I.2.1 and the field axioms it follows that $(a - x) + (b - y) > 0 \Leftrightarrow x + y < a + b$. In alternatives b) and c) too the claim holds in this form, and in alternative d) in the form '=', as is easily seen. $\square$

Subfields and field extensions

A field $(\mathbb{A}, +, \cdot)$ is called a *subfield* of a field $(\mathbb{K}, +, \cdot)$ if $\mathbb{A} \subset \mathbb{K}$ and the fields have the same operations, i.e. the operations of the field $(\mathbb{A}, +, \cdot) =$ the operations of the field $(\mathbb{K}, +, \cdot)$ restricted to the subset $\mathbb{A}$. Then the fields also have common zero and unit elements, i.e. for the zero and unit elements of the field $(\mathbb{K}, +, \cdot)$,

$0, 1 \in \mathbb{A}$ holds. For if the zero element of $\mathbb{K}$ is $0 \in \mathbb{K}$ and the zero element of $\mathbb{A}$ is $\theta \in \mathbb{A}$, then for every $x \in \mathbb{A}$ (e.g. $x = \theta$) one can conclude

$$x = x + \theta \Rightarrow -x + x = -x + x + \theta \Rightarrow 0 = 0 + \theta = \theta + 0 = \theta.$$

Here $-x \in \mathbb{K}$ is the negative element of x in $\mathbb{K}$, so the reasoning first relied on axiom (K6) of the field $(\mathbb{A}, +, \cdot)$ and then on axioms (K1), (K3), (K6), and (K7) of the field $(\mathbb{K}, +, \cdot)$. In a similar way one sees that the unit element is the same in both fields.

If a field $(\mathbb{A}, +, \cdot)$ is a subfield of a field $(\mathbb{K}, +, \cdot)$, then, correspondingly, the latter is called a *field extension* of the former. In later chapters several extensions of the number domain of the type $\mathbb{Q} \hookrightarrow \mathbb{K}$ are made. Common to these is that the extensions produce new fields, all of which have $(\mathbb{Q}, +, \cdot)$ as a common subfield. Below is a very cautious example of such an extension.

EXAMPLE 3 Suppose there exists a number a with the property

$$a^2 = a \cdot a = 2.$$

(We take it as known that $a \notin \mathbb{Q}$.) Concerning the operations between the number a and the rational numbers, we agree that $1 \cdot a = a$ and $0 \cdot a = 0 \in \mathbb{Q}$, and in addition we agree that the commutative, associative, and distributive laws (K1)–(K5) hold also when the number a is involved in the operations. On the basis of these agreements one can form the set of numbers

$$\mathbb{K} = \{ z = x + ya \mid x, y \in \mathbb{Q} \}.$$

This is a proper extension of $\mathbb{Q}$, for $x \in \mathbb{Q} \Rightarrow x = x + 0a \in \mathbb{K}$, but $0 + 1a = a \notin \mathbb{Q}$. By the agreements made, in particular $0 + 0a = 0$. The number 0 has no other representations in $\mathbb{K}$, for if $x + ya = 0$, then one must have $y = 0 \Rightarrow x = 0$, since otherwise $a = -x/y \in \mathbb{Q}$. (More generally, one can conclude that the representation $z = x + ya$ of every number $z \in \mathbb{K}$ is unique.)

If $z_1 = x_1 + y_1a \in \mathbb{K}$ and $z_2 = x_2 + y_2a \in \mathbb{K}$ ($x_1, x_2, y_1, y_2 \in \mathbb{Q}$), then by the assumed axioms (K1)–(K5) and the rule $a^2 = 2$,

$$\begin{aligned} z_1 + z_2 &= (x_1 + x_2) + (y_1 + y_2)a, \\ z_1 z_2 &= (x_1 x_2 + y_1 y_2 a^2) + (x_1 y_2 + x_2 y_1)a = (x_1 x_2 + 2y_1 y_2) + (x_1 y_2 + x_2 y_1)a, \end{aligned}$$

so $z_1 + z_2 \in \mathbb{K}$ and $z_1 z_2 \in \mathbb{K}$. The basic assumption concerning the operations (the first assumption above) thus holds. Further, one sees that of the field axioms, in addition to the assumed ones, (K6), (K8), and (K10) ($0, 1 \in \mathbb{Q}$) and (K7) also

hold (if $z = x + ya$, then $-z = -x - ya$). Finally, one concludes that (K9) also holds. Indeed, $zz^{-1} = 1$ when one defines

$$z^{-1} = (x^2 - 2y^2)^{-1} (x - ya) = \frac{x}{x^2 - 2y^2} - \frac{y}{x^2 - 2y^2} a.$$

Here $x^2 - 2y^2 = 0$ only when $x = y = 0$ (since $(x/y)^2 \neq 2$ when $x, y \in \mathbb{Q}$ and $y \neq 0$), so every $z \in \mathbb{K}$, $z \neq 0$ has a reciprocal z^{-1} . We have concluded that $(\mathbb{K}, +, \cdot)$ is a field and thus a proper extension of the field $(\mathbb{Q}, +, \cdot)$ of rational numbers.

The field $(\mathbb{K}, +, \cdot)$ can also be ordered. First we agree that $a > 0$ (possible, since $a^2 = (-a)^2$). More generally, one sees that the comparison $z_1 < z_2$ ($z_1, z_2 \in \mathbb{K}$) always reduces, by means of field algebra, ultimately to a comparison between the number a and a rational number. (E.g. $7 - 2a > 1 - a \Leftrightarrow 6 - 3a > 0 \Leftrightarrow 3/2 > a$.) If $x \in \mathbb{Q}$, then evidently $x < a$ whenever $x \leq 0$. If, on the other hand, $x > 0$, then also $x + a > 0 \Leftrightarrow (x + a)^{-1} > 0$, so that the comparison question $x ? a$ is settled by using axiom (J4) of the order relation:

$$x > a \quad \Leftrightarrow \quad x - a = \frac{x^2 - a^2}{x + a} > 0 \quad \Leftrightarrow \quad x^2 - a^2 > 0 \quad \Leftrightarrow \quad x^2 > 2.$$

In the ordered field $(\mathbb{K}, +, \cdot, <)$ the number a thus falls 'between' the rational numbers: For every number $x \in \mathbb{Q}$ one can decide whether $a < x$ or $a > x$, and this comparison is based only on the comparison of the rational numbers $0, x, 2$, and x^2 . $\square$

Radicals and fractional powers

A considerably more useful field extension than the field of Example 3 is obtained when the set of rational numbers is extended to a set of numbers in which, in addition to the field operations $x, y \mapsto x + y, x - y, xy, x/y$, the operations

$$x \mapsto \sqrt[m]{x}, \quad x > 0, \quad m \in \mathbb{N}, \quad m \geq 2$$

are allowed. Here the number $\sqrt[m]{x}$ denotes the *radical* y satisfying the conditions

$$y^m = x \quad \& \quad y > 0.$$

With the convention $\sqrt[1]{x} = x$, the radical $\sqrt[m]{x}$ becomes defined $\forall m \in \mathbb{N}$. The number $\sqrt[m]{x}$ is called the *square root* of x if $m = 2$ (abbreviated notation $\sqrt{x}$), the *cube root* if $m = 3$, or more generally the *mth root*, read 'mth root of x '. Such a number can simply be agreed to 'exist' in the same way as was done in Example

3 for the number $a (= \sqrt{2})$. The additional condition $y > 0$ (which already refers to the order relation) is needed to guarantee the uniqueness of the number y , for when m is even, $(-y)^m = y^m$. Uniqueness under the said additional condition follows from the identity (see Proposition I.2.2)

$$(y_1 - y_2)(y_1^{m-1} + y_1^{m-2}y_2 + \dots + y_2^{m-1}) = y_1^m - y_2^m.$$

If here one assumes that $y_1^m = y_2^m = x$ and $y_1, y_2 > 0$, then the right-hand side $= 0$ and on the left-hand side the latter factor is > 0 , so one must have $y_1 - y_2 = 0$.

The set of numbers extended by the operations $x \mapsto \sqrt[m]{x}$, starting from the rational numbers, will be denoted by the symbol $\mathbb{J}$. The set $\mathbb{J}$ is defined simply as consisting of the numbers obtained by a finite number of field operations and operations $x \mapsto \sqrt[m]{x}$ starting from the numbers $0, 1 \in \mathbb{Z}$. It is then evident that if $x, y \in \mathbb{J}$, then also $x \pm y, xy, x/y \in \mathbb{J}$ ($y \neq 0$), so the basic assumption concerning the operations holds. The 'appearance' of the numbers $x \in \mathbb{J}$ can, admittedly, be intricate.

EXAMPLE 4

$$\left(3 + \sqrt{2} - \sqrt[8]{\frac{\sqrt[6]{5 - \sqrt[4]{3}}}{\sqrt[4]{3 + \sqrt[3]{2}}} + \frac{1}{\sqrt[7]{7}}} \right) / \left(2 + \sqrt{3} + \sqrt[8]{\frac{\sqrt[6]{3 - \sqrt[4]{5}}}{\sqrt[4]{2 + \sqrt[3]{3}}} + \sqrt[7]{7}} \right) \in \mathbb{J}. \quad \square$$

When computing with radicals, the computational rules of field algebra are assumed to hold, i.e. it is assumed that $(\mathbb{J}, +, \cdot)$ is a field. From this assumption and the definition of the radical, the following general computational rules can be derived (assuming $x, y > 0$, $m, n \in \mathbb{N}$):

$$\text{a) } \sqrt[m]{x} = \sqrt[mn]{x^n}. \quad \text{b) } \sqrt[m]{x} \sqrt[n]{y} = \sqrt[mn]{xy}. \quad \text{c) } (\sqrt[m]{x})^{-1} = \sqrt[m]{x^{-1}}.$$

To justify the rules, write $a = \sqrt[m]{x}$, $b = \sqrt[n]{y}$, so that $a, b > 0 \Rightarrow ab, a^{-1} > 0$. Appealing to the definition of the radical and to field algebra, one can then conclude:

$$\begin{aligned} \text{a) } a^m = x &\Rightarrow (a^m)^n = a^{mn} = x^n \Rightarrow a = \sqrt[mn]{x^n}. \\ \text{b) } a^m = x \ \&\ \ b^n = y \Rightarrow a^m b^n = (ab)^{mn} = xy \Rightarrow ab = \sqrt[mn]{xy}. \\ \text{c) } a^m = x &\Rightarrow (a^m)^{-1} = (a^{-1})^m = x^{-1} \Rightarrow a^{-1} = \sqrt[m]{x^{-1}}. \end{aligned}$$

The rules for computing with radicals can be conveniently combined with the rules for powers (cf. the previous section) by defining the *fractional power* of a number $x > 0$ as

$$x^{p/q} = \sqrt[q]{x^p}, \quad p \in \mathbb{Z}, \quad q \in \mathbb{N}, \quad q \geq 2.$$

This uniquely defines the number x^r for every $r \in \mathbb{Q}$, for if r is expressed in standard form $r = p/q$, $q \in \mathbb{N}$, then by the definition and rule a), $x^{(pn)/(qn)} = \sqrt[q]{x^{pn}} = \sqrt[q]{(x^p)^n} = \sqrt[n]{x^p} = x^{p/q} \forall n \in \mathbb{N}$, i.e. x^r does not depend on the form in which r is expressed. From rules a)–c) one can now derive for fractional powers the same computational rules as for integer powers (Exercise 8):

$$x^r x^s = x^{r+s}, \quad x^r y^r = (xy)^r, \quad (x^r)^s = x^{rs}, \quad x, y > 0, \quad r, s \in \mathbb{Q}.$$

The numbers $x \in \mathbb{J}$, $x \notin \mathbb{Q}$ can also be placed into the 'gaps' between the rational numbers so that an ordered field $(\mathbb{J}, +, \cdot, <)$ arises. Defining the order relation in the general case would, however, require more serious number-theoretic considerations than hitherto, so the matter is not pursued for now. Let it only be noted that in favorable special cases the comparison can be reduced directly to a comparison of rational numbers in the same way as in Example 3 above. The idea is then to unwind the radical expressions by the reasoning: if $x, y > 0$ and $m \in \mathbb{N}$, then $x < y \Leftrightarrow x^m < y^m$ (Exercise 6).

EXAMPLE 5 Which is greater, $x = \sqrt{\sqrt{2} + 5}$ or $y = \sqrt[4]{41}$?

Solution Since $x < y \Leftrightarrow x^4 < y^4$ ($x, y > 0$), we conclude

$$x < y \Leftrightarrow (\sqrt{2} + 5)^2 < 41 \Leftrightarrow 10\sqrt{2} + 27 < 41 \Leftrightarrow 10\sqrt{2} < 14.$$

Continuing from this by the reasoning $a < b \Leftrightarrow a^2 < b^2$ ($a, b > 0$), one sees that

$$10\sqrt{2} < 14 \Leftrightarrow 200 < 196.$$

Since the last inequality does not hold, we reverse the directions of the inequalities and conclude: $x > y \Leftrightarrow 200 > 196$. Thus x is greater. $\square$

EXERCISES

1. The set of numbers $\mathbb{K}$ contains only the numbers 0, 1, and $\diamond = 1 + 1$ (all distinct from one another). Define (if possible) the computational rules of $\mathbb{K}$ (addition, multiplication, negatives, reciprocals) so that $(\mathbb{K}, +, \cdot)$ is a field.
2. Appealing to the field axioms or the statements of Theorem 1.2.1, show that in every field: a) $x \cdot y = 0 \Leftrightarrow x = 0 \vee y = 0$ (the *zero-product rule*), b) $(-x) \cdot (y) = -(x \cdot y)$, c) $(-x) \cdot (-y) = x \cdot y$, d) 0 has no inverse element.

3. Show that in every field
 - a) $x^n - y^n = (x - y) \sum_{k=0}^{n-1} x^{n-1-k} y^k$, $n \in \mathbb{N}$
 - b) $x^n + y^n = (x + y) \sum_{k=0}^{n-1} (-1)^k x^{n-1-k} y^k$, if $n \in \mathbb{N}$ and n is odd
 - c) $1 + x + \dots + x^n = (x^{n+1} - 1)(x - 1)^{-1}$, $n \in \mathbb{N}$, $x \neq 1$
4. Show that axiom (J3) of an ordered field can be replaced by the axiom (J3')
 $x < y \Leftrightarrow x - y < 0$.
5. Show that in every ordered field:
 - a) $x > 1 \Rightarrow 0 < x^{-1} < 1$
 - b) $x > 0 \ \& \ y > 1 \Rightarrow 0 < x/y < x$
 - c) $|x \cdot y| = |x| \cdot |y|$
 - d) $|x^{-1}| = |x|^{-1}$
 - e) $x^2 < y^2 \Leftrightarrow |x| < |y|$
 - f) $0 \leq x \leq a \ \& \ 0 \leq y \leq b \Rightarrow 0 \leq xy \leq ab$
6. Starting from formula a) of Exercise 3, show that in an ordered field: If $x > 0$, $y > 0$, and $n \in \mathbb{N}$, $n \geq 2$, then $x < y \Leftrightarrow x^n < y^n$.
7. Arrange the numbers a , $17/12$, and $72/51$ in order of magnitude in the field of Example 3. Carry out the comparisons exactly!
8. Justify the computational rules for fractional powers a) $x^r x^s = x^{r+s}$, b) $x^r y^r = (xy)^r$, c) $(x^r)^s = x^{rs}$ ($x, y > 0$, $r, s \in \mathbb{Q}$).
9. Without a calculator, determine the order of magnitude of the numbers $x, y \in \mathbb{J}$ by reducing the comparison to a comparison of rational numbers:
 - a) $x = \sqrt[3]{1020}$, $y = \sqrt{102}$, b) $x = \sqrt[3]{3}$, $y = \sqrt[4]{43/10}$,
 - c) $x = 2 - \sqrt{3}$, $y = 1/\sqrt{4\sqrt{3} + 7}$.
10. Let $(\mathbb{K}, +, \cdot, <)$ be an ordered field. Show: a) $x + 1 > x \ \forall x \in \mathbb{K}$.
 b) The set $\mathbb{K}$ has infinitely many distinct elements.
11. (*) Show by hand computation that
 - a) $\frac{577}{408} - \frac{1}{400000} < \sqrt{2} < \frac{577}{408}$, b) $\sqrt{2} < \frac{665857}{470832}$.
12. (*) Show that axiom (J2) of an ordered field can be replaced by the axiom (J2') $x > 0 \ \& \ y > 0 \Rightarrow x + y > 0$.

13. (*) We wish to define the smallest possible extension $(\mathbb{K}, +, \cdot)$ of the field of rational numbers such that $\sqrt{2} \in \mathbb{K}$ and $\sqrt{3} \in \mathbb{K}$. Show that a) $\mathbb{K}$ consists of the numbers of the form $x + y\sqrt{2} + z\sqrt{3} + u\sqrt{6}$, $x, y, z, u \in \mathbb{Q}$, b) in the field $(\mathbb{K}, +, \cdot)$ an order relation can be defined that is based only on the comparison of rational numbers.
14. (*) (Big Ben) Consider the set $\mathbb{K} = \{\text{clock hands}\}$. Each hand $v \in \mathbb{K}$ is a *number pair* (r, θ) , where r = the length of the hand (unit m) and θ = the direction of the hand in degrees, measured from 12 o'clock, positive clockwise or negative counterclockwise. We agree that $(r_1, \theta_1) = (r_2, \theta_2)$ if either (i) $r_1 = r_2 = 0$ or (ii) $r_1 = r_2 > 0$ and $\theta_1 - \theta_2 = k \cdot 360$, $k \in \mathbb{Z}$. If $r = 0$, the hand (r, θ) is called the *zero hand*, denoted 0_v . The multiplication of hands is defined as follows:

$$(r_1, \theta_1) \cdot (r_2, \theta_2) = (r_1 r_2, \theta_1 + \theta_2).$$

- a) Show that both the commutative law and the associative law hold for the multiplication.
- b) Define the unit hand and the inverse hand of a hand $(r, \theta) \neq 0_v$.
- c) Big Ben, whose minute hand has length 4 and hour hand has length 2, shows a time of about 3 o'clock. What time is it more precisely (to the nearest second!), given that Big Ben's inverse clock, Small Ben, also shows at the same moment a perfectly sensible (albeit different) time? — Note that Small Ben's minute hand is also the longer one!

I.3 On the notations of logic and set theory

In *logic* (and more generally in mathematics) one considers statements, i.e. *propositions*^{*}, which have a certain *truth value*, either 'true' (truth value = 1, or T = True) or 'false' (truth value = 0, or F = False).

EXAMPLE 1 Of the utterances

A = 'it rained yesterday'
 B = 'yesterday was dry'
 C = 'it will rain tomorrow'
 D = 'it really ought to rain tomorrow'
 E = 'three over four is less than two over three'

the first three can be regarded as propositions, insofar as the utterances are restricted to a particular day and place (and possible measurement problems are ignored). Likewise E is a proposition. The utterance D is not a proposition. $\square$

Of the example statements, A and B are each other's *negations*, i.e. *complements*. We write $B = \neg A$, $A = \neg B$, or less formally $B = \text{not}(A)$, $A = \text{not}(B)$. The truth value of statement C is not (today) known — but this is not a problem in logic.

By means of the *logical operators* $\wedge$ ('and'), $\vee$ ('or'), $\Rightarrow$ ('implies'), $\Leftrightarrow$ ('exactly when'), new propositions can be derived from propositions. In particular, if A and B are propositions, then $A \wedge B$, $A \vee B$, $A \Rightarrow B$, and $A \Leftrightarrow B$ are also propositions. The interpretation of the first two of these is evident: $A \wedge B$ is true when A and B are both true, otherwise false, and $A \vee B$ is true exactly when at least one of the statements A, B is true. The statement $A \wedge \neg A$ is *identically false* (false for every A), in natural language 'impossible'. Such a statement is the basic form of a *logical contradiction*. The statement $A \vee \neg A$, on the other hand, is *identically true* (true for every A), i.e. 'says nothing' about the statement A .

EXAMPLE 2 If A is false and B and C are both true, then $(A \wedge B) \vee C$ is true and $A \wedge (B \vee C)$ is false. The latter two statements are thus not in general equivalent (i.e. the parentheses cannot be removed). $\square$

^{*}Instead of 'proposition', Finnish-language texts often use the term *väitelause* ('assertion') or simply 'lause'. In this text the word 'lause' has a special meaning (= 'theorem'); the term 'proposition', on the other hand, is used in both a general and a special sense, see the footnote in Section I.1.

Implication

The logical interpretation of the implication statement $A \Rightarrow B$ is not entirely obvious. Possible readings are, first:

$A \Rightarrow B$: B follows from A
 A *implies* B
 if A , then B
 whenever A , then B
 always B , when A
 A only if B
 A is a *sufficient condition* for B
 B is a *necessary condition* for A

In applications, and also in natural language, an implication statement can often be interpreted so that A is the 'cause' and B the 'effect'. In logic, however, no 'causality' is required; the implication can even be pure coincidence ('by chance always B when A '). In the logical calculus, the truth value of the statement $A \Rightarrow B$ can be computed from the interpretations (see also Exercise 4a)

$$A \Rightarrow B \quad \Leftrightarrow \quad \neg(A \wedge \neg B) \quad (a)$$

$$\Leftrightarrow \quad \neg A \vee (A \wedge B). \quad (b)$$

Here $P \Leftrightarrow Q$ is read ' P and Q have the same truth value', see the interpretations of the equivalence arrow below. By interpretation (b) of implication, $A \Rightarrow B$ is true exactly when either A is false or A and B are both true. In particular, then, 'from the impossible anything follows', i.e. the statement $A \Rightarrow B$ is true for every B (i.e. 'says nothing' with respect to B) if A is false.

EXAMPLE 3 From interpretations (a)–(b) one sees that the statement $A \Rightarrow A$ is identically true, i.e. the statement 'says nothing' about A . $\square$

EXAMPLE 4 If 0 and 1 are the zero and unit elements of some field, then the proposition $[0 = 1 \Rightarrow \text{the author is a splendid mathematician}]$ is (logically) true. $\square$

Theorems, propositions, lemmas, and corollaries in mathematics are typically of the form 'If .. [assumptions], then .. [claim]', i.e. of the form $A \Rightarrow B$, where A is the assumption (assumptions) and B is the claim. Logical reasoning that shows a theorem or the like to be true (for the theorems of mathematics are true!) is called a *proof*. If a theorem is of the form $A \Rightarrow B$, then proving it (i.e.

showing the proposition $A \Rightarrow B$ to be true) proceeds by assuming that A is true and showing that then also B is true (this suffices, since $A \Rightarrow B$ is true if A is false!). Besides this so-called *direct proof*, another possible method of proof is the *indirect proof*. The idea of an indirect proof is to show that if the negation of the statement to be proved is true, then a logical contradiction follows (of the form ' C true and false', where C is a proposition), from which one concludes that the negation must be false and the statement therefore true (see Exercise 3). In the case of an implication statement, the structure of the indirect proof is as follows, cf. interpretation (a) above.

- (1) Assume that A is true and B false (i.e. $A \Rightarrow B$ false).
- (2) Show that assumption (1) leads to a logical contradiction. Conclude that the assumption was false and hence $A \Rightarrow B$ true.

Assumption (1) (or the sub-assumption ' B false') is called the *counter-assumption*, and the whole method of proof is also called *proof by contradiction*. — Note that the counter-assumption, despite its name, is not the negation of the assumption but rather a 'counter-claim'.

The indirect method of proof was in fact already resorted to above, in part (g) of Theorem I.2.1. Here the statement was of the form $A \Rightarrow B$, where

$$\begin{aligned} A &= \text{axioms (K1)–(K10) and (J1)–(J4),} \\ B &= 0 < 1. \end{aligned}$$

The counter-assumption that B is false led to the logical contradiction ' C true and false', where $C = (K10) \wedge (J1)$.

Equivalence

If A and B are propositions, then $A \Leftrightarrow B$ is a proposition stating that A and B are of equal value, i.e. *equivalent* statements. This means simply that the truth values of the statements A and B are the same, i.e. either both are true or both are false. Common readings of the equivalence are

$$\begin{aligned} A \Leftrightarrow B : \quad & A \text{ and } B \text{ equivalent, of equal value} \\ & A \text{ and } B \text{ agree} \\ & A \text{ if and only if } B \\ & A \text{ exactly when } B \\ & A \text{ iff } B \end{aligned}$$

By means of the equivalence arrow one can, e.g., express some statement in other words or notation, or just in another form. This may be, e.g., a matter of the *definition* of some notation, or of a universally valid restatement, i.e. a *tautology*.

EXAMPLE 5

$$\begin{aligned}
 0 < x < 1 &\Leftrightarrow 0 < x \wedge x < 1 \\
 A \Rightarrow B &\Leftrightarrow \neg B \Rightarrow \neg A \\
 A \Rightarrow B \Rightarrow C &\Leftrightarrow A \Rightarrow B \wedge B \Rightarrow C \\
 A \Leftrightarrow B &\Leftrightarrow A \Rightarrow B \wedge B \Rightarrow A \quad \square
 \end{aligned}$$

The first example defines the notation $0 < x < 1$. The second example concerns a general (and useful!) tautology (Exercise 4b). The third example defines the *chain of reasoning* formed by two implication statements.*. The last example interprets the statement $A \Leftrightarrow B$ itself. According to this interpretation, an equivalence statement is proved by showing separately $\boxed{\Rightarrow}$ ($A \Rightarrow B$) and $\boxed{\Leftarrow}$ ($A \Leftarrow B$), i.e. by showing the implications true in both directions. If it is a simple tautology, one can also form a *truth(-value) table*, in which all possibilities are gone through.

EXAMPLE 6 Show to be identically true the **de Morgan laws**:

$$\text{a) } \neg(A \vee B) \Leftrightarrow \neg A \wedge \neg B \quad \text{b) } \neg(A \wedge B) \Leftrightarrow \neg A \vee \neg B$$

Solution In the truth table (below), we have written $P = A \vee B$, $Q = A \wedge B$ and gone through all truth-value combinations of the propositions A, B (4 of them). From the table one sees that the truth values of the propositions $\neg P$ and $\neg A \wedge \neg B$, as well as of the propositions $\neg Q$ and $\neg A \vee \neg B$, are the same in all cases. This proves the claims. $\square$

A	B	$\neg A$	$\neg B$	P	$\neg P$	$\neg A \wedge \neg B$	Q	$\neg Q$	$\neg A \vee \neg B$
1	1	0	0	1	0	0	1	0	0
1	0	0	1	1	0	0	0	1	1
0	1	1	0	1	0	0	0	1	1
0	0	1	1	0	1	1	0	1	1

In connection with logical operators, as well as order and identification relations, etc., the sign of negation is generally an overbar.

*If a mathematical theorem etc. is an implication statement of the form $A \Rightarrow B$, then the proof is typically a chain of reasoning of the form $A \Rightarrow C_1 \Rightarrow C_2 \Rightarrow \dots \Rightarrow C_n \Rightarrow B$, where the sub-statements $A \Rightarrow C_1$, $C_1 \Rightarrow C_2$, $\dots$ $C_n \Rightarrow B$ are either evidently true, or true on the basis of other known theorems (or auxiliary results to be proved separately). The definition of the chain of reasoning here follows mathematical practice — in formal logic there is no such convention.

EXAMPLE 7

$$\begin{aligned}
A \not\Rightarrow B &\Leftrightarrow \neg(A \Rightarrow B) \Leftrightarrow A \wedge \neg B \\
A \not\Rightarrow B &\not\Rightarrow A \Rightarrow \neg B \\
x \neq y &\Leftrightarrow \neg(x = y) \quad \square
\end{aligned}$$

Predicates and quantifiers

In logic, a *predicate* is an utterance containing one or more free *variables*. A predicate becomes a proposition when the variables are *bound* — as such a predicate is not a proposition. For example, if we consider the rational numbers and write

$$P(x) : x > 2, \quad Q(x, y) : x = y^2,$$

then $P(x)$ is a *unary* (i.e. one-variable) predicate, called an *inequality*, and $Q(x, y)$ is a *binary* predicate, called an *equation*. For every value of the variable x , $P(x)$ is a proposition, and likewise $Q(x, y)$ when both variables are fixed. E.g. $P(5/2)$ and $Q(4, 2)$ are true, $P(1)$ and $Q(0, 1)$ false. Binding of variables can also occur by means of *quantifiers*. The quantifiers are the symbols ' $\forall$ ' and ' $\exists$ ', which are read

$$\begin{aligned}
\forall &\text{ 'for all', 'for every',} \\
\exists &\text{ 'there exists'}.
\end{aligned}$$

For example, if $X \subset \mathbb{Q}$, then propositions associated with the above predicate $P(x)$ are

$$A : \exists x \in X (P(x)), \quad B : \forall x \in X (P(x)).$$

Here the scope of the quantifiers is marked by parentheses. If one wishes to avoid parentheses, then a less formal form of A is

$$A : \exists x \in X \text{ such that } P(x),$$

where 'such that' can, if desired, be abbreviated to 's.t.'. In both propositions one can also do without parentheses by writing more simply

$$A : P(x) \text{ for some } x \in X, \quad B : P(x) \forall x \in X.$$

EXAMPLE 8 If $X = \mathbb{Q}$, then the proposition derived from the predicate $P(x) : x^2 \neq 2$, $B : x^2 \neq 2 \forall x \in \mathbb{Q}$, is true (Exercise 8). $\square$

The negations of the above propositions A, B are brought into a more understandable form by unpacking the negation as follows:

$$\begin{aligned}\neg [\exists x \in X (P(x))] &\Leftrightarrow \forall x \in X (\neg P(x)), \\ \neg [\forall x \in X (P(x))] &\Leftrightarrow \exists x \in X (\neg P(x)).\end{aligned}$$

By the latter rule, to show the proposition $B : P(x) \forall x \in X$ to be false it suffices to find a single counterexample $x \in X$ for which $P(x)$ is false.

EXAMPLE 9 If $X = \{x \in \mathbb{Q} \mid x < 1\}$, then the proposition

$$A : \forall x \in X \exists y \in X (y > x)$$

is true. If $X = \{x \in \mathbb{Q} \mid x \leq 1\}$, then the negation of A ,

$$\neg A : \exists x \in X [\neg \exists y \in X (y > x)] \Leftrightarrow \exists x \in X [\forall y \in X (y \leq x)]$$

is true. $\square$

The formal quantifier notation $\forall x \in X$ is often left unwritten in mathematical texts when it is clear that the reasoning concerns an entire particular set X (e.g. $X = \mathbb{Q}$).

EXAMPLE 10 The conclusion concerning the rational numbers

$$2x^2 + x < 1 \Leftrightarrow -1 < x < \frac{1}{2}$$

solves (correctly!) the inequality $2x^2 + x < 1$ in $\mathbb{Q}$. In form the conclusion is a predicate, but it can be interpreted as a proposition in which the binding $\forall x \in \mathbb{Q}$ is, for reasons of fluency, 'invisible'. $\square$

Set theory

Alongside logic, the ultimate foundations of abstract thought are treated by a branch of mathematics called *set theory*. Here we note only briefly the connection between certain basic ideas of logic and set theory. First, the set-theoretic counterpart of the 'impossible statement' of the form $A \wedge \neg A$ is the *empty set*, whose symbol is ' $\emptyset$ '. The empty set has no elements, i.e. $x \in \emptyset$ is always false.

The set-theoretic counterpart of the logical negation $\neg A$ is the *complement* of the set A , which is denoted $\mathbb{C}(A)$ and defined by

$$x \in \mathbb{C}(A) \Leftrightarrow x \notin A.$$

In practice the complement must be defined with respect to some *universal set** U . By this is understood a set in which all the sets under consideration (including the complements of the sets) are contained as subsets.

EXAMPLE 11 Let $P(x)$ be some predicate associated with the set $\mathbb{Q}$ of rational numbers, e.g. $P(x) = x < 2$. Then the condition ' $P(x)$ true' defines a subset A of $\mathbb{Q}$, denoted $A = \{x \in \mathbb{Q} \mid P(x)\}$. It is natural here to take as the universal set $U = \mathbb{Q}$, whereupon the complement of A is

$$\mathbb{C}(A) = \{x \in \mathbb{Q} \mid \neg P(x)\} = \{x \in \mathbb{Q} \mid x \geq 2\}. \quad \square$$

The set-theoretic counterparts of the logical combinations $A \vee B$, $A \wedge B$, $A \Rightarrow B$, and $A \Leftrightarrow B$ are $A \cup B$, $A \cap B$, $A \subset B$, and $A = B$, as is seen from the definitions:

$$\begin{aligned} x \in A \cup B &\Leftrightarrow x \in A \vee x \in B \\ x \in A \cap B &\Leftrightarrow x \in A \wedge x \in B \\ A \subset B &\Leftrightarrow x \in A \Rightarrow x \in B \\ A = B &\Leftrightarrow x \in A \Leftrightarrow x \in B \end{aligned}$$

The set $A \cup B$ is called the *union* of A and B , and $A \cap B$ is the *intersection* of A and B . If two sets A, B have no common elements, i.e. $x \in A \wedge x \in B$ is false for every x , then this can be expressed briefly by the notation $A \cap B = \emptyset$. In that case A and B are said to be *separate*, i.e. *disjoint*.

EXAMPLE 12 The reasoning of Example 10 was interpreted as a proposition by imagining the quantifier notation $\forall x \in \mathbb{Q}$ to be added. Even more natural is the set-theoretic interpretation: $\{x \in \mathbb{Q} \mid 2x^2 + x < 1\} = \{x \in \mathbb{Q} \mid -1 < x < \frac{1}{2}\}$. $\square$

EXAMPLE 13 Prove the de Morgan laws of set theory (cf. Example 6)

$$\begin{aligned} \mathbb{C}(A \cup B) &= \mathbb{C}(A) \cap \mathbb{C}(B), \\ \mathbb{C}(A \cap B) &= \mathbb{C}(A) \cup \mathbb{C}(B). \end{aligned}$$

*The term 'universal set' may evoke the image of a set containing literally 'everything'. Ideas of this type did exist in set theory in the early stages of the theory, but they were found to lead to paradoxes, i.e. to logical impossibilities. From the subsequently made-precise axioms of set theory it indeed follows that 'nothing contains everything'.

Solution In the chain of reasoning

$$\begin{aligned}
 x \in \mathbb{C}(A \cup B) &\Leftrightarrow x \notin (A \cup B) \\
 &\Leftrightarrow \neg(x \in A \vee x \in B) \\
 &\Leftrightarrow x \notin A \wedge x \notin B && [\text{Example 6 a)}] \\
 &\Leftrightarrow x \in \mathbb{C}(A) \wedge x \in \mathbb{C}(B) \\
 &\Leftrightarrow x \in \mathbb{C}(A) \cap \mathbb{C}(B)
 \end{aligned}$$

one can proceed in both directions, so the sets $\mathbb{C}(A \cup B)$ and $\mathbb{C}(A) \cap \mathbb{C}(B)$ have the same elements, and the first statement is thus proved. The proof of the second statement is obtained from this by the substitutions $\cup \leftrightarrow \cap$, $\cap \leftrightarrow \cup$, $\vee \leftrightarrow \wedge$, and $\wedge \leftrightarrow \vee$. $\square$

Equivalence relation

A *relation* is the set-theoretic notion of a 'relationship'. If R is a relation defined on a set A and $x, y \in A$, then the notation $x R y$ is read ' x is in the relation R with y ', or more fluently just ' x [name of the relation] y '. The practically most important relations are already familiar, namely the order and identification relations. An identification relation must always be an *equivalence relation*, whose axioms are the following:

$$(E1) \quad x R x \quad \forall x$$

$$(E2) \quad x R y \Rightarrow y R x$$

$$(E3) \quad x R y \wedge y R z \Rightarrow x R z$$

The required properties are called *reflexivity*, *symmetry*, and *transitivity*. For example, the relation ' $\leq$ ' is reflexive and transitive but not symmetric, and the relation ' $\neq$ ' is only symmetric. The identification relation 'the same as', on the other hand, is most evidently an equivalence relation. For example, when one writes $x = y = z$ (meaning: $x = y$ and $y = z$), it is evident that $x = z$. This is thus a matter of transitivity.*

EXAMPLE 14 If $A = \{\text{Finns}\}$, then the following relations defined on A are equivalence relations:

$$\begin{aligned}
 x \sim y &\Leftrightarrow x \text{ and } y \text{ were born in the same year,} \\
 x \heartsuit y &\Leftrightarrow y = x \text{ or } y \text{ is the spouse of } x. \quad \square
 \end{aligned}$$

*The general axioms of an identification relation are the same as those of an equivalence relation, cf. the footnote in the previous section.

An equivalence relation defined on a set A has the property that it partitions A into *equivalence classes*. These are subsets of A whose contained elements are all in the relation with one another. By the reflexivity property (E1), every element of A belongs to at least one equivalence class (possibly alone). On the other hand, it follows from the transitivity property (E3) that two equivalence classes are either completely the same or they are disjoint — thus every element of A belongs to exactly one equivalence class. If it is an identification relation, then the elements contained in each equivalence class are 'classified as the same', i.e. identified with one another. In that case one can also speak of *identification classes*.

EXAMPLE 14 (continued) The equivalence classes determined by the relation ' $\sim$ ' are called age groups. The relation ' $\heartsuit$ ' partitions A into equivalence classes containing either one element (minors, singles, etc.) or two elements (registered couples). $\square$

EXERCISES

- Let A be the proposition 'It is raining', B the proposition 'I go for a run', and C the proposition 'I have time'. Using these and the logical operators, write as concisely as possible:
 - 'I'm tired, it's dark, and BB is on TV, so I don't go for a run.'
 - 'I go for a run, rain or shine!'
 - 'It is raining, so I don't go for a run.'
 - 'It is raining and I don't go for a run.'
 - 'I go for a run only if I have time.'
 - 'I have time and it isn't raining, but I don't go for a run.'
 - 'If it isn't raining and I have time, then I go for a run.'
 - 'I have time only if, because of the rain, I don't go for a run.'
- Let A = 'It is raining today', B = 'Tomorrow is dry', and C = 'Today is dry'.
 - If the rain that began today continues into tomorrow, what are the truth values of the propositions $A \Rightarrow B$ and $B \Rightarrow A$?
 - Does the truth value of the proposition $\neg(\neg A \wedge B) \Rightarrow \neg C$ depend on which day is 'today'?
- Show, either by means of a truth table or by other reasoning, that the following proposition (the so-called *modus tollens*) is identically true: $[(\neg P \Rightarrow Q) \wedge \neg Q] \Rightarrow P$.
How is this result appealed to in an indirect proof?

4. Show, by means of a truth table or by other reasoning, to be identically true:

$$\begin{aligned} \text{a)} \quad & \neg(A \wedge \neg B) \Leftrightarrow \neg A \vee (A \wedge B) & \text{b)} \quad & (A \Rightarrow B) \Leftrightarrow (\neg B \Rightarrow \neg A) \\ \text{c)} \quad & [(A \Rightarrow B) \wedge (B \Rightarrow C)] \Rightarrow (A \Rightarrow C) \\ \text{d)} \quad & [(A \Rightarrow C) \wedge (B \Rightarrow C)] \Rightarrow [(A \vee B) \Rightarrow C] \\ \text{e)} \quad & [(A \wedge B) \Rightarrow C] \Leftrightarrow [A \Rightarrow (B \Rightarrow C)] \end{aligned}$$

Compare by means of a truth table:

$$\text{f)} \quad (A \wedge B) \vee C \text{ and } A \wedge (B \vee C) \quad \text{g)} \quad A \not\Rightarrow B \text{ and } A \Rightarrow \neg B$$

Investigate, by means of a truth table, whether the following propositions can be expressed in some simpler (equivalent) way:

$$\text{h)} \quad (A \Rightarrow B) \wedge B \quad \text{i)} \quad (A \Rightarrow B) \Rightarrow A \quad \text{j)} \quad [(A \Rightarrow B) \wedge B] \Rightarrow A$$

5. If the following predicates $P(x)$ are interpreted as the statements $P(x) \forall x \in \mathbb{Q}$, then which of the statements are true and which false?

$$\begin{aligned} \text{a)} \quad & 2x > 3 \Rightarrow x > 4/3 & \text{b)} \quad & x < -2 \Leftrightarrow x^2 > 4 \\ \text{c)} \quad & 3x < 4 \Rightarrow x < 1 \vee 2x < 3 & \text{d)} \quad & x > 1 \not\Leftrightarrow x > 0 \wedge x > 1 \\ \text{e)} \quad & x = 4/3 \wedge 3x^2 < x + 4 \Rightarrow x^2 + 2x + 1 < 0 \\ \text{f)} \quad & 2x^2 + x \leq 1 \leq 3x - 2x^2 \Leftrightarrow x = 4/3 \end{aligned}$$

6. Let x and y be rational numbers. Which of the following propositions are true?

$$\begin{aligned} \text{a)} \quad & \forall x \exists y (x \cdot y = 0) & \text{b)} \quad & \forall x \exists y (x \cdot y = 1) \\ \text{c)} \quad & \exists y \forall x (x \cdot y = 0) & \text{d)} \quad & \exists y \forall x (x \cdot y = 1) \\ \text{e)} \quad & \exists y \forall x (x \cdot y = x) & \text{f)} \quad & \exists y \forall x (x \cdot y = y) \\ \text{g)} \quad & \forall x \exists y (y < x) & \text{h)} \quad & \exists x \forall y (y \geq x) \end{aligned}$$

How does the situation change if x and y are positive rational numbers?

7. Let $x \in \mathbb{Q}$. Form the negations of the following two propositions, fully unpacked. Which of the four propositions thus arising are true?

$$\text{a)} \quad \forall(\varepsilon > 0) \exists(x \neq 1)(|x - 1| < \varepsilon) \quad \text{b)} \quad \exists(x \neq 1) \forall(\varepsilon > 0)(|x - 1| < \varepsilon)$$

8. Show by the indirect method of proof: a) There is no smallest positive rational number. b) $x^2 \neq 2 \forall x \in \mathbb{Q}$. c) $x^2 \neq 3 \forall x \in \mathbb{Q}$.

9. Let the universal set U be the set formed by the first ten natural numbers $\{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$. Let $A = \{2, 5, 7, 8, 10\}$ and $B = \{2, 4, 6, 8, 10\}$. Determine $A \cup B$, $A \cap B$, $\mathcal{C}(A)$, $\mathcal{C}(A \cup B)$, and $\mathcal{C}(A) \cap \mathcal{C}(B)$.

10. Prove the following set-theoretic statements:

$$\text{a)} \quad A \subset B \Leftrightarrow A \cup B = B \Leftrightarrow A \cap B = A$$

- b) $A \subset B \wedge A \subset C \Leftrightarrow A \subset B \cap C$
- c) $A \subset C \wedge B \subset C \Leftrightarrow A \cup B \subset C$
- d) $A \subset B \Leftrightarrow A \cap C \subset B \cap C \forall C$
- e) $\emptyset \subset A \forall A$

11. Formulate and prove the set-theoretic counterparts of statements b)–d) of Exercise 4.
12. Let $A = \{(p, q) \mid p, q \in \mathbb{Z}\}$ (the set of integer pairs) and $Q = \{(p, q) \in A \mid q \neq 0\}$. Define on Q the relation $(p_1, q_1) \sim (p_2, q_2) \Leftrightarrow p_1 q_2 = p_2 q_1$. Show that this is an equivalence relation on Q . How is this result related to the rational numbers?
13. Which properties of an equivalence relation do the following relations, defined on the given sets A , have?
 - a) $A = \mathbb{N} : xRy \Leftrightarrow x + y$ is even.
 - b) $A = \mathbb{N} : xRy \Leftrightarrow x + y$ is odd.
 - c) $A = \mathbb{Q} : xRy \Leftrightarrow |x - y| \leq 10^{-100}$.
 - d) $A = \{\text{Finns}\} : xRy \Leftrightarrow x$ and y are related to each other directly in an ascending or descending line.
14. (*) A natural number $k \in \mathbb{N}$ is a *factor* of a number $n \in \mathbb{N}$ if $n = k \cdot m$ for some $m \in \mathbb{N}$. The number n is a *prime number* if n has no factors other than $k = 1$ and $k = n$. Show that there is no end to the prime numbers, i.e. there is no largest prime number.

I.4 The concept of a sequence

A *sequence* (Swedish *följd*) is an object of the form

$$\{a_1, a_2, a_3, \dots\}.$$

What distinguishes a sequence from a set is above all order: the members of the sequence, which are called the *terms* of the sequence (cf. the element of a set), have been 'put in a row', i.e. in order. In a mathematical sequence, order means a correspondence between the set of natural numbers and the sequence:

$$\begin{aligned} 1 &\mapsto a_1 \\ 2 &\mapsto a_2 \\ &\vdots \end{aligned}$$

Here ' $\mapsto$ ' again denotes 'assignment': to each $n \in \mathbb{N}$ is assigned a unique term of the sequence (the n th term) a_n , and then n is said to be the ordinal number, i.e. *index*, of that term. The index set is thus all of $\mathbb{N}$, which means that the sequence is nonterminating*.

Into a mathematical sequence, as into a set, one can in principle place anything. In what follows the objects of study are above all *number sequences*, for which abbreviated notations are used

$$\begin{aligned} &\{a_n, n = 1, 2, \dots\}, \\ &\{a_n\}_{n=1}^{\infty}, \\ &\{a_n\}. \end{aligned}$$

If the terms of a sequence are determined by a known expression depending on the index, that expression can be written in place of a_n .

EXAMPLE 1

$$\begin{aligned} \{1, -1, 1, -1, \dots\} &= \{(-1)^{n+1}, n = 1, 2, \dots\} \\ \left\{\frac{1}{9}, \frac{1}{16}, \frac{1}{25}, \dots\right\} &= \left\{\frac{1}{(k+2)^2}, k = 1, 2, \dots\right\} \\ \{1, 1, 2, 6, 24, 120, 720, 5040, 40320, \dots\} &= \{n!\}_{n=0}^{\infty} \quad \square \end{aligned}$$

*For a 'terminating sequence' of the form

$$(a_1, a_2, \dots, a_n), \quad n \in \mathbb{N}$$

this text uses the general name (finite) *ordered set* (or 'ordered set of n elements'), and in the cases $n = 2, 3$ the names *pair* and *triple*. Unlike the sequence notation, round brackets are used here as a rule. — In the literature, round brackets are sometimes also used to denote nonterminating sequences, and one also sees the notation $\langle a_n \rangle$ used for sequences.

The last example is also an example of a more general sequence of the form

$$\{a_m, a_{m+1}, a_{m+2}, \dots\} = \{a_n\}_{n=m}^{\infty}, \quad m \in \mathbb{Z}.$$

By a change of index this can be reduced to the normal form:

$$\{a_n\}_{n=m}^{\infty} = \{a_{m+k-1}\}_{k=1}^{\infty}.$$

EXAMPLE 2 The number sequence

$$a_0 = \frac{1}{2}, \quad a_n = 1 - a_{n-1}^2, \quad n = 1, 2, \dots$$

is an example of a *recursive* number sequence that 'defines itself'. Here $a_1 = 3/4$, $a_2 = 7/16$, $a_3 = 207/256$, ... $\square$

Series

If $\{a_k, k = 1, 2, \dots\}$ is a number sequence and $\{s_n\}$ is another number sequence, defined by

$$s_n = a_1 + \dots + a_n = \sum_{k=1}^n a_k,$$

then $\{s_n\}$ is called a *series*. The usual abbreviated notation for a series is

$$\sum_{k=1}^{\infty} a_k \quad \left(= \left\{ \sum_{k=1}^n a_k, \quad n = 1, 2, \dots \right\} \right).$$

The numbers a_k are called the *terms of the series* and the numbers s_n the *partial sums of the series*. A series is thus interpreted as the number sequence formed by the partial sums of the series.*

EXAMPLE 3

$$\sum_{k=0}^{\infty} (-1)^k = \left\{ \sum_{k=0}^n (-1)^k, \quad n = 0, 1, \dots \right\} = \{1, 0, 1, 0, \dots\} \quad \square$$

'The mother of all series' is the (basic-form) *geometric series*, whose terms are $a_k = q^k$, $k = 0, 1, \dots$, i.e. the series is of the form $\sum_{k=0}^{\infty} q^k = \{1, 1+q, 1+q^2, \dots\}$.

*The interpretation of a series in this text 'cuts a corner' slightly on the formal definition of a series, which reads: A series is the sequence-pair formed by the sequences $\{a_k\}$ (the terms of the series) and $\{s_n\}$ (the partial sums).

$q + q^2, \dots\}^*$. For the partial sums one obtains in this case the formula (see Exercise I.2:3 c)

$$s_n = \sum_{k=0}^n q^k = \frac{q^{n+1} - 1}{q - 1}, \quad q \neq 1.$$

Induction

Closely connected with the natural numbers and the concept of a sequence is also the mathematical principle of proof called *induction*. Let $P(n)$, $n \in \mathbb{N}$, be a predicate. Then the proposition

$$\mathcal{P} : \quad P(n) \quad \forall n \in \mathbb{N}$$

can be interpreted as referring to the sequence of statements $\{P(1), P(2), \dots\}$. The idea of induction is to transform this sequence into a recursive one, whereupon the sequence 'proves itself' in the same way as a recursive number sequence 'computes itself' (cf. Example 2 above). The idea is realized by considering two propositions:

$$\begin{aligned} \mathcal{P}_1 &: \quad P(1), \\ \mathcal{P}_2 &: \quad P(n) \Rightarrow P(n+1) \quad \forall n \in \mathbb{N}. \end{aligned}$$

If $\mathcal{P}_1$ and $\mathcal{P}_2$ are both true, then $\mathcal{P}_1$ starts the 'proof machine' based on the proposition $\mathcal{P}_2$, so the implication

$$\mathcal{P}_1 \wedge \mathcal{P}_2 \quad \Rightarrow \quad \mathcal{P},$$

which states the so-called *induction principle*, seems evidently true.[†] By the induction principle, then, to prove the statement $\mathcal{P}$ it suffices to show that $\mathcal{P}_1$ and $\mathcal{P}_2$ are both true. Proving the proposition $\mathcal{P}_2$ is called the *induction step*, and the initial assumption of that part of the proof, ' $n \in \mathbb{N}$ and $P(n)$ true', the *induction hypothesis*.

Let us prove the following familiar result by induction:

PROPOSITION I.4.1 (Binomial formula) In the field of rational numbers or its extensions,

$$(x + y)^n = \sum_{k=0}^n \binom{n}{k} x^{n-k} y^k, \quad n \in \mathbb{N},$$

*For convenience, let us agree that in the geometric series $\sum_{k=0}^{\infty} q^k$ the first term is $q^0 = 1$ also when $q = 0$.

[†]The induction principle follows from Peano's axiom (P5) by setting $S = \{n \in \mathbb{N} \mid P(n) \text{ true}\}$, see the footnote in Section I.1.

where the *binomial coefficients* are defined by

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}.$$

Proof The statement is of the form $P(n) \forall n \in \mathbb{N}$. Here $P(1)$ is evidently true, so it suffices to carry out the induction step. So assume that $P(n)$ is true, i.e. that the binomial formula holds for a certain (but arbitrarily chosen) value of n . Then, by the field axioms and the assumption,

$$\begin{aligned} (x+y)^{n+1} &= (x+y)(x+y)^n = (x+y) \sum_{k=0}^n \binom{n}{k} x^{n-k} y^k \\ &= \sum_{k=0}^n \binom{n}{k} x^{n-k+1} y^k + \sum_{k=0}^n \binom{n}{k} x^{n-k} y^{k+1} \\ &= \sum_{k=0}^n \binom{n}{k} x^{n+1-k} y^k + \sum_{l=1}^{n+1} \binom{n}{l-1} x^{n+1-l} y^l. \end{aligned}$$

Here a change of index $l = k + 1$ has been made in the latter sum. Writing again $l = k$, both sums can be combined into the result

$$(x+y)^{n+1} = \sum_{k=0}^{n+1} c_k x^{n+1-k} y^k,$$

where

$$c_0 = \binom{n}{0} = 1 = \binom{n+1}{0}, \quad c_{n+1} = \binom{n}{n} = 1 = \binom{n+1}{n+1},$$

and for the index values $k = 1 \dots n$

$$\begin{aligned} c_k &= \binom{n}{k} + \binom{n}{k-1} = \frac{n!}{k!(n-k)!} + \frac{n!}{(k-1)!(n-k+1)!} \\ &= \frac{n!}{(k-1)!(n-k)!} \left(\frac{1}{k} + \frac{1}{n-k+1} \right) \\ &= \frac{n!}{(k-1)!(n-k)!} \cdot \frac{n+1}{k(n-k+1)} \\ &= \frac{(n+1)!}{k!(n-k+1)!} = \binom{n+1}{k}. \end{aligned}$$

Consequently $P(n+1)$ is true and the induction step is thus carried out. Besides the general field operations, the proof required arithmetic operations between

natural numbers (including division), so the proof is valid in the field of rational numbers or its extensions. $\square$

As is clear from the proof of Proposition I.4.1, the binomial coefficients can be computed recursively from the formula

$$\binom{n+1}{0} = \binom{n+1}{n+1} = 1, \quad \binom{n+1}{k} = \binom{n}{k} + \binom{n}{k-1}, \quad k = 1 \dots n.$$

The formula is illustrated in *Pascal's triangle*, by means of which the coefficients can be determined conveniently for small values of n :

$$\begin{array}{cccccccc}
 & & & & 1 & & & \\
 & & & 1 & & 1 & & \\
 & & 1 & & 2 & & 1 & \\
 & 1 & & 3 & & 3 & & 1 \\
 1 & & 4 & & 6 & & 4 & & 1 \\
 & 1 & & 5 & & 10 & & 10 & & 5 & & 1 \\
 & & 1 & & 6 & & 15 & & 20 & & 15 & & 6 & & 1 \\
 & & & 1 & & 7 & & 21 & & 35 & & 35 & & 21 & & 7 & & 1 \\
 1 & & & & 8 & & 28 & & 56 & & 70 & & 56 & & 28 & & 8 & & 1
 \end{array}
 \begin{array}{l}
 (n=1) \\
 (n=2) \\
 (n=3) \\
 (n=4) \\
 (n=5) \\
 (n=6) \\
 (n=7) \\
 (n=8)
 \end{array}$$

By means of Pascal's triangle the binomial coefficients are determined by mere addition of natural numbers.

As a second application of induction, we prove a frequently used inequality.

PROPOSITION I.4.2 (Bernoulli's inequality) In the field of rational numbers or its ordered extensions,

$$(1+x)^n > 1+nx \quad \text{when } n \in \mathbb{N}, n \geq 2 \text{ and } x \geq -1, x \neq 0.$$

Proof If $x > 0$, then the inequality follows most easily from the binomial formula:

$$(1+x)^n = 1 + \binom{n}{1}x + [\dots] = 1 + nx + [\dots],$$

where $[\dots] > 0$ when $x > 0$ and $n \geq 2$. A more generally valid proof rests on induction, and it too is not long: first,

$$(1+x)^2 = 1 + 2x + x^2 > 1 + 2x, \quad \text{when } x \neq 0,$$

so the statement is true when $n = 2$. On the other hand, if the claim is true for any value of n (induction hypothesis, $n \geq 2$), then

$$\begin{aligned}(1+x)^{n+1} &= (1+x)(1+x)^n \\ &\geq (1+x)(1+nx) \\ &= 1 + (n+1)x + nx^2 \\ &> 1 + (n+1)x,\end{aligned}$$

when $x \geq -1 \Rightarrow 1+x \geq 0$ and $x \neq 0 \Rightarrow x^2 > 0$. By the induction principle the statement is thus proved. $\square$

Countability, cardinality

Two sets A and B are said to be *of the same cardinality* if there exists a one-to-one correspondence between the sets such that to each element of A there corresponds a unique element of B and vice versa. The correspondence will henceforth be denoted by the double arrow ' $\leftrightarrow$ ':

$$A \leftrightarrow B \quad (\text{one-to-one correspondence}).$$

If in particular $A \leftrightarrow \{1, 2, \dots, n\}$ for some $n \in \mathbb{N}$, then A is said to be a *finite* set. Then A has exactly n elements, and A and B are of the same cardinality exactly when B too has n elements. In the case of a finite set, the correspondence $A \leftrightarrow \{1, 2, \dots, n\}$ is called a *numbering* (indexing) of the elements of A . A numbering makes A into an ordered set.

EXAMPLE 4

$$\begin{aligned}A &= \{1, 2, 3, 1000\} \\ B &= \{\text{goat G, horse H, student S, professor P}\} \\ C &= \{\text{the atoms of the Milky Way}\}\end{aligned}$$

All three are finite sets. The sets A and B are of the same cardinality. $\square$

If a set is not finite, then it is *infinite*.

EXAMPLE 5 The sets

$$\begin{aligned}A &= \{1, 4, 9, 16, 25, 36, 49, \dots\}, \\ B &= \{1, 100, 10000, 1000000, \dots\}\end{aligned}$$

are both infinite. They are also of the same cardinality, for from the correspondences

$$\begin{aligned} n \in \mathbb{N} &\leftrightarrow n^2 \in A \\ n \in \mathbb{N} &\leftrightarrow 10^{2n-2} \in B \end{aligned}$$

one sees that $A \leftrightarrow \mathbb{N} \leftrightarrow B$. $\square$

On the basis of the example, one cannot trust the 'feeling' of smaller or larger when comparing the cardinalities of infinite sets. Expressions such as 'as many' or 'the same number' are also most clearly restricted to the comparison of finite sets.

The elements of the sets A, B of Example 5 can be numbered on the basis of the correspondences given. In general, a set A is said to be *countable* (or 'countably infinite') if $A \leftrightarrow \mathbb{N}$. Countability thus means that the elements of the set can be arranged into a sequence.

EXAMPLE 6 One sequence form of the set of integers $\mathbb{Z} = \{0, \pm 1, \pm 2, \dots\}$ is

$$\{0, 1, -1, 2, -2, \dots\} = \{a_n, n = 1, 2, \dots\},$$

so $\mathbb{Z}$ is countable. For example, the number -777 is the 1555th term of the sequence. $\square$

EXAMPLE 7 The set of rational numbers $\mathbb{Q}$ can be expressed in the form

$$\mathbb{Q} = A_1 \cup A_2 \cup \dots = \bigcup_{m=1}^{\infty} A_m,$$

where

$$\begin{aligned} A_1 &= \{0\}, \\ A_m &= \{x = p/q \in \mathbb{Q} \mid |p| + |q| = m \text{ \& } x \notin A_k \text{ when } k < m\}, \quad m = 2, 3, \dots \end{aligned}$$

The subsets A_m are, by definition, pairwise disjoint ($A_m \cap A_k = \emptyset$ when $k < m$), so every $x \in \mathbb{Q}$ is an element of at most one subset. On the other hand, it also follows from the definition that if $x \in \mathbb{Q}$, then $x \in A_m$, where $m \in \mathbb{N}$ is the smallest number for which $m = |p| + |q|$ and $x = p/q$. Consequently every $x \in \mathbb{Q}$ is an element of exactly one subset A_m . The sets A_m are also finite, so each of them can be numbered separately. If now the number of elements of A_m is N_m , then for an arbitrary $x \in \mathbb{Q}$ a unique ordinal number n is obtained by the rule

$$n = \begin{cases} 1, & \text{if } x = 0, \\ N_1 + \dots + N_{m-1} + k, & \text{if } x \text{ is the } k\text{th element of } A_m, \quad m \geq 2. \end{cases}$$

All the rational numbers are thus arranged into a sequence, and one can therefore state that $\mathbb{Q}$ is countable. $\square$

If a set is infinite but not countable, then it is said to be *uncountable*. Such 'powerful' sets — number sets too — do exist, as we shall see.

EXERCISES

1. Show by induction that the sum formulas hold ($n \in \mathbb{N}$)

$$\begin{array}{ll} \text{a)} \quad \sum_{k=1}^n k = \frac{1}{2}n(n+1) & \text{b)} \quad \sum_{k=1}^n k^2 = \frac{1}{6}n(n+1)(2n+1) \\ \text{c)} \quad \sum_{k=1}^n k^3 = \frac{1}{4}n^2(n+1)^2 & \text{d)} \quad \sum_{k=1}^n k^4 = \frac{1}{30}n(n+1)(2n+1)(3n^2+3n-1) \end{array}$$

2. Show by induction that the following sum formulas hold for every $n \in \mathbb{N}$:

$$\text{a)} \quad 1 \cdot 3 + 3 \cdot 5 + \dots + (2n-1)(2n+1) = \frac{n}{3}(4n^2 + 6n - 1)$$

$$\text{b)} \quad \sum_{k=1}^n k^2 2^k = (n^2 - 2n + 3)2^{n+1} - 6$$

$$\text{c)} \quad \sum_{k=1}^n kq^{k-1} = \frac{1 - (n+1)q^n + nq^{n+1}}{(1-q)^2}, \quad q \in \mathbb{Q}, \quad q \neq 1$$

3. Define the recursive number sequences

$$\text{a)} \quad a_0 = 1, \quad a_{n+1} = qa_n + 1, \quad n = 0, 1 \dots$$

$$\text{b)} \quad a_0 = 2, \quad a_{n+1} = \frac{a_n}{2} + \frac{1}{a_n}, \quad n = 0, 1 \dots$$

$$\text{c)} \quad a_1 = 3, \quad a_2 = 6, \quad a_{n+1} = \frac{na_n + a_{n-1} + 3}{n}, \quad n = 2, 3 \dots$$

$$\text{d)} \quad a_0 \in \mathbb{Q}, \quad a_0 \notin \{-1/n \mid n \in \mathbb{N}\}, \quad a_{n+1} = \frac{a_n}{1 + a_n}, \quad n = 0, 1 \dots$$

Show by induction that a) $\{a_n\}$ is a geometric series, b) $1 \leq a_n \leq 2 \forall n$,
c) $a_n < 4n \forall n$, d) $a_n = a_0/(1 + na_0) \forall n$.

4. Compute the values of the following sum expressions ($n \in \mathbb{N}$):

$$\text{a)} \quad \sum_{k=0}^n \binom{n}{k} \quad \text{b)} \quad \sum_{k=0}^n (-1)^k \binom{n}{k} \quad \text{c)} \quad \sum_{k=0}^n 2^k \binom{n}{k} \quad \text{d)} \quad \sum_{k=0}^n 3^{-k} \binom{n}{k}$$

5. (*) Let $x \in \mathbb{Q}$, $x > 0$. Show that if for some $a \in \mathbb{Q}$ and $n \in \mathbb{N}$, $n \geq 2$ one has $1 < x^n \leq a$, then $1 < x < 1 + (a - 1)/n$. *Hint:* Write $x = 1 + y$.
6. (*) a) Show that if $x \in \mathbb{Q}$, $x > 0$ and $n \in \mathbb{N}$, $n \geq 3$, then $(1 + x)^n > 1 + nx + \frac{1}{2}n(n-1)x^2$. b) What kind of even better estimates are obtained if $n \geq k$, $k = 4, 5, \dots$? c) Show that there exists $n \in \mathbb{N}$ such that

$$\frac{(1 + 10^{-100})^n}{n^{100}} > 10^{100}.$$

7. (*) Show that if the sets $A_1, A_2, A_3, \dots$ are pairwise disjoint and countable, then the set

$$A = A_1 \cup A_2 \cup \dots = \bigcup_{n=1}^{\infty} A_n$$

is also countable. Verify the claim directly (by determining A) when $A_n = \{x \in \mathbb{Q} \mid n-1 \leq |x| < n\}$.

I.5 Infinite decimals

A sequence very commonly encountered in everyday mathematics is the *infinite decimal**. In outward form this is a string of the form

$$ed_{-m}d_{-m+1}\dots d_0.d_1d_2d_3\dots = x_0.d_1d_2d_3\dots,$$

where

- 'e' is the *sign*, either $e = +$ or $e = -$ ('empty' = +)
- '.' is the *decimal point* (or 'decimal comma', between d_0 and d_1)
- $m \in \mathbb{N} \cup \{0\}$
- $x_0 = ed_{-m}d_{-m+1}\dots d_0 \in \mathbb{Z}$ = the *integer part* of the decimal
- $d_n, n = 1, 2, \dots$ = the *decimals* of the decimal number ($d_n \in \{0, \dots, 9\}$)

EXAMPLE 1

$$\begin{array}{ll} 141.42135623730950488\dots & (e = +, m = 2, x_0 = 141) \\ -0.00314159265358979\dots & (e = -, m = 0, x_0 = 0) \end{array} \quad \square$$

An infinite decimal is thus essentially a sequence of digits which, besides the digits, carries a separate sign and a decimal point whose position within the digit string determines the integer part x_0 . Such a string is interpreted to mean the sequence of rational numbers

$$\{x_0, x_1, \dots\} = \{x_n\}_{n=0}^\infty,$$

where x_0 is the integer part and

$$x_n = x_0 \pm (d_1 \cdot 10^{-1} + \dots + d_n \cdot 10^{-n}) = x_0 \pm \sum_{k=0}^n d_k \cdot 10^{-k}, \quad n \in \mathbb{N},$$

where the sign of the summation expression is chosen according to e . Under this interpretation, an infinite decimal is thus in fact a series whose first term is x_0 and the others $\pm d_k \cdot 10^{-k}$, $k = 1, 2, \dots$. From the index $n = 1$ onward, the partial

*An alternative name is *nonterminating decimal*.

sums of the series are so-called *finite* (or terminating, see the footnote) *decimals*, which from now on are denoted

$$x_n = x_0.d_1 \dots d_n \quad (n \in \mathbb{N}).$$

This denotes the number $\pm(|x_0| + \sum_{k=0}^n d_k \cdot 10^{-k})$, where the sign is set separately (the same as in x_0 , if $x_0 \neq 0$). Writing the number in the form

$$x_n = \pm 10^{-n} \left(|x_0| \cdot 10^n + \sum_{k=0}^n d_k \cdot 10^{n-k} \right)$$

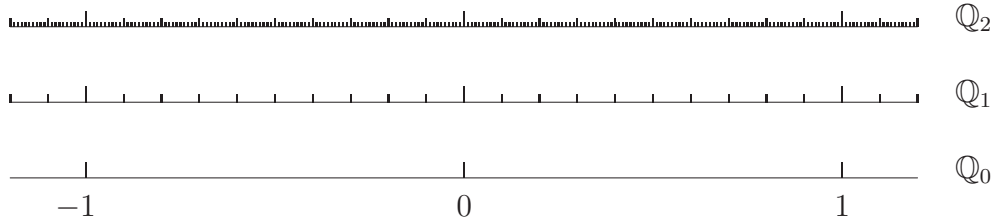
one sees that $x_n = p/10^n$, where $p = ed_{-m} \dots d_0 d_1 \dots d_n \in \mathbb{Z}$ is an integer. Conversely, if one starts from a number $x = p/10^n$, $p \in \mathbb{Z}$, then by expressing p in the decimal system one sees that x can be brought into the form $x = x_0.d_1 \dots d_n$. Thus the finite decimals of this form constitute exactly the subset of $\mathbb{Q}$

$$\mathbb{Q}_n = \{ p/10^n \mid p \in \mathbb{Z} \}.$$

When accordingly one sets $\mathbb{Q}_0 = \mathbb{Z}$, then

$$\mathbb{Z} = \mathbb{Q}_0 \subset \mathbb{Q}_1 \subset \mathbb{Q}_2 \subset \dots \subset \mathbb{Q}.$$

In the number set $\mathbb{Q}_n$ the numbers are equally spaced, so that the distance between consecutive numbers x, y is $|x - y| = 10^{-n}$. The figure illustrates the number sets $\mathbb{Q}_0, \mathbb{Q}_1, \mathbb{Q}_2$ geometrically.*



From now on, infinite decimals are referred to either by the usual number symbols x, y, z etc., or alternatively by the symbols $\check{x}, \check{y}, \check{z}$ etc., when one wishes to distinguish them from the 'genuine' numbers, in particular the rational numbers. The set of infinite decimals is denoted by the symbol $\mathbb{D}$:

$$\mathbb{D} = \{\text{infinite decimals}\}.$$

*The 'geometrization' of numbers is examined later in Section II.1. At this stage the basic ideas of geometric illustration are taken as known.

In the case of an individual decimal, the string representation is regarded as the 'name' of the decimal in question, so that in the notation

$$\check{x} = x_0.d_1d_2\dots = \{x_n\}_{n=0}^{\infty}$$

the first '=' names it and the second interprets it (gives it its meaning).

EXAMPLE 1 (continued) With the notational conventions adopted, one can write

$$\begin{aligned} x &= 141.421356\dots \\ &= \{141, 141.4, 141.42, \dots\} \\ &= \left\{141, \frac{1414}{10}, \frac{14142}{100}, \dots\right\} \\ &= \{x_0, x_1, x_2, \dots\} \\ y &= -0.00314159\dots \\ &= \{0, 0.0, 0.00, -0.003, -0.0031, \dots\} \\ &= \left\{0, 0, 0, -\frac{3}{1000}, -\frac{31}{10000}, \dots\right\} \\ &= \{y_0, y_1, y_2, \dots\} \quad \square \end{aligned}$$

Every infinite decimal $x \in \mathbb{D}$ is thus, in meaning, a sequence $\{x_n\}_{n=0}^{\infty}$, where $x_n \in \mathbb{Q}_n \forall n$. By its definition the sequence $\{x_n\}$ can be determined by first setting $x_0 \in \mathbb{Z}$ and then $x_1, x_2, \dots$ recursively in the form

$$x_n = \begin{cases} x_{n-1} + d_n \cdot 10^{-n}, & \text{if } e = +, \\ x_{n-1} - d_n \cdot 10^{-n}, & \text{if } e = -, \end{cases}$$

where e is the sign of x and $d_n \in \{0, \dots, 9\} \forall n \in \mathbb{N}$. Accordingly, either $x_n \geq x_{n-1} \forall n \in \mathbb{N}$ or $x_n \leq x_{n-1} \forall n \in \mathbb{N}$, depending on the sign e . Such 'monotonous' sequences are a remarkably important special case of sequences, so for them we set down

DEFINITION I.5.1 The sequence $\{a_n, n = m, m+1, \dots\}$ is

- *increasing*, if $a_{n+1} \geq a_n \forall n \geq m$
- *decreasing*, if $a_{n+1} \leq a_n \forall n \geq m$

The sequence is *strictly* increasing if $a_{n+1} > a_n \forall n \geq m$ and strictly decreasing if $a_{n+1} < a_n \forall n \geq m$. If a sequence is (strictly) increasing or decreasing, it is (strictly) *monotonic*.

An infinite decimal is thus, as a sequence, monotonic: increasing when the sign is $e = +$ and decreasing when the sign is $e = -$. According to the following result, however, this increase or decrease 'fades' rapidly as n grows.

PROPOSITION I.5.2 If $x = \{x_n\}_{n=0}^\infty \in \mathbb{D}$, then for every $k \in \mathbb{N} \cup \{0\}$ and every $n \in \mathbb{N}$, $n > k$, one has $|x_n - x_k| < 10^{-k}$, more precisely

$$|x_n - x_k| \leq 10^{-k} - 10^{-n}.$$

Proof When $n > k \geq 0$,

$$\begin{aligned} |x_n - x_k| &= \sum_{j=k+1}^n d_j \cdot 10^{-j} \leq \sum_{j=k+1}^n 9 \cdot 10^{-j} \\ &= 9 \cdot 10^{-k-1} \sum_{i=0}^{n-k-1} \left(\frac{1}{10}\right)^i \\ &= 10^{-k} \cdot \frac{9}{10} \cdot \frac{1 - \left(\frac{1}{10}\right)^{n-k}}{1 - \frac{1}{10}} \\ &= 10^{-k} - 10^{-k} \left(\frac{1}{10}\right)^{n-k} \\ &= 10^{-k} - 10^{-n}. \quad \square \end{aligned}$$

When, alongside the result of Proposition I.5.2, one takes into account the monotonicity of the sequence $\{x_n\}$, one sees that for every infinite decimal $\check{x} = \{x_n\}$

$$0 \leq k < n \quad \Rightarrow \quad \begin{cases} x_k \leq x_n < x_k + 10^{-k}, & \text{if the sign } e = +, \\ x_k - 10^{-k} < x_n \leq x_k, & \text{if the sign } e = -. \end{cases}$$

Accordingly, the range of variation of the terms x_n shrinks rapidly as the index n grows.

The long-division algorithm

It was noted above that infinite decimals, as sequences $\{x_n\}$, are both monotonic and 'fading'. It may also happen that the sequence $\{x_n\}$ not only 'fades' but *approaches a rational number* $x \in \mathbb{Q}$ in the sense that $|x_n - x|$ becomes ever smaller as n grows. When such 'approaching' is possible in general is clarified in later sections. At this stage we pose the question the other way around: given $x \in \mathbb{Q}$, how does one find an infinite decimal $\check{x} = \{x_n\}$ such that $|x_n - x|$

becomes ever smaller as n grows? The answer is given by the well-known *long-division algorithm**. In the algorithm one constructs the decimal counterpart of the number $|x|$ and equips it with the sign of x , so one may restrict to the case $x \geq 0$. One then chooses $x_n \in \mathbb{Q}_n$, $n = 0, 1, 2, \dots$ so that for every n the condition holds

$$x_n \leq x < x_n + 10^{-n}. \quad (1)$$

If x is not a finite decimal, i.e. $x \notin \mathbb{Q}_n \forall n$, then by condition (1) one has $x_n < x < x_n + 10^{-n} \forall n$. If $x \in \mathbb{Q}_k$ for some k , then $x_n \in \mathbb{Q}_n$ also when $n > k$, so condition (1) implies that $x_n = x$ for every $n \geq k$. In both cases $x_n + 10^{-n}$ is the number in $\mathbb{Q}_n$ nearest to x that is strictly greater than x . It is evident that the sequence $\{x_n\}$ constructed in this way is unique, satisfies

$$|x_n - x| < 10^{-n}, \quad n = 0, 1, \dots \quad (2)$$

and is also uniquely determined by this condition, independently of the sign of x .

If $x \geq 0$, then the construction, by the long-division algorithm, of a sequence satisfying condition (1) proceeds as follows: Let $x = r/s$, where $r \in \mathbb{N} \cup \{0\}$ and $s \in \mathbb{N}$. First, one can write

$$\frac{r}{s} = x_0 + \frac{r_1}{s},$$

where $x_0 \in \mathbb{N} \cup \{0\}$ and $r_1 \in \{0, \dots, s-1\}$. The numbers x_0 and r_1 are evidently unique. Since $0 \leq r_1/s < 1 \Rightarrow 0 \leq 10r_1/s < 10$, one can further find unique $d_1 \in \{0, \dots, 9\}$ and $r_2 \in \{0, \dots, s-1\}$ such that

$$\frac{10r_1}{s} = d_1 + \frac{r_2}{s}.$$

Continuing in this way, the general algorithm becomes

$\frac{10r_k}{s} = d_k + \frac{r_{k+1}}{s}, \quad k = 1, 2, \dots \quad (\text{long-division algorithm})$

Here $d_k \in \{0, \dots, 9\}$ and the numbers $r_{k+1} \in \{0, \dots, s-1\}$ are the so-called *remainders*. If the algorithm is cut off at the index $k = n$, one obtains for the

*An *algorithm* is a precise and executable (programmable) set of instructions for achieving some computational goal. If the goal is a sequence $\{a_n, n = 1, 2, \dots\}$, then by following the algorithm one must obtain, with a finite number of operations, an answer to the question $a_n = ?$, whatever given number the index $n \in \mathbb{N}$ may be.

Infinite binary numbers

By *infinite binary numbers* one means sign/digit strings of the form

$$x = eb_mb_{m+1}\dots b_0.b_1b_2\dots = \{x_n\}_{n=0}^\infty,$$

where the b_n are digits of the binary system, i.e. *bits*, $b_n \in \{0, 1\}$, and

$$x_n = x_0 \pm \sum_{k=1}^n b_k \cdot 2^{-k},$$

where $x_0 = eb_mb_{m+1}\dots b_0$ is the integer part (in binary form) and the sign of the summation expression is the same as in x . If the base of the number system is 8 or 16, one speaks correspondingly of (infinite) *octal* and *hexadecimal numbers*. As number systems used by computers and calculators, these are common. — Everything said above about infinite decimals is easily adapted to the number systems mentioned (and to others).

Scaling of a decimal

For now, no operations other than *sign change* and *scaling* = multiplication by the numbers 10^m , $m \in \mathbb{Z}$ are defined for infinite decimals. Scaling by 10^m simply means shifting the position of the decimal point either m places forward ($m > 0$) or $|m|$ digits backward ($m < 0$). In the latter case, enough 'leading zeros' are added to the integer part as needed. Scaling by $10^0 = 1$ leaves the decimal unchanged. Scaling obeys the commutative and associative laws

$$10^m \cdot (10^n \cdot x) = 10^{m+n} \cdot x = 10^n \cdot (10^m \cdot x), \quad m, n \in \mathbb{Z}.$$

When sign change is interpreted as the same as multiplication by -1 (as in field algebra), then, by combining sign change and scaling, scaling by the numbers $\pm 10^m$, $m \in \mathbb{Z}$ becomes defined.

EXAMPLE 1 (continued)

$$\begin{aligned} 10^{-4} \cdot 141.421356\dots &= 0.0141421356\dots \\ -10^4 \cdot (-0.00314159\dots) &= 31.4159\dots \end{aligned}$$

Floating-point numbers

Floating-point numbers (Swedish *flyttal*) is the name given to the finite decimals in the decimal system of the form

$$\pm 10^\beta \cdot 0.d_1 \dots d_n,$$

where $\beta \in \mathbb{Z}$ and $n \in \mathbb{N}$ is fixed. In numerical computation, floating-point numbers are used as *approximations* of infinite decimals, i.e. as approximate counterparts. The approximation is carried out by first writing the decimal, with the help of a scaling factor, in the form $x = \pm 10^\beta \cdot 0.d_1 d_2 \dots$, where $\beta \in \mathbb{Z}$ and $d_1 \neq 0$ (always possible when $x \neq 0$), and truncating the number thus obtained at the digit d_n . The digits $d_1 \dots d_n$ thus obtained are said to be the first n *significant digits* of the decimal, and this floating-point number is said to be the *truncated* form of the decimal. As an alternative to truncation, *rounding* can be used in the approximation, usually to the nearest floating-point number of the same type (so-called normal rounding). The information about the position of the decimal point relative to the first significant digit is contained in the number $\beta \in \mathbb{Z}$. When an infinite decimal is given to n *significant digits of accuracy*, one means the (truncated or rounded) floating-point approximation of the number.

EXAMPLE 3 To nine ($n = 9$) significant digits,

$$\begin{aligned} 0.0001414213562373095 \dots &\approx 10^{-3} \cdot 0.141421356 \\ -172/3 = -57.333333333 \dots &\approx -10^2 \cdot 0.573333333 \end{aligned}$$

Here truncation and normal rounding give the same result. $\square$

Computers and calculators always operate with floating-point numbers in numerical computations, e.g. ones based on the binary system. Then both the number of significant digits and the number of digits contained in β are limited in a machine-specific way. The possible *machine numbers* are thus always finite in number.

EXAMPLE 4 The machine numbers are of the form $\pm 2^\beta \cdot 0.b_1 \dots b_{30}$, where $\beta = \pm \beta_1 \dots \beta_{10}$, $b_k, \beta_k \in \{0, 1\}$. Storing one number requires 42 bits (digits and signs), and the number of distinct numbers is

$$2^{30} \cdot (2^{11} - 1) + 1 = 2197949513729 \text{ in all.} \quad \square$$

The basic arithmetic operations (addition, subtraction, multiplication, and division) performed on machine numbers by a computer/calculator are called *floating-point operations* (abbreviated flop). These differ slightly from ordinary arithmetic

EXAMPLE 5 Let a and b be given numbers. We wish to compute

$$x = \frac{1}{a} \left(1 - \frac{1}{1+b} \right).$$

Solution The algorithm based directly on the given expression is:

$$y = 1 + b, \quad z = 1/y, \quad u = 1 - z, \quad x = u/a.$$

In the example, a rounding that looks harmless leads to a large error in the final result. The problem disappears if the computations are carried out in *double precision*, i.e. using a floating-point representation that stores 30 significant digits instead of 15. One then obtains the result $x = 2$, which is quite close to the exact value $x = 1.9999999999999996000000000000000000799999\dots$. Switching to double (or multiple) precision is a common procedure when roundoff errors threaten. Another option is a better design of the algorithm.

$$x = \frac{b}{a(1+b)}.$$

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EXERCISES

1. a) Show that if the numbers $p, q \in \mathbb{N}$ have no common factors, then $x = p/q \in \mathbb{Q}$ is a finite decimal exactly when there exist $m, n \in \mathbb{N} \cup \{0\}$ such that $q = 2^m \cdot 5^n$. b) Express the number $x = (2/5)^{10}$ as a finite decimal. c) Which rational numbers are the counterparts of the finite decimals in the number system with base 60? d) The number $1111111/5400$ has, in the base-60 system, a representation corresponding to a finite decimal. Determine it, using for the numbers $n = 0 \dots 59$ the symbols (n) in the base-60 system.
2. Show that if the sequences $\{a_n\}$ and $\{b_n\}$ ($n \in \mathbb{N}$) are increasing, then a) also the sequence $\{a_n + b_n\}$ is increasing, b) if in addition $a_n \geq 0 \forall n$ and $b_n \geq 0 \forall n$, then the sequence $\{a_n b_n\}$ is increasing. c) Show by an example that if the additional condition $b_n \geq 0 \forall n$ in part b) is removed, then the sequence $\{a_n b_n\}$ can be strictly decreasing.
3. Concerning the decimals $x = \{x_n\} \in \mathbb{D}$ and $y = \{y_n\} \in \mathbb{D}$ it is known that $x = 23.4569\dots$ and $y = 23.4567\dots$. On the basis of this information, how small at least and how large at most is the number $x_n - y_n$ when $n = 10$?
4. Formulate and prove the counterpart of Proposition I.5.2 for infinite binary numbers.
5. Using the long-division algorithm, compute the counterparts of the following rational numbers as periodic decimals: a) $1/11$, b) $31/13$, c) $19/17$.
6. Let the counterpart of a number $x \in \mathbb{Q}$ as an infinite decimal, according to the long-division algorithm, be periodic with period length $= m$ and periodicity beginning at the digit d_k . Give an example of a number $x = r/s$, $r, s \in \mathbb{N}$ for which a) $m = 1$, $k = 5$, b) $m = s - 1$, $k = s$, c) $m = s - 1$, $k = 1$.
7. The decimal number $7/11$ is identified, in the base-2 number system, with an infinite binary number, and in the base-3 system with an infinite ternary number. Determine these by the long-division algorithm. *Note* that in the algorithm the remainder is multiplied not by ten but by the base, i.e. by 2 or 3. Continue the computation until the periodicity appears!
8. We wish to compute $y = (x+a)^2 - (x+b)^2$ when $x = 1/7$, $a = 0.1457 \cdot 10^{-6}$ and $b = 0.6973 \cdot 10^{-6}$. a) Determine y to four significant digits by first rewriting the expression for y in a form in which no loss of significant digits

occurs. b) If the original expression is used, to what accuracy must the computations be carried out in order to reach the stated accuracy in the final result?

9. If $x = 7 \cdot 10^{-8}$, then to how many significant digits are

$$\text{a) } \frac{1}{8+3x} \approx 0.125 \quad \text{b) } (1+x)^3 \approx 1+3x ?$$

10. Rewrite the following expressions so that the susceptibility to loss of significant digits is removed. Then compute the values of the expressions to ten significant digits.

$$\begin{aligned} \text{a) } & 10^{20} \left(\frac{1}{1+x} - \frac{1}{1+2x} \right), \quad x = \frac{6}{7} \cdot 10^{-20} \\ \text{b) } & 10^{40} \left(\frac{1}{1+x} - \frac{1}{1-2x} + 3x \right), \quad x = \frac{2}{7} \cdot 10^{-20} \\ \text{c) } & 10^{60} \left(\frac{1+x}{1-x} - \frac{1-x}{1+x} - 4x \right), \quad x = \frac{5}{7} \cdot 10^{-20} \\ \text{d) } & 10^{80} \left(\frac{1+x}{1-x} + \frac{1-x}{1+x} + \frac{2}{1+2x^2} - 4 \right), \quad x = \frac{3}{7} \cdot 10^{-20} \\ \text{e) } & 10^{120} \left(\frac{1+x}{1-x} + \frac{1-x}{1+x} - \frac{2}{1-2x^2+2x^4} \right), \quad x = \frac{4}{7} \cdot 10^{-20} \end{aligned}$$

11. (*) (Grandma's four-function calculator) A four-function calculator can display a number with eight digits, a sign, and a possible decimal point between the digits, e.g. -0.0000999 . How many distinct rational numbers can the calculator display exactly?
12. (*) Let $x = \{x_n\} \in \mathbb{D}$ and $y = \{y_n\} \in \mathbb{D}$. Show that if $x_k < y_k$ for some $k \in \mathbb{N} \cup \{0\}$, then a) $x_n < y_n$ for every $n \geq k$, b) $x_n \leq y_n$ for every n .
13. (*) A forever-computing computer produces the bit string $\{b_n\}_{n=1}^{\infty}$ by the following algorithm:
1. Set $A = 0$, $B = 1$, $n = 0$.
 2. Set

$$n \leftarrow n + 1 \text{ (new value=old value} + 1)$$

$$C = \frac{1}{2}(A + B) \quad \text{and}$$

$$b_n = 1 \text{ and } A = C, \quad \text{if } 3 \cdot C \leq 1$$

$$b_n = 0 \text{ and } B = C, \quad \text{if } 3 \cdot C > 1$$

Output b_n and return to step 2.

- a) What numbers b_n , $n = 1 \dots 4$, does the machine output?
- b) If the final result of the computation (after eternal waiting) is interpreted as the infinite binary number $\check{x} = 0.b_1b_2\dots$, then the binary representation of which rational number x is it? Justify!

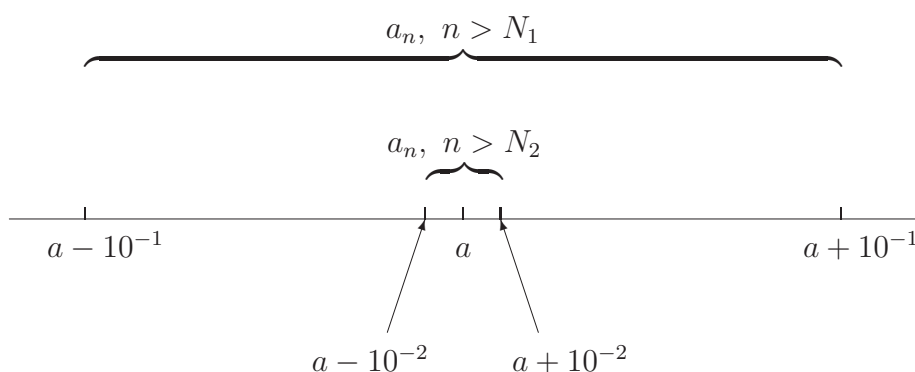
I.6 The limit of a sequence

Here and in the following sections the objects of study are the sequences $\{a_n\}$ of an ordered field. We thus assume that $a_n \in \mathbb{K} \forall n$, where $(\mathbb{K}, +, \cdot, <)$ is an ordered field; this may be either the field of rational numbers or some extension of it in which an order relation is also defined. For example one may have $\mathbb{K} = \mathbb{J}$, cf. Section I.2.

The sequence $\{a_n\}$ is said to *converge* to the number a if 'as n grows, a_n comes ever closer to a '. A more precise definition follows a little later; at this stage it suffices to note that convergence means, for example, the following:

$$\begin{aligned} |a_n - a| &< 10^{-1} && \text{starting from the index } n = N_1, \\ |a_n - a| &< 10^{-2} && \text{starting from the index } n = N_2, \\ &\vdots \\ |a_n - a| &< 10^{-k} && \text{starting from the index } n = N_k, \\ &\vdots \end{aligned}$$

Here each N_k is finite, i.e. $N_k \in \mathbb{N}$ for every $k \in \mathbb{N}$. Convergence is illustrated graphically in the figures below*.



If a sequence converges to a , the number a is called

the *limit* of the sequence. The term *limes* is also used (Latin *limes* = boundary). Notations are

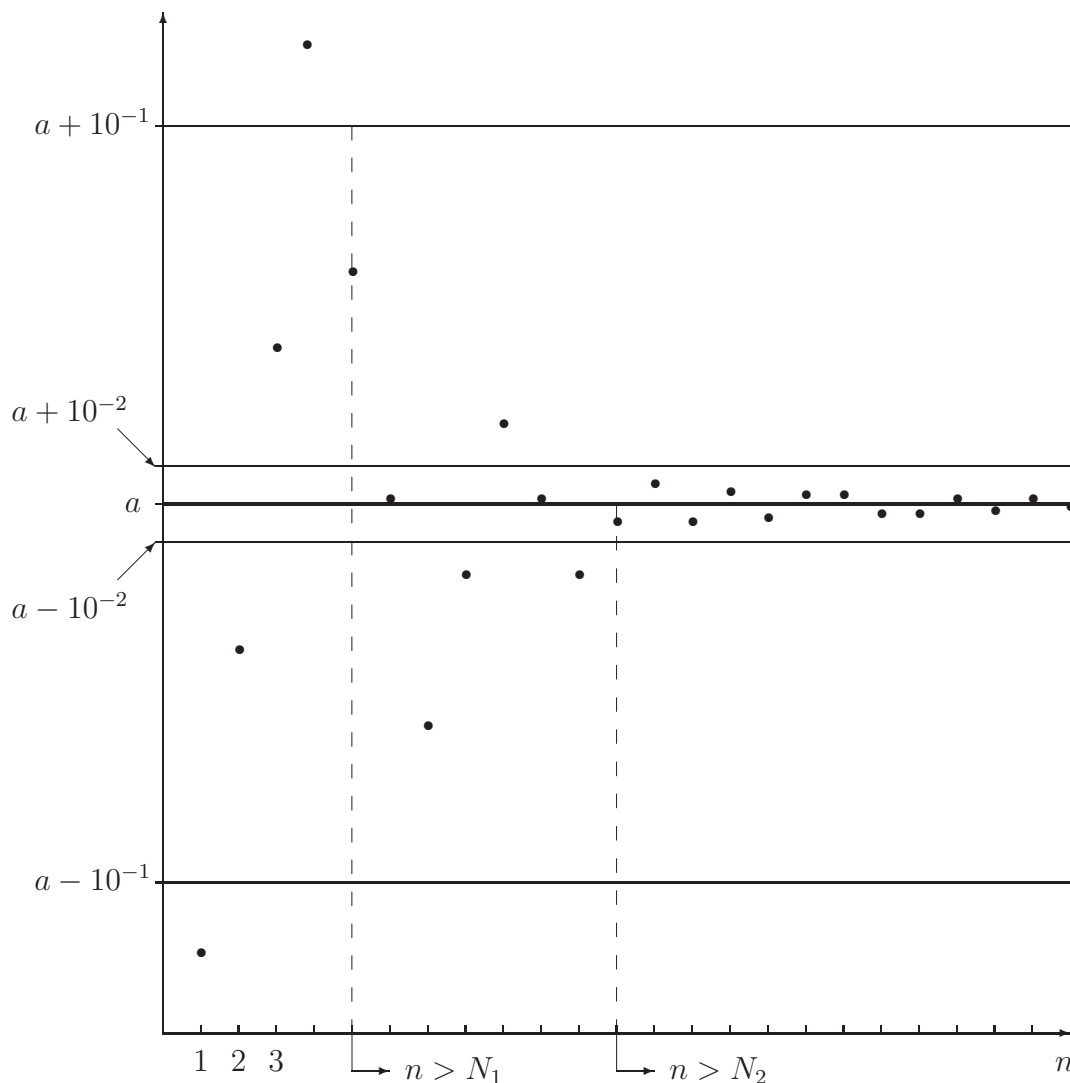
$$\lim_{n \rightarrow \infty} a_n = a \quad \text{or} \quad \lim_n a_n = a$$

*The methods for illustrating numbers graphically (geometrically) are again taken as familiar. We return to the matter later in Chapter II.

or, as is more common in what follows:

$$a_n \rightarrow a \quad (\text{as } n \rightarrow \infty).$$

In the notations 'lim' is read 'limes' and the symbol ' $\rightarrow$ ' is read 'converges to', 'goes to', or 'approaches'.



Convergence is a rather strong requirement for a sequence, and it is indeed easy to give examples of sequences that do not converge to any number. Such is, for instance, the *unboundedly increasing* sequence $\{a_n\} = \{n\}$. In this case it is customary to write

$$a_n \rightarrow \infty \quad \text{as } n \rightarrow \infty \quad \text{or even:} \quad \lim_n a_n = \infty.$$

Permissible readings and turns of phrase are ' a_n approaches infinity' or 'the limit is infinity'. In accordance with these phrasings, the notation ' $\lim_{n \rightarrow \infty}$ ' is usually read 'limes n approaches infinity', meaning the limit as n grows without bound. Correspondingly, if a sequence is *unboundedly decreasing*, one may write $a_n \rightarrow -\infty$ or $\lim_n a_n = -\infty$, read 'the limit is minus infinity'. Despite these notations and turns of phrase, an unboundedly increasing or unboundedly decreasing sequence is not convergent.

EXAMPLE 1 Find the possible limit of the rational sequence $\{a_n\}_{n=1}^{\infty}$, when

- a) $a_n = n^2$ b) $a_n = 100 - n$ c) $a_n = (-1)^n$
 d) $a_n = (n+1)/n$ e) $a_n = (10^{100} + 2n)/(10^{100} + n)$

Solution a) This is an unboundedly increasing sequence: $a_n \rightarrow \infty$. So there is no limit.

b) This sequence is unboundedly decreasing: $a_n \rightarrow -\infty$. There is no limit.

c) The terms of the sequence alternately take the values $+1$ and -1 , but do not 'approach' either of these or any other number. We conclude that this sequence too has no limit.

d) Here is the first convergent sequence: Writing

$$a_n = 1 + \frac{1}{n}$$

one sees that as n grows the terms come ever closer to the number 1, so we conclude that $\lim_n a_n = 1$.

e) This sequence is also convergent, but the true limit is revealed only for very large values of the index: If, for example, one examines the terms of the sequence at the index values $n = N, N+1, \dots, N+10^{10}$, one obtains the following results:

$$N \leq n \leq N + 10^{10} \quad \Rightarrow \quad \begin{cases} a_n \approx 1, & \text{when } N = 1, \\ a_n \approx 1, & \text{when } N = 10^{50}, \\ a_n \approx 3/2, & \text{when } N = 10^{100}, \\ a_n \approx 2, & \text{when } N = 10^{120}. \end{cases}$$

Only the last test tells the truth: $\lim_n a_n = 2$. $\square$

Example 1e) reveals that the limit of a sequence cannot necessarily be 'seen' at once, nor found even by experimenting. This kind of uncertainty can be removed only by a more precise definition of the limit.

DEFINITION I.6.1 (**The limit of a sequence**) The sequence $\{a_n\}$ converges to the limit a if for every $\varepsilon > 0^*$ there exists $N \in \mathbb{N}$ such that

$$|a_n - a| < \varepsilon, \quad \text{when } n > N.$$

The most 'compressed' form of the definition is

$$\lim_n a_n = a \Leftrightarrow \forall \varepsilon > 0 \exists N \in \mathbb{N} (|a_n - a| < \varepsilon \forall n > N).$$

If ∞ takes the place of a , then in the definition the condition $|a_n - a| < \varepsilon$ must be replaced by the condition $a_n > M$, where M is the symbol for an arbitrarily large number. The concise form of the definition is then

$$\lim_n a_n = \infty \Leftrightarrow \forall M \exists N \in \mathbb{N} (a_n > M \forall n > N).$$

In Definition I.6.1, N generally (almost always!) depends on ε . The strength of the dependence does not matter, as long as for every positive ε a finite index N (i.e. a number $N \in \mathbb{N}$) can be found such that the imposed condition holds. The condition, in turn, means that from the index N onward all the terms of the sequence remain at most ε away from a . When the condition is tightened by giving ε ever smaller but positive values, N generally grows, but always remains finite. Correspondingly, if $\lim_n a_n = \infty$, then from the index N onward all the terms of the sequence are greater than the number M . As M grows, N generally grows too, but remains finite for every (finite) value of M .

THEOREM I.6.2 If a sequence $\{a_n\}$ is convergent, then the limit $a = \lim_n a_n$ is unique.

Proof If $a_n \rightarrow a$ and $b \neq a$, then by the triangle inequality (Theorem I.2.3) and the definition of convergence, for every $\varepsilon > 0$ there exists $N \in \mathbb{N}$ such that

$$\begin{aligned} |a_n - b| &= |(a_n - a) + (a - b)| \\ &\geq |a - b| - |a_n - a| \\ &> |a - b| - \varepsilon, \quad \text{when } n > N. \end{aligned}$$

If here one chooses $\varepsilon = \frac{1}{2}|a - b|$ (possible, since $a \neq b \Rightarrow |a - b| > 0$), then for this value of ε

$$|a_n - b| > \varepsilon, \quad \text{when } n > N,$$

*The Greek letter ε (read 'epsilon') is a very well-established symbol for a small positive number in mathematics. Another frequently used symbol, also belonging to the Greek alphabet, is δ (read 'delta').

so by the definition of convergence $a_n \not\rightarrow b$. The claim of the theorem has been proved, in the form

$$a_n \rightarrow a \text{ and } b \neq a \Rightarrow a_n \not\rightarrow b. \quad \square$$

EXAMPLE 1 (continued) c) $a_n = (-1)^n$. If any number a is chosen, then $\forall n \in \mathbb{N}$

$$\max\{|a_n - a|, |a_{n+1} - a|\} = \max\{|1 - a|, |1 + a|\} \geq 1.$$

Consequently, if one chooses $\varepsilon < 1$, then the condition of the convergence definition

$$|a_n - a| < \varepsilon, \text{ when } n > N$$

does not hold for any value of the index N . Thus there is no limit in the sense of Definition I.6.1.

d) $a_n = (n + 1)/n$. When $a = 1$,

$$|a_n - a| = \frac{1}{n} < \varepsilon, \text{ when } n > \frac{1}{\varepsilon}.$$

Consequently, if one chooses, for example, $N = (\text{the smallest integer greater than } 1/\varepsilon)$, then the condition imposed in the definition of the limit holds for this value of N . Thus $a_n \rightarrow 1$.

e) $a_n = (10^{100} + 2n)/(10^{100} + n)$. When one chooses $a = 2$,

$$a_n - a = \frac{10^{100} + 2n}{10^{100} + n} - 2 = -\frac{10^{100}}{10^{100} + n} \Rightarrow |a_n - a| < \frac{10^{100}}{n}.$$

Consequently

$$|a_n - a| < \varepsilon, \text{ when } n \geq 10^{100} \cdot \varepsilon^{-1}.$$

Thus if N is chosen, for example, so that

$$10^{100} \cdot \varepsilon^{-1} \leq N < 10^{100} \cdot \varepsilon^{-1} + 1,$$

then the requirement $|a_n - a| < \varepsilon$ imposed in the definition of convergence holds when $n > N$. Although N is very large (already when $\varepsilon = 1$), it is nonetheless finite for every $\varepsilon > 0$, so by the definition $a_n \rightarrow 2$. $\square$

The following limit result is often needed in practice (and in theory too).

$$\text{PROPOSITION I.6.3 } \lim_{n \rightarrow \infty} q^n = \begin{cases} 0, & \text{if } -1 < q < 1, \\ \infty, & \text{if } q > 1. \end{cases}$$

Proof If $q = 0$, then $q^n = 0 \forall n \in \mathbb{N}$, so $q^n \rightarrow 0$ by the definition of the limit. If $0 < q < 1$, then $q = 1/(1+x)$, where $x > 0$. Then by Bernoulli's inequality (Proposition I.4.2)

$$q^n = \frac{1}{(1+x)^n} < \frac{1}{1+nx} < \frac{1}{nx}, \quad n \geq 2.$$

Here, for every $\varepsilon > 0$,

$$\frac{1}{nx} < \varepsilon, \quad \text{when } n > \frac{1}{x\varepsilon},$$

so by choosing $N \in \mathbb{N}$ such that $N \geq (x\varepsilon)^{-1}$ (always possible when $x > 0$ and $\varepsilon > 0$), it follows that

$$0 < q^n < \varepsilon, \quad \text{when } n > N.$$

By the definition of the limit, $q^n \rightarrow 0$. If finally $-1 < q < 0$, then $q^n = (-1)^n(-q)^n$, where $0 < -q < 1$, so by choosing N in the same way as above it follows that

$$-\varepsilon < q^n < \varepsilon, \quad \text{when } n > N.$$

Again $q^n \rightarrow 0$ by the definition of the limit, so the first claim is proved. The second claim, concerning the case $q > 1$, is proved in a corresponding way, by first writing $q = 1+x$ and using Bernoulli's inequality. $\square$

Regarding the definition of convergence of a sequence, let it be noted further that the condition 'for every $\varepsilon > 0$ ' imposed there can be relaxed without changing the definition. For example, it suffices to choose ε from the sequence $\{10^{-k}, k = 0, 1, 2, \dots\}$, whereupon the definition becomes (cf. the introductory discussion at the start of the section): For every $k = 0, 1, 2, \dots$ there exists an index $N_k \in \mathbb{N}$ such that $|a_n - a| < 10^{-k}$ when $n > N_k$, i.e.:

$$\lim_n a_n = a \quad \Leftrightarrow \quad \forall k \in \mathbb{N} \cup \{0\} \exists N_k \in \mathbb{N} (|a_n - a| < 10^{-k} \quad \forall n > N_k).$$

Here q^k , $0 < q < 1$ could equally well take the place of 10^{-k} , or, still more generally, ε may be picked from any strictly decreasing (see Definition I.5.1) sequence $\{b_k\}$, provided one has first ensured, appealing to Definition I.6.1, that $\lim_k b_k = 0$. Stated more precisely, the result is the following. The proof is left as an exercise (Exercise 7a).

PROPOSITION I.6.4 Let $\{b_k, k = 1, 2, \dots\}$ be a strictly decreasing sequence whose limit is $\lim_k b_k = 0$. Then the limit of the sequence $\{a_n\}$ equals a exactly when for every $k \in \mathbb{N}$ there exists an index $N_k \in \mathbb{N}$ such that $|a_n - a| < b_k$ when $n > N_k$.

The sum of a series

If the series $\sum_{n=m}^{\infty} a_n$ converges (as the sequence of its partial sums, cf. Section I.4), then the limit is called the *sum of the series*. The limit is denoted

$$s = \lim_{n \rightarrow \infty} \sum_{k=m}^n a_k = \sum_{k=m}^{\infty} a_k.^*$$

EXAMPLE 2 Investigate the convergence of the basic geometric series.

Solution The basic geometric series is the sequence

$$\{s_n\} = \left\{ \sum_{k=0}^n q^k \right\} = \begin{cases} \{n+1, n=0, 1, \dots\}, & \text{if } q = 1, \\ \{(q^{n+1} - 1)/(q - 1), n=0, 1, \dots\}, & \text{if } q \neq 1. \end{cases}$$

If $q = 1$, then the series does not converge ($s_n \rightarrow \infty$). If $q = -1$, then $\{s_n\} = \{1, 0, 1, 0, \dots\}$, so the series does not converge in this case either. If $|q| > 1$, then by the triangle inequality we estimate

$$|s_n| = \frac{1}{|q-1|} |q^{n+1} - 1| \geq \frac{1}{|q-1|} (|q|^{n+1} - 1).$$

By means of Proposition I.6.3 we conclude from this that $|s_n| \rightarrow \infty$, so the series does not converge in this case either. In the remaining case $-1 < q < 1$ we choose as the limit candidate $s = 1/(1 - q)$, whereupon

$$|s_n - s| = \frac{|q|^{n+1}}{|1 - q|}.$$

Let $\varepsilon > 0$, whereupon $|1 - q|\varepsilon > 0$ as well. Since $|q|^{n+1} \rightarrow 0$ (Proposition I.6.3) and $|1 - q|\varepsilon > 0$, then by the definition of the limit of a sequence there exists an index $N \in \mathbb{N}$ such that

$$|q|^{n+1} < |1 - q|\varepsilon \quad \Rightarrow \quad |s_n - s| < \varepsilon, \quad \text{when } n > N.$$

Here $\varepsilon > 0$ was arbitrary and $N \in \mathbb{N}$, so by the definition of the limit of a sequence $\lim_n s_n = s$. Thus the geometric series converges exactly when $|q| < 1$, and the limit is then

$$\lim_n s_n = \lim_n \left(\sum_{k=0}^n q^k \right) = \frac{1}{1 - q}. \quad \square$$

*If the series is convergent, then it must be clear from the context whether $\sum_{n=m}^{\infty} a_n$ means the series itself or its sum. For example, the notation $\sum_{k=m}^{\infty} a_k = \sum_{k=m}^{\infty} b_k$ may mean either that the series are the same as sequences (in which case they are also the same term by term: $a_k = b_k \forall k$), or only that the sums of the series are the same.

Using the series-sum notation, the result of the example is

$$\sum_{k=0}^{\infty} q^k = \frac{1}{1-q}, \quad |q| < 1.$$

A rational sequence whose limit is not rational

The limit of a rational sequence need not be a rational number, for according to Definition I.6.1 it suffices that the limit can be found in a number set $\mathbb{K} \supset \mathbb{Q}$ in which the operations and the order relation are defined so that $(\mathbb{K}, +, \cdot, <)$ is an extension of the field of rational numbers. What follows is an example of a rational sequence that converges, provided the number set $\mathbb{K} \supset \mathbb{Q}$ also contains the number $a = \sqrt{2}$ (cf. Section I.2, Example 3).

EXAMPLE 3 Consider the recursive rational sequence

$$a_0 = 2, \quad a_{n+1} = \frac{a_n}{2} + \frac{1}{a_n}, \quad n = 0, 1, \dots$$

From the recursion formula it follows that $a_n > 0 \forall n$ (induction!) and that

$$a_{n+1}^2 - 2 = \frac{1}{4a_n^2} (a_n^2 - 2)^2, \quad n = 0, 1, \dots$$

From this one sees (induction!) that necessarily $a_n^2 - 2 > 0 \forall n$, whereupon one can further conclude:

$$a_{n+1}^2 - 2 = \frac{a_n^2 - 2}{4a_n^2} (a_n^2 - 2) < \frac{a_n^2}{4a_n^2} (a_n^2 - 2) = \frac{1}{4} (a_n^2 - 2), \quad n = 0, 1, \dots$$

Since $a_0^2 - 2 = 2$, it follows (induction!) that

$$0 < a_n^2 - 2 < 2 \cdot 4^{-n}, \quad n = 0, 1, \dots$$

By Propositions I.6.3 and I.6.4 it follows from this that $a_n^2 \rightarrow 2$.

We have concluded that $a_n > 0 \forall n$ and, in addition, that the sequence $\{a_n^2\}$ is convergent and $a_n^2 \rightarrow 2$. We now show that, according to Definition I.6.1, necessarily $\lim_n a_n = a = \sqrt{2}$. With this in view, we first write

$$a_n^2 - 2 = a_n^2 - a^2 = (a_n + a)(a_n - a).$$

Since $a_n > 0$ and $a_n^2 > 2 = a^2$ for every n , we have $a_n > a \forall n$ (see Section 1.2). Also $a > 1$, so it follows that

$$\begin{aligned} a_n^2 - 2 &= (a_n + a)(a_n - a) > 2a(a_n - a) > 2(a_n - a) \\ \Rightarrow a_n - a &< 2^{-1}(a_n^2 - 2) < 4^{-n}, \quad n = 0, 1, \dots \end{aligned}$$

Thus $0 < a_n - a < 4^{-n} \forall n$, so $a_n \rightarrow a$ (cf. the reasoning above). A non-rational limit $a = \sqrt{2}$ has thus been found for the sequence $\{a_n\}$. $\square$

The sequence $\{a_n\}$ of the example cannot have a rational limit, since $\sqrt{2} \notin \mathbb{Q}$ and, by Theorem 1.6.2, the limit is unique. Among rational sequences there are thus ones that 'appear to converge' but nonetheless have no limit in the set of rational numbers. When a rational sequence can, more generally, be given a limit by extending the number set (the field of rational numbers) and when it cannot is a central question in what follows.

EXERCISES

1. In the following examples the sequence $\{a_n\}$ either converges to a number a or grows without bound. Determine the smallest $N \in \mathbb{N}$ such that either $|a_n - a| < 1/100$ or $a_n > 100$ holds when $n > N$:

$$\begin{array}{lll} \text{a) } a_n = \frac{4n-1}{2n+1} & \text{b) } a_n = \frac{4n+100}{2n+100} & \text{c) } a_n = \frac{3n-10000}{2n+10000} \\ \text{d) } a_n = \frac{n^2}{2n+1} & \text{e) } a_n = \frac{n^2}{3n+100} & \text{f) } a_n = \frac{n^2}{4n+10000} \end{array}$$

2. Let

$$a_n = \begin{cases} 1/k, & \text{when } n = 10^k, \ k \in \mathbb{N}, \\ 0, & \text{otherwise } (n \in \mathbb{N}), \end{cases} \quad b_n = \begin{cases} 10^{-100}, & \text{when } n = 10^k, \ k \in \mathbb{N}, \\ 0, & \text{otherwise } (n \in \mathbb{N}). \end{cases}$$

Show that a) $a_n \rightarrow 0$, b) $\{b_n\}$ does not converge to any number.

3. Prove that for sequences:

- a) $\lim_n a_n = a \Rightarrow \lim_n |a_n| = |a|$.
- b) $\lim_n a_n = a$ and $a_n \geq b_n \geq a \forall n \Rightarrow \lim_n b_n = a$.
- c) $a_n \rightarrow a$ and $b_n \rightarrow b \Leftrightarrow c_n = \max\{|a_n - a|, |b_n - b|\} \rightarrow 0$.
- d) $a_n \rightarrow a$ and $b_n \rightarrow b \Leftrightarrow (a_n - a)^2 + (b_n - b)^2 \rightarrow 0$.
- e) $a_n \rightarrow a$ and $b_n \rightarrow b$ and $c_n \rightarrow c \Leftrightarrow (a_n - a)^2 + (b_n - b)^2 + (c_n - c)^2 \rightarrow 0$.

4. Concerning the sequences $\{a_n\}$ and $\{b_n\}$ it is known that $a_n \rightarrow a$ and $b_n \rightarrow a$, and concerning the sequence $\{c_n\}$, that for every n either $c_n = a_n$ or $c_n = b_n$ (it is not known which). Show that $c_n \rightarrow a$.
5. Let $q \in \mathbb{Q}$, $q > 0$. Compute $\lim_n q^{-n} \sum_{k=0}^n 2^k$ for different values of q .
6. Determine the smallest numbers N_k , $k = 1, 2, 3$ in the sense of Proposition I.6.4, when $b_k = 10^{-k}$ and $a_n = 0.9^n$, $n \in \mathbb{N}$.
7. (*) a) Prove Proposition I.6.4. b) Formulate and prove the counterpart of Proposition I.6.4 for a sequence $\{a_n\}$ that grows without bound.
8. (*) Prove: $a_n > 0 \forall n$ and $\lim_n a_n = 0 \Rightarrow \exists \max_n \{a_n\}$. Show also, by a counterexample, that $\lim_n a_n = 0 \not\Rightarrow \exists \max_n \{a_n\}$.
9. (*) Let $k \in \mathbb{N}$, $\mathbb{Q}_k = \{\text{finite decimals of the form } x_0.d_1 \dots d_k\}$, and $a_n \in \mathbb{Q}_k \forall n \in \mathbb{N}$. Show that if $\lim_n a_n = a$, then $a \in \mathbb{Q}_k$ and for some $N \in \mathbb{N}$: $a_n = a$ when $n > N$.
10. (*) Let $\{a_k, k = 1, 2, \dots\}$ be a rational sequence and $\{N_k, k = 1, 2, \dots\}$ a strictly increasing index sequence (a sequence of natural numbers) such that $N_1 = 1$. Define the sequence $\{b_n\}$ by setting

$$b_n = a_k, \quad \text{when } N_k \leq n < N_{k+1}, \quad k = 1, 2, \dots$$

Prove: $\lim_k a_k = a \Rightarrow \lim_n b_n = a$.

11. (*) Consider, in the field $(\mathbb{J}, +, \cdot, <)$ (see Section I.2), the recursively defined sequence

$$a_0 = x, \quad a_{n+1} = \frac{1}{2} \left(a_n + \frac{x}{a_n} \right), \quad n = 0, 1, \dots$$

where $x \in \mathbb{J}$, $x > 0$. Show:

- a) $a_{n+1}^2 - x = \frac{1}{4a_n^2} (a_n^2 - x)^2, \quad n = 0, 1, \dots$
- b) $a_n \geq \sqrt{x}, \quad n = 1, 2, \dots$
- c) $\lim_n a_n^2 = x$.
- d) $\lim_n a_n = \sqrt{x}$.

I.7 Properties of convergent sequences

In this section we examine the properties of convergent sequences from a general standpoint — ‘number’ (math.) is understood as in the previous section.

DEFINITION I.7.1 The sequence $\{a_n\}$ is *bounded* if there exists a number C such that $|a_n| \leq C \quad \forall n$.

THEOREM I.7.2 A convergent sequence is bounded.

Proof Let the sequence be of the general form $\{a_n\}_{n=m}^{\infty}$, $m \in \mathbb{Z}$. If $a_n \rightarrow a$, then by the definition of convergence $\exists N \in \mathbb{N}$ such that

$$|a_n - a| < 1, \quad \text{when } n > N$$

($\varepsilon = 1$ chosen in Definition I.6.1). Then by the triangle inequality

$$\begin{aligned} |a_n| &= |(a_n - a) + a| \\ &\leq |a_n - a| + |a| < 1 + |a|, \quad \text{when } n > N, \end{aligned}$$

so

$$|a_n| \leq \max\{|a_m|, \dots, |a_N|, |a| + 1\} = C \quad \forall n \geq m. \quad \square$$

The following general rules of calculation are remarkably useful both in computing limits and in more theoretical considerations.

THEOREM I.7.3 (Rules for combining limits) Let $\{a_n\}$ and $\{b_n\}$ be sequences and a, b, λ numbers. Then if $\lim_n a_n = a$ and $\lim_n b_n = b$:

- (1) $\lim_n (a_n + b_n) = a + b$.
- (2) $\lim_n (\lambda a_n) = \lambda a$.
- (3) $\lim_n (a_n b_n) = ab$.
- (4) $\lim_n (a_n / b_n) = a/b$ with the additional conditions: $b_n \neq 0 \quad \forall n$ and $b \neq 0$.

We prove only the claims (3) and (4) (the others are easier: Exercise 1).

Claim (3) First we write

$$\begin{aligned} a_n b_n - ab &= (a_n b_n - ab_n) + (ab_n - ab) \\ &= b_n (a_n - a) + a(b_n - b). \end{aligned}$$

Since $\{b_n\}$ is a convergent sequence, then by Theorem I.7.2 there is some C with

$$|b_n| \leq C \quad \forall n,$$

whereupon by the triangle inequality

$$|a_n b_n - ab| \leq C|a_n - a| + |a||b_n - b|.$$

In what follows we assume that $C > 0$ and $|a| > 0$ (if $C = 0$ or $a = 0$, the reasoning simplifies). Choose an arbitrary $\varepsilon > 0$, whereupon

$$\varepsilon_1 = \frac{\varepsilon}{2C} > 0 \quad \text{and} \quad \varepsilon_2 = \frac{\varepsilon}{2|a|} > 0.$$

Since the sequences $\{a_n\}$ and $\{b_n\}$ converge, indices $N_1 \in \mathbb{N}$ and $N_2 \in \mathbb{N}$ can then be found such that

$$\begin{aligned} |a_n - a| &< \varepsilon_1, & \text{when } n > N_1, \\ |b_n - b| &< \varepsilon_2, & \text{when } n > N_2. \end{aligned}$$

Here both inequalities hold from the index $N = \max\{N_1, N_2\} \in \mathbb{N}$ onward, so

$$|a_n b_n - ab| < C\varepsilon_1 + |a|\varepsilon_2 = \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon, \quad \text{when } n > N.$$

Since here $\varepsilon > 0$ was arbitrary and $N \in \mathbb{N}$ for every $\varepsilon > 0$, then by the definition of the limit of a sequence $a_n b_n \rightarrow ab$. $\square$

Claim (4) Here we begin by writing

$$\begin{aligned} \frac{a_n}{b_n} - \frac{a}{b} &= \frac{a_n b - ab_n}{b_n b} = \frac{(a_n b - ab) + (ab - ab_n)}{b_n b} \\ &= \frac{a_n - a}{b_n} - \frac{a(b_n - b)}{b_n b}. \end{aligned}$$

Since $b_n \rightarrow b \neq 0$, then $\exists N_0 \in \mathbb{N}$ such that

$$|b_n - b| < \frac{1}{2}|b|, \quad \text{when } n > N_0$$

($\varepsilon = \frac{1}{2}|b|$, $N = N_0$ imposed in the definition of the limit). Then by the triangle inequality

$$|b_n| \geq |b| - |b_n - b| > \frac{1}{2}|b| \quad \forall n > N_0,$$

so

$$\begin{aligned} \left| \frac{a_n}{b_n} - \frac{a}{b} \right| &\leq \frac{|a_n - a|}{|b_n|} + \left| \frac{a}{b_n b} \right| |b_n - b| \\ &< \frac{2}{|b|} |a_n - a| + \frac{2|a|}{b^2} |b_n - b| \quad \forall n > N_0. \end{aligned}$$

Here the right-hand side is $< \varepsilon$ ($\varepsilon > 0$) when

$$\frac{2}{|b|} |a_n - a| < \frac{\varepsilon}{2} \quad \text{and} \quad \frac{2|a|}{b^2} |b_n - b| < \frac{\varepsilon}{2},$$

i.e. when

$$|a_n - a| < \frac{|b|}{4} \varepsilon = \varepsilon_1 \quad \text{and} \quad |b_n - b| < \frac{b^2}{4|a|} \varepsilon = \varepsilon_2,$$

assuming here that $a \neq 0$. (If $a = 0$, the reasoning simplifies.) Since $a_n \rightarrow a$ and $b_n \rightarrow b$ and since $\varepsilon_1, \varepsilon_2 > 0$, the first condition holds from some index N_1 and the second from some index N_2 onward, whereupon

$$\left| \frac{a_n}{b_n} - \frac{a}{b} \right| < \varepsilon, \quad \text{when } n > N = \max\{N_0, N_1, N_2\} \in \mathbb{N}.$$

Here $\varepsilon > 0$ was arbitrary, so by the definition $a_n/b_n \rightarrow a/b$. $\square$

EXAMPLE 1 Compute $\lim_n a_n$, when

$$a_n = \frac{n^2 + 2n + 2}{3n^2 + n}.$$

Solution By writing a_n in the form

$$a_n = \frac{1 + 2 \cdot \frac{1}{n} + 2 \cdot \frac{1}{n} \cdot \frac{1}{n}}{3 + \frac{1}{n}}$$

one can directly apply rules (1)–(4) of Theorem I.7.3:

$$\begin{aligned} \lim_n a_n &= \frac{\lim_n (1 + 2 \cdot \frac{1}{n} + 2 \cdot \frac{1}{n} \cdot \frac{1}{n})}{\lim_n (3 + \frac{1}{n})} \\ &= \frac{\lim_n (1) + 2 \lim_n (\frac{1}{n}) + 2 \lim_n (\frac{1}{n}) \cdot \lim_n (\frac{1}{n})}{3 + \lim_n (\frac{1}{n})} \\ &= \frac{1 + 2 \cdot 0 + 2 \cdot 0 \cdot 0}{3 + 0} = \frac{1}{3}. \end{aligned}$$

Here, then, only one limit result provable directly from the definition was needed, $\lim_n (\frac{1}{n}) = 0$ (cf. Example 1d in the previous section). $\square$

EXAMPLE 2 The general geometric series is the sequence

$$\sum_{k=0}^{\infty} aq^k = \left\{ \sum_{k=0}^n aq^k, \quad n = 0, 1, \dots \right\}.$$

When the sum of the basic series is known (Example 2 of the previous section), then by rule (2) of Theorem I.7.3 the sum of the general geometric series is

$$\sum_{k=0}^{\infty} aq^k = a \sum_{k=0}^{\infty} q^k = \frac{a}{1-q}, \quad |q| < 1.$$

If $|q| \geq 1$, then the general geometric series converges only when $a = 0$. $\square$

EXAMPLE 3 In Example 3 of the previous section the recursive sequence

$$a_0 = 2, \quad a_{n+1} = \frac{a_n}{2} + \frac{1}{a_n}, \quad n = 0, 1, \dots$$

was considered. If one assumes that $a_n > 0 \forall n$ and $a_n \rightarrow a > 0$ (cf. the considerations of the previous section), then by Theorem I.7.3 necessarily

$$a = \lim_n a_{n+1} = \lim_n \left(\frac{a_n}{2} + \frac{1}{a_n} \right) = \frac{a}{2} + \frac{1}{a}.$$

Thus

$$a = \frac{a}{2} + \frac{1}{a} \quad \Rightarrow \quad a^2 = 2.$$

Since $a > 0$, it follows that $a = \sqrt{2}$. $\square$

EXAMPLE 4 If the recursive sequence

$$a_{n+1} = qa_n - 1, \quad n = 0, 1, \dots$$

converges (for some a_0) to a limit a , then from this assumption it follows that:

$$\begin{aligned} q = 1 & \Rightarrow a = a - 1 \Rightarrow 0 = 1. \\ q \neq 1 & \Rightarrow a = qa - 1 \Rightarrow a = 1/(q - 1). \end{aligned}$$

We thus conclude that the sequence does not converge (for any a_0) if $q = 1$, and that in the other cases the only possible limit is $a = 1/(q - 1)$. A closer study of the sequence reveals that

$$a_n = a_0 q^n - \sum_{k=0}^{n-1} q^k, \quad n = 1, 2, \dots,$$

so the sequence converges, independently of the initial value a_0 , exactly when $|q| < 1$, and then $a_n \rightarrow 1/(q - 1)$. — Note that the reasoning in the case $q = 1$ is in fact an indirect proof of the claim: $\{a_n\}$ does not converge. (Cf. Section I.3.)

$\square$

Next, three more claims that rest directly on the definition of the limit of a sequence. The middle of these is known as the '*sandwich theorem*'.

THEOREM I.7.4 For sequences:

V1. If $a_n \leq b \ \forall n$ ($a_n \geq b \ \forall n$) and $a_n \rightarrow a$, then $a \leq b$ ($a \geq b$).

V2. If $a_n \rightarrow c$ and $b_n \rightarrow c$ and $a_n \leq c_n \leq b_n \ \forall n$, then $c_n \rightarrow c$.

V3. If $a_n \rightarrow 0$ and the sequence $\{b_n\}$ is bounded, then $a_n b_n \rightarrow 0$.

Proof V1. Assume that $a_n \leq b \ \forall n$, and make the counter-assumption: $\lim_n a_n = a > b$. Set $\varepsilon = a - b$, whereupon by the counter-assumption $\varepsilon > 0$. Since $a_n \leq b \ \forall n$ (assumption) and $b = a - \varepsilon$, $\varepsilon > 0$ (consequence of the counter-assumption), it follows that $a_n \leq a - \varepsilon \ \forall n$ and hence $|a_n - a| \geq \varepsilon > 0 \ \forall n$. From the definition of convergence it then follows that $a_n \not\rightarrow a$. But by assumption $a_n \rightarrow a$, so we have arrived at a logical contradiction: $a_n \rightarrow a$ and $a_n \not\rightarrow a$. The counter-assumption $a > b$ is thus false, i.e. necessarily $a \leq b$. The first part of the claim is thus proved. The second part follows when the already proved claim is applied to the sequence $\{-a_n\}$.

V2. Since $a_n \leq c_n \leq b_n \ \forall n$,

$$\begin{aligned} a_n - c &\leq c_n - c \leq b_n - c \quad \text{and} \quad c - b_n \leq c - c_n \leq c - a_n \\ &\Rightarrow |c - c_n| \leq \max\{|a_n - c|, |b_n - c|\} \ \forall n. \end{aligned}$$

Since $a_n \rightarrow c$ and $b_n \rightarrow c$, then for every $\varepsilon > 0$ there exist indices $N_1, N_2 \in \mathbb{N}$ such that $|a_n - c| < \varepsilon$ when $n > N_1$ and $|b_n - c| < \varepsilon$ when $n > N_2$. Then

$$|c - c_n| \leq \max\{|a_n - c|, |b_n - c|\} < \varepsilon, \quad \text{when } n > N = \max\{N_1, N_2\}.$$

By the definition, $c_n \rightarrow c$.

V3. Exercise (Exercise 6a). $\square$

EXAMPLE 5 The claim (cf. claim V1 of Theorem I.7.4)

$$a_n < b \ \forall n \quad \text{and} \quad a_n \rightarrow a \quad \Rightarrow \quad a < b$$

is shown to be false by the counterexample $a_n = 1 - 1/n$, $n \in \mathbb{N}$, $b = 1$. $\square$

EXAMPLE 6 The recursive sequence defined in the field $(\mathbb{J}, +, \cdot, <)$ (see Section I.2)

$$a_0 = 0, \quad a_{n+1} = \sqrt{a_n + 1}, \quad n = 0, 1, \dots$$

is stated to be convergent. What is the limit?

Solution From the recursion formula it follows that $a_n \geq 0 \forall n$ and that

$$a_{n+1}^2 = a_n + 1, \quad n = 0, 1, \dots$$

Setting $\lim_n a_n = a$, it follows by Theorem I.7.3 that

$$\lim_n a_{n+1}^2 = a^2 = \lim_n (a_n + 1) = a + 1.$$

Thus $a^2 - a - 1 = 0 \Rightarrow a = \frac{1}{2}(1 \pm \sqrt{5})$. Since $a_n \geq 0 \forall n$, necessarily $a \geq 0$ (Theorem I.7.4, claim V1). Thus $\lim_n a_n = \frac{1}{2}(1 + \sqrt{5})$. $\square$

Periodic decimals

In Section I.5 it was noted that to every rational number $x \in \mathbb{Q}$ one can associate an infinite, periodic decimal $\check{x} = \{x_n\} \in \mathbb{D}_p$ by the criterion

$$|x - x_n| < 10^{-n} \quad \forall n.$$

This means (cf. Proposition I.6.4) that

$$\lim_n x_n = x.$$

In what follows it is natural to loosen slightly the identification relation agreed on in Section I.5 between rational numbers and infinite decimals. We set

DEFINITION I.7.5 If $x \in \mathbb{Q}$ and $\check{x} = \{x_n\} \in \mathbb{D}$, then

$$x = \check{x} \Leftrightarrow \lim_n x_n = x.$$

By the criterion of Definition I.7.5, then, every rational number is still identified with at least one periodic decimal, namely the one given by the long-division algorithm. And conversely: if one starts from a periodic decimal, is it always identifiable with some rational number in the sense of Definition I.7.5? To investigate the matter, suppose the decimal $\check{x}$ is of the form

$$\check{x} = x_0.d_1 \dots d_k d_{k+1} \dots d_{k+m} d_{k+1} \dots d_{k+m} \dots,$$

i.e. the periodicity begins (at the latest) at digit no. $k+1$ and the period length is m . Consider the sequence $\{x_k, x_{k+m}, x_{k+2m}, \dots\} = \{x_{k+lm}, l = 0, 1, \dots\} = \{y_l\}$. For it,

$$y_{l+1} - y_l = \sum_{i=k+1}^{k+m} d_i \cdot 10^{-lm-i} = 10^{-lm} (y_1 - y_0), \quad l = 0, 1, \dots$$

so

$$\begin{aligned} y_{l+1} &= y_0 + (y_1 - y_0) + \dots + (y_{l+1} - y_l) \\ &= y_0 + (y_1 - y_0) \sum_{j=0}^l 10^{-jm}, \quad l = 0, 1, \dots \end{aligned}$$

Consequently the sequence $\{y_l\}$ converges as a geometric series:

$$\lim_{l \rightarrow \infty} y_l = y_0 + \frac{y_1 - y_0}{1 - 10^{-m}} = x_k + \frac{x_{k+m} - x_k}{1 - 10^{-m}} = x \in \mathbb{Q}.$$

From this and the monotonicity of the sequence $\{x_n\}$ (cf. Section I.5) it follows that the whole sequence $\{x_n\}$ converges as well. Namely, if for example $x > 0$, then the sequence $\{x_n\}$ is increasing, whereupon for every $n > k$

$$x_{k+lm} = y_l \leq x_n \leq y_{l+1} = x_{k+(l+1)m},$$

where $l \in \mathbb{N} \cup \{0\}$ is chosen (depending on n) so that

$$k + lm < n \leq k + (l + 1)m.$$

Here, when $n \rightarrow \infty$, then $l \rightarrow \infty$, whereupon $y_l \rightarrow x$ and $y_{l+1} \rightarrow x$ and hence $x_n \rightarrow x$ (cf. Theorem I.7.4, claim V2). By the identification convention I.7.5 it has thus been proved (see also Theorem I.6.2)

THEOREM I.7.6 Every periodic decimal is identified, in the sense of Definition I.7.5, with a unique rational number.

EXAMPLE 7

$$5.5027027027\dots = 5.5 + 10^{-1} \cdot \frac{0.027}{1 - 10^{-3}} = \frac{55}{10} + \frac{27}{9990} = \frac{1018}{185}. \quad \square$$

EXAMPLE 8 Which decimals are identified with the numbers 0, 10, and $3/125$?

Solution In every case there are two alternatives:

$$\begin{aligned} 0 &= +0.0000000\dots = -0.0000000\dots \\ 10 &= +10.000000\dots = +9.9999999\dots \\ 3/125 &= +0.0080000\dots = +0.0079999\dots \quad \square \end{aligned}$$

Identification of decimals

By Example 8, it is possible that two decimals $\check{x}, \check{y} \in \mathbb{D}$ that are different as strings (i.e. non-identical) are identified with the same rational number x . In that case it is natural to identify $\check{x}$ and $\check{y}$ with each other as well, i.e. to write $\check{x} = \check{y}$. If $\check{x} = \{x_n\}$ and $\check{y} = \{y_n\}$, then by Definition 1.7.5 in the situation mentioned $\lim_n x_n = \lim_n y_n = x$, whereupon by Theorem 1.7.3 $\lim_n (x_n - y_n) = 0$. We take this limit condition as the general identification criterion when comparing infinite decimals directly with each other:

DEFINITION 1.7.7 Two decimals $\check{x} = \{x_n\} \in \mathbb{D}$ and $\check{y} = \{y_n\} \in \mathbb{D}$ are identified by the criterion

$$\check{x} = \check{y} \quad \Leftrightarrow \quad \lim_n (x_n - y_n) = 0.$$

The identification of two 'different-looking' decimals is thus possible at least if both are identified with a finite decimal. — This turns out to be the only special case:

THEOREM 1.7.8 If the decimals $\check{x} = \{x_n\} \in \mathbb{D}$ and $\check{y} = \{y_n\} \in \mathbb{D}$ are identified with each other in the sense of Definition 1.7.7, then either $\check{x}$ and $\check{y}$ are identical as strings or $\check{x} = x = \check{y}$, where x is a finite decimal.

Proof If $\{x_n\} = \{0\}$, then also $\{y_n\} = \{0\}$, for otherwise $|x_n - y_n| = |y_n| \not\rightarrow 0$, since the sequence $\{|y_n|\}$ is increasing. Then $\check{x}, \check{y} = \pm 0.000\dots$ and $\check{x} = 0 = \check{y}$. In what follows we assume that $\{x_n\} \neq \{0\}$ and $\{y_n\} \neq \{0\}$. Then the signs of $\check{x}$ and $\check{y}$ must be the same, otherwise $|x_n - y_n| = |x_n| + |y_n| \not\rightarrow 0$. Since, moreover, $\check{x}$ and $\check{y}$ can each be scaled by a factor $\pm 10^m$, $m \in \mathbb{Z}$ without changing the identification criterion, we may assume that $0 \leq x_n < 1$ and $0 \leq y_n < 1$ for every n , whereupon $\check{x}$ and $\check{y}$ are, for some $k \in \mathbb{N}$, of the form

$$\begin{aligned} \check{x} &= 0.d_1 \dots d_{k-1} d_k \dots \dots d_m \dots \dots, \\ \check{y} &= 0.d_1 \dots d_{k-1} \tilde{d}_k \dots \dots \tilde{d}_m \dots \dots, \end{aligned}$$

where $\tilde{d}_k \neq d_k$. Then $x_n = y_n$ when $n < k$ and $x_k - y_k = (d_k - \tilde{d}_k) 10^{-k}$. Without loss of generality we may further assume that $\tilde{d}_k < d_k$ ($d_k \neq 0$). Then one sees that the difference $x_n - y_n$ attains its smallest possible value for every $n > k$ when $d_n = 0$, $\tilde{d}_n = 9 \forall n > k$, i.e. when $\check{x}$ and $\check{y}$ are of the form

$$\begin{aligned} \check{x} &= 0.d_1 \dots d_{k-1} d_k 000\dots \\ \check{y} &= 0.d_1 \dots d_{k-1} \tilde{d}_k 999\dots \end{aligned}$$

In this case (cf. Proposition I.5.2)

$$\begin{aligned} x_n - y_n &= (x_k + 0 \cdot 10^{-k-1} + \dots + 0 \cdot 10^{-n}) - (y_k + 9 \cdot 10^{-k-1} + \dots + 9 \cdot 10^{-n}) \\ &= (x_k - y_k) - (10^{-k} - 10^{-n}) \\ &= (d_k - \tilde{d}_k - 1) 10^{-k} + 10^{-n}, \quad n > k. \end{aligned}$$

Thus the smallest possible value of the difference $x_n - y_n$ is 10^{-n} when $n \geq k$, and this is attained when $d_k = \tilde{d}_k + 1$ and $d_n = 0$ and $\tilde{d}_n = 9$ for every $n > k$. In this special case $x_n = x_k \forall n \geq k$ and $\lim_n y_n = x_k$, whereupon $\check{x} = x_k = \check{y}$ in the sense of Definition I.7.5.

In cases other than the aforementioned special case, either $d_k - \tilde{d}_k \geq 2$ (case 1), $d_m \geq 1$ for some $m > k$ (case 2), or $\tilde{d}_m \leq 8$ for some $m > k$ (case 3). In case 1 it follows from the above result that $x_n - y_n > 10^{-k} = \varepsilon > 0 \forall n > k$, whereupon $x_n - y_n \not\rightarrow 0$. In cases 2 and 3 necessarily $x_m - y_m \geq 2 \cdot 10^{-m}$ (since the smallest value of the difference was 10^{-m}). Then, since the sequence $\{x_n\}$ is increasing and since $y_n < y_m + 10^{-m} \forall n > m$ (Proposition I.5.2), it follows that $x_n - y_n > 10^{-m} = \varepsilon > 0$ when $n > m$. Thus in this case too $x_n - y_n \not\rightarrow 0$. We have concluded that if $\check{x}$ and $\check{y}$ are not identical as strings, then the assumption $\lim_n (x_n - y_n) = 0$ holds only in the special case described above, in which $\check{x}$ and $\check{y}$ are identified with a finite decimal. $\square$

As a summary of the identification of rational numbers and infinite decimals, we can state:

COROLLARY I.7.9 Every rational number x is identified with at least one and at most two infinite decimals, which are periodic. There are two identification alternatives exactly when x is a finite decimal.

EXAMPLE 9 With which infinite binary numbers are the decimal numbers a) $x = 3/4$, b) $y = 4/3$ identified?

Solution a) Since $3/4 = 2^{-2}(1 \cdot 2 + 1) = 1 \cdot 2^{-1} + 1 \cdot 2^{-2}$,

$$x = 0.11000000\dots = 0.10111111\dots$$

There are two alternatives, since x is a finite binary number.

b) Since

$$\frac{4}{3} = \frac{1}{1 - \frac{1}{4}} = 1 + \frac{1}{4} + \frac{1}{4^2} + \dots = 1 \cdot 2^0 + 0 \cdot 2^{-1} + 1 \cdot 2^{-2} + 0 \cdot 2^{-3} + \dots,$$

we have $y = 1.010101\dots$. Here there is only one alternative. $\square$

EXERCISES

1. Prove claims (1) and (2) of Theorem I.7.3.

2. Using the rules for combining limits, determine

$$\begin{array}{ll} \text{a) } \lim_n \frac{2n^3 - 100n^2 - 5000n}{(n + 200)^3} & \text{b) } \lim_n \frac{1}{n^3} [(n + 3)^4 - (n - 2)^4] \\ \text{c) } \lim_n \left(\frac{1 + 2 + \dots + n}{n + 2} - \frac{n}{2} \right) & \text{d) } \lim_n \left[\frac{(n + 2)! - n!}{(n + 1)! + (n - 1)!} - n \right] \end{array}$$

3. Compute the limits of the sequences (insofar as they are *number* sequences)

$$a_n = \frac{x^n}{1 + x^n}, \quad b_n = \frac{x^n}{1 + x^{2n}}, \quad n = 1, 2, \dots,$$

for all rational values of x .

4. Determine the possible limits of the following recursive sequences, assuming that the sequences are convergent:

$$\begin{array}{ll} \text{a) } a_{n+1} = -\frac{1}{3}a_n + \frac{2n^2}{n^2 + 1} & \text{b) } a_{n+1} = \frac{na_n - 8n + 3}{2n + 5} \\ \text{c) } a_{n+2} = \frac{2n}{3n + 4} - a_{n+1} - \frac{1}{4}a_n & \text{d) } a_{n+1} = \frac{1}{2 + a_n} \\ \text{e) } a_{n+1} = \sqrt{a_n + 2} & \text{f) } a_{n+1} = \sqrt[3]{a_n + 6} \\ \text{g) } a_{n+1} = \frac{3}{4 - a_n^2} & \text{h) } a_{n+1} = \frac{1}{\sqrt{4 - a_n^2}} \end{array}$$

5. Let $a_n + a \neq 0 \forall n$ and $\lim_n (a_n + a)^{-1} = b \neq 0$. Show that the sequence $\{a_n\}$ converges and determine $\lim_n a_n$.

6. a) Prove claim V3 of Theorem I.7.4.

b) Let $\{a_n\}$ be a bounded sequence. Prove that $\lim_n \frac{2n + a_n}{n} = 2$.

7. a) Which of the following numbers are different and which are the same? $1.285714285714285714\dots$, $\frac{11}{9}$, $\frac{1287}{1250}$, $\frac{9}{7}$, $1.029599999\dots$, $1.02949999\dots$, $1.029600000\dots$, $a = \text{one and two ninths}$.

b) Which of the following decimals, compared pairwise, are certainly different and which possibly the same?

$$\check{x} = 2.41789\dots, \quad \check{y} = 2.41799\dots, \quad \check{a} = 2.41788\dots, \quad \check{b} = 2.41798\dots, \quad \check{c} = 2.4179\dots$$

8. Determine a rational number x such that $x = \check{x}$, when
 a) $\check{x} = -0.46127127127\dots$, b) $\check{x} = 2015.201520152015\dots$
9. In a certain number system, in which one writes $1 + 1 = 2$, the counterparts of the infinite decimals form the set $\mathbb{L}$. In this set the number $\check{x} = 0.222\dots \in \mathbb{L}$ has a second representation as well. What is the base of this number system, and with which rational number is $\check{x}$ identified?
10. Let $x, y \in \mathbb{Q}$ and $\check{x}, \check{y} \in \mathbb{D}$. Starting from Definitions I.7.5 and I.7.7 and using only the rules for combining limits (Theorem I.7.3), show that a) $\check{x} = x$ and $x = \check{y} \Rightarrow \check{x} = \check{y}$, b) $\check{x} = \check{y}$ and $\check{y} = y \Rightarrow \check{x} = y$.
11. (*) Prove: $\lim_n a_n = a \Rightarrow \lim_n \frac{a_1 + a_2 + \dots + a_n}{n} = a$.
12. (*) Define the recursive sequence

$$a_1 \in \mathbb{Q}, \quad a_{n+1} = \frac{a_n}{1 + 10^{-n}a_n}, \quad n = 1, 2, \dots$$

Show that $\{a_n\}$ converges and

$$\lim_n a_n = \frac{9a_1}{a_1 + 9},$$

except for certain exceptional values of a_1 — which?

Hint: Investigate the sequence $\{b_n\} = \{a_n^{-1}\}$.

13. (*) a) Determine the possible limits $a = \lim_n a_n$ of the recursive sequence

$$a_{n+1} = (2 - a_n)^2, \quad n = 0, 1, \dots$$

b) By investigating the sequence $\{b_n\} = \{a_n - a\}$, conclude that $\lim_n a_n = a$ only when $a_n = a$ for some n . c) Conclude that if $a_0 \in \mathbb{Q}$, then $\{a_n\}$ converges exactly when a_0 is one of the numbers 0, 1, 2, 3, 4.

I.8 Monotonic and bounded sequences

In this section the objects of study are sequences that are both monotonic (increasing or decreasing, Definition 1.5.1) and bounded (Definition 1.7.1). Such sequences are common in practice, even though they form only a small 'minority' of all sequences. The main result of this section is that for a monotonic and bounded sequence a limit can always be found — provided the limit and convergence toward it are defined suitably.

EXAMPLE 1 Every infinite decimal $\check{x} = \{x_n\} \in \mathbb{D}$ is, as a sequence, monotonic, as was noted earlier (see Section 1.5). The sequence $\{x_n\}$ is also bounded, since $|x_n| \leq C = |x_0| + 1 \ \forall n$ (Proposition 1.5.2). $\square$

EXAMPLE 2 Show that the series a) $\sum_{k=0}^{\infty} 1/k!$, b) $\sum_{k=1}^{\infty} 1/k^2$ is, as a sequence, monotonic and bounded.

Solution In both cases we have a *positive-term* series, i.e. a series of the form $\sum_k a_k$ where $a_k > 0 \ \forall k$. A positive-term series is, as the sequence of its partial sums, even strictly increasing, so it suffices to show that the sequences

$$\begin{aligned} \text{a) } \{s_n\} &= \left\{ \sum_{k=0}^n \frac{1}{k!}, \ n = 0, 1, 2, \dots \right\} = \left\{ 1, 2, \frac{5}{2}, \dots \right\}, \\ \text{b) } \{s_n\} &= \left\{ \sum_{k=1}^n \frac{1}{k^2}, \ n = 1, 2, 3, \dots \right\} = \left\{ 1, \frac{5}{4}, \frac{23}{18}, \dots \right\} \end{aligned}$$

are bounded. a) When $n \geq 4$, we can estimate

$$\begin{aligned} s_n &= \sum_{k=0}^n \frac{1}{k!} = 1 + 1 + \frac{1}{2!} \left(1 + \frac{1}{3} + \frac{1}{3 \cdot 4} + \dots + \frac{1}{3 \cdot 4 \dots n} \right) \\ &\leq 2 + \frac{1}{2} \left(1 + \frac{1}{3} + \frac{1}{3^2} + \dots + \frac{1}{3^{n-2}} \right) \\ &< 2 + \frac{1}{2} \cdot \frac{1}{1 - \frac{1}{3}} = 2.75. \end{aligned}$$

Thus $0 < s_n < 2.75 \ \forall n$, so $\{s_n\}$ is bounded.

b) Here a little more ingenuity is needed. For example: Since

$$\frac{1}{k^2} < \frac{1}{(k-1)k} = \frac{1}{k-1} - \frac{1}{k}, \quad k \geq 2,$$

the partial sums can be estimated by means of telescoping sums (cf. Exercise I.1:13) as follows:

$$\begin{aligned}
 s_n &= 1 + \sum_{k=2}^n \frac{1}{k^2} < 1 + \sum_{k=2}^n \left(\frac{1}{k-1} - \frac{1}{k} \right) \\
 &= 1 + \left(1 - \frac{1}{2} \right) + \left(\frac{1}{2} - \frac{1}{3} \right) + \dots + \left(\frac{1}{n-1} - \frac{1}{n} \right) \\
 &= 1 + 1 + \left(-\frac{1}{2} + \frac{1}{2} \right) + \dots + \left(-\frac{1}{n-1} + \frac{1}{n-1} \right) - \frac{1}{n} \\
 &= 2 - \frac{1}{n} < 2, \quad n \geq 3.
 \end{aligned}$$

Thus $0 < s_n < 2 \forall n$, so $\{s_n\}$ is bounded. $\square$

Convergence toward a decimal

Neither of the series in Example 2 has a limit within the set of numbers known so far, as we shall see. Nonetheless these series 'appear to converge' when the (rational) partial sums of the series are converted into infinite decimals.

EXAMPLE 2 (continued) a) The partial sums $s_1 \dots s_{12}$ are

$s_1 = 2$	$= 2.0000000000 \dots$	$s_7 = \frac{685}{252}$	$= 2.7182539682 \dots$
$s_2 = \frac{5}{2}$	$= 2.5000000000 \dots$	$s_8 = \frac{109601}{40320}$	$= 2.7182787698 \dots$
$s_3 = \frac{8}{3}$	$= 2.6666666666 \dots$	$s_9 = \frac{98641}{36288}$	$= 2.7182815255 \dots$
$s_4 = \frac{65}{24}$	$= 2.7083333333 \dots$	$s_{10} = \frac{9864101}{3628800}$	$= 2.7182818011 \dots$
$s_5 = \frac{163}{60}$	$= 2.7166666666 \dots$	$s_{11} = \frac{13563139}{4989600}$	$= 2.7182818261 \dots$
$s_6 = \frac{1957}{720}$	$= 2.7180555555 \dots$	$s_{12} = \frac{260412269}{95800320}$	$= 2.7182818282 \dots$

b) The partial sums at the index values $n = 10^k$, $k = 1 \dots 12$ are

$s_{10} = 1.5497677311 \dots$	$s_{10000000} = 1.6449339668 \dots$
$s_{100} = 1.6349839001 \dots$	$s_{100000000} = 1.6449340568 \dots$
$s_{1000} = 1.6439345666 \dots$	$s_{1000000000} = 1.6449340658 \dots$
$s_{10000} = 1.6448340718 \dots$	$s_{10000000000} = 1.6449340667 \dots$
$s_{100000} = 1.6449240668 \dots$	$s_{100000000000} = 1.6449340668 \dots$
$s_{1000000} = 1.6449330668 \dots$	$s_{1000000000000} = 1.6449340668 \dots$

$\square$

On the basis of the partial sums it appears that the limit $s = \lim_n s_n$ (i.e. the sum of the series) can, in the example, be interpreted as an infinite decimal:

$$\text{a) } \sum_{k=0}^{\infty} \frac{1}{k!} = 2.71828182 \dots \quad \text{b) } \sum_{k=1}^{\infty} \frac{1}{k^2} = 1.64493406 \dots$$

But what does it mean, more precisely, that an infinite decimal is the limit of a sequence? — The original definition of the limit (Definition I.6.1) cannot be appealed to directly, because the subtraction and comparison of numbers needed in the condition $|a_n - a| < \varepsilon$ of the definition are not (at least so far) defined when $a \in \mathbb{D}$. We solve the problem by laying down a separate definition.

DEFINITION I.8.1 The sequence $\{a_n\}$ *converges toward* the infinite decimal $\check{x} = \{x_n\} \in \mathbb{D}$ exactly when

$$\lim_n (a_n - x_n) = 0.$$

In that case $\check{x}$ is called the *limit* of the sequence $\{a_n\}$, and one writes $\lim_n a_n = \check{x}$ or $a_n \rightarrow \check{x}$. If a sequence has a limit $\check{x} \in \mathbb{D}$, then it is said to *converge*; otherwise the sequence *diverges*.

Note that in the condition $\lim_n (a_n - x_n) = 0$ we are dealing with the ordinary (rational) limit of the sequence $\{a_n - x_n\}$ in the sense of Definition I.6.1. If the sequence $\{a_n\}$ also has a rational limit, then Definitions I.8.1 and I.6.1 are in agreement, for if $a_n \rightarrow x \in \mathbb{Q}$ (Definition I.6.1) and $x = \check{x} = \{x_n\} \in \mathbb{D}$ (Definition I.7.5), then $a_n - x_n = (a_n - x) + (x - x_n) \rightarrow 0 + 0 = 0$ (Theorem I.7.3), i.e. $a_n \rightarrow \check{x}$ in the sense of Definition I.8.1. The following also holds (cf. Theorems I.7.2 and I.6.2)

THEOREM I.8.2 If the sequence $\{a_n\}$ converges in the sense of Definition I.8.1, then

(i) $\{a_n\}$ is a bounded sequence, and (ii) the limit $\lim_n a_n$ is unique.

Proof (i) By writing $a_n = (a_n - x_n) + x_n$ and applying the triangle inequality we conclude

$$|a_n| \leq |a_n - x_n| + |x_n| \leq C_1 + C_2 = C \quad \forall n,$$

since $\{a_n - x_n\}$ is bounded, being convergent (Theorem I.7.2), and $\{x_n\}$ is also bounded (see Example 1 above). Thus $\{a_n\}$ is a bounded sequence.

(ii) Assume that $a_n \rightarrow \check{x} = \{x_n\} \in \mathbb{D}$ and $a_n \rightarrow \check{y} = \{y_n\} \in \mathbb{D}$. Then $a_n - x_n \rightarrow 0$ and $a_n - y_n \rightarrow 0$ (Definition I.8.1), whereupon by Theorem I.7.3

$$x_n - y_n = (a_n - y_n) - (a_n - x_n) \rightarrow 0.$$

Thus $\lim_n (x_n - y_n) = 0$, which by Definition I.7.7 means: $\check{x} = \check{y}$. $\square$

EXAMPLE 3 By Definition I.8.1, every infinite decimal $\check{x} = \{x_n\}$ 'converges to itself' as the sequence $\{a_n\} = \{x_n\}$ (!). $\square$

In what follows a central question is: if no obvious (e.g. rational) limit can be exhibited for a sequence $\{a_n\}$, then under what conditions can one be sure that a limit in the sense of Definition I.8.1, as an infinite decimal, exists? A precise answer to this question is given later in Section I.10; in this connection we content ourselves with a partial answer. First, by Theorem I.8.2 a necessary condition for the existence of a limit is that $\{a_n\}$ be bounded. By the following theorem, a sufficient condition is that $\{a_n\}$ be both monotonic and bounded.

*THEOREM I.8.3 * (**A monotonic and bounded sequence converges**) Every monotonic and bounded sequence has a limit, in the sense of Definition I.8.1, as an infinite decimal.[†]

Proof From Definition I.8.1 one sees that if $a_n \rightarrow \check{x}$, then $-a_n \rightarrow -\check{x}$. We may therefore restrict to the cases in which either (i) $\{a_n\}$ is increasing and for some $N \in \mathbb{N}$ one has $a_n \geq 0 \forall n \geq N$, or (ii) $\{a_n\}$ is decreasing and $a_n \geq 0 \forall n \in \mathbb{N}$, since in the other cases the sequence $\{-a_n\}$ is of type (i) or (ii). Let the sequence first be of type (i), and define, in connection with the sequence, the predicate (cf. Section I.3)

$$Q(x) : a_n < x \quad \forall n \quad (x \in \mathbb{Q}).$$

Since $\{a_n\}$ is bounded, an $M \in \mathbb{N}$ can be found such that $a_n < M \forall n$. By assumption (i), then, $0 \leq a_n < M \forall n \geq N$. One can then choose uniquely $x_0 \in \{0, \dots, M-1\}$ such that $Q(x_0)$ is false and $Q(x_0+1)$ is true. Since $Q(x_0)$ is false, $a_n \geq x_0$ for some n . Let $n = N_0$ be the smallest such index. Then, since the sequence $\{a_n\}$ is increasing and since $Q(x_0+1)$ is true, $x_0 \leq a_n < x_0+1 \forall n \geq N_0$.

Next choose $x_1 = x_0 + d_1 \cdot 10^{-1}$, $d_1 \in \{0, 1, \dots, 9\}$ so that $Q(x_1)$ is false and $Q(x_1 + 10^{-1})$ is true. Since $Q(x_1)$ is false, $a_n \geq x_1$ for some n . Let $n = N_1$ be the smallest such index ($N_1 \geq N_0$). Then, since the sequence $\{a_n\}$ is increasing and since $Q(x_1 + 10^{-1})$ is true, $x_1 \leq a_n < x_1 + 10^{-1} \forall n \geq N_1$.

Continuing in the same way, one constructs an increasing sequence $\{x_k\}$ and an increasing index sequence $\{N_0, N_1, \dots\}$ such that for every k

$$x_k \leq a_n < x_k + 10^{-k}, \quad \text{when } n \geq N_k. \quad (\star)$$

Here and in what follows, the mark () attached to a theorem means that the proof of the theorem is a more demanding logical construction than usual.

[†]Theorem I.8.3 is an example of an *existence theorem*, in which something (here a limit) is asserted to exist. — The emphasis in the theorem is precisely on the word 'exists'.

In case (ii) one arrives at the same result when the predicate $Q(x)$ is defined

$$Q(x) : a_n < x \text{ for some } n \quad (x \in \mathbb{Q}).$$

In both cases (i) and (ii), then, an infinite decimal $\check{x} = \{x_k, k = 0, 1, \dots\}$ and an index sequence $\{N_k, k = 0, 1, \dots\}$ have been constructed such that

$$|a_n - x_k| < 10^{-k}, \quad \text{when } n \geq N_k.$$

Taking into account that for the sequence $\{x_k\}$ (Proposition I.5.2)

$$|x_n - x_k| < 10^{-k}, \quad \text{when } n \geq k,$$

then, by writing $a_n - x_n = (a_n - x_k) + (x_k - x_n)$ and applying the triangle inequality, it follows from the last two inequalities written that

$$|a_n - x_n| \leq |a_n - x_k| + |x_n - x_k| < 2 \cdot 10^{-k}, \quad \text{when } n \geq \max\{k, N_k\} = N'_k.$$

By Proposition I.6.4 this condition is the same as the condition $\lim_n (a_n - x_n) = 0$. Thus $\lim_n a_n = \check{x} = \{x_n\}$ in the sense of Definition I.8.1. $\square$

When the sequence $\{a_n\}$ is known, the proof construction of Theorem I.8.3 can be followed as an algorithm that determines the limit starting from the integer part and proceeding one digit at a time.

EXAMPLE 2 (continued) The limit $\lim_n s_n = \{x_k\} \in \mathbb{D}$ is determined by choosing the numbers $x_k \in \mathbb{Q}_k$, $k = 0, 1, \dots$ so that $s_n \geq x_k$ for some n and $s_n < x_k + 10^{-k} \forall n$, whereupon condition $(\star)$ holds for some $N_k \in \mathbb{N}$ (for every k), when $a_n = s_n$.

a) Since $s_n < 3 \forall n$ and $s_1 = 2 \Rightarrow s_n \geq 2 \forall n \geq 1$, we have $x_0 = 2$ and the smallest value of N_0 for which condition $(\star)$ holds when $k = 0$ is $N_0 = 1$. Similarly, since $s_3 < 2.7$, $s_4 > 2.7$ and evidently $s_n < 2.8 \forall n$ (this must be verified separately by estimating the partial sums, see Exercise 1), we have $x_1 = 2.7$ and the smallest value of the index N_1 is $N_1 = 4$. In the same way one sees from the computed partial sums that $x_2 = 2.71$, $x_3 = 2.718$, $\dots$, $x_8 = 2.71828182$ and that the smallest values of the indices N_k , $k = 2 \dots 8$ in condition $(\star)$ are $N_2 = 5$, $N_3 = 6$, $N_4 = 7$, $N_5 = N_6 = 9$, $N_7 = 10$ and $N_8 = 11$.

b) From the partial sums

$$\begin{array}{ll} s_{21} = 1.59843081\dots & s_{1070} = 1.64399992\dots \\ s_{22} = 1.60049693\dots & s_{1071} = 1.64400079\dots \\ s_{202} = 1.63999580\dots & s_{29353} = 1.64489999\dots \\ s_{203} = 1.64002007\dots & s_{29354} = 1.64490000\dots \end{array}$$

one can deduce (and verify, see Exercise 1) that the smallest values of the index N_k for which conditions $(\star)$ hold when $k = 1 \dots 4$ are $N_1 = 22$, $N_2 = 203$, $N_3 = 1071$ and $N_4 = 29354$. $\square$

EXAMPLE 4 In Example 3 of Section I.6 it was shown that for the recursive rational sequence

$$a_0 = 2, \quad a_{n+1} = \frac{a_n}{2} + \frac{1}{a_n}, \quad n = 0, 1, \dots$$

one has $a_n > 0 \forall n$ and $a_n^2 > 2 \forall n$. From the recursion formula and the results mentioned it also follows that

$$a_{n+1} - a_n = \frac{a_n}{2} + \frac{1}{a_n} - a_n = \frac{1}{2a_n} (2 - a_n^2) < 0,$$

so $\{a_n\}$ is strictly decreasing and also bounded: $0 < a_n \leq 2 \forall n$. Thus $\{a_n\}$ converges, in the sense of Definition I.8.1, toward an infinite decimal. The convergence is in fact very rapid:

$$\begin{aligned} a_0 &= 2.000\dots \\ a_1 &= 1.5000\dots \\ a_2 &= 1.41666\dots \\ a_3 &= 1.41421568\dots \\ a_4 &= 1.41421356237468\dots \\ a_5 &= 1.41421356237309504880168962\dots \\ a_6 &= 1.4142135623730950488016887242096980785696\dots \\ a_7 &= 1.4142135623730950488016887242096980785696\dots \quad \square \end{aligned}$$

It is known of the sequence of the example that it also converges, in the sense of Definition I.6.1 toward the number $\sqrt{2}$ (see Section I.6). It was also noted above that the limits in the senses of Definitions I.6.1 and I.8.1 are the same if the limit is rational. It is to be expected that this is the case more generally as well, if both limits exist. Assuming this (the matter is confirmed in the next section), the abstract number $\sqrt{2}$ acquires a concrete 'outward form' as an infinite decimal:

$$\sqrt{2} = 1.4142135623730950488016887242096980785696\dots$$

As an algorithm (when one wishes to compute the digits of $\sqrt{2}$), the sequence $\{a_n\}$ of the example is among the fastest known.*

*The algorithm of the example was already known in ancient Babylonia 3000 years ago.

Napier's number

PROPOSITION I.8.4 The sequences

$$\{a_n\} = \left\{ \left(1 + \frac{1}{n}\right)^n, n = 1, 2, \dots \right\}, \quad \{b_n\} = \left\{ \left(1 - \frac{1}{n}\right)^n, n = 1, 2, \dots \right\}$$

are increasing and bounded. Moreover $\lim_n a_n b_n = 1$.

Proof Applying Bernoulli's inequality (Proposition I.4.2), we obtain

$$\begin{aligned} \frac{a_{n+1}}{a_n} &= \frac{\left(1 + \frac{1}{n+1}\right)^{n+1}}{\left(1 + \frac{1}{n}\right)^n} = \left(1 + \frac{1}{n}\right) \left(\frac{1 + \frac{1}{n+1}}{1 + \frac{1}{n}}\right)^{n+1} \\ &= \left(1 + \frac{1}{n}\right) \left[1 - \frac{1}{(n+1)^2}\right]^{n+1} > \left(1 + \frac{1}{n}\right) \left(1 - \frac{1}{n+1}\right) = 1. \end{aligned}$$

Thus $a_{n+1} > a_n \ \forall n$, i.e. $\{a_n\}$ is a (strictly) increasing sequence. By a corresponding argument one shows (Exercise 4) that the sequence $\{b_n\}$ is also strictly increasing. Then, since for every $n \geq 2$

$$1 - \frac{1}{n^2} < 1 \quad \Leftrightarrow \quad 1 + \frac{1}{n} < \frac{1}{1 - \frac{1}{n}},$$

we conclude

$$a_n < \frac{1}{b_n} \leq \frac{1}{b_2} = 4 \quad \text{when } n \geq 2.$$

Thus $1 < a_n < 4$ and $0 \leq b_n < 1$ for all n , i.e. the sequences $\{a_n\}$ and $\{b_n\}$ are, besides being increasing, also bounded. From Bernoulli's inequality it further follows that

$$1 > a_n b_n = \left(1 - \frac{1}{n^2}\right)^n > 1 - \frac{1}{n} \quad (n \geq 2),$$

so $0 < 1 - a_n b_n < 1/n \ \forall n \geq 2$. From this and the definition of the limit of a sequence the last part of the claim follows. $\square$

On the clay tablet no. 7289 preserved at Yale University, an approximation x_3 of $\sqrt{2}$ computed with the algorithm, starting from the initial value x_1 , is given. The computations were carried out in the base-60 number system to four significant digits, giving the result

$$x_3^* = 1 + 24 \cdot 60^{-1} + 51 \cdot 60^{-2} + 10 \cdot 60^{-3} = 1.41421296 \dots$$

Owing to the truncation error, the error of the result (about $-6 \cdot 10^{-7}$) happens to be even smaller than the error of the exact value of x_3 (about $+21 \cdot 10^{-7}$).

Since the sequences of Proposition I.8.4 are increasing and bounded, for them one can compute a limit, in the sense of Definition I.8.1, as an infinite decimal. In the case of the sequence $\{a_n\}$ the limit is the so-called *Napier's number*, whose established symbol is e :

$$\lim_n \left(1 + \frac{1}{n}\right)^n = e = 2.71828182845905 \dots$$

The number e is aperiodic, i.e. it is not identified with any rational number.

EXAMPLE 5 Determine the smallest indices N_k , $k = 0 \dots 3$ in the proof construction of Theorem I.8.3, when $a_n = (1 + 1/n)^n$.

Solution The comparison numbers of the proof construction are

$$x_0 = 2, \quad x_1 = 2.7, \quad x_2 = 2.71, \quad x_3 = 2.718, \quad \dots$$

By computing the early terms of the sequence $\{a_n\}$ as decimals one finds

$$\begin{array}{ll} a_1 = 2.00000000 \dots & a_{163} = 2.70999015 \dots \\ a_2 = 2.25000000 \dots & a_{164} = 2.71004043 \dots \\ a_{73} = 2.69989423 \dots & a_{4821} = 2.71799996 \dots \\ a_{74} = 2.70013967 \dots & a_{4822} = 2.71800001 \dots \end{array}$$

Thus $N_0 = 1$, $N_1 = 74$, $N_2 = 164$, $N_3 = 4822$. $\square$

Let us return once more to the positive-term series $\sum_{k=0}^{\infty} 1/k!$ of Example 2. From the partial sums of this series computed above, one may surmise that the series in fact converges toward Napier's number (and, moreover, rapidly). To confirm this surmise it suffices to show that if $s_n = \sum_{k=0}^n 1/k!$ and $a_n = (1 + 1/n)^n$, then $\lim_n (s_n - a_n) = 0$, cf. the definition of Napier's number above. We prove this claim, which at the same time gives a fast algorithm for computing the digits of Napier's number.

PROPOSITION I.8.5 $\lim_n \left[\sum_{k=0}^n \frac{1}{k!} - \left(1 + \frac{1}{n}\right)^n \right] = 0.$

Proof By the binomial formula

$$\begin{aligned}
 \left(1 + \frac{1}{n}\right)^n &= 1 + n \cdot \frac{1}{n} + \frac{n(n-1)}{1 \cdot 2} \left(\frac{1}{n}\right)^2 + \frac{n(n-1)(n-2)}{1 \cdot 2 \cdot 3} \left(\frac{1}{n}\right)^3 + \dots + \left(\frac{1}{n}\right)^n \\
 &= 1 + 1 + \frac{1}{2!} \left(1 - \frac{1}{n}\right) + \frac{1}{3!} \left(1 - \frac{1}{n}\right) \left(1 - \frac{2}{n}\right) + \dots \\
 &\quad \dots + \frac{1}{n!} \left(1 - \frac{1}{n}\right) \left(1 - \frac{2}{n}\right) \dots \left(1 - \frac{n-1}{n}\right) \\
 &= \sum_{k=0}^n a_k^{(n)} \frac{1}{k!},
 \end{aligned}$$

where

$$a_k^{(n)} = \begin{cases} 1, & \text{when } k = 0, 1, \\ \prod_{j=1}^{k-1} \left(1 - \frac{j}{n}\right), & \text{when } k = 2 \dots n. \end{cases}$$

One sees that for every $1 \leq m \leq n$

$$\begin{aligned}
 1 &\geq a_k^{(n)} \geq a_m^{(n)} \geq \left(1 - \frac{m-1}{n}\right)^{m-1}, \quad k = 0 \dots m \\
 \Rightarrow \quad 0 &\leq 1 - a_k^{(n)} \leq 1 - \left(1 - \frac{m-1}{n}\right)^{m-1}, \quad k = 0 \dots m.
 \end{aligned}$$

On the basis of this and the estimate (see Exercise 1)

$$\sum_{k=0}^n \frac{1}{k!} < \sum_{k=0}^m \frac{1}{k!} + \frac{2}{(m+1)!}, \quad 0 \leq m < n$$

one can estimate

$$\begin{aligned}
 0 &\leq \sum_{k=0}^n \frac{1}{k!} - \left(1 + \frac{1}{n}\right)^n = \sum_{k=0}^n \left(1 - a_k^{(n)}\right) \frac{1}{k!} \\
 &= \sum_{k=0}^m \left(1 - a_k^{(n)}\right) \frac{1}{k!} + \sum_{k=m+1}^n \left(1 - a_k^{(n)}\right) \frac{1}{k!} \\
 &< \left[1 - \left(1 - \frac{m-1}{n}\right)^{m-1}\right] \sum_{k=0}^m \frac{1}{k!} + \sum_{k=m+1}^n \frac{1}{k!} \\
 &< \left[1 - \left(1 - \frac{m-1}{n}\right)^{m-1}\right] \cdot 3 + \frac{2}{(m+1)!},
 \end{aligned}$$

where $0 \leq m < n$. Now let $\varepsilon > 0$ and choose $m \in \mathbb{N}$ so that

$$\frac{2}{(m+1)!} < \frac{\varepsilon}{2}$$

(possible, since $2/(m+1)! \rightarrow 0$ as $m \rightarrow \infty$). Once m is fixed in this way, choose further $N \in \mathbb{N}$, $N \geq m$ so that

$$1 - \left(1 - \frac{m-1}{n}\right)^{m-1} < \frac{\varepsilon}{6}, \quad \text{when } n > N$$

(possible, since $\left(1 - \frac{m-1}{n}\right)^{m-1} \rightarrow 1 \Rightarrow 1 - \left(1 - \frac{m-1}{n}\right)^{m-1} \rightarrow 0$, as $n \rightarrow \infty$). Then it follows that

$$0 \leq \sum_{k=0}^n \frac{1}{k!} - \left(1 + \frac{1}{n}\right)^n < \varepsilon, \quad \text{when } n > N.$$

Since here $\varepsilon > 0$ was arbitrary and $N \in \mathbb{N}$ (N depends only on ε), then by the definition of the limit of a sequence the claim follows. $\square$

EXERCISES

- By computing the sums a) $s_n = \sum_{k=0}^n 1/k!$, b) $s_n = \sum_{k=1}^n 1/k^2$ one finds:
a) $s_n = 2.718281826..$ when $n = 11$, b) $s_n = 1.644934065..$ when $n = 10^9$.
We draw the conclusion from the results: $\lim_n s_n = s \in \mathbb{D}$ truncated to eight significant digits is a) $s = 2.71828182..$ b) $s = 1.64493406..$
Confirm the conclusion by first showing: If a) $m \in \mathbb{N} \cup \{0\}$, b) $m \in \mathbb{N}$, then for every $n > m$

$$\text{a) } s_n < s_m + \frac{m+2}{m+1} \cdot \frac{1}{(m+1)!}, \quad \text{b) } s_n < s_m + \frac{1}{m}.$$

- Show that the sequence of partial sums of the positive-term series

$$\sum_{k=0}^{\infty} \frac{10^{-k}}{k+1}$$

is bounded. Compute the sum $x \in \mathbb{D}$ of the series to five significant digits and confirm the result, i.e. prove the computed digits correct.

3. What decimal form does the proof construction of Theorem I.8.3 give for the limits $a = \lim_n(1 - 1/n)$, $b = \lim_n(1 + 1/n)$, $c = \lim_n(-1 + 1/n)$ and $d = \lim_n(-1 - 1/n)$?
4. Show that the sequence $\{b_n\}$ of Proposition I.8.4 is strictly increasing.
5. Show that the recursive sequence

$$a_0 = \frac{1}{10}, \quad a_{n+1} = \frac{a_n}{1 + a_n^6}, \quad n = 0, 1, \dots$$

is decreasing and bounded. The sequence also has a rational limit — which?

6. Show that the recursive rational sequence

$$a_0 \in \mathbb{Q}, \quad a_{n+1} = \frac{a_n}{1 + 10^{-n}a_n^2}, \quad n = 0, 1, \dots$$

is monotonic and bounded. Compute the limit $\lim_n a_n = a \in \mathbb{D}$ to six significant digits and justify the accuracy, when a) $a_0 = 1$, b) $a_0 = 3$.

7. Consider the recursive rational sequence

$$a_0 = 1, \quad a_{n+1} = 1 - \frac{1}{a_n + 3}, \quad n = 0, 1, \dots$$

- a) Show by induction: $\frac{5}{7} < a_n \leq \frac{3}{4}$, when $n = 1, 2, \dots$
- b) Show: for the sequence $\{b_n\} = \{a_{n+1} - a_n\}$

$$b_{n+1} = \frac{b_n}{(a_n + 3)(a_{n+1} + 3)}, \quad n = 0, 1, \dots$$

- c) Show by induction: $\{a_n\}$ is a strictly decreasing sequence.
 - d) Compute the limit $\lim_n a_n$ according to Definition I.6.1.
 - e) Conclude: the limit in the sense of Definition I.8.1, truncated to six significant digits, is $\lim_n a_n = 0.732050..$ (you may rely on a calculator!).
8. (*) One algorithm for computing the well-known number $\pi = 3.1415926535897932384626433832..$ is based on the formula

$$\pi = \sum_{k=0}^{\infty} \frac{1}{16^k} \left(\frac{4}{8k+1} - \frac{2}{8k+4} - \frac{1}{8k+5} - \frac{1}{8k+6} \right).$$

Verify that the series is positive-term and estimate how many correct digits of π can be extracted from the decimal form of the partial sum s_n of the series when a) $n = 10$, b) $n = 15$, c) $n = 20$. (The formula was discovered in 1995.)

9. (*) Show that the following sequences (in the field $(\mathbb{J}, +, \cdot, <)$, see Section I.2) are monotonic and bounded (*hint*: induction!). Compute the limits of the sequences as well (*hint*: the definition of the root number and Theorem I.7.3).

a) $a_0 = \sqrt{3}, \quad a_{n+1} = \sqrt{3 + a_n}, \quad n = 0, 1, \dots$

b) $a_0 = \sqrt{3}, \quad a_{n+1} = \sqrt{2a_n}, \quad n = 0, 1, \dots$

c) $a_0 = \sqrt{5}, \quad a_{n+1} = \sqrt{2a_n}, \quad n = 0, 1, \dots$

d) $a_0 = 3, \quad a_{n+1} = \sqrt[4]{6 + a_n^2}, \quad n = 0, 1, \dots$

10. (*) Show that the recursive sequence

$$a_{n+1} = \frac{a_n}{5 - a_n}, \quad n = 0, 1, \dots$$

is bounded and, from some index onward, monotonic, insofar as

$$a_0 \neq \frac{4}{1 - 5^{-n}} \quad \forall n \in \mathbb{N}.$$

Determine also, under this condition, the (a_0 -dependent) limit $\lim_n a_n$.

Hint: Investigate the sequence $\{b_n\} = \{a_n^{-1}\}$.

I.9 Real numbers

We begin with a definition.

DEFINITION I.9.1 (Real numbers) The set $\mathbb{R}$ of *real numbers* consists of the infinite decimals, i.e. $\mathbb{R} = \mathbb{D}$.

Definition I.9.1 of the real numbers is not the only possible one, for it is evident that, for example, the infinite binary numbers could equally well be taken as the starting point of the definition. There are many other approaches too, as will be seen later. Definition I.9.1 in any case gives one possible interpretation of the real numbers, which is taken as the starting point in what follows.

Together with the previously agreed identification relation (Definition I.7.7), Definition I.9.1 tells us only what real numbers 'look like' and how they are distinguished. In order also to be able to compute with real numbers, one must define the operations among them.

The operations of $\mathbb{R}$

Referring to Definition I.8.1, we set

DEFINITION I.9.2 (The operations of the real numbers) If $\check{x} = \{x_n\}$ and $\check{y} = \{y_n\}$ are real numbers, then one defines

$$\begin{aligned} \text{addition:} \quad & \check{x} + \check{y} = \lim_n (x_n + y_n) = \check{z} \in \mathbb{R}. \\ \text{multiplication:} \quad & \check{x}\check{y} = \lim_n x_n y_n = \check{z} \in \mathbb{R}. \\ \text{division:} \quad & \check{x}/\check{y} = \lim_n x_n/y_n = \check{z} \in \mathbb{R}. \end{aligned}$$

Here in division one must assume, in the usual way, that $\check{y} \neq 0$, i.e. $y_n \not\rightarrow 0$ (Definition I.7.5). The condition means that for some m one has $y_m \neq 0$, whereupon from that index onward $|y_n| \geq |y_m| > 0$, $n \geq m$ (since $\{|y_n|\}$ is an increasing sequence). The definition of division is thus formally in order, provided $\check{y} \neq 0$ and one restricts, in the sequence $\{x_n/y_n\}$, to indices $n \geq m$. From the definition of addition one sees that the negative of the number $\check{x} = \{x_n\}$ is $-\check{x} = \{-x_n\}$, i.e. the operation $\check{x} \mapsto -\check{x}$ corresponds to a change of sign of the decimal, as was already agreed in Section I.5. Subtraction is then defined in the normal way, i.e. $\check{x} - \check{y} = \check{x} + (-\check{y}) = \lim_n (x_n - y_n) \in \mathbb{R}$.

Definition I.9.2 is practical in that it is based directly on the representations of the numbers $\check{x}$ and $\check{y}$ in a given number system, here the decimal system. Computing

the terms of the sequences $\{x_n + y_n\}$, $\{x_n y_n\}$ and $\{x_n/y_n\}$ can be thought of as corresponding to approximate operations with floating-point numbers (cf. Section I.5). The definition, however, contains a theoretical problem as yet unresolved: the sequences $\{x_n + y_n\}$, $\{x_n y_n\}$ and $\{x_n/y_n\}$ are assumed to converge toward a real number (i.e. an infinite decimal), but so far this has been confirmed only for a monotonic and bounded sequence (Theorem I.8.3). The sequences of Definition I.9.2 are indeed bounded (with the additional conditions mentioned concerning the sequence $\{x_n/y_n\}$) but not necessarily monotonic. To support the definition, the following theorem is needed.

THEOREM I.9.3 If $\{a_n\}$ and $\{b_n\}$ are monotonic and bounded sequences, then the sequences $\{a_n + b_n\}$ and $\{a_n b_n\}$ converge, in the sense of Definition I.8.1, toward a real number. If further $|b_n| \geq b > 0 \forall n$, then the sequence $\{a_n/b_n\}$ also converges toward a real number.

Proof Addition. If $\{a_n\}$ and $\{b_n\}$ are both increasing or both decreasing, then $\{a_n + b_n\}$ is correspondingly increasing/decreasing (Exercise I.5:2), whereupon $\lim_n (a_n + b_n) \in \mathbb{R}$ exists by Theorem I.8.3. Let one of the sequences be increasing and the other decreasing, e.g. $\{a_n\}$ increasing. Choose $b \in \mathbb{Q}$ such that $b_n + b \geq 0 \forall n$ (possible, since $\{b_n\}$ is bounded) and write

$$a_n + b_n = (a_n - b) + (b_n + b).$$

Here $\{b_n + b\}$ is a bounded and decreasing sequence, so by Theorem I.8.3 it converges: $\lim_n (b_n + b) = \check{x} = \{x_n\} \in \mathbb{R}$. Since $b_n + b \geq 0 \forall n$, the sign of $\check{x}$ is + (possibly $\check{x} = 0$), so the sequence $\{x_n\}$ is increasing. Since $\{a_n - b\}$ is also an increasing sequence, so is $\{a_n - b + x_n\}$, whereupon by Theorem I.8.3 $\lim_n (a_n - b + x_n) = \check{y} = \{y_n\} \in \mathbb{R}$. The $\check{y}$ thus determined is the limit of the sequence $\{a_n + b_n\}$ in the sense of Definition I.8.1, since by that definition and Theorem I.7.3

$$(a_n + b_n) - y_n = [(a_n - b + x_n) - y_n] + (b_n + b - x_n) \rightarrow 0 + 0 = 0.$$

Multiplication. Let $\lim_n a_n = \check{x} = \{x_n\} \in \mathbb{R}$ and $\lim_n b_n = \check{y} = \{y_n\} \in \mathbb{R}$. If $\check{x}$ and $\check{y}$ have the same sign, then the sequence $\{x_n y_n\} = \{|x_n||y_n|\}$ is increasing (since $\{|x_n|\}$ and $\{|y_n|\}$ are increasing, cf. Exercise I.5:2), otherwise $\{x_n y_n\} = \{-|x_n||y_n|\}$ is a decreasing sequence. By Theorem I.8.3 there thus exists a limit $\lim_n x_n y_n = \check{z} = \{z_n\} \in \mathbb{R}$. This is also the limit of the sequence $\{a_n b_n\}$ in the sense of Definition I.8.1, since by that definition and Theorem I.7.4 (V3)

$$a_n b_n - z_n = (a_n - x_n)b_n + (b_n - y_n)x_n + (x_n y_n - z_n) \rightarrow 0 + 0 + 0 = 0.$$

Division. Since $\{b_n\}$ is a monotonic sequence, then, under the additional assumptions made, from some index $n = m$ onward one has either $b_n \geq b$ or $b_n \leq -b$.

Then the sequence $\{1/b_n\}$ is monotonic from the index $n = m$ onward. From that index onward we write $a_n/b_n = a_n(1/b_n)$ and appeal to the multiplication already treated. $\square$

Since the sequences $\{x_n\}$ and $\{y_n\}$ of Definition I.9.2 are monotonic and bounded, Theorem I.9.3 guarantees the workability of the definition. Let us illustrate the reasoning used in the proof of the theorem with a further computational example.

EXAMPLE 1 Let

$$\begin{aligned}\tilde{\pi} &= 3.141592653 \dots = \{3, 3.1, 3.14, 3.141 \dots\} = \{p_n\}, \\ \check{e} &= 2.718281828 \dots = \{2, 2.7, 2.71, 2.718 \dots\} = \{e_n\}.\end{aligned}$$

Determine a) $\tilde{\pi} + \check{e}$ and $\tilde{\pi}\check{e}$, b) $\tilde{\pi} - \check{e}$, c) $\tilde{\pi}/\check{e}$ in the sense of Definition I.9.2 to five decimals, using the proof of Theorem I.8.3 as an algorithm.

Solution a) The sequences

$$\{p_n + e_n\} = \{5, 5.8, 5.85, 5.859 \dots\}, \quad \{p_n e_n\} = \{6, 8.37, 8.5094, 8.537238 \dots\}$$

are both increasing and bounded, so they converge toward a real number (Theorem I.8.3). Using the proof construction of Theorem I.8.3 as an algorithm, one obtains

$$\tilde{\pi} + \check{e} = 5.85987 \dots, \quad \tilde{\pi}\check{e} = 8.53974 \dots$$

b) The sequence $\{p_n - e_n\} = \{0, 0.4, 0.43, 0.423, \dots\}$ is not monotonic, so Theorem I.8.3 does not apply directly. We therefore first write

$$p_n - e_n = (p_n - 3) + (3 - e_n).$$

Here $\{3 - e_n\}$ is a bounded and decreasing sequence. Its limit is obtained by the proof construction of Theorem I.8.3

$$\lim_n (3 - e_n) = 3 - \check{e} = 0.28171 \dots$$

When this intermediate result is interpreted as the sequence $\{0, 0.2, 0.28, 0.281, \dots\} = \{x_n\}$, then $\{p_n - 3\}$ and $\{x_n\}$ are both increasing sequences. Thus $\{(p_n - 3) + x_n\}$ is also increasing, and one can compute (the algorithm is the same as above)

$$\tilde{\pi} - \check{e} = \lim_n [(p_n - 3) + x_n] = 0.42331 \dots$$

c) The sequence $\{p_n/e_n\}$ is not monotonic, so Theorem I.8.3 does not apply directly here either. We first compute, as an intermediate result, the limit of the decreasing sequence $\{1/e_n\}$:

$$\lim_n (1/e_n) = 1/\check{e} = 0.36787 \dots$$

When the intermediate result is interpreted as the sequence $\{x_n\} = \{0, 0.3, 0.36, 0.367, \dots\}$, then $\{p_n x_n\}$ is an increasing sequence. To this the proof construction of Theorem I.8.3 applies and gives

$$\check{\pi}/\check{e} = \check{\pi}(1/\check{e}) = \lim_n p_n x_n = 1.15572 \dots \quad \square$$

The order relation of $\mathbb{R}$

The definition of the order relation for real numbers is also straightforward.

DEFINITION I.9.4 (The order relation of $\mathbb{R}$) If $\check{x} = \{x_n\} \in \mathbb{R}$, $\check{y} = \{y_n\} \in \mathbb{R}$, then $\check{x} < \check{y}$ exactly when $x_n < y_n$ for some n and $\check{x} \neq \check{y}$.

By Definition I.9.4, the order of the real numbers $\check{x} = \{x_n\}$ and $\check{y} = \{y_n\}$ can be resolved computationally by comparing the numbers x_n, y_n , $n = 0, 1, \dots$. The comparison is continued until the first index is encountered at which $x_n \neq y_n$. (If no such index is encountered, then $\check{x} = \check{y}$.) If $|x_n - y_n| \geq 2 \cdot 10^{-n}$, the order is resolved. Otherwise, i.e. if $x_n - y_n = \pm 10^{-n}$, there is still the possibility that $\check{x}$ and $\check{y}$ are each identified with the number x_n or y_n , in which case $\check{x} = \check{y}$ (see the considerations of Section I.7 concerning Theorem I.7.8).

From Definitions I.9.1–I.9.4 the following significant result can be derived. The proof is presented at the end of the section (supported by exercises).

***THEOREM I.9.5 ($(\mathbb{R}, +, \cdot, <)$ is an ordered field)** The set of real numbers, equipped with the operations of Definition I.9.2 and the order relation of Definition I.9.4, is an ordered field.

From Definition I.9.4 it follows immediately that if $\check{x} = \{x_n\} \in \mathbb{R}$, then

$$\begin{aligned} \check{x} = 0 &\Leftrightarrow x_n = 0 \quad \forall n \\ \check{x} < 0 &\Leftrightarrow x_n < 0 \quad \text{for some } n \quad (\text{sign } e = -) \\ \check{x} > 0 &\Leftrightarrow x_n > 0 \quad \text{for some } n \quad (\text{sign } e = +) \end{aligned}$$

This comparison is thus resolved solely on the basis of the sign of $\check{x}$, unless $\check{x} = 0$. The definition of the order relation of $\mathbb{R}$ can alternatively be based on this comparison combined with subtraction (cf. the order relation of $\mathbb{Q}$, Section I.1):

$$\check{x} < \check{y} \Leftrightarrow \check{x} - \check{y} < 0.$$

It is evident that, of the axioms of the order relation, at least (J1) (cf. Section I.1) holds under either definition.

Real numbers as abstract numbers

Once both the operations and the order relation have been defined for the real numbers, the nature of real numbers as sequences can, if desired, be 'forgotten', and the numbers treated merely as abstract numbers with which one can compute and which one can compare. In accordance with this way of thinking, from now on the real numbers too are regarded as 'genuine' numbers, no longer distinguished from the rational numbers in symbolic notation.

EXAMPLE 2 On the basis of the results of Section I.8 and Definitions I.9.1–I.9.2, we can now write

$$\begin{aligned}\lim_n \left(1 + \frac{1}{n}\right)^n &= \sum_{k=0}^{\infty} \frac{1}{k!} = 2.71828182845905\dots = e \in \mathbb{R}, \\ \lim_n \left(1 - \frac{1}{n}\right)^n &= 0.36787944117144\dots = \frac{1}{e}.\end{aligned}$$

Introducing another very well-known real number, $\pi = 3.1415926535897\dots$, one can also show that (cf. Section I.8)

$$\sum_{k=1}^{\infty} \frac{1}{k^2} = 1.64493406684822\dots = \frac{\pi^2}{6}.$$

Now that the real numbers have been defined as an ordered field, the original definition of the limit of a sequence (Definition I.6.1) also becomes valid when the terms of the sequence and/or the limit are real numbers. Since convergence of a sequence toward a real limit was defined earlier separately (Definition I.8.1), to close the circle the following result is needed (proof at the end of the section).

*THEOREM I.9.6 Definitions I.6.1 and I.8.1 for the convergence of a sequence toward a real number are equivalent.

Decimal-division and bisection constructions

By Definition I.9.1, a real number can, computationally, only be 'approached, not reached', except when the number is rational. Algorithmically, the 'approaching' can take place e.g. directly on the basis of Definition I.9.1, whereby one first computes the integer part of the number and then the digits one at a time. We now consider this problem of computation somewhat more generally.

Let a real number a be uniquely defined, for example by imposing some algebraic condition on the number, by defining the number as the limit of some known

sequence, or in some other way. We wish to construct a as a decimal in the sense of Definition I.9.1, i.e. in the form $a = x_0.d_1d_2\dots$. Assume in what follows that the sign of a is $+$ (otherwise one constructs the number $-a$). Then the integer part and the digits of a in the numbers $x_n = x_0.d_1\dots d_n$ are determined by following the instruction:

Find $x_n \in \mathbb{Q}_n$ such that $x_n \leq a$ and $x_n + 10^{-n} > a$, $n = 0, 1, \dots$

If a is a rational number, then the instruction is the same as in the long-division algorithm (cf. Section I.5). In the case of a more general real number $a > 0$, the instruction may be regarded as a thought-construction that defines the number a . We give it the name *decimal-division* construction. — The decimal-division construction has already been used in the proof of Theorem I.8.3. As a definition of a real number, the construction may be regarded as a variant or refinement of Definition I.9.1 resting on the order relation.

The decimal-division construction has already been noted to work also as an algorithm (cf. the examples and exercises of the previous section). More generally, the algorithm works if one assumes that the comparisons involved in the construction are computationally realizable. This is so at least if the number $a \in \mathbb{R}$ to be computed is assumed to be such that, for any rational number $x \in \mathbb{Q}$, one can determine by a finite number of rational operations (operations and comparisons) which of the alternatives $a < x$, $a = x$, $a > x$ holds. Then one first finds the sign of a by comparing with $x = 0$, after which the integer part of a or $-a$ and the digits d_n can be computed by following the above instruction, in principle up to any desired index. Such a *decimal-division algorithm* is thus a generalization of the long-division algorithm.

EXAMPLE 3 Construct the real number $a = \sqrt{2}$ by the decimal-division algorithm.

Solution The comparison of the number a with a positive rational number x reduces to a rational one by the equivalence

$$a < x \Leftrightarrow 2 < x^2.$$

Since $1^2 < 2$ and $2^2 > 2$, one must first have $a = 1.d_1d_2d_3\dots$. Further, since $1.4^2 = 1.96 < 2$, $1.5^2 = 2.25 > 2$, $1.41^2 = 1.9881 < 2$, $1.42^2 = 2.0164 > 2$, $1.414^2 = 1.999396 < 2$ and $1.415^2 = 2.002225 > 2$, we have $d_1 = 4$, $d_2 = 1$ and $d_3 = 4$. Continuing in this way, the result is the (aperiodic) decimal $a = 1.4142135623730950488016887\dots$ * $\square$

*Before calculators, the hand computation of square roots was practiced in schools too, in the manner of the example. — Nowadays calculators and computers compute square roots by algorithms much more efficient than decimal division, based on recursive sequences, cf. Example 4 of the previous section.

EXAMPLE 4 Compute the number $a = \sqrt{\sqrt{2} + 1}$ to 20 significant digits using the decimal-division algorithm.

Solution When comparing the number a with a rational number $x \geq 1$,

$$a < x \Leftrightarrow \sqrt{2} + 1 < x^2 \Leftrightarrow \sqrt{2} < x^2 - 1 \Leftrightarrow 2 < (x^2 - 1)^2,$$

so the comparison reduces to a rational operation. Following the algorithm, one obtains

$$a = 1.5537739740300373073.. \quad \square$$

In the examples, the reduction of the comparison $a < x$ to a rational one rests on *symbolic* (non-numerical) *computation*, here algebra, which relies ultimately on the field axioms of the real numbers, the axioms of the order relation, and the symbolic definition of the square root $(\sqrt{a})^2 = a$. As in these examples, the task of symbolic computation is generally to simplify or clarify a computational task before the actual numerical computations.

The counterpart of decimal division can naturally also be constructed based on other number systems. In the case of the binary system the name *bisection* construction (or bisection method) is used. In the bisection construction a real number $a \geq 0$ is constructed in binary form $a = x_0.b_1b_2b_3..$, where the integer part x_0 and the bits $b_n \in \{0, 1\}$ are chosen so that the numbers x_0 and $x_n = x_0 + \sum_{k=1}^n b_k \cdot 2^{-k}$, $n \in \mathbb{N}$, satisfy

$$x_n \leq a \quad \text{and} \quad x_n + 2^{-n} > a, \quad n = 0, 1, \dots$$

Because of its simple logic, the bisection method is very often used in the thought-constructions of mathematical proofs. (The bisection construction would have been an alternative in the proof of Theorem I.8.3 as well.) When the additional assumptions above hold, the construction can also be used as a workable algorithm.

Sequences of real numbers

'Forgetting' the sequence-nature of the real numbers is especially advisable when one studies *sequences of real numbers* and their convergence. Definition I.6.1, which appeals to the order relation and (to a small extent) to field algebra, is valid as such also as a definition of the convergence of a sequence of real numbers. If a sequence of real numbers has a limit in the sense of this definition (as a real number), then the sequence is said to *converge*, otherwise to *diverge* (cf. Definition I.8.1).

EXAMPLE 5 Determine the sum of the basic geometric series when $q = 1/\sqrt{2}$.

Solution The summation formula for the geometric series derived in Section I.6 is valid when $q \in \mathbb{R}$ and $|q| < 1$. Here $0 < q < 1$, so

$$\sum_{n=0}^{\infty} \left(\frac{1}{\sqrt{2}}\right)^n = \frac{1}{1 - \frac{1}{\sqrt{2}}} = \frac{\sqrt{2}}{\sqrt{2} - 1} = 2 + \sqrt{2} = 3.4142135623730950488 \dots \quad \square$$

The computation of the example is again an example of symbolic computation as well, in which the number $\sqrt{2}$ is treated as an abstract number using only the general properties of the field $(\mathbb{R}, +, \cdot)$ (the field axioms) and the symbolic definition $(\sqrt{2})^2 = 2$.

Let it be noted that all the definitions and claims concerning sequences presented in Sections I.6–I.8 are valid in the field of real numbers $(\mathbb{R}, +, \cdot, <)$ — for the reason that they rest only on the algebra of an ordered field and the assumption that the field contains the rational numbers.

As an example of a convergence consideration for a sequence of real numbers, let us prove

PROPOSITION I.9.7 $\lim_n \sqrt[n]{a} = 1 \quad \forall a \in \mathbb{R}, a > 0.$

Proof Set $b_n = \sqrt[n]{a}$. In the case $a = 1$ the claim is evidently true, and in the case $0 < a < 1$ we have: $b_n = 1/(1/a)^{1/n} \rightarrow 1$, if $(1/a)^{1/n} \rightarrow 1$ (Theorem I.7.3 for sequences of real numbers). It thus suffices to consider the case $a > 1$. Then necessarily $b_n > 1 \forall n$ (since $b_n^n = a > 1$), whereupon

$$\left(\frac{b_{n+1}}{b_n}\right)^n = \frac{1}{b_{n+1}} \frac{b_{n+1}^{n+1}}{b_n^n} = \frac{1}{b_{n+1}} \frac{a}{a} = \frac{1}{b_{n+1}} < 1.$$

Thus $b_{n+1}/b_n < 1 \forall n$, so $\{b_n\}$ is a strictly decreasing sequence. Since $\{b_n\}$ is also evidently a bounded sequence, $b_n \rightarrow b \in \mathbb{R}$ (Theorem I.8.3), and since $b_n > 1 \forall n$, then $b \geq 1$ (Proposition I.7.4 (V1) for sequences of real numbers). Here the alternative $b > 1$ would lead to a logical contradiction: $a = b_n^n \geq b^n \rightarrow \infty$. Thus $\lim_n b_n = 1$. $\square$

*Proofs of Theorems I.9.5 and I.9.6

Theorem I.9.5. Of the field axioms we prove, as an example, only the validity of axioms (K4) (associative law of multiplication) and (K9) (inverse element); the others are left as an exercise (Exercise 1).

[K4] Let $\check{x} = \{x_n\}$, $\check{y} = \{y_n\}$, $\check{z} = \{z_n\}$ be real numbers and let, further, $\check{x}\check{y} = \check{a} = \{a_n\} \in \mathbb{R}$ and $\check{y}\check{z} = \check{b} = \{b_n\} \in \mathbb{R}$. Then the content of axiom (K4) is, on the basis of Definitions I.9.2 and I.8.1,

$$\check{a}\check{z} = \check{x}\check{b} \Leftrightarrow \lim_n (a_n z_n - x_n b_n) = 0.$$

We write here

$$a_n z_n - x_n b_n = (a_n - x_n y_n) z_n + x_n (y_n z_n - b_n).$$

Since here $a_n - x_n y_n \rightarrow 0$ and $y_n z_n - b_n \rightarrow 0$ (Definitions I.9.2 and I.8.1) and the sequences $\{x_n\}$ and $\{z_n\}$ are bounded, the claim $a_n z_n - x_n b_n \rightarrow 0$ follows by Theorem I.7.3 and Proposition I.7.4 (V3).

[K9] To construct the inverse $\check{x}^{-1}$ of the number $\check{x} = \{x_n\} \in \mathbb{R}$, let $\check{x} \neq 0$, whereupon for some $m \in \mathbb{N}$ one must have $|x_n| \geq |x_m| > 0 \quad \forall n \geq m$. Then the sequence $\{x_m^{-1}, x_{m+1}^{-1}, \dots\}$ is monotonic and bounded, whereupon there exists a real number $\check{y} = \{y_n\}$ such that $x_n^{-1} - y_n \rightarrow 0$ (Theorem I.8.3). By Proposition I.7.4 (V3) and Definition I.9.2 we then conclude

$$x_n^{-1} - y_n \rightarrow 0 \Leftrightarrow 1 - x_n y_n \rightarrow 0 \Leftrightarrow \check{x}\check{y} = 1.$$

Thus $\check{y} = \check{x}^{-1}$, and axiom (K9) is therefore valid. $\square$

The validity of axiom (J1) of the order relation has already been established. The verification of axioms (J2) and (J4) is left as an exercise (Exercise 2cd). We thus prove only the validity of axiom (J3).

PROPOSITION I.9.8 For real numbers: If $\check{x} < \check{y}$, then $\check{x} + \check{z} < \check{y} + \check{z} \quad \forall \check{z} \in \mathbb{R}$.

Proof Let $\check{x} = \{x_n\} \in \mathbb{R}$, $\check{y} = \{y_n\} \in \mathbb{R}$, $\check{z} = \{z_n\} \in \mathbb{R}$ and set $\check{x} + \check{z} = \check{a} = \{a_n\} \in \mathbb{R}$ and $\check{y} + \check{z} = \check{b} = \{b_n\} \in \mathbb{R}$, whereupon $x_n + z_n - a_n \rightarrow 0$ and $y_n + z_n - b_n \rightarrow 0$ as $n \rightarrow \infty$ (Definition I.9.2). Claim: If $\check{x} < \check{y}$, then $\check{a} < \check{b}$. Make the counter-assumption: $\check{a} = \check{b}$ or $\check{a} > \check{b}$. In the first alternative $\lim_n (a_n - b_n) = 0$ (Definition I.7.7), whereupon by the assumptions and Theorem I.7.3 it follows that

$$x_n - y_n = (x_n + z_n - a_n) - (y_n + z_n - b_n) + (a_n - b_n) \rightarrow 0.$$

Thus $\check{x} = \check{y}$ (Definition I.7.7), but this is a logical contradiction, since (J1) holds and it was assumed that $\check{x} < \check{y}$. In the second alternative ($\check{a} > \check{b}$) one has $a_n \geq b_n \quad \forall n$ (Exercise 2a) and likewise $x_n \leq y_n \quad \forall n$ (since $\check{x} < \check{y}$), so it follows that

$$\begin{aligned} c_n &= (x_n + z_n - a_n) - (y_n + z_n - b_n) \\ &= (x_n - y_n) + (b_n - a_n) \leq x_n - y_n \leq 0 \quad \forall n. \end{aligned}$$

Since here $c_n \rightarrow 0$, then $x_n - y_n \rightarrow 0$ (Proposition I.7.4 (V2)), so we have again arrived at a logical contradiction: $\check{x} < \check{y}$ and $\check{x} = \check{y}$. The alternatives $\check{a} = \check{b}$ and $\check{a} > \check{b}$ are thus ruled out, so necessarily $\check{a} < \check{b}$. $\square$

Theorem I.9.6. We base the proof on the following claim, which follows easily from Definition I.9.4 (Exercise 2b).

LEMMA I.9.9 If $\check{x} = \{x_n\} \in \mathbb{R}$, then $|\check{x} - x_n| \leq 10^{-n} \forall n$.

Assume that $a_n \rightarrow \check{x}$, $\check{x} = \{x_n\} \in \mathbb{R}$ in the sense of Definition I.6.1 and let $\varepsilon > 0$. Then there exist indices $N_1, N_2 \in \mathbb{N}$ such that $|a_n - \check{x}| < \varepsilon/2$ when $n > N_1$ and $10^{-n} < \varepsilon/2$ when $n > N_2$, whereupon by the triangle inequality (of the ordered field) and Lemma I.9.9

$$|a_n - x_n| \leq |a_n - \check{x}| + |\check{x} - x_n| < |a_n - \check{x}| + 10^{-n} < \varepsilon, \quad \text{when } n > \max\{N_1, N_2\} = N.$$

Here $\varepsilon > 0$ was arbitrary, so by Definition I.6.1 $a_n - x_n \rightarrow 0$, i.e. $a_n \rightarrow \check{x}$ in the sense of Definition I.8.1. This proves the first part of the claim. The second part is proved by a similar argument. $\square$

EXERCISES

- Show, on the basis of Definition I.9.2, that for the real numbers the following hold:
 - the commutative laws of addition and multiplication (K1,K2)
 - the associative law of addition (K3)
 - the distributive law of addition and multiplication (K5)
- Let $\check{x} = \{x_n\} \in \mathbb{R}$, $\check{y} = \{y_n\} \in \mathbb{R}$ and $z = \{z_n\} \in \mathbb{R}$. Show that, on the basis of Definitions I.9.4 and I.9.2,
 - $\check{x} < \check{y} \Leftrightarrow \check{x} \neq \check{y}$ and $x_n \leq y_n \forall n$
 - $|\check{x} - x_n| \leq 10^{-n} \forall n$
 - $\check{x} < \check{y}$ and $\check{y} < \check{z} \Rightarrow \check{x} < \check{z}$
 - $\check{x} > 0$ and $\check{y} > 0 \Rightarrow \check{x}\check{y} > 0$
- Investigate how large an error is made when, in computing
 - $e + \pi$,
 - e^2/π ,
 - $\pi^2 - e^2$
 - $e^5\pi^6$, e and π are first truncated to 9 significant digits and, in the operations, the result likewise to 9 significant digits.
- Let the base of the number system be $= k$. Using the k -division algorithm corresponding to the decimal-division algorithm, construct to four significant digits
 - $\sqrt[3]{11}$ in the binary system,
 - $\sqrt{7}$ in the base-3 system.

5. (*) It is known that $\sum_{k=1}^{\infty} k^{-2} = \pi^2/6$. Show, on the basis of this,

$$\text{a) } \sum_{k=0}^{\infty} \frac{1}{(2k+1)^2} = \frac{\pi^2}{8} \quad \text{b) } \sum_{k=0}^{\infty} \frac{(-1)^k}{(k+1)^2} = \frac{\pi^2}{12}$$

6. (*) Let $a_k \rightarrow a$ ($k \rightarrow \infty$), where, in decimal notation,

$$a_k = x_0^{(k)}.d_1^{(k)}d_2^{(k)}\dots = \{x_n^{(k)}\} \in \mathbb{R}, \quad a = x_0.d_1d_2\dots = \{x_n\} \in \mathbb{R}.$$

Prove the following claims:

- a) If a is not a finite decimal, then for every $n \in \mathbb{N}$ there exists an index $N_n \in \mathbb{N}$ such that $a_k = x_0.d_1\dots d_n\dots \forall k > N_n$.
- b) The claim of part a) is not true for every $a \in \mathbb{R}$ (counterexample!).
- c) If a is a finite decimal, then there exists $m \in \mathbb{N}$ and, for every $n \geq m$, an index $N_n \in \mathbb{N}$ such that $d_n^{(k)} \in \{0, 9\} \forall k > N_n$.
- d) For every $a \in \mathbb{R}$: $\lim_k x_k^{(k)} = a$.

I.10 Cauchy sequences

In this section, 'sequence' means a sequence of real numbers. The object of study is a still-open question in the theory of sequences (also of rational-number sequences), namely: Under exactly what conditions—conditions on the sequence itself—does a sequence $\{a_n\}$ converge, i.e., does the limit $\lim_n a_n = a$ exist as a real number in the sense of Definition I.6.1? The convergence conditions known so far, in Theorems I.8.3 and I.9.3, are sufficient but not necessary. In seeking a complete solution to the convergence question, i.e., conditions both necessary and sufficient for convergence, the key turns out to be

DEFINITION I.10.1 (Cauchy* sequence) A sequence $\{a_n\}$ is *Cauchy*, or a *Cauchy sequence*, if for every $\varepsilon > 0$ there exists $N \in \mathbb{N}$ such that

$$|a_n - a_m| < \varepsilon \quad \text{when } n, m > N.$$

EXAMPLE 1 Show that the sequence of partial sums of the series $\sum_{k=0}^{\infty} (-1)^k / (k+1)^2$,

$$\{s_n\} = \left\{1, 1 - \frac{1}{2^2}, 1 - \frac{1}{2^2} + \frac{1}{3^2}, \dots\right\} = \left\{1, \frac{3}{4}, \frac{7}{9}, \dots\right\}$$

is a Cauchy sequence.

Solution If $0 \leq m < n$, then applying the triangle inequality we can estimate (see Exercise I.8:1)

$$|s_n - s_m| = \left| \sum_{k=m+1}^n \frac{(-1)^k}{(k+1)^2} \right| \leq \sum_{k=m+1}^n \frac{1}{(k+1)^2} < \frac{1}{m+1}.$$

Consequently, if $\varepsilon > 0$ and we choose $N \in \mathbb{N}$ such that $1/(N+1) < \varepsilon$ (possible for every $\varepsilon > 0$), then

$$|s_n - s_m| < \max \left\{ \frac{1}{n+1}, \frac{1}{m+1} \right\} < \varepsilon, \quad \text{when } n, m > N.$$

Thus $\{s_n\}$ is Cauchy. $\square$

EXAMPLE 2 If $a \neq 0$, then the sequence $\{a_n\} = \{(-1)^n a\}$ is not Cauchy. Indeed, if $0 < \varepsilon \leq 2|a|$, then the condition of Definition I.10.1 fails for every $N \in \mathbb{N}$, since $|a_n - a_m| = 2|a| \geq \varepsilon$ whenever $n - m$ is odd. $\square$

*The French mathematician **Augustin Louis Cauchy** (1789-1857) ranks among the greatest mathematicians of all time. Cauchy substantially sharpened the conceptual apparatus of mathematics and laid the groundwork for new lines of research. His extensive output extended to physics as well, among other things to the elasticity theory of solids.

Cauchy sequences have properties similar to those of convergent sequences. First we show:

THEOREM I.10.2 A convergent sequence is Cauchy.

Proof If $\lim_n a_n = a \in \mathbb{R}$ in the sense of Definition I.6.1, then for every $\varepsilon > 0$ there exists $N \in \mathbb{N}$ such that $|a_n - a| < \varepsilon/2$ when $n > N$. Then, by the triangle inequality,

$$\begin{aligned} |a_n - a_m| &= |(a_n - a) + (a - a_m)| \\ &\leq |a_n - a| + |a_m - a| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon, \quad \text{when } n, m > N. \quad \square \end{aligned}$$

Theorems I.7.2 and I.7.3 concerning convergent sequences have the following counterparts for Cauchy sequences:

THEOREM I.10.3 A Cauchy sequence is bounded.

THEOREM I.10.4 (Combination rules for Cauchy sequences) If the sequences $\{a_n\}$ and $\{b_n\}$ are Cauchy sequences and $\lambda \in \mathbb{R}$, then the sequences $\{a_n + b_n\}$, $\{\lambda a_n\}$ and $\{a_n b_n\}$ are also Cauchy sequences. If in addition $|b_n| \geq \delta > 0 \ \forall n$, then $\{a_n/b_n\}$ is a Cauchy sequence as well.

The proofs of these theorems (which we omit; see Exercise 1) are very similar to those of the counterpart theorems mentioned above. For instance, the claim of Theorem I.10.3 is reached by taking as the point of comparison, instead of the limit, a term a_m of the sequence, where the index m is chosen so that $|a_n - a_m| < 1$ when $n > m$; cf. Definition I.10.1 and the proof of Theorem I.7.2.

The main result in what follows is

***THEOREM I.10.5 (Cauchy's convergence criterion)** A real-number sequence $\{a_n\}$ converges to a real number if and only if $\{a_n\}$ is Cauchy.

EXAMPLE 1 (continued) By Theorem I.10.5, the series of the example converges. Computing partial sums far enough shows that the sum of the series, truncated to 10 significant figures, is

$$\sum_{k=0}^{\infty} \frac{(-1)^k}{(k+1)^2} = 0.8224670334 \dots \quad \square$$

Since we already know that every convergent sequence is Cauchy (Theorem I.10.2), to prove Theorem I.10.5 it suffices to establish the implication

$$\{a_n\} \text{ is Cauchy} \Rightarrow \{a_n\} \text{ converges.} \quad (\star)$$

The proof has several stages. First we must examine the definition of the limit of a sequence more closely, on the basis of the concept of a *subsequence*. In what follows, these considerations and the proof of claim (★) based on them are set apart as a separate subsection*.

*Subsequences

DEFINITION I.10.6 A sequence $\{b_k\}_{k=1}^{\infty}$ is a *subsequence* of the sequence $\{a_n\}_{n=m}^{\infty}$ if there exist indices $m \leq n_1 < n_2 < \dots$ such that

$$b_k = a_{n_k}, \quad k = 1, 2, \dots$$

EXAMPLE 3 If $\check{x} = \{x_n\}_{n=p}^{\infty}$ is a periodic decimal, then for some $m \in \mathbb{N}$ and sufficiently large $k \in \mathbb{N}$ the subsequence

$$\{x_k, x_{k+m}, x_{k+2m}, \dots\} = \{x_{k+(l-1)m}\}_{l=1}^{\infty}$$

is a geometric series; cf. Section I.6. Here the index of the subsequence is l , and $n_l = k + (l-1)m$. $\square$

EXAMPLE 4 The sequence $\{a_n = (-1)^n, n = 0, 1, 2, \dots\}$ does not converge, but it has subsequences that converge (to a rational number), e.g. $\{a_0, a_2, a_4, \dots\}$. $\square$

EXAMPLE 5 Every subsequence of the unboundedly increasing sequence $\{n\} = \{1, 2, 3, \dots\}$ is also unboundedly increasing. $\square$

The concept of a subsequence is illuminated a little by the following 'warm-up theorem'.

THEOREM I.10.7 $\lim_n a_n = a \in \mathbb{R}$ if and only if $\lim_k b_k = a$ for every subsequence $\{b_k\}$ of the sequence $\{a_n\}$.

Proof The part $\Rightarrow$ of the claim is obvious, since by Definition I.6.1 a subsequence converges at least as fast as the sequence itself. Even so, the part $\Leftarrow$ is even more obvious, since the sequence $\{a_n\}$ itself also qualifies as a subsequence. $\square$

The following result, in which the concept of a subsequence is central, is the key to what follows. In it, the negation of the proposition $a_n \rightarrow a$ (namely

The mark () in the title of a section or subsection indicates (here and henceforth) that the part of the text in question broadens the perspective—and is often more challenging than the rest of the text.

$a_n \not\rightarrow a$) is interpreted on the basis of Definition I.6.1. — Note that this is not the negation 'does not converge' (to any number), because $a_n \not\rightarrow a$ is also true when $a_n \rightarrow b \neq a$.

*THEOREM I.10.8 If $\{a_n\}$ is a sequence and $a \in \mathbb{R}$, then $a_n \not\rightarrow a$ if and only if there exist $\varepsilon > 0$ and a subsequence $\{a_{n_1}, a_{n_2}, \dots\}$ such that $|a_{n_k} - a| \geq \varepsilon \quad \forall k \in \mathbb{N}$.

Proof We start from the definition of the limit (Definition I.6.1)

$$a_n \rightarrow a \quad \Leftrightarrow \quad \forall \varepsilon > 0 \text{ it holds that } [\dots]$$

and set about unfolding the negation: first (cf. Section I.3)

$$a_n \not\rightarrow a \quad \Leftrightarrow \quad \text{for some } \varepsilon > 0 \text{ it does not hold that } [\dots]$$

Here $[\dots] = [\exists N \in \mathbb{N} \text{ such that } \{\dots\}]$, so

$$\begin{aligned} \text{it does not hold that } [\dots] &\Leftrightarrow \nexists N \in \mathbb{N} \text{ such that } \{\dots\} \\ &\Leftrightarrow \text{not } \{\dots\} \quad \forall N \in \mathbb{N}. \end{aligned}$$

Here $\{\dots\} = \{|a_n - a| < \varepsilon \quad \forall n > N\}$, so

$$\text{not } \{\dots\} \quad \Leftrightarrow \quad |a_n - a| \geq \varepsilon \text{ for some } n > N.$$

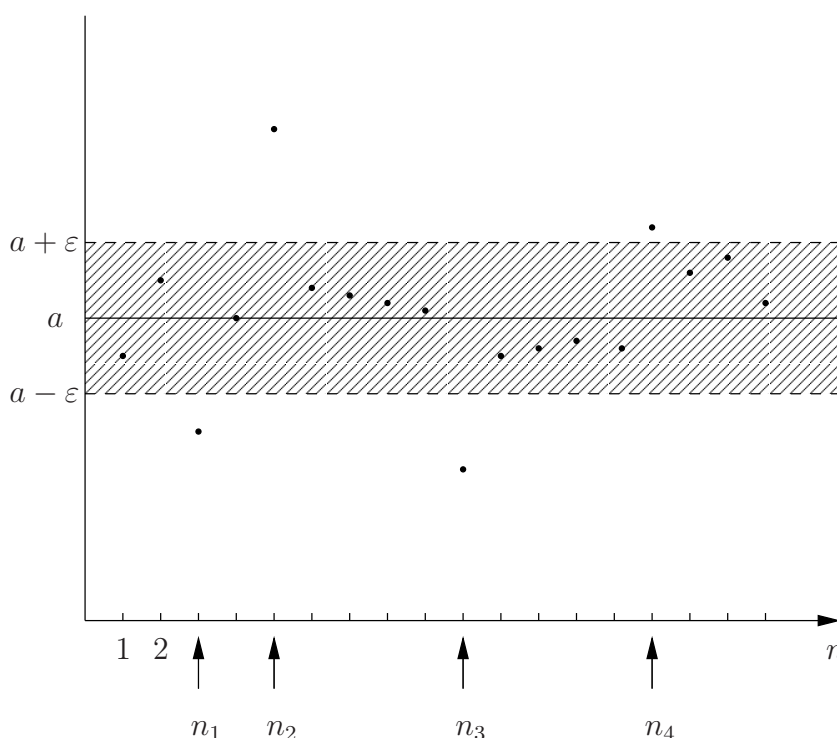
Thus we have deduced:

$$a_n \not\rightarrow a \quad \Leftrightarrow \quad \text{for some } \varepsilon > 0 \text{ it holds: } \forall N \in \mathbb{N} (|a_n - a| \geq \varepsilon \text{ for some } n > N).$$

With this ε we may now proceed as follows:

1. Choose $N = 1$ and $n_1 > N$ such that $|a_{n_1} - a| \geq \varepsilon$.
2. Choose $N = n_1$ and $n_2 > N$ such that $|a_{n_2} - a| \geq \varepsilon$.
- $\vdots$

The construction can be continued indefinitely, so we obtain a subsequence $\{a_{n_1}, a_{n_2}, \dots\}$ (see the figure below). This has the required property, so the part $\Rightarrow$ of the claim is proved. The part $\Leftarrow$ is obvious from the definition of the limit. $\square$



Applying Theorem I.10.8 to a Cauchy sequence yields an interesting result.

*THEOREM I.10.9 If a sequence $\{a_n\}$ is a Cauchy sequence, then for each $x \in \mathbb{R}$ exactly one of the following alternatives holds:

1. $a_n \rightarrow x$.
2. $\exists \varepsilon > 0$ and $N \in \mathbb{N}$ such that $a_n < x - \varepsilon$ whenever $n > N$.
3. $\exists \varepsilon > 0$ and $N \in \mathbb{N}$ such that $a_n > x + \varepsilon$ whenever $n > N$.

Proof First, either $a_n \rightarrow x$ (alternative 1) or $a_n \not\rightarrow x$. In the latter case, by Theorem I.10.8 we can find a number $\varepsilon > 0$ and a subsequence $\{b_k = a_{n_k}, k = 1, 2, \dots\}$ such that $|b_k - x| \geq 2\varepsilon \forall k$. (Here 2ε has been written in place of ε for convenience.) Since $\{a_n\}$ is a Cauchy sequence, there also exists, corresponding to ε , an index N such that $|a_n - a_m| < \varepsilon$ when $n, m > N$. Now choose in the subsequence $\{b_k\}$ an index k such that $n_k > N$, and write

$$a_n = b_k + (a_n - b_k).$$

Here $b_k = a_{n_k}$, $m = n_k > N$, so $|a_n - b_k| < \varepsilon$ when $n > N$. Also $|b_k - x| \geq 2\varepsilon$, so we must have either $b_k \geq x + 2\varepsilon$ or $b_k \leq x - 2\varepsilon$. If $b_k \geq x + 2\varepsilon$ (as in the figure

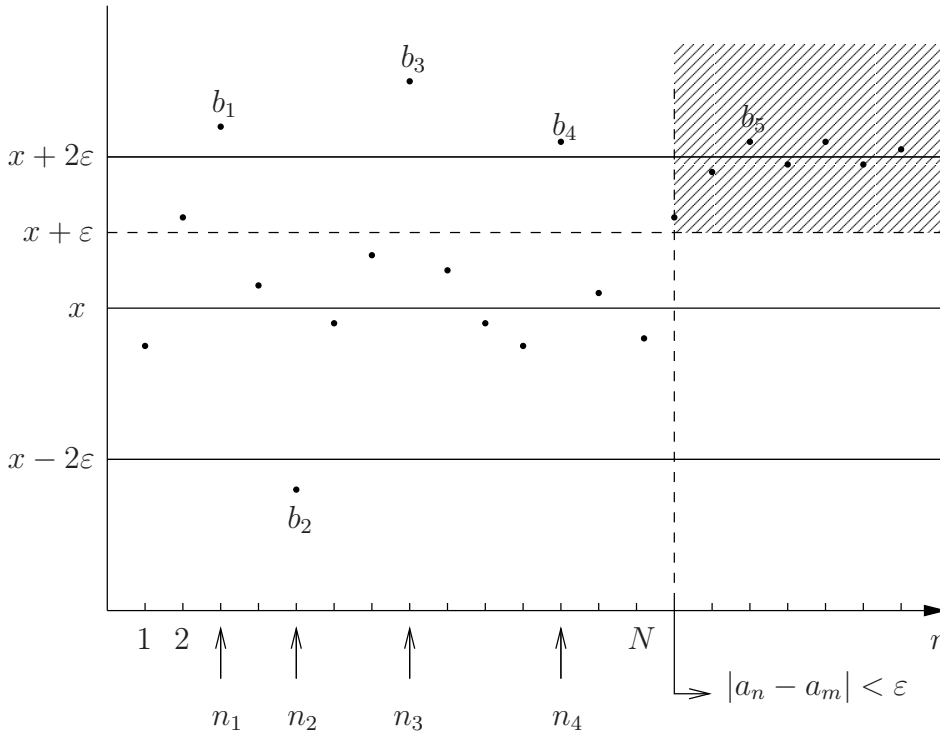
below), then we estimate

$$a_n \geq x + 2\varepsilon - |a_n - a_k| > x + 2\varepsilon - \varepsilon = x + \varepsilon, \quad \text{when } n > N.$$

If $b_k \leq x - 2\varepsilon$, then we estimate

$$a_n \leq x - 2\varepsilon + |a_n - a_k| < x - 2\varepsilon + \varepsilon = x - \varepsilon, \quad \text{when } n > N.$$

The theorem is thus proved. $\square$



By Theorem I.10.9, a Cauchy sequence must 'choose its side' with respect to a given number—for example a rational number—unless that number is the limit of the sequence. Using this result as a 'workhorse', we can now prove claim $(\star)$, and hence also Theorem I.10.5. The proof is otherwise essentially the same as the proof of Theorem I.8.3, except that instead of monotonicity we appeal to Theorem I.10.9.

*THEOREM I.10.10 A Cauchy sequence converges.

Proof Let $\{a_n\}$ be a Cauchy sequence. If $a_n \rightarrow x \in \mathbb{Q}$, then the claim of the theorem is true, so we may assume that $\{a_n\}$ does not converge to a rational

number. Then, first of all, for every integer $x \in \mathbb{Z}$ either alternative 2 or 3 of Theorem I.10.9 holds. Since a Cauchy sequence is bounded (Theorem I.10.3), there is then a unique $x_0 \in \mathbb{Z}$ such that $a_n > x_0 + \varepsilon_1$ when $n > N_1$ and $a_n < x_0 + 1 - \varepsilon_2$ when $n > N_2$ (here $\varepsilon_1, \varepsilon_2 > 0$ and $N_1, N_2 \in \mathbb{N}$), so that

$$x_0 < a_n < x_0 + 1, \quad \text{when } n > N = \max\{N_1, N_2\}.$$

Assume henceforth that $x_0 \geq 0$ (otherwise consider the sequence $\{-a_n\}$). Continuing the reasoning similarly, we can find unique $d_k \in \{0, \dots, 9\}$, $k = 1, 2, \dots$ and corresponding finite decimals $x_k = x_0.d_1 \dots d_k$ and indices $N_k \in \mathbb{N}$ such that

$$x_k < a_n < x_k + 10^{-k}, \quad \text{when } n > N_k, \quad k = 0, 1, 2, \dots$$

This means that $\lim_n(a_n - x_n) = 0$, i.e., $\lim_n a_n = \check{x} = \{x_n\} \in \mathbb{R}$ in the sense of Definitions I.8.1 and I.9.1. By Theorem I.9.6, $\lim_n a_n = \check{x}$ also in the sense of Definition I.6.1. $\square$

The real numbers as Cauchy sequences

Definition I.8.1 interprets an infinite decimal as a sequence 'converging to itself'. This viewpoint can be taken more generally toward Cauchy sequences of rational numbers, yielding the following alternative definition of the real numbers (cf. Definitions I.9.1 and I.7.7).

DEFINITION I.10.11 (The real numbers as Cauchy sequences) The real numbers are the Cauchy sequences of rational numbers. Two real numbers $a = \{a_n\}$ and $b = \{b_n\}$ are equal if and only if $\lim_n (a_n - b_n) = 0$.

By this definition, the real numbers are the equivalence classes (or identification classes, cf. Section I.3) formed by the Cauchy sequences of rational numbers, within which different sequences are identified by the criterion of the definition. Each identification class contains infinitely many different sequences; e.g., the sequences $\{a_n\}$ and $\{a_n + n^{-k}\}$ belong to the same class for every $k \in \mathbb{N}$. Unless it is a finite decimal, exactly one of the Cauchy sequences belonging to a given identification class is a 'representative' of the real number in the sense of Definition I.9.1, i.e., an infinite decimal.

On the basis of Definition I.10.11, defining the arithmetic operations for real numbers is quite simple: If $a = \{a_n\}$ and $b = \{b_n\}$ are real numbers (i.e., $\{a_n\}$ and $\{b_n\}$ are Cauchy sequences of rational numbers), then the sequences $\{a_n + b_n\}$ and $\{a_n b_n\}$ are also Cauchy sequences (Theorem I.10.4) and, as such, represent the numbers $a + b$ and ab . Likewise, if $b \neq 0$ (i.e., $b_n \not\rightarrow 0$), then from some index

$n = m$ onward $|b_n| \geq \delta > 0$ (Theorem I.10.9), so that $\{a_n/b_n\}$ is Cauchy from that index onward (Theorem I.10.4) and represents the number a/b .

The order relation, too, is easily defined by means of Cauchy sequences, since if $a = \{a_n\}$, then the alternatives $a = 0$, $a > 0$ and $a < 0$ are determined by setting $x = 0$ in Theorem I.10.9: $a = 0$ in alternative 1, $a < 0$ in alternative 2, and $a > 0$ in alternative 3. More generally, the ordering of numbers a and b is decided by comparing the number $a - b = \{a_n - b_n\}$ with 0 in the manner described.

If Definition I.10.11 is interpreted computationally, then an individual Cauchy sequence can be regarded as an algorithm that produces for a real number a sequence of rational approximations. In general, the algorithm is the better, the faster the sequence converges (or 'zips along') relative to the number of (rational) arithmetic operations required. So one can also impose quality criteria on Cauchy sequences!

EXAMPLE 6 One equivalence class of Cauchy sequences of rational numbers is $\sqrt{2}$. The infinite decimal belonging to this class, i.e., the sequence $\{1, 1.4, 1.414, \dots\}$, corresponds as an algorithm to the base-ten subdivision algorithm of Example 3 in the previous section. Another sequence belonging to the same equivalence class is the one considered in Example 4 of Section I.8. As an algorithm, this is clearly more efficient than base-ten subdivision. $\square$

*The Bolzano–Weierstrass theorem

For the sake of context, let us present one more theorem connected with the concept of a subsequence, one that belongs to the cornerstones of modern mathematical analysis.

*THEOREM I.10.12 (**Bolzano–Weierstrass***) Every bounded real-number sequence has a convergent subsequence.

Proof The basic idea of the proof is a bisection construction, in which the logical reasoning rests on a two-place predicate

$$Q(x, y) : \quad x \leq a_n \leq y \quad \text{for infinitely many values of the index } n.$$

*The Bolzano–Weierstrass theorem has many forms. The basic idea was first presented by the Czech mathematician, philosopher and priest **Bernard Bolzano** (1781–1848) in 1817, but the result attracted little attention. The German **Karl Weierstrass** (1815–1897), regarded as the 'father of modern mathematical analysis', later proved the theorem without knowing of Bolzano's work.

Since $\{a_n\}$ is a bounded sequence, we can find numbers a and $b = a + L$ such that $Q(a, b)$ is true. Choose some index n_1 such that $a \leq a_n \leq b$ when $n = n_1$, and next examine whether $Q(a, (a + b)/2)$ is true. If it is, set the new value of b to $(a + b)/2$. If not, then the proposition $Q((a + b)/2, b)$ must be true (since $Q(a, b)$ was true); in this case set the new value of a to $(a + b)/2$. Once one of the numbers a, b has thus been redefined, we know that $Q(a, b)$ is again true. Now find an index $n_2 > n_1$ such that $a \leq a_n \leq b$ when $n = n_2$ (possible, since $Q(a, b)$ was true). Proceeding in the same way, we can construct a subsequence $\{b_k = a_{n_k}, k = 1, 2, \dots\}$ of the original sequence, which by construction has the property $|b_k - b_l| \leq 2^{-N}L$ when $k, l > N$. Consequently, if for every $\varepsilon > 0$ we choose $N \in \mathbb{N}$ such that $2^{-N}L < \varepsilon$, then $|b_k - b_l| < \varepsilon$ when $k, l > N$. By Definition I.10.1, $\{b_k\}$ is a Cauchy sequence and therefore converges (Theorem I.10.10). $\square$

*Nonconstructive reasoning

Hidden in the proof of Theorem I.10.12 is a philosophical problem that occurs more generally in mathematical thinking: If in the predicate $Q(x, y)$ we think of x and y as fixed (bound), then to determine the truth value of the proposition $P = Q(x, y)$ one must go literally through every term of the sequence $\{a_n\}$, i.e., one must carry out infinitely many comparisons of numbers. This is of course not possible in practice. In mathematical analysis, however, such so-called *nonconstructive* reasoning is nevertheless deemed permissible. (Ultimately the 'permission' is granted by the basic axioms of set theory.) A nonconstructive proof is characterized by the fact that the proof cannot be followed by 'computing', unless some additional information is available, i.e., unless additional assumptions are made. For example, the proof of Theorem I.8.3 given earlier is also nonconstructive, for it too involves the notion of comparing all the terms of the sequence with a fixed number. In fact, the proofs of Theorems I.10.8 and I.10.9 are nonconstructive as well, since the number ε claimed to 'exist' is constructed in the proofs only as a logical necessity. No computationally workable idea for determining the number is given—nor, within the general assumptions made, would it be possible to give one.

In all the proofs mentioned, the philosophical problems disappear (or at least diminish) if the sequence under consideration is assumed to be such that the logical steps contained in the proof can be carried out (one at a time) with a finite number of (ultimately) rational arithmetic operations or comparisons. Such a sequence might be called, say, *predictable*. For example, the example sequences of Sections I.4 — I.9, or more generally all 'predictably defined' individual sequences, are of this kind. When this additional assumption holds, the thought-

constructions presented in the proofs can also be followed computationally. In the case of Theorem I.10.8, this is done, e.g., by searching for a valid value of ε in the sequence $\{\varepsilon_n = 10^{-n}, n = 1, 2, \dots\}$. By the additional assumption, for any fixed $n \in \mathbb{N}$ one can decide, with a finite amount of computation, whether $\varepsilon = \varepsilon_n$ is a valid choice. If it is not, we set $n \leftarrow n + 1$ and continue computing. By the predictability assumption, the computation is known to terminate at some index, even though one does not know in advance which.

EXERCISES

1. a) Prove Theorem I.10.3.
 b) Prove the claims of Theorem I.10.4 concerning the sequences $\{a_n + b_n\}$ and $\{\lambda a_n\}$.
 c) Prove the claim of Theorem I.10.4 concerning the sequence $\{a_n b_n\}$.
 d) Prove: If $\{a_n\}$ is Cauchy and $|a_n| \geq \delta > 0 \forall n$, then $\{a_n^{-1}\}$ is Cauchy.
 e) Prove the claim of Theorem I.10.4 concerning the sequence $\{a_n/b_n\}$.
2. Let $\{a_k, k = 0, 1, \dots\}$ be a sequence and $b_n = \sum_{k=0}^n 2^{-k} a_k, n = 0, 1, \dots$. Show that if $\{a_k\}$ is bounded, then $\{b_n\}$ is Cauchy.
3. Let $\{a_n\}$ be a sequence of rational numbers. Prove true or false:
 a) $a_n^3 \rightarrow 4 \Rightarrow \{a_n\}$ is Cauchy, b) $a_n^4 \rightarrow 3 \Rightarrow \{a_n\}$ is Cauchy.
4. If $\{b_k\}$ is a convergent subsequence of the sequence $\{a_n, n = 1, 2, \dots\}$, then what are the possible limits of the subsequence in the following cases?

$$\text{a) } a_n = \frac{(n+1)^n}{(-n)^n} \quad \text{b) } a_n = \frac{n^3 + (-1)^n(n-1)^3}{(n+2)^3 + (-1)^n n^3}$$

5. a) Let the n th decimal digit of $1/7$ be a_n and the n th decimal digit of $1/17$ be b_n . Determine the possible limits of the convergent subsequences of the sequence $\{a_n + b_n\}$. b) Let d_n be the n th decimal digit of the known non-periodic decimal $\pi = 3.14159628\dots$. Present ten sequences, at least two of which are convergent subsequences of the sequence $\{d_n\}$. Justify your answer!
6. a) Show that Theorem I.10.7 holds also when $a = \infty$. b) Show that a sequence is unbounded if and only if it has a subsequence that is strictly monotonic and either unboundedly increasing or unboundedly decreasing.

7. (*) Prove: If $\{a_n\}$ and $\{b_n\}$ are bounded sequences, then there exists a strictly increasing index sequence $\{n_k\}$ such that the sequences $\{a_{n_k}, k = 1, 2, \dots\}$ and $\{b_{n_k}, k = 1, 2, \dots\}$ are both convergent.
8. (*) Using the definition of a real number based on Cauchy sequences, show that $(\mathbb{R}, +, \cdot, <)$ is an ordered field.
9. (*) A *random-number* sequence is a sequence of rational numbers $\{a_n, n = 1, 2, \dots\}$ satisfying $0 \leq a_n \leq 1 \forall n$ and having the property: If $0 \leq x < y \leq 1$ and $N_n =$ the number of indices $k \in \mathbb{N}$ for which $k \leq n$ and $x \leq a_k \leq y$ ($n \in \mathbb{N}$), then $\lim_n N_n/n = y - x$. By the Bolzano-Weierstrass theorem, the sequence $\{a_n\}$ has a convergent subsequence. Show that a stronger result holds: If $0 \leq x \leq 1$, then there is a subsequence whose limit is x .
10. (*) Show that every real-number sequence has a monotonic subsequence.

I.11 Properties of the real numbers

In this section we introduce some basic concepts of mathematical analysis connected with sets of real numbers. At the same time, we bring out general properties of the real numbers, alternative definitions, and ways of classifying them.

Supremum and infimum

DEFINITION I.11.1 A number $a \in \mathbb{R}$ is the *least upper bound*, or *supremum*, of a set $A \subset \mathbb{R}$ if

- (1) $x \leq a \quad \forall x \in A.$
- (2) $b < a \Rightarrow \exists x \in A (x > b).$

We write $a = \sup A$.

In the definition, condition (1) means that a is an *upper bound* of the set A , and condition (2) says that if $b < a$, then b is not an upper bound. Together, then, the conditions mean that a is the smallest possible upper bound. For such a least upper bound to exist, the set A must be at least *bounded above* (condition (1) holding for some $a \in \mathbb{R}$) and nonempty (condition (2) holding for some $a \in \mathbb{R}$). Insofar as $\sup A$ exists, it is quite evident from the definition that it is unique. If $a = \sup A$ and $a \in A$, then a is said to be the *maximum* (= greatest number) of A .

If in Definition I.11.1 the directions of the inequalities are reversed, then the *greatest lower bound*, or *infimum*, of the set A is defined, denoted $\inf A$. An infimum can exist only if A is nonempty and *bounded below*, i.e., if for some $a \in \mathbb{R}$ we have $x \geq a \quad \forall x \in A$. If $a = \inf A$ and $a \in A$, then a is the *minimum* (= smallest number) of A .

EXAMPLE 1 If $A = \{x \in \mathbb{R} \mid x \leq 1\}$, then by Definition I.11.1 $\sup A = 1$. Since $1 \in A$, this is also the maximum of A . Since A is not bounded below, the number $\inf A$ does not exist.

EXAMPLE 2 The set of real numbers $A = \{x \in \mathbb{R} \mid x^2 < 2\}$ and the set of rational numbers $B = \{x \in \mathbb{Q} \mid x^2 \leq 2\}$ both have the same least upper bound and greatest lower bound: $\sup A = \sup B = \sqrt{2}$ and $\inf A = \inf B = -\sqrt{2}$. Neither set has a maximum or a minimum. $\square$

By Example 2, the least upper bound or greatest lower bound of a bounded set of rational numbers need not be rational. The analogous phenomenon is already familiar from sequences: the limit of a sequence of rational numbers need not be rational. It is also known of sequences that for an increasing and bounded sequence of rational numbers a limit can always be constructed as a real number (i.e., as an infinite decimal; see the proof-construction of Theorem I.8.3). In the same way, for every nonempty and bounded-above set of real numbers a least upper bound can be constructed as a real number.

***THEOREM I.11.2 (Supremum theorem)** Every nonempty, bounded-above set of real numbers has a supremum.

Proof Consider a set $A \subset \mathbb{R}$ satisfying the assumptions. Since A is nonempty, there exists $c_1 \in \mathbb{R}$ such that $x > c_1$ for some $x \in A$, i.e., c_1 is not an upper bound of A . Since A is bounded above, A has an upper bound $c_2 = c_1 + L \in \mathbb{R}$. Without loss of generality we may further assume that c_1 and c_2 are rational numbers, or even integers. We now construct an increasing sequence of rational numbers $\{a_n\}$ and a decreasing sequence of rational numbers $\{b_n\}$ by a bisection construction as follows:

1. Set $a_0 = c_1$, $b_0 = c_2$ and $n = 1$.
2. Set $c = (c_1 + c_2)/2$. If c is an upper bound of A , set $c_2 = c$; otherwise set $c_1 = c$.
3. Set $a_n = c_1$, $b_n = c_2$.
4. Increase the value of the index n by one ($n \leftarrow n + 1$) and go to step 2.

By the construction, the sequence $\{a_n\}$ is increasing and bounded and the sequence $\{b_n\}$ is decreasing and bounded, and moreover $b_n - a_n = 2^{-n}L$. Thus $\{a_n\}$ and $\{b_n\}$ converge to a common limit: $a_n \rightarrow a$ and $b_n \rightarrow a$ for some $a \in \mathbb{R}$. By the construction, every b_n is an upper bound of A and no a_n is an upper bound of A , so the obvious candidate for the least upper bound of A is the common limit a . First, this is an upper bound of A , for if $x \in A$, then $x \leq b_n \forall n$ (since every b_n was an upper bound), whence $x \leq \lim_n b_n = a$ (Proposition I.7.4 (V1)). Thus $x \in A \Rightarrow x \leq a$, i.e., a is an upper bound of A .

It remains to show that a is the smallest of the upper bounds, so let $b < a$. Then, since $a_n \rightarrow a$ and $a > b$, we have $a_n > b$ for some n . In that case b cannot be an upper bound of A , since not even a_n is. Thus no number smaller than a is an upper bound of A , i.e., $a = \sup A$. $\square$

If A is a nonempty, bounded-below set of real numbers, then it is evident that $B = \{-x \mid x \in A\}$ is bounded above and that $\inf A = -\sup B$. From this and Theorem I.11.2 it follows immediately that every nonempty, bounded-below set of real numbers has an infimum.

The axiomatic real numbers

The definitions of a real number given in the previous section, based on decimals or Cauchy sequences, are called *constructive*. A characteristic feature of constructive definitions is that no 'outside world' relative to the rational numbers is assumed. One simply decides to call certain types of sequences of rational numbers real numbers, and defines the arithmetic operations among these numbers as well as the identification and order relations by reducing all operations ultimately to rational ones. The good side of a constructive definition is that it resembles the actual, computational 'definition' of real numbers by means of calculators and computers. On the other hand, when one wants to compute with real numbers symbolically, it is often easier to conceive of the real numbers directly as abstract numbers than as sequences of more concrete numbers (cf. Example 5 in Section I.9). Abstract thinking is carried furthest in the *axiomatic* definition of the real numbers, in which one simply agrees that numbers other than the rationals exist. The procedure is then similar to that in Example 3 of Section I.2, where the 'existence' of the number $\sqrt{2}$ was treated as a matter of convention. As with constructive definitions, there are also many equivalent axiomatic approaches to the real numbers. The common starting point of these definitions is the assumption that the real numbers form an ordered field that is an extension of the field of rational numbers. These assumptions leave open the possibility that $\mathbb{R} = \mathbb{Q}$, so one more axiom is needed to distinguish the sets from each other. A very commonly used procedure is to make an axiom out of either Theorem I.11.2 or Theorem I.10.5, i.e., to postulate one of the following.

- (A) (**Supremum axiom**) Every bounded-above set of real numbers has a least upper bound.*
- (B) (**Completeness axiom**) Every Cauchy sequence of real numbers converges to a real number.

*In connection with the supremum axiom or the supremum theorem, the real numbers can also be defined as bounded-above sets of rational numbers. This constructive definition, resting on the so-called *Dedekind cuts*, was presented by the German mathematician **Richard Dedekind** (1831-1916). Dedekind's definition was in its day popular in teaching as well.

In the aforementioned constructive definition of the real numbers, these are thus true statements. As axioms they are also interchangeable with each other: if (A) is assumed, then (B) is a true statement, and vice versa.

As in the constructive definition, in the axiomatic definition of the real numbers too the starting point is at first the rational numbers. On the basis of axiom (A) or (B), the set of rational numbers is *completed* by adjoining to it either the least upper bounds of bounded-above sets of rational numbers or the limits of rational Cauchy sequences, which by the assumed axiom exist as real numbers. Since completion is the underlying idea in the constructive definition as well, this is customarily called the *completion construction*. The rational numbers can thus be completed into the real numbers either constructively, by 'building', or axiomatically, by 'assuming'.

$\mathbb{R}$ is uncountable

If $a, b \in \mathbb{R}$ and $a < b$, then it is easily seen that a rational number can always be placed between a and b , i.e., there exists $x \in \mathbb{Q}$ with $a < x < b$. Right away, infinitely (countably) many such numbers are found, and in fact the construction does not even need the whole set $\mathbb{Q}$; e.g., the finite decimals (or finite binary numbers) suffice (Exercise 4). On the basis of this result, $\mathbb{Q}$ (or a subset of $\mathbb{Q}$ of the type mentioned) is said to be *dense* in $\mathbb{R}$. Despite this result, $\mathbb{R}$ has greater cardinality than $\mathbb{Q}$, since (cf. Section I.4)

THEOREM I.11.3 $\mathbb{R}$ is an uncountable set.

Proof Let $\{x^{(n)}\}$ be a sequence of real numbers with $0 < x^{(n)} < 1 \quad \forall n$. Write the terms of the sequence as infinite decimals:

$$\begin{aligned} x^{(1)} &= 0.d_1^{(1)}d_2^{(1)}d_3^{(1)} \dots \\ x^{(2)} &= 0.d_1^{(2)}d_2^{(2)}d_3^{(2)} \dots \\ x^{(3)} &= 0.d_1^{(3)}d_2^{(3)}d_3^{(3)} \dots \\ &\vdots \end{aligned}$$

Then define an infinite decimal $y = 0.d_1d_2\dots$ by choosing

$$d_n \in \{1, \dots, 8\}, \quad d_n \neq d_n^{(n)}, \quad n = 1, 2, \dots$$

Then $0 < y < 1$ and y is not a finite decimal, so y is not identified with any other decimal that differs at all as a string. It then follows from the construction that

$y \neq x^{(n)} \forall n$. We conclude that no attempt to enumerate the set $A = \{x \in \mathbb{R} \mid 0 < x < 1\}$ covers all the numbers of A , so not even A is countable. $\square$

Let $a, b \in \mathbb{R}$, $a < b$ and $B = \{x \in \mathbb{R} \mid a < x < b\}$. Since

$$B = \{x = a + (b - a)t \mid t \in \mathbb{R}, 0 < t < 1\},$$

it can be inferred from the proof of Theorem I.11.3 that B is uncountable. In other words: between two distinct real numbers there is always an uncountable number of real numbers.

Open and closed interval

A set $A \subset \mathbb{R}$ is an *interval* if A has more than one element and $x, y \in A$ & $x < z < y \Rightarrow z \in A$. An interval can be either *finite*, i.e., bounded above and below as a set, or *infinite* (unbounded above and/or below). The main types of finite interval are the *open* and *closed* interval, whose notation and definitions are:

Open interval: $(a, b) = \{x \in \mathbb{R} \mid a < x < b\}$.

Closed interval: $[a, b] = \{x \in \mathbb{R} \mid a \leq x \leq b\}$.

Here $a = \inf A$ and $b = \sup A$ are the *endpoints* of the interval ($a, b \in \mathbb{R}$, $a < b$). An alternative notation for the open interval is $]a, b[$. An interval can also be of the form $(a, b]$ or $[a, b)$ (with alternative notations $]a, b]$ and $[a, b[$), in which case it is said to be *half-open*—the definitions being obvious. Points of an interval other than the endpoints are called *interior points* of the interval. All points of an open interval are thus interior points. A closed interval is always finite. Intervals of other types can also be infinite, in which case the absence of an endpoint is denoted by the symbol $\pm\infty$:

$$(a, \infty) = \{x \in \mathbb{R} \mid x > a\},$$

$$(-\infty, b] = \{x \in \mathbb{R} \mid x \leq b\},$$

$$(-\infty, \infty) = \mathbb{R}.$$

Let us mention in this connection also the commonly used notations

$$(0, \infty) = \mathbb{R}_+ \quad (\text{the positive real numbers, 'R plus'}),$$

$$(-\infty, 0) = \mathbb{R}_- \quad (\text{the negative real numbers, 'R minus'}).$$

A sequence of closed intervals $[a_n, b_n]$, $n = 1, 2, \dots$ is said to be *nested* if $[a_n, b_n] \supset [a_{n+1}, b_{n+1}] \forall n$. Connected with such a sequence, the real numbers have the following pleasant property:

- (C) If $\{[a_n, b_n], n = 1, 2, \dots\}$ is a sequence of nested closed intervals and $b_n - a_n \rightarrow 0$, then there exists a unique real number x such that $x \in [a_n, b_n] \forall n \in \mathbb{N}$.

By the constructive definition of the real numbers, (C) is a true statement, since x is determined as the common limit of the sequences $\{a_n\}$ (increasing and bounded) and $\{b_n\}$ (decreasing and bounded). In the axiomatic definition of the real numbers this result can also be taken as a basic axiom (assuming $a_n, b_n \in \mathbb{Q}$), in which case the aforementioned axioms (A), (B) become true statements.

If $x \in \mathbb{R}$ and $\delta \in \mathbb{R}$, $\delta > 0$, then the δ -neighborhood of x is defined as the open interval

$$U_\delta(x) = (x - \delta, x + \delta).$$

The terms 'endpoint' and 'neighborhood' allude to the idea that a real number can also be conceived of as a 'point' in some 'place'. This idea does not originate in algebra but in *geometry*, the other great main branch of mathematical thinking. The foundations of geometry are treated more closely in Chapter II.

Algebraic and transcendental numbers

By their definition, the real numbers divide naturally into rational and non-rational, i.e., *irrational*, numbers. Another, also commonly used, way of classifying is to divide the real numbers into *algebraic* and non-algebraic, i.e., *transcendental*, numbers. A number $x \in \mathbb{R}$ is said to be algebraic if it satisfies an equation of the form

$$\sum_{k=0}^n a_k x^k = 0,$$

where $n \in \mathbb{N}$ and $a_k \in \mathbb{Z}$, $k = 0, 1, \dots, n$. A number is thus algebraic if it is a zero of some polynomial with integer coefficients.

EXAMPLE 3 Radicals, and as a special case of them rational numbers, satisfy an equation of the form $px^m - q = 0$, where $m \in \mathbb{N}$ and $p, q \in \mathbb{Z}$, so such numbers are algebraic. $\square$

If the set of algebraic numbers is denoted by the symbol $\mathbb{A}$, then by the example we have

$$\mathbb{Q} \subset \mathbb{A} \subset \mathbb{R}.$$

Like $\mathbb{Q}$, the set $\mathbb{A}$ too can be shown to be countable, so the 'majority' of the real numbers are transcendental.*

EXERCISES

- Determine the supremum, infimum, maximum and minimum of the following sets, insofar as they exist :
 - $\{x \in \mathbb{R} \mid (x+1)(x-2)(x+4) > 0\}$
 - $\{x \in \mathbb{R} \mid |x| + |x+2| < 5\}$
 - $\{x \in \mathbb{R} \mid |x||x+2| < 5\}$
 - $\{n/(n+1) \mid n \in \mathbb{N}\}$
 - $\{(2-3n)/(5+n) \mid n \in \mathbb{N}\}$
 - $\{n^3 2^{-n} \mid n \in \mathbb{N}\}$
- For what nonempty sets $A \subset \mathbb{R}$ does $\inf A = \sup A$?
 - Let $A \subset \mathbb{R}$ be a bounded-above set and $a = \sup A$. Show that for every $\varepsilon > 0$ there exists $x \in A$ such that $x > a - \varepsilon$.
 - Show that if $A, B \subset \mathbb{R}$ are both bounded above (below) and $A \subset B$, then $\sup A \leq \sup B$ ($\inf A \geq \inf B$).
- Let
 - $A = \{x \in \mathbb{R} \mid x^3 < 200\}$,
 - $A = \{x \in \mathbb{R} \mid x^3 < -400\}$.
 Follow the proof-construction of Theorem I.11.2 as an algorithm up to the index $n = 3$, by choosing the initial values of c_0 and c_1 so that for the sequences produced by the algorithm:
 - $\{a_n\} = a$,
 - $\{b_n\} = a$,
 where $a = \sup A$ as an infinite binary number.
- Let $a, b \in \mathbb{R}$ and $a < b$. Show that there exist infinitely many distinct finite decimals x with $a < x < b$.
- Determine all intervals with the property: the endpoints of the interval—insofar as it has any—belong to the set $\{-1, 2\}$.

*The branch of mathematics that studies the properties of numbers, such as the real numbers or the natural numbers, from the viewpoint of various classifications among other things, is called *number theory*. Among the well-known results of number theory is, for example, that Napier's number e is transcendental (**Charles Hermite**, 1873), as is the number π (**Ferdinand von Lindemann**, 1882). The irrationality of both numbers had already been established in the 18th century. By contrast, statements as simple as

P1: The number $e + \pi$ is not rational.

P2: The number $e\pi$ is not rational.

had not yet, at the time of this writing (2015), been shown to be true (let alone false). — Number theory is known for its many mathematical puzzles that are easy to state but often hard to solve.

6. Give the following sets a simpler definition as intervals, i.e., in the form (a, b) , $[a, b]$, $(a, b]$ or $[a, b)$.

$$\begin{array}{ll} \text{a) } \bigcup_{n=1}^{\infty} \left(\frac{1}{3n}, \frac{n}{n+1} \right) & \text{b) } \bigcap_{n=1}^{\infty} \left(\frac{1}{3n}, \frac{n+1}{n} \right) \\ \text{c) } \bigcap_{n=1}^{\infty} \left[\frac{1}{3n}, \frac{n}{n+1} \right] & \text{d) } \bigcup_{n=1}^{\infty} \left[\frac{1}{3n}, \frac{n+1}{n} \right] \end{array}$$

7. Show: a) If x is irrational, then so is $y = (3+x)/(x-2)$. b) If $x \neq 0$ is an algebraic number, then so is x^{-1} . c) If $x > 0$ is an algebraic number, then so is $\sqrt[m]{x} \forall m \in \mathbb{N}, m \geq 2$. d) Every zero of a polynomial with rational coefficients is an algebraic number.
8. (*) Let $\{[a_n, b_n], n = 1, 2, \dots\}$ be a sequence of closed intervals. Show that $A = \bigcap_{n=1}^{\infty} [a_n, b_n]$ is either the empty set, contains exactly one element, or is a closed interval.

I.12 Classical theory of series. Power series

Let us recall from earlier sections that the notation

$$\sum_{k=1}^{\infty} a_k = a_1 + a_2 + a_3 + \dots$$

means either a series, i.e., the sequence $\{a_1, a_1 + a_2, \dots\} = \{s_1, s_2, \dots\}$, or, insofar as $\{s_n\}$ converges (to a real number), also the *sum of the series*, i.e., the limit $s = \lim_n s_n$. Let us also recall that the numbers a_k are called the *terms* of the series and the terms of the sequence $\{s_n\}$ the *partial sums* of the series. Series have much use in applications and also in mathematical thinking. In this section we examine the convergence theory of series from a more general point of view than before, and study some common special types of series. The terms a_k of a series are assumed henceforth to be real numbers.

Positive-term series

From the standpoint of convergence theory, the 'friendliest' series is a *positive-term* series $\sum_k a_k$, whose terms are positive, or more generally nonnegative: $a_k \geq 0 \forall k$. Examples have already been considered in Section I.8. The sequence of partial sums of a positive-term series is increasing, so the series converges if and only if that sequence is bounded (Theorem I.8.3). The convergence analysis can often be further eased by comparing the terms of the series with a simpler series already known to be convergent or divergent. The following comparison principles are useful.

THEOREM I.12.1 If $0 \leq a_k \leq b_k \forall k$, then

$$\begin{aligned} \text{Majorant principle: } \sum_k b_k \text{ converges} &\Rightarrow \sum_k a_k \text{ converges} \\ \text{Minorant principle: } \sum_k a_k \text{ diverges} &\Rightarrow \sum_k b_k \text{ diverges} \end{aligned}$$

Proof If the sequences of partial sums of the series $\sum_k a_k$ and $\sum_k b_k$ are $\{s_n\}$ and $\{t_n\}$ respectively, then by assumption $s_n \leq t_n \forall n$. The majorant principle then follows by the reasoning

$$\{t_n\} \text{ converges} \Rightarrow \{t_n\} \text{ bounded} \Rightarrow \{s_n\} \text{ bounded} \Rightarrow \{s_n\} \text{ converges}.$$

The minorant principle follows by analogous reasoning—or simply by noting that the comparison principles are logically equivalent (cf. Section I.3). $\square$

EXAMPLE 1 Does the series

$$\text{a) } \sum_{k=1}^{\infty} \frac{(k+3)^2}{k^4}, \quad \text{b) } \sum_{k=0}^{\infty} \frac{2k+50}{k^2+1}$$

converge or diverge?

Solution a) Since

$$0 < a_k = \frac{(k+3)^2}{k^4} = \frac{1}{k^2} + \frac{6}{k^3} + \frac{9}{k^4} \leq \frac{16}{k^2}, \quad k \geq 1,$$

and since the comparison series $\sum_{k=1}^{\infty} 16k^{-2} = 16 \sum_{k=1}^{\infty} k^{-2}$ is known to converge (Section I.8, Example 2), by the majorant principle the series converges.

b) Writing

$$a_k = \frac{2k+50}{k^2+1} = \frac{1}{k} \cdot \frac{2+50k^{-1}}{1+k^{-2}} = \frac{1}{k} \cdot c_k,$$

we see that $c_k \rightarrow 2$, so from some index $k = m$ onward (in fact when $k \geq 50$) we have $1 \leq c_k \leq 3$. Then

$$\frac{1}{k} \leq a_k \leq \frac{3}{k}, \quad k = m, m+1, \dots,$$

so by the majorant and minorant principles we conclude that the series under consideration converges/diverges if and only if the series $\sum_{k=m}^{\infty} 1/k$ converges/diverges. This comparison series turns out to be divergent (Theorem I.12.2 below), so the series under consideration diverges as well. $\square$

Harmonic, subharmonic and superharmonic series

The positive-term series

$$\sum_{k=1}^{\infty} \frac{1}{k^{\alpha}} = 1 + \frac{1}{2^{\alpha}} + \frac{1}{3^{\alpha}} + \dots,$$

where α is a rational number*, is called *harmonic* if $\alpha = 1$, *subharmonic* if $\alpha < 1$, and *superharmonic* if $\alpha > 1$. When using the majorant and minorant principles, this type of series is often a good point of comparison.

THEOREM I.12.2 The harmonic and subharmonic series diverge. The superharmonic series converges.

*So far x^{α} ($x \in \mathbb{R}$, $x > 0$) has been defined only when $\alpha \in \mathbb{Q}$.

Proof Consider first the harmonic series

$$\sum_{k=1}^{\infty} \frac{1}{k} = 1 + \frac{1}{2} + \frac{1}{3} + \dots$$

Estimate the partial sums s_4, s_8, s_{16} , etc., as follows:

$$\begin{aligned} s_4 &= 1 + \frac{1}{2} + \left(\frac{1}{3} + \frac{1}{4} \right) > \frac{3}{2} + 2 \cdot \frac{1}{4} = 2, \\ s_8 &= s_4 + \left(\frac{1}{5} + \dots + \frac{1}{8} \right) > 2 + 4 \cdot \frac{1}{8} = \frac{5}{2}, \\ &\vdots \end{aligned}$$

We conclude that if $n = 2^m$, $m \in \mathbb{N}$, $m \geq 2$, then $s_n > 1 + m/2$, so the subsequence $\{s_4, s_8, s_{16}, \dots\}$ of partial sums of the harmonic series is divergent. Thus the series diverges: $s_n \rightarrow \infty$. Then the subharmonic series diverges as well by the minorant principle, since from the definition of the numbers k^α , $\alpha \in \mathbb{Q}$ (see Section I.2) it easily follows that $k > k^\alpha \Leftrightarrow k^{-\alpha} > k^{-1}$ when $\alpha < 1$ and $k \geq 2$.

In the case of the superharmonic series, set $\alpha = 1 + \delta$, $\delta > 0$ and estimate the partial sums s_3, s_7 , etc., as follows:

$$\begin{aligned} s_3 &= 1 + \left(\frac{1}{2^\alpha} + \frac{1}{3^\alpha} \right) < 1 + 2 \cdot \frac{1}{2^\alpha} = 1 + \frac{1}{2^\delta}, \\ s_7 &= s_3 + \left(\frac{1}{4^\alpha} + \dots + \frac{1}{7^\alpha} \right) < 1 + \frac{1}{2^\delta} + 4 \cdot \frac{1}{4^\alpha} = 1 + \frac{1}{2^\delta} + \left(\frac{1}{2^\delta} \right)^2, \\ &\vdots \end{aligned}$$

(Here we have used the estimate $(k/j)^\alpha > 1 \Leftrightarrow k^{-\alpha} < j^{-\alpha}$ when $k/j > 1$ and $\alpha > 0$.) We conclude that if $n = 2^m - 1$, $m \in \mathbb{N}$, $m \geq 2$, then

$$s_n < 1 + \frac{1}{2^\delta} + \dots + \left(\frac{1}{2^\delta} \right)^{m-1} < \sum_{k=0}^{\infty} \left(\frac{1}{2^\delta} \right)^k = \frac{2^\delta}{2^\delta - 1}.$$

Thus the sequence of partial sums of the series is bounded, so the series converges. $\square$

Let us remark that in the case $\alpha = 2$ the claim of Theorem I.12.2 has already been proved earlier by other means (see Section I.8).

EXAMPLE 2 The superharmonic series $\sum_{k=1}^{\infty} k^{-5/4}$ converges, but the convergence is so slow that the sum of the series cannot be determined accurately merely by computing partial sums. With better algorithms (see Exercise VIII.9:11) the result is

$$s = \sum_{k=1}^{\infty} \frac{1}{k^{5/4}} = 1 + \frac{1}{2^{5/4}} + \frac{1}{3^{5/4}} + \dots = 4.5951118258\dots \quad \square$$

EXAMPLE 3 For the partial sums of the divergent subharmonic series $\sum_{k=1}^{\infty} 1/\sqrt{k}$, the following estimate holds for large values of n (see Exercise 16)

$$s_n = \sum_{k=1}^n \frac{1}{\sqrt{k}} \approx 2\sqrt{n}. \quad \square$$

Alternating series

A series is said to be *alternating* if the sign of its terms changes each time one passes from one index to the next. An alternating series is thus of the form $\sum_k (-1)^k a_k$, where the terms of the sequence $\{a_k\}$ have the same sign.

EXAMPLE 4 The series

$$\sum_{k=0}^{\infty} (-1)^k \frac{1}{\sqrt{k+1}} = \left\{ 1, 1 - \frac{1}{\sqrt{2}}, 1 - \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{3}}, \dots \right\}$$

is alternating. $\square$

The following more general statement applies to the example series.

THEOREM I.12.3 If an alternating series $\sum_{k=0}^{\infty} (-1)^k a_k$ satisfies (i) $a_k > 0 \forall k$, (ii) the sequence $\{a_k\}$ is decreasing, i.e., $a_0 \geq a_1 \geq a_2 \geq \dots$, and (iii) $\lim_k a_k = 0$, then the series converges. Moreover, when the sum s of the series is approximated by the partial sums s_n , the estimate

$$|s - s_n| \leq a_{n+1}.$$

holds.

Proof Consider the partial sums $s_n = \sum_{k=0}^n (-1)^k a_k$ of the series separately for even and odd values of the index. Since

$$\begin{aligned} s_{2n} &= (a_0 - a_1) + \dots + (a_{2n-2} - a_{2n-1}) + a_{2n} \\ &= a_0 - (a_1 - a_2) - \dots - (a_{2n-1} - a_{2n}), \\ s_{2n+1} &= (a_0 - a_1) + \dots + (a_{2n} - a_{2n+1}) \\ &= a_0 - (a_1 - a_2) - \dots - (a_{2n-1} - a_{2n}) - a_{2n+1}, \end{aligned}$$

it follows from assumptions (i)–(ii) that $0 \leq s_n \leq a_0$ for both even and odd values of n . Thus $\{s_n\}$ is a bounded sequence. Similarly one sees that the sequence $\{s_{2n}\}$ is decreasing and $\{s_{2n+1}\}$ is increasing, so both converge. Since, furthermore, $s_{2n} - s_{2n+1} = a_{2n+1} \rightarrow 0$ by assumption (iii), the sequences $\{s_{2n}\}$ and $\{s_{2n+1}\}$ have a common limit. From this it is easily inferred that the whole sequence $\{s_n\}$ converges.

To prove the second part of the claim, let n first be odd. Then

$$\begin{aligned} s - s_n &= a_{n+1} - (a_{n+2} - a_{n+3}) - \dots \\ &= (a_{n+1} - a_{n+2}) + (a_{n+3} - a_{n+4}) + \dots, \end{aligned}$$

so by the assumptions $0 \leq s - s_n \leq a_{n+1}$. If n is even, then one concludes similarly that $-a_{n+1} \leq s - s_n \leq 0$. Thus in both cases the claimed estimate holds. $\square$

EXAMPLE 4 (continued) By Theorem I.12.3, the example series is convergent. Its sum is found (by numerical means) to be $s = 0.6048986434..$ In this case the estimate $|s - s_n| \leq a_{n+1}$ sharpens, for large values of n , to the form $|s - s_n| \approx \frac{1}{2}a_{n+1}$; cf. the table below. (More precisely, one can show: $\lim_n |s - s_n|/a_{n+1} = \frac{1}{2}$.)

$s - s_{100} = -0.049628..$	$a_{101} = 0.099503..$
$s - s_{101} = +0.049386..$	$a_{102} = 0.099014..$
$s - s_{1000} = -0.015799..$	$a_{1001} = 0.031606..$
$s - s_{1001} = +0.015791..$	$a_{1002} = 0.031591.. \quad \square$

Cauchy's criterion for series

If a series is neither positive-term nor alternating, then in convergence theory one must generally resort to Theorem I.10.5. By that theorem, the series $\sum_{k=1}^{\infty} a_k$ converges (to a real number) if and only if the sequence formed by its partial sums $s_n = \sum_{k=1}^n a_k$ is a Cauchy sequence. Since $s_m - s_n = \sum_{k=n+1}^m a_k$ when $m > n$, by the definition of a Cauchy sequence we thus have

THEOREM I.12.4 (Cauchy's criterion for series) The series $\sum_{k=1}^n a_k$ converges if and only if for every $\varepsilon > 0$ there exists $N \in \mathbb{N}$ such that

$$\left| \sum_{k=n+1}^m a_k \right| < \varepsilon, \quad \text{when } m > n > N.$$

Choosing $m = n+1$ in the criterion, one arrives at the following obvious (mutually logically equivalent) conclusions:

COROLLARY I.12.5 If the series $\sum_k a_k$ converges, then $a_k \rightarrow 0$.

COROLLARY I.12.6 If $a_k \not\rightarrow 0$, then the series $\sum_k a_k$ diverges.

If the signs of the terms of a series vary, then one can try to settle the convergence question of the series simply by comparing it with the positive-term series obtained by removing the sign variation. When it succeeds, this attempt rests on the following theorem. First a definition:

DEFINITION I.12.7 A series $\sum_k a_k$ converges *absolutely* if the series $\sum_k |a_k|$ converges.

THEOREM I.12.8 If a series converges absolutely, then it converges.

Proof By the triangle inequality,

$$\left| \sum_{k=n+1}^m a_k \right| \leq \sum_{k=n+1}^m |a_k| \quad (m > n).$$

Since the series $\sum_k |a_k|$ converges, by Theorem I.12.4 for every $\varepsilon > 0$ one can find $N \in \mathbb{N}$ such that the right-hand side is $< \varepsilon$ when $n > N$, whence by the same criterion the series $\sum_k a_k$ converges as well. $\square$

EXAMPLE 5 If $\{b_k\}$ is a sequence with $b_k \in [-1, 1] \forall k$, then the series $\sum_{k=1}^{\infty} b_k/k^2$ converges absolutely (majorant principle and Theorem I.12.2). The series thus converges, no matter how the numbers b_k are chosen from $[-1, 1]$. $\square$

EXAMPLE 6 The series $\sum_{k=1}^{\infty} (-1)^k/\sqrt{k}$ converges (Theorem I.12.3) but not absolutely (Theorem I.12.2). $\square$

Power series

A series of the form

$$\sum_{k=0}^{\infty} a_k x^k = a_0 + a_1 x + a_2 x^2 + \dots$$

is called a *power series*. The numbers a_k are called the *coefficients of the power series*, and the symbol x , which also represents a real number, is called the

variable of the series. The term 'variable' indicates that the numerical value of x may vary. Thus this is not merely a sequence but rather a set of sequences, in which the coefficients a_k are thought of as fixed and the variable x is allowed various (real) values.*

As we shall see, the power series is a very central concept and tool in mathematics. Here we study for now only the convergence question of the series, whose natural formulation is: For what values of x does the series converge? Since in the case of a convergent series the sum generally depends on the value of x , this is expressed by writing the sum of the series as $s(x)$, read ' s of x '.

EXAMPLE 7 The basic geometric series $\sum_{k=0}^{\infty} x^k$ is an example of a power series. The series converges if and only if $x \in (-1, 1)$, and the sum of the series is then $s(x) = 1/(1 - x)$. $\square$

By the example, a power series can be regarded as a generalization of the geometric series. It turns out that in convergence analysis the geometric series is a good point of comparison. On such a comparison rests, for instance, the following, the most central result of the convergence theory of power series.

THEOREM I.12.9 (Convergence of a power series) For a power series $\sum_{k=0}^{\infty} a_k x^k$, one of the following alternatives holds:

- (a) The series converges only when $x = 0$.
- (b) $\exists \rho \in \mathbb{R}_+$ such that the series converges, also absolutely, when $x \in (-\rho, \rho)$ and diverges whenever $|x| > \rho$. Then ρ is the *radius of convergence* of the series.
- (c) The series converges absolutely $\forall x \in \mathbb{R}$.

Proof It suffices to show that (b) or (c) holds when (a) does not. So assume that the series converges for some $x_0 \in \mathbb{R}$, $x_0 \neq 0$. Then we must have (see Corollary I.12.5 and Theorem I.7.2)

$$a_n x_0^n \rightarrow 0 \Rightarrow |a_n x_0^n| \leq C \quad \forall n \quad (C \in \mathbb{R}_+).$$

It follows that if $|x| < |x_0|$, then

$$|a_n x^n| \leq C |x/x_0|^n,$$

*In the case $x = 0$ the power series is interpreted as the sequence $\{a_0, a_0, \dots\}$, i.e., in the series notation one agrees that $0^0 = 1$.

whence by the majorant principle the series converges absolutely, like the geometric series ($q = |x/x_0| < 1$). Thus it has been established: If the series converges when $x = x_0 \neq 0$, then it converges absolutely $\forall x \in (-|x_0|, |x_0|)$. It follows further that if the series diverges when $x = x_1$, then it diverges whenever $|x| > |x_1|$, for otherwise, by what has already been established, it would be both convergent and divergent when $x = x_1$. So it has been deduced: If (a) does not hold, then either (c) the series converges for every $x \in \mathbb{R}$, in which case it converges for every $x \in \mathbb{R}$ also absolutely, or (b) the series converges when $|x| < \rho$ and diverges when $|x| > \rho$, where ρ is defined by (cf. Theorem I.11.2)

$$\rho = \sup \left\{ x \in \mathbb{R} \mid \text{the series } \sum_{k=0}^{\infty} a_k x^k \text{ converges} \right\}.$$

The theorem is thus proved. $\square$

In alternatives (a), (c) of the theorem one may agree on the notations

$$(a) \quad \rho = 0, \quad (c) \quad \rho = \infty.$$

Insofar as the radius of convergence of a power series can be determined, Theorem I.12.9 thus settles the convergence question of the series completely, except for the variable values $x = \rho$ and $x = -\rho$ when $\rho \in \mathbb{R}_+$. These must be resolved case by case; cf. Example 9 below. In the most central computational problem, namely determining the radius of convergence, the following result often helps:

THEOREM I.12.10 (Radius of convergence of a power series) If in the power series

$\sum_{k=0}^{\infty} a_k x^k$ we have $a_k \neq 0 \forall k$ and the limit

$$\rho = \lim_{k \rightarrow \infty} \left| \frac{a_k}{a_{k+1}} \right|,$$

exists, where $\rho \in \mathbb{R}_+$, $\rho = 0$ or $\rho = \infty$, then the radius of convergence of the series $= \rho$.

Proof Let first $\rho > 0$ and $|x| < \rho$. Choose q such that $|x| < q < \rho$ (e.g., $q = (|x| + \rho)/2$). Then, since $|a_k|/|a_{k+1}| \rightarrow \rho$ and $q < \rho$, from some index $k = m$ onward we have $|a_k|/|a_{k+1}| \geq q$ (in the definition of convergence of a sequence, choosing $\varepsilon = \rho - q$ and $m > N$). Then we can conclude

$$|a_{k+1}| \leq q^{-1} |a_k|, \quad k = m, m+1, \dots \quad \Rightarrow \quad |a_k| \leq q^{m-k} |a_m|, \quad k \geq m,$$

so that

$$|a_k x^k| \leq q^{m-k} |a_m| |x|^k = q^m |a_m| \left(\frac{|x|}{q} \right)^k = C \left(\frac{|x|}{q} \right)^k, \quad k \geq m.$$

Here $|x|/q < 1$, so by the majorant principle we conclude that the series $\sum_{k=0}^{\infty} a_k x^k$ converges absolutely, from the index $k = m$ onward like the geometric series. Thus if $\rho \in \mathbb{R}_+$, then the series converges absolutely when $|x| < \rho$. If the assumed limit is $\rho = \infty$, then in the above reasoning $\rho \in \mathbb{R}_+$ can be chosen arbitrarily, so in this case the radius of convergence $= \infty$ as well.

If $\rho \in \mathbb{R}_+$ and $|x| > \rho$, then one sees in the same way as above that, for m large enough, for every $k \geq m$

$$|a_k x^k| \geq q^{m-k} |a_m| |x|^k = q^m |a_m| \left(\frac{|x|}{q}\right)^k = C \left(\frac{|x|}{q}\right)^k, \quad k \geq m.$$

where now $\rho < q < |x|$. Consequently $|a_k x^k| \rightarrow \infty$ as $k \rightarrow \infty$, so the series $\sum_k a_k x^k$ diverges (Corollary I.12.6). Thus if $\rho \in \mathbb{R}_+$, then the series converges when $|x| < \rho$ and diverges when $|x| > \rho$, i.e., the radius of convergence of the series $= \rho$. If $\rho = 0$, then by the same reasoning the series is found to diverge whenever $x \neq 0$, i.e., in this case the radius of convergence $= 0$. The theorem is thus proved. $\square$

EXAMPLE 8 Determine the radius of convergence of the power series when the coefficients are

$$\text{a) } a_k = k! \quad \text{b) } a_k = (-1)^k (k+1)^{-1} \quad \text{c) } a_k = k^2 \cdot 3^{-k} \quad \text{d) } a_k = 1/k!$$

Solution By Theorem I.12.10,

$$\begin{aligned} \text{a) } \rho &= \lim_k 1/(k+1) = 0 & \text{b) } \rho &= \lim_k (k+2)/(k+1) = 1 \\ \text{c) } \rho &= \lim_k 3k^2/(k+1)^2 = 3 & \text{d) } \rho &= \lim_k (k+1) = \infty \end{aligned} \quad \square$$

If a power series converges when $x \in A$ and diverges when $x \notin A$, then the set A ($A \subset \mathbb{R}$) is called the *interval of convergence* of the series. By Theorem I.12.9, A is one of the alternatives $(-\rho, \rho)$, $[-\rho, \rho)$, $(-\rho, \rho]$, $[-\rho, \rho]$, where $\rho =$ radius of convergence.

EXAMPLE 9 Determine the interval of convergence of the power series $\sum_{k=0}^{\infty} a_k x^k$ when

$$\text{a) } a_k = 1 \quad \text{b) } a_k = 1/(k+1) \quad \text{c) } a_k = (-1)^k/(k+1) \quad \text{d) } a_k = 1/(k+1)^2$$

Solution The formula of Theorem I.12.10 gives the radius of convergence $\rho = 1$ in all cases, so the series converge when $|x| < 1$ and diverge when $|x| > 1$. In the cases $x = \pm 1$ the example series represent the already familiar types

of geometric, harmonic, alternating and superharmonic series. Combining the results, we obtain the answer

$$\text{a) } (-1, 1), \quad \text{b) } [-1, 1), \quad \text{c) } (-1, 1], \quad \text{d) } [-1, 1]. \quad \square$$

EXAMPLE 10 Determine the interval of convergence of the power series

$$\sum_{k=0}^{\infty} (-1)^k (k^2 + 1) \frac{x^{2k}}{2^k} = 1 - x^2 + \frac{5}{4}x^4 - \dots$$

Solution The radius-of-convergence formula of Theorem I.12.10 does not apply directly, since the series is of the form $\sum_{k=0}^{\infty} a_k x^k$ with $a_k = 0$ at odd values of the index. The problem is nevertheless solved simply by changing the variable to $t = x^2$, whereupon the series takes the form $\sum_{k=0}^{\infty} b_k t^k$. Theorem I.12.10 applies to this series, since

$$\lim_k \left| \frac{b_k}{b_{k+1}} \right| = \lim_k \frac{(k^2 + 1) 2^{-k}}{[(k+1)^2 + 1] 2^{-k-1}} = 2.$$

Thus the series converges when $|t| = x^2 < 2$ and diverges when $|t| = x^2 > 2$. The radius of convergence with respect to the variable x is therefore $\rho = \sqrt{2}$. At the variable values $x = \pm\sqrt{2}$ the series is seen to diverge, so the interval of convergence is $(-\sqrt{2}, \sqrt{2})$. $\square$

EXAMPLE 11 For what values $x \in \mathbb{R}_+$ does the series

$$\sum_{k=0}^{\infty} \frac{x^{2k-1/2}}{k!} = \frac{1}{\sqrt{x}} + x\sqrt{x} + \frac{x^3\sqrt{x}}{2} + \dots$$

converge?

Solution Insofar as the series converges, the scaling factor $1/\sqrt{x}$ can be factored out of both the partial sums and the limit $s(x)$ (cf. Theorem I.7.3), so that

$$s(x) = \frac{1}{\sqrt{x}} \sum_{k=0}^{\infty} \frac{(x^2)^k}{k!}.$$

Here $\sum_{k=0}^{\infty} x^{2k}/k!$ is an ordinary power series that converges $\forall x \in \mathbb{R}$, so the series in question converges for every $x \in \mathbb{R}_+$. $\square$

We conclude this first presentation of power series with a result that will be of later use.

THEOREM I.12.11 The power series $\sum_{k=1}^{\infty} k^m a_k x^k$ have the same radius of convergence $\forall m \in \mathbb{Z}$.

Proof Let the radius of convergence of the series $\sum_k a_k x^k$ ($m = 0$) be either $\rho \in \mathbb{R}_+$ or $\rho = \infty$, and let $0 < |x| < \rho$. Choose $q \in (0, 1)$ such that $|x|/q < \rho$ (e.g., $q = 2|x|/(|x| + \rho)$ if $\rho \in \mathbb{R}_+$, or $q = 1/2$ if $\rho = \infty$), and write

$$k^m |a_k| |x|^k = (k^m q^k) |a_k| \left(\frac{|x|}{q} \right)^k, \quad k \geq 1, \quad m \in \mathbb{Z}.$$

Here the sequence $\{k^m q^k, k = 1, 2, \dots\}$ is bounded (since it is convergent, see Exercise 8), so for some $C \in \mathbb{R}_+$

$$k^m |a_k| |x|^k \leq C |a_k| \left(\frac{|x|}{q} \right)^k, \quad k = 1, 2, \dots$$

By assumption the series $\sum_k |a_k| (|x|/q)^k$ converges (since $|x|/q < \rho$), so by the majorant principle the series $\sum_k k^m |a_k| |x|^k$ converges as well. We have deduced that if the radius of convergence of the series $\sum_k a_k x^k$ is $\rho \neq 0$, then the series $\sum_k k^m a_k x^k$ too converges absolutely whenever $|x| < \rho$ and for every $m \in \mathbb{Z}$.

Writing $k^m a_k = b_k$, $a_k = k^{-m} b_k$, one sees that the above reasoning is symmetric with respect to the series considered, $\sum_k a_k x^k$ and $\sum_k k^m a_k x^k$ ($m \in \mathbb{Z}$), i.e., if the radius of convergence of either one is $\rho \neq 0$, then the other too converges absolutely when $|x| < \rho$. It follows that the radii of convergence must be the same in all cases (including when $\rho = 0$), for otherwise a logical contradiction with the conclusions drawn would arise. $\square$

EXERCISES

1. It is known that $a_1 = 3$, $a_2 = 5$, $a_3 = 7$, $a_4 = -6$ and $\sum_{k=1}^{\infty} a_k = 4$. Compute $\sum_{k=1}^{\infty} (a_k + a_{k+1} + a_{k+2} + a_{k+3})$.
2. Classify the following series as convergent or divergent. If a series contains a parameter α , assume $\alpha \in \mathbb{Q}$ and classify the series for different values of

α .

$$\begin{array}{lll}
 \text{a)} \sum_{k=1}^{\infty} \frac{1}{k^2 - k + 3} & \text{b)} \sum_{k=1}^{\infty} \frac{k-1}{k^2 + 1} & \text{c)} \sum_{k=1}^{\infty} \frac{100+k}{9901+k-k^3} \\
 \text{d)} \sum_{k=1}^{\infty} \frac{\sqrt{k+1} - \sqrt{k}}{k} & \text{e)} \sum_{k=0}^{\infty} \frac{1}{\sqrt{k^2 + 100}} & \text{f)} \sum_{k=1}^{\infty} \left(\sqrt{\frac{k+1}{k}} - 1 \right) \\
 \text{g)} \sum_{k=1}^{\infty} \frac{\sqrt{k+1} \sqrt[3]{k+2}}{k^2 + k + 1} & \text{h)} \sum_{k=1}^{\infty} \frac{1+k!}{(1+k)!} & \text{i)} \sum_{k=1}^{\infty} \frac{2+k!}{(2+k)!} \\
 \text{j)} \sum_{k=0}^{\infty} (-1)^k \left(\sqrt{k+7} - \sqrt{k+1} \right) & \text{k)} \sum_{k=1}^{\infty} \frac{3(-1)^k + 1}{k} & \text{l)} \sum_{k=1}^{\infty} \frac{(-1)^k}{\sqrt[100]{k}} \\
 \text{m)} \sum_{k=1}^{\infty} \left(\frac{1}{k} - \frac{1}{k+|\alpha|} \right) & \text{n)} \sum_{k=1}^{\infty} \frac{1}{k^\alpha + k^{2\alpha}} & \text{o)} \sum_{k=1}^{\infty} \frac{\sqrt{k^\alpha + 1} - 1}{k} \\
 \text{p)} \sum_{k=1}^{\infty} (\sqrt{k^{2\alpha} + 1} - k^\alpha) & \text{q)} \sum_{k=1}^{\infty} \frac{k^\alpha + 1}{k^{2\alpha} + 1} & \text{r)} \sum_{k=1}^{\infty} \frac{(-1)^k k^{4\alpha+3} + 1}{k^{\alpha+2}}
 \end{array}$$

3. Estimate how many terms of the series must be taken in the following formulas in order to obtain π correct to 10 decimal places:

$$\pi = \sum_{k=0}^{\infty} \frac{4(-1)^k}{2k+1} = 2\sqrt{3} \sum_{k=0}^{\infty} \frac{(-1)^k}{3^k(2k+1)} = 2\sqrt{3} \sqrt{\sum_{k=0}^{\infty} \frac{(-1)^k}{(k+1)^2}}$$

4. a) Show that the alternating series $\sum_{k=0}^{\infty} (-1)^k / \sqrt{k+1}$ is convergent by means of the majorant principle, grouping consecutive terms of the series together so that the series becomes positive-term.
 b) Consider the series $\sum_{k=1}^{\infty} a_k$, where

$$a_{2p-1} = \frac{1}{\sqrt{p+1} - 1}, \quad a_{2p} = -\frac{1}{\sqrt{p+1} + 1}, \quad p = 1, 2, \dots$$

Which of the assumptions of Theorem 1.12.3 hold for this series? Settle the convergence question of the series by grouping consecutive terms of the series together in pairs, i.e., by examining the series $\sum_{p=1}^{\infty} b_p = \sum_{p=1}^{\infty} (a_{2p-1} + a_{2p})$.

5. Prove true or false:

- a) $\sum_k a_k$ converges $\Rightarrow \{a_k\}$ is a bounded sequence
 b) $\sum_k a_k$ converges and $\sum_k b_k$ converges $\Rightarrow \sum_k (a_k + b_k)$ converges

- c) $\sum_k a_k$ diverges and $\sum_k b_k$ diverges $\Rightarrow \sum_k (a_k + b_k)$ diverges
- d) $\sum_k a_k$ converges and $\sum_k b_k$ diverges $\Rightarrow \sum_k (a_k + b_k)$ diverges
- e) $\sum_k a_k$ converges and $\{b_k\}$ bounded $\Rightarrow \sum_k a_k b_k$ converges
- f) $\sum_k |a_k|$ converges and $\{b_k\}$ bounded $\Rightarrow \sum_k |a_k b_k|$ converges
- g) $\sum_k a_k$ converges $\Rightarrow \sum_k a_k^2$ converges
- h) $\sum_k a_k^2$ converges $\Rightarrow \sum_k a_k$ converges
- i) $\sum_k a_k$ converges absolutely $\Rightarrow \sum_k a_k^4$ converges

6. The *ratio test* of classical series theory for positive-term series is the statement: If the limit

$$\lim_k \frac{a_{k+1}}{a_k} = L \in \mathbb{R},$$

exists, then the series $\sum_k a_k$ converges if $L < 1$ and diverges if $L > 1$. Prove it!

7. The *root test* of classical series theory for positive-term series is the statement: If the limit

$$\lim_k \sqrt[k]{a_k} = L \in \mathbb{R},$$

exists, then the series $\sum_k a_k$ converges if $L < 1$ and diverges if $L > 1$. Prove it!

8. Using the theory of power series (Theorems I.12.9 and I.12.10) and Corollary I.12.5, prove the limit result

$$\lim_k k^m q^k = 0, \quad -1 < q < 1, \quad m \in \mathbb{N}.$$

9. Let the radii of convergence of the power series $\sum_k a_k x^k$ and $\sum_k b_k x^k$ be ρ_1 and ρ_2 . What can be said with certainty about the radius of convergence of the series $\sum_k (a_k + b_k) x^k$ if a) $\rho_1 < \rho_2$, b) $\rho_2 = \rho_1 \in \mathbb{R}_+$, c) $\rho_1 = \rho_2 = \infty$, d) $\rho_1 = \rho_2 = 0$?
10. a) It is known of the coefficients of a power series that $k \leq a_k \leq k^2 \forall k \in \mathbb{N}$. What is the interval of convergence of the series?
 b) The coefficients of a power series are $a_k = 1$ when $k = m!$ and $m \in \mathbb{N}$, and $a_k = 0$ otherwise. What is the sum of the series to 20 significant figures when $x = 1/10$, and what is the interval of convergence of the series?

11. Determine the intervals of convergence of the following power series:

$$\begin{array}{lll} \text{a)} \sum_{k=0}^{\infty} \frac{k^2}{k+1} x^k & \text{b)} \sum_{k=0}^{\infty} (k-2) 2^{k+1} x^k & \text{c)} \sum_{k=1}^{\infty} (-1)^k \left(\frac{x}{k}\right)^k \\ \text{d)} \sum_{k=1}^{\infty} \frac{1}{\sqrt{k}} x^k & \text{e)} \sum_{k=1}^{\infty} \frac{(-1)^k}{\sqrt{k}} 3^k x^k & \text{f)} \sum_{k=1}^{\infty} \frac{\sqrt{k+1}}{k^2} 2^k x^k \\ \text{g)} \sum_{k=1}^{\infty} \frac{2^k}{k} x^{2k} & \text{h)} \sum_{k=1}^{\infty} \frac{1}{k^3 3^k} x^{3k} & \text{i)} \sum_{k=1}^{\infty} \frac{1}{k^{1/3} 3^k} x^{3k} \end{array}$$

12. Determine the following series to be either convergent or divergent by appealing to the theory of power series:

$$\text{a)} \sum_{k=1}^{\infty} (-1)^k k^{-10} \pi^k e^{-k} \quad \text{b)} \sum_{k=1}^{\infty} \frac{k!}{(3k)^k} \quad \text{c)} \sum_{k=1}^{\infty} \frac{(2k)^k}{k!}$$

13. By a change of variable, determine for the following series a set $A \subset \mathbb{R}$ such that the series converges if and only if $x \in A$:

$$\text{a)} \sum_{k=1}^{\infty} \frac{1}{k} (2x+1)^k \quad \text{b)} \sum_{k=1}^{\infty} (x^2 - 3x + 1)^k \quad \text{c)} \sum_{k=1}^{\infty} \frac{(-1)^k}{k} (x^2 - 4x + 3)^k$$

14. Prove Theorem I.12.11 by means of Theorem I.12.10 in the case where the limit $\lim_k |a_k/a_{k+1}|$ exists.

15. Show that the series $\sum_{k=0}^{\infty} a_k x^k$, $\sum_{k=0}^{\infty} (k-2)^2 a_k x^k$ and $\sum_{k=1}^{\infty} k^{-2} (k+2)^3 a_k x^k$ all have the same radius of convergence.

16. (*) Show that the partial sums s_n of the series $\sum_{k=1}^{\infty} 1/\sqrt{k}$ satisfy

$$2\sqrt{n+1} - 2 < s_n \leq 2\sqrt{n} - 1, \quad n \in \mathbb{N}.$$

Hint: Compare with telescoping sums!

17. (*) Prove: If the sequence of terms $\{a_k\}$ of a positive-term series $\sum_k a_k$ is decreasing and the series converges, then $\lim_k k a_k = 0$.

18. (*) Give some power series with rational coefficients whose interval of convergence is $(-\sqrt[3]{e}, \sqrt[3]{e}]$ (e = Napier's number).

Chapter II

Vectors and analytic geometry

Among the cornerstones of the mathematical culture created by humankind is a mathematical model called the *Euclidean plane*. The term refers to the work *Stoikheia*, i.e. 'Elements', written around 300 BC by the Greek **Euclid**. Euclid's aim was to lay down clear axiomatic foundations for the mathematics of his day, and in particular for the branch of mathematics known as *geometry*. Geometry is — unlike the algebra of numbers considered above — a kind of mathematics that appeals very strongly to visual perception. Algebra and geometry represent, in present-day mathematics too, the two great main branches of mathematical thought.*

Euclid's work is the most successful mathematics textbook of all time, and its influence as an early model of axiomatic thinking has reached even beyond mathematics. In practical terms the work did not, of course, arise 'from nothing'; it was in fact preceded by a millennia-long tradition of computational methods in both Egypt and Babylonia. In Egypt, for instance, geometric methods had been developed (if the historians are to be believed) by both land surveyors and priests.

In this chapter we set out from the foundations of Euclidean geometry in the plane and in space, and devote ourselves to examining the concept of a *vector* that arises on a geometric basis, together with computation with vectors, i.e. *vector algebra*. By means of vectors it is possible to 'algebraize' geometric problems, that is, to formulate and solve them as problems of vector algebra. Such an approach to geometry — historically rather late in arising — is called *analytic geometry*.

*Drawing the line between algebra and geometry in mathematics is not always easy. The problem is illustrated, e.g., by a modern branch of mathematics called *algebraic geometry*.

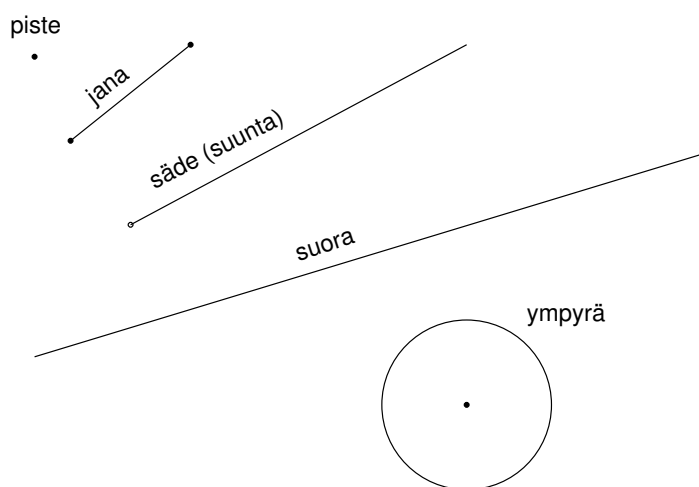
II.1 The Euclidean plane. Geometric numbers

We take as basically familiar such geometric concepts as, among others,

- *point, line, line segment*
- *angle, triangle, n -gon, polygon, similarity*
- *direction (half-line, ray), parallelism, parallelogram*
- *right angle, Pythagorean theorem, rectangle*
- *circle, the length measure of a segment ('measuring stick')*

Behind these lie, in Euclidean geometry, the basic axioms assumed as a starting point, which are known as *Hilbert's axioms*^{*}.

The basic objects of the *Euclidean plane* are the points, of which the other objects (such as segment, line, etc.) are thought to consist. The Euclidean plane is accordingly called a *point space*. Its symbolic sign is E^2 , read 'E two'.

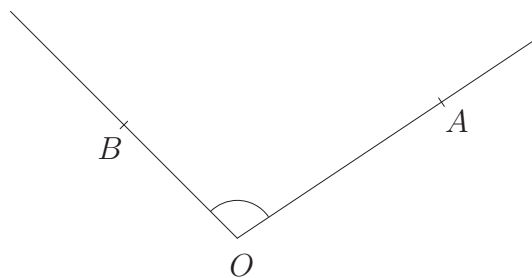


^{*}The German **David Hilbert** (1862-1943) was a trailblazer of modern mathematics and one of the most notable mathematicians of all time. Hilbert is known, among other things, for the unsolved mathematical problems he posed in 1900, which have since been named *Hilbert's problems*. The basic axioms of Euclidean geometry come from Hilbert's small classic "Grundlagen der Geometrie" of 1899. The work was the first comprehensive and systematic account of the axiomatic foundations of geometry — including certain non-Euclidean geometries — as well.

Geometric operations such as drawing parallel lines, constructing a right angle, or the *translation* and *rotation* of a given geometric object (point set) — the so-called *Euclidean motion* (e.g. moving a segment while preserving its length) — are carried out in Euclidean geometry by means of circular arcs (in practice, a compass). A line too can be constructed using a compass alone, since a line is the set of points — the *locus* — that are equidistant from two given points. — A straightedge is thus needed in geometry only as a 'mass-production device' for points.

Angle

In geometry an angle is defined by two directions. More concretely, an angle may be thought of as the point set formed by two half-lines issuing from the same point, or alternatively as the *sector* of the Euclidean plane bounded by these half-lines. In the latter interpretation (not in the former!) one must distinguish between the *interior angle* and the *exterior angle*. If A and B are points on the half-lines and O is the common point of the half-lines, i.e. the *vertex* of the angle ($A \neq O$, $B \neq O$), then the angle is denoted $\angle AOB$. In the figure the interior angle (sector interpretation!) is marked with a circular arc.



In comparing angles and in arithmetic operations on them (also in the measurement of an angle later on) an angle is generally interpreted as a sector. One then says that two angles are *equal* if they can be transformed into each other by a Euclidean motion.* Equality is thus verifiable geometrically. The *sum* and *difference* of two angles can likewise be determined geometrically by moving one angle by a Euclidean motion so that the angles come to lie 'side by side' (sum) or 'on top of each other' (difference). By geometric addition one can, for example, prove the well-known assertion about the angles of a triangle: the sum of the angles = a *straight angle* (Exercise 1b).

*Equality of angles is an equivalence relation defined on the set of angles, cf. Section 1.3.

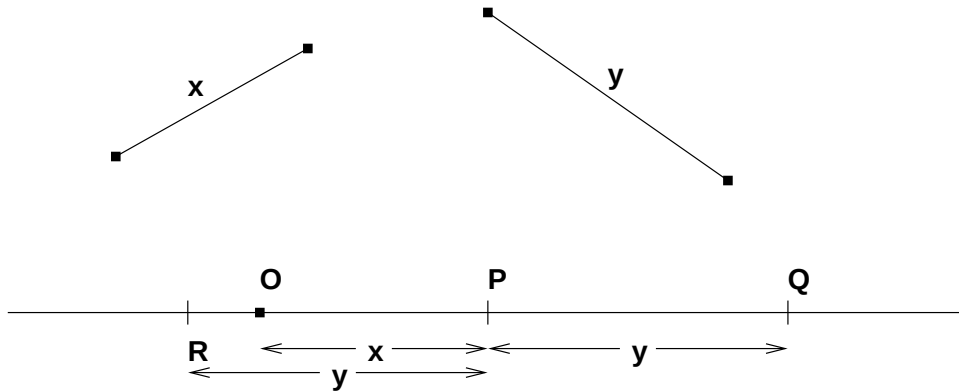
If in two triangles the (corresponding) angles are equal, then the triangles are *similar*. Among the basic assertions of geometry is the statement that in similar triangles the ratios of the lengths of corresponding sides are the same. This assertion plays a central role in many geometric proofs (see, e.g., Exercise 2).

Geometric numbers

Measures of angles are not needed in the basic operations of Euclidean geometry. Measuring the length of a segment with a compass and a measuring segment, on the other hand, is basic geometry. The operation brings into geometry the real numbers (or at least some of them) and also the arithmetic operations associated with numbers. As a starting point one takes the numbers 0 and 1:

$$0 \leftrightarrow \text{point}, \quad 1 \leftrightarrow \text{measuring segment}.$$

When we consider a fixed line on which a reference point O is chosen as the counterpart of the number 0, then the sum and difference of two numbers given (as segment lengths) can be determined geometrically:

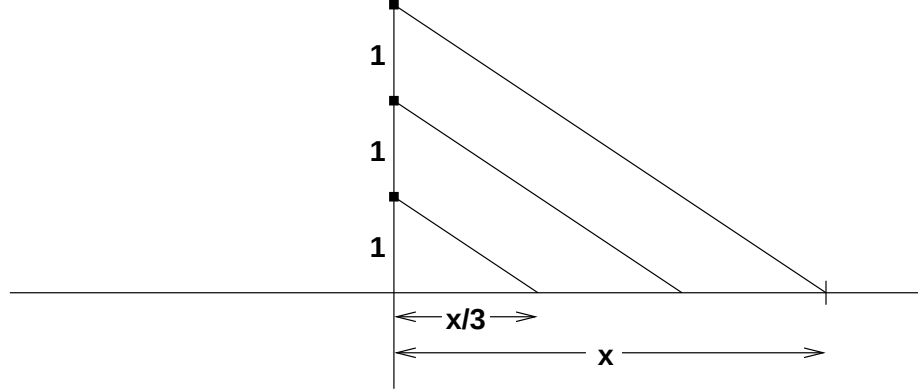


In the figure the sum and difference have geometric counterparts as segments and as points of the line:

$$\begin{aligned} x + y &\leftrightarrow OQ \leftrightarrow Q, \\ x - y &\leftrightarrow OR \leftrightarrow R. \end{aligned}$$

When the endpoint of the segment lies to the left of O (recall that the direction was defined!), OR can be interpreted as the negative number $x - y$. In this way both addition and subtraction have been defined in the number system. Applying the rules $+$, $-$ in succession, starting from the number 1 (= the measuring

segment placed on the line), one sees that the geometric counterpart of any integer $m \in \mathbb{Z}$ is obtained by a finite number of geometric operations. Thus $\mathbb{Z}$ has been constructed geometrically. Division by any integer likewise succeeds by 'slicing' the number with parallel lines, which can be produced in the plane with a compass and straightedge (see the figure below).

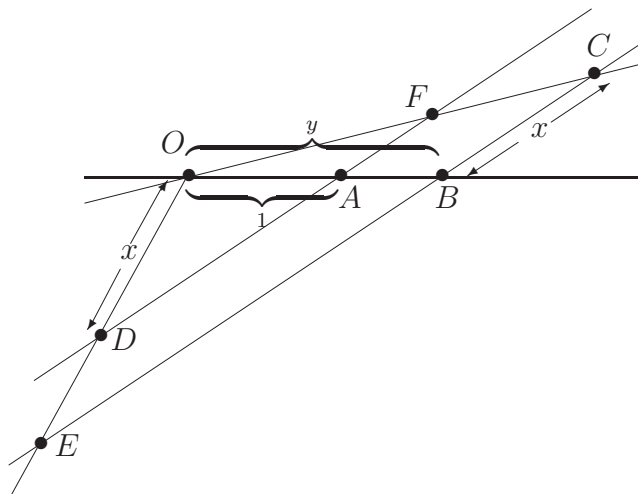


More generally, if one starts from two measure numbers x, y , then their multiplication and division can be directly geometrized by means of similar triangles, using the ratios

$$\begin{aligned} \frac{z}{x} &= \frac{y}{1} & (z = xy), \\ \frac{z}{1} &= \frac{x}{y} & (z = x/y). \end{aligned}$$

In the figure (below) we have

$$\begin{aligned} OA &\leftrightarrow 1, & OB &\leftrightarrow y, & BC &\leftrightarrow OD &\leftrightarrow x, \\ OE &\leftrightarrow x \cdot y, & AF &\leftrightarrow x/y. \end{aligned}$$



In what follows, the numbers that are constructible (as segment lengths) by a finite number of geometric operations will be named *geometric numbers*^{*}, and for this set of numbers we use the symbol $\mathbb{G}$:

$$\mathbb{G} = \{\text{geometric numbers}\}.$$

We already know that $\mathbb{G}$ contains all rational numbers: $\mathbb{G} \supset \mathbb{Q}$. Moreover, we know that $\mathbb{G}$ is closed under the basic arithmetic operations:

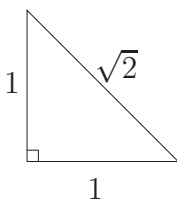
$$x, y \in \mathbb{G} \Rightarrow x + y, x - y, xy, x/y \in \mathbb{G} \quad (y \neq 0).$$

This means that $(\mathbb{G}, +, \cdot)$ is a field.

In forming the number set $\mathbb{G}$ the possibilities of geometry do not stop at the basic arithmetic operations, for by the Pythagorean theorem the operations

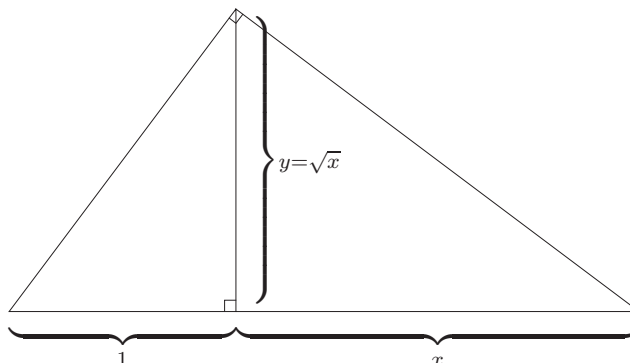
$$\begin{aligned} x, y &\mapsto \sqrt{x^2 + y^2}, & x, y &\in \mathbb{G}, \\ x, y &\mapsto \sqrt{x^2 - y^2}, & x, y &\in \mathbb{G}, \quad x \geq y \end{aligned}$$

are also geometrically realizable. In particular, then, the number $\sqrt{2}$ — which caused headaches for thousands of years — belongs to the set of geometric numbers:



^{*}The English term is 'constructible numbers'. There is no established Finnish rendering.

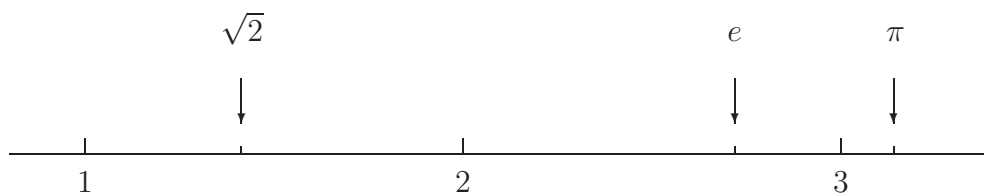
In fact the square-root operation $x \mapsto \sqrt{x}$ is geometrically possible for any measure number $x \in \mathbb{G}$ (thought of as $x \geq 0$) whatsoever:



Here, however, the possibilities of geometry come to an end, so $\mathbb{G}$ consists altogether of the numbers obtained, starting from the measure number 1, by a finite number of five different operations: $+$, $-$, $\cdot$, $/$ and $\sqrt{}$. When $\mathbb{G}$ is further equipped with a geometric order relation (comparison of segment lengths by means of a compass), the result is an ordered field $(\mathbb{G}, +, \cdot, <)$ based solely on geometric operations. This is the smallest possible extension of the field of rational numbers that satisfies the condition $x \in \mathbb{G} \ \& \ x > 0 \Rightarrow \sqrt{x} \in \mathbb{G}$ (cf. the construction of the number set $\mathbb{J}$ in Section I.2). It can be shown that all geometric numbers are algebraic (!), so that

$$\mathbb{Q} \subset \mathbb{G} \subset \mathbb{A} \subset \mathbb{R}.$$

When the geometric numbers are placed on a fixed line in the manner described above, the other real numbers can be imagined as also placed 'in between' these numbers. Thought of in this way, every real number acquires a geometric counterpart as a point of the *number line*.

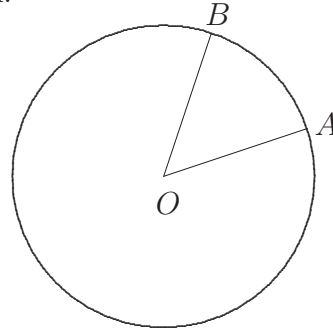


The measure of an angle

In dealing with angles the possibilities of geometry are rather limited. Addition and subtraction of angles succeed geometrically, as was noted, and an angle can

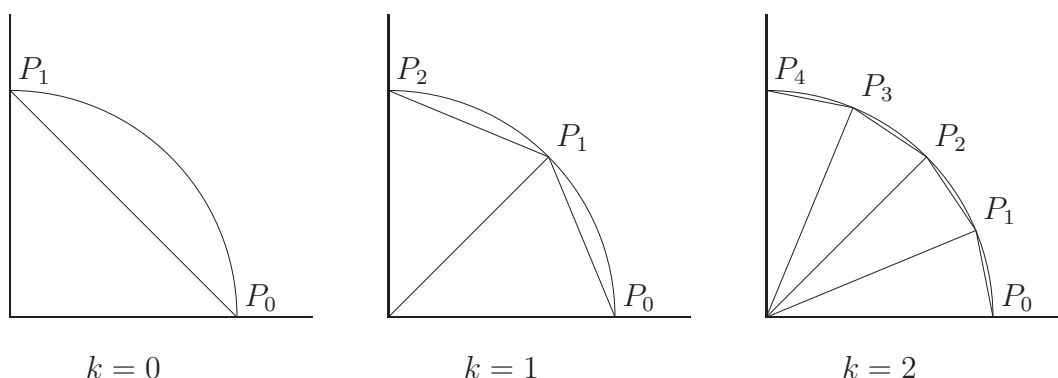
also easily be bisected, i.e. divided into two equal parts, by geometric means. On the other hand, already the division of an angle into three equal parts by a (finite) geometric construction is impossible, unless it is a special angle (such as a straight angle). The means of geometry also give out when one tries to assign to an angle a measure number describing its size. Geometrically one can construct only a sequence whose limit is the measure number; that is, the measure number is obtained only by a nonterminating construction.

Consider the angle $\angle AOB$, where the points A and B are placed on the circumference of the *unit circle* centered at O , i.e. the segments OA and OB have the length of the chosen measuring segment (corresponding to the number 1 on the number line).



In the situation of the figure one takes as the measure number of the angle $\angle AOB$ the *arc length* of the unit circle between the points A and B . Since there are two alternatives, one must decide whether the interior or the exterior angle is being measured. Next one has to solve a more serious problem: how to define the length of a circular arc? In the solution of the problem that follows, the means of geometry and algebra are combined.

Since the bisection of the angle $\angle OAB$ succeeds geometrically, the angle can at once be divided into $n = 2^k$ equal parts for every $k \in \mathbb{N}$. Let $k \in \mathbb{N}$ be given and denote the subangles arising in the division by $\angle P_{i-1}OP_i$, $i = 1 \dots n = 2^k$, where $P_0 = A$, $P_n = B$ and the other points P_i lie on the same circular arc centered at O . Now join the successive points P_i by segments into a *polygonal line* $P_0 \dots P_n$ and set $a_k =$ the length of that polygonal line, i.e. $a_k =$ the sum of the lengths of the segments $P_{i-1}P_i$, $i = 1 \dots n$. It is evident that each term of the sequence $\{a_k\}$ is determinable geometrically, starting from the number $a_0 =$ the length of the segment AB , so that $a_k/a_0 \in \mathbb{G} \forall k$ (possibly $a_0 \notin \mathbb{G}$). In the figure the polygonal line $P_0 \dots P_n$ is constructed for the indices $k = 0, 1, 2$, when the angle being measured is a right angle.



Appealing to the Pythagorean theorem it is easy to show that the sequence $\{a_k\}$ is strictly increasing (Exercise 5a), and on geometric grounds it can also be shown that $\{a_k\}$ is a bounded sequence (Exercise 5b). After this one must fall back on algebra: since $\{a_k\}$ is thus an increasing and bounded sequence, there exists a limit $\lim_k a_k = a \in \mathbb{R}$ (Theorem I.8.3, Definition I.9.1). We define the length of the arc AB , i.e. the measure number of the angle $\angle AOB$, to be this limit a .

The unit of measurement in the arc-length measurement of an angle is called the *radian*. As is well known, the number $\pi = 3.14159\dots$ is defined as the arc length of a half of the unit circle, i.e.

$$\pi = \text{measure number of a straight angle.}$$

Since π is a transcendental number (see Section I.11), the measure number of a right angle ($= \pi/2$) is also transcendental, as is, for example, the measure number of an angle of an equilateral triangle ($= \pi/3$).

Angles have traditionally also been measured in *degrees* ($^\circ$), the measure of a straight angle then being fixed at 180° . More generally, measure numbers in degrees and in radians are obtained from one another by scaling:

$$\text{angle in degrees} = \frac{180}{\pi} \cdot (\text{angle in radians}).$$

EXERCISES

1. a) When a line crosses two parallel lines, eight angles arise. Show that each of these is equal to three of the others.
 b) Prove: the sum of the angles of a triangle = a straight angle.

2. a) A segment divides a right triangle into two parts so that three similar triangles arise. Starting from this idea and from the theory of similar triangles, prove the Pythagorean theorem!
 b) A *median* of a triangle joins a vertex of the triangle to the midpoint of the opposite side. Starting from the theory of similar triangles, prove: the median AD of a triangle ABC bisects every segment parallel to BC whose endpoints lie on the sides AB and AC .
3. Give a geometric construction (with compass and straightedge as your weapons) whose result is
 - a) a circle passing through three given points,
 - b) the (largest possible) circle inscribed in a given triangle,
 - c) a line passing through a given point and tangent to a given circle, d) the *golden section* of a given segment AB , i.e. the point C on the segment such that if a, b, c are the lengths of the segments AB , AC and CB , then $a/b = b/c$.
4. Two of the following numbers are not geometric. Which three are?
 a) $\sqrt[1024]{17}$ b) $\sqrt[3]{512}$ c) $\sqrt[6]{343}$ d) $\sqrt[3]{7\sqrt{2}+5}$ e) $\sqrt[3]{5\sqrt{2}+7}$
5. (*) The measure number of a right angle in radians is defined as the limit $a = \lim_k a_k$, where $4a_k =$ the perimeter of the regular 2^{k+2} -gon whose vertices lie on the unit circle.
 - a) Show that the sequence $\{a_k\}_{k=0}^{\infty}$ is strictly increasing.
 - b) Show that $a_k < 2 \ \forall k$.
 - c) Show that $\{a_k\}$ is computable as a recursive sequence

$$a_0 = \sqrt{2}, \quad a_{k+1} = \frac{2a_k}{\sqrt{2 + \sqrt{4 - (2^{-k}a_k)^2}}}, \quad k = 0, 1, \dots$$

Compute a_k , $k = 1 \dots 5$ and compare with the number $\frac{\pi}{2}$.

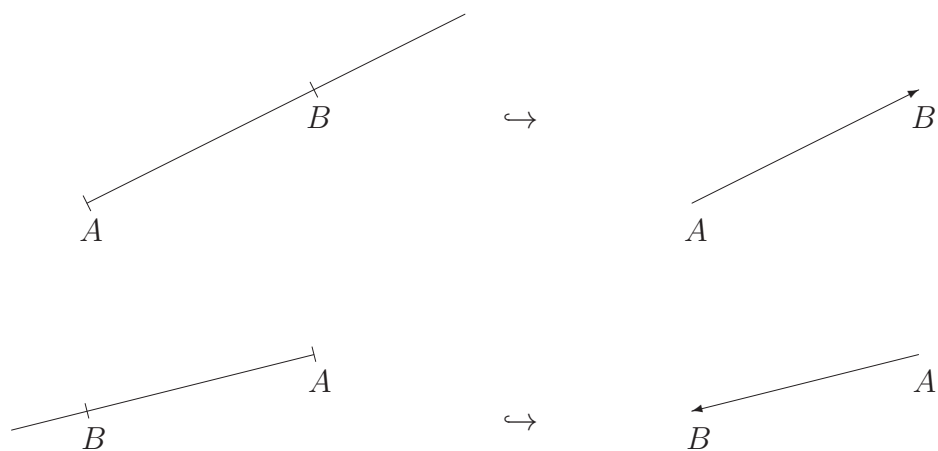
II.2 Plane vectors

As a mathematical concept, a *vector* is somewhat two-sided. It may be, to begin with, a geometric object, in which case in applications it is often thought of as physical, 'acting in the world'. Such 'physics vectors' include, among others,

- position, velocity, acceleration, angular velocity
- force, moment (torque)
- the strength of an X-field, X=gravitational, electric, magnetic...

On the other hand, a vector can also be interpreted algebraically, which leads to a vector concept more general (and also more abstract) than the geometric-physical vector. Below we shall see the first examples of such 'mathematics vectors' as well (many more examples will come later). In what follows, however, the starting point for the vector concept is at first geometric.

In the Euclidean plane E^2 a vector is defined geometrically as a *directed segment*, i.e. a segment that has a direction. The idea is then that the segment is contained in the half-line issuing from one of its endpoints. The direction expressed by the half-line is depicted by an arrowhead marked at the end of the segment:



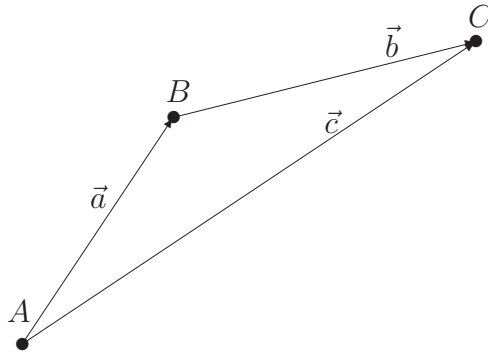
More precisely, by a vector one means only the partial information contained in a directed segment

$$\{(\text{segment}) \text{ length, direction}\}.$$

The vector associated with a particular directed segment is denoted $\overrightarrow{AB}$; more generally one uses the vector notations $\vec{a}, \vec{b}, \vec{v}$ etc.

Note that the vector $\overrightarrow{AB}$ thus does not 'know' where the point A lies in the Euclidean plane. Consequently 'a vector may be moved', as long as one 'neither rotates nor stretches' it. (Physical counterpart: the effect of a force is the same regardless of the point of application.)

If the vectors $\vec{a} = \overrightarrow{AB}$ and $\vec{b} = \overrightarrow{BC}$ are known, then $\vec{c} = \overrightarrow{AC}$ is also known:



We say that $\vec{c}$ is the *sum vector* of the vectors $\vec{a}, \vec{b}$ and write

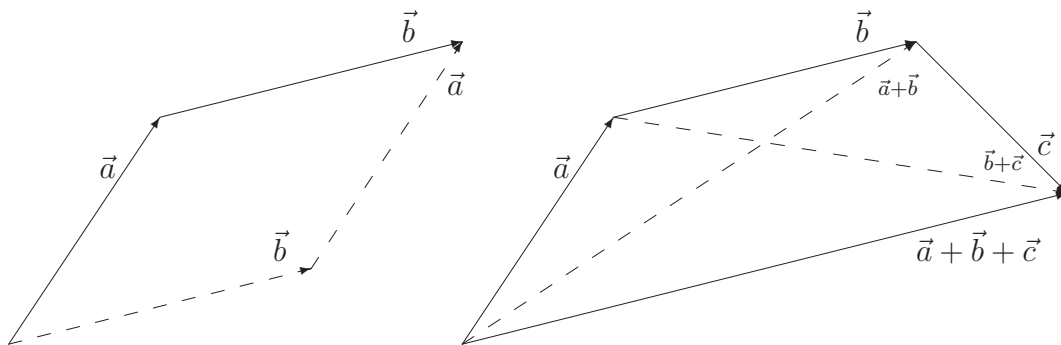
$$\vec{c} = \vec{a} + \vec{b}.$$

The sum is thus determined geometrically by a triangle diagram. For the vector addition defined in this way the usual commutative and associative laws hold:

$$(V1) \quad \vec{a} + \vec{b} = \vec{b} + \vec{a}$$

$$(V2) \quad (\vec{a} + \vec{b}) + \vec{c} = \vec{a} + (\vec{b} + \vec{c})$$

These are not axioms (cf. the corresponding field axioms, Section I.2) but assertions whose proof is geometric:



The *absolute value* (more colloquially, the 'length') of the vector $\vec{a} = \overrightarrow{AB}$ is

$$|\vec{a}| = |\overrightarrow{AB}| = |AB| = \text{length of segment } AB.$$

For this the *triangle inequality* holds (cf. Theorem I.2.3)

$$|\vec{a} + \vec{b}| \leq |\vec{a}| + |\vec{b}|.$$

If $\lambda \in \mathbb{R}$, then $\lambda\vec{a}$ denotes the *scaled* vector, for which:

- (1) $|\lambda\vec{a}| = |\lambda||\vec{a}|$
- (2) $\lambda\vec{a} \uparrow\uparrow \vec{a}$, if $\lambda > 0$, and $\lambda\vec{a} \uparrow\downarrow \vec{a}$, if $\lambda < 0$

Here $\uparrow\uparrow$ denotes parallelism and $\uparrow\downarrow$ opposite direction. The scaling of a vector $\vec{a} \mapsto \lambda\vec{a}$ ($\lambda \in \mathbb{R}$) is called, more generally in vector algebra, *multiplication by a scalar*. The scaling factors, i.e. the real numbers, are thus called *scalars* in this operation. From the definitions of scalar multiplication and addition one can deduce (Exercise 1a) that the following 'natural' rules hold (cf. the field axioms in Section I.2):

- (V3) $(\lambda + \mu)\vec{a} = \lambda\vec{a} + \mu\vec{a}$
- (V4) $\lambda(\mu\vec{a}) = (\lambda\mu)\vec{a}$
- (V5) $\lambda(\vec{a} + \vec{b}) = \lambda\vec{a} + \lambda\vec{b}$
- (V6) $1\vec{a} = \vec{a}$

By the definition of the operation $\vec{a} \rightarrow \lambda\vec{a}$ we have $|0\vec{a}| = 0$. Such a vector is called the *zero vector* and denoted $0\vec{a} = \vec{0}$. The direction of the zero vector is indeterminate (undefined). The zero vector is the zero element of vector addition, i.e. the following holds

$$(V7) \quad \vec{a} + \vec{0} = \vec{a} \quad \forall \vec{a}$$

Every vector $\vec{a}$ also has an *opposite vector* $-\vec{a}$, for which

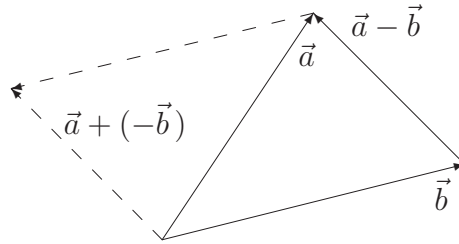
$$(V8) \quad \vec{a} + (-\vec{a}) = \vec{0}$$

Namely, choosing $\lambda = 1, \mu = -1$ in rule (V3) and using the definition of the zero vector and rule (V6), it follows that

$$-\vec{a} = (-1)\vec{a}.$$

By means of the opposite vector, subtraction of vectors is also defined:

$$\vec{a} - \vec{b} = \vec{a} + (-\vec{b}).$$

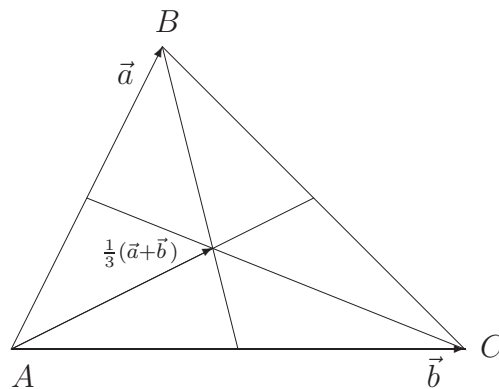


Rules (V1)–(V8), together with the definitions of the zero vector and the opposite vector, constitute the computational laws of the *vector algebra* (of plane vectors).

EXAMPLE 1 The identity following from the computational laws (V1)–(V8)

$$\vec{a} + \frac{2}{3} \left(\frac{1}{2} \vec{b} - \vec{a} \right) = \vec{b} + \frac{2}{3} \left(\frac{1}{2} \vec{a} - \vec{b} \right) = \frac{2}{3} \left[\vec{b} + \frac{1}{2} (\vec{a} - \vec{b}) \right] = \frac{1}{3} (\vec{a} + \vec{b})$$

can be read as: the medians of a triangle intersect at the same point, which divides the medians in the ratio 2 : 1 (cf. the figure). The point of intersection is called the *centroid* (center) of the triangle. $\square$



Vector space

If the set of plane vectors is denoted by the symbol V , then with the definitions above an algebra has arisen, which we shall denote $(V, \mathbb{R})$. When the computational rules (V1)–(V8) (more generally: the axioms) hold, one says that this is a (linear) *vector space*. The concept thus includes

- the set V formed by the vectors (on which one operates)
- the addition of vectors $(+)$
- the so-called *coefficient field* formed by the *scalars*, here $= \mathbb{R}$
- the multiplication of a scalar and a vector

The abbreviated phrase ' V is a vector space' refers to this whole algebra, with the coefficient field being precisely $\mathbb{R}$, or otherwise clear from the context. The phrases *real-coefficient* vector space or, more generally, $\mathbb{K}$ -*coefficient* vector space fix the coefficient field more precisely. The coefficient field thus need not be $\mathbb{R}$; it could be, e.g., $\mathbb{K} = \mathbb{Q}$ or $\mathbb{K} = \mathbb{G} = \{\text{geometric numbers}\}$. The linear vector space is a very central concept in mathematics, and it also plays a central role in many applications of mathematics.

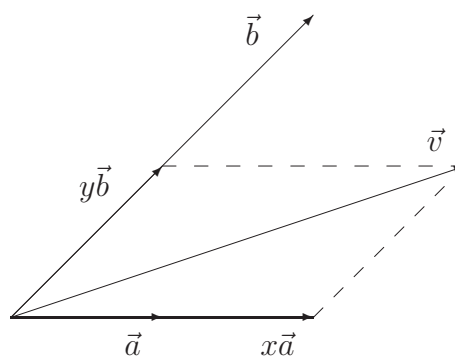
Basis and coordinate system

Let $\vec{v} \in V$ be an arbitrary plane vector and $\vec{a}, \vec{b} \in V$ two vectors for which

$$\vec{a}, \vec{b} \neq \vec{0}, \quad \vec{a} \neq \lambda \vec{b} \quad \forall \lambda \in \mathbb{R}.$$

Then, by a geometric construction (see the figure), one can find unique $x, y \in \mathbb{R}$ such that

$$\vec{v} = x\vec{a} + y\vec{b}.$$



On this basis one says that $\vec{v}$ is a *linear combination* of $\vec{a}$ and $\vec{b}$ and that $x\vec{a}$ and $y\vec{b}$ are the (vector) *components* of $\vec{v}$ in the directions of $\vec{a}$ and $\vec{b}$.

Thus, once two vectors satisfying the above conditions have been chosen, the whole of V can be represented in the form

$$V = \{x\vec{a} + y\vec{b} \mid x, y \in \mathbb{R}\}.$$

One says that the vector pair $\{\vec{a}, \vec{b}\}$ is a *basis* of V^* and that x, y are the *coordinates* of the vector $\vec{v} = x\vec{a} + y\vec{b} \in V$ in the basis $\{\vec{a}, \vec{b}\}$. If the coordinates are represented as a number pair (x, y) (a pair = an ordered set of two elements), then a connection has been created between V and the set of such number pairs. We denote the latter set by the symbol $\mathbb{R}^2$, pronounced 'R two':

$$\mathbb{R}^2 = \{(x, y) \mid x \in \mathbb{R} \ \& \ y \in \mathbb{R}\}.\dagger$$

By the above conventions we have in fact created a *one-to-one* (= unique in both directions) *correspondence* between V and $\mathbb{R}^2$. We denote this:

$$V \leftrightarrow \mathbb{R}^2.$$

In connection with this correspondence, $\mathbb{R}^2$ is named the *coordinate space* of V associated with the basis $\{\vec{a}, \vec{b}\}$. The name already suggests that $\mathbb{R}^2$ too can be interpreted as a linear vector space. Namely, all the computational laws of a vector space hold when addition and scalar multiplication in $\mathbb{R}^2$ are defined by

$$\begin{aligned} (x_1, y_1) + (x_2, y_2) &= (x_1 + x_2, y_1 + y_2), \\ \lambda(x, y) &= (\lambda x, \lambda y). \end{aligned}$$

These operations also correspond to the arithmetic operations of V , since

$$\begin{aligned} \vec{v}_1 = x_1\vec{a} + y_1\vec{b}, \ \vec{v}_2 = x_2\vec{a} + y_2\vec{b} &\Rightarrow \vec{v}_1 + \vec{v}_2 = (x_1 + x_2)\vec{a} + (y_1 + y_2)\vec{b}, \\ \vec{v} = x\vec{a} + y\vec{b} &\Rightarrow \lambda\vec{v} = (\lambda x)\vec{a} + (\lambda y)\vec{b}. \end{aligned}$$

*A basis is an ordered set of vectors, so the notation $(\vec{a}, \vec{b})$ would be more logical. To avoid various confusions, however, braces are more commonly used in this context.

†The set $\mathbb{R}^2$ can also be denoted $\mathbb{R} \times \mathbb{R}$, in which case one uses the set-theoretic notation for the *Cartesian product*

$$A \times B = \{(x, y) \mid x \in A \ \& \ y \in B\}.$$

This is read 'the Cartesian product of A and B ', or simply ' A cross B '. The elements of the set $A \times B$ are thus pairs. The identification relation between them is

$$(x_1, y_1) = (x_2, y_2) \quad \Leftrightarrow \quad x_1 = x_2 \ \wedge \ y_1 = y_2.$$

The vector space $(\mathbb{R}^2, \mathbb{R})$, with the operations mentioned, is an example (the first of its kind) of a vector space formed by algebraic vectors. This space is thus free of geometric interpretations. On the other hand, by interpreting $(\mathbb{R}^2, \mathbb{R})$ as the coordinate space of the plane vectors (in the chosen basis), the vector operations are converted into algebraic form. For example, in addition such a computational scheme is

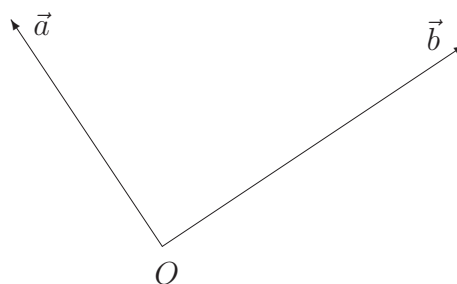
$$\begin{array}{ccc} \vec{v}_1 & \vec{v}_2 & \vec{v}_1 + \vec{v}_2 \\ \downarrow & \downarrow & \uparrow \\ (x_1, y_1) & (x_2, y_2) & \mapsto (x_1 + x_2, y_1 + y_2) \end{array}$$

The advantage of the scheme is that in the operation itself no information about the 'appearance' of the vectors is needed. Many results of geometry can in this way be derived by algebraic means (cf. Example 1 above).

The coordinate space $\mathbb{R}^2$ can equally well be associated with the original point space E^2 from which the whole vector idea originated. Namely, if vectors are thought of as directed segments of E^2 'that can be moved about', then one can equally well think that each vector corresponds to a unique directed segment whose initial point is a fixed point O . In this way a one-to-one correspondence between E^2 and V has been created:

$$P \in E^2 \leftrightarrow \overrightarrow{OP} \in V.$$

When the previously chosen basis vectors $\vec{a}, \vec{b}$ are now also interpreted as directed segments starting from the point O , a *coordinate system*^{*} has been created in the Euclidean plane, denoted $\{O, \vec{a}, \vec{b}\}$. The point O is called the *origin* of the coordinate system.



^{*}The idea of arithmetizing geometry by means of a coordinate system was introduced into mathematics by the French philosopher-mathematician **René Descartes** (1596-1650) in his treatise "La géométrie", which appeared in 1637 as an appendix to a broader philosophical work. The treatise marked the beginning of *analytic geometry*, and more generally foreshadowed the 'algebraization' of geometry — a trend that, especially in the age of computers, has grown even stronger.

If $P \in E^2$ and $\overrightarrow{OP} = x\vec{a} + y\vec{b}$, then one says that the *coordinates* of the *point* P in the chosen coordinate system $\{O, \vec{a}, \vec{b}\}$ are (x, y) . When the coordinate system is fixed, the coordinate pair (x, y) may, if desired, be regarded as the 'name' of P , in which case one is permitted to write, without further ado,

$$P = (x, y).$$

In this way a one-to-one correspondence $E^2 \leftrightarrow \mathbb{R}^2$ has been created.

EXAMPLE 2 Let T be a (proper) triangle of the plane whose vertices are O, A, B . Write $\vec{a} = \overrightarrow{OA}$, $\vec{b} = \overrightarrow{OB}$. Then the vertices of T , the midpoints of the sides of T , and the centroid of T in the coordinate system $\{O, \vec{a}, \vec{b}\}$ are (cf. Example 1)

vertices: $O = (0, 0)$, $A = (1, 0)$, $B = (0, 1)$,

midpoints of the sides: $O' = (1/2, 1/2)$, $A' = (0, 1/2)$, $B' = (1/2, 0)$,

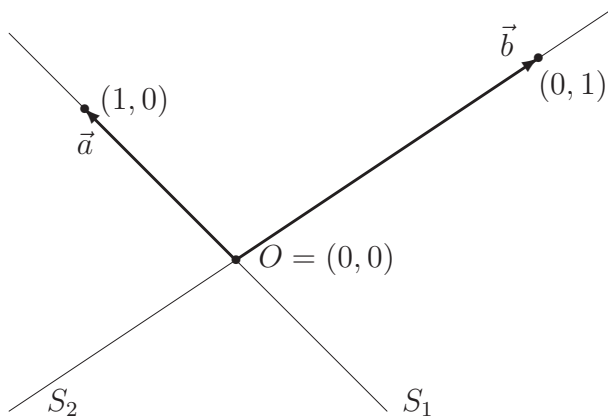
centroid: $C = (1/3, 1/3)$. $\square$

A coordinate system thus requires, first of all, a reference point O . From this so-called origin one chooses two directions, which determine the directions of the vectors $\vec{a}, \vec{b}$. It is further decided that when measuring distance from O , the unit of length is α when going in the direction of $\vec{a}$ and β when going in the direction of $\vec{b}$. Here $\alpha, \beta \in \mathbb{R}$ and $\alpha, \beta > 0$. When $\vec{a}$ and $\vec{b}$ are now chosen so that $|\vec{a}| = \alpha$ and $|\vec{b}| = \beta$, the vectors $\vec{a}$ and $\vec{b}$ are uniquely determined and the coordinate system is thus complete. The lines through the origin

$$S_1 = \{P \in E^2 \mid \overrightarrow{OP} = \lambda \vec{a}, \lambda \in \mathbb{R}\},$$

$$S_2 = \{P \in E^2 \mid \overrightarrow{OP} = \mu \vec{b}, \mu \in \mathbb{R}\},$$

whose *direction vectors* are $\vec{a}, \vec{b}$, are called the *coordinate axes*.



Linear independence

The assumptions made above about the basis vectors $\vec{a}, \vec{b}$ of the coordinate system ($\vec{a}, \vec{b} \neq \vec{0}$ and $\vec{a}, \vec{b}$ not parallel or antiparallel) can be stated more briefly as the condition

$$x\vec{a} + y\vec{b} = \vec{0} \quad \Rightarrow \quad x = y = 0. \quad (\star)$$

If the plane vectors have property $(\star)$, one says that $\{\vec{a}, \vec{b}\}$ is a *linearly independent* (vector) *system* (= set) or that $\vec{a}$ and $\vec{b}$ are linearly independent. Thus any linearly independent vector system $\{\vec{a}, \vec{b}\}$ qualifies as a basis of the plane vector space V .

If the vectors $\vec{a}, \vec{b}$ are not linearly independent, then they are *linearly dependent*. By condition $(\star)$ this is so if

$$x\vec{a} + y\vec{b} = \vec{0} \quad \text{for some } (x, y) \neq (0, 0).$$

This condition, in turn, is satisfied exactly when either (i) $\vec{a} = \vec{0} \Rightarrow x\vec{a} + 0\vec{b} = \vec{0} \forall x \in \mathbb{R}$, (ii) $\vec{b} = \vec{0} \Rightarrow 0\vec{a} + y\vec{b} = \vec{0} \forall y \in \mathbb{R}$, or (iii) $\vec{a}$ and $\vec{b}$ are parallel (= parallel or antiparallel, assuming $\vec{a} \neq \vec{0}$ and $\vec{b} \neq \vec{0}$), in which case $\exists \lambda \in \mathbb{R}, \lambda \neq 0$ such that $\vec{a} + \lambda\vec{b} = \vec{0}$.

EXAMPLE 3 Concerning the plane vectors $\vec{a}, \vec{b}$ it is known that $\vec{a} - \vec{b}$ and $\vec{a} + 2\vec{b}$ are linearly dependent. Can one conclude that $\vec{a}$ and $\vec{b}$ are also linearly dependent?

Solution By the given information, for some $(x, y) \neq (0, 0)$

$$x(\vec{a} - \vec{b}) + y(\vec{a} + 2\vec{b}) = \vec{0}.$$

By the rules of vector algebra this can be brought to the form

$$\vec{0} = (x + y)\vec{a} + (-x + 2y)\vec{b} = x'\vec{a} + y'\vec{b}.$$

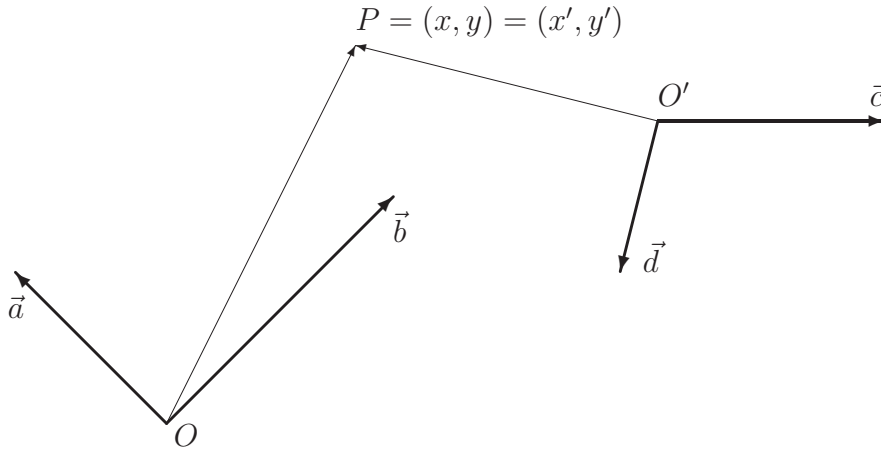
Since

$$\begin{cases} x' = 0 \\ y' = 0 \end{cases} \Leftrightarrow \begin{cases} x + y = 0 \\ -x + 2y = 0 \end{cases} \Leftrightarrow \begin{cases} x = 0 \\ y = 0 \end{cases}$$

and it is known that $(x, y) \neq (0, 0)$, we conclude that $(x', y') \neq (0, 0)$. Thus, since for some $(x', y') \neq (0, 0)$ we have $x'\vec{a} + y'\vec{b} = \vec{0}$, the answer is: Yes! $\square$

Change of coordinate system

When plane geometric problems are solved by the means of vector algebra, choosing a coordinate system suited to the problem is usually the first step of the solution (cf. Example 1 above). If the chosen coordinate system turns out to be inconvenient during the solution, one can perform a *change of coordinate system*. A change of coordinate system consists of a *translation of the origin* and a *change of basis* of the vector space, or of just one of the two. In the figure the coordinates of the point P are (x, y) in the coordinate system $\{O, \vec{a}, \vec{b}\}$ and (x', y') in the coordinate system $\{O', \vec{c}, \vec{d}\}$.



By the vector addition diagram we have

$$\overrightarrow{OP} = \overrightarrow{OO'} + \overrightarrow{O'P} \Leftrightarrow x\vec{a} + y\vec{b} = \overrightarrow{OO'} + (x'\vec{c} + y'\vec{d}).$$

Let here $\overrightarrow{OO'} = \alpha\vec{a} + \beta\vec{b}$, i.e. $O' = (\alpha, \beta)$ in the coordinate system $\{O, \vec{a}, \vec{b}\}$, and

$$\vec{c} = \lambda_1\vec{a} + \mu_1\vec{b}, \quad \vec{d} = \lambda_2\vec{a} + \mu_2\vec{b},$$

i.e. the coordinates of the vectors $\vec{c}, \vec{d}$ in the basis $\{\vec{a}, \vec{b}\}$ are (λ_1, μ_1) and (λ_2, μ_2) . Then the above equation simplifies, by the rules of vector algebra, to the form

$$(x - \lambda_1x' - \lambda_2y' - \alpha)\vec{a} + (y - \mu_1x' - \mu_2y' - \beta)\vec{b} = \vec{0}.$$

Since the vectors $\vec{a}$ and $\vec{b}$ are linearly independent, it follows that

$$\begin{cases} x - \lambda_1x' - \lambda_2y' - \alpha = 0 \\ y - \mu_1x' - \mu_2y' - \beta = 0 \end{cases} \Leftrightarrow \begin{cases} \lambda_1x' + \lambda_2y' = x - \alpha \\ \mu_1x' + \mu_2y' = y - \beta \end{cases}$$

Solving this for (x', y') in terms of the coordinates (x, y) — the system of equations is always solvable when $\vec{c}$ and $\vec{d}$ are linearly independent — yields the computational formulas for the *coordinate transformation* $(x, y) \mapsto (x', y')$. From the system of equations one also sees directly how the inverse transformation $(x', y') \mapsto (x, y)$ is to be computed.

EXAMPLE 4 Let

$$\overrightarrow{OO'} = \vec{a} - \vec{b}, \quad \vec{c} = 2\vec{a} + \vec{b}, \quad \vec{d} = \vec{a} - 2\vec{b}.$$

The course of the above computation is then

$$\begin{aligned} x\vec{a} + y\vec{b} &= (\vec{a} - \vec{b}) + x'(2\vec{a} + \vec{b}) + y'(\vec{a} - 2\vec{b}) \\ \Leftrightarrow (x - 2x' - y' - 1)\vec{a} + (y - x' + 2y' + 1)\vec{b} &= \vec{0} \\ \Leftrightarrow \begin{cases} 2x' + y' = x - 1 \\ x' - 2y' = y + 1 \end{cases} \end{aligned}$$

Solving, we obtain as the computational formulas of the coordinate transformation

$$\begin{cases} x = 2x' + y' + 1 \\ y = x' - 2y' - 1 \end{cases} \Leftrightarrow \begin{cases} x' = \frac{1}{5}(2x + y - 1) \\ y' = \frac{1}{5}(x - 2y - 3) \end{cases} \quad \square$$

Dimension. Subspace

Since the basis of the plane vector space must contain two linearly independent vectors, one says that V is a *2-dimensional* vector space or that the *dimension* of V is 2. We write

$$\dim V = 2.$$

The coordinate space $\mathbb{R}^2$ too is 2-dimensional as a vector space. Namely, if we write

$$\mathbf{a} = (1, 0), \quad \mathbf{b} = (0, 1),$$

then by the vector algebra of $\mathbb{R}^2$ we have $(x, y) = x\mathbf{a} + y\mathbf{b}$, i.e. every $\mathbf{v} = (x, y) \in \mathbb{R}^2$ can be represented uniquely as a linear combination of $\mathbf{a}$ and $\mathbf{b}$. Thus $\{\mathbf{a}, \mathbf{b}\}$ is a basis of $\mathbb{R}^2$, and since the basis contains two vectors, $\dim(\mathbb{R}^2) = 2$.

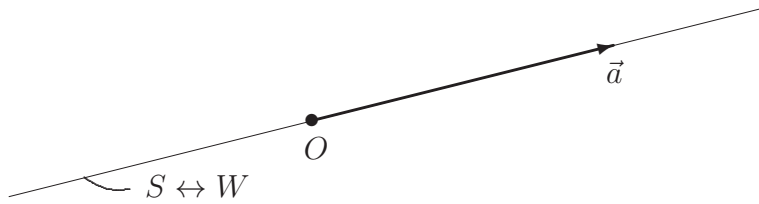
From the vectors of the plane or of $\mathbb{R}^2$ one can also form *1-dimensional* vector spaces. In the case of plane vectors these can all be expressed by means of some vector $\vec{a} \in V$, $\vec{a} \neq \vec{0}$ in the form

$$W = \{\vec{v} = x\vec{a}, \ x \in \mathbb{R}\}.$$

Since evidently

$$\begin{aligned}\vec{v}_1, \vec{v}_2 \in W &\Rightarrow \vec{v}_1 + \vec{v}_2 \in W, \\ \vec{v} \in W &\Rightarrow \lambda \vec{v} \in W \quad \forall \lambda \in \mathbb{R},\end{aligned}$$

W is itself a vector space. Its basis requires only one vector, e.g. $\vec{a}$, so $\dim W = 1$. Since also $W \subset V$, one says that W is a (proper) *subspace* of V and that $\vec{a}$ *spans* W . The geometric counterpart of the subspace W is a line $S \subset E^2$ passing through the origin. This is a *1-dimensional point space*.



The correspondence

$$P \in S \leftrightarrow \overrightarrow{OP} = x\vec{a} \in W \leftrightarrow x \in \mathbb{R}$$

generates a one-to-one correspondence between $\mathbb{R}$ and the point space S . The geometrization of numbers as points of the number line (cf. Section II.1) rested precisely on this correspondence.

EXERCISES

1. a) Justify the rules (V4)–(V6) of vector algebra geometrically.
 b) For vectors the triangle inequality $|\vec{a} + \vec{b}| \leq |\vec{a}| + |\vec{b}|$ holds. Deduce, without resorting to geometry any further, that $|\vec{a}| - |\vec{b}| \leq |\vec{a} + \vec{b}|$ also holds.
2. Concerning the plane vectors $\vec{a}, \vec{b}$ assume that $\vec{a}, \vec{b} \neq \vec{0}$ and that $\vec{a}$ and $\vec{b}$ are not parallel. For which values of the vector $\vec{c}$ can the vectors $\vec{a} + \vec{b}$, $\vec{a} + \vec{c}$ and $\vec{b} + 2\vec{c}$ be made into a triangle by 'moving them about'?
3. Prove the assertion of Exercise II.1:2b by means of vector algebra.

4. In the quadrilateral $ABCD$ we have $\overrightarrow{AB} = \vec{a}$, $\overrightarrow{AD} = \vec{b}$ and $\overrightarrow{BC} = (1/2)(\vec{a} + \vec{b})$. Compute, in terms of the vectors $\vec{a}$ and $\vec{b}$, the vector $\overrightarrow{AE}$, where E is the intersection point of the diagonals of the quadrilateral.
5. In the parallelogram $ABCD$ the vertex A is joined to the midpoint P of the side CD and the vertex B to the midpoint R of the side AD . Let the joining segments intersect at the point X . Express the vector $\overrightarrow{AX}$ in terms of the vectors $\vec{u} = \overrightarrow{AB}$ and $\vec{v} = \overrightarrow{AD}$.
6. In the triangle ABC write $\vec{a} = \overrightarrow{AB}$, $\vec{b} = \overrightarrow{AC}$. a) Reason geometrically that the vector $\vec{c} = |\vec{b}|\vec{a} + |\vec{a}|\vec{b}$ bisects the angle BAC . b) The point D lies on the segment BC and the segment AD bisects the angle BAC . Prove, by vector computation, the *angle bisector theorem*: the ratio of the lengths of the segments BD and $DC = |\vec{a}|/|\vec{b}|$.
7. Let the points M and N be the centroids of the triangles ABC and DEF . Show that $\overrightarrow{AD} + \overrightarrow{BE} + \overrightarrow{CF} = 3\overrightarrow{MN}$.
8. The sides BC , CA and AB of the triangle ABC are divided at the points A^* , B^* and C^* in the ratio $m : n$. Prove that the centroids of the triangles ABC and $A^*B^*C^*$ coincide.
9. Let $\{\vec{a}, \vec{b}\}$ be a basis of the plane vector space and let $\vec{u} = 2\vec{a} + 3\vec{b}$, $\vec{v} = -3\vec{a} + 2\vec{b}$ and $\vec{w} = -\vec{a} - 2\vec{b}$. Determine the coordinates of the vector $2\vec{u} - \vec{v} + \vec{w}$ in the basis $\{\vec{a}, \vec{b}\}$. Draw a figure!
10. The vectors $\vec{a}, \vec{b}$ form a basis of the plane vector space V . Show that the vectors $\vec{u} = \vec{a} + \vec{b}$, $\vec{v} = \vec{a} - 2\vec{b}$ also form a basis. Compute the coordinates of the vector $\vec{w} = \vec{a} - \vec{b}$ in this basis.
11. In the triangle ABC the point O bisects the side AB , the point E_1 divides the side BC in the ratio $1 : 2$ and the point E_2 divides the side CA in the ratio $1 : 3$. Determine the coordinates of the vertices of the triangle in the coordinate system whose origin is the point O and whose basis vectors are $\overrightarrow{OE_1}$ and $\overrightarrow{OE_2}$.
12. If $\vec{a}, \vec{b}$ are linearly independent plane vectors, then for which values of t ($t \in \mathbb{R}$) are the vectors $\vec{a} - t\vec{b}$ and $t\vec{a} - 2\vec{b}$ also linearly independent?
13. The coordinates of the point P are (x, y) in the coordinate system $\{O, \vec{a}, \vec{b}\}$ and (x', y') in the coordinate system $\{O', \vec{c}, \vec{d}\}$. Derive the computational formulas of the coordinate transformation in both directions, when

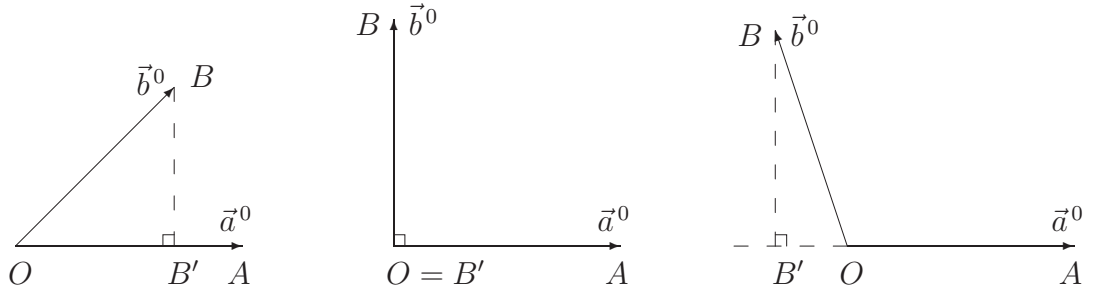
- a) $O' = O$, $\vec{c} = \vec{a} + \vec{b}$ and $\vec{d} = \vec{a} - \vec{b}$.
b) $\overrightarrow{O'O} = \vec{c} + 2\vec{d}$, $\vec{c} = 2\vec{a} + 3\vec{b}$ and $\vec{d} = -\vec{a} + 2\vec{b}$.
14. Let $\vec{a} = \overrightarrow{OA}$ and $\vec{b} = \overrightarrow{OB}$. Derive the computational formulas of the coordinate transformation from the coordinate system $\{O, \vec{a}, \vec{b}\}$ to the coordinate system $\{O', \vec{c}, \vec{d}\}$, when O' = the centroid of the triangle OAB and $\vec{c} = \overrightarrow{O'A}$, $\vec{d} = \overrightarrow{O'B}$. What are the coordinates of the vertices of the triangle and of the midpoints of the sides in the latter coordinate system?

II.3 Inner product

Let $\vec{a} \neq \vec{0}$ and $\vec{b} \neq \vec{0}$ be two plane vectors. The *unit vectors* (vectors of unit length) parallel to them are

$$\vec{a}^0 = \frac{1}{|\vec{a}|} \vec{a} = \frac{\vec{a}}{|\vec{a}|}, \quad \vec{b}^0 = \frac{1}{|\vec{b}|} \vec{b} = \frac{\vec{b}}{|\vec{b}|}.$$

In what follows we associate with the vectors $\vec{a}, \vec{b}$ a number $\gamma(\vec{a}, \vec{b}) \in \mathbb{R}$ that depends only on the directions of $\vec{a}$ and $\vec{b}$, i.e. only on the unit vectors $\vec{a}^0, \vec{b}^0$. The association is made by a geometric construction as follows: let $\vec{a}^0 = \overrightarrow{OA}$, $\vec{b}^0 = \overrightarrow{OB}$ and, on the line passing through the points O and A , choose the point B' nearest to the point B . Then the angle $\angle OB'B$ (or $\angle AB'B$, if $B' = O$) is a right angle.



In connection with the figure we set:

$$\gamma(\vec{a}, \vec{b}) = \gamma(\vec{a}^0, \vec{b}^0) = \begin{cases} |\overrightarrow{OB'}|, & \text{if } \overrightarrow{OB'} \uparrow \vec{a}^0, \\ 0, & \text{if } \overrightarrow{OB'} = \vec{0}, \\ -|\overrightarrow{OB'}|, & \text{if } \overrightarrow{OB'} \updownarrow \vec{a}^0. \end{cases}$$

The number $\gamma(\vec{a}, \vec{b})$ is thus determined geometrically as a segment length or its negative, once the vectors $\vec{a}$ and $\vec{b}$ are known. The construction is evidently symmetric with respect to these vectors, i.e. $\gamma(\vec{a}, \vec{b}) = \gamma(\vec{b}, \vec{a})$. It is also evident that $\gamma(\vec{a}, \vec{b})$ does not change if the geometric figure associated with the construction is altered by a Euclidean motion. From this one can conclude that $\gamma(\vec{a}, \vec{b})$ depends only on the measure number of the interior angle determined by the directions of the vectors $\vec{a}, \vec{b}$. From the construction one sees that the same holds conversely as well: the number $\gamma(\vec{a}, \vec{b})$ determines the said measure number uniquely.

In what follows, the angle determined by the directions of the vectors $\vec{a}, \vec{b}$ is denoted by the symbol $\angle(\vec{a}, \vec{b})$.^{*} From now on the number $\gamma(\vec{a}, \vec{b})$ is called the *cosine* of the *angle* $\angle(\vec{a}, \vec{b})$ and denoted

$$\gamma(\vec{a}, \vec{b}) = \cos \angle(\vec{a}, \vec{b}).$$

Since the number $\cos \angle(\vec{a}, \vec{b})$ determines, in connection with the angle $\angle(\vec{a}, \vec{b})$, the measure number of the interior angle, it also determines the geometric 'appearance' of the angle. In particular,

$$\begin{aligned} \cos \angle(\vec{a}, \vec{b}) = 1 & \Leftrightarrow \vec{a} \uparrow \uparrow \vec{b} & (\text{zero angle}), \\ \cos \angle(\vec{a}, \vec{b}) = -1 & \Leftrightarrow \vec{a} \uparrow \downarrow \vec{b} & (\text{straight angle}), \\ \cos \angle(\vec{a}, \vec{b}) = 0 & \Leftrightarrow \vec{a} \perp \vec{b} & (\text{right angle}). \end{aligned}$$

Here the notation $\vec{a} \perp \vec{b}$ is used to express that the vectors $\vec{a}$ and $\vec{b}$ are *perpendicular*, i.e. *orthogonal* ($\angle(\vec{a}, \vec{b}) = \text{a right angle}$). By the definition of the number $\cos \angle(\vec{a}, \vec{b})$ and the Pythagorean theorem, in all cases

$$-1 \leq \cos \angle(\vec{a}, \vec{b}) \leq 1. \quad (1)$$

From the definition one also sees that

$$\cos \angle(\lambda \vec{a}, \vec{b}) = \cos \angle(\vec{a}, \lambda \vec{b}) = \begin{cases} \cos \angle(\vec{a}, \vec{b}), & \text{if } \lambda > 0, \\ -\cos \angle(\vec{a}, \vec{b}), & \text{if } \lambda < 0. \end{cases} \quad (2)$$

DEFINITION II.3.1 (Inner product) The *inner product*, or *dot product*, of the vectors $\vec{a}, \vec{b}$ is a real number, denoted $\vec{a} \cdot \vec{b}$, read 'a dot b', and defined by

$$\vec{a} \cdot \vec{b} = \begin{cases} |\vec{a}| |\vec{b}| \cos \angle(\vec{a}, \vec{b}), & \text{if } \vec{a} \neq \vec{0} \text{ and } \vec{b} \neq \vec{0}, \\ 0, & \text{if } \vec{a} = \vec{0} \text{ or } \vec{b} = \vec{0}. \end{cases}$$

From the definition it follows first that

$$\vec{a} \cdot \vec{a} = |\vec{a}|^2. \quad (3)$$

Second, the symmetry law holds

$$\vec{a} \cdot \vec{b} = \vec{b} \cdot \vec{a}. \quad (4)$$

^{*}In the notation $\angle(\vec{a}, \vec{b})$ there is no need to distinguish between the interior and exterior angle, i.e. the angle need not be interpreted as a sector. Cf. Section II.1.

Third, from the definition and property (2) there follows the distributive law between scalar multiplication and the dot product

$$(\lambda \vec{a}) \cdot \vec{b} = \vec{a} \cdot (\lambda \vec{b}) = \lambda(\vec{a} \cdot \vec{b}), \quad \lambda \in \mathbb{R}. \quad (5)$$

By the last-mentioned law, omitting the parentheses in the notation $\lambda \vec{a} \cdot \vec{b}$ causes no confusion.

A somewhat less evident consequence of the definition of the inner product is the distributive law between the dot product and vector addition

$$\vec{a} \cdot (\vec{b} + \vec{c}) = \vec{a} \cdot \vec{b} + \vec{a} \cdot \vec{c}. \quad (6)$$

To prove this, consider the figures below.

Figure 1:

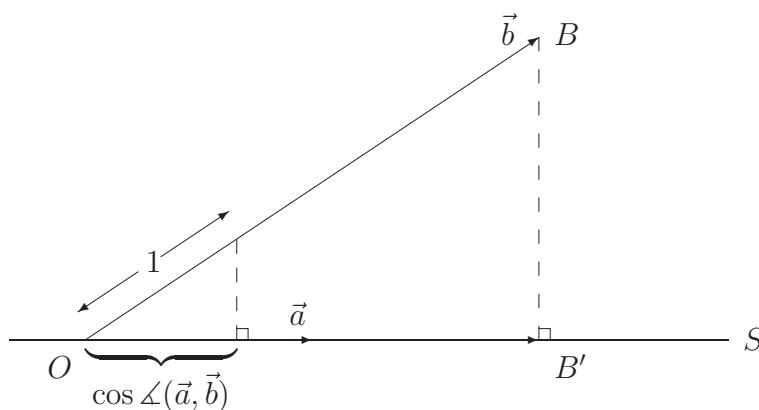
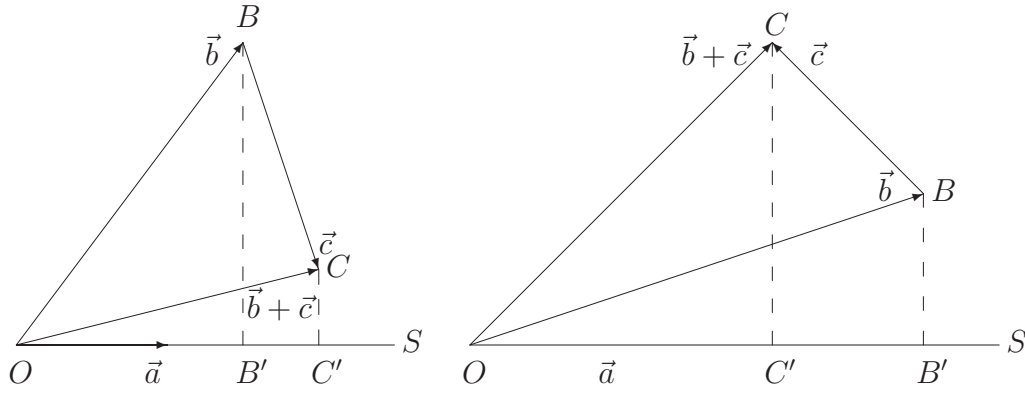


Figure 2:



$$\begin{aligned} |\vec{a}||\overrightarrow{OC'}| &= |\vec{a}||\overrightarrow{OB'}| + |\vec{a}||\overrightarrow{B'C'}| \\ \Rightarrow \vec{a} \cdot (\vec{b} + \vec{c}) &= \vec{a} \cdot \vec{b} + \vec{a} \cdot \vec{c} \end{aligned}$$

$$\begin{aligned} |\vec{a}||\overrightarrow{OC'}| &= |\vec{a}||\overrightarrow{OB'}| - |\vec{a}||\overrightarrow{B'C'}| \\ \Rightarrow \vec{a} \cdot (\vec{b} + \vec{c}) &= \vec{a} \cdot \vec{b} + \vec{a} \cdot \vec{c} \end{aligned}$$

In Figure 1 we have $|\overrightarrow{OB'}| = |\cos \angle(\vec{a}, \vec{b})| |\vec{b}|$ by the similarity of triangles, so from the definition of the inner product it follows that

$$\vec{a} \cdot \vec{b} = \begin{cases} |\vec{a}||\overrightarrow{OB'}|, & \text{if } \overrightarrow{OB'} \uparrow\uparrow \vec{a}, \\ -|\vec{a}||\overrightarrow{OB'}|, & \text{if } \overrightarrow{OB'} \uparrow\downarrow \vec{a}. \end{cases}$$

Based on this geometric interpretation, in Figure 2 the distributive law (6) has been derived in the case $\cos \angle(\vec{a}, \vec{b}) > 0$ and $\cos \angle(\vec{a}, \vec{b} + \vec{c}) > 0$. In the other cases too ($\cos \angle(\vec{a}, \vec{b}) \leq 0$ and/or $\cos \angle(\vec{a}, \vec{b} + \vec{c}) \leq 0$) one can reason in a corresponding way that the distributive law (6) holds.

By combining the properties (4), (5), (6) of the inner product one obtains the so-called *bilinearity properties*

$$\begin{aligned} (\lambda \vec{a} + \mu \vec{b}) \cdot \vec{c} &= \lambda \vec{a} \cdot \vec{c} + \mu \vec{b} \cdot \vec{c}, \\ \vec{a} \cdot (\lambda \vec{b} + \mu \vec{c}) &= \lambda \vec{a} \cdot \vec{b} + \mu \vec{a} \cdot \vec{c}, \quad \lambda, \mu \in \mathbb{R}. \end{aligned} \tag{7}$$

The inner product, which was defined above on the basis of geometric ideas, is in fact a very general concept related to vector spaces. In more general contexts one does not assume for the inner product, to begin with, any properties other than (i) *symmetry*, i.e. the property corresponding to the symmetry law (4), (ii) *bilinearity*, i.e. the counterparts of the computational rules (7), and (iii) *positive definiteness*, whose formulation for plane vectors is

$$\vec{a} \cdot \vec{a} \geq 0 \quad \forall \vec{a} \quad \text{and} \quad \vec{a} \cdot \vec{a} = 0 \Rightarrow \vec{a} = \vec{0}.$$

This is evident by the geometric definition above (property (3)). From the definition and property (1) it also follows that the inner product of plane vectors satisfies the inequality

$$\boxed{|\vec{a} \cdot \vec{b}| \leq |\vec{a}| |\vec{b}|.} \quad (8)$$

This inequality has more general — less evident — algebraic dimensions, which are examined more closely in the next section.

EXAMPLE 1 Concerning the plane vectors $\vec{a}, \vec{b}$ and $\vec{u}$ it is known that

$$|\vec{a}| = 1, \quad |\vec{b}| = 3, \quad \vec{a} \cdot \vec{b} = 2, \quad \vec{a} \cdot \vec{u} = -2, \quad \vec{b} \cdot \vec{u} = 1.$$

Determine the coordinates of $\vec{u}$ in the basis $\{\vec{a}, \vec{b}\}$.

Solution If we write $\vec{u} = x\vec{a} + y\vec{b}$, then by taking the dot product with $\vec{a}$ and $\vec{b}$ and using the bilinearity rules (7) and rules (3),(4), we obtain

$$x\vec{a} + y\vec{b} = \vec{u} \quad \Rightarrow \quad \begin{cases} |\vec{a}|^2 x + (\vec{a} \cdot \vec{b})y = \vec{a} \cdot \vec{u} \\ (\vec{a} \cdot \vec{b})x + |\vec{b}|^2 y = \vec{b} \cdot \vec{u} \end{cases}$$

Thus

$$\begin{cases} x + 2y = -2 \\ 2x + 9y = 1 \end{cases} \quad \Rightarrow \quad \begin{cases} x = -4 \\ y = 1 \end{cases} \quad \square$$

EXAMPLE 1 (continued) Split the vector $\vec{u}$ of the example into two components so that one component is parallel to $\vec{a}$ and the other is perpendicular to a) $\vec{a}$, b) $\vec{b}$.

Solution We wish to compute $\vec{u}_1, \vec{u}_2$ such that $\vec{u} = \vec{u}_1 + \vec{u}_2$, $\vec{u}_1 = t\vec{a}$, $t \in \mathbb{R}$ and

$$\text{a) } \vec{u}_2 \cdot \vec{a} = (\vec{u} - t\vec{a}) \cdot \vec{a} = 0, \quad \text{b) } \vec{u}_2 \cdot \vec{b} = (\vec{u} - t\vec{a}) \cdot \vec{b} = 0.$$

With the data of the example we obtain

$$\text{a) } t = \frac{\vec{a} \cdot \vec{u}}{|\vec{a}|^2} = -2, \quad \text{b) } t = \frac{\vec{b} \cdot \vec{u}}{\vec{a} \cdot \vec{b}} = \frac{1}{2}.$$

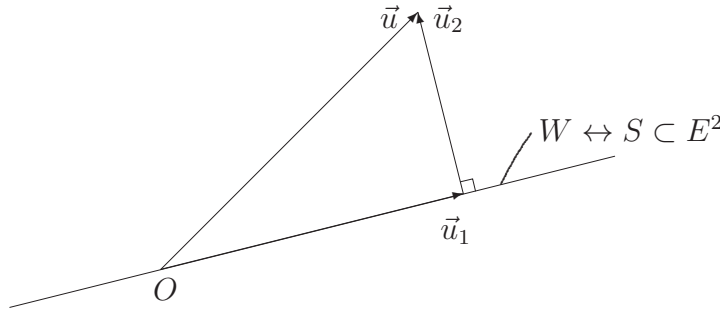
Thus $\vec{u} = \vec{u}_1 + \vec{u}_2$, where

$$\begin{aligned} \text{a) } \vec{u}_1 &= -2\vec{a}, \quad \vec{u}_2 = \vec{u} - \vec{u}_1 = (-4\vec{a} + \vec{b}) + 2\vec{a} = -2\vec{a} + \vec{b}, \\ \text{b) } \vec{u}_1 &= \frac{1}{2}\vec{a}, \quad \vec{u}_2 = \vec{u} - \vec{u}_1 = (-4\vec{a} + \vec{b}) - \frac{1}{2}\vec{a} = -\frac{9}{2}\vec{a} + \vec{b}. \quad \square \end{aligned}$$

In part a) of the example we have a basic task in which a given vector $\vec{u} \in V$ is to be split into two components $\vec{u}_1, \vec{u}_2$ so that $\vec{u}_1$ is parallel to the given vector $\vec{a}$ and $\vec{u}_2 \perp \vec{u}_1$. One then says that $\vec{u}_1$ is the *orthogonal projection* of $\vec{u}$ in the *direction* of $\vec{a}$, i.e. onto the one-dimensional subspace of V spanned by $\vec{a}$

$$W = \{\vec{v} = \lambda \vec{a} \mid \lambda \in \mathbb{R}\}.$$

In the Euclidean plane the subspace W corresponds to a line through the origin.



Orthonormal basis

In Section II.2 a basis for the space V was constructed using two linearly independent vectors. Once the inner product has been defined, practical computations can be considerably simplified by choosing the basis vectors so that they are orthogonal, i.e. the inner product between the basis vectors = 0. If, in addition, the basis vectors are *normalized* to unit vectors, one says that the basis so obtained is *orthonormal* (otherwise merely *orthogonal*). An orthonormal basis $\{\vec{i}, \vec{j}\}$ is thus one for which

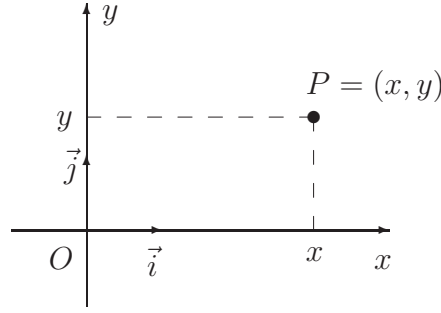
$$\begin{cases} \vec{i} \cdot \vec{j} = 0 & \text{(orthogonality condition),} \\ |\vec{i}| = |\vec{j}| = 1 & \text{(normalization condition).} \end{cases}$$

The $\{\vec{i}, \vec{j}\}$ satisfying these conditions is linearly independent, since

$$x\vec{i} + y\vec{j} = \vec{0} \quad \Rightarrow \quad \begin{cases} x = \vec{i} \cdot (x\vec{i} + y\vec{j}) = \vec{i} \cdot \vec{0} = 0, \\ y = \vec{j} \cdot (x\vec{i} + y\vec{j}) = \vec{j} \cdot \vec{0} = 0. \end{cases}$$

The corresponding coordinate system of the Euclidean plane is called *Cartesian*^{*} coordinate system.

^{*}The term was formed from the Latinized name Cartesius of the Frenchman **René Descartes**,
cf. the footnote in the preceding section.



The inner product of vectors given in an orthonormal basis $\{\vec{i}, \vec{j}\}$ is easily computed: if

$$\vec{v}_1 = x_1\vec{i} + y_1\vec{j}, \quad \vec{v}_2 = x_2\vec{i} + y_2\vec{j},$$

then by the bilinearity and symmetry of the inner product

$$\begin{aligned} \vec{v}_1 \cdot \vec{v}_2 &= (x_1\vec{i} + y_1\vec{j}) \cdot (x_2\vec{i} + y_2\vec{j}) \\ &= x_1x_2\vec{i} \cdot \vec{i} + (x_1y_2 + x_2y_1)\vec{i} \cdot \vec{j} + y_1y_2\vec{j} \cdot \vec{j}. \end{aligned}$$

Since here $\vec{i} \cdot \vec{i} = \vec{j} \cdot \vec{j} = 1$ and $\vec{i} \cdot \vec{j} = 0$, the computational formula becomes

$$\boxed{\vec{v}_1 \cdot \vec{v}_2 = x_1x_2 + y_1y_2.} \quad (9)$$

According to this, in an orthonormal basis the inner product is determined by the computational scheme (cf. the corresponding scheme for vector addition in the preceding section)

$$\begin{array}{ccc} \vec{v}_1 & & \vec{v}_2 \\ \downarrow & & \downarrow \\ (x_1, y_1) & (x_2, y_2) & \mapsto (x_1x_2 + y_1y_2) = \vec{v}_1 \cdot \vec{v}_2 \end{array}$$

The computation of the absolute value of a vector is also easy in an orthonormal basis, for by the formulas (3) and (9)

$$\boxed{|\vec{v}|^2 = \vec{v} \cdot \vec{v} = x^2 + y^2, \quad \vec{v} = x\vec{i} + y\vec{j}.} \quad (10)$$

EXAMPLE 2 Compute $\cos \angle(\vec{a}, \vec{b})$, when $\vec{a} = 4\vec{i} + 3\vec{j}$ and $\vec{b} = 2 - 3\vec{j}$.

Solution By Definition II.3.1 and formulas (9)–(10)

$$\cos \angle(\vec{a}, \vec{b}) = \frac{\vec{a} \cdot \vec{b}}{|\vec{a}| |\vec{b}|} = \frac{4 \cdot 2 + 3 \cdot (-3)}{\sqrt{4^2 + 3^2} \sqrt{2^2 + 3^2}} = -\frac{1}{5\sqrt{13}}. \quad \square$$

EXERCISES

1. Compute
 - a) $|4\vec{a} - 5\vec{b}|$, when $|\vec{a}| = 1$, $|\vec{b}| = 2$ and $\vec{a} \cdot \vec{b} = -\frac{1}{3}|\vec{a}||\vec{b}|$
 - b) $\vec{a} \cdot \vec{b}$, when $|\vec{a} + 3\vec{b}| = 16$ and $|\vec{a} - 3\vec{b}| = 2\sqrt{58}$
2. a) In the triangle ABC the lengths of the sides AB , AC and BC are a , b and c . Show, by means of the inner product, that $c^2 = a^2 + b^2 - 2ab \cos \angle BAC$ holds.
 b) In the quadrilateral $ABCD$ we have $\cos \angle BAD = \gamma$ and the lengths of the sides AB , AD , BC and CD are a , b , c and d . Compute $\cos \angle BCD$.
3. When $\vec{a}$ and $\vec{b}$ are plane vectors and $|\vec{a}| = |\vec{b}|$, the inner product $(\vec{a} + \vec{b}) \cdot (\vec{a} - \vec{b})$ takes a simple form. What form? What does the result mean geometrically if $\vec{a} = \overrightarrow{OA}$ and $\vec{b} = \overrightarrow{OB}$, where a) OA and OB are two sides of a parallelogram, b) O is the center of a circle and A , B points of its circumference? A figure!
4. In the right triangle ABC there is an altitude AD issuing from the vertex of the right angle. The lengths of the sides AB and AC are 5 and 12. Compute the inner products $\overrightarrow{AB} \cdot \overrightarrow{DC}$, $\overrightarrow{BD} \cdot \overrightarrow{CA}$ and $\overrightarrow{AC} \cdot \overrightarrow{CD}$.
5. Concerning the plane vectors $\vec{a}$ and $\vec{b}$ it is known that $|\vec{a}| = |\vec{b}| = 1$ and $\vec{a} \cdot \vec{b} = \frac{1}{7}$. It is further known about the vector $\vec{v}$ that $\vec{a} \cdot \vec{v} = 3$ and $\vec{b} \cdot \vec{v} = -1$. Determine the coordinates of $\vec{v}$ in the basis $\{\vec{a}, \vec{b}\}$.
6. In the plane coordinate system $\{O, \vec{a}, \vec{b}\}$ the coordinates of the point P are $(2, 1)$ and the coordinates of the point O' are $(-1, 3)$. Determine the coordinates of the point P in the coordinate system $\{O', \vec{a} + \vec{b}, \vec{a} - 2\vec{b}\}$, given that $|\vec{a}| = |\vec{b}| = 1$ and $\vec{a} \cdot \vec{b} = -\frac{1}{3}$.
7. Compute $|\vec{a}|$, $|\vec{b}|$ and $\cos \angle(\vec{a}, \vec{b})$, when
 - a) $\vec{a} = \vec{i} - \vec{j}$, $\vec{b} = \vec{i} + 2\vec{j}$
 - b) $\vec{a} = -2\vec{i} + 3\vec{j}$, $\vec{b} = 3\vec{i} - \vec{j}$
 - c) $\vec{a} = -68\vec{i} + 51\vec{j}$, $\vec{b} = 3\vec{i} + 4\vec{j}$
 - d) $\vec{a} = 76\vec{i} - 57\vec{j}$, $\vec{b} = 92\vec{i} + 69\vec{j}$
 - e) $\vec{a} = (3\sqrt{2} + 5)\vec{i} + (5\sqrt{2} - 3)\vec{j}$, $\vec{b} = (5\sqrt{2} + 3)\vec{i} + (3\sqrt{2} - 5)\vec{j}$
8. Concerning the plane vectors $\vec{a}, \vec{b} \in V$ it is known that $|\vec{a}| = 1$, $|\vec{b}| = 3$ and $\vec{a} \cdot \vec{b} = -2$. For which values of the coefficients λ, μ is $\{\vec{a}, \lambda\vec{a} + \mu\vec{b}\}$ an orthonormal basis of V ?

9. (*) Show, using the inner product, that the altitudes of a triangle lie on three lines passing through the same point, i.e. the altitudes or their extensions intersect at the same point.
10. (*) The set $S \subset E^2$ consists of the points P whose Cartesian coordinates satisfy the condition

$$(x-1)^2 - (y+2)^2 = 1.$$

Let (ξ, η) be the coordinates of the point P in another orthonormal coordinate system whose origin is at the point $(x, y) = (1, -2)$ and whose basis vectors are

$$\vec{e}_1 = \frac{1}{\sqrt{2}}(\vec{i} + \vec{j}), \quad \vec{e}_2 = \frac{1}{\sqrt{2}}(\vec{i} - \vec{j}).$$

If the coordinates of the point $P = (x, y)$ in this coordinate system are (x', y') , then under what condition on the coordinates x', y' is $P = (x', y') \in S$? Draw a figure showing both coordinate systems, and sketch the set S in the figure.

II.4 *Abstract inner product and norm

Assume as a starting point a given real-coefficient vector space $(U, \mathbb{R})$ together with an *inner product* defined on it, denoted by the symbol $\langle \cdot, \cdot \rangle$. The elements (vectors) of U are denoted $\mathbf{u}, \mathbf{v}$, etc. — No hypotheses are made about their ‘appearance’. Of the inner product it is assumed that $\langle \mathbf{u}, \mathbf{v} \rangle \in \mathbb{R}$ is defined for all pairs $(\mathbf{u}, \mathbf{v})$, $\mathbf{u}, \mathbf{v} \in U$ (i.e. $(\mathbf{u}, \mathbf{v}) \in U \times U$, cf. the footnote in Section II.2) and that the following laws hold:

1. Symmetry

$$\langle \mathbf{u}, \mathbf{v} \rangle = \langle \mathbf{v}, \mathbf{u} \rangle \quad \forall \mathbf{u}, \mathbf{v} \in U.$$

2. Bilinearity

$$\begin{aligned} \text{(a)} \quad & \langle \alpha \mathbf{u} + \beta \mathbf{v}, \mathbf{w} \rangle = \alpha \langle \mathbf{u}, \mathbf{w} \rangle + \beta \langle \mathbf{v}, \mathbf{w} \rangle, \\ \text{(b)} \quad & \langle \mathbf{u}, \alpha \mathbf{v} + \beta \mathbf{w} \rangle = \alpha \langle \mathbf{u}, \mathbf{v} \rangle + \beta \langle \mathbf{u}, \mathbf{w} \rangle \\ & \forall \mathbf{u}, \mathbf{v}, \mathbf{w} \in U, \alpha, \beta \in \mathbb{R}. \end{aligned}$$

3. Positive definiteness

$$\begin{aligned} \text{(a)} \quad & \langle \mathbf{u}, \mathbf{u} \rangle \geq 0 \quad \forall \mathbf{u} \in U, \\ \text{(b)} \quad & \langle \mathbf{u}, \mathbf{u} \rangle = 0 \Leftrightarrow \mathbf{u} = \mathbf{0}. \end{aligned}$$

Here the zero vector of U (the zero element of vector addition) has been denoted by $\mathbf{0}$.

Assumptions 1–3 are the general *inner product axioms* of a real-coefficient vector space.* If $\langle \cdot, \cdot \rangle$ is an inner product of U satisfying axioms 1–3, then one says that the *norm* of the vector $\mathbf{u} \in U$ associated with that inner product is

$$|\mathbf{u}| = \langle \mathbf{u}, \mathbf{u} \rangle^{1/2}.$$

EXAMPLE 1 Let $U = (\mathbb{R}^2, \mathbb{R})$ be the coordinate space corresponding to an orthonormal basis $\{\vec{i}, \vec{j}\}$ of the plane vector space $(V, \mathbb{R})$. Then if $\mathbf{u} = (x_1, y_1) \in U$ and $\mathbf{v} = (x_2, y_2) \in U$, and the corresponding plane vectors are $\vec{u} = x_1\vec{i} + y_1\vec{j}$ and $\vec{v} = x_2\vec{i} + y_2\vec{j}$, then by the conclusions of the preceding section axioms 1–3 hold, when the inner product of U is defined by

$$\langle \mathbf{u}, \mathbf{v} \rangle = \vec{u} \cdot \vec{v} = x_1x_2 + y_1y_2. \quad \square$$

*The bilinearity requirement (b) is unnecessary as an axiom, since it follows from condition (a) together with symmetry. Part (b) of the positive-definiteness axiom may also be stated in the form ‘ $\Rightarrow$ ’, for ‘ $\Leftarrow$ ’ follows from bilinearity.

With reference to the example, we set

DEFINITION II.4.1 If $\mathbf{u} = (x_1, y_1) \in \mathbb{R}^2$ and $\mathbf{v} = (x_2, y_2) \in \mathbb{R}^2$, then one says that the inner product

$$\langle \mathbf{u}, \mathbf{v} \rangle = x_1 x_2 + y_1 y_2$$

is the *Euclidean inner product* of $\mathbb{R}^2$ (of the vector space $(\mathbb{R}^2, \mathbb{R})$). The norm

$$|\mathbf{u}| = \langle \mathbf{u}, \mathbf{u} \rangle^{1/2} = (x^2 + y^2)^{1/2}, \quad \mathbf{u} = (x, y) \in \mathbb{R}^2$$

is the *Euclidean norm* of $\mathbb{R}^2$.

EXAMPLE 1 (continued) The inequality $|\vec{u} \cdot \vec{v}| \leq |\vec{u}| |\vec{v}|$ valid for the inner product of plane vectors can be written, by means of the Euclidean inner product and norm of $\mathbb{R}^2$, in the form

$$|\langle \mathbf{u}, \mathbf{v} \rangle| \leq |\mathbf{u}| |\mathbf{v}| \quad \forall \mathbf{u}, \mathbf{v} \in \mathbb{R}^2. \quad \square$$

Written out, the inequality of the example means the assertion

$$|x_1 x_2 + y_1 y_2| \leq \sqrt{x_1^2 + y_1^2} \sqrt{x_2^2 + y_2^2} \quad \forall x_1, x_2, y_1, y_2 \in \mathbb{R}.$$

This is thus true on geometric grounds — but as an assertion it does not look at all dependent on geometry. The following theorem shows that it is in fact not a geometric but an algebraic assertion: the inequality is a special case of the more general *Cauchy* or *Cauchy–Schwarz inequality*, valid for all inner products. Alongside the triangle inequality, this is the most commonly used inequality in mathematical analysis.

THEOREM II.4.2 (**Cauchy–Schwarz**) For every inner product of a real-coefficient vector space U satisfying axioms 1–3, the inequality

$$|\langle \mathbf{u}, \mathbf{v} \rangle| \leq |\mathbf{u}| |\mathbf{v}|,$$

holds, where $|\mathbf{u}| = \sqrt{\langle \mathbf{u}, \mathbf{u} \rangle}$.

Proof If $\mathbf{u} = \mathbf{0}$, then $\langle \mathbf{u}, \mathbf{v} \rangle = \langle \mathbf{u}, \mathbf{u} \rangle = 0$ by the bilinearity conditions, so in this case the assertion is true in the form $0 \leq 0$. Assume, then, that $\mathbf{u} \neq \mathbf{0}$, so that by axiom 3 $\langle \mathbf{u}, \mathbf{u} \rangle > 0$. Then by the same axiom

$$\langle \beta \mathbf{u} + \mathbf{v}, \beta \mathbf{u} + \mathbf{v} \rangle \geq 0 \quad \forall \beta \in \mathbb{R}.$$

By axioms 1–2 this can be written equivalently in the form

$$\beta^2 \langle \mathbf{u}, \mathbf{u} \rangle + 2\beta \langle \mathbf{u}, \mathbf{v} \rangle + \langle \mathbf{v}, \mathbf{v} \rangle \geq 0 \quad \forall \beta \in \mathbb{R}.$$

Choosing here

$$\beta = -\frac{\langle \mathbf{u}, \mathbf{v} \rangle}{\langle \mathbf{u}, \mathbf{u} \rangle}$$

it follows that

$$\begin{aligned} -\frac{\langle \mathbf{u}, \mathbf{v} \rangle^2}{\langle \mathbf{u}, \mathbf{u} \rangle} + \langle \mathbf{v}, \mathbf{v} \rangle \geq 0 &\Leftrightarrow \langle \mathbf{u}, \mathbf{v} \rangle^2 \leq \langle \mathbf{u}, \mathbf{u} \rangle \langle \mathbf{v}, \mathbf{v} \rangle \\ &\Leftrightarrow |\langle \mathbf{u}, \mathbf{v} \rangle| \leq \|\mathbf{u}\| \|\mathbf{v}\|. \quad \square \end{aligned}$$

Norm

Above, the real number $\|\mathbf{u}\| = \langle \mathbf{u}, \mathbf{u} \rangle^{1/2}$ defined by means of the inner product was already called the norm of the vector $\mathbf{u}$. More generally, the *norm of a vector space* does not require the definition of an inner product; rather, it is a more general way of assigning a 'length' to a vector. The more general norm of a vector $\mathbf{u}$ is denoted by the symbol $\|\mathbf{u}\|$. 'Norming' a vector is a matter of an operation (or measurement operation) $\mathbf{u} \in U \mapsto \|\mathbf{u}\| \in \mathbb{R}$ that must satisfy the following *norm axioms*:

1. Positive definiteness

- (a) $\|\mathbf{u}\| \geq 0 \quad \forall \mathbf{u} \in U$,
- (b) $\|\mathbf{u}\| = 0 \Rightarrow \mathbf{u} = \mathbf{0}$.

2. Scalability

$$\|\alpha \mathbf{u}\| = |\alpha| \|\mathbf{u}\| \quad \forall \mathbf{u} \in U, \alpha \in \mathbb{R}.$$

3. Triangle inequality

$$\|\mathbf{u} + \mathbf{v}\| \leq \|\mathbf{u}\| + \|\mathbf{v}\| \quad \forall \mathbf{u}, \mathbf{v} \in U.$$

By choosing $\alpha = 0$ in axiom 2 one sees that $\mathbf{u} = \mathbf{0} \Rightarrow \|\mathbf{u}\| = 0$. By axiom 1b this implication also holds conversely, so for the norm

$$\|\mathbf{u}\| = 0 \Leftrightarrow \mathbf{u} = \mathbf{0}.$$

EXAMPLE 2 For the norm $\|\mathbf{u}\| = \|\mathbf{u}\| = \langle \mathbf{u}, \mathbf{u} \rangle^{1/2}$ defined by means of the inner product, the validity of axioms 1–2 is evident from the positive definiteness and bilinearity of the inner product (inner product axioms 2 and 3). The triangle inequality also holds (Exercise 5a), so this is indeed a norm. $\square$

EXAMPLE 3 The so-called *maximum norm* of the space $\mathbb{R}^2$ is defined by

$$\|\mathbf{u}\| = \|(x, y)\| = \max\{|x|, |y|\}.$$

The validity of axioms 1–2 for this norm is easily verified. To demonstrate axiom 3, let $\mathbf{u} = (x_1, y_1) \in \mathbb{R}^2$ and $\mathbf{v} = (x_2, y_2) \in \mathbb{R}^2$, so that

$$\|\mathbf{u} + \mathbf{v}\| = \max\{|x_1 + x_2|, |y_1 + y_2|\}.$$

If here $|x_1 + x_2| \leq |y_1 + y_2|$, then by the triangle inequality valid in the field of real numbers (Theorem 1.2.3) one can conclude:

$$\begin{aligned} \|\mathbf{u} + \mathbf{v}\| &= |y_1 + y_2| \leq |y_1| + |y_2| \\ &\leq \max\{|x_1|, |y_1|\} + \max\{|x_2|, |y_2|\} = \|\mathbf{u}\| + \|\mathbf{v}\|. \end{aligned}$$

If $|x_1 + x_2| > |y_1 + y_2|$, then one concludes correspondingly

$$\begin{aligned} \|\mathbf{u} + \mathbf{v}\| &= |x_1 + x_2| \leq |x_1| + |x_2| \\ &\leq \max\{|x_1|, |y_1|\} + \max\{|x_2|, |y_2|\} = \|\mathbf{u}\| + \|\mathbf{v}\|. \end{aligned}$$

Thus the triangle inequality holds. $\square$

A real-coefficient vector space in which a norm is defined is called a (real-coefficient) *normed space*. If an inner product is also defined in the space and the norm is derived from it, the space is called an *inner product space*.

EXAMPLE 4 The vector space $U = (\mathbb{R}^2, \mathbb{R})$ equipped with the Euclidean inner product (Definition II.4.1) is an inner product space called *Euclidean space* $\mathbb{R}^2$. Equipped with the maximum norm (Example 3), U is only a normed space, since the maximum norm cannot be derived from any inner product (Exercise 5b).

EXAMPLE 5 A very simple example of an inner product space is the one-dimensional space $(U, \mathbb{R}, \cdot)$, where $U = \mathbb{R}$. In this space $\mathbf{u} + \mathbf{v}$ is interpreted as addition of real numbers and $\alpha\mathbf{u}$ and $\langle \mathbf{u}, \mathbf{v} \rangle$ both as multiplication of real numbers. Then the norm $|\mathbf{u}|$ = the absolute value of $\mathbf{u}$, and the Cauchy–Schwarz inequality is realized as the equation $|\mathbf{u}\mathbf{v}| = |\mathbf{u}||\mathbf{v}|$. $\square$

EXERCISES

1. Show, without appealing to geometry, that the inner product axioms hold for the Euclidean inner product of $\mathbb{R}^2$.
2. Investigate whether this is an inner product of $\mathbb{R}^2$, when $\mathbf{u} = (x_1, y_1)$, $\mathbf{v} = (x_2, y_2)$ and one defines
 - a) $\langle \mathbf{u}, \mathbf{v} \rangle = x_1 y_1 + x_2 y_2$
 - b) $\langle \mathbf{u}, \mathbf{v} \rangle = 2x_1 x_2 + 5y_1 y_2$
 - c) $\langle \mathbf{u}, \mathbf{v} \rangle = x_1 x_2 - y_1 y_2$
 - d) $\langle \mathbf{u}, \mathbf{v} \rangle = (x_1 + y_1)(x_2 + y_2)$
 - e) $\langle \mathbf{u}, \mathbf{v} \rangle = (x_1 + y_1)(x_2 + y_2) + y_1 y_2$
 What form does the Cauchy–Schwarz inequality take (in the affirmative case)?
3. Let $\{\vec{a}, \vec{b}\}$ be a basis of the plane vector space $U = (V, \mathbb{R})$. Show that if the coordinates of a vector $\vec{v} \in V$ in that basis are (x, y) , then $\|\vec{v}\| = |x| + |y|$ defines a norm of U .
4. Let $\mathbf{u} = (x, y) \in \mathbb{R}^2$. Which of the following are norms of the vector space $U = (\mathbb{R}^2, \mathbb{R})$?
 - a) $\|\mathbf{u}\| = |x + y|$
 - b) $\|\mathbf{u}\| = |x| + |y|$
 - b) $\|\mathbf{u}\| = 2|x| - |y|$
 - c) $\|\mathbf{u}\| = 0.01|x| + 0.003|y|$
 - d) $\|\mathbf{u}\| = x + |y|$
 - e) $\|\mathbf{u}\| = |x| + y^2$
 - f) $\|\mathbf{u}\| = \sqrt{2x^2 + 3y^2}$
 - g) $\|\mathbf{u}\| = \sqrt{|xy|}$
5. a) Prove the triangle inequality valid for the norm of an inner product space in the form $\|\mathbf{u} + \mathbf{v}\|^2 \leq (\|\mathbf{u}\| + \|\mathbf{v}\|)^2$. b) Show that the *parallelogram law* $\|\mathbf{u} + \mathbf{v}\|^2 + \|\mathbf{u} - \mathbf{v}\|^2 = 2\|\mathbf{u}\|^2 + 2\|\mathbf{v}\|^2$ holds for the norm of an inner product space. Conclude that the maximum norm $\|\mathbf{u}\| = \|(x, y)\| = \max\{|x|, |y|\}$ of $\mathbb{R}^2$ cannot be derived from any inner product.
6. Investigate what kind of geometric deformation the unit circle $S = \{\mathbf{u} = (x, y) \in \mathbb{R}^2 \mid |\mathbf{u}| = 1\}$ of $\mathbb{R}^2$ undergoes when the distance of a point from the origin is measured, instead of by the Euclidean norm, by the following norms:

$$\text{a) } \|\mathbf{u}\| = \max\{|x|, |y|\} \quad \text{b) } \|\mathbf{u}\| = |x| + |y| \quad \text{c) } \|\mathbf{u}\| = \sqrt{4x^2 + y^2}$$

7. (*) Let $a, b, c, d \in \mathbb{R}$ and define

$$\langle \mathbf{u}, \mathbf{v} \rangle = ax_1x_2 + bx_1y_2 + cx_2y_1 + dy_1y_2,$$

where $\mathbf{u} = (x_1, y_1) \in \mathbb{R}^2$ and $\mathbf{v} = (x_2, y_2) \in \mathbb{R}^2$. Under exactly what conditions on the numbers a, b, c, d is $\langle \cdot, \cdot \rangle$ an inner product of the vector space $U = (\mathbb{R}^2, \mathbb{R})$?

II.5 The concept of a function. Trigonometric functions

This section introduces the concept of a *function*, central to mathematics, its derived counterpart the *real function*, and, as the first real functions, the *trigonometric functions*. It turns out that, understood in a sufficiently general and abstract way, functions have already appeared earlier under various 'aliases'. As for real functions, none other than the trigonometric functions are treated for now; the more general theory of real functions is presented later in Chapters IV–VI.

The concept of a function

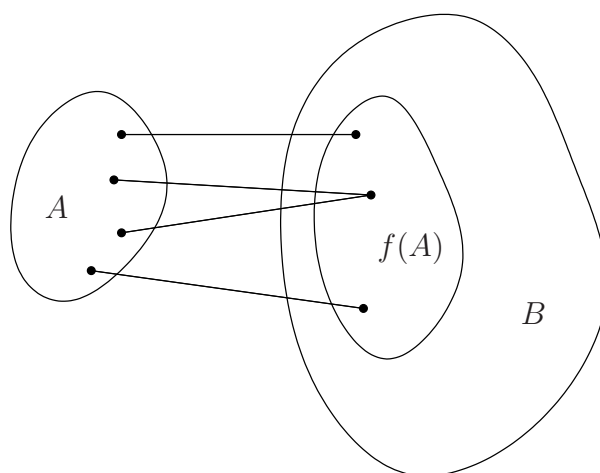
A *function*, or *map*, is most commonly understood in mathematics as a triple of the form

$$\{\text{set } A, \text{ rule } f, \text{ set } B\}.$$

By this we mean that there is some rule f that determines a unique $y \in B$ for every $x \in A$. The relationship between x and y is written $y = f(x)$ and read ' y is f of x '. One may also use the maps-to arrow and write $x \mapsto f(x)(= y)$. It is said that A is the *domain* of f (or the source set) and B its *codomain*, and one writes $f : A \rightarrow B$. The elements of the domain of a function are referred to by the generic name *variable*. If $y = f(x)$, then y is said to be the *value* of f *at* x (or at the value x of the variable); more briefly one says ' f at x ' or, often, ' f at the point x '. The subset of the codomain

$$\{y \in B \mid y = f(x) \text{ for some } x \in A\}$$

is called the *range* of f , or the *image* of A under f . This is denoted briefly by $f(A)$.



EXAMPLE 1 If x and y are real numbers and one writes $f(x, y) = x^2y - y^3$, this means (unless otherwise agreed about the domain) the function

$$\{\mathbb{R}^2, f(x, y) = x^2y - y^3, \mathbb{R}\},$$

which thus assigns to any pair of numbers $(x, y) \in \mathbb{R}^2$ the real number expressed by the computational rule. Determining the values of the function requires in this case only the multiplication and subtraction operations of the real numbers. E.g.

$$\begin{aligned} f(3, -2) &= 3^2 \cdot (-2) - (-2)^3 = -10, \\ f(2, \sqrt{2}) &= 2^2 \cdot \sqrt{2} - (\sqrt{2})^3 = 0, \\ f(\pi, e) &= \pi^2 e - e^3 = 6.74282937.. \quad \square \end{aligned}$$

Functions in set theory

Thought of very generally and abstractly, a function may be regarded, instead of as a 'rule', rather as a 'list' that assigns to each element $x \in A$ a unique element $y \in B$ (depending on x). Thought of this way, the function becomes a purely set-theoretic notion: a function is defined by any subset $F \subset A \times B$ of the set $A \times B = \{(x, y) \mid x \in A \text{ \& } y \in B\}$ (= the Cartesian product of A and B , cf. the footnote in Section II.2) that satisfies the condition

$$\forall x \in A \ [(x, y) \in F \text{ for exactly one } y \in B].$$

Indeed, such a set corresponds to a function f defined on A when one interprets

$$y = f(x) \Leftrightarrow (x, y) \in F.$$

For the concept of a function it thus ultimately makes no difference at all whether the function is known as an 'expression' (a computational rule) or merely as 'data'.^{*}

EXAMPLE 2 A sequence of real numbers $\{a_1, a_2, \dots\}$ can be interpreted as a function of type $f : \mathbb{N} \rightarrow \mathbb{R}$. Indeed, the sequence can be thought of as a 'list' $1 \mapsto a_1, 2 \mapsto a_2, \dots$, i.e. as the set

$$F = \{(1, a_1), (2, a_2), \dots\} \subset \mathbb{N} \times \mathbb{Q}. \quad \square$$

All the operations that have appeared so far are in fact functions. A function can then be conceived of as a 'calculating machine' that gives the result of an operation from the given input.

EXAMPLE 3 The addition and multiplication of real numbers are functions of type $f : \mathbb{R}^2 \mapsto \mathbb{R}$ with the computational rules

$$\text{addition: } f(x, y) = x + y, \quad \text{multiplication: } f(x, y) = xy.$$

Division is a function $f : \{(x, y) \in \mathbb{R}^2 \mid y \neq 0\} \rightarrow \mathbb{R}$ with the rule $f(x, y) = x/y$.
 $\square$

EXAMPLE 4 In the previous section, to an angle $\angle(\vec{a}, \vec{b})$ of the Euclidean plane there was associated a real number, determined by geometric means, which was denoted $\cos \angle(\vec{a}, \vec{b})$. Since the number in question is unique, we have a function $\cos : A \rightarrow \mathbb{R}$, where A = the set of all angles. The range of the function is $[-1, 1] \subset \mathbb{R}$. In the domain the angles may, if desired, be interpreted as pairs of plane vectors $(\vec{a}, \vec{b})$, where $\vec{a}, \vec{b} \neq \vec{0}$, or as pairs of unit vectors ($|\vec{a}| = |\vec{b}| = 1$).
 $\square$

The real function

The 'mother of functions' is the *real function*, i.e. a real-valued function of a real variable of the form $f : A \rightarrow \mathbb{R}$, where $A \subset \mathbb{R}$. The term 'function of one variable' is also often used in this restricted sense. A function of a real variable can conveniently be illustrated by means of its *graph*. The graph of a function $f : A \rightarrow \mathbb{R}$ on a set $B \subset A$ ($A \subset \mathbb{R}$) is the point set of the Euclidean space E^2

$$G = \{P = (x, y) \mid y = f(x) \text{ \& } x \in B\},$$

^{*}In practice 'function data' may be based e.g. on direct measurements, or it may be a computed prediction based on computer models, as in weather forecasts.

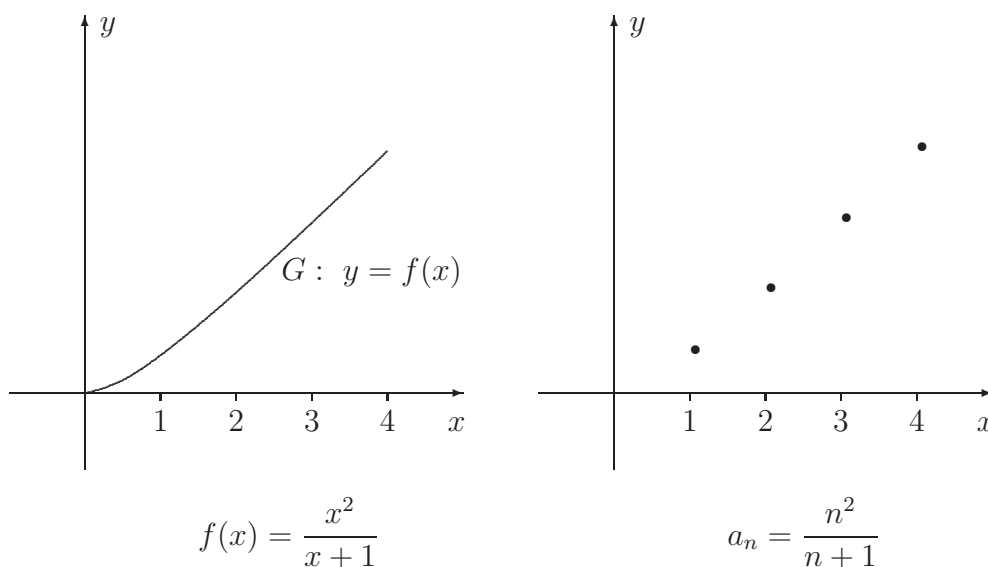
where the coordinates (x, y) of P refer to a Cartesian coordinate system (unless otherwise agreed). Note that if the domain of f is restricted to the set B , then the set-theoretic definition of the function is

$$F = \{(x, y) \in \mathbb{R}^2 \mid y = f(x) \text{ \& } x \in B\}.$$

The interpretation of a function as a 'curve' (i.e. as the graph G) thus simply geometrizes the abstract set-theoretic definition of the function.

When a sequence of real numbers is interpreted as a function $f: \mathbb{N} \rightarrow \mathbb{R}$, then the graph of f is a sequence of points of E^2 . This way of depicting a sequence has already been used above in Sections I.6 and I.10.

EXAMPLE 5 Depicted below is the function with computational rule $f(x) = x^2/(x+1)$, on the sets $B = [0, 4]$ and $B = [0, 4] \cap \mathbb{N}$. In the latter case the graph represents the initial portion of the sequence $\{a_n\} = \{n^2/(n+1), n = 1, 2, \dots\}$. $\square$



If a real function has an easily expressible computational rule, then the function is most commonly stated as the rule alone. In that case it is assumed (unless otherwise agreed) that the domain is the largest subset of $\mathbb{R}$ on which the computational rule can be applied. A single use of the computational rule, i.e. the operation $x \mapsto f(x)$, is called a *function evaluation*. Often this can be done only numerically (i.e. approximately), in which case in practice one can only compute a finite number of terms of a sequence whose limit $= f(x)$.

EXAMPLE 6 The domain of the real function $f(x) = x^2/(x^2 - 2)$ is $A = \{x \in \mathbb{R} \mid x^2 \neq 2\} = (-\infty, -\sqrt{2}) \cup (-\sqrt{2}, \sqrt{2}) \cup (\sqrt{2}, \infty)$. $\square$

EXAMPLE 7 A power series $\sum_k a_k x^k$ can be interpreted as a real function whose value at x is the sum of the series and whose domain is the interval of convergence of the series. For example, the domain of the function

$$f(x) = \sum_{k=1}^{\infty} \frac{x^k}{k}$$

is the interval $[-1, 1)$, cf. Section I.12. Unless $x = 0$, the function evaluation $x \mapsto f(x)$ must in this case be carried out numerically, e.g. by computing the partial sums of the series. $\square$

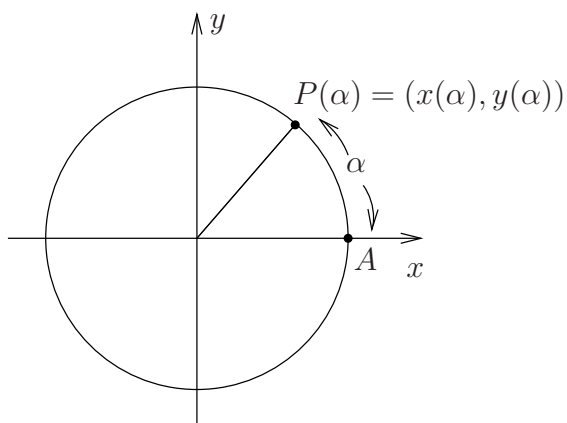
The functions cosine and sine

Consider the unit circle whose center is at the origin O of a Cartesian coordinate system and on which the point $A = (1, 0)$ has been chosen as the reference point. Starting from A , the unit circle can be traversed either in the *positive direction of rotation*, i.e. counterclockwise, or in the *negative direction of rotation*, i.e. clockwise. In the former case the measure of the *rotation angle* is taken to be the arc length (= the distance traveled along the circumference), in the latter its negative. In this way the measure of the rotation angle can take any real value, and to each measure value α there corresponds a unique point on the circumference

$$P(\alpha) = (x(\alpha), y(\alpha)), \quad \alpha \in \mathbb{R}.$$

Here P thus does not mean a single point, but the unique dependence $\alpha \mapsto P(\alpha)$, i.e. a 'point-valued' function of a real variable. Correspondingly, x and y above denote functions of the type

$$x : \mathbb{R} \rightarrow [-1, 1], \quad y : \mathbb{R} \rightarrow [-1, 1].$$



According to the earlier definition of the cosine of an angle (see Section II.3), we have

$$\cos \angle AOP(\alpha) = x(\alpha).$$

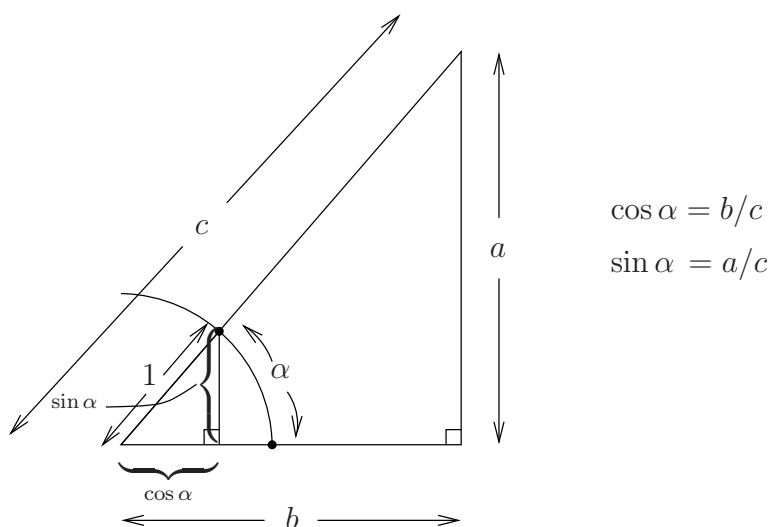
We now define the real function $\cos$, the *cosine*, by replacing here the 'angle variable' with the measure value α of the rotation angle:

$$\cos \alpha = x(\alpha), \quad \alpha \in \mathbb{R}.$$

We define correspondingly the real function $\sin$, the *sine*, by setting

$$\sin \alpha = y(\alpha), \quad \alpha \in \mathbb{R}.$$

The real functions cosine and sine (with domain $\mathbb{R}$) are the basic types of *trigonometric functions*. By the principle of similar triangles one sees that these definitions correspond to the familiar notations of trigonometry when $0 < \alpha < \pi/2$.



A few properties of the cosine and sine that follow directly from the definitions, or are easily justified by symmetry, are listed below. These hold $\forall \alpha \in \mathbb{R}$.

$$\cos(\alpha + 2\pi) = \cos \alpha, \quad \sin(\alpha + 2\pi) = \sin \alpha. \quad (1)$$

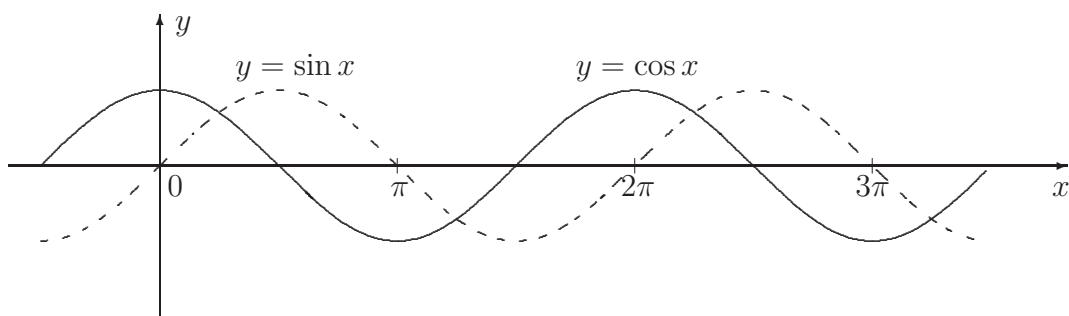
$$\cos(-\alpha) = \cos \alpha, \quad \sin(-\alpha) = -\sin \alpha. \quad (2)$$

$$\cos(\alpha + \pi) = -\cos \alpha, \quad \sin(\alpha + \pi) = -\sin \alpha. \quad (3)$$

$$\sin \alpha = \cos(\pi/2 - \alpha), \quad \cos \alpha = \sin(\pi/2 - \alpha). \quad (4)$$

$$\cos^2 \alpha + \sin^2 \alpha = 1. \quad (5)$$

Of these, property (1) (which also follows from (3)) tells us that $\cos$ and $\sin$ are *periodic* functions, with period 2π = the length of the circumference of the unit circle. Property (2) tells us that the cosine is an *even* (symmetric) function and the sine an *odd* (antisymmetric) function. Property (5) is the Pythagorean theorem.



In the trigonometric function evaluations $x \mapsto \cos x$ and $x \mapsto \sin x$ one must, except for exceptional values of x , resort to calculators. A calculator is also

needed when one wants to find out at what angle a trigonometric function takes a given value, i.e. one wants to solve α from the equation $\cos \alpha = y$ or $\sin \alpha = y$, given $y \in [-1, 1]$. A calculator generally gives an approximate value of one solution (e.g. by the command $\cos^{-1} y$ or $\arccos y$). The other solutions can then be inferred without further recourse to the calculator, since by the definitions of cosine and sine the following holds

$$\begin{aligned} \cos \alpha = \cos \beta &\Leftrightarrow \beta = \alpha + n \cdot 2\pi \vee \beta = -\alpha + n \cdot 2\pi, & n \in \mathbb{Z}, \\ \sin \alpha = \sin \beta &\Leftrightarrow \beta = \alpha + n \cdot 2\pi \vee \beta = \pi - \alpha + n \cdot 2\pi, & n \in \mathbb{Z}. \end{aligned}$$

EXAMPLE 8 For which degree values of the angle α in the interval $[0^\circ, 360^\circ]$ is $\cos 2\alpha = -2/3$ (to one decimal place)?

Solution The calculator gives for the equation $\cos x = -2/3$ the solution $x = 2.300523.. \hat{=} 131.8^\circ$, so the possible values of α are

$$\begin{cases} 2\alpha = 131.8^\circ + n \cdot 360^\circ & \Rightarrow \alpha = 65.9^\circ + n \cdot 180^\circ, & n \in \mathbb{Z}, \\ 2\alpha = -131.8^\circ + m \cdot 360^\circ & \Rightarrow \alpha = -65.9^\circ + m \cdot 180^\circ, & m \in \mathbb{Z}. \end{cases}$$

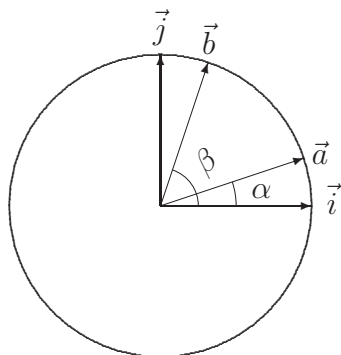
The requested values of α in increasing order ($n = 0, m = 1, n = 1, m = 2$):

$$65.9^\circ, 114.1^\circ, 245.9^\circ, 294.1^\circ. \quad \square$$

In addition to the basic properties (1)–(5), one often needs the so-called *addition formulas*

$$\begin{aligned} \cos(\alpha + \beta) &= \cos \alpha \cos \beta - \sin \alpha \sin \beta, \\ \sin(\alpha + \beta) &= \sin \alpha \cos \beta + \cos \alpha \sin \beta. \end{aligned} \tag{6}$$

These hold $\forall \alpha, \beta \in \mathbb{R}$. The formulas are most easily justified by vector calculation, see the figure.



$$\vec{a} = \cos \alpha \vec{i} + \sin \alpha \vec{j}$$

$$\vec{b} = \cos \beta \vec{i} + \sin \beta \vec{j}$$

By the figure and the definition of the dot product, we have

$$\vec{a} \cdot \vec{b} = \cos \alpha \cos \beta + \sin \alpha \sin \beta = \cos(\beta - \alpha).$$

By replacing $\alpha \rightarrow -\alpha$ and using the symmetry properties (2), the first of formulas (6) follows. The second then follows from the first together with formulas (4), (2):

$$\begin{aligned} \sin(\alpha + \beta) &= \cos\left(\frac{\pi}{2} - \alpha - \beta\right) \\ &= \cos\left(\frac{\pi}{2} - \alpha\right) \cos(-\beta) - \sin\left(\frac{\pi}{2} - \alpha\right) \sin(-\beta) \\ &= \sin \alpha \cos \beta + \cos \alpha \sin \beta. \end{aligned}$$

By choosing $\beta = \alpha$ in the above formulas, one obtains the often-needed *double-angle formulas*:

$$\begin{aligned} \cos 2\alpha &= \cos^2 \alpha - \sin^2 \alpha \\ &= 2 \cos^2 \alpha - 1 \\ &= 1 - 2 \sin^2 \alpha, \\ \sin 2\alpha &= 2 \sin \alpha \cos \alpha. \end{aligned}$$

(7)

EXAMPLE 9 By formulas (6), (7) and (5),

$$\begin{aligned} \sin 3\alpha &= \sin(\alpha + 2\alpha) \\ &= \sin \alpha \cos 2\alpha + \cos \alpha \sin 2\alpha \\ &= \sin \alpha (1 - 2 \sin^2 \alpha) + 2 \cos^2 \alpha \sin \alpha \\ &= \sin \alpha - 2 \sin^3 \alpha + 2(1 - \sin^2 \alpha) \sin \alpha \\ &= 3 \sin \alpha - 4 \sin^3 \alpha. \quad \square \end{aligned}$$

EXAMPLE 10 Compute $x = \sin \frac{\pi}{18} = \sin 10^\circ$.

Solution By the previous example, x satisfies the third-degree equation

$$3x - 4x^3 = \sin 30^\circ = \frac{1}{2}.$$

By numerical means one finds the solution $x = 0.173648177669303 \dots$ * $\square$

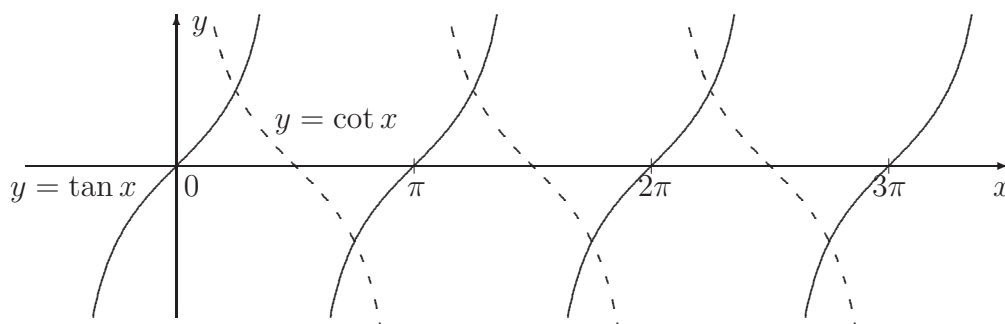
*The number x in the example is algebraic but not geometric, i.e. $x \notin \mathbb{G}$ (cf. Section II.1). The angle $\frac{\pi}{18} = 10^\circ$ therefore cannot be constructed geometrically. In general, too, the geometrically constructible angles form a rather sparse set. For example, the angle 3° can be constructed (see Exercise 18), whereas the angles 2° and 1° cannot.

Tangent and cotangent

Of the other trigonometric functions, the most important are the *tangent* ($\tan$) and the *cotangent* ($\cot$), which are defined by

$$\tan \alpha = \frac{\sin \alpha}{\cos \alpha} \quad (\alpha \in \mathbb{R}, \alpha \neq (n + \tfrac{1}{2})\pi, n \in \mathbb{Z}),$$

$$\cot \alpha = \frac{\cos \alpha}{\sin \alpha} \quad (\alpha \in \mathbb{R}, \alpha \neq n\pi, n \in \mathbb{Z}).$$



From the definition of the tangent one can infer that the equation $\tan \alpha = y$ has a solution for every $y \in \mathbb{R}$ and that the solution is unique, e.g. under the additional condition $\alpha \in (-\pi/2, \pi/2)$ or $\alpha \in [0, \pi)$. Except for exceptional values of y , finding an approximate value of such a solution requires a calculator (the command $\tan^{-1} y$ or $\arctan y$). The other solutions of the equation are obtained from the basic solution by adding multiples of π , since by its definition the tangent has the property

$$\tan \alpha = \tan \beta \quad \Leftrightarrow \quad \beta = \alpha + n \cdot \pi, \quad n \in \mathbb{Z}.$$

Directly from formula (5) follows the frequently used formula involving the tangent

$$\frac{1}{\cos^2 \alpha} = 1 + \tan^2 \alpha. \quad (8)$$

By means of the tangent one also obtains the sometimes-needed *half-angle formulas*: Starting from formulas (7) and using formula (8), one gets

$$\begin{aligned} \sin \alpha &= 2 \sin \frac{\alpha}{2} \cos \frac{\alpha}{2} = 2 \tan \frac{\alpha}{2} \cos^2 \frac{\alpha}{2} = \frac{2 \tan \frac{\alpha}{2}}{1 + \tan^2 \frac{\alpha}{2}}, \\ \cos \alpha &= 2 \cos^2 \frac{\alpha}{2} - 1 = \frac{2}{1 + \tan^2 \frac{\alpha}{2}} - 1 = \frac{1 - \tan^2 \frac{\alpha}{2}}{1 + \tan^2 \frac{\alpha}{2}}. \end{aligned}$$

Thus we obtained the formulas

$$\left. \begin{aligned} \sin \alpha &= \frac{2t}{1+t^2} \\ \cos \alpha &= \frac{1-t^2}{1+t^2} \\ \tan \alpha &= \frac{2t}{1-t^2} \end{aligned} \right\} t = \tan \frac{\alpha}{2}. \quad (9)$$

According to these, to compute the values of the trigonometric functions it suffices to carry out only one 'genuinely trigonometric' operation $\alpha \mapsto \tan \frac{\alpha}{2} = t$, after which only the arithmetic operations of the field of real numbers are needed.*

Also worth mentioning among the trigonometric functions are the functions derived from cosine and sine, the *secant* (sec) and *cosecant* (csc):

$$\begin{aligned} \sec \alpha &= \frac{1}{\cos \alpha} & (\alpha \in \mathbb{R}, \alpha \neq (n + \tfrac{1}{2})\pi, n \in \mathbb{Z}), \\ \csc \alpha &= \frac{1}{\sin \alpha} & (\alpha \in \mathbb{R}, \alpha \neq n\pi, n \in \mathbb{Z}). \end{aligned}$$

Application example: Harmonic oscillation

The combined function of sine and cosine $f(x) = A \sin x + B \cos x$ ($A, B \in \mathbb{R}$) is common in applications. Typically it is a matter of harmonic oscillation, in which some physical quantity y (e.g. an electric voltage) varies with the time variable t so that $y(t) = A \sin \omega t + B \cos \omega t$. The quantity ω (unit = 1/s) is called the angular frequency of the oscillation. One arrives at the mentioned function $f(x)$ by introducing the (dimensionless) variable $x = \omega t$.

The function under consideration is put into a clearer form by first setting

$$R = \sqrt{A^2 + B^2}.$$

*Concerning the computational formulas (1)–(9) for the trigonometric functions, it should be noted that all the operations appearing in these formulas, such as $\alpha \mapsto \sin \alpha$, $\alpha \mapsto \pi/2 - \alpha$, $(\alpha, \beta) \mapsto \alpha + \beta$ or $\alpha \mapsto \alpha/2$, can be carried out geometrically if the underlying angles are known as geometric objects. Understood in this way, formulas (1)–(9) are thus valid even if the measure of the angle were not defined at all.

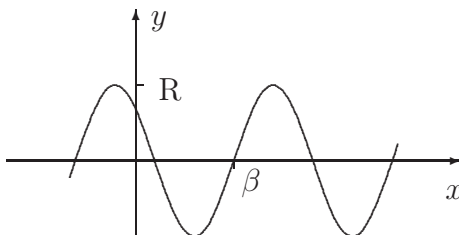
Then $(A/R)^2 + (B/R)^2 = 1$, i.e. the point $P = (A/R, B/R)$ lies on the unit circle (assuming $(A, B) \neq (0, 0)$). Consequently, for some $\alpha \in \mathbb{R}$ we have

$$\frac{A}{R} = \cos \alpha, \quad \frac{B}{R} = \sin \alpha \quad \Rightarrow \quad \tan \alpha = \frac{B}{A}.$$

Of these, the last condition determines α uniquely up to a multiple of π . Taking into account also the sign of $\sin \alpha$ or $\cos \alpha$ (from the other two conditions), one sees that α is unique up to a multiple of 2π . By the addition formula (7), $f(x)$ can now be written in the form

$$\begin{aligned} f(x) &= R(\cos \alpha \sin x + \sin \alpha \cos x) \\ &= R \sin(x + \alpha) \\ &= R \sin(x - \beta), \quad \beta = -\alpha. \end{aligned}$$

One then says that R is the *amplitude* (e.g. of the oscillation) and α (or $\beta = -\alpha$) is the *phase angle* (phase shift). The graph of the function f is thus obtained by scaling the graph of the function $\sin x$ by the amplitude R and shifting the scaled graph by the phase angle in the direction of the x -axis.



EXAMPLE 11 Write the following functions in the standard form $f(x) = R \sin(x - \beta)$:

a) $f(x) = \sin x - \sqrt{3} \cos x$ b) $f(x) = -3 \sin x - 4 \cos x$

Solution a) Here $R = 2$, so α is solved from the conditions

$$\cos \alpha = \frac{1}{2}, \quad \sin \alpha = -\frac{\sqrt{3}}{2} \quad \Rightarrow \quad \alpha = -\frac{\pi}{3} + n \cdot 2\pi, \quad n \in \mathbb{Z}.$$

Thus $f(x) = 2 \sin(x - \frac{\pi}{3})$.

b) Here $R = 5$ and $\tan \alpha = 4/3$. The calculator gives $\alpha = 0.927295..$, i.e. $\alpha \approx 53.1^\circ$, but this solution is not valid, because $\cos \alpha > 0$. We choose $\alpha = -\pi + 0.927295.. = -2.214297..$, i.e. $\alpha \approx 53.1^\circ - 180^\circ = -126.9^\circ$, for which we still have $\tan \alpha = 4/3$ and $\cos \alpha < 0$. Thus $f(x) = 5 \sin(x - \beta)$, where $\beta = 2.214297.. \hat{=} 126.9^\circ$. $\square$

EXERCISES

1. A psychiatrist's consultation can be described mathematically as a triple (psychiatrist, patient, diagnosis). In the following cases (each separately), consider under what assumptions we have a function:
 - a) psychiatrist : patients $\rightarrow$ diagnoses
 - b) patient : psychiatrists $\rightarrow$ diagnoses
 - c) psychiatrist : diagnoses $\rightarrow$ patients
 - d) patient : diagnoses $\rightarrow$ psychiatrists
2. The functions f and g are defined by the computational rules $f(x, y) = x^5 + 2x^4y - 2x^2y^3$ and $g(x, y, z) = (x + y + z)/(1 + x^2 + y^2 + z^2)$, where $x, y, z \in \mathbb{R}$. Compute $f(2, -3)$, $f(\sqrt{2}, \sqrt{2})$, $g(0, 0, 0)$ and $g(1, -2, 3)$.
3. What are the domains of the following functions ($x, y \in \mathbb{R}$)?

$$\text{a) } f(x) = x^0 \quad \text{b) } f(x) = \frac{x^2}{x} \quad \text{c) } f(x, y) = \frac{x + y}{x^2 - y^2}$$

Can these functions be expressed on their domain by some simpler computational rule?

4. What do the following real functions have in common and how do they differ?
 - a) $f(x) = \frac{x^4}{x^2}$ and $g(x) = x^2$
 - b) $f(x) = \sum_{k=1}^{\infty} \left(\frac{x}{2}\right)^k$ and $g(x) = \frac{x}{2-x}$
5. Which of the following subsets F of the set $\mathbb{R}^2 = \mathbb{R} \times \mathbb{R}$ are functions?
 - a) $F = \{(1, 2), (2, 1), (3, 3), (\pi, e), (e, \pi)\}$
 - b) $F = \{(1, 2), (2, 3), (3, 3), (3, 2), (2, 1)\}$
 - c) $F = \{(n, n^2) \mid n \in \mathbb{Z}\}$
 - d) $F = \{(n^2, n) \mid n \in \mathbb{Z}\}$
 - e) $F = \{(n^2, n) \mid n \in \mathbb{N}\}$
6. Interpret as functions (domain and range!) the operations on plane vectors: addition, scalar multiplication and the dot product.
7. a) Is there a function f whose domain $= \mathbb{R}$ and for which $f(x) = \sum_{k=0}^{\infty} x^k$ holds $\forall x \in \mathbb{R}$? If the answer is yes, define f !
 b) A predicate containing one real variable (Section I.3) can be interpreted as a function, and even as a real function. How?

8. Let $\sin \alpha = 7/25$ and $\cot \beta = -5/12$. Compute the possible values of the expression $\sin(\alpha - \beta)$.
9. The side lengths of a quadrilateral are 1, 2, 3 and 4, and the measure of one angle in degrees is 160° . Compute, to 0.1 degree accuracy, the measures of the other three angles of all quadrilaterals satisfying these conditions. Draw the figures!
10. Solve the following trigonometric equations:
 - a) $\sin 2x = \cos 7x$ b) $\tan 2x = 3 \tan x$ c) $4 \sin^2 x = \tan x$
 - d) $|\sin x + |\sin x|| = \cos x + |\cos x|$ e) $\cos 2x = \sin x + \cos x$
11. Solve the trigonometric inequality, i.e. determine the set $A \subset \mathbb{R}$ or $A \subset \mathbb{R}^2$ such that the inequality holds exactly when $x \in A$ or $(x, y) \in A$:
 - a) $\sin |x| < |\sin x|$ b) $|\sin 2x| \geq |\sin 3x|$ c) $\sin 4x > \cot x - \tan x$
 - d) $2 \sin(x - y^2) > 1$ e) $\sin(x - y) + \cos x > 0$
12. a) Derive the addition formula for the tangent $\tan(\alpha + \beta) = \frac{\tan \alpha + \tan \beta}{1 - \tan \alpha \tan \beta}$.
 b) Given $t = \cos \alpha$. Express $\cos n\alpha$ in terms of t for $n = 2, 3, 4, 5$.
 c) Derive the inverse addition formulas, expressing $\cos x \cos y$, $\sin x \sin y$ and $\cos x \sin y$ in terms of $\cos(x \pm y)$ and $\sin(x \pm y)$.
13. a) Show that

$$\sin \frac{\alpha}{2} = \pm \sqrt{\frac{1}{2}(1 - \cos \alpha)}, \quad \cos \frac{\alpha}{2} = \pm \sqrt{\frac{1}{2}(1 + \cos \alpha)}.$$

On which intervals does the sign $+$ hold in both formulas?

b) Derive the formulas

$$\tan \frac{\alpha}{2} = \frac{\sin \alpha}{1 + \cos \alpha}, \quad \cot \frac{\alpha}{2} = \frac{\sin \alpha}{1 - \cos \alpha}.$$

14. Determine the amplitude and phase angle:
 - a) $f(x) = 3 \cos x - 4 \sin x$ b) $f(x) = -4 \sin x + \cos x$
 - c) $f(x) = 76 \cos x + 57 \sin x$ d) $f(x) = \sin 2x(\sec x - 2 \csc x)$
15. The voltages of the three phases of an alternating current are

$$V_i(t) = V_0 \sin(\omega t + \varphi_i), \quad i = 1, 2, 3, \quad \varphi_2 = \varphi_1 + \frac{2\pi}{3}, \quad \varphi_3 = \varphi_1 + \frac{4\pi}{3}.$$

Determine the amplitude and phase angle of the voltage $V_1 - V_2$ between phases 1 and 2. What is the sum of the voltages of all the phases?

16. Put the function

$$f(x) = \sin x + 2 \sin\left(x + \frac{2\pi}{3}\right) + 3 \sin\left(x + \frac{4\pi}{3}\right)$$

into the form $f(x) = R \sin(x + \alpha)$.

17. (*) When speaking of functions, $f(x)$ is often read ' f at the point x '. Now let $x \in \mathbb{R}$ be a given 'point' and define the 'point function' x as follows:

$$x(f) = f(x) \quad \forall f \in A,$$

where $A = \{\text{real functions that are defined at the point } x\}$. Is such a definition of a function really possible, and if so, how should $x(f)$ be read?

18. (*) An isosceles triangle with apex angle $= 36^\circ$ can be divided into two triangles such that both sub-triangles are isosceles. Using this idea as a starting point, compute $x = \sin 3^\circ$ exactly in terms of radicals. Conclude that the angle 3° is geometrically constructible.

19. (*) Prove: $\sin \frac{\pi}{2^{n+1}} \leq \left(\frac{1}{\sqrt{2}}\right)^n$, $n = 0, 1, 2, \dots$

20. (*) Determine the smallest and largest value of the function $f(x) = \cos^2 x + 4 \sin x \cos x + 3 \sin^2 x$, as well as the minimum and maximum points, by putting the function into the form $f(x) = R \sin(2x + \alpha) + C$.

II.6 Space vectors. The cross product

In considerations of spatial geometry the classical starting point is again the *Euclidean point space*, this time called E^3 , 'E three'. The following spatial concepts are assumed known henceforth (also in later sections):

- point, (space) segment, *space triangle*
- *space line*, *space plane*, a ray in space = *space direction*
- *tetrahedron*, *parallelepiped*, *polyhedron*
- *rectangular parallelepiped*
- the *volume* of a rectangular parallelepiped, a parallelepiped and a tetrahedron

The starting point for defining the *space vector* is the basic assumption of Euclidean spatial geometry (or a consequence of the basic axioms) that the geometry on every space plane of E^3 is the same as in E^2 . In that case one can, first of all, measure the length of a space segment (on a space plane containing the segment). When a space direction is attached to the length, the space vector becomes defined as a directed segment. A vector is thus again combined information about length (= the absolute value of the vector) and direction.

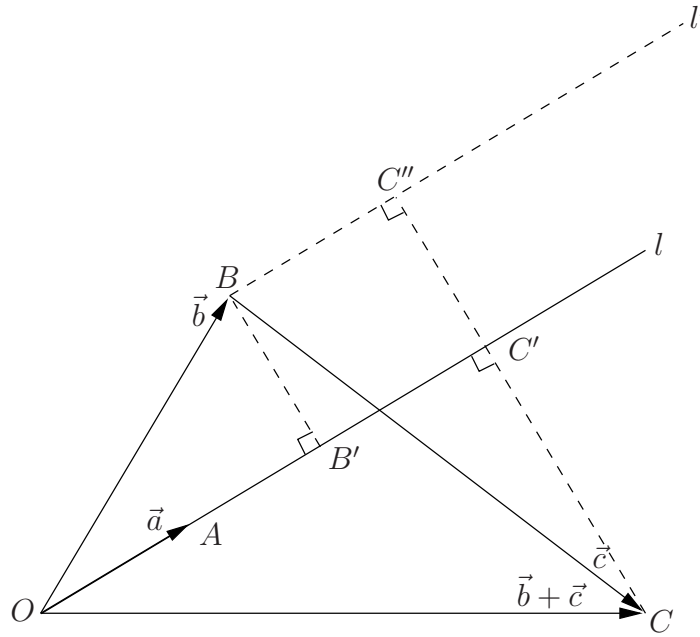
Multiplication of a vector by a scalar (a real number) is defined as in the plane, i.e. the length of the vector is scaled and the direction is either preserved or reversed, depending on the sign of the multiplier. To define the addition of vectors, let O be the reference point of E^3 = the *origin* and $\vec{a} = \overrightarrow{OA}$ and $\vec{b} = \overrightarrow{OB}$ two space vectors. Then there exists a space plane T passing through the points O, A, B (exactly one, if the points are not on the same space line). Moving to the plane T , one can define $\vec{a} + \vec{b}$ by the triangle diagram of T , i.e. by means of a space triangle. The *dot product* of space vectors is also defined as in the plane:

$$\vec{a} \cdot \vec{b} = |\vec{a}| |\vec{b}| \cos \angle(\vec{a}, \vec{b}).$$

Here the geometric construction $\angle(\vec{a}, \vec{b}) \mapsto \cos \angle(\vec{a}, \vec{b})$ works on the plane T , by assumption, as in E^2 , cf. Section II.3. The same laws hold for the dot product as in the plane. Of these, however, the distributive law

$$\vec{a} \cdot (\vec{b} + \vec{c}) = \vec{a} \cdot \vec{b} + \vec{a} \cdot \vec{c}$$

requires further justification, since three space vectors cannot in general be placed in the same plane.



In the figure, C is not in general in the same plane as the points O, A, B . The lines l and l' are, however, parallel (direction vector $= \vec{a}$), so they can be cut perpendicularly by space planes passing through the points B and C (the parallel dashed lines in the figure). The space triangles $OB'B$ and $OC'C$ formed from the intersection points and the points O, A, B (see the figure) are then right triangles. When the plane-geometric definition of the dot product is now applied on the planes containing these triangles, and taking into account that in the space rectangle $BB'C'C''$ the sides BC'' and $B'C'$ are of equal length, it follows (cf. Section II.3)

$$\begin{aligned} \vec{a} \cdot \vec{b} + \vec{a} \cdot \vec{c} &= |\vec{a}| \left(|\overrightarrow{OB'}| + |\overrightarrow{BC''}| \right) \\ &= |\vec{a}| \left(|\overrightarrow{OB'}| + |\overrightarrow{B'C'}| \right) = |\vec{a}| |\overrightarrow{OC'}| = \vec{a} \cdot (\vec{b} + \vec{c}). \end{aligned}$$

Thus the distributive law is valid.

Once the addition, scalar multiplication and dot product have been defined on the set $V = \{\text{space vectors}\}$, $(V, \mathbb{R})$ is again a vector space, and, equipped with the dot product (inner product) $\vec{u}, \vec{v} \mapsto \vec{u} \cdot \vec{v}$, an inner product space, cf. Sections II.3–II.4. If $\vec{a}$, $\vec{b}$ and $\vec{c}$ are three space vectors that differ from $\vec{0}$ and are not parallel to the same space plane, then one can infer by spatial geometry (cf. the corresponding plane-geometric reasoning in Section II.2) that every $\vec{v} \in V$ can

be expressed uniquely in the form $\vec{v} = x\vec{a} + y\vec{b} + z\vec{c}$, where $x, y, z \in \mathbb{R}$. The stated conditions on the vectors $\vec{a}, \vec{b}, \vec{c}$ can be reduced to the condition

$$x\vec{a} + y\vec{b} + z\vec{c} = \vec{0} \quad \Rightarrow \quad x = y = z = 0.$$

One then says that the vectors $\vec{a}, \vec{b}, \vec{c}$ are *linearly independent*, and that the vector system $\{\vec{a}, \vec{b}, \vec{c}\}$ is a *basis* of V . Thus the vector space formed by space vectors is 3-dimensional: $\dim V = 3$.

When every space vector $\vec{v} \in V$ is expressed as a linear combination of the given basis vectors in the form $\vec{v} = x\vec{a} + y\vec{b} + z\vec{c}$, there arises a one-to-one correspondence $V \leftrightarrow \mathbb{R}^3$, where $\mathbb{R}^3$ ('R three') is the set of triples of real numbers:

$$\mathbb{R}^3 = \{(x, y, z) \mid x \in \mathbb{R}, y \in \mathbb{R}, z \in \mathbb{R}\} = \mathbb{R} \times \mathbb{R} \times \mathbb{R}.$$

$\mathbb{R}^3$ too is a vector space, in which the vector operations are defined (cf. the operations of $\mathbb{R}^2$)

$$\begin{aligned} (x_1, y_1, z_1) + (x_2, y_2, z_2) &= (x_1 + x_2, y_1 + y_2, z_1 + z_2), \\ \lambda(x, y, z) &= (\lambda x, \lambda y, \lambda z) \quad (\lambda \in \mathbb{R}). \end{aligned}$$

An *orthonormal basis* of the vector space formed by space vectors is a system of three vectors $\{\vec{i}, \vec{j}, \vec{k}\}$ satisfying

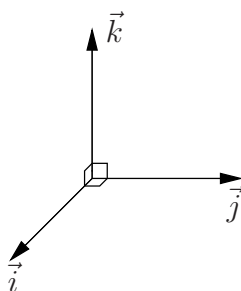
$$|\vec{i}| = |\vec{j}| = |\vec{k}| = 1, \quad \vec{i} \cdot \vec{j} = \vec{j} \cdot \vec{k} = \vec{i} \cdot \vec{k} = 0.$$

In addition, it is usually assumed that $\{\vec{i}, \vec{j}, \vec{k}\}$ is a so-called *right-handed system*.* By this is meant the (functional) correspondence associated with the right hand

$$(\text{thumb, index finger, middle finger}) \leftrightarrow (\vec{i}, \vec{j}, \vec{k})$$

or, alternatively,

$$(\text{thumb, index finger, middle finger}) \leftrightarrow (\vec{k}, \vec{i}, \vec{j}).$$



*A right-handed vector system becomes *left-handed* (and correspondingly a left-handed one right-handed) if the direction of one vector is reversed.

Under the above assumptions, the coordinate system $\{O, \vec{i}, \vec{j}, \vec{k}\}$ of E^3 is called *Cartesian*. Lines and planes on which two of the coordinates (a line) or one (a plane) takes the value 0 are called *coordinate axes* and *coordinate planes*. These are named after the coordinates that vary on the respective lines or planes, e.g. the x -axis, the xy -plane.

Corresponding to the dot product of space vectors is the *Euclidean dot product* of $\mathbb{R}^3$ (cf. Definition II.4.1)

$$\mathbf{u}_1 = (x_1, y_1, z_1), \mathbf{u}_2 = (x_2, y_2, z_2) : \quad \mathbf{u}_1 \cdot \mathbf{u}_2 = x_1x_2 + y_1y_2 + z_1z_2.$$

The axioms of the dot product hold for this (see Section II.4), so the Cauchy–Schwarz inequality (Theorem II.4.2) also holds

$$|\mathbf{u}_1 \cdot \mathbf{u}_2| \leq |\mathbf{u}_1| |\mathbf{u}_2|,$$

i.e.

$$|x_1x_2 + y_1y_2 + z_1z_2| \leq (x_1^2 + y_1^2 + z_1^2)^{1/2} (x_2^2 + y_2^2 + z_2^2)^{1/2},$$

where

$$|\mathbf{u}| = (x^2 + y^2 + z^2)^{1/2}, \quad \mathbf{u} = (x, y, z) \in \mathbb{R}^3$$

is the *Euclidean norm* of $\mathbb{R}^3$.

EXAMPLE 1 The vertices of a space triangle are $A = (-1, 1, 2)$, $B = (4, -2, 3)$ and $C = (-3, 3, 3)$. Compute the measures of the angles of the triangle in degrees (to one decimal place).

Solution If $\vec{a} = \overrightarrow{AB}$ and $\vec{b} = \overrightarrow{AC}$, then by the definition of the dot product

$$\cos \angle BAC = \frac{\vec{a} \cdot \vec{b}}{|\vec{a}| |\vec{b}|}.$$

Here $\vec{a} = 5\vec{i} - 3\vec{j} + \vec{k}$ and $\vec{b} = -2\vec{i} + 2\vec{j} + \vec{k}$, so we get

$$\cos \angle BAC = \frac{5 \cdot (-2) + (-3) \cdot 2 + 1 \cdot 1}{(5^2 + 3^2 + 1^2)^{1/2} (2^2 + 2^2 + 1^2)^{1/2}} = -\sqrt{\frac{5}{7}}.$$

Computing in the same way, we get

$$\cos \angle ABC = \sqrt{\frac{250}{259}}, \quad \cos \angle ACB = \sqrt{\frac{32}{37}}.$$

With the aid of a calculator we get the answer:

$$\angle BAC \approx 147.7^\circ, \quad \angle ABC \approx 10.7^\circ, \quad \angle ACB \approx 21.6^\circ. \quad \square$$

The cross product of vectors

For space vectors there is defined, besides the dot product, another operation of the nature of multiplication, called the *cross product* or *vector product*. If $V = \{\text{space vectors}\}$, then the cross product is a map (function) of type $V \times V \rightarrow V$, i.e. the result is a vector (hence the name vector product). The cross product is denoted $\vec{a} \times \vec{b}$, read ' a cross b '.

DEFINITION II.6.1 (Cross product) The *cross product* or *vector product* of space vectors $\vec{a}, \vec{b}$ is the vector $\vec{a} \times \vec{b}$ satisfying

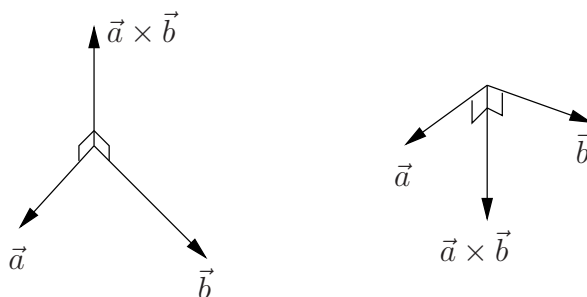
1. $|\vec{a} \times \vec{b}| = |\vec{a}||\vec{b}| \sin \angle(\vec{a}, \vec{b})$
2. $\vec{a} \times \vec{b} \perp \vec{a}$ & $\vec{a} \times \vec{b} \perp \vec{b}$
3. $\{\vec{a}, \vec{b}, \vec{a} \times \vec{b}\}$ is a right-handed system.

The sine of the angle appearing in rule (1) is interpreted as non-negative, i.e.

$$\sin \angle(\vec{a}, \vec{b}) = \sin \alpha,$$

where α is the measure of the interior angle ($0 \leq \alpha \leq \pi$). Rule (2) leaves two possibilities for the direction of the cross product, between which the choice is made by the right-hand rule expressed by rule (3)

$$(\text{thumb, index finger, middle finger}) \mapsto (\vec{a}, \vec{b}, \vec{a} \times \vec{b}).$$



From Definition II.6.1 there follows the 'skew' commutative law for the vector product

$$\boxed{(\vec{b} \times \vec{a}) = -(\vec{a} \times \vec{b}).} \quad (1)$$

In particular,

$$\vec{a} \times \vec{a} = \vec{0}$$

and, more generally,

$$\vec{a} \times \vec{b} = \vec{0} \quad \Leftrightarrow \quad \vec{a} \parallel \vec{b} \quad \vee \quad \vec{a} = \vec{0} \quad \vee \quad \vec{b} = \vec{0}.$$

(Here $\vec{a} \parallel \vec{b}$ means: $\vec{a} \uparrow\uparrow \vec{b}$ or $\vec{a} \uparrow\downarrow \vec{b}$.) From the cross product (if not otherwise) one can thus determine whether two vectors differing from the zero vector are parallel.

The vector product is not associative:

$$(\vec{a} \times \vec{b}) \times \vec{c} \neq \vec{a} \times (\vec{b} \times \vec{c}).$$

E.g. if $\vec{a} = \vec{i}$, $\vec{b} = \vec{i}$ and $\vec{c} = \vec{j}$, the left side $= \vec{0}$, but the right side $= -\vec{j}$.

If $\lambda \in \mathbb{R}$ and $\lambda \geq 0$, then from the definition of the cross product

$$(\lambda \vec{a}) \times \vec{b} = \vec{a} \times (\lambda \vec{b}) = \lambda(\vec{a} \times \vec{b}).$$

From the orientation rule it also follows that if the direction of $\vec{a}$ or $\vec{b}$ is reversed, then the direction of $(\vec{a} \times \vec{b})$ is also reversed, i.e.

$$(-\vec{a}) \times \vec{b} = \vec{a} \times (-\vec{b}) = -(\vec{a} \times \vec{b}).$$

Combining these results, one finds that the cross product and scalar multiplication can be combined by the usual distributive laws:

$$(\lambda \vec{a}) \times \vec{b} = \vec{a} \times (\lambda \vec{b}) = \lambda(\vec{a} \times \vec{b}), \quad \lambda \in \mathbb{R}. \quad (2)$$

The usual distributive laws also hold between the cross product and vector addition:

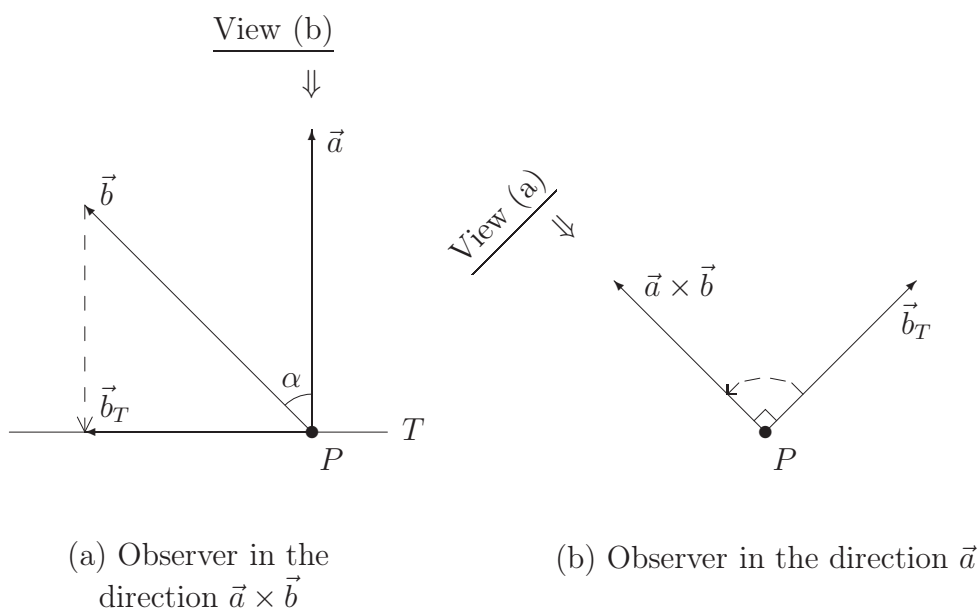
$$\begin{aligned} \vec{a} \times (\vec{b} + \vec{c}) &= \vec{a} \times \vec{b} + \vec{a} \times \vec{c}, \\ (\vec{a} + \vec{b}) \times \vec{c} &= \vec{a} \times \vec{c} + \vec{b} \times \vec{c}. \end{aligned} \quad (3)$$

These laws are, however, not obvious from the definition; a little geometric analysis is needed: Let T be the plane whose normal vector $= \vec{a}$. Then $\vec{a} \times \vec{b}$ can be understood as a three-stage geometric operation:

1. Project the vector $\vec{b}$ onto the plane T (*orthogonal projection*). Result $\vec{b}_T$.
2. Perform a *rotation* of the vector $\vec{b}_T$ in the plane T by the angle $\pi/2$ counter-clockwise as seen from the direction $\vec{a}$ (the length of the vector is preserved).

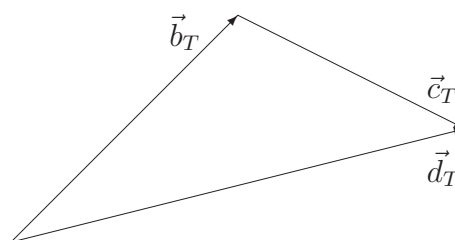
3. Scale the result of stage 2 by multiplying by the number $|\vec{a}|$.

Stages 1–2 are illustrated by two figures below. In the first figure the observer is in the direction $\vec{a} \times \vec{b}$ from the point P , in the second in the direction $\vec{a}$. The viewing direction of the second figure is also marked in the figures.



The first figure shows the projection operation $\vec{b} \mapsto \vec{b}_T$. The result of the rotation operation (stage 2) points toward the viewer. In the second figure the vector $\vec{a}$ is not visible, and of $\vec{b}$ only its projection $\vec{b}_T$ is seen. The rotation operation is visible in this figure.

Now let $\vec{b}, \vec{c}$ be space vectors. Then the addition diagram $\vec{b} + \vec{c} = \vec{d}$ is seen as in the figure below in the aforementioned observation situation (b):



The figure shows only the projection onto the plane T of the actual addition diagram (a triangle in space). The figure nonetheless shows that the projected

diagram is the addition diagram of plane vectors, i.e. $\vec{d}_T = \vec{b}_T + \vec{c}_T$, that is

$$\text{projection}(\vec{b} + \vec{c}) = \text{projection}(\vec{b}) + \text{projection}(\vec{c}).$$

On the other hand, the rotation in stage 2 amounts merely to a rotation of the entire projected addition diagram, so

$$\text{rotation}(\vec{b}_T + \vec{c}_T) = \text{rotation}(\vec{b}_T) + \text{rotation}(\vec{c}_T).$$

Combining these two results, it follows that:

$$\begin{aligned} \text{rotation}[\text{projection}(\vec{b} + \vec{c})] &= \text{rotation}[\text{projection}(\vec{b}) + \text{projection}(\vec{c})] \\ &= \text{rotation}[\text{projection}(\vec{b})] + \text{rotation}[\text{projection}(\vec{c})]. \end{aligned}$$

When both sides here are multiplied by the number $|\vec{a}|$ and one notes that

$$|\vec{a}| \text{rotation}[\text{projection}(\vec{\cdot})] = \vec{a} \times (\vec{\cdot}),$$

then the first part of the distributive law (3) has been proved. The second part follows from this together with the commutative law (1):

$$(\vec{a} + \vec{b}) \times \vec{c} = -\vec{c} \times (\vec{a} + \vec{b}) = -\vec{c} \times \vec{a} - \vec{c} \times \vec{b} = \vec{a} \times \vec{c} + \vec{b} \times \vec{c}.$$

The determinant formula for the cross product

In practice the cross product is computed in the orthonormal basis $\{\vec{i}, \vec{j}, \vec{k}\}$. Since

$$\begin{aligned} \vec{i} \times \vec{i} &= \vec{j} \times \vec{j} = \vec{k} \times \vec{k} = \vec{0}, \\ \vec{i} \times \vec{j} &= \vec{k}, \quad \vec{j} \times \vec{k} = \vec{i}, \quad \vec{k} \times \vec{i} = \vec{j}, \\ \vec{j} \times \vec{i} &= -\vec{k}, \quad \vec{k} \times \vec{j} = -\vec{i}, \quad \vec{i} \times \vec{k} = -\vec{j}, \end{aligned}$$

by rules (2)–(3) we have

$$\begin{aligned} \vec{a} \times \vec{b} &= (x_1\vec{i} + y_1\vec{j} + z_1\vec{k}) \times (x_2\vec{i} + y_2\vec{j} + z_2\vec{k}) \\ &= (y_1z_2 - y_2z_1)\vec{i} - (x_1z_2 - x_2z_1)\vec{j} + (x_1y_2 - x_2y_1)\vec{k}. \end{aligned}$$

We express this by a tabular mnemonic

$$\vec{a} \times \vec{b} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ x_1 & y_1 & z_1 \\ x_2 & y_2 & z_2 \end{vmatrix}.$$

Here the second and third rows consist of the coordinates of $\vec{a}$ and $\vec{b}$. This is a so-called *three-row determinant*, which is first expanded by the rule

$$\vec{a} \times \vec{b} = \begin{vmatrix} y_1 & z_1 \\ y_2 & z_2 \end{vmatrix} \vec{i} - \begin{vmatrix} x_1 & z_1 \\ x_2 & z_2 \end{vmatrix} \vec{j} + \begin{vmatrix} x_1 & y_1 \\ x_2 & y_2 \end{vmatrix} \vec{k},$$

where the new tabular objects are *two-row determinants*.* These are obtained from the original three-row determinant by deleting the rows and columns at whose intersection the coefficient $\vec{i}$, $\vec{j}$ or $\vec{k}$ stands — i.e. the first row together with the column containing that vector — note also the sign change in the middle term of the formula! The two-row determinants are finally expanded by the rule

$$\begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix} = a_1 b_2 - a_2 b_1.$$

EXAMPLE 2 Jos $\vec{a} = 3\vec{i} - 6\vec{j} + 9\vec{k}$, $\vec{b} = 2\vec{i} + 5\vec{j} - 3\vec{k}$, $\vec{c} = -\vec{i} + 2\vec{j} - 3\vec{k}$, niin

$$\begin{aligned} \vec{a} \times \vec{b} &= \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 3 & -6 & 9 \\ 2 & 5 & -3 \end{vmatrix} \\ &= [(-6) \cdot (-3) - 5 \cdot 9]\vec{i} - [3 \cdot (-3) - 2 \cdot 9]\vec{j} + [3 \cdot 5 - 2 \cdot (-6)]\vec{k} \\ &= -27\vec{i} - 27\vec{j} + 27\vec{k}, \\ \vec{a} \times \vec{c} &= \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 3 & -6 & 9 \\ -1 & 2 & -3 \end{vmatrix} = \vec{0}. \end{aligned}$$

The latter result means that the vectors $\vec{a}$ and $\vec{c}$ are parallel (in fact antiparallel).
□

The scalar triple product $\vec{a} \times \vec{b} \cdot \vec{c}$

The *scalar triple product* of three space vectors is defined by

$$\vec{a} \times \vec{b} \cdot \vec{c} = (\vec{a} \times \vec{b}) \cdot \vec{c}.$$

Here $\vec{a} \times \vec{b}$ must be computed first, so parentheses need not be written. Computed in component form in a Cartesian coordinate system,

$$\vec{a} \times \vec{b} \cdot \vec{c} = \vec{c} \cdot (\vec{a} \times \vec{b}) = \begin{vmatrix} x_3 & y_3 & z_3 \\ x_1 & y_1 & z_1 \\ x_2 & y_2 & z_2 \end{vmatrix},$$

*The theory of determinants is a peculiar branch of mathematics of its own, formerly much used in hand computation. From it we borrow here only the mnemonics needed in handling the cross product.

where $\vec{a} \leftrightarrow (x_1, y_1, z_1)$, $\vec{b} \leftrightarrow (x_2, y_2, z_2)$ and $\vec{c} \leftrightarrow (x_3, y_3, z_3)$. From the definition of the determinant it follows that the rows can be 'cycled' so that the first row goes to the bottom without the value of the determinant changing:

$$\begin{vmatrix} x_3 & y_3 & z_3 \\ x_1 & y_1 & z_1 \\ x_2 & y_2 & z_2 \end{vmatrix} = \begin{vmatrix} x_1 & y_1 & z_1 \\ x_2 & y_2 & z_2 \\ x_3 & y_3 & z_3 \end{vmatrix}.$$

Consequently, the 'dot-cross interchange rule' holds

$$\boxed{\vec{a} \times \vec{b} \cdot \vec{c} = \vec{a} \cdot \vec{b} \times \vec{c}.$$

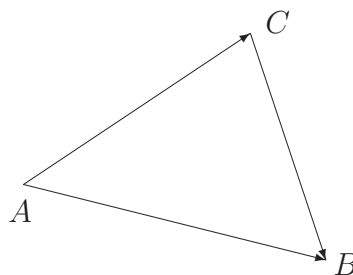
Area of a triangle. Volume of a tetrahedron

Besides physical applications, the cross product and the scalar triple product also have amusing geometric uses. For example, the area of a space triangle (also of a plane triangle) can be conveniently determined by means of the cross product if the vertices of the triangle are known in a Cartesian coordinate system. Let an example illustrate the matter.

EXAMPLE 3 The vertices of a space triangle are at the points $A = (1, 0, 0)$, $B = (1, 1, -1)$ and $C = (0, 2, 2)$. What is the area of the triangle?

Solution By the well-known area formula for a triangle and the definition of the vector product:

$$\begin{aligned} \text{area} &= \frac{1}{2} \cdot \text{base} \cdot \text{height} \\ &= \frac{1}{2} |\vec{AB}| |\vec{AC}| \sin \angle BAC \\ &= \frac{1}{2} |\vec{AB} \times \vec{AC}|. \end{aligned}$$



Here $\vec{AB} = \vec{j} - \vec{k}$, $\vec{AC} = -\vec{i} + 2\vec{j} + 2\vec{k}$, so

$$\begin{aligned} \vec{AB} \times \vec{AC} &= \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 0 & 1 & -1 \\ -1 & 2 & 2 \end{vmatrix} = \begin{vmatrix} 1 & -1 \\ 2 & 2 \end{vmatrix} \vec{i} - \begin{vmatrix} 0 & -1 \\ -1 & 2 \end{vmatrix} \vec{j} + \begin{vmatrix} 0 & 1 \\ -1 & 2 \end{vmatrix} \vec{k} \\ &= (1 \cdot 2 + 1 \cdot 2) \vec{i} - (0 \cdot 2 - 1 \cdot 1) \vec{j} + (0 \cdot 2 + 1 \cdot 1) \vec{k} \\ &= 4\vec{i} + \vec{j} + \vec{k}. \end{aligned}$$

Thus the area $= \frac{1}{2}\sqrt{4^2 + 1^2 + 1^2} = \frac{3}{2}\sqrt{2}$. $\square$

The three-dimensional counterpart of a triangle is the tetrahedron. Its volume is conveniently computed by means of the scalar triple product when the vertices are known in a Cartesian coordinate system. Let an example again illustrate the matter.

EXAMPLE 4 The vertices of the tetrahedron K are the points $O = (0, 0, 0)$, $A = (1, 0, 0)$, $B = (1, 1, -1)$ and $C = (0, 2, 2)$. Determine the volume V of the tetrahedron.

Solution The well-known formula for the volume of a tetrahedron is

$$V = \frac{1}{3} \cdot \text{area of the base} \cdot \text{height}.$$

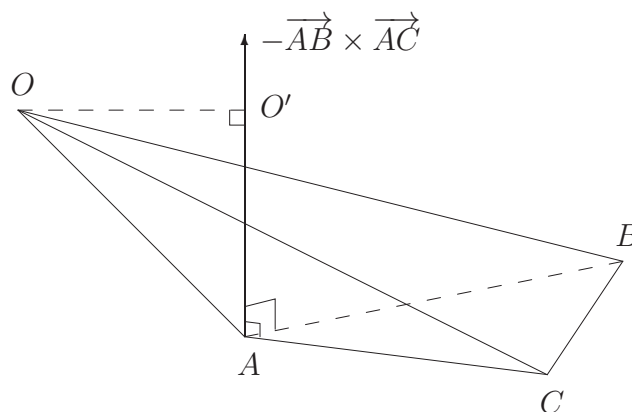
If the triangle ABC is taken here as the base, then

$$V = \frac{1}{6} |\vec{AO} \cdot \vec{AB} \times \vec{AC}|,$$

since

$$\text{area of the base} = \frac{1}{2} |\vec{AB} \times \vec{AC}|, \quad \text{height} = |\vec{AO'}| = |\vec{AO}| |\cos \angle OAO'|,$$

where $\vec{AO'}$ is the orthogonal projection of the vector $\vec{AO}$ onto the direction of the vector $\vec{AB} \times \vec{AC}$, see the figure.



Since

$$\vec{AO} \cdot \vec{AB} \times \vec{AC} = \begin{vmatrix} -1 & 0 & 0 \\ 0 & 1 & -1 \\ -1 & 2 & 2 \end{vmatrix} = (-1) \cdot \begin{vmatrix} 1 & -1 \\ 2 & 2 \end{vmatrix} = -4,$$

we have $V = \frac{1}{6} \cdot |-4| = \frac{2}{3}$ $\square$

Area of a parallelogram. Volume of a parallelepiped

A parallelogram is a plane quadrilateral whose opposite sides are parallel. In space the three-dimensional counterpart of a parallelogram is the parallelepiped, i.e. a 6-faced solid whose faces are parallelograms (on space planes). A special case of it is the rectangular parallelepiped (its faces rectangles). If the segments OA , OB and OC are three edges of a parallelepiped, then the vectors

$$\vec{a} = \overrightarrow{OA}, \quad \vec{b} = \overrightarrow{OB}, \quad \vec{c} = \overrightarrow{OC}$$

span that parallelepiped. Correspondingly, the vectors $\vec{a}, \vec{b}$, likewise the vectors $\vec{a}, \vec{c}$ and $\vec{b}, \vec{c}$, span a parallelogram (on a space plane). The formula for the volume of a parallelepiped is: $V = \text{area of the base} \cdot \text{height}$. Here the base is a parallelogram whose area is obtained by means of the cross product (area = base $\cdot$ height), so the volume of the parallelepiped is obtained by means of the scalar triple product, as in the case of the tetrahedron. The formulas, when the spanning vectors are known:

$$\begin{aligned} \text{Area of the parallelogram} &= |\vec{a} \times \vec{b}|. \\ \text{Volume of the parallelepiped} &= |\vec{a} \times \vec{b} \cdot \vec{c}|. \end{aligned}$$

The vector triple product $\vec{a} \times (\vec{b} \times \vec{c})$

If T is the plane whose direction vectors are $\vec{b}, \vec{c}$ (assuming $\vec{b} \nparallel \vec{c}$), then the vector $\vec{b} \times \vec{c}$ is the normal vector of the plane. Consequently $\vec{a} \times (\vec{b} \times \vec{c})$ is again parallel to the plane, i.e.

$$\vec{a} \times (\vec{b} \times \vec{c}) = \lambda \vec{b} + \mu \vec{c}$$

for some $\lambda, \mu \in \mathbb{R}$. On the other hand, this vector is also perpendicular to $\vec{a}$, i.e.

$$\vec{a} \cdot (\lambda \vec{b} + \mu \vec{c}) = \lambda(\vec{a} \cdot \vec{b}) + \mu(\vec{a} \cdot \vec{c}) = 0.$$

This holds exactly when $\lambda = \gamma \vec{a} \cdot \vec{c}$ and $\mu = -\gamma \vec{a} \cdot \vec{b}$ for some $\gamma \in \mathbb{R}$, so it has been inferred that

$$\vec{a} \times (\vec{b} \times \vec{c}) = \gamma [(\vec{a} \cdot \vec{c})\vec{b} - (\vec{a} \cdot \vec{b})\vec{c}].$$

In the case $\vec{a} = \vec{b} = \vec{i}$, $\vec{c} = \vec{j}$ this holds when $\gamma = 1$, so we conclude that the expansion formula holds for the vector triple product*

$$\vec{a} \times (\vec{b} \times \vec{c}) = (\vec{a} \cdot \vec{c})\vec{b} - (\vec{a} \cdot \vec{b})\vec{c}.$$

*In a more careful justification of the expansion rule one would also have to show that γ does not depend on the vectors $\vec{a}, \vec{b}, \vec{c}$. The justification is omitted here, since the expansion rule can also be verified directly from the component form of the vectors.

EXERCISES

- The coordinates of the vector $\vec{u}$ in the basis $\{\vec{i}, \vec{j}, \vec{k}\}$ are (x, y, z) . What are the coordinates of $\vec{u}$ in the basis $\{\vec{i} + \vec{j}, \vec{j} + \vec{k}, \vec{i} + \vec{k}\}$?
- Compute, in degrees, the angles $\angle(\vec{a}, \vec{b})$ formed by the following space vectors:
 a) $\vec{a} = \vec{i}$, $\vec{b} = \vec{i} + \vec{j} + \vec{k}$ b) $\vec{a} = \vec{i} + \vec{j}$, $\vec{b} = \vec{j} + \vec{k}$
- a) Of the space vectors $\vec{a}, \vec{b}, \vec{c}$ it is known that $|\vec{a}| = |\vec{b}| = 1$, $|\vec{c}| = 2$, $\vec{a} \perp \vec{b}$ and $\angle(\vec{a}, \vec{c}) = \angle(\vec{b}, \vec{c}) = 60^\circ$. What is the length of the vector $\vec{u} = \vec{a} + \vec{b} + \vec{c}$?

 b) Determine the unit vectors that form equal angles with the vectors $\vec{k}$, $\vec{j} + \vec{k}$ and $\vec{i} + \vec{j} + \vec{k}$.
 c) Of the segment OP ($O = \text{origin}$) it is known that its length is 5 and that the segment forms an angle $\alpha = 32^\circ$ with the positive x -axis and an angle $\beta = 73^\circ$ with the positive y -axis. Compute the coordinates of P .
- a) The edge vectors emanating from a vertex of a regular tetrahedron are $\vec{a}, \vec{b}$ and $\vec{c}$. Compute $\cos \angle(\vec{a}, \vec{b} + \vec{c})$ and $\cos \angle(\vec{a} + \vec{b} + 2\vec{c}, 2\vec{a} - \vec{b})$.
 b) An A4-sized sheet of paper lying on a table is a rectangle $ABCD$ whose ratio of side lengths is $|AB|/|BC| = \sqrt{2}$. The sheet is folded along the diagonal AC so that the triangular part ABC remains on the table and the part ACD turns into a triangle ACD' on a plane perpendicular to the surface of the table. Compute $\cos \angle D'AB$.
- a) Determine all space vectors $\vec{u}$ satisfying the conditions (1) $|\vec{u}| = 1$, (2) $\vec{u} = \lambda(\vec{i} + \vec{j}) + \mu(\vec{j} + \vec{k})$ for some $(\lambda, \mu) \in \mathbb{R}^2$ and (3) $\vec{u} \perp \vec{i} - 2\vec{j} - \vec{k}$.
 b) Decompose the vector $\vec{u} = \vec{i} + 2\vec{j} - 3\vec{k}$ into three components, one of which is parallel to the vector $\vec{a} = 2\vec{i} + \vec{j} + \vec{k}$, another parallel to the vector $\vec{b} = 2\vec{i} + 3\vec{j} - 3\vec{k}$, and the third perpendicular to the vectors $\vec{a}$ and $\vec{b}$. What is the smallest value of $\sin \angle(\vec{u}, \vec{v})$ when $\vec{v} = \lambda\vec{a} + \mu\vec{b}$, $(\lambda, \mu) \in \mathbb{R}^2$ ($\vec{v} \neq \vec{0}$)?

 c) Form an orthonormal, right-handed basis of space vectors $\{\vec{a}, \vec{b}, \vec{c}\}$ such that $\vec{a}$ is parallel to the vector $\vec{i} + \vec{j} + \vec{k}$ and $\vec{b}$ is parallel to the xy -plane. Is the basis uniquely determined by these conditions?
- Place a regular octahedron (= an eight-faced polyhedron whose faces are equilateral triangles) in a convenient position in the coordinate system so

that one vertex is at the origin. Compute the vector representations of the edges starting from this vertex and, using these, the normal vectors of the faces. Using these, further compute the *dihedral angle* between adjacent faces, i.e. the angle that the faces form as seen from the direction of the common edge.

7. Show: $\vec{a} + \vec{b} + \vec{c} = \vec{0} \Rightarrow \vec{a} \times \vec{b} = \vec{b} \times \vec{c} = \vec{c} \times \vec{a}$.
8. a) The points $(-1, -2, 4)$, $(5, -1, 0)$, $(2, -3, 6)$ and $(1, -1, 1)$ are the vertices of both a tetrahedron and a parallelepiped. Compute the volume and surface area of each.
 b) Two vertices of a plane triangle are at the points $(1, -2)$ and $(3, 3)$. In what point set of E^2 must the third vertex lie exactly, so that the area of the triangle = 10? Figure!
9. Deduce that the space vectors $\vec{a}$, $\vec{b}$, $\vec{c}$ are linearly dependent exactly when the volume of the parallelepiped spanned by the vectors = 0. Using this criterion, investigate whether the following vector systems are linearly independent:
 a) $\{\vec{i} - \vec{j} + \vec{k}, 3\vec{i} + 2\vec{j} + \vec{k}, \vec{i} + \vec{j} - 5\vec{k}\}$ b) $\{\vec{i} - 2\vec{k}, 2\vec{i} - \vec{j} + 3\vec{k}, 5\vec{i} - 2\vec{j} + 4\vec{k}\}$
10. Let $\vec{n}$ be a unit vector in space. We wish to express a space vector $\vec{F}$ in the form $\vec{F} = \vec{F}_1 + \vec{F}_2$, where $\vec{F}_1 \parallel \vec{n}$ and $\vec{F}_2 \perp \vec{n}$. Show that $\vec{F}_1 = (\vec{F} \cdot \vec{n})\vec{n}$ and $\vec{F}_2 = -\vec{n} \times (\vec{n} \times \vec{F})$.
11. Prove:
 a) $|\vec{a} \times \vec{b}|^2 = |\vec{a}|^2|\vec{b}|^2 - (\vec{a} \cdot \vec{b})^2$.
 b) $|\vec{a} \times (\vec{a} \times \vec{b})|^2 = |\vec{a}|^2[|\vec{a}|^2|\vec{b}|^2 - (\vec{a} \cdot \vec{b})^2]$.
 c) $(\vec{a} \times \vec{b}) \times (\vec{b} \times \vec{c}) \cdot (\vec{c} \times \vec{a}) = (\vec{a} \times \vec{b} \cdot \vec{c})^2$.
12. a) The vectors $\vec{a}$, $\vec{b}$, $\vec{c}$ are linearly independent. Simplify the expression $[(\vec{a} \times \vec{b}) \times (\vec{b} \times \vec{c})] \times (\vec{c} \times \vec{a})$. Under what condition is the expression = $\vec{0}$?
 b) Determine the length of the vector $(\vec{a} \times (\vec{a} \times (\vec{a} \times (\vec{a} \times (\vec{a} \times (\vec{a} \times \vec{b}))))))$ when $|\vec{a}| = 3$, $|\vec{b}| = 1$ and $\vec{a} \cdot \vec{b} = -2$.
 c) Determine the vectors $\vec{u}$ satisfying $\vec{u} \cdot \vec{j} = 0$ and $\vec{u} \times (\vec{k} \times \vec{u}) = \vec{k}$.
13. (*) a) Show that the medians of a tetrahedron, i.e. the segments joining a vertex to the centroid of the opposite face, intersect at a single point (= the centroid of the tetrahedron). In what ratio are the medians divided at the intersection point?
 c) Give two examples of a tetrahedron all four of whose altitude segments have a common point.

14. (*) The points $A = (1, 1, 4)$, $B = (1, -1, 3)$, $C = (-1, -1, 2)$ and $D = (0, -2, 2)$ lie on a certain space plane T . Investigate what kind of figure arises in that plane when the points are joined in the stated order by segments into a closed polygonal line. Is it a quadrilateral? Choose a viewing angle (coordinate system) of your choice in the plane T and draw the figure!
15. (*) The points $(-3, -1, 4)$, $(0, -1, -2)$, $(2, 5, 1)$, $(3, 2, 7)$ and $(5, 1, -2)$ are the vertices of a 5-faced solid K . Compute the volume and surface area of K .

II.7 Geometry of lines, planes and surfaces

When geometric problems are solved by algebraic means using a (usually Cartesian) coordinate system, one speaks of *analytic geometry*. Vectors, with their dot and cross products, provide a convenient tool for many problems.

If O is the origin of a Euclidean space (E^2 or E^3) and P is a point of that space, then the vector $\overrightarrow{OP}$ is henceforth called the *position vector* of the point P and is denoted

$$\vec{r} = \overrightarrow{OP} \quad (\text{position vector}).$$

The correspondence notation is also used

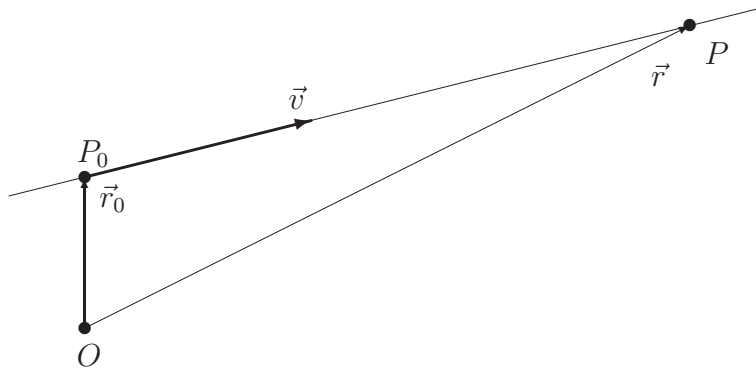
$$\vec{r} \hat{=} P \Leftrightarrow \vec{r} = \overrightarrow{OP}.$$

The line

If $P_0 = (x_0, y_0) \hat{=} \vec{r}_0 = x_0\vec{i} + y_0\vec{j}$ is a point of the Euclidean plane and $\vec{v} = \alpha\vec{i} + \beta\vec{j}$ is a plane vector, $\vec{v} \neq \vec{0}$, then the point set

$$S = \{P \in E^2 \mid P \hat{=} \vec{r} = \vec{r}_0 + t\vec{v} \text{ for some } t \in \mathbb{R}\}$$

is a line whose *direction vector* $= \vec{v}$ and which passes through the point P_0 .



It is said that the equation

$$\vec{r} = \vec{r}_0 + t\vec{v} \tag{1}$$

is the *parametric equation* or *parametrization* of the line S , and that t is the *parameter*. More generally, by these terms one means that every point of the point

set under consideration (here the line) is obtained from the given representation for some value of the parameter. In this case each point of the line corresponds to a unique value of the parameter t and conversely, i.e. the parametrization (1) creates a one-to-one correspondence $S \leftrightarrow \mathbb{R}$.

Writing $\vec{r} = x\vec{i} + y\vec{j}$, one obtains for equation (1) the *coordinate form*

$$\begin{cases} x = x_0 + \alpha t, \\ y = y_0 + \beta t. \end{cases} \quad (2)$$

Eliminating t , there follows the relation between x and y

$$ax + by + c = 0, \quad (3)$$

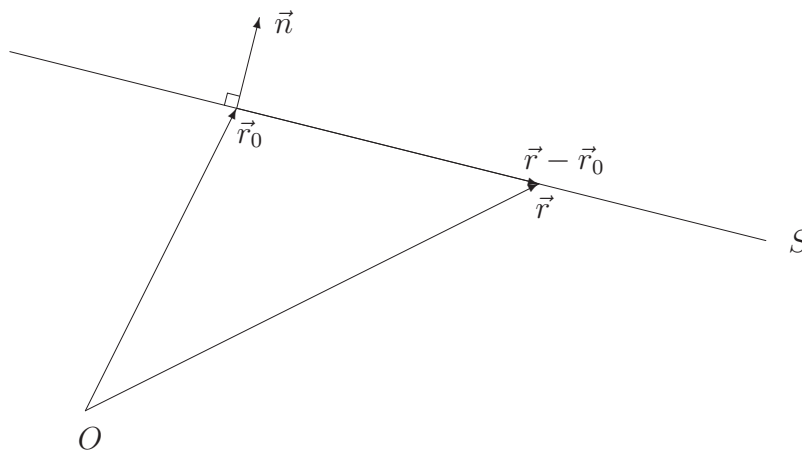
where $a = \beta$, $b = -\alpha$ and $c = -\beta x_0 + \alpha y_0$. This is the *standard form* of the equation of the line (the non-parametric form). The standard-form equation (3) can also be expressed by means of vectors, for when one sets

$$\vec{n} = a\vec{i} + b\vec{j} \quad (= \beta\vec{i} - \alpha\vec{j}),$$

the equation is the same as

$$\vec{n} \cdot (\vec{r} - \vec{r}_0) = 0. \quad (4)$$

Here $\vec{n}$ is the *normal vector* of S , i.e. a vector perpendicular to the direction vector $\vec{v}$ of S (and different from $\vec{0}$). By means of the normal vector the equation of the line can thus be written directly as the condition $\vec{r} - \vec{r}_0 \perp \vec{n}$, as is also easy to infer geometrically.



EXAMPLE 1 From the line's equation $3x - 2y + 5 = 0$ one sees at once that the normal vector of the line is $\vec{n} = 3\vec{i} - 2\vec{j}$. Thus one direction vector of the line is

$\vec{v} = 2\vec{i} + 3\vec{j}$. Since, for example, $P_0 = (1, 4)$ is on the line, one obtains for the line a parametrized representation (1), where

$$\vec{r}_0 = \vec{i} + 4\vec{j}, \quad \vec{v} = 2\vec{i} + 3\vec{j}.$$

The coordinate form (2) of the parametrization is in this case

$$\begin{cases} x = 1 + 2t, \\ y = 4 + 3t. \end{cases}$$

The mentioned point P_0 thus corresponds here to the parameter value $t = 0$.

□

The parametrization (1) works also for a space line. If $\vec{r}_0 = x_0\vec{i} + y_0\vec{j} + z_0\vec{k}$ and $\vec{v} = \alpha\vec{i} + \beta\vec{j} + \gamma\vec{k}$, then the coordinate-form equations of the line are

$$\begin{cases} x = x_0 + \alpha t, \\ y = y_0 + \beta t, \\ z = z_0 + \gamma t. \end{cases} \quad (5)$$

Eliminating t , two non-parametric equations remain.

EXAMPLE 2 A line passes through the points $A = (1, 1, 1)$ and $B = (1, -2, 3)$. What is the equation of the line?

Solution One direction vector of the line is

$$\vec{v} = \overrightarrow{AB} = -3\vec{j} + 2\vec{k}$$

and thus one parametric representation is of the form (1), where, for example,

$$\vec{r}_0 = \overrightarrow{OA} = \vec{i} + \vec{j} + \vec{k}.$$

In coordinate form

$$\begin{cases} x = 1, \\ y = 1 - 3t, \\ z = 1 + 2t. \end{cases}$$

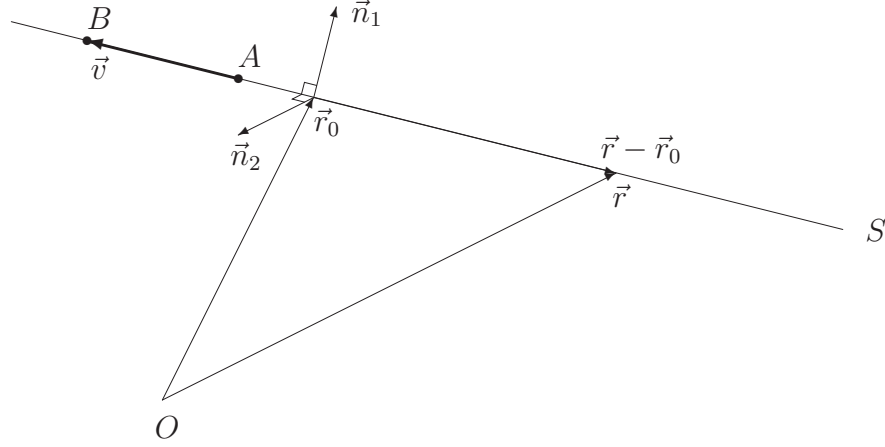
Eliminating t , the non-parametric equations of the line become

$$\begin{cases} x &= 1, \\ 2y + 3z &= 5. \end{cases}$$

These can also be written in the form

$$\begin{cases} \vec{n}_1 \cdot (\vec{r} - \vec{r}_0) = 0, \\ \vec{n}_2 \cdot (\vec{r} - \vec{r}_0) = 0, \end{cases}$$

where $\vec{n}_1 = \vec{i}$ and $\vec{n}_2 = 2\vec{j} + 3\vec{k}$ are normal vectors of the line. This corresponds to the equation (4) of a plane line. $\square$



EXAMPLE 3 Fighter A takes off from an airfield at the point $(5, 7, 0)$ (unit=km) in the direction $-\vec{i} - 2\vec{j} + \vec{k}$, and fighter B takes off from another airfield at the point $(1, -9, 0)$ in the direction $-\vec{i} + \vec{j} + 2\vec{k}$. How close to each other can the aircraft come (the worst possible scenario)?

Solution This is a matter of the so-called two-lines problem, in which the shortest distance d between the lines must be found. The lines are

$$\begin{aligned} S_1 : \vec{r} &= 5\vec{i} + 7\vec{j} + t(-\vec{i} - 2\vec{j} + \vec{k}) = \vec{r}_1 + t\vec{v}_1, \\ S_2 : \vec{r} &= \vec{i} - 9\vec{j} + s(-\vec{i} + \vec{j} + 2\vec{k}) = \vec{r}_2 + s\vec{v}_2. \end{aligned}$$

The shortest distance is $d = |\overrightarrow{PQ}|$, where $P \in S_1$, $Q \in S_2$ and the vector $\overrightarrow{PQ}$ is perpendicular to both lines, i.e.

$$\overrightarrow{PQ} \cdot \vec{v}_1 = \overrightarrow{PQ} \cdot \vec{v}_2 = 0.$$

When $P \hat{=} \vec{r}(t)$ and $Q \hat{=} \vec{r}(s)$ here are expressed according to the parametrizations, then

$$\begin{aligned} \overrightarrow{PQ} &= \vec{r}_2 - \vec{r}_1 + s\vec{v}_2 - t\vec{v}_1 \\ &= (-4 + t - s)\vec{i} + (-16 + 2t + s)\vec{j} + (-t + 2s)\vec{k}, \end{aligned}$$

so we obtain the system of equations

$$\begin{cases} -1 \cdot (-4 + t - s) - 2 \cdot (-16 + 2t + s) + 1 \cdot (-t + 2s) = 0 \\ -1 \cdot (-4 + t - s) + 1 \cdot (-16 + 2t + s) + 2 \cdot (-t + 2s) = 0 \end{cases}$$

$$\Leftrightarrow \begin{cases} 6t - s = 36 \\ t - 6s = -12 \end{cases}$$

Solving for t, s , we get

$$t = \frac{228}{35}, \quad s = \frac{108}{35}.$$

For these values,

$$\overrightarrow{PQ} = \frac{1}{35}(-20\vec{i} + 4\vec{j} - 12\vec{k}).$$

Since the coefficient of the component $\vec{k}$ is negative, aircraft A flies higher at the encounter. The distance between the aircraft is thus, according to the worst scenario,

$$d = |\overrightarrow{PQ}| = 4/\sqrt{35} \approx 0.676 = 676 \text{ m},$$

and the points P and Q are obtained, separately as well, from the parametric representations of the lines:

$$\begin{aligned} S_1 : \vec{r} \left(t = \frac{228}{35} \right) &\hat{=} P \approx (-1.514, -6.029, 6.514), \\ S_2 : \vec{r} \left(s = \frac{108}{35} \right) &\hat{=} Q \approx (-2.086, -5.914, 6.171). \quad \square \end{aligned}$$

The plane

Consider a space plane T passing through the points $P_0 = (x_0, y_0, z_0)$, $P_1 = (x_1, y_1, z_1)$ and $P_2 = (x_2, y_2, z_2)$. Assume that the points are not on the same line, so that the vectors

$$\begin{aligned} \vec{v}_1 &= \overrightarrow{P_0P_1} = \alpha_1\vec{i} + \beta_1\vec{j} + \gamma_1\vec{k}, \\ \vec{v}_2 &= \overrightarrow{P_0P_2} = \alpha_2\vec{i} + \beta_2\vec{j} + \gamma_2\vec{k} \end{aligned}$$

are linearly independent. By means of these, the *direction vectors* of the plane T , one obtains for the plane a parametrized representation

$$\vec{r} = \vec{r}_0 + t_1\vec{v}_1 + t_2\vec{v}_2, \quad t_1, t_2 \in \mathbb{R}. \quad (6)$$

Here there are thus two real parameters. If the parameter is to be seen as 'a single one', then the parameter must be interpreted as $\mathbf{t} = (t_1, t_2) \in \mathbb{R}^2$. The parametrization (6) indeed creates a one-to-one correspondence $T \leftrightarrow \mathbb{R}^2$, and $\{P_0, \vec{v}_1, \vec{v}_2\}$ can be interpreted as a coordinate system of T , cf. Section II.2.

The coordinate form of equation (6) is

$$\begin{cases} x &= x_0 + \alpha_1 t_1 + \alpha_2 t_2, \\ y &= y_0 + \beta_1 t_1 + \beta_2 t_2, \\ z &= z_0 + \gamma_1 t_1 + \gamma_2 t_2. \end{cases}$$

Eliminating the parameters is, however, easier from the vector form (6), upon introducing the vector

$$\vec{n} = \vec{v}_1 \times \vec{v}_2.$$

This is perpendicular to both vectors $\vec{v}_1, \vec{v}_2$, i.e. it is the *normal vector* of the plane. Multiplying equation (6) scalarly by the vector $\vec{n}$ and noting that $\vec{n} \cdot \vec{v}_1 = \vec{n} \cdot \vec{v}_2 = 0$, there follows for the plane the equation

$$\vec{n} \cdot (\vec{r} - \vec{r}_0) = 0, \quad (7)$$

which is thus of exactly the same form as the equation (4) of a plane line. If

$$\vec{n} = a\vec{i} + b\vec{j} + c\vec{k}, \quad d = -\vec{n} \cdot \vec{r}_0,$$

then (7) is again the same as

$$ax + by + cz + d = 0. \quad (8)$$

This is the *standard form* of the equation of the plane. From the standard form one thus sees directly what the normal vector of the plane is.*

EXAMPLE 4 The plane T_1 passes through the points $A = (1, 1, 2)$, $B = (0, 3, -1)$ and $C = (-2, -2, -3)$. The plane T_2 passes through the point $(-1, -1, -1)$ and its normal vector is $\vec{n}_2 = \vec{i} - \vec{j} - \vec{k}$. Determine the equations of the planes in standard form (8) and the parametric representation of the intersection line of the planes in the forms (1) and (5).

Solution The equation of the plane T_2 is

$$\vec{n}_2 \cdot (\vec{r} + \vec{i} + \vec{j} + \vec{k}) = 0,$$

*A *normal* of the plane is any space line whose direction vector = the normal vector of the plane. The term is often used also when speaking of the normal vector. E.g.: 'The unit vector $\vec{n}$ is a normal of the plane.'

i.e.

$$T_2 : x - y - z - 1 = 0.$$

The normal vector of the plane T_1 is

$$\vec{n}_1 = \overrightarrow{AB} \times \overrightarrow{AC} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ -1 & 2 & -3 \\ -3 & -3 & -5 \end{vmatrix} = -19\vec{i} + 4\vec{j} + 9\vec{k},$$

so choosing for example $\vec{r}_0 \hat{=} A$, the equation of the plane T_1 becomes

$$T_1 : \vec{n}_1 \cdot (\vec{r} - \vec{i} - \vec{j} - 2\vec{k}) = 0 \quad \Leftrightarrow \quad -19x + 4y + 9z - 3 = 0.$$

The direction vector of the intersection line S is

$$\vec{v} = \vec{n}_1 \times \vec{n}_2 = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ -19 & 4 & 9 \\ 1 & -1 & -1 \end{vmatrix} = 5\vec{i} - 10\vec{j} + 15\vec{k}.$$

The vector $\frac{1}{5}\vec{v} = \vec{i} - 2\vec{j} + 3\vec{k}$ also serves as a direction vector. One more point $P_0 \hat{=} \vec{r}_0$ of the line is still needed. This is found e.g. as the intersection point of the planes T_1, T_2 and a (suitably chosen) third plane T_3 . We choose the yz -plane as the third plane

$$T_3 : x = 0,$$

whereupon $P_0 = (x_0, y_0, z_0)$ is solved (if it is solvable) from the system of equations

$$\begin{cases} -19x_0 + 4y_0 + 9z_0 &= 3 \\ x_0 - y_0 - z_0 &= 1 \\ x_0 &= 0 \end{cases}$$

The solution is $P_0 = (0, -\frac{12}{5}, \frac{7}{5})$, and the parametric representation of the line S is accordingly

$$\vec{r} = \frac{1}{5}(-12\vec{j} + 7\vec{k}) + t(\vec{i} - 2\vec{j} + 3\vec{k}),$$

i.e. in coordinate form

$$S : \begin{cases} x &= t, \\ y &= -\frac{12}{5} - 2t, \\ z &= \frac{7}{5} + 3t. \end{cases} \quad \square$$

EXAMPLE 5 Determine the distance of a given point $P = (x_0, y_0, z_0)$ from the plane

$$T : ax + by + cz + d = 0.$$

Solution If Q is the point on the plane T nearest to the point P , then evidently for some $t \in \mathbb{R}$ we have $\overrightarrow{PQ} = t\vec{n}$, where $\vec{n} = a\vec{i} + b\vec{j} + c\vec{k}$ is the normal vector of the plane. Thus $Q = (x_0 + ta, y_0 + tb, z_0 + tc)$ for some t . Since $Q \in T$, by the equation of T

$$0 = a(x_0 + ta) + b(y_0 + tb) + c(z_0 + tc) + d = ax_0 + by_0 + cz_0 + d + t|\vec{n}|^2.$$

Solving for t from this, the formula for the distance becomes

$$h = |t||\vec{n}| = \frac{|ax_0 + by_0 + cz_0 + d|}{\sqrt{a^2 + b^2 + c^2}}.$$

The distance of a point (x_0, y_0) of the plane from the line $S : ax + by + c = 0$ is obtained from the same formula by first setting $c = 0$ and then putting c in place of d . $\square$

The circle and the sphere

The circle is (in the form of a compass) one of the original objects of the Euclidean plane. If the center of the circle is $P_0 \hat{=} \vec{r}_0$ and the radius is R , then the *equation* of the *circle* (the circular curve) is

$$|\vec{r} - \vec{r}_0| = R.$$

This is also the equation of the *sphere* (the spherical surface)*. Squared and written in terms of coordinates, the equations are

$$\begin{aligned} \text{Circle: } & (x - x_0)^2 + (y - y_0)^2 = R^2. \\ \text{Sphere: } & (x - x_0)^2 + (y - y_0)^2 + (z - z_0)^2 = R^2. \end{aligned}$$

The *tangent to a circle* is a line that has exactly one point in common with the circle. It is said that the tangent *touches* the circle at that point. If the point of tangency is a given point Q , then the tangent is constructed geometrically by first drawing a line S_1 through the point Q and the center of the circle, whereupon the tangent S_2 is perpendicular to it. It is said that the line S_1 is the *normal to the circle* at the point Q , or that the line S_1 intersects the circle *perpendicularly*.

*By circle and sphere one may also mean the 'solid' set of Euclidean space

$$K = \{P \hat{=} \vec{r} \mid |\vec{r} - \vec{r}_0| \leq R\}.$$

Its more precise name in the plane is *disk* (the circle being the curve), in space *ball* (the sphere being the surface).

More generally, the *angle of intersection* between a circular curve and a line can be defined as the angle between the direction vectors of that line and of the tangent drawn at the intersection point.

In the same way as the tangent to a circle, the *tangent plane to a sphere* is defined. A line passing through the center of the sphere is a *normal to the sphere*, i.e. it is perpendicular to the tangent plane at the intersection point of the sphere and the line. If the sphere and the plane T have more than one point in common, the set of common points is a *space circle* (a circle on the plane T). If T passes through the center of the sphere, the intersection curve is called a *great circle* of the sphere. The *tangent to a space circle* is a space line lying in the same plane as the circle and touching the circle.

EXAMPLE 6 The center of a sphere is (x_0, y_0, z_0) and its radius $= R$. Under what condition does the plane $T : ax + by + cz + d = 0$ touch the sphere?

Solution The plane T touches the sphere exactly when the distance of the center of the sphere from the plane $= R$. By Example 5, the condition is

$$|ax_0 + by_0 + cz_0 + d| = R\sqrt{a^2 + b^2 + c^2}. \quad \square$$

EXAMPLE 7 The system of equations

$$\begin{cases} x^2 + y^2 + z^2 = R^2 \\ x + y - z = 1 \end{cases}$$

defines the intersection curve S of an origin-centered sphere and a plane. Since the distance of the plane from the origin is $h = 1/\sqrt{3}$ (Example 5), one concludes that S is a space circle if $R > 1/\sqrt{3}$, a point if $R = 1/\sqrt{3}$, and the empty set if $R < 1/\sqrt{3}$. In the first-mentioned case the center of S lies on the line through the origin whose direction vector is the normal vector of the plane $\vec{n} = \vec{i} + \vec{j} - \vec{k}$. $\square$

The cylinder and the cone

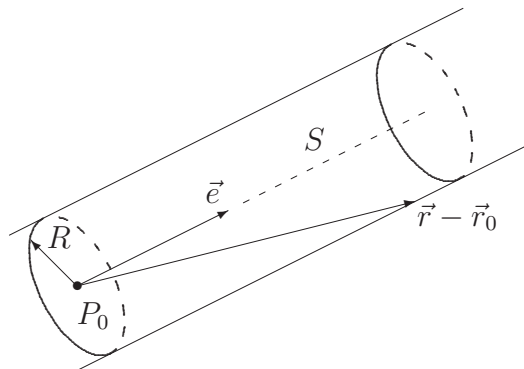
Let S be a space line passing through the point $P_0 \hat{=} \vec{r}_0$ and whose direction vector is a unit vector $\vec{e}$. Then the distance of the point $P \hat{=} \vec{r}$ from the line S can be computed by the formula

$$d = |\vec{r} - \vec{r}_0| \sin \angle(\vec{r} - \vec{r}_0, \vec{e}) = |(\vec{r} - \vec{r}_0) \times \vec{e}|.$$

The surface on which this distance is constant $= R$ ($R > 0$) is called a *cylinder*, more precisely a *circular cylinder*. The line S is the *axis* of the cylinder and

R = the *radius* of the cylinder. Setting $d = R$ in the formula and squaring, one obtains for the equation of the cylinder the form

$$|(\vec{r} - \vec{r}_0) \times \vec{e}|^2 = R^2. \quad (9)$$



EXAMPLE 8 Let the axis of the cylinder be the line through the origin in the direction of the unit vector $\vec{e} = \frac{1}{3}(\vec{i} + 2\vec{j} - 2\vec{k})$, and the radius $R = 2$. Then $\vec{r}_0 = \vec{0}$ and

$$\vec{r} \times \vec{e} = \frac{1}{3} \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ x & y & z \\ 1 & 2 & -2 \end{vmatrix} = \frac{1}{3} [(-2y - 2z)\vec{i} + (2x + z)\vec{j} + (2x - y)\vec{k}],$$

so the equation of the cylinder is

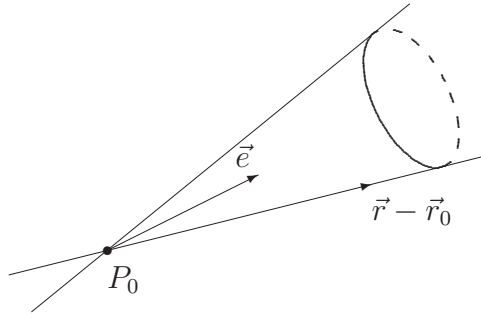
$$\begin{aligned} \frac{1}{9} [(2y + 2z)^2 + (2x + z)^2 + (2x - y)^2] &= 4 \\ \Leftrightarrow 8x^2 + 5y^2 + 5z^2 - 4xy + 4xz + 8yz &= 36. \quad \square \end{aligned}$$

If in equation (9) the cross product is replaced by the dot product and $\gamma|\vec{r} - \vec{r}_0|$ is written in place of R , where $\gamma \in (0, 1)$, then the equation takes the form

$$|(\vec{r} - \vec{r}_0) \cdot \vec{e}|^2 = \gamma^2 |\vec{r} - \vec{r}_0|^2 \quad \Leftrightarrow \quad |\cos \angle(\vec{r} - \vec{r}_0, \vec{e})| = \gamma. \quad (10)$$

This equation defines a *cone*, more precisely a *circular cone*. The point $P_0 \hat{=} \vec{r}_0$ is the *apex* of the cone. The cone consists of two *half-cones*, whose equations are

$$\cos \angle(\vec{r} - \vec{r}_0, \vec{e}) = \pm \gamma.$$



Expressed in terms of the coordinates x, y, z , the equations (9) and (10) can each be written in the form

$$Ax^2 + By^2 + Cz^2 + Dxy + Exz + Fyz + Gx + Hy + Iz + J = 0,$$

where A, B , etc. are real numbers (cf. the example above). This means that the cylinder and the cone, and also the sphere, are so-called *second-degree surfaces*. The intersection curve of a second-degree surface and a space plane (e.g. a coordinate plane) is a *second-degree curve* of that plane. Such is, for example, the circle. The general equation of a second-degree curve in the xy -plane is of the form

$$Ax^2 + By^2 + Cxy + Dx + Ey + F = 0.$$

The more general classification of second-degree curves and surfaces is a mathematical problem to which we return later in another context.

EXERCISES

1. a) Show that the equations represent the same line:

$$\begin{cases} x = 3 + 2t, \\ y = -11 - 6t \end{cases} \quad \text{and} \quad \begin{cases} x = -1 - t, \\ y = 1 + 3t \end{cases}$$

- b) Determine the intersection point of the lines:

$$\begin{cases} x = 3 + 2t, \\ y = -1 - 3t \end{cases} \quad \text{and} \quad \begin{cases} x = -1 - t, \\ y = 2(1 + t) \end{cases}$$

- c) Determine the intersection points of the line $x = 3 + t$, $y = -2 - t$, $z = 4 - 2t$ with the coordinate planes.

2. Determine α so that the vector $\vec{i} + 2\vec{j} + \alpha\vec{k}$ and the space lines

$$2(x-1) = 1-y = 2z-3 \quad \text{and} \quad \begin{cases} x = 17, \\ y = 7 + 3t, \\ z = 2t \end{cases}$$

are parallel to the same plane.

3. Determine points $P_1 \in S_1$ and $P_2 \in S_2$ on the lines $S_1 : x = -y = z$ and $S_2 : x + y - 1 = 0, z = 0$ such that the vector $\overrightarrow{P_1P_2}$ is parallel to the vector $2\vec{i} - \vec{j} - \vec{k}$.
4. The line S passes through the point $(-1, 1, 3)$ and through the midpoint of the segment that the xy - and xz -planes cut from the line $x - 1 = 2(y + 1) = z + 3$. Determine the direction vector of S .
5. Determine α so that the lines $2(x - 1) = y + 1 = 2\alpha(z - 1)$ and $x + 1 = y - 1 = z$ intersect. What is the shortest distance between the lines if $\alpha = 1$?
6. Determine the shortest distance between the lines

$$\vec{r} = 2\vec{i} + 5\vec{k} + t(\vec{i} - 2\vec{j} + 2\vec{k}) \quad \text{and} \quad \begin{cases} 3x - 2y = 12 \\ x + 2z = 6 \end{cases}$$

and the points closest to each other.

7. Determine the parametric equation of the line that perpendicularly intersects the lines

$$\begin{cases} x = 1 + t, \\ y = 1 - 2t, \\ z = t \end{cases} \quad \text{and} \quad \begin{cases} x = -1, \\ y = 2s, \\ z = 1 - 2s. \end{cases}$$

8. Compute the distance of the point $(2, 3, -1)$ from the line

$$\text{a) } \vec{r} = \vec{i} + 2\vec{j} + 3\vec{k} + t(\vec{i} - 2\vec{j} + 2\vec{k}) \quad \text{b) } \begin{cases} 3x - y + z = 0 \\ x - 2y + 8 = 0 \end{cases}$$

9. a) Express the equation of the plane T in standard form $ax + by + cz + d = 0$, given the parametric equations

$$T : \begin{cases} x = 2 + 2t_1 + t_2, \\ y = -1 + 3t_1 + 2t_2, \\ z = 3 - t_1 + t_2. \end{cases}$$

- b) A plane passes through the point $P = (1, -13, -5)$ and its direction vectors are $\vec{v}_1 = \vec{i} - \vec{j}$ and $\vec{v}_2 = \vec{i} + \vec{j} + \vec{k}$. Is the point $Q = (3, -1, 2)$ in the plane?
10. Determine the equation of the plane (standard form!), given that the plane contains the line $x = 3 + t$, $y = 1 - 2t$, $z = -2 + t$ and a) passes through the point $(0, 2, 1)$, b) is parallel to the vector $3\vec{i} + \vec{j} - 2\vec{k}$
11. Determine the common points of the following planes:
 a) $x + y - z + 2 = 0$, $2x + y + 2z - 4 = 0$, $x - y + 3z - 2 = 0$
 b) $x + y + z - 6 = 0$, $x + 2y - z - 2 = 0$, $x + 4y - 5z + 5 = 0$
 c) $x + 2y - 2z - 1 = 0$, $x - y + z - 2 = 0$, $x + 5y - 5z = 0$
12. A plane contains the line $S_1 : \vec{r} = (1 + t)\vec{i} + (1 + 2t)\vec{j} + (1 + 3t)\vec{k}$ and is parallel to the line $S_2 : \vec{r} = (1 + t)\vec{i} + (-1 + t)\vec{j} + \vec{k}$. Derive the equation of the plane in standard form. Likewise derive the equation of the plane that contains the line S_2 and is parallel to S_1 .
13. Derive the equation of the plane that bisects the segments between the point $P_0 = (x_0, y_0, z_0)$ and the plane $T : ax + by + cz + d = 0$.
14. Determine the orthogonal projection of the point $(3, 4, -2)$ onto the plane whose normal vector is $\vec{i} - 2\vec{j} + \vec{k}$ and which passes through the point $(1, 1, 1)$.
15. Determine the mirror-image point of the point $(3, 2, -4)$ with respect to the plane $x + y - 2z + 5 = 0$.
16. Determine the greatest possible distance of the origin from a plane through the points $(0, 1, 0)$ and $(2, 2, -1)$. Which plane gives the maximum distance?
17. Determine the planes passing through the intersection line of the planes $x + y + z = 3$ and $3x - 2y - z = 1$ that bisect the angle between the planes.
18. The (original) height of the Pyramid of Cheops is 147 m and the side length of its square base is 230 m. Place the pyramid in a coordinate system so that it feels as comfortable as possible, and determine, in this coordinate system, the equations of the side planes of the pyramid, with 100 m as the unit of measurement. Also compute the dihedral angle between adjacent sides (cf. Exercise II.6:6).

19. A ray of light traveling in the direction of the vector $2\vec{i} - \vec{j} + 3\vec{k}$ is reflected from the plane T at the point $(1, 2, -1)$. The reflected ray passes through the point $(2, 5, -3)$. What is the equation of the plane?
20. A ray of light coming from the direction of the positive z -axis strikes, at the point $(1, 2, 3)$, a mirror lying on the plane $3x + 2y + z = 10$. Determine the direction of the reflected ray. At what point does the reflected ray intersect (if it does) the xy -plane?
21. Gravity acts in the direction of the negative z -axis. A drop falls from the point $(1, 1, 3)$ onto the plane $3x - 4y + 12z = 12$ and begins to run down along the plane. At what point does the drop meet the xy -plane?
22. From the plain $z = 0$ a mountainside rises along the plane $x + 2y + 4z = 0$. From a point P of the slope, at height $h = 10$, a point-like avalanche starts moving. It proceeds in a straight line down the slope according to the laws of gravity. Upon reaching the plain, it continues its straight-line motion horizontally until it hits a cabin at the point $Q = (10, 40, 0)$. What was the point P ?
23. (The hawk and two hens) In the universal coordinate system the ground surface is the plane $x + 2y - 3z = 0$. On the ground surface two point-like hens A and B are strolling about. The hens are stalked by a point-like hawk flying on a plane parallel to the ground surface at height $h = 5$. A point-like sun shines in the direction $-4\vec{i} + \vec{k}$. At the instant H the following happens: hen A lets out a startled 'cluck' upon noticing the hawk's shadow on itself. The clucking is heard by hen B , which at the same moment glances up at the sky and sees the hawk in the direction $\vec{i} - \vec{j} + \vec{k}$. Determine the vector $\overrightarrow{AB}$ at that instant H .
24. At what angle does the plane line $y = 4x$ intersect the circular curve $x^2 + y^2 - 2x - 4y + 4 = 0$?
25. The vertices of a space triangle are $(0, 0, 0)$, $(3, 2, 1)$ and $(2, -1, 3)$. Compute the center and radius of the circle inscribed in the triangle (i.e. tangent to all sides).
26. The position vectors of three points of the space E^3 are $\vec{a}$, $\vec{b}$ and $\vec{c}$. Present a procedure for determining the position vector of the center of the circle passing through the points. Apply the procedure when the points are $(1, 2, 3)$, $(2, -5, 3)$ and $(-1, 3, -6)$. Also determine the radius of the circle and the normal vector of the plane of the circle.

27. Determine the tangent to the space circle

$$\begin{cases} x^2 + y^2 + z^2 = 49 \\ x + 2y - z = 10 \end{cases}$$

at the point $(-2, 3, -6)$.

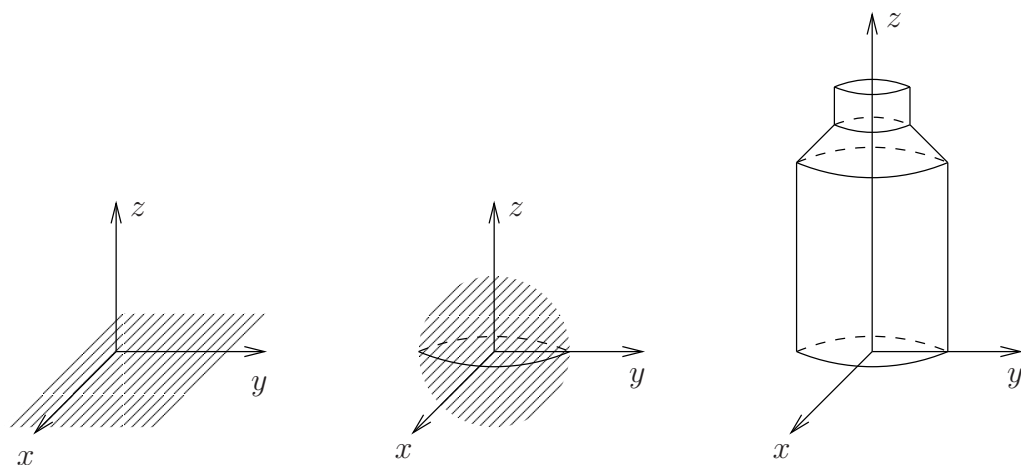
28. The radius of a cylinder is $R = 2$ and its axis is the line through the point $P_0 = (1, 1, 1)$ with direction vector $-\vec{i} + \vec{j} + \vec{k}$. Determine the equation of the cylinder in the standard form of a second-degree surface equation. What are the intersection points of the cylinder and the line $x = y = z$, and which point of the cylinder is closest to the origin?
29. A cylinder and a cone have as common axis the line $S : x = 2y = -2z$, and each passes through the point $(1, 0, 0)$. The apex of the cone is the point $(-2, -1, 1)$. Compute the radius of the cylinder and the opening angle of the cone (of its side profile), and put the equations of both surfaces into the standard form of a second-degree surface equation. Further determine the equations of the intersection curves of both surfaces with the xy -plane and sketch the shape of these curves.
30. (*) One vertex of a cube with side length $= 4$ is at the origin and three others on the positive coordinate axes. The cube is viewed from afar from the direction indicated by the vector $\vec{i} + 2\vec{j} + 3\vec{k}$, at a viewing angle in which the z -axis appears vertical. Compute how the cube appears from this viewing angle. Figure!
31. (*) In what way can one find the space planes that are tangent to three given spheres? How many such planes are there, if the spheres do not intersect or touch one another and none of the spheres is inside another?
32. (*) Show that the equation $K : xy + yz + xz = 0$ defines a cone, and determine the point of K that is closest to the point $(-1, 2, 3)$.
33. (*) The apex of a cone is at the origin, its axis of symmetry is in the direction of the vector $\vec{i} - 2\vec{j} + 2\vec{k}$, and the y -axis is on the conical surface. A plane T passes through the point $(1, 1, 1)$ and touches the cone along a space line. Determine the equation of T (two solutions!).

II.8 Curvilinear coordinate systems

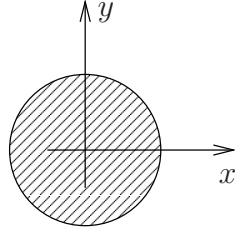
The Euclidean plane and space are, as mathematical concepts, immaterial or *isotropic* in the sense that no point or direction holds a special position in them. In particular, the origin of a Cartesian coordinate system, and also the directions of the coordinate axes (within the limits of the orthogonality conditions), may be chosen arbitrarily, without the geometric laws of space changing as a result. When geometric reasoning is applied to the physical world, however, one runs into the reality that the world generally contains matter, which is not evenly distributed. There may, for instance, be interfaces between different materials. A physical problem may also be non-isotropic because of the external forces (such as gravitation) acting in it. Even in such situations, however, the problem may possess *symmetries*, by which we mean that, although the whole of space is not isotropic, the geometry of the physical problem is nonetheless the same when one moves in some direction or directions. For example, in a 'pancake world', where the half-space

$$E_-^3 = \{P = (x, y, z) \in E^3 \mid z \leq 0\}$$

is filled with homogeneous matter and the rest of space is empty (or filled with another homogeneous substance), the geometry remains the same when one moves in the planes on which $z = \text{constant}$. In other words: the geometry is independent of the coordinates x, y . This is a *plane-symmetric* situation. In the somewhat more realistic (though still geocentric) 'ball world', where a ball filled with matter is surrounded by homogeneous space, the geometry is in turn *spherically symmetric*. Finally, in the 'world of engineering', where the homogeneity of space is broken by a single object with a circular cross section (a milk churn, a pipe, a coaxial cable, etc.), the situation is *cylindrically symmetric*.



The counterpart of cylindrical symmetry in E^2 is *circular symmetry* (rotational symmetry).



In the plane-symmetric situation mentioned above, a Cartesian coordinate system is well suited to the geometry of the problem, provided one of the basis vectors (e.g. $\vec{k}$) is chosen as the normal of the plane of symmetry. In the other example situations, on the other hand, it becomes advantageous to pass to a *curvilinear* coordinate system in which one of the coordinates is constant on the material interface. A coordinate system is called curvilinear if the *coordinate curves*, i.e. the sets of points of E^2 or E^3 on which only one of the coordinates varies, are not all straight lines. In the following we introduce the three most common curvilinear coordinate systems of physics, which are related to the example situations mentioned above. These are:

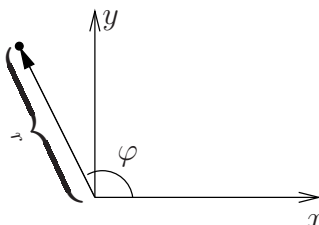
- the *polar coordinate system* (E^2)
- the *cylindrical coordinate system* (E^3)
- the *spherical coordinate system* (E^3)

When applying such coordinate systems (as when applying mathematics in general), one usually makes simplifying assumptions – sometimes strong ones – i.e. idealizations, about the practical problem situation.

Polar coordinate system

In the polar coordinate system, the coordinates of a point P of the Euclidean plane are given by the distance r between the origin (O) and P and by the so-called *polar angle* φ , which measures the angle between the vector $\overrightarrow{OP}$ and the positive x -axis (of the Cartesian coordinate system), taken as positive counterclockwise from the positive x -axis and negative clockwise. All points of the plane are obtained when the polar angle is chosen, e.g. from the interval $[0, 2\pi)$

or $(-\pi, \pi]$. Since the polar angle φ is not defined at the origin, we agree that $O = (0, \varphi) \forall \varphi$.



The conversion formulas from polar to Cartesian coordinates are

$$x = r \cos \varphi, \quad y = r \sin \varphi$$

and in the reverse direction, for example,

$$r = \sqrt{x^2 + y^2}, \quad \sin \varphi = y / \sqrt{x^2 + y^2}.$$

Here, to determine φ , one must be able to 'invert' the trigonometric sine. When the signs of x and y are also taken into account, the angle φ is uniquely determined, e.g. in the interval $[0, 2\pi)$, when $r > 0$.

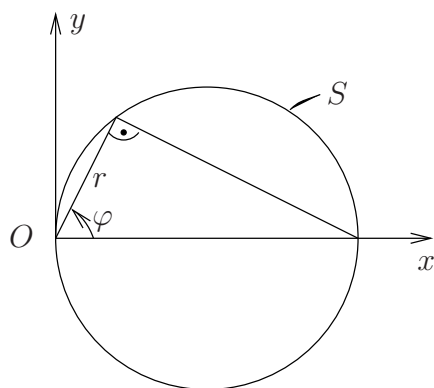
EXAMPLE 1 What sort of point set is

$$S = \{P = (r, \varphi) \in E^2 \mid r = 2 \cos \varphi, \varphi \in [-\frac{\pi}{2}, \frac{\pi}{2}]\}?$$

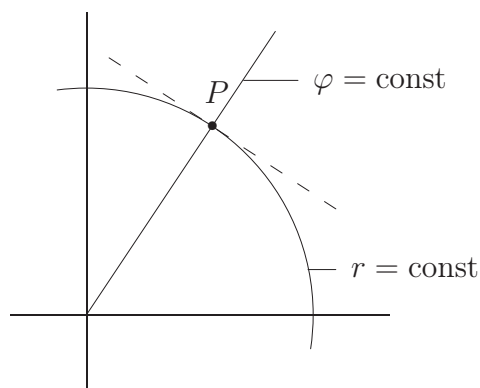
Solution If $r > 0$, then

$$\begin{aligned} r = 2 \cos \varphi &\Leftrightarrow r^2 = 2r \cos \varphi \\ &\Leftrightarrow x^2 + y^2 = 2x \\ &\Leftrightarrow (x - 1)^2 + y^2 = 1. \end{aligned}$$

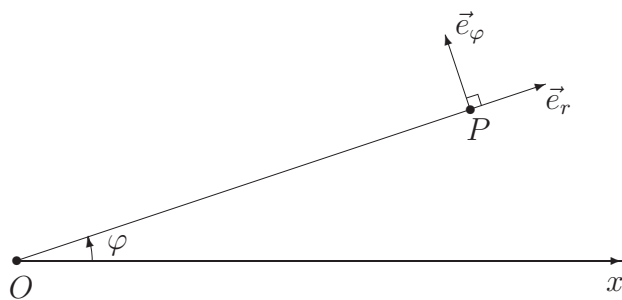
The origin is also included in the curve ($r = 0, \varphi = \pm \frac{\pi}{2}$), so it is a circle with radius = 1 and center = (1, 0). $\square$



In the polar coordinate system the coordinate curves on which $r = \text{const}$ are circles centered at the origin, and the curves on which $\varphi = \text{const}$ are rays emanating from the origin. Since these intersect one another orthogonally (see the figure), the polar coordinate system may be called *orthogonal*, though not a rectilinear coordinate system.



When vectors are considered in the polar coordinate system, they are usually expressed in a basis $\{\vec{e}_r, \vec{e}_\varphi\}$ that 'rotates' along with the observation point P , where the basis vectors are the unit tangent vectors of the coordinate curves at P . The directions are chosen so that the vectors point in the direction of increase of the coordinate that varies along the coordinate curve in question:



If the polar coordinates of the point P are (r, φ) , then the basis vectors are (cf.

the figure)

$$\vec{e}_r = \cos \varphi \vec{i} + \sin \varphi \vec{j}, \quad \vec{e}_\varphi = -\sin \varphi \vec{i} + \cos \varphi \vec{j}.$$

If P is thought of as fixed, any vector of the plane can be represented in the basis $\{\vec{e}_r, \vec{e}_\varphi\}$, i.e. $\{P, \vec{e}_r, \vec{e}_\varphi\}$ is a coordinate system of the plane. On the other hand, if P is changed so that the coordinate φ changes, then $\vec{e}_r$ and $\vec{e}_\varphi$ change as well. In this sense the coordinate system $\{P, \vec{e}_r, \vec{e}_\varphi\}$ thus differs from the corresponding rectilinear coordinate system $\{P, \vec{i}, \vec{j}\}$, in which the basis vectors $\vec{i}, \vec{j}$ remain constant as the reference point P varies.

A typical physical application of the coordinate system $\{P, \vec{e}_r, \vec{e}_\varphi\}$ that 'rotates' along with the point P arises when P represents a body moving in a central force field (e.g. a planet, a spacecraft).

EXAMPLE 2 Ceres and Dawn. At the origin of a Cartesian coordinate system there is a dwarf planet, and at the point P located from the planet in the direction $3\vec{i} + 4\vec{j}$ there is a space probe of mass $m = 1200$ kg. The probe orbits the planet on a circular orbit in the plane $z = 0$, counterclockwise as seen from the direction of the positive z -axis. At the moment considered, the probe's ion engine is switched on, so that in addition to the planet's gravitational attraction $\vec{G} = -78\vec{e}_r$ the probe experiences the engine's thrust $\vec{T} = 450(-\vec{i} + \vec{j})$ (units N=kg m/s²). In which direction does the probe's trajectory curve, and what is the probe's acceleration in the direction of the trajectory?

Solution At the point P we have $\vec{e}_r = \frac{1}{5}(3\vec{i} + 4\vec{j})$, and the probe's direction of motion is $\vec{e}_\varphi = \frac{1}{5}(-4\vec{i} + 3\vec{j})$. Since

$$\begin{cases} (\vec{T} + \vec{G}) \cdot \vec{e}_r = \frac{450}{5}[(-1) \cdot 3 + 1 \cdot 4] - 78 = 12, \\ (\vec{T} + \vec{G}) \cdot \vec{e}_\varphi = \frac{450}{5}[(-1) \cdot (-4) + 1 \cdot 3] + 0 = 630, \end{cases}$$

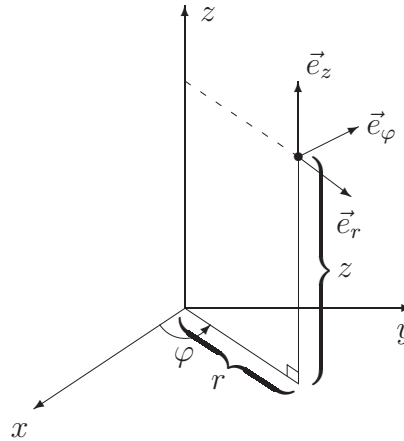
the total force acting on the probe in the coordinate system $\{P, \vec{e}_r, \vec{e}_\varphi\}$ is

$$\vec{F} = \vec{T} + \vec{G} = 12\vec{e}_r + 630\vec{e}_\varphi = F_r\vec{e}_r + F_\varphi\vec{e}_\varphi.$$

Since $F_r > 0$, the trajectory curves to the right (away from the planet). The along-track acceleration is determined by Newton's law of motion: $a_\varphi = F_\varphi/m \approx \underline{\underline{0.53 \text{ m/s}^2}}$. $\square$

Cylindrical coordinate system

The cylindrical coordinate system is the three-dimensional counterpart of the polar coordinate system. Of the Cartesian coordinates x, y, z , the coordinate z is left unchanged and $(x, y) \mapsto (r, \varphi)$ is transformed in the same way as when passing to polar coordinates. The 'rotating' basis vectors of the cylindrical coordinate system are denoted $\vec{e}_r, \vec{e}_\varphi, \vec{e}_z$. Here $\vec{e}_r, \vec{e}_\varphi$ are the same as in the polar coordinate system and $\vec{e}_z = \vec{k}$ as in the Cartesian coordinate system. The basis $\{\vec{e}_r, \vec{e}_\varphi, \vec{e}_z\}$ is an orthonormal right-handed system.



EXAMPLE 3 A *helix* is the set of points of the space E^3

$$S = \{P = (r, \varphi, z) \mid r = R, z = a\varphi\} \quad (a \neq 0). \quad \square$$

The coordinate curves of the cylindrical coordinate system are the sets of points of E^3 on which two of the coordinates r, φ, z take constant values. For these we obtain the following geometric descriptions (cf. the figure above):

- $S_r (\varphi, z \text{ const})$: a ray emanating from the z -axis, parallel to the xy -plane,
- $S_\varphi (r, z \text{ const})$: a circle in a plane parallel to the xy -plane,
- $S_z (r, \varphi \text{ const})$: a line parallel to the z -axis.

Sets of points of E^3 on which one of the coordinates takes a constant value are called *coordinate surfaces*. In the cylindrical coordinate system their geometric descriptions are

- $r = \text{const}$: a cylinder with the z -axis as its axis,
- $\varphi = \text{const}$: a half-plane bounded by the z -axis,
- $z = \text{const}$: a plane parallel to the xy -plane.

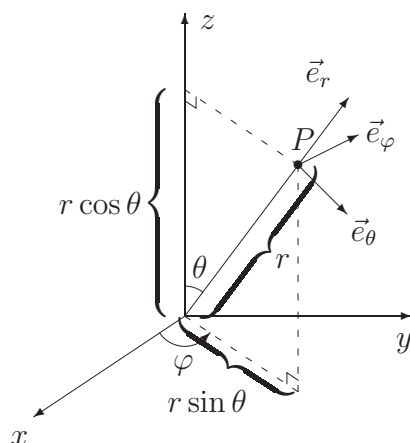
Spherical coordinate system

The *spherical coordinates* of the point $P = (x, y, z)$ are (see the figure)

- the distance of P from the origin: $r = \sqrt{x^2 + y^2 + z^2}$

- the angle θ formed by the position vector $\vec{r} = \overrightarrow{OP}$ and the positive z -axis (the vector $\vec{k}$), $0 \leq \theta \leq \pi$
- the angle φ , $0 \leq \varphi < 2\pi$, formed by the positive x -axis (the vector $\vec{i}$) and the orthogonal projection of the vector $\vec{r} = \overrightarrow{OP}$ onto the xy -plane

The coordinates θ, φ are called *spherical surface coordinates*.



The conversion formulas are

$$\begin{aligned} x &= r \sin \theta \cos \varphi, \\ y &= r \sin \theta \sin \varphi, \\ z &= r \cos \theta. \end{aligned}$$

The coordinate curves of the spherical coordinate system are of the following types:

- $S_r (\theta, \varphi \text{ const})$: a ray emanating from the origin,
- $S_\theta (r, \varphi \text{ const})$: a semicircle centered at the origin, bounded by the z -axis,
- $S_\varphi (r, \theta \text{ const})$: a circle in a plane parallel to the xy -plane.

The coordinate surfaces can be classified as follows:

- $r = \text{const}$: a sphere centered at the origin,
- $\theta = \text{const}$: a half-cone with apex at the origin,
- $\varphi = \text{const}$: a half-plane bounded by the z -axis.

In the spherical coordinate system, too, the coordinate curves intersect one another orthogonally; that is, the tangents of the coordinate curves (see the previous section) are pairwise orthogonal at the point of intersection. The basis of space vectors formed from the unit tangent vectors of the coordinate curves (the direction vectors of the tangents) is denoted $\{\vec{e}_r, \vec{e}_\theta, \vec{e}_\varphi\}$ and defined by (cf. the figure above)

$$\begin{aligned}\vec{e}_r &= \sin \theta \cos \varphi \vec{i} + \sin \theta \sin \varphi \vec{j} + \cos \theta \vec{k}, \\ \vec{e}_\theta &= \cos \theta \cos \varphi \vec{i} + \cos \theta \sin \varphi \vec{j} - \sin \theta \vec{k}, \\ \vec{e}_\varphi &= -\sin \varphi \vec{i} + \cos \varphi \vec{j}.\end{aligned}$$

Here $\vec{e}_\varphi$ is the same as in the polar and cylindrical coordinate systems, and the expression for $\vec{e}_r$ follows immediately from the coordinate conversion formulas given above:

$$\begin{aligned}\vec{e}_r &= \frac{1}{|\vec{r}|} \vec{r} = \frac{1}{r} (x\vec{i} + y\vec{j} + z\vec{k}) \\ &= \sin \theta \cos \varphi \vec{i} + \sin \theta \sin \varphi \vec{j} + \cos \theta \vec{k}.\end{aligned}$$

After this, $\vec{e}_\theta$ can be computed from the fact that $\{\vec{e}_r, \vec{e}_\theta, \vec{e}_\varphi\}$ is an orthonormal, right-handed system:

$$\vec{e}_\theta = \vec{e}_\varphi \times \vec{e}_r.$$

EXAMPLE 4 The Cartesian coordinates of the point P are $(x, y, z) = (-2, -3, -6)$. Determine the spherical coordinates of P as well as the coordinates $(v_r, v_\theta, v_\varphi)$ of the vector $\vec{v} = \vec{i} - \vec{j}$ in the basis $\{\vec{e}_r, \vec{e}_\theta, \vec{e}_\varphi\}$ of the spherical coordinate system at that point.

Solution If we set $\vec{r} = \overrightarrow{OP} = -2\vec{i} - 3\vec{j} - 6\vec{k}$, then $r = |\vec{r}| = 7$ and

$$\vec{e}_r = \frac{1}{r} \vec{r} = \frac{1}{7} (-2\vec{i} - 3\vec{j} - 6\vec{k}).$$

Then

$$\begin{aligned}\cos \theta &= \vec{e}_r \cdot \vec{k} = -\frac{6}{7}, \quad \sin \theta = \frac{\sqrt{13}}{7}, \\ \cos \varphi &= \frac{\vec{i} \cdot (-2\vec{i} - 3\vec{j})}{\sqrt{2^2 + 3^2}} = -\frac{2}{\sqrt{13}}, \quad \sin \varphi = -\frac{3}{\sqrt{13}},\end{aligned}$$

so the spherical coordinates of P are

$$(r, \theta, \varphi) \approx (7, 149.0^\circ, 236.3^\circ).$$

The exact values of the basis vectors $\vec{e}_\theta, \vec{e}_\varphi$ are

$$\begin{aligned}\vec{e}_\theta &= -\frac{6}{7} \left(-\frac{2}{\sqrt{13}} \right) \vec{i} - \frac{6}{7} \left(-\frac{3}{\sqrt{13}} \right) \vec{j} - \frac{\sqrt{13}}{7} \vec{k} \\ &= \frac{1}{7\sqrt{13}} (12\vec{i} + 18\vec{j} - 13\vec{k}), \\ \vec{e}_\varphi &= \frac{1}{\sqrt{13}} (3\vec{i} - 2\vec{j}),\end{aligned}$$

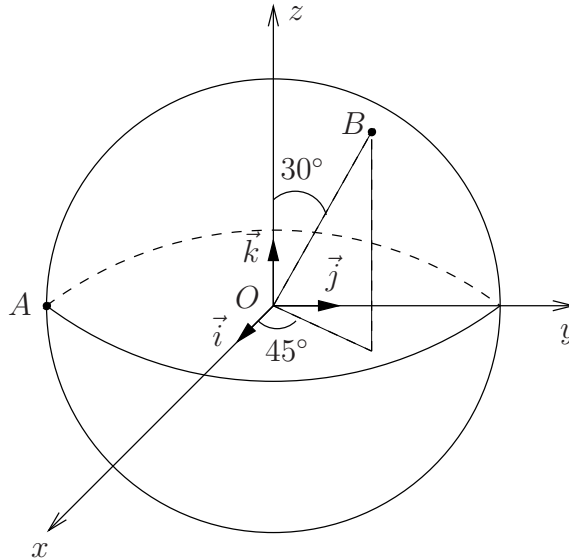
so the requested coordinates of $\vec{v}$ are

$$\begin{aligned}v_r &= \vec{v} \cdot \vec{e}_r = \frac{1}{7} \approx 0.143, \\ v_\theta &= \vec{v} \cdot \vec{e}_\theta = -\frac{6}{7\sqrt{13}} \approx -0.238, \\ v_\varphi &= \vec{v} \cdot \vec{e}_\varphi = \frac{5}{\sqrt{13}} \approx 1.387.\end{aligned}$$

Check:

$$v_r^2 + v_\theta^2 + v_\varphi^2 = \frac{1274}{13 \cdot 49} = 2 = |\vec{v}|^2. \quad \text{OK!} \quad \square$$

EXAMPLE 5 : Flight direction. An airplane departs from the point A on the equator at 90° west longitude and flies along the great circle passing through the point B at 45° east longitude, 60° north latitude. Determine the airplane's flight direction at the point B , both in the basis $\{\vec{i}, \vec{j}, \vec{k}\}$ and relative to the compass directions at B .



Solution Assume the radius of the sphere is $R = 1$; then (see the figure)

$$\begin{aligned}\overrightarrow{OA} &= -\vec{j}, \\ \overrightarrow{OB} &= \sin 30^\circ \cos 45^\circ \vec{i} + \sin 30^\circ \sin 45^\circ \vec{j} + \cos 30^\circ \vec{k} \\ &= \frac{1}{2\sqrt{2}}\vec{i} + \frac{1}{2\sqrt{2}}\vec{j} + \frac{\sqrt{3}}{2}\vec{k}.\end{aligned}$$

The airplane's flight path lies in the plane passing through the points O, A, B , so the normal vector of the plane is

$$\vec{n} = \overrightarrow{OA} \times \overrightarrow{OB} = -\frac{\sqrt{3}}{2}\vec{i} + \frac{1}{2\sqrt{2}}\vec{k}.$$

Since the flight path at B is also perpendicular to the vector $\overrightarrow{OB}$, the flight direction must be the same as that of the vector

$$\vec{n} \times \overrightarrow{OB} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ -\frac{\sqrt{3}}{2} & 0 & \frac{1}{2\sqrt{2}} \\ \frac{1}{2\sqrt{2}} & \frac{1}{2\sqrt{2}} & \frac{\sqrt{3}}{2} \end{vmatrix} = -\frac{1}{8}\vec{i} + \frac{7}{8}\vec{j} - \frac{1}{4}\sqrt{\frac{3}{2}}\vec{k}.$$

The length of this vector is $\sqrt{56}/8$, so the unit vector indicating the flight direction is

$$\underline{\underline{\vec{v}^0 = \frac{1}{\sqrt{56}}(-\vec{i} + 7\vec{j} - \sqrt{6}\vec{k}).}}$$

The unit vector pointing east at the point B is

$$\vec{e}_\varphi = -\sin 45^\circ \vec{i} + \cos 45^\circ \vec{j} = \frac{1}{\sqrt{2}}(-\vec{i} + \vec{j}),$$

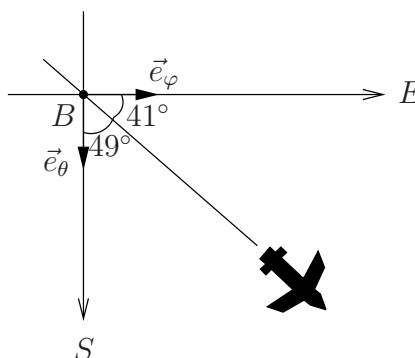
and the unit vector pointing south is

$$\begin{aligned}\vec{e}_\theta &= \cos 30^\circ \cos 45^\circ \vec{i} + \cos 30^\circ \sin 45^\circ \vec{j} - \sin 30^\circ \vec{k} \\ &= \frac{1}{2}\sqrt{\frac{3}{2}}(\vec{i} + \vec{j}) - \frac{1}{2}\vec{k}.\end{aligned}$$

Since

$$\begin{aligned}\vec{v}^0 \cdot \vec{e}_\varphi &= \frac{2}{\sqrt{7}} \approx \cos 41^\circ, \\ \vec{v}^0 \cdot \vec{e}_\theta &= \sqrt{\frac{3}{7}} \approx \cos 49^\circ,\end{aligned}$$

the flight direction at B is toward the southeast, slightly toward the east. $\square$



EXERCISES

- In each of the following, a point P is given in either the Cartesian, cylindrical, or spherical coordinate system. Compute the coordinates of P in the other two coordinate systems (exactly if possible, otherwise as approximate values).

$$\begin{array}{llll}
 P = (x, y, z) : & \text{a) } (1, 1, 1) & \text{b) } (2, 3, -1) & \text{c) } (-1, -\sqrt{3}, -5) \\
 P = (r, \varphi, z) : & \text{d) } (\sqrt{2}, \frac{\pi}{2}, 1) & \text{e) } (1, \frac{\pi}{3}, -1) & \text{f) } (6\sqrt{2}, \frac{7\pi}{4}, 3) \\
 P = (r, \theta, \varphi) : & \text{g) } (1, \frac{\pi}{2}, \pi) & \text{h) } (\sqrt{3}, \frac{3\pi}{4}, \frac{3\pi}{4}) & \text{i) } (3, \frac{2\pi}{3}, \frac{5\pi}{6})
 \end{array}$$

- a) Investigate the nature of the plane curve expressed in polar coordinates

$$S = \{P = (r, \varphi) \mid r = \cos \varphi + \sin \varphi, \varphi \in [-\frac{\pi}{4}, \frac{3\pi}{4}]\}.$$

- A circle S in the plane has center (x_0, y_0) and radius $= R$. Show that $P = (x, y) \in S$ exactly when, for some $\varphi \in [0, 2\pi)$,

$$\begin{cases} x = x_0 + R \cos \varphi, \\ y = y_0 + R \sin \varphi. \end{cases}$$

What is the corresponding statement for a sphere in space with center (x_0, y_0, z_0) and radius R ?

- The spherical coordinates $(r_1, \theta_1, \varphi_1)$ and $(r_2, \theta_2, \varphi_2)$ of two points P_1 and P_2 are compared as elements of the set $[0, \infty) \times [0, \pi] \times [0, 2\pi) \subset \mathbb{R}^3$. List the cases in which the comparison yields $P_1 = P_2$ even though the coordinates are not the same.

4. At a point P , whose Cartesian coordinates are known, a force $\vec{F}$ acts. Convert $\vec{F}$ to the given basis at that point.
 - a) $P = (-2, 3, 6)$, $\vec{F} = \vec{i} - \vec{j} + \vec{k}$, $\{\vec{e}_r, \vec{e}_\varphi, \vec{e}_z\}$
 - b) $P = (-2, 3, 6)$, $\vec{F} = \vec{i} + \vec{j} + \vec{k}$, $\{\vec{e}_r, \vec{e}_\theta, \vec{e}_\varphi\}$
 - c) $P = (6, -6, -3)$, $\vec{F} = -9\vec{e}_r$ (spher.), $\{\vec{i}, \vec{j}, \vec{k}\}$
 - d) $P = (-6, -6, 3)$, $\vec{F} = -9\vec{e}_r$ (cyl.), $\{\vec{e}_r, \vec{e}_\theta, \vec{e}_\varphi\}$
 - e) $P = (-1, \sqrt{3}, -2\sqrt{3})$, $\vec{F} = \vec{e}_\theta - \vec{e}_\varphi$ (spher.), $\{\vec{i}, \vec{j}, \vec{k}\}$
5. The origin of a Cartesian coordinate system is at the center of the Earth and the vector $\vec{k}$ points toward the North Pole. At a point on the Earth's surface in the southern hemisphere, whose spherical surface coordinates are $\theta = 135^\circ$, $\varphi = 120^\circ$, a ship is heading northwest and an airplane northeast. Determine the directions of motion of the ship and the airplane as unit vectors a) in the spherical coordinate system, using the vectors $\{\vec{e}_\theta, \vec{e}_\varphi\}$, b) in the Cartesian coordinate system, using the vectors $\{\vec{i}, \vec{j}, \vec{k}\}$.
6. One flies from Helsinki ($60^\circ N, 25^\circ E$) to Tokyo ($36^\circ N, 140^\circ E$) along the shortest route. How long is the journey, and in which compass direction must one set out from Helsinki? The radius of the Earth is 6370 km.
7. (*) The half-cone $\theta = 30^\circ$, expressed by means of the spherical coordinate θ , is cut by the plane $T : y + z - 4 = 0$. First express the resulting curve of intersection $S \subset E^3$ by means of the spherical coordinates (r, θ, φ) . Then transform the curve into the polar coordinate system of the plane T , whose origin is at the point $(0, 0, 4)$ and whose polar angle is measured from the direction $\vec{i}$, and further into the corresponding Cartesian coordinate system of T . Sketch the curve in the last-mentioned coordinate system.

Chapter III

Complex numbers

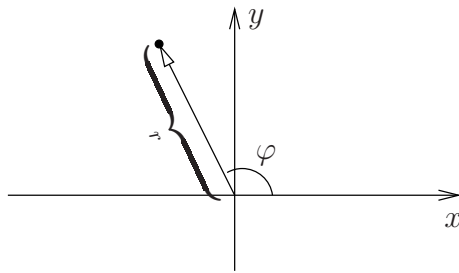
The passage from the real numbers to the complex numbers is one of the most significant and at the same time most remarkable conquests of mathematical analysis. It is a matter of extending the number domain, i.e., of moving once again to a new level of 'reality'. Although the extension may be regarded as a mere fiction, this fiction has simplified mathematical thinking to such a degree that it has ultimately had a profound influence on all of mathematics, including practical methods of computation. In the tradition of mathematical analysis, the extension of the number domain is reflected in the notions of *real analysis* and *complex analysis*. Both are nowadays very broad (and somewhat ill-defined) areas of mathematics. Among the subfields of complex analysis, one worth mentioning is the theory of functions of a complex variable, or *function theory*^{*}.

^{*}The research tradition in function theory is strong in Finland. This branch of mathematics was also represented by the best-known mathematician in Finnish history, the academician **Rolf Nevanlinna** (1895-1980).

III.1 The field of phasors

The starting point is the Euclidean plane together with a Cartesian coordinate system erected on it, with coordinates (x, y) . This time we call a *phasor* a directed segment having a certain length and direction. The length is denoted by the symbol r , and the direction is measured from the direction of the x -axis, rotating counterclockwise, by the *phase angle* (polar angle) φ . We write

$$\vec{p} = r\angle\varphi.$$



So for the time being a phasor is exactly the same kind of object as a vector: it has a length and a direction. Phasors are also identified in the same way as vectors. One need only note that φ and $\varphi + 2\pi$ correspond to the same direction, so that the identification rules are

$$r_1\angle\varphi_1 = r_2\angle\varphi_2 \quad \Leftrightarrow \quad \begin{cases} r_1 = r_2 = 0 & \text{or} \\ r_1 = r_2 > 0 \quad \wedge \quad \varphi_1 - \varphi_2 = k \cdot 2\pi, & k \in \mathbb{Z}. \end{cases}$$

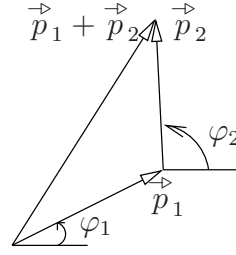
In the case $r = 0$ all directions are thus identified, as with the zero vector. In this case the phasor is accordingly called the *zero phasor* and is denoted

$$0\angle\varphi = \vec{0}.$$

The set of phasors will henceforth be denoted by $\mathbb{P}$:

$$\mathbb{P} = \{\vec{p} = r\angle\varphi \mid r \in [0, \infty), \varphi \in \mathbb{R}\}.$$

The addition operation $(+)$ of phasors is defined in the same way as for vectors, i.e., by means of the triangle diagram:



Multiplication by scalars, i.e., by real numbers, is likewise carried out just as for vectors. In particular, if $\lambda \in \mathbb{R}$ and $\lambda > 0$, then

$$\lambda(r\angle\varphi) = \lambda r\angle\varphi, \quad \lambda > 0.$$

In the case $\lambda = 0$ the product is the zero phasor. If $\lambda \in \mathbb{R}$ and $\lambda < 0$, then one defines (cf. vectors)

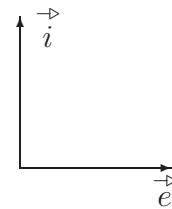
$$\begin{aligned} \lambda(r\angle\varphi) &= |\lambda| \cdot (-1)(r\angle\varphi) \\ &= |\lambda|r\angle(\varphi + \pi), \quad \lambda < 0. \end{aligned}$$

Here $r\angle(\varphi + \pi)$ is, by the definition of addition, the *opposite phasor* of the phasor $\vec{p} = r\angle\varphi$:

$$-\vec{p} = (-1)\vec{p} = r\angle(\varphi + \pi).$$

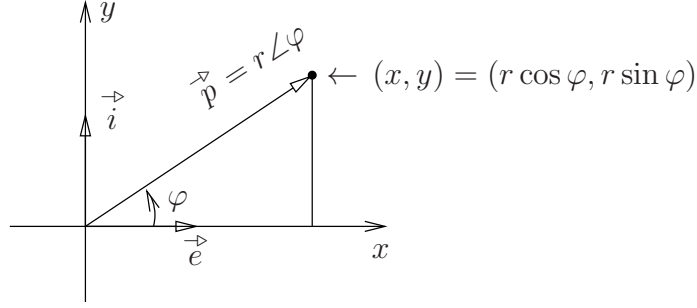
On the basis of the foregoing, the algebra $(\mathbb{P}, +, \mathbb{R})$ formed by the addition and scaling of phasors looks simply like the vector space of the plane. And so it is for the time being, and the vector analogy can be carried a little further: we introduce the phasor basis corresponding to the orthonormal basis $\{\vec{i}, \vec{j}\}$ of the plane vector space, in which the basis phasors are denoted

$$\begin{aligned} \vec{i} &\leftrightarrow \vec{e} = 1\angle 0, \\ \vec{j} &\leftrightarrow \vec{i} = 1\angle \pi/2. \end{aligned}$$



If now $\vec{p} = r\angle\varphi \in \mathbb{P}$, then $\vec{p}$ can be expressed uniquely in terms of the basis phasors in the form

$$\begin{aligned} \vec{p} = r\angle\varphi &= r \cos \varphi \vec{e} + r \sin \varphi \vec{i} \\ &= x\vec{e} + y\vec{i}. \end{aligned} \tag{1}$$



On the other hand, if the coordinates x, y of $\vec{p}$ in the phasor basis are known, then the representation $\vec{p} = r \angle \varphi$ can be computed by the formulas

$$r = \sqrt{x^2 + y^2}, \quad \begin{cases} \cos \varphi = x/r, \\ \sin \varphi = y/r. \end{cases} \quad (2)$$

According to this, φ is determined uniquely up to a multiple of 2π if $r > 0$ (i.e., $(x, y) \neq (0, 0)$), so that the phasor is determined uniquely by virtue of the agreed identification rules. In what follows, the different representations of a phasor are given the names

$$\begin{aligned} \vec{p} &= r \angle \varphi && \text{polar form (polar representation),} \\ \vec{p} &= x \vec{e} + y \vec{i} && \text{component form.} \end{aligned}$$

Passing from one representation to the other is thus done by the rules (1) and (2). Since the component representation creates a one-to-one correspondence $\mathbb{P} \leftrightarrow \mathbb{R}^2$, phasors can also be thought of abstractly as pairs of real numbers:

$$\vec{p} = x \vec{e} + y \vec{i} \leftrightarrow (x, y) \in \mathbb{R}^2.$$

The component form of a phasor is especially convenient in addition:

$$\begin{aligned} \vec{p}_1 &= r_1 \angle \theta_1 \quad \& \quad \vec{p}_2 = r_2 \angle \theta_2 \\ \Rightarrow \vec{p}_1 + \vec{p}_2 &= (r_1 \cos \theta_1 + r_2 \cos \theta_2) \vec{e} + (r_1 \sin \theta_1 + r_2 \sin \theta_2) \vec{i}. \end{aligned}$$

One returns from this to the polar form by the rules (2) — it is not worth writing out the final result in general form.

Up to this point the analogy between the phasor space and the vector space of the plane has been complete. When it comes to the multiplication of phasors,

however, the paths diverge. For phasors one defines neither an inner product nor a cross product, but a product of a different type, which will henceforth be called the *phasor product*. The definition is:

$$\vec{p}_1 = r_1 \angle \varphi_1 \quad \& \quad \vec{p}_2 = r_2 \angle \varphi \quad \Rightarrow \quad \vec{p}_1 \cdot \vec{p}_2 = r_1 r_2 \angle (\varphi_1 + \varphi_2).$$

The phasor product is thus a function of type $\mathbb{P} \times \mathbb{P} \rightarrow \mathbb{P}$ — it is a somewhat peculiar combination of the multiplication (lengths) and addition (directions) of real numbers (cf. Exercise I.2:14).

The commutative law follows directly from the definition of the phasor product:

$$\vec{p}_1 \vec{p}_2 = \vec{p}_2 \vec{p}_1$$

(here the multiplication sign has been dropped, as usual). The associative law also holds (Exercise 1). Furthermore, one sees that the *unit phasor*, i.e., the identity element of multiplication, is

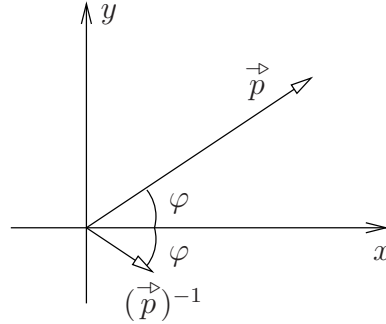
$$\vec{1} = \vec{e} = 1 \angle 0.$$

Every $\vec{p} \in \mathbb{P}$, $\vec{p} \neq \vec{0}$, also has an *inverse phasor* $(\vec{p})^{-1}$, which satisfies

$$\vec{p} \cdot (\vec{p})^{-1} = \vec{1} = 1 \angle 0.$$

Namely, if $\vec{p} = r \angle \varphi$, $r \neq 0$, then

$$(\vec{p})^{-1} = r^{-1} \angle -\varphi.$$



From the definition of the phasor product it follows immediately that for every $n \in \mathbb{N}$ the rule for raising to a power holds:

$$(\vec{p})^n = r^n \angle n\varphi.$$

In the case $\vec{p} \neq \vec{0}$ this is valid for every $n \in \mathbb{Z}$, provided one agrees, as usual, that

$$(\vec{p})^0 = \vec{1}, \quad (\vec{p})^{-n} = [(\vec{p})^{-1}]^n, \quad n \in \mathbb{N} \quad (\vec{p} \neq \vec{0}).$$

EXAMPLE 1 Find all solutions of the phasor equation $(\vec{p})^3 = -\vec{p}$.

Solution By the definition of the phasor product and the identification rules, one deduces

$$\begin{aligned} (\vec{p})^3 &= -\vec{p} \\ \Leftrightarrow r^3 \angle 3\varphi &= r \angle (\varphi + \pi) \\ \Leftrightarrow r^3 = r = 0 \quad \text{or} \quad r^3 = r > 0 \quad \& \quad 3\varphi = (\varphi + \pi) + k \cdot 2\pi, \quad k \in \mathbb{Z} \\ \Leftrightarrow r = 0 \quad \text{or} \quad r = 1 \quad \& \quad \varphi = \frac{\pi}{2} + k \cdot \pi, \quad k \in \mathbb{Z}. \end{aligned}$$

In the case $r = 1$ distinct phasors are obtained only for the values 0, 1 of k , so the solutions are

$$\vec{p} = \vec{0}, \quad \vec{p} = \pm i. \quad \square$$

From the definitions of the phasor product and of phasor addition it further follows that the distributive law also holds:

$$\vec{p} \cdot (\vec{q}_1 + \vec{q}_2) = \vec{p} \cdot \vec{q}_1 + \vec{p} \cdot \vec{q}_2.$$

This is a consequence of the fact that if $\vec{p} = r \angle \varphi$, then the product $\vec{p} \cdot \vec{q}$ can be written in the form

$$\vec{p} \cdot \vec{q} = r(1 \angle \varphi) \cdot \vec{q}.$$

Here multiplication by the phasor $(1 \angle \varphi)$ is the same as a rotation by the angle φ . The distributive law with respect to addition holds both for this operation and for multiplication by the scalar r (cf. the proof of the corresponding distributive law for the cross product in Section II.6), so the asserted distributive law follows.

Now that both an addition and a multiplication have been defined in $\mathbb{P}$, satisfying the usual commutative, associative and distributive laws, and moreover the zero and opposite elements of addition and the identity and inverse elements of multiplication have been constructed, it has been shown that $(\mathbb{P}, +, \cdot)$ is in fact a field. We thus see that the inclusion of the phasor product changes the phasor algebra quite radically. Phasors can indeed be characterized as 'rotating numbers' that are simply added together like vectors. — A separate scalar multiplication operation is no longer needed once the phasor product has been defined, since a scalar $\lambda \in \mathbb{R}$ can be interpreted as the phasor

$$\vec{\lambda} = \begin{cases} \lambda \angle 0, & \text{if } \lambda \geq 0, \\ |\lambda| \angle \pi, & \text{if } \lambda < 0, \end{cases}$$

so that scaling is a special case of the phasor product: $\lambda \vec{p} = \vec{\lambda} \vec{p}$.

The phasor product also has a rather simple representation in component form. Namely, if

$$\begin{aligned}\vec{p}_1 &= r_1 \cos \varphi_1 \vec{e} + r_1 \sin \varphi_1 \vec{i} = x_1 \vec{e} + y_1 \vec{i}, \\ \vec{p}_2 &= r_2 \cos \varphi_2 \vec{e} + r_2 \sin \varphi_2 \vec{i} = x_2 \vec{e} + y_2 \vec{i},\end{aligned}$$

then from the definition of the phasor product and the addition formulas for the trigonometric functions (see Section II.5) it follows that

$$\begin{aligned}\vec{p}_1 \cdot \vec{p}_2 &= r_1 r_2 \cos(\varphi_1 + \varphi_2) \vec{e} + r_1 r_2 \sin(\varphi_1 + \varphi_2) \vec{i} \\ &= r_1 r_2 (\cos \varphi_1 \cos \varphi_2 - \sin \varphi_1 \sin \varphi_2) \vec{e} \\ &\quad + r_1 r_2 (\sin \varphi_1 \cos \varphi_2 + \cos \varphi_1 \sin \varphi_2) \vec{i} \\ &= (x_1 x_2 - y_1 y_2) \vec{e} + (x_1 y_2 + y_1 x_2) \vec{i}.\end{aligned}$$

Consequently the phasor product corresponds to the product defined in $\mathbb{R}^2$

$$(x_1, y_1) \cdot (x_2, y_2) = (x_1 x_2 - y_1 y_2, x_1 y_2 + y_1 x_2). \quad (3)$$

Together with the (vector) addition already defined earlier, i.e.,

$$(x_1, y_1) + (x_2, y_2) = (x_1 + x_2, y_1 + y_2) \quad (4)$$

the pairs of numbers give rise to an algebra $(\mathbb{R}^2, +, \cdot)$, which is thus a field (!). The zero and identity elements of the field $(\mathbb{R}^2, +, \cdot)$ are

$$(0, 0) \hat{=} \vec{0}, \quad (1, 0) \hat{=} \vec{1}.$$

EXERCISES

1. Prove the associative law for the phasor product, $(\vec{p}_1 \vec{p}_2) \vec{p}_3 = \vec{p}_1 (\vec{p}_2 \vec{p}_3)$.
2. Show directly from the definition of the phasor product that the following hold:
 - a) $-\vec{p} = (-\vec{1}) \vec{p}$ b) $\vec{0} \vec{p} = \vec{0}$ c) $\vec{p} \vec{q} = \vec{0} \Rightarrow \vec{p} = \vec{0} \vee \vec{q} = \vec{0}$

3. Find all solutions of the following phasor equations:

a) $(\vec{p})^2 = -\vec{1}$ b) $2(\vec{p})^3 = \vec{1}$ c) $(\vec{p})^4 + 3\vec{p} = \vec{0}$ d) $(\vec{p})^4 = 4\angle\pi$

4. Let $(x, y) \in \mathbb{R}^2$, $(x, y) \neq (0, 0)$. Starting from the definition of the product of pairs of numbers, find $(a, b) \in \mathbb{R}^2$ such that $(x, y) \cdot (a, b) = (1, 0)$, i.e., $(a, b) = (x, y)^{-1}$.

III.2 Complex numbers and computing with them

As the starting point for defining the *complex numbers*, mathematics often takes the field $(\mathbb{R}^2, +, \cdot)$, in which the operations are defined by the rules (4) and (3) of the previous section. The most straightforward definition is simply to agree that $(\mathbb{R}^2, +, \cdot)$ itself is the field of complex numbers, i.e., that the complex numbers are the pairs of numbers of $\mathbb{R}^2$, with the operations between them defined in the stated way. In practical computational routines, however, more concrete representations of complex numbers are usually used. These are the polar form (phasor form) described above, or the even more general third representation, derived from the component form, which in what follows is taken as the starting point for the practical definition of complex numbers.

We begin by denoting the ' x -axis' of $\mathbb{R}^2$ by the symbol $\mathbb{R}_0^2$:

$$\mathbb{R}_0^2 = \{(x, 0) \mid x \in \mathbb{R}\}.$$

One observes that $\mathbb{R}_0^2$ contains the zero element $(0, 0)$ of $(\mathbb{R}^2, +, \cdot)$, as well as the identity element $(1, 0)$. Moreover, if $(x, 0)$, $(y, 0)$ are any two elements of $\mathbb{R}_0^2$, then by the arithmetic rules of the previous section their sum, difference, product and quotient (if $y \neq 0$) are also in $\mathbb{R}_0^2$. These facts together mean that $(\mathbb{R}_0^2, +, \cdot)$ is itself a field and hence a subfield of $(\mathbb{R}^2, +, \cdot)$ (cf. Section I.2). On the other hand, one observes that the evident correspondence between $\mathbb{R}_0^2$ and $\mathbb{R}$

$$(x, 0) \in \mathbb{R}_0^2 \leftrightarrow x \in \mathbb{R}$$

extends in the same way to the operations: if an operation is carried out in $\mathbb{R}_0^2$ by the rules of the field $(\mathbb{R}^2, +, \cdot)$, then this corresponds to an operation of $(\mathbb{R}, +, \cdot)$ in the ordinary field of real numbers. For example, the multiplication

$$(x_1, 0) \cdot (x_2, 0) = (x_1 x_2, 0)$$

(applying rule (3) of the previous section) corresponds to multiplication in $\mathbb{R}$, when the aforementioned correspondence principle is used:

$$\begin{array}{ccccc} \mathbb{R}_0^2 : & (x_1, 0) & \cdot & (x_2, 0) & = & (x_1 x_2, 0) \\ & \downarrow & & \downarrow & & \downarrow \\ \mathbb{R} : & x_1 & \cdot & x_2 & = & x_1 x_2 \end{array}$$

The same holds for addition, so that the fields $(\mathbb{R}_0^2, +, \cdot)$ and $(\mathbb{R}, +, \cdot)$ differ in no way as far as practical computation is concerned. On this basis we make, in the field $(\mathbb{R}^2, +, \cdot)$, the identification convention

$$(x, 0) = x \quad \forall x \in \mathbb{R}.$$

By this convention the field of real numbers $(\mathbb{R}, +, \cdot)$ becomes a subfield of the field $(\mathbb{R}^2, +, \cdot)$. We further write $(0, 1) = i$, whereupon

$$\begin{aligned}(x, y) &= (x, 0) + (0, y) = (x, 0) + (y, 0)(0, 1) \\ &= x + yi = x + iy.\end{aligned}$$

By the definition of multiplication in the field $(\mathbb{R}^2, +, \cdot)$ and the identification convention made, we have

$$i^2 = i \cdot i = (-1, 0) = -1.$$

DEFINITION III.2.1 (Complex numbers) The set of complex numbers $\mathbb{C}$ contains the real numbers and in addition the so-called *imaginary number* i , which satisfies

$$i^2 = i \cdot i = -1 \in \mathbb{R}.$$

The complex numbers form a field $(\mathbb{C}, +, \cdot)$, in which every $z \in \mathbb{C}$ has a unique representation

$$z = x + iy, \quad x, y \in \mathbb{R}.$$

In particular, $x + 0i = x \in \mathbb{R}$ and $0 + 1i = i$.

By this definition, $(\mathbb{C}, +, \cdot)$ arises as an extension of the field of real numbers $(\mathbb{R}, +, \cdot)$, when the imaginary number i is adjoined to the number set $\mathbb{R}$ and the basic field axiom (K0) is observed — cf. Example 3 of Section I.2, where the field $(\mathbb{Q}, +, \cdot)$ was extended in a corresponding way by adjoining the number $a = \sqrt{2}$. The condition $i^2 = -1$ imposed on the number i is satisfied by no $i \in \mathbb{R}$, so i is a 'genuinely imaginary' number. In the field of phasors $(\mathbb{P}, +, \cdot)$ the arithmetic rule $i^2 = -1$ corresponds to the result

$$\vec{i} \cdot \vec{i} = 1\angle\pi = -(1\angle 0) = -\vec{1}.$$

By Definition III.2.1, one can operate with complex numbers more or less normally, i.e., in the manner familiar from the algebra of real numbers. The only departure from the normal is the rule $i^2 = -1$. In computation the representations of a complex number can be varied freely within the conversion rules of the previous section. Most commonly one uses either the *standard form* of a complex number, as in Definition III.2.1, or else the polar form. Between the different representations of the same number we shall henceforth use either the correspondence symbol ' $\hat{=}$ ' or, where natural, simply the identification '='.

EXAMPLE 1 Put the complex number $(2 + 3i) \cdot (5 - 2i)$ into standard form $x + iy$.

Solution By Definition III.2.1,

$$(2 + 3i) \cdot (5 - 2i) = 2 \cdot 5 - 2 \cdot 2i + 5 \cdot 3i - 3 \cdot 2 \cdot i^2 = 16 + 11i. \quad \square$$

EXAMPLE 2 Let $z = a + ib \neq 0$. Compute z^{-1} in standard form.

Solution If $z^{-1} = x + iy$, then one must have

$$\begin{aligned} (a + ib)(x + iy) &= 1 \\ \Leftrightarrow (ax - by) + i(bx + ay) &= 1 = 1 + 0i \\ \Leftrightarrow \begin{cases} ax - by = 1 \\ bx + ay = 0 \end{cases} \\ \Leftrightarrow x = \frac{a}{a^2 + b^2}, \quad y &= -\frac{b}{a^2 + b^2}. \end{aligned}$$

Thus

$$z^{-1} = \frac{1}{a^2 + b^2}(a - ib) = \frac{a}{a^2 + b^2} - \frac{b}{a^2 + b^2}i. \quad \square$$

DEFINITION III.2.2 For a complex number

$$z = x + iy \in \mathbb{C} \hat{=} (x, y) \in \mathbb{R}^2 \hat{=} r\angle\varphi \in \mathbb{P}$$

- the *real part* is $\operatorname{Re} z = x = r \cos \varphi$.
- the *imaginary part* is $\operatorname{Im} z = y = r \sin \varphi$.
- the *absolute value* (modulus) is $|z| = \sqrt{x^2 + y^2} = r$.
- the *phase angle* (argument, polar angle) is $\arg z = \varphi$.
- the *complex conjugate*, or *conjugate*, is $\bar{z} = x - iy \hat{=} (x, -y) \hat{=} r\angle -\varphi$.

When computing with complex numbers, the following formulas connected with Definition III.2.2 are often needed:

- (1) $\overline{z_1 + z_2} = \bar{z}_1 + \bar{z}_2$
- (2) $\overline{z_1 z_2} = \bar{z}_1 \bar{z}_2$
- (3) $z\bar{z} = |z|^2$
- (4) $|z_1 z_2| = |z_1| |z_2|$
- (5) $|z^{-1}| = |z|^{-1}$
- (6) $|\bar{z}| = |z|$
- (7) $z + \bar{z} = 2 \operatorname{Re} z, \quad z - \bar{z} = 2i \operatorname{Im} z$
- (8) $\overline{\bar{z}} = z$

These can easily be justified from the definitions (formulas (2)–(5) most directly from the polar form).

EXAMPLE 2 (continued) In the Example the reciprocal z^{-1} of a complex number $z \neq 0$ was computed. Since $z \neq 0 \Leftrightarrow \bar{z} \neq 0$, one can also reason as follows, using formula (3):

$$\begin{aligned} zz^{-1} &= 1 \\ \Leftrightarrow \bar{z}(zz^{-1}) &= \bar{z} \\ \Leftrightarrow (\bar{z}z)z^{-1} &= \bar{z} \\ \Leftrightarrow |z|^2 z^{-1} &= \bar{z} \\ \Leftrightarrow z^{-1} &= |z|^{-2} \bar{z}. \end{aligned}$$

This corresponds to the usual procedure of multiplying numerator and denominator by the conjugate:

$$z^{-1} = \frac{1}{a+ib} = \frac{(a-ib)}{(a+ib)(a-ib)} = \frac{1}{a^2+b^2}(a-ib).$$

Even faster, however, is the phasor computation:

$$z = r\angle\varphi \Rightarrow z^{-1} = r^{-1}\angle -\varphi = r^{-2}(r\angle -\varphi) = |z|^{-2}\bar{z}. \quad \square$$

EXAMPLE 3 Put $z = (1+i)^7$ into standard form $x+iy$.

Solution Here too the polar representation is the most efficient: since

$$1+i = r\angle\varphi, \quad \text{where} \quad r = \sqrt{1+1} = \sqrt{2}, \quad \varphi = \frac{\pi}{4},$$

we have

$$\begin{aligned} (1+i)^7 &= (\sqrt{2})^7 \angle 7\pi/4 \\ &= 8\sqrt{2} \angle -\pi/4 \\ &= 8\sqrt{2} (\cos \frac{\pi}{4} - i \sin \frac{\pi}{4}) \\ &= 8(1-i). \quad \square \end{aligned}$$

Among the formulas of complex algebra, one worth mentioning is also *de Moivre's formula*

$$(9) \quad (\cos \varphi + i \sin \varphi)^n = \cos n\varphi + i \sin n\varphi.$$

This follows immediately when the result of the phasor computation

$$(1\angle\varphi)^n = 1\angle n\varphi$$

is expressed in the standard form of complex numbers.

EXAMPLE 4 By de Moivre's formula and the binomial formula,

$$\begin{aligned}
 \cos 3\varphi + i \sin 3\varphi &= (\cos \varphi + i \sin \varphi)^3 \\
 &= \cos^3 \varphi + 3i \cos^2 \varphi \sin \varphi + 3i^2 \cos \varphi \sin^2 \varphi + i^3 \sin^3 \varphi \\
 &= (\cos^3 \varphi - 3 \cos \varphi \sin^2 \varphi) + i(3 \cos^2 \varphi \sin \varphi - \sin^3 \varphi) \\
 \Rightarrow \begin{cases} \cos 3\varphi &= \cos^3 \varphi - 3 \cos \varphi \sin^2 \varphi, \\ \sin 3\varphi &= 3 \cos^2 \varphi \sin \varphi - \sin^3 \varphi \end{cases}
 \end{aligned}$$

(cf. Example 9 of Section II.5). $\square$

The triangle inequality in $\mathbb{C}$

The addition of complex numbers is the same operation as the addition of plane vectors, and the absolute value of a complex number likewise corresponds to the absolute value of a vector. For this reason the triangle inequality holds for complex numbers in the same form as for vectors:

$$||z_1| - |z_2|| \leq |z_1 + z_2| \leq |z_1| + |z_2|, \quad z_1, z_2 \in \mathbb{C}.$$

The triangle inequality is a very central tool in the mathematical analysis based on complex numbers, i.e., in *complex analysis*. As a typical example of the use of the triangle inequality, let us consider an assertion concerning a general *polynomial in a complex variable* of the form

$$p(z) = c_0 + c_1 z + \cdots + c_n z^n = \sum_{k=0}^n c_k z^k, \quad c_n \neq 0.$$

Here, then, $z \in \mathbb{C}$ is a *complex variable*, and the numbers c_k , i.e., the *coefficients* of the polynomial, are also complex numbers. The number $n \in \mathbb{N} \cup \{0\}$ is the *degree* of the polynomial.

When the triangle inequality and the aforementioned formulas (4), (5) are applied to a polynomial in a complex variable, one arrives at the assertion that the absolute value $|p(z)|$ of the polynomial grows, for sufficiently large values of $|z|$, qualitatively in the same way as the absolute value of the highest-degree term of the polynomial, i.e., for sufficiently large values of $|z|$

$$|p(z)| \sim |c_n z^n| = |c_n| |z|^n$$

(here formula (4) has been used). The assertion can be formulated more precisely, e.g., as follows (see also Exercise 10):

PROPOSITION III.2.3 (Growth of a polynomial) If $p(z) = \sum_{k=0}^n c_k z^k$, where $c_k \in \mathbb{C}$, $n \in \mathbb{N}$ and $c_n \neq 0$, then there exists $R \in \mathbb{R}_+$ such that

$$\frac{1}{2}|c_n||z|^n \leq |p(z)| \leq \frac{3}{2}|c_n||z|^n, \quad \text{when } z \in \mathbb{C} \text{ and } |z| \geq R.$$

Proof Since $c_n \neq 0$, one can write

$$p(z) = c_n z^n (1 + b_{n-1} z^{-1} + \cdots + b_0 z^{-n}),$$

where $b_k = c_k/c_n$. Using the first and second parts of the triangle inequality in the forms

$$|z_1 + z_2| \geq |z_1| - |z_2|, \quad -|z_1 + z_2| \geq -|z_1| - |z_2|$$

together with formulas (4), (5), one deduces

$$\begin{aligned} |p(z)| &= |c_n||z|^n |1 + b_{n-1} z^{-1} + \cdots + b_0 z^{-n}| \\ &\geq |c_n||z|^n (1 - |b_{n-1} z^{-1} + \cdots + b_0 z^{-n}|) \\ &\geq |c_n||z|^n (1 - |b_{n-1} z^{-1}| - \cdots - |b_0 z^{-n}|) \\ &= |c_n||z|^n \left(1 - \frac{|b_{n-1}|}{|z|} - \cdots - \frac{|b_0|}{|z|^n} \right). \end{aligned}$$

When

$$|z| \geq \max\{1, 2(|b_{n-1}| + \cdots + |b_0|)\} = R,$$

then

$$\frac{|b_{n-1}|}{|z|} + \cdots + \frac{|b_0|}{|z|^n} \leq \frac{|b_{n-1}| + \cdots + |b_0|}{|z|} \leq \frac{1}{2},$$

so the first part of the assertion follows:

$$|p(z)| \geq \frac{1}{2}|c_n||z|^n, \quad \text{when } |z| \geq R.$$

The latter part follows correspondingly by applying the latter part of the triangle inequality. $\square$

EXAMPLE 5 If $p(z) = z^5 - 4iz^3 + 10z + (30 + 40i)$, then by following the logic of the above proof one sees that

$$\begin{aligned} |p(z)| &\geq |z|^5 \left(1 - \frac{|5i|}{|z|^2} - \frac{10}{|z|^4} - \frac{|30 + 40i|}{|z|^5} \right) \\ &= |z|^5 \left(1 - \frac{5}{|z|^2} - \frac{10}{|z|^4} - \frac{50}{|z|^5} \right) \\ &\geq |z|^5 \left(1 - \frac{5}{|z|} - \frac{10}{|z|} - \frac{50}{|z|} \right) \\ &= |z|^5 \left(1 - \frac{65}{|z|} \right), \quad \text{when } |z| \geq 1. \end{aligned}$$

Correspondingly one deduces that $|p(z)| \leq |z|^5(1 + 65|z|^{-1})$, when $|z| \geq 1$, so it follows that

$$\frac{1}{2}|z|^5 \leq |p(z)| \leq \frac{3}{2}|z|^5, \quad \text{when } |z| \geq 130 = R. \quad \square$$

EXERCISES

1. Convert the following complex numbers into standard form $x + iy$ from the given phasor form, or into phasor form from the given standard form. Use approximate values where necessary.

$$\text{a) } \sqrt{2} \angle -\frac{5\pi}{4} \quad \text{b) } \sqrt{3} - 2i \quad \text{c) } 2 \angle 700^\circ \quad \text{d) } -3 - 4i$$

2. Put the following complex numbers into standard form $x + iy$ and into polar form:

$$\text{a) } (1 + i)(1 - i)^5 \quad \text{b) } (1 - i)(1 + i\sqrt{3})^{-1} \quad \text{c) } (\sqrt{2} + 1 + i)^8$$

3. Compute the absolute value and phase angle using the polar form:

$$\text{a) } (1 + i)^6 \quad \text{b) } (1 - i\sqrt{3})(1 - i)^2 \quad \text{c) } (2 - 2i)(\sqrt{3} + i)^{-2}$$

4. Let $z = \frac{1}{2}(1 + i\sqrt{3})$. For which $n \in \mathbb{Z}$ does a) $z^n = z$, b) $z^n = -z$ hold?

5. For the absolute values of complex numbers, $|z_1 z_2|^2 = |z_1|^2 |z_2|^2$ holds. Verify the validity of the formula directly from the standard form of complex numbers, i.e., by writing

$$z_1 = x_1 + iy_1, \quad z_2 = x_2 + iy_2.$$

6. Using de Moivre's formula, derive:

$$\text{a) an expression for } \sin 5x \text{ as a polynomial in } \sin x,$$

$$\text{b) an expression for } \sin^5 x \text{ in the form } a \sin x + b \sin 3x + c \sin 5x, \quad a, b, c \in \mathbb{R}.$$

7. If an order relation is defined in the field of complex numbers, then by the first order axiom (J1) one must have either $i > 0$, $i = 0$ or $i < 0$. Deduce that no order relation can be defined.

8. Find some R such that

$$|z|^5 \leq |2z^5 + 1000z^4 - 10^4(3 + 4i)| \leq 3|z|^5, \quad \text{when } |z| \geq R.$$

9. (*) Prove that for all $z_1, z_2 \in \mathbb{C}$, $(1 + |z_1|^2)(1 + |z_2|^2) \geq |1 + z_1 z_2|^2$ holds.

10. (*) Show that under the assumptions of Proposition [III.2.3](#) there exists $R \in \mathbb{R}_+$ such that

$$0.999|c_n||z|^n \leq |p(z)| \leq 1.001|c_n||z|^n, \quad \text{when } z \in \mathbb{C} \text{ and } |z| \geq R.$$

Of what still more general assertion is this a special case?

III.3 The fundamental theorem of algebra

Function theory, a subfield of complex analysis, studies complex-valued functions of a complex variable of type

$$f : \mathbb{C} \rightarrow \mathbb{C} \quad \text{or} \quad f : A \rightarrow \mathbb{C}, \quad A \subset \mathbb{C}.$$

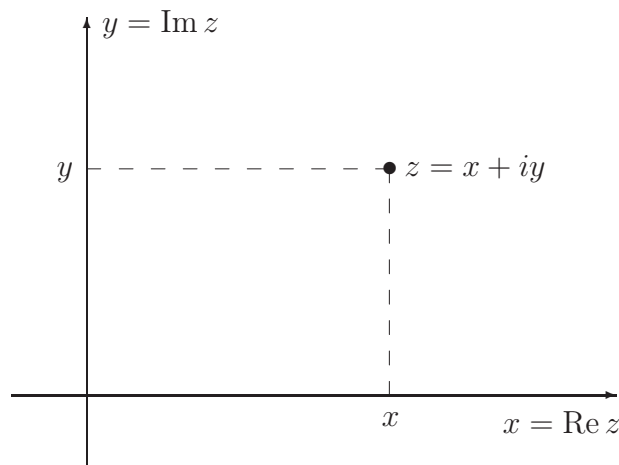
The simplest (but not the least important) example of such a function is the polynomial function, defined on all of $\mathbb{C}$, considered in the previous section (Proposition III.2.3)

$$p(z) = \sum_{k=0}^n c_k z^k, \quad c_k \in \mathbb{C}, \quad c_n \neq 0. \quad (1)$$

Every complex-valued function of a complex variable can be expressed in the form

$$f(z) = \operatorname{Re} f(z) + i \operatorname{Im} f(z) = u(x, y) + iv(x, y), \quad z = x + iy,$$

where u, v are functions of type $u, v : \mathbb{R}^2 \rightarrow \mathbb{R}$ (cf. Section II.5). Thus f itself can also be interpreted as a function of type $f : \mathbb{R}^2 \rightarrow \mathbb{C}$. It is indeed natural to think of functions of a complex variable geometrically as being defined in the plane. In such considerations the plane is called the *complex plane*. In the complex plane the point (x, y) thus means the complex number $z = x + iy$.



A *root* of a complex function f , or simply a *zero*, is any $z \in \mathbb{C}$ for which

$$f(z) = 0.$$

From the form decomposed into real and imaginary parts, the roots $z = x + iy$ are obtained from the solutions of the system of equations

$$\begin{cases} u(x, y) = 0 \\ v(x, y) = 0 \end{cases}$$

.

EXAMPLE 1 $f(z) = z^3 + 1$. Roots?

Solution Here

$$\begin{aligned} f(x + iy) &= (x + iy)^3 + 1 \\ &= (x^3 - 3xy^2 + 1) + (3x^2y - y^3)i, \end{aligned}$$

so the zeros of f are found from the system of equations

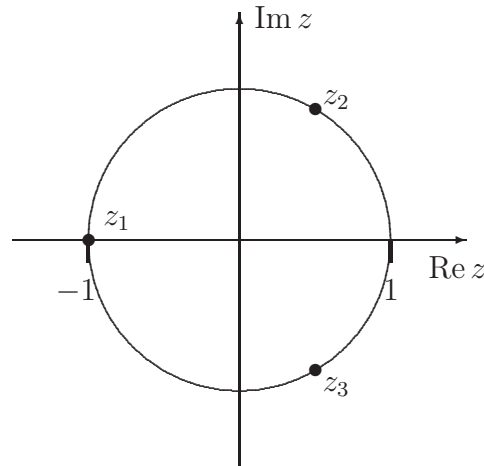
$$\begin{cases} x^3 - 3xy^2 + 1 = 0, \\ 3x^2y - y^3 = 0. \end{cases}$$

To each real solution $(x, y) \in \mathbb{R}^2$ of the system corresponds a root $x + iy \in \mathbb{C}$ of f . The roots are easily found here, because the latter equation of the system can be written in the form $(3x^2 - y^2)y = 0$. Hence one must have either $y = 0$ or $y^2 = 3x^2 \Leftrightarrow y = \pm\sqrt{3}x$. Substituting these into the former equation gives the roots

$$(x_1, y_1) = (-1, 0) \quad \hat{=} \quad z_1 = -1,$$

$$(x_2, y_2) = \left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right) \quad \hat{=} \quad z_2 = \frac{1}{2}(1 + i\sqrt{3}),$$

$$(x_3, y_3) = \left(\frac{1}{2}, -\frac{\sqrt{3}}{2}\right) \quad \hat{=} \quad z_3 = \frac{1}{2}(1 - i\sqrt{3}).$$



In the Example f was a polynomial of degree 3 and three roots were found for it. The result is a special case of a theorem that is both far-reaching and profound:

*THEOREM III.3.1 (**Fundamental theorem of algebra**) If p is a polynomial of the form (1) and of degree $n \geq 1$, then there exist n complex numbers $z_1, \dots, z_n$ such that

$$\begin{aligned} p(z) &= c_n(z - z_1) \cdots (z - z_n) \\ &= c_n \prod_{i=1}^n (z - z_i). \end{aligned}$$

In the product decomposition of Theorem III.3.1, the numbers z_i are not necessarily distinct, so the polynomial may have a factor of the form

$$(z - z_k)^m, \quad 1 \leq m \leq n,$$

associated with the root z_k . If p has this factor but not the factor $(z - z_k)^{m+1}$, then the root z_k is said to be m -fold, or that the *order* of the root is m . If the order is $m = 1$, the root is said to be *simple*. If the factors representing the same root are collected together, the product decomposition of Theorem III.3.1 takes the form

$$p(z) = c_n \prod_{i=1}^{\nu} (z - z_i)^{m_i}, \quad (2)$$

where now $z_i \neq z_j$ when $i \neq j$, $m_i \in \mathbb{N}$ and

$$\sum_{i=1}^{\nu} m_i = n.$$

In particular, if in the decomposition (2) we have $\nu = n$, i.e., the polynomial has n distinct roots, then all the roots must be simple. This was the situation in Example 1.

In Example 1 the genuinely complex roots ($\operatorname{Im} z \neq 0$) occurred as a *conjugate pair*. This is more generally the case when (as in the example) the polynomial is *real-coefficient*, i.e., $c_k \in \mathbb{R}$ in the expression (1). This rests on the fact that in the case of a real-coefficient polynomial one has (cf. formulas (1)–(2) of the previous section)

$$\overline{p(z)} = p(\bar{z}) \quad (\text{real-coefficient polynomial}).$$

Then

$$p(z) = 0 \Rightarrow p(\bar{z}) = 0,$$

and the roots thus 'pair up'. If the roots forming a conjugate pair are

$$z = a \pm ib,$$

then the polynomial has as a factor

$$(z - a - ib)(z - a + ib) = (z - a)^2 + b^2.$$

A real-coefficient polynomial can thus always be decomposed into real-coefficient factors that are either of the form

$$(a) \quad (z - a_k)^m, \quad a_k \in \mathbb{R},$$

corresponding to an m -fold real root $z_k = a_k$, or of the form

$$(b) \quad [(z - a_k)^2 + b_k^2]^m \quad a_k, b_k \in \mathbb{R}, \quad b_k \neq 0,$$

corresponding to an m -fold conjugate root pair $z_k = a_k \pm ib_k$.

EXAMPLE 1 (continued) The polynomial of the Example decomposes into the form

$$\begin{aligned} z^3 + 1 &= (z + 1)(z^2 - z + 1) \\ &= (z + 1) \left[\left(z - \frac{1}{2} \right)^2 + \left(\frac{\sqrt{3}}{2} \right)^2 \right]. \quad \square \end{aligned}$$

The 'fundamental theorem of algebra' in the form presented above is in fact a technical consequence of the true *fundamental theorem of algebra*. The true fundamental theorem is formulated as follows:

*THEOREM III.3.2 (**Fundamental theorem of algebra – short form**) If $n \geq 1$, then the polynomial (1) has a root in the complex plane.

By Theorem III.3.2, the polynomial immediately factors according to Theorem III.3.1. Namely, if one root z_1 can be found, then $p(z)$ can be written in the form

$$p(z) = p(z) - p(z_1) = \sum_{k=1}^n c_k (z^k - z_1^k).$$

Here, by field algebra (Proposition I.2.2),

$$z^k - z_1^k = (z - z_1)(z^{k-1} + z_1 z^{k-2} + \dots + z_1^{k-1}), \quad k = 1 \dots n,$$

so one sees that $p(z)$ can be written in the form

$$p(z) = (z - z_1) p_1(z),$$

where p_1 is a polynomial of degree $n - 1$ and of the form

$$p_1(z) = c_n z^{n-1} + [\text{lower-degree terms}].$$

If $n \geq 2$, then p_1 in turn again has a root by Theorem III.3.2, whereupon a factor $(z - z_2)$ can be split off from p_1 , and so on, and the assertion of Theorem III.3.1 thus follows.

The consequences of the fundamental theorem of algebra are so extensive that mathematics — even applied mathematics — is nowadays hard to imagine without it. The theorem is also very elegant when presented in the aforementioned ‘short form’. Even peeking behind the principle of the proof does not diminish the elegance, rather the opposite: the fundamental theorem of algebra can nowadays be seen as a consequence of the more general theory of complex functions. More precisely, it is a matter of the theory of so-called *analytic* functions. The concept ‘analytic’ is defined later in Section V.8; in the present context let it merely be noted that these are extremely regular complex functions, for example just such as the polynomials.

The basic idea of the proof of the fundamental theorem of algebra, presented by means of the theory of analytic functions, is as follows*: We make the counterassumption that the polynomial p has no zeros. In that case the function

$$f(z) = \frac{1}{p(z)}$$

is regular on the whole complex plane. In fact f is ‘extremely regular’, i.e., analytic. When one additionally uses the result of Proposition III.2.3, one can state: if the degree of p is $n \geq 1$ and p has no zeros, then the function $f = 1/p$ satisfies

1. f is analytic on the whole complex plane.
2. $f(z) \rightarrow 0$ as $|z| \rightarrow \infty$.

There is at least one function that satisfies both of these conditions, namely $f(z) = 0$. The question is: are there others? The answer given by the theory of analytic functions to this question — somewhat surprisingly — is:

*Among those who devoted themselves to proving the fundamental theorem of algebra was the German **Carl Friedrich Gauss** (1777-1855), called the ‘prince of mathematicians’. Gauss presented a very plausible proof in his doctoral dissertation in 1799. A logically fully rigorous proof could not be presented until later in the 19th century, once real and complex analysis had developed sufficiently. — Later, in Section V.10, a fairly straightforward proof of the fundamental theorem of algebra is presented, based only on real analysis, Proposition III.2.3, and another assertion of complex algebra presented in Exercise 13.

No!

The counterassumption has thus led to a logical contradiction, and the theorem is therefore true.

Roots of a complex number

By the fundamental theorem of algebra, the polynomial equation

$$z^n = a, \quad a \in \mathbb{C}, \quad n \in \mathbb{N}, \quad n \geq 2 \quad (3)$$

has at least one root. In fact it turns out that if $a \neq 0$ (which is assumed), then there are exactly n roots, i.e., all the roots are simple. They are all denoted by the symbol

$$z = a^{1/n} = \sqrt[n]{a},$$

and are called the *roots of the complex number a* . There will thus be n of them, and we wish to compute them in standard form

$$z_k = x_k + iy_k, \quad k = 0 \dots n-1.$$

This is most easily achieved via the polar representation: we write

$$a = r \angle \varphi = r(\cos \varphi + i \sin \varphi),$$

where $r = |a| \neq 0$. Writing $z = |z| \angle \psi$, we have

$$\begin{aligned} z^n = a &\Leftrightarrow |z|^n = r \quad \& \quad n\psi = \varphi + k \cdot 2\pi, \quad k \in \mathbb{Z} \\ &\Leftrightarrow |z| = \sqrt[n]{r} \quad \& \quad \psi = \frac{\varphi}{n} + k \cdot \frac{2\pi}{n}, \quad k \in \mathbb{Z}. \end{aligned}$$

Distinct solutions are obtained from this for the index values $k = 0 \dots n-1$. These can be expressed in the form

$$\begin{aligned} z_k &= \sqrt[n]{r} \angle \frac{\varphi}{n} + k \cdot \frac{2\pi}{n} \\ &= \left(\sqrt[n]{r} \angle \frac{\varphi}{n} \right) \left(1 \angle k \cdot \frac{2\pi}{n} \right) \\ &= \left(\sqrt[n]{r} \angle \frac{\varphi}{n} \right) \left(1 \angle \frac{2\pi}{n} \right)^k \\ &= z_0 \rho^k, \quad k = 0 \dots n-1, \end{aligned}$$

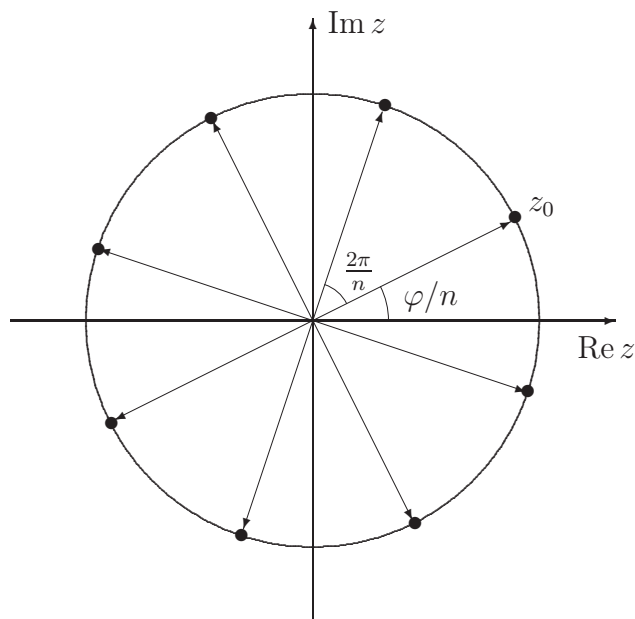
where

$$\begin{aligned} z_0 &= \sqrt[n]{r} \angle \frac{\varphi}{n} = \sqrt[n]{r} \left(\cos \frac{\varphi}{n} + i \sin \frac{\varphi}{n} \right), \quad r = |a|, \quad \varphi = \arg a, \\ \rho &= 1 \angle \frac{2\pi}{n} = \cos \frac{2\pi}{n} + i \sin \frac{2\pi}{n}. \end{aligned}$$

In standard form the solutions of equation (3) are

$$z_k = \sqrt[n]{r} \left[\cos \left(\frac{\varphi}{n} + k \cdot \frac{2\pi}{n} \right) + i \sin \left(\frac{\varphi}{n} + k \cdot \frac{2\pi}{n} \right) \right], \quad k = 0 \dots n-1.$$

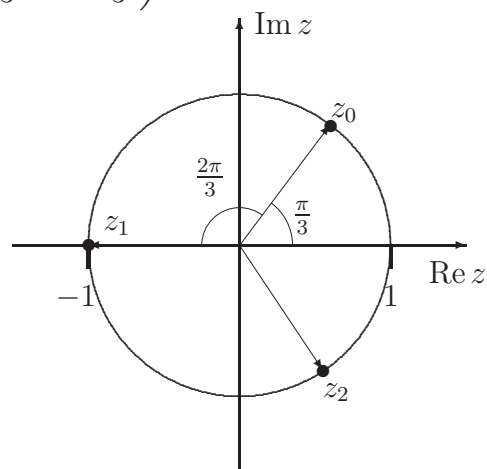
Starting from the principal root z_0 , the roots are located at equal intervals on a circle in the complex plane whose radius $= \sqrt[n]{|a|}$. The figure shows the case $n = 8$.



EXAMPLE 1 (continued) Here $a = -1 = 1\angle\pi$, so $r = 1$, $\varphi = \pi$. The roots are

$$z_k = \cos \left(\frac{\pi}{3} + k \cdot \frac{2\pi}{3} \right) + i \sin \left(\frac{\pi}{3} + k \cdot \frac{2\pi}{3} \right), \quad k = 0, 1, 2,$$

$$\text{i.e. } \begin{cases} z_0 = \frac{1}{2} + \frac{\sqrt{3}}{2}i, \\ z_1 = -1, \\ z_2 = \frac{1}{2} - \frac{\sqrt{3}}{2}i. \end{cases} \quad \square$$



EXAMPLE 2 $\sqrt{i} = ?$

Solution This amounts to solving the equation $z^2 = i$. Since $i = 1\angle\pi/2$, the solutions are

$$z_k = \cos\left(\frac{\pi}{4} + k\pi\right) + i \sin\left(\frac{\pi}{4} + k\pi\right), \quad k = 0, 1$$

$$\Rightarrow \sqrt{i} = \pm \frac{1}{\sqrt{2}}(1 + i).$$

Check:

$$(\sqrt{i})^2 = \frac{1}{2}(1 + i)^2 = \frac{1}{2}(1 - 1 + 2i) = i. \quad \text{OK!} \quad \square$$

EXAMPLE 3 Solve the equation $z^2 + (2 + 2i)z - i = 0$.

Solution The quadratic formula rests only on field algebra, so it is valid also for a complex-coefficient polynomial. The solutions are thus

$$z_{1,2} = -(1 + i) \pm \sqrt{(1 + i)^2 + i} = -(1 + i) \pm \sqrt{3i},$$

i.e., by the previous example,

$$z_1 = -(1 + i) + \sqrt{\frac{3}{2}}(1 + i) = \left(\sqrt{\frac{3}{2}} - 1\right)(1 + i),$$

$$z_2 = -(1 + i) - \sqrt{\frac{3}{2}}(1 + i) = -\left(\sqrt{\frac{3}{2}} + 1\right)(1 + i). \quad \square$$

EXERCISES

- Find all roots in the complex plane of the following functions by splitting the functions into real and imaginary parts.
 - $f(z) = z^2 + i$
 - $f(z) = z^3 + 8i$
 - $f(z) = z^2 + 4\bar{z} - 1$
- If $\operatorname{Re} z = \operatorname{Im} z = a$, for which values of a does $|z - i| < |z - 3|$ hold? Draw a figure!
- Determine what kinds of point sets in the complex plane are defined by the following conditions. Draw the figures.
 - $|z + 1 + i|^2 = 2$
 - $|z - i| \leq 2$
 - $|z + i| = |z - 1 - i|$
 - $\operatorname{Re}[(z - i)/(z + i)] = 0$

4. Factor the polynomial into real-coefficient factors of degree at most two:
 - a) $z^3 + 2z^2 + 3z + 2$ b) $z^4 + 2$ c) $z^4 + 2z^3 + z^2 - 2z - 2$
 - d) $z^8 - 256$ e) $(z^4 + 1)^2 - z^4$
5. Find a) all real-coefficient polynomials, b) all polynomials p of degree ≤ 5 , having the properties: $p(0) = 1$ and the zeros of p are $1, i$ and $-i$.
6. Find all values of the following roots:
 - a) $\sqrt[4]{-4}$ b) $\sqrt[6]{-64}$ c) $\sqrt[3]{i-1}$ d) $\sqrt{3+4i}$ e) $\sqrt{-7+24i}$
7. One approximate value of the root $\sqrt[6]{2+3i}$ is $1.2217 + 0.2019i$. Draw a figure placing all values of the root in the complex plane (without separately computing the approximate values of the other roots).
8. Find all solutions of the following equations in standard form $x + iy$.
 - a) $z^2 + 2iz - i - 1 = 0$ b) $z^2 - 4iz - 4 + i = 0$ c) $z^2 - (3+5i)z = 4 - 7i$
 - d) $z^4 - 2z^2 + 4 = 0$ e) $z^4 + (1 - 2i\sqrt{3})z^2 - 3 - i\sqrt{3} = 0$
9. Find $a \in \mathbb{C}$ such that $z = 1+i$ is a root of the polynomial $p(z) = z^3 + az + 1$, and then compute the other roots.
10. (*) a) The roots $\sqrt[4]{3+4i}$ can be computed exactly in standard form $z_k = x_k + iy_k$ such that x_k and y_k are geometric numbers. Compute them!
 b) Determine exactly those complex numbers whose fifth power = 1. Draw a figure of the locations of the numbers in the complex plane.
11. (*) Let $p(z) = c_n z^n + \dots + c_0$ be a real-coefficient polynomial, $c_n \neq 0$. Show that the polynomial has at least one real zero if either n is odd or $c_n c_0 \leq 0$.
12. (*) Prove that if two polynomials of degree $\leq n$ have the same values at $n+1$ distinct points of the complex plane, then the polynomials are equal.
13. (*) Let $p(z)$ be a polynomial in a complex variable that is not constant, and let $c \in \mathbb{C}$ and $p(c) = a \in \mathbb{C}$. Show:
 - a) There exist $m \in \mathbb{N}$ and $b \in \mathbb{C}$, $b \neq 0$ as well as a polynomial $q(z)$ such that

$$p(z) = a + b(z - c)^m + (z - c)^{m+1}q(z) \quad \forall z \in \mathbb{C}.$$
 - b) If in the decomposition of part a) we have $a \neq 0$ and $q(z) = 0$, then $|p(z)|$ decreases in some direction on setting out from c , i.e., there exist $\rho \in \mathbb{C}$, $|\rho| = 1$ and $\delta > 0$ such that

$$|p(c + t\rho)| < |p(c)|, \quad \text{when } 0 < t < \delta.$$

- c) If in the decomposition of part a) we have $a \neq 0$ and $q(z) \neq 0$, then the assertion of part b) is still true.
- d) If $|p(z)|$ attains a local minimum value at c , then $p(c) = a = 0$.

III.4 *The complex vector space

In Section II.2 the algebra called a (linear) vector space was introduced, symbolically $(U, \mathbb{K})$, where U is the set formed by the vectors and $\mathbb{K}$ is the so-called field of scalars of the vector space. In a vector space two operations are defined, the addition of vectors and multiplication by a scalar, which are functions of type

$$\begin{aligned} \text{addition:} & & U \times U &\rightarrow U, \\ \text{multiplication by a scalar:} & & \mathbb{K} \times U &\rightarrow U. \end{aligned}$$

An inner product space is a vector space in which an inner product is also defined (see Section II.4) as a function of type

$$\text{inner product:} \quad U \times U \rightarrow \mathbb{K}.$$

Up to now we have mainly treated the case $\mathbb{K} = \mathbb{R}$, in which case one speaks of a *real* space. The extension from real numbers to complex numbers now makes possible also the *complex* vector space, in which $\mathbb{K} = \mathbb{C}$. As an algebra, a complex vector space does not differ essentially from a real one as long as only the addition of vectors and multiplication by a scalar are defined. On the other hand, the definition of the inner product as a function of type $U \times U \rightarrow \mathbb{C}$ introduces additional features into the theory that must be taken into account. The inner product $\langle \cdot, \cdot \rangle$ of a complex inner product space $(U, \mathbb{C})$ must satisfy the following conditions (cf. the conditions for the real case in Section II.4):

1. *Conjugate symmetry*

$$\langle \mathbf{u}, \mathbf{v} \rangle = \overline{\langle \mathbf{v}, \mathbf{u} \rangle} \quad \forall \mathbf{u}, \mathbf{v} \in U.$$

2. *Sesquilinearity*

$$\begin{aligned} \text{(a)} \quad & \langle \alpha \mathbf{u} + \beta \mathbf{v}, \mathbf{w} \rangle = \alpha \langle \mathbf{u}, \mathbf{w} \rangle + \beta \langle \mathbf{v}, \mathbf{w} \rangle, \\ \text{(b)} \quad & \langle \mathbf{u}, \alpha \mathbf{v} + \beta \mathbf{w} \rangle = \overline{\alpha} \langle \mathbf{u}, \mathbf{v} \rangle + \overline{\beta} \langle \mathbf{u}, \mathbf{w} \rangle \\ & \forall \mathbf{u}, \mathbf{v}, \mathbf{w} \in U, \alpha, \beta \in \mathbb{C}. \end{aligned}$$

3. *Positive definiteness*

$$\begin{aligned} \text{(a)} \quad & \langle \mathbf{u}, \mathbf{u} \rangle \geq 0 \quad \forall \mathbf{u} \in U, \\ \text{(b)} \quad & \langle \mathbf{u}, \mathbf{u} \rangle = 0 \Leftrightarrow \mathbf{u} = \mathbf{0}. \end{aligned}$$

One observes that the only conditions differing from the real case are the symmetry condition (1) and the sesquilinearity condition (2b) (sesquilinear = $1\frac{1}{2}$ —linear

– cf. bilinear = doubly linear). In fact only the difference in the symmetry condition is essential, since (1) & (2a) $\Rightarrow$ (2b) (Exercise 1). Note that the symmetry condition (1) contains as a special case the condition $\langle \mathbf{u}, \mathbf{v} \rangle = \langle \mathbf{v}, \mathbf{u} \rangle$ when $\langle \mathbf{u}, \mathbf{v} \rangle \in \mathbb{R}$, so the conditions imposed earlier for the real case are no longer needed separately. Since condition (1) also guarantees that $\langle \mathbf{u}, \mathbf{u} \rangle \in \mathbb{R} \ \forall \mathbf{u} \in U$, condition (3a) is meaningful. Of the new conditions, the ones needed as axioms are (1), (2a), (3a) and the $[\Rightarrow]$ part of (3b), since (1) & (2a) $\Rightarrow$ (2b), as stated, and (2a) $\Rightarrow$ the $[\Leftarrow]$ part of (3b).

A central property of the inner product is the Cauchy–Schwarz inequality, which was freed from geometry in Section II.4. We now verify that the inequality holds also in the complex case (cf. Theorem II.4.2).

THEOREM III.4.1 For every inner product of a complex vector space U satisfying axioms 1–3, the Cauchy–Schwarz inequality holds:

$$|\langle \mathbf{u}, \mathbf{v} \rangle| \leq |\mathbf{u}| |\mathbf{v}|,$$

where

$$|\mathbf{u}| = \langle \mathbf{u}, \mathbf{u} \rangle^{1/2}.$$

Proof The case $\mathbf{u} = \mathbf{0}$ is again clear, so we may assume $\mathbf{u} \neq \mathbf{0}$, in which case $\langle \mathbf{u}, \mathbf{u} \rangle > 0$ by axioms (3a,b). We again start from the inequality

$$\langle \beta \mathbf{u} + \mathbf{v}, \beta \mathbf{u} + \mathbf{v} \rangle \geq 0,$$

which now holds $\forall \beta \in \mathbb{C}$ (axiom (3a)).

Using the sesquilinearity conditions (2a,b), the symmetry condition (1) and formulas (2), (3), (7) of Section III.2, this unfolds into the form

$$\begin{aligned} & \langle \beta \mathbf{u}, \beta \mathbf{u} \rangle + \langle \beta \mathbf{u}, \mathbf{v} \rangle + \langle \mathbf{v}, \beta \mathbf{u} \rangle + \langle \mathbf{v}, \mathbf{v} \rangle \geq 0 \\ \Leftrightarrow & |\beta|^2 \langle \mathbf{u}, \mathbf{u} \rangle + 2 \operatorname{Re} \{ \beta \langle \mathbf{u}, \mathbf{v} \rangle \} + \langle \mathbf{v}, \mathbf{v} \rangle \geq 0. \end{aligned}$$

Since this holds $\forall \beta \in \mathbb{C}$ and since $\langle \mathbf{u}, \mathbf{u} \rangle > 0$, one can choose

$$\beta = -\frac{\overline{\langle \mathbf{u}, \mathbf{v} \rangle}}{\langle \mathbf{u}, \mathbf{u} \rangle},$$

whereupon, by formulas (3), (6) of Section III.2, it follows that

$$\begin{aligned} & -\frac{|\langle \mathbf{u}, \mathbf{v} \rangle|^2}{\langle \mathbf{u}, \mathbf{u} \rangle} + \langle \mathbf{v}, \mathbf{v} \rangle \geq 0 \\ \Leftrightarrow & |\langle \mathbf{u}, \mathbf{v} \rangle|^2 \leq \langle \mathbf{u}, \mathbf{u} \rangle \langle \mathbf{v}, \mathbf{v} \rangle \\ \Leftrightarrow & |\langle \mathbf{u}, \mathbf{v} \rangle| \leq |\mathbf{u}| |\mathbf{v}|. \quad \square \end{aligned}$$

EXAMPLE 1 The simplest example of a complex inner product space is $\mathbb{C}$ itself, i.e., $(U, \mathbb{C})$, where $U = \mathbb{C}$. Namely, since the addition of complex numbers is the same operation as the addition of plane vectors, and multiplication can also be interpreted as multiplication by a scalar, $(\mathbb{C}, \mathbb{C})$ is a vector space. When the inner product

$$\langle u, v \rangle = u\bar{v}, \quad u, v \in \mathbb{C},$$

is defined in this space, the aforementioned conditions 1–3 are satisfied (cf. the formulas of Section III.2). The Cauchy–Schwarz inequality reduces in this special case to an equality:

$$|\langle u, v \rangle| = |u\bar{v}| = |u||v|, \quad u, v \in \mathbb{C}. \quad \square$$

EXAMPLE 2 Complex inner product spaces also include the extensions $(\mathbb{C}^2, \mathbb{C})$ and $(\mathbb{C}^3, \mathbb{C})$ of the spaces $(\mathbb{R}^2, \mathbb{R})$ and $(\mathbb{R}^3, \mathbb{R})$. For example, in $(\mathbb{C}^3, \mathbb{C})$ the inner product is defined as

$$\begin{aligned} \mathbf{u} &= (u_1, u_2, u_3) \in \mathbb{C}^3, \quad \mathbf{v} = (v_1, v_2, v_3) \in \mathbb{C}^3 : \\ \langle \mathbf{u}, \mathbf{v} \rangle &= \sum_{i=1}^3 u_i \bar{v}_i. \end{aligned}$$

The *Euclidean norm* of the space $(\mathbb{C}^n, \mathbb{C})$, $(n = 2, 3)$ is

$$|\mathbf{u}| = \langle \mathbf{u}, \mathbf{u} \rangle^{1/2} = \left(\sum_{i=1}^n |u_i|^2 \right)^{1/2}. \quad \square$$

EXERCISES

1. Show that the sesquilinearity property (2b) of the inner product follows from conditions (1) and (2a).
2. Are the vectors of the space $(\mathbb{C}^2, \mathbb{C})$

$$\mathbf{u} = (3 - 4i, 7 - i), \quad \mathbf{v} = (2 - 11i, 13 - 9i)$$

linearly independent?

3. Split the vector $\mathbf{u} = (1 + i, 2 - 3i)$ of the inner product space $(\mathbb{C}^2, \mathbb{C})$ into two components such that one component is parallel to the vector $\mathbf{v} = (1 - i, 2 + i)$ and the other is perpendicular to $\mathbf{v}$.

Chapter IV

Functions of real variables

The main types of mathematical functions are

- real-valued *functions* of *one* real *variable*, i.e. real functions
- real-valued *functions* of *several* real *variables*
- *vector-valued functions* of one or several real variables
- *complex functions*, i.e. complex-valued functions of a complex variable

In this chapter we begin the study of functions by considering real-valued functions of one or several—for now two or three—real variables, as well as vector-valued functions of one or two variables. What the function types under consideration have in common is the 'visible' form they take when numbers, pairs of numbers, and triples of numbers as variables, or vectors as function values, are understood as objects of Euclidean point spaces or of the corresponding vector spaces.

The study of functions is very central in mathematics, and consequently the associated body of concepts and repertoire of tools is remarkably broad. In this chapter the whole 'technology' is not yet brought into use; for now we restrict ourselves to the very simplest tools of algebra and geometry. On the other hand, with applications in mind (especially applications in physics), the selection of function types considered in this chapter is already fairly representative. The aim is, within this selection, to go through such concepts of the algebra of functions as the *algebraic combinations* of functions, the *composite function*, the *inverse function*, and the *implicit function*. The most clearly 'geometric functions' are the *parametric curves* and *parametric surfaces* presented in Section IV.4.

IV.1 Functions of one variable

By a (real-valued) function of one (real) variable, i.e. a real function, we mean a function of the type

$$f : \mathcal{D}_f \rightarrow \mathbb{R}, \quad \mathcal{D}_f \subset \mathbb{R}.$$

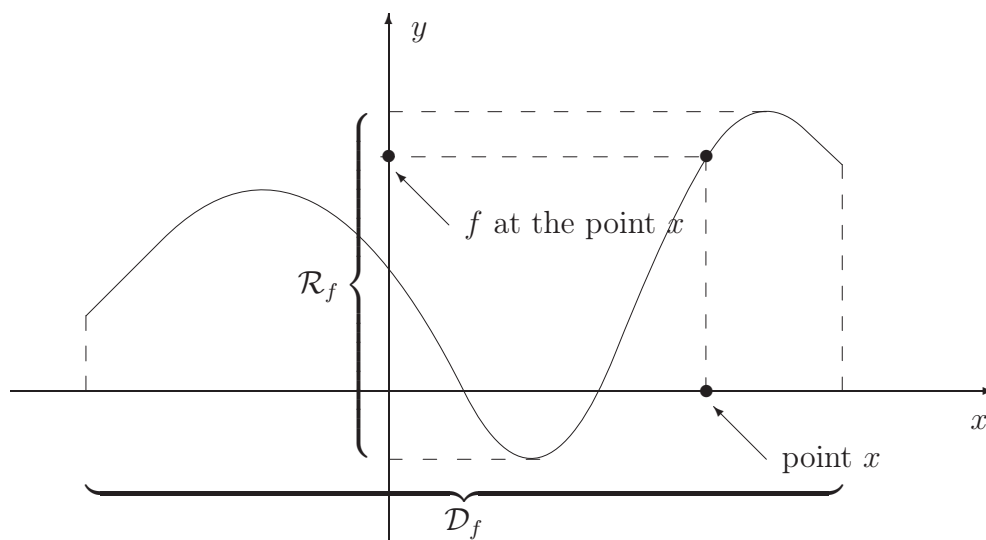
Here $\mathcal{D}_f$ is the *domain* of f (the source set) and the set

$$\mathcal{R}_f = \{y \in \mathbb{R} \mid y = f(x) \text{ for some } x \in \mathcal{D}_f\}$$

is the *range* of f .^{*} The *graph* of the function f on a set $A \subset \mathcal{D}_f$ is the point set defined by means of a (usually Cartesian) coordinate system of the Euclidean plane,

$$G_{f,A} = \{P \hat{=} (x, y) \in \mathbb{R}^2 \mid x \in A \text{ \& } y = f(x)\} \subset E^2.$$

The purpose of the graph is to 'geometrize' the function so that visual observations become possible.[†] For example, the range $\mathcal{R}_f$ is easy to discern from the graph, and an expression that often feels natural, such as ' f at the point x ' ($= f(x)$), also carries the geometrizing idea ($x \hat{=} P \in E^1$, cf. Section II.2).



A function f of a real variable is most often studied on some open, closed, or half-open interval, to which the domain of f may, if desired, be thought of as

^{*}In Finnish-language texts the notations M_f, A_f are also used for the domain and range.

[†]The graph is the geometric counterpart of the set-theoretic definition of a real function as a subset of $\mathbb{R}^2$, see Section II.5.

restricted in the analysis in question. Let us recall from Section I.9 the notations

$$\begin{aligned}\text{open interval: } (a, b) &= \{x \in \mathbb{R} \mid a < x < b\}, \\ \text{closed interval: } [a, b] &= \{x \in \mathbb{R} \mid a \leq x \leq b\}, \\ \text{half-open interval: } (a, b] \text{ or } [a, b)\end{aligned}$$

and also the notations for an 'infinite interval', which are in general use,

$$\begin{aligned}(0, \infty) &= \{x \in \mathbb{R} \mid x > 0\} = \mathbb{R}_+, \\ (-\infty, 0) &= \{x \in \mathbb{R} \mid x < 0\} = \mathbb{R}_-, \\ (-\infty, \infty) &= \mathbb{R}.\end{aligned}$$

Often a real function of one variable is specified only by its rule of computation (assignment rule) in the form 'the function $f(x)$ '. In that case it is assumed (unless stated otherwise) that the domain $\mathcal{D}_f$ is the largest possible one, i.e.

$$\mathcal{D}_f = \{x \in \mathbb{R} \mid y = f(x) \text{ is uniquely defined and } y \in \mathbb{R}\}.$$

EXAMPLE 1

$$\begin{aligned}\text{a) } f(x) &= x^4 + x^2 + 1 & \text{b) } f(x) &= x^2/(x-1)^2 & \text{c) } f(x) &= \cot x \\ \text{d) } f(x) &= x^{3/4} & \text{e) } f(x) &= \begin{cases} \cos x, & \text{when } x < 0 \\ x - x^2, & \text{when } 0 \leq x \leq 1 \\ 2 - \sqrt{x}, & \text{when } x > 1 \end{cases} & \text{f) } f(x) &= \sum_{k=1}^{\infty} \frac{x^k}{k^2}\end{aligned}$$

a) Polynomial: $\mathcal{D}_f = \mathbb{R}$, $\mathcal{R}_f = [1, \infty)$.

b) Rational function: $\mathcal{D}_f = \{x \in \mathbb{R} \mid x \neq 1\}$, $\mathcal{R}_f = [0, \infty)$.

c) Trigonometric function: $\mathcal{D}_f = \{x \in \mathbb{R} \mid x/\pi \notin \mathbb{Z}\}$, $\mathcal{R}_f = \mathbb{R}$.

d) Power function: $\mathcal{D}_f = \mathcal{R}_f = [0, \infty)$.

e) Piecewise defined function: $\mathcal{D}_f = \mathbb{R}$, $\mathcal{R}_f = (-\infty, 1]$.

f) Power series: The domain is $\mathcal{D}_f = [-1, 1]$ (cf. Section I.12). The range is harder to determine, but it turns out that $\mathcal{R}_f = [-\frac{\pi^2}{12}, \frac{\pi^2}{6}]$. (We omit the justification.) $\square$

The more general form of the rational function in part b) of the example is $f(x) = p(x)/q(x)$, where p and q are (real-coefficient) polynomials. The domain is then $\mathcal{D}_f = \{x \in \mathbb{R} \mid q(x) \neq 0\}$. The general form of the power function (part d) of the example) is $f(x) = x^\alpha$, where (for now) $\alpha \in \mathbb{Q}$. The domain is either $\mathcal{D}_f = [0, \infty)$ ($\alpha > 0$, $\alpha \notin \mathbb{N}$), $\mathcal{D}_f = (0, \infty) = \mathbb{R}_+$ ($\alpha < 0$, $\alpha \notin \mathbb{Z}$), $\mathcal{D}_f = \{x \in \mathbb{R} \mid x \neq 0\}$ ($\alpha \in \mathbb{Z}$, $\alpha \leq 0$), or $\mathcal{D}_f = \mathbb{R}$ ($\alpha \in \mathbb{N}$).

EXAMPLE 2 Determine the range of the function $f(x) = x^2 - 7x + 11$ on the interval $[1, 4]$.

Solution In accordance with the way the problem is posed, we restrict the domain of the function here to the interval $A = [1, 4]$, so that the natural notation for the range is $f(A)$. Since

$$f(x) = \left(x - \frac{7}{2}\right)^2 - \frac{49}{4} + 11 = \left(x - \frac{7}{2}\right)^2 - \frac{5}{4},$$

the minimum of $f(A)$ is

$$f_{\min} = \min \{f(x) \mid x \in [1, 4]\} = f(7/2) = -5/4,$$

and the maximum of $f(A)$ is attained as far as possible from the point $x = 7/2$, i.e. at the point $x = 1$:

$$f_{\max} = \max \{f(x) \mid x \in [1, 4]\} = f(1) = 5.$$

So far we have deduced that $y \in f(A) \Rightarrow -5/4 \leq y \leq 5$, i.e. $f(A) \subset [-5/4, 5]$. On the other hand, if $y \in [-5/4, 5]$, then the equation $f(x) = y$ has the solution

$$x = \frac{7}{2} - \sqrt{y + \frac{5}{4}},$$

which lies in the interval $[1, 4]$. Thus $y \in [-5/4, 5] \Rightarrow y \in f(A)$, i.e. $[-5/4, 5] \subset f(A)$. Since both $f(A) \subset [-5/4, 5]$ and $[-5/4, 5] \subset f(A)$ hold, we have

$$f(A) = [-5/4, 5]. \quad \square$$

The example concerns a typical 'study of a function', in which a set $A \subset \mathcal{D}_f$ is given (in the case of a real function often an interval) and one must determine $B = f(A)$, i.e. the range of the function when the domain is restricted to A . The problem may also be posed in reverse, so that a set B is known and one must determine the set $A = \{x \in \mathcal{D}_f \mid f(x) \in B\}$. For example, in determining the zeros of a function one must solve the reverse problem with $B = \{0\}$, i.e. one must solve the equation $f(x) = 0$. If $B = [0, \infty)$, then one must solve the inequality $f(x) \geq 0$.

EXAMPLE 2 (continued) If in the example we set $\mathcal{D}_f = \mathbb{R}$ and $B = \{0, 1\}$, then

$$\begin{aligned} A &= \{x \in \mathbb{R} \mid f(x) \in B\} \\ &= \{x \in \mathbb{R} \mid x^2 - 7x + 11 = 0 \vee x^2 - 7x + 11 = 1\} \\ &= \{2, \tfrac{1}{2}(7 - \sqrt{5}), \tfrac{1}{2}(7 + \sqrt{5}), 5\}. \quad \square \end{aligned}$$

Monotonic functions

In solving equations and inequalities, and in studying functions generally, the following definition, which gives functions of one variable a 'character', is of great practical use (cf. Definition I.5.1 for sequences).

DEFINITION IV.1.1 A function of one real variable is *increasing* on the interval $A \subset \mathcal{D}_f$ if

$$\forall x_1, x_2 \in A \left[x_1 < x_2 \Rightarrow f(x_1) \leq f(x_2) \right]$$

and *strictly increasing* if

$$\forall x_1, x_2 \in A \left[x_1 < x_2 \Rightarrow f(x_1) < f(x_2) \right].$$

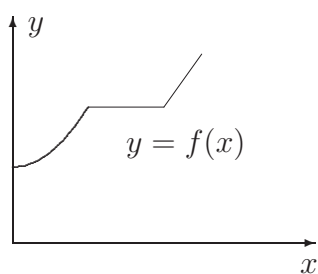
Correspondingly, f is *decreasing* on the interval $A \subset \mathcal{D}_f$ if

$$\forall x_1, x_2 \in A \left[x_1 < x_2 \Rightarrow f(x_1) \geq f(x_2) \right]$$

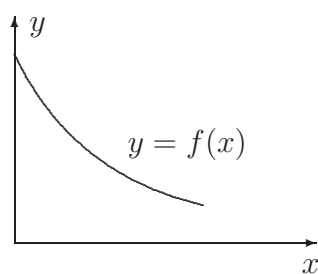
and *strictly decreasing* if

$$\forall x_1, x_2 \in A \left[x_1 < x_2 \Rightarrow f(x_1) > f(x_2) \right].$$

If f is of either type on an interval A , then f is said to be (strictly) *monotonic* on that interval.



f increasing



f strictly decreasing

EXAMPLE 3 From the identity

$$\sqrt{x_1} - \sqrt{x_2} = \frac{x_1 - x_2}{\sqrt{x_1} + \sqrt{x_2}}, \quad x_1, x_2 \geq 0, \quad x_1 \neq x_2$$

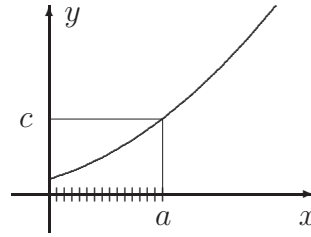
one sees that the function $f(x) = \sqrt{x}$ is strictly increasing on the interval $[0, \infty)$ (i.e. on its domain). $\square$

If f is strictly increasing on the interval $A \subset \mathcal{D}_f$ and $c \in A$, then the inequality

$$f(x) \leq c$$

has, when restricted to the interval A , an easy solution:

$$x \in A \cap (-\infty, a], \text{ where } f(a) = c.$$



EXAMPLE 3 (continued) If $a > 0$, then the solution of the inequality $\sqrt{x} \leq a$ is

$$\{x \in \mathbb{R} \mid \sqrt{x} \leq a\} = \{x \in [0, \infty) \mid x \leq a^2\} = [0, a^2]. \quad \square$$

If a function is not monotonic on the whole interval under consideration, the first step in studying the function is often to divide the interval into subintervals on which monotonicity holds.

EXAMPLE 4 $f(x) = \sqrt{|x|}$ ($\mathcal{D}_f = \mathbb{R}$) is strictly decreasing on the interval $(-\infty, 0]$ and strictly increasing on the interval $[0, \infty)$. $\square$

EXAMPLE 5 If $f(x) = -x^2 + 6x + 1$, then by writing

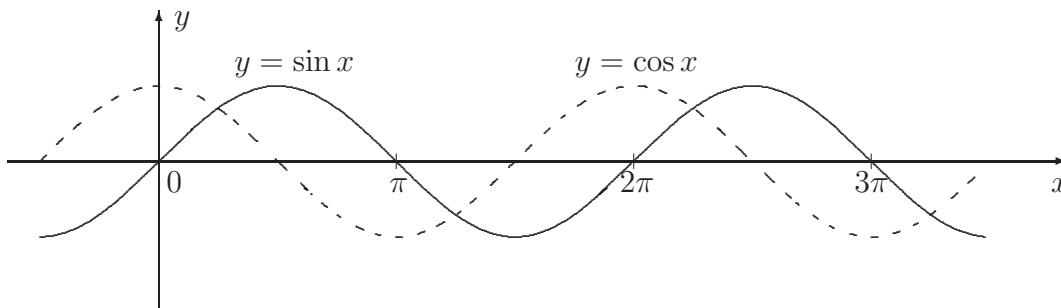
$$f(x) = -(x - 3)^2 + 10$$

one sees that f is strictly increasing on the interval $(-\infty, 3]$ and strictly decreasing on the interval $[3, \infty)$. $\square$

EXAMPLE 6 On the basis of the definitions of the trigonometric functions (Section II.5), the following claims are plausible (and also true).

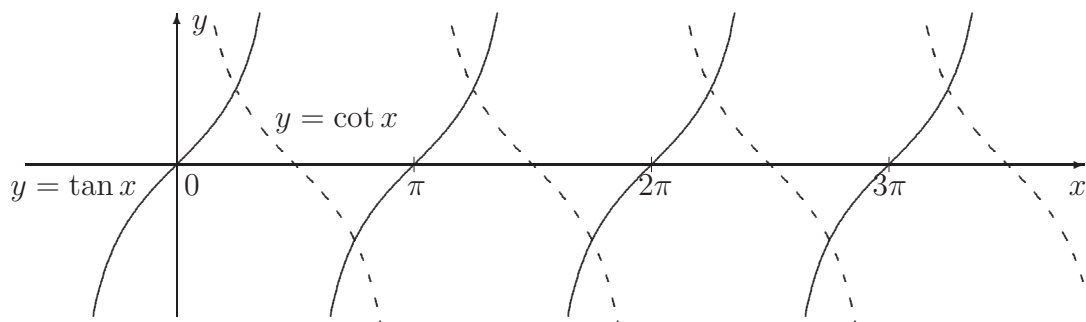
$\sin x$: Strictly monotonic on the intervals $[(k - \frac{1}{2})\pi, (k + \frac{1}{2})\pi]$, $k \in \mathbb{Z}$: increasing when k is even and decreasing when k is odd.

$\cos x$: Strictly monotonic on the intervals $[k\pi, (k + 1)\pi]$, $k \in \mathbb{Z}$: decreasing when k is even and increasing when k is odd.



$\tan x$: Strictly increasing on the intervals $((k - \frac{1}{2})\pi, (k + \frac{1}{2})\pi)$, $k \in \mathbb{Z}$.

$\cot x$: Strictly decreasing on the intervals $((k\pi, (k + 1)\pi)$, $k \in \mathbb{Z}$. $\square$



Composite function

The *composite function* of two functions f, g is denoted $f \circ g$ and defined by the rule

$$(f \circ g)(x) = f(g(x)).$$

The usual way of pronouncing it is the somewhat colloquial ' f circle g '.

EXAMPLE 7 A recursive sequence of the form

$$x_0 \in \mathbb{R}, \quad x_n = f(x_{n-1}), \quad n = 1, 2, \dots$$

can be interpreted by means of 'nested', i.e. composite, functions:

$$\begin{aligned} x_1 &= f(x_0), \\ x_2 &= f(x_1) = f(f(x_0)) = (f \circ f)(x_0), \\ x_3 &= f(f(f(x_0))) = (f \circ f \circ f)(x_0), \quad \text{etc.} \quad \square \end{aligned}$$

The domain of a composite function must always be restricted so that the functions do not 'conflict'. The largest possible domain (which is generally assumed unless stated otherwise) is then

$$\mathcal{D}_{f \circ g} = \{x \in \mathcal{D}_g \mid g(x) \in \mathcal{D}_f\} \subset \mathcal{D}_g.$$

If the set $\mathcal{D}_{f \circ g}$ so defined is empty, the composite function cannot be defined.

EXAMPLE 8 Define the composite functions $f \circ g$ and $g \circ f$ when $f(x) = \sqrt{a - x}$, $g(x) = \sqrt{x - b}$ ($a, b \in \mathbb{R}$).

Solution The rules of computation are

$$(f \circ g)(x) = \sqrt{a - \sqrt{x - b}}, \quad (g \circ f)(x) = \sqrt{\sqrt{a - x} - b}.$$

Since $\mathcal{D}_f = (-\infty, a]$, $\mathcal{D}_g = [b, \infty)$, we have

$$\mathcal{D}_{f \circ g} = \{x \in [b, \infty) \mid \sqrt{x - b} \in (-\infty, a]\}.$$

If $a < 0$, then $\mathcal{D}_{f \circ g} = \emptyset$, otherwise

$$\mathcal{D}_{f \circ g} = \{x \in \mathbb{R} \mid x \geq b \ \& \ x - b \leq a^2\} = [b, b + a^2] \quad (a \geq 0).$$

Thus

$$\mathcal{D}_{f \circ g} = \begin{cases} \emptyset, & \text{if } a < 0, \\ [b, b + a^2], & \text{if } a \geq 0. \end{cases}$$

In the same way one deduces

$$\begin{aligned} \mathcal{D}_{g \circ f} &= \{x \in (-\infty, a] \mid \sqrt{a - x} \in [b, \infty)\} \\ &= \begin{cases} (-\infty, a], & \text{if } b \leq 0, \\ (-\infty, a - b^2], & \text{if } b > 0. \end{cases} \quad \square \end{aligned}$$

Change of variable

If the function f under study on a set $A \subset \mathcal{D}_f$ can be expressed as a composite function $f(x) = g(v(x))$, then a *change of variable*, i.e. a *substitution* $t = v(x)$, often helps. One can then pass to studying the (possibly simpler) function $g(t)$ on the set $B = v(A)$.

EXAMPLE 9 Determine the range $f(A)$ of the function $f(x) = x - 7\sqrt{x} + 11$ on the interval $A = [1, 16]$.

Solution With the substitution $t = \sqrt{x}$ the object of study becomes the function $g(t) = t^2 - 7t + 11$ on the interval $B = [1, 4]$. Thus $f(A) = g(B) = [-5/4, 5]$ (Example 2). $\square$

Combining functions by arithmetic operations

Real-valued functions can be combined by the basic arithmetic operations in the same way as sequences. Here one thinks that, just as sequences are added,

multiplied, and divided termwise, functions are combined in the corresponding way *pointwise*. Thinking this way, one obtains a definition of the basic arithmetic operations between functions. For functions of one real variable a more precise definition is the following:

DEFINITION IV.1.2 (Combining functions) If $f : \mathcal{D}_f \rightarrow \mathbb{R}$, $g : \mathcal{D}_g \rightarrow \mathbb{R}$, $\mathcal{D}_f, \mathcal{D}_g \subset \mathbb{R}$, then the functions λf ($\lambda \in \mathbb{R}$), $f + g$, fg , and f/g are defined by

$$\begin{aligned} (1) \quad (\lambda f)(x) &= \lambda f(x), & x \in \mathcal{D}_f \\ (2) \quad (f + g)(x) &= f(x) + g(x), & x \in \mathcal{D}_f \cap \mathcal{D}_g \\ (3) \quad (fg)(x) &= f(x)g(x), & x \in \mathcal{D}_f \cap \mathcal{D}_g \\ (4) \quad (f/g)(x) &= f(x)/g(x), & x \in \mathcal{D}_f \cap \mathcal{D}_g \wedge g(x) \neq 0 \end{aligned}$$

The only new feature compared with sequences is that in combining functions the domain must be restricted, unless $\mathcal{D}_f = \mathcal{D}_g$. In the case of the basic 'sequence functions' this problem did not arise, since the domain was always the same ($= \mathbb{N}$). Note in particular that, according to the above definition, the domain of the function f/g is

$$\mathcal{D}_{f/g} = \{x \in \mathbb{R} \mid x \in \mathcal{D}_f \wedge x \in \mathcal{D}_g \wedge g(x) \neq 0\}.$$

Here the requirement $g(x) \neq 0$ already appeared in connection with sequences, cf. Theorem I.7.3.

EXAMPLE 10 By rule (4) of Definition IV.1.2,

$$\tan = \sin / \cos, \quad \cot = \cos / \sin. \quad \square$$

Even and odd functions

DEFINITION IV.1.3 If $f : \mathcal{D}_f \rightarrow \mathbb{R}$ and the domain $\mathcal{D}_f \subset \mathbb{R}$ of f is symmetric with respect to the origin, i.e. if $-x \in \mathcal{D}_f \forall x \in \mathcal{D}_f$, then f is said to be *even* if

$$f(-x) = f(x) \quad \forall x \in \mathcal{D}_f,$$

and *odd* if

$$f(-x) = -f(x) \quad \forall x \in \mathcal{D}_f.$$

EXAMPLE 11 Among the trigonometric functions the cosine is even and the sine odd. The power function $f(x) = x^n$, $n \in \mathbb{Z}$, is even/odd when n is even/odd. The function $f(x) = 0 \ \forall x \in \mathcal{D}_f$ (e.g. $\mathcal{D}_f = \mathbb{R}$) is the only function that is both even and odd. $\square$

When functions are combined according to Definition IV.1.2, the following rules are easily verified:

- (1) f and g even/odd $\Rightarrow f + g, \lambda f$, and $1/f$ even/odd
- (2a) f and g even/odd $\Rightarrow fg$ even
- (2b) f even and g odd $\Rightarrow fg$ odd

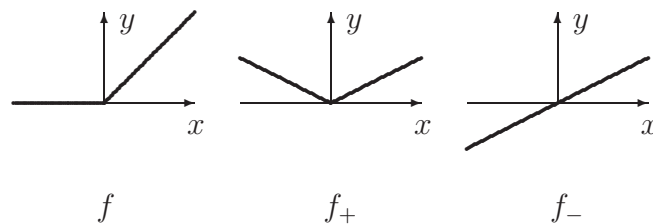
EXAMPLE 11 (continued) On the basis of the above rules (and otherwise) one can deduce: a) The trigonometric functions tan and cot are odd. b) The polynomial $f(x) = \sum_{k=0}^n a_k x^k$ is even/odd precisely when $a_k = 0$ for every odd/even value of the index k . $\square$

If a function is defined symmetrically with respect to the origin but is 'impure' as regards parity, then it can always be expressed as the sum of an even and an odd function. Namely, $f = f_+ + f_-$, where

$$f_+(x) = \frac{1}{2}[f(x) + f(-x)], \quad f_-(x) = \frac{1}{2}[f(x) - f(-x)].$$

EXAMPLE 12

$$f(x) = \begin{cases} 0, & \text{when } x \leq 0 \\ x, & \text{when } x > 0 \end{cases} = \frac{1}{2}|x| + \frac{1}{2}x \quad \forall x \in \mathbb{R}. \quad \square$$



Periodic functions

DEFINITION IV.1.4 A real function f is *periodic* if for some $a \in \mathbb{R}_+$ one has $x \pm a \in \mathcal{D}_f \forall x \in \mathcal{D}_f$ and

$$f(x+a) = f(x) \quad \forall x \in \mathcal{D}_f.$$

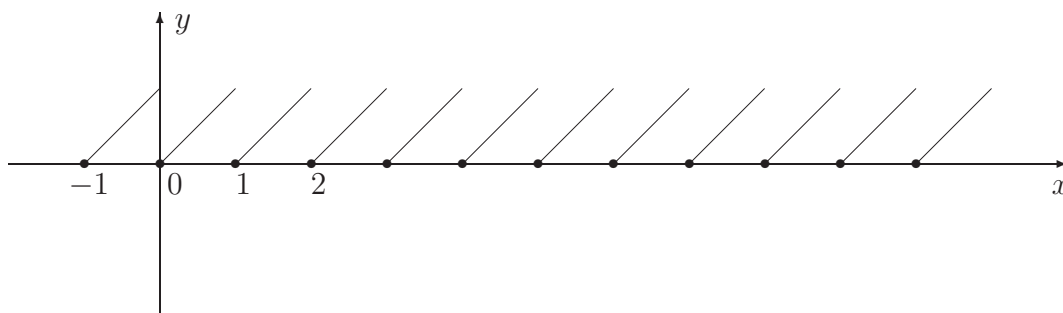
Then a is a *period* of f .

By the definition, every multiple of a period is also a period. The smallest of the periods is called the *fundamental period* (often simply called the 'period').

EXAMPLE 13 The functions $|\sin x|$, $|\cos x|$, $\tan x$, and $\cot x$ are periodic, with (fundamental) period $a = \pi$. $\square$

EXAMPLE 14 If $\mathcal{D}_f = \mathbb{R}$, the period of f is $a = 1$, and $f(x) = x$ when $x \in [0, 1)$, then

$$f(x) = x - k, \quad \text{when } x \in [k, k+1), \quad k \in \mathbb{Z}.$$



EXERCISES

- Using the tools of algebra, determine the ranges of the following real functions:
 - $x^2 + 2x + 8$,
 - $1 - x - 4x^2$
 - $1/(2 + x + x^2)$
 - $1/(1 - \sqrt{x})$
 - $x^2/(1 - x^2)$
 - $(x + 1)/(x + 2)$
 - $|x| + |x + 1|$
 - $|x| - |x + 2|$
 - $\sum_{k=0}^{\infty} x^k$
 - $\sum_{k=0}^{\infty} (-1)^k 2^k x^k$
 - $\sum_{k=0}^{\infty} x^{2k}$
 - $\sum_{k=0}^{\infty} (-1)^k x^{2k}$
- Using the tools of algebra, determine on which intervals the following functions are strictly increasing and on which strictly decreasing:
 - $1/x^4$,
 - $x/(x + 1)$,
 - $|x^2 + x - 2|$,
 - $1/(x^2 + 2x + 2)$,
 - $1/(x^2 + 3x + 2)$,
 - $x^4/(2 - x^4)$
 - $\max\{0, \sin x\}$

3. What are the domains and ranges of the following composite functions?
 a) $\sqrt{8-2x}$ b) $\sqrt{1-x-x^2}$ c) $1/(1-\sqrt{2-x})$ d) $\cos(\sin x)$
4. Determine the rules of computation and the domains of the composite functions $f \circ g$ and $g \circ f$ in the following cases:
 a) $f(x) = 1/\sqrt{x+1}$, $g(x) = \sqrt{x-1}$
 b) $f(x) = \sqrt{x+1}$, $g(x) = x/(1-2x)$
 c) $f(x) = \sqrt{2x+1}$, $g(x) = \sin x$
5. a) Let $f(x) = 1 - 4\sqrt{x}$. What are the domains of the functions $f \circ f$ and $f \circ f \circ f$? b) Let $g(x) = \frac{1}{2}x + 1$. What kind of real function is $f(x) = \lim_n g_n(x)$, where $g_1 = g$, $g_2 = g \circ g$, $g_3 = g \circ g \circ g$, etc.?
6. Using the tools of algebra, determine the ranges of the following functions:
 a) $1 - \sqrt[4]{17x} + \sqrt{x}$ b) $2 + 5x^{48} - x^{96}$ c) $x - 4\sqrt{x} + 3 - 4|\sqrt{x} - 1|$
 d) $x^4/(2-x^4)$ e) $\cos x - 7\sin x$ f) $\sin x + \cos 2x$ g) $2\sin x - |\cos 2x|$
7. a) Show that if g is increasing on the interval $A \subset \mathcal{D}_g$ and f is increasing on the interval $B = g(A) \subset \mathcal{D}_f$, then $f \circ g$ is increasing on the interval A . Apply the claim to the function $f(x) = \sqrt{x^2 + x - 2}$ on the interval $[1, \infty)$.
 b) Show that if f and g are (strictly) increasing on the interval $A \subset \mathcal{D}_f \cap \mathcal{D}_g$, then so is the function $f + g$. What about the function fg ?
8. Define $f + g$, fg , and f/g (the domain and the simplified rule of computation) when
 a) $f(x) = g(x) = x + 1$, b) $f(x) = |x| + x$, $g(x) = |x| - x$,
 c) $f(x) = \sin x \sin 2x$, $g(x) = 2\cos^3 x$.
9. Show: a) The decomposition of a function into an even and an odd part is unique. b) If g is even, then $f \circ g$ is even or undefined. c) If f is even/odd and g is odd, then $f \circ g$ is even/odd or undefined.
10. Decompose the following functions into an even and an odd part:
 a) $f(x) = 2 + x - 3x^2 + x^4 + x^5 + \sin x - 3\cos x$
 b) $f(x) = |x+1| + |x-1| - x + 2x^2$
 c) $f(x) = \sin(x + \frac{\pi}{4}) + \cos(x + \frac{\pi}{3})$
11. a) Show that if f is (strictly) increasing/decreasing on the intervals A_1 and A_2 and $A_1 \cap A_2 \neq \emptyset$, then f is (strictly) increasing/decreasing on the interval $A = A_1 \cup A_2$.
 b) Let $\mathcal{D}_f = \mathbb{R}$ and let f be (strictly) increasing on the interval $[0, \infty)$. Show that if in addition f is even/odd, then f is (strictly) decreasing on the interval $(-\infty, 0]$ / (strictly) increasing on $\mathbb{R}$.

12. The function $f(x) = \sin \frac{x}{5} + \cos \frac{x}{7}$ is periodic. What is the fundamental period?
13. The period of the function g is $a = 2$ and $g(x) = 1 - |x|$ when $x \in [-1, 1]$. Determine the zeros of the function $f(x) = g(x) + cx - 5$ when a) $c = 1$, b) $c = 1/2$.
14. Define the step function

$$f(x) = k, \quad \text{when } 2k - 2 \leq x < 2k, \quad k \in \mathbb{Z}.$$

Determine $a \in \mathbb{R}$ and a periodic function g such that $f(x) = ax + g(x)$, $x \in \mathbb{R}$.

15. (*) Prove using the tools of algebra alone:
- a) The range of the function $f(x) = (3x^2 + 3)/(x^2 + x + 1)$ is $\mathcal{R}_f = [2, 6]$.
- b) The function $f(x) = x^2/(x + 1)$ is strictly increasing on the intervals $(-\infty, -2]$ and $[0, \infty)$ and strictly decreasing on the intervals $[-2, -1)$ and $(-1, 0]$.

IV.2 Inverse function. Implicit functions

A real function $f : \mathcal{D}_f \rightarrow \mathbb{R}$ is said to be 1-1 ('one to one') or *invertible* or an *injection*, i.e. *injective*, if

$$\forall x_1, x_2 \in \mathcal{D}_f [x_1 \neq x_2 \Rightarrow f(x_1) \neq f(x_2)].$$

If f is 1-1 and $y \in \mathcal{R}_f$, then the equation

$$f(x) = y \quad (x \in \mathcal{D}_f)$$

has a unique solution. Since to each $y \in \mathcal{R}_f$ there corresponds in this way a unique $x \in \mathcal{D}_f$, we have a functional dependence $y \mapsto x$, which is denoted

$$x = f^{-1}(y).$$

The function f^{-1} (read ' f minus 1', or ' f inverse') is called the *inverse function* of f . The domain of the inverse function is thus $\mathcal{D}_{f^{-1}} = \mathcal{R}_f$ and its range is $\mathcal{R}_{f^{-1}} = \mathcal{D}_f$.

EXAMPLE 1 For the function $f(x) = 1/x$ ($\mathcal{D}_f = \{x \in \mathbb{R} \mid x \neq 0\}$) we have

$$f(x_1) - f(x_2) = \frac{1}{x_1} - \frac{1}{x_2} = \frac{x_2 - x_1}{x_1 x_2}, \quad x_1, x_2 \in \mathcal{D}_f.$$

From this, $f(x_1) - f(x_2) \neq 0$ whenever $x_1, x_2 \neq 0$ and $x_1 \neq x_2$, so f is 1-1. The inverse function is found by solving the equation $f(x) = y$:

$$x^{-1} = y \quad \Rightarrow \quad x = y^{-1}, \quad \text{if } y \neq 0.$$

If $y = 0$, the equation has no solution, so the domain of the inverse function ($=$ the range of f) $= \{y \in \mathbb{R} \mid y \neq 0\}$, and $f^{-1}(y) = 1/y$. Since therefore $\mathcal{D}_{f^{-1}} = \mathcal{D}_f$ and $f^{-1}(y) = f(y) \forall y \in \mathcal{D}_{f^{-1}} = \mathcal{D}_f$, we have $f^{-1} = f$. $\square$

By Definition IV.1.1 it is obvious that if the domain of a function f is an interval and f is strictly increasing or strictly decreasing (i.e. strictly monotonic) on that interval, then f is 1-1. Strict monotonicity is indeed in practice the most common form of injectivity when the domain of a function is an interval (or is restricted to an interval).

EXAMPLE 2 Show that the function $f(x) = x^m$, $m \in \mathbb{N}$ is an injection when the domain is restricted to the interval $[0, \infty)$. Determine the inverse function.

Solution We write (see Proposition 1.2.2)

$$f(x_1) - f(x_2) = x_1^m - x_2^m = (x_1 - x_2)(x_1^{m-1} + x_1^{m-2}x_2 + \cdots + x_2^{m-1}).$$

If here $x_1, x_2 \geq 0$ and $x_1 \neq x_2$, then in the second factor of the product just written every summand is non-negative and at least one is positive (since either $x_1 > 0$ or $x_2 > 0$), so that factor is positive. We deduce that f is strictly increasing on the interval $[0, \infty)$:

$$\forall x_1, x_2 \geq 0 \ [x_1 < x_2 \Rightarrow f(x_1) < f(x_2)].$$

Thus f is an injection, so the equation $f(x) = y$ ($y \in \mathbb{R}$) has at most one solution x on the interval $[0, \infty)$. If $y < 0$, there is no solution. If $y = 0$, the solution is $x = 0$. Finally, if $y > 0$, a solution also exists and we denote it $x = \sqrt[m]{y}$. (The number $\sqrt[m]{y} \in \mathbb{R}_+$ can be computed e.g. by the decimal-division algorithm, cf. Section 1.9.) The sought inverse function is thus $f^{-1}(x) = \sqrt[m]{x}$, with domain $\mathcal{D}_{f^{-1}} = \mathcal{R}_f = [0, \infty)$. $\square$

EXAMPLE 2 (continued) If m is odd, then $f(x) = x^m$, $\mathcal{D}_f = \mathbb{R}$, is odd and hence strictly increasing on all of $\mathbb{R}$ (see Exercise IV.1:11b). Thus f is an injection. The equation $f(x) = y$ then has a (unique) solution for every $y \in \mathbb{R}$, since if $y < 0$, the solution is $x = -\sqrt[m]{-y}$. The inverse function is usually denoted simply $f^{-1}(x) = \sqrt[m]{x}$, whereby it is agreed that

$$\sqrt[m]{-x} = -\sqrt[m]{x}, \quad \text{when } x > 0 \text{ and } m \text{ is odd.}^* \quad \square$$

The following general rule is easily derived from Definition IV.1.1 and the definition of the inverse function (Exercise 1a):

$\begin{aligned} &f \text{ strictly increasing/decreasing on the interval } A \subset \mathcal{D}_f \\ \Rightarrow &f^{-1} \text{ strictly increasing/decreasing on the intervals } B \subset f(A). \end{aligned}$
--

EXAMPLE 2 (continued) Since $f(x) = x^m$ ($m \in \mathbb{N}$) is strictly increasing on the interval $A = [0, \infty)$, the function $f^{-1}(x) = \sqrt[m]{x}$ is likewise strictly increasing on the interval $[0, \infty) = f(A)$. $\square$

*Extending the definition of roots in the manner mentioned is not entirely unproblematic, as is seen from the computation

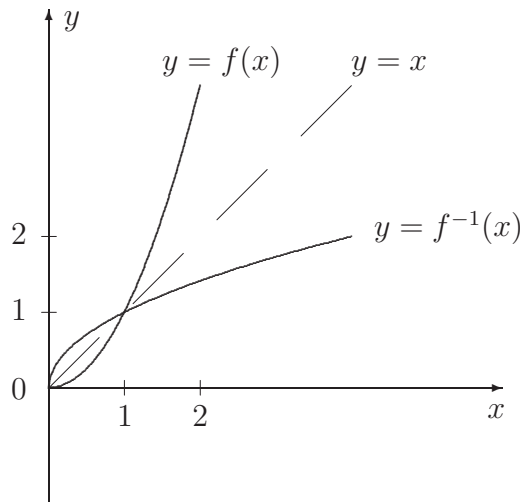
$$1 = \sqrt[6]{1} = \sqrt[6]{(-1)^2} = (-1)^{2/6} = (-1)^{1/3} = \sqrt[3]{-1} = -1.$$

To avoid contradictions of this type, it is clearest to agree that the real power function $f(x) = x^\alpha$ is defined on the interval $(-\infty, 0)$ only when $\alpha \in \mathbb{Z}$.

The general laws of the algebra of functions are also easily verified from the definition of the inverse function:

$$\begin{aligned} (f^{-1} \circ f)(x) &= x \quad \forall x \in \mathcal{D}_f, \\ (f \circ f^{-1})(y) &= y \quad \forall y \in \mathcal{R}_f, \\ (f^{-1})^{-1} &= f. \end{aligned}$$

The graph of the inverse function f^{-1} is obtained by interchanging x and y in the graph of f , i.e. by reflecting the graph of f across the line $y = x$. In the figure $f(x) = x^2$, $\mathcal{D}_f = [0, \infty)$.



EXAMPLE 3 Investigate the (possible) inverse function of the function $f(x) = x^5 + 3x$, $\mathcal{D}_f = \mathbb{R}$.

Solution Since f is odd and

$$f(x_1) - f(x_2) = (x_1 - x_2)(x_1^4 + x_1^3x_2 + x_1^2x_2^2 + x_1x_2^3 + x_2^4 + 3),$$

we deduce, as in Example 2, that f is strictly increasing on all of $\mathbb{R}$. The inverse function $f^{-1} : \mathcal{R}_f \rightarrow \mathbb{R}$ therefore exists. When $y \in \mathcal{R}_f$, the function evaluation $y \mapsto x = f^{-1}(y)$ means solving the equation

$$x^5 + 3x = y.$$

Solvability for every $y > 0$ can be shown (Exercise 12), and since f is odd, the equation is also solvable for every $y \leq 0$, i.e. $\mathcal{R}_f = \mathcal{D}_{f^{-1}} = \mathbb{R}$. One also sees

that, for example, $f^{-1}(0) = 0$, $f^{-1}(4) = 1$, and $f^{-1}(-38) = -2$. On the other hand, a number such as $a = f^{-1}(1)$ cannot be determined 'exactly' even by means of roots; rather it is a more general (algebraic, cf. Section I.11) number. In defining such a number one has to be content with the general definition of a real number, e.g. as an infinite decimal (see Exercise 12). Symbolic computation with the number a is of course also possible, but such computation based on the definition $a^5 + 3a = 1$ is quite limited. — As small consolation, we have the result

$$a^{-1} = a^4 + 3. \quad \square$$

Injection, surjection, and bijection

As in Example 2 above, in studying inverse functions it is quite common that f is not necessarily injective on its whole domain, but is nevertheless an injection if the domain is suitably restricted. As the starting point for the study of inverse functions one can then take a restricted map

$$f : A \rightarrow B,$$

where $A \subset D_f$ and $B \supset f(A)$. Usually A is some subinterval of D_f . A function restricted to the set A may, if desired, be denoted $f|_A$, and one says that it is the *restriction of f to A* . If moreover B has been chosen so that $B = f(A)$, then $f : A \rightarrow B$ is said to be a *surjection onto B* (French *sur jeter* = to throw over; the English 'f onto B' carries the same idea). The term is also used when f is not an injection. Thus from a function $f : A \rightarrow B$ one obtains a surjection simply by restricting the target set to $f(A)$. If $f : A \rightarrow B (= f|_A)$ is both an injection and a surjection, then $f : A \rightarrow B$ is said to be a *bijection* (one to one and onto). Then the inverse function $f^{-1} : B \rightarrow A$ (here, that is, $f^{-1} = (f|_A)^{-1}$) is also a bijection.*

EXAMPLE 4 If $f(x) = x^m$, then $f : [0, \infty) \rightarrow [0, \infty)$ is a bijection for every $m \in \mathbb{N}$, and so, for example, are $f : (0, \infty) \rightarrow (0, \infty)$, $f : [0, 1] \rightarrow [0, 1]$, and $f : [1, \infty) \rightarrow [1, \infty)$ (cf. Example 2). $\square$

EXAMPLE 5 If $f(x) = x^2$, then

*The terms injection, surjection, and bijection are not restricted to real functions, but can equally well be attached to the general set-theoretic notion of a function (see Section II.5). If A and B are sets and there exists a bijection $f : A \rightarrow B$, then one says that there is a *one-to-one correspondence* between A and B (created by the bijection f), and one writes $A \leftrightarrow B$. For example, the notation $E^2 \leftrightarrow \mathbb{R}^2$ of Section II.2 means such a correspondence in a given coordinate system of the plane.

$f : [-2, 3] \rightarrow [0, 10]$ is a function, but neither an injection nor a surjection.
 $f : [-2, 3] \rightarrow [0, 9]$ is a surjection, not an injection.
 $f : [0, 3] \rightarrow [0, 10]$ is an injection, not a surjection: inverse function $f^{-1}(x) = \sqrt{x}$.
 $f : [0, 3] \rightarrow [0, 9]$ is a bijection: inverse function $f^{-1}(x) = \sqrt{x}$.
 $f : [-2, -1] \rightarrow [1, 4]$ is a bijection: inverse function $f^{-1}(x) = -\sqrt{x}$. $\square$

Inverse trigonometric functions

The trigonometric functions are not injective on their whole domain, so they are not 'invertible' in the ordinary sense. By inverse trigonometric functions one most commonly means the functions obtained by restricting a trigonometric function either to the interval $[-\frac{\pi}{2}, \frac{\pi}{2}]$ or $(-\frac{\pi}{2}, \frac{\pi}{2})$ (sin, tan) or to the interval $[0, \pi]$ or $(0, \pi)$ (cos, cot), whereby the function is strictly monotonic on that interval. The inverse functions so obtained are called *arcus functions* or *cyclometric functions*. The definitions are

$$\begin{aligned}
 x = \sin y \quad \& \quad y \in [-\frac{\pi}{2}, \frac{\pi}{2}] &\Leftrightarrow y = \operatorname{Arcsin} x, \\
 x = \cos y \quad \& \quad y \in [0, \pi] &\Leftrightarrow y = \operatorname{Arccos} x, \\
 x = \tan y \quad \& \quad y \in (-\frac{\pi}{2}, \frac{\pi}{2}) &\Leftrightarrow y = \operatorname{Arctan} x, \\
 x = \cot y \quad \& \quad y \in (0, \pi) &\Leftrightarrow y = \operatorname{Arccot} x.
 \end{aligned}$$

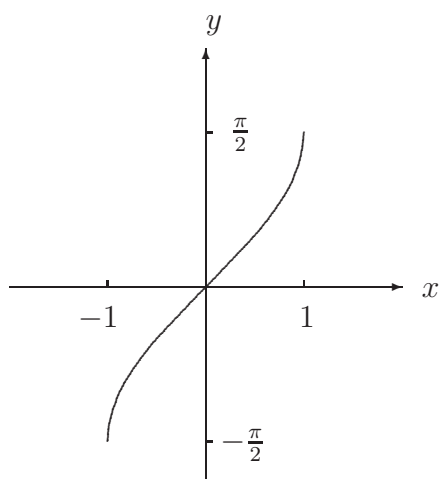
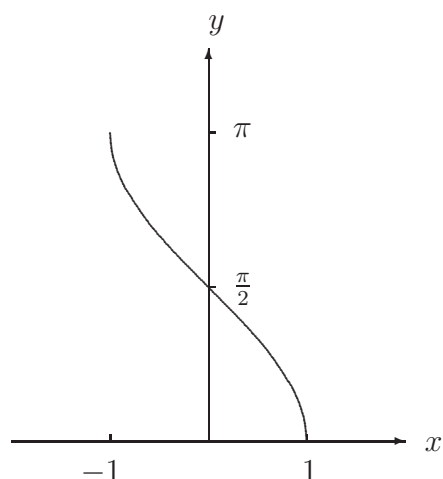
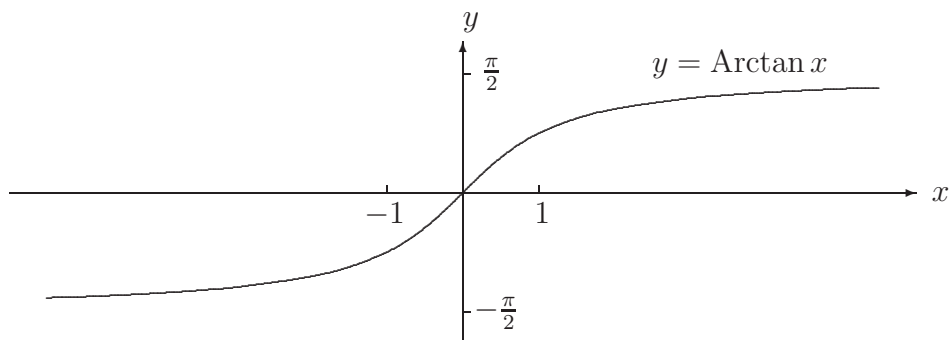
These are read 'arc sine', etc. They are the so-called *principal branches* of the inverse trigonometric functions.*

By the definitions, the domains and ranges of the arcus functions are the following:

$$\begin{aligned}
 \operatorname{Arcsin} x &: \mathcal{D}_f = [-1, 1], & \mathcal{R}_f &= [-\frac{\pi}{2}, \frac{\pi}{2}]. \\
 \operatorname{Arccos} x &: \mathcal{D}_f = [-1, 1], & \mathcal{R}_f &= [0, \pi]. \\
 \operatorname{Arctan} x &: \mathcal{D}_f = \mathbb{R}, & \mathcal{R}_f &= (-\frac{\pi}{2}, \frac{\pi}{2}). \\
 \operatorname{Arccot} x &: \mathcal{D}_f = \mathbb{R}, & \mathcal{R}_f &= (0, \pi).
 \end{aligned}$$

The functions Arcsin and Arctan are strictly increasing on their interval of definition (like sin and tan), the other two strictly decreasing.

*For the inverse functions of trigonometric functions restricted to other intervals, this text uses the common notation arcsin, etc. — see the more precise definition at the end of the section. In the literature the notations for the inverse trigonometric functions are somewhat varied. Older notations for the principal branches are $\overline{\operatorname{arcsin}}$, $\overline{\operatorname{arccos}}$, $\overline{\operatorname{arctan}}$, $\overline{\operatorname{arccot}}$. The notations $\sin^{-1}$, $\cos^{-1}$, and $\tan^{-1}$, common on calculators, are also seen used in the literature. — In this text we define $\sin^n x = (\sin x)^n \ \forall n \in \mathbb{Z}$, so that $\sin^{-1} x = 1/\sin x$.

 $y = \text{Arcsin } x$  $y = \text{Arccos } x$ 

From the pairs (Arcsin, Arccos) and (Arctan, Arccot) it suffices for applications to choose one representative from each pair (most commonly Arcsin and Arctan are chosen), since the 'inter-member' relation of the pairs reduces to the simple identities

$$\begin{aligned} \text{Arcsin } x + \text{Arccos } x &= \frac{\pi}{2} \quad \forall x \in [-1, 1], \\ \text{Arctan } x + \text{Arccot } x &= \frac{\pi}{2} \quad \forall x \in \mathbb{R}. \end{aligned}$$

The first of these, for example, follows by the reasoning (cf. the figures above)

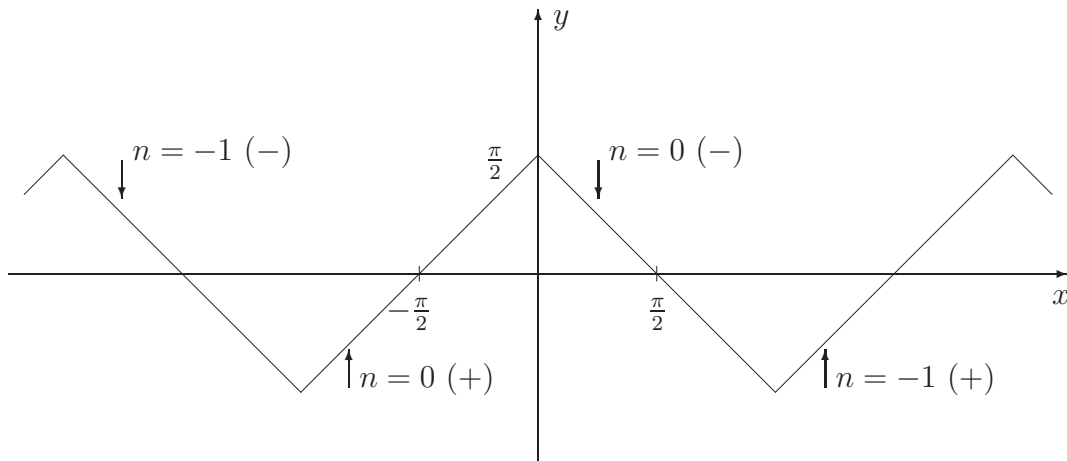
$$\begin{aligned} &\begin{cases} \frac{\pi}{2} - \text{Arccos } x \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right] & \forall x \in [-1, 1] \\ \sin\left(\frac{\pi}{2} - \text{Arccos } x\right) = \cos(\text{Arccos } x) = x & \forall x \in [-1, 1] \end{cases} \\ &\Rightarrow \frac{\pi}{2} - \text{Arccos } x = \text{Arcsin } x \quad \forall x \in [-1, 1]. \end{aligned}$$

EXAMPLE 6 Sketch the graph of the function $\text{Arcsin}(\cos x)$.

Solution First we simplify $y = \text{Arcsin}(\cos x)$:

$$\begin{aligned} y &= \text{Arcsin}(\cos x) \\ \Leftrightarrow y &\in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right] \quad \& \quad \sin y = \cos x \\ \Leftrightarrow y &\in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right] \quad \& \quad \sin y = \sin\left(\frac{\pi}{2} - x\right) \\ \Leftrightarrow y &\in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right] \quad \& \quad y = \frac{\pi}{2} \pm x + n \cdot 2\pi, \quad n \in \mathbb{Z}. \end{aligned}$$

From this, the graph consists of a set of lines restricted to the interval $y \in [-\frac{\pi}{2}, \frac{\pi}{2}]$ (see the figure). $\square$



Implicit function $y(x)$

If from an equation of the form

$$F(x, y) = 0$$

y can be solved uniquely for every $x \in A$ ($A \subset \mathbb{R}$), then the equation is said to define on A an *implicit function* $y = f(x)$. The term means that the function is defined indirectly, i.e. *implicitly*. In that case it is not required that the function can be written directly (i.e. explicitly) as a known expression. In the equation, F is a real-valued function of two real variables — in practice an expression in which x and y occur. (Functions of two variables as independent objects are treated in the following section.)

In denoting an implicit function one often wishes to preserve the 'sense of place' by using, instead of the general notation $f(x)$, the notation $y(x)$. Accordingly, the functional dependence $x \mapsto y(x)$ is defined by the indirect rule

$$F(x, y(x)) = 0, \quad x \in A.$$

A simple example of an implicit function is the inverse function, for if in the definition of the inverse function one interchanges x and y , the definition is

$$F(x, y) = x - f(y) = 0, \quad y \in \mathcal{D}_f \quad \Leftrightarrow \quad y(x) = f^{-1}(x).$$

— In fact an ordinary function also becomes 'implicit' when $y = f(x)$ is written in the form $F(x, y) = f(x) - y = 0$.

EXAMPLE 7 If $b \neq 0$, then the equation of a line in the plane in standard form

$$ax + by + c = 0$$

can be interpreted as the definition of the implicit function $y(x) = -(ax + c)/b$.
□

EXAMPLE 8 In considering the equation

$$F(x, y) = x + y + \cos x \sin y = 0$$

it turns out (we omit the more precise justification) that the equation has a unique solution $y(x)$ for every $x \in \mathbb{R}$, i.e. the equation defines an implicit function $y(x)$ on all of $\mathbb{R}$. From the equation one can deduce that $y(x) = -x$ when $x = (n + \frac{1}{2})\pi$ and likewise when $x = n\pi$, $n \in \mathbb{Z}$. For other values of x the function evaluation $x \mapsto y(x)$ is more laborious, since it involves a so-called *transcendental equation*, which can be solved only by numerical means. From the equation x can also be solved uniquely for every $y \in \mathbb{R}$. This determines (though not as an expression) a function $y \mapsto x(y)$, which is the inverse function of the function $x \mapsto y(x)$. □

Multi-branch implicit functions

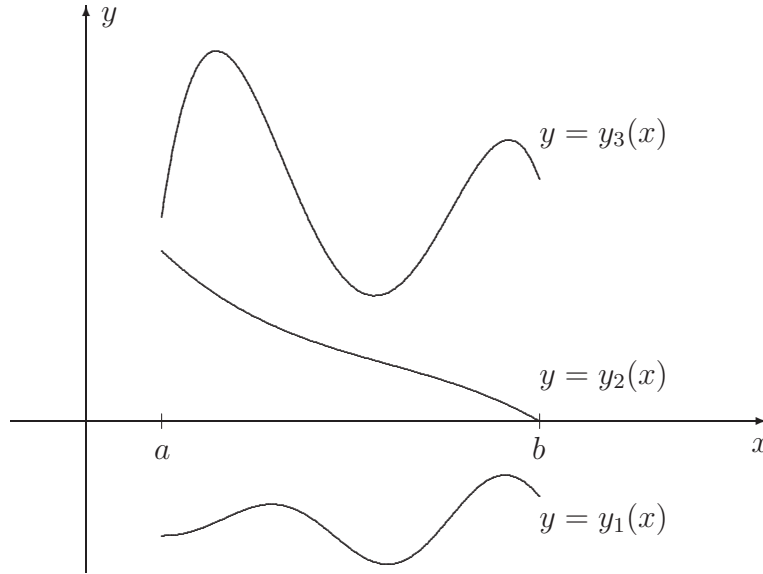
In attempts to define an implicit function, a quite common situation is that the equation $F(x, y) = 0$ is solvable for y at a given x , but not uniquely. In such a situation, let $A \subset \mathbb{R}$ be an interval (closed, open, or half-open) and consider the equation for $x \in A$. Suppose there exist an index set $\Lambda \subset \mathbb{N}$ characteristic of the interval A (possibly also $\Lambda = \mathbb{N}$) and a set of functions

$$\mathcal{Y} = \{y_i, \quad i \in \Lambda\}$$

such that

$$\forall x \in A [F(x, y) = 0 \Leftrightarrow y \in \{y_i(x), i \in \Lambda\}].$$

One then says that $\mathcal{Y}$ is the *multi-branch* (multi-valued) *implicit function* defined by the equation $F(x, y) = 0$ on the interval A . A multi-branch implicit function is thus not a function but a set of such. The elements y_i of the set $\mathcal{Y}$ are called the *branches* of $\mathcal{Y}$.^{*} In the figure there are three branches on the interval $A = [a, b]$.[†]



EXAMPLE 9 The equation $x^2 + y^2 = 1$ defines on the interval $[-1, 1]$ a two-branch implicit function $\mathcal{Y} = \{y_+, y_-\}$, where $y_{\pm} = \pm\sqrt{1 - x^2}$. $\square$

EXAMPLE 10 What kind of multi-branch implicit function is defined by the equation $y^4 - 2y^2 + x^2 - x = 0$ on the various subintervals of $\mathbb{R}$?

^{*}A multi-branch function $\mathcal{Y}$ can also be interpreted as *set-valued* by defining

$$\mathcal{Y}(x) = \{y_i(x), i \in \Lambda\},$$

where $y_i, i \in \Lambda$ are the branches of $\mathcal{Y}$ on the interval A under consideration. Since $\mathcal{Y}(x)$ is (as a set) unique for every $x \in A$, $\mathcal{Y}$ is a function of the type $\mathcal{Y} : A \rightarrow \mathcal{B}$, where the target set $\mathcal{B}$ is a 'set of sets' $\mathcal{B} = \{\{t_i, i \in \Lambda\} \mid t_i \in \mathbb{R} \ \forall i \in \Lambda\}$.

[†]In order for the branches y_i of a multi-branch function $\mathcal{Y} = \{y_i\}$ to be unambiguous, the definition must prevent 'jumping from one branch to another'. The condition can be interpreted geometrically-intuitively as requiring that the graphs $G_i = \{P = (x, y) \in E^2 \mid y = y_i(x) \ \& \ x \in A\}$ be *connected* point sets of the plane, i.e. that no 'breaks' be allowed in the graphs. (More precisely, this means that the functions y_i must be *continuous*, see Section V.1 below.)

Solution If $x \in \mathbb{R}$, then the possible solutions $y \in \mathbb{R}$ of the equation are

$$y = \pm \sqrt{1 \pm \sqrt{1 + x - x^2}}.$$

Here

$$\begin{aligned} 1 + x - x^2 \geq 0 &\Leftrightarrow \frac{1}{2}(1 - \sqrt{5}) \leq x \leq \frac{1}{2}(1 + \sqrt{5}), \\ 1 + x - x^2 \geq 1 &\Leftrightarrow 0 \leq x \leq 1. \end{aligned}$$

We deduce that on the interval $[0, 1]$ the function has two branches: $\mathcal{Y} = \{y_1, y_2\}$, where

$$y_1(x) = \sqrt{1 + \sqrt{1 + x - x^2}}, \quad y_2(x) = -\sqrt{1 + \sqrt{1 + x - x^2}}.$$

On the intervals $[\frac{1}{2}(1 - \sqrt{5}), 0]$ and $[1, \frac{1}{2}(1 + \sqrt{5})]$ we have $0 \leq 1 + x - x^2 \leq 1$, so on these intervals the function has four branches: $\mathcal{Y} = \{y_1, y_2, y_3, y_4\}$, where y_1 and y_2 are the same as above and

$$y_3(x) = \sqrt{1 - \sqrt{1 + x - x^2}}, \quad y_4(x) = -\sqrt{1 - \sqrt{1 + x - x^2}}.$$

If $x \in (-\infty, \frac{1}{2}(1 - \sqrt{5}))$ or $x \in (\frac{1}{2}(1 + \sqrt{5}), \infty)$, the equation is not satisfied by any $y \in \mathbb{R}$, so on these intervals the equation defines no (real) function. $\square$

The functions arcsin, arccos, arctan

By the function notations arcsin, arccos, and arctan one means the infinitely multi-branch functions defined by the equations $x - \sin y = 0$, $x - \cos y = 0$, and $x - \tan y = 0$, with domain either the interval $[-1, 1]$ (arcsin, arccos) or $\mathbb{R}$ (arctan). For example, arctan is defined as

$$\arctan = \{y_n, n \in \mathbb{Z}\},$$

where

$$y_n(x) = \operatorname{Arctan} x + n\pi.$$

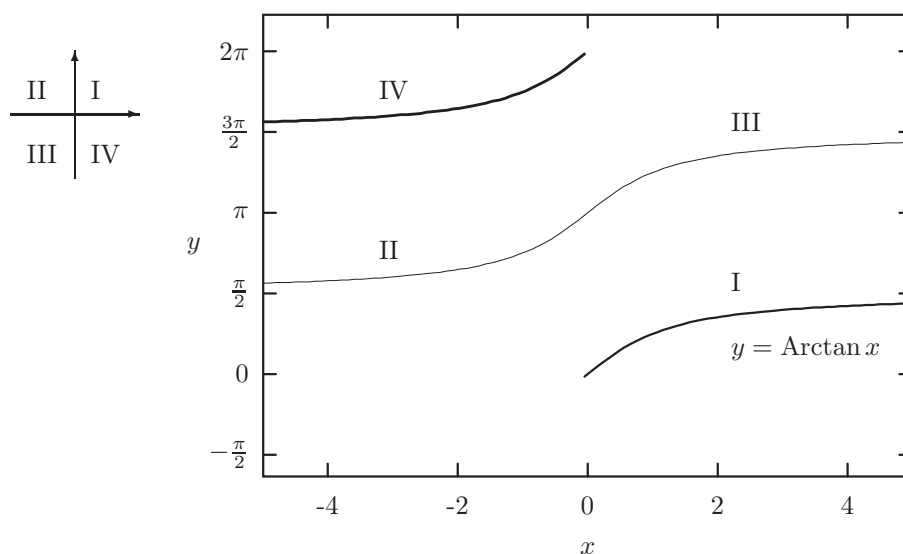
Here $y_0(x) = \operatorname{Arctan} x$ is the *principal branch* of arctan.

EXAMPLE 11 If $\varphi(x, y)$ is the polar angle corresponding to the point (x, y) of the Cartesian coordinate system, $\varphi \in [0, 2\pi)$, then the functional dependence

$(x, y) \mapsto \varphi(x, y)$ is

$$\varphi(x, y) = \begin{cases} \operatorname{Arctan}(y/x), & \text{if } x > 0 \text{ \& } y \geq 0, \\ \operatorname{Arctan}(y/x) + \pi, & \text{if } x < 0, \\ \operatorname{Arctan}(y/x) + 2\pi, & \text{if } x > 0 \text{ \& } y < 0, \\ \pi/2, & \text{if } x = 0 \text{ \& } y > 0, \\ 3\pi/2, & \text{if } x = 0 \text{ \& } y < 0. \end{cases}$$

Here, then, as many as three branches of arctan were needed (!). $\square$



EXERCISES

1. Prove: a) If f is strictly increasing or decreasing on the interval $A \subset \mathcal{D}_f$, then so is f^{-1} on the intervals $B \subset f(A)$. b) If f is 1-1 and odd, then so is f^{-1} .
2. a) The function $f(x) = x^2 + x + 2$ is invertible when it is restricted to the interval $(-\infty, a]$ or to the interval $[a, \infty)$. Determine a and the inverse function on each interval.
b) Show using the tools of algebra that the function $f(x) = x^3 + 2x$ is strictly

increasing on its whole domain ($= \mathbb{R}$). Determine $f^{-1}(-1)$, $f^{-1}(3)$, $f^{-1}(10)$, $f^{-1}(12)$, and $f^{-1}(4\sqrt{2})$ exactly if possible, otherwise approximately to one decimal place.

c) Let $f(x) = (1 - x^n)^{1/n}$, $\mathcal{D}_f = [0, 1]$, $n \in \mathbb{N}$. Show that $f = f^{-1}$.

d) Let $f(x) = (ax + b)/(cx + d)$. For which values of the constants a, b, c, d is $f = f^{-1}$?

3. Consider functions $f : A \rightarrow \mathbb{R}$. In the following cases, determine B such that $f : A \rightarrow B$ is a surjection. Is $f : A \rightarrow B$ then also a bijection?

a) $f(x) = -x^5$, $A = (-4, -3]$

b) $f(x) = x^2$, $A = [\sqrt{2} - \sqrt{3}, \sqrt{2} + \sqrt{3}]$

c) $f(x) = -6x - x^2$, $A = (-5, -2)$

d) $f(x) = (x + 5)/(x + 2)$, $A = [-1, 1]$

e) $f(x) = 3 + 2 \sin x$, $A = [3\pi/2, 11\pi/6]$

f) $f(x) = x + \operatorname{Arcsin} x$, $A = [-1, 1]$

4. Compute the exact value of the expression, or solve the equation:

$$\text{a) } \sin \left(\operatorname{Arccos} \left(-\frac{3}{5} \right) \right) \qquad \text{b) } \cos \left(\operatorname{Arcsin} \left(-\frac{3}{5} \right) \right)$$

$$\text{c) } \sin \left(\operatorname{Arctan} \left(-\sqrt{3} \right) \right) \qquad \text{d) } \sin \left(\operatorname{Arccot} \left(-\sqrt{3} \right) \right)$$

$$\text{e) } \sin (\operatorname{Arctan} 2 - \operatorname{Arctan} 3) \qquad \text{f) } \cos \left(\operatorname{Arccot} \frac{2}{5} + \operatorname{Arctan} \frac{3}{7} \right)$$

$$\text{g) } \operatorname{Arctan} x = \frac{\pi}{4} - \operatorname{Arctan} 3 \qquad \text{h) } \operatorname{Arccos} x = \operatorname{Arctan} 2 + \operatorname{Arccos} \frac{3}{4}$$

5. Put the following functions into the form $R \sin(x + \alpha)$:

$$\text{a) } f(x) = \sin x + \sin \left(x - \operatorname{Arctan} \frac{4}{3} \right)$$

$$\text{b) } f(x) = \sqrt{5} \sin \left(x + \operatorname{Arctan} \frac{1}{2} \right) + 2\sqrt{2} \sin \left(x + \frac{3\pi}{4} \right) + \sqrt{3} \sin(x - \pi)$$

6. Determine R and α such that

$$2 \sin \left(x + \operatorname{Arctan} 2\sqrt{2} \right) + R \sin(x + \alpha) = 6 \sin x, \quad x \in \mathbb{R}.$$

7. Simplify the expression and draw the graph:

$$\text{a) } y = \operatorname{Arccos}(\sin x)$$

$$\text{b) } y = \operatorname{Arctan}(\tan x)$$

$$\text{c) } y = \operatorname{Arcsin} (2 \cos^2 x - 1) \qquad \text{d) } y = \operatorname{Arctan} \frac{2 \tan x}{1 - \tan^2 x}$$

8. Show that the implicit function of Example 8 can be expressed in the form $y(x) = u(x) - x$, where u is periodic. What is the fundamental period? Is $u(x)$ even/odd? Sketch approximately the graph of the function $y(x)$.
9. The following define implicit functions, possibly multi-branch. Determine their branches as functions $x \mapsto y(x)$ on the various subintervals of $\mathbb{R}$. Sketch the functions graphically as well!

$$\begin{array}{ll} \text{a)} & x^2 + y^2 + 6x = 0 \\ \text{b)} & x^2 - y^2 - 4x = 0 \\ \text{c)} & xy^4 + 3y^2 - x = 0 \\ \text{d)} & |x|^{2/3} + |y|^{2/3} = 1 \\ \text{e)} & x^2 - 4y^2 + y^4 = 0 \\ \text{f)} & xy^4 + 3x^3 + 6y^2 - 12x = 0 \\ \text{g)} & x^2 + \frac{\sin y}{3 + \sin y} = 0 \\ \text{h)} & \arctan x + \arctan y = \frac{\pi}{4} \end{array}$$

10. What are the values of the multi-branch function $\arccos(\cos x)$ when $x = 13\pi/4$?
11. a) Define the branches of the multi-branch function $\arcsin = \{y_n, n \in \mathbb{Z}\}$ by means of the principal branch y_0 . b) Express the dependence $\varphi(x, y)$ between the points of the Cartesian coordinate system and the polar angle by means of the functions y_n .
12. (*) Let $f(x) = x^3 + 3x$ and $y \in (0, \infty)$. Prove the solvability of the equation $f(x) = y$ using the decimal-division algorithm (see Section 1.9). *Hint:* Start from the results: f is strictly increasing on the interval $[0, \infty)$ and $\lim_n f(n) = \infty$.
13. (*) Investigate for which values of $x \in \mathbb{R}$ or $(x, y) \in \mathbb{R}^2$ the following holds:
- $$\begin{array}{ll} \text{a)} & 2 \operatorname{Arcsin} x = \operatorname{Arccos}(1 - 2x^2) \\ \text{b)} & \operatorname{Arctan} x + \operatorname{Arctan} y = \operatorname{Arctan} \frac{x + y}{1 - xy} \end{array}$$

14. (*) Investigate which of the following functions of the type $f : \mathbb{N} \rightarrow \mathbb{Q}$ are injections:
- a) $f(n) = n^2 2^{-n}$ b) $f(n) = n^{333} 2^{-n}$ c) $f(n) = n^{243} 3^{-n}$ d) $f(n) = n^{80} 3^{-n}$

IV.3 Functions of two and three variables

A function of *two real variables* is of the type

$$f : \mathbb{R}^2 \rightarrow \mathbb{R} \quad \text{or} \quad f : A \rightarrow \mathbb{R}, \quad A \subset \mathbb{R}^2.$$

The *graph* of such a function is the point set of the Euclidean space E^3

$$G_{f,A} = \{P \hat{=} (x, y, z) \in \mathbb{R}^3 \mid z = f(x, y), (x, y) \in A\} \subset E^3,$$

whose general name is a *surface*. A function can be 'viewed', for example, by generating (usually with the aid of a computer) a perspective picture of $G_{f,A}$. Another way to make a function of two variables geometrically vivid is to draw the function's *level curves* in the Euclidean plane. A level curve is the point set of the plane

$$S_{f,c} = \{P \hat{=} (x, y) \in \mathbb{R}^2 \mid f(x, y) = c\} \subset E^2 \quad (c \in \mathbb{R}).$$

On a map, the level curves are the contour lines (f = the elevation function of the terrain).*

EXAMPLE 1 An island rising from the sea floor is described by the function

$$h(x, y) = -\frac{1}{10} (x^2 + 2y^2 + 2xy + 4x + 14y),$$

where $h(x, y)$ = the elevation of the terrain above sea level (in the underwater parts $h(x, y) < 0$). The unit of length = 100 m. Sketch the location of the island and some contour lines. Where and how high is the highest point of the island?

Solution Denote $f(x, y) = -10 h(x, y) = x^2 + 2y^2 + 2xy + 4x + 14y$ and let us first try to write f in a slightly clearer form

$$f(x, y) = (x - x_0)^2 + 2(y - y_0)^2 + 2(x - x_0)(y - y_0) + c,$$

*The geometric notion of a *curve* is, understood intuitively, a 'connected line without width'. The concept is surprisingly hard to define precisely and at the same time in full generality, and the definition does not even succeed by the tools of algebra and geometry alone. In applications one typically encounters only special cases of curves, such as lines, second-degree curves, etc., in which case conceptual problems are generally absent. In this and the following sections the term 'curve' is used intuitively, thinking mainly of the unproblematic special cases. — The term 'level curve', on the other hand, is deliberately careless with applications too in mind, for it may well be a point set with 'blotches' or isolated points, or the set may be empty. In fact, if A is any subset of $\mathbb{R}^2$, it is easy to define a function f such that $S_{f,c} = A$ for a given c .

i.e.

$$f(x, y) = g(\xi, \eta) = \xi^2 + 2\eta^2 + 2\xi\eta + c, \quad \begin{cases} \xi = x - x_0, \\ \eta = y - y_0, \end{cases}$$

where $x_0, y_0, c \in \mathbb{R}$ are to be determined. The transformation $(x, y) \mapsto (\xi, \eta)$ thus corresponds to shifting the origin of the coordinate system to the (as yet unknown) point (x_0, y_0) . Comparing the expressions for f , we see that x_0, y_0, c must be chosen so that

$$\begin{aligned} f(x, y) &= x^2 + 2y^2 + 2xy - (2x_0 + 2y_0)x - (2x_0 + 4y_0)y + (x_0^2 + 2y_0^2 + 2x_0y_0 + c) \\ &= x^2 + 2y^2 + 2xy + 4x + 14y \quad \forall (x, y) \in \mathbb{R}^2, \end{aligned}$$

i.e. we must have

$$\begin{cases} 2x_0 + 2y_0 = -4 \\ 2x_0 + 4y_0 = -14 \\ x_0^2 + 2y_0^2 + 2x_0y_0 + c = 0 \end{cases} \Leftrightarrow \begin{cases} x_0 = 3 \\ y_0 = -5 \\ c = -29 \end{cases}$$

Thus $\xi = x - 3$, $\eta = y + 5$, and

$$\begin{aligned} f(x, y) &= g(\xi, \eta) = \xi^2 + 2\eta^2 + 2\xi\eta - 29 \\ &= (\xi + \eta)^2 + \eta^2 - 29. \end{aligned}$$

From the last expression we see that the minimum point of g = the maximum point of h is at the point

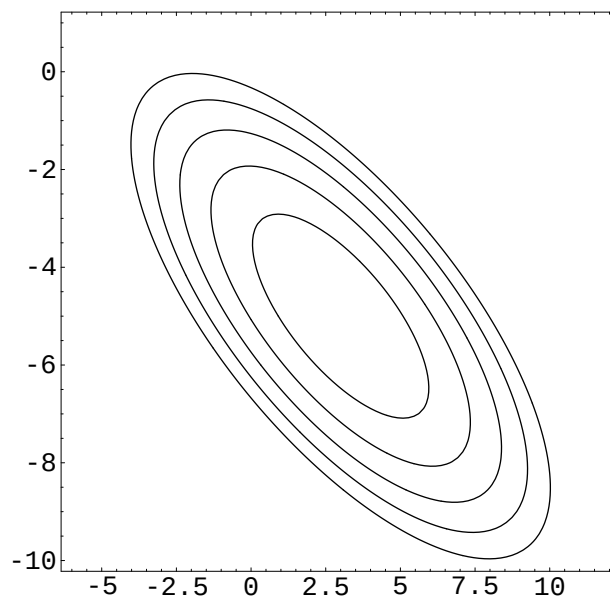
$$(\xi, \eta) = 0 \Leftrightarrow (x, y) = (3, -5),$$

and the highest point of the island is 2.9 km above sea level. The shoreline of the island is the point set

$$S = \{P \hat{=} (x, y) \in \mathbb{R}^2 \mid f(x, y) = 0\}.$$

By sketching further contour lines one obtains an overall picture of the island.

□



EXAMPLE 2 Determine the range of the function $f(x, y) = x + 2y + 2$ on the unit square

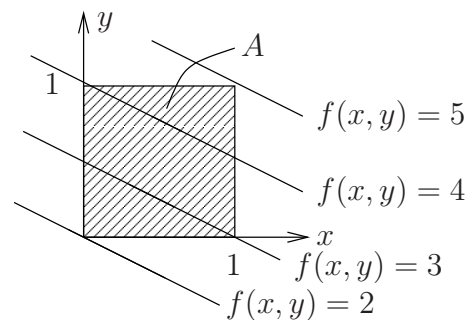
$$A = \{(x, y) \in \mathbb{R}^2 \mid 0 \leq x \leq 1, 0 \leq y \leq 1\}.$$

Solution The level curves $S : f(x, y) = c$ are the lines

$$x + 2y + 2 = c,$$

so the range is (see the figure!)

$$f(A) = [2, 5]. \quad \square$$



Functions of three variables

A function of three real variables is of the type

$$f : \mathbb{R}^3 \rightarrow \mathbb{R} \quad \text{or} \quad f : A \rightarrow \mathbb{R}, \quad A \subset \mathbb{R}^3.$$

Functions of three variables can be visualized, for example, by slicing: one chooses a finite set of values z_i of the variable z and studies the functions of two variables

$$g_i(x, y) = f(x, y, z_i).$$

EXAMPLE 3 : Tomography. (Computed) tomography, much used in medicine, determines a so-called attenuation function f , whose domain is the human body or a part of it. The values of the attenuation function are real numbers that indicate the darkness of the shadow. From the function f one obtains, by measurement, approximate values of the sliced functions $g_i(x, y) = f(x, y, z_i)$ at chosen (finitely many) values of z_i . The functions g_i are determined by combining a large collection of X-ray (shadow) images taken from different directions by computational means (with the aid of a computer).

Let the object of study be an (idealized) human head

$$A = \{P \triangleq (x, y, z) \in \mathbb{R}^3 \mid x^2 + y^2 + z^2 \leq 7^2\}$$

(unit of length = cm). Suppose the tomographic imaging produces the attenuation function

$$f(x, y, z) = 1 + 0.02(2x - 3y + 6z).$$

Where is it brightest and where darkest?

Solution The function f has in practice been obtained by combining a collection of slice functions $g_i(x, y) = f(x, y, z_i)$, where $-7 < z_i < 7$. Insofar as the measurements and the functions g_i and f computed from them are regarded as exact, we must have

$$g_i(x, y) = 0.02(2x - 3y) + c_i, \quad (x, y) \in A_i,$$

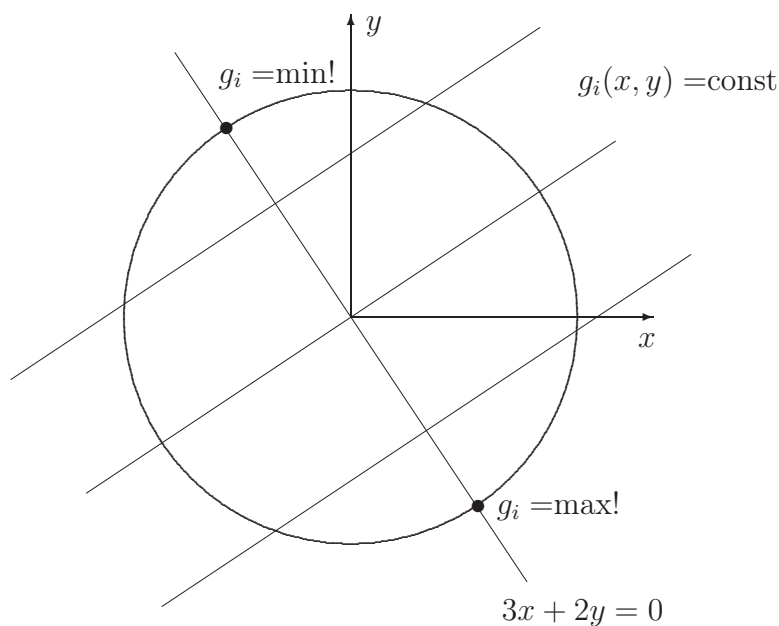
where $c_i = 0.12z_i + 1$, and A_i is the disk-shaped cross-sectional figure

$$A_i = \{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 \leq r_i^2\}, \quad r_i = \sqrt{7^2 - z_i^2}.$$

The level curves of the function g_i are the lines

$$2x - 3y = \text{const},$$

so the maximum and minimum of g_i are found at the intersection points of the normal $3x + 2y = 0$ to these lines and the boundary of A_i .



By studying the various slice images, the maximum and minimum values of f are also found approximately. Here, however, f is known exactly, so one can directly study the *level surfaces* of f . These are the planes

$$2x - 3y + 6z = \text{const},$$

so one can deduce geometrically that the maximum and minimum of f are found at the intersection points of the common normal to the level surfaces passing through the origin

$$\begin{cases} x = 2t, \\ y = -3t, \\ z = 6t \end{cases}$$

and the boundary surface of the body

$$x^2 + y^2 + z^2 = 7^2.$$

These points correspond to the values of t

$$(2t)^2 + (-3t)^2 + (6t)^2 = 7^2 \quad \Leftrightarrow \quad t = \pm 1.$$

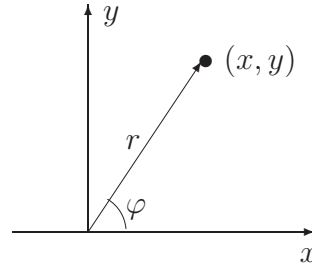
The maximum and minimum values of the shadow are thus deduced to be

$$\begin{aligned} f(2, -3, 6) &= 1.98 = f_{\max}, \\ f(-2, 3, -6) &= 0.02 = f_{\min}. \end{aligned} \quad \square$$

Functions in curvilinear coordinate systems

When studying functions of two and three real variables, it can be helpful to pass to a curvilinear polar, cylindrical, or spherical coordinate system when the geometry of the function's domain is suited to such a transformation. For example, if a function of two real variables is defined on a rotationally symmetric set $A \subset \mathbb{R}^2$ (possibly also $A = \mathbb{R}^2$), passing to polar coordinates may help. The passage is effected by the transformation (cf. Section II.8)

$$\begin{aligned} f(x, y) &= f(r \cos \varphi, r \sin \varphi) \\ &= g(r, \varphi). \end{aligned}$$



Note that the transformed function g is in fact a composite function

$$g(r, \varphi) = f(x(r, \varphi), y(r, \varphi)),$$

where $x(r, \varphi) = r \cos \varphi$, $y(r, \varphi) = r \sin \varphi$. The passage from polar coordinates to Cartesian is effected by the inverse transformation

$$f(x, y) = g(r(x, y), \varphi(x, y)),$$

where $r(x, y) = \sqrt{x^2 + y^2}$ and $\varphi(x, y)$ is the polar angle corresponding to the point $(x, y)^*$ (the formula for φ is somewhat tricky, see Example 11 of the previous section).

When considering functions of three variables, one can pass to a cylindrical or spherical coordinate system by the transformations

$$\text{Cylindrical: } f(x, y, z) = f(r \cos \varphi, r \sin \varphi, z) = g(r, \varphi, z).$$

$$\text{Spherical: } f(x, y, z) = f(r \sin \theta \cos \varphi, r \sin \theta \sin \varphi, r \cos \theta) = g(r, \theta, \varphi).$$

In the case of the cylindrical coordinate system, the transformation $(x, y) \hookrightarrow (r, \varphi)$, and likewise the inverse transformation $(r, \varphi) \hookrightarrow (x, y)$, is the same as in

*In the notations $x(r, \varphi)$, $y(r, \varphi)$, $r(x, y)$, and $\varphi(x, y)$, function symbols have been made out of what were originally independent variables x, y or r, φ . In coordinate transformations (and also in implicit functions, cf. the previous section) such notations are common, because they make the computation clearer.

polar coordinates. In the spherical coordinate system the inverse transformation is

$$\begin{aligned} g(r, \theta, \varphi) &= g(r(x, y, z), \theta(x, y, z), \varphi(x, y)) \\ &= f(x, y, z), \end{aligned}$$

where

$$\begin{aligned} r(x, y, z) &= \sqrt{x^2 + y^2 + z^2}, \\ \theta(x, y, z) &= \operatorname{Arccos} \left(\frac{z}{\sqrt{x^2 + y^2 + z^2}} \right), \end{aligned}$$

and $\varphi(x, y)$ is the same as in polar coordinates. The polar angle θ is not defined at the origin, nor is the angle φ on the z -axis.

EXAMPLE 4 The elevation of a mountainous terrain above sea level is, in the neighborhood of the origin,

$$f(x, y) = x^2 - 2xy - 3y^2 + 0.2x - 0.1y + 10$$

(unit = 100 m). At the origin there is an intersection of two roads. The roads are lines on the map and their side profile is also a straight line. In which directions do the roads run and what is the slope of the roads?

Solution We pass to polar coordinates:

$$f(x, y) = h(r, \varphi) = r^2(\cos^2 \varphi - 2 \cos \varphi \sin \varphi - 3 \sin^2 \varphi) + r(0.2 \cos \varphi - 0.1 \sin \varphi) + 10.$$

Starting from the origin, the roads follow the half-lines $\varphi = \varphi_1$ and $\varphi = \varphi_2$ (since they are lines on the map). Since the side profile of the roads is also a straight line, we must have

$$\cos^2 \varphi - 2 \cos \varphi \sin \varphi - 3 \sin^2 \varphi = 0.$$

Here $\cos \varphi = 0$ is not a solution, so we can divide by $\cos^2 \varphi$:

$$1 - 2 \tan \varphi - 3 \tan^2 \varphi = 0 \quad \Leftrightarrow \quad \begin{cases} \tan \varphi_1 = 1/3 & \Rightarrow \varphi_1 \approx 18^\circ \text{ or } 198^\circ, \\ \tan \varphi_2 = -1 & \Rightarrow \varphi_2 = 135^\circ \text{ or } 315^\circ. \end{cases}$$

The slope angles in the directions $\varphi_1 \approx 18^\circ$ and $\varphi_2 = 135^\circ$ are obtained by solving

$$\tan \alpha_i = 0.2 \cos \varphi_i - 0.1 \sin \varphi_i \quad \Rightarrow \quad \begin{cases} \alpha_1 \approx 9^\circ, \\ \alpha_2 \approx -12^\circ. \end{cases} \quad \square$$

EXAMPLE 5 Find (if possible) the points at which the function

$$f(x, y, z) = xyz(3x^2 + 3y^2 - z^2)/(x^2 + y^2)$$

attains its maximum or minimum value on the set

$$A = \{(x, y, z) \in \mathbb{R}^3 \mid x^2 + y^2 + z^2 \leq 96, (x, y) \neq (0, 0)\}.$$

Solution We pass to spherical coordinates. Since in spherical coordinates

$$x^2 + y^2 = r^2 \sin^2 \theta (\cos^2 \varphi + \sin^2 \varphi) = r^2 \sin^2 \theta,$$

we have (cf. Section II.5)

$$\begin{aligned} f(x, y, z) &= g(r, \theta, \varphi) = r^3 \cos \theta (3 \sin^2 \theta - \cos^2 \theta) \cos \varphi \sin \varphi \\ &= r^3 (3 \cos \theta - 4 \cos^3 \theta) \cos \varphi \sin \varphi \\ &= -\frac{1}{2} r^3 \cos 3\theta \sin 2\varphi. \end{aligned}$$

The function g attains its maximum value $192\sqrt{6}$ on the sphere ($r = \sqrt{96} = 4\sqrt{6}$), when either $\cos 3\theta = -1$, $\sin 2\varphi = 1$ or $\cos 3\theta = 1$, $\sin 2\varphi = -1$. There are four maximum points:

$(r,$	$\theta,$	$\varphi)$	$(x,$	$y,$	$z)$
$(4\sqrt{6},$	$\pi/3,$	$\pi/4)$	$(6,$	$6,$	$2\sqrt{6})$
$(4\sqrt{6},$	$\pi/3,$	$5\pi/4)$	$(-6,$	$-6,$	$2\sqrt{6})$
$(4\sqrt{6},$	$2\pi/3,$	$3\pi/4)$	$(-6,$	$6,$	$-2\sqrt{6})$
$(4\sqrt{6},$	$2\pi/3,$	$7\pi/4)$	$(6,$	$-6,$	$-2\sqrt{6})$

There are likewise four minimum points (at which $g = g_{\min} = -192\sqrt{6}$). These are obtained by changing the sign of one coordinate in the Cartesian representation of the maximum points. The function g also attains its maximum or minimum value at the values of the spherical surface coordinates correspond to the points $(0, 0, z)$ of the Cartesian coordinate system, which lie outside the domain of the original function f . $\square$

Combining functions of several variables

Real-valued functions of several real variables can be combined into composite functions by the same principles as in the one-variable case. An additional condition, however, is that the functions be compatible in type. For example, the composite function $F = f \circ g$ of the functions $f(x)$ and $g(x, y)$ can be defined as

$$F(x, y) = (f \circ g)(x, y) = f(g(x, y)), \quad \mathcal{D}_F = \{(x, y) \in \mathcal{D}_g \mid g(x, y) \in \mathcal{D}_f\}.$$

One can also define the composite functions

$$\begin{aligned} G_1(x) &= g(x, f(x)), & \mathcal{D}_{G_1} &= \{x \in \mathcal{D}_f \mid (x, f(x)) \in \mathcal{D}_g\}, \\ G_2(y) &= g(f(y), y), & \mathcal{D}_{G_2} &= \{y \in \mathcal{D}_f \mid (f(y), y) \in \mathcal{D}_g\}, \\ G_3(x, y) &= g(f(x), f(y)), & \mathcal{D}_{G_3} &= \{(x, y) \in \mathcal{D}_f \times \mathcal{D}_f \mid (f(x), f(y)) \in \mathcal{D}_g\}. \end{aligned}$$

The function $g \circ f$, on the other hand, cannot be defined, nor can the function $g \circ g$.

EXAMPLE 6 If $f(x) = \sqrt{x}$, $\mathcal{D}_f = [0, \infty)$ and $g(x, y) = x - y^2$, $\mathcal{D}_g = \mathbb{R}^2$, then

$$\begin{aligned} F(x, y) &= (f \circ g)(x, y) = \sqrt{x - y^2}, & \mathcal{D}_F &= \{(x, y) \in \mathbb{R}^2 \mid x \geq y^2\}, \\ G(x) &= g(x, f(x)) = 0, & \mathcal{D}_G &= [0, \infty). \quad \square \end{aligned}$$

EXAMPLE 7 Determine the minimum value of the function $f(x, y) = x^2 + 2y^2 + 2xy + 4x + 14y$ on the line $S : y = 2x - 5$. (Cf. Example 1.)

Solution Using the notation $y(x)$ for the functional dependence $x \mapsto y$ prevailing on the line S , we have a composite function $F(x) = f(x, y(x))$, $\mathcal{D}_F = \mathbb{R}$. Since

$$\begin{aligned} F(x) &= f(x, 2x - 5) = x^2 + 2(2x - 5)^2 + 2x(2x - 5) + 4x + 14(2x - 5) \\ &= 13x^2 - 18x - 20 \\ &= 13 \left(x - \frac{9}{13} \right)^2 - \frac{341}{13}, \end{aligned}$$

we see that the sought minimum value is

$$F_{\min} = -\frac{341}{13} \approx -26.2.$$

The minimum point is at the point $(9/13, -47/13) \approx (0.69, -3.62)$ of the line. $\square$

Functions of several variables can also be combined by arithmetic operations in the same way as functions of one variable (Definition IV.1.2). If the number of variables of the functions being combined does not match, one must think of the functions as having 'invisible' variables.

EXAMPLE 8 If $f(x) = \tan x$ and $g(x, y) = \sin(x + y^2)$, then in writing

$$F(x, y) = \tan x + \sin(x + y^2)$$

one means the function F expressed by this rule, with domain

$$\mathcal{D}_F = \{(x, y) \in \mathbb{R}^2 \mid x \neq (n + 1/2)\pi \ \forall n \in \mathbb{Z}\}.$$

The same result is reached also via the combination of functions by thinking that $F = \tilde{f} + g$, $\mathcal{D}_F = \mathcal{D}_{\tilde{f}} \cap \mathcal{D}_g$, where $\tilde{f}$ is obtained from f by adding a second variable:

$$\tilde{f}(x, y) = \tan x, \quad \mathcal{D}_{\tilde{f}} = \mathcal{D}_f \times \mathbb{R}. \quad \square$$

Implicit functions of two variables

If F is a function of three variables and the equation

$$F(x, y, z) = 0$$

is solvable for z when $(x, y) \in A \subset \mathbb{R}^2$, then the equation defines on A an implicit function of two variables (possibly multi-branch). If for this one uses the function notation $z(x, y)$, then the definition is

$$F(x, y, z(x, y)) = 0, \quad (x, y) \in A.$$

EXAMPLE 9 The equation $x^2 + y^2 + z^2 = R^2$ defines the two-branch implicit function $z(x, y) = \pm\sqrt{R^2 - x^2 - y^2}$ on the disk $A : x^2 + y^2 \leq R^2$. $\square$

EXAMPLE 10 If $m \in \mathbb{N}$, $m \geq 2$, then the equation

$$y, z \in \mathbb{C} : y^m = z$$

defines the m -branch complex function $y = \sqrt[m]{z}$, cf. Section III.3. If we write $z = x + iy$ and $y = u(x, y) + iv(x, y)$, then u and v are m -branch functions of the type $u, v : \mathbb{R}^2 \rightarrow \mathbb{R}$. $\square$

EXERCISES

1. Determine the ranges of the following functions on the given line segment AB :
 - a) $f(x, y) = x^2 + 2xy - y^2$, $A = (1, 2)$, $B = (-1, 3)$
 - b) $f(x, y, z) = x + xy + yz + z^2$, $A = (1, 1, 1)$, $B = (-1, 2, -3)$

2. Determine the ranges of the following functions on the given set A :
- a) $f(x, y) = x + 3y - 1$, A = the triangle with vertices $(0, 1)$, $(4, 0)$, and $(3, 4)$
 - b) $f(x, y, z) = x + 2y - 3z$, $A = \{(x, y, z) \mid (x-1)^2 + (y+1)^2 + (z-2)^2 \leq 9\}$
 - c) $f(x, y, z) = x + xy + yz + z^2$, A = the line $2x - 2 = y + 1 = 4 - 2z$
3. On the plane $z = 0$ there is a lake $A = \{(x, y) \in \mathbb{R}^2 \mid h(x, y) < 0\}$, where

$$h(x, y) = 4x^2 + y^2 + 24x + 8y$$

is the elevation profile of the lake bottom. Here h is expressed in meters and x, y in kilometers. A highway crosses the lake along the line $y = x + 2$. Where is the deepest point of the lake and what is the depth at that point? Sketch the shoreline of the lake and compute at which points the highway intersects the shoreline.

4. Convert the following functions to polar or spherical coordinates:

$$\begin{array}{lll} \text{a) } f(x, y) = x + 2y & \text{b) } f(x, y) = xy^2 & \text{c) } f(x, y) = \frac{xy}{x^2 + y^2} \\ \text{d) } f(x, y) = \max\{0, x, y\} & \text{e) } f(x, y) = \begin{cases} x^2/y, & \text{when } x > 0 \text{ and } y > 0 \\ 0, & \text{otherwise} \end{cases} \\ \text{f) } f(x, y) = \begin{cases} xy^2 - y^3, & \text{when } 0 \leq x \leq y \\ 0, & \text{otherwise} \end{cases} & \text{g) } f(x, y, z) = \frac{xyz}{x^2 + y^2 + z^2} \\ \text{h) } f(x, y, z) = \frac{xy^2z^3}{x^2 + y^2} & \text{i) } f(x, y, z) = \begin{cases} z^2 - xyz, & \text{when } z > 0 \\ 0, & \text{when } z \leq 0 \end{cases} \end{array}$$

5. Convert the following functions expressed in polar coordinates to Cartesian coordinates:

$$\begin{array}{ll} \text{a) } g(r, \varphi) = r^3(\cos \varphi - \sin \varphi) & \text{b) } g(r, \varphi) = \sin 2\varphi - r^2 \cos 2\varphi \\ \text{c) } g(r, \varphi) = r^2(\tan \varphi + \cot \varphi) & \text{c) } g(r, \varphi) = \frac{\sin \varphi}{2 + \cos \varphi} \end{array}$$

6. Let $f(x, y) = x + y$ and $g(x, y) = xy$, $(x, y) \in \mathbb{R}^2$. Define, as simplified expressions (rules of computation), the following composite functions:
- a) $f(x, g(x, y))$ b) $f(g(x, y), y)$ c) $f(g(x, y), g(x, y))$
 - d) $g(x, f(x, y))$ e) $g(f(x, y), y)$ f) $g(f(x, y), f(x, y))$
 - g) $f(f(x, y), f(x, y))$ h) $g(g(x, y), g(x, y))$

7. Let $f(x) = \sqrt{2-x}$ and $g(x, y) = \sqrt{x-y^2}$ ($x, y \in \mathbb{R}$). Define the composite function $F = f \circ g$ (rule of computation and domain).
8. Let $f(x) = \sqrt{25-x^2}$ and $g(x, y) = 3x + 4y - 1$ ($x, y \in \mathbb{R}$). Determine the smallest and largest value of the composite function $F = f \circ g$ on the quarter of the unit disk $A = \{(x, y) \in \mathbb{R}^2 \mid x \geq 0 \ \& \ y \geq 0 \ \& \ x^2 + y^2 \leq 1\}$.
9. a) Express the real part, imaginary part, and absolute value of the complex function $f(z) = (z+1)^2 + (z+i)^2$ as functions of the type $g: \mathbb{R}^2 \rightarrow \mathbb{R}$.
b) Define the two-branch functions $u = \{u_1, u_2\}$ and $v = \{v_1, v_2\}$ such that $u(x, y) + iv(x, y) = \sqrt{z} \ \forall z = x + iy \in \mathbb{C}$.
10. The equation $x + 3xyz^2 + z^4 = 0$ ($x, y, z \in \mathbb{R}$) defines an implicit function $z = f(x, y)$. a) Compute the values of f at the points $(0, 0)$, $(2, -1)$, and $(1, 1)$. b) What are the branches of f (domains and rules of computation) more generally?
11. (*) The elevation of a mountainous terrain above sea level is, near the origin O , the function

$$f(x, y) = \frac{1}{20}(3x^2 - 5xy - 2y^2) + 2.$$

At the origin there is an intersection of two roads S_1, S_2 . The roads are fitted to the terrain and horizontal, and moreover they are lines on the map as well. A straight railway crosses the roads S_1, S_2 at the points $A \in S_1$, $B \in S_2$. Both points are at distance 1 from O (unit = km). What is the greatest drop into the valley from a railway bridge whose ends are at the points A, B ?

12. (*) We want to determine at which points the function $f(x, y) = x^2 - 2xy + 3y^2$ attains its greatest and smallest value a) on the circle $S: x^2 + y^2 = 9$, b) on the disk $A = \{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 \leq 9\}$. Solve the problem using polar coordinates (cf. Exercise [II.5:20](#)).
13. (*) It is known about the function $f(x) = x - x^3$ that on the interval $[0, 1]$ f attains its greatest value at the point $x = 1/\sqrt{3}$. In which direction, starting from the origin, does the function a) $f(x, y) = xy^2$, b) $f(x, y, z) = xy^2z^3$ increase fastest relative to the distance traveled?
14. (*) (Function in a keyhole) Let

$$A = \{(x, y, z) \in \mathbb{R}^3 \mid (x, y) \in B_1 \cup B_2, \ z \in [0, 10]\},$$

where

$$B_1 = \{ (x, y) \in \mathbb{R}^2 \mid x^2 + (y - 1)^2 \leq 2 \text{ \& } y \geq 0 \},$$

$$B_2 = \{ (x, y) \in \mathbb{R}^2 \mid x \in [-1, 1] \text{ \& } y \in [-4, 0] \}.$$

At which points of A does the function $f(x, y, z) = x - 3y + 2z$ attain its greatest and at which its smallest value?

IV.4 Parametric curves and surfaces

A *parametric curve* in the plane or in space is a function of the form

$$t \in A \mapsto P(t) \in E^d,$$

where $A \subset \mathbb{R}$ is an interval (often a closed interval) and $d = 2$ or $d = 3$. The variable t is here called the *parameter*. The curve is a *plane curve* if $d = 2$, a *space curve* if $d = 3$. Using the geometric correspondences already familiar to us, we can write

$$P(t) \cong \overrightarrow{OP}(t) = \begin{cases} x(t)\vec{i} + y(t)\vec{j} & \widehat{=} (x(t), y(t)), & (d = 2) \\ x(t)\vec{i} + y(t)\vec{j} + z(t)\vec{k} & \widehat{=} (x(t), y(t), z(t)), & (d = 3) \end{cases}$$

where x, y, z are functions of the type $f : A \rightarrow \mathbb{R}$. Accordingly, a parametric curve can be interpreted as a *vector-valued* function of a real variable, in which case a natural way of representing it is also the vector notation

$$\vec{r}(t) = \begin{cases} x(t)\vec{i} + y(t)\vec{j}, & t \in A, & \text{(plane curve)} \\ x(t)\vec{i} + y(t)\vec{j} + z(t)\vec{k}, & t \in A. & \text{(space curve)} \end{cases}$$

In connection with the correspondence $E^d \leftrightarrow \mathbb{R}^d$, a parametric curve can equally well be presented as a system of equations, e.g. in the case of a plane curve

$$(x, y) = (x(t), y(t)) \Leftrightarrow \begin{cases} x = x(t), \\ y = y(t). \end{cases} \quad \text{(plane curve)}^*$$

EXAMPLE 1 A function of a real variable $f : [a, b] \rightarrow \mathbb{R}$ can be interpreted as a parametric plane curve

$$\vec{r} = \vec{r}(t) = t\vec{i} + f(t)\vec{j} \Leftrightarrow \begin{cases} x = x(t) = t, \\ y = y(t) = f(t), \end{cases} \quad t \in [a, b]. \quad \square$$

EXAMPLE 2 Parametric space curves familiar from Sections II.7 and II.8 are

line in space: $\vec{r}(t) = (x_0 + \alpha t)\vec{i} + (y_0 + \beta t)\vec{j} + (z_0 + \gamma t)\vec{k} = \vec{r}_0 + t\vec{v}, \quad t \in \mathbb{R},$

$$\text{helix: } \begin{cases} x = x(\varphi) = R \cos \varphi, \\ y = y(\varphi) = R \sin \varphi, \\ z = z(\varphi) = a\varphi, \end{cases} \quad \varphi \in \mathbb{R}. \quad \square$$

*In the notations $x = x(t)$ and $y = y(t)$ the symbols x, y are used in two different meanings: on the right as the symbol of a function, on the left as the symbol of an element of that function's range. Inconsistencies of this type are considered tolerable in mathematical practice, on the grounds that they simplify the notation.

The 'geometric trace' that a parametric curve leaves in Euclidean space is the range $S = \{P(t) \hat{=} \vec{r}(t) \mid t \in A\} \subset E^d$, which may be referred to by such terms as *curve* or (the curve's) *arc*. In its simplest form S is a connected, non-self-intersecting line that begins at some point A and ends at another point B , i.e. a so-called *simple arc*. Examples of these are a line segment or a circular arc. The line may also be unending, like a line, or closed, a *closed curve*, like a circle in the plane or in space. The set S may also be self-intersecting, i.e. it may have 'loops'.^{*} If we take as our starting point only such a 'line' S , i.e. a purely geometric object, then a function $\vec{r}(t)$, $t \in A$ whose range $= S$ is called a *parametric representation* or *parametrization* of S . The parametrizing function $t \in A \mapsto \vec{r}(t) \in S$ need not be injective, i.e. the same point $P \in S$ may be reached with many (even infinitely many) parameter values.

EXAMPLE 3 A natural parametrization of the plane circle

$$S = \{P \hat{=} (x, y) \in \mathbb{R}^2 \mid (x - x_0)^2 + (y - y_0)^2 = R^2\}$$

is

$$\begin{cases} x = x_0 + R \cos t, \\ y = y_0 + R \sin t, \quad t \in [0, 2\pi). \end{cases}$$

Here the interval $[0, 2\pi)$ may also be replaced e.g. by $A = (-\pi, \pi]$, $A = [0, 2\pi]$ or $A = \mathbb{R}$. In the two latter alternatives the parametrization is not an injection. $\square$

EXAMPLE 4 If S is a line in space, then its most common parametrization is $\vec{r}(t) = \vec{r}_0 + t\vec{v}$, $t \in \mathbb{R}$, where $P_0 \hat{=} \vec{r}_0$ is a point of the line and $\vec{v} \neq \vec{0}$ is the direction vector of the line (cf. Section II.7). There are already infinitely many parametrizations of this kind, but the possibilities do not end here: If $\vec{r}_0$ and $\vec{v}$ satisfy the stated conditions, then more generally one can choose as the parametrization

$$\vec{r}(t) = \vec{r}_0 + f(t)\vec{v}, \quad t \in A,$$

where the choice of the function $f : A \rightarrow \mathbb{R}$ is restricted only by the condition $\mathcal{R}_f = \mathbb{R}$. For example one can choose $f(t) = \tan t$, $A = (-\pi/2, \pi/2)$ (f injective), or $f(t) = t \sin t$, $A = \mathbb{R}$ (f not injective). $\square$

EXAMPLE 4 (continued) If one wishes to parametrize the line segment lying on the line S whose endpoints are $A \hat{=} \vec{r}_1$ and $B \hat{=} \vec{r}_2$, then this can be done in the

^{*}The geometric classification of plane or space curves defined in the form $S = \{P(t) \hat{=} \vec{r}(t) \mid t \in A\}$ cannot in reality be made precise without additional conditions imposed on the coordinate functions x, y, z of the function $\vec{r}$. Cf. the footnote concerning curves in the previous section.

form

$$\vec{r}(t) = f(t)\vec{r}_1 + [1 - f(t)]\vec{r}_2, \quad t \in A,$$

where f is a function of the type $f : A \rightarrow \mathbb{R}$, $A \subset \mathbb{R}$, and $\mathcal{R}_f = [0, 1]$. The simplest parametrization is obtained by choosing $A = [0, 1]$ and $f(t) = t$. $\square$

If the parameter t can be eliminated from the equations $x = x(t)$, $y = y(t)$, $t \in A$ of a plane curve, the result is an equation of the form

$$F(x, y) = 0,$$

where F is thus a function of two variables for which $F(x(t), y(t)) = 0 \quad \forall t \in A$. If $S = \{P \in E^2 \mid P \hat{=} (x(t), y(t)) \text{ for some } t \in A\}$, then the above equation is said to be the *equation* of the *curve* S (or of the point set S). In the case of a space curve, elimination of the parameter (when it succeeds) leads to a system of equations of the form

$$\begin{cases} F_1(x, y, z) = 0, \\ F_2(x, y, z) = 0. \end{cases}$$

EXAMPLE 5 If $a, b > 0$, then the parametric plane curve

$$S : \begin{cases} x = a \cos t, \\ y = b \sin t, \quad t \in [0, 2\pi] \end{cases}$$

is called an *ellipse* (in the case $a = b$ a circle). By eliminating t we obtain as the equation of S

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1. \quad \square$$

Trajectory

In a typical physical application of a parametric curve, the parameter t is a time variable, A is the time interval under consideration, and $P(t) \hat{=} \vec{r}(t)$ is the position of a moving point (e.g. a particle thought of as point-like, or a point of a moving rigid body) at the instant t . Then the range S of the function $P(t)$, $t \in A$ is the trajectory of that point over the time interval A . The function $t \mapsto \vec{r}(t)$, $t \in A$ is a parametrization of S that tells the whole history of motion.

EXAMPLE 6 : **Projectile parabola.** A stone is thrown from a tower at height h with initial speed v_0 and at an angle α to the horizontal. What is the trajectory if air resistance is neglected?

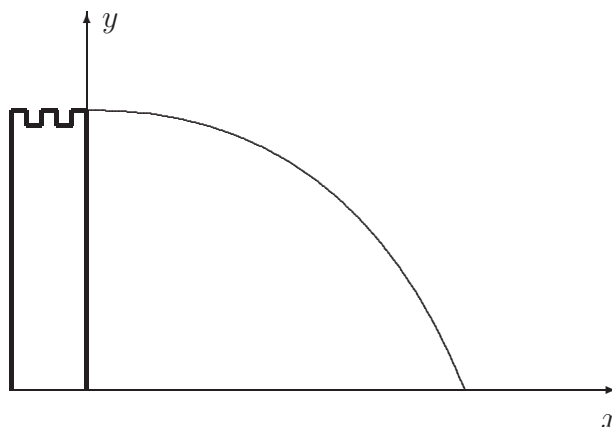
Solution Consider the motion in a (plane) coordinate system in which x measures the horizontal distance from the launch point and y the height above the ground level. By the laws of motion, the position $P(t) = (x(t), y(t))$ of the stone, as a function of the flight time t , is the parametric curve

$$\begin{cases} x(t) = v_0 t \cos \alpha, \\ y(t) = h + v_0 t \sin \alpha - \frac{1}{2}gt^2, \end{cases}$$

where g = the acceleration of gravity. By eliminating t and noting that $1/\cos^2 \alpha = 1 + \tan^2 \alpha$ we obtain the equation of the trajectory in the form

$$y = h + kx - (1 + k^2) \frac{x^2}{2a},$$

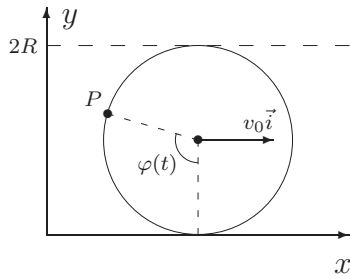
where $k = \tan \alpha$ and $a = v_0^2/g$. The trajectory is an arc of a *parabola*. In the case $\alpha = 0$ shown in the figure, the stone hits the ground at the instant $t = \sqrt{2h/g}$. $\square$



EXAMPLE 7 : Cycloid. A wheel of radius R rolls without slipping along a plane such that the velocity of the wheel's center is $v_0 \vec{i}$, $v_0 = \text{constant}$. Determine the position of a point P on the wheel's outer rim as functions of time t .

Solution Assume that the wheel rolls along the x -axis and that P is at the origin when $t = 0$. Then the parametric representation of the trajectory is

$$\begin{cases} x(t) = v_0 t - R \sin \varphi(t), \\ y(t) = R - R \cos \varphi(t), \end{cases} \quad \text{where } \varphi(t) \text{ is the rolling angle.}$$



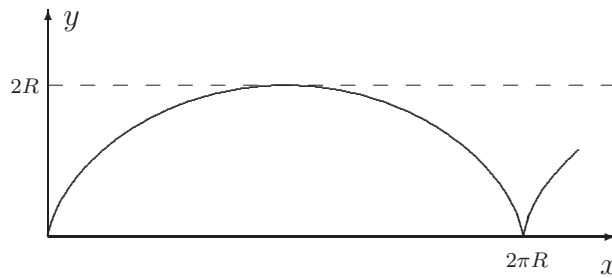
Since no slipping occurs, we must have $R\varphi(t) = v_0 t$, so the position vector of P at the instant t is

$$\vec{r}(t) = \left[v_0 t - R \sin\left(\frac{v_0 t}{R}\right) \right] \vec{i} + R \left[1 - \cos\left(\frac{v_0 t}{R}\right) \right] \vec{j}.$$

If the rolling angle φ is taken as the parameter, then the parametric representation of the trajectory is

$$\begin{cases} x = x(\varphi) = R(\varphi - \sin \varphi), \\ y = y(\varphi) = R(1 - \cos \varphi). \end{cases}$$

This is called a *cycloid*. $\square$



EXAMPLE 8 A point-like particle is at the instant $t = 0$ (unit of t : s) at the point $(1, 1, 1)$ (unit: m) and moves in a straight line at constant velocity $v = 10$ (unit: m/s) such that at some instant of time the particle is at the point $(2, -1, 0)$. Determine the position $(x(t), y(t), z(t))$ of the particle for $t \geq 0$.

Solution The particle moves on a line whose direction vector is $(2\vec{i} - \vec{j}) - (\vec{i} + \vec{j} + \vec{k}) = \vec{i} - 2\vec{j} - \vec{k}$. The unit vector pointing in the direction of motion is thus

$$\vec{e} = \frac{1}{\sqrt{6}}(\vec{i} - 2\vec{j} - \vec{k}),$$

and the position vector of the particle at the instant $t \geq 0$ is therefore

$$\vec{r}(t) = \vec{i} + \vec{j} + \vec{k} + (vt) \vec{e} \Leftrightarrow \begin{cases} x(t) = 1 + \frac{10t}{\sqrt{6}}, \\ y(t) = 1 - \frac{20t}{\sqrt{6}}, \\ z(t) = 1 - \frac{10t}{\sqrt{6}}. \end{cases} \quad \square$$

Parametric surfaces

A *parametric surface* in Euclidean space E^3 is a map (function) of the type

$$(u, v) \in A \mapsto P(u, v) \in E^3,$$

where $A \subset \mathbb{R}^2$ and the variables u, v are called parameters. In connection with the correspondences $P \in E^3 \leftrightarrow \vec{r} \in V \leftrightarrow (x, y, z) \in \mathbb{R}^3$ ($V = \{\text{space vectors}\}$) the codomain of the map can equally well be thought of as V or $\mathbb{R}^3$. For the map one can then use either the vector notation

$$\vec{r} = \vec{r}(u, v), \quad (u, v) \in A,$$

or the corresponding coordinate-form representation

$$\begin{cases} x = x(u, v), \\ y = y(u, v), \\ z = z(u, v), \end{cases} \quad (u, v) \in A.$$

The range $S \subset E^3$ of the function $(u, v) \in A \mapsto P(u, v) \in E^3$ is a *surface* as a geometric object.* The function itself is then one *parametrization* of S . If the starting point is a surface S , then the parametrization is often chosen so that the source set A is geometrically as simple as possible, e.g. a rectangle. The *equation* of the *surface* S is an equation of the form

$$F(x, y, z) = 0,$$

which holds for every $(x, y, z) \hat{=} P \in S$. One arrives at the equation if the parameters u, v can be eliminated from the above coordinate-form representation. If a function F of three real variables is known from the outset, then more generally the equation $F(x, y, z) = c$ ($c \in \mathbb{R}$) is said to define a *level surface* of F (insofar as it is a surface, see the footnote).

*In a more precise definition of surfaces there are the same problems as with curves, cf. the footnotes above. In this connection we do not undertake any attempts at precision, but rely on geometric intuition.

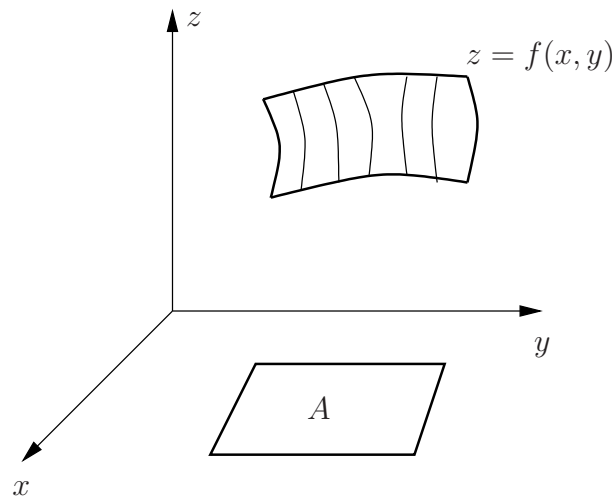
EXAMPLE 9 'The mother of all surfaces' is the *plane*, whose general parametric representation is of the form (cf. Section II.7)

$$\begin{cases} x(u, v) = x_0 + \alpha_1 u + \alpha_2 v, \\ y(u, v) = y_0 + \beta_1 u + \beta_2 v, \\ z(u, v) = z_0 + \gamma_1 u + \gamma_2 v. \end{cases}$$

Defined in this way, the plane passes through the point $\vec{r}_0 \hat{=} (x_0, y_0, z_0)$ and its direction vectors are $\vec{v}_1 = \alpha_1 \vec{i} + \beta_1 \vec{j} + \gamma_1 \vec{k}$ and $\vec{v}_2 = \alpha_2 \vec{i} + \beta_2 \vec{j} + \gamma_2 \vec{k}$. By eliminating the parameters (assuming $\vec{v}_1$ and $\vec{v}_2$ linearly independent) one can derive for the plane an equation of the form $F(x, y, z) = ax + by + cz + d = 0$ (cf. Section II.7). $\square$

EXAMPLE 10 If $f : \mathcal{D}_f \rightarrow \mathbb{R}$, $\mathcal{D}_f \subset \mathbb{R}^2$ is a function of two real variables, then the *graph* of f over the set $A \subset \mathcal{D}_f$ is a surface whose equation is

$$z = f(x, y), \quad (x, y) \in A.$$



The natural parametrization of the surface is in this case

$$x = u, \quad y = v, \quad z = f(u, v), \quad (u, v) \in A. \quad \square$$

EXAMPLE 11 The equation of a general sphere in space is

$$(x - x_0)^2 + (y - y_0)^2 + (z - z_0)^2 = R^2.$$

The most natural parametrization is based on spherical-surface coordinates:

$$\begin{cases} x = x(\theta, \varphi) = x_0 + R \sin \theta \cos \varphi, \\ y = y(\theta, \varphi) = y_0 + R \sin \theta \sin \varphi, \\ z = z(\theta, \varphi) = z_0 + R \cos \theta, \quad (\theta, \varphi) \in [0, \pi] \times [0, 2\pi]. \end{cases}$$

In the spherical coordinate system whose origin is at the point (x_0, y_0, z_0) , the equation of the surface is simplest of all: $r = R$. $\square$

EXAMPLE 12 If $a, b, c > 0$, then the equation

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$$

defines a surface called an *ellipsoid*. A parametrization based on spherical-surface coordinates is

$$\begin{cases} x = a \sin \theta \cos \varphi, \\ y = b \sin \theta \sin \varphi, \\ z = c \cos \theta, \quad (\theta, \varphi) \in [0, \pi] \times [0, 2\pi]. \end{cases} \quad \square$$

Surfaces of revolution

A *surface of revolution* is generated when a plane curve rotates about a line lying in the plane. Let the curve be given in the form

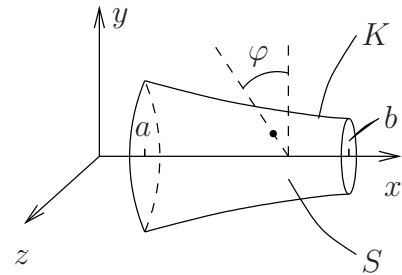
$$K = \{(x, y) \in \mathbb{R}^2 \mid y = f(x) \text{ \& } x \in [a, b]\},$$

where $f(x) \geq 0 \forall x \in [a, b]$. Then, as the curve rotates about the x -axis, a surface S is generated whose natural parameters are $u = x$ and $v = \varphi = \text{rotation angle}$, so that the parametrization of the surface becomes

$$\begin{cases} x = u, \\ y = f(u) \cos \varphi, \\ z = f(u) \sin \varphi, \end{cases}$$

where

$$(u, \varphi) \in A = [a, b] \times [0, 2\pi].$$



By eliminating the parameters u, φ we obtain the *equation of the surface of revolution*

$$y^2 + z^2 = [f(x)]^2, \quad x \in [a, b].$$

EXAMPLE 13 Parametrize the surface of revolution

$$S: \quad x^2 + y^2 = z, \quad z \geq 0.$$

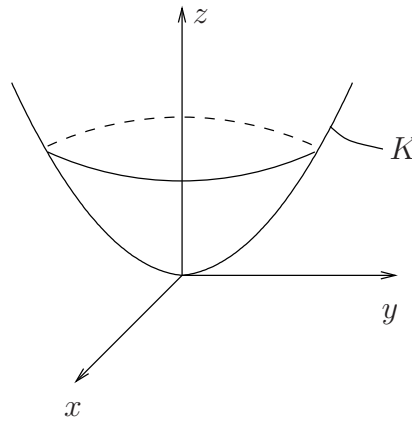
Solution The surface S is generated when the curve $K = \{(y, z) \in \mathbb{R}^2 \mid z = y^2, y \geq 0\}$ in the yz -plane rotates about the z -axis. A natural parametrization is obtained by means of cylindrical coordinates:

$$\begin{cases} x = r \sin \varphi, \\ y = r \cos \varphi, \\ z = r^2, \end{cases}$$

where

$$(r, \varphi) \in A = [0, \infty) \times [0, 2\pi].$$

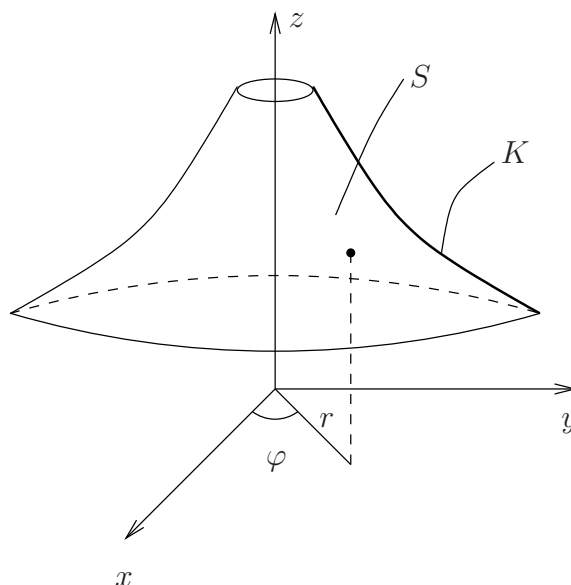
The surface is a *paraboloid* (of revolution). $\square$



The example is a special case of a more general surface of revolution, generated when the curve $K: z = f(y), y \in B \subset [0, \infty)$ in the yz -plane rotates about the z -axis. The (most natural) parametrization of the surface, based on cylindrical coordinates, is

$$\begin{cases} x = r \cos \varphi, \\ y = r \sin \varphi, \\ z = f(r), \end{cases} \quad (r, \varphi) \in A = B \times [0, 2\pi].$$

The last of these equations is in fact the equation of the surface in cylindrical coordinates (!).



Ruled surfaces

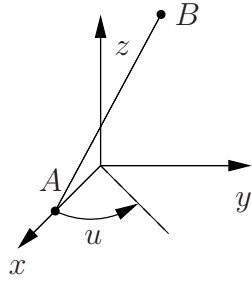
A *ruled surface* is generated when a line or line segment moves in space such that a point $P_0 \hat{=} \vec{r}_0$ of the line/segment and the direction vector $\vec{t}$ depend on a single parameter (u). The natural parametrization of the surface is then

$$\begin{aligned}\vec{r}(u, v) &= \vec{r}_0(u) + v\vec{t}(u) \\ &= x(u, v)\vec{i} + y(u, v)\vec{j} + z(u, v)\vec{k}.\end{aligned}$$

EXAMPLE 14 : **Cooling tower.** A line segment whose endpoints are $A = (2, 0, 0)$ and $B = (0, 1, 3)$ rotates about the z -axis. What kind of parametrized surface is generated? It is also a surface of revolution — what kind?

Solution The direction segment rotated by the angle u (see figure) is

$$\begin{aligned}\vec{t}(u) &= \overrightarrow{A'B'} \\ &= (-\sin u \vec{i} + \cos u \vec{j} + 3\vec{k}) - (2\cos u \vec{i} + 2\sin u \vec{j}) \\ &= -(2\cos u + \sin u)\vec{i} + (\cos u - 2\sin u)\vec{j} + 3\vec{k}.\end{aligned}$$



Thus for the surface we obtain the parametrization

$$\begin{aligned}
 \vec{r} &= \vec{r}(u, v) = \vec{r}_0(u) + v\vec{t}(u) \\
 &= 2 \cos u \vec{i} + 2 \sin u \vec{j} + v\vec{t}(u) \\
 &= [(2 - 2v) \cos u - v \sin u] \vec{i} + [v \cos u + (2 - 2v) \sin u] \vec{j} + 3v \vec{k} \\
 \Leftrightarrow \begin{cases} x = (2 - 2v) \cos u - v \sin u, \\ y = v \cos u + (2 - 2v) \sin u, \\ z = 3v, \end{cases} & \quad (u, v) \in [0, 2\pi] \times [0, 1].
 \end{aligned}$$

From the parametric representation we see that

$$\begin{aligned}
 [x(u, v)]^2 + [y(u, v)]^2 &= (2 - 2v)^2 + v^2 \\
 &= 5v^2 - 8v + 4.
 \end{aligned}$$

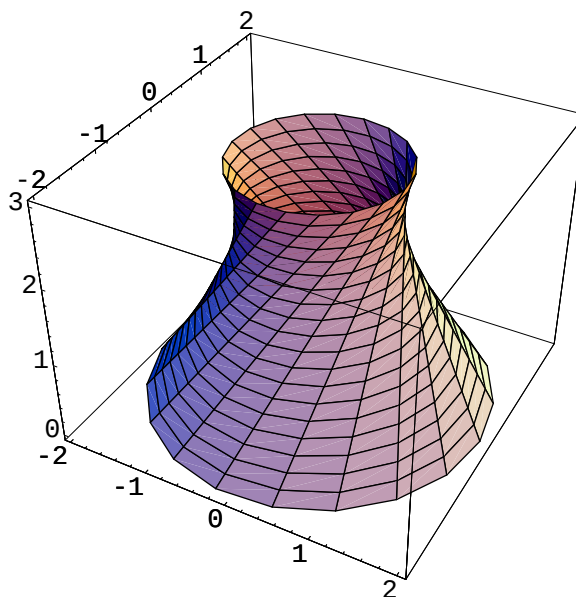
Since here $v = z(u, v)/3$, we see that the surface can be represented in the cylindrical coordinate system in the form

$$\begin{aligned}
 r^2 &= \frac{5}{9} z^2 - \frac{8}{3} z + 4 \\
 &= \frac{5}{9} \left(z - \frac{12}{5} \right)^2 + \frac{4}{5}.
 \end{aligned}$$

This is the surface of revolution generated when the curve

$$K: \quad y^2 - \frac{5}{9} \left(z - \frac{12}{5} \right)^2 = \frac{4}{5}, \quad z \in [0, 3]$$

in the yz -plane rotates about the z -axis. The curve K is an arc of a *hyperbola*, and the surface of revolution is part of a *hyperboloid of one sheet*.



The narrowest point of the surface is at the height $z = 12/5$. $\square$

EXERCISES

- Sketch the course of the following parametric plane curves. By eliminating the parameter, also derive the equation of the curve in Cartesian coordinates.
 - $x = 2 - t, y = t + 1, t \in \mathbb{R}$
 - $x = t^2, y = 2 - t, t \in [0, \infty)$
 - $x = \frac{1}{t}, y = t - 1, t \in (0, 4)$
 - $x = \frac{1}{1 + t^2}, y = \frac{t}{1 + t^2}, t \in \mathbb{R}$
 - $\vec{r} = 3 \sin \pi t \vec{i} + 4 \cos \pi t \vec{j}, t \in [-1, 1]$
 - $x = 1 - \sqrt{4 - t^2}, y = 2 + t, t \in [-2, 2]$
 - $\vec{r} = t \cos t \vec{i} + t \sin t \vec{j}, t \in [0, 4\pi]$
- One parametrization of a plane curve is $\vec{r} = \cos 2t \vec{i} + \sin^2 t \vec{j}, t \in \mathbb{R}$. Give the curve an alternative, as simple as possible parametrization.
 - Verify that $\vec{r} = (t - 1)\vec{i} + \sqrt{2t - t^2} \vec{j}, t \in [0, 2]$ and $\vec{r} = t\sqrt{2 - t^2} \vec{i} + (1 - t^2)\vec{j}, t \in [-1, 1]$ are parametrizations of the same curve. What curve is it?

3. The plane curve $S : x^3 + y^3 = 3xy$ is called the *folium of Descartes*. Derive a parametric representation for the curve by writing $y = tx$. Sketch the course of the curve from the parametric form and mark in the figure which parts of the curve correspond to parameter values in the intervals $(-\infty, -1)$, $(-1, 0)$ and $[0, \infty)$. Why does $t = -1$ not correspond to any point of the curve?
4. A circle of radius $R = 1$ rolls without slipping along the positive x -axis. Attached to the circle and rotating with it is a line segment, one endpoint of which is at the center of the circle and whose midpoint is on the circumference of the circle. Determine the position of the other endpoint of the segment (the one outside the circle) as a parametric curve $x = x(t)$, $y = y(t)$, where t is the rolling angle of the circle measured from the initial situation in which the endpoints of the segment are $(0, 1)$ and $(0, 3)$. Sketch the curve graphically. At what point does the curve first intersect itself? (The curve is called a *prolate cycloid*.)
5. Parametrize the intersection curve of the plane $T : x + y = 4$ and the cone $K : xy + yz + xz = 0$ by taking as the parameter a) $t = x$, b) $t = x - y$.
6. a) A parametrization of a space curve S is $\vec{r} = \vec{r}_0 + \cos t \vec{a} + \sin t \vec{b}$, $t \in [0, 2\pi]$, where $\vec{a}$, $\vec{b}$ and $\vec{r}_0$ are space vectors. Under exactly what conditions is S a circle in space?
 b) A circle in space has center $(1, 1, 2)$, radius $R = 3$ and lies in the plane $x - y - 3z + 6 = 0$. Derive for the circle some parametrization of the form $\vec{r} = \vec{r}_0 + \cos t \vec{a} + \sin t \vec{b}$, $t \in [0, 2\pi]$.
7. Show that the system of equations
$$\begin{cases} x^2 + y + z = 2 \\ xy + z = 1 \end{cases}$$
 defines two intersecting space curves, one of which has the parametrization $\vec{r} = t\vec{i} + (1+t)\vec{j} + (1-t-t^2)\vec{k}$, $t \in \mathbb{R}$. What is the other curve like?
8. (Angry bird and shot-putters) a) An angry bird is launched from the starting point $(x, y) = (0, 1)$ (unit of length = cm) by stretching, in the projectile parabola (see Example 6), the constant a to the value 8 cm and aiming at a pig located at the point $(4, 3)$. For what values of k is there a hit?
 b) The engineering students Yrjölä and Ståhlberg compete in the shot put. Determine which one won, given that in the competitors' best puts the parameters of the projectile parabola are
 Yrjölä: $h = 2.00$ m, $\alpha = 60.0^\circ$, $a = 8.00$ m
 Ståhlberg: $h = 1.80$ m, $\alpha = 30.0^\circ$, $a = 6.65$ m

9. Give some parametrization for the surfaces determined by the following equations:
 a) $x^3y^2z = 5$, b) $(x - z)(x + z) + y + 2z = 0$, c) $x \sin z + xy^5 + y = 1$
10. a) Derive for the surface S an equation of the form $F(x, y, z) = 0$ from the parametrization

$$S : \begin{cases} x = 3 + 2 \sin \theta \cos \varphi, \\ y = -1 + \sin \theta \sin \varphi, \\ z = 2 + 3 \cos \theta, \end{cases} \quad (\theta, \varphi) \in \mathbb{R}^2.$$

- b) A sphere K has center $(1, 1, 1)$ and radius $R = 2$. For drawing a figure, one wishes to parametrize the part of the sphere above the xy -plane ($z \geq 0$). Give a parametrization!
- c) The equation of a surface S in cylindrical coordinates is $r = \varphi$, $(\varphi, z) \in A$, where $A = [0, 4\pi] \times [-5, 5]$. Parametrize S as a ruled surface. What is the intersection curve of S and the xy -plane like?
- d) The equation of a surface in cylindrical coordinates is $r = z^2 |\cos \varphi|$. Give the equation of the surface in Cartesian coordinates. What are the intersection curves of the surface with the planes $z = c$ ($c \in \mathbb{R}$) like?
11. a) Determine the equation (in the form $F(x, y, z) = 0$) of the ruled surface generated when the line $S : x = z, y = 1$ rotates about the x -axis. Verify that the same surface (called a hyperboloid of one sheet) is also generated when a certain curve K in the xy -plane rotates about the x -axis. Sketch K graphically.
- b) When the plane curve $K : x^2 - y^2 = 1$ rotates about the x -axis, a surface called a *hyperboloid of two sheets* is generated. Determine the equation of that surface. At what points does the line $x = y = z$ intersect the surface?
12. a) A half-cone K has its apex at the origin, its axis of symmetry is the positive z -axis, and the half-line $x = 2y = 3z, x \geq 0$ lies on the surface K . Parametrize K as a surface of revolution and as a ruled surface. What is the equation of K in the cylindrical coordinate system?
- b) Parametrize the cone $K : xy + yz + xz = 0$ as a ruled surface.
13. (*) A circle has its center at the origin and radius a . Along the inside of this circle rolls without slipping another circle of radius $b < a$. Then a fixed point P of the rolling circle traces a plane curve called a *hypocycloid*. a) Show that the parametric representation of the hypocycloid passing through

the point $(a, 0)$ is

$$x = (a - b) \cos t + b \cos \left(\frac{a - b}{b} t \right), \quad y = (a - b) \sin t - b \sin \left(\frac{a - b}{b} t \right).$$

- b) Deduce that in the case $a = 2b$ the point P moves along a line segment.
 c) Show that in the case $a = 4b$ the parametric representation simplifies to the form

$$x = a \cos^3 t, \quad y = a \sin^3 t.$$

Sketch the course of this curve — called an *astroid*. What is the equation of the astroid in Cartesian coordinates?

14. (*) (Military exercise 1) The trajectory of a cannonball fired from the origin is, as a function of time t (units km and s)

$$\begin{cases} x(t) = (\sin \theta \cos \varphi + a) t, \\ y(t) = (\sin \theta \sin \varphi + b) t, \\ z(t) = (\cos \theta) t - 0.005 t^2, \end{cases}$$

where θ, φ are the aiming angles and a, b are wind parameters. Terrain obstacles impose on the aim the restriction $\tan \theta > 0.2$. How must the aim be chosen in windless weather ($a = b = 0$) so that the shell hits a target located at the point $(10, 20, 0)$? How high does the shell travel? How far from the target does the shell land with this aim, if $a = 0.002$ and $b = -0.001$?

15. (*) A line segment of length 20 moves as follows: The midpoint of the segment moves along the z -axis in the positive direction at constant speed. As it moves, the segment stays parallel to the xy -plane and rotates uniformly (constant angular velocity) such that as the midpoint moves 30 units of length, the segment makes a full revolution counterclockwise as seen from the direction of the positive z -axis. Give a parametrization of the ruled surface S that the segment traces in space, given additionally that the point $(1, 0, 0)$ is on this surface. Does the line $z = 25$, $x + y = 4$ intersect the surface S ?

IV.5 *Function space

Consider real-valued functions of one, two, or three real variables, all defined on the same set A , and denote their set by V :

$$V = \{ \text{functions } f : A \rightarrow \mathbb{R} \}.$$

On the set V , addition of functions $f, g \mapsto f + g$ and multiplication by a scalar $f \mapsto \lambda f$ are defined in the manner described earlier (rules (1) and (2) of Definition IV.1.2, with $A \subset \mathbb{R}$). On the basis of these arithmetic operations, V can be interpreted as a vector space. Functions defined on the same set can thus be conceived of as 'vectors', in which case one speaks of a *function space*. The zero element of such a space is the so-called (algebraic) *zero function*, defined by

$$\mathbf{0}(x) = 0 \quad \forall x \in A.$$

A finite set of functions $\{f_i, i = 1 \dots n\} \subset V$ is (algebraically) *linearly independent* if

$$\sum_{i=1}^n \lambda_i f_i = \mathbf{0} \quad \Rightarrow \quad \lambda_i = 0, \quad i = 1 \dots n.$$

EXAMPLE 1 The polynomials of degree at most three (domain = $\mathbb{R}$) form, by the combination rules of the function algebra, the function space

$$V = \{f = \lambda_1 f_1 + \lambda_2 f_2 + \lambda_3 f_3 + \lambda_4 f_4 \mid \lambda_i \in \mathbb{R}\},$$

where $f_1(x) = 1$, $f_2(x) = x$, $f_3(x) = x^2$ and $f_4(x) = x^3$. From the fundamental theorem of algebra (see Section III.3) it is easily deduced that the functions f_i are linearly independent, so $\{f_i, i = 1 \dots 4\}$ is a basis of V . Thus V is a four-dimensional vector space: $\dim V = 4$. $\square$

EXAMPLE 2 Functions of the form $f(x) = c_1 \sin x + c_2 \cos x$, $c_1, c_2 \in \mathbb{R}$ form a 2-dimensional function space, with natural basis $\{\sin x, \cos x\}$. $\square$

EXAMPLE 3 The function space

$$V = \{f(x) = c_1 + c_2 \cos^2 x + c_3 \sin^2 x, \quad c_i \in \mathbb{R}\}$$

is not 3-dimensional, since the function system $\{f_1, f_2, f_3\} = \{1, \cos^2 x, \sin^2 x\}$ is linearly dependent:

$$f_1 - f_2 - f_3 = \mathbf{0}.$$

Any pair of the three functions mentioned, on the other hand, is linearly independent, so V is 2-dimensional, with basis e.g. $\{1, \cos^2 x\}$. One 1-dimensional subspace of V is

$$W = \{\lambda \cos 2x \mid \lambda \in \mathbb{R}\},$$

since $\cos 2x = 2f_2 - f_1 \in V$. $\square$

EXAMPLE 4 The function space

$$V = \{f(x, y) = c_1 + c_2x + c_3y + c_4x^2 + c_5xy + c_6y^2, c_i \in \mathbb{R}\}$$

consists of two-variable polynomials of degree at most 2, and its subspace

$$W = \{f(x, y) = c_1 + c_2x + c_3y, c_i \in \mathbb{R}\}$$

of polynomials of degree at most one. The function system $\{1, x, y, x^2, xy, y^2\}$ is linearly independent (Exercise 4), so $\dim V = 6$ and $\dim W = 3$. $\square$

EXERCISES

1. Show that the functions $f_1(x) = x$ and $f_2(x) = |x|$ are linearly independent if the domain is the interval $[-1, 1]$, and linearly dependent if the domain is the interval $[0, 1]$.
2. How many elements must the set $A \subset \mathbb{R}$ have in order for the functions $f_i(x) = x^{i-1}$, $i = 1 \dots n$ defined on A to be linearly independent?
3. a) Show that $\{1, \sin x, \sin^2 x, \sin^3 x\}$ is a basis of a certain 4-dimensional function space V (domain of the functions $= \mathbb{R}$). b) Insofar as $f(x) = 2 + \cos 2x - \sin 3x \in V$, determine the coordinates (coefficients) of f in the said basis.
4. Show that the polynomial space V of Example 4 is 6-dimensional, i.e. show that $c_1 + c_2x + c_3y + c_4x^2 + c_5xy + c_6y^2 = 0 \ \forall (x, y) \in \mathbb{R}^2 \Rightarrow c_1 = \dots = c_6 = 0$.
5. (*) Let $V = \{f(x) = c_1 + c_2x + c_3x^2 \mid c_i \in \mathbb{R}\}$. Show that

$$\langle f, g \rangle = f(0)g(0) + f(1)g(1) + f(2)g(2), \quad f, g \in V$$

defines an inner product on V . Compute the orthogonal projection w of the function $u(x) = 2 + 3x - 2x^2$ (in the sense of that inner product) onto the subspace W of V whose basis is $\{1, x\}$. Draw in the same figure the graphs of the functions u and w on the interval $[0, 2]$. Verify that $u - w \perp W$, i.e. $\langle u - w, v \rangle = 0 \ \forall v \in W$.

Chapter V

Continuity and differentiability

The *continuity* and *differentiability* of a function are the fundamental conceptual starting points of the branch of mathematics loosely called *analysis*. When we speak of continuity, or more generally of the *regularity* of a function, we are concerned with the predictability of the function's values as the value (or values) of the variable varies. The concepts of continuity and differentiability bring an entirely distinct 'flavor' to the theory of mathematical functions compared with the algebra of functions considered up to now.

In this chapter the objects of study are mainly functions of a single real variable only. For such functions we first define continuity as a basic concept, together with the closely related concept of the *limit of a function*. As kinds of regularity stronger than mere continuity we define, among other things, differentiability (Section V.3) and the kinds of *smoothness* stronger than differentiability (Section V.5). On the basis of these concepts we characterize functions, presenting among other things the two central *mean value theorems* of the analysis of a real variable, and we examine the consequences of these theorems in the solvability theory of equations and also of simple *differential equations*.

The derivative also brings with it the rules of differentiation, that is, a new dimension to the algebra of functions. In Sections V.3–V.4 we derive all the central rules of differentiation, including implicit functions, functions defined as sums of power series, and (separately, in Section V.4) the trigonometric functions.

In Section V.7 the objects of study are the operating principles and convergence theory of the algorithms commonly used in the approximate solution of equations, namely *fixed-point iterations*. In Section V.8 we briefly examine the extension of the concepts of continuity and differentiability to complex functions, and in this connection we define the concept of an *analytic* complex function. In the final section we prove some of the most central assertions of the theory of continuous

functions, such as the *Weierstrass theorem*. In this connection we also present briefly how the fundamental theorem of algebra can be proved by combining results of complex algebra and analysis.

V.1 The concept of continuity

The problem of the continuity of a function arises even in a task as simple as the numerical determination of the value of a function, that is, numerical function evaluation. Consider the inverse function of Example 3 in Section IV.2, which we now write in the form

$$y = f(x) \Leftrightarrow y^5 + 3y = x, \quad x \in \mathbb{R}.$$

Let the task be to compute approximately the number $a = f(\pi)$, that is, to evaluate f numerically at π . This actually involves two numerical problems: First, there is no 'exact' numerical value for the number π , and second, the equation in general cannot be solved exactly for y , even if x is a rational number. In practice one proceeds (a calculator or computer proceeds) as follows: For π one chooses a representative x_n from a sequence of rational numbers $\{x_n\}$ satisfying $x_n \rightarrow \pi$. One computes $a_n = f(x_n)$ approximately using some numerical algorithm for solving the equation $y^5 + y = x_n$. In practice the algorithm produces a sequence of rational numbers $\{b_k\}$ satisfying $b_k \rightarrow a_n$ as $k \rightarrow \infty$. From this sequence one picks b_m for some m (e.g. $m = 10$) as an approximation of the number a_n , and reports as the final result an approximation of this number as a finite decimal (a truncated or rounded floating-point number).

If in the above computation the numerical rounding errors in floating-point arithmetic are not taken into account, then the error of the final result b_m consists of two parts:

$$b_m - f(\pi) = [b_m - f(x_n)] + [f(x_n) - f(\pi)].$$

Here the first part is the error of the approximation $b_m \approx f(x_n)$, that is, it concerns the accuracy of the algorithm for solving the equation $y^5 + 3y = x_n$. The latter part of the error, on the other hand, concerns, besides the accuracy of the approximation $x_n \approx \pi$, also the function f : it is a matter of the continuity of f . Qualitatively, the assertion ' f is continuous at x ' means that the function evaluation $x \mapsto f(x)$ is reliable when the variable is allowed a small variation, i.e. that

$$x_n \approx x \Rightarrow f(x_n) \approx f(x).$$

In the case of the above function, the question of continuity is settled as follows: Since

$$\begin{cases} a^5 + a = \pi, \\ a_n^5 + a_n = x_n, \end{cases}$$

we obtain by subtraction (cf. the considerations in Section IV.2)

$$(a - a_n)(a^4 + a^3a_n + a^2a_n^2 + aa_n^3 + a_n^4 + 3) = \pi - x_n.$$

Here we may safely assume that $a, a_n \geq 0$, so it follows that

$$|a_n - a| \leq \frac{1}{3}|x_n - \pi| \Leftrightarrow |f(x_n) - f(\pi)| \leq \frac{1}{3}|x_n - \pi|.$$

We have thus obtained even a quantitative confirmation of continuity: The error of the approximation $f(x_n) \approx f(\pi)$ is at most one third of the error of the approximation $x_n \approx \pi$.

This example leads to the following definition of the continuity of a function (an alternative definition follows below).

DEFINITION V.1.1 (Continuity: sequential criterion) A function $f : \mathcal{D}_f \rightarrow \mathbb{R}$, $\mathcal{D}_f \subset \mathbb{R}$, is *continuous* (Swedish *kontinuerlig*) at the point $a \in \mathcal{D}_f$ (or at a) if for every sequence of real numbers $\{x_n\}$

$$x_n \in \mathcal{D}_f \ \forall n \quad \& \quad x_n \rightarrow a \Rightarrow f(x_n) \rightarrow f(a).$$

If f is not continuous at the point $a \in \mathcal{D}_f$, then f is *discontinuous* at the point a .

EXAMPLE 1 The function $f(x) = \begin{cases} x - 1, & \text{if } x < \pi \\ x, & \text{if } x \geq \pi \end{cases}$

is discontinuous at the point $x = \pi$, since if $x_n \rightarrow \pi$ and $x_n < \pi \ \forall n$, then $f(x_n) = x_n - 1 \rightarrow \pi - 1 \neq f(\pi) = \pi$. At the other points f is continuous, since if e.g. $a < \pi$ and $x_n \rightarrow a$, then from some index N onward $|x_n - a| < \pi - a$ (taking $\varepsilon = \pi - a$ in the definition of convergence of a sequence), so that in particular $x_n - a < \pi - a \Leftrightarrow x_n < \pi$ when $n > N$. Then $f(x_n) = x_n - 1$, $n > N$ and hence $f(x_n) \rightarrow a - 1 = f(a)$. $\square$

EXAMPLE 2 Let $\mathcal{D}_f = \{a\}$ ($a \in \mathbb{R}$) and $f(a) = c \in \mathbb{R}$. According to Definition V.1.1, f is continuous at the point a (!). More generally, a real function is continuous at every so-called *isolated point* of its domain, see Exercise 4. $\square$

Continuity can also be defined directly without appealing to sequences, in which case the definition comes to resemble the definition of convergence of a sequence (Definition I.6.1). The alternative definition is the following.

DEFINITION V.1.2 (Continuity: (ε, δ) -criterion) A function $f : \mathcal{D}_f \rightarrow \mathbb{R}$, $\mathcal{D}_f \subset \mathbb{R}$, is continuous at the point $a \in \mathcal{D}_f$ if for every $\varepsilon > 0$ there exists $\delta > 0$ such that for every $x \in \mathbb{R}$

$$x \in \mathcal{D}_f \quad \& \quad |x - a| < \delta \Rightarrow |f(x) - f(a)| < \varepsilon.$$

When Definitions V.1.1 and V.1.2 are compared, it does not seem entirely obvious that (see the proof at the end of the section)

*THEOREM V.1.3 The definitions of continuity V.1.1 and V.1.2 are equivalent.

EXAMPLE 1 (continued) If $a \neq \pi$, then for the function of the example

$$|f(x) - f(a)| = |x - a|, \quad \text{when } |x - a| < |\pi - a|,$$

so the condition of Definition V.1.2 holds when we choose $\delta = \min\{\varepsilon, |\pi - a|\} > 0$. Hence f is continuous at the points $a \neq \pi$ according to Definition V.1.2. $\square$

EXAMPLE 3 Show that $f(x) = x^2$ is continuous at every point $a \in \mathbb{R}$ using the a) sequential criterion, b) (ε, δ) -criterion of continuity.

Solution a) Continuity according to Definition V.1.1 is a consequence of Theorem I.7.3:

$$x_n \rightarrow a \Rightarrow x_n^2 \rightarrow a^2 \Rightarrow f(x_n) \rightarrow f(a).$$

b) If $a \in \mathbb{R}$ and $|x - a| \leq 1$, then by field algebra and the triangle inequality

$$\begin{aligned} |f(x) - f(a)| &= |x - a||x + a| = |x - a||2a + (x - a)| \\ &\leq |x - a|(2|a| + |x - a|) \leq |x - a|(2|a| + 1). \end{aligned}$$

Consequently, if $\varepsilon > 0$, then $|f(x) - f(a)| < \varepsilon$ whenever $|x - a| < \delta = \min\{1, \varepsilon/(2|a| + 1)\}$ (so that also $|x - a| < 1$). Since here $\delta > 0$ whenever $\varepsilon > 0$, f is continuous at a according to Definition V.1.2. $\square$

Defining continuity by the sequential criterion (Definition V.1.1) is convenient in many theoretical considerations, which in this way reduce to the theory of convergent sequences. (That theory must, admittedly, be known, which can also be regarded as a drawback.) Definition V.1.2 is the more traditional 'school definition' of continuity. It is often clearer than Definition V.1.1 when one wants to find out what continuous functions 'look like'. For example, the following often-used result, which tells of the 'rigidity' of a continuous function, is easy to derive from this definition. The proof is left as an exercise (Exercise 3a).

PROPOSITION V.1.4 If f is continuous at the point a and $f(a) > 0$ ($f(a) < 0$), then $\exists \delta > 0$ such that $f(x) > 0$ ($f(x) < 0$) whenever $x \in (a - \delta, a + \delta) \cap \mathcal{D}_f$.

Continuity at a single point can be extended to a set $A \subset \mathcal{D}_f$: A function is continuous on A if it is continuous at every point of A . It is, however, typical of considerations of continuity that the consideration is restricted to an interval $A \subset \mathcal{D}_f$ in such a way that the properties of the function outside the interval have no significance whatever. For this reason it is customary to make the following definition, which deviates slightly from the general rule mentioned above.

DEFINITION V.1.5 (Continuity on an interval) A function $f : \mathcal{D}_f \rightarrow \mathbb{R}$, $\mathcal{D}_f \subset \mathbb{R}$, is *continuous on the interval* $A \subset \mathcal{D}_f$ if for every $x \in A$ and for every sequence of real numbers $\{x_n\}$

$$x_n \in A \ \& \ x_n \rightarrow x \Rightarrow f(x_n) \rightarrow f(x).$$

According to this, continuity on the interval A means the same as continuity in the sense of Definition V.1.1 at every point of A in the case where the domain of the function is restricted to the interval A . (In the alternative definition of continuity V.1.2, the condition $x \in \mathcal{D}_f$ is replaced by the condition $x \in \mathcal{D}_f \cap A$.) Continuity on an interval is thus defined as continuity 'from within' the set in question. If the interval is open, this restriction is not needed, since if $x \in (a, b) = A$ and $x_n \rightarrow x$, then for some N we have $x_n \in A \ \forall n > N$ (Definition I.6.1, $\varepsilon = \min\{x - a, b - x\} > 0$), that is, the condition of the definition is satisfied from index N onward in any case. If, on the other hand, the interval is closed, then the condition $x_n \in A$ is significant at the endpoints of the interval (not at the interior points).

EXAMPLE 1 (continued) If $b > \pi$, the function f of the example is continuous on the interval $[\pi, b]$ (Definition V.1.5) even though f is not continuous at the point π (Definition V.1.1). If $a < \pi$, then f is continuous on the interval $[a, \pi]$ but not on the interval $(a, \pi]$. $\square$

EXAMPLE 4 The function $f(x) = \begin{cases} 1, & \text{if } x \in \mathbb{Q}, \\ 0, & \text{if } x \in \mathbb{R}, x \notin \mathbb{Q} \end{cases}$

is an example of a 'pathological' function that is defined on all of $\mathbb{R}$ but is not continuous at any point. The function is called the *Dirichlet function*. $\square$

Combining continuous functions

As a counterpart to Example 4 one may ask what kinds of 'normal' functions are typically continuous. A first sense of this is obtained by combining simple functions through the basic arithmetic operations in the sense of Definition IV.1.2. Since the continuity of a function is ultimately a matter of the convergence of the sequence $\{f(x_n)\}$, the following theorem is an immediate consequence of Theorem I.7.3 (Exercise 3b).

THEOREM V.1.6 (Rules for combining continuous functions) If $f : \mathcal{D}_f \rightarrow \mathbb{R}$ is continuous at the point $x \in \mathcal{D}_f$, then λf is continuous at $x \ \forall \lambda \in \mathbb{R}$. If in addition $x \in \mathcal{D}_g$ and $g : \mathcal{D}_g \rightarrow \mathbb{R}$ is continuous at the point x , then $f + g$ and fg are continuous at x . If furthermore $g(x) \neq 0$, then also f/g is continuous at x .

EXAMPLE 5 Every polynomial or rational function can be defined recursively as an algebraic combination of the basic functions $f_0(x) = 1$ and $f_1(x) = x$. Since these are evidently continuous on $\mathbb{R}$, we conclude from Theorem V.1.6 that every polynomial is continuous on $\mathbb{R}$ and every rational function is continuous on its domain, that is, everywhere except at the zeros of its denominator. $\square$

EXAMPLE 6 If we take as known that the trigonometric functions $\sin$ and $\cos$ are continuous on $\mathbb{R}$, then by Theorem V.1.6 the functions $\tan = \sin / \cos$ and $\cot = \cos / \sin$ are likewise continuous on their entire domains. $\square$

THEOREM V.1.7 (**Continuity of a composite function**) If g is continuous at the point $x \in \mathcal{D}_g$, $g(x) \in \mathcal{D}_f$ and f is continuous at the point $g(x)$, then the composite function $f \circ g$ is continuous at x .

Proof Since $x_n \in \mathcal{D}_{f \circ g} \Leftrightarrow x_n \in \mathcal{D}_g \wedge g(x_n) \in \mathcal{D}_f$, we conclude by the sequential criterion:

$$\begin{aligned} x_n \in \mathcal{D}_{f \circ g} \ \& \ x_n \rightarrow x &\Rightarrow g(x_n) \in \mathcal{D}_f \ \& \ x_n \in \mathcal{D}_g \ \& \ x_n \rightarrow x \\ &\Rightarrow g(x_n) \in \mathcal{D}_f \ \& \ g(x_n) \rightarrow g(x) \\ &\Rightarrow f(g(x_n)) \rightarrow f(g(x)). \quad \square \end{aligned}$$

EXAMPLE 7 Show that if f is continuous at x , then also $|f|$ is continuous at x .

Solution The function $g(x) = |x|$ is easily shown to be continuous on $\mathbb{R}$ directly from the definitions of continuity (or by appeal to Proposition V.1.8 below), so by Theorem V.1.7 the composite function $(g \circ f)(x) = |f(x)|$ is continuous at every point where f is continuous. $\square$

The inheritance of continuity for a function defined piecewise (i.e. by different computational rules on different intervals) is settled by the following (Exercise 3c; cf. also Example 1 above).

PROPOSITION V.1.8 Let f_1 and f_2 be continuous on the open interval (a, b) and

$$f(x) = \begin{cases} f_1(x), & \text{when } x \in (a, c), \\ k, & \text{when } x = c, \\ f_2(x), & \text{when } x \in (c, b), \end{cases}$$

where $a < c < b$. Then f is continuous at c precisely when $f_1(c) = f_2(c) = k$.

EXAMPLE 8 The function

$$f(x) = \begin{cases} x + a, & \text{when } x \leq a, \\ -(x + 1)^2, & \text{when } x > a \end{cases}$$

is the polynomial $f_1(x) = x + a$ on the interval $(-\infty, a)$ and the polynomial $f_2(x) = -(x + 1)^2$ on the interval (a, ∞) , so on these intervals the function is continuous. By Proposition V.1.8 the condition for the continuity of the function at the point $x = a$ (and hence the condition for continuity on all of $\mathbb{R}$) is

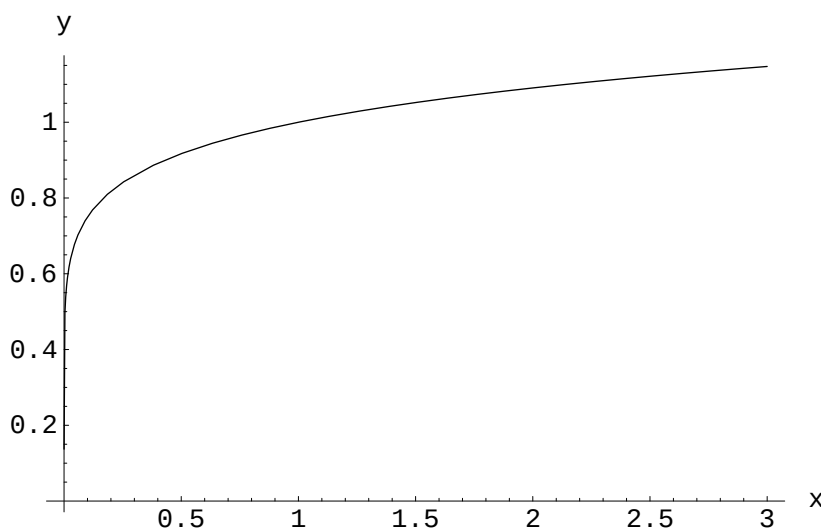
$$f_1(a) = f_2(a) \quad \Leftrightarrow \quad a^2 + 4a + 1 = 0 \quad \Leftrightarrow \quad a = -2 \pm \sqrt{3}. \quad \square$$

The function $f(x) = \sqrt[m]{x}$

We show next that the function $f : [0, \infty) \rightarrow [0, \infty)$ defined by

$$f(x) = x^{1/m} = \sqrt[m]{x}, \quad m \in \mathbb{N}, \quad m \geq 2$$

is continuous on its entire domain (the figure shows $f(x)$ for $m = 8$).



We begin from the point $x = 0$, which is the most critical for continuity (cf. the figure). Since f is strictly increasing (see Section IV.2, Example 2), for every $a > 0$ we have

$$0 \leq x < a^m \Leftrightarrow 0 \leq f(x) < f(a^m) = a.$$

If now $x_n \geq 0$ and $x_n \rightarrow 0$, then for every $\varepsilon > 0 \exists N \in \mathbb{N}$ such that

$$n > N \Rightarrow 0 \leq x_n < \varepsilon^m \Rightarrow 0 \leq f(x_n) < \varepsilon.$$

Consequently the sequence $\{f(x_n)\}$ is convergent and $f(x_n) \rightarrow 0 = f(0)$, so f is continuous at 0.

Away from the origin, the continuity of f can be deduced by means of field algebra: If $x_n \geq 0$ and $x_n \rightarrow x > 0$, then writing $y = f(x)$, $y_n = f(x_n)$ we obtain

$$x - x_n = y^m - y_n^m = (y - y_n)(y^{m-1} + y^{m-1}y_n + \cdots + y_n^{m-1}).$$

Here $y_n \geq 0$, $y > 0$, so we conclude

$$|x - x_n| \geq |y - y_n|y^{m-1} \Leftrightarrow |y - y_n| \leq y^{1-m}|x - x_n|.$$

Consequently $x_n \rightarrow x \Rightarrow y_n \rightarrow y$, that is, f is continuous at the point x .

EXAMPLE 9 Let $\varepsilon = 10^{-6}$. Determine the largest δ such that the condition of Definition V.1.2 holds for the function $f(x) = \sqrt[8]{x}$ at the point $x = 0$.

Solution Since f is strictly increasing on the interval $[0, \infty) = \mathcal{D}_f$, for all $x \in \mathcal{D}_f$ we have $|f(x) - f(0)| = \sqrt[8]{x} < \varepsilon \Leftrightarrow 0 \leq x < \varepsilon^8 \Leftrightarrow x \in (-\varepsilon^8, \varepsilon^8) \cap \mathcal{D}_f$. Hence $\delta_{\max} = \varepsilon^8 = 10^{-48}$. $\square$

Once the continuity of the function $f(x) = x^{1/m}$ has been established, it follows from Theorem V.1.7 and the continuity of the function $g(x) = |x|$ that the composite function $(f \circ g)(x) = |x|^{1/m}$ is continuous on its entire domain ($= \mathbb{R}$). Taking also Theorem V.1.6 into account, we obtain

PROPOSITION V.1.9 The function $f(x) = |x|^\alpha$, $\alpha = p/q \in \mathbb{Q}$ is continuous on its entire domain ($\mathcal{D}_f = \mathbb{R}$ if $\alpha > 0$, $\mathcal{D}_f = \{x \in \mathbb{R} \mid x \neq 0\}$ if $\alpha \leq 0$).

On the basis of Theorems V.1.6–V.1.7, Proposition V.1.9 and Examples 6 and 7, we can draw the more general conclusion that all ‘single-expression’ functions known so far are continuous on their entire domains (!).

EXAMPLE 10 Without a case-by-case examination, for instance the following functions can be concluded to be continuous at every point of their domains:

$$\sqrt[3]{|1 - \sqrt{x}|}, \quad \sqrt[6]{\frac{1 + \sqrt{x}}{2 - x^2}}, \quad \frac{|\sin x|^{3/4}}{x - \pi}, \quad \frac{\cot x}{\sqrt[4]{x + 50 \cos(\tan x)}}. \quad \square$$

The main theorems on continuous functions

In this section we present three central theorems of mathematical analysis, all concerning functions continuous on a closed interval. The first of the theorems is also the first of the *mean value theorems* for functions of a single real variable, of which there are three in all. (The other two are presented later.) Of the

main theorems presented here, the latter two are special cases of more general assertions that rest on a deeper logic of continuity. Their proofs are not yet presented; instead the theorems are formulated and proved later in Section V.9 in a form more general than that presented here.

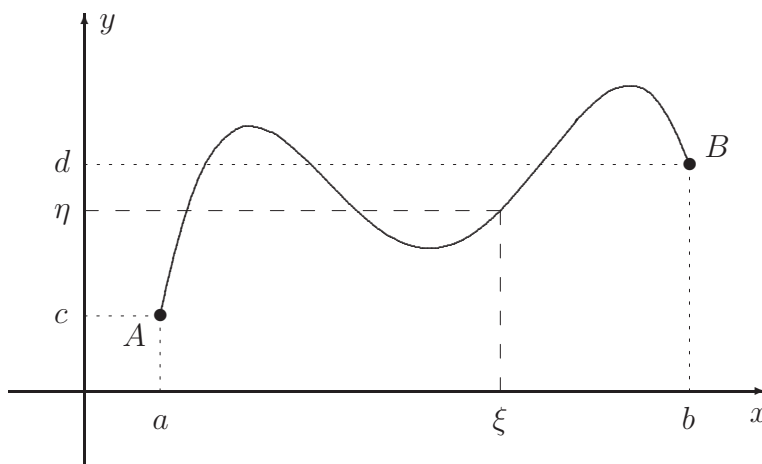
Consider a function f continuous on a closed interval $[a, b]$. Let $f(a) \neq f(b)$ and $c < \eta < d$, where

$$c = \min\{f(a), f(b)\}, \quad d = \max\{f(a), f(b)\}.$$

Since f is continuous, it seems natural to think that the graph of f on the interval $[a, b]$ is a 'continuous thread' joining the points $A = (a, f(a))$ and $B = (b, f(b))$. According to this intuition it seems clear that the graph must intersect the line $y = \eta$ at least once on the interval (a, b) . In other words, the problem

$$x \in [a, b] : \quad f(x) = \eta$$

must have at least one solution $x = \xi \in (a, b)$ for every $\eta \in (c, d)$ (figure).



Confirmation of the above geometric intuition is given by

THEOREM V.1.10 (First mean value theorem — Intermediate value theorem for continuous functions) If $f : \mathcal{D}_f \rightarrow \mathbb{R}$, $\mathcal{D}_f \subset \mathbb{R}$, is continuous on the interval $[a, b] \subset \mathcal{D}_f$ and $c = \min\{f(a), f(b)\} < d = \max\{f(a), f(b)\}$ (i.e. $f(a) \neq f(b)$), then for every $\eta \in (c, d)$ there exists $\xi \in (a, b)$ such that $f(\xi) = \eta$.*

*Theorem V.1.10 confirms that the graph of a function f continuous on the interval $[a, b]$,

$$G_f = \{P = (x, y) \mid x \in [a, b] \text{ \& } y = f(x)\},$$

If f and η satisfy the conditions of Theorem V.1.10, then one sees that the assertion of the theorem follows when the following more reduced result is applied to the function $g(x) = f(x) - \eta$ or $g(x) = \eta - f(x)$.

THEOREM V.1.11 (Bolzano) If $f(a) < 0$, $f(b) > 0$, and f is continuous on the interval $[a, b]$, then $f(\xi) = 0$ for some $\xi \in (a, b)$.

Proof is purely constructive and works in practice as an algorithm that searches for one of the zeros of f on the interval (a, b) . The most common way of proving it is by so-called bracketing, i.e. the bisection method (the decimal-subdivision algorithm would also work, cf. Section I.9). First, on the basis of the function evaluations (assumed exact!) one can determine that either with the choice $a_1 = a$, $b_1 = \frac{1}{2}(a + b)$ or with the choice $a_1 = \frac{1}{2}(a + b)$, $b_1 = b$ we have

$$f(a_1) \leq 0 \quad \& \quad f(b_1) \geq 0.$$

If here $f(a_1) = 0$ or $f(b_1) = 0$, then ξ has been found. Otherwise $f(a_1) < 0$ and $f(b_1) > 0$, in which case we can continue the 'bracketing' of ξ on the interval (a_1, b_1) by the same principle. Provided the construction does not terminate with the finding of ξ , two sequences $\{a_n\}$, $\{b_n\}$ arise. By the construction, $\{a_n\}$ is increasing, $\{b_n\}$ is decreasing and

$$b_n - a_n = \left(\frac{1}{2}\right)^n (b - a) \rightarrow 0,$$

so the sequences have a common limit: $a_n \rightarrow \xi$ and $b_n \rightarrow \xi$ for some $\xi \in \mathbb{R}$. Since $a \leq a_n \leq b \quad \forall n$, we must also have $a \leq \xi \leq b$ (Proposition I.7.4: V1), that is, $\xi \in [a, b]$. Then, since f is continuous on the interval $[a, b]$ (Definition V.1.5), we have

$$f(a_n) \rightarrow f(\xi) \quad \text{and} \quad f(b_n) \rightarrow f(\xi).$$

But by the construction of the sequences $\{a_n\}$ and $\{b_n\}$,

$$\begin{aligned} f(a_n) < 0 \quad \forall n &\Rightarrow f(\xi) \leq 0, \\ f(b_n) > 0 \quad \forall n &\Rightarrow f(\xi) \geq 0 \end{aligned}$$

(Proposition I.7.4: V1), so we must have $f(\xi) = 0$. Here $\xi \in [a, b]$, but since $f(a) < 0$ and $f(b) > 0$, we have $\xi \in (a, b)$. Bolzano's theorem is thus proved. $\square$

is a *curve* in the sense of geometric intuition, i.e. a 'connected line without width'. G_f is 'without width' because $f(x)$ is unique $\forall x$ (i.e. f is a function), and 'connected' because f is continuous. — Continuity (as a guarantor of connectedness) is associated with the concept of a curve more generally as well. For example, in a general parametric curve $\vec{r}(t) = x(t)\vec{i} + y(t)\vec{j} + z(t)\vec{k}$ the coordinate functions $x(t)$, $y(t)$, $z(t)$ are assumed continuous on the interval under consideration. Cf. the footnotes in Sections IV.2–IV.4.

EXAMPLE 11 For the function $f(x) = x^5 + 3x$ we have $f(0) = 0$ and $f(1) = 4$, so by Theorem V.1.10 the equation

$$x^5 + 3x = 2$$

has a solution $x = \xi$ on the interval $(0, 1)$. The solution is in fact unique, since f is one-to-one (cf. Example 3 in Section IV.2). $\square$

The following theorem, which also has more general forms, is one of the most significant results of real analysis (proof in a more general form in Section V.9).

*THEOREM V.1.12 (**Weierstrass**) A function f continuous on a closed interval $[a, b]$ attains its smallest and largest value on that interval, i.e. there exist $x_1 \in [a, b]$ and $x_2 \in [a, b]$ such that

$$f(x_1) = \min\{f(x) \mid x \in [a, b]\}, \quad f(x_2) = \max\{f(x) \mid x \in [a, b]\}.$$

EXAMPLE 12 The function

$$f(x) = \begin{cases} x, & \text{when } x \leq 0, \\ 1 - x, & \text{when } x > 0 \end{cases}$$

is continuous on the interval $[a, b]$ if either $a < b \leq 0$ or $0 < a < b$. In that case the function attains its maximum and minimum values at the endpoints of the interval. Otherwise, that is, if $a \leq 0$ and $b > 0$, then f is not continuous on the interval $[a, b]$, since if $x_n \in (0, b]$ and $x_n \rightarrow 0$, then $f(x_n) \rightarrow 1 \neq f(0)$. In this case f also does not attain a maximum value on the interval $[a, b]$, only a minimum value ($= \min\{f(a), f(b)\}$). $\square$

By combining the assertions of Theorems V.1.10 and V.1.12 we arrive at the following conclusion, which is worth noting:

COROLLARY V.1.13 If f is continuous on the interval $A = [a, b]$ and f is not constant, then $f(A)$ is a closed interval.

The following theorem gives a sufficient criterion for the continuity of an inverse function. The proof (for a somewhat more generally formulated theorem) is presented in Section V.9.

*THEOREM V.1.14 (**Continuity of an inverse function**) If $f : [a, b] \rightarrow [c, d]$ is a continuous bijection, then also $f^{-1} : [c, d] \rightarrow [a, b]$ is a continuous bijection.

EXAMPLE 13 The function $f(x) = x^{1/m}$, $m \in \mathbb{N}$, $m \geq 2$ was concluded above to be continuous on the interval $[0, \infty)$. This also follows from Theorem V.1.14, since $f = g^{-1}$, where $g(x) = x^m$ is a continuous bijection $g : [0, a] \rightarrow [0, a^m]$ for every $a > 0$. $\square$

*Proof of Theorem V.1.3

The assertion of Theorem V.1.3 is twofold: It is claimed that if f is continuous at the point $a \in \mathcal{D}_f$ by the sequential criterion of Definition V.1.1, then f is continuous at a also by the (ε, δ) -criterion of Definition V.1.2, and that this implication holds also conversely.

Proof $\Rightarrow$ Assume that f is continuous at the point $a \in \mathcal{D}_f$ in the sense of Definition V.1.1. Make the counter-assumption: f is not continuous at a by the criterion of Definition V.1.2. Then, in the same way as in the proof of Theorem I.10.8, one can conclude that for some $\varepsilon > 0$

$$\forall \delta > 0 \exists x \in \mathcal{D}_f \text{ such that } |x - a| < \delta \ \& \ |f(x) - f(a)| \geq \varepsilon.$$

Now if from the values of δ we form a sequence $\{\delta_n\}$ that converges to zero (e.g. $\delta_n = 1/n$), then correspondingly a sequence $\{x_n\}$ can be found such that for every n

$$x_n \in \mathcal{D}_f \ \& \ |x_n - a| < \delta_n \ \& \ |f(x_n) - f(a)| \geq \varepsilon.$$

We have thus found a sequence for which

$$x_n \in \mathcal{D}_f \ \& \ x_n \rightarrow a \ \& \ f(x_n) \not\rightarrow f(a).$$

This is a logical contradiction, since it was assumed that f is continuous at a in the sense of Definition V.1.1. The counter-assumption made is therefore false, i.e. f is continuous at a by the criterion of Definition V.1.2.

$\Leftarrow$ Assume that f is continuous at the point $a \in \mathcal{D}_f$ by the criterion of Definition V.1.2. Let $\varepsilon > 0$ and choose $\delta > 0$ such that the condition of Definition V.1.2 is satisfied. Further, let $\{x_n\}$ be a sequence for which $x_n \in \mathcal{D}_f \ \forall n$ and $x_n \rightarrow a$. Then from some index $n = N$ onward $|x_n - a| < \delta$ (since $\delta > 0$), whereupon by the assumptions made we also have $|f(x_n) - f(a)| < \varepsilon$. We have thus found an index $N \in \mathbb{N}$ such that

$$|f(x_n) - f(a)| < \varepsilon, \quad \text{when } n > N.$$

Here $\varepsilon > 0$ was arbitrary, so $f(x_n) \rightarrow f(a)$ (Definition I.6.1). The sequence $\{x_n\}$ was also freely chosen within the assumptions made, so for every such sequence $f(x_n) \rightarrow f(a)$ holds. Hence f is continuous at the point a by the criterion of Definition V.1.1. $\square$

EXERCISES

1. Prove directly, on the basis of Definition V.1.1 and results on sequences, that each of the following functions is either continuous or discontinuous at the point $x = 0$. Sketch also the graphs of the functions on the set $A = \mathcal{D}_f \cap [-1, 1]$.

$$\text{a) } f(x) = x^7 + 7x + 49 \quad \text{b) } f(x) = x^5 - 6x^3 + \frac{x^3 + 20}{100x^2 - 1}$$

$$\text{c) } f(x) = \begin{cases} \frac{1}{x} \left(\frac{1}{1+7x} - \frac{1}{1+5x} \right), & \text{when } x \neq 0, \\ 2, & \text{when } x = 0 \end{cases}$$

$$\text{d) } f(x) = \begin{cases} x \cos \frac{1}{x}, & \text{when } x \neq 0, \\ 0, & \text{when } x = 0 \end{cases} \quad \text{e) } f(x) = \begin{cases} \sin \frac{1}{x}, & \text{when } x \neq 0, \\ 0, & \text{when } x = 0 \end{cases}$$

2. Show that the function $f(x) = 1/x$ is continuous on its domain by appealing directly to a) the sequential criterion of continuity, b) the (ε, δ) -criterion of continuity.
3. Using the (ε, δ) -criterion of continuity in part a), the sequential criterion in part b), and either one in part c), prove:
 a) Proposition V.1.4. b) Theorem V.1.6. c) Proposition V.1.8.
4. We say that $a \in A$ is an *isolated point* of the set $A \subset \mathbb{R}$ if for some $\delta > 0$ we have $(a - \delta, a + \delta) \cap A = \{a\}$. Show that a real function is continuous at every isolated point of its domain.
5. Find the values of the parameter a for which the function becomes continuous on all of $\mathbb{R}$, and sketch the graph for these values of a :

$$\text{a) } f(x) = \begin{cases} x - 1, & \text{when } x \leq a, \\ 1 - x^2, & \text{when } x > a \end{cases} \quad \text{b) } f(x) = \begin{cases} a - x, & \text{when } x \leq -2, \\ x^3 - 4x, & \text{when } x > -2 \end{cases}$$

$$\text{c) } f(x) = \begin{cases} \frac{x^2 - 4}{x^3 - 8}, & \text{when } x < 2, \\ a, & \text{when } x \geq 2 \end{cases} \quad \text{d) } f(x) = \begin{cases} \sin x, & \text{when } x \leq \pi/3, \\ \cos ax, & \text{when } x > \pi/3 \end{cases}$$

$$\text{e) } f(x) = \begin{cases} \frac{x^3 + 4x^2 + 5x + 2}{x^2 + 3x + 2}, & \text{when } x < a, \\ ax - 10, & \text{when } x \geq a \end{cases}$$

6. Using Theorems V.1.6–V.1.7 and Proposition V.1.9, verify that the following functions are continuous on their entire domains. — What is the domain?

$$\text{a) } f(x) = \sqrt{1 + \frac{1}{2-x}} \quad \text{b) } f(x) = \sqrt[4]{1 - \frac{1}{\sqrt{2-x}}}$$

7. The function $f(x) = \sin x$ is strictly increasing on the interval $[-\frac{\pi}{2}, \frac{\pi}{2}]$, strictly decreasing on the interval $[\frac{\pi}{2}, \frac{3\pi}{2}]$ and continuous on all of $\mathbb{R}$. On the basis of this information and Theorem V.1.10, exactly how many solutions does the equation $\sin x = y \in [-1, 1]$ have on the interval $[-\frac{\pi}{2}, \frac{\pi}{6}]$ for different values of y ?
8. a) The transcendental equation $x = \cos x$ has a unique solution on the interval $[0, 1]$. Following the constructive proof of Bolzano's theorem, compute for the solution the nearest lower and upper approximations as finite binary numbers of the form $p/64$, $p \in \mathbb{N}$.
b) How many zeros does the function $f(x) = x - \cos 49x$ have? If a zero is sought on the interval $[0, 1]$ by the algorithm used in the proof of Bolzano's theorem, then which of the zeros is found?
9. a) Show that all the roots of the polynomial $p(x) = x^3 - 15x + 1$ are real and lie on the interval $[-4, 4]$.
b) Show that the polynomial $f(x) = (x - a)^2(x - b)^2 + x$ takes the value $(a + b)/2$ at some point.
10. Show that if in the constructive proof of Bolzano's theorem it is assumed only that f is defined on the interval $[a, b]$ and $f(a) < 0$, $f(b) > 0$ (i.e. continuity is not assumed), then in this case the construction produces a point $\xi \in [a, b]$ at which either $f(\xi) = 0$ or — the other alternative? Give an example of a function for which the result of the construction is $\xi = a$.
11. Prove, with the aid of Theorem V.1.10 and the Weierstrass theorem, the assertion: If f is continuous on the interval $[a, b]$ and $f(x) \neq 0 \forall x \in [a, b]$, then $\exists c > 0$ such that either $f(x) \geq c \forall x \in [a, b]$ or $f(x) \leq -c \forall x \in [a, b]$.
12. (*) The function f is continuous on all of $\mathbb{R}$ and

$$f(x) = \frac{1}{x} \cdot \frac{x - 1 + \sqrt{x^2 + 1}}{x + 1 + \sqrt{x^2 + 1}}, \quad \text{when } x \neq 0.$$

Show that a simpler definition of f is

$$f(x) = \frac{1}{1 + \sqrt{x^2 + 1}}, \quad x \in \mathbb{R}.$$

13. (*) Let f be continuous on the interval $A = [a, b]$. Show that
- a) if $f(A) \subset A$, then $f(c) = c$ for some $c \in A$.
 - b) if $f(a) = f(b)$, then $f(c) = f(c + \frac{b-a}{2})$ for some $c \in [a, \frac{a+b}{2}]$.
14. (*) Let $\{a_k, k = 0, 1, 2, \dots\}$ be a sequence of real numbers for which $\lim_k a_k = \infty$ and let

$$H(x) = \begin{cases} 1 - |x|, & \text{when } |x| \leq 1 \\ 0, & \text{when } |x| > 1 \end{cases}$$

Show that the function

$$f(x) = \sum_{k=0}^{\infty} H(x - a_k)$$

is continuous on all of $\mathbb{R}$. What is the maximum value of f and where is it attained when $a_k = (5/4)^k$?

V.2 The limit of a function

Closely related to continuity is the concept of the *limit of a function*. The limit describes the behavior of the function's values as some point $a \in \mathbb{R}$ is approached from within the function's domain. The point a need not be in the domain; it suffices that it can be approached arbitrarily closely. A typical situation of application is indeed of just this kind. On the other hand, if the point is in the domain, the limit still does not depend on the value of the function at this point.

There are two types of limits of a function, proper limits, that is, limits without further qualification, and *one-sided* limits. The definitions are as follows.

DEFINITION V.2.1 (Limit of a function) A function $f : \mathcal{D}_f \rightarrow \mathbb{R}$, $\mathcal{D}_f \subset \mathbb{R}$, has *at the point* a (or *at* a) the *limit* $A \in \mathbb{R}$ if for every sequence of real numbers $\{x_n\}$

$$(x_n \in \mathcal{D}_f \ \& \ x_n \neq a) \forall n \ \& \ x_n \rightarrow a \Rightarrow f(x_n) \rightarrow A,$$

and the hypothesis holds for some sequence $\{x_n\}$. The function has at the point a a *left-hand limit* $A_- \in \mathbb{R}$ if for every sequence of real numbers $\{x_n\}$

$$(x_n \in \mathcal{D}_f \ \& \ x_n < a) \forall n \ \& \ x_n \rightarrow a \Rightarrow f(x_n) \rightarrow A_-,$$

and the hypothesis holds for some sequence $\{x_n\}$. The function has at the point a a *right-hand limit* $A_+ \in \mathbb{R}$ if for every sequence of real numbers $\{x_n\}$

$$(x_n \in \mathcal{D}_f \ \& \ x_n > a) \forall n \ \& \ x_n \rightarrow a \Rightarrow f(x_n) \rightarrow A_+,$$

and the hypothesis holds for some sequence $\{x_n\}$.

The additional condition in the definition of a limit, 'the hypothesis holds for some sequence $\{x_n\}$ ', means that approaching ($x_n \rightarrow a$) in the assumed manner is possible, i.e. that a is not an isolated point of the set $\mathcal{D}_f \cup \{a\}$ (see Exercise V.1:4). For one-sided limits the additional condition means correspondingly that a is not an isolated point of the set $[\mathcal{D}_f \cap (-\infty, a)] \cup \{a\}$ (left-hand limit) or of the set $[\mathcal{D}_f \cap (a, \infty)] \cup \{a\}$ (right-hand limit). At an isolated point the limit is therefore not defined, whether or not the point is in the domain $\mathcal{D}_f$.

For the limits appearing in the definition we use the notations

$$A = \lim_{x \rightarrow a} f(x), \quad A_{\pm} = \lim_{x \rightarrow a^{\pm}} f(x),$$

or we write (cf. the notations for the limit of a sequence)

$$f(x) \rightarrow A, \quad \text{when } x \rightarrow a, \quad f(x) \rightarrow A_{\pm}, \quad \text{when } x \rightarrow a^{\pm}.$$

A convenient notation for one-sided limits is also

$$A_+ = f(a^+), \quad A_- = f(a^-).$$

EXAMPLE 1 Determine $\lim_{x \rightarrow 1} f(x)$ when $f(x) = (x - 1)/(x^2 + x - 2)$.

Solution The domain is $\mathcal{D}_f = \{x \in \mathbb{R} \mid x \neq 1 \ \& \ x \neq -2\}$ and $f(x) = 1/(x + 2) \ \forall x \in \mathcal{D}_f$. Consequently, if $x_n \in \mathcal{D}_f \ \forall n$ ($\Rightarrow x_n \neq 1 \ \forall n$) and $x_n \rightarrow 1$, then $f(x_n) = 1/(x_n + 2) \rightarrow 1/3$. Hence $\lim_{x \rightarrow 1} f(x) = 1/3$ (Definition V.2.1). $\square$

As with the limit of a sequence, in the notation for the limit of a function too, $\infty/-\infty$ may stand in place of A or $A_{\pm}$, in which case it is meant, in accordance with Definition V.2.1, that $f(x)$ increases/decreases without bound as $x \rightarrow a$ or $x \rightarrow a^{\pm}$.

EXAMPLE 1 (continued) For the function of the example

$$\lim_{x \rightarrow -2^-} f(x) = -\infty, \quad \lim_{x \rightarrow -2^+} f(x) = \infty. \quad \square$$

Insofar as the limit $\lim_{x \rightarrow a} f(x)$ exists, the one-sided limits (insofar as they are definable) also coincide with it. On the other hand, it is possible that both one-sided limits exist at the point a but are unequal, in which case there is no limit at the point a . As for the connection between continuity and limit, it is seen from Definitions V.1.1 and V.2.1 that, insofar as the limit $\lim_{x \rightarrow a} f(x)$ is definable (i.e. a is not an isolated point of the set $\mathcal{D}_f \cup \{a\}$),

$$f \text{ continuous at the point } a \in \mathcal{D}_f \Leftrightarrow \lim_{x \rightarrow a} f(x) = f(a) \\ (a \text{ not an isolated point}).$$

If f has a one-sided limit $f(a^+)$ or $f(a^-)$ and either $f(a) = f(a^+)$ or $f(a) = f(a^-)$ holds, then we say correspondingly that f is *right-continuous* or *left-continuous* at the point a . If $a \in \mathcal{D}_f$ and the point a can be approached from both directions from within $\mathcal{D}_f$, then it is evident that f is continuous at a precisely when f is both left-continuous and right-continuous at a . It is further seen that f is continuous on the closed interval $[a, b]$ (Definition V.1.5) precisely when f is continuous on the open interval (a, b) and in addition right-continuous at a and left-continuous at b .

Like continuity, the limit too can be defined by an alternative (ε, δ) -criterion, cf. Definition V.1.2. The alternative definition is formulated here as a theorem for the limit $\lim_{x \rightarrow a} f(x) = A \in \mathbb{R}$. The proof is omitted, since it is very similar to the proof of Theorem V.1.3 in the preceding section. (For one-sided limits a corresponding result can be formulated, and likewise for the cases $\lim_{x \rightarrow a} f(x) = \pm\infty$.)

*THEOREM V.2.2 (**(ε, δ) -criterion for a limit**) A function $f : \mathcal{D}_f \rightarrow \mathbb{R}$, $\mathcal{D}_f \subset \mathbb{R}$, has the limit $\lim_{x \rightarrow a} f(x) = A \in \mathbb{R}$ precisely when $\mathcal{D}_f \cap [(a - \delta, a) \cup (a, a + \delta)] \neq \emptyset$ $\forall \delta > 0$ and for every $\varepsilon > 0$ there exists $\delta > 0$ such that for every $x \in \mathbb{R}$

$$x \in \mathcal{D}_f \ \& \ 0 < |x - a| < \delta \Rightarrow |f(x) - A| < \varepsilon.$$

From sequences it is known that a convergent sequence is bounded (Theorem I.7.2). For a function that has a limit, a corresponding result holds by Theorem V.2.2:

THEOREM V.2.3 If $\lim_{x \rightarrow a} f(x) = A \in \mathbb{R}$, then f is *bounded* in some neighborhood of the point, i.e. $\exists \delta > 0$ and $C \in \mathbb{R}_+$ such that

$$|f(x)| \leq C \quad \forall x \in (a - \delta, a + \delta) \cap \mathcal{D}_f.$$

Proof Choose $\varepsilon = 1$ and a corresponding $\delta > 0$ so that the condition of Theorem V.2.2 holds. Then it follows from the triangle inequality that the assertion is true when $C = |A| + 1$. $\square$

Approximating a function by a limit

If the limit $A = \lim_{x \rightarrow a} f(x)$ of a function is known, then by Theorem V.2.2 the approximation $f(x) \approx A$ can be used near the limit point. The theorem does not, however, give (nor, under the assumptions made, can it give) any quantitative information about the accuracy of the approximation, since the dependence of δ on ε is not known. In typical example cases, however, more is usually known about the function f than the theorem assumes, in which case it may be possible to derive a quantitative error estimate for the approximation on the basis of this additional information. In approximate function evaluation, knowledge of the limit can help especially when the computational formula for the function, used directly, is susceptible to numerical rounding errors near the limit point.

EXAMPLE 2 The function $f(x) = (1 - \cos x)/x^2$ is defined when $x \neq 0$. Later (Section V.4) it is shown that $\lim_{x \rightarrow 0} f(x) = 1/2$. The approximation based on this result,

$$f(0.000000003) \approx \lim_{x \rightarrow 0} f(x) = 0.5,$$

is considerably safer than the direct use of the computational formula for f , since in the operation $x \mapsto 1 - \cos x$ a considerable loss of significant digits occurs when $|x|$ is small (cf. Section I.5). Besides being safe, the limit-based approximation is in this case also very accurate: the error is less than 10^{-18} . $\square$

Combining limits

The computation of limits is facilitated by the following theorems, which are counterparts of Theorems V.1.6 and V.1.7. The proofs are, on the basis of Definition V.2.1, straightforward (Exercise 1; see also Exercise V.1:4). The theorems hold, with obvious changes, also for one-sided limits.

THEOREM V.2.4 If $\lim_{x \rightarrow a} f(x) = A \in \mathbb{R}$, then

$$\lim_{x \rightarrow a} (\lambda f)(x) = \lambda A \quad \forall \lambda \in \mathbb{R}.$$

If in addition $\lim_{x \rightarrow a} g(x) = B \in \mathbb{R}$ and a is not an isolated point of the set $(\mathcal{D}_f \cap \mathcal{D}_g) \cup \{a\}$, then

$$\begin{aligned} \lim_{x \rightarrow a} (f + g)(x) &= A + B, \\ \lim_{x \rightarrow a} (fg)(x) &= AB. \end{aligned}$$

If in addition $B \neq 0$ and a is not an isolated point of the set $\mathcal{D}_{f/g} \cup \{a\} = \{x \in \mathcal{D}_f \cap \mathcal{D}_g \mid g(x) \neq 0\} \cup \{a\}$, then

$$\lim_{x \rightarrow a} (f/g)(x) = A/B.$$

THEOREM V.2.5 If $\lim_{x \rightarrow a} g(x) = A \in \mathbb{R}$, f is continuous at the point $x = A$ and a is not an isolated point of the set $\mathcal{D}_{f \circ g} \cup \{a\}$, then $\lim_{x \rightarrow a} (f \circ g)(x) = f(A)$.

EXAMPLE 2 (continued) Since $g(x) = \sqrt{|x|}$ is continuous on $\mathbb{R}$ (Proposition V.1.9), the limit result of the example and Theorem V.2.5 give

$$\lim_{x \rightarrow 0} \frac{\sqrt{1 - \cos x}}{|x|} = \lim_{x \rightarrow 0} \sqrt{\frac{1 - \cos x}{x^2}} = \frac{1}{\sqrt{2}}. \quad \square$$

EXAMPLE 3 For the functions $f_0(x) = 1$ and $f_1(x) = x$ the obvious limit results hold

$$\lim_{x \rightarrow a} f_0(x) = 1, \quad \lim_{x \rightarrow a} f_1(x) = a, \quad a \in \mathbb{R}.$$

From these and Theorem V.2.4 it follows that

$$\lim_{x \rightarrow a} \frac{(x+1)^2}{x+2} = \frac{(a+1)^2}{a+2}, \quad \text{when } a \neq -2.$$

The result is also clear from Theorem V.1.6, since by this a rational function f is continuous on its entire domain, whereupon $\lim_{x \rightarrow a} f(x) = f(a)$ when $a \in \mathcal{D}_f$. $\square$

Computing a limit by substitution

The computation of the limit $\lim_{x \rightarrow a} f(x)$ can often be facilitated by making a *substitution*, that is, a *change of variable* $x = u(t)$. In changing the variable one relies on the following theorem, which is easily formulated also for a one-sided limit (Exercise 1c).

THEOREM V.2.6 Let $u : U \rightarrow [a - \delta, a + \delta]$ ($\delta > 0$) be a continuous bijection, where U is a closed interval. Then if $u(\alpha) = a$,

$$\lim_{x \rightarrow a} f(x) = \lim_{t \rightarrow \alpha} f(u(t)) = \lim_{t \rightarrow \alpha} F(t) = A,$$

insofar as the limit on the right exists ($A \in \mathbb{R}$ or $A = \pm\infty$).

Proof By Theorem V.1.14 the inverse function $u^{-1} : [a - \delta, a + \delta] \rightarrow U$ is also a continuous bijection. Now let $x_n \in \mathcal{D}_f$, $x_n \neq a \forall n$ and $x_n \rightarrow a$. Then from some index onward $|x_n - a| < \delta$, whereupon we can write $x_n = u(t_n)$, $t_n \in U$. Since $u : U \rightarrow [a - \delta, a + \delta]$ is injective, $x_n \neq a \Rightarrow t_n \neq \alpha$. Since u^{-1} is continuous at the point a , $x_n \rightarrow a \Rightarrow t_n = u^{-1}(x_n) \rightarrow u^{-1}(a) = \alpha$. Then by the assumption $\lim_{t \rightarrow \alpha} F(t) = A$ we have $f(x_n) = f(u(t_n)) = F(t_n) \rightarrow A$. It has been shown that $\lim_{x \rightarrow a} f(x) = A$ (Definition V.2.1, when $A \in \mathbb{R}$; the argument works also when $A = \pm\infty$). $\square$

EXAMPLE 4 Determine the limit $A = \lim_{x \rightarrow 81} \frac{\sqrt{x} - 9}{\sqrt[4]{x} - 3}$.

Solution Here a suitable substitution is $\sqrt[4]{x} = t$, so that $x = t^4 = u(t)$, and the value $x = 81 (= a)$ corresponds to $t = 3 (= \alpha)$. Since

$$F(t) = f(t^4) = \frac{t^2 - 9}{t - 3} = t + 3 = G(t), \quad \text{when } t \neq 3,$$

by the continuity of the function G we have $\lim_{t \rightarrow 3} F(t) = G(3) = 6$. The assumptions of Theorem V.2.6 hold ($\delta \leq 3$), so the requested limit is $A = 6$. $\square$

EXAMPLE 5 By the substitution $x = 2t$ we obtain (cf. Section II.5)

$$\lim_{x \rightarrow 0^+} \frac{\sin \frac{x}{2}}{\sqrt{1 - \cos x}} = \lim_{t \rightarrow 0^+} \frac{\sin t}{\sqrt{1 - \cos 2t}} = \lim_{t \rightarrow 0^+} \frac{\sin t}{\sqrt{2 \sin^2 t}} = \lim_{t \rightarrow 0^+} \frac{1}{\sqrt{2}} = \frac{1}{\sqrt{2}}.$$

Here the final simplification was based on the reasoning $t \in (0, \pi] \Rightarrow \sqrt{\sin^2 t} = \sin t$. $\square$

In the examples a four-step computational operation very typical of limit calculations was carried out:

$$\lim_{x \rightarrow a} f(x) = \lim_{t \rightarrow \alpha} f(u(t)) = \lim_{t \rightarrow \alpha} F(t) = \lim_{t \rightarrow \alpha} G(t) = G(\alpha).$$

Here

1. Substitute $x = u(t)$ [or $v(x) = t \Rightarrow x = u(t)$] and compute $\alpha = u^{-1}(a)$.
2. Simplify $f(u(t))$ to the expression $F(t)$.
3. Reduce $F(t)$ to the expression $G(t)$ for $t \neq \alpha$.
4. Compute the limit by appeal to the continuity of G at the point $t = \alpha$.

Insofar as the variable is not changed, the computation reduces to steps 3–4 ($F = f$, $t = x$), as in Example 1 above.

Extending a function

If $a \in \mathbb{R}$ is not in the domain of the function f , but there exists a (proper) limit $\lim_{x \rightarrow a} f(x) = A \in \mathbb{R}$, then it is natural to include a in the domain by setting $f(a) = A$. Proceeding in this way, f becomes, in the sense of Definition V.1.1, continuous at the point a . Enlarging the domain of a function in this way is called *extending the function*.

EXAMPLE 6 In Example 1 it was found that the function $f(x) = (x - 1)/(x^2 + x - 2)$ has the limit $\lim_{x \rightarrow 1} f(x) = 1/3$. When the point $x = 1$ is included in the domain of f by setting $f(1) = 1/3$, then f becomes extended to the function $f(x) = 1/(x + 2)$. The domain cannot be enlarged any further by extension, since f has no real (nor any more general) limit as $x \rightarrow -2$. $\square$

EXAMPLE 7 On the basis of the limit result of Example 2 (and the information of Example V.1:6) the function

$$f(x) = \begin{cases} \frac{1 - \cos x}{x^2}, & \text{when } x \neq 0, \\ \frac{1}{2}, & \text{when } x = 0 \end{cases}$$

is continuous on $\mathbb{R}$. The value of the function at 0 is determined by extension. $\square$

Piecewise continuity

The one-sided limits of a function come into use especially in application situations where a function (often of physical origin) is continuous everywhere except at isolated points, at which it has a simple so-called *jump discontinuity*. With such practical needs in mind, we make the following definition.

DEFINITION V.2.7 A function $f : \mathcal{D}_f \rightarrow \mathbb{R}$, $\mathcal{D}_f \subset \mathbb{R}$, is *piecewise continuous* on the interval $[a, b]$ if $\exists$ points c_k , $k = 0 \dots n$, $n \in \mathbb{N}$ such that

$$a = c_0 < c_1 < \dots < c_n = b$$

and

- (i) $(c_{k-1}, c_k) \subset \mathcal{D}_f$ and f is continuous on the interval (c_{k-1}, c_k) , $k = 1 \dots n$,
- (ii) the one-sided limits exist:

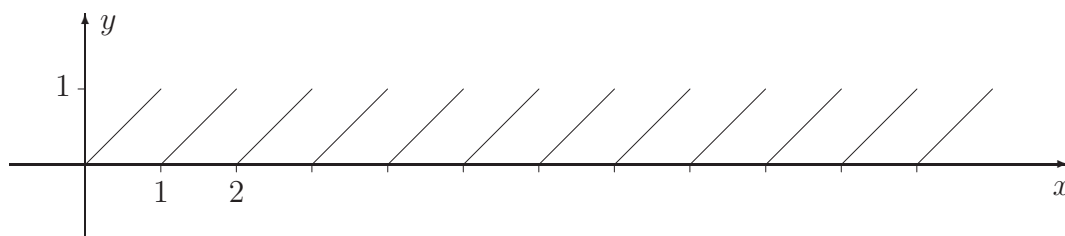
$$\begin{aligned} \lim_{x \rightarrow c_k^+} f(x) &= f(c_k^+), & k = 0 \dots n-1, \\ \lim_{x \rightarrow c_k^-} f(x) &= f(c_k^-), & k = 1 \dots n. \end{aligned}$$

Note that the limits $f(c_k^\pm)$ do not, by the definition of a limit, depend on the possible values of f at the points c_k . If both one-sided limits exist but do not coincide, there is a jump discontinuity (jump = $f(c_k^+) - f(c_k^-)$). If they coincide, the function is at the point c_k either continuous or definable as continuous by means of the limit (the extension procedure).

EXAMPLE 8 Let $[x]$ = the largest integer for which $[x] \leq x$. The function $f(x) = x - [x]$, that is,

$$f(x) = x - k, \quad \text{when } x \in [k, k+1), \quad k \in \mathbb{Z},$$

is piecewise continuous on every interval $[a, b] \subset \mathbb{R}$. At the points $k \in \mathbb{Z}$ the function could be defined in any way whatsoever (or left undefined) without affecting the one-sided limits $f(k^+) = 0$ and $f(k^-) = 1$. $\square$

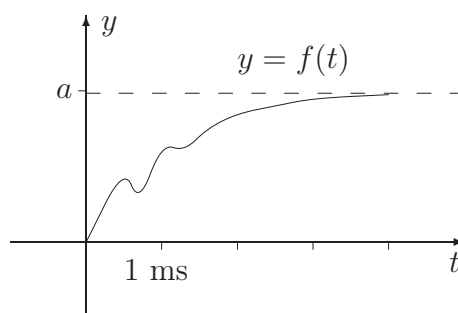
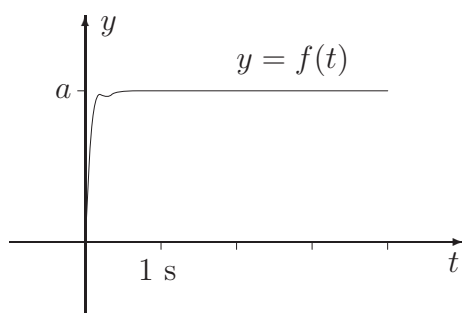


EXAMPLE 9 Acceleration. A car is at rest with its engine running. At the instant of time $t = 0$ the accelerator pedal is stamped to the floor. The acceleration $f(t) = ?$

Solution According to an idealized mathematical model,

$$f(t) = \begin{cases} 0, & \text{when } t < 0, \\ a = \text{constant}, & \text{when } t \geq 0. \end{cases}$$

In reality no jump discontinuity occurs at the instant $t = 0$. The mathematical model (if a good one) does indeed depend essentially in this example on the time scale on which the events are considered. On different scales the function f may look very different. $\square$



Limits $\lim_{x \rightarrow \pm\infty} f(x)$

The limit $\lim_{x \rightarrow \infty} f(x)$ is defined in the same way as above, that is, by substituting $a = \infty$ in the general definition. Then $x_n \rightarrow \infty$ means that $\{x_n\}$ increases without bound. The limit $\lim_{x \rightarrow -\infty} f(x)$ is defined correspondingly. Theorem V.2.4 is valid also for the limits $\lim_{x \rightarrow \pm\infty} f(x)$, as is Theorem V.2.5 when the assumption concerning the domain $\mathcal{D}_{f \circ g}$ is changed either to the condition $\mathcal{D}_{f \circ g} \cap (M, \infty) \neq \emptyset \forall M \in \mathbb{R}$ (if $x \rightarrow \infty$) or to the condition $\mathcal{D}_{f \circ g} \cap (-\infty, M) \neq \emptyset \forall M \in \mathbb{R}$ (if $x \rightarrow -\infty$).

In determining limits $\lim_{x \rightarrow \infty} f(x)$ a often convenient substitution is $x = t^{-1}$, that is, passing to the function $F(t) = f(1/t)$. Since $x \rightarrow \pm\infty \Leftrightarrow t \rightarrow 0^\pm$, it is seen from the definitions of limits that (cf. Theorem V.2.6)

$$\lim_{x \rightarrow \pm\infty} f(x) = \lim_{t \rightarrow 0^\pm} F(t) = F(0^\pm), \quad F(t) = f(1/t).$$

EXAMPLE 10 (continued) By the change of variable $x = t^{-1}$ we conclude, appealing to Proposition V.1.9 and to Theorems V.1.6 and V.1.7:

$$\lim_{x \rightarrow \infty} \frac{\sqrt[3]{x} + \sqrt[4]{x}}{\sqrt[3]{8x+3}} = \lim_{t \rightarrow 0^+} \frac{1 + \sqrt[12]{t}}{\sqrt[3]{8+3t}} = \frac{1+0}{\sqrt[3]{8+0}} = \frac{1}{2}. \quad \square$$

From the theory of sequences it is known that a monotonic and bounded sequence converges. The corresponding assertion concerning limits $\lim_{x \rightarrow \infty} f(x)$ is

THEOREM V.2.8 If a function f is monotonic and bounded on the interval $[a, \infty)$, then the limit $\lim_{x \rightarrow \infty} f(x) \in \mathbb{R}$ exists.

Proof Let f be, say, increasing on the interval $[a, \infty)$. Then if we write $y_k = f(k)$, $k \in \mathbb{N}$ and $k \geq a$, then $\{y_k\}$ is an increasing and bounded sequence, so $y_k \rightarrow y \in \mathbb{R}$. Now let $\{x_n\}$ be any sequence for which $x_n \geq a \forall n$ and $x_n \rightarrow \infty$. Then if $x_n \in [k, k+1)$, $k \in \mathbb{N}$, then $y_k \leq f(x_n) \leq y_{k+1}$, since f is increasing. Here $k \rightarrow \infty$ as $n \rightarrow \infty$ (since $x_n \rightarrow \infty$), so we conclude (Proposition I.7.4[V2]) that $\lim_n f(x_n) = \lim_k y_k = y$. Since this holds for every sequence $\{x_n\}$ satisfying the stated conditions, $\lim_{x \rightarrow \infty} f(x) = y$. If f is decreasing on the interval $[a, \infty)$, the argument is analogous. $\square$

Asymptotes

We say that a function $g(x)$ is an *asymptote* of a function $f(x)$ if

$$\lim_{x \rightarrow \infty} [f(x) - g(x)] = 0, \quad \text{or} \quad \lim_{x \rightarrow -\infty} [f(x) - g(x)] = 0.$$

The idea of an asymptote is to approximate a function by some (preferably) simple function $g(x)$ when either x is very large (notation $x \gg 1$) or $-x$ is very large (notation $x \ll -1$):

$$f(x) \approx g(x), \quad \text{when } x \gg 1 \text{ or } x \ll -1.*$$

This is a matter of Theorem V.2.2, which can be generalized also to limits $\lim_{x \rightarrow \pm\infty} f(x)$: For example, if $\mathcal{D}_f \cap \mathcal{D}_g \supset [a, \infty)$ and $\lim_{x \rightarrow \infty} [f(x) - g(x)] = 0$, then for every $\varepsilon > 0$ there exists $M \in [a, \infty)$ such that $|f(x) - g(x)| < \varepsilon \forall x > M$.

*The more traditional and more restricted geometric definition of an asymptote, in connection with a plane curve, is a line that 'the curve approaches at infinity'. For example, the asymptotes of the curve $S : y = x^2/(x+1)$ are, according to this interpretation, the lines $y = x - 1$ and $x = -1$. The latter is a so-called vertical asymptote, which has no function counterpart.

EXAMPLE 11 One asymptote of the function $f(x) = \sqrt{x^2 + 4x}$, when $|x| \gg 1$, is $g(x) = |x + 2|$, since at the limits $x \rightarrow \pm\infty$

$$\begin{aligned} f(x) - g(x) &= \sqrt{x^2 + 4x} - |x + 2| = \frac{x^2 + 4x - (x + 2)^2}{\sqrt{x^2 + 4x} + |x + 2|} \\ &= -\frac{4}{\sqrt{x^2 + 4x} + |x + 2|} \rightarrow 0. \quad \square \end{aligned}$$

EXERCISES

1. a) Prove Theorem V.2.4. b) Prove Theorem V.2.5. c) Formulate and prove the counterpart of Theorem V.2.6 concerning the one-sided limit $\lim_{x \rightarrow a^+} f(x)$.
2. Of a function f it is known that $[-1, 1] \subset \mathcal{D}_f$, $f(0) = 0$ and $\sqrt{2 - x^2} \leq f(x) \leq \sqrt{2 + 9|x|}$ when $0 < |x| \leq 1$. Show that $\lim_{x \rightarrow 0} f(x) = \sqrt{2}$.
3. a) Of a function f it is known that $\lim_{x \rightarrow 0^+} f(x) = A$. Show that if f is even, then $\lim_{x \rightarrow 0^-} f(x) = A$, and if odd, then $\lim_{x \rightarrow 0^-} f(x) = -A$.
b) Of a function f it is known that $\lim_{x \rightarrow 0^+} f(x) = A$ and $\lim_{x \rightarrow 0^-} f(x) = B$. Compute the limits $\lim_{x \rightarrow 0^+} f(x^2 - x)$ and $\lim_{x \rightarrow 0^-} f(x^2 + x^3)$.
4. Determine the following limits, either as a real number or in the form $\pm\infty$,

or else verify that there is no limit. Change the variable if necessary.

- a) $\lim_{x \rightarrow 4} (x^2 - 4x + 1)$ b) $\lim_{x \rightarrow 3} \frac{x+3}{x+6}$ c) $\lim_{x \rightarrow \pi} \frac{(x+\pi)^2}{\pi x}$
- d) $\lim_{x \rightarrow -2} \frac{x^2 + 2x}{x^2 - 4}$ e) $\lim_{x \rightarrow 2} \left(\frac{1}{x-2} - \frac{4}{x^2 - 4} \right)$ f) $\lim_{x \rightarrow \frac{\pi}{2}} \frac{\sin(2x + \pi)}{\cot x}$
- g) $\lim_{t \rightarrow 0} \frac{(t+1)^2 - (t-1)^2}{t}$ h) $\lim_{s \rightarrow 0} \frac{s^2 + 3s}{(s+2)^2 - (s-2)^2}$
- i) $\lim_{x \rightarrow -3} |x-3|$ j) $\lim_{x \rightarrow 2} \frac{|x^2 - 4x + 3|}{x^2 + 2x - 3}$ k) $\lim_{x \rightarrow 1} \frac{|x^2 - 4x + 3|}{x^2 + 2x - 3}$
- l) $\lim_{x \rightarrow 0} \frac{\sqrt{1+2x+3x^2} - \sqrt{1-x}}{x}$ m) $\lim_{y \rightarrow 1} \frac{y - 4\sqrt{y} + 3}{y^2 - 1}$
- n) $\lim_{x \rightarrow 0} \frac{\sqrt{2-x} - \sqrt{2+x}}{x\sqrt{x}}$ o) $\lim_{x \rightarrow 2} \frac{\sqrt{4-4x+x^2}}{x-2}$ p) $\lim_{t \rightarrow 8} \frac{t^{2/3} - 4}{t^{1/3} - 2}$
- q) $\lim_{x \rightarrow 3^+} \frac{|x-3|}{3-x}$ r) $\lim_{x \rightarrow 2^-} \frac{\sqrt{4-4x+x^2}}{x-2}$ s) $\lim_{x \rightarrow \pi^+} \frac{\sqrt{1+\cos x}}{\cos \frac{x}{2}}$
- t) $\lim_{x \rightarrow -0.4^+} \frac{2x+5}{5x+2}$ u) $\lim_{x \rightarrow 1^+} \frac{x}{\sqrt{x^2-1}}$ v) $\lim_{x \rightarrow 1^+} \frac{\sqrt{x^2-x}}{x-x^2}$
- x) $\lim_{x \rightarrow \infty} \frac{1+x+x^4}{2+30x+200x^3}$ y) $\lim_{x \rightarrow \infty} \left(\frac{x^2}{x+1} - \frac{x^2}{x-1} \right)$
- z) $\lim_{x \rightarrow -\infty} \frac{2x-1}{\sqrt{3x^2+x+1}}$ ä) $\lim_{x \rightarrow \infty} \frac{x\sqrt{x+1}(1-\sqrt{2x+3})}{7-6x+4x^2}$
- ä) $\lim_{x \rightarrow \infty} \left(\sqrt{x^2+9x} - \sqrt{x^2-5x} \right)$ ö) $\lim_{x \rightarrow -\infty} \left(\sqrt{x^2-4x} - \sqrt{x^2+10x} \right)$

5. Prove, with the aid of Definition V.2.1 and the theory of sequences:

- a) If f is monotonic and bounded on the interval $(a, b) \subset \mathcal{D}_f$, then the limits $f(a^+)$ and $f(b^-)$ exist.
- b) If $f(x) \leq g(x) \forall x \in (a, b) \subset \mathcal{D}_f \cap \mathcal{D}_g$, then $f(a^+) \leq g(a^+)$ and $f(b^-) \leq g(b^-)$ insofar as the limits exist as real numbers.

6. Enlarge the domain of the function $f(x) = \frac{x^2 - 1}{\sqrt{x+3} - 2}$ to the interval $[-3, \infty)$ by extension. What is the simplified computational rule of the extended function?

7. Let f be defined on the interval $[1, \infty)$ and let $y_n = f(n)$, $n \in \mathbb{N}$. Show that $\lim_{x \rightarrow \infty} f(x) = A \Rightarrow \lim_n y_n = A$. Show by a counterexample that the converse implication is not true.

8. For the following functions $f(x)$ or $y(x)$, determine an asymptote $g(x)$ of the given form $(a, b \in \mathbb{R})$ that is as accurate as possible (though not unique). Consider separately the cases $x \rightarrow \infty$ and $x \rightarrow -\infty$.

a) $f(x) = \frac{|x+3|}{2x-1}, \quad g(x) = a$

b) $f(x) = \frac{(x+3)^2}{|3x+1|}, \quad g(x) = ax + b$

c) $f(x) = \sqrt{2x^2 + 3x + \cos x}, \quad g(x) = ax + b$

d) $f(x) = \frac{1}{\sqrt{3x^2 + 4x} + \sqrt{x^2 + x + 1}}, \quad g(x) = \frac{a}{x}$

e) $f(x) = \frac{|3x+2|}{2x+3}, \quad g(x) = a + \frac{b}{x}$

f) $9(x-1)^2 - 16(y+2)^2 = 25, \quad y(x) \geq -2, \quad g(x) = ax + b$

g) $(2x^2 + 1)y + x \cos y = (2x + 1)^3, \quad g(x) = ax + b$

9. The function $f(x) = \sqrt{x^4 + 8x^3 + 35x^2 + 78x + 98}$ has second-degree polynomials as asymptotes. Try to find as accurate an asymptote $p(x)$ of this kind as possible and estimate the error of the approximation $f(x) \approx p(x)$ when $|x| \geq 100$.

10. (*) Let $a > 0$ and $m, n \in \mathbb{N}$. Compute the limit $\lim_{x \rightarrow a} \frac{\sqrt[n]{x} - \sqrt[n]{a}}{\sqrt[m]{x} - \sqrt[m]{a}}$.

11. (*) Of a real function f it is known that $f(x) = x + 1 \quad \forall x \in \mathbb{R}, \quad x \notin X$, where $X \subset \mathbb{R}$ is a finite set. The set X is unknown, and of the points $x \in X$ nothing is known except that at these points f is either defined in some unknown manner (real-valued) or left undefined. Show that $\lim_{x \rightarrow 0} f(x) = 1$.

V.3 The derivative

Consider the approximation of a function f near a point $a \in \mathcal{D}_f$, based on various assumptions about the predictability of the function. First, if f is continuous at the point a , then by Definition V.1.2 f is approximately constant near a : $f(x) \approx f(a)$ when $x \approx a$. The error of this approximation is near a typically of the form $f(x) - f(a) \approx L(x - a)$, where L is a constant depending on f and a . Except in special cases (such as $f(x) = \text{constant}$ or $f(x) = x^2$, $a = 0$), the error estimate mentioned is in general the best possible.

EXAMPLE 1 If $f(x) = x^3$ and $a \neq 0$, then

$$f(x) - f(a) = x^3 - a^3 = (x - a)(x^2 + ax + a^2) = (x - a)g(x).$$

Here $g(x) = x^2 + ax + a^2$ is (as a polynomial) continuous, so near the point $x = a$ we have $f(x) - f(a) \approx L(x - a)$, where $L = g(a) = 3a^2$. $\square$

If, for a continuous function, one wants near the point a an essentially more accurate approximation than $f(x) \approx f(a)$, then the approximation must be of some type other than $f(x) \approx A = \text{constant}$, since among these alternatives the choice $A = f(a)$ is the best. Since a constant = a polynomial of degree $n = 0$, a natural next attempt is to use in the approximation a polynomial p of degree $n = 1$. When the prior condition $p(a) = f(a)$ is imposed, the approximation takes the form

$$f(x) \approx f(a) + k(x - a).$$

Here the coefficient $k \in \mathbb{R}$ is determined (if possible) so that for the error $g(x) = f(x) - f(a) - k(x - a)$ an essentially better estimate is obtained near a than $|g(x)| \leq L|x - a|$. As a minimal condition we impose

$$\lim_{x \rightarrow a} \frac{g(x)}{x - a} = 0 \Leftrightarrow \lim_{x \rightarrow a} \left[\frac{f(x) - f(a)}{x - a} - k \right] = 0 \Leftrightarrow \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a} = k.$$

Depending on the function f , the coefficient k is thus either uniquely determined by this condition (the limit on the right existing as a real number) or else the coefficient cannot be determined, in which case the above attempt at approximation is considered to have failed.

EXAMPLE 1 (continued) For the function of the example

$$\begin{aligned} f(x) &= f(a) + (x - a)(x^2 + ax + a^2) \\ &= f(a) + 3a^2(x - a) + (x - a)(x^2 + ax - 2a^2) \\ &= f(a) + 3a^2(x - a) + (x - a)^2(x + 2a). \end{aligned}$$

So if we choose $k = 3a^2$, then $g(x)/(x - a) = (x - a)(x + 2a) \rightarrow 0$ as $x \rightarrow a$, so the approximation succeeded. $\square$

DEFINITION V.3.1 (Derivative) A function $f : \mathcal{D}_f \rightarrow \mathbb{R}$, $\mathcal{D}_f \subset \mathbb{R}$, is *differentiable* at the point $a \in \mathcal{D}_f$ if $(a - \delta, a + \delta) \subset \mathcal{D}_f$ for some $\delta > 0$ and the limit

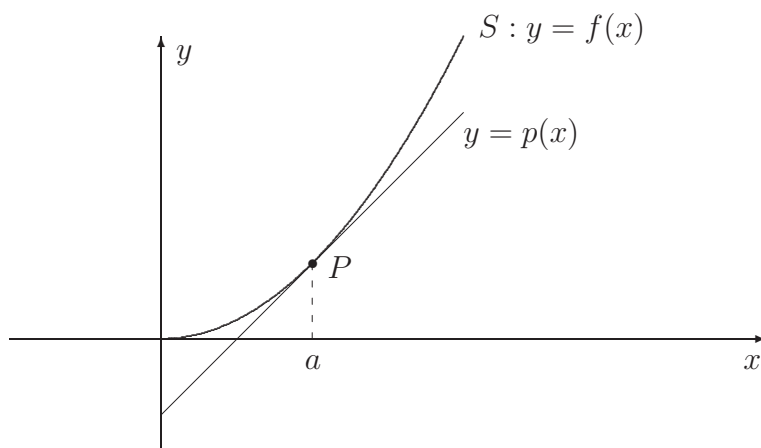
$$\lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a} = k \in \mathbb{R}$$

exists. The number k is called the *derivative* of f at the point a , and is denoted $k = f'(a)$. If f is differentiable at the point a , then the polynomial

$$p(x) = f(a) + f'(a)(x - a)$$

is called the *linearizing approximation* or *linearization* of f at the point a .

The geometric counterpart of the linearizing approximation is the line through the point $P = (a, f(a))$ whose *slope* is $k = f'(a)$ ($= p'(a)$). We say that this line is the *tangent* of the curve $S : y = f(x)$ at the point P . Accordingly, 'the derivative is the slope of the tangent'.



EXAMPLE 1 (continued) On the basis of the result of the example, the function $f(x) = x^3$ is differentiable at every point $a \in \mathbb{R}$ and $f'(a) = 3a^2$. The equation of the tangent of the curve $S : y = x^3$ at the point $x = a$ is thus $y = p(x) = a^3 + 3a^2(x - a)$. $\square$

EXAMPLE 2 The function $f(x) = |x - a|$ is not differentiable at the point a , since

$$\frac{f(x) - f(a)}{x - a} = \begin{cases} 1, & \text{when } x > a, \\ -1, & \text{when } x < a. \end{cases} \quad \square$$

If f is differentiable at the point a , then by Definition V.3.1 and Theorem V.2.4

$$\lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} \left[f(a) + (x - a) \frac{f(x) - f(a)}{x - a} \right] = f(a) + 0 \cdot f'(a) = f(a).$$

Thus differentiability is a stronger property than continuity:

$$f \text{ differentiable at the point } a \Rightarrow f \text{ continuous at the point } a.$$

The derivative as an operator

When in the definition of the derivative we write $x - a = \Delta x$ and replace a by x , the definition takes the form

$$f'(x) = \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x) - f(x)}{\Delta x}.$$

Other notations for the derivative are

$$f'(x) = \frac{df}{dx} = \frac{d}{dx}f = Df(x).$$

In the last two notations written, the derivative is interpreted as an *operator* separate from its object, that is, as a 'function of a function':

$$D : f \mapsto f', \quad D = \frac{d}{dx}.$$

More generally, operators are functions whose domain and codomain are both (e.g. real-variable) functions. The 'function of a function' D that carries out differentiation is called the *differential operator*.

By its definition, the differential operator can act only at points where f is differentiable, that is, on the domain of f'

$$\mathcal{D}_{f'} = \{x \in \mathcal{D}_f \mid f \text{ differentiable at the point } x\}.$$

Many common functions are differentiable 'almost everywhere', so that $\mathcal{D}_{f'}$ is obtained from $\mathcal{D}_f$ by removing at most a finite or countable number of points.

EXAMPLE 3 The function $f(x) = x - [x]$, where $[x]$ = the largest integer for which $[x] \leq x$, is differentiable everywhere except at the points $k \in \mathbb{Z}$, that is, $\mathcal{D}_{f'} = \{x \in \mathbb{R} \mid x \notin \mathbb{Z}\}$. On this set $f'(x) = 1$. (Cf. Example 8 in the preceding section.) $\square$

If the derivative f' of f is in turn differentiable at the point x , then the *second derivative* of f at the point x can be defined:

$$f''(x) = \lim_{\Delta x \rightarrow 0} \frac{f'(x + \Delta x) - f'(x)}{\Delta x} = Df'(x) = D(Df(x)) = D^2f(x).$$

At points where f'' is in turn differentiable, the third derivative $f'''(x)$ can be defined, and so on. In general, the *nth derivative* of a function at the point x is

$$f^{(n)} = \frac{d^n}{dx^n} f(x) = D^n f(x).$$

We say that n is the *order* of the derivative $f^{(n)}$, and we agree that $f^{(0)} = f$.

Rules of differentiation

In the general *rules of differentiation* we present formulas for computing the derivatives of the algebraic combinations of functions

$$f, g, \lambda \mapsto \lambda f, f + g, fg, f/g, f \circ g, f^{-1}$$

in terms of f' and g' . Of the rules of differentiation the simplest are

$$\begin{aligned} D(\lambda f) &= \lambda f', \quad \lambda \in \mathbb{R}, \\ D(f + g) &= f' + g', \end{aligned}$$

which can be combined into the rule

$$\boxed{D(\alpha f + \beta g) = \alpha Df + \beta Dg, \quad \alpha, \beta \in \mathbb{R}.} \quad (1)$$

Rule (1) means that D is a *linear* operator. The rule is valid at every point where f and g are both differentiable, as can easily be verified on the basis of the rules for combining limits (Theorem V.2.4). Under the same conditions the product rule of differentiation holds

$$\boxed{D(fg) = f'g + fg'.} \quad (2)$$

To justify this we write

$$\begin{aligned} f(x + \Delta x)g(x + \Delta x) - f(x)g(x) \\ = [f(x + \Delta x) - f(x)]g(x + \Delta x) + f(x)[g(x + \Delta x) - g(x)], \end{aligned}$$

divide both sides by Δx , and apply the rules for combining limits mentioned. We also note that g is (being differentiable) continuous at x .

EXAMPLE 4 Starting from the obvious differentiation rule $Dx = 1$ and applying rule (2) to the sequence of functions $f_1(x) = x$, $f_n(x) = xf_{n-1}(x)$, $n = 2, 3, \dots$, one sees by induction the correctness of the rule

$$Dx^m = mx^{m-1}, \quad m \in \mathbb{N} \cup \{0\}.$$

Another way to derive this result is to start directly from the definition of the derivative and use the decomposition based on the binomial formula:

$$\begin{aligned} \frac{(x + \Delta x)^m - x^m}{\Delta x} &= mx^{m-1} + \binom{m}{2} x^{m-2} \Delta x + \dots + (\Delta x)^{m-1} \\ &\rightarrow mx^{m-1}, \quad \text{when } \Delta x \rightarrow 0. \quad \square \end{aligned}$$

When, alongside the result of the example, rule (1) is also taken into account, one sees that the derivative of a polynomial is another polynomial (of one degree lower).

EXAMPLE 5

$$\begin{aligned} f(x) &= x^5 - 4x^4 + 6x^3 + x^2 - 2x + 3 \\ f'(x) &= 5x^4 - 16x^3 + 18x^2 + 2x - 2 \\ f''(x) &= 20x^3 - 48x^2 + 36x + 2 \\ f'''(x) &= 60x^2 - 96x + 36 \\ f^{(4)}(x) &= 120x - 96 \\ f^{(5)}(x) &= 120 \\ f^{(n)}(x) &= 0, \quad n \geq 6 \quad \square \end{aligned}$$

If f and g are n times differentiable at the point x , then by applying the same rule again on the right-hand side of rule (2) together with rule (1), one derives the *Leibniz rule*, which resembles the binomial formula (Proposition I.4.1):

$$D^n(fg) = \sum_{k=0}^n \binom{n}{k} f^{(k)} g^{(n-k)}.$$

If $h(x) = (f/g)(x)$, then the product rule of differentiation (2) applied to the identity $h(x)g(x) = f(x)$ gives the result $h'g + hg' = f'$. At points where $g(x) \neq 0$, this can be solved for the derivative of h , that is, of the quotient:

$$D\left(\frac{f}{g}\right) = \frac{f'}{g} - \frac{fg'}{g^2} = \frac{f'g - fg'}{g^2}. \quad (3)$$

From rule (3) together with the differentiation rule for a polynomial one sees that every rational function $f = p/q$ (p and q polynomials) is differentiable arbitrarily many times on its entire domain, that is, on the set $\{x \in \mathbb{R} \mid q(x) \neq 0\}$, and that the derivatives are rational functions defined on the same set.

EXAMPLE 6 By the first computational formula of rule (3)

$$\frac{d}{dx} \left(\frac{x}{1-x^2} \right) = \frac{1}{1-x^2} + \frac{2x^2}{(1-x^2)^2} = \frac{1+x^2}{(1-x^2)^2}. \quad \square$$

EXAMPLE 7 By applying rule (3) to the case $f(x) = 1$, $g(x) = x^m$, $m \in \mathbb{N}$ one sees that the rule $Dx^m = mx^{m-1}$ of Example 4 is valid $\forall m \in \mathbb{Z}$. $\square$

THEOREM V.3.2 (**Derivative of a composite function**) If g is differentiable at the point x and f is differentiable at the point $g(x)$, then $f \circ g$ is differentiable at the point x and

$$Df(g(x)) = f'(g(x)) g'(x).$$

Proof Write $g(x + \Delta x) - g(x) = \Delta g$. Then, since f is differentiable at the point $g(x)$, we have (cf. the definition of the derivative above)

$$f(g(x + \Delta x)) = f(g(x) + \Delta g) = f(g(x)) + f'(g(x))\Delta g + h(\Delta g)\Delta g,$$

where $h(\Delta g) \rightarrow 0$ as $\Delta g \rightarrow 0$. Since g is differentiable at the point x and since $\Delta x \rightarrow 0 \Rightarrow \Delta g \rightarrow 0$, it follows that

$$\begin{aligned} D(f \circ g)(x) &= \lim_{\Delta x \rightarrow 0} \frac{f(g(x + \Delta x)) - f(g(x))}{\Delta x} \\ &= \lim_{\Delta x \rightarrow 0} \left[f'(g(x)) \frac{\Delta g}{\Delta x} + h(\Delta g) \frac{\Delta g}{\Delta x} \right] \\ &= f'(g(x))g'(x) + 0 \cdot g'(x) = f'(g(x))g'(x). \quad \square \end{aligned}$$

By applying the rule of Theorem V.3.2 and the result of Example 7 to the function $g \circ f$, where $g(x) = x^m$, we obtain the differentiation rule

$$D[f(x)]^m = m[f(x)]^{m-1} f'(x), \quad m \in \mathbb{Z}. \quad (4)$$

EXAMPLE 8 Continuing from Example 6 with rules (2), (4) and (1) we obtain

$$\begin{aligned} \frac{d^2}{dx^2} \left(\frac{x}{1-x^2} \right) &= \frac{d}{dx} \left(\frac{1+x^2}{(1-x^2)^2} \right) = \frac{2x}{(1-x^2)^2} + (1+x^2) \frac{(-2)(-2x)}{(1-x^2)^3} \\ &= \frac{6x+2x^3}{(1-x^2)^3}. \quad \square \end{aligned}$$

If $f(x) = x^{m/n}$, $m \in \mathbb{Z}$, $n \in \mathbb{N}$, $x > 0$, then $[f(x)]^n = x^m$. Differentiating both sides here by rule (4) we obtain

$$n[f(x)]^{n-1}f'(x) = mx^{m-1} \Rightarrow f'(x) = (m/n)x^{m/n-1}.$$

Thus we have derived the differentiation rule for the general power function

$$\boxed{Dx^\alpha = \alpha x^{\alpha-1}, \quad x \in \mathbb{R}_+, \alpha \in \mathbb{Q}.}$$

EXAMPLE 9 Differentiate $f(x) = \sqrt[3]{1 + \sqrt{x}}$.

Solution By the differentiation rules for the power function and for a composite function

$$\begin{aligned} f'(x) &= \frac{1}{3}(1 + \sqrt{x})^{-2/3} \cdot \frac{1}{2}x^{-1/2} = \frac{(1 + \sqrt{x})^{1/3}}{6\sqrt{x}(1 + \sqrt{x})} \\ &= \frac{\sqrt[3]{1 + \sqrt{x}}}{6(x + \sqrt{x})}, \quad x > 0. \quad \square \end{aligned}$$

From the differentiation rule for a composite function (Theorem V.3.2) one can also derive the general differentiation rule for an inverse function: Assume that f is differentiable at the point x and that the inverse function $g = f^{-1}$ is differentiable at the point $f(x)$. Then from the identity $g(f(x)) = x$ it follows by differentiating both sides that $g'(f(x))f'(x) = 1$. Thus, under the assumptions made and the additional assumption $f'(x) \neq 0$,

$$\boxed{Df^{-1}(y) = \frac{1}{f'(x)}, \quad y = f(x).} \quad (5)$$

The differentiability of the inverse function is justified more precisely by the following theorem.

THEOREM V.3.3 (Derivative of an inverse function) Let $f : [a - \delta, a + \delta] \rightarrow U$ be a continuous bijection for some $\delta > 0$ and let f be differentiable at the point a . Then $f^{-1} : U \rightarrow [a - \delta, a + \delta]$ is differentiable at the point $b = f(a)$ precisely when $f'(a) \neq 0$, in which case $Df^{-1}(b) = 1/f'(a)$.

Proof Since f is continuous and 1-1 on the interval $[a - \delta, a + \delta]$ and the range of f on that interval $= U$, by Theorem V.1.10 we have $[b - \varepsilon, b + \varepsilon] \subset U$, where

$$\varepsilon = \min\{|f(a - \delta) - f(a)|, |f(a + \delta) - f(a)|\} > 0.$$

Then if $f'(a) \neq 0$, then by making on the interval $[b - \varepsilon, b + \varepsilon]$ the change of variable $y = f(x)$, it follows by Theorems V.2.6 and V.2.4 that

$$\lim_{y \rightarrow b} \frac{f^{-1}(y) - f^{-1}(b)}{y - b} = \lim_{x \rightarrow a} \frac{x - a}{f(x) - f(a)} = \left[\lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a} \right]^{-1} = \frac{1}{f'(a)},$$

so f^{-1} is differentiable at b and $Df^{-1}(b) = 1/f'(a)$ (Definition V.3.1). If $f'(a) = 0$, it follows from Theorem V.2.6 correspondingly that

$$\lim_{y \rightarrow b} \left| \frac{f^{-1}(y) - f^{-1}(b)}{y - b} \right| = \lim_{x \rightarrow a} \left| \frac{x - a}{f(x) - f(a)} \right| = \infty,$$

so in this case f^{-1} is not differentiable at b . $\square$

EXAMPLE 10 For the function $f(x) = x^5 + 3x$ the assumptions of Theorem V.3.3 hold for every $a \in \mathbb{R}$ (and for every $\delta > 0$, see Example 3 in Section IV.2), so rule (5) is applicable:

$$Df^{-1}(y) = \frac{1}{5x^4 + 3}, \quad y = x^5 + 3x.$$

For example $Df^{-1}(0) = 1/3$, since $y = 0 \Rightarrow x = 0$. If one wants to compute $Df^{-1}(1)$, one must first solve (numerically) the equation $x^5 + 3x = 1$. If the solution is denoted $x = a$, then $Df^{-1}(1) = 1/(5a^4 + 3)$. $\square$

Implicit differentiation

If a function $y(x)$ is defined implicitly in the form (cf. Section IV.2)

$$F(x, y) = 0 \quad \Leftrightarrow \quad y = y(x),$$

then the derivative $y'(x)$ can be computed by differentiating both sides of the equation

$$F(x, y(x)) = 0.$$

Such an indirect procedure is called *implicit differentiation*. With the differentiation rules known so far, implicit differentiation succeeds e.g. when the function F is a (two-variable) polynomial or rational function.

EXAMPLE 10 (continued) If in the example we write $f^{-1}(x) = y(x)$, then $y(x)$ can be interpreted as an implicit function determined by the equation $y^5 + 3y = x$, that is,

$$[y(x)]^5 + 3y(x) = x, \quad x \in \mathbb{R}.$$

When both sides here are differentiated, applying the differentiation rule for a composite function and rule (1), it follows that

$$[5[y(x)]^4 + 3] y'(x) = 1 \Rightarrow y'(x) = \frac{1}{5y^4 + 3} \quad (y = y(x)). \quad \square$$

EXAMPLE 11 The equation

$$x^2 + y^2 = 1$$

defines on the interval $[-1, 1]$ a two-branch implicit function, each branch of which is differentiable on the interval $(-1, 1)$. If $x \in (-1, 1)$, then $y'(x)$ is easily computed by implicit differentiation:

$$D[x^2 + (y(x))^2] = 2x + 2y(x)y'(x) = D1 = 0 \Rightarrow y'(x) = -\frac{x}{y}, \quad y = y(x).$$

The result is valid on either branch of the implicit function

$$y(x) = \pm\sqrt{1-x^2},$$

as can also be verified by direct differentiation:

$$y(x) = \pm\sqrt{1-x^2} \Rightarrow y'(x) = \pm\left(\frac{-x}{\sqrt{1-x^2}}\right) = -\frac{x}{y(x)}, \quad x \in (-1, 1). \quad \square$$

Differentiation of a power series

On the basis of the considerations so far, one may take as a general rule that ordinary functions (defined by algebraic means as expressions) are differentiable on their domains, except at possible isolated points, of which there may be a finite or sometimes countable number. We show at the end of the section that functions defined as sums of power series are also no exception to the general rule. With such functions the set of differentiable functions in fact expands significantly, as will be seen later.

Let a real function f be defined in the form

$$f(x) = \sum_{k=0}^{\infty} a_k x^k.$$

Assume that here the radius of convergence of the power series is either $\rho \in \mathbb{R}_+$ or $\rho = \infty$. Then the largest open interval contained in the domain of f is $(-\rho, \rho)$ (cf. Section I.12). In what follows it is shown that f is differentiable on this entire

interval and that the derivative can be computed simply by differentiating the series term by term, that is,

$$f'(x) = \sum_{k=1}^{\infty} k a_k x^{k-1}, \quad x \in (-\rho, \rho).$$

The radius of convergence of the series resulting from the differentiation is also ρ (Theorem I.12.11), so f' is in turn differentiable on the interval $(-\rho, \rho)$, and the result of the differentiation is thus

$$f''(x) = \sum_{k=2}^{\infty} k(k-1) a_k x^{k-2}, \quad x \in (-\rho, \rho).$$

The radius of convergence of this series too $= \rho$ (Theorem I.12.11), so f'' is in turn differentiable on the interval $(-\rho, \rho)$, and so on. We conclude that a function defined as the sum of a power series is in fact, on its open interval of convergence, an arbitrarily many times differentiable function (!). The conclusion thus rested on the theory of convergence of power series and on the following assertion, which can be proved directly from the definition of the derivative (that is, without appeal to the earlier differentiation rules).

THEOREM V.3.4 (Derivative of a power series) If the radius of convergence of the power series $\sum_{k=0}^{\infty} a_k x^k$ is $\rho > 0$, then the function $f(x)$ defined as the sum of the series is differentiable on the interval $(-\rho, \rho)$ and on that interval $f'(x) = \sum_{k=1}^{\infty} k a_k x^{k-1}$.

Proof Let $x \in (-\rho, \rho)$ and choose $\Delta x \neq 0$ such that

$$|x| + |\Delta x| \leq \rho_0 < \rho.$$

By the binomial formula

$$\begin{aligned} (x + \Delta x)^k &= \sum_{l=0}^k \binom{k}{l} (\Delta x)^l x^{k-l} \\ &= x^k + k x^{k-1} \Delta x + \sum_{l=2}^k \binom{k}{l} (\Delta x)^l x^{k-l}. \end{aligned}$$

Here we write the last term, by a change of summation index, in the form

$$\begin{aligned} \sum_{l=2}^k \binom{k}{l} (\Delta x)^l x^{k-l} &= \sum_{i=0}^{k-2} \binom{k}{i+2} (\Delta x)^{i+2} x^{k-2-i} \\ &= (\Delta x)^2 \sum_{i=0}^{k-2} \binom{k}{i+2} (\Delta x)^i x^{k-2-i}, \end{aligned}$$

and further

$$\sum_{i=0}^{k-2} \binom{k}{i+2} (\Delta x)^i x^{k-2-i} = \sum_{i=0}^{k-2} c_i \binom{k-2}{i} (\Delta x)^i x^{k-2-i},$$

where

$$c_i = \binom{k}{i+2} \binom{k-2}{i}^{-1} = \frac{k!}{(k-i-2)!(i+2)!} \cdot \frac{(k-i-2)!i!}{(k-2)!} = \frac{k(k-1)}{(i+1)(i+2)}.$$

Since

$$c_i \leq \frac{1}{2} k(k-1) < \frac{1}{2} k^2, \quad i = 0 \dots k-2,$$

and $|x| + |\Delta x| \leq \rho_0$, we obtain for every $k \geq 2$ the estimate

$$\begin{aligned} \left| \frac{(x + \Delta x)^k - x^k}{\Delta x} - kx^{k-1} \right| &= |\Delta x| \left| \sum_{i=0}^{k-2} c_i \binom{k-2}{i} (\Delta x)^i x^{k-2-i} \right| \\ &\leq |\Delta x| \sum_{i=0}^{k-2} c_i \binom{k-2}{i} |\Delta x|^i |x|^{k-2-i} \\ &\leq \frac{1}{2} k^2 |\Delta x| \sum_{i=0}^{k-2} \binom{k-2}{i} |\Delta x|^i |x|^{k-2-i} \\ &= \frac{1}{2} k^2 |\Delta x| (|x| + |\Delta x|)^{k-2} \\ &\leq \frac{1}{2} k^2 |\Delta x| \rho_0^{k-2}. \end{aligned}$$

Consequently

$$\begin{aligned} \left| \frac{f(x + \Delta x) - f(x)}{\Delta x} - \sum_{k=1}^{\infty} k a_k x^{k-1} \right| &= \left| \sum_{k=2}^{\infty} a_k \left[\frac{(x + \Delta x)^k - x^k}{\Delta x} - kx^{k-1} \right] \right| \\ &\leq \sum_{k=2}^{\infty} |a_k| \left| \frac{(x + \Delta x)^k - x^k}{\Delta x} - kx^{k-1} \right| \\ &\leq |\Delta x| \sum_{k=2}^{\infty} \frac{1}{2} k^2 |a_k| \rho_0^{k-2}. \end{aligned}$$

Here the series on the right converges by Theorem I.12.11, since $\rho_0 < \rho$, so

$$\left| \frac{f(x + \Delta x) - f(x)}{\Delta x} - \sum_{k=1}^{\infty} k a_k x^{k-1} \right| \leq C |\Delta x|, \quad C = \sum_{k=2}^{\infty} \frac{1}{2} k^2 |a_k| \rho_0^{k-2}.$$

The estimate obtained is valid when $x \in (-\rho, \rho)$ and $|\Delta x| < \delta$, where e.g.

$$\delta = \frac{1}{2}(\rho - |x|) > 0.$$

Thus f is differentiable at every point $x \in (-\rho, \rho)$ and

$$f'(x) = \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x) - f(x)}{\Delta x} = \sum_{k=1}^{\infty} k a_k x^{k-1}. \quad \square$$

EXAMPLE 12 Compute $\sum_{k=1}^{\infty} k q^k$ when $|q| < 1$.

Solution Differentiate the function

$$f(x) = \sum_{k=0}^{\infty} x^k = \frac{1}{1-x}, \quad x \in (-1, 1)$$

on the one hand by Theorem V.3.4 and on the other hand as a rational function:

$$f'(x) = \sum_{k=1}^{\infty} k x^{k-1} = \frac{1}{(1-x)^2}, \quad x \in (-1, 1).$$

Choose $x = q$ and multiply both sides by q :

$$q \sum_{k=1}^{\infty} k q^{k-1} = \sum_{k=1}^{\infty} k q^k = \frac{q}{(1-q)^2}. \quad \square$$

EXERCISES

1. Compute directly from the definition (without using differentiation rules) the linearizing approximation for the following functions at the point $a = 1$. Draw a figure as well!

$$\text{a) } f(x) = \frac{x}{1+2x} \quad \text{b) } f(x) = \frac{x}{2+x^2} \quad \text{c) } f(x) = \sqrt{10+6x}$$

2. Compute the derivatives of the following functions at the point $a \in \mathcal{D}_f$ directly from the definition of the derivative:

$$\text{a) } f(x) = x^3 + 3x^2 \quad \text{b) } f(x) = \frac{1}{x} \quad \text{c) } f(x) = \frac{1}{\sqrt{x}}$$

3. Let f be differentiable at the point a . Determine the following limits in terms of $f(a)$ and $f'(a)$:

$$\begin{aligned} \text{a) } \lim_{h \rightarrow 0} \frac{f(a+h^2) - f(a-h)}{h} & \quad \text{b) } \lim_{x \rightarrow a} \frac{xf(a) - af(x)}{x-a} \\ \text{c) } \lim_{t \rightarrow 0^+} \frac{f(a+\alpha t) - f(a+\beta t)}{t}, \quad \alpha, \beta \in \mathbb{R} \end{aligned}$$

4. Starting from the definition of the derivative, determine $f'(0)$ or show that f is not differentiable at 0:

$$\begin{aligned} \text{a) } f(x) &= \begin{cases} x(1 + \sqrt{|x|} \cos \frac{1}{x}), & \text{when } x \neq 0, \\ 0, & \text{when } x = 0 \end{cases} \\ \text{b) } f(x) &= \begin{cases} x \sin^2 \frac{1}{x}, & \text{when } x \neq 0, \\ 0, & \text{when } x = 0 \end{cases} \\ \text{c) } f(x) &= \begin{cases} 1 + x \cos \frac{1}{x} \sqrt[100]{x \left| \sin \frac{1}{x} \right|}, & \text{when } x > 0, \\ 1 + x^2, & \text{when } x \leq 0 \end{cases} \end{aligned}$$

5. Reduce the quotient rule of differentiation to the product rule of differentiation by first deriving the differentiation rule for the function $1/f$ directly from the definition of the derivative.
6. In parts a)–c) compute $f^{(n)}(x)$ for $n = 1, 2$ and in parts d)–f) the general expression for the derivative $f^{(n)}(x)$, $n \in \mathbb{N}$:

$$\begin{aligned} \text{a) } f(x) &= \frac{x}{\sqrt{a^2 - x^2}} & \text{b) } f(x) &= \frac{1}{\sqrt{x+1}} & \text{c) } f(x) &= \sqrt{x + \sqrt{x}} \\ \text{d) } f(x) &= \frac{1}{1+x} & \text{e) } f(x) &= \sqrt{x+1} & \text{f) } f(x) &= \frac{1-x}{1+x} \end{aligned}$$

7. It is known that $f(2) = 1$, $f'(2) = 3$ and $f''(2) = -1$. Compute

$$\begin{aligned} \text{a) } \left[\frac{d}{dx} \left(\frac{x^2}{f(x)} \right) \right]_{x=2} & \quad \text{b) } \left[\frac{d}{dx} \left(\frac{f(x)}{x^2} \right) \right]_{x=2} & \quad \text{c) } \left[\frac{d^2}{dx^2} \left(\frac{f(x)}{x^2} \right) \right]_{x=2} \end{aligned}$$

8. The function f is defined as the two-branch inverse function of $g(x) = x^2 - 3x + 4$. Compute $f'(2)$ on both branches a) by first solving the equation $g(x) = 2$ and using the differentiation rule for an inverse function, b) by solving $g(y) = x \Rightarrow y = f(x)$, differentiating and substituting $x = 2$.

9. The following equations define an implicit function $y(x)$. Compute by implicit differentiation $y'(x)$ at the given point $(x, y) = (x, y(x))$:
- $2x^2 + 3y^2 = 5$, $(x, y) = (1, 1)$
 - $2x^2 + 3y^2 = 5$, $(x, y) = (1, -1)$
 - $x^2y^3 - x^3y^2 = 12$, $(x, y) = (-1, 2)$
 - $(x-1)(x+2y+1) = y^2$, $(x, y) = (2, -1)$
10. The function g is the inverse function of the function f , that is, $y = f(x) \Leftrightarrow x = g(y)$. Derive the differentiation rule

$$g''(y) = -\frac{f''(x)}{[f'(x)]^3}, \quad y = f(x)$$

and use the rule to compute $g''(1)$ when $f(x) = x^5 + 2x + 1$.

11. Compute y'' in terms of x and y by implicit differentiation:

$$\text{a) } xy = x + y \quad \text{b) } x^2 + y^2 = 1 \quad \text{c) } x^3 - y^2 + y^3 = x$$

12. a) Show that $xy'' = y$ when we define

$$y(x) = \sum_{k=1}^{\infty} \frac{x^k}{k!(k-1)!}, \quad x \in \mathbb{R}.$$

- b) Show that $u' = -v$ and $v' = u$ when we define

$$u(x) = \sum_{k=0}^{\infty} (-1)^k \frac{x^{2k}}{(2k)!}, \quad v(x) = \sum_{k=0}^{\infty} (-1)^k \frac{x^{2k+1}}{(2k+1)!}, \quad x \in \mathbb{R}.$$

13. The following functions are rational functions on the interval $(-1, 1)$. Compute the expressions for the functions by differentiating the power series $\sum_{k=0}^{\infty} x^k$.

$$\begin{array}{lll} \text{a) } f(x) = \sum_{k=2}^{\infty} kx^k & \text{b) } f(x) = \sum_{k=3}^{\infty} (-1)^k kx^k & \text{c) } f(x) = \sum_{k=1}^{\infty} k^2 x^k \\ \text{d) } f(x) = \sum_{k=0}^{\infty} (k+2)^2 x^k & \text{e) } f(x) = \sum_{k=1}^{\infty} k^3 x^k & \text{f) } f(x) = \sum_{k=0}^{\infty} (k+1)^3 x^k \end{array}$$

14. (*) Prove the Leibniz rule.
15. (*) Let $a \in \mathbb{R}$ be a zero of a real-coefficient polynomial p and let $m \in \mathbb{N}$, $m \geq 2$. Show that a is a zero of multiplicity m precisely when $p^{(k)}(a) = 0$, $k = 1 \dots m-1$ and $p^{(m)}(a) \neq 0$.

16. (*) a) Show that if a power series $\sum_k a_k x^k$ converges on the interval $(-\rho, \rho)$ and the function g is differentiable on the interval $(a, b) \subset (-\rho, \rho)$, then the derivative on the interval (a, b) of the function

$$f(x) = \sum_{k=0}^{\infty} x^k g(x), \quad x \in (a, b)$$

can be computed by differentiating the series term by term.

- b) Determine $f'(x)$ and $f''(x)$ on the interval $(0, \infty)$ when

$$f(x) = \sum_{k=0}^{\infty} \frac{1}{k!} x^{k+\frac{1}{2}}, \quad x \geq 0.$$

V.4 Differentiation of the trigonometric functions

The trigonometric functions, as real functions, are based on the concept of arc length (on the unit circle, see Section II.5). The concept was earlier defined briefly in Section II.1. In what follows we first consider arc length from a somewhat more general point of view and derive, on the basis of these considerations, the limit result

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1.$$

All the differentiation rules for the trigonometric functions can be derived from this fundamental result and the rules of Section V.3.

Arc length

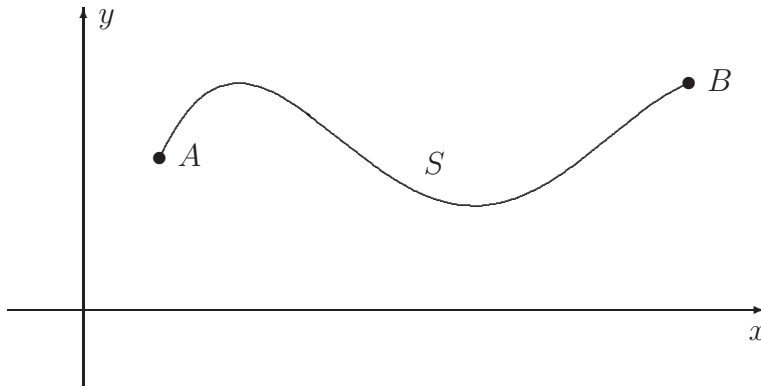
Let $f : \mathcal{D}_f \rightarrow \mathbb{R}$, $\mathcal{D}_f \subset \mathbb{R}$, be continuous on the closed interval $[a, b] \subset \mathcal{D}_f$. Write

$$A = (a, f(a)), \quad B = (b, f(b))$$

and say that the curve in the Euclidean plane

$$S = \{P = (x, y) \in E^2 \mid x \in [a, b] \text{ \& } y = f(x)\}$$

is the (simple) *arc* joining the points A, B (or the arc of the curve).



Let further $\{P_0, \dots, P_n\}$, $n \in \mathbb{N}$, be a finite, ordered set of points on the arc S such that

$$P_i = (x_i, f(x_i)), \quad i = 0, \dots, n,$$

where

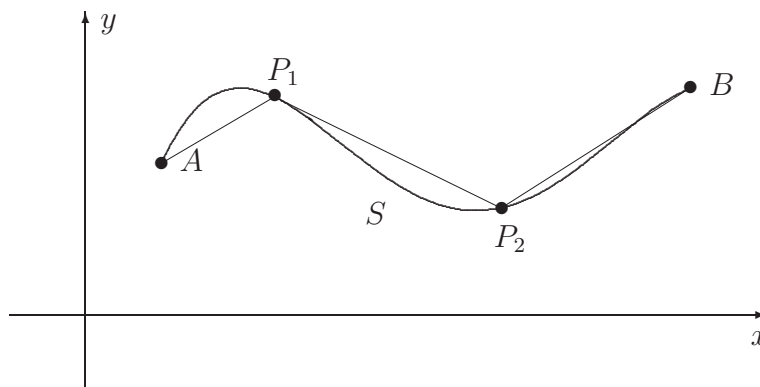
$$a = x_0 < x_1 < \dots < x_n = b,$$

so that $P_0 = A$, $P_n = B$. We denote such a set of points symbolically by $\mathcal{P}$,

$$\mathcal{P} = \{P_0, \dots, P_n\} \subset S,$$

and the set of all such sets of points by $\mathcal{M}$. The set $\mathcal{M}$ is uncountable, since already the point P_1 has an uncountable number of alternatives.

The polygonal line passing through the set of points $\mathcal{P} \in \mathcal{M}$, with endpoints A and B , will be called the *shortcut path* of the arc S through the point set $\mathcal{P}$.



Every shortcut path has a (geometric) *length*, denoted by the symbol $s_{\mathcal{P}}$ and defined by

$$s_{\mathcal{P}} = \sum_{i=1}^n \sqrt{(x_i - x_{i-1})^2 + [f(x_i) - f(x_{i-1})]^2}.$$

We say that the arc S is *rectifiable* if the set

$$\mathcal{S} = \{s_{\mathcal{P}} \mid \mathcal{P} \in \mathcal{M}\}$$

is (as a set of real numbers) bounded. In this case an *arc length* can be defined for the arc S . Namely, since a bounded set has a least upper bound, that is, a supremum (cf. Section I.11), the definition that feels natural is

$$\text{length of } S = s = \sup \mathcal{S}.$$

The continuity of f alone does not imply that the arc S is rectifiable (even though counterexamples are not quite easy to find). In what follows we therefore make a somewhat stronger assumption about the function f , which guarantees the existence of arc length. The assumption can be weakened to concern subintervals of the interval $[a, b]$ (see the footnote below), so for ordinary functions the additional assumption is fairly innocuous.

THEOREM V.4.1 If f is continuous and, in addition, monotonic on the interval $[a, b]$, then the arc $S = \{P = (x, y) \in E^2 \mid x \in [a, b] \text{ \& } y = f(x)\}$ is rectifiable, and the arc length s satisfies $b - a \leq s \leq b - a + |f(b) - f(a)|$.

Proof Since f is monotonic, in the expression for $s_{\mathcal{P}}$ the numbers $f(x_i) - f(x_{i-1})$ are of the same sign (or zero), whereupon, first applying the inequality $\sqrt{x^2 + y^2} \leq |x| + |y|$, it follows that

$$\begin{aligned} s_{\mathcal{P}} &\leq \sum_{i=1}^n [(x_i - x_{i-1}) + |f(x_i) - f(x_{i-1})|] \\ &= \sum_{i=1}^n (x_i - x_{i-1}) + \left| \sum_{i=1}^n [f(x_i) - f(x_{i-1})] \right| = b - a + |f(b) - f(a)|. \end{aligned}$$

We also note that $s_{\mathcal{P}} \geq b - a \ \forall \ \mathcal{P}$, since the smallest possible $s_{\mathcal{P}}$ (for each $\mathcal{P}$) is obtained when f is constant. Under the assumptions made, we therefore have

$$b - a \leq s_{\mathcal{P}} \leq b - a + |f(b) - f(a)| \quad \forall \ s_{\mathcal{P}} \in \mathcal{S}.$$

The set $\mathcal{S}$ has thus been given a lower bound and an upper bound, so the number $s = \sup \mathcal{S}$ exists and the bounds obtained hold also for this number. $\square$

Unless S itself happens to be a polygonal line (in which case $s \in \mathcal{S}$), the arc length must in practice be determined by constructing a sequence $\mathcal{P}_n \in \mathcal{M}$, $n = 1, 2, \dots$ such that $s_{\mathcal{P}_n} = s_n \rightarrow s$ as $n \rightarrow \infty$. For example, if we choose $\mathcal{P}_n = \{a + k(b - a)/n, k = 0 \dots n\}$, then the algorithm does work in ordinary cases (cf. Exercise 1).*

If instead of the whole interval $[a, b]$ we consider the interval $[a, x]$, $a < x \leq b$, and define

$$S_x = \{P = (t, f(t)) \mid a \leq t \leq x\},$$

then the arc length $s(x)$ of S_x becomes a function defined on the interval $[a, b]$, once we further set $s(a) = 0$. Since $s(x_2) - s(x_1)$ then denotes the arc length on the interval $[x_1, x_2]$ (see the footnote), by Theorem V.4.1

$$s(x_2) - s(x_1) \geq x_2 - x_1, \quad a \leq x_1 < x_2 \leq b,$$

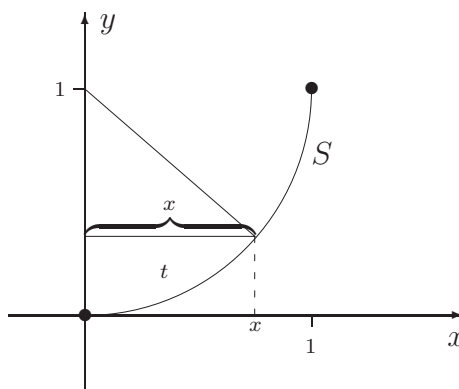
*We return to the computation of arc length later in another context. There it is also shown that arc length is *additive* with respect to the interval under consideration, i.e. if $a < c < b$, then $s = s_1 + s_2$, where s , s_1 and s_2 are respectively the arc lengths on the intervals $[a, b]$, $[a, c]$ and $[c, b]$. By additivity and Theorem V.4.1, for the rectifiability of the arc of the curve $S: y = f(x)$ on the interval $[a, b]$ it suffices, in addition to the assumption of continuity, that the interval $[a, b]$ can be divided into finitely many subintervals on which f is monotonic.

so $s(x)$ is strictly increasing on the interval $[a, b]$. Moreover, it is seen from the upper-bound estimate of Theorem V.4.1 that $s(x)$ is continuous on the interval $[a, b]$.

We now turn to consider a special case important for what follows, in which $[a, b] = [0, 1]$ and the function f is defined by

$$f(x) = 1 - \sqrt{1 - x^2}, \quad x \in [0, 1].$$

The assumptions of Theorem V.4.1 are then satisfied, and S is the arc of the circle of radius 1 tangent to the x -axis at the origin.



Writing $t =$ the arc length on the interval $[0, x]$, then (see the figure)

$$x = \sin t.$$

When this relation between x and t is assumed on the interval $x \in [0, 1] \Leftrightarrow t \in [0, \frac{\pi}{2}]$, then on that interval we can write

$$f(x) = 1 - \sqrt{1 - x^2} = 1 - \cos t$$

and estimate, by Theorem V.4.1,

$$\begin{cases} t \geq x, \\ t \leq x + f(x) \end{cases} \Leftrightarrow \begin{cases} t \geq \sin t, \\ t \leq \sin t + 1 - \cos t \end{cases}$$

that is,

$$t - (1 - \cos t) \leq \sin t \leq t, \quad t \in [0, \frac{\pi}{2}].$$

Here we have referred to the notation agreed upon earlier (Section II.1) (the definition of π !)

$$\text{length of } S = \frac{\pi}{2}.$$

When on the left-hand side of the inequality obtained we use the trigonometric formula $1 - \cos t = 2 \sin^2 \frac{t}{2}$ (see Section II.5) and apply the latter part of the inequality, it follows that

$$t - (1 - \cos t) = t - 2 \sin^2 \frac{t}{2} \geq t - 2 \left(\frac{t}{2}\right)^2 = t - \frac{1}{2}t^2, \quad t \in [0, \pi].$$

Combining the inequalities, we have arrived at the conclusion

$$\begin{aligned} t - \frac{1}{2}t^2 &\leq \sin t \leq t, \quad t \in [0, \frac{\pi}{2}] \\ \Rightarrow \quad 1 - \frac{1}{2}t &\leq \frac{\sin t}{t} \leq 1, \quad t \in (0, \frac{\pi}{2}] \\ \Rightarrow \quad 1 - \frac{1}{2}|t| &\leq \frac{\sin t}{t} \leq 1, \quad 0 < |t| \leq \frac{\pi}{2}. \end{aligned}$$

Here the last step was based on the evenness of the function $\sin t/t$.

If now $t_n \in [-\frac{\pi}{2}, \frac{\pi}{2}]$, $t_n \neq 0$ and $t_n \rightarrow 0$, then by the last inequality written

$$1 \geq \frac{\sin t_n}{t_n} \geq 1 - \frac{1}{2}|t_n| \rightarrow 1.$$

By the definition of the limit of a function (Definition V.2.1), we have derived the limit result

$$\boxed{\lim_{t \rightarrow 0} \frac{\sin t}{t} = 1.} \quad (1)$$

Applying this result, the formula $1 - \cos t = 2 \sin^2 \frac{t}{2}$ and the rules for combining limits (Theorem V.2.4), it follows that

$$\boxed{\lim_{t \rightarrow 0} \frac{1 - \cos t}{t^2} = \frac{1}{2}.} \quad (2)$$

In the above we essentially also derived the following inequalities, valid for all $t \in \mathbb{R}$:

$$\boxed{|\sin t| \leq |t|, \quad 0 \leq 1 - \cos t \leq \frac{1}{2}t^2, \quad t \in \mathbb{R}.}$$

When the limit results (1), (2) are combined with the trigonometric addition formulas

$$\begin{aligned} \sin(x + t) &= \sin x \cos t + \cos x \sin t, \\ \cos(x + t) &= \cos x \cos t - \sin x \sin t \end{aligned}$$

(see Section II.5) we obtain the more general limit results (Exercise 3)

$$\begin{aligned} \lim_{t \rightarrow 0} \frac{\sin(x+t) - \sin x}{t} &= \cos x, \\ \lim_{t \rightarrow 0} \frac{\cos(x+t) - \cos x}{t} &= -\sin x. \end{aligned} \tag{3}$$

The derivatives of the trigonometric functions

By the limit results (3) the differentiation rules for the functions $\sin$ and $\cos$ are

$$\begin{aligned} D \sin x &= \cos x, & x \in \mathbb{R}, \\ D \cos x &= -\sin x, & x \in \mathbb{R}. \end{aligned}$$

By these rules and the rules of Section V.3

$$D \left(\frac{\sin x}{\cos x} \right) = 1 + \frac{\sin^2 x}{\cos^2 x}.$$

This and the corresponding differentiation rule for the $\cot$ function are usually given in the form

$$\begin{aligned} D \tan x &= 1 + \tan^2 x = \frac{1}{\cos^2 x}, \\ D \cot x &= -(1 + \cot^2 x) = -\frac{1}{\sin^2 x}. \end{aligned}$$

The derivatives of the inverse trigonometric functions can be computed by rule (5) of the preceding section. E.g. if $y \in (-1, 1)$ and $\operatorname{Arcsin} y = x \in (-\frac{\pi}{2}, \frac{\pi}{2})$, then that rule together with the trigonometric formulas gives

$$\frac{d}{dy} \operatorname{Arcsin} y = \frac{1}{D \sin x} = \frac{1}{\cos x} = \frac{1}{\sqrt{1 - \sin^2 x}} = \frac{1}{\sqrt{1 - y^2}}.$$

When Arccos and Arctan are differentiated in a corresponding way, the result is (Exercise 6):

$$\begin{aligned} D \operatorname{Arcsin} x &= \frac{1}{\sqrt{1 - x^2}}, & x \in (-1, 1), \\ D \operatorname{Arccos} x &= -\frac{1}{\sqrt{1 - x^2}}, & x \in (-1, 1), \\ D \operatorname{Arctan} x &= \frac{1}{1 + x^2}, & x \in \mathbb{R}. \end{aligned}$$

By further differentiating the functions given by the above differentiation rules, one sees that all the functions mentioned in the rules are arbitrarily many times differentiable on their entire domains, if these consist of open intervals, and otherwise on all open subintervals of the domain. The same holds also for more general trigonometric functions obtained by combining the basic functions through arithmetic operations or as composite functions.

EXAMPLE 1 By the functions $\sin$ and $\cos$ and the differentiation rules of the preceding section

$$D \frac{\sin x}{1 + \cos x} = \frac{\cos x}{1 + \cos x} + \frac{\sin^2 x}{(1 + \cos x)^2} = \frac{\cos x + \cos^2 x + \sin^2 x}{(1 + \cos x)^2} = \frac{1}{1 + \cos x}.$$

The computation is valid for every $x \in \mathbb{R}$ with $\cos x \neq -1$, that is, on the entire domain of the function being differentiated. $\square$

EXAMPLE 2 The functions $u(x) = \cos x$ and $v(x) = \sin x$ are (at least one) solution to the problem

$$\begin{cases} u' = -v, & v' = u, & x \in \mathbb{R} \\ u(0) = 1, & v(0) = 0 \end{cases}$$

(The problem is solvable in another way as well, see Exercise [V.3:12b](#).) $\square$

EXERCISES

1. Investigate how accurately the length of the arc

$$S = \{P = (x, y) \mid y = x^2 \text{ \& } x \in [0, 1]\}$$

can be computed using the point set $\{(x_i, f(x_i)), i = 0 \dots 10\}$, where $x_i = i/10$. The exact value is $s = 1.47894..$

2. Starting from the origin, one moves along the curve $r = 4 \cos \varphi$ (polar coordinates) to the point whose x -coordinate = 3. What is the length of the journey by the shortest route?
3. Derive the limit results [\(3\)](#).
4. Show that if $a, b \in \mathbb{R}$ and $b \neq 0$, then $\lim_{x \rightarrow 0} \frac{\sin ax}{\sin bx} = \frac{a}{b}$.

5. Determine the limits

$$\begin{array}{lll} \text{a) } \lim_{x \rightarrow 0} \frac{x \sin x}{1 - \cos x} & \text{b) } \lim_{x \rightarrow 0} \frac{x \sin 2x}{1 - \cos 2x} & \text{c) } \lim_{x \rightarrow 0^+} \frac{x - \sqrt{x}}{\sqrt{\sin x}} \\ \text{d) } \lim_{x \rightarrow \frac{\pi}{2}} (1 + \cos 2x) \tan^2 x & \text{e) } \lim_{x \rightarrow \infty} x \sin \frac{1}{x} & \end{array}$$

6. Derive the differentiation rules for the functions Arccos and Arctan.

7. Differentiate and simplify the result:

$$\begin{array}{lll} \text{a) } \frac{1 - \cos x}{\sin x} & \text{b) } \frac{1 + \cos x}{1 - \cos x} & \text{c) } \frac{\sin x - \cos x}{\sin x + \cos x} \\ \text{d) } \operatorname{Arcsin}(\cos x) & \text{e) } \operatorname{Arccos}(\sin x) & \text{f) } \operatorname{Arctan}(\cot x) \end{array}$$

8. Determine the derivative of the implicit function $y(x)$ at the given point:

$$\begin{array}{ll} \text{a) } y + 2 \sin y + \cos y = x, & (x, y) = (1, 0) \\ \text{b) } 2x + y - \sqrt{2} \sin(xy) = \frac{\pi}{2}, & (x, y) = \left(\frac{\pi}{4}, 1\right) \\ \text{c) } x \sin(xy - y^2) = x^2 - 1, & (x, y) = (1, 1) \\ \text{d) } \tan(xy^2) = \frac{2xy}{\pi}, & (x, y) = \left(-\pi, \frac{1}{2}\right) \end{array}$$

9. Show by induction that $D^n \tan x = p(\tan x)$, where p is a polynomial of degree $n + 1$ and of the form $p(t) = n! t^{n+1} + c_{n-1} t^{n-1} + c_{n-3} t^{n-3} + \dots$

10. Verify that the function

$$f(x) = \begin{cases} x^2 \sin \frac{1}{x^2}, & \text{when } x \neq 0, \\ 0, & \text{when } x = 0 \end{cases}$$

is, for instance, a function that is differentiable at every point $x \in \mathbb{R}$ but whose derivative is not continuous at every point. Sketch the graphs of f and f' !

11. (*) Let s_n be the arc length of the curve $S : y = x^n$ on the interval $[0, 1]$, where $n \in \mathbb{N}$. Show that $\lim_n s_n = 2$.

12. (*) Show, by a suitable change of variable:

$$\text{a) } \lim_{x \rightarrow \infty} x \left(\frac{\pi}{2} - \operatorname{Arctan} x \right) = 1 \quad \text{b) } \lim_{x \rightarrow 1^-} \frac{\pi - 2 \operatorname{Arcsin} x}{\sqrt{1 - x}} = 2\sqrt{2}$$

V.5 Extrema. Smoothness

The derivative is a most excellent tool when one wishes to characterize ordinary real functions, which are most often differentiable 'almost everywhere'. In this and the next section we examine the use of the derivative in the study of functions.

Local extrema

We begin with a definition.

DEFINITION V.5.1 A function $f : \mathcal{D}_f \rightarrow \mathbb{R}$, $\mathcal{D}_f \subset \mathbb{R}$, has at a point $c \in \mathcal{D}_f$ a *local* or *relative extremum*, i.e. c is a *local extremum point* of f , if for some $\delta > 0$ and $\forall x \in \mathbb{R}$ one has $(c - \delta, c + \delta) \subset \mathcal{D}_f$ and

$$\begin{aligned} \text{either: } & 0 < |x - c| < \delta \Rightarrow f(x) \leq f(c), \\ \text{or: } & 0 < |x - c| < \delta \Rightarrow f(x) \geq f(c). \end{aligned}$$

In the former case it is a *local maximum*, in the latter a *local minimum*. If in addition, for some $\delta > 0$, $f(x) \neq f(c)$ whenever $0 < |x - c| < \delta$, then the extremum (extremum point) is *essential*, otherwise *inessential*.

EXAMPLE 1 The function

$$f(x) = \begin{cases} \sin^2 \frac{1}{x}, & \text{if } x \neq 0, \\ 0, & \text{if } x = 0 \end{cases}$$

attains its smallest value $f_{\min} = 0$ at the point $x = 0$ and at the points

$$a_n = \frac{1}{n\pi}, \quad n \in \mathbb{Z}, \quad n \neq 0.$$

The function attains its largest value $f_{\max} = 1$ at the points

$$b_n = \frac{1}{(n + \frac{1}{2})\pi}, \quad n \in \mathbb{Z}.$$

The points a_n and b_n are also essential local minimum and maximum points in the sense of Definition V.5.1, since by the definition of f and the known properties of the sin function, $0 < f(x) < 1$ at all points other than those mentioned or at the point $x = 0$. Then if, for example, $c = b_n$, $n \in \mathbb{Z}$, the local-maximum condition of Definition V.5.1 holds when δ is chosen so that $0 < \delta < |b_{n\pm 1} - b_n|$. At the point $x = 0$, f has an inessential local minimum in the sense of Definition V.5.1, since for any $\delta > 0$ we have $a_n \in (0, \delta)$ when $n > (\pi\delta)^{-1}$, so that the interval $(0, \delta)$ always contains points where $f(x) = f(0)$. $\square$

According to the following theorem, a local extremum point of a differentiable function is necessarily also a zero of the derivative. In this connection the zeros of the derivative are indeed called the *critical points* of the function.

THEOREM V.5.2 If $c \in \mathcal{D}_f$ is a local extremum point of f and f is differentiable at c , then $f'(c) = 0$.

Proof The logically equivalent form of the claim is: If f is differentiable at c and $f'(c) \neq 0$, then c is not a local extremum point of f . We prove the claim in this form, i.e. we assume that $f'(c) = k \neq 0$. By the definition of the derivative (cf. Section V.3), f is then defined on an interval $(c - \delta, c + \delta)$ for some $\delta > 0$ and on that interval one has

$$f(x) = f(c) + k(x - c) + g(x),$$

where $\lim_{x \rightarrow c} g(x)/(x - c) = 0$. Then, since $k \neq 0$, by Theorem V.2.2, for some $\delta > 0$ (corresponding to the choice $\varepsilon = |k|/2 > 0$ in that theorem) one also has

$$|g(x)/(x - c)| \leq \frac{1}{2}|k| \Rightarrow |g(x)| \leq \frac{1}{2}|k||x - c|, \quad \text{when } 0 < |x - c| < \delta.$$

Consequently, if $k > 0$, then by the triangle inequality

$$\begin{cases} f(x) - f(c) \geq k(x - c) - \frac{k}{2}|x - c| = \frac{k}{2}(x - c) > 0 & \forall x \in (c, c + \delta), \\ f(x) - f(c) \leq k(x - c) + \frac{k}{2}|x - c| = \frac{k}{2}(x - c) < 0 & \forall x \in (c - \delta, c). \end{cases}$$

If $k < 0$, one concludes in the same way that $f(x) - f(c) < 0$ when $x \in (c, c + \delta)$ and $f(x) - f(c) > 0$ when $x \in (c - \delta, c)$. By Definition V.5.1, in neither case is c a local extremum point of f . $\square$

EXAMPLE 1 (continued) For the function of the example we have

$$f'(x) = -\frac{2}{x^2} \sin \frac{1}{x} \cos \frac{1}{x}, \quad \text{when } x \neq 0,$$

so that $f'(a_n) = f'(b_n) = 0$ in accordance with the claim of Theorem V.5.2. At the point $x = 0$, f is not differentiable — this too is consistent with Theorem V.5.2. $\square$

EXAMPLE 2 The rational function $f(x) = x^2/(x+1)$ is differentiable throughout its domain, so the possible local extremum points of f are necessarily critical points. Since

$$f'(x) = \frac{2x}{1+x} - \frac{x^2}{(1+x)^2} = \frac{x^2 + 2x}{(x+1)^2}, \quad x \neq -1,$$

the critical points are $x = -2$ and $x = 0$. It turns out that at the former point f has a local maximum, at the latter a local minimum. (See Example 3 in the next section.) $\square$

Extremum problems

The question of the local extremum points of a function usually arises as a sub-problem when one wishes to solve an *extremum problem*: given a function f and a set $A \subset \mathcal{D}_f$, one wants to determine the largest and/or smallest value of f on A . Insofar as the problem is solvable, i.e. the largest/smallest value can be found, that value is called the *absolute* maximum/minimum of f on A . The absolute maximum and minimum always exist if the set A is finite, since then it is merely a matter of choosing among a finite set. If A is an infinite set, e.g. an interval, the situation is more problematic. Here we restrict ourselves to a basic problem that occurs often in applications, in which A is a closed interval and f is continuous on that interval in the sense of Definition V.1.5. In this case the solvability of the extremum problem for both extrema is guaranteed by Weierstrass's theorem (Theorem V.1.12).

Solving an extremum problem for a continuous function f and a closed interval $[a, b]$ becomes considerably easier in practice if f is not only continuous but also differentiable or 'almost differentiable' on the interval (a, b) . The result can be formulated as a corollary of Theorem V.5.2 as follows.

COROLLARY V.5.3 Let f be continuous on the interval $[a, b]$ and let $X \subset [a, b]$ be a set for which

1. $a \in X$ and $b \in X$,
2. $\{c \in (a, b) \mid f \text{ differentiable at } c \text{ and } f'(c) = 0\} \subset X$,
3. $\{c \in (a, b) \mid f \text{ not differentiable at } c\} \subset X$.

Then $\max_{x \in [a, b]} f(x) = \max_{x \in X} f(x)$ and $\min_{x \in [a, b]} f(x) = \min_{x \in X} f(x)$.

A set X satisfying the conditions of Corollary V.5.3 thus contains the endpoints of the interval $[a, b]$ and, in addition, all the critical points of f in the interval (a, b) as well as the points where f is not differentiable. Usually the set X can be chosen so that it is finite, in which case the search for the maximum and minimum values of f reduces, by the corollary, to a choice within the finite set X . — Note that the set X (insofar as it is finite) may, by its definition, also include points at which the differentiability of f is merely in doubt. If one only wishes to determine the maximum and minimum points and values of the function on $[a, b]$, no closer examination of the 'suspects' is needed. One may also leave open the question of whether the points to be included in X are local extremum points of f or not.

EXAMPLE 3 Determine the maximum and minimum points and values of the function $f(x) = \max\{2 - x, 6x - 3x^2\}$ on the interval $[0, 3]$.

Solution This is a continuous and piecewise polynomial-valued function. Since $2 - x = 6x - 3x^2 \Leftrightarrow 3x^2 - 7x + 2 = 0 \Leftrightarrow x = \frac{1}{3} \vee x = 2$, we conclude that

$$f(x) = \begin{cases} 6x - 3x^2, & \text{if } \frac{1}{3} \leq x \leq 2, \\ 2 - x, & \text{if } x \leq \frac{1}{3} \text{ or } x \geq 2. \end{cases}$$

On the intervals $(0, \frac{1}{3})$ and $(2, 3)$ we have $f'(x) = -1 \neq 0$. On the interval $(\frac{1}{3}, 2)$ we have $f'(x) = 6 - 6x$, so on this interval f has a critical point $c = 1$. At the points $x = \frac{1}{3}$ and $x = 2$, f is possibly (and in reality also) not differentiable, while at the other points of the interval $(0, 3)$ it is, so in Corollary V.5.3 we may choose $X = \{0, \frac{1}{3}, 1, 2, 3\}$. Computing the values of f at these points, we find that $f_{\max} = f(1) = 3$, $f_{\min} = f(3) = -1$. $\square$

In the example, the possible local extremum points in the interval $(0, 3)$ are, by Theorem V.5.2, the points $\frac{1}{3}, 1, 2$. A closer examination of the function in the neighborhood of these points (by means of algebra or by the methods presented later) shows that $c = \frac{1}{3}$ is an essential local minimum, $c = 1$ is an essential local maximum, and $c = 2$ is not a local extremum point.

One-sided derivatives

If in Example 3 we look at the points where f is not differentiable, we see that at those points too f has *one-sided* derivatives in the sense of the following definition (cf. one-sided limits in Section V.2).

DEFINITION V.5.4 A function $f : \mathcal{D}_f \rightarrow \mathbb{R}$, $\mathcal{D}_f \subset \mathbb{R}$, is at a point $x \in \mathcal{D}_f$ *differentiable from the left* if $(x - \delta, x] \subset \mathcal{D}_f$ for some $\delta > 0$ and the limit

$$D_-f(x) = \lim_{\Delta x \rightarrow 0^-} \frac{f(x + \Delta x) - f(x)}{\Delta x}$$

exists, and *differentiable from the right* if $[x, x + \delta) \subset \mathcal{D}_f$ for some $\delta > 0$ and the limit

$$D_+f(x) = \lim_{\Delta x \rightarrow 0^+} \frac{f(x + \Delta x) - f(x)}{\Delta x}$$

exists.

By Definitions V.5.4 and V.3.1, f is differentiable at a point x exactly when f is differentiable at x both from the left and from the right and $D_-f(x) = D_+f(x)$ ($= f'(x)$).

If f is continuous at a point c and, in addition, differentiable both from the left and from the right, then (cf. the introduction to the derivative in Section V.3)

$$\begin{cases} f(x) = f(c) + D_-f(c)(x-c) + g(x), & \text{when } x \in (c-\delta, c], \\ f(x) = f(c) + D_+f(c)(x-c) + g(x), & \text{when } x \in [c, c+\delta), \end{cases}$$

where $g(x)/(x-c) \rightarrow 0$ as $x \rightarrow c^-$ or $x \rightarrow c^+$. From this we see the correctness of

THEOREM V.5.5 If f is continuous at a point c and, in addition, both differentiable from the left and from the right, then

(a) $D_-f(c) D_+f(c) < 0 \Rightarrow f$ has at the point c a local extremum, which is

- an essential maximum, if $D_-f(c) > 0$ and $D_+f(c) < 0$,
- an essential minimum, if $D_-f(c) < 0$ and $D_+f(c) > 0$.

(b) $D_-f(c) D_+f(c) > 0 \Rightarrow c$ is not a local extremum point of f .

EXAMPLE 3 (continued) In the example $D_-f(\frac{1}{3}) = -1$, $D_+f(\frac{1}{3}) = 4$ and $D_-f(2) = -6$, $D_+f(2) = -1$, so by Theorem V.5.5 f has at the point $x = \frac{1}{3}$ an essential local minimum, and $x = 2$ is not a local extremum point. $\square$

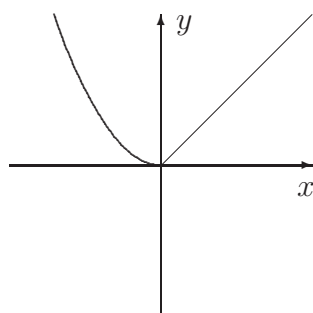
If f is differentiable at a point c and $f'(c) \neq 0$, then $D_-f(c) D_+f(c) = [f'(c)]^2 > 0$, so claim (b) of Theorem V.5.5 also contains the claim of Theorem V.5.2 in the form

$$f'(c) \neq 0 \Rightarrow f \text{ has no local extremum at } c.$$

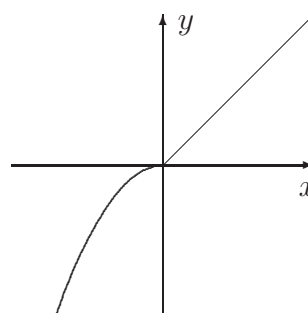
If $D_-f(c) = 0$ or $D_+f(c) = 0$ (also when $f'(c) = 0$), then c may or may not be a local extremum point, depending on the case.

EXAMPLE 4 For the functions $f(x) = \begin{cases} x^2, & x < 0, \\ x, & x \geq 0 \end{cases}$ and $g(x) = \begin{cases} -x^2, & x < 0, \\ x, & x \geq 0 \end{cases}$

we have $D_-f(0) = D_-g(0) = 0$ and $D_+f(0) = D_+g(0) = 1$. The function f has at the origin an (essential) local minimum, whereas g has no local extremum at the origin.



$$y = f(x)$$



$$y = g(x)$$

Smoothness

The following definition combines the concepts of continuity on a closed interval (Definition V.1.5) and differentiability.

DEFINITION V.5.6 A function $f : \mathcal{D}_f \rightarrow \mathbb{R}$, $\mathcal{D}_f \subset \mathbb{R}$ is *continuously differentiable* on a *closed interval* $[a, b] \subset \mathcal{D}_f$ if f is differentiable on (a, b) , differentiable from the right at the point a , differentiable from the left at the point b , and the derivative f' , defined as a one-sided derivative at the endpoints of $[a, b]$, is a continuous function on $[a, b]$. More generally, if $m \in \mathbb{N}$, then f is *m times continuously differentiable* on $[a, b]$ if f is m times differentiable on (a, b) , m times differentiable from the right at the point a , m times differentiable from the left at the point b , and the derivatives $f^{(k)}$, defined as one-sided derivatives at the endpoints of the interval, are continuous on $[a, b]$ for $k = 1 \dots m$. Furthermore, if $A \subset \mathcal{D}_f$ is a half-open or open interval, then f is said to be *m times continuously differentiable on A* if f is m times continuously differentiable on every closed subinterval of A .

When the definition is applied in the case of a closed interval with $m \geq 2$, the derivatives $f^{(k)}$, $k = 1 \dots m - 1$ are thought of as defined at the endpoints of the interval recursively (as one-sided derivatives), starting from the index $k = 1$. The term '0 times continuously differentiable' corresponding to the index $m = 0$ is interpreted as mere continuity.

EXAMPLE 5 Let $n \in \mathbb{N}$ and consider the function

$$f(x) = \begin{cases} 0, & \text{if } x \leq 0, \\ x^n, & \text{if } x > 0. \end{cases}$$

By differentiating one-sidedly at the point $x = 0$, we find that f is at this point (and hence on all of $\mathbb{R}$) $n - 1$ times differentiable, and

$$f^{(k)}(x) = \begin{cases} 0, & \text{if } x \leq 0, \\ \frac{n!}{(n-k)!} x^{n-k}, & \text{if } x > 0, \quad k = 1 \dots n-1. \end{cases}$$

Since the derivatives $f^{(k)}(0) = 0$, $k = 1 \dots n-1$ are continuous on $\mathbb{R}$ and since $f^{(n-1)}$ is not differentiable at the point $x = 0$, we conclude that f is on $\mathbb{R}$ exactly $n - 1$ times (not n times) continuously differentiable. $\square$

If f is m times but not $m + 1$ times continuously differentiable on an interval A , then the index m is often called the *degree of smoothness* of f , and one may also say that f is *smooth of degree m* on that interval. Thus 'smooth of degree zero' means mere continuity. In the example the degree of smoothness of f (on $\mathbb{R}$) is $m = n - 1$. If the degree of smoothness has no upper bound, i.e. f is 'infinitely smooth', then the degree of smoothness may be denoted $m = \infty$. The term *smooth* (without further qualification) often has precisely this meaning. In the example $m = \infty$, e.g. on the interval $[0, \infty)$.

EXAMPLE 6 A polynomial is smooth (= 'infinitely smooth') on every interval, i.e. on $\mathbb{R}$, and so are the trigonometric functions $\sin$, $\cos$ and Arctan . The trigonometric function $\tan$ is smooth on the intervals $((n - \frac{1}{2})\pi, (n + \frac{1}{2})\pi)$, $n \in \mathbb{Z}$. $\square$

EXAMPLE 7 A function defined as the sum of a power series is smooth on the interval $(-\rho, \rho)$ (i.e. on intervals $[a, b] \subset (-\rho, \rho)$), where ρ = the radius of convergence of the series, cf. Section V.3. $\square$

The function f defined above in Example 5 is *piecewise smooth* as measured by any index $m \in \mathbb{N}$. This means that, whatever $m \in \mathbb{N}$ may be, f satisfies the conditions of the following definition.

DEFINITION V.5.7 A function $f : \mathcal{D}_f \rightarrow \mathbb{R}$, $\mathcal{D}_f \subset \mathbb{R}$ is *m times piecewise continuously differentiable* on the interval $[a, b]$ if there exist points c_k , $k = 0 \dots n$, $n \in \mathbb{N}$ and functions f_k , $k = 1 \dots n$ such that

- (i) $a = c_0 < c_1 < \dots < c_n = b$ and $(c_{k-1}, c_k) \subset \mathcal{D}_f$, $k = 1 \dots n$,
- (ii) $[c_{k-1}, c_k] \subset \mathcal{D}_{f_k}$ and f_k is m times continuously differentiable on the interval $[c_{k-1}, c_k]$, $k = 1 \dots n$,
- (iii) $f(x) = f_k(x)$, when $x \in (c_{k-1}, c_k)$, $k = 1 \dots n$.

EXAMPLE 8 If $a < 0$ and $b > 0$, then for the function of Example 5 the conditions of Definition V.5.7 hold for every $m \in \mathbb{N}$ when one sets $n = 1$, $c_1 = 0$, $f_1(x) = 0$ and $f_2(x) = x^n$. $\square$

If f is piecewise smooth of degree m on $[a, b]$ for every $m \in \mathbb{N}$, as in Example 5, then one may say simply that f is *piecewise smooth* without further qualification. The function of Example 3 is also of this type.

EXAMPLE 9 The function

$$f(x) = \begin{cases} x^3, & x < 0, \\ x^5\sqrt{x}, & x \geq 0 \end{cases}$$

is on the interval $[-1, 1]$ smooth of degree $m = 2$ (exactly) and piecewise smooth of degree $m = 5$ (exactly). Here the piecewise smoothness is limited by the fact that f is only five times differentiable from the right at the origin. $\square$

EXERCISES

- Investigate the nature of the origin as a possible local extremum point:

$$\text{a) } f(x) = 100 + x^{99} - x^{98} \quad \text{b) } f(x) = x^4 - x^3 \cos x$$

$$\text{c) } f(x) = \begin{cases} x^3 \sin \frac{1}{x}, & \text{if } x \neq 0, \\ 0, & \text{if } x = 0 \end{cases} \quad \text{d) } f(x) = \begin{cases} 0.000001, & \text{if } x = 0, \\ x \cos \frac{1}{x}, & \text{if } x \neq 0 \end{cases}$$

- Determine the maximum and minimum values of the following functions on the given interval, and the points where the maximum/minimum is

attained:

a) $f(x) = (x-1)^3(x+1)^3$, interval $[-2, 4]$

b) $f(x) = |x^5 - 80x + 1|$, interval $[-1, 1]$

c) $f(x) = \frac{2-x}{5-4x+x^2}$, interval $[-100, 2]$

d) $f(x) = \begin{cases} 8x^2 - 16x - 14, & \text{if } x \leq 2, \\ 3x^2 - 24x + 22, & \text{if } x > 2, \end{cases}$ interval $[0, 5]$

e) $f(x) = \min\{2x^3 - 2x, 3 - 2x\}$, interval $[-1, 2]$

f) $f(x) = |\sin x| - \cos x$, interval $[0, 2\pi]$

g) $f(x) = x + 5 \sin x$, interval $[0, 2\pi]$

3. What is the smallest value of the expression $\sqrt{x} + 2\sqrt{y}$ when $x, y \geq 0$ and $x + y = 5/6$?
4. Beside a straight road, perpendicular to the road, there is a billboard whose width is 10 meters and whose nearest distance from the roadway of those traveling on the road is 20 m. What is the largest angle at which someone moving along the road sees the board?
5. The point $P = (x, y)$ lies on the folium of Descartes $S : x^3 + y^3 = 3xy$. What is the largest possible value of x if $y \geq 0$?
6. Determine
 - a) the maximum and minimum values of the function $f(x, y) = (2x - y)(x + y - 1)(x - y - 1)$ on the line segment with endpoints $(-1, 0)$ and $(3, 1)$,
 - b) the smallest value of the function $f(x, y) = |x^2 + 2y| + 2x^2$ on the curve $S : xy = 1$,
 - c) the smallest value of the function $f(x, y, z) = x + y + z(2x + y + z)(x + 3y + z)(x + y + 4z)$ on the line $S : x = 2y = 3z$.
7. A function f is defined on the quarter of the unit disk $A = \{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 \leq 1 \text{ \& } x \geq 0 \text{ \& } y \geq 0\}$ as follows (polar coordinates!): $f(r, \phi) = (3r^2 - 2r)(2\phi^2 - 3\phi + 1)$. At which points of A does f attain its largest and at which its smallest value?

8. By passing to polar or spherical coordinates, determine the maximum and minimum values of the following functions on the given set A . Also give, in Cartesian coordinates, the points where these values are attained.

$$\begin{aligned} \text{a) } f(x, y) &= x^3 y^4, \quad A = \{(x, y) \mid x^2 + y^2 \leq 1\} \\ \text{b) } f(x, y) &= x^2 y^4, \quad A = \{(x, y) \mid x^2 + y^2 \leq 25\} \\ \text{c) } f(x, y, z) &= xyz^3, \quad A = \{(x, y, z) \mid x^2 + y^2 + z^2 \leq 1\} \end{aligned}$$

9. What is the degree of smoothness of the function

$$f(x) = |x - 3|^{11/11} + (x + 1)^3 |x + 1| + \max\{1, x^2 - 2x + 2\}$$

on the interval a) $[-2, 0]$, b) $[0, 2]$, c) $[2, 4]$?

10. Determine the degrees of smoothness of the following functions on the interval $[-1, 1]$:

$$\begin{aligned} \text{a) } x^2 |\sin x| \quad \text{b) } |x|^3 (1 - \cos x) \quad \text{c) } \sin^2 x |\tan x| \\ \text{d) } f(x) = \begin{cases} 1 - \frac{1}{2}x^2, & \text{if } x < 0, \\ \cos x, & \text{if } x \geq 0 \end{cases} \quad \text{e) } f(x) = \begin{cases} \sin x, & \text{if } x \leq 0, \\ x - \frac{1}{6}x^3, & \text{if } x > 0 \end{cases} \end{aligned}$$

11. Let $f_1(x) = 2x^3 - 7x^2 + 9x$ and $f_2(x) = ax^2 + bx + c$. Determine the coefficients a, b, c so that the function

$$f(x) = \begin{cases} f_1(x), & \text{if } x \leq 1, \\ f_2(x), & \text{if } x > 1 \end{cases}$$

is twice continuously differentiable on the interval $[0, 2]$. In the same figure, draw the graph of f on that interval and, with a dashed line, the graph of f_2 on $[0, 1]$ and the graph of f_1 on $[1, 2]$.

12. (*) a) It is known about a function $f : \mathbb{R} \rightarrow \mathbb{R}$ that f is differentiable on $\mathbb{R}$ and that

$$\lim_{x \rightarrow -\infty} f(x) = A_-, \quad \lim_{x \rightarrow \infty} f(x) = A_+,$$

where $A_- \in \mathbb{R}$ and $A_+ \in \mathbb{R}$. Show that f attains an absolute maximum value on $\mathbb{R}$ exactly when at some critical point c of f one has $f(c) \geq \max\{A_-, A_+\}$. b) For which values of a does the function

$$f(x) = \frac{x^4 + ax^2}{(x^2 + 7)^2}$$

attain an absolute maximum value at some $x \in \mathbb{R}$?

13. (*) Determine the critical points, the discontinuities of the derivative, the local extremum points, and the absolute maximum and minimum values of the function $f(x) = 4|\cos x| - 3\sin x + 2\cos x$ on the interval $[0, 2\pi]$. Sketch the course of the curve $y = f(x)$.
14. (*) On the interval $(0, \infty)$, define a function g such that the smallest value of the function $f(x) = x + a\sin x$ on the interval $[0, a]$ is $f_{\min} = g(a)$. Sketch the graph of g .
15. (*) a) Let p_1 and p_2 be polynomials of degree $\leq n$. Show that if the function

$$f(x) = \begin{cases} p_1(x), & \text{if } x < a, \\ p_2(x), & \text{if } x \geq a \end{cases}$$

is n times continuously differentiable on the interval $[a - 1, a + 1]$, then $p_1 = p_2$.

b) Let the function f be m times continuously differentiable on the interval $[a, b]$. Show that there exist unique polynomials p_1 and p_2 of degree $\leq m$ such that the function

$$g(x) = \begin{cases} p_1(x), & \text{if } x < a, \\ f(x), & \text{if } x \in [a, b], \\ p_2(x), & \text{if } x > b \end{cases}$$

is m times continuously differentiable on $\mathbb{R}$.

16. (*) (Calf pen) A farmer wants to fence off, next to his barn, a rectangular enclosure as grazing land for calves. The dimensions of the enclosure are chosen as a function of the total length of the fence (x m) so that the area of the pasture is as large as possible and the 40-m-long wall of the barn is used as effectively as possible. In an enclosure designed in this way, let $f(x)$ be the ratio of the lengths of the side parallel to the barn wall and the side perpendicular to the wall. Determine $f(x)$ on the interval $(0, \infty)$. What is the degree of smoothness of f on the interval $[0, \infty)$ when one sets $f(0) = f(0^+)$?
17. (*) (Longest throw?) If air resistance is ignored, the trajectory of a body thrown into the air follows the projectile parabola

$$y = h + kx - (1 + k^2)\frac{x^2}{2a},$$

where x measures the horizontal distance from the launch point, y the height above ground level, and the constants h , k and a are characteristic

of the individual throw (see Section [IV.4](#), Example [6](#)). If h and a are fixed, for which value of k does the body fly the farthest — and how far? — before it hits the ground?

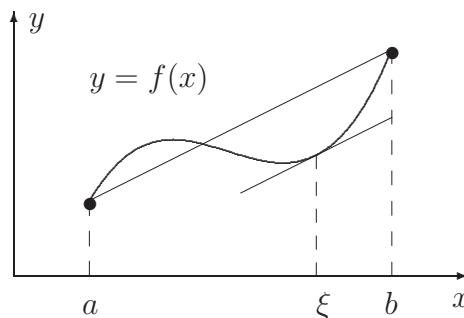
V.6 The mean value theorem of differential calculus

The following theorem is among the most notable and also very frequently used fundamental results of mathematical analysis. It is the second in the series of mean value theorems — cf. the First mean value theorem in Section V.1 (Theorem V.1.10).

THEOREM V.6.1 (Second mean value theorem – Mean value theorem of differential calculus) If $f : \mathcal{D}_f \rightarrow \mathbb{R}$, $\mathcal{D}_f \subset \mathbb{R}$, is continuous on the interval $[a, b] \subset \mathcal{D}_f$ and differentiable on the interval (a, b) , then for some $\xi \in (a, b)$

$$f(b) - f(a) = f'(\xi)(b - a).$$

The claim of the theorem is geometrically very plausible (cf. the figure), but in the proof one must nevertheless deal with the deeper logic of continuity.



To prove Theorem V.6.1, we first reduce the claim to a simpler one by defining

$$g(x) = f(x) - f(a) \cdot \frac{b-x}{b-a} - f(b) \cdot \frac{x-a}{b-a}, \quad x \in [a, b].$$

Then $g(a) = g(b) = 0$ and one has

$$f(b) - f(a) = f'(\xi)(b - a) \quad \Leftrightarrow \quad g'(\xi) = 0.$$

The claim of Theorem V.6.1 thus takes the following more reduced form (cf. the First mean value theorem and its reduced form, Bolzano's theorem, in Section V.1):

THEOREM V.6.2 (Rolle's theorem) If f is continuous on the interval $[a, b]$ and differentiable on the interval (a, b) and $f(a) = f(b) = 0$, then $f'(\xi) = 0$ for some $\xi \in (a, b)$.

Proof Since f is continuous on the interval $[a, b]$, by Weierstrass's theorem (Theorem V.1.12) f attains its minimum and maximum values on the interval $[a, b]$. Excluding the obvious case $f(x) = 0 \forall x \in [a, b]$ (in which $f'(\xi) = 0 \forall \xi \in (a, b)$), we must have

$$\begin{aligned} \text{either: } & f(\xi) = f_{\min} < 0, \quad \xi \in (a, b), \\ \text{or: } & f(\xi) = f_{\max} > 0, \quad \xi \in (a, b). \end{aligned}$$

In either case we must have $f'(\xi) = 0$ (Theorem V.5.2). Theorem V.6.2 is thus proved, and as an immediate consequence so is Theorem V.6.1. — Note that the proof was relatively straightforward only because it had a 'hard core' (Weierstrass's theorem). $\square$

EXAMPLE 1 The function $f(x) = \sqrt{|x|}$ satisfies the conditions of Theorem V.6.1 on the interval $[0, 1]$, and the asserted ξ is unique: $\xi = \frac{1}{4} \in (0, 1)$. On the interval $[-1, 1]$ the conditions of Theorem V.6.1 are not satisfied, since f is not differentiable at the point $x = 0$. Nor, in this case, does the asserted point ξ (where one would have $f'(\xi) = 0$) exist. $\square$

The mean value theorem of differential calculus has very many uses in studying the properties of differentiable (or 'almost differentiable') functions. In what follows we present the most central of these uses.

Monotonicity of a function

In simple cases one can determine, by means of algebra alone, on which intervals a given function is monotonic (increasing or decreasing, cf. Section IV.1). More generally, the task becomes considerably easier when we introduce the following claim, which can be derived from Theorem V.6.1.

THEOREM V.6.3 Let f be continuous on the interval $[a, b]$ and let $X \subset (a, b)$ be a finite set of points such that

- (1) f is differentiable at every point $x \in (a, b)$, $x \notin X$.
- (2) $(\star)$ holds $\forall x \in (a, b)$, $x \notin X$, where $(\star)$ is one of the following alternatives:
 - (a) $f'(x) \geq 0$, (b) $f'(x) > 0$,
 - (c) $f'(x) \leq 0$, (d) $f'(x) < 0$.

Then f is (a) increasing, (b) strictly increasing, (c) decreasing, (d) strictly decreasing on the interval $[a, b]$.

Proof Assume $x_1, x_2 \in [a, b]$, $x_1 < x_2$. Since the set X is finite, we can choose a set of points $\{t_1, \dots, t_{n-1}\} \subset X$ (possibly the empty set) such that

$$x_1 = t_0 < t_1 < \dots < t_n = x_2 \quad \text{and} \quad (t_{k-1}, t_k) \cap X = \emptyset, \quad k = 1 \dots n.$$

Then, writing $f(x_2) - f(x_1)$ as a telescoping sum and applying Theorem V.6.1, it follows that

$$f(x_2) - f(x_1) = \sum_{k=1}^n [f(t_k) - f(t_{k-1})] = \sum_{k=1}^n f'(\xi_k)(t_k - t_{k-1}), \quad \xi_k \in (t_{k-1}, t_k),$$

whereupon, by the assumptions made, we conclude

$$f(x_2) - f(x_1) = \begin{cases} \geq 0 & \text{(a),} \\ > 0 & \text{(b),} \\ \leq 0 & \text{(c),} \\ < 0 & \text{(d).} \end{cases} \quad \square$$

When applying Theorem V.6.3, the zeros of f' can always be included in the set X , insofar as they are finite in number. A finite number of zeros of f' thus does not disturb the (even strict) monotonicity of the function, as long as the sign of f' does not change at the zeros.

EXAMPLE 2 The function $f(x) = \begin{cases} x^3, & \text{if } -1 \leq x \leq 1, \\ x, & \text{if } 1 < x \leq 2 \end{cases}$

is continuous on the interval $[-1, 2]$ and differentiable on the interval $(-1, 2)$ except at the point $x = 1$. When we choose $X = \{0, 1\}$, assumption (b) of Theorem V.6.3 holds, so f is strictly increasing on the interval $[-1, 2]$. $\square$

EXAMPLE 3 By Example 2 of the previous section, the function $f(x) = x^2/(x+1)$ satisfies: $f'(x) > 0$ on the intervals $(-\infty, -2)$ and $(0, \infty)$ and $f'(x) < 0$ on the intervals $(-2, -1)$ and $(-1, 0)$. By Theorem V.6.3 we conclude that f is strictly increasing on the intervals $(-\infty, -2]$ and $[0, \infty)$ (i.e. on the closed subintervals of these intervals) and correspondingly strictly decreasing on the intervals $[-2, -1)$ and $(-1, 0]$. (The same result can be reached by algebra alone, but more laboriously: Exercise IV.1:15b.) $\square$

The nature of a critical point

With the monotonicity criteria of Theorem V.6.3 one can usually easily determine whether a critical point of a function is a local extremum point (and of what

nature) or not. Namely, the matter is settled (unless f is exceptionally 'ill-behaved') by examining the sign of the derivative in the immediate neighborhood of the critical point. For example, if $f'(x) < 0$ on the interval $(c - \delta_1, c)$ and $f'(x) > 0$ on the interval $(c, c + \delta_2)$ for some $\delta_1, \delta_2 > 0$, then f is strictly decreasing on the interval $[c - \delta_1, c]$ and strictly increasing on the interval $[c, c + \delta_2]$, so c must be a local minimum point. In the argument it suffices that f is continuous at the point c , i.e. differentiability need not be assumed (cf. Theorem V.5.5).

EXAMPLE 3 (continued) In the example, the critical points of f are -2 and 0 . From the sign of the derivative we conclude that $x = -2$ is a local maximum point of f and $x = 0$ is a local minimum point. $\square$

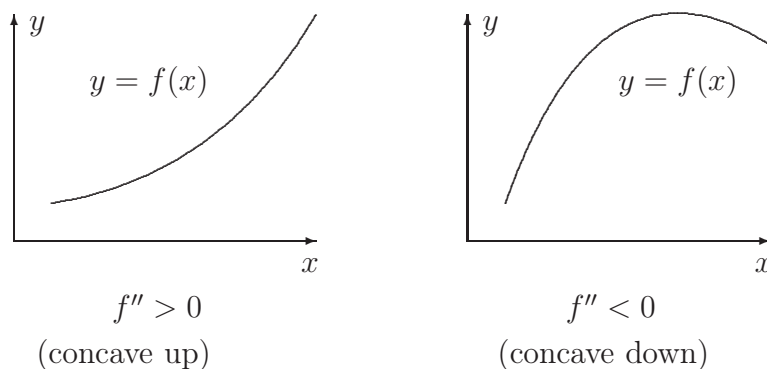
Concavity of a function (curve)

If the sign of $f'(x)$ tells whether f is increasing or decreasing, then the sign of $f''(x)$ in turn tells the *direction of concavity* of the function f (or the curve $y = f(x)$). If f is differentiable on the open interval (a, b) , then f is said to be *concave up* on that interval if f' is strictly increasing on that interval, and *concave down* if f' is strictly decreasing on the interval (a, b) . If f is twice differentiable, except possibly on a finite set of points, then the direction of concavity of f can be inferred from the sign of f'' in accordance with Theorem V.6.3. The geometric interpretation of concavity is obvious, cf. the figure.*

*As a synonym for concavity, mathematical texts also use the term *convexity* (up or down), but then it may remain unclear which direction of concavity is meant. The concept of concavity (convexity) can also be defined more generally, independently of derivatives, in which case the terms *convex* and *concave* are more often used. A function f is called convex on the interval $[a, b] \in \mathcal{D}_f$ if f has the property

$$f(tx_1 + (1-t)x_2) \leq tf(x_1) + (1-t)f(x_2), \quad \text{when } x_1, x_2 \in [a, b] \text{ and } t \in [0, 1].$$

If in this condition the inequality holds in the form ' $<$ ' whenever $x_1 \neq x_2$ and $t \in (0, 1)$, then f is *strictly convex* on the interval $[a, b]$. Geometrically, strict convexity means that if $x_1, x_2 \in [a, b]$ and $x_1 < x_2$, then the graph of the function f lies below the line through the points $(x_1, f(x_1))$ and $(x_2, f(x_2))$ on the open interval (x_1, x_2) . A function that is continuous on $[a, b]$ and concave up on (a, b) (sufficiently differentiable) is, by the definitions, strictly convex. The corresponding concept for downward curving, concaveness, is defined analogously.



A point where the direction of concavity changes is called an *inflection point* (of the function/curve). If f'' is continuous at an inflection point c , then $f''(c) = 0$.

EXAMPLE 4 By differentiating, we find that the third-degree polynomial $f(x) = x^3 + ax^2 + bx + c$ ($a, b, c \in \mathbb{R}$) has exactly one inflection point: $x = -\frac{1}{3}a$. On the interval $(-\infty, -\frac{1}{3}a)$ f is concave down and on the interval $(-\frac{1}{3}a, \infty)$ concave up. $\square$

Lipschitz continuity

The continuity of a function defined in Section V.1 can be interpreted as a minimal assumption about the function when one wishes to guarantee the reliability of the function evaluation $x \mapsto f(x)$. In what follows we present a kind of continuity stronger than 'mere continuity', which will occasionally be of use later (especially theoretical use).

DEFINITION V.6.4 A function f is *Lipschitz continuous*, or *Lipschitz**, on the interval $[a, b] \subset \mathcal{D}_f$ if for some $L \in \mathbb{R}_+$

$$|f(x_1) - f(x_2)| \leq L|x_1 - x_2| \quad \forall x_1, x_2 \in [a, b].$$

The number L in the definition is called the *Lipschitz constant* of f on the interval $[a, b]$. The constant L is not unique, since if L is a Lipschitz constant of f , then by the definition so is every $L_1 > L$. If f is Lipschitz on the interval $[a, b]$, then the smallest value of the Lipschitz constant of f is the least upper bound of the (by the definition bounded) set of real numbers

$$A = \left\{ y = \frac{|f(x_1) - f(x_2)|}{|x_1 - x_2|} \mid x_1, x_2 \in [a, b] \text{ \& } x_1 \neq x_2 \right\},$$

***Rudolf Lipschitz** (1832-1903) was a German mathematician.

i.e. $L_{\min} = \sup A$ (cf. Section I.11).

Lipschitz continuity is conceptually close to differentiability, and Lipschitz continuous functions may indeed be characterized as 'almost differentiable'.

EXAMPLE 5 The function $f(x) = |x|$ is Lipschitz continuous on every interval with constant $L = 1$ (= the smallest value of L), since by the triangle inequality

$$|f(x_1) - f(x_2)| = ||x_1| - |x_2|| \leq |x_1 - x_2|, \quad x_1, x_2 \in \mathbb{R}. \quad \square$$

The function of the example is also piecewise continuously differentiable on every interval (Definition V.5.7, $m = 1$). Together with continuity, this is more generally a sufficient guarantee of Lipschitz continuity:

THEOREM V.6.5 If f is continuous and piecewise continuously differentiable on the interval $[a, b]$, then f is Lipschitz continuous on the interval $[a, b]$.

As an introduction to the proof (Exercise 3), let us prove the following claim, whose part (ii) Theorem V.6.5 generalizes. The proof is a typical example of applying the mean value theorem of differential calculus.

THEOREM V.6.6 If f is continuously differentiable on the interval $[a, b]$, then

- (i) $D_+f(a) = f'(a^+)$, $D_-f(b) = f'(b^-)$,
- (ii) f is Lipschitz continuous on the interval $[a, b]$ with a constant whose smallest value is $L = \max_{x \in [a, b]} |f'(x)|$.

Proof (i) Let $\{x_n\}$ be a sequence for which $a < x_n \leq b \forall n$ and $x_n \rightarrow a^+$. Then by Theorem V.6.1, for every n there exists $\xi_n \in (a, x_n)$ such that

$$\frac{f(x_n) - f(a)}{x_n - a} = f'(\xi_n).$$

Here $\xi_n \rightarrow a^+$ as $x_n \rightarrow a^+$, so by the definition of the one-sided derivative $D_+f(a)$ and the assumed one-sided continuity of f' it follows that

$$D_+f(a) = \lim_{x_n \rightarrow a^+} \frac{f(x_n) - f(a)}{x_n - a} = \lim_{x_n \rightarrow a^+} f'(\xi_n) = f'(a^+).$$

The second part of claim (i) is proved analogously.

(ii) If $x_1, x_2 \in [a, b]$ and $x_1 < x_2$, then by Theorem V.6.1

$$f(x_2) - f(x_1) = f'(\xi)(x_2 - x_1), \quad \xi \in (x_1, x_2).$$

Since f' is continuous on the interval $[a, b]$, by Weierstrass's theorem (Theorem V.1.12) $|f'(x)|$ attains its maximum value on the interval $[a, b]$. Denoting this $= L$, we see that L qualifies as a value of the Lipschitz constant of f . By choosing x_1 and x_2 near the maximum point of $|f'(x)|$ (cf. the proof of part (i)), one also easily sees that this value of L is the smallest possible. $\square$

EXAMPLE 6 If $0 < a < b$, then by Theorem V.6.6 the function $f(x) = \sqrt{x}$ is Lipschitz continuous on the interval $[a, b]$ with constant $L = 1/(2\sqrt{a})$. On intervals $[0, b]$, f is not Lipschitz continuous (only continuous), since if we choose $x_1 = 0$ and $x_2 > 0$, then

$$\frac{|f(x_1) - f(x_2)|}{|x_1 - x_2|} = \frac{1}{\sqrt{x_2}} \rightarrow \infty, \quad \text{as } x_2 \rightarrow 0^+. \quad \square$$

The differential equation $y' = f(x)$. Antiderivative

By the rules of differentiation, 'the derivative of a constant is zero', i.e. if $f(x) = C$, $x \in (a, b)$ for some $C \in \mathbb{R}$, then $f'(x) = 0$, $x \in (a, b)$. From the mean value theorem of differential calculus it follows that the claim also holds conversely, in the form: only the derivative of a constant $= 0$. This simple result turns out to be notable in its consequences.

THEOREM V.6.7 (**Fundamental theorem of integral* calculus**) If f is differentiable on the interval (a, b) and $f'(x) = 0 \ \forall x \in (a, b)$, then for some $C \in \mathbb{R}$ we have $f(x) = C$, $x \in (a, b)$.

Proof If $c \in (a, b)$ and $x \in (a, b)$, $x \neq c$, then the assumptions of the mean value theorem of differential calculus hold on the interval $[c, x]$ ($x > c$) or the interval $[x, c]$ ($x < c$). Consequently $f(x) - f(c) = f'(\xi)(x - c)$ for some $\xi \in (c, x) \subset (a, b)$ or $\xi \in (x, c) \subset (a, b)$. Since $f'(\xi) = 0 \ \forall \xi \in (a, b)$, it follows that $f(x) = f(c) = C$, $x \in (a, b)$. $\square$

Theorem V.6.7 can be interpreted as a claim concerning a very simple (in fact the very simplest) *differential equation*

$$y' = 0.$$

Here $y(x)$ is an (unknown) function differentiable on the interval (a, b) under consideration[†]. A *solution* of the differential equation is every function $y(x)$

*Integral calculus is a branch of mathematics examined more thoroughly in Chapter VIII.

[†]In differential equations the function symbol y (not f) is usually used, because it is an implicitly defined function (or set of functions), cf. implicit functions (Section IV.2).

that satisfies the equation on that interval. By Theorem V.6.7, every solution of the differential equation $y' = 0$ is a constant, i.e. the *general solution* of that differential equation on the interval (a, b) is

$$y(x) = C, \quad x \in (a, b),$$

where C is a so-called *arbitrary constant* ($C \in \mathbb{R}$).

The above result is easily generalized to the differential equation

$$y'(x) = f(x), \quad x \in (a, b),$$

where f is a known function on the interval (a, b) .

COROLLARY V.6.8 If F is a differentiable function on the interval (a, b) and $F'(x) = f(x)$, $x \in (a, b)$, then the general solution of the differential equation $y' = f(x)$ on the interval (a, b) is

$$y(x) = F(x) + C, \quad x \in (a, b).$$

Proof If y is a solution of the differential equation and we set $u(x) = y(x) - F(x)$, then $u'(x) = f(x) - f(x) = 0$, $x \in (a, b)$, whereupon by Theorem V.6.7 one must have

$$u(x) = C \Leftrightarrow y(x) = F(x) + C, \quad x \in (a, b). \quad \square$$

EXAMPLE 7 To solve the differential equation

$$y' = x^3 - 2x, \quad x \in \mathbb{R}$$

(here $(a, b) = (-\infty, \infty)$), we must find a function F for which $F'(x) = x^3 - 2x$, $x \in \mathbb{R}$. By the rules of differentiation we see that we may choose $F(x) = \frac{1}{4}x^4 - x^2$, so the general solution is

$$y(x) = \frac{1}{4}x^4 - x^2 + C. \quad \square$$

EXAMPLE 8 What is the general (twice differentiable on $\mathbb{R}$) solution of the differential equation

$$y'' = 0, \quad x \in \mathbb{R}?$$

Solution If we set $u(x) = y'(x)$, the differential equation reduces to the form $u' = 0$, so $u(x) = C_1$, $x \in \mathbb{R}$ (Theorem V.6.7), and hence $y' = C_1$, $x \in \mathbb{R}$. By Corollary V.6.8 and the rules of differentiation, one must then have

$$y(x) = C_1x + C_2, \quad x \in \mathbb{R}.$$

Here C_1 and C_2 are both arbitrary (mutually independent) constants, so the general solution consists of all polynomials of degree ≤ 1 . $\square$

Solutions of the differential equation $y' = f(x)$ are called *antiderivatives* of the function f . An antiderivative is thus unique up to an arbitrary constant, the so-called *constant of integration*. In applications, the constant of integration is often determined by an *initial condition* (arising from the application) of the form $y(x_0) = y_0$. This is then a so-called *initial value problem*

$$\begin{cases} y' = f(x), & x \in (a, b), \\ y(x_0) = y_0, \end{cases}$$

where $x_0 \in (a, b)$ and $y_0 \in \mathbb{R}$ are given. The solution is

$$y(x) = F(x) - F(x_0) + y_0,$$

where F is (any) antiderivative of f . The solution is unique, since $F(x) - F(x_0)$ does not change if a constant is added to F .

EXAMPLE 8 (continued) The unique solution of the initial value problem

$$\begin{cases} y'' = 0, & x \in \mathbb{R}, \\ y(1) = 0, \quad y'(1) = 2 \end{cases}$$

is $y(x) = 2x - 2$. $\square$

The 'integration' of a given function f , i.e. finding an antiderivative, is a very traditional mathematical skill whose broader treatment belongs to integral calculus (see Sections VIII.1–VIII.3 later). For now, let us note only that, by the rules of differentiation, easily integrable functions include, among others, the trigonometric functions $\sin$ and $\cos$, as well as polynomials (cf. Example 7). Also, a function defined on the interval $(-\rho, \rho)$ as the sum of a convergent power series is (surprisingly) easily integrable on that interval. Let an example illuminate the matter.

EXAMPLE 9 The antiderivative of the function $f(x) = x^2/(1-x)$ cannot be expressed (as a formula) in terms of the functions known so far. On the interval $(-1, 1)$, however, the antiderivative can be determined, since on this interval

$$f(x) = \frac{x^2}{1-x} = x^2 \sum_{k=0}^{\infty} x^k = \sum_{k=0}^{\infty} x^{k+2}, \quad x \in (-1, 1).$$

By the differentiation rule for power series (Theorem V.3.4), we conclude that the antiderivative of f on the interval $(-1, 1)$ is

$$F(x) = C + \sum_{k=0}^{\infty} \frac{x^{k+3}}{k+3} = C + \frac{x^3}{3} + \frac{x^4}{4} + \dots \quad \square$$

l'Hospital's rules

By *l'Hospital's rules* we mean rules for computing limits, based on differentiation, of the form

$$\lim_{x \rightarrow a^\pm} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a^\pm} \frac{f'(x)}{g'(x)}.$$

In computing difficult limits, these rules are a notable — and also easily remembered — alternative to, for example, a change of variable (cf. Section V.2). The rules are valid, under mild additional conditions, in cases where $f(a)/g(a)$ is either of the form $0/0$ or ∞/∞ .

Behind l'Hospital's rules is again the mean value theorem of differential calculus, more precisely the following so-called *generalized mean value theorem*, which too can be derived from Rolle's theorem (Exercise 9).

THEOREM V.6.9 If f and g are both continuous on the interval $[a, b]$ and differentiable on the interval (a, b) and, in addition, $g'(x) \neq 0 \forall x \in (a, b)$, then there exists $\xi \in (a, b)$ such that

$$\frac{f(b) - f(a)}{g(b) - g(a)} = \frac{f'(\xi)}{g'(\xi)}.$$

THEOREM V.6.10 (l'Hospital's* rules)

1. Let f and g be differentiable on the interval $(a, a + \delta)$, $\delta > 0$, and let $g'(x) \neq 0$ on this interval. Further, let $\lim_{x \rightarrow a^+} f(x) = \lim_{x \rightarrow a^+} g(x) = 0$. Then

$$\lim_{x \rightarrow a^+} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a^+} \frac{f'(x)}{g'(x)} = A$$

insofar as the limit on the right exists ($A \in \mathbb{R}$ or $A = \pm\infty$).

2. Let f and g be differentiable on the interval (M, ∞) , $M \in \mathbb{R}$, and let $g'(x) \neq 0$ on this interval. Further, let $\lim_{x \rightarrow \infty} f(x) = \lim_{x \rightarrow \infty} g(x) = 0$. Then

$$\lim_{x \rightarrow \infty} \frac{f(x)}{g(x)} = \lim_{x \rightarrow \infty} \frac{f'(x)}{g'(x)} = A$$

insofar as the limit on the right exists ($A \in \mathbb{R}$ or $A = \pm\infty$).

3. Rules 1–2 are also valid when the assumptions concerning the limits of the functions f and g are changed to the form: $\lim |f(x)| = \lim |g(x)| = \infty$.

*The French marquis and mathematician **Guillaume de l'Hospital** (1661–1704) is known above all for the first textbook on differential calculus, which he published (1696).

Proof Rule 1. We set $f(a) = g(a) = 0$, whereupon, by the assumptions, f and g are continuous on the interval $[a, b]$ when $a < b < a + \delta$. Consequently, if $a < x < a + \delta$, then by Theorem V.6.9 one has, for some $\xi \in (a, x)$,

$$\frac{f(x)}{g(x)} = \frac{f(x) - f(a)}{g(x) - g(a)} = \frac{f'(\xi)}{g'(\xi)}.$$

Here $\xi \rightarrow a^+$ as $x \rightarrow a^+$, so the rule follows.

Rule 2. We make the change of variable $x = 1/t$ and apply Rule 1:

$$\lim_{x \rightarrow \infty} \frac{f(x)}{g(x)} = \lim_{t \rightarrow 0^+} \frac{f(\frac{1}{t})}{g(\frac{1}{t})} = \lim_{t \rightarrow 0^+} \frac{-\frac{1}{t^2} f'(\frac{1}{t})}{-\frac{1}{t^2} g'(\frac{1}{t})} = \lim_{t \rightarrow 0^+} \frac{f'(\frac{1}{t})}{g'(\frac{1}{t})} = \lim_{x \rightarrow \infty} \frac{f'(x)}{g'(x)}.$$

Rule 3. This too is based on Theorem V.6.9, but the proof is fairly laborious. The proof is omitted (see Exercise 14). $\square$

The rules of Theorem V.6.10 have obvious counterparts for the limits $\lim_{x \rightarrow a^-} f(x)/g(x)$ and $\lim_{x \rightarrow -\infty} f(x)/g(x)$. The rules are likewise valid for the limit $\lim_{x \rightarrow a} f(x)/g(x)$ ($a \in \mathbb{R}$), assuming that f and g are differentiable on the intervals $(a - \delta, a)$ and $(a, a + \delta)$ and $g'(x) \neq 0$ on these intervals.

EXAMPLE 10 Computing with l'Hospital's Rule 1, we obtain (cf. Example V.2:4)

$$\lim_{x \rightarrow 81} \frac{\sqrt{x} - 9}{\sqrt[4]{x} - 3} = \lim_{x \rightarrow 81} \frac{\frac{1}{2}x^{-1/2}}{\frac{1}{4}x^{-3/4}} = \lim_{x \rightarrow 81} 2x^{1/4} = 6.$$

Applying the same rule, we also obtain

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} = \lim_{x \rightarrow 0} \frac{\cos x}{1} = 1.$$

This computation, however, conceals a circular argument: in computing the limit we use a differentiation rule that was based on that very limit (!) (see Section V.4). $\square$

EXERCISES

1. Determine the critical points of the following functions and the intervals on which the functions are strictly increasing or decreasing. Also determine the

absolute minimum and maximum values of the functions on their domains, insofar as they exist. Sketch the graphs of the functions.

$$\begin{array}{lll}
 \text{a) } f(x) = 1 - 6x + 9x^2 - 5x^3 & \text{b) } f(x) = x^4 + x & \text{c) } f(x) = \frac{9 + x^2}{1 + x} \\
 \text{d) } f(x) = \frac{x^4 + x + 1}{x^4 + 1} & \text{e) } f(x) = x\sqrt{4 - x^2} & \text{f) } f(x) = \sqrt{3x^2 - x^3} \\
 \text{g) } f(x) = \frac{x^2}{\sqrt{4 - x^2}} & \text{h) } f(x) = \frac{x}{\sqrt{x^4 + 1}} & \text{i) } f(x) = x - \sin x \\
 \text{j) } f(x) = x + 2 \cos x & \text{k) } f(x) = x - \frac{2}{\sin x} & \text{l) } f(x) = x - 2 \tan x \\
 \text{m) } f(x) = 1 - \frac{1}{2}x^2 - \cos x
 \end{array}$$

2. a) Determine the smallest Lipschitz constant of the function $f(x) = 3x^4 - 2x^3 + 15x^2 + 10x - 20$ on the interval $[0, 6]$. b) Show that if f is Lipschitz continuous on the interval $[a, b]$ with constant L , then the arc of the curve $S : y = f(x)$ over the interval $[a, b]$ is rectifiable and its arc length satisfies the estimate $s \leq \sqrt{1 + L^2} (b - a)$. c) Show that if f and g are Lipschitz continuous on the interval $[a, b]$ with constants L_1 and L_2 , then $\alpha f + \beta g$ is Lipschitz continuous on the interval $[a, b]$ with constant $L = |\alpha|L_1 + |\beta|L_2$ ($\alpha, \beta \in \mathbb{R}$).
3. Show that if f is continuous and piecewise continuously differentiable on the interval $[a, b]$, then f is Lipschitz continuous on the interval $[a, b]$ with a constant whose smallest value is $L = \max_{x \in [a, b]} g(x)$, where $g(a) = |D_+ f(a)|$, $g(b) = |D_- f(b)|$ and $g(x) = \max\{|D_- f(x)|, |D_+ f(x)|\}$, $x \in (a, b)$. Applying the result, compute the smallest Lipschitz constant of the functions $f(x) = |x - 1| + |x| + |x + 1|$ and $g(x) = \max\{2x^3 - 2x, -2x^2 - 4x, 3 - 2x\}$ on the interval $[-2, 2]$.
4. Determine the inflection points and directions of concavity of the following functions on the various subintervals of $\mathbb{R}$:
 - a) $f(x) = 3x^5 + 35x^4 + 100x^2 - 200x$
 - b) $f(x) = 3x^5 - 10x^4 + 10x^2$
 - c) $f(x) = 2x^3 + x + 1 - 1/x$
 - d) $f(x) = x/(x^2 + 1)$
5. Derive the formula $\operatorname{Arcsin} x + \operatorname{Arccos} x = \frac{\pi}{2}$, $x \in (-1, 1)$ from the rules of differentiation and Theorem V.6.7.
6. Solve (the general solution or the solution of the initial value problem),

either as a formula or by means of a power series:

- a) $y' = 4x^7 - 5x^3 + 3$, $x \in \mathbb{R}$ b) $y' = \frac{x^5 - 1}{x - 1}$, $x \in (1, \infty)$, $y(1) = 1$
c) $y' = \sqrt{x} - \sqrt[3]{x}$, $x \in (0, \infty)$ d) $y' = \frac{1}{\sqrt{x}}$, $x \in (0, \infty)$, $y(1) = -1$
e) $y' = \sin x + \cos x$, $x \in \mathbb{R}$ f) $y' = \frac{\cot x}{\sin x}$, $x \in (0, \pi)$, $y(\frac{\pi}{2}) = 1$
g) $y' = \sum_{k=1}^{\infty} \frac{x^k}{k}$, $x \in (-1, 1)$ h) $y' = \sum_{k=0}^{\infty} \frac{x^k}{k!}$, $x \in \mathbb{R}$, $y(0) = 1$
i) $y' = \sum_{k=0}^{\infty} \frac{(-1)^k}{k!} x^k$, $x \in \mathbb{R}$ j) $y' = \frac{1}{1+x}$, $x \in (-1, 1)$, $y(0) = 2$
k) $y' = \frac{1}{1-x^2}$, $x \in (-1, 1)$ l) $y' = \frac{2+3x}{1-x^2}$, $x \in (-1, 1)$, $y(0) = 0$

7. Show that if $y(x)$ is n times differentiable on $\mathbb{R}$ ($n \in \mathbb{N}$) and $y^{(n)} = 0$, then y is a polynomial of degree $\leq n$. What is the expression for $y(x)$ with the initial conditions $y(x_0) = c_0$, $y'(x_0) = c_1$, $\dots$, $y^{(n-1)}(x_0) = c_{n-1}$?
8. Using Corollary [V.6.8](#), show that on the interval $(-1, 1)$:

$$\operatorname{Arctan} x = \sum_{k=0}^{\infty} \frac{(-1)^k}{2k+1} x^{2k+1} = x - \frac{x^3}{3} + \frac{x^5}{5} - \dots$$

9. Prove Theorem [V.6.9](#) by applying Rolle's theorem to the function

$$h(x) = [f(b) - f(a)][g(x) - g(a)] - [g(b) - g(a)][f(x) - f(a)].$$

Why can one not have $g(a) = g(b)$?

10. Try to compute the limit $\lim_{x \rightarrow \infty} x/\sqrt{x^2+1}$ using l'Hospital's rule.

11. Compute using l'Hospital's rules:

$$\begin{array}{lll}
 \text{a) } \lim_{x \rightarrow -3} \frac{x^2 + 3x}{x^2 - 9} & \text{b) } \lim_{x \rightarrow 3^-} \frac{|x^2 - 4x + 3|}{2x^2 - 5x - 3} & \text{c) } \lim_{t \rightarrow 8} \frac{t^{2/3} - 4}{t^{1/3} - 2} \\
 \text{d) } \lim_{y \rightarrow 1} \frac{y - 4\sqrt{y} + 3}{y^2 - 1} & \text{e) } \lim_{x \rightarrow \infty} \frac{2x^2 + x + 3}{1 - x^2} & \text{f) } \lim_{x \rightarrow 0} \frac{\sin 2x}{\sin 3x} \\
 \text{g) } \lim_{x \rightarrow 0} \frac{1 - \cos 4x}{1 - \cos 3x} & \text{h) } \lim_{x \rightarrow 0} \frac{x - \sin x}{x^3} & \text{i) } \lim_{x \rightarrow 0} \frac{2 - x^2 - 2 \cos x}{x^4} \\
 \text{j) } \lim_{x \rightarrow 0} \frac{x - \sin x}{x - \tan x} & \text{k) } \lim_{x \rightarrow 0^+} \frac{\sin^2 x}{|x - \tan x|} & \text{l) } \lim_{t \rightarrow 0} \frac{3 \sin t - \sin 3t}{3 \tan t - \tan 3t} \\
 \text{m) } \lim_{x \rightarrow 1^-} \frac{\operatorname{Arccos} x}{\sqrt{1 - x^2}} & \text{n) } \lim_{x \rightarrow \infty} x(2 \operatorname{Arctan} x - \pi) &
 \end{array}$$

12. (*) Let f and g be Lipschitz continuous on the interval $[a, b]$. Prove: a) fg is Lipschitz on the interval $[a, b]$. b) If, in addition, $g(x) \neq 0 \forall x \in [a, b]$, then f/g is Lipschitz on the interval $[a, b]$.

13. (*) a) The function $y(x)$ is the solution of the initial value problem $y' = \frac{1}{1+x^7}$, $x \in (-1, 1)$, $y(0) = 1$. Compute $y(\frac{1}{2})$ to eight significant figures.
 b) Determine the general solution of the differential equation $y' = \frac{1}{\sqrt{x}(1+x)}$ in series form on the interval $(0, 1)$.

14. (*) We wish to prove that the rule

$$\lim_{x \rightarrow a^+} \frac{f'(x)}{g'(x)} = A \in \mathbb{R} \quad \Rightarrow \quad \lim_{x \rightarrow a^+} \frac{f(x)}{g(x)} = A$$

(Theorem V.6.10: Rule 1, when $A \in \mathbb{R}$) is also valid when the assumption concerning the limits of the functions f and g is changed to the form: $\lim_{x \rightarrow a^+} |f(x)| = \infty$ and $\lim_{x \rightarrow a^+} |g(x)| = \infty$. Check and complete the following reasoning into a proof:

If $a < x < t < a + \delta$, then for some $\xi \in (x, t)$

$$\begin{aligned}
 & \frac{f(x) - f(t)}{g(x) - g(t)} = \frac{f'(\xi)}{g'(\xi)} \\
 \Rightarrow & \frac{f(x)}{g(x)} - A = \frac{f'(\xi)}{g'(\xi)} - A + \frac{1}{g(x)} \left(f(t) - g(t) \frac{f'(\xi)}{g'(\xi)} \right) \\
 \Rightarrow & \left| \frac{f(x)}{g(x)} - A \right| \leq \left| \frac{f'(\xi)}{g'(\xi)} - A \right| + \frac{1}{|g(x)|} \left(|f(t)| + |g(t)| \left| \frac{f'(\xi)}{g'(\xi)} \right| \right).
 \end{aligned}$$

If now $\varepsilon > 0$, then in the last inequality written one can choose first t and then x so that the right-hand side of the inequality is smaller than ε .

V.7 Fixed-point iteration. Newton's method

Earlier, in Sections I.4–I.8, we examined examples of recursive sequences determined by an initial value $x_0 \in \mathbb{R}$ and (in relation to some real function f) a recursion formula

$$x_{n+1} = f(x_n), \quad n = 0, 1, \dots \quad (\star)$$

In what follows we examine such sequences and the conditions for their convergence from a more general standpoint than before.

Assume first that f is continuous on the closed interval $[a, b]$ and that $x_n \in [a, b] \forall n$. Then if the sequence $\{x_n\}$ is convergent, we must have

$$x_n \rightarrow c \in [a, b].$$

Since f is continuous on the interval $[a, b]$,

$$x_n \rightarrow c \Rightarrow f(x_n) \rightarrow f(c),$$

so by formula $(\star)$

$$c = f(c).$$

We say that c is a *fixed point* of f , and computing the numbers x_n from the recursion formula $(\star)$ is therefore called *fixed-point iteration* (Latin *itero* = to repeat, to do again).

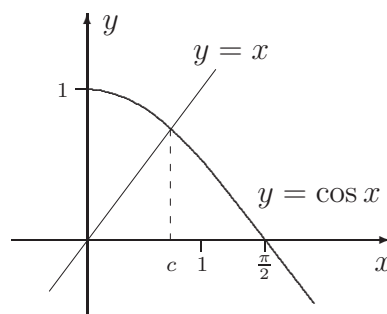
If one wishes to find a fixed point of a given function f , a natural algorithm (though not always working, see the discussions later) is the fixed-point iteration $(\star)$ from some initial guess x_0 .

EXAMPLE 1 Solve the (transcendental) equation $x = \cos x$ by fixed-point iteration.

Solution We choose the initial guess $x_0 = 0$, giving the iteration

$$x_0 = 0, \quad x_{n+1} = \cos x_n, \quad n = 0, 1, \dots$$

This converges rather slowly toward the (unique) fixed point of the function $f(x) = \cos x$, $c = 0.7390851332..$



$x_1 = 1$	$x_{10} = 0.73140404..$
$x_2 = 0.5403..$	$x_{20} = 0.73893775..$
$x_3 = 0.8575..$	$x_{30} = 0.73908229..$
$\vdots$	$\vdots \quad \square$

The example raises the question of under what conditions the fixed-point iteration $(\star)$ converges in general, and how fast, if $x_0 \neq c$. (The case $x_0 = c$ is trivial, since then $x_n = c \forall n$.) Let the direction for the convergence considerations be given by

EXAMPLE 2 Let f be a first-degree polynomial. Then if $f(c) = c$, i.e. c is a fixed point, then for some $k \in \mathbb{R}$ we have $f(x) = c + k(x - c)$, whereupon by the iteration formula $(\star)$

$$x_{n+1} - c = k(x_n - c), \quad n = 0, 1, \dots \Rightarrow x_n - c = k^n(x_0 - c), \quad n = 1, 2, \dots$$

Thus $x_n \rightarrow c$ regardless of the initial value x_0 if $|k| < 1$. If $|k| \geq 1$, then $x_n \rightarrow c$ only when $x_0 = c$. $\square$

By the example, the regularity (even smoothness) of f is not a suitable criterion for the convergence of the fixed-point iteration; rather, conditions of a different type are needed. In the following two theorems we set, following the result of the example, sufficient conditions both for the convergence of the fixed-point iteration toward the desired fixed point c and for the failure of the iteration ($x_n \not\rightarrow c$).

THEOREM V.7.1 If (i) the function f has a fixed point c , and (ii) f is defined on the interval $[c - a, c + a]$, $a > 0$ and satisfies the condition

$$|f(x) - f(c)| \leq L|x - c|, \quad x \in [c - a, c + a],$$

where $0 \leq L < 1^*$, then

- (1) c is the only fixed point of f on the interval $[c - a, c + a]$,
- (2) the iteration $(\star)$ converges toward the fixed point for every $x_0 \in [c - a, c + a]$,
and

$$|x_n - c| \leq L^n |x_0 - c|, \quad n = 1, 2, \dots$$

Proof (1) If $c_1 \in [c - a, c + a]$ is also a fixed point, then

$$\begin{aligned} c = f(c) \ \& \ c_1 = f(c_1) \quad \Rightarrow \quad |c - c_1| = |f(c) - f(c_1)| \leq L|c - c_1| \\ & \Rightarrow \quad (1 - L)|c - c_1| \leq 0. \end{aligned}$$

Since $0 \leq L < 1$, we must have $c - c_1 = 0$.

*Without the additional condition $L < 1$, condition (ii) of Theorem V.7.1 is called the *Lipschitz condition at the point c* , cf. the Lipschitz continuity condition on a closed interval: Definition V.6.4.

(2) By the assumptions, we first have

$$|f(x) - c| = |f(x) - f(c)| \leq L|x - c| \leq |x - c| \leq a, \quad \text{when } x \in [c - a, c + a].$$

Consequently, if $x_0 \in [c - a, c + a]$, then for the iteration $(\star)$ we have $x_n \in [c - a, c + a] \forall n$, so by the assumptions we can estimate

$$|x_n - c| = |f(x_{n-1}) - f(c)| \leq L|x_{n-1} - c| \leq \dots \leq L^n|x_0 - c|.$$

Since $L < 1$, $x_n \rightarrow c$. $\square$

THEOREM V.7.2 If (i) the function f has a fixed point c , and (ii) f is defined on the interval $[c - a, c + a]$, $a > 0$ and satisfies the condition

$$|f(x) - f(c)| \geq L|x - c|, \quad x \in [c - a, c + a],$$

where $L > 1$, then the fixed-point iteration $(\star)$ converges toward the fixed point c only in the case that $x_n = c$ for some n .

Proof If $x_k = c$, then by the iteration formula $(\star)$ $x_n = c \forall n \geq k$, so $x_n \rightarrow c$. Assume then that $x_n \neq c \forall n$, so that the claim is that $x_n \not\rightarrow c$. If $|x_n - c| > a \forall n$, then this is true. We may therefore assume that $x_n \in [c - a, c + a]$ for some n . Then by the assumptions

$$|x_{n+1} - c| = |f(x_n) - f(c)| \geq L|x_n - c|.$$

If $x_{n+1} \in [c - a, c + a]$, we can further estimate $|x_{n+2} - c| \geq L^2|x_n - c|$, etc. Since we assumed that $x_n \neq c$ and $L > 1$, we conclude that for some $m \in \mathbb{N}$, $m > n$

$$|x_m - c| \geq L^{m-n}|x_n - c| > a.$$

It has been concluded above that if $x_n \neq c \forall n$, then from any index N onward some index $n > N$ can always be found such that $|x_n - c| > a$. Then $x_n \not\rightarrow c$. $\square$

If f is differentiable in a neighborhood of the fixed point, or at least at the fixed point (as is often the case), then the validity of the conditions set in Theorems V.7.1 and V.7.2 can be easily investigated by means of the derivative. First, if we assume that f is continuously differentiable on the interval $[c - a, c + a]$, then by the mean value theorem of differential calculus $f(x) - f(c) = f'(\xi)(x - c)$ for some $\xi \in (c - a, c + a)$ when $x \in [c - a, c + a]$. Consequently we can conclude:

$$\begin{aligned} \max_{x \in [c-a, c+a]} |f'(x)| = L < 1 &\Rightarrow \text{the conditions of Theorem V.7.1 hold.} \\ \min_{x \in [c-a, c+a]} |f'(x)| = L > 1 &\Rightarrow \text{the conditions of Theorem V.7.2 hold.} \end{aligned}$$

EXAMPLE 1 (continued) In the example, $|f'(x)| = |\sin x| \leq L < 1$, e.g. on the interval $[c-0.5, c+0.5]$. For the iteration of the example, therefore, the conditions of Theorem V.7.1 hold (with $a = 0.5$) starting from the index $n = 1$. $\square$

If the conditions of Theorems V.7.1 and V.7.2 are set in the form 'for some $a > 0$ ', i.e. the interval $[c-a, c+a]$ is not fixed in advance, then the following simple criterion is obtained for the conditions to hold:

PROPOSITION V.7.3 If f is differentiable at the fixed point c , then:

$$|f'(c)| < 1 \quad \Leftrightarrow \quad \text{the conditions of Theorem V.7.1 hold for some } a > 0.$$

$$|f'(c)| > 1 \quad \Leftrightarrow \quad \text{the conditions of Theorem V.7.2 hold for some } a > 0.$$

Proof If $|f'(c)| < 1$, then by the definition of the derivative and the (ε, δ) -criterion for the limit (Theorem V.2.2), for every $\varepsilon > 0$ there exists $\delta > 0$ such that

$$\left| \frac{f(x) - f(c)}{x - c} - f'(c) \right| < \varepsilon, \quad \text{when } |x - c| < \delta \text{ \& } x \neq c.$$

Choosing here $\varepsilon = (1 - |f'(c)|)/2 > 0$ and using the triangle inequality, we see that the assumptions of Theorem V.7.1 hold for every $a \in (0, \delta)$ (e.g. $a = \delta/2$) when we choose $L = (1 + |f'(c)|)/2 < 1$. This proves part $\Rightarrow$ of the first claim. Part $\Leftarrow$ follows when, from assumption (ii) of Theorem V.7.1, we conclude that we must have $|f'(c)| \leq L < 1$ (Definitions V.3.1 and V.2.1 and Theorem I.7.4 [V1]). The second claim is proved analogously. $\square$

By Proposition V.7.3 and Theorem V.7.1, the fixed-point iteration $(\star)$ converges toward the fixed point c if $|f'(c)| < 1$ and, in addition, x_0 is sufficiently close to the fixed point. If, on the other hand, $|f'(c)| > 1$, then $x_n \rightarrow c$ is, by Theorem V.7.2, true only in the (rather lucky) case that $x_n = c$ for some n . On the basis of these results, a fixed point is called *attractive* (i.e. attracting) if $|f'(c)| < 1$, and *repelling* or *repulsive* if $|f'(c)| > 1$. Outside the classification (to be examined case by case) thus remain only those fixed points where $f'(c) = \pm 1$.

EXAMPLE 3 The only fixed point c of the function $f(x) = \sqrt{x+1}$ is

$$c \geq 0 \text{ \& } c = \sqrt{c+1} \Leftrightarrow c = \frac{1}{2}(\sqrt{5} + 1).$$

Since

$$f'(x) = \frac{1}{2\sqrt{1+x}} \Rightarrow 0 < f'(c) < \frac{1}{2},$$

this is an attractive fixed point. A closer examination reveals that the fixed-point iteration $x_{n+1} = f(x_n)$ converges for every $x_0 \in D_f = [-1, \infty)$. $\square$

EXAMPLE 4 The function $f(x) = 1 - x^2$ has two fixed points:

$$c = 1 - c^2 \Leftrightarrow c = \frac{1}{2}(-1 \pm \sqrt{5}).$$

Since $f'(c) = -2c = 1 \pm \sqrt{5}$, we see that both fixed points are repelling. Consequently we conclude (Theorem V.7.2) that the fixed-point iteration

$$x_0 \in \mathbb{R}, \quad x_{n+1} = 1 - x_n^2, \quad n = 0, 1, \dots$$

can be convergent only if $x_n = \frac{1}{2}(1 \pm \sqrt{5})$ for some n . This possibility is excluded, for example, if $x_0 \in \mathbb{Q}$, since then $\{x_n\}$ is a sequence of rational numbers. $\square$

Asymptotic rate of convergence

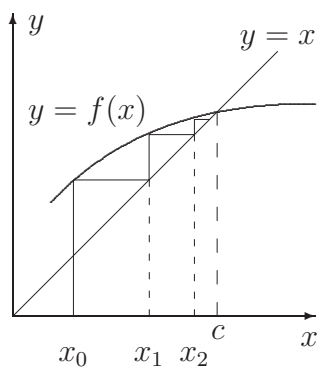
Consider the fixed-point iteration $(\star)$, assuming that (i) f is differentiable at the fixed point c and $|f'(c)| < 1$, (ii) $x_n \rightarrow c$, and (3) $x_n \neq c \forall n$. Then from the iteration formula $(\star)$ and the definition of the derivative it follows that

$$\lim_{n \rightarrow \infty} \frac{x_{n+1} - c}{x_n - c} = \lim_{n \rightarrow \infty} \frac{f(x_n) - f(c)}{x_n - c} = f'(c).$$

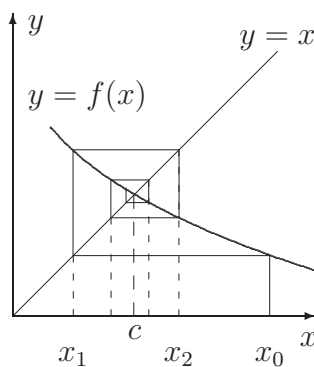
On this basis, one may say that the number $q = f'(c)$ determines the *asymptotic rate of convergence* of the fixed-point iteration: for large values of n , approximately

$$x_n - c \sim \text{constant} \times q^n \quad (n \text{ large}).$$

From this we also see that if $q > 0$, then the sequence $\{x_n\}$ is asymptotically (i.e. for large n) monotonic. If $q < 0$, then the sequence is asymptotically 'oscillating', cf. the figure.



$$0 < q < 1$$



$$-1 < q < 0$$

EXAMPLE 5 In Example 1, $f'(c) = -\sin c \approx -0.67$. The slowness and 'oscillation' of the convergence thus found their explanation. $\square$

The fixed-point iteration $x_{n+1} = f(x_n)$ converges asymptotically especially fast when $f'(c) = 0$ at the fixed point. In order that this case too be examined more closely, let us assume more generally that for some $m \in \mathbb{N}$ and $A \in \mathbb{R}$, $A \neq 0$

$$\lim_{x \rightarrow c} \frac{f(x) - f(c)}{(x - c)^m} = A.$$

If we assume, as above, that for the fixed-point iteration $x_n \rightarrow c$ and $x_n \neq c \forall n$, then by the assumption

$$\lim_{n \rightarrow \infty} \frac{x_{n+1} - c}{(x_n - c)^m} = \lim_{n \rightarrow \infty} \frac{f(x_n) - f(c)}{(x_n - c)^m} = A,$$

whereupon for large values of n , approximately

$$x_{n+1} - c \approx A(x_n - c)^m \quad (n \text{ large}).$$

On this basis, m is called the (asymptotic) *order* of the rate of convergence. — Note that if $f'(c) \neq 0$, then the above assumption holds (by the definition of the derivative, cf. Section V.3) with $m = 1$ and $A = f'(c)$. The convergence in this case (when $|A| < 1$) is indeed called *linear* (order = 1), and in other cases *superlinear*. The most common of the superlinear cases is $m = 2$, in which case one says that the fixed-point iteration converges *quadratically*.

EXAMPLE 6 In Example 4 of Section I.8, the recursive sequence was examined in the case $a = 2$

$$x_0 = a, \quad x_{n+1} = \frac{1}{2} \left(x_n + \frac{a}{x_n} \right), \quad n = 0, 1, \dots$$

Now let more generally $a > 0, x_0 > 0$ and interpret the sequence as a fixed-point iteration for the function $f(x) = \frac{1}{2}(x + \frac{a}{x})$. There are two fixed points:

$$c = \frac{1}{2} \left(c + \frac{a}{c} \right) \Leftrightarrow c^2 = a \Leftrightarrow c = \pm\sqrt{a}.$$

Of these, only $c = \sqrt{a}$ is a possible limit of the sequence $\{x_n\}$ when $x_0 > 0$, since then $x_n > 0 \forall n$. With this value of c , we see that

$$\begin{aligned} f(x) - f(c) &= \frac{1}{2} \left(x + \frac{a}{x} \right) - \sqrt{a} = \frac{1}{2x} (x^2 - 2\sqrt{a}x + a) \\ &= \frac{1}{2x} (x - c)^2. \end{aligned}$$

On this basis $(x - c)^{-2}[f(x) - f(c)] \rightarrow 1/(2c)$ as $x \rightarrow c = \sqrt{a}$. Thus if the sequence $\{x_n\}$ converges, then it converges quadratically. — A closer examination shows that $x_n \rightarrow \sqrt{a}$ whenever $x_0 > 0$. $\square$

Newton's method

If a function is differentiable at a point c , then it can be approximated in a neighborhood of the point c based on the linearization (cf. Section V.3)

$$f(x) \approx f(c) + f'(c)(x - c).$$

On this linearization idea is based a very commonly used and efficient method for solving nonlinear equations, *Newton's method*. In Newton's method one seeks a solution of the equation

$$f(x) = 0$$

near the point x_0 (the initial guess). In the algorithm, f is linearized at the point x_n (initially $n = 0$) and the linearized equation is solved

$$f(x_n) + f'(x_n)(x - x_n) = 0.$$

This is solvable if $f'(x_n) \neq 0$. Denoting the solution $x = x_{n+1}$, we obtain the algorithm

$$\boxed{x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}, \quad n = 0, 1, 2, \dots} \quad (\star\star)$$

Behind the 'Solve' or 'FindRoot' commands of calculators and computer programs is usually either this method or some variation of it, such as the *secant method*, see the comments later.

EXAMPLE 7 If $f(x) = x^2 - a$, $a > 0$, then Newton's algorithm for determining the zero $c = \sqrt{a}$ of f is

$$x_0 > 0, \quad x_{n+1} = x_n - \frac{x_n^2 - a}{2x_n} = \frac{1}{2} \left(x_n + \frac{a}{x_n} \right), \quad n = 0, 1, 2, \dots$$

Thus Example 6 was about Newton's method. $\square$

By the iteration formula ($\star\star$), Newton's algorithm is a fixed-point iteration applied to the function

$$F(x) = x - \frac{f(x)}{f'(x)}.$$

The popularity of the algorithm is based on the fact that, insofar as the iteration converges, the convergence is under fairly general conditions quadratic. The condition for quadratic convergence is that f is sufficiently regular near the zero c and that the zero is *simple*, i.e. $f'(c) \neq 0$. The proof of the following precise convergence theorem is based on a claim of differential calculus that is not yet available. For that reason the proof is restricted for now to the special case in which f is a polynomial. (For a more general proof, see Exercise VII.4:16.)

THEOREM V.7.4 If f is twice continuously differentiable on the interval $[c - a, c + a]$ for some $a > 0$ and $f(c) = 0$ and $f'(c) \neq 0$, then Newton's iteration (★★) converges toward c when x_0 is sufficiently close to c , and

$$\lim_{n \rightarrow \infty} \frac{x_{n+1} - c}{(x_n - c)^2} = \frac{f''(c)}{2f'(c)}.$$

Proof, when f is a polynomial: Since $f(c) = 0$, $f(x) = (x - c)g(x)$, where g is a polynomial, and by the same reasoning $g(x) - g(c) = (x - c)h(x)$ and $f'(x) - f'(c) = (x - c)r(x)$, where h and r are polynomials (Theorem III.3.1). By differentiating, we see that $f'(c) = g(c)$, so we obtain the decompositions

$$\begin{aligned} f(x) &= (x - c)[g(c) + (x - c)h(x)] = f'(c)(x - c) + (x - c)^2h(x), \\ f'(x) &= f'(c) + (x - c)r(x). \end{aligned}$$

Substituting these into the expression for F and noting that $F(c) = c$, it follows that

$$\begin{aligned} F(x) - F(c) &= x - c - \frac{f(x)}{f'(x)} \\ &= x - c - \frac{f'(c)(x - c) + (x - c)^2h(x)}{f'(c) + (x - c)r(x)} \\ &= \frac{(x - c)^2[r(x) - h(x)]}{f'(c) + (x - c)r(x)}. \end{aligned}$$

On this basis

$$\lim_{x \rightarrow c} \frac{F(x) - F(c)}{(x - c)^2} = \frac{r(c) - h(c)}{f'(c)}.$$

By differentiating the above decompositions of f and f' , we further see that $h(c) = \frac{1}{2}f''(c)$ and $r(c) = f''(c)$, so the claim follows. $\square$

By Theorem V.7.4, the convergence of Newton's iteration is, under the assumptions of the theorem, quadratic, or even 'superquadratic' (if $f''(c) = 0$). If f satisfies the regularity condition of Theorem V.7.4 but the zero c is *double*, i.e.

$$f(c) = f'(c) = 0, \quad f''(c) \neq 0,$$

then Newton's algorithm converges in this case too (from sufficiently close to c), but the convergence slows to linear. More precisely, in this case: $\lim_n (x_{n+1} - c)/(x_n - c) = 1/2$ (Exercise 16; cf. also Example 6, when $a = 0$).

EXAMPLE 8 Solve the equation $x = \cos x$ of Example 1 by Newton's method.

Solution When we choose $f(x) = x - \cos x$, the conditions of Theorem V.7.4 hold and $f''(c) = \cos c \neq 0$, so Newton's iteration converges (insofar as it converges) quadratically. The iteration formula is

$$x_{n+1} = x_n - \frac{x_n - \cos x_n}{1 + \sin x_n} = \frac{x_n \sin x_n + \cos x_n}{1 + \sin x_n}, \quad n = 0, 1, 2, \dots$$

With the initial guess $x_0 = 0$, the result is (cf. Example 1)

$$\begin{aligned} x_0 &= 0 \\ x_1 &= 1 \\ x_2 &= 0.7503638678.. \\ x_3 &= 0.7391128909.. \\ x_4 &= 0.7390851333.. \\ x_5 &= 0.7390851332.. \\ &\vdots \quad \square \end{aligned}$$

If the derivative f' varies rapidly near the zero of f , Newton's method can be very sensitive to the initial guess, and the iteration may not converge at all. The computation program then gives the result 'failed to converge'. In that case one usually simply varies the initial value x_0 until luck turns. Another possibility is to use some more reliable method to find a good initial guess, leaving Newton's iteration the task of 'final acceleration'. For example, the bisection construction used in the proof of Bolzano's theorem (Theorem V.1.11) is an excellent starting method — slow but sure.

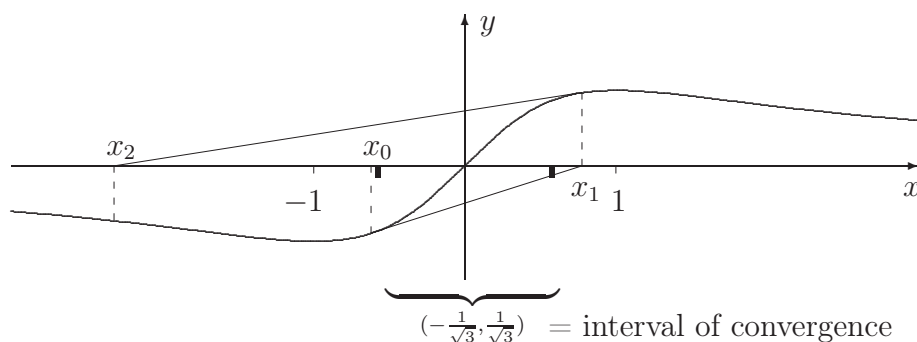
EXAMPLE 9 If the zero of the function $f(x) = x/(1+x^2)$ is sought by Newton's method, the iteration formula becomes

$$x_{n+1} = F(x_n) = -\frac{2x_n^3}{1-x_n^2}, \quad n = 0, 1, 2, \dots$$

One reaches the boundary of the region of convergence if one chooses $x_0 = a \neq 0$ such that $x_1 = -a$, in which case by the iteration formula $x_n = (-1)^n a$. This happens when

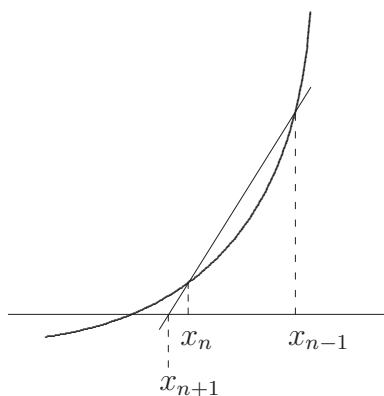
$$a = \frac{2a^3}{1-a^2} \quad \& \quad a \neq 0 \Leftrightarrow a = \pm \frac{1}{\sqrt{3}}.$$

If $|x_0| < 1/\sqrt{3}$, then the iteration converges: $x_n \rightarrow 0$. (The order of the rate of convergence is $m = 3$, since $\lim_{x \rightarrow 0} x^{-3}F(x) = -2$.) If $|x_0| \geq 1/\sqrt{3}$, the result is 'failed to converge'. Geometrically too one sees that if $|x_0| > 1/\sqrt{3}$, then in fact $|x_n| \rightarrow \infty$ (in the figure $x_n \rightarrow -\infty$).

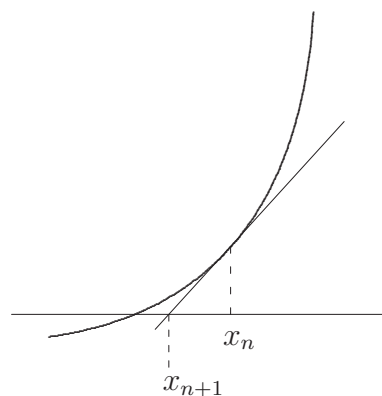


The secant method

If there are problems with the differentiability of the function, or if the derivatives are difficult to determine, one can use a close relative of Newton's method, the *secant method*. Here the idea is to use the line through the points $(x_{n-1}, f(x_{n-1}))$ and $(x_n, f(x_n))$, i.e. the *secant* of the curve $y = f(x)$, to approximate the function when determining the next point x_{n+1} . Under the assumptions of Theorem V.7.4, the iteration of the secant method converges almost as fast as Newton's.* To get the algorithm going, in the secant method two initial guesses x_0, x_1 must be given.



Secant method



Newton

*The asymptotic rate of convergence of the secant method is intermediate between linear and quadratic convergence; more precisely, it can be shown that under the assumptions of Theorem V.7.4

$$\lim_n \frac{|x_{n+1} - c|}{|x_n - c|^\alpha} = \left| \frac{f''(c)}{2f'(c)} \right|,$$

where $\alpha = \frac{1}{2}(\sqrt{5} + 1) \approx 1.62$. (The power function $f(x) = x^\alpha$, $\alpha \notin \mathbb{Q}$ is defined later in Section VI.1.)

EXERCISES

1. The following equations can be solved by fixed-point iteration. Determine the asymptotic rates of convergence exactly (if possible) or to one decimal place:

$$\begin{array}{lll} \text{a)} \ x = \frac{12}{1+x} & \text{b)} \ x = \sqrt{3+x} & \text{c)} \ x = \frac{1}{2+x^2} \\ \text{d)} \ x = \sqrt[3]{x+9} & \text{e)} \ x = \cos \frac{x}{3} & \text{f)} \ x = 1 + \frac{1}{4} \sin x \end{array}$$

2. Investigate which zeros of the polynomial $p(x) = x^3 + 8x^2 - 44x - 10$ can be refined by the fixed-point iteration

$$x_{n+1} = \frac{1}{44}(x_n^3 + 8x_n^2 - 10), \quad n = 0, 1, \dots$$

assuming that a sufficiently good initial guess x_0 is available. How was this iteration method arrived at?

3. The real solution of the equation $x^3 + x = 1$ can be sought by fixed-point iteration by rewriting the equation in the form

$$\text{a)} \ x = 1 - x^3 \quad \text{b)} \ x = x^{-2} - x^{-1} \quad \text{c)} \ x = \sqrt[3]{1-x} \quad \text{d)} \ x = \frac{1}{1+x^2}$$

Investigate how the iterations (asymptotically) converge or diverge, assuming that the initial guess is very close to the fixed point.

4. a) The equation $x = \cos x$ can be written in the form $y = \operatorname{Arccos} y$ and one can try to solve it by the fixed-point iteration $y_{n+1} = \operatorname{Arccos} y_n$. Does the method work?

b) If the equation is solved by the iteration $x_0 = 0$, $x_{n+1} = \cos x_n$, then toward what number and how fast does the iteration converge if in the function evaluations $x_n \mapsto \cos x_n$ the unit of the variable is the degree, i.e. $\cos x = \cos x^\circ$? Try it by selecting degree mode on your calculator and pressing the cos key!

5. Show that if x_0 is a rational number, then the fixed-point iteration

$$x_{n+1} = (x_n - 2)^2, \quad n = 0, 1, \dots$$

converges only if x_0 is one of the numbers 0, 1, 2, 3, 4.

6. Derive Newton's iteration formula for determining the number $\sqrt[m]{a}$ as a zero of the function $f(x) = x^m - a$ ($a > 0$, $m \in \mathbb{N}$, $m \geq 2$). Conclude that the convergence is quadratic. Also conclude geometrically that the iteration converges whenever $x_0 > 0$.
7. Find the zeros of the following functions on the given interval to four decimal places using Newton's method:

- a) $f(x) = x^3 + 2x - 1$, $c \in [0, 1]$
 b) $f(x) = x^4 - 8x^2 - x + 16$, $c \in [1, 3]$
 c) $f(x) = \cos x - x^2$, $c \in (-\infty, \infty)$
 d) $f(x) = 3 \sin x - x - 1$, $c \in [0, \infty)$

8. Compute the maximum and minimum values of the following functions exactly, if possible, otherwise to six decimal places:

$$\text{a) } \frac{\sin x}{1+x^2} \quad \text{b) } \frac{\cos x}{1+x^2} \quad \text{c) } f(x) = \begin{cases} \frac{\sin x}{x}, & \text{if } x \neq 0 \\ 1, & \text{if } x = 0 \end{cases}$$

9. For which values of a does the equation $\cos x = ax$ have exactly two solutions?
10. Compute (approximately) a) the maximum value of the function $f(x, y) = xy^2 + y^4$ on the circle $S: x^2 + y^2 = 1$, b) the smallest and largest value of the function $f(x, y) = (x - y)(x + y)^2$ on the circle $x = 2 \cos t$, $y = 1 + 2 \sin t$, $t \in [0, 2\pi)$.
11. What algorithmic form does the division operation $a \mapsto a^{-1}$ take when it is carried out by applying Newton's iteration to the function $f(x) = x^{-1} - a$? Are divisions needed in the algorithm? Try it with $a = 3$.
12. The function evaluation $y \mapsto \text{Arctan } y$ is to be implemented by an algorithm based on Newton's method that, in addition to the field operations of $\mathbb{R}$, contains only the function evaluations $x \mapsto \cos x$ and $x \mapsto \sin x$. Present such an algorithm.
13. Compute an approximation for the number $\sqrt{2}$ by iterating four times with the secant method from the initial guesses $x_0 = 2$, $x_1 = 1.5$ (function $f(x) = x^2 - 2$). Compare with Newton's method.

14. (*) A function f is called a *contraction* on the interval $[a, b]$ if f is Lipschitz continuous on the interval $[a, b]$ with constant $L < 1$. Prove the *Contraction mapping theorem*: If f is a contraction on the interval $A = [a, b]$ and, in addition, $f(A) \subset A$, then:
- (i) f has exactly one fixed point c on the interval A .
 - (ii) The fixed-point iteration $x_{n+1} = f(x_n)$, $n = 0, 1, \dots$ converges toward c for every $x_0 \in A$.

Hint: First apply Bolzano's theorem to the function $g(x) = f(x) - x$.

15. (*) It is known about a function f that f is continuously differentiable near the (unknown) zero c and that $f'(c) \neq 0$. The zero is sought by the fixed-point iteration

$$x_{n+1} = x_n - kf(x_n), \quad n = 0, 1, \dots$$

Show that if x_0 is sufficiently close to c and one chooses $k = 1/f'(x_0)$, then the iteration converges toward c at least linearly. Show further that in the limit $x_0 \rightarrow c$ the convergence becomes superlinear, i.e. for large n , $x_n - c \sim q^n$, where $q \rightarrow 0$ as $x_0 \rightarrow c$.

16. (*) Show that if f is a polynomial and c is an m -fold zero of f , $m \geq 2$, then Newton's iteration converges toward c whenever the initial guess is sufficiently close to c , and the convergence is asymptotically linear, more precisely

$$q = \lim_n \frac{x_{n+1} - c}{x_n - c} = \frac{1}{m}.$$

17. (*) The number $\sqrt{a}$ can be determined by the iterations

$$\begin{aligned} \text{a) } x_{n+1} &= \frac{x_n^3 + 3ax_n}{3x_n^2 + a}, \quad n = 0, 1, \dots \\ \text{b) } x_{n+1} &= \frac{3x_n}{8} + \frac{3a}{4x_n} - \frac{a^2}{8x_n^3}, \quad n = 0, 1, \dots \end{aligned}$$

Show that if $x_n \rightarrow \sqrt{a}$, then the convergence is in both cases *cubic* (order=3). Do the iterations have other fixed points, and of what nature? Try the methods in practice with $a = 2$, $x_0 = 1.5$, and compare with the quadratic method of Example 6.

18. (*) (Shrovetide, descent 1) A snowboarder wants to build a hill down which one can slide from the origin of the xy -plane to the point $(5, -1)$ along

the fastest possible route (direction of gravity $-\vec{j}$, no friction). Become a consultant, using old knowledge*, according to which the correct profile of the hill is a cycloid arc

$$\begin{cases} x = R(t - \sin t), \\ y = -R(1 - \cos t). \end{cases}$$

So compute R and present graphically your proposal for the optimal hill.

*The problem of the curve of shortest time, i.e. the *brachistochrone*, was solved by the Swiss mathematician **Johann Bernoulli** (1667-1748) in 1697. Also competing in its solution was Johann B.'s brother **Jakob** (1654-1705), after whom, among other things, Bernoulli's inequality (Proposition [I.4.2](#)) is named.

V.8 Analytic complex functions

By a complex function we mean a complex-valued function of a complex variable, i.e. a function of the type $f : \mathcal{D}_f \rightarrow \mathbb{C}$, $\mathcal{D}_f \subset \mathbb{C}$. An example is the polynomial (defined on all of $\mathbb{C}$) examined in Section III.3. Since z can be interpreted as a point of the plane (the complex plane), a complex function can be understood also as a complex-valued function of two real variables, by writing (cf. Example 1 of Section III.3)

$$f(x + iy) = u(x, y) + iv(x, y),$$

where $u(x, y) = \operatorname{Re} f(x + iy)$ and $v(x, y) = \operatorname{Im} f(x + iy)$. Despite this difference relative to functions of a single real variable, the concepts of continuity, limit and differentiability can be defined for complex functions in exactly the same way as for real functions. For example, f is continuous at a point $z \in \mathcal{D}_f$ exactly when for all sequences of complex numbers $\{z_n\}$ (cf. Definition V.1.1)

$$z_n \in \mathcal{D}_f \ \& \ z_n \rightarrow z \quad \Rightarrow \quad f(z_n) \rightarrow f(z).$$

Here the convergence of the sequences $\{z_n\}$ and $\{f(z_n)\}$ refers to Definition I.6.1, which works as such also for sequences of complex numbers, as long as the notation $|\cdot|$ is interpreted to mean the absolute value of a complex number. By the definition, in particular (as for sequences of real numbers)

$$z_n \rightarrow z \quad \Leftrightarrow \quad |z_n - z| \rightarrow 0.$$

If here $z = 0$, then

$$z_n = r_n \angle \varphi_n \rightarrow 0 \quad \Leftrightarrow \quad r_n \rightarrow 0.$$

According to this, approach toward a point of the complex plane (here the origin) can occur from infinitely many different directions ($\varphi_n = \varphi \in [0, 2\pi) \ \forall n$) or with varying directions (e.g. a spiral-like approach). In what follows we shall see that this qualitative difference relative to a sequence of real numbers (for which there are only two possible directions of approach toward the limit) can produce surprises in the study of limits of complex functions.

EXAMPLE 1 The function $f(z) = \bar{z}$ (i.e. $f(x + iy) = x - iy$) is continuous on $\mathbb{C}$, since by complex algebra (see Section III.2)

$$|f(z_n) - f(z)| = |\bar{z}_n - \bar{z}| = |\overline{z_n - z}| = |z_n - z| \rightarrow 0$$

whenever $z_n \rightarrow z$. $\square$

The alternative definition of continuity (Definition V.1.2) also works as such for complex functions, as does the definition of the limit of a function. The derivative of a function, based on the concept of limit, is also defined in the same way as for real functions:

$$f'(z) = \lim_{\Delta z \rightarrow 0} \frac{f(z + \Delta z) - f(z)}{\Delta z}.$$

If the limit exists, then f is said to be *differentiable* at the point z .

EXAMPLE 2 Investigate the differentiability of the complex functions

$$\text{a) } f(z) = \bar{z}, \quad \text{b) } f(x + iy) = x^2 + iy^2, \quad \text{c) } f(x + iy) = x^2 - y^2 + 2ixy.$$

Solution a) If $\Delta z_n = \Delta r_n \angle \varphi_n = \Delta r_n(\cos \varphi_n + i \sin \varphi_n) \neq 0$ ($\Delta r_n \neq 0$), then

$$\frac{f(z + \Delta z_n) - f(z)}{\Delta z_n} = \frac{\cos \varphi_n - i \sin \varphi_n}{\cos \varphi_n + i \sin \varphi_n}.$$

We see that the required limit does not exist, i.e. f is not differentiable at any point.

b) If here we set $\Delta z_n = \Delta x_n + i\Delta y_n$, then

$$\frac{f(z + \Delta z_n) - f(z)}{\Delta z_n} = 2 \frac{x\Delta x_n + iy\Delta y_n}{\Delta x_n + i\Delta y_n} + \frac{(\Delta x_n)^2 + (\Delta y_n)^2}{\Delta x_n + i\Delta y_n}.$$

By passing to the polar representation, we see that the latter term on the right $\rightarrow 0$ whenever $|\Delta z_n| = r_n \rightarrow 0$. The first term, on the other hand, has only directional limits depending on the direction of approach, unless $x = y = t$, in which case that term simplifies to the form $2t$. We conclude, therefore, that f is differentiable only on the line $z = t(1 + i)$, $t \in \mathbb{R}$ of the complex plane, and on this line $f'(z) = 2\operatorname{Re} z$.

c) In this example we are luckier, since $f(x + iy) = (x + iy)^2$, whereupon

$$\frac{f(z + \Delta z) - f(z)}{\Delta z} = \frac{(z + \Delta z)^2 - z^2}{\Delta z} = 2z + \Delta z, \quad \Delta z \neq 0.$$

Thus f is differentiable at every point $z \in \mathbb{C}$ and $f'(z) = 2z$. $\square$

The example shows that the differentiability of a complex function is anything but a foregone conclusion, even in the case where the real and imaginary parts of the function are regular functions (as in the example, polynomials). Differentiability is indeed a much more demanding property for a complex function than for a real function.

In the following definition, the differentiability condition is extended to an *open* subset $G \subset \mathbb{C}$ of the complex plane.

DEFINITION V.8.1 (**Analytic complex function**) A subset $G \subset \mathbb{C}$ of the complex plane is *open* if every $z_0 \in G$ has a *neighborhood*

$$U_\delta(z_0) = \{z \in \mathbb{C} \mid |z - z_0| < \delta\}, \quad \delta > 0$$

such that $U_\delta \subset G$. A complex function that is differentiable on a non-empty, open set $G \subset \mathbb{C}$ is *analytic* on G .

The analytic functions form an important and much-studied 'minority' among complex functions.*

EXAMPLE 2 (continued) In the example, the function $f(z) = z^2$ of part c is analytic on the whole complex plane ($G = \mathbb{C}$), whereas the functions of parts a and b are not analytic anywhere. $\square$

If f is analytic on the whole complex plane, then f is said to be an *entire* function. All polynomials $p(z)$ (including complex-coefficient ones) are entire functions, since the differentiation rule for a polynomial is based only on field algebra, which does not change at all when passing from a real variable to a complex variable (cf. Example 2, part c). The differentiation rules for the sum, product, quotient and composite of functions for real functions are also based on algebra, so in them too the complex variable z can be substituted for the real variable x without the rule changing. By these rules, in particular, every (complex-coefficient) *rational function of a complex variable* can be differentiated in the same way as in the real-variable case. Rational functions are indeed, by Definition V.8.1, analytic throughout their domain, since if $f(z) = p(z)/q(z)$ (p and q polynomials) and $z_0 \in \mathcal{D}_f$, then $q(z_0) \neq 0$, whereupon $q(z) \neq 0$ also in some neighborhood $U_\delta(z_0)$. Then $U_\delta(z_0) \subset \mathcal{D}_f$, i.e. $\mathcal{D}_f$ is an open set.

EXAMPLE 3 The rational functions

$$f(z) = \frac{1}{z^2 + 1}, \quad g(z) = \frac{i}{z^2 + iz}$$

are defined on the whole complex plane except at the points $\pm i$ (f) and $0, -i$ (g). Their domains are open, and both functions are differentiable on their domains, hence analytic. The derivatives are computed as in the real-variable case:

$$f'(z) = -\frac{2z}{(z^2 + 1)^2}, \quad g'(z) = -\frac{i(2z + i)}{(z^2 + iz)^2}. \quad \square$$

*A more detailed presentation of the theory of analytic complex functions belongs to other contexts. Let us mention, however, among the results of the theory, that if f is analytic on G , then so is f' , whereupon f is in fact differentiable arbitrarily many times on G . — This result indicates, for its part, that the analyticity of a complex function is a much stronger property than the differentiability of a real function.

If a complex function is analytic in a neighborhood of its zero, then the zero can be sought by Newton's iteration

$$z_{n+1} = z_n - \frac{f(z_n)}{f'(z_n)}, \quad n = 0, 1, \dots$$

For example, when seeking a simple zero of a polynomial, the algorithm works excellently, as long as the initial guess is sufficiently good (see Exercises 3–5).

EXERCISES

1. Show directly from the definition of the derivative of a complex function that

$$\text{a) } Dz^{-1} = -z^{-2}, \quad \text{b) } D(z+i)^{-2} = -2(z+i)^{-3}.$$

2. On what subset of the complex plane are the following functions analytic?

$$\begin{array}{lll} \text{a) } f(z) = z^3 + iz & \text{b) } f(z) = \frac{z+1}{z^2+z+1} & \text{c) } f(z) = \frac{1}{z^3-8} \\ \text{d) } f(z) = \frac{z}{|z|^2} & \text{e) } f(z) = \frac{\bar{z}}{|z|^2} & \text{f) } f(z) = (z+\bar{z})^2 - 2|z|^2 - \bar{z}^2 \end{array}$$

3. Show that if $a \in \mathbb{C}$, $a \neq 0$, then Newton's iteration

$$z_0 \in \mathbb{C}, \quad z_{n+1} = \frac{1}{2} \left(z_n + \frac{a}{z_n} \right), \quad n = 0, 1, \dots$$

(cf. Section V.7, Example 6) converges quadratically toward a zero of the function $f(z) = z^2 - a$ if the initial guess z_0 is sufficiently close to a zero (either one). Test the workability of the iteration in the case $a = i$ with the choices a) $z_0 = 1$, b) $z_0 = i$, c) $z_0 = -1 - i$.

4. Find the zeros of the polynomial $f(z) = z^4 + z + 4$ approximately by Newton's method. Note that Newton's iteration does not converge in this case with real initial guesses — why not?
5. (*) Prove the counterpart of Theorem V.7.4 for an analytic complex function $f(z)$ in the case where f is a polynomial.

V.9 *The logic of continuity

In this section we prove the two main theorems on continuous functions presented in Section V.1: Weierstrass's theorem (Theorem V.1.12) and Theorem V.1.14 concerning the continuity of the inverse function. The proofs are based on the theory of subsequences presented in Section I.10, and they are fairly demanding. In particular, the Bolzano–Weierstrass theorem concerning bounded sequences of real numbers (Theorem I.10.12) is in frequent use in the proofs. As a continuation of the proof of Weierstrass's theorem, we show that the same proof technique, combined with certain more technical claims concerning polynomials, leads even to a proof of the fundamental theorem of algebra. At the end of the section we also define the concept of *uniform continuity*, which may be regarded as a minimalist counterpart of Lipschitz continuity. In connection with uniform continuity, we prove one of the most astonishing theorems of real analysis.

Since the intention is to prove Theorems V.1.12 and V.1.14 in a slightly more general form, we first define the concept of a *compact* set, which is more general than a closed interval. For the sake of context, we also present at the same time other so-called *topological* basic concepts of $\mathbb{R}$.

Open, closed and compact sets

The open interval

$$U_\delta(x) = (x - \delta, x + \delta) \quad (\delta > 0)$$

is called the (open) *neighborhood* of a point $x \in \mathbb{R}$ (or more precisely the δ -neighborhood; cf. the corresponding concept in the complex plane in Definition V.8.1).

DEFINITION V.9.1 A set $A \subset \mathbb{R}$ is

- *open* if $\forall x \in A$ one has $(x - \delta, x + \delta) \subset A$ for some $\delta > 0$,
- *closed* if for all sequences of real numbers $\{x_n\}$:
 $x_n \in A \ \forall n \ \& \ x_n \rightarrow x \in \mathbb{R} \Rightarrow x \in A$,
- *bounded* if $\exists C \in \mathbb{R}_+$ such that $|x| \leq C \ \forall x \in A$,
- *compact* if A is closed and bounded.

EXAMPLE 1 The open interval (a, b) is an open set, since if $x \in (a, b)$, then $U_\delta(x) \subset (a, b)$ when $0 < \delta \leq \min\{x-a, b-x\}$. A closed interval is correspondingly

a closed set, since if $x_n \in [a, b] \forall n$ and $x_n \rightarrow x \in \mathbb{R}$, then $a \leq x \leq b$ (Theorem I.7.4 [V1]), i.e. $x \in [a, b]$. Since a closed interval is also bounded ($|x| \leq C = \max\{|a|, |b|\} \forall x \in [a, b]$), a closed interval is a compact set. $\square$

EXAMPLE 2

$[0, 1] \cup [2, 3]$	is compact (not open),	
$(0, 1) \cup (2, 3)$	is open and bounded (not closed),	
$\mathbb{R}$	is open and closed (not bounded),	
$[0, 1)$	is bounded (neither open nor closed),	
$[0, \infty)$	is closed (neither open nor bounded),	
$(0, \infty)$	is open (neither closed nor bounded),	
$\mathbb{Q}$	is neither open, closed nor bounded,	
$\emptyset$	(empty) is open and compact.	$\square$

DEFINITION V.9.2 For a set $A \subset \mathbb{R}$:

- the *complement* is $\mathcal{C}(A) = \{x \in \mathbb{R} \mid x \notin A\}$.

- the *closure* is

$$\overline{A} = \{x \in \mathbb{R} \mid \exists \text{ a sequence } \{x_n\} \text{ such that } x_n \in A \forall n \text{ \& } x_n \rightarrow x\}.$$

- the *boundary* is $\partial A = \overline{A} \cap \overline{\mathcal{C}(A)}$.

- the *set of interior points* is $A_0 = \mathcal{C}(\overline{\mathcal{C}(A)})$.

The closure of a set is, as its name suggests, a closed set, and an alternative definition is indeed: $\overline{A}$ is the smallest closed set that contains A . The set of interior points A_0 is open (see Theorem V.9.3 below); alternative definitions of this are: A_0 is the largest open subset of A , or

$$A_0 = \{x \in A \mid x \notin \partial A\}.$$

It also holds that $\overline{A} = A_0 \cup \partial A$, $A_0 \cap \partial A = \emptyset$, and likewise

$$A \text{ closed} \Leftrightarrow A = \overline{A} \Leftrightarrow \partial A \subset A.$$

EXAMPLE 3 The closures, boundaries and sets of interior points of the sets

$$A = (0, 1], \quad B = (-1, 0) \cup (0, 1), \quad C = \mathbb{Q}$$

are

$$\begin{array}{lll} \overline{A} = [0, 1], & \overline{B} = [-1, 1], & \overline{C} = \mathbb{R}, \\ \partial A = \{0, 1\}, & \partial B = \{-1, 0, 1\}, & \partial C = \mathbb{R}, \\ A_0 = (0, 1), & B_0 = B, & C_0 = \emptyset. \end{array} \quad \square$$

When one wishes to determine whether a given set is open, closed or compact, the task is often considerably eased by making use of the following logical claims. The first of the claims can also be interpreted as the definition of a closed or open set, provided that only one of the concepts is defined separately.

THEOREM V.9.3 It holds that

$$A \text{ open} \Leftrightarrow \mathbb{C}(A) \text{ closed}, \quad A \text{ closed} \Leftrightarrow \mathbb{C}(A) \text{ open}.$$

THEOREM V.9.4 It holds that

$$\begin{aligned} &A, B \text{ open/closed/compact} \\ &\Rightarrow A \cup B \text{ and } A \cap B \text{ open/closed/compact.} \end{aligned}$$

Proof The claims include ten implication claims in all, whose proofs are all fairly short. As an example we prove only the claim $A \text{ open} \Rightarrow \mathbb{C}(A) \text{ closed}$, i.e.: If A is open, then for every sequence of real numbers $\{x_n\}$

$$x_n \in \mathbb{C}(A) \ \forall n \quad \& \quad x_n \rightarrow x \quad \Rightarrow \quad x \in \mathbb{C}(A).$$

We make the counter-assumption: $x \notin \mathbb{C}(A) \Leftrightarrow x \in A$. Then $(x - \delta, x + \delta) \subset A$ for some $\delta > 0$, since A was open. Then, since $x_n \in \mathbb{C}(A) \ \forall n$, we must have

$$|x_n - x| \geq \delta \ \forall n \quad \Rightarrow \quad x_n \not\rightarrow x.$$

This is a logical contradiction, since we assumed that $x_n \rightarrow x$. $\square$

EXAMPLE 4 The set $A = (-\infty, a) \cup (b, \infty)$ is open by Theorem V.9.4 (or directly by Definition V.9.1). By Theorem V.9.3 the complement

$$\mathbb{C}(A) = \begin{cases} [a, b], & \text{if } a < b, \\ \{a\}, & \text{if } a = b, \\ \emptyset, & \text{if } a > b \end{cases}$$

is closed. $\square$

EXAMPLE 5 A set $A = \{a\}$ containing a single element is compact by Definition V.9.1, so by Theorem V.9.4 so is every set of two, three, etc. elements. Thus every finite set is compact. $\square$

EXAMPLE 6 The domain of the rational function $f(x) = p(x)/q(x)$ (p and q polynomials) is $\mathcal{D}_f = \mathbb{C}(A)$, where $A = \{\text{zeros of } q\}$. As a finite set, A is closed (even compact), so by Theorem V.9.3 $\mathcal{D}_f$ is open. $\square$

Weierstrass's theorem

The continuity of a function f on a closed interval $[a, b] \subset \mathcal{D}_f$ meant, by Definition V.1.5, continuity 'from the inside' on that interval. We set a more general definition in a corresponding way.

DEFINITION V.9.5 A real function f is *continuous on a compact set* $K \subset \mathcal{D}_f$ if for every sequence of real numbers $\{x_n\}$

$$x_n \in K \ \forall n \quad \& \quad x_n \rightarrow x \Rightarrow f(x_n) \rightarrow f(x).^*$$

*THEOREM V.9.6 If f is continuous on a compact set $K \subset \mathcal{D}_f$, then $f(K)$ is a bounded set.

Proof We use an indirect proof, i.e. we make the counter-assumption: $f(K)$ is not bounded. Then there exists a sequence $\{x_n\}$ such that

$$x_n \in K \ \forall n \quad \& \quad |f(x_n)| \rightarrow \infty.$$

Since A is bounded (as a compact set), the sequence $\{x_n\}$ is bounded. Then, by the Bolzano–Weierstrass theorem (Theorem I.10.12), the sequence has a convergent subsequence. Indexing this subsequence as a sequence $\{x_k\}$, we then have $x_k \rightarrow x \in \mathbb{R}$. Since K is closed (as a compact set), we must have $x \in K$. Thus we have found a sequence $\{x_k\}$ for which $x_k \in K \ \forall k$ and $x_k \rightarrow x \in K$. Then, by the continuity assumption and Definition V.9.5, we must have $f(x_k) \rightarrow f(x)$. On the other hand, since for the original sequence $\{f(x_n)\}$ we have $|f(x_n)| \rightarrow \infty$, also $|f(x_k)| \rightarrow \infty$. From the assumptions and the counter-assumption it thus followed that there exists a sequence $\{x_k\}$ for which both $f(x_k) \rightarrow f(x) \in \mathbb{R}$ and $|f(x_k)| \rightarrow \infty$. This is a logical contradiction, so the theorem is proved. $\square$

The following result contains, as a special case, Weierstrass's theorem V.1.12.

*THEOREM V.9.7 If a real function f is continuous on a compact set $K \subset \mathcal{D}_f$, then f attains its smallest and largest value on K .

Proof Since $f(K)$ is bounded by Theorem V.9.6, this set has a supremum by Theorem I.11.2: $\sup f(K) = y \in \mathbb{R}$. By the definition of the supremum (see Section I.11), there exists a sequence $\{y_n\}$ for which $y_n \in f(K) \ \forall n$ and $y_n \rightarrow y$. Since $y_n \in f(K)$, $f(x_n) = y_n$ for some $x_n \in K$. Since K is bounded (as a compact set), the sequence $\{x_n\}$ is bounded, so by Theorem I.10.12 it has a convergent

*Let us note that the condition of the definition is satisfied for any real function f if $K \subset \mathcal{D}_f$ is a finite (and hence compact) set. Cf. the original definition of continuity (Definition V.1.1), according to which a function is continuous at the isolated points of its domain.

subsequence. We re-index this as a sequence $\{x_k\}$, whereupon $x_k \rightarrow x \in \mathbb{R}$. Since K is closed (as a compact set), we must have $x \in K$. Then, by the assumption and Definition V.9.5, $f(x_k) \rightarrow f(x)$. But the sequence $\{f(x_k)\}$ is a subsequence of the original sequence $\{f(x_n)\}$, so also $f(x_k) \rightarrow y$ (Theorem I.10.7). Thus $f(x) = y$ (since the limit of a sequence is unique). Since y is an upper bound of $f(K)$, $f(t) \leq y \ \forall t \in K$ — thus f attains its largest value on the set K at the point x . The theorem is thus proved for the maximum value. For the minimum value, the theorem is proved either by a corresponding argument or by applying the already-proved claim to the function $-f$. $\square$

EXAMPLE 7 The function

$$f(x) = \begin{cases} 1/x, & \text{if } x > 0, \\ 0, & \text{if } x = 0 \end{cases}$$

is continuous on the (compact) set $K = [a, b]$ if $0 < a < b$. Then the maximum value of f on $K = f(a)$ and the minimum value $= f(b)$. On the compact interval $[0, 1] \subset \mathcal{D}_f$, f is not continuous in the sense of Definition V.9.5, nor does f attain its maximum value on this interval (minimum value $= 0$). At every point of the interval $A = (0, 1]$ (bounded, not closed), f is continuous, but f does not attain its maximum value on A (minimum value $= 1$). On the interval $A = (1, 2)$ (bounded, not closed), f attains neither of its extrema. On the interval $A = [1, \infty)$ (closed, not bounded), f is continuous at every point, but f does not attain its minimum value on A (maximum value $= 1$). $\square$

The fundamental theorem of algebra

The concepts of open, closed and compact set are defined in the complex plane in exactly the same way as in $\mathbb{R}$. The only difference is that the neighborhood needed in the definition of an open set is disk-shaped in $\mathbb{C}$: $U_\delta(c) = \{z \in \mathbb{C} \mid |z - c| < \delta\}$ (Definition V.8.1). Theorems V.9.3 and V.9.4 also hold in the complex plane.

EXAMPLE 8 The disk $K = \{z \in \mathbb{C} \mid |z| \leq R\}$ is closed, i.e. for every sequence of complex numbers $\{z_n\}$: $z_n \in K \ \forall n$ & $z_n \rightarrow z \Rightarrow z \in K$. K is also bounded ($|z| \leq C = R \ \forall z \in K$), so K is a compact set. The complement $\mathbb{C}(K) = \{z \in \mathbb{C} \mid |z| > R\}$ is an open set. $\square$

The continuity of a complex function was defined in the previous section in a way corresponding to that for real functions, and continuity on a compact set (Definition V.9.5) likewise generalizes correspondingly to complex functions. The counterpart of Weierstrass's theorem for complex functions is the following claim,

whose proof follows very closely the logic of the proof of Theorem V.9.7. (The proof is omitted.)

*THEOREM V.9.8 If f is a complex-valued function continuous on a compact set $K \subset \mathbb{C}$, then $|f|$ attains its smallest and largest value on K .

The proof of the fundamental theorem of algebra (Theorem III.3.2) can be based on this result and on the considerations of Section III.3: Let $p(z)$ be a polynomial of degree $n \geq 1$ (not constant). Then, by Theorem V.9.8, $|p(z)|$ attains its minimum value on the compact set

$$K = \{z \in \mathbb{C} \mid |z| \leq R\}$$

for every $R \in \mathbb{R}_+$. Since $|p(z)| \sim |z|^n$ as $|z| \rightarrow \infty$ (see Proposition III.2.3), we conclude that when R is sufficiently large, the minimum point of $|p|$ on K is at the same time the absolute minimum point of $|p|$ on $\mathbb{C}$. Thus $|p|$ attains its absolute minimum value at some point $c \in \mathbb{C}$. Then, by the following claim, we must have $p(c) = 0$, which proves the fundamental theorem of algebra.

THEOREM V.9.9 If a polynomial p of a complex variable is not constant, then $p(c) = 0$ at every point $c \in \mathbb{C}$ where $|p|$ has a local minimum.

Proof See Exercise III.3:13. $\square$

Continuity of the inverse function

The following theorem guaranteeing the continuity of the inverse function is a generalization of Theorem V.1.14. In the proof, the Bolzano–Weierstrass theorem again plays a central role.

*THEOREM V.9.10 If f is continuous on a compact set $A \subset \mathcal{D}_f$ and $f : A \rightarrow B$ is a bijection, then B is compact and $f^{-1} : B \rightarrow A$ is likewise continuous on the compact set B .

Proof We first show that B is a compact set. By Theorem V.9.6, B is bounded, so it suffices to show that B is closed. Assume then that $y_n \in B \forall n$ and that $y_n \rightarrow y \in \mathbb{R}$. Since $y_n \in B = f(A)$, for every n we have $y_n = f(x_n)$ for some $x_n \in A$. Since A is bounded (as a compact set), the sequence $\{x_n\}$ is bounded, so this sequence has a subsequence $\{x_k\}$ that converges: $x_k \rightarrow x \in \mathbb{R}$ (Theorem I.10.12). Since A is closed (as a compact set), $x \in A$. Then, since f is continuous at x in the sense of Definition V.9.5, $f(x_k) \rightarrow f(x)$. But $\{f(x_k)\} = \{y_k\}$ is a

subsequence of the sequence $\{y_n\}$, so $f(x_k) = y_k \rightarrow y$. By the uniqueness of the limit of a sequence, we then have $f(x) = y$, so $y \in f(A) = B$. It has been shown that for every sequence $\{y_n\}$: $y_n \in B \ \forall n \ \& \ y_n \rightarrow y \Rightarrow y \in B$. Thus B is closed.

To show the asserted continuity of the inverse function f^{-1} , it must be shown: If $y_n \in B$, $n = 1, 2, \dots$ and $f(x_n) = y_n$, $x_n \in A$, then

$$y_n \rightarrow y \in B \quad \Rightarrow \quad x_n \rightarrow x \in A \ \& \ x = f^{-1}(y).$$

Since $x_n \in A$ and A is compact, the sequence $\{x_n\}$ is bounded, so the sequence has a convergent subsequence $\{x_k\}$: $x_k \rightarrow x \in \mathbb{R}$. Since A is compact, $x \in A$, whereupon by the continuity of f , $f(x_k) \rightarrow f(x)$. But the sequence $\{f(x_k)\}$ is a subsequence of the sequence $\{f(x_n)\} = \{y_n\}$, so by the assumption $y_n \rightarrow y$ we must also have $f(x_k) \rightarrow y$. Thus $f(x) = y \Leftrightarrow x = f^{-1}(y)$. We have concluded that the sequence $\{x_n\}$ has at least a subsequence $\{x_k\}$ for which $x_k \rightarrow x \in A$ and $x = f^{-1}(y)$. We now show that the original sequence too satisfies $x_n \rightarrow x$, whereupon the theorem is proved. We make the counter-assumption: $x_n \not\rightarrow x$. Then, by Theorem I.10.8, there exist another subsequence $\{x_l\}$ and $\varepsilon > 0$ such that

$$|x_l - x| \geq \varepsilon \quad \forall l \in \mathbb{N}.$$

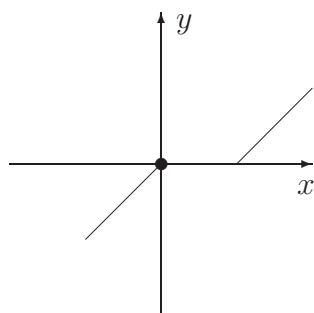
This subsequence too, however, has a convergent subsequence $\{x_\nu\}$, for which $x_\nu \rightarrow x' \neq x$ as $\nu \rightarrow \infty$. Then again $x' \in A$, since A is closed, so by the continuity of f , $f(x_\nu) \rightarrow f(x')$. On the other hand, since the sequence $\{f(x_\nu)\}$ is still a subsequence of the original sequence $\{f(x_n)\} = \{y_n\}$, it follows from the assumption $y_n \rightarrow y$ that also $f(x_\nu) \rightarrow y$. Thus both $f(x_\nu) \rightarrow f(x')$ and $f(x_\nu) \rightarrow y$, whereupon we must have $f(x') = y = f(x)$. From the assumptions and the counter-assumption, the conclusion has thus been derived: there exist $x \in A$ and $x' \in A$ such that $x \neq x'$ and $f(x) = f(x')$. But by the (so-far-unused) assumption, f is an injection, so $x \neq x' \Rightarrow f(x) \neq f(x')$. Thus $f(x) = f(x')$ and $f(x) \neq f(x')$ — a logical contradiction, which shows the counter-assumption to be false. Thus $x_n \rightarrow x = f^{-1}(y)$. $\square$

EXAMPLE 9 We define a function f on the set

$$D_f = A = [-1, 0] \cup (1, 2]$$

as follows:

$$f(x) = \begin{cases} x & , \text{ if } x \in [-1, 0] \\ x - 1 & , \text{ if } x \in (1, 2] \end{cases}$$



Then f is continuous throughout its domain and $f : A \rightarrow [-1, 1]$ is a bijection, so also $f^{-1} : [-1, 1] \rightarrow A$ is a bijection (cf. Section IV.2). But f^{-1} is not continuous at the point $x = 0$. $\square$

In the example, the problem for the continuity of the inverse function is that the subinterval $(1, 2]$ of the domain is not closed. The problem cannot be removed by including $x = 1$ in the domain, since if one sets $f(1) = 0$, then $f(0) = f(1)$, whereupon f is not an injection, and if one sets $f(1) = c \neq 0$, then f is not (right-)continuous at the point $x = 1$.

Uniform continuity

We consider one more new concept of continuity, which may be regarded as a minimalist counterpart of the Lipschitz continuity defined in Section V.6 (cf. Definition V.6.4 and Theorem V.9.12 below).

DEFINITION V.9.11 A real function f is *uniformly continuous* on a set $A \subset \mathcal{D}_f$ if for all sequences of real numbers $\{x_n\}$ and $\{t_n\}$

$$x_n, t_n \in A \ \& \ x_n - t_n \rightarrow 0 \Rightarrow f(x_n) - f(t_n) \rightarrow 0.$$

If f is uniformly continuous on the interval $(c - \delta, c + \delta)$, $\delta > 0$, then by choosing $y_n = c$ in the above definition we see that f is continuous at the point c . Uniform continuity is thus, in this sense, a stronger property than the continuity of Definition V.1.1, which is also commonly called *pointwise*.

EXAMPLE 10 $f(x) = 1/x$ is (pointwise) continuous throughout its domain, but not uniformly continuous: e.g. if $x_n = 1/n$ and $t_n = 2/n$, then $x_n - t_n \rightarrow 0$, but $f(x_n) - f(t_n) = n/2 \not\rightarrow 0$.

The significance of uniform continuity is seen more clearly from the following result, which may also be regarded as an alternative definition of uniform continuity, cf. Theorem V.2.2. The proof (which is omitted) is in idea the same as the proof of Theorem V.1.3.

*THEOREM V.9.12 A function $f : \mathcal{D}_f \rightarrow \mathbb{R}$, $\mathcal{D}_f \subset \mathbb{R}$, is uniformly continuous on a set $A \subset \mathcal{D}_f$ exactly when for every $\varepsilon > 0$ there exists $\delta > 0$ such that $\forall x_1, x_2 \in \mathbb{R}$

$$x_1, x_2 \in A \quad \& \quad |x_1 - x_2| < \delta \quad \Rightarrow \quad |f(x_1) - f(x_2)| < \varepsilon.$$

EXAMPLE 11 A function that is Lipschitz continuous on a closed interval $A = [a, b]$ (Definition V.6.4) is also uniformly continuous on that interval, since the condition of Theorem V.9.12 is satisfied for every $\varepsilon > 0$ when one chooses $\delta = \varepsilon/L$, where L = the Lipschitz constant. $\square$

The following theorem — which again relies on the Bolzano–Weierstrass theorem — is among the most astonishing results of ordinary real analysis.

*THEOREM V.9.13 If a real function f is continuous on a compact set $K \subset \mathcal{D}_f$, then f is uniformly continuous on K .

Proof We make the counter-assumption: f is not uniformly continuous on K . Then there exist sequences $\{x_n\}$ and $\{t_n\}$ such that

$$x_n, t_n \in K \quad \forall n \quad \& \quad x_n - t_n \rightarrow 0 \quad \& \quad f(x_n) - f(t_n) \not\rightarrow 0.$$

Since $f(x_n) - f(t_n) \not\rightarrow 0$, the sequence $\{f(x_n) - f(t_n)\}$ has, by Theorem I.10.8, a subsequence $\{f(x_k) - f(t_k)\}$ such that for some $\varepsilon > 0$

$$|f(x_k) - f(t_k)| \geq \varepsilon \quad \forall k \in \mathbb{N}.$$

But the sequences $\{x_k\}$ and $\{t_k\}$ are bounded (since $x_k, t_k \in K$ and K is bounded), so first the sequence $\{x_k\}$ has a convergent subsequence $\{x_l\}$ (Theorem I.10.12). When we now turn to consider the corresponding subsequence $\{t_l\}$ of the sequence $\{t_k\}$, this in turn has a convergent subsequence $\{t_\nu\}$. But then the corresponding subsequence $\{x_\nu\}$ of the sequence $\{x_l\}$ also converges (since it is a subsequence of a convergent sequence). Thus, from the sequence formed by the pairs (x_k, t_k) , a subsequence $\{(x_\nu, t_\nu)\}$ has been found for which:

$$x_\nu \rightarrow x \in \mathbb{R} \quad \& \quad t_\nu \rightarrow t \in \mathbb{R}.$$

6. Show that the equation

$$8 - 63y^5 + 90y^7 - 35y^9 = x$$

defines a function $y = f(x)$ on the interval $[0, 8]$ and that $f : [0, 8] \rightarrow [0, a]$ is a continuous bijection for a certain — which? — value of a .

7. A function f is *Hölder continuous with index α* on the interval $[a, b]$, $\alpha \in (0, 1] \cap \mathbb{Q}$, if for some $C \in \mathbb{R}_+$

$$|f(x_1) - f(x_2)| \leq C|x_1 - x_2|^\alpha, \quad x_1, x_2 \in [a, b].$$

a) Show that a function that is Hölder continuous on $[a, b]$ is uniformly continuous on that interval. b) Show that the function $f(x) = \sqrt[n]{x}$, $n \in \mathbb{N}$ is Hölder continuous on $[0, 1]$ exactly for the indices $0 < \alpha \leq 1/n$.

8. (*) Let f be continuous on the interval $[0, 1]$ and form the sequence $\{y_n\}$ as follows:

$$y_n = \max_i \{f(i/2^n), \quad i = 0 \dots 2^n\}, \quad n = 0, 1, \dots$$

Show that $\{y_n\}$ is a monotonically increasing sequence and that

$$\max_{x \in [0, 1]} f(x) = \lim_n y_n.$$

Can the maximum point of f also be found in this way (algorithmically)?

Chapter VI

The exponential function

The *exponential function* to be defined in this chapter may with good reason be regarded as the most common function in physics. The exponential function appears in the most diverse mathematical models describing endless growth or decay/attenuation. Besides physics, there are many such models of *exponential growth* or *exponential decay* in biology, the economic sciences, etc.

Even amid the multitude of mathematical functions, the exponential function stands out as so distinctive that it deserves a chapter of its own. In what follows we examine the mathematical definition and properties of this function and of its inverse function, the *logarithm function*, first as real functions (Sections [VI.1](#)–[VI.2](#)). After that the exponential function is extended to a complex function, and we examine the derived functions to which the extension gives rise as well as — somewhat surprising — connections to the trigonometric functions (Section [VI.3](#)). In the last subsection we briefly go through some examples of applications of the real exponential function in physics.

VI.1 The general exponential function $E(x)$

DEFINITION VI.1.1 A real function E is an *exponential function* if

- (E1) $E(x)$ is defined $\forall x \in \mathbb{R}$,
- (E2) $E(0) \neq 0$,
- (E3) E is continuous at the point $x = 0$,
- (E4) $E(x + y) = E(x)E(y) \quad \forall x, y \in \mathbb{R}$.

The most central axiom of the exponential function, E4, can in applications often be regarded even as a law of nature. Physical reasons can be seen behind axiom E3 as well; cf. the application examples later in Section VI.4.

THEOREM VI.1.2 Every exponential function E has the following properties:

- (a) $E(x) > 0 \quad \forall x \in \mathbb{R}$.
- (b) $E(0) = 1$.
- (c) E is continuous on all of $\mathbb{R}$.
- (d) If $x \in \mathbb{Q}$, then $E(x)$ is uniquely determined by the number $E(1) = b$ and

$$E(x) = b^x \quad \forall x \in \mathbb{Q}.$$

- (e) E is strictly increasing on $\mathbb{R}$ when $b = E(1) > 1$, and strictly decreasing when $b < 1$. If $b = 1$, then $E(x) = 1 \quad \forall x \in \mathbb{R}$.

Proof (a) Since $E(x) \cdot E(-x) = E(x - x) = E(0) \neq 0$ (E4, E2), we must have $E(x) \neq 0 \quad \forall x \in \mathbb{R}$. On the other hand we also have $E(x) = E(\frac{x}{2} + \frac{x}{2}) = [E(\frac{x}{2})]^2 \geq 0$ (E4), so we must have $E(x) > 0 \quad \forall x \in \mathbb{R}$.

(b) Since $E(0) \neq 0$ (E2) and $E(0) = E(0 + 0) = [E(0)]^2$ (E4), we must have $E(0) = 1$.

(c) If $x_n \rightarrow x \in \mathbb{R}$, then $x_n - x \rightarrow 0 \Rightarrow E(x_n - x) \rightarrow E(0) = 1$ (E3, (b)). By axiom E4 it then holds that

$$E(x_n) = E(x)E(x_n - x) \rightarrow E(x) \cdot 1 = E(x).$$

Since this is true for every real sequence for which $\lim_n x_n = x \in \mathbb{R}$, E is continuous at x and hence on all of $\mathbb{R}$.

(d) If $E(1) = b$, then by axiom E4

$$\begin{aligned} b &= E\left(\sum_{k=1}^n \frac{1}{n}\right) = \left[E\left(\frac{1}{n}\right)\right]^n \quad \forall n \in \mathbb{N} \\ \Rightarrow E\left(\frac{1}{n}\right) &= b^{1/n} \quad \forall n \in \mathbb{N} \\ \Rightarrow E\left(\frac{m}{n}\right) &= E\left(\sum_{k=1}^m \frac{1}{n}\right) = (b^{1/n})^m = b^{m/n} \quad \forall m, n \in \mathbb{N}. \end{aligned}$$

Since $E(0) = 1 = b^{0/n}$ and

$$E\left(\frac{m}{n}\right) \cdot E\left(-\frac{m}{n}\right) = E(0) = 1 \Rightarrow E\left(-\frac{m}{n}\right) = \left[E\left(\frac{m}{n}\right)\right]^{-1} = b^{-m/n},$$

it has been shown that

$$E\left(\frac{m}{n}\right) = b^{m/n} \quad \forall m \in \mathbb{Z}, n \in \mathbb{N} \Leftrightarrow \text{claim (d)}.$$

(e) Let $b > 1$ and $x_1 < x_2$. Suppose first that $x_1, x_2 \in \mathbb{Q}$, so that $x = x_2 - x_1 \in \mathbb{Q}$ and $x > 0$, whence $x = p/q$ for some $p, q \in \mathbb{N}$. Then, by claim (d) and the rules for computing with fractional powers,

$$E(x_1) - E(x_2) = b^{x_1} - b^{x_2} = b^{x_1} (1 - b^x) = b^{x_1} \left(1 - \sqrt[q]{b^p}\right).$$

Here $\sqrt[q]{b^p} > 1$, since $b > 1$, so $E(x_1) - E(x_2) < 0$. We have deduced that if $b > 1$, then $\forall x_1, x_2 \in \mathbb{Q}$ it holds that $x_1 < x_2 \Rightarrow E(x_1) < E(x_2)$.

Next let $x_1, x_2 \in \mathbb{R}$ and $x_1 < x_2$ (still $b > 1$). Choose $t_1, t_2 \in \mathbb{Q}$ such that $x_1 < t_1 < t_2 < x_2$ and rational sequences $\{\alpha_n\}$ and $\{\beta_n\}$ such that $\lim_n \alpha_n = x_1$, $\lim_n \beta_n = x_2$ and moreover $\alpha_n < x_1 \forall n$ and $\beta_n > x_2 \forall n$. Then, since $\alpha_n, t_1, t_2, \beta_n \in \mathbb{Q}$ and $\alpha_n < t_1 < t_2 < \beta_n$, it follows from the earlier reasoning that $E(\alpha_n) < E(t_1) < E(t_2) < E(\beta_n) \forall n$. On the other hand, since $\alpha_n \rightarrow x_1$ and $\beta_n \rightarrow x_2$, it follows by claim (c) that $E(\alpha_n) \rightarrow E(x_1)$ and $E(\beta_n) \rightarrow E(x_2)$. Since here $E(\alpha_n) < E(t_1) \forall n$ and $E(\beta_n) > E(t_2) \forall n$, it follows that $E(x_1) \leq E(t_1)$ and $E(x_2) \geq E(t_2)$ (Theorem 1.7.4 [V1]), so it has been shown that

$$E(x_1) = \lim_n E(\alpha_n) \leq E(t_1) < E(t_2) \leq \lim_n E(\beta_n) = E(x_2).$$

Thus $E(x_1) < E(x_2)$. This relied only on the assumption that $x_1, x_2 \in \mathbb{R}$ and $x_1 < x_2$, so claim (e) is proved in the case $b > 1$. In the case $b < 1$ the proof is

analogous. In the case $b = 1$, by claims (d) and (c) we have $E(x) = 1 \ \forall x \in \mathbb{Q} \Rightarrow E(x) = 1 \ \forall x \in \mathbb{R}$. $\square$

Regarding Theorem VI.1.2, it is worth noting that it does not take any closer stance on what the set of exponential functions looks like. It is only shown that the function $E(x) = 1 \ \forall x \in \mathbb{R}$ belongs to the set, which even leaves open the possibility that this is the only exponential function (!). The question of the existence of an exponential function under the condition $E(1) = b \neq 1$ is indeed a problem of its own, whose solution requires deeper considerations from the theory of sequences. These considerations are presented at the end of the chapter, whereupon there is proved

*THEOREM VI.1.3 An exponential function $E(x)$ satisfying the axioms E1–E4 exists and is uniquely determined by the condition $E(1) = b$ for every $b \in \mathbb{R}_+$.

In what follows we use the notation $E(x) = b^x$ for the exponential function also when $x \notin \mathbb{Q}$. In that case we mean the function that, in addition to the axioms E1–E4, satisfies the condition $E(1) = b$. By this notation the general power function $f(x) = x^\alpha$, $x > 0$, $\alpha \in \mathbb{R}$ also becomes defined, that is, the restriction $\alpha \in \mathbb{Q}$ can be dropped.

EXAMPLE 1 The number π^π is the common value at π of the exponential function $E(x) = \pi^x$ and the power function $f(x) = x^\pi$. By claim (c) of Theorem VI.1.2, π^π can be computed, for example, as the limit $\pi^\pi = \lim_n a_n$, where

$$\begin{aligned} \{a_n\} &= \{\pi^3, \pi^{3.1}, \pi^{3.14}, \dots\} \\ &= \{\pi^3, \sqrt[10]{\pi^{31}}, \sqrt[100]{\pi^{314}}, \dots\}. \end{aligned}$$

As an algorithm this is on the slow side (exact value 36.46215960..):

$$\begin{aligned} a_0 &= 31.00627668.. \\ a_1 &= 34.76679088.. \\ a_2 &= 36.39574388.. \\ a_3 &= 36.43743103.. \\ a_4 &= 36.45829251.. \\ a_5 &= 36.46204884.. \quad \square \end{aligned}$$

The central rules for computing with the exponential function b^x are

$$b^x b^y = b^{x+y}, \quad a^x b^x = (ab)^x, \quad (b^x)^y = b^{xy}, \quad a, b \in \mathbb{R}_+, \quad x, y \in \mathbb{R}.$$

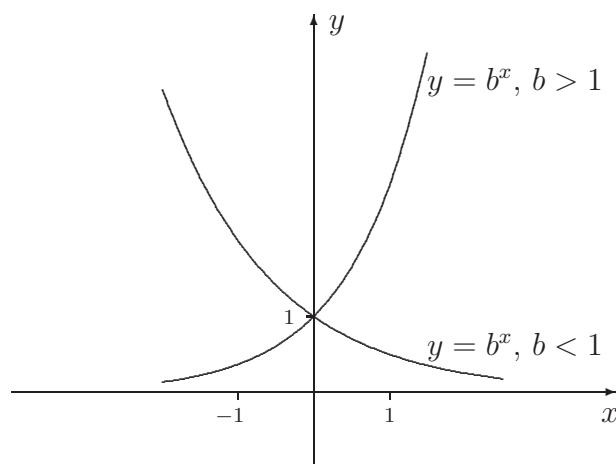
The first of these rules is the basic axiom E4, which generalizes the rule $b^x b^y = b^{x+y} \forall x, y \in \mathbb{Q}$ familiar for fractional powers. The other rules too can be interpreted as generalizations of the corresponding rules for fractional powers (cf. Section I.2). Their justification is left as exercises (Exercises 4–5).

Since the exponential function $E(x)$ is continuous and strictly monotonic when $E(1) = b \neq 1$, the following can be deduced from the first mean value theorem (Theorem V.1.10) and the obvious limit results

$$\lim_{x \rightarrow \infty} b^x = \begin{cases} \infty, & \text{if } b > 1, \\ 0, & \text{if } b < 1 \end{cases}$$

:

$E(x) = b^x$ is a bijection $E : \mathbb{R} \rightarrow \mathbb{R}_+$ for every $b \in \mathbb{R}_+, b \neq 1$.



*Construction of the exponential function

The proof of Theorem VI.1.3 presented below is constructive, i.e. in the proof a number $E(x) \in \mathbb{R}$ is constructed for every $x \in \mathbb{R}$ so that the axioms E1–E4 and the additional condition $E(1) = b \in \mathbb{R}_+$ are satisfied. Since, by Theorem VI.1.2, it is already known that $E(x) = b^x \forall x \in \mathbb{Q}$, it suffices to define $E(x)$ also at irrational values of x and to verify that for the function so defined the axioms E3 and E4 hold. Let us first prove an auxiliary result.

LEMMA VI.1.4 If $b \in \mathbb{R}_+$, then the function $F(x) = b^x, \mathcal{D}_F = \mathbb{Q}$, is continuous at the point $x = 0$, i.e. for every rational sequence $\{x_n\}$ it holds that $x_n \rightarrow 0 \Rightarrow F(x_n) \rightarrow 1$.

Proof If $\{x_n\}$ is a rational sequence for which $x_n \rightarrow 0$, then $\forall k \in \mathbb{N}$ there exists an index $N_k \in \mathbb{N}$ such that $|x_n| < 1/k \ \forall n > N_k$ (Definition I.6.1, $\varepsilon = 1/k$). Since the function $F(x) = b^x$ is monotonic on $\mathbb{Q}$ (even strictly, if $b \neq 1$, see the proof of claim (c) of Theorem VI.1.2), it then holds that

$$|b^{x_n} - 1| \leq |b^{1/k} - 1|, \quad \text{when } n > N_k.$$

Here $b^{1/k} \rightarrow 1$ as $k \rightarrow \infty$ (Proposition I.9.7), so $\forall \varepsilon > 0$ there exists an index m such that $|b^{1/m} - 1| < \varepsilon$. Choosing $k = m$ it thus follows that

$$|b^{x_n} - 1| < \varepsilon, \quad \text{when } n > N_m.$$

Here $\varepsilon > 0$ is arbitrary and $N_m \in \mathbb{N}$, so by the definition of convergence of a sequence $b^{x_n} \rightarrow 1$. $\square$

Construction. We wish to define $E(x)$ when $x \in \mathbb{R}$, $x \notin \mathbb{Q}$. Since it is known that E (insofar as it exists) is continuous on $\mathbb{R}$ (Theorem VI.1.2 (c)), the basis for the definition can be taken to be continuation (cf. Section V.2): starting from the original domain $\mathbb{Q}$ of the function b^x , the domain is extended to $\mathbb{R}$ on the basis of continuity. This means that for every rational sequence $\{x_n\}$ it must hold that

$$x_n \rightarrow x \in \mathbb{R} \quad \Rightarrow \quad \lim_n b^{x_n} = \lim_n E(x_n) = E(x). \quad (1)$$

In order for this condition to serve as the definition of $E(x)$, it must first be verified that $\forall x \in \mathbb{R}$ and for every rational sequence $\{x_n\}$ it holds that

$$x_n \rightarrow x \quad \Rightarrow \quad \{E(x_n)\} \text{ is Cauchy.} \quad (2)$$

Insofar as this is true, then by the Cauchy convergence criterion (Theorem I.10.5) $E(x_n) \rightarrow y \in \mathbb{R}$. This, however, does not yet establish the functional dependence $x \mapsto y$, since it is possible that y depends on the sequence $\{x_n\}$ and not only on its limit x . Therefore it must still be shown that for every $x \in \mathbb{R}$ and all rational sequences $\{x_n\}$ and $\{x'_n\}$ it holds that

$$x_n, x'_n \in \mathbb{Q} \quad \& \quad x_n \rightarrow x \quad \& \quad x'_n \rightarrow x \quad \Rightarrow \quad E(x_n) - E(x'_n) \rightarrow 0. \quad (3)$$

If conditions (2) and (3) are satisfied for all rational sequences $\{x_n\}$ and $\{x'_n\}$, then it has been shown that for every $x \in \mathbb{R}$ (also when $x \in \mathbb{Q}$) there exists a unique $y \in \mathbb{R}$ such that for every rational sequence $\{x_n\}$ it holds that

$$x_n \rightarrow x \in \mathbb{R} \quad \Rightarrow \quad E(x_n) \rightarrow y.$$

Then y depends only on the limit x , so the dependence $x \mapsto y$ can be expressed by the function symbol E , i.e. one can write $y = E(x)$ in accordance with the continuation principle (1).

The claims (2) and (3) on which the continuation is based can be proved by appealing to the already established validity of axiom E4 as a rule for computing with fractional powers (i.e. at rational values of x and y) and to Lemma VI.1.4. First, if $\{x_n\}$ is a rational sequence, then by (E4)

$$E(x_n) - E(x_m) = E(x_m)[E(x_n - x_m) - 1].$$

If here $x_n \rightarrow x \in \mathbb{R}$, then $\{x_n\}$ is Cauchy, so $x_n - x_m \rightarrow 0$ as $n, m \rightarrow \infty$. Then also $E(x_n - x_m) \rightarrow 1$ (Lemma VI.1.4). Since $\{E(x_n)\} = \{b^{x_n}\}$ is undoubtedly a bounded sequence, we deduce (see Theorem I.7.4 [V3])

$$\begin{aligned} n, m \rightarrow \infty &\Rightarrow [E(x_n - x_m) - 1] \rightarrow 0 \\ &\Rightarrow E(x_m)[E(x_n - x_m) - 1] = E(x_n) - E(x_m) \rightarrow 0, \end{aligned}$$

which proves claim (2). Claim (3) is established in a similar way, starting from the decomposition

$$E(x_n) - E(x'_n) = E(x'_n)[E(x_n - x'_n) - 1].$$

The function $E(x)$ is thus defined uniquely on all of $\mathbb{R}$ as the continuation of the known function b^x , $x \in \mathbb{Q}$, $b \in \mathbb{R}_+$, whereby the condition $E(1) = b$ is also satisfied. It still remains to be shown that the E so defined is an exponential function.

Axiom E3. In accordance with the definition of continuity it must be shown that for every real sequence $\{x_n\}$ it holds that

$$x_n \rightarrow 0 \quad \Rightarrow \quad E(x_n) \rightarrow E(0) = 1.$$

So let $\{x_n\}$ be a real sequence for which $\lim_n x_n = 0$. Choose a rational sequence $\{\alpha_n\}$ such that $\lim_n \alpha_n = 0$ and moreover for each n either (i) $0 \leq x_n \leq \alpha_n$ or (ii) $\alpha_n \leq x_n \leq 0$. By claim (e) of Theorem VI.1.2 the function E constructed above is monotonic on $\mathbb{R}$, since the proof relied only on the properties of the function $F(x) = b^x$, $x \in \mathbb{Q}$ together with the continuity condition (1). By assumption (i) or (ii) and monotonicity,

$$|E(x_n) - 1| \leq |E(\alpha_n) - 1| \quad \forall n.$$

Here $\alpha_n \in \mathbb{Q} \quad \forall n$ and $\alpha_n \rightarrow 0$, so $E(\alpha_n) \rightarrow 1$ (Lemma VI.1.4). Thus also $E(x_n) \rightarrow 1$, i.e. E is continuous at the point $x = 0$.

Axiom E4. Let $x, y \in \mathbb{R}$ and let $\{x_n\}, \{y_n\}$ be rational sequences for which $x_n \rightarrow x$ and $y_n \rightarrow y$. Then for each n it holds that $E(x_n + y_n) = E(x_n)E(y_n)$, so

by the definition (1) of E and the rules for combining limits of sequences,

$$\begin{aligned} E(x+y) - E(x)E(y) &= \lim_n E(x_n + y_n) - \lim_n E(x_n) \cdot \lim_n E(y_n) \\ &= \lim_n [E(x_n + y_n) - E(x_n)E(y_n)] \\ &= \lim_n 0 = 0. \quad \square \end{aligned}$$

EXERCISES

1. Check which of the exponential function axioms are satisfied by

$$\begin{aligned} \text{a) } & E(x) = 0, \quad x \in \mathbb{R}, \\ \text{b) } & E(x) = \begin{cases} b^x, & \text{if } x \in \mathbb{Q}, \\ 0, & \text{if } x \in \mathbb{R}, \quad x \notin \mathbb{Q}. \end{cases} \end{aligned}$$

Which familiar function is contained in the latter set?

2. Let $\alpha \in \mathbb{R}$. Using the properties of the exponential function, show that the function $f(x) = x^\alpha$ is strictly increasing on the interval $(0, \infty)$ if $\alpha > 0$, and strictly decreasing if $\alpha < 0$.
3. Show that for the relative error of the approximation

$$\pi^e \approx \sqrt[100]{\pi^{271}} = a$$

the estimate

$$0 < \frac{\pi^e - a}{\pi^e} < \sqrt[100]{\pi} - 1$$

holds.

4. Appealing to the continuity of the exponential function, justify the rule $a^x b^x = (ab)^x$, $x \in \mathbb{R}$, $a, b \in \mathbb{R}_+$. Assume the rule known when $x \in \mathbb{Q}$.
5. (*) We wish to prove the rule $(b^x)^y = b^{xy}$ ($b \in \mathbb{R}_+$, $x, y \in \mathbb{R}$), i.e. the rule

$$E(x)^y = E(xy), \quad x, y \in \mathbb{R},$$

where $E(x) = b^x$. a) Show that if $y \in \mathbb{Q}$, then the rule follows from axiom E4. b) Appealing to the continuity of the exponential function, show that the rule is valid also when $y \in \mathbb{R}$, $y \notin \mathbb{Q}$.

VI.2 The functions e^x and $\ln x$

In the previous section the exponential function $E(x)$ was constructed by continuing the known function b^x , $x \in \mathbb{Q}$. In this way it was verified that exponential functions exist, that the number $b = E(1)$ determines $E(x)$ uniquely, and that the values of the exponential function can be computed numerically directly from the definition. What was not verified is, for example, whether $E(x)$ is possibly also differentiable, and not merely continuous. More generally, it remained open just how regular a function is at issue. In the following we seek answers to these questions by approaching the exponential function from another direction.

As a starting point we take the strengthening of axiom E3 of the exponential function (see the previous section) to the form

(E3') E is differentiable at the point $x = 0$.

Since it is known that $E(0) = 1$ (Theorem VI.1.2 (b)), by axiom E3' there exists a number $a \in \mathbb{R}$ such that

$$E'(0) = \lim_{x \rightarrow 0} \frac{E(x) - 1}{x} = a.$$

Combining this result with axiom E4, it follows that

$$\lim_{\Delta x \rightarrow 0} \frac{E(x + \Delta x) - E(x)}{\Delta x} = \lim_{\Delta x \rightarrow 0} E(x) \frac{E(\Delta x) - 1}{\Delta x} = aE(x), \quad x \in \mathbb{R}.$$

Thus E is differentiable at every point $x \in \mathbb{R}$ and

$$E'(x) = aE(x), \quad x \in \mathbb{R}.$$

We have deduced that if a function $E(x)$ satisfying the axioms E1, E2, E3', E4 exists, then it is differentiable on $\mathbb{R}$ and, for some $a \in \mathbb{R}$, a solution of the problem

$$(P) \quad \begin{cases} y' = ay, & x \in \mathbb{R}, \\ y(0) = 1. \end{cases}$$

This is an initial value problem, in which one must solve the differential equation $y' = ay$ with the initial condition $y(0) = 1$ (cf. the analogous problem for the differential equation $y' = f(x)$ considered in Section V.6).

THEOREM VI.2.1 The problem (P) has the solution $y(x) = E(x)$, where

$$E(x) = \sum_{k=0}^{\infty} \frac{(ax)^k}{k!} = 1 + ax + \frac{(ax)^2}{2} + \dots$$

This solution is unique, and $E(x)$ is an exponential function.

Proof From the rule for differentiating a power series (Theorem V.3.4) one sees immediately that the claimed $y(x)$ is a solution of the problem (P). One also sees that the function $E(x)$ has the properties (E1)–(E3) required of an exponential function, since $E(0) = 1$ and E is defined on all of $\mathbb{R}$ as the sum of a convergent power series, hence as a differentiable and even smooth function.

We next show that $E(x)$ also satisfies axiom (E4) of the exponential function. With this in view we first consider the function

$$u(x) = E(x)E(-x), \quad x \in \mathbb{R}.$$

Since $E' = aE$, using the product rule for differentiation one finds that

$$u'(x) = 0, \quad x \in \mathbb{R}. \quad (1)$$

Since $u(0) = 1$ and $u' = 0$, it follows that $u(x) = 1 \quad \forall x \in \mathbb{R}$ (Theorem V.6.7). Thus $E(x)E(-x) = 1 \quad \forall x \in \mathbb{R}$, and consequently $E(x) \neq 0 \quad \forall x \in \mathbb{R}$ and

$$E(-x) = [E(x)]^{-1}, \quad x \in \mathbb{R}. \quad (2)$$

Next let $t \in \mathbb{R}$ be fixed and consider the function

$$u(x) = E(t+x)E(-t)E(-x), \quad x \in \mathbb{R}.$$

Differentiating with respect to the variable x and using the differential equation $E' = aE$, one finds that (1) again holds. Since by the result (2) we have $u(0) = 1$, we deduce, as above and using the result (2) once more, that

$$u(x) = E(t+x)E(-t)E(-x) = 1 \Leftrightarrow E(t+x) = E(t)E(x), \quad x \in \mathbb{R}.$$

This holds for every $t \in \mathbb{R}$, so axiom E4 is valid for the function $E(x)$. Thus $E(x)$ is an exponential function (for every $a \in \mathbb{R}$).

It remains only to prove the uniqueness of the solution of the problem (P). With this in view let $y(x)$ be (any) solution of (P) and consider the function

$$u(x) = y(x)E(-x), \quad x \in \mathbb{R}.$$

Differentiating and using the differential equations $y' = ay$ and $E' = aE$, one finds that (1) again holds. Since $u(0) = 1$, it follows that

$$u(x) = y(x)E(-x) = 1 \Leftrightarrow y(x) = E(x), \quad x \in \mathbb{R}. \quad \square$$

In what follows we call the solution of the problem (P) when $a = 1$ the *basic exponential function*, with symbol $\exp$. Its definition as a power series is thus

$$\exp(x) = \sum_{k=0}^{\infty} \frac{x^k}{k!}.$$

If we write $e = \exp(1)$, then in the notation of the previous section we can write

$$\exp(x) = e^x.$$

Here e is indeed Napier's number, since it holds that (see Propositions I.8.4 and I.8.5)

$$\exp(1) = \sum_{k=0}^{\infty} \frac{1}{k!} = \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n = e = 2.71828182845905..$$

Here the latter limit result is a special case of the more general result (see Exercise 16)

$$\lim_n \left(1 + \frac{x}{n}\right)^n = e^x, \quad x \in \mathbb{R}.$$

From the definition of the basic exponential function e^x there follows the differentiation rule

$$De^x = e^x.$$

Many of the properties of the function e^x can be easily derived from this rule or from the power series representation. For example, from the power series representation it follows that

$$e^x > x^m/m! \quad \forall m \in \mathbb{N}, \quad \text{when } x > 0,$$

from which the limit results

$$\lim_{x \rightarrow \infty} x^{-\alpha} e^x = \infty \quad \Leftrightarrow \quad \lim_{x \rightarrow \infty} x^{\alpha} e^{-x} = 0 \quad \forall \alpha \in \mathbb{R}. \quad (3)$$

can further be easily deduced. According to these, the function e^x grows faster, and the function e^{-x} decreases faster, than any power function as $x \rightarrow \infty$. On the basis of the power series the numerical values of e^x can also be computed easily, since for small values of $|x|$ (e.g. when $|x| \leq 1$) the series converges rapidly, while for larger values one can use the formula $e^x = (e^{x/n})^n$. From the differentiation rule (also from the power series) one sees that the function e^x is on $\mathbb{R}$ an arbitrarily many times differentiable (smooth) function.

EXAMPLE 1 Extend the function $f(x) = (e^x - 1 - x)/x^2$ to the point $x = 0$.

Solution From the power series of e^x one sees that

$$f(x) = \frac{1}{2!} + \frac{x}{3!} + \frac{x^2}{4!} + \dots = \sum_{k=0}^{\infty} \frac{x^k}{(k+2)!}, \quad x \neq 0.$$

Setting $f(0) = 1/2$, then for all $x \in \mathbb{R}$ it holds that

$$f(x) = \sum_{k=0}^{\infty} \frac{x^k}{(k+2)!} = \begin{cases} \frac{e^x - 1 - x}{x^2}, & \text{when } x \neq 0, \\ \frac{1}{2}, & \text{when } x = 0. \end{cases}$$

Here the radius of convergence of the power series is $\rho = \infty$, so f is not only continuous but arbitrarily many times differentiable on $\mathbb{R}$ (also at the point $x = 0$). $\square$

In the previous section it was found that the general exponential function is of the form $E(x) = b^x$, where $b = E(1) \in \mathbb{R}_+$. In computations the alternative representation derived from the basic exponential function is often more convenient:

$$E(x) = e^{ax}, \quad E(1) = b = e^a.$$

The number $a \in \mathbb{R}$ is uniquely determined by the equation $e^a = b$, since $e^x : \mathbb{R} \rightarrow \mathbb{R}_+$ is a bijection. Every exponential function is thus of the form $E(x) = e^{ax}$ for some $a \in \mathbb{R}$.

Let us note further, concerning the problem (P) that was our starting point, that its solutions are thus all exponential functions $y(x) = e^{ax}$ as the value of a is varied. If a particular exponential function $E(x)$ is wanted as the solution, then a is determined either in the aforementioned way via the number $b = E(1)$ or from the condition $E'(0) = a$. If a is fixed, then the general solution of the differential equation $y' = ay$ (= the set of all solutions) is (Exercise 1)

$$y(x) = Ce^{ax}, \quad C \in \mathbb{R}.$$

The solution satisfying the initial condition $y(0) = y_0$ is accordingly unique and is obtained by choosing $C = y_0$.

The logarithm function

Since the exponential function $E : \mathbb{R} \rightarrow \mathbb{R}_+$ is a bijection when $E(1) = b \neq 1$, it has an inverse function, which is called the *logarithm function*. The inverse function is denoted by the symbol $\log_b$:

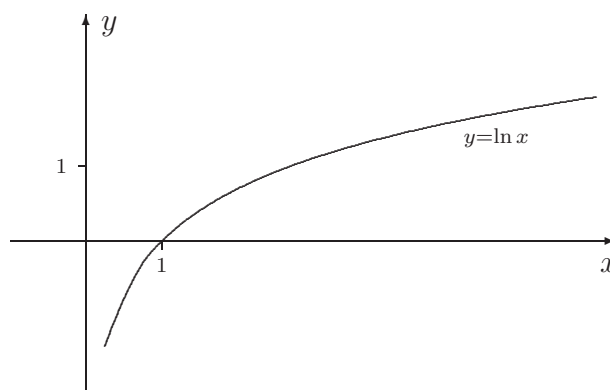
$$y = b^x \Leftrightarrow x = \log_b y.$$

The inverse function of the basic exponential function e^x is called the *natural logarithm*, with symbol $\ln$ (sometimes $\log$):

$$y = e^x \Leftrightarrow x = \ln y.$$

The natural logarithm, like the other logarithm functions, is on its whole domain ($= \mathbb{R}_+$) continuous and strictly increasing, and $\ln x \rightarrow \infty$ as $x \rightarrow \infty$ and $\ln x \rightarrow -\infty$ as $x \rightarrow 0^+$. By the substitutions $x = \pm \ln t \Leftrightarrow t = e^{\pm x}$ one can deduce from the limit results (3) that at the limits $x \rightarrow \infty$ and $x \rightarrow 0^+$ the function $|\ln x|$ grows more slowly than any power function:

$$\lim_{x \rightarrow \infty} x^{-\alpha} \ln x = \lim_{x \rightarrow 0^+} x^{\alpha} \ln x = 0 \quad \forall \alpha \in \mathbb{R}_+. \quad (4)$$



When computing with exponential and logarithm functions, one most commonly uses the basic exponential function e^x and its inverse function $\ln x$. The most frequently needed formulas, which follow directly from the definition and from the properties of the exponential function, are (Exercise 2):

$$e^{\ln x} = x, \quad x \in \mathbb{R}_+. \quad (5)$$

$$\ln(xy) = \ln x + \ln y, \quad x, y \in \mathbb{R}_+. \quad (6)$$

$$\ln(1/x) = -\ln x, \quad x \in \mathbb{R}_+. \quad (7)$$

$$\ln x^y = y \ln x, \quad x \in \mathbb{R}_+, y \in \mathbb{R}. \quad (8)$$

Combining formulas (5) and (8) yields the formula

$$x^y = e^{y \ln x}, \quad x \in \mathbb{R}_+, y \in \mathbb{R}. \quad (9)$$

From formula (5), the differentiation rule for the exponential function, and the chain rule for differentiating a composite function, the differentiation rule for the logarithm function $\ln x$ can be derived:

$$1 = Dx = De^{\ln x} = e^{\ln x} D \ln x = x D \ln x \quad \Rightarrow \quad D \ln x = 1/x, \quad x > 0.$$

On this basis $D \ln(-x) = (-1)/(-x) = 1/x$, $x < 0$, so we obtain the more general rule

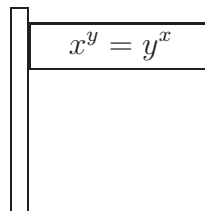
$$\boxed{D \ln |x| = \frac{1}{x}, \quad x \neq 0.} \quad (10)$$

EXAMPLE 2 By formula (9) and rule (10),

$$Dx^x = De^{x \ln x} = e^{x \ln x} D(x \ln x) = (\ln x + 1) x^x, \quad x > 0.$$

From the result one sees that $f(x) = x^x$ is strictly decreasing on the interval $(0, 1/e]$ and strictly increasing on the interval $[1/e, \infty)$. Thus the absolute minimum value of f is $f_{\min} = f(1/e) = e^{-1/e}$. At the limit $x \rightarrow 0^+$ we have $f(0^+) = 1$ (Exercise 3e). $\square$

EXAMPLE 3 The mathematician's cottage is at the intersection of two roads. Determine the location of the cottage, given that on each road there is the sign shown alongside.



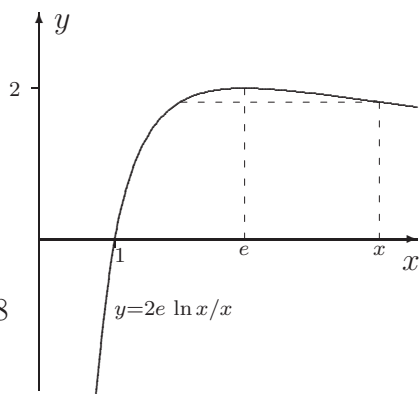
Solution Road 1 is evidently the half-line $y = x$, $x > 0$. On road 2, on the other hand, lie e.g. the points $(2, 4)$ and $(4, 2)$. To determine the intersection point of the roads, we first use formulas (5) and (8) (assume $x, y \in \mathbb{R}_+$):

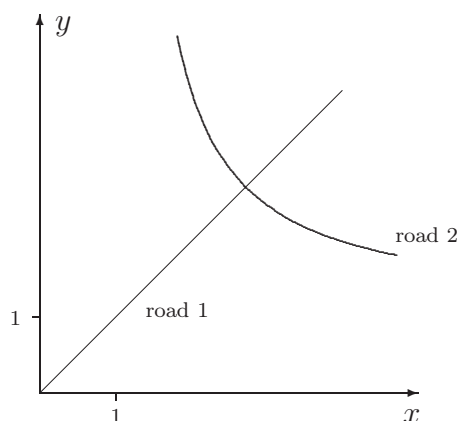
$$x^y = y^x \Leftrightarrow e^{y \ln x} = e^{x \ln y} \Leftrightarrow y \ln x = x \ln y \Leftrightarrow \ln y/y = \ln x/x.$$

From rule (10) and the differentiation rules of Section V.3 it follows that

$$D \ln x/x = x^{-2}(1 - \ln x), \quad x > 0,$$

so the function $f(x) = \ln x/x$ is strictly increasing on the interval $(0, e]$ and strictly decreasing on the interval $[e, \infty)$. Consequently, if $x \in \mathbb{R}_+$ and $x > 1$, $x \neq e$, then the equation $f(y) = f(x)$ has two solutions y , one of which is $y = x$ (see the figure). When $x = e$ or $x \leq 1$, the only solution is $y = x$. We conclude that the cottage is at the point (e, e) . $\square$





EXAMPLE 4 $f(x) = \ln \left| \frac{1 - \cos x}{\sin x} \right|$, $f'(x) = ?$

Solution By rule (10) and the differentiation rules of Sections V.3–V.4,

$$\begin{aligned} Df(x) &= \frac{\sin x}{1 - \cos x} \cdot \left[1 - \frac{(1 - \cos x) \cos x}{\sin^2 x} \right] \\ &= \frac{\sin x}{1 - \cos x} \cdot \frac{\sin^2 x + \cos^2 x - \cos x}{\sin^2 x} \\ &= \frac{1}{\sin x}. \quad \square \end{aligned}$$

The result of the example, and the corresponding result obtained by the transformation $x \mapsto \pi/2 - x$, is worth keeping in mind:

$D \ln \left \frac{1 - \cos x}{\sin x} \right = \frac{1}{\sin x}, \quad x \in \mathbb{R}, \quad \sin x \neq 0,$ $D \ln \left \frac{1 - \sin x}{\cos x} \right = -\frac{1}{\cos x}, \quad x \in \mathbb{R}, \quad \cos x \neq 0.$
--

Here in fact $\frac{1 - \cos x}{\sin x} = \frac{\sin x}{1 + \cos x} = \tan \frac{x}{2}$ (see Exercise II.5:13b).

EXERCISES

1. Show that the general solution of the differential equation $y' = ay$ on $\mathbb{R}$ is $y(x) = Ce^{ax}$. *Hint:* Study the function $u(x) = e^{-ax}y(x)$.

2. a)–d) Justify the formulas (5)–(8).

3. Determine or verify the limits

$$\begin{array}{ll} \text{a) } \lim_{x \rightarrow 0} \frac{e^{2x} - 1 - 2x}{x^2} & \text{b) } \lim_{x \rightarrow 0} \frac{1 - 6x + 18x^2 - e^{-6x}}{x^3} \\ \text{c) } \lim_{x \rightarrow 0} \frac{3e^x - e^{-x} - 2e^{2x}}{x^2} & \text{d) } \lim_{x \rightarrow \infty} x(e^{4/x} - e^{1/x}) \\ \text{e) } \lim_{x \rightarrow 0^+} x^x = 1 & \text{f) } \lim_{x \rightarrow 0^+} x^{x^x} = 0 \quad \text{g) } \lim_{x \rightarrow 0^+} x^{\sqrt{x}} = 1 \end{array}$$

4. Differentiate the following functions:

$$\text{a) } 2^x \quad \text{b) } e^{\sqrt{x}} \quad \text{c) } \pi^{1/x} \quad \text{d) } \ln(\ln x) \quad \text{e) } x^{2x} \quad \text{f) } x^{x^x}$$

5. Determine the local extremum points and the absolute maximum and minimum values of the following functions, insofar as they exist. Also sketch the graphs.

$$\text{a) } x \ln x, \quad \text{b) } \sqrt[10]{|x|} \ln |x|, \quad \text{c) } |x|e^{1/x^2}, \quad \text{d) } x^{-\pi} \ln x, \quad \text{e) } x^{1/x}.$$

6. a) For which values of a is the function $f(x) = 2 \ln x + x^2 - ax + 1$ strictly increasing on the interval $(0, \infty)$?

$$\text{b) Show that if } 0 < a < b, \text{ then } 1 - \frac{a}{b} < \ln \frac{b}{a} < \frac{b}{a} - 1.$$

$$\text{c) Prove the formulas } \log_a b \cdot \log_b a = 1 \text{ and } \log_b a \cdot \log_c b = \log_c a.$$

7. Determine the local extremum points and the absolute maximum and minimum values of the function $f(t) = e^{-x} \sin x$ on the interval $[0, \infty)$, and sketch the graph of f .

8. Deduce that the fixed-point iteration $x_{n+1} = e^{-x_n}$, $n = 0, 1, \dots$ converges for every $x_0 \in \mathbb{R}$. Determine the fixed point by first iterating three times from the initial guess $x_0 = 0$ and then accelerating with Newton's method.

9. Show that the general ($\mathbb{R}$ -twice-differentiable) solution of the differential equation $y'' - y' = 0$ is $y(x) = C_1 e^x + C_2$, $C_1, C_2 \in \mathbb{R}$.

10. a) Show that for the function

$$f(x) = \begin{cases} e^{-\frac{1}{x^2}}, & \text{when } x \neq 0 \\ 0, & \text{when } x = 0 \end{cases}$$

it holds that $f^{(k)}(0) = 0 \forall k \in \mathbb{N}$. Sketch the graph of the function.

b) We wish to define a function f by the conditions

- (i) $f(x) > 0$ when $x \in (0, 1)$
- (ii) $f(x) = 0$ when $x \leq 0$ or $x \geq 1$
- (iii) f is smooth on all of $\mathbb{R}$, i.e. arbitrarily many times differentiable

Show that one function meeting the requirements is

$$f(x) = \begin{cases} e^{-\frac{1}{x(1-x)}}, & \text{when } x \in (0, 1) \\ 0, & \text{otherwise} \end{cases}$$

Sketch its graph.

c) Show that the function

$$f(x) = \begin{cases} 0, & \text{when } x/\pi \in \mathbb{Z} \\ e^{-\cot^2 x}, & \text{otherwise} \end{cases}$$

is an arbitrarily many times differentiable (smooth) function on all of $\mathbb{R}$. Sketch the graph of f on the interval $[0, \pi]$.

11. (*) For which values of k do the curve $y = \ln |x|$ and the line $y = k(x-1)+1$ have exactly one common point?
12. (*) For which values of a does the equation $e^x = x+ax^2$ have real solutions
a) none, b) only one, c) exactly two, d) three or more? Determine the solutions by numerical means in the cases $a = 1$ and $a = 2$.
13. (*) a) Show that

$$1+x \leq e^x \leq 1+x+(e-2)x^2 \quad \forall x \in [0, 1].$$

b) Let $a \geq 1$ and $n \in \mathbb{N}$, $n \geq \ln a$. Show that

$$1 + \frac{\ln a}{n} \leq \sqrt[n]{a} \leq 1 + \frac{\ln a}{n} + (e-2) \left(\frac{\ln a}{n} \right)^2.$$

How large is the error of the approximation $\sqrt[n]{a} \approx 1 + \ln a/n$ in reality, when $a = \pi$ and $n = 100$?

14. (*) Show that the general solution of the differential equation $y' = 2xy$ on $\mathbb{R}$ is $y(x) = Ce^{x^2}$, $C \in \mathbb{R}$.
15. (*) The *Hermite polynomial* $H_n(x)$ is defined by the differentiation formula

$$D^n e^{-x^2} = H_n(x) e^{-x^2}, \quad n \in \mathbb{N} \cup \{0\}.$$

Show that the recurrence formula

$$H_{n+1}(x) = -2xH_n(x) - 2nH_{n-1}(x), \quad n \in \mathbb{N},$$

holds.

16. (*) Consider the functions

$$f(x) = \left(1 + \frac{1}{x}\right)^x, \quad x \in (0, \infty), \quad g(x) = \left(1 - \frac{1}{x}\right)^x, \quad x \in [1, \infty).$$

- a) Show that f and g are strictly increasing on their intervals of definition.
b) Show that

$$\begin{aligned} \lim_{x \rightarrow \infty} \left(1 + \frac{1}{x}\right)^x &= \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n = e, \\ \lim_{x \rightarrow \infty} \left(1 - \frac{1}{x}\right)^x &= \lim_{n \rightarrow \infty} \left(1 - \frac{1}{n}\right)^n = \frac{1}{e}. \end{aligned}$$

- c) Show that $\lim_{n \rightarrow \infty} \left(1 + \frac{x}{n}\right)^n = e^x \quad \forall x \in \mathbb{R}$.

17. (*) (The mathematician's cottage) The mathematician's cottage is in a coordinate system whose origin is at Helsinki, the positive x -axis points east, and the unit of length = 100 km. The cottage is by a certain road at a spot where the road runs in the east-west direction. Determine the location of the cottage, given that the equation of the road is

$$(2x)^y = y^{3x} \quad (x, y > 0)$$

and that, in addition, the endpoint of the road a) is Helsinki, b) is not Helsinki.

18. (*) (Little wonders) Let $A \in \mathbb{R}$. We wish to find functions u and v , differentiable on $\mathbb{R}$, that solve the initial value problem

$$(P) \quad \begin{cases} u' = -v, & v' = u, & x \in \mathbb{R}, \\ u(0) = A, & v(0) = 0. \end{cases}$$

- a) Verify: one solution is $u(x) = A \cos x$, $v(x) = A \sin x$.
b) Prove: if u and v are any solution of (P), then

$$[u(x)]^2 + [v(x)]^2 = A^2 \quad \forall x \in \mathbb{R}.$$

- c) Show: if $A = 0$, then the only solution of (P) is $u(x) = v(x) = 0$.
d) Deduce: the solution of part a) is the only solution of (P).
d) Show: a solution of (P) is also

$$u(x) = A \sum_{k=0}^{\infty} (-1)^k \frac{x^{2k}}{(2k)!}, \quad v(x) = A \sum_{k=0}^{\infty} (-1)^k \frac{x^{2k+1}}{(2k+1)!}.$$

- e) Deduce: for every $x \in \mathbb{R}$ it holds that

$$\cos x = \sum_{k=0}^{\infty} (-1)^k \frac{x^{2k}}{(2k)!}, \quad \sin x = \sum_{k=0}^{\infty} (-1)^k \frac{x^{2k+1}}{(2k+1)!}.$$

VI.3 The complex exponential function.

Hyperbolic functions

The basic exponential function can be extended to a function of a complex variable $\exp : \mathbb{C} \rightarrow \mathbb{C}$ in such a way that its central properties are preserved. The definition on the complex domain — which also works when restricted to a real variable — is the following.

DEFINITION VI.3.1 The basic exponential function $\exp(z) = e^z$ is the function having the properties (axioms)

- (C1) $\exp(z)$ is defined $\forall z \in \mathbb{C}$,
- (C2) $\exp(x + i0) = e^x \forall x \in \mathbb{R}$,
- (C3) $\exp(z)$ is differentiable at the origin,
- (C4) $\exp(z_1 + z_2) = \exp(z_1) \exp(z_2) \quad \forall z_1, z_2 \in \mathbb{C}$.

By axiom C2, e^z is the extension of the real exponential function to the complex plane. Axiom C3 (cf. Section V.8) corresponds to axiom E3' of the previous section. For the complex exponential function this form of the axiom is necessary, for a mere continuity assumption (corresponding to axiom E3 of the real exponential function) would not guarantee the uniqueness of the function $\exp(z)$ as a complex function (see Exercise 1). With the axioms C1–C4, on the other hand, $\exp(z)$ becomes uniquely defined. Here we show only that a function $\exp(z)$ satisfying the axioms exists, and at the same time we give for the function a computation rule valid on all of $\mathbb{C}$.

THEOREM VI.3.2 The function

$$e^{x+iy} = e^x(\cos y + i \sin y), \quad z = x + iy \in \mathbb{C} \quad (1)$$

satisfies the axioms C1–C4.

Proof (C4) If $z_1 = x_1 + iy_1$, $z_2 = x_2 + iy_2$, then by the definition (1) and the rules of complex algebra (see Section III.2),

$$\begin{aligned} e^{z_1+z_2} &= e^{x_1+x_2} [1\angle(y_1+y_2)] \\ &= e^{x_1} \cdot e^{x_2} \cdot (1\angle y_1) \cdot (1\angle y_2) \\ &= [e^{x_1} \cdot (1\angle y_1)] [e^{x_2} \cdot (1\angle y_2)] \\ &= e^{z_1} \cdot e^{z_2}. \end{aligned}$$

(C3) We use the decompositions

$$\begin{aligned} e^x &= 1 + x + r_1(x), \\ \cos y &= 1 + r_2(y), \\ \sin y &= y + r_3(y), \end{aligned}$$

where $\lim_{t \rightarrow 0} t^{-1}r_i(t) = 0$ (cf. Sections V.4 and VI.2). By these and rule (1) we have

$$\begin{aligned} e^z &= [1 + x + r_1(x)] [1 + iy + r_2(y) + ir_3(y)] \\ &= 1 + x + iy + r(z) \\ &= 1 + z + r(z), \quad z = x + iy, \end{aligned}$$

where the remainder term satisfies $\lim_{z \rightarrow 0} z^{-1}r(z) = 0$. Consequently

$$\lim_{z \rightarrow 0} \frac{e^z - e^0}{z} = \lim_{z \rightarrow 0} \frac{e^z - 1}{z} = \lim_{z \rightarrow 0} \left(1 + \frac{r(z)}{z} \right) = 1.$$

Thus e^z is differentiable at the origin and the value of the derivative = 1.

(C2) and (C1) Obvious. $\square$

The function e^z is differentiable not only at the origin (axiom C3), but on the whole complex plane, for combining the properties C3–C4 gives

$$\lim_{\Delta z \rightarrow 0} \frac{e^{z+\Delta z} - e^z}{\Delta z} = e^z \lim_{\Delta z \rightarrow 0} \frac{e^{\Delta z} - 1}{\Delta z} = e^z, \quad z \in \mathbb{C}.$$

The differentiation rule is thus the same as for the real (basic) exponential function:

$$\boxed{D e^z = e^z.} \quad (2)$$

According to this, the complex function e^z is analytic on the whole complex plane, i.e. an entire function (cf. Section V.8). With reference to this and to axiom C2, one says that e^z is the *analytic continuation* of the real exponential function to the whole complex plane.

From the differentiation rule (2) (which thus follows directly from the axioms C3–C4) it can also be deduced that the function satisfying the axioms C1–C4 is unique. Namely, if a function $E(z)$ satisfies these axioms, then $DE(z) = E(z)$, so that by differentiating the function $u(z) = E(z)e^{-z}$ one finds that $Du(z) = 0$ (cf. the proof of Theorem VI.2.1). Since by axiom C2 we have $u(0) = 1$, it can be deduced (details omitted) that $u(z) = 1 \Leftrightarrow E(z) = e^z \forall z \in \mathbb{C}$.

From the definition (1) one sees that

$$|e^z| = e^x > 0 \quad \forall z = x + iy \in \mathbb{C},$$

so the complex exponential function too has no zeros. Choosing in the definition $x = \operatorname{Re} z = 0$ yields *Euler's formula*

$$e^{iy} = \cos y + i \sin y, \quad y \in \mathbb{R}. \quad (3)$$

Using Euler's formula (3), the trigonometric functions $\cos$ and $\sin$ can be expressed by means of the complex exponential function:

$$\cos x = \frac{1}{2}(e^{ix} + e^{-ix}), \quad \sin x = \frac{1}{2i}(e^{ix} - e^{-ix}), \quad x \in \mathbb{R}. \quad (4)$$

On this basis $\cos$ and $\sin$ can also be extended to functions of a complex variable. The definitions are

$$\cos z = \frac{1}{2}(e^{iz} + e^{-iz}), \quad \sin z = \frac{1}{2i}(e^{iz} - e^{-iz}), \quad z \in \mathbb{C}. \quad (5)$$

All the computation rules characteristic of the trigonometric functions (cf. Section II.5) extend to the complex domain as well. For example, the rules

$$\begin{aligned} \cos^2 z + \sin^2 z &= 1, \\ \cos 2z &= \cos^2 z - \sin^2 z, \quad \sin 2z = 2 \sin z \cos z, \\ D \sin z &= \cos z, \quad D \cos z = -\sin z \end{aligned}$$

can be verified directly from the definitions (5) and the differentiation rule (2) of e^z . One also sees that $\sin z$ and $\cos z$ are (like e^z) analytic on the whole complex plane, i.e. entire functions.

EXAMPLE 1 Find all solutions of the equation $\sin z = 2$ in the complex plane.

Solution Writing $t = e^{iz}$, then by the definition (5)

$$\begin{aligned} \sin z = 2 &\Leftrightarrow \frac{1}{2i}(t - t^{-1}) = 2 \Leftrightarrow t^2 - 4it - 1 = 0 \\ &\Leftrightarrow t = (2 \pm \sqrt{3})i. \end{aligned}$$

By the definition (1) and axiom C3,

$$t = e^{iz} = e^{-y+ix} = e^{-y}e^{ix}, \quad z = x + iy,$$

so we must have

$$\begin{cases} e^{-y} = |t| = 2 \pm \sqrt{3}, \\ e^{ix} = i \end{cases} \Leftrightarrow \begin{cases} y = -\ln(2 \pm \sqrt{3}), \\ x = \frac{\pi}{2} + 2k\pi, \quad k \in \mathbb{Z}. \end{cases}$$

Here $\ln(2 - \sqrt{3}) = -\ln(2 + \sqrt{3})$, so the solutions are

$$z = \frac{\pi}{2} + 2k\pi \pm \ln(\sqrt{3} + 2)i, \quad k \in \mathbb{Z}. \quad \square$$

Hyperbolic functions

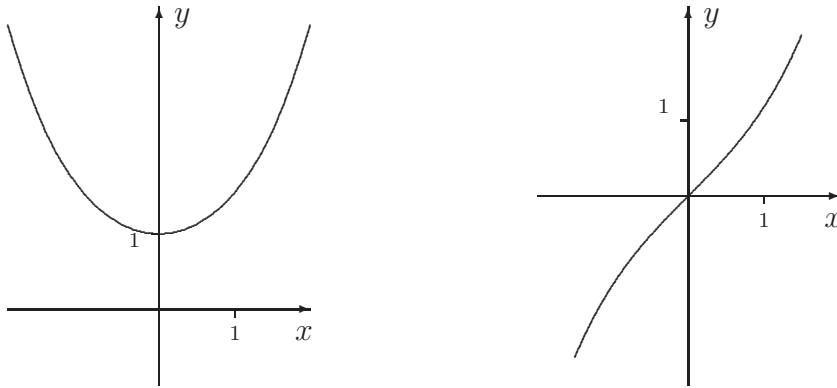
The hyperbolic functions are derived from the exponential function and have properties of a type similar to those of the trigonometric functions. The *hyperbolic cosine*, symbol $\cosh$ (cosinus hyperbolicus), and the *hyperbolic sine*, symbol $\sinh$ (sinus hyperbolicus), are defined (cf. the definition (5) of the trigonometric functions) by

$$\cosh z = \frac{1}{2}(e^z + e^{-z}), \quad \sinh z = \frac{1}{2}(e^z - e^{-z}), \quad z \in \mathbb{C}. \quad (6)$$

For example, the following laws can be verified from the definition (cf. the corresponding ones for the trigonometric functions):

$$\begin{aligned} \cosh^2 z - \sinh^2 z &= 1, \\ \cosh 2z &= \cosh^2 z + \sinh^2 z, \quad \sinh 2z = 2 \sinh z \cosh z, \\ D \sinh z &= \cosh z, \quad D \cosh z = \sinh z. \end{aligned}$$

For real values of the variable, $\cosh x = \frac{1}{2}(e^x + e^{-x})$ is an even and $\sinh x = \frac{1}{2}(e^x - e^{-x})$ an odd function. The former is strictly increasing on the interval $[0, \infty)$ and a bijection as a map $\cosh : [0, \infty) \rightarrow [1, \infty)$. The latter is strictly increasing on all of $\mathbb{R}$ and a bijection as a map $\sinh : \mathbb{R} \rightarrow \mathbb{R}$.



The inverse functions of the real functions $\cosh$ and $\sinh$ ($\cosh$ restricted to the interval $[0, \infty)$ or $(-\infty, 0]$) are called *area functions* and denoted arcosh , arsinh . Their computation rules are obtained by solving

$$\begin{aligned} \frac{1}{2}(e^y + e^{-y}) &= x \Leftrightarrow y = \operatorname{arcosh} x, \\ \frac{1}{2}(e^y - e^{-y}) &= x \Leftrightarrow y = \operatorname{arsinh} x. \end{aligned}$$

These are quadratic equations in the unknown $t = e^y$, so y can be expressed by means of logarithms:

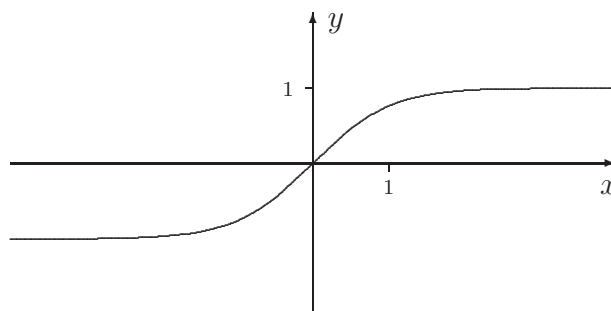
$$\begin{aligned} y = \operatorname{arcosh} x &= \pm \ln \left(x + \sqrt{x^2 - 1} \right), \quad x \geq 1, \\ y = \operatorname{arsinh} x &= \ln \left(x + \sqrt{x^2 + 1} \right), \quad x \in \mathbb{R}. \end{aligned}$$

Here the principal branch Arcosh of arcosh is obtained with the sign $+$. The other branch can also be written in the form

$$y = -\ln \left(x + \sqrt{x^2 - 1} \right) = \ln \frac{1}{x + \sqrt{x^2 - 1}} = \ln \left(x - \sqrt{x^2 - 1} \right).$$

The *hyperbolic tangent* $\tanh$ (tangens hyperbolicus) is defined (assume a real variable) by

$$\tanh x = \frac{\sinh x}{\cosh x} = \frac{e^x - e^{-x}}{e^x + e^{-x}} = \frac{1 - e^{-2x}}{1 + e^{-2x}}.$$



The hyperbolic tangent is a bijection: $\tanh : \mathbb{R} \rightarrow (-1, 1)$. Its inverse function is

$$\operatorname{artanh} x = \frac{1}{2} \ln \left(\frac{1+x}{1-x} \right), \quad |x| < 1.$$

For later use let us note further the differentiation formulas for the inverse hy-

perbolic functions

$$\begin{aligned} D \ln(x + \sqrt{x^2 + 1}) &= \frac{1}{\sqrt{x^2 + 1}}, \quad x \in \mathbb{R}, \\ D |\ln(x + \sqrt{x^2 - 1})| &= \frac{1}{\sqrt{x^2 - 1}}, \quad |x| > 1, \\ D \frac{1}{2} \ln \left| \frac{1+x}{1-x} \right| &= \frac{1}{1-x^2}, \quad |x| \neq 1. \end{aligned}$$

The complex logarithm function

The complex (basic) logarithm function is defined by

$$e^w = z \Leftrightarrow w = \log z.$$

Writing $w = u + iv$ and taking the polar representation $z = r(\cos \varphi + i \sin \varphi)$ for z , from the equation

$$e^w = z \Leftrightarrow e^u(\cos v + i \sin v) = r(\cos \varphi + i \sin \varphi)$$

one sees that we must have

$$\begin{cases} e^u = r \Leftrightarrow u = \ln r = \ln |z|, \\ v = \varphi + 2k\pi, \quad k \in \mathbb{Z}. \end{cases}$$

This is thus an infinitely many-valued function. As the definition of the logarithm function $\log z$ we agree on the principal branch of this function, where $k = 0$:

$$\log z = \ln |z| + i \arg z, \quad 0 \leq \arg z < 2\pi.$$

The function so defined is not continuous on the positive real axis, for if $z_n = x + iy_n$, $x > 0$, then $z_n \rightarrow x$ as $y_n \rightarrow 0$, but

$$\begin{aligned} y_n \rightarrow 0^+ &\Rightarrow \log z_n \rightarrow \ln x, \\ y_n \rightarrow 0^- &\Rightarrow \log z_n \rightarrow \ln x + 2\pi i. \end{aligned}$$

At the other points of the domain $\log z$ is not only continuous but also differentiable (see Exercise 12), i.e. $\log z$ is analytic in the so-called *slit complex plane*, from which the origin and the positive real axis have been removed.

Owing to the multi-valuedness of the complex logarithm function, the computation rules of the real domain do not generalize as such. For example, the logarithm of a product is, by the aforementioned definition (cf. Section VI.2),

$$\log(z_1 z_2) = \begin{cases} \log z_1 + \log z_2, & \text{if } 0 \leq \arg z_1 + \arg z_2 < 2\pi, \\ \log z_1 + \log z_2 - 2\pi i, & \text{otherwise.} \end{cases}$$

EXAMPLE 2 By the definition of the function $\log$,

$$\log(-1) = \log(1 \angle \pi) = \ln 1 + \pi i = \pi i,$$

$$\log(1+i) = \log(\sqrt{2} \angle \pi/4) = \frac{1}{2} \ln 2 + \frac{\pi}{4} i,$$

$$\log(-1-i) = \log(\sqrt{2} \angle 5\pi/4) = \frac{1}{2} \ln 2 + \frac{5\pi}{4} i. \quad \square$$

By means of the complex logarithm function one can define (cf. formula (9) of the previous section)

$$z^y = e^{y \log z}, \quad z, y \in \mathbb{C}, \quad z \neq 0.$$

EXAMPLE 3 By the previous example,

$$(-1)^\pi = e^{\pi \log(-1)} = e^{i\pi^2} = \cos \pi^2 + i \sin \pi^2. \quad \square$$

EXERCISES

1. Define an extension of the function e^x to the complex plane by the formula

$$\exp(z) = \exp(x + iy) = e^x (\cos ay + i \sin ay),$$

where $a \in \mathbb{R}$, $a \neq 1$. Which of the axioms of the complex exponential function hold for this function?

2. Check the validity of the following formulas for $z \in \mathbb{C}$:

$$\text{a) } \sin^2 z + \cos^2 z = 1, \quad \sin 2z = 2 \sin z \cos z, \quad \cos 2z = 2 \cos^2 z - 1$$

$$\text{b) } D \sin z = \cos z, \quad D \cos z = -\sin z$$

3. a) Express $\sinh \frac{x}{2}$ and $\cosh \frac{x}{2}$ in terms of $\cosh x$.
 b) For which numbers n does $(\cosh x + \sinh x)^n = \cosh nx + \sinh nx$ hold?
4. a) Define $\coth x = 1/\tanh x$ ($x \in \mathbb{R}$, $x \neq 0$). What is the range of the function $f(x) = 4 \tanh x + \coth x$?
 b) At which point does the function $f(x) = \frac{\sinh x}{1 - a \cosh x}$, $x \in \mathbb{R}$ attain its greatest value, when $a \in \mathbb{R}$, $a > 1$?
5. Show that the equation $\cos x \cosh x + 1 = 0$ has infinitely many real solutions.
6. a) Show that the functions Arcosh and arsinh have the function $\ln(2x) = \ln x + \ln 2$ as an asymptote as $x \rightarrow \infty$.
 b) Compute the derivatives of the functions $\operatorname{arsinh} x$ and $\operatorname{Arcosh} x$, both by implicit differentiation and directly from the expressions for these functions.
7. Simplify:
 - a) $\operatorname{Arcosh}(\cosh x)$ b) $\tanh(\operatorname{Arcosh} x)$ c) $\operatorname{artanh} \frac{x}{\sqrt{x^2 + 1}}$
8. Compute the following complex numbers in the standard form $x + iy$ (exact values!):
 - a) e^z , $\sinh z$, $\cosh z$ and $\tanh z$, when $z = 2 + 3i$
 - b) e^z , $\sin z$, $\cos z$ and $\tan z$, when $z = -1 - i$
 - c) e^z , $\sin z$, $\cos z$ and $\tan z$, when $z = 3 - 2i$
 - d) $\cosh(\operatorname{arsinh} \frac{4}{3})$, $\tanh(\operatorname{arsinh} \pi)$, $\operatorname{Arcosh} \sqrt{2}$, $\operatorname{artanh} 1$
 - e) e^i , i^e , $\log i$, 5^{-i} , $i^{-\pi}$, i^i
9. A computer program computes the numerical value of the number $(-\pi)^\pi$ as $-32.9139 - 15.6897i$. Check the computation!
10. Determine (exactly, in the standard form $x + iy$) all solutions of the following equations in the complex plane:
 - a) $e^z = e$, b) $\cos z = -2$, c) $\cosh z = -1$, d) $\sin z = i$.
11. For the following functions determine the largest open subset G of the complex plane in which the function is analytic:

$$\text{a) } f(z) = \frac{1}{e^z + 1} \quad \text{b) } f(z) = \frac{z}{\cos z} \quad \text{c) } f(z) = \frac{1}{z \cosh z + 2z}$$

12. (*) Derive directly from the definition of the derivative the differentiation rule

$$D \log z = \frac{1}{z}, \quad z = r \angle \varphi, \quad r > 0, \quad 0 < \varphi < 2\pi.$$

Hint: Write $\Delta z = \Delta r \angle (\varphi + \psi)$.

VI.4 Application examples of the exponential function

In physics there occurs a whole array of so-called exponential phenomena, in which the change of a physical quantity as a function of position (x) or time (t) is described by an exponential function. In the following, five examples.

EXAMPLE 1 : Radioactive decay. Let $A(t)$ be the number of radioactive nuclei of a certain species at time t . If we write

$$A(t) = A_0 E(t), \quad t \geq 0,$$

where $A_0 = A(0)$, then $E(0) = 1$. If we assume that the same law is applicable at every instant of time, then we must have

$$A(t_1 + t_2) = A_0 E(t_1 + t_2) = A(t_1)E(t_2) = A_0 E(t_1)E(t_2), \quad t_1, t_2 \geq 0.$$

Thus the function $E(t)$ has the properties

$$E(0) = 1, \quad E(t_1 + t_2) = E(t_1)E(t_2), \quad t_1, t_2 \geq 0.$$

We conclude that E is an exponential function and $A(t)$ can therefore be represented in the form

$$A(t) = A_0 e^{-at}, \quad t \geq 0,$$

where a (unit of measure 1/s) is a constant characteristic of the nuclei (positive as a dimensionless quantity). The half-life $t_{1/2}$ is defined by the condition $A(t_{1/2}) = A_0/2$, from which we must have

$$e^{-at_{1/2}} = \frac{1}{2} \quad \Leftrightarrow \quad at_{1/2} = \ln 2 \quad \Leftrightarrow \quad t_{1/2} = \frac{\ln 2}{a}. \quad \square$$

EXAMPLE 2 : Attenuation of radiation. Let $I(x)$ be the intensity of radioactive radiation as a function of the distance x traveled in an attenuating (homogeneous) medium. If we write

$$I(x) = I_0 E(x), \quad x \geq 0,$$

where $I_0 = I(0)$, then we conclude, just as in the previous example, that E is an exponential function, i.e.

$$I(x) = I_0 e^{-ax}, \quad x \geq 0,$$

where a is the attenuation constant characteristic of the medium (unit of measure $1/\text{m}$). The distance over which the intensity of the radiation has attenuated to half of the initial intensity, i.e. the so-called half-value distance, is

$$d_{1/2} = \frac{\ln 2}{a}. \quad \square$$

In the above examples we relied directly on the basic axiom E4 of the exponential function. More often one arrives at the exponential function in applications by finding that the physical quantity $y(x)$ (or $y(t)$) under consideration satisfies the differential equation of the exponential function

$$y' = ay, \tag{1}$$

where a is a (dimensional) constant. If we assume that the physically relevant values of the variable (x) are (as dimensionless quantities) nonnegative and that the value $y(0) = y_0$ is known, then this is an initial value problem: one must find a function $y = y(x)$ that is continuous on the interval $[0, \infty)$, differentiable on the interval $(0, \infty)$, and satisfies

$$\begin{cases} y' = ay, & x > 0, \\ y(0) = y_0. \end{cases}$$

The solution is $y(x) = y_0 e^{ax}$, $x \geq 0$.

EXAMPLE 3 : Electric circuit: RC. A capacitor has capacitance C and charge q_0 . At time $t = 0$ (unit of time s) the capacitor begins to discharge through a resistor R , so that the charge at time $t \geq 0$ is $q(t)$. Mathematical model?

Solution If the current through the resistor at time t is $i(t)$, then the circuit equations are

$$q(t)/C = Ri(t), \quad i(t) = -q'(t), \quad t > 0.$$

Eliminating $i(t)$ leads to the initial value problem

$$\begin{cases} q' = -aq, & t > 0, \\ q(0) = q_0, \end{cases}$$

where $a = RC$. The solution is

$$q(t) = q_0 e^{-t/\tau}, \quad t \geq 0,$$

where $\tau = 1/(RC)$ is the so-called time constant characteristic of the circuit (unit of measure s). $\square$

In applications there also frequently occurs a variation of the differential equation (1) of the exponential function, namely

$$y' = ay + b, \quad (2)$$

where a and b are constants. Assuming $a \neq 0$, one solution of this is $y_0(x) = -b/a = \text{constant}$. When a more general solution is sought in the form $y(x) = y_0(x) + v(x)$, one finds that the function v satisfies the differential equation (1) of the exponential function. The general solution of the differential equation (2), when $a \neq 0$, is thus

$$y(x) = -b/a + Ce^{ax}, \quad x \in \mathbb{R}.$$

In an application the constant C is determined from the initial condition.

EXAMPLE 4 : Electric circuit: LR. An inductance L and a resistor R are connected in series. At time $t = 0$ s the circuit is connected to a constant voltage E , so that a current $i(t)$ arises in the circuit ($i(0) = 0$). Mathematical model for the voltage $y(t)$ across the resistor?

Solution The circuit equations are

$$L i'(t) + R i(t) = E, \quad R i(t) = y(t), \quad t > 0.$$

Eliminating $i(t)$ leads to the differential equation (2), where $a = -R/L$, $b = ER/L$. The initial value is $y(0) = 0$. The solution of the initial value problem is obtained by choosing $C = b/a$ in the aforementioned general solution, giving

$$y(t) = (b/a)(e^{at} - 1) = E(1 - e^{-t/\tau}), \quad t \geq 0,$$

where $\tau = L/R$ (time constant, unit s). $\square$

EXAMPLE 5 : Law of cooling. At time t (unit s) a body has temperature $u(t)$ and heat energy $U(t) = cu(t)$ (c = heat capacity). How does the body cool from the initial temperature $u(0) = u_0$, if it is assumed that the heat flow into the surrounding air is $Q(t) = k[u(t) - u_1]$, where $u_1 < u_0$ is the temperature of the outside air?

Solution The law of conservation of energy is

$$\begin{aligned} U' = -Q &\Leftrightarrow cu' = -k(u - u_1) \\ &\Leftrightarrow u' = au + b, \quad a = -k/c, \quad b = ku_1/c. \end{aligned}$$

The general solution $u(t) = -b/a + Ce^{at} = u_1 + Ce^{at}$ of the differential equation satisfies the initial condition $u(0) = u_0$ when $C = u_0 - u_1$, so

$$u(t) = u_1 + (u_0 - u_1)e^{-t/\tau}, \quad t \geq 0,$$

where $\tau = -a^{-1} = c/k$ is the time constant of the cooling (unit s). $\square$

EXERCISES

1. The half-life of plutonium Pu^{239} is 25400 years. How much of 1000 kg of plutonium is left after a million years?
2. Of radiation propagating in the direction of the positive x -axis, 3% of the radiation intensity passes through the radiation shielding located in the interval $[0, 3]$. The shielding is built of two material layers such that in material 1, located in the interval $[0, 1]$, the radiation attenuation constant is three times as large as in material 2, located in the interval $[1, 3]$. By how much must the latter material layer be reinforced so that only 1% of the radiation passes through?
3. The annual interest rate is 5 percent and the interest is added to the principal a) according to continuous compounding, b) semiannually. What is the ratio of the principals obtained in the different ways after 50 years?
4. The initial temperature of a cooking pot is 96°C . The pot is first allowed to cool at room temperature (20°C) until its temperature is 40°C . After this the pot is placed in a refrigerator, where the temperature is 6°C . If the duration of the first cooling stage is 32 min, how long must the pot remain in the refrigerator until it has cooled to the temperature 7°C ? Assume that the same exponential law of cooling holds in the room and in the refrigerator.
5. A cup of coffee, whose temperature is initially 80°C , cools in the outside air so that after 5 minutes the coffee's temperature is 60°C and after 10 minutes 44°C . What is the temperature of the outside air?
6. On horizontal ground there is a cube-shaped water tank whose edge length is 10 m. The tank is full of water. At time $t = 0$ the bottom valve of the tank is opened, so that the tank begins to empty at the rate $kH(t)$, where $H(t)$ is the height of the water in the tank (unit m) at time t (unit h = hour) and k is a coefficient depending on the position of the valve (unit m^2/h). One hour after the valve is opened it is observed that 8% of the water has flowed out. At that point the valve is opened further, so that the flow rate instantaneously doubles. The valve is then left in its new position. Determine the amount of water $m(t)$ in the tank (in cubic meters) as a function of time t on the interval $[0, \infty)$.
7. At time $t = 0$ s a body is at rest on the surface of the water and begins to sink, obeying the law of motion

$$v'(t) = 0.8g - kv(t),$$

where v is the sinking speed, $g = 10 \text{ m/s}^2$ and k is a constant. Determine $v(t)$, $t \geq 0$ (unit of time s), given that the body sinks in very deep water at the (asymptotic) speed 3.2 m/s .

8. (*) In the previous exercise the sinking depth of the body before reaching the bottom satisfies

$$\begin{cases} h'(t) = v(t), & t > 0, \\ h(0) = 0. \end{cases}$$

At what depth is the body's sinking speed 99.9 % of the asymptotic value of the speed?

Chapter VII

Differential calculus of one variable

When the derivative is used as a tool in computation, one speaks of *differential calculus* (literally 'the calculus of small differences'). The derivative was brought into mathematics independently by the English physicist-mathematician **Isaac Newton** (1642-1727) and the German philosopher-mathematician **Gottfried Wilhelm Leibniz** (1646-1716) in the late 17th century. Leibniz has had a considerable influence on the notation of differential calculus (and of the integral calculus considered later as well) that is still in use today.

Since the derivative was from the outset a concept motivated by physics (especially in Newton's investigations), there is nothing surprising in its applicability to physics. Well into the 19th century the development of differential calculus and of the branches of mathematics arising from it was strongly tied to physics, and even today the connection has remained strong in many areas of mathematics.

In this chapter we examine the application of one-variable differential calculus to the theory of curves, to physics (among others, kinematics) and to the approximation of functions. The central result on the approximation of a function is *Taylor's theorem*, presented and proved in Section VII.4. This is a matter of generalizing the linearizing approximation into a more general polynomial approximation based on *Taylor polynomials*. Under favorable conditions a function can also be represented exactly as a *Taylor series*. Taylor series are power series — and as the beautiful conclusion of the theory of power series it turns out that every power series whose radius of convergence is positive is in fact the Taylor series of the function defined as the sum of the series. In the last section we also introduce the theory and uses of the *interpolation polynomials* commonly employed in numerical computation methods.

VII.1 Differential and rate of change

If f is differentiable at the point x , then according to the linearizing approximation principle we have approximately

$$\Delta f(x) = f(x + \Delta x) - f(x) \approx f'(x)\Delta x,$$

when $|\Delta x|$ is small. The expression $f'(x)\Delta x$ is called the *differential* of f at the point x and is denoted

$$df(x, \Delta x) = f'(x)\Delta x.$$

By means of the differential one can thus estimate approximately the change Δf in the value of the function corresponding to a small change Δx in the variable. If the error of the approximation is denoted

$$r(x, \Delta x) = f(x + \Delta x) - f(x) - f'(x)\Delta x,$$

then by the definition of the derivative

$$\lim_{\Delta x \rightarrow 0} \frac{r(x, \Delta x)}{\Delta x} = 0.$$

EXAMPLE 1 A square-shaped cabin, whose outer dimensions are $6 \text{ m} \times 6 \text{ m}$, is marketed as a 36 m^2 cabin. Estimate the cabin's true floor area using the differential, assuming a wall thickness of 20 cm .

Solution If $f(x) = x^2$, then the floor area, estimated using the differential, is

$$A = f(x - \Delta x) \approx f(x) - f'(x)\Delta x = x^2 - 2x\Delta x.$$

With the values $x = 6 \text{ m}$, $\Delta x = 0.4 \text{ m}$ (note!) we obtain

$$A \approx 31.2 \text{ m}^2.$$

The error of the approximation is in this case exactly

$$r(x, \Delta x) = (\Delta x)^2 = 0.16 \text{ m}^2,$$

i.e. the 'true' floor area is $A \approx 31.4 \text{ m}^2$. $\square$

Later, in Section VII.4, it will be shown that the error of the approximation based on the differential is in general of order $\sim (\Delta x)^2$ when the function is sufficiently regular (as in the example).

EXAMPLE 2 In the adiabatic (sudden) variations of pressure and density of an ideal gas the equation of state

$$p\rho^{-\gamma} = K = \text{const.}$$

holds, where γ is a constant characteristic of the gas ($\gamma > 1$). If the density changes by 2%, by how much does the pressure change?

Solution Since $p = K\rho^\gamma = f(\rho)$, we have

$$\begin{aligned}\Delta p &\approx f'(\rho)\Delta\rho = \gamma K\rho^{\gamma-1}\Delta\rho = \gamma p\rho^{-1}\Delta\rho \\ \Rightarrow \frac{\Delta p}{p} &\approx \gamma \frac{\Delta\rho}{\rho} = \underline{\underline{2\gamma\%}}. \quad \square\end{aligned}$$

Rate of change

The most common interpretation of the derivative in physics and other applications is

Derivative = (instantaneous) rate of change.
--

If m is some physical quantity that changes with time (t) (unit of measure s), i.e. $m = m(t)$, then the instantaneous rate of change is

$$\lim_{\Delta t \rightarrow 0} \frac{m(t + \Delta t) - m(t)}{\Delta t} = \lim_{\Delta t \rightarrow 0} \frac{\Delta m}{\Delta t} = m'(t).$$

If the unit of m is M, then the unit of the rate of change is M/s.

EXAMPLE 3 If A = national economy (unit E), then economic growth is the instantaneous rate of change of A (unit E/s). When one says 'economic growth accelerates' or 'economic growth slows down', one means the rate of change of $A'(t)$, i.e. the second derivative $A''(t)$ (unit E/s²). $\square$

If a body moves so that its position at time t is $x(t)$ (one-dimensional motion), then as is well known

$$x'(t) = \underline{\text{velocity}}, \quad x''(t) = \underline{\text{acceleration}}.$$

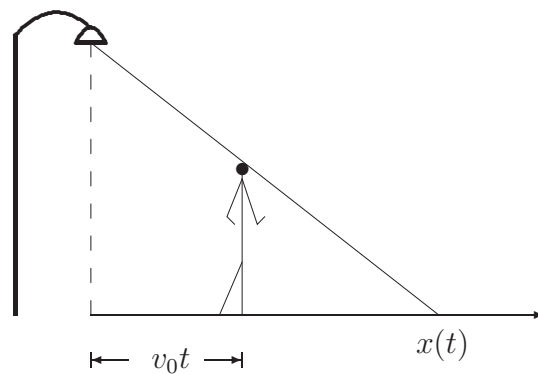
EXAMPLE 4 A two-meter-tall man walks under a street lamp at a height of four meters at time $t = 0$. Determine the position, velocity and acceleration of the tip of the man's shadow at time $t \geq 0$, given that the man walks at constant speed $v_0 = 2$ m/s. (Assume the street is horizontal.)

Solution The position of the tip of the shadow is determined by the condition (see figure below)

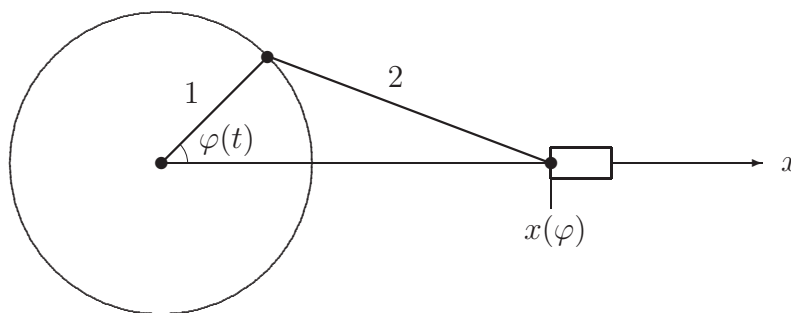
$$\frac{x(t) - v_0 t}{2} = \frac{x(t)}{4},$$

so that

$$\begin{aligned} x(t) &= 2v_0 t, \\ x'(t) &= 2v_0 = 4 \text{ m/s}, \\ x''(t) &= 0. \quad \square \end{aligned}$$



EXAMPLE 5 : **Engine.** The end of the connecting rod, whose length = 2 (unit of measure = 10 cm), attached to the crankshaft moves along the unit circle so that the polar angle at time t is $\varphi(t) = at$ ($a = \text{const.}$), and the other end (to which the piston is attached) slides along the x -axis. a) Using the differential, estimate how much the piston moves as the crankshaft rotates from angle $\varphi = 30^\circ$ to angle $\varphi + \Delta\varphi = 31^\circ$. b) What is the maximum value of the piston's velocity when the rotational speed of the crankshaft is 3600 revolutions/min?



Solution a) When the crankshaft's angle of rotation is φ , the position of the piston is

$$x(\varphi) = \cos \varphi + \sqrt{4 - \sin^2 \varphi} = \cos \varphi + \sqrt{3 + \cos^2 \varphi}.$$

When the angle changes by $\Delta\varphi$, we have

$$\Delta x(\varphi) \approx x'(\varphi)\Delta\varphi = - \left(\sin \varphi + \frac{\cos \varphi \sin \varphi}{\sqrt{3 + \cos^2 \varphi}} \right) \Delta\varphi.$$

With the values $\varphi = 30^\circ$, $\Delta\varphi = 1^\circ \hat{=} \pi/180$ we obtain

$$|\Delta x| \approx \frac{1}{2} \left(1 + \frac{1}{\sqrt{5}} \right) \frac{\pi}{180} \cdot 10 \text{ cm} \approx \underline{\underline{1.3 \text{ mm}}}.$$

b) Writing $x(t) = x(\varphi(t)) = x(at)$, we have

$$\begin{aligned} x'(t) &= a \left(-\sin at - \frac{\cos at \sin at}{\sqrt{3 + \cos^2 at}} \right), \\ x''(t) &= a^2 \left(-\cos at + \frac{\sin^2 at - \cos^2 at}{\sqrt{3 + \cos^2 at}} - \frac{\cos^2 at \sin^2 at}{(3 + \cos^2 at)^{3/2}} \right). \end{aligned}$$

At the extremum points of the velocity $x'(t)$ we must have

$$x''(t) = 0 \Leftrightarrow \cos at (3 + \cos^2 at)^{3/2} = 3 - 6 \cos^2 at - \cos^4 at.$$

Squaring both sides and setting $u = \cos^2 at$, this simplifies to the polynomial equation

$$u^3 + u^2 - 21u + 3 = 0.$$

On the interval $[0, 1]$ this has a unique solution

$$u = 0.143987.. \Rightarrow \cos \varphi(t) = \sqrt{0.143987..} = 0.379456.. \Rightarrow \varphi(t) \approx 68^\circ.$$

The corresponding maximum value of the speed is

$$|x'(t)| = a\sqrt{1-u} \left(1 + \sqrt{\frac{u}{3+u}} \right) \approx 1.12 a.$$

At the given rotational speed $a = 60 \cdot 2\pi/\text{s}$, so the numerical value is

$$v_{\max} \approx 1.12 \cdot 120\pi \text{ s}^{-1} \cdot 10 \text{ cm} \approx \underline{\underline{42 \text{ m/s}}}. \quad \square$$

Differentials and differential equations

Thinking based on differentials is very common when the mathematical model describing a natural phenomenon (or other phenomenon) is a differential equation, and one wants to derive this equation. In the following examples we arrive

at the differential equation $y' = ay$ of the exponential function from a physical (or other) law of the form

$$\Delta y(x) = y(x + \Delta x) - y(x) \approx ay(x)\Delta x, \quad \text{when } |\Delta x| \text{ small.}$$

When this is interpreted to mean that

$$\Delta y(x) = ay(x)\Delta x + r(x, \Delta x),$$

where $\lim_{\Delta x \rightarrow 0} r(x, \Delta x)/\Delta x = 0$, then dividing by Δx and letting $\Delta x \rightarrow 0$ we obtain the law, valid for 'infinitely small' changes,

$$\lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x} = \frac{dy}{dx} = ay(x).$$

EXAMPLE 6 : Radioactive decay. In a radioactive material the probability that an individual nucleus decays (and thereby 'disappears') during the time interval $[t, t + \Delta t]$ is proportional to the length Δt of the interval, when Δt is small. If the number of radioactive (as yet undecayed) nuclei at time t is $A(t)$, then we obtain as the approximate decay law

$$\frac{A(t + \Delta t) - A(t)}{\Delta t} = \frac{\Delta A}{\Delta t} \approx -a\Delta t,$$

where a is a constant characteristic of the nuclei (positive; unit of measure 1/s). In the limit $\Delta t \rightarrow 0$ we obtain the exact decay law (cf. Example VI.4:1)

$$A'(t) = -aA(t) \quad \Rightarrow \quad A(t) = A(0)e^{-at}, \quad t \geq 0. \quad \square$$

EXAMPLE 7 : Attenuation of radiation. As radiation passes through a homogeneous medium, the probability that an individual particle collides with an atomic nucleus of the medium is proportional to the distance Δx traveled, when $|\Delta x|$ is small. If the intensity of the radiation is $I(x)$ (particles/m²/s), then owing to collisions the radiation attenuates by the amount

$$I(x + \Delta x) - I(x) = \Delta I \approx -aI(x)\Delta x,$$

where a is a constant characteristic of the material (positive; unit of measure 1/m). If here we assume that the error of the approximation is $r(x, \Delta x)$, where $\lim_{\Delta x \rightarrow 0} r(x, \Delta x)/\Delta x = 0$, then dividing by Δx and letting $\Delta x \rightarrow 0$ we obtain the exact attenuation law for radiation (cf. Example VI.4:2)

$$I'(x) = -aI(x) \quad \Rightarrow \quad I(x) = I(0)e^{-ax}, \quad x \geq 0. \quad \square$$

EXAMPLE 8 : **Compound interest.** When capital grows at continuous compound interest, the amount of capital $A(t)$ obeys the law

$$A(t + \Delta t) - A(t) = \Delta A \approx aA(t)\Delta t \quad (\Delta t \text{ small}),$$

where $a = \frac{p}{100T}$, the interest rate being p percent over time T (e.g. $T = 1$ year). In the limit $\Delta t \rightarrow 0$ we arrive at the growth law of the capital

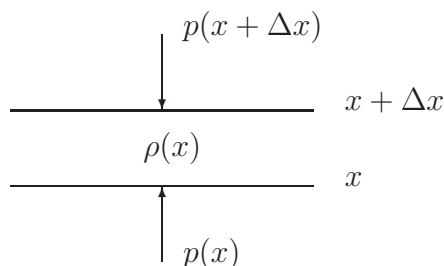
$$A'(t) = aA(t) \quad \Rightarrow \quad A(t) = A(0)e^{at}, \quad t \geq 0. \quad \square$$

EXAMPLE 9 : **Atmospheric pressure.** Determine the atmospheric pressure at height x above the earth's surface. Assume constant temperature.

Solution If the density of air at height x is $\rho(x)$, then from force balance we obtain the relation between the pressure change and the density

$$\Delta p = p(x + \Delta x) - p(x) = -g\rho(x)\Delta x + r(x, \Delta x),$$

where $g \approx 10 \text{ m/s}^2$ is the acceleration of the earth's gravity and $r(x, \Delta x)/\Delta x \rightarrow 0$ as $\Delta x \rightarrow 0$.



The equation of state of the gas at constant temperature is

$$p/\rho = K = \text{const.}$$

Substituting $\rho(x) = p(x)/K$ from this into the equilibrium equation and letting $\Delta x \rightarrow 0$ we arrive at the pressure-decrease law

$$p' = -p/a \quad \Rightarrow \quad p(x) = p(0)e^{-x/a},$$

where

$$a = \frac{K}{g} = \frac{p(0)}{\rho(0)g} \approx \frac{10^5 \text{ N/m}^2}{1 \text{ kg/m}^3 \cdot 10 \text{ m/s}^2} = 10 \text{ km.}$$

The half-value height of the pressure is $h_{1/2} = (\ln 2)a \approx 6.9 \text{ km.}$ $\square$

EXERCISES

1. Compute $df(x, \Delta x) = f'(x)\Delta x$ and $f(x + \Delta x) - f(x)$ to six decimals, when $x = 1$, $\Delta x = 0.02$ and a) $f(x) = x^2 + 200x + 700$, b) $f(x) = x^{-2} - x^{-1}$, c) $f(x) = e^x$, d) $f(x) = \ln(1 + x)$, e) $f(x) = \tan \frac{\pi x}{2}$, f) $f(x) = x^x$.
2. Using the differential, estimate how much a cube's a) volume increases when the edge lengthens by $p\%$, b) surface area decreases when the volume decreases by $p\%$.
3. An airplane flies straight over an observer on the ground at a height of 10 km. When the airplane is seen at an angle of 60° relative to the horizontal, the angle is observed to change by 4.3° in five seconds. Estimate the plane's speed using the differential.
4. A skier weighing 80 kg descends a slope of constant incline. The skier's speed v rises to 50 km/h, at which point gravity, friction and air resistance reach equilibrium. Gravity and the friction force are proportional to the skier's mass m , and the air-resistance force is $F_v = \text{const.} \times Av^\gamma$, where A is the skier's effective cross-sectional area and $\gamma \geq 1$ is a constant. Assume further that $m = \text{const.} \times L^{2.4}$ and $A = \text{const.} \times L^{1.2}$, where L is the skier's waist circumference. In the cases a) $\gamma = 1$, b) $\gamma = 2$, estimate by how many percent the skier's descending speed increases or decreases after gaining 2 kg. Use the differential! (Other conditions, such as snow conditions, are assumed constant.)
5. a) In manufacturing x refrigerators the production cost is

$$K(x) = 8000 + 200x + \max\{0, 200x - 0.5x^2\}.$$

Using the differential, estimate the so-called marginal production cost of one refrigerator, i.e. the number $K(x + 1) - K(x)$, when $x = 200$.

b) In a certain state's taxation the marginal (i.e. of a small additional income) tax percentage, as a function of the annual income x (unit 10^5 euros), is

$$p(x) = 20 + 10 \min\{x + 3x^2, 4\}.$$

At what value of annual income is the marginal tax percentage = 50? And at what value of annual income is the tax half of the annual income?

6. a) A cube's surface area grows at $50 \text{ cm}^2/\text{s}$. What is the rate of growth of the volume when the edge length = 20 cm?

b) A water tank has a volume of 400 l. Water is drained from the tank so that the amount of water (unit l) in the tank at time t (unit min) is

$$V(t) = (20 - t)^2, \quad 0 \leq t \leq 20.$$

What is the instantaneous flow rate of water from the tank when the tank is half full of water?

7. A point P moves along the x -axis in the negative direction at constant speed v_1 , and a point Q moves along the y -axis in the negative direction at constant speed v_2 . At time $t = 0$ we have $P = (a, 0)$ and $Q = (0, b)$, where $a, b > 0$. If $s(t)$ is the length of the segment PQ at time t , at what instant is $s'(t) = 0$, and what is the smallest value of s ?
8. (IKEA) During the time interval $[0, T]$, $Q_{in}(t)$ cars/h arrive at a store's parking area and $Q_{out}(t)$ cars/h leave the area (instantaneous values). Derive a differential equation (= the conservation law for cars!) for the number of cars $m(t)$ in the parking area at time t , and solve $m(t)$ on the interval $[0, T]$ when $T = 6$ h, there are 120 cars in the parking area at time $t = 0$ and

$$Q_{in}(t) = 200(1 - t/T), \quad Q_{out}(t) = 270(t/T)^2.$$

If one only wants to find out at what instant the parking area has the most cars, how is this most simply determined?

9. (*) (Lusi) A motorway narrows to a two-lane road at L. Along the motorway $Q(t)$ cars/h arrive at L (instantaneous value), and at most $Q_0 = 1800$ cars/h can drive onto the narrower road beginning at L. Let $m(t)$ = the number of cars congested at L at time t (time unit = h). Form a mathematical model of the congestion and use it to solve $m(t)$ on the interval $t \in [0, 20]$, given that $m(0) = 0$ and

$$Q(t) = 540 + \max \{0, 1440t - 180t^2\}.$$

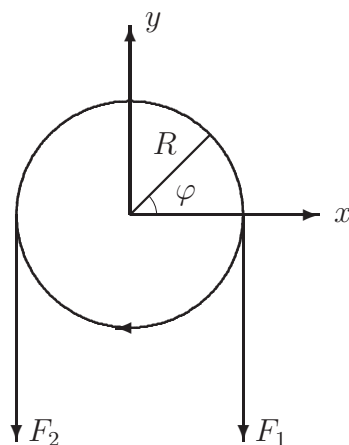
10. (*) (Threshing machine) The center of a belt pulley is at the origin and its radius is $R = 0.3$ (unit m). The pulley is rotated clockwise by a belt that touches the pulley at polar angles $\varphi \in [0, \pi]$. The belt is driven by another motor-driven pulley of the same radius, located on the negative y -axis.

Let $F(\varphi)$ be the tension force in the belt at polar angle φ , $\varphi \in [0, \pi]$. By examining the force balance and let μ be the coefficient of friction of the piece of belt on the interval

$[\varphi - \Delta\varphi/2, \varphi + \Delta\varphi/2]$, derive the differential equation

$$F'(\varphi) = -\mu F(\varphi), \quad \varphi \in (0, \pi)$$

and compute the tension forces F_1 and F_2 in the belt portions between the pulleys (see figure), when $\mu = 0.2$ and the torque twisting the pulley is $M = (F_1 - F_2)R = 1200 \text{ Nm}$.

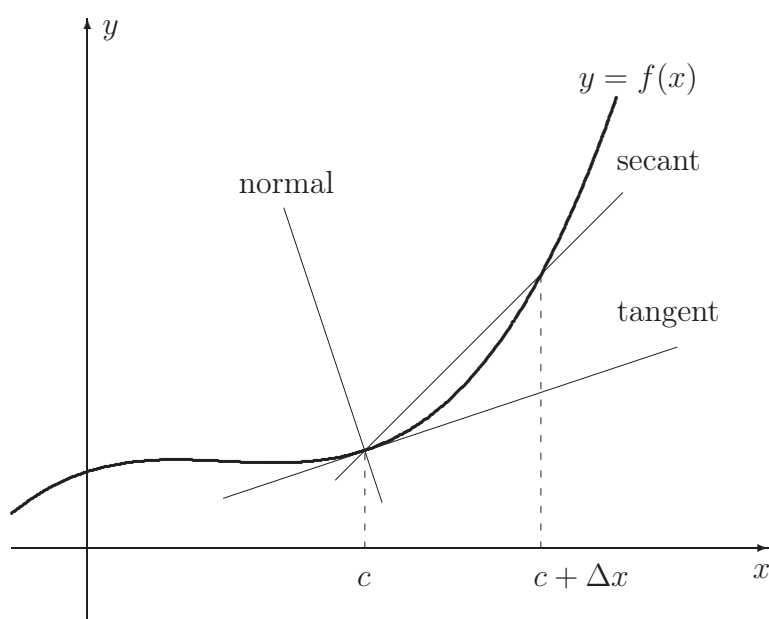


VII.2 Tangent and normal of a curve

The usual geometric interpretation of the derivative in the Euclidean plane is:

Derivative = slope of the tangent.

The line touching the curve $y = f(x)$, i.e. its *tangent*, at the point $(c, f(c))$ is the 'limit' of the line through the points $(c, f(c))$ and $(c + \Delta x, f(c + \Delta x))$, i.e. of the *secant*, as $\Delta x \rightarrow 0$. The line through the point $(c, f(c))$ that is perpendicular to the tangent is the *normal* of the curve $y = f(x)$ at that point.



The equations of the tangent and normal are

$$\text{Tangent : } y - f(c) = f'(c)(x - c).$$

$$\text{Normal : } x - c = -f'(c)(y - f(c)).$$

Two curves are said to *intersect orthogonally* if at the point of intersection the tangents are perpendicular to each other.

EXAMPLE 1 A line passes through the point $(4, 0)$ and intersects the curve $y = x^2$ orthogonally. The equation of the line?

Solution The line is the normal of the curve at the point of intersection. If the intersection point is (t, t^2) , then the equation of the line is

$$x - t = -2t(y - t^2).$$

This passes through the point $(4, 0)$ on the condition

$$2t^3 + t - 4 = 0 \Rightarrow t \approx 1.12817390.$$

Equation of the line: $y = k(x - 4)$, $k \approx -0.44319409$. $\square$

EXAMPLE 2 Show that the *families of curves*

$$xy = a, \quad x^2 - y^2 = b,$$

where $a, b \in \mathbb{R}$, $a \neq 0$, intersect each other orthogonally.

Solution At common points (x, y) we must have $x \neq 0$, $y \neq 0$, since $xy = a \neq 0$. Setting

$$xy = a \Leftrightarrow y = y(x) \quad (x \neq 0),$$

then by implicit differentiation we obtain

$$\begin{aligned} \frac{d}{dx}[xy(x)] &= y(x) + xy'(x) = \frac{d}{dx}a = 0 \\ \Rightarrow y'(x) &= -y(x)/x = -y/x. \end{aligned}$$

Similarly we can write

$$x^2 - y^2 = b \Leftrightarrow y = y(x),$$

which is a two-branched implicit function. For both branches the differentiation rule holds

$$\begin{aligned} \frac{d}{dx}(x^2 - [y(x)]^2) &= 2x - 2y(x)y'(x) = \frac{d}{dx}b = 0 \\ \Rightarrow y'(x) &= x/y(x) = x/y. \end{aligned}$$

At the common points (x, y) of the curves the product of the slopes of the tangents is thus

$$-\frac{y}{x} \cdot \frac{x}{y} = -1,$$

so the tangents are perpendicular at all intersection points. $\square$

EXAMPLE 3 : Law of reflection. We wish to determine, on an as yet unknown interval $A \subset \mathbb{R}$, a function $x \in A \mapsto y(x) \geq 0$ and the corresponding curve arc

$$S = \{ (x, y) \in \mathbb{R}^2 \mid x \in A \ \& \ y = y(x) \geq 0 \}$$

such that every ray of light emanating from the point $(-c, 0)$ and reflected from the curve passes through the point $(c, 0)$ ($c > 0$). Formulate the problem as a differential equation of the form

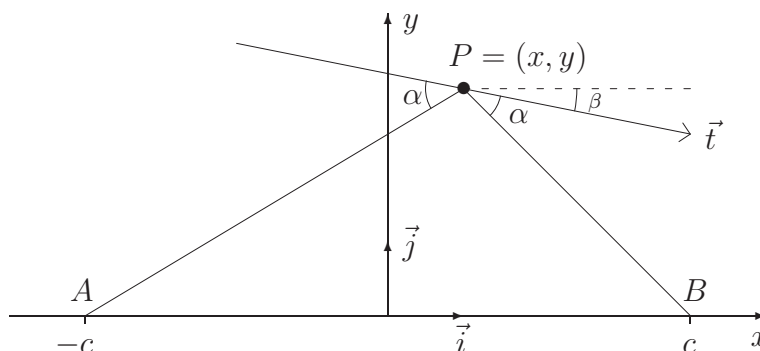
$$F(x, y, y') = 0,$$

and look for solutions among the arcs of the *ellipse*

$$y = y(x) \Leftrightarrow \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

$(a, b \in \mathbb{R}_+)$.

Solution When a ray of light reflects at the point $P = (x, y)$, the angle of incidence equals the angle of reflection with respect to the tangent of the curve.



A vector in the direction of the curve's tangent at the point of reflection is (cf. figure)

$$\vec{t} = \vec{i} + y'\vec{j},$$

and the vectors $\overrightarrow{AP}$ and $\overrightarrow{PB}$ are (figure)

$$\overrightarrow{AP} = (x + c)\vec{i} + y\vec{j}, \quad \overrightarrow{PB} = (c - x)\vec{i} - y\vec{j}.$$

The required reflection condition is

$$\begin{aligned} \frac{\overrightarrow{AP} \cdot \vec{t}}{|\overrightarrow{AP}|} &= \frac{\overrightarrow{PB} \cdot \vec{t}}{|\overrightarrow{PB}|} \\ \Leftrightarrow \frac{(c+x) + yy'}{\sqrt{(c+x)^2 + y^2}} &= \frac{(c-x) - yy'}{\sqrt{(c-x)^2 + y^2}}. \end{aligned}$$

By squaring and simplifying, this reduces to the equation

$$F(x, y, y') = xy^2(y')^2 - (c^2 - x^2 + y^2)yy' - xy^2 = 0.$$

Let us test whether this is satisfied when

$$y^2 = b^2 - \frac{b^2}{a^2}x^2 \Rightarrow y' = -\frac{b^2}{a^2}x/y.$$

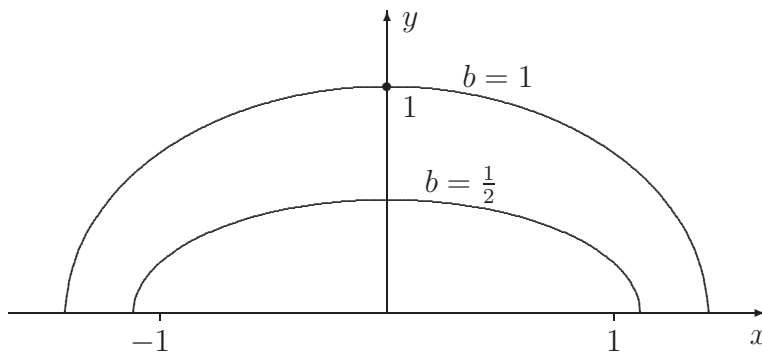
With these substitutions the differential equation reduces to the equation

$$\left(\frac{c^2 + b^2}{a^2} - 1\right)b^2x = 0,$$

which is satisfied $\forall x$ on the condition $a^2 = c^2 + b^2$. Here $b \in \mathbb{R}_+$ can be chosen freely, so the solutions of the problem are

$$y = y(x) = b\sqrt{1 - \frac{x^2}{a^2}}, \quad x \in [-a, a] = A, \quad a = \sqrt{c^2 + b^2}, \quad b \in \mathbb{R}_+.$$

These are half-arcs of ellipses of various shapes whose *foci* are $(\pm c, 0)$. In the figure $c = 1$. $\square$



Tangent of a parametric curve

The derivative $\vec{r}'(t)$ of a parametric curve, i.e. of a vector-valued function $t \in A \mapsto \vec{r}(t) \in \mathbb{R}^d$, where $A \subset \mathbb{R}$ and $d = 2$ or $d = 3$ (cf. Section IV.4), is defined

in the usual way, namely

$$\vec{r}'(t) = \lim_{\Delta t \rightarrow 0} \frac{1}{\Delta t} [\vec{r}(t + \Delta t) - \vec{r}(t)] = \begin{cases} x'(t)\vec{i} + y'(t)\vec{j} & (\text{plane curve}) \\ x'(t)\vec{i} + y'(t)\vec{j} + z'(t)\vec{k} & (\text{space curve}) \end{cases}$$

at those points where the functions $x(t)$, $y(t)$ and $z(t)$ are differentiable. By the definition, the derivative vector is the limit of the direction vector of the line through the points $P \hat{=} \vec{r}(t)$ and $Q \hat{=} \vec{r}(t + \Delta t)$, insofar as the points P and Q are distinct when $|\Delta t|$ is sufficiently small. The last-mentioned condition holds at least if

$$|\vec{r}'(t)| \neq 0, \quad (1)$$

for according to the linearizing approximation principle

$$|\vec{r}(t + \Delta t) - \vec{r}(t)| = |\vec{r}'(t)||\Delta t| + g(t, \Delta t), \quad \lim_{\Delta t \rightarrow 0} \frac{g(t, \Delta t)}{\Delta t} = 0,$$

so that under condition (1) the left-hand side here is $\neq 0$ when $\Delta t \in (-\delta, \delta)$ for some (sufficiently small) $\delta > 0$. Under condition (1) the vector $\vec{r}'(t) \neq \vec{0}$ can thus be interpreted as the *tangent vector* of the curve $S = \{P(t) \hat{=} \vec{r}(t) \mid t \in A\}$ at the point $P(t) \hat{=} \vec{r}(t)$. The unit vector $\vec{\tau}(t)$ in the direction of the vector $\vec{r}'(t)$ is called the *unit tangent vector*:

$$\boxed{\vec{\tau}(t) = \frac{\vec{r}'(t)}{|\vec{r}'(t)|} \quad (\text{unit tangent vector, } |\vec{r}'(t)| \neq 0).} \quad (2)$$

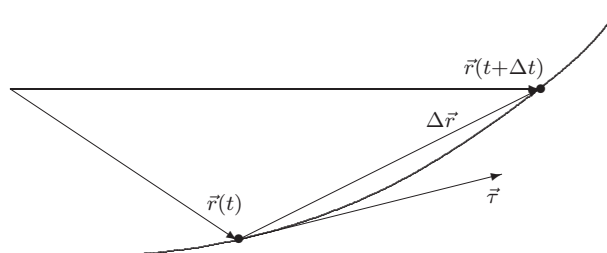
Note that the unit tangent vector at a given point P of the curve is a purely geometric concept: there are only two possible values of $\vec{\tau}$ (insofar as $\vec{\tau}$ is defined at all). When condition (1) holds, the parametrization selects one of these according to formula (2); otherwise $\vec{\tau}$ does not depend on the parametrization.

The unit tangent vector can be interpreted more clearly geometrically when we set

$$\Delta \vec{r} = \vec{r}(t + \Delta t) - \vec{r}(t).$$

If the points $P \hat{=} \vec{r}(t)$ and $Q \hat{=} \vec{r}(t + \Delta t)$ are distinct, then $\Delta \vec{r}$ is the direction vector of the secant of the curve through the points P and Q . When condition (1) holds, the approximation $|\Delta \vec{r}| \approx |\vec{r}'(t)||\Delta t|$ becomes more accurate in the limit $\Delta t \rightarrow 0$, whereby both values of the unit tangent vector are obtained as one-sided limits

$$\vec{\tau}_{\pm} = \lim_{\Delta t \rightarrow 0^{\pm}} \frac{\Delta \vec{r}}{|\Delta \vec{r}|}.$$



When studying the geometry of a curve by means of a parametrization, condition (1) is often imposed at the outset as a 'convenience condition' on the whole interval under consideration, so that formula (2) is available. Typically one assumes that the coordinate functions $x(t)$, $y(t)$, $z(t)$ of $\vec{r}(t)$ are continuously differentiable on the closed interval $A = [a, b]$ and that condition (1) holds on the whole interval, including the endpoints, where the derivative $\vec{r}'$ is interpreted as one-sided.

EXAMPLE 4 The unit tangent vector of the helix $\vec{r}(\varphi) = R \cos \varphi \vec{i} + R \sin \varphi \vec{j} + a\varphi \vec{k}$ ($R > 0$, $a \neq 0$) at the point $P(\varphi) = (x(\varphi), y(\varphi), z(\varphi))$ is

$$\vec{\tau}(\varphi) = \pm \frac{-R \sin \varphi \vec{i} + R \cos \varphi \vec{j} + a \vec{k}}{\sqrt{R^2 + a^2}} = \pm \frac{1}{\sqrt{R^2 + a^2}} (-y \vec{i} + x \vec{j} + a \vec{k}).$$

Here $|\vec{r}'(\varphi)| = \sqrt{R^2 + a^2} = \text{const.}$ $\square$

EXAMPLE 5 The curve $y = x^2$, $z = x^3$ intersects the plane T orthogonally at the point $(1, 1, 1)$. The equation of the plane?

Solution One parametrization of the curve is

$$x(t) = t, \quad y(t) = t^2, \quad z(t) = t^3 \Leftrightarrow \vec{r}(t) = t \vec{i} + t^2 \vec{j} + t^3 \vec{k}.$$

The plane T passes through the point $(1, 1, 1) \hat{=} \vec{r}(1)$ and its normal vector is $\vec{n} = \vec{r}'(1) = \vec{i} + 2\vec{j} + 3\vec{k}$, so the equation of the plane is (cf. Section II.7)

$$(x - 1) + 2(y - 1) + 3(z - 1) = 0 \Leftrightarrow x + 2y + 3z - 6 = 0. \quad \square$$

Velocity vector

If in a parametric curve $t \mapsto \vec{r}(t)$ the parameter t is a physical time variable and $\vec{r}(t)$ is the position at time t of a point (particle, etc.) moving in space, then the velocity (vector) and the path speed, i.e. the speed, of that point are defined:

<u>Velocity vector:</u> $\vec{v}(t) = \vec{r}'(t) = x'(t)\vec{i} + y'(t)\vec{j} + z'(t)\vec{k}.$
<u>Speed:</u> $v(t) = \vec{r}'(t) .$

In a time parametrization, condition (1) is thus a 'no-stopping rule'.

EXAMPLE 6 A particle moves along a circle of radius R centered at the origin so that the polar angle at time t is $\varphi(t)$. Velocity and speed at time t ?

Solution The position vector of the particle at time t is $\vec{r}(t) = R \cos \varphi(t) \vec{i} + R \sin \varphi(t) \vec{j}$, so

$$\vec{v}(t) = R\varphi'(t) [-\sin \varphi(t) \vec{i} + \cos \varphi(t) \vec{j}] = \begin{cases} v(t) \vec{\tau}_+(t), & \text{if } \varphi'(t) \geq 0, \\ v(t) \vec{\tau}_-(t), & \text{if } \varphi'(t) < 0, \end{cases}$$

where $v(t) = R|\varphi'(t)|$ is the speed and $\vec{\tau}_{\pm}(t) = \pm[-\sin \varphi(t) \vec{i} + \cos \varphi(t) \vec{j}]$ is the unit vector in the direction of motion = the unit tangent vector of the trajectory. $\square$

EXAMPLE 7 : Cycloid. A wheel of radius R rolls without slipping along the x -axis so that the velocity of the wheel's center is $v_0 \vec{i}$, $v_0 = \text{const}$. A point P on the wheel's rim is at the origin at time $t = 0$. What is the velocity of P as a function of time?

Solution

The trajectory of the point P is a cycloid whose time parametrization is (cf. Section IV.4)

$$\begin{aligned} x(t) &= v_0 t - R \sin \varphi(t), \\ y(t) &= R - R \cos \varphi(t), \end{aligned}$$

where

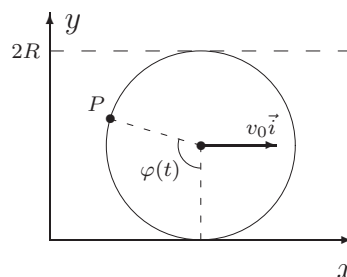
$$R\varphi(t) = v_0 t,$$

so the position vector of P at time t is

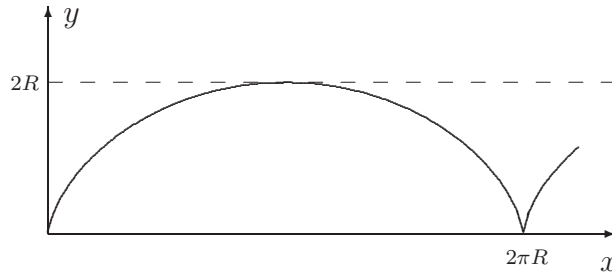
$$\vec{r}(t) = \left[v_0 t - R \sin \frac{v_0 t}{R} \right] \vec{i} + R \left[1 - \cos \frac{v_0 t}{R} \right] \vec{j}.$$

The velocity and speed at time t are

$$\begin{aligned} \vec{v}(t) &= \vec{r}'(t) = v_0 \left[\left(1 - \cos \frac{v_0 t}{R} \right) \vec{i} + \sin \frac{v_0 t}{R} \vec{j} \right], \\ v(t) &= |\vec{r}'(t)| = |v_0| \sqrt{2 - 2 \cos \left(\frac{v_0 t}{R} \right)} = 2|v_0| \left| \sin \frac{v_0 t}{2R} \right|. \end{aligned}$$



The minimum value of the speed $v_{\min} = 0$ is reached whenever the point P touches the x -axis, and the maximum value $v_{\max} = 2|v_0|$ whenever P is at height $2R$. $\square$



EXERCISES

- Determine the equations of the tangent and normal of the following plane curves in the standard form $ax + by + c = 0$ at the given point:
 - $y = x^3 - 2x^2 + x + 1$, $(x, y) = (2, 3)$
 - $y = \ln x$, $(x, y) = (e, 1)$
 - $x^2y^3 - x^3y^2 = 12$, $(x, y) = (-1, 2)$
 - $x \sin(xy - y^2) = x^2 - 1$, $(x, y) = (1, 1)$
 - $x^y = y^x$, $(x, y) = (2, 4)$
- Determine a, b, c so that the curves $y = x^2 + ax + b$ and $y = cx - x^2$ are tangent to each other (i.e. have a common tangent) at the point $(1, 3)$.
 - Determine the equations of the normals of the curve $9y = x^2$ passing through the point $(-4, 23/2)$.
 - At which point of the curve $y = x^3$ does the normal placed there intersect the x -axis at the point $(4, 0)$?
 - Determine the intersection points of the tangents and normals of the curves $y = \sinh x$ and $y = \cosh x$ placed at the point $x = a$.
 - A line passes through the point $(3, 0)$ and intersects the curve $y = e^x$ orthogonally. Determine the equation of the line, by numerical means if necessary.
 - Which line intersects the curves $y = e^x$ and $y = \ln x$ orthogonally? What is the shortest distance between the curves?
- Determine the equations of the tangent and normal of the *logarithmic spiral* $r = e^\varphi$, $\varphi \in \mathbb{R}$ (polar coordinates) at the point $(r, \varphi) = (1, 0)$.

- b) Show by implicit differentiation that the curve $S : x^5 + x^2y^3 + y^5 = 1$ has at least one horizontal (i.e. parallel to the x -axis) and one vertical (parallel to the y -axis) tangent. At which points of the curve are these located?
4. a) Under what condition do the curves $y = e^{ax}$ and $y = e^{bx}$ intersect orthogonally? At what angle do the curves intersect if $a = 1$ and $b = 2$?
- b) Compute the angle between the curves $y = \operatorname{Arcsin} x$ and $y = \operatorname{Arccos} x$ (their tangents) at the intersection point of the curves.
- c) Show that, insofar as the curves $y = ae^x$ and $y = \sqrt{b - 2x}$ ($a, b \in \mathbb{R}$) intersect, they intersect orthogonally. Under what condition do the curves intersect?
5. The space line S is the tangent of the space curve

$$\vec{r}(t) = 7\sqrt{2} \cos t \vec{i} + \sqrt{2} \sin t (3\vec{i} - 2\vec{j} + 6\vec{k})$$

at the point $(4, 2, -6)$. Determine the equation of S in parametric form.

6. Determine the unit tangent vector of the following space curves at the given point P of the curve, as well as the equation of the plane that the curve intersects orthogonally at P :
- a) $\vec{r} = t^3 \vec{i} + (2t - t^2) \vec{j} + (3t - 2t^4) \vec{k}$, $P = (1, 1, 1)$
- b) $\vec{r} = e^t \vec{i} - \ln(t + 1) \vec{j} - \cos t \vec{k}$, $P = (1, 0, -1)$
- c) $x = 2\sqrt{2} \cos t$, $y = \sqrt{2} \sin t$, $z = 4t$, $P = (-2, 1, 3\pi)$
7. Show that the intersection points of the tangents of the helix $\vec{r}(t) = \cos t \vec{i} + \sin t \vec{j} + t \vec{k}$, $t \in \mathbb{R}$ with the xy -plane form the plane curve

$$x = \cos t + t \sin t, \quad y = \sin t - t \cos t, \quad t \in \mathbb{R}.$$

8. A point-like body moves in the plane so that at time t the body is at the point $P(t) = (x(t), y(t))$, where $x(t) = \sin f(t)$, $y(t) = 1 - \cos f(t)$, and further

$$f(t) = \begin{cases} t^2(3 - t)^2, & \text{when } t \in [0, 3], \\ 0, & \text{otherwise.} \end{cases}$$

At which instants is the body's speed (= the absolute value of velocity) greatest, and what kind of geometric trace does the body's motion leave?

9. (*) Determine (by numerical means if necessary) the equation of the line S from the data:
- a) S is tangent to the curves $y = x^2$ and $y = \ln x$.
- b) S intersects the curves $y = x^2$ and $y = \ln x$ orthogonally.

10. (*) The space curve

$$S : \begin{cases} x^3 + 2y^3 = 3x^2yz, \\ y^3 + z^3 = 2xy \end{cases}$$

intersects the plane T orthogonally at the point $(1, 1, 1)$. Determine the equation of T in the standard form $ax + by + cz + d = 0$.

11. (*) (Spherical aberration) A ray of light propagating in space along the
- xy
- plane in the direction of the negative
- y
- axis reflects at the point
- P
- from the spherical mirror

$$S : x^2 + (y - R)^2 + z^2 = R^2, \quad 0 \leq y \leq R,$$

whereby the reflected ray intersects the y -axis at the point $y = c$. Compute an expression for c as a function of the x -coordinate of P . Verify that for small values of the ratio $|x|/R$ we have approximately $c(x) \approx R/2$ and that the deviation (the so-called spherical aberration) is approximately

$$c(x) - \frac{R}{2} \approx -\frac{3R}{4} \left(\frac{x}{R}\right)^2.$$

12. (*) (Radar) Radio waves emanating from the origin reflect from a surface of revolution whose axis of symmetry is the
- y
- axis and whose profile in the
- xy
- plane is the curve
- $y = y(x)$
- ,
- $x \in \mathbb{R}$
- . The profile is chosen so that the reflected waves travel in the direction of the positive
- y
- axis regardless of the point of reflection. Show that this condition can be expressed as the differential equation

$$(xy' - y)^2 = x^2 + y^2.$$

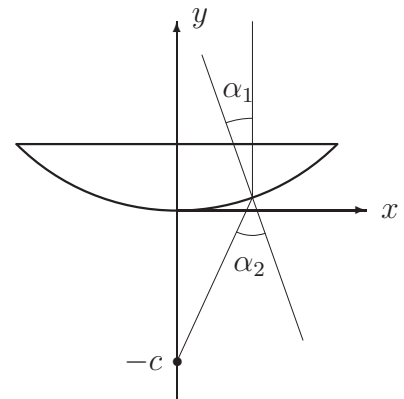
Find the possible polynomial solutions of the form $y(x) = ax^2 + b$.

13. (*) (Burning glass) A lens is made of glass, whose optical axis (axis of symmetry) is the
- y
- axis and whose profile in the
- xy
- plane is

$$A = \{(x, y) \in \mathbb{R}^2 \mid x \in [-a, a] \ \& \ f(x) \leq y \leq b\},$$

where f is an even function and $f(0) = 0$. The shape of the curve $y = f(x)$ is to be chosen so that the lens gathers to the point $(0, -c)$ all rays of light traveling in the direction of the negative y -axis that have passed through the lens. The law of refraction at the curved surface of the lens is $k \sin \alpha_1 = \sin \alpha_2$, where α_1 is the ray's angle of incidence (in the glass) with respect to the surface normal, α_2 is the angle of departure,

and $k > 1$ is the refractive index between the glass and air.



Show that the function $y = f(x)$ must satisfy the differential equation

$$\left[k^2 x^2 + (k^2 - 1)(y + c)^2 \right] (y')^2 - 2x(y + c)y' = x^2.$$

VII.3 Curvature of a curve

Consider a (parametrized) curve in the plane or in space

$$S = \{P(t) \in \mathbb{R}^d \mid P(t) \hat{=} \vec{r}(t), \quad t \in [a, b]\},$$

where

$$\vec{r}(t) = \begin{cases} x(t)\vec{i} + y(t)\vec{j}, & (\mathbb{R}^2) \\ x(t)\vec{i} + y(t)\vec{j} + z(t)\vec{k}. & (\mathbb{R}^3) \end{cases}$$

In what follows we assume that $x(t)$, $y(t)$ and $z(t)$ are twice continuously differentiable on the interval $[a, b]$. We denote further $|\vec{r}'(t)| = v(t)$ and impose the 'no-stopping rule' (cf. the previous section)

$$v(t) = |\vec{r}'(t)| > 0 \quad \forall t \in [a, b]. \quad (1)$$

The following differentiation rules for vector-valued functions will also come into use.

PROPOSITION VII.3.1 If f is differentiable at the point t and

$$\vec{u}(t) = u_1(t)\vec{i} + u_2(t)\vec{j} + u_3(t)\vec{k}, \quad \vec{v}(t) = v_1(t)\vec{i} + v_2(t)\vec{j} + v_3(t)\vec{k},$$

where the functions u_i and v_i are differentiable at the point t , then

$$\begin{aligned} \frac{d}{dt} [f(t)\vec{u}(t)] &= f'(t)\vec{u}(t) + f(t)\vec{u}'(t), \\ \frac{d}{dt} [\vec{u}(t) \cdot \vec{v}(t)] &= \vec{u}'(t) \cdot \vec{v}(t) + \vec{u}(t) \cdot \vec{v}'(t), \\ \frac{d}{dt} [\vec{u}(t) \times \vec{v}(t)] &= \vec{u}'(t) \times \vec{v}(t) + \vec{u}(t) \times \vec{v}'(t). \end{aligned}$$

Proof These are generalizations of the product rule for differentiation (Section V.3). As an example let us derive the second of the rules (the others are justified in a similar way):

$$\begin{aligned} \frac{d}{dt} \vec{u}(t) \cdot \vec{v}(t) &= \frac{d}{dt} \sum_{i=1}^3 u_i(t) v_i(t) = \sum_{i=1}^3 \frac{d}{dt} u_i(t) v_i(t) \\ &= \sum_{i=1}^3 [u'_i(t) v_i(t) + u_i(t) v'_i(t)] \\ &= \sum_{i=1}^3 u'_i(t) v_i(t) + \sum_{i=1}^3 u_i(t) v'_i(t) = \vec{u}'(t) \cdot \vec{v}(t) + \vec{u}(t) \cdot \vec{v}'(t). \quad \square \end{aligned}$$

Let us recall from the previous section that when we set

$$\Delta \vec{r} = \vec{r}(t + \Delta t) - \vec{r}(t) \approx \vec{r}'(t) \Delta t,$$

then, when condition (1) holds, the unit tangent vector $\vec{\tau}(t)$ of the curve S at the point $t \in [a, b]$ can be determined as one-sided limits

$$\vec{\tau}_{\pm} = \lim_{\Delta t \rightarrow 0^{\pm}} \frac{\Delta \vec{r}}{|\Delta \vec{r}|}, \quad t \in [a, b].$$

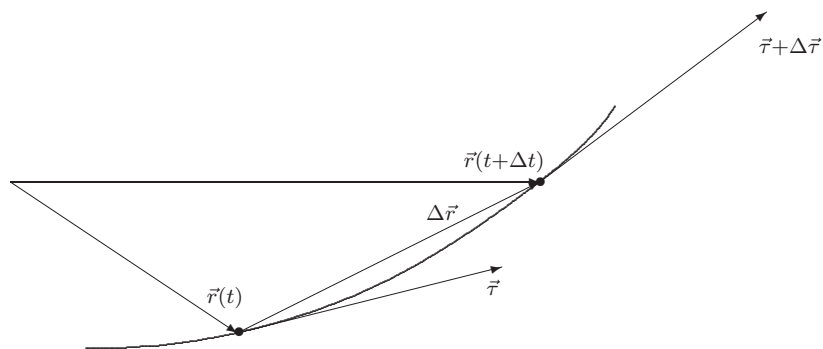
(At the endpoints of the interval only one of the limits is possible.) When the function $t \mapsto \vec{r}(t)$ is twice continuously differentiable on the interval $[a, b]$, the derivative $\vec{\tau}'(t)$ of the unit tangent vector can further be computed for $t \in [a, b]$ (a one-sided derivative at the endpoints of the interval). If the curve S is a line segment, then $\vec{\tau}(t)$ is constant, so that $\vec{\tau}'(t) = \vec{0}$, $t \in [a, b]$. More generally, the absolute value $|d\vec{\tau}/dt|$ tells how curved the curve is at that point. When the change in the tangent vector $\vec{\tau}$ is denoted

$$\Delta \vec{\tau} = \vec{\tau}(t + \Delta t) - \vec{\tau}(t),$$

then the *curvature* of the curve is defined as the limit

$$\kappa(t) = \frac{1}{R(t)} = \lim_{\Delta t \rightarrow 0} \frac{|\Delta \vec{\tau}|}{|\Delta \vec{r}|} = \frac{1}{v(t)} \left| \frac{d\vec{\tau}}{dt} \right| \quad (\text{curvature}). \quad (2)$$

Here $v(t) = |\vec{r}'(t)|$, and $R = 1/\kappa$ is called the *radius of curvature*.



Just like the unit tangent vector, the curvature too depends only on the geometry of the curve, not on the parametrization.

EXAMPLE 1 A particle moves in the plane along a circular path of radius R centered at the origin so that the polar angle at time t is $\varphi(t)$. Determine the curvature of the trajectory at the point $P(t) \hat{=} \vec{r}(t)$, where $\varphi'(t) \neq 0$.

Solution Here we have (cf. Example 6 in the previous section)

$$\vec{r}(t) = R[\cos \varphi(t)\vec{i} + \sin \varphi(t)\vec{j}], \quad \vec{\tau}(t) = \pm[-\sin \varphi(t)\vec{i} + \cos \varphi(t)\vec{j}],$$

so $v(t) = |\vec{r}'(t)| = R|\varphi'(t)|$ and $|d\vec{\tau}/dt| = |\varphi'(t)|$. By formula (2) the curvature is $\kappa(t) = 1/R = \text{const.}$, and thus the radius of curvature $= R$, as one would expect. $\square$

The unit vector $\vec{n}$ in the direction of the vector $d\vec{\tau}/dt$ is called the *principal normal vector* of the curve. This is indeed a normal vector of the curve, i.e. perpendicular to the tangent vector. Namely, since $|\vec{\tau}(t)| = 1 \ \forall t$, by the rules of Proposition VII.3.1 we deduce

$$\vec{\tau}(t) \cdot \vec{\tau}(t) = 1 \quad \Rightarrow \quad \frac{d}{dt} \vec{\tau}(t) \cdot \vec{\tau}(t) = 2 \vec{\tau}(t) \cdot \vec{\tau}'(t) = \frac{d}{dt} 1 = 0.$$

(More generally, the derivative of a unit vector is always perpendicular to the vector.) Taking formula (2) into account, we have thus arrived at the vector-form definition of curvature

$$\boxed{\frac{1}{v(t)} \frac{d\vec{\tau}}{dt} = \frac{1}{R} \vec{n}, \quad \begin{cases} 1/R = \text{curvature,} \\ \vec{n} = \text{principal normal vector.} \end{cases}} \quad (3)$$

EXAMPLE 1 (continued) Here formula (3) gives, as expected,

$$\vec{n}(t) = \frac{R}{v(t)} \frac{d\vec{\tau}}{dt} = -\cos \varphi(t)\vec{i} - \sin \varphi(t)\vec{j} = -\vec{r}(t)/R. \quad \square$$

The principal normal vector $\vec{n}$ is thus, of the curve's normal vectors (two choices!), the one that points in the direction in which the curve bends. Differentiating both sides of the equations $\vec{n}(t) \cdot \vec{n}(t) = 1$ and $\vec{n}(t) \cdot \vec{\tau}(t) = 0$ and applying formula (3), we obtain

$$\frac{d\vec{n}(t)}{dt} \cdot \vec{n}(t) = 0, \quad \frac{d\vec{n}(t)}{dt} \cdot \vec{\tau}(t) = -\vec{n}(t) \cdot \frac{d\vec{\tau}(t)}{dt} = -\frac{v(t)}{R}.$$

Accordingly, the definition of curvature (3) can also be expressed in the form

$$\boxed{\frac{1}{v(t)} \frac{d\vec{n}}{dt} = -\frac{1}{R} \vec{\tau}.} \quad (4)$$

Let us further derive an expression for the curvature whose value is determined directly from the derivatives $\vec{r}'$ and $\vec{r}''$. By the definition and the first rule of Proposition VII.3.1 we have

$$\frac{1}{R}\vec{n} = \frac{1}{|\vec{r}'|} \frac{d}{dt} \left(\frac{1}{|\vec{r}'|} \vec{r}' \right) = \frac{1}{|\vec{r}'|^2} \vec{r}'' + \frac{d}{dt} \left(\frac{1}{|\vec{r}'|} \right) \vec{r}.$$

Cross-multiplying this by the vector $\vec{r}$ and taking absolute values of both sides, we obtain

$$\frac{1}{R} = \frac{|\vec{r}'' \times \vec{r}|}{|\vec{r}'|^2}.$$

Substituting $\vec{r} = \vec{r}'/|\vec{r}'|$ into this, the computational formula becomes

$$\kappa = \frac{1}{R} = \frac{|\vec{r}' \times \vec{r}''|}{|\vec{r}'|^3}. \quad (5)$$

Let us interpret this for the plane curve $y = f(x)$. When the parameter is $t = x$, then $\vec{r}(x) = x\vec{i} + f(x)\vec{j}$, $\vec{r}' = \vec{i} + f'(x)\vec{j}$, $\vec{r}'' = f''(x)\vec{j}$, so $\vec{r}' \times \vec{r}'' = f''(x)\vec{k}$. Substituting this into formula (5) we find that the curvature of the plane curve at the point $(x, f(x))$ is

$$\kappa = \frac{1}{R} = \frac{|f''(x)|}{[1 + (f'(x))^2]^{3/2}} \quad (\text{plane curve } y = f(x)). \quad (6)$$

Without the absolute-value signs this is called the *signed* curvature.

EXAMPLE 2 The radius of curvature of the curve $S : y = x^2$ at the point (x, x^2) is, by formula (6),

$$R(x) = \frac{1}{2} (1 + 4x^2)^{3/2}. \quad \square$$

Center of curvature. Evolute

When one travels from the point $P(t) \hat{=} \vec{r}(t)$ of a parametric curve a distance equal to the radius of curvature R in the direction of the principal normal vector $\vec{n}$, one arrives at the *center of curvature*. Its position vector is thus

$$\vec{r}_0(t) = \vec{r}(t) + R(t)\vec{n}(t) \quad (\text{center of curvature}).$$

Geometrically this can be interpreted so that in a neighborhood of the point $P(t) \hat{=} \vec{r}(t)$ the curve is approximately a circle whose radius = radius of curvature R and whose center = the center of curvature. Moreover, this so-called *circle of curvature* lies in the plane whose direction vectors are $\vec{r}$ and $\vec{n}$. As the circle of curvature 'rolls' along the curve, the center of curvature traces another curve, which is called the *evolute* of the original curve.

EXAMPLE 3 Determine the evolute of the curve $y = 1/x$, $x > 0$.

Solution Since $f'(x) = -1/x^2$, the (principal) normal vector of the curve at the point $(x, 1/x)$ is

$$\vec{n} = (1 + x^{-4})^{-1/2}(x^{-2}\vec{i} + \vec{j}) = (x^4 + 1)^{-1/2}(\vec{i} + x^2\vec{j}).$$

The curvature at that point is

$$\frac{1}{R} = \frac{2x^{-3}}{(1 + x^{-4})^{3/2}} = \frac{2x^3}{(x^4 + 1)^{3/2}}.$$

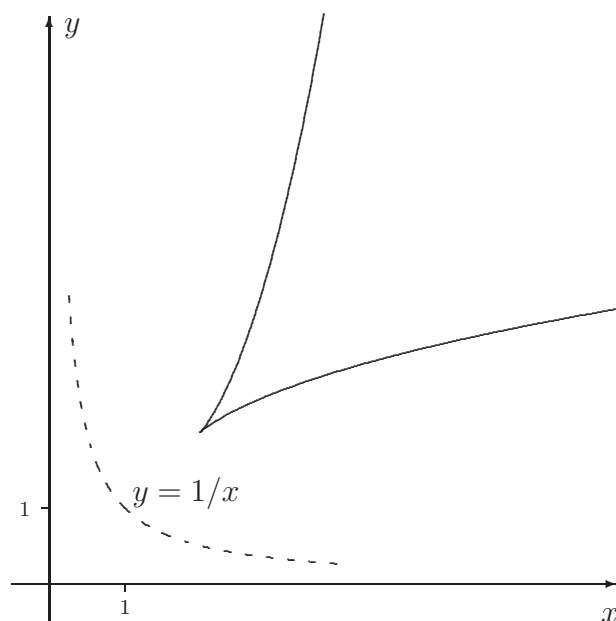
The center of curvature $(x(t), y(t))$ corresponding to the point $(t, 1/t)$ of the curve is therefore

$$\begin{aligned} x(t) &= t + \frac{(t^4 + 1)^{3/2}}{2t^3} \cdot (t^4 + 1)^{-1/2} = \frac{1}{2}(3t + t^{-3}), \\ y(t) &= t^{-1} + \frac{(t^4 + 1)^{3/2}}{2t^3} \cdot t^2(t^4 + 1)^{-1/2} = \frac{1}{2}(3t^{-1} + t^3). \end{aligned}$$

This is the parametric representation of the evolute ($t > 0$). For large and small values of t we have approximately

$$\begin{cases} t \gg 1 \Rightarrow y \approx \frac{4}{27}x^3, \\ t \ll 1 \Rightarrow x \approx \frac{4}{27}y^3. \end{cases}$$

At the point $(x, y) = (2, 2)$ ($t = 1$) the evolute has a so-called *turning point*. At the turning point $x'(t) = y'(t) = 0$, so it is also a 'stopping point'. $\square$



Curvature in physics: acceleration

When a parametric curve concerns the position of a moving point $P(t) \hat{=} \vec{r}(t)$ as a function of time, then $\vec{v}(t) = \vec{r}'(t)$ is the velocity vector. Then the vector

$$\vec{a}(t) = \vec{v}'(t) = \vec{r}''(t) = x''(t)\vec{i} + y''(t)\vec{j} + z''(t)\vec{k}$$

is called the acceleration at time t . On a curving trajectory the acceleration vector can always be represented in the form

$$\vec{a}(t) = a_\tau \vec{\tau} + a_n \vec{n},$$

where a_τ is the tangential acceleration in the direction of the trajectory and a_n is the normal acceleration in the direction of the principal normal vector. This result is arrived at by writing

$$\vec{v}(t) = v(t) \vec{\tau}(t), \quad v(t) = |\vec{v}(t)| = |\vec{r}'(t)|$$

and using the first rule of Proposition VII.3.1 and formula (3):

$$\begin{aligned} \vec{a}(t) &= v'(t) \vec{\tau}(t) + v(t) \frac{d\vec{\tau}}{dt} \\ &= v'(t) \vec{\tau}(t) + \frac{[v(t)]^2}{R} \vec{n}(t). \end{aligned}$$

Accordingly we have

$$a_\tau(t) = v'(t), \quad a_n(t) = \frac{[v(t)]^2}{R} \quad (\text{tangential and normal acceleration}).$$

From the standpoint of acceleration, the motion can thus be thought of instantaneously as circular motion in which the trajectory coincides with the circle of curvature.

EXAMPLE 4 : Free-fall motion. The (Newtonian) equation of motion of free fall beginning at time $t = 0$ in the earth's gravitational field is

$$\vec{a}(t) = \vec{r}''(t) = -g\vec{k}, \quad t > 0,$$

where $g \approx 9.81 \text{ m/s}^2$. The equation contains three scalar differential equations:

$$x''(t) = 0, \quad y''(t) = 0, \quad z''(t) = -g.$$

According to these, first of all we must have

$$x'(t) = \alpha, \quad y'(t) = \beta, \quad z'(t) = -gt + \gamma,$$

where α, β, γ are (undetermined) constants. From this we see that we must further have (cf. Section V.6)

$$x(t) = \alpha t + x_0, \quad y(t) = \beta t + y_0, \quad z(t) = -\frac{1}{2}gt^2 + \gamma t + z_0,$$

where x_0, y_0, z_0 are again undetermined constants. Thus

$$\begin{aligned} \vec{r}(t) &= \vec{r}_0 + \vec{v}_0 t - \frac{1}{2}gt^2 \vec{k}, \quad t > 0, \quad \text{where} \\ \vec{r}_0 &= x_0 \vec{i} + y_0 \vec{j} + z_0 \vec{k} = \vec{r}(0^+), \quad \vec{v}_0 = \alpha \vec{i} + \beta \vec{j} + \gamma \vec{k} = \vec{r}'(0^+) = \vec{v}(0^+). \end{aligned}$$

If the limits $\vec{r}(0^+) = \vec{r}_0$ and $\vec{v}(0^+) = \vec{v}_0$ are known as initial conditions, then $\vec{r}(t)$ is uniquely determined for $t \geq 0$. The trajectory is then a parabola in the plane whose direction vectors are $\vec{v}_0$ and $\vec{k}$ (cf. Example IV.4:6). $\square$

EXAMPLE 5 : Detachment. A body, on which gravity

$$\vec{G} = -mg\vec{j} \quad (m = \text{mass}, g = \text{acceleration of the earth's gravity})$$

acts, is at rest at the origin and starts from there in frictionless sliding along the curve

$$y = -\frac{1}{3}x^3, \quad x \geq 0.$$

Determine the trajectory of the body in the form $y = f(x)$, $x \geq 0$.

Solution The path initially follows the curve $y = -x^3/3$, but detaches from it when the normal acceleration corresponding to this path exceeds the earth's gravitational acceleration in the normal direction. The detachment condition is thus

$$\frac{v^2}{R} = -g\vec{j} \cdot \vec{n},$$

where $\vec{n}$ is the (principal) normal vector

$$\vec{n} = -\frac{1}{\sqrt{1+x^4}}(x^2\vec{i} + \vec{j}).$$

By the energy principle we have

$$\frac{1}{2}m[v(x)]^2 = mg \cdot \frac{1}{3}x^3,$$

and the curvature at the point x is

$$\frac{1}{R} = \frac{2x}{(1+x^4)^{3/2}},$$

so detachment occurs when

$$\begin{aligned} \frac{2x}{(1+x^4)^{3/2}} \cdot \frac{2}{3}gx^3 &= \frac{g}{(1+x^4)^{1/2}} \\ \Leftrightarrow x^4 &= 3 \Leftrightarrow \underline{\underline{x = \sqrt[4]{3} = 1.316074..}} \end{aligned}$$

After detachment the flight path is a parabola (free-fall motion), so the entire trajectory is of the form

$$y = f(x) = \begin{cases} -x^3/3, & \text{when } 0 \leq x \leq \sqrt[4]{3}, \\ Ax^2 + Bx + C, & \text{when } x > \sqrt[4]{3}. \end{cases}$$

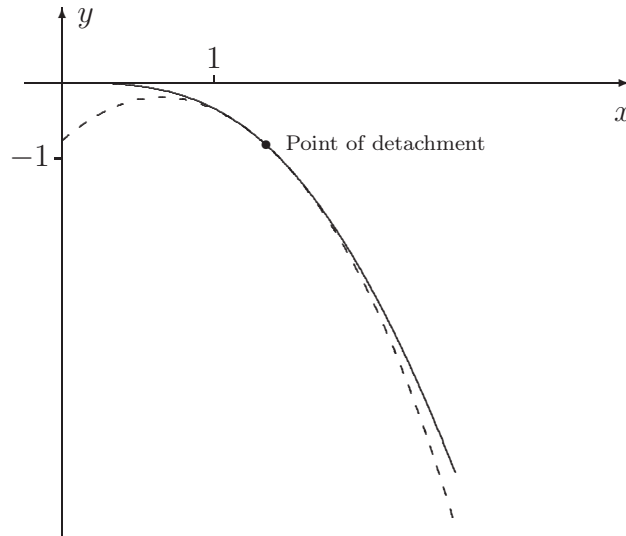
At the point of detachment gravity is the only external force acting on the body. Since this force is continuous, and the support force acting on the body can also be assumed continuous at the point of detachment (from detachment onward the support force = 0), we conclude that the acceleration vector is continuous. This means that the function f is twice continuously differentiable also at the point of detachment, i.e. we must have

$$f^{(k)}(x_0^+) = f^{(k)}(x_0^-), \quad k = 0, 1, 2, \quad x_0 = \sqrt[4]{3}.$$

From these conditions the constants A, B, C can be solved. The result (trajectory) is

$$y = f(x) = \begin{cases} -x^3/3, & x \in [0, \sqrt[4]{3}], \\ -\sqrt[4]{3}x^2 + \sqrt{3}x - 1/\sqrt[4]{3}, & x \in (\sqrt[4]{3}, \infty). \end{cases}$$

In the figure the slide/flight path is drawn with a solid line. The second-degree curve before the point of detachment and the curve $y = -x^3/3$ after the point of detachment are marked with a dashed line. $\square$



EXERCISES

- Let $\vec{u}(t)$ and $\vec{v}(t)$ be differentiable vector-valued functions. Prove:
 - $\vec{u}(t) \parallel \vec{v}(t) \forall t \Rightarrow \vec{u}' \times \vec{v} = \vec{v}' \times \vec{u}$
 - $\vec{u}(t) \perp \vec{v}(t) \forall t \Rightarrow \vec{u}' \cdot \vec{v} = -\vec{v}' \cdot \vec{u}$
- Determine the circle of curvature (radius R and center of curvature) of the following plane curves at the given point P of the curve:
 - $y = x^2, \quad P = (1, 1)$
 - $y = x^3 - 2x^2, \quad P = (2, 0)$
 - $y = e^x, \quad P = (0, 1)$
 - $\vec{r} = t \cos t \vec{i} + t \sin t \vec{j}, \quad P = (-\pi, 0)$
 - $x = t - \sin t, \quad y = 1 - \cos t, \quad P = (\pi, 2)$
 - $2x^2 + 3y^2 = 5, \quad P = (1, -1)$
 - $x^3 - y^3 + y^2 - 3x + 2 = 0, \quad P = (2, 2)$
 - $r = e^\varphi, \quad P = (r, \varphi) = (1, 0)$
- Determine the radius of curvature of the following space curves at the point of the curve corresponding to the given parameter value. Also determine

the equation of the plane in which the circle of curvature lies, as well as the center of curvature.

a) $\vec{r} = t\vec{i} + t^2\vec{j} + t^3\vec{k}, \quad t = 1$

b) $x = \cos t, \quad y = \sin t, \quad z = (4/\pi)t, \quad t = \pi/4$

c) $x = e^t \cos t, \quad y = e^t \sin t, \quad z = e^t, \quad t = 0$

4. Two plane curves S_1 and S_2 are said to *osculate* (literally 'to kiss') at the point (x_0, y_0) if the curves have a common circle of curvature at that point. a) Deduce that if the equations of the curves are $S_1 : y = f(x)$ and $S_2 : y = g(x)$, then the osculation conditions are

$$f(x_0) = g(x_0) = y_0, \quad f^{(k)}(x_0) = g^{(k)}(x_0), \quad k = 1, 2.$$

b) Find a second-degree polynomial curve $S_1 : y = ax^2 + bx + c$ that osculates the curve $S_2 : 2x^3 + 6y^3 + xy = 0$ at the point $(3, -2)$.

5. Determine the evolute of the curve $y = x^2$. Draw a figure!
6. A point-like body moves along the helix

$$x = \cos \varphi, \quad y = \sin \varphi, \quad z = \varphi, \quad \varphi \in [0, \infty),$$

so that its path speed is constant v_0 . Determine the body's velocity $\vec{v}$ and acceleration $\vec{a}$ as a function of time t ($t \geq 0$), given that the body is at the point $(1, 0, 0)$ at time $t = 0$. Also determine the radius of curvature R of the path and verify that $|\vec{a}| = |\vec{v}|^2/R$ holds.

7. The elevation profile of the terrain in a fell landscape is

$$h(x, y) = 40 - 0.005xy,$$

where the unit of length = m. A hiker, from a campsite located at the point $(x, y) = (0, 0)$, throws a spoiled tomato so that the tomato's initial velocity is

$$\vec{v}_0 = (10 \text{ m/s})(\vec{i} - 2\vec{j} + 2\vec{k}).$$

At what point does the tomato hit the ground, and what is its speed then? (Assume $g = 10 \text{ m/s}^2$, no air resistance and no wind correction.)

8. (*) Show that the evolute of the cycloid $S : x = R(t - \sin t), \quad y = R(1 - \cos t), \quad t \in \mathbb{R}$ is another cycloid congruent with S , obtained by translating S by the vector $-\pi R\vec{i} - 2R\vec{j}$. Figure!
9. (*) A body at the origin starts from rest, sliding frictionlessly along the curve $y = 1 - \cosh x, \quad x \geq 0$, with gravity acting in the direction $-\vec{j}$. Determine the body's trajectory.

10. (*) The *Frenet frame* associated with the point $P(t) \hat{=} \vec{r}(t)$ of a space curve is the system of vectors $\{\vec{\tau}, \vec{n}, \vec{\nu}\}$, where $\vec{\tau}$ and $\vec{n}$ are the unit tangent and principal normal vectors at that point and $\vec{\nu} = \vec{\tau} \times \vec{n}$ is the *binormal vector*. The *torsion* ω of a space curve is then defined by the formula

$$\frac{1}{v(t)} \frac{d\vec{\nu}}{dt} = -\omega(t)\vec{n}, \quad v(t) = |\vec{r}'(t)|.$$

- a) Show that $d\vec{\nu}/dt$ is indeed in the direction of the principal normal vector.
 b) Deduce that if a space curve is a plane curve in some plane, then its torsion = 0.
 c) Compute the torsion of the helix $S : x = a \cos t, y = a \sin t, z = bt$ at the point $P(t)$ of the curve.
11. (*) (Military exercise 2) An artillery shell is fired from the origin with initial velocity $\vec{v}_0 = (150 \text{ m/s})(\vec{i} + 2\vec{j} + \vec{k})$. On the flight path the shell is acted upon, in addition to gravity, by wind and by a resistance force proportional to velocity, so that the equations of motion are

$$\vec{v}' = c\vec{i} - g\vec{k} - k\vec{v}, \quad \vec{r}' = \vec{v},$$

where $\vec{r}(t)$ and $\vec{v}(t)$ are the position and velocity vectors of the shell at time t , $g = 10 \text{ m/s}^2$, $k = 0.01 \text{ s}^{-1}$ and $c = 0.10 \text{ m/s}^2$. Compute at which point of the xy -plane (to 10 meter accuracy!) the shell lands. *Hint*: Solve the equations of motion separately in the directions $\vec{i}, \vec{j}, \vec{k}$ (first $\vec{v}$, then $\vec{r}$). Start with the vertical direction to find the flight time!

VII.4 Taylor polynomials and Taylor's theorem

In this section we consider functions that are sufficiently smooth in a neighborhood of a given point x_0 , i.e. sufficiently many times (continuously) differentiable.

DEFINITION VII.4.1 The *Taylor polynomial of degree n at the point x_0* of a function $f : \mathcal{D}_f \rightarrow \mathbb{R}$, $\mathcal{D}_f \subset \mathbb{R}$, which is n times differentiable at the point $x_0 \in \mathcal{D}_f$, is

$$T_n(x, x_0) = \sum_{k=0}^n \frac{f^{(k)}(x_0)}{k!} (x - x_0)^k.$$

The first three Taylor polynomials according to the definition are

$$\begin{aligned} T_0(x, x_0) &= f(x_0), \\ T_1(x, x_0) &= f(x_0) + f'(x_0)(x - x_0), \\ T_2(x, x_0) &= f(x_0) + f'(x_0)(x - x_0) + \frac{1}{2}f''(x_0)(x - x_0)^2. \end{aligned}$$

In particular, then, $T_1(x, x_0)$ is the linearizing approximation of f at the point x_0 .

PROPOSITION VII.4.2 The Taylor polynomial $T_n(x, x_0)$ of a function f is uniquely determined by the conditions

$$\frac{d^k}{dx^k} T_n(x, x_0)|_{x=x_0} = f^{(k)}(x_0), \quad k = 0 \dots n.$$

Proof It is easily seen that $T_n(x, x_0)$ satisfies the stated conditions. If some other polynomial $p(x)$ satisfies the same conditions, i.e.

$$p^{(k)}(x_0) = f^{(k)}(x_0), \quad k = 0 \dots n,$$

then the polynomial

$$q(x) = T_n(x, x_0) - p(x) = \sum_{k=0}^n a_k x^k$$

satisfies

$$q^{(k)}(x_0) = 0, \quad k = 0 \dots n.$$

Then, since $q^{(n)}(x_0) = n!a_n$, it follows that $a_n = 0$, whereby $q^{(n-1)}(x_0) = (n-1)!a_{n-1} \Rightarrow a_{n-1} = 0$, etc. Thus $q = 0$, and consequently the polynomial determined by the above conditions is unique. $\square$

EXAMPLE 1 Determine the following Taylor polynomials at the point $x_0 = 0$:

- a) $f(x) = \sqrt[3]{1+x}$, $n = 2$ b) $f(x) = \tan x$, $n = 5$
 c) $f(x) = \ln(1+x)$, $n \in \mathbb{N}$

Solution a) $f'(x) = \frac{1}{3}(1+x)^{-2/3}$, $f''(x) = -\frac{2}{9}(1+x)^{-5/3}$

$$\Rightarrow f(0) = 1, \quad f'(0) = \frac{1}{3}, \quad f''(0) = -\frac{2}{9}$$

$$\Rightarrow T_2(x, 0) = \underline{\underline{1 + \frac{x}{3} - \frac{1}{9}x^2.}}$$

b) $f'(x) = 1/\cos^2 x$, $f''(x) = 2 \sin x / \cos^3 x$,
 $f'''(x) = 2/\cos^2 x + 6 \sin^2 x / \cos^4 x = -4/\cos^2 x + 6/\cos^4 x$
 $f^{(4)}(x) = -8 \sin x / \cos^3 x + 24 \sin x / \cos^5 x$
 $f^{(5)}(x) = -16/\cos^2 x + 120 \sin^2 x / \cos^6 x$

$$\Rightarrow \begin{cases} f(0) = f''(0) = f^{(4)}(0) = 0, \\ f'(0) = 1, \quad f'''(0) = 2, \quad f^{(5)}(0) = 16 \end{cases}$$

$$\Rightarrow T_5(x, 0) = \underline{\underline{x + \frac{1}{3}x^3 + \frac{2}{15}x^5.}}$$

c) $f'(x) = (1+x)^{-1}$, $f''(x) = -(1+x)^{-2}$, $f'''(x) = 2(1+x)^{-3}$,
 $\dots$, $f^{(k)}(x) = (-1)^{k-1}(k-1)!(1+x)^{-k}$
 $\Rightarrow f(0) = 0$, $f^{(k)}(0) = (-1)^{k-1}(k-1)!$, $k = 1, 2, \dots$,
 $\Rightarrow T_n(x, 0) = \underline{\underline{x - \frac{1}{2}x^2 + \frac{1}{3}x^3 + \dots + (-1)^{n-1}\frac{x^n}{n}.}} \quad \square$

The derivative of the Taylor polynomial is

$$\frac{d}{dx}T_n(x, x_0) = f'(x_0) + f''(x_0)(x - x_0) + \dots + \frac{f^{(n-1)}(x_0)}{(n-1)!}(x - x_0)^{n-1}.$$

The result of differentiation = the Taylor polynomial of f' of degree $n-1$ at the point x_0 , i.e. briefly: the derivative of the Taylor polynomial = the Taylor polynomial of the derivative (of one lower degree).

EXAMPLE 1 (continued). From the results of the example, differentiation yields the following Taylor polynomials:

- a) $f(x) = (1+x)^{-2/3}$: $T_1(x, 0) = 1 - \frac{2}{3}x$.
 b) $f(x) = 1/\cos^2 x$: $T_4(x, 0) = 1 + x^2 + \frac{2}{3}x^4$.
 c) $f(x) = 1/(1+x)$: $T_{n-1}(x, 0) = 1 - x + \cdots + (-1)^{n-1}x^{n-1}$. $\square$

Taylor's theorem

The following remarkable approximation theorem is based on Taylor polynomials. The proof is presented at the end of the section.

THEOREM VII.4.3 (Taylor's theorem) If f is continuous on the interval $[a, b]$ and $n+1$ times differentiable on the interval (a, b) , then for every $x_0 \in (a, b)$ and $x \in [a, b]$, $x \neq x_0$

$$f(x) = T_n(x, x_0) + R_n(x),$$

where

$$R_n(x) = \frac{f^{(n+1)}(\xi)}{(n+1)!} (x - x_0)^{n+1} \quad \text{for some } \xi \in (x_0, x) \text{ or } \xi \in (x, x_0).$$

By Taylor's theorem, a function that is regular in a neighborhood of a certain point x_0 can be approximated in this neighborhood by a polynomial — namely by the Taylor polynomial — and the approximation is in general the more accurate the higher the degree of the polynomial and the closer one is to the point x_0 . The result can be stated qualitatively in the form:

A smooth function $\approx$ a polynomial on a short interval.

The result of Taylor's theorem is of fundamental importance in almost all numerical computation that involves the approximation of functions. For the error term $R_n(x, x_0)$, i.e. the so-called *remainder* of the Taylor polynomial approximation, many forms are known. The one given in Theorem VII.4.3 is called the *Lagrange** form of the remainder.

*The Italian-French **Joseph Louis** (born Giovanni Luigi) **Lagrange** (1736-1813) was among the most notable mathematicians of his time. Lagrange's contribution was significant especially in the development of differential calculus (and of integral calculus as well). Besides mathematics, Lagrange studied mechanics and achieved a lasting name in that field too.

The Taylor polynomials, Taylor's theorem, and especially the *Taylor series* presented later, refer to the English mathematician **Brook Taylor** (1685-1731). Taylor did not actually know the theorem that bears his name.

EXAMPLE 2 Applying the differentiation rules

$$\begin{cases} D^k e^x = e^x, & k = 0, 1, 2, \dots \\ D^{2k} \cos x = (-1)^k \cos x, & D^{2k+1} \cos x = (-1)^{k+1} \sin x, & k = 0, 1, 2, \dots \\ D^{2k} \sin x = (-1)^k \sin x, & D^{2k+1} \sin x = (-1)^k \cos x, & k = 0, 1, 2, \dots \end{cases}$$

and Theorem VII.4.3, we see that if $x \in \mathbb{R}$, $x \neq 0$ and $n \in \mathbb{N}$, then for the functions e^x , $\cos x$, $\sin x$ the following holds for some $\xi \in (0, x)$ ($x > 0$) or $\xi \in (x, 0)$ ($x < 0$)

$$\begin{aligned} e^x &= \left(1 + x + \frac{x^2}{2!} + \dots + \frac{x^n}{n!} \right) + \frac{e^\xi}{(n+1)!} x^{n+1}, \\ \cos x &= \left(1 - \frac{x^2}{2!} + \dots + (-1)^n \frac{x^{2n}}{(2n)!} \right) + (-1)^{n+1} \frac{\cos \xi}{(2n+2)!} x^{2n+2}, \\ \sin x &= \left(x - \frac{x^3}{3!} + \dots + (-1)^n \frac{x^{2n+1}}{(2n+1)!} \right) + (-1)^{n+1} \frac{\cos \xi}{(2n+3)!} x^{2n+3}. \end{aligned}$$

Here the Taylor polynomials are enclosed in parentheses

$$\begin{aligned} e^x &: T_n(x, 0), \\ \cos x &: T_{2n}(x, 0) = T_{2n+1}(x, 0), \\ \sin x &: T_{2n+1}(x, 0) = T_{2n+2}(x, 0). \quad \square \end{aligned}$$

Taylor's theorem can also be formulated so that, instead of $(n+1)$ -fold differentiability of the function, only n -fold differentiability and the continuity of the derivative $f^{(n)}$ is assumed.

THEOREM VII.4.4 If f is n times differentiable on the interval (a, b) and $f^{(n)}$ is continuous on that interval, then the remainder $R_n(x) = f(x) - T_n(x, x_0)$ satisfies, for every $x_0 \in (a, b)$,

$$\lim_{x \rightarrow x_0} \frac{R_n^{(k)}(x)}{(x - x_0)^{n-k}} = 0, \quad k = 0 \dots n.$$

Proof In the case $n = 0$ the claim is true by the definition of continuity. If $n \geq 1$,

then by the assumptions and Taylor's theorem

$$\begin{aligned} f(x) &= T_{n-1}(x, x_0) + \frac{f^{(n)}(\xi)}{n!} (x - x_0)^n \\ &= \left[T_{n-1}(x, x_0) + \frac{f^{(n)}(x_0)}{n!} (x - x_0)^n \right] + \frac{1}{n!} [f^{(n)}(\xi) - f^{(n)}(x_0)] (x - x_0)^n \\ &= T_n(x, x_0) + \frac{1}{n!} [f^{(n)}(\xi) - f^{(n)}(x_0)] (x - x_0)^n, \quad x \in (a, b), \end{aligned}$$

where $\xi = \xi(x) = x_0$ if $x = x_0$, otherwise $\xi(x) \in (x_0, x)$ or $\xi \in (x, x_0)$. Thus

$$R_n(x) = \frac{1}{n!} [f^{(n)}(\xi(x)) - f^{(n)}(x_0)] (x - x_0)^n,$$

where $\xi(x) \rightarrow x_0$ as $x \rightarrow x_0$. Since $f^{(n)}$ is continuous at x_0 , it follows that

$$\lim_{x \rightarrow x_0} \frac{R_n(x)}{(x - x_0)^n} = \lim_{x \rightarrow x_0} \frac{1}{n!} [f^{(n)}(\xi(x)) - f^{(n)}(x_0)] = 0.$$

The other claimed limit results follow from this by differentiation: Since

$$f^{(k)}(x) = \left(\frac{d}{dx} \right)^k T_n(x, x_0) + R_n^{(k)}(x), \quad x \in (x_0, x_0 + a), \quad k = 1 \dots n,$$

and since here $(d/dx)^k T_n(x, x_0)$ is the Taylor polynomial of $f^{(k)}$ of degree $n - k$, then by what was already proved

$$\lim_{x \rightarrow x_0} \frac{R_n^{(k)}(x)}{(x - x_0)^{n-k}} = 0, \quad k = 1 \dots n. \quad \square$$

Rapid computation of Taylor polynomials

Sometimes the derivatives of f are so difficult to compute that the Taylor polynomial can be determined more directly by other methods, whereby, conversely, the derivatives $f^{(k)}(x_0)$, $k = 0 \dots n$, can be determined by means of the polynomial (!). When determining the polynomial by other means, it suffices that the remainder be made sufficiently small, since by this criterion too the polynomial is unique:

PROPOSITION VII.4.5 Let the function f be n times differentiable on the interval (a, b) and let $f^{(n)}$ be continuous on the interval (a, b) . Then if p is a polynomial of degree $\leq n$ such that for some $x_0 \in (a, b)$

$$\lim_{x \rightarrow x_0} \frac{f(x) - p(x)}{(x - x_0)^n} = 0,$$

then $p(x) =$ the Taylor polynomial $T_n(x, x_0)$ of f .

Proof Setting $q(x) = p(x) - T_n(x, x_0)$, then by the combination rules for limits (Theorem V.2.4), by Theorem VII.4.4 and by the assumption,

$$\lim_{x \rightarrow x_0} \frac{q(x)}{(x - x_0)^n} = \lim_{x \rightarrow x_0} \left[\frac{f(x) - T_n(x, x_0)}{(x - x_0)^n} - \frac{f(x) - p(x)}{(x - x_0)^n} \right] = 0 - 0 = 0.$$

This is possible only when $q(x) = 0$, since q is a polynomial of degree $\leq n$. $\square$

In the following examples the abbreviation $[x^m]$ is used for a function of the form $x^m g(x)$, where g is bounded in a neighborhood of the point $x = 0$.

EXAMPLE 3 $f(x) = (x + 1)/\cos x$, $T_5(x, 0) = ?$

Solution Since $\cos x = 1 - \frac{x^2}{2} + \frac{x^4}{24} + [x^6]$ and $1/(1 - t) = 1 + t + t^2 + [t^3]$, then

$$\begin{aligned} f(x) &= (x + 1) \left(1 - \frac{x^2}{2} + \frac{x^4}{24} \right)^{-1} (1 + [x^6])^{-1} \\ &= (x + 1) \left[1 - \left(\frac{x^2}{2} - \frac{x^4}{24} \right) \right]^{-1} + [x^6] \\ &= (x + 1) \left[1 + \left(\frac{x^2}{2} - \frac{x^4}{24} \right) + \left(\frac{x^2}{2} - \frac{x^4}{24} \right)^2 \right] + [x^6] \\ &= (x + 1) \left(1 + \frac{1}{2}x^2 + \frac{5}{24}x^4 \right) + [x^6] \\ &= \left(1 + x + \frac{1}{2}x^2 + \frac{1}{2}x^3 + \frac{5}{24}x^4 + \frac{5}{24}x^5 \right) + [x^6]. \end{aligned}$$

By Proposition VII.4.5 the polynomial in parentheses = the Taylor polynomial $T_5(x, 0)$ of f . $\square$

EXAMPLE 4 $f(x) = 1/(1 + x^4 e^{x^2})$, $f^{(8)}(0) = ?$

$$\begin{aligned} \text{Solution } f(x) &= 1 - (x^4 e^{x^2}) + (x^4 e^{x^2})^2 + [x^{12}] \\ &= 1 - x^4 \left(1 + x^2 + \frac{1}{2}x^4 + [x^6] \right) + x^8 (1 + [x^2])^2 + [x^{12}] \\ &= 1 - x^4 - x^6 + \frac{1}{2}x^8 + [x^{10}] = T_8(x, 0) + [x^{10}] \\ \Rightarrow f^{(8)}(0) &= \frac{1}{2} \cdot 8! = \underline{\underline{20160}}. \quad \square \end{aligned}$$

Taylor series

When the remainder in Taylor's theorem VII.4.3 is estimated for the functions $\cos x$ and $\sin x$, we see that (see Example 2 above)

$$|\cos x - T_{2n}(x, 0)| \leq \frac{1}{(2n+2)!} |x|^{2n+2}, \quad x \in \mathbb{R},$$

$$|\sin x - T_{2n+1}(x, 0)| \leq \frac{1}{(2n+3)!} |x|^{2n+3}, \quad x \in \mathbb{R}.$$

Since $|x|^n/n! \rightarrow 0 \quad \forall x \in \mathbb{R}$ as $n \rightarrow \infty$, then for every $x \in \mathbb{R}$

$$\cos x = \lim_{n \rightarrow \infty} T_{2n}(x, 0), \quad \sin x = \lim_{n \rightarrow \infty} T_{2n+1}(x, 0),$$

i.e. (cf. Exercise VI.2:18)

$$\begin{aligned} \cos x &= \sum_{k=0}^{\infty} (-1)^k \frac{x^{2k}}{(2k)!}, \quad x \in \mathbb{R}, \\ \sin x &= \sum_{k=0}^{\infty} (-1)^k \frac{x^{2k+1}}{(2k+1)!}, \quad x \in \mathbb{R}. \end{aligned}$$

DEFINITION VII.4.6 If f is arbitrarily many times differentiable at x_0 , then the series

$$\sum_{k=0}^{\infty} \frac{f^{(k)}(x_0)}{k!} (x - x_0)^k$$

is the *Taylor series* of f at x_0 .*

In the theory of Taylor series the most central question is quite obviously: When does the series converge to $f(x)$, i.e. when does

$$f(x) = \lim_{n \rightarrow \infty} T_n(x, x_0) = \sum_{k=0}^{\infty} \frac{f^{(k)}(x_0)}{k!} (x - x_0)^k ?$$

For the functions $\sin x$ and $\cos x$ the answer is: Always, i.e. for every $x \in \mathbb{R}$ (and for every $x_0 \in \mathbb{R}$). For the exponential function e^x the answer is the same; here Taylor's theorem confirms the previously known result (cf. Section VI.2)

$$e^x = \sum_{k=0}^{\infty} \frac{x^k}{k!}, \quad x \in \mathbb{R}.$$

Strictly speaking, the above question about the convergence of a Taylor series contains two separate questions, as is seen from the following example.

*In the case $x_0 = 0$ the Taylor series is also called the *Maclaurin series*.

EXAMPLE 5 The function

$$f(x) = \begin{cases} e^{-1/x^2}, & \text{when } x \neq 0, \\ 0, & \text{when } x = 0 \end{cases}$$

is arbitrarily many times differentiable at the point $x = 0$ (and elsewhere) and

$$f^{(k)}(0) = 0, \quad k = 0, 1, 2, \dots,$$

so $T_n(x, 0) = 0 \forall n$. In this case, then, the Taylor series converges $\forall x \in \mathbb{R}$, but $\lim_n T_n(x, 0) = 0 \neq f(x)$ when $x \neq 0$. $\square$

According to the example, two different functions can have the same Taylor series (in the example the functions f and $g = 0$), so from the coefficients of a Taylor series (or from the sum of the series) one cannot infer the function from which the series was derived. For many 'normal' functions f , however, it holds that the sum of the Taylor series of $f = f(x)$ whenever the series converges. Such normal cases are, e.g. rational functions.

EXAMPLE 6 The Taylor polynomials of the function $f(x) = 1/(1 + 4x^2)$ at the origin are

$$T_{2n}(x, 0) = T_{2n+1}(x, 0) = \sum_{k=0}^n (-4)^k x^{2k}, \quad n = 0, 1, \dots$$

The Taylor series, i.e. the power series $\{T_n(x, 0), n = 0, 1, 2, \dots\}$, converges in this case exactly when $|x| < 1/2$ (cf. Section I.12), and then the sum $= f(x)$:

$$\sum_{k=0}^{\infty} (-4)^k x^{2k} = \frac{1}{1 + 4x^2} = f(x), \quad x \in \left(-\frac{1}{2}, \frac{1}{2}\right). \quad \square$$

When studying the convergence of Taylor series, one can always make the change of variable $x - x_0 \mapsto x$, so that it suffices to consider a general power series of the form

$$f(x) = \sum_{k=0}^n a_k x^k.$$

The convergence question for such a series has been solved in Section I.12: By Theorem I.12.9 the series converges either (a) only when $x = 0$ or (b) on the interval $(-\rho, \rho)$ (possibly also when $x = \pm\rho$), where ρ is the radius of convergence of the series ($\rho \in \mathbb{R}_+$ or $\rho = \infty$). In Section V.3 it was shown that a function defined as the sum of a power series is arbitrarily many times differentiable on the

interval $(-\rho, \rho)$ and that the derivatives can be computed by differentiating the series term by term (Theorem V.3.4). Consequently, if the radius of convergence of the power series $\sum_{k=0}^{\infty} a_k x^k$ is $\rho > 0$ and $x_0 \in \mathbb{R}$, then the function

$$f(x) = \sum_{k=0}^{\infty} a_k (x - x_0)^k$$

is defined and arbitrarily many times differentiable on the interval $(x_0 - \rho, x_0 + \rho)$ (on all of $\mathbb{R}$ if $\rho = \infty$) and the derivatives of f can be computed by differentiating the series term by term. Choosing the point of differentiation to be x_0 in particular, we obtain the result

$$f^{(k)}(x_0) = k! a_k \quad \Leftrightarrow \quad a_k = \frac{f^{(k)}(x_0)}{k!}, \quad k = 0, 1, 2, \dots$$

Thus $f(x)$ can be represented in the form

$$f(x) = \sum_{k=0}^{\infty} \frac{f^{(k)}(x_0)}{k!} (x - x_0)^k.$$

We have arrived at the following noteworthy result:

THEOREM VII.4.7 If the series $\sum_{k=0}^{\infty} a_k (x - x_0)^k$ converges on the interval $(x_0 - \rho, x_0 + \rho)$, $\rho > 0$, then that series = the Taylor series, at the point x_0 , of the function defined as the sum of the series.

EXAMPLE 7 From the Taylor series of the function $\sin x$ we see that

$$\sum_{k=0}^{\infty} (-1)^k \frac{x^{2k}}{(2k+1)!} = f(x) = \begin{cases} \sin x/x, & x \neq 0, \\ 1, & x = 0. \end{cases}$$

Since the series converges $\forall x \in \mathbb{R}$, this is the Taylor series at the origin of the function defined as the sum of the series. The function f is thus arbitrarily many times differentiable at every point $x \in \mathbb{R}$, including the origin (!). $\square$

EXAMPLE 8 Solve, using Taylor series, the initial value problem

$$\begin{cases} y' = e^{-x^2}, & x \in \mathbb{R}, \\ y(0) = 0. \end{cases}$$

Solution Since the Taylor series of the exponential function e^x converges everywhere,

$$e^{-x^2} = \sum_{k=0}^{\infty} \frac{(-x^2)^k}{k!} = \sum_{k=0}^{\infty} \frac{(-1)^k}{k!} x^{2k}, \quad x \in \mathbb{R}.$$

Choosing

$$y(x) = \sum_{k=0}^{\infty} \frac{(-1)^k}{(2k+1)k!} x^{2k+1} = x - \frac{x^3}{18} + \frac{x^5}{600} - \dots,$$

then by differentiating term by term we see that $y'(x) = e^{-x^2}$, $x \in \mathbb{R}$. Since also $y(0) = 0$, this is the solution. $\square$

Proof of Taylor's theorem

There are many different ways to prove Taylor's theorem. In the following proof, which follows the 'short formula', the cornerstone of the reasoning is Rolle's theorem (Theorem V.6.2).

Let $x_0 < x$ (the case $x_0 > x$ is handled similarly) and consider, on the interval $[x_0, x]$, the function $g(t)$ defined by

$$g(t) = f(t) - T_n(t, x_0) - H(t - x_0)^{n+1}, \quad H = \frac{f(x) - T_n(x, x_0)}{(x - x_0)^{n+1}}.$$

For this we have $g^{(k)}(x_0) = 0$, $k = 0 \dots n$ and $g(x) = 0$. Since therefore $g(x_0) = g(x) = 0$, by Rolle's theorem $g'(\xi_1) = 0$ for some $\xi_1 \in (x_0, x) \subset (a, b)$. If $n \geq 1$, then also $g'(x_0) = 0$, whereby by the same theorem $g''(\xi_2) = 0$ for some $\xi_2 \in (x_0, \xi_1)$. If further $n \geq 2$, then also $g''(x_0) = 0$, so by the same theorem $g'''(\xi_2) = 0$ for some $\xi_2 \in (x_0, \xi_1)$. Continuing in the same way, we deduce that in general $g^{(n)}(x_0) = g^{(n)}(\xi_n) = 0$ for some $\xi_n \in (x_0, \xi_{n-1})$, whereby by Rolle's theorem $g^{(n+1)}(\xi_{n+1}) = 0$ for some $\xi_{n+1} \in (x_0, \xi_n) \subset (x_0, x) \subset (a, b)$. But $g^{(n+1)}(t) = f^{(n+1)}(t) - H(n+1)!$ — Thus $f^{(n+1)}(\xi_{n+1}) - H(n+1)! = 0 \Rightarrow$ the claim. $\square$

EXERCISES

1. Compute all the Taylor polynomials $T_n(x, 2)$, $n = 0, 1, 2, \dots$ of the function $f(x) = x^4 + 3x^3 + x^2 + 2x + 8$, and put them into the standard polynomial form (in powers of x). Draw the graphs of those polynomials and compare with the function f .
2. Compute the Taylor polynomial $T_3(x, 0)$ of the function $f(x) = (x-1)/(x-2)$ and draw the graphs of the function, the Taylor polynomial and the remainder.

3. For the following functions determine the Taylor polynomial $T_n(x, 0)$ of the given degree n and, from the Lagrange form of the remainder, estimate as accurately as possible the number $r_n = \max_{x \in [-1, 1]} |R_n(x)|$:

- | | |
|---------------------------------------|---|
| a) $f(x) = \cosh x$, $n = 4$ | b) $f(x) = e^{-0.2x}$, $n = 3$ |
| c) $f(x) = \sqrt{5+x}$, $n = 3$ | d) $f(x) = \sqrt[5]{5-x}$, $n = 3$ |
| e) $f(x) = 2^x$, $n = 5$ | f) $f(x) = \ln(e+x)$, $n = 6$ |
| g) $f(x) = \cot(x + \pi/3)$, $n = 4$ | h) $f(x) = \sin x - 1/\cos x$, $n = 3$ |

4. a) The Taylor polynomials $T_n(x, 0)$ of the functions $\cosh x$ and $\sinh x$ can be computed either directly from the definition or by means of the Taylor polynomials of the functions e^x and e^{-x} . Verify that the result is the same either way.

- b) Show that in the Taylor polynomial $T_n(x, 0)$ of an even (respectively odd) function there are only even (odd) powers.

5. Prove, using Taylor's theorem:

$$1 - \frac{1}{2}x^2 \leq \cos x \leq 1 - \frac{1}{2}x^2 + \frac{1}{24}x^4, \quad \text{when } x \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right].$$

Are these inequalities also true outside the interval $[-\pi/2, \pi/2]$?

6. In Exercise [V.5:15b](#) it is assumed that the function f is m times continuously differentiable on the interval $[a, b]$. Show that the solution of the exercise is $p_1(x) = T_m^+(a, x)$, $p_2(x) = T_m^-(b, x)$, where $T_m^+(a, x)$ and $T_m^-(b, x)$ are the Taylor polynomials of f defined by means of the one-sided derivatives $D_+^k f(a)$ and $D_-^k f(b)$.

7. Compute the Taylor polynomial $T_n(x, 0)$ of the following functions of the given degree using the fastest possible shortcuts:

- | | |
|---|---|
| a) $f(x) = 2/(4+x^3)$, $n = 12$ | b) $f(x) = \cos x^4$, $n = 16$ |
| c) $f(x) = \operatorname{Arcsin} x^3$, $n = 6$ | d) $f(x) = (x^3 - x^5) \operatorname{Arctan} x^2$, $n = 7$ |

8. Compute the following derivatives making use of Taylor polynomials:

- | | |
|--|--|
| a) $f(x) = \sin x^8$, $f^{(40)}(0)$ | b) $f(x) = e^{-x^4}$, $f^{(20)}(0)$ |
| c) $f(x) = x^2/(1+x^4)$, $f^{(100)}(0)$ | d) $f(x) = x^3 \ln(2+x^2)$, $f^{(87)}(0)$ |
| e) $f(x) = e^{x^3}/(1+e^{x^3})$, $f^{(9)}(0)$ | f) $f(x) = \cos(x^2 \sin^2 x)$, $f^{(12)}(0)$ |

9. The following curves have a turning point (cf. Example [3](#) in the previous section) at the given point P . In which direction does the curve leave the point P ?

- | |
|--|
| a) $x(t) = 4t + 1/t$, $y(t) = t^2 + 16/t$, $P = (17/2, 12)$ |
| b) $\vec{r}(t) = t^4 \vec{i} + (2 - 2 \cos t - t^2) \vec{j} + (t^2 - t \sinh t) \vec{k}$, $P = (0, 0, 0)$ |

10. What is the sum of the series

$$\text{a) } 1 + 4 + \frac{16}{2!} + \frac{64}{3!} + \frac{256}{4!} + \dots \quad \text{b) } 1 + \frac{4}{3!} + \frac{16}{5!} + \frac{64}{7!} + \dots$$

?

11. Give the Taylor series of the following functions at the point $x_0 = 0$:

$$\begin{array}{llllll} \text{a) } e^{3x+1} & \text{b) } \cos(2x^3) & \text{c) } \sin(x - \frac{\pi}{4}) & \text{d) } \cos(2x - \pi) & \text{e) } x^2 \sin 3x \\ \text{f) } \sin x \cos x & \text{g) } (1 + x^3)/(1 + x^2) & \text{h) } \ln(2 + x^2) & \text{i) } x^2 \ln(1 + x) \end{array}$$

12. Determine the Taylor series of the following functions at the given point, as well as the interval of convergence of the series:

$$\begin{array}{lll} \text{a) } e^{-2x}, x_0 = -1 & \text{b) } \sin x, x_0 = \frac{\pi}{2} & \text{c) } \ln x, x_0 = 1 \\ \text{d) } \cos x, x_0 = \pi & \text{e) } \cos^2 x, x_0 = \frac{\pi}{8} & \text{f) } x/(4 + x), x_0 = 3 \end{array}$$

13. Let

$$f(x) = \begin{cases} \frac{2 \cos x - 2}{x^2}, & \text{when } x \neq 0, \\ -1, & \text{when } x = 0. \end{cases}$$

Compute $f'(0)$ and $f''(0)$ a) directly from the definition of the derivative, b) by means of the Taylor series of f .

14. The following functions f are each defined to be continuous at the point $x = 0$, whereby the functions are arbitrarily many times differentiable at this point (and thus on all of $\mathbb{R}$). Determine the Taylor series of the functions at the point $x_0 = 0$ and, by means of these, $f(0)$, $f'(0)$ and $f''(0)$.

$$\begin{array}{lll} \text{a) } \frac{e^x - 1}{x} & \text{b) } \frac{e^{-2x} - 1 + 2x - 2x^2}{x^3} & \text{c) } \frac{\sinh x - x}{x^3} \\ \text{d) } \frac{\ln(1 + x) - \sin x}{x^2} & \text{e) } \frac{e^{2x} - 4e^{-x} - 6x + 3}{x^2} \end{array}$$

15. It is known that

$$\sum_{k=0}^{\infty} a_k (x + \pi)^k = x \cos x, \quad x \in \mathbb{R}.$$

Compute the coefficients a_0 , a_1 and a_2 of the series.

16. (*) Using Taylor's theorem, prove Theorem V.7.4 concerning the convergence of Newton's method.

17. (*) The power series $\sum_{k=0}^{\infty} a_k(x+1)^k$ converges near the point $x = -1$, and its sum is

$$\sum_{k=0}^{\infty} a_k(x+1)^k = \frac{x+2}{x^2+2x+3}.$$

What are the values of the coefficients a_0 , a_1 and a_2 , and what is the interval of convergence of the series?

VII.5 Applications of Taylor polynomials

In this section we examine some common ways of using Taylor polynomials and Taylor's theorem in the study of functions and in approximating functions. When computing Taylor polynomials or estimating the remainder term, the calculations can often be considerably shortened and clarified by using so-called *order-of-magnitude notation*. A foretaste of such notation was already given in Examples 3 and 4 of the previous section, where the shorthand $[x^m]$ was used. In its place it is more common to use the notation $\mathcal{O}(|x|^m)$, read 'of order x^m ' or just 'big-oh x^m '. Another order-of-magnitude symbol, slightly stronger in meaning, is $o(|x|^m)$, read 'little-oh x^m '. In these notations $|x|^m$ is the so-called comparison function. In its place there may be a more general, non-negative valued function, e.g. $|x - x_0|^\alpha$, $\alpha \in \mathbb{Q}$.

DEFINITION VII.5.1 (Order-of-magnitude notation $\mathcal{O}$ and o) If f and g are defined on the interval $[x_0 - a, x_0 + a]$ ($a > 0$) and $g(x) \geq 0 \ \forall x \in [x_0 - a, x_0 + a]$, then one writes

$$f(x) = \mathcal{O}(g(x)), \quad x \in [x_0 - a, x_0 + a],$$

if there exists a constant $C \in \mathbb{R}_+$ such that

$$|f(x)| \leq Cg(x) \quad \forall x \in [x_0 - a, x_0 + a].$$

If $g(x) > 0 \ \forall x \neq x_0$ and $\lim_{x \rightarrow x_0} (f/g)(x) = 0$, then one writes

$$f(x) = o(g(x)), \quad \text{as } x \rightarrow x_0.$$

If the notation $\mathcal{O}(\cdot)$ is meant only to convey that the estimate is valid in some neighborhood of x_0 , then this can be expressed by writing, as in the corresponding notation $o(\cdot)$,

$$f(x) = \mathcal{O}(g(x)), \quad \text{as } x \rightarrow x_0.$$

In these notations the limit point x_0 may be replaced by $\pm\infty$, in which case one is estimating f for large values of $|x|$. The comparison function is then typically $g(x) = |x|^\alpha$ for some $\alpha \in \mathbb{Q}$.

In calculations it is convenient to treat order-of-magnitude symbols like functions, so that they can be combined in the same way as functions. For example, $f(x) + o(x^2)$ means the function $f(x) + g(x)$, where $g(x) = o(x^2)$, and $\mathcal{O}(x^2) + \mathcal{O}(y^2)$ means the same as $\mathcal{O}(x^2 + y^2)$. From Definition VII.5.1 the following rules of order-of-magnitude algebra are easily verified (Exercise 1). In the

rules it is assumed that $x, y \geq 0$.

$$\begin{aligned}\mathcal{O}(x) + \mathcal{O}(y) &= \mathcal{O}(\max\{x, y\}), \\ \mathcal{O}(x)\mathcal{O}(y) &= \mathcal{O}(xy), \\ o(x)\mathcal{O}(y) &= o(xy).\end{aligned}$$

In applications the latter two rules are most commonly used in the form

$$\mathcal{O}(x^\alpha)\mathcal{O}(x^\beta) = \mathcal{O}(x^{\alpha+\beta}), \quad o(x^\alpha)\mathcal{O}(x^\beta) = o(x^{\alpha+\beta}), \quad x \geq 0, \quad \alpha, \beta \in \mathbb{R}.$$

EXAMPLE 1 Using the change of variable $x^2 = t$ and Taylor's theorem, one deduces that near the point $x = 0$ (e.g. on the interval $[-1, 1]$)

$$\sqrt{x^2 + 1} = \sqrt{1 + t} = 1 + \frac{1}{2}t + \mathcal{O}(t^2) = 1 + \frac{1}{2}x^2 + \mathcal{O}(x^4).$$

For large values of x ($x \rightarrow \infty$) the change of variable $x^{-2} = t$ similarly gives the estimate

$$\sqrt{x^2 + 1} = x\sqrt{1 + x^{-2}} = x \left(1 + \frac{1}{2x^2} + \mathcal{O}(x^{-4}) \right) = x + \frac{1}{2x} + \mathcal{O}(x^{-3}). \quad \square$$

EXAMPLE 2 Show that for small values of $|x|$

$$f(x) = \frac{1 - x}{1 + 2x - x^2 + \mathcal{O}(|x|^3)} = 1 - 3x + 7x^2 + \mathcal{O}(|x|^3).$$

Solution Since for small values of $|t|$ we have $1/(1+t) = 1 - t + t^2 + \mathcal{O}(|t|^3)$, order-of-magnitude algebra gives (cf. Examples 3–4 of the previous section)

$$\begin{aligned}f(x) &= (1 - x) \left\{ 1 - [2x - x^2 + \mathcal{O}(|x|^3)] + [2x - x^2 + \mathcal{O}(|x|^3)]^2 + \mathcal{O}(|x|^3) \right\} \\ &= (1 - x)[1 - (2x - x^2) + 4x^2] + \mathcal{O}(|x|^3) = 1 - 3x + 7x^2 + \mathcal{O}(|x|^3). \quad \square\end{aligned}$$

Approximating a function

By means of Taylor polynomials, the differential-based approximation

$$f(x + \Delta x) \approx f(x) + f'(x)\Delta x$$

(cf. Section VII.1) can be both generalized and refined: if f is $n + 1$ times continuously differentiable in a neighborhood of the point x , then

$$\begin{aligned}f(x + \Delta x) &= f(x) + f'(x)\Delta x + \frac{1}{2!}f''(x)(\Delta x)^2 \\ &\quad + \cdots + \frac{1}{n!}f^{(n)}(x)(\Delta x)^n + \mathcal{O}(|\Delta x|^{n+1}).\end{aligned}$$

A good estimate for the error of the approximation, or at least its order of magnitude, is usually

$$\text{error} = \text{exact} - \text{approximation} \approx \frac{1}{(n+1)!} f^{(n+1)}(x) (\Delta x)^{n+1}.$$

EXAMPLE 3 $\sqrt[10]{1000} \approx ?$

Solution

$$\sqrt[10]{1000} = \sqrt[10]{1024 - 24} = 2 \sqrt[10]{1 - \frac{3}{128}}.$$

Setting $f(x) = \sqrt[10]{1-x}$, we have

$$\begin{aligned} f'(x) &= -\frac{1}{10}(1-x)^{-9/10}, & f''(x) &= -\frac{9}{100}(1-x)^{-19/10}, \\ f'''(x) &= -\frac{171}{1000}(1-x)^{-29/10}, & f^{(4)}(x) &= -\frac{4959}{10000}(1-x)^{-39/10} \end{aligned}$$

$$\begin{aligned} \Rightarrow \sqrt[10]{1000} &= 2f\left(\frac{3}{128}\right) \\ &\approx 2 \left[1 + \left(-\frac{1}{10}\right) \cdot \frac{3}{128} + \frac{1}{2} \cdot \left(-\frac{9}{100}\right) \left(\frac{3}{128}\right)^2 + \frac{1}{6} \cdot \left(-\frac{171}{1000}\right) \left(\frac{3}{128}\right)^3 \right] \\ &\approx 2 (1 - 0.00234275 - 0.0000247192 - 0.000000366926) \\ &= 1.995262327671. \end{aligned}$$

The error (exact value – approximation) is of order

$$+\frac{1}{4!} f^{(4)}(0) \cdot (\Delta x)^4 = -\frac{1653}{80000} \cdot \left(\frac{3}{128}\right)^4 \approx -6 \cdot 10^{-9}.$$

The correct value to 12 decimals is

$$\sqrt[10]{1000} \approx 1.995262314969. \quad \square$$

EXAMPLE 4 Estimate $\sin 31^\circ$ starting from the value $\sin 30^\circ = \frac{1}{2}$.

Solution With $f(x) = \sin x$, $x = \frac{\pi}{6}$, $\Delta x = \frac{\pi}{180}$, we have

$$\begin{aligned} f(x + \Delta x) &\approx \sin x + \cos x \Delta x - \frac{1}{2} \sin x (\Delta x)^2 \\ &= \frac{1}{2} + \frac{\sqrt{3}}{2} \cdot \frac{\pi}{180} - \frac{1}{2} \cdot \frac{1}{2} \cdot \left(\frac{\pi}{180}\right)^2 \approx 0.51503882. \end{aligned}$$

The error (exact value – approximation) is of order

$$-\frac{1}{3!} \cos x (\Delta x)^3 \approx -8 \cdot 10^{-7}.$$

The correct value to 8 decimals is: $\sin 31^\circ \approx 0.51503807$. $\square$

Local extrema

By Theorem V.5.2, at a local extremum of a differentiable function the derivative also has a zero. If the function is sufficiently regular in a neighborhood of the zero of the derivative, then Taylor polynomials can be used to determine whether it really is an extremum and, if so, of what kind.

THEOREM VII.5.2 If f is n times continuously differentiable in a neighborhood of the point c , $n \geq 2$, and $f^{(k)}(c) = 0$, $k = 1 \dots n-1$ and $f^{(n)}(c) \neq 0$, then

- a) if n is odd, then f has no local extremum at c ,
- b) if n is even, then f has at c
 - a local minimum if $f^{(n)}(c) > 0$,
 - a local maximum if $f^{(n)}(c) < 0$.

Proof By Theorem VII.4.4 and the hypotheses,

$$f(x) = f(c) + \frac{1}{n!}f^{(n)}(c)(x-c)^n + o(|x-c|^n), \quad \text{as } x \rightarrow c.$$

Since $f^{(n)}(c) \neq 0$, this can also be written in the form

$$f(x) = f(c) + \frac{1}{n!}f^{(n)}(c) [1 + o(1)] (x-c)^n, \quad \text{as } x \rightarrow c,$$

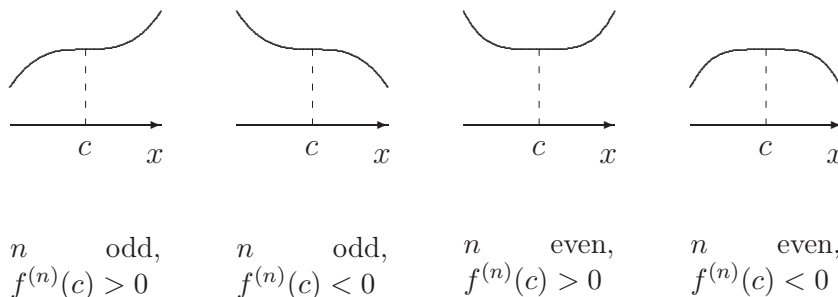
from which one sees that for a sufficiently small δ ($\delta > 0$)

$$f(x) = f(c) + \frac{1}{n!}f^{(n)}(c)k(x)(x-c)^n, \quad x \in [c-\delta, c+\delta],$$

where, for example,

$$\frac{1}{2} \leq k(x) \leq \frac{3}{2}.$$

Thus if n is even and $f^{(n)}(c) > 0$, then $f(x) > f(c)$ when $x \in [c-\delta, c)$ or $x \in (c, c+\delta]$, so that c is a local minimum point (Definition V.5.1). In the other cases the reasoning is analogous (cf. the figure). $\square$



EXAMPLE 5 Determine the type of the possible extremum in the following cases:

- a) $f(x) = \sin x + \cos x$, $c = \pi/4$
 b) $f(x) = e^{-x} + \sin x + \sqrt{1-x^2}$, $c = 0$
 c) $f(x) = 1 - \cos x - \sqrt{1+x^2}$, $c = 0$

Solution a) $f'(\pi/4) = 0$, $f''(\pi/4) = -\sqrt{2} \Rightarrow$ loc. maximum.
 b) $f'(0) = f''(0) = 0$, $f'''(0) = -2 \Rightarrow$ no extremum.
 c) $f^{(k)}(0) = 0$, $k = 1 \dots 3$, $f^{(4)}(0) = 5 \Rightarrow$ loc. minimum. $\square$

The function $\frac{f(x)}{g(x)} = \frac{0}{0}$ at $x = a$

If the functions f and g are regular in a neighborhood of the point a and $f(a) = g(a) = 0$, then Taylor's theorem can be used to study what the function $F(x) = f(x)/g(x)$ is like in a neighborhood of a . First, if one wants to resolve the limit question

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = ?$$

then the procedure based on Taylor's theorem is as follows: assume that f and g are n times continuously differentiable in a neighborhood of the point a and that

$$\frac{f(a)}{g(a)} = \frac{0}{0}, \quad \frac{f'(a)}{g'(a)} = \frac{0}{0}, \quad \dots, \quad \frac{f^{(n-1)}(a)}{g^{(n-1)}(a)} = \frac{0}{0}, \quad \frac{f^{(n)}(a)}{g^{(n)}(a)} = \frac{A}{B},$$

where $A \neq 0$ or $B \neq 0$. Then by Theorem VII.4.4

$$\frac{f(x)}{g(x)} = \frac{\frac{A}{n!}(x-a)^n + o(|x-a|^n)}{\frac{B}{n!}(x-a)^n + o(|x-a|^n)} = \frac{A + o(1)}{B + o(1)}, \quad \text{as } x \rightarrow a.$$

One concludes that if $A \neq 0$ and $B = 0$, then $|f(x)/g(x)| \rightarrow \infty$ as $x \rightarrow a$, so that the limit does not exist (as a real number — it may be that $\lim = \pm\infty$). Otherwise, i.e. if $B \neq 0$,

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \frac{A}{B} = \frac{f^{(n)}(a)}{g^{(n)}(a)},$$

and then, more generally,

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)} = \dots = \lim_{x \rightarrow a} \frac{f^{(n)}(x)}{g^{(n)}(x)}.$$

This result thus holds (within the regularity assumptions made), provided the last limit in the chain exists. — The result is already familiar as l'Hospital's rule (Theorem V.6.10).

EXAMPLE 6 Determine the limit $\lim_{x \rightarrow 0} \frac{\sin 2x - 2x}{\sinh x - x}$.

Solution The functions $f(x) = \sin 2x - 2x$ and $g(x) = \sinh x - x$ are smooth on $\mathbb{R}$. Differentiating gives

$$\frac{f'(x)}{g'(x)} = \frac{2 \cos 2x - 2}{\cosh x - 1}, \quad \frac{f''(x)}{g''(x)} = \frac{-4 \sin 2x}{\sinh x}, \quad \frac{f'''(x)}{g'''(x)} = \frac{-8 \cos 2x}{\cosh x}, \dots$$

We observe that $f^{(k)}(0) = g^{(k)}(0) = 0$ for $k = 0, 1, 2$, and $f'''(0) = -8$, $g'''(0) = 1$. Hence

$$\lim_{x \rightarrow 0} \frac{\sin 2x - 2x}{\sinh x - x} = \frac{-8}{1} = -8. \quad \square$$

By Taylor's theorem one thus obtains for the limit $\lim_{x \rightarrow a} f(x)/g(x)$ the same formula as with l'Hospital's rule. However, Taylor polynomials reveal much more about the function $f(x)/g(x)$. The following is an example of a more demanding problem setting.

EXAMPLE 7 Determine polynomials p and q such that on the intervals $[-1, 0)$ and $(0, 1]$

$$F(x) = \frac{\sin 2x - 2x}{\sinh x - x} = p(x) + \mathcal{O}(x^4), \quad G(x) = \frac{\sin x}{2x^2} + \frac{\cos x - 1}{x^3} = q(x) + \mathcal{O}(|x|^5).$$

$$\begin{aligned} \text{Solution} \quad \sin 2x - 2x &= \left(2x - \frac{1}{6}(2x)^3 + \frac{1}{120}(2x)^5 + \mathcal{O}(|x|^7)\right) - 2x \\ &= -\frac{4}{3}x^3 + \frac{4}{15}x^5 + \mathcal{O}(|x|^7), \\ \sinh x - x &= \left(x + \frac{1}{6}x^3 + \frac{1}{120}x^5 + \mathcal{O}(|x|^7)\right) - x \\ &= \frac{1}{6}x^3 + \frac{1}{120}x^5 + \mathcal{O}(|x|^7) \\ \Rightarrow \frac{\sin 2x - 2x}{\sinh x - x} &= \frac{-8 + \frac{8}{5}x^2 + \mathcal{O}(x^4)}{1 + \frac{1}{20}x^2 + \mathcal{O}(x^4)} \\ &= -8 + 2x^2 + \mathcal{O}(x^4) = p(x) + \mathcal{O}(x^4). \end{aligned}$$

$$\begin{aligned} \frac{\sin x}{2x^2} + \frac{\cos x - 1}{x^3} &= \frac{1}{2x^2} \left(x - \frac{x^3}{6} + \frac{x^5}{120} + \mathcal{O}(|x|^7) \right) \\ &\quad + \frac{1}{x^3} \left(-\frac{1}{2}x^2 + \frac{1}{24}x^4 - \frac{1}{720}x^6 + \mathcal{O}(x^8) \right) \\ &= -\frac{1}{24}x + \frac{1}{360}x^3 + \mathcal{O}(|x|^5) = q(x) + \mathcal{O}(|x|^5). \end{aligned}$$

From the results one can deduce, for example, the limits

$$\lim_{x \rightarrow 0} \left(\frac{\sin 2x - 2x}{x^2 \sinh x - x^3} + \frac{8}{x^2} \right) = 2, \quad \lim_{x \rightarrow 0} \left(\frac{\sin x}{2x^3} + \frac{\cos x - 1}{x^4} \right) = -\frac{1}{24}.$$

One also sees that if we set (by continuous extension) $F(0) = -8$ and $G(0) = 0$, then F has a local minimum at $x = 0$ and G is strictly decreasing in a neighborhood of the origin. $\square$

Difference approximations

Taylor's theorem often plays a central role when one wants to derive error estimates for numerical (approximate) computational methods that rely on an assumed regularity of the function. As an example, consider the so-called *difference approximations* used in the numerical computation of derivatives. In these the function is not assumed to be known as an expression; it suffices to know the value of the function at finitely many (typically only a few) points near the point under consideration. The following three are the most common difference approximations (assume $h > 0$):

$$f'(a) \approx \frac{f(a+h) - f(a)}{h}, \tag{a}$$

$$f'(a) \approx \frac{f(a+h) - f(a-h)}{2h}, \tag{b}$$

$$f''(a) \approx \frac{f(a+h) - 2f(a) + f(a-h)}{h^2}. \tag{c}$$

Of these, (a) is based directly on the definition of the derivative. Approximations (b) and (c) are called *central* difference approximations, because in them the function is evaluated at a set of points symmetric about a .

For the above approximations an error estimate can be derived from Taylor's theorem, assuming that f is sufficiently regular. As an example, consider approximation (b); the others are left as exercises (Exercises [14](#), [15](#)).

PROPOSITION VII.5.3 If f is three times continuously differentiable on the interval $[a-h, a+h]$, then the difference approximation (b) satisfies the error estimate

$$\left| f'(a) - \frac{f(a+h) - f(a-h)}{2h} \right| \leq \frac{1}{6} M h^2, \quad M = \max_{x \in [a-h, a+h]} |f'''(x)|.$$

Proof By the hypotheses, for some $\xi_1 \in (a, a+h)$ and $\xi_2 \in (a-h, a)$

$$\begin{aligned} f(a+h) &= f(a) + f'(a)h + \frac{1}{2}f''(a)h^2 + \frac{1}{6}f'''(\xi_1)h^3, \\ f(a-h) &= f(a) - f'(a)h + \frac{1}{2}f''(a)h^2 - \frac{1}{6}f'''(\xi_2)h^3 \end{aligned}$$

(Theorem VII.4.3: with the interval $[a, b]$ replaced by $[a-h, a+h]$, $x_0 = a$, $x = a \pm h$, $n = 2$). Subtracting these gives

$$f'(a) - \frac{f(a+h) - f(a-h)}{2h} = -\frac{1}{12} [f'''(\xi_1) + f'''(\xi_2)] h^2.$$

Using on the right the estimate $|f'''(\xi_1) + f'''(\xi_2)| \leq |f'''(\xi_1)| + |f'''(\xi_2)| \leq 2M$ the claim follows. $\square$

By Proposition VII.5.3, approximation (b) is exact for second-degree polynomials (for which $M = 0$). Choosing $f(x) = \frac{1}{6}M(x-a)^3$ also shows that, under the assumptions made, the error estimate is the sharpest possible.

In general, a difference approximation is said to have (accuracy) *order* r if its error is $\mathcal{O}(h^r)$ but not $o(h^r)$ for a general, sufficiently regular function. Thus approximation (b) has order $r = 2$, i.e. it is a *second-order* approximation. Approximation (c) likewise has order $r = 2$ (Exercise 15), whereas (a) is a first-order ($r = 1$) approximation (Exercise 14).

EXERCISES

1. Justify the rules of order-of-magnitude algebra (assume $x, y \geq 0$)

$$\mathcal{O}(x) + \mathcal{O}(y) = \mathcal{O}(\max\{x, y\}), \quad \mathcal{O}(x)\mathcal{O}(y) = \mathcal{O}(xy), \quad o(x)\mathcal{O}(y) = o(xy).$$

2. What properties of the function f are meant by the following notations?

- a) $f(x) = f(a) + o(1)$, as $x \rightarrow a$.

- b) $f(x_1) - f(x_2) = \mathcal{O}(|x_1 - x_2|)$, for $x_1, x_2 \in [a, b]$.

- c) $f(x) = f(a) + k(x-a) + o(|x-a|)$, as $x \rightarrow a$.

3. Give order-of-magnitude estimates for the errors of the following approximations:

- a) $\sqrt{x^4 + 2} \approx \sqrt{2}$, as $x \rightarrow 0$

- b) $\sqrt{x^4 + 3x} \approx \sqrt{3x}$, as $x \rightarrow 0^+$

- c) $\sqrt{x^4 + 4x} \approx x^2$, as $x \rightarrow \pm\infty$
 d) $\sqrt{x^4 + 4x} \approx x^2 + 2x^{-1}$ as $x \rightarrow \infty$
 e) $x/\sin x \approx 1$, as $x \rightarrow 0$
 f) $\ln(1 + e^x) \approx x$, as $x \rightarrow \infty$
4. Estimate the absolute value of the error of the approximation $f(x) \approx T_2(x, 0)$ on the interval $[-1/2, 1/2]$ for the following functions:
 a) $\sqrt{1+x}$ b) $\sqrt[10]{1+x}$ c) $\tan x$ d) $\ln(1+x)$ e) $e^{\sin x}$
5. Estimate the error of the following function approximations:

a) $\sqrt[3]{8+x} \approx 2 + \frac{x}{12} - \frac{x^2}{288}$, for $|x| \leq 1$
 b) $\sin x \approx x - \frac{1}{6}x^3$, when $x \in [0^\circ, 10^\circ]$
 c) $\frac{1}{x^4} \left[\frac{1}{\sqrt{1+x^2}} - \sqrt{1-x^2} \right] \approx \frac{1}{2}$, for $|x| \leq 0.15$ and $x \neq 0$

6. If $T_n(x, 0)$ is the Taylor polynomial of the function $f(x) = \cos x$, for which values of a and n can one guarantee that
 a) $|\cos x - T_2(x, 0)| \leq 10^{-4}$ on the interval $[-a, a]$,
 b) $|\cos x - T_n(x, 0)| \leq 10^{-4}$ on the interval $[-\pi/2, \pi/2]$?
7. a) Compute a rational approximation for the number $1/\sqrt[4]{e}$ by approximating the function e^x with its second-degree Taylor polynomial. Estimate the error of the approximation using Taylor's theorem and compare it with the exact value of the error.
 b) Compute a rational approximation for the number $a = \sqrt[12]{4000}$ by writing $4000 = 4096(1-x)$ and approximating the function $f(x) = \sqrt[12]{1-x}$ with its second-degree Taylor polynomial. Estimate the error of the approximation using Taylor's theorem and compare it with the exact value of the error.
8. According to the theory of relativity, the kinetic energy of a particle moving with speed v is

$$E_k = \frac{mc^2}{\sqrt{1 - \frac{v^2}{c^2}}} - mc^2,$$

where m is the mass of the particle and $c \approx 3 \cdot 10^8$ m/s is the speed of light. For what values of the speed v is the relative error of the approximation $E_k \approx \frac{1}{2}mv^2$ at most 0.01%?

9. For the following functions, the point $x = 0$ is a possible local minimum or maximum point. Investigate this by differentiation!

a) $(1-x)e^x$ b) $(2-x^2)\cos x$ c) $(x-x^4)\sin x$ d) $\sqrt{1-x^2}-1/\sqrt{1+x^2}$

10. For which values of a and b does the function

a) $e^x + ax + bx^2$ b) $x \sin x + ax^2 + bx^4$ c) $\sin x + \cos x + ax + bx^2$

have a local minimum at the point $x = 0$?

11. The equation $y(\cos y - \sin y) = 2 \sin x + \cos x + ax + b$ defines a function $y(x)$ in a neighborhood of the point $x = 0$. Determine the constants a and b , given that $y(x)$ attains at $x = 0$ a local extremum $y(0) = 0$. Is it a maximum or a minimum?

12. Determine the following limits:

a) $\lim_{x \rightarrow 0} \frac{\sqrt[7]{1+x} - 1}{x}$ b) $\lim_{x \rightarrow 0} \frac{3x}{\tan 4x}$ c) $\lim_{x \rightarrow 0} \frac{2 \cos x - 2 + x^2}{x^4}$
d) $\lim_{x \rightarrow 0} \frac{x - \sin x}{x - \tan x}$ e) $\lim_{x \rightarrow 0} \frac{10^x - e^x}{x}$ f) $\lim_{x \rightarrow 0} \frac{1 - \cos ax}{1 - \cos bx}$
g) $\lim_{x \rightarrow \pi/2} \frac{\cos 3x}{\pi - 2x}$ h) $\lim_{t \rightarrow \pi/2} \frac{\ln \sin t}{\cos t}$ i) $\lim_{x \rightarrow 0^+} \frac{\sin^2 x}{\tan x - x}$
j) $\lim_{x \rightarrow 1} \frac{\ln(ex) - 1}{\sin \pi x}$ k) $\lim_{x \rightarrow -\infty} x \sin \frac{1}{x}$ l) $\lim_{x \rightarrow \infty} x^2 \ln(\cos \frac{\pi}{x})$
m) $\lim_{x \rightarrow \infty} [\sqrt{x^2 + 154x} - \ln(5 + e^x)]$ n) $\lim_{x \rightarrow \infty} (\sqrt[6]{x^6 + 42x^5 + 77x} - x)$

13. For the following functions f , determine a polynomial p of the lowest possible degree such that, for the given m and some $\delta > 0$, $f(x) = p(x) + \mathcal{O}(|x|^m)$ when $0 < |x| < \delta$. Also investigate the type of the point $x = 0$ as a possible local extremum of f , when we set $f(0) = \lim_{x \rightarrow 0} f(x)$.

a) $\frac{1 - \cos x}{x^2}$, $m = 7$ b) $\frac{\sin x - x}{x^3}$, $m = 8$ c) $\frac{x^3}{\sin x - x}$, $m = 3$
d) $\frac{x}{e^x - 1}$, $m = 3$ e) $\frac{\ln(1+x)}{x}$, $m = 5$ f) $\frac{\sqrt{1+x} - 1}{\sqrt[3]{1-x} - 1}$, $m = 3$

14. Show that if f is continuous on the interval $[a, a+h]$ and twice differentiable on the interval $(a, a+h)$, then for some $\xi \in (a, a+h)$

$$f'(a) - \frac{f(a+h) - f(a)}{h} = -\frac{1}{2} f''(\xi) h.$$

15. (*) Using Taylor's theorem, show that if f is four times continuously differentiable on the interval $[a - h, a + h]$, then

$$\left| f''(a) - \frac{f(a+h) - 2f(a) + f(a-h)}{h^2} \right| \leq \frac{1}{12} \left(\max_{x \in [a-h, a+h]} |f^{(4)}(x)| \right) h^2.$$

How does the estimate hold for the function $f(x) = (x - a)^4$?

16. (*) a) Show that for certain (which?) values of A and B

$$\arccos x = A\sqrt{1-x} + B(1-x)^{3/2} + \mathcal{O}((1-x)^{5/2}), \quad \text{as } x \rightarrow 1^-.$$

b) Show that if the equations of the cycloid $x = R(t - \sin t)$, $y = R(1 - \cos t)$ are eliminated to the form $y = y(x)$ (by solving $t = t(x)$ from the first equation), then in a neighborhood of the origin

$$y(x) = 3\sqrt[3]{\frac{Rx^2}{6}} + \mathcal{O}\left(\sqrt[3]{\frac{x^4}{R}}\right).$$

17. (*) Let c be a simple zero of the function f and let f be three times continuously differentiable in a neighborhood of c . One wants to determine c by an iteration modified from Newton's method, $x_{n+1} = F(x_n)$, where

$$F(x) = x - \frac{f(x)}{f'(x)} + [f(x)]^2 g(x).$$

How should $g(x)$ be chosen so that the convergence of the iteration toward c is at least cubic? Under what condition is the convergence exactly cubic? Apply the method to the function $f(x) = x^2 - a$, $a > 0$ (cf. Exercise V.7:17b).

VII.6 Interpolation polynomials

In ordinary *polynomial interpolation* the starting idea is that only a finite number of point values of the function f are known:

$$f(x_i) = f_i, \quad i = 0 \dots n.$$

On the basis of this data one wants to represent f approximately as a polynomial also elsewhere than at the points x_i , e.g. on some interval. One speaks of interpolation in particular when $f(x)$ is to be computed at 'interior points' $x \in [\min\{x_i\}, \max\{x_i\}]$; otherwise (i.e. when $x < \min\{x_i\}$ or $x > \max\{x_i\}$) one says that it is a matter of *extrapolation*. The given data may be, for example, measured 'data'. Another common application is one in which the function f is known indirectly, e.g. as the solution of a differential equation, and one wants to compute the values f_i at chosen points x_i . In that case polynomial interpolation is useful in designing the computational algorithm itself.

In polynomial approximation of a function the basic assumption (or at least the hope) is usually that the function is sufficiently regular, so that by Taylor's theorem one knows that the function can (at least on a short interval) be well approximated by a polynomial, namely its Taylor polynomial. Since the Taylor polynomial itself cannot be determined directly from the above data, it is natural to choose as the approximating polynomial p one that fits the given data, i.e. satisfies

$$p(x_i) = f(x_i), \quad i = 0 \dots n. \quad (1)$$

Since there are $n+1$ conditions here, it is further natural to choose n as the degree of p , so that p likewise has $n+1$ free coefficients. The polynomial p defined in this way is called the *Lagrange interpolation polynomial* of f at the points x_i . In this connection the points x_i are called *interpolation points* and the conditions (1) *interpolation conditions*. These conditions do indeed define a unique polynomial of degree n :

PROPOSITION VII.6.1 The Lagrange interpolation polynomial is unique.

Proof The existence of an interpolation polynomial satisfying conditions (1) follows from the computational formula (2) presented below, so it suffices to show uniqueness. So let p_1 and p_2 both be polynomials of degree n satisfying conditions (1). Then $q(x) = p_1(x) - p_2(x)$ is a polynomial of degree at most n and

$$q(x_i) = 0, \quad i = 0 \dots n,$$

i.e. q has $n + 1$ real zeros. By the fundamental theorem of algebra we must then have $q = 0$. Hence $p_1 = p_2$, i.e. the polynomial satisfying conditions (1) (insofar as it exists) is unique. $\square$

For the approximation error of the Lagrange interpolation polynomial the following result holds, resembling the error formula for Taylor polynomials (Theorem VII.4.3). Indeed, it will be seen later (Theorem VII.6.4) that these two results are special cases of a more general error formula for interpolation polynomials. In what follows, the closed interval $[a, b]$, where $a = \min\{x \mid x \in X\}$ and $b = \max\{x \mid x \in X\}$, is called the interval *spanned* by a finite point set X .

THEOREM VII.6.2 Let f be continuous on the interval $[a, b]$ and $n + 1$ times differentiable on the interval (a, b) , where $[a, b]$ is the interval spanned by the point x and the distinct points $x_0, \dots, x_n$, $n \in \mathbb{N}$. Then if p is the Lagrange interpolation polynomial of f at the points $x_0, \dots, x_n$, the error formula

$$f(x) - p(x) = \frac{1}{(n+1)!} f^{(n+1)}(\xi) \prod_{i=0}^n (x - x_i).$$

holds for some $\xi \in (a, b)$.

Proof is based on the same idea as the proof of Taylor's theorem in Section VII.4. First, if $x \in \{x_0, \dots, x_n\}$, the claim is true for every $\xi \in (a, b)$, so we may assume that $x \notin \{x_0, \dots, x_n\}$. Denote

$$w(t) = \prod_{i=0}^n (t - x_i)$$

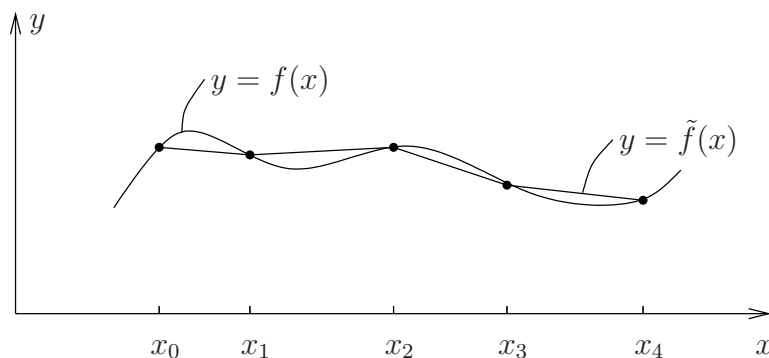
and consider the function

$$g(t) = f(t) - p(t) - Hw(t), \quad H = [f(x) - p(x)]/w(x).$$

The zeros of the function g are the interpolation points $x_0, \dots, x_n$ together with the point x . Since there are therefore at least $n + 2$ zeros on the interval $[a, b]$, and since between any two of them there is always a zero of the derivative (Theorem V.6.2), $g'(t)$ has at least $n + 1$ zeros on the open interval (a, b) . Then g'' has at least n zeros on this interval, and finally $g^{(n+1)}$ has at least one zero $\xi \in (a, b)$. But then

$$\begin{aligned} 0 &= g^{(n+1)}(\xi) = f^{(n+1)}(\xi) - p^{(n+1)}(\xi) - Hw^{(n+1)}(\xi) \\ &= f^{(n+1)}(\xi) - H(n+1)! \\ \Rightarrow \quad H &= \frac{f(x) - p(x)}{w(x)} = \frac{1}{(n+1)!} f^{(n+1)}(\xi). \quad \square \end{aligned}$$

Of the interpolation principles, *linear interpolation*, based on degree $n = 1$, is very widely used because of its simplicity. For example, approximating the curve $y = f(x)$ by the polygonal line through the points $(x_i, f(x_i))$ (say, to draw a figure or to estimate arc length) amounts to *piecewise linear* interpolation of f .



If f is twice continuously differentiable on the interpolation interval $[x_0, x_1]$, then by Theorem VII.6.2 the error of linear interpolation on that interval is at most

$$\begin{aligned} \max_{x \in [x_0, x_1]} |f(x) - p(x)| &\leq \max_{x \in [x_0, x_1]} \left| \frac{1}{2}(x - x_0)(x - x_1) \right| \max_{\xi \in [x_0, x_1]} |f''(\xi)| \\ &= \frac{1}{8} (x_1 - x_0)^2 \max_{x \in [x_0, x_1]} |f''(x)|. \end{aligned}$$

If the interpolation is done piecewise on $[a, b]$ and the spacing of the interpolation points is at most h , then the maximum error of the approximation is thus at most

$$\max_{x \in [a, b]} |f(x) - \tilde{f}(x)| \leq \frac{1}{8} h^2 \max_{x \in [a, b]} |f''(x)|.$$

This estimate cannot be improved, since if $f''(x) = \text{constant}$ (i.e. f is a second-degree polynomial) and the interpolation points are equally spaced, then the upper bound of the error estimate is attained at the midpoint between successive interpolation points.

If in *quadratic* (= second-degree) interpolation the points x_0, x_1, x_2 are equally spaced with spacing $= h$, then

$$\max_{x \in [x_0, x_2]} |(x - x_0)(x - x_1)(x - x_2)| = \max_{x \in [-h, h]} |x|(h^2 - x^2) = \frac{2}{3\sqrt{3}} h^3,$$

so that the quadratic interpolation error, on the interval spanned by equally spaced interpolation points, is at most

$$\max_{x \in [x_0, x_2]} |f(x) - p(x)| \leq \frac{1}{9\sqrt{3}} h^3 \max_{x \in [x_0, x_2]} |f'''(x)| \quad (h = x_1 - x_0 = x_2 - x_1).$$

EXAMPLE 1 A function E defined on the interval $(0, \infty)$ satisfies the conditions

$$E'(x) = -\frac{e^{-x}}{x}, \quad x > 0, \quad \lim_{x \rightarrow \infty} E(x) = 0.*$$

One wants to determine the values of E approximately on the interval $[1, 2]$ by first computing E at the points $x_i = 1 + (i-1)h$, $i = 0 \dots n$, $h = 1/n$ (assume that this can be done very accurately) and using interpolation at the other points. How large must n be if one uses a) linear, b) quadratic interpolation and wants the interpolation error to be at most $5 \cdot 10^{-9}$?

Solution Since $E'(x) = -e^{-x}/x$, $x > 0$, we have

$$\begin{aligned} E''(x) &= e^{-x} \left(\frac{1}{x} + \frac{1}{x^2} \right) \Rightarrow |E''(x)| \leq 2/e, \quad x \in [1, 2], \\ E'''(x) &= -e^{-x} \left(\frac{1}{x} + \frac{2}{x^2} + \frac{2}{x^3} \right) \Rightarrow |E'''(x)| \leq 5/e, \quad x \in [1, 2]. \end{aligned}$$

Thus it suffices to choose $N = n + 1$ such that

$$\begin{aligned} \text{a) } \frac{1}{8} \cdot \frac{2}{e} \cdot \left(\frac{1}{N-1} \right)^2 &\leq 5 \cdot 10^{-9} \Leftrightarrow \underline{\underline{N \geq 4290}}, \\ \text{b) } \frac{1}{9\sqrt{3}} \cdot \frac{5}{e} \cdot \left(\frac{1}{N-1} \right)^3 &\leq 5 \cdot 10^{-9} \Leftrightarrow \underline{\underline{N \geq 288}}. \quad \square \end{aligned}$$

Lagrange basis polynomials

In higher-degree polynomial interpolations one is often dealing with *polynomial fitting* to the points $(x_i, f(x_i))$, where the values $f(x_i)$ may, for example, be measured. Advanced numerical and symbolic analysis software has its own commands for this task (e.g. Mathematica: `Fit`, Matlab: `Polyfit`). When computing by hand, or especially when one wants to study the properties of the interpolation polynomial from a theoretical standpoint, one can use the representation

$$p(x) = \sum_{i=0}^n f(x_i) L_i(x), \quad (2)$$

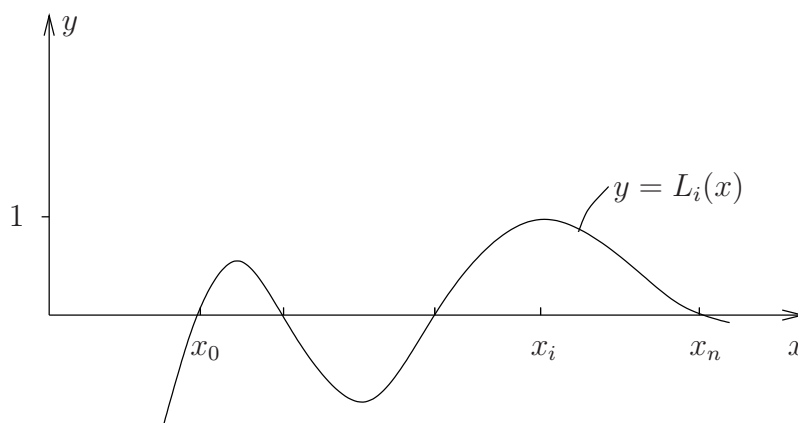
where the so-called (Lagrange) *basis polynomials* $L_i(x)$ (of degree n) are determined by the interpolation conditions

$$L_i(x_j) = \begin{cases} 1, & \text{when } j = i, \\ 0, & \text{when } j \neq i \end{cases} \quad (j \in \{0, \dots, n\}). \quad (3)$$

*The function is called the *exponential integral function*.

It is easily checked that the polynomial satisfying these conditions is (cf. the figure)

$$L_i(x) = \frac{(x - x_0) \cdots (x - x_{i-1})(x - x_{i+1}) \cdots (x - x_n)}{(x_i - x_0) \cdots (x_i - x_{i-1})(x_i - x_{i+1}) \cdots (x_i - x_n)} = \frac{\prod_{j \neq i} (x - x_j)}{\prod_{j \neq i} (x_i - x_j)}.$$



From conditions (3) one also sees immediately that the polynomial (2) satisfies the imposed interpolation conditions (1). Thus the solvability of the Lagrange interpolation problem has now been established.

EXAMPLE 2 For a regular function the following measurement results are known:

$$f_1 = f(0.1) = 1.4491, \quad f_2 = f(0.2) = 1.4832, \quad f_3 = f(0.3) = 1.5166.$$

Estimate $f(0)$.

Solution Fit a second-degree Lagrange interpolation polynomial p to the measurement results and estimate $f(0) \approx p(0)$. Here $\{x_0, x_1, x_2\} = \{0.1, 0.2, 0.3\}$, so

$$\begin{aligned} p(0) &= 1.4491 \cdot \frac{(0 - 0.2) \cdot (0 - 0.3)}{(0.1 - 0.2) \cdot (0.1 - 0.3)} \\ &\quad + 1.4832 \cdot \frac{(0 - 0.1) \cdot (0 - 0.3)}{(0.2 - 0.1) \cdot (0.2 - 0.3)} \\ &\quad + 1.5166 \cdot \frac{(0 - 0.1) \cdot (0 - 0.2)}{(0.3 - 0.1) \cdot (0.3 - 0.2)} \\ &= 3 \cdot 1.4491 - 3 \cdot 1.4832 + 1 \cdot 1.5166 = \underline{\underline{1.4143}}. \end{aligned}$$

Here the 'measurement results' were in fact obtained from the function $f(x) = \sqrt{2+x}$, for which $f(0) = 1.4142$. $\square$

The representation (2) based on basis polynomials is especially convenient when one wants to estimate the effect of the function's evaluation errors (e.g. measurement errors) on the interpolation result. Namely, if an interpolation polynomial $\tilde{p}$ is computed from erroneous values $\tilde{f}_i$ (correct value = f_i), then by formula (2)

$$p(x) - \tilde{p}(x) = \sum_{i=0}^n (f_i - \tilde{f}_i) L_i(x).$$

If in particular it is known that $|f_i - \tilde{f}_i| \leq \delta$, $i = 0, \dots, n$, then

$$|p(x) - \tilde{p}(x)| \leq \delta \sum_{i=0}^n |L_i(x)| = \delta K(x).$$

The error amplification factor $K(x) = \sum_{i=0}^n |L_i(x)|$ defined here thus tells how much the evaluation errors can, at worst, be amplified at the point x .

EXAMPLE 2 (continued) In the example, $L_1(0) = 3$, $L_2(0) = -3$ and $L_3(0) = 1$. Hence $K(0) = 3 + 3 + 1 = 7$, i.e. the effect of the errors at the point $x = 0$ is at worst 7 times the individual evaluation errors. If we assume that the errors are at most 10^{-4} in absolute value, then the worst case occurs when $f_1 - \tilde{f}_1 = -(f_2 - \tilde{f}_2) = f_3 - \tilde{f}_3 = \pm 10^{-4}$. $\square$

Extrapolation

In Example 2 the function was approximated outside the interval spanned by the interpolation points, in which case one says that it is a matter of *extrapolation*. Extrapolation is, from of old, a very popular and fairly universally applicable way of improving the accuracy of numerical computations. Extrapolation can be used whenever the quantity to be computed can be assumed to depend in a regular (i.e. smooth) way on some parameter connected with the computation. Suppose, for example, that the quantity to be computed is a real number a determined as a limit

$$a = \lim_{h \rightarrow 0^+} f(h),$$

where each $f(h)$, $h > 0$, can be computed, but the computation becomes more and more laborious as h decreases. If it can now be assumed that the function $f(x)$ is regular on some interval $[0, b]$, $b > 0$, then the approximations

$$a \approx a_n = f(x_n), \quad n = 1, 2, \dots \quad (x_n \rightarrow 0^+)$$

given by the numerical algorithm can be improved by extrapolation. This gives rise to a so-called *extrapolation table*, in which polynomials $p(x)$ of ever higher degree are fitted to the computed results, and each polynomial is used to estimate $a \approx p(0)$:

	degree = 0	degree = 1	degree = 2	degree = 3
x_1	$f(x_1)$			
x_2	$f(x_2)$	$p^{(1,2)}(0)$		
x_3	$f(x_3)$	$p^{(2,3)}(0)$	$p^{(1,3)}(0)$	
x_4	$f(x_4)$	$p^{(3,4)}(0)$	$p^{(2,4)}(0)$	$p^{(1,4)}(0)$

Here $p^{(i,j)}(x)$ denotes the interpolation polynomial of degree $j - i$ fitted to the points $x_i \dots x_j$ ($p^{(i,i)}(x) = f(x_i) = \text{constant}$). It turns out that the columns of the table are determined recursively from the previous column. Namely

$$p^{(i,j)}(x) = \frac{(x_j - x)p^{(i,j-1)}(x) + (x - x_i)p^{(i+1,j)}(x)}{x_j - x_i}. \quad (4)$$

This recursion formula (justification by induction: Exercise 10) greatly facilitates constructing the table.* The following is an example of the 'art of computation' based on formula (4).

EXAMPLE 3 By *Stirling's formula*, $\phi(n) = n! / (\sqrt{2\pi n} e^{-n} n^n) \approx 1$ for large values of n . Extrapolate $\phi(100)$ from the values $\phi(n)$, $n = 5 \dots 9$, given in addition that $\phi(n) = f(1/n)$, where $f(x)$ is a smooth function on the interval $[0, 1]$.

Solution Construct the extrapolation table

x_i	$f(x_i)$	degree = 1	degree = 2	degree = 3	degree=4
1/5	1.0167..				
1/6	1.0139..	1.00076..			
1/7	1.0119..	1.00078..	1.000823..		
1/8	1.0104..	1.00079..	1.000827..	1.000833565..	
1/9	1.0092..	1.00080..	1.000830..	1.000833632..	1.000833708..

*The extrapolation table based on formula (4) is called *Neville's scheme*. Before the age of computers, such algorithmic inventions (facilitating hand computation) were of considerable practical importance.

Presumably the fourth-degree interpolation gives the most accurate result, so $\phi(100) \approx \underline{\underline{1.00083371}}$. (The correct value is 1.000833677..) $\square$

The essential thing in the example was the additional information that made it possible to choose the right variable ($x_i = 1/n_i$) in the extrapolation.

*Generalized polynomial interpolation

If p is a polynomial of degree n and satisfies the conditions

$$\begin{aligned} p^{(k)}(x_i) &= f^{(k)}(x_i), \quad k = 0 \dots \nu_i - 1, \quad i = 1 \dots m, \\ \text{where } \nu_i &\in \mathbb{N} \quad \text{and} \quad \sum_{i=1}^m \nu_i = n + 1, \end{aligned} \tag{5}$$

then p is said to be the *generalized interpolation polynomial* of the function f at the points x_i . Here and in what follows it is assumed that the derivatives of f appearing in conditions (5) (if $\nu_i \geq 2$) exist at the points x_i . According to conditions (5), the number m of interpolation points in a generalized polynomial interpolation of degree n can be anything in the range $1 \leq m \leq n + 1$. In the case $m = n + 1$ one must have $\nu_i = 1 \forall i$, which is Lagrange interpolation. At the other extreme ($m = 1$) one must have $\nu_1 = n + 1$, in which case $p(x) =$ the Taylor polynomial $T_n(x, x_1)$ of f .

PROPOSITION VII.6.3 If p is a polynomial of degree n , then p is uniquely determined by conditions (5).

Proof The solvability of the interpolation problem (5) can be established in the same way as in the case of Lagrange interpolation, see Exercise 12. To establish uniqueness it suffices to show that if $f^{(k)}(x_i) = 0 \forall i, k$, then $p = 0$ (cf. the proof of Proposition VII.6.1). In this case x_i is a ν_i -fold zero of the polynomial p by the interpolation conditions (5), so that

$$p(x) = q(x) \prod_{i=1}^m (x - x_i)^{\nu_i} = q(x) w(x),$$

where q is a polynomial. But p is of degree n , and by conditions (5) w is of degree $n + 1$, so the only possibility is $q = 0$, whence $p = 0$ as well. $\square$

The following general polynomial approximation result, whose complete proof is not given, contains as special cases both Theorem VII.6.2 and Taylor's theorem VII.4.3.

THEOREM VII.6.4 (Multi-point Taylor theorem) Let f be continuous on the interval $[a, b]$ and $n + 1$ times differentiable on the interval (a, b) , where $[a, b]$ is the interval spanned by the point x and the distinct points $x_0, \dots, x_m$. Then if p is the generalized interpolation polynomial of degree n defined by conditions (5), the error formula

$$f(x) - p(x) = \frac{1}{(n+1)!} f^{(n+1)}(\xi) \prod_{i=1}^m (x - x_i)^{\nu_i}.$$

holds for some $\xi \in (a, b)$.

Proof (idea) If $x \in \{x_1, \dots, x_m\}$, the claim is true $\forall \xi$. So let $x \neq x_i \forall i$, and consider the function

$$g(t) = f(t) - p(t) - Hw(t), \quad H = [f(x) - p(x)]/w(x), \quad w(x) = \prod_{i=1}^m (x - x_i)^{\nu_i}.$$

Since $w(t) = t^{n+1} + (\text{polynomial of degree } n)$, we have $w^{(n+1)}(t) = (n+1)! \forall t$, so the error formula claims that $g^{(n+1)}(\xi) = 0$ for some $\xi \in (a, b)$. This has already been shown in the cases $m = 1, \nu_1 = n + 1$ (Taylor's theorem) and $m = n + 1, \nu_i = 1$ (Theorem VII.6.2). In the more general case the reasoning is analogous. For example, let

$$\nu_i = 2, \quad i = 1 \dots m, \quad m \geq 2, \quad n = 2m - 1.$$

Then, since $g(x_i) = 0, i = 1 \dots m$, and $g(x) = 0$, on each of the open intervals between the point x and the points $x_i, i = 1 \dots m$, there is a zero of g' (m in total). On the other hand, $g'(x_i) = 0, i = 1 \dots m$ as well, so g' has at least $2m$ zeros on $[a, b]$. It follows (cf. the proof of Theorem VII.6.2) that at at least one point $\xi \in (a, b)$ we have $g^{(2m)}(\xi) = g^{(n+1)}(\xi) = 0$, i.e. the claim is true in the assumed case. (The general case is omitted.) $\square$

In the case considered above, where $\nu_i = 2, i = 1 \dots m$, the polynomial p is called the *Hermite* interpolation polynomial of the function f at the points x_i .

EXAMPLE 1 (continued) How many equally spaced nodes x_i are needed if third-degree Hermite interpolation is used between the points?

Solution By Theorem VII.6.4 the interpolation error on the interval $[x_i, x_{i+1}]$ is at most

$$\begin{aligned} |E(x) - p(x)| &\leq \max_{x \in [x_i, x_{i+1}]} \frac{1}{4!} (x - x_i)^2 (x - x_{i+1})^2 \max_{\xi \in [x_i, x_{i+1}]} |E^{(4)}(\xi)| \\ &= \frac{1}{384} h^4 \max_{x \in [x_i, x_{i+1}]} |E^{(4)}(x)|. \end{aligned}$$

On the interval $[1, 2]$ we have $|E^{(4)}(x)| \leq 16/e$, so the required accuracy is achieved by:

$$\frac{1}{384} \cdot \frac{16}{e} \cdot \left(\frac{1}{N-1}\right)^4 \leq 5 \cdot 10^{-9} \quad \Leftrightarrow \quad \underline{\underline{N \geq 43}}. \quad \square$$

EXERCISES

1. The following is known about the function $f(x)$: $f(-0.1) = 1.70 \pm 0.05$, $f(0.2) = 1.80 \pm 0.03$ and $-1 \leq f''(x) \leq 0$ on the interval $[-0.1, 0.2]$. Determine the narrowest possible interval in which $f(0)$ is certain to lie.
2. The values of the function e^x have been computed to five significant figures (ordinary rounding) at the points $x_i = i/100$, $i = 0 \dots 200$ of the interval $[0, 2]$. Estimate the maximum error, on the interval $[0, 2]$, of a) linear, b) quadratic interpolation computed from these values. Estimate separately the effect of the rounding errors.
3. It is known about a certain regular function F defined on the interval $(0, \infty)$ that F attains its absolute minimum value on the interval $[1, 2]$. It is further known that $F(0.5) = \sqrt{\pi}$, $F(1) = 1$, $F(1.5) = \frac{1}{2}\sqrt{\pi}$ and $F(2) = 1$. Compute the minimum point and value of F approximately using a) quadratic interpolation at the points $\{0.5, 1, 1.5\}$, b) quadratic interpolation at the points $\{1, 1.5, 2\}$, c) third-degree interpolation at all four points. (The assumptions are satisfied by the Γ function, see Exercise VIII.7:5 and Proposition XIII.4.1.)
4. A parametric curve $\vec{r} = \vec{u}(t) = x(t)\vec{i} + y(t)\vec{j}$, $t \in [a, b]$, is approximated by the polygonal line $\vec{r} = \vec{v}(t) = \hat{x}(t)\vec{i} + \hat{y}(t)\vec{j}$ ($a = t_0 < t_1 < \dots < t_n = b$) through the points $(x(t_i), y(t_i))$, $i = 0 \dots n$. If $x(t)$ and $y(t)$ are twice continuously differentiable on $[a, b]$, $|x''(t)| \leq M$, $|y''(t)| \leq M$ and $t_i - t_{i-1} \leq h$, how large at most is $\delta_h = \max_{t \in [a, b]} |\vec{u}(t) - \vec{v}(t)|$? Compare the estimate with reality in the case $x(t) = R \cos t$, $y(t) = R \sin t$, $t \in [0, \pi]$.
5. A pendulum wall clock loses 5 min 24 s per day. The clock is regulated by tightening the screw at the end of the pendulum by 5 full turns (which shortens the pendulum rod slightly). After the adjustment, the clock is observed to gain 3 min 36 s per day. a) How should the clock be regulated next? b) Let the relative daily gain of the clock be $f(x)$, where x = the rotation of the screw (in turns) from the correct setting in the tightening direction. Estimate how many seconds the clock gains or loses per day after the adjustment of part a), if one assumes that $f''(0) = +2 \cdot 10^{-5}$.

6. From the numbers $7! \dots 10!$ extrapolate the limit $a = \lim_{n \rightarrow \infty} n!e^n n^{-n-\frac{1}{2}}$ and compare with the exact value $a = \sqrt{2\pi}$. (Cf. Example 3.)
7. Prove Theorem VII.6.4 in the case $m = n = 3$, $\nu_1 = \nu_3 = 1$, $\nu_2 = 2$.
8. The values of the functions $\sin x$ and $\cos x$ are to be determined on the interval $[0, \pi/4]$ by first computing the function values sufficiently accurately on an equally spaced set of points of that interval and then using third-degree Hermite interpolation between the points. How many interpolation points are needed if the error is allowed to be at most $5 \cdot 10^{-9}$?
9. One wants to compute numerically the derivative $f^{(k)}(a)$ of a function f by the approximation $f^{(k)}(a) \approx p^{(k)}(a)$, where $p(x)$ is an interpolation polynomial of f . Show that in the following cases the alternative interpolations lead to the same difference approximation — which one? (Cf. the previous section.)
 - a) $k = 1$: Linear interpolation at the points $a \pm h$, or quadratic interpolation at the points a and $a \pm h$.
 - b) $k = 2$: Quadratic interpolation at the points a and $a \pm h$, or generalized third-degree interpolation at the same points with the additional condition $p'(a) = f'(a)$.
10. (*) Prove the recursion formula (4) by induction.
11. (*) (Limit data) About the sequence $\{a_n\}$, where $a_n = \sum_{k=1}^n k^{-5/4}$, $n = 1, 2, \dots$ it is known that there exist (unknown) $a, b, c \in \mathbb{R}$ such that for large values of n

$$a_n = a + n^{-1/4}(b + cn^{-1}) + \mathcal{O}(n^{-9/4}).$$

Extrapolate the limit $\lim_n a_n = a$ from this information as accurately as possible from the results $a_{100} = 3.331779$, $a_{200} = 3.532117$, $a_{400} = 3.700964$. (The exact value to six decimals: $a = 4.595112$.)

12. (*) Show that if in the interpolation problem (5) $f^{(k)}(x_i) = 1$ when $i = j$ and $k = l$ ($1 \leq j \leq m$, $0 \leq l \leq \nu_j - 1$) and $f^{(k)}(x_i) = 0$ otherwise, then the problem has the solution

$$p(x) = L_{j,l}(x) = q(x) \prod_{\substack{i=1 \\ i \neq j}}^m (x - x_i)^{\nu_i},$$

where q is a polynomial of the form $q(x) = \sum_{r=l}^{\nu_j-1} c_r(x - x_j)^r$. Deduce further that the solution of the interpolation problem (5) in the general

case is

$$p(x) = \sum_{i=1}^m \sum_{k=0}^{\nu_i-1} f^{(k)}(x_i) L_{i,k}(x).$$

Chapter VIII

Integral

In the analysis of one real variable the concept of the *integral* takes two forms: the *indefinite* and the *definite* integral. The former refers to the *antiderivative* of a given function, that is, to the inverse operation of differentiation, which was already introduced in a preliminary way in Section V.6. In this chapter, more precisely in Sections VIII.1–VIII.3, we present more systematically than before the methods used to find antiderivatives and the types of functions that these methods can handle. This is a very traditional mathematical skill, in which various tools of functional algebra play a central role, such as *integration by parts*, change of variable or *substitution*, and *partial fraction decompositions*.

In Sections VIII.4–VIII.6 we examine the various ways of defining the definite integral—more precisely the *Riemann integral*—as well as its properties and its connection to the indefinite integral through the *fundamental theorem of calculus*. In Section VIII.7 we examine extensions of the concept of the Riemann integral, the *improper* integrals, and, in connection with these, the comparison of integrals and series. In the last two sections the objects of study are the simplest formulas for area and arc length based on the definite integral, together with their applications (Section VIII.8), and finally the methods of *numerical integration* (Section VIII.9).

VIII.1 Antiderivative

Recall from Section V.6 that a function $F(x)$ is called an *antiderivative* of a function $f(x)$ on an open interval (a, b) if F is differentiable on (a, b) and

$$F'(x) = f(x), \quad x \in (a, b).$$

The antiderivative is denoted by

$$F(x) = \int f(x) dx.$$

This somewhat peculiar notation refers to the computational definition of the antiderivative, which is presented later (Section VIII.4). The antiderivative is also called the *indefinite integral*. Here 'indefinite' refers to the fact that the antiderivative is not unique; rather, an arbitrary constant may be added to it: if one antiderivative F is known, then all antiderivatives can be expressed by means of the so-called *constant of integration* C in the form

$$\int f(x) dx = F(x) + C.$$

The indefinite integral is thus in fact a set of functions. It is the general solution of the differential equation $y' = f(x)$ on the interval (a, b) (see Corollary V.6.8).

EXAMPLE 1 Determine the antiderivative on $\mathbb{R}$ of the function

$$f(x) = \begin{cases} 1, & \text{when } x \leq 2, \\ \frac{10}{x} - 2x, & \text{when } x > 2. \end{cases}$$

Solution Since $f(2^-) = f(2^+) = 1 = f(2)$, the function f is continuous on $\mathbb{R}$. By the known differentiation rules (Section V.3) we must have

$$F(x) = \begin{cases} x + C_1, & \text{when } x \in (-\infty, 2), \\ 10 \ln x - x^2 + C_2, & \text{when } x \in (2, \infty), \end{cases}$$

so that $F'(x) = f(x)$ when $x \neq 2$. In order for F to be differentiable also at the point $x = 2$, F must at least be continuous there, that is, we must have

$$2 + C_1 = 10 \ln 2 - 4 + C_2 \quad \Rightarrow \quad C_2 = C_1 - 10 \ln 2 + 6.$$

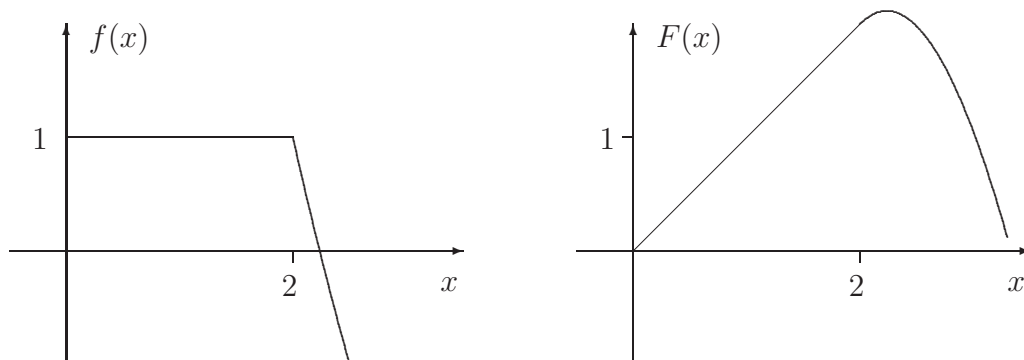
Writing $C_1 = C$ we obtain

$$F(x) = \begin{cases} x + C, & \text{when } x \leq 2, \\ 10 \ln(x/2) - x^2 + 6 + C, & \text{when } x \geq 2. \end{cases}$$

The one-sided derivatives of F at the point $x = 2$ are

$$D_-F(2) = f(2^-), \quad D_+F(2) = f(2^+).$$

Since $f(2^-) = f(2^+)$, the function F is differentiable also at the point $x = 2$ and $F'(2) = f(2)$. The antiderivative has thus been found. The figure shows f and F for $C = 0$. $\square$



'Integration', understood as finding an antiderivative, is thus the inverse operation of differentiation. Both the integration formulas for individual functions and the more general computational rules for antiderivatives are in fact differentiation rules read 'backwards'.

EXAMPLE 2 Since (cf. Section VI.3)

$$D \ln(x + \sqrt{x^2 + 1}) = \frac{1}{\sqrt{x^2 + 1}}, \quad x \in \mathbb{R},$$

we obtain the integration rule (valid on all of $\mathbb{R}$)

$$\int \frac{1}{\sqrt{x^2 + 1}} dx = \ln(x + \sqrt{x^2 + 1}) + C. \quad \square$$

The simplest of the integration methods is the 'table method', i.e. a collection of formulas based directly on the known differentiation rules. The following table gathers the most central of such integration formulas (including the result of the example). These can be read directly from the differentiation rules of Sections V.3, V.4, VI.2 and VI.3.

[l]	(1)	$\int x^\alpha dx$	$= \frac{1}{\alpha+1} x^{\alpha+1} + C, \quad \alpha \in \mathbb{R}, \alpha \neq -1.$
	(2)	$\int \frac{1}{x} dx$	$= \ln x + C.$
	(3)	$\int \cos x dx$	$= \sin x + C.$
	(4)	$\int \sin x dx$	$= -\cos x + C.$
	(5)	$\int e^x dx$	$= e^x + C.$
	(6)	$\int \frac{1}{\cos^2 x} dx$	$= \tan x + C.$
	(7)	$\int \frac{1}{\sin^2 x} dx$	$= -\cot x + C.$
	(8)	$\int \frac{1}{\sqrt{1-x^2}} dx$	$= \operatorname{Arcsin} x + C.$
	(9)	$\int \frac{1}{\sqrt{x^2+1}} dx$	$= \ln(x + \sqrt{x^2+1}) + C.$
	(10)	$\int \frac{1}{\sqrt{x^2-1}} dx$	$= \ln x + \sqrt{x^2-1} + C.$
	(11)	$\int \frac{1}{1+x^2} dx$	$= \operatorname{Arctan} x + C.$
	(12)	$\int \frac{1}{1-x^2} dx$	$= \frac{1}{2} \ln \left \frac{1+x}{1-x} \right + C.$

The table could easily be extended on the basis of the known differentiation rules. For example, the following holds (see Section VI.2):

[l]	(13)	$\int \frac{1}{\sin x} dx$	$= \ln \left \frac{1 - \cos x}{\sin x} \right + C.$
	(14)	$\int \frac{1}{\cos x} dx$	$= -\ln \left \frac{1 - \sin x}{\cos x} \right + C.$

Rules (1)–(14) may more precisely be used on those open intervals on which the given function f and the antiderivative F are both defined and the latter is differentiable. For example, rules (3)–(5) are valid on all of $\mathbb{R}$, rule (2) is valid

on the intervals $(-\infty, 0)$ and $(0, \infty)$, rule (8) on the interval $(-1, 1)$, and rule (7) on the intervals $(k\pi, (k+1)\pi)$, $k \in \mathbb{Z}$. When the validity of an integration rule is restricted to several disjoint subintervals of $\mathbb{R}$ (as in rules (2) and (7)), the constant of integration need not be the same on the different subintervals.

The table formulas (9), (10) and (12) can also be expressed by means of the inverse hyperbolic functions arsinh , Arcosh and artanh , since (see Section VI.3)

$$\begin{aligned}\ln(x + \sqrt{x^2 + 1}) &= \operatorname{arsinh} x, \quad x \in \mathbb{R}, \\ \ln|x + \sqrt{x^2 - 1}| &= \begin{cases} \operatorname{Arcosh} x, & \text{when } x > 1, \\ -\operatorname{Arcosh} |x|, & \text{when } x < -1, \end{cases} \\ \frac{1}{2} \ln \left| \frac{1+x}{1-x} \right| &= \begin{cases} \operatorname{artanh} x, & \text{when } |x| < 1, \\ \operatorname{artanh}(1/x), & \text{when } |x| > 1. \end{cases}\end{aligned}$$

Finding an antiderivative—when the function is not found directly from the simplest differentiation rules—is a skill that has been practiced a great deal in traditional mathematics. The pursuit has a sporting interest, among other reasons because determining the antiderivatives of many quite simple functions, such as

$$\frac{1}{x^4 + 1}, \quad \frac{\sqrt{x^2 + 1}}{x}$$

is laborious. There are also many functions, e.g.

$$e^{x^2}, \quad \cos x^2, \quad \frac{e^x}{x}, \quad \frac{\sin x}{x}, \quad \frac{1}{\ln x}, \quad \sqrt{x} \sin x, \quad \sqrt{x^4 + 1}, \quad \sqrt{1 + \sin^2 x},$$

whose antiderivatives are not so-called *elementary functions*, that is, they are not expressions that can be written by means of polynomials or rational functions, root expressions, exponential or logarithmic functions, or trigonometric functions or their inverses. One then says that the function cannot be integrated *in closed form*. Such antiderivatives used to be spoken of in a reverent tone, being called, among other things, 'higher transcendental functions'. Nowadays calculators and computers, and with them the ease of numerical computation, have dropped even such functions into the company of ordinary mortals.

With computers has also come the possibility of symbolic computation, which has reduced the importance of the traditional handicraft in finding antiderivatives. It is still worth knowing the most common integration rules and methods, among other reasons because this computational technique has more general uses as well. In this and the two following sections we go through the most common methods of integration technique.

Three general integration rules

By reading general differentiation rules 'backwards' one obtains general integration rules. Of the following three rules, the first states that integration, like differentiation, is a linear operation. The second and third rules are based directly on the differentiation rule for a composite function.

$$\int f(x) dx = F(x) + C \quad \& \quad \int g(x) dx = G(x) + C$$

$$\Rightarrow \begin{cases} \int [\alpha f(x) + \beta g(x)] dx = \alpha F(x) + \beta G(x) + C, & \alpha, \beta \in \mathbb{R}, & \text{(I-1)} \\ \int f(ax + b) dx = \frac{1}{a} F(ax + b) + C, & a, b \in \mathbb{R}, a \neq 0, & \text{(I-2)} \\ \int f(g(x))g'(x) dx = F(g(x)) + C. & & \text{(I-3)} \end{cases}$$

In these rules (as in rules (1)–(14)) it must be noted that the constant of integration C is a so-called *generic* constant, which in different contexts (such as the expressions $F(x) + C$ and $G(x) + C$) may always take different values, unless otherwise agreed.

EXAMPLE 3 $\int \frac{1}{x^2 + x} dx = ?$

Solution We first decompose the function to be integrated into a sum of two simpler functions:

$$\begin{aligned} \frac{1}{x^2 + x} &= \frac{1}{x(x+1)} = \frac{A}{x} + \frac{B}{x+1} = \frac{(A+B)x + A}{x(x+1)} \\ \Rightarrow \begin{cases} A+B=0 \\ A=1 \end{cases} &\Leftrightarrow \begin{cases} A=1 \\ B=-1 \end{cases} \Rightarrow \frac{1}{x^2 + x} = \frac{1}{x} - \frac{1}{x+1} \\ \Rightarrow \int \frac{1}{x^2 + x} dx &\stackrel{\text{(I-1)}}{=} \int \frac{1}{x} dx - \int \frac{1}{x+1} dx \\ &\stackrel{(2), \text{(I-2)}}{=} \ln|x| - \ln|x+1| + C = \underline{\underline{\ln \left| \frac{x}{x+1} \right| + C}}. \quad \square \end{aligned}$$

EXAMPLE 4 Derive table formula (12).

Solution Following the previous example, we compute

$$\begin{aligned} \frac{1}{1-x^2} &= \frac{1}{(1+x)(1-x)} = \frac{1}{2} \frac{1}{1+x} + \frac{1}{2} \frac{1}{1-x} \\ \Rightarrow \int \frac{1}{1-x^2} dx &= \frac{1}{2} \ln|1+x| - \frac{1}{2} \ln|1-x| + C = \frac{1}{2} \ln \left| \frac{1+x}{1-x} \right| + C. \quad \square \end{aligned}$$

In these examples the antiderivative was found by means of a so-called *partial fraction decomposition*. This is a method that applies to more general rational functions as well, as will be seen later in Section VIII.3.

EXAMPLE 5 By rule (I-3), table formula (1), and the differentiation rules $D \ln |x| = 1/x$ and $D \sin x = \cos x$,

$$\begin{aligned}\int \frac{1}{x} \ln |x| dx &= \frac{1}{2} (\ln |x|)^2 + C, \\ \int \sin^4 x \cos x dx &= \frac{1}{5} \sin^5 x + C. \quad \square\end{aligned}$$

EXAMPLE 6 $\int \frac{1}{x^2 + x + 1} dx = ?$

Solution Since

$$\frac{1}{x^2 + x + 1} = \frac{1}{(x + \frac{1}{2})^2 + \frac{3}{4}} = \frac{4}{3} \frac{1}{(\frac{2x+1}{\sqrt{3}})^2 + 1},$$

we have

$$\begin{aligned}\int \frac{1}{x^2 + x + 1} dx &\stackrel{(I-1)}{=} \frac{4}{3} \int \frac{1}{(\frac{2x+1}{\sqrt{3}})^2 + 1} dx \\ &\stackrel{(I-2), (11)}{=} \frac{4}{3} \cdot \frac{\sqrt{3}}{2} \operatorname{Arctan} \left(\frac{2x+1}{\sqrt{3}} \right) + C \\ &= \underline{\underline{\frac{2}{\sqrt{3}} \operatorname{Arctan} \left(\frac{2x+1}{\sqrt{3}} \right) + C.}} \quad \square\end{aligned}$$

EXAMPLE 7 a) $\int \cos^2 x dx = ?$ b) $\int \cos^3 x dx = ?$

Solution Since

$$\cos^2 x = \frac{1}{2} + \frac{1}{2} \cos 2x, \quad \cos^3 x = (1 - \sin^2 x) \cos x,$$

we have

$$\begin{aligned}\text{a) } \int \cos^2 x dx &= \frac{1}{2} x + \frac{1}{4} \sin 2x + C = \underline{\underline{\frac{1}{2} (x + \cos x \sin x) + C}}, \\ \text{b) } \int \cos^3 x dx &= \int (\cos x - \sin^2 x \cos x) dx = \underline{\underline{\sin x - \frac{1}{3} \sin^3 x + C.}} \quad \square\end{aligned}$$

EXERCISES

- Determine the antiderivative of the function $2x - 3$ whose graph is tangent to the line $x + y = 0$.
- Solve the following initial value problems on the interval $(-\infty, \infty)$:
 - $F'(x) = 2x^2 - |x|$, $F(0) = 1$.
 - $F'(x) = |x^2 + 4x|$, $F(-4) = 0$.
 - $F'(x) = \max\{x, 8x - x^2\}$, $F(4) = -1$.
 - $F'(x) = \min\{4 - 4x + x^2, 40 + 2x - x^2\}$, $F(0) = 180$.
- Determine the antiderivatives of the following functions, appealing to the table formulas (1)–(12) and the rules (I-1)–(I-3). If necessary, first transform the function into a form more suitable for integration.

$$\begin{array}{llll}
 \text{a)} & (2x^2 - 5x + 7)(4x - 4) & \text{b)} & \frac{1}{\sqrt{2x+3}} & \text{c)} & \frac{3}{3-\pi x} & \text{d)} & \frac{x^2}{4x^3-1} \\
 \text{e)} & \frac{2x^3+x}{(x^4+x^2+1)^3} & \text{f)} & \frac{x}{(x^2+1)\sqrt{x^2+1}} & \text{g)} & \frac{1}{x \ln|x|} & \text{h)} & \frac{e^x}{e^x+1} \\
 \text{i)} & \frac{1}{x(\ln|x|)^2} & \text{j)} & \frac{1}{\sqrt{x^2-3}} & \text{k)} & \frac{1}{\sqrt{x^2+3}} & \text{l)} & \frac{x+2}{\sqrt{x^2-1}} \\
 \text{m)} & \frac{3x-2}{\sqrt{2x^2+5}} & \text{n)} & \frac{1}{x^2+14x+50} & \text{o)} & \frac{1}{x^2+14x+48} & & \\
 \text{p)} & \frac{1}{x^2+2x+10} & \text{q)} & \frac{1}{x^2+3x-10} & \text{r)} & \frac{2x+1}{x^2+3x-10} & & \\
 \text{s)} & \sin 2x \cos^4 x & \text{t)} & \frac{\sin x}{\cos^3 x} & \text{u)} & \tan x & \text{v)} & \tan^2 x & \text{x)} & \tan^3 x \\
 \text{y)} & \frac{e^{\tan x}}{\cos^2 x} & \text{z)} & \sin^5 x & \text{å)} & \cosh^2 x & \text{ä)} & \tanh x & \text{ö)} & \frac{\operatorname{arsinh} x}{\sqrt{x^2+1}}
 \end{array}$$

- (*) Determine $a \in \mathbb{R}$ and a function u periodic on $\mathbb{R}$ such that $y(x) = ax + u(x)$ is the solution of the initial value problem $y' = |\sin x|$, $y(0) = 0$.
- (*) Let $f(0) = 0$ and let the definition of f elsewhere than at the origin be

$$\text{a)} \ f(x) = \frac{x}{\sqrt{|x|}}, \quad \text{b)} \ f(x) = \frac{x}{|x|}, \quad \text{c)} \ f(x) = \frac{1}{x} \left(\sin \frac{1}{x^2} + x^2 \cos \frac{1}{x^2} \right).$$

Construct an antiderivative of f on the interval $(-1, 1)$ or show that f has no antiderivative on that interval.

VIII.2 Integration by parts and substitution

The two most central general methods of integration technique, i.e. of finding antiderivatives, are *integration by parts* and the *substitution* (procedure). As 'formulas of mathematics' these methods have more general uses as well.

The formula for integration by parts is simply the product rule for differentiation written in another form:

$$\frac{d}{dx}(fg) = f'g + fg' \quad \Leftrightarrow \quad f'g = \frac{d}{dx}(fg) - fg'$$

$$\Rightarrow \quad \boxed{\int f'(x)g(x) dx = f(x)g(x) - \int f(x)g'(x) dx.}$$

EXAMPLE 1 a) $\int xe^x dx = ?$ b) $\int e^x \sin x dx = ?$

Solution

$$\text{a) } \int xe^x dx = \int \underbrace{e^x}_{f'(x)} \underbrace{x}_{g(x)} dx = e^x \cdot x - \int e^x dx = \underline{\underline{(x-1)e^x + C.}}$$

$$\text{Check: } \frac{d}{dx}((x-1)e^x) = e^x + (x-1)e^x = xe^x. \quad \text{OK!} \quad \square$$

b) Choose $f'(x) = f(x) = e^x$ and integrate by parts twice:

$$\begin{aligned} F(x) &= \int e^x \sin x dx = e^x \sin x - \int e^x \cos x dx \\ &= e^x \sin x - e^x \cos x - \int e^x \sin x dx \\ &= e^x(\sin x - \cos x) - F(x) \end{aligned}$$

$$\Rightarrow F(x) = \underline{\underline{\frac{1}{2} e^x(\sin x - \cos x) + C.}} \quad \square$$

EXAMPLE 2 Determine the antiderivative $\int \sqrt{a^2 + x^2} dx$, where $a > 0$.

Solution By integration by parts ($f'(x) = 1$, $g(x) = \sqrt{a^2 + x^2}$) we first obtain

$$\int \sqrt{a^2 + x^2} dx = x\sqrt{a^2 + x^2} - \int \frac{x^2}{\sqrt{a^2 + x^2}} dx.$$

Here

$$\begin{aligned} - \int \frac{x^2}{\sqrt{a^2 + x^2}} dx &= - \int \frac{(x^2 + a^2) - a^2}{\sqrt{a^2 + x^2}} dx \\ &= - \int \sqrt{a^2 + x^2} dx + \int \frac{a^2}{\sqrt{a^2 + x^2}} dx, \end{aligned}$$

so that it follows (see formulas (9) and (I-2) of the previous section)

$$\begin{aligned} 2 \int \sqrt{a^2 + x^2} dx &= x\sqrt{a^2 + x^2} + \int \frac{a}{\sqrt{(x/a)^2 + 1}} dx \\ &= x\sqrt{a^2 + x^2} + a^2 \ln[(x/a) + \sqrt{(x/a)^2 + 1}] + C. \end{aligned}$$

Writing here $a^2 \ln a + C$ in place of the constant C (which is possible, since C is in any case indeterminate), we bring the result to the form

$$\int \sqrt{a^2 + x^2} dx = \underline{\underline{\frac{1}{2}[x\sqrt{a^2 + x^2} + a^2 \ln(x + \sqrt{a^2 + x^2})]} + C. \quad \square}$$

Reduction formulas

By integration by parts one can derive *reduction formulas* for a whole family of integrals containing an integer parameter. For example, the following integrals can be computed in closed form in this way:

$$\begin{aligned} \int x^n e^x dx, \quad \int x^n \cos x dx, \quad x^n \sin x dx, \quad \int (1 + x^2)^{-n} dx, \quad n \in \mathbb{N}, \\ \int \cos^n x dx, \quad \int \sin^n x dx, \quad n \in \mathbb{Z}. \end{aligned}$$

As an example, consider the integral

$$I_n(x) = \int \cos^n x dx.$$

By integration by parts we obtain

$$\begin{aligned} I_n(x) &= \int \cos x \cdot \cos^{n-1} x dx \quad [f'(x) = \cos x, g(x) = \cos^{n-1} x] \\ &= \sin x \cos^{n-1} x + (n-1) \int \sin^2 x \cos^{n-2} x dx \\ &= \sin x \cos^{n-1} x + (n-1) \int (1 - \cos^2 x) \cos^{n-2} x dx \\ &= \sin x \cos^{n-1} x + (n-1)I_{n-2}(x) - (n-1)I_n(x). \end{aligned}$$

On this basis the reduction formula

$$nI_n(x) - (n-1)I_{n-2}(x) = \sin x \cos^{n-1} x. \quad (1)$$

holds for the integral $I_n(x)$. The formula in fact holds when $n \mapsto \alpha$, $\alpha \in \mathbb{R}$, but it is mainly useful when $n \in \mathbb{Z}$, in which case I_n can be reduced by means of the formula to the cases $n = 0, \pm 1$ and hence integrated in closed form (for the case $n = -1$, table formula (14) of the previous section applies).

EXAMPLE 3 a) $\int \cos^3 x \, dx = ?$ b) $\int \cos^{-3} x \, dx = ?$

Solution For the indices $n = 3$ and $n = -1$ the reduction formula (1) gives

$$\begin{aligned} I_3(x) &= \frac{1}{3} \sin x \cos^2 x + \frac{2}{3} I_1(x), \\ I_{-3}(x) &= \frac{1}{2} \sin x \cos^{-2} x + \frac{1}{2} I_{-1}(x), \end{aligned}$$

so that (cf. Example 7 of the previous section)

$$\begin{aligned} \text{a) } \int \cos^3 x \, dx &= \frac{1}{3} \sin x \cos^2 x + \frac{2}{3} \int \cos x \, dx \\ &= \underline{\underline{-\frac{1}{3} \sin^3 x + \sin x + C}} \end{aligned}$$

and by formula (14) of the previous section

$$\begin{aligned} \text{b) } \int \cos^{-3} x \, dx &= \frac{1}{2} \sin x \cos^{-2} x + \frac{1}{2} \int \cos^{-1} x \, dx \\ &= \underline{\underline{\frac{1}{2} \ln \left| \frac{\cos x}{1 - \sin x} \right| + \frac{1}{2} \frac{\sin x}{\cos^2 x} + C.}} \quad \square \end{aligned}$$

As a second example, consider the integral

$$I_n(x) = \int \frac{1}{(1+x^2)^n} \, dx, \quad n \geq 2.$$

We aim to reduce this to the case $n = 1$, in which the antiderivative is known (table formula (11) of the previous section). Choose $f'(x) = 1$, $g(x) = (1+x^2)^{-n}$ and integrate by parts:

$$\begin{aligned} I_n(x) &= \frac{x}{(1+x^2)^n} + 2n \int \frac{x^2}{(1+x^2)^{n+1}} \, dx \quad [x^2 = (x^2+1) - 1] \\ &= \frac{x}{(1+x^2)^n} + 2nI_n(x) - 2nI_{n+1}(x). \end{aligned}$$

Writing $n - 1$ in place of n we arrive at the reduction formula

$$I_n(x) = \frac{1}{2n-2} \frac{x}{(1+x^2)^{n-1}} + \frac{2n-3}{2n-2} I_{n-1}(x). \quad (2)$$

EXAMPLE 4 Applying formula (2) twice we obtain

$$\begin{aligned} \int \frac{1}{(1+x^2)^3} dx &= \frac{1}{4} \frac{x}{(1+x^2)^2} + \frac{3}{4} \int \frac{1}{(1+x^2)^2} dx \\ &= \frac{1}{4} \frac{x}{(1+x^2)^2} + \frac{3}{4} \left(\frac{1}{2} \frac{x}{1+x^2} + \frac{1}{2} \int \frac{1}{1+x^2} dx \right) \\ &= \frac{1}{4} \frac{x}{(1+x^2)^2} + \frac{3}{8} \frac{x}{1+x^2} + \frac{3}{8} \operatorname{Arctan} x + C. \quad \square \end{aligned}$$

The substitution procedure

By substitution one means, in integration, a change of variable (as in computing limits, cf. Section V.2). The substitution procedure for integration is based on the following result, which in turn is based on the differentiation rules for a composite function and for an inverse function.

PROPOSITION VIII.2.1 Let $u : (c, d) \longrightarrow (a, b)$ be a bijection, let u be differentiable on the interval (c, d) , and let the inverse function $v = u^{-1}$ be differentiable on the interval (a, b) . Then, if $G(t)$ is an antiderivative of the function $g(t) = f(u(t))u'(t)$ on the interval (c, d) , i.e.

$$\int f(u(t))u'(t) dt = G(t) + C, \quad t \in (c, d),$$

then an antiderivative of the function $f(x)$ on the interval (a, b) is

$$\int f(x) dx = G(v(x)) + C, \quad x \in (a, b).$$

Proof If $x \in (a, b)$ and $t = v(x) \Leftrightarrow x = u(t)$, then by the differentiation rule for a composite function and by the assumptions,

$$\begin{aligned} \frac{d}{dx} G(v(x)) &= G'(v(x))v'(x) = G'(t)v'(x) \\ &= f(u(t))u'(t)v'(x). \end{aligned}$$

Here $u(t) = x$, so $f(u(t)) = f(x)$ and, by the differentiation rule for an inverse function (Section V.3, formula (5)),

$$u'(t)v'(x) = 1 \quad \Rightarrow \quad \text{the claim.} \quad \square$$

In applications the substitution is often (perhaps more often) made in the form $v(x) = t$. Mechanically, then, the substitution procedure works in the following way:

Problem: $\int f(x) dx = ?$

1. Make the substitution: $x = u(t)$ or $v(x) = t$.

2. Solve

$$x = u(t) \Rightarrow t = v(x) \quad \text{or} \quad v(x) = t \Rightarrow x = u(t).$$

3. Substitute into the integral

$$x = u(t) \text{ (or } v(x) = t) \quad \text{and} \quad dx = u'(t) dt.$$

4. Write $f(u(t))u'(t) = g(t)$ (with possible simplification) and solve the transformed problem:

$$\int g(t) dt = G(t) + C.$$

5. Substitute $t = v(x)$ into the solution, whereby one obtains the solution of the original problem:

$$\int f(x) dx = G(v(x)) + C.$$

In step 2 it does no harm if the inverse function $v = u^{-1}$ or $u = v^{-1}$ is multi-branched (the solution of the equation $u(t) = x$ or $v(x) = t$ being multivalued), as long as the assumptions of Proposition VIII.2.1 are satisfied on the chosen branch. One must only take care not to jump from one branch to another during the various stages of the computation.

EXAMPLE 5 $\int \frac{1}{\sqrt{x}+1} dx = ?$

Solution Here $(a, b) = (0, \infty)$.

1. Make the substitution: $v(x) = \sqrt{x} = t \in (0, \infty)$.

2. Solve: $\sqrt{x} = t \Rightarrow x = u(t) = t^2$.
3. Substitute into the integral: $\sqrt{x} = t, dx = 2t dt$.
4. Solve the transformed problem:

$$\int \frac{2t}{t+1} dt = \int \left(2 - \frac{2}{t+1}\right) dt = 2t - 2\ln(t+1) + C.$$

5. The solution of the original problem by the substitution $t = \sqrt{x}$:

$$\int \frac{1}{\sqrt{x}+1} dx = \underline{\underline{2\sqrt{x} - 2\ln(\sqrt{x}+1) + C}}.$$

EXAMPLE 6 $\int \frac{1}{e^x+1} dx = ?$

Solution Substituting $e^x = t \in (0, \infty) \Rightarrow x = \ln t, dx = \frac{1}{t} dt$ we obtain

$$\begin{aligned} \int \frac{1}{e^x+1} dx &= \int \frac{1}{(t+1)t} dt = \ln \left| \frac{t}{t+1} \right| + C \quad (\text{Example VIII.1:3}) \\ &\Rightarrow \int \frac{1}{e^x+1} dx = \ln \left(\frac{e^x}{e^x+1} \right) + C = \underline{\underline{x - \ln(e^x+1) + C}}. \quad \square \end{aligned}$$

EXAMPLE 7 $\int \frac{1}{(1+x^2)^2} dx = ?$

Solution This can be computed with the reduction formula (2). Another option is the trigonometric substitution

$$x = \tan t, \quad dx = \frac{1}{\cos^2 t} dt, \quad t \in (-\pi/2, \pi/2).$$

Since $1 + \tan^2 t = 1/\cos^2 t$, the transformed integral becomes

$$\begin{aligned} \int \frac{1}{(1+\tan^2 t)^2} \cdot \frac{1}{\cos^2 t} dt &= \int \cos^2 t dt = \frac{1}{2}(t + \sin t \cos t) + C \\ &= \frac{1}{2} \left(t + \frac{\sin t}{\cos t} \cdot \cos^2 t \right) + C = \frac{1}{2} \left(t + \frac{\tan t}{1+\tan^2 t} \right) + C. \end{aligned}$$

Finally, solving

$$x = \tan t \quad \& \quad t \in (-\pi/2, \pi/2) \quad \Leftrightarrow \quad t = \operatorname{Arctan} x$$

we obtain the result

$$\int \frac{1}{(1+x^2)^2} dx = \frac{1}{2} \left(\operatorname{Arctan} x + \frac{x}{1+x^2} \right) + C.$$

Comparison: As a method, the reduction formula (2) is more straightforward. $\square$

EXAMPLE 8 $\int \frac{1}{x+1+\sqrt[3]{x+1}} dx = ?$

Solution By the substitution

$$\sqrt[3]{x+1} = t \Leftrightarrow x = t^3 - 1, \quad dx = 3t^2 dt \quad (x > -1, t > 0)$$

we obtain

$$\begin{aligned} \int \frac{1}{x+1+\sqrt[3]{x+1}} dx &= \int \frac{3t^2}{t^3+t} dt = \int \frac{3t}{t^2+1} dt \\ &= \frac{3}{2} \ln(t^2+1) + C \\ &= \frac{3}{2} \ln[(x+1)^{2/3}+1] + C. \end{aligned}$$

If on the interval $x \in (-\infty, -1)$ one interprets $\sqrt[3]{x+1} = -\sqrt[3]{|x+1|}$, then the result is valid on this interval as well. $\square$

EXAMPLE 9 $\int \frac{e^x}{\sqrt{x}} dx = ?$

Solution By the substitution $\sqrt{x} = t$, $x = t^2$, $dx = 2t dt$ the problem is transformed into the form

$$\int 2e^{t^2} dt = ?$$

This has no elementary-function solution (only a series solution, see the next section), so we settle for the result

$$\int \frac{e^x}{\sqrt{x}} dx = 2G(\sqrt{x}), \quad G(t) = \int e^{t^2} dt. \quad \square$$

EXERCISES

1. Compute by integration by parts

$$\begin{array}{lll} \text{a)} \int x \cos x \, dx & \text{b)} \int x^3 e^{-x^2} \, dx & \text{c)} \int x^3 \cos(x^2) \, dx \\ \text{d)} \int x^2 e^{ax} \, dx, \, a \in \mathbb{R} & \text{e)} \int x^\alpha \ln x \, dx, \, \alpha \in \mathbb{R} & \\ \text{f)} \int \operatorname{Arctan} x \, dx & \text{g)} \int \operatorname{Arcsin} x \, dx & \text{h)} \int x \operatorname{Arctan} x \, dx \end{array}$$

2. Derive by integration by parts a reduction formula for the given integral $I_n(x)$ and use the formula to compute $I_n(x)$ for the given values of n .

- a) $\int x^n e^x \, dx, \, n \in \mathbb{N} \cup \{0\}, \, n = 1, 2, 3.$
- b) $\int x^n \sin x \, dx, \, n \in \mathbb{N} \cup \{0\}, \, n = 1, 2, 3.$
- c) $\int x^n \cos x \, dx, \, n \in \mathbb{N} \cup \{0\}, \, n = 1, 2, 3.$
- d) $\int \sin^n x \, dx, \, n \in \mathbb{Z}, \, n = -3, 4.$
- e) $\int (\ln x)^n \, dx, \, n \in \mathbb{N} \cup \{0\}, \, n = 1, 2.$
- f) $\int (x^2 + 1)^{n-1/2} \, dx, \, n \in \mathbb{Z}, \, n = -1, 2.$
- g) $\int (x^2 - 1)^{n-1/2} \, dx, \, n \in \mathbb{Z}, \, n = -1, 2.$
- h) $\int (1 - x^2)^{n-1/2} \, dx, \, n \in \mathbb{Z}, \, n = -1, 2.$

3. Compute using the given substitution:

$$\begin{array}{ll} \text{a)} \int \frac{1}{\sqrt{5-x^2}} \, dx, \, x = \sqrt{5}t & \text{b)} \int \frac{1}{x^2+2x+3} \, dx, \, x+1 = \sqrt{2}t \\ \text{c)} \int e^{\sqrt{x}} \, dx, \, x = t^2 & \text{d)} \int e^{\sqrt[3]{x}} \, dx, \, \sqrt[3]{x} = t \end{array}$$

4. Transform the following integrals into another form using the given substitution.

$$\begin{array}{ll} \text{a)} \int \frac{1}{\sqrt{1+x^4}} \, dx, \, x^4 = t & \text{b)} \int e^{x^2} \, dx, \, x^2 = t \\ \text{c)} \int \sin(\ln x) \, dx, \, x = e^t & \text{d)} \int \ln(\tan x) \, dx, \, \tan x = t \end{array}$$

5. The antiderivatives of the functions $x^\alpha \sin \beta x$, $x^\alpha \cos \beta x$, $x^\alpha e^{\beta x}$ ($\alpha, \beta \in \mathbb{R}$) are elementary functions only when either $\beta = 0$ or $\alpha \in \mathbb{N} \cup \{0\}$. On the

basis of this information, which of the following are elementary functions and which are not?

$$\begin{array}{lll} \text{a)} \int \cos(x^2) dx & \text{b)} \int x^{13} \sqrt{x} \cos \sqrt{x} dx & \text{c)} \int x^5 e^{x^4} dx \\ \text{d)} \int x^7 e^{x^4} dx & \text{e)} \int \sqrt{\ln x} dx & \text{f)} \int \frac{1}{\ln x} dx \\ \text{g)} \int \frac{1}{x \sqrt{\ln x}} dx & \text{h)} \int \sin(e^x) dx & \text{i)} \int \cos x \ln x dx \end{array}$$

6. (*) Let $n \in \mathbb{N} \cup \{0\}$ and $a, b \in \mathbb{R}$. Derive reduction formulas for the integrals

$$I_n(x) = \int x^n e^{ax} \cos bx, \quad J_n(x) = \int x^n e^{ax} \sin bx.$$

VIII.3 Partial fraction decompositions. The series method

An antiderivative of a general rational function can be found using the so-called *partial fraction decomposition* together with the techniques of the previous section. Consider a general (real-coefficient) rational function

$$f(x) = \frac{p(x)}{q(x)} = \frac{a_mx^m + \cdots + a_0}{b_nx^n + \cdots + b_0},$$

where $m \geq 0$, $n \geq 1$ and $a_m, b_n \neq 0$. If $m \geq n$, one first carries out the division:

$$\frac{p(x)}{q(x)} = p_0(x) + \frac{p_1(x)}{q(x)},$$

where p_0 is a polynomial of degree $m - n$,

$$p_0(x) = c_{m-n}x^{m-n} + \cdots + c_0,$$

and the remainder $p_1(x)$ has degree $n - 1$:

$$p_1(x) = d_{n-1}x^{n-1} + \cdots + d_0.$$

The division can be carried out, for example, simply by identifying the coefficients of the polynomials in the equation

$$p(x) = p_0(x)q(x) + p_1(x),$$

whereby one obtains a (solvable) linear system of equations for the unknown coefficients $c_{m-n}, \dots, c_0$ and $d_{n-1}, \dots, d_0$. In the following example an alternative (also universally valid) method is used, the *polynomial division algorithm*, which is especially convenient in hand computation. The general principle of the algorithm can be guessed from the example.

EXAMPLE 1 Compute p_0 and p_1 when $f(x) = \frac{x^3 + 2x^2}{x^2 - 2x + 3}$.

Solution

$$\begin{aligned} x^3 + 2x^2 &= [x(x^2 - 2x + 3) + 2x^2 - 3x] + 2x^2 \\ &= x(x^2 - 2x + 3) + 4x^2 - 3x \\ &= x(x^2 - 2x + 3) + [4(x^2 - 2x + 3) + 8x - 12] - 3x \\ &= (x + 4)(x^2 - 2x + 3) + 5x - 12 \\ \Rightarrow \frac{x^3 + 2x^2}{x^2 - 2x + 3} &= x + 4 + \frac{5x - 12}{x^2 - 2x + 3} = p_0(x) + \frac{p_1(x)}{q(x)}. \quad \square \end{aligned}$$

Once the division has been carried out, the linearity of the integral gives

$$\int \frac{p(x)}{q(x)} dx = \int p_0(x) dx + \int \frac{p_1(x)}{q(x)} dx.$$

Here the antiderivative of the polynomial part is immediately determinable (as a polynomial), so in what follows we may restrict ourselves to studying the latter term, in which the degree of the numerator is smaller than the degree of the denominator. Assume that all the (real and complex) zeros of q are known—these are, after all, at least numerically computable. If the zeros $x_1 \dots x_n$ are all real and simple, then one obtains the partial fraction decomposition (see Theorem VIII.3.1 below)

$$\frac{p_1(x)}{q(x)} = \frac{A_1}{x - x_1} + \dots + \frac{A_n}{x - x_n}.$$

The coefficients A_i can be determined from the linear system of equations that arises when this equation is multiplied on both sides by $q(x)$ and the polynomial coefficients on the left and right are identified. The coefficients are, however, more easily computed by writing $q(x) = (x - x_i)q_i(x)$, whereby multiplying the equation on both sides by $(x - x_i)$ gives

$$A_i = \lim_{x \rightarrow x_i} \left[(x - x_i) \frac{p_1(x)}{q(x)} \right] = \frac{p(x_i)}{q_i(x_i)}.$$

If $q(x)$ has a factor $(x - a)^k$ corresponding to a k -fold real root $x_i = a$, then in the partial fraction decomposition this factor corresponds, more generally, to the terms

$$(x - a)^k \hookrightarrow \frac{A_1}{x - a} + \frac{A_2}{(x - a)^2} + \dots + \frac{A_k}{(x - a)^k}.$$

Each of these terms is directly integrable. If, finally, $q(x)$ has a factor $(x^2 + ax + b)^k$ corresponding to a k -fold complex conjugate root pair (in which case $b - a^2/4 > 0$), then this factor contributes to the partial fraction decomposition the terms

$$(x^2 + ax + b)^k \hookrightarrow \frac{B_1x + C_1}{x^2 + ax + b} + \frac{B_2x + C_2}{(x^2 + ax + b)^2} + \dots + \frac{B_kx + C_k}{(x^2 + ax + b)^k}.$$

To integrate these, one notes first that

$$\frac{Ax + B}{(x^2 + ax + b)^l} = \frac{A}{2} \frac{2x + a}{(x^2 + ax + b)^l} + \frac{B - \frac{1}{2}aA}{(x^2 + ax + b)^l},$$

so that

$$\int \frac{Ax + B}{(x^2 + ax + b)^l} dx = \frac{A}{2} \cdot \begin{cases} \ln(x^2 + ax + b), & \text{if } l = 1, \\ -\frac{1}{l-1} \frac{1}{(x^2 + ax + b)^{l-1}}, & \text{if } l > 1 \end{cases} + (B - \frac{1}{2}aA) \int \frac{1}{(x^2 + ax + b)^l} dx.$$

The integral remaining here can be handled by the methods of the previous section, since when one writes

$$x^2 + ax + b = \left(x + \frac{a}{2}\right)^2 + D^2, \quad D = \sqrt{b - \frac{a^2}{4}},$$

then by the substitution

$$x + \frac{a}{2} = Dt, \quad dx = Ddt$$

one obtains

$$\int \frac{1}{(x^2 + ax + b)^l} dx = D^{-2l+1} \int \frac{1}{(t^2 + 1)^l} dt, \quad t = D^{-1} \left(x + \frac{a}{2}\right).$$

When the above results are combined and the results of the previous section are taken into account, one can conclude that the antiderivative of every rational function is of the form

$$\int \frac{p(x)}{q(x)} dx = R(x) + \sum A_i F_i(x) + C,$$

where R is a rational function (or polynomial), the coefficients A_i are real numbers, and each F_i (at most n in number, $n =$ the degree of q) is one of the following function forms:

$$\ln|x + a_i|, \quad \ln(x^2 + a_i x + b_i), \quad \text{Arctan}(a_i x + b_i).$$

It has thus been concluded that a rational function can always be integrated in closed form when all the zeros (with their multiplicities) of the denominator polynomial are known. The conclusion rested on the assumption that a partial fraction decomposition in the manner described above is always possible, i.e. that the following holds.

THEOREM VIII.3.1 (Partial fraction decomposition of a rational function) Let $f(x) = p(x)/q(x)$ be a real rational function in which the degree of the numerator polynomial p is $<$ the degree of q . Let the real roots of the denominator polynomial q be x_i , $i = 1 \dots n_1$ and the complex roots $c_j = a_j \pm ib_j$, $j = 1 \dots n_2$ ($a_j, b_j \in \mathbb{R}$, $b_j \neq 0$). Further, let the multiplicities of these roots be m_i , $i = 1 \dots n_1$ and ν_j , $j = 1 \dots n_2$. Then there exist coefficients $A_{ik}, B_{jk}, C_{jk} \in \mathbb{R}$ such that for every $x \in \mathcal{D}_f$

$$f(x) = \sum_{i=1}^{n_1} \sum_{k=1}^{m_i} \frac{A_{ik}}{(x-x_i)^k} + \sum_{j=1}^{n_2} \sum_{k=1}^{\nu_j} \frac{B_{jk}x + C_{jk}}{[(x-a_j)^2 + b_j^2]^k}.$$

Proof Omitted, see Exercise 8.

EXAMPLE 2 $\int \frac{x^2}{x^3 + x^2 + x - 3} dx = ?$

Solution One root of the denominator polynomial is $x_1 = 1$: $q(x) = x^3 + x^2 + x - 3 = (x-1)(x^2 + 2x + 3)$. The other two roots are complex (a conjugate pair), so the partial fraction decomposition is of the form

$$\frac{x^2}{x^3 + x^2 + x - 3} = \frac{A}{x-1} + \frac{Bx + C}{x^2 + 2x + 3}.$$

Multiplying both sides here by $q(x)$, we first determine the coefficients A, B, C , after which the integration succeeds:

$$\begin{aligned} x^2 &= A(x^2 + 2x + 3) + (Bx + C)(x-1) \quad \forall x \in \mathbb{R} \\ \Leftrightarrow (A+B-1)x^2 + (2A-B+C)x + (3A-C) &= 0 \quad \forall x \in \mathbb{R} \\ \Leftrightarrow \begin{cases} A+B-1 &= 0 \\ 2A-B+C &= 0 \\ 3A-C &= 0 \end{cases} &\Leftrightarrow \begin{cases} A &= 1/6 \\ B &= 5/6 \\ C &= 1/2 \end{cases} \\ \Rightarrow \int \frac{x^2}{x^3 + x^2 + x - 3} dx &= \frac{1}{6} \int \frac{1}{x-1} dx + \frac{1}{6} \int \frac{5x+3}{x^2 + 2x + 3} dx \\ &= \frac{1}{6} \ln|x-1| + \frac{1}{6} \int \left(\frac{5}{2} \frac{2x+2}{x^2 + 2x + 3} - \frac{2}{x^2 + 2x + 3} \right) dx \\ &= \frac{1}{6} \ln|x-1| + \frac{5}{12} \int \frac{2x+2}{x^2 + 2x + 3} dx - \frac{1}{6} \int \frac{1}{(\frac{x+1}{\sqrt{2}})^2 + 1} dx \\ &= \frac{1}{6} \ln|x-1| + \frac{5}{12} \ln(x^2 + 2x + 3) - \frac{\sqrt{2}}{6} \operatorname{Arctan} \left(\frac{x+1}{\sqrt{2}} \right) + C. \end{aligned}$$

The coefficient A could also have been determined more directly by computing

$$A = \lim_{x \rightarrow 1} \left[(x-1) \frac{x^2}{x^3 + x^2 + x - 3} \right] = \lim_{x \rightarrow 1} \frac{x^2}{x^2 + 2x + 3} = \frac{1}{6}. \quad \square$$

Integrals reducible to rational ones

If $R(x, y)$ is a rational function of two variables, then, among others, the following integrals can be reduced to rational ones by suitable substitutions:

- a) $\int R(\sin x, \cos x) dx$, b) $\int R(e^x) dx$,
 c) $\int R(x, \sqrt{1-x^2}) dx$, d) $\int R(x, \sqrt{x^2+1}) dx$, e) $\int R(x, \sqrt{x^2-1}) dx$,
 f) $\int R(x, \sqrt[n]{ax+b}) dx$ ($a \neq 0$, $m \in \mathbb{N}$).

Case a) Here the trigonometric 'guru substitution' works (cf. Section II.5)

$$\begin{aligned} \tan \frac{1}{2}x = t &\Rightarrow \begin{cases} \sin x = \frac{2t}{1+t^2}, & \cos x = \frac{1-t^2}{1+t^2}, \\ \frac{dt}{dx} = \frac{1}{2} \left(1 + \tan^2 \frac{1}{2}x \right) & \Rightarrow dx = \frac{2 dt}{1+t^2} \end{cases} \\ \Rightarrow \int R(\sin x, \cos x) dx &= \int R\left(\frac{2t}{1+t^2}, \frac{1-t^2}{1+t^2}\right) \frac{2}{1+t^2} dt \\ &= \int Q(t) dt, \end{aligned}$$

where Q is a rational function.

Case b) By the substitution (cf. Example 6 of the previous section)

$$e^x = t \quad \Rightarrow \quad x = \ln t, \quad dx = \frac{1}{t} dt$$

the integral is brought to the form

$$\int R(e^x) dx = \int R(t) \frac{1}{t} dt = \int Q(t) dt,$$

where Q is again a rational function.

Case c) This reduces to case a) by the substitutions

$$x = \cos t \quad \text{or} \quad x = \sin t.$$

Cases d) and e) These reduce to case b) by the substitutions

$$\text{d) } x = \sinh t, \quad \text{e) } x = \cosh t.$$

Case f) Here the substitution works (cf. Example 8 of the previous section)

$$\begin{aligned} \sqrt[m]{ax+b} = t &\Rightarrow x = \frac{1}{a}(t^m - b), \quad dx = \frac{m}{a}t^{m-1}dt \\ \Rightarrow \int R(x, \sqrt[m]{ax+b}) dx &= \int R\left(\frac{t^m - b}{a}, t\right) \frac{m}{a}t^{m-1}dt = \int Q(t) dt. \end{aligned}$$

EXAMPLE 3 $\int \frac{1}{\sin x} dx = ?$

Solution By the substitution $\tan \frac{1}{2}x = t$ we obtain

$$\begin{aligned} \int \frac{1}{\sin x} dx &= \int \frac{1+t^2}{2t} \cdot \frac{2}{1+t^2} dt = \int \frac{1}{t} dt \\ &= \ln |t| + C \\ &= \ln \left| \tan \frac{1}{2}x \right| + C = \ln \left| \frac{1 - \cos x}{\sin x} \right| + C. \quad \square \end{aligned}$$

EXAMPLE 4 $\int \frac{\sqrt{x^2+1}}{x} dx = ?$

Solution We first make the substitution

$$x = \sinh t, \quad dx = \cosh t dt, \quad \sqrt{x^2+1} = \cosh t$$

$$\begin{aligned} \Rightarrow \int \frac{\sqrt{x^2+1}}{x} dx &= \int \frac{\cosh^2 t}{\sinh t} dt = \int \frac{\sinh^2 t + 1}{\sinh t} dt \\ &= \int \sinh t dt + \int \frac{1}{\sinh t} dt = \cosh t + \int \frac{2}{e^t - e^{-t}} dt. \end{aligned}$$

We make a new substitution

$$e^t = u, \quad t = \ln u, \quad dt = \frac{1}{u} du$$

$$\begin{aligned}\Rightarrow \int \frac{2}{e^t - e^{-t}} dt &= \int \frac{2}{u - u^{-1}} \cdot \frac{1}{u} du = \int \frac{2}{u^2 - 1} du \\ &= \ln \left| \frac{u-1}{u+1} \right| + C = \ln \left| \frac{e^t - 1}{e^t + 1} \right| + C.\end{aligned}$$

Since $x = \sinh t \Leftrightarrow e^t = x + \sqrt{x^2 + 1}$, we conclude that

$$\int \frac{\sqrt{x^2 + 1}}{x} dx = \sqrt{x^2 + 1} + \ln \left| \frac{x - 1 + \sqrt{x^2 + 1}}{x + 1 + \sqrt{x^2 + 1}} \right| + C.$$

Noting further that (cf. Exercise V.1:12)

$$\frac{x - 1 + \sqrt{x^2 + 1}}{x + 1 + \sqrt{x^2 + 1}} = \frac{x}{1 + \sqrt{x^2 + 1}}, \quad x \in \mathbb{R},$$

we obtain a slightly neater form for the final result:

$$\int \frac{\sqrt{x^2 + 1}}{x} dx = \sqrt{x^2 + 1} + \ln \frac{|x|}{1 + \sqrt{x^2 + 1}} + C. \quad \square$$

The series method of integration

When a function can be represented as the sum of a convergent power series (a Taylor series, see Section VII.4), its antiderivative can also be represented in this form. Assume that

$$f(x) = \sum_{k=0}^{\infty} a_k (x - x_0)^k, \quad |x - x_0| < \rho \quad (\rho > 0).$$

Then the series

$$F(x) = \sum_{k=0}^{\infty} \frac{a_k}{k+1} (x - x_0)^{k+1}$$

also converges when $|x - x_0| < \rho$ (Theorem I.12.11), and by differentiating the series term by term (which by Theorem V.3.4 is permitted) one finds that F is an antiderivative of f on the interval $(x_0 - \rho, x_0 + \rho)$.

EXAMPLE 5 $\int e^{x^2} dx = ?$

Solution

$$e^{x^2} = \sum_{k=0}^{\infty} \frac{1}{k!} x^{2k}, \quad x \in (-\infty, \infty),$$

so that

$$\begin{aligned}\int e^{x^2} dx &= \sum_{k=0}^{\infty} \frac{1}{(2k+1)k!} x^{2k+1} + C \\ &= C + x + \frac{1}{3} x^3 + \frac{1}{10} x^5 + \frac{1}{42} x^7 + \dots, \quad x \in \mathbb{R}. \quad \square\end{aligned}$$

EXAMPLE 6 Determine, making use of series,

$$\text{a) } \int \frac{e^{-x}}{x} dx, \quad \text{b) } \int \frac{\sin x}{x^4} dx.$$

Solution Starting from the Taylor series

$$e^{-x} = \sum_{k=0}^{\infty} \frac{(-1)^k}{k!} x^k, \quad \sin x = \sum_{k=0}^{\infty} \frac{(-1)^k}{(2k+1)!} x^{2k+1}$$

we obtain

$$\begin{aligned}\int \frac{e^{-x}}{x} dx &= \int \frac{1}{x} \left(1 - x + \frac{1}{2!} x^2 - \frac{1}{3!} x^3 + \dots \right) dx \\ &= \int \left(x^{-1} - 1 + \frac{1}{2!} x - \frac{1}{3!} x^2 + \dots \right) dx \\ &= \left(\ln |x| - x + \frac{1}{2 \cdot 2!} x^2 - \frac{1}{3 \cdot 3!} x^3 + \dots \right) + C \\ &= \ln |x| + C + \sum_{k=1}^{\infty} \frac{(-1)^k}{k \cdot k!} x^k,\end{aligned}$$

$$\begin{aligned}\int \frac{\sin x}{x^4} dx &= \int \frac{1}{x^4} \left(x - \frac{1}{3!} x^3 + \frac{1}{5!} x^5 - \dots \right) dx \\ &= \int \left(x^{-3} - \frac{1}{3!} x^{-1} + \frac{1}{5!} x - \dots \right) dx \\ &= \left(-\frac{1}{2} x^{-2} - \frac{1}{6} \ln |x| + \frac{1}{2 \cdot 5!} x^2 - \dots \right) + C \\ &= -\frac{1}{2x^2} - \frac{1}{6} \ln |x| + C + \sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{2k(2k+3)!} x^{2k}.\end{aligned}$$

Both results are valid on the intervals $(-\infty, 0)$ and $(0, \infty)$. $\square$

EXERCISES

1. Integrate the following rational functions:

$$\begin{array}{llll}
 \text{a)} \quad \frac{x^3 + 2x^2 - x + 5}{x + 2} & \text{b)} \quad \frac{x^2}{x^2 + x - 2} & \text{c)} \quad \frac{1}{x^3 - 4x^2 + 3x} \\
 \text{d)} \quad \frac{24}{x(x^2 - 1)(x^2 - 4)} & \text{e)} \quad \frac{1}{(1 - x)^2(1 + x)} & \text{f)} \quad \frac{3 + x - 2x^3}{1 - x^3} \\
 \text{g)} \quad \frac{x^8 + 1}{x^6 + x^4} & \text{h)} \quad \frac{3x^2 + 1}{(x^2 - 1)^2} & \text{i)} \quad \frac{1}{x^4 + 3x^2 + 2} & \text{j)} \quad \frac{1}{x^4 + 1} \\
 \text{k)} \quad \frac{1}{x(x^2 + a^2)} & \text{l)} \quad \frac{1}{x^2(x^2 - a^2)} & \text{m)} \quad \frac{x^3}{x^3 - a^3} & \text{n)} \quad \frac{1}{x^4 - a^4}
 \end{array}$$

2. Transform the integral

$$\int \frac{1}{x^4} \sqrt{x^2 + a^2} dx, \quad a > 0$$

using the substitutions a) $x = a \sinh t$, b) $x = a \tan t$, c) $x = a/t$. Choose the easiest of the options and compute the integral in closed form.

3. Integrate using a suitable substitution:

$$\begin{array}{llll}
 \text{a)} \quad \frac{e^{3x}}{e^x + 2} & \text{b)} \quad \frac{1}{\cosh x} & \text{c)} \quad \frac{1}{1 + \sqrt[3]{x}} & \text{d)} \quad \frac{1}{\sqrt{x+1} + \sqrt[4]{x+1}} \\
 \text{e)} \quad x^2 \sqrt{a^2 - x^2} & \text{f)} \quad \frac{1}{x\sqrt{1-x^2}} & \text{g)} \quad \frac{1}{x(x^2 + 1)^{3/2}} \\
 \text{h)} \quad \frac{1}{5 + 4 \sin x} & \text{i)} \quad \frac{2 - \sin x}{2 + \cos x} & \text{j)} \quad \frac{1}{5 - 4 \sin x + 3 \cos x}
 \end{array}$$

4. Compute the following integrals using the substitution $\tan x = t$:

$$\text{a)} \quad \int \frac{\tan x}{1 + 2 \tan x} dx \quad \text{b)} \quad \int \frac{1}{a \cos^2 x + b \sin^2 x} dx, \quad (a, b) \neq (0, 0)$$

5. If $R(x, y)$ is a rational expression in two variables, then by what kind of substitution does the integral

$$\int R\left(x, \sqrt[m]{\frac{x+a}{x+b}}\right) dx, \quad m \in \mathbb{N}, \quad m \geq 2, \quad a, b \in \mathbb{R}$$

become rational? Compute $\int \frac{1}{x} \sqrt{\frac{x+1}{x-1}} dx$ in this way.

6. Compute the antiderivatives of the following functions by means of power series:

$$\begin{array}{llll} \text{a)} & \frac{\sin x}{x} & \text{b)} & \frac{\cosh x}{x} \\ \text{c)} & \frac{1 - \cos x}{x^3} & \text{d)} & \frac{\sin x}{x^6} \\ \text{e)} & x^3 e^{x^3} & \text{f)} & \sin x^2 \\ \text{g)} & \frac{\sin x^2}{x^3} & \text{h)} & \frac{\cos x^3}{\sqrt{x}} \end{array}$$

7. (*) Express as elementary functions (if possible) or as power series the functions F and G such that

$$\begin{array}{ll} \text{a)} & \int e^x \ln x \, dx = F(x) \ln x + G(x) + C, \quad x \in (0, \infty), \\ \text{b)} & \int \frac{1}{\sqrt{x}} \ln x \sin x \, dx = x\sqrt{x} [F(x) \ln x + G(x)] + C, \quad x \in (0, \infty), \\ \text{c)} & \int \frac{\cos x}{x^2 + 1} \, dx = F(x) \operatorname{Arctan} x + G(x), \quad x \in \mathbb{R}, \\ \text{d)} & \int \frac{\sin x}{x^2 + 1} \, dx = F(x) \ln(x^2 + 1) + G(x), \quad x \in \mathbb{R}. \end{array}$$

8. (*) To prove Theorem VIII.3.1, consider a rational function of a complex variable $f(z) = p(z)/q(z)$, where q is a real-coefficient polynomial of degree $n \in \mathbb{N}$ and p is a real-coefficient polynomial of degree $\leq n - 1$.

a) Let $c \in \mathbb{C}$ be a simple zero of the polynomial q and let $q(z) = (z - c) q_1(z)$ and $g(z) = p(z)/q_1(z)$. Show that

$$f(z) = \frac{g(c)}{z - c} + \frac{r(z)}{q_1(z)}, \quad z \in \mathcal{D}_f,$$

where r is a polynomial of degree $\leq n - 2$. Conclude that if $c = a \in \mathbb{R}$, then r is real-coefficient.

b) Let $c = a + ib$ and $\bar{c} = a - ib$ be simple complex roots of the polynomial q ($a, b \in \mathbb{R}$, $b \neq 0$) and write $q(z) = (z - c)(z - \bar{c}) q_1(z)$. Show that there exist $A \in \mathbb{C}$ and $B, C \in \mathbb{R}$ such that

$$f(z) = \frac{A}{z - c} + \frac{\bar{A}}{z - \bar{c}} + \frac{r(z)}{q_1(z)} = \frac{Bz + C}{(z - a)^2 + b^2} + \frac{r(z)}{p_1(z)}, \quad z \in \mathcal{D}_f,$$

where r is a real-coefficient polynomial of degree $\leq n - 3$.

c) Let $c \in \mathbb{C}$ be an m -fold root of q and let $q(z) = (z - c)^m q_1(z)$ and $g(z) = p(z)/q_1(z)$. Show that

$$f(z) = \sum_{k=1}^m \frac{A_k}{(z - c)^k} + \frac{r(z)}{p_1(z)}, \quad z \in \mathcal{D}_f,$$

where $A_k = g^{(m-k)}(c)/(m-k)!$ and r is a polynomial of degree $\leq n-m-1$. Conclude that if $c = a \in \mathbb{R}$, then the coefficients A_k are real and r is real-coefficient.

d) Let $c = a + ib$ and $\bar{c} = a - ib$ be m -fold complex roots of the polynomial q ($a, b \in \mathbb{R}$, $b \neq 0$) and write $q(z) = (z-c)^m(z-\bar{c})^m q_1(z)$. Show that there exist coefficients $A_k \in \mathbb{C}$ and $B_k, C_k \in \mathbb{R}$ such that

$$\begin{aligned} f(z) &= \sum_{k=1}^m \frac{A_k}{(z-c)^k} + \sum_{k=1}^m \frac{\bar{A}_k}{(z-\bar{c})^k} + \frac{r(z)}{q_1(z)} \\ &= \sum_{k=1}^m \frac{B_k z + C_k}{[(z-a)^2 + b^2]^k} + \frac{r(z)}{q_1(z)}, \quad z \in \mathcal{D}_f, \end{aligned}$$

where r is a real-coefficient polynomial of degree $\leq n-2m-1$.

e) Based on the results of parts a) and b) and the fundamental theorem of algebra, derive the assertion of Theorem VIII.3.1 in the case where all roots of q are simple.

f) Prove Theorem VIII.3.1 in the general case, based on the results of parts c) and d) and the fundamental theorem of algebra.

VIII.4 Numerical computation of the antiderivative.

The definite integral

To determine the antiderivative of a given function, we have so far examined various tools of functional algebra. In these the function must be known as a computational rule, i.e. as some algebraic expression on the interval under consideration or on its subintervals. With applications in mind, such an assumption is rather restrictive, since a function may just as well be a mere pointwise-defined 'list'. This is the case, for example, if the values of the function can in practice only be measured, or if the definition of the function is indirect, so that the function evaluation $x \mapsto f(x)$ succeeds only numerically. If such a function has an antiderivative, then naturally the values of the antiderivative too cannot in practice be computed other than numerically. But how?—In what follows we solve this problem by deriving a universally valid computational formula for the numerical function evaluation $x \mapsto F(x)$, where F is an antiderivative of the given function. At the same time the formula leads to the concept of the *definite integral*, which is independent of the antiderivative.

Assume that f is defined (as a real function) on the interval $[a, b]$, i.e. to each $x \in [a, b]$ there corresponds a unique number $f(x) \in \mathbb{R}$. Let us make precise the problem of determining an antiderivative of f on the interval (a, b) as follows:

- (P) Find a function $y(x)$ that is continuous on the interval $[a, b]$, differentiable on the interval (a, b) , and satisfies

$$\begin{cases} y'(x) = f(x), & x \in (a, b), \\ y(a) = 0. \end{cases}$$

According to this, one seeks a function that is an antiderivative of f on the interval (a, b) and that additionally satisfies the imposed continuity condition and initial condition. Since differentiability already guarantees continuity on the interval (a, b) , the continuity condition means only the additional conditions: $y(x)$ is right-continuous at a and left-continuous at b .

Insofar as problem (P) has a solution, the imposed initial condition guarantees the uniqueness of the solution (cf. Corollary V.6.8). To the more difficult question, which concerns the existence of a solution, the following remarkable theorem gives one answer.

*THEOREM VIII.4.1 If f is continuous on the interval $[a, b]$, then problem (P) has a solution.

Theorem VIII.4.1 is a central sub-assertion of a more general theorem known as the *fundamental theorem of calculus*. This is formulated and (partly) proved later in Section VIII.6. In this section we pose the question differently: if one simply assumes that (P) is solvable, then how can the number $y(x) \in \mathbb{R}$ be computed in practice, given $x \in (a, b]$? In what follows we consider this question first in the case $x = b$, so that the task is to compute the number $y(b)$ when $y(x)$ is defined indirectly as the solution of (P). Since $y(b)$ is a real number, this number cannot in general be computed 'exactly'; rather, in practice one must be content to construct $y(b)$ as the limit of some (ultimately rational) sequence $\{Y_n\}$. The task is thus to determine an algorithm by which, for each $n \in \mathbb{N}$, it is possible to compute a number Y_n such that $\lim_n Y_n = y(b)$.

Let $n \in \mathbb{N}$ be given. In determining the number Y_n , we take as our starting point a *partition into subintervals* $[x_{k-1}, x_k]$, $k = 1 \dots n$, of the interval $[a, b]$, where $a = x_0 < x_1 < \dots < x_n = b$. The partition may be, e.g. *uniform*, in which case $x_k = a + kh_n$, where $h_n = (b - a)/n$. More generally, the subinterval partitions are restricted only by the condition

$$\lim_n h_n = 0, \quad \text{where } h_n = \max_{k=1 \dots n} (x_k - x_{k-1}), \quad n \in \mathbb{N}. \quad (\star)$$

Here h_n is the so-called *mesh parameter* of the subinterval partition. For uniform partitions the limit condition $(\star)$ is of course satisfied.

Once the points x_k , $k = 0 \dots n$ have been chosen in the manner described above, one can first write, for the solution $y(x)$ of (P),

$$y(b) = [y(x_n) - y(x_{n-1})] + \dots + [y(x_1) - y(x_0)] + y(x_0)$$

and hence, by the initial condition $y(x_0) = y(a) = 0$,

$$y(b) = \sum_{k=0}^n [y(x_k) - y(x_{k-1})].$$

The function $y(x)$ satisfies the conditions of the mean value theorem of differential calculus on the subintervals $[x_{k-1}, x_k]$, and in addition $y'(x) = f(x)$, $x \in (x_{k-1}, x_k)$. By the mean value theorem there then exist points $\xi_k \in (x_{k-1}, x_k)$, $k = 1 \dots n$ such that

$$y(x_k) - y(x_{k-1}) = f(\xi_k)(x_k - x_{k-1}), \quad k = 1 \dots n.$$

Combining the last two results, we have concluded that for the solution of (P)

$$y(b) = \sum_{k=1}^n f(\xi_k)(x_k - x_{k-1}).$$

This does not work directly as a computational formula, because the points ξ_k are not known, but the formula is a natural starting point for an approximation: we replace the unknown points ξ_k by some chosen points $\xi_k \in [x_{k-1}, x_k]$, e.g. $\xi_k = x_{k-1}$ or $\xi_k = (x_{k-1} + x_k)/2$. Once the choice has been made on each subinterval, we compute

$$Y_n = \sum_{k=1}^n f(\xi_k)(x_k - x_{k-1}).$$

In this way a sequence $\{Y_n\}$ is formed, whose n th term is thus computed as follows:

- A1. Choose the *partition points* x_k , $k = 0 \dots n$ so that $a = x_0 < x_1 < \dots < x_n = b$.
- A2. Choose the *sample points* $\xi_k \in [x_{k-1}, x_k]$, $k = 1 \dots n$.
- A3. Compute $Y_n = \sum_{k=1}^n f(\xi_k)(x_k - x_{k-1})$.

Accordingly, to compute Y_n it suffices to perform a finite number of arithmetic operations: n (in practice usually approximate) function evaluations $x \mapsto f(x)$ and, in addition, n multiplications and $n - 1$ additions.

In comparing the number Y_n computed by algorithm A1–A3 with the number $y(b)$, one must compare the exact expression for $y(b)$ above with the corresponding expression for Y_n (A3). In these the sample points ξ_k are not the same, so let us write η_k in place of ξ_k in expression (A3). For the approximation $Y_n \approx y(b)$ one then obtains the error formula

$$y(b) - Y_n = \sum_{k=1}^n [f(\xi_k) - f(\eta_k)](x_k - x_{k-1}).$$

To estimate the error, one must therefore be able to estimate the differences $f(\xi_k) - f(\eta_k)$, when all that is known about the points ξ_k and η_k is that they lie in the same subinterval $[x_{k-1}, x_k]$. For the estimate to succeed in the desired way, one evidently must make some regularity assumption about the function f .— In general, if f is sufficiently regular, then $|f(\xi) - f(\eta)|$ is at most proportional to the number $|\xi - \eta|$, i.e. $f(\xi) - f(\eta) = \mathcal{O}(|\xi - \eta|)$. The minimal assumption guaranteeing this is that f is Lipschitz continuous on the interval $[a, b]$, i.e. $\exists L \in \mathbb{R}_+$ (= the Lipschitz constant of f) such that

$$|f(\xi) - f(\eta)| \leq L|\xi - \eta| \quad \forall \xi, \eta \in [a, b]$$

(see Definition V.6.4 and also Theorem V.6.5).

Assuming the Lipschitz continuity of f , we have

$$|f(\xi_k) - f(\eta_k)| \leq L|\xi_k - \eta_k|, \quad k = 1 \dots n.$$

Here $|\xi_k - \eta_k| \leq x_k - x_{k-1}$, because $\xi_k, \eta_k \in [x_{k-1}, x_k]$, and further $x_k - x_{k-1} \leq h_n$, so that $|f(\xi_k) - f(\eta_k)| \leq Lh_n$, $k = 1 \dots n$. Using these estimates together with the triangle inequality in the error formula above, it follows that

$$\begin{aligned} |y(b) - Y_n| &\leq \sum_{k=1}^n |f(\xi_k) - f(\eta_k)|(x_k - x_{k-1}) \\ &\leq Lh_n \sum_{k=1}^n (x_k - x_{k-1}) = L(b-a)h_n. \end{aligned}$$

We have arrived at the following result.

THEOREM VIII.4.2 Let f be Lipschitz continuous on the interval $[a, b]$ with constant L and let $y(x)$ be the solution of problem (P). Then if $\{Y_n\}$ is any sequence computed by algorithm A1–A3 with the limit condition $(\star)$, then $\lim_n Y_n = y(b)$, more precisely $|y(b) - Y_n| \leq L(b-a)h_n \forall n \in \mathbb{N}$.*

Above, we have so far computed the solution of (P) only at the point $x = b$. The result is, however, easily generalized. Namely, if $y(x)$ is the solution of (P) and we fix $x \in (a, b]$, then the function $y(t)$, $t \in [a, x]$ is continuous on the interval $[a, x]$ (because it is continuous on the interval $[a, b]$) and is the solution of the initial value problem

$$\begin{cases} y'(t) = f(t), & t \in (a, x), \\ y(a) = 0. \end{cases}$$

This is thus problem (P), where the variable is t instead of x and x is in place of b . Accordingly, when the same substitutions are made in algorithm A1–A3, one can compute $y(x)$ at any desired point $x \in (a, b]$.

EXAMPLE 1 Solve the initial value problem $y'(x) = x$, $x > 0$, $y(0) = 0$ by algorithm A1–A3.

*As will be seen later in Section VIII.6, the limit assertion of Theorem VIII.4.2 (without the refinement) is true also if f is assumed only to be continuous on the interval $[a, b]$. No quantitative error estimate for the approximation $Y_n \approx y(b)$ is obtained on the basis of the continuity assumption alone.

Solution Fix $x > 0$ and consider the interval $[0, x]$. Place the partition points t_k uniformly, i.e. $t_k = kx/n$, $k = 0 \dots n$, so that by the computational formula (A3)

$$Y_n = \sum_{k=1}^n f(\xi_k)(t_k - t_{k-1}) = \sum_{k=1}^n \xi_k \frac{x}{n}.$$

In the choice of the sample points let us try the options a) $\xi_k = t_k$, b) $\xi_k = t_{k-1}$ and c) $\xi_k = (t_{k-1} + t_k)/2$, $k = 1 \dots n$, whereby we obtain:

$$\text{a) } Y_n = \sum_{k=1}^n k \frac{x}{n} \cdot \frac{x}{n} = \frac{x^2}{n^2} \sum_{k=1}^n k = \frac{x^2}{n^2} \cdot \frac{1}{2} n(n+1) = \frac{x^2}{2} + \frac{x^2}{2n}.$$

$$\text{b) } Y_n = \sum_{k=1}^n (k-1) \frac{x}{n} \cdot \frac{x}{n} = \frac{x^2}{n^2} \sum_{k=1}^n (k-1) = \frac{x^2}{n^2} \cdot \frac{1}{2} n(n-1) = \frac{x^2}{2} - \frac{x^2}{2n}.$$

$$\text{c) } Y_n = \sum_{k=1}^n \frac{1}{2} \left[(k-1) \frac{x}{n} + k \frac{x}{n} \right] \cdot \frac{x}{n} = \frac{x^2}{n^2} \sum_{k=1}^n \left(k - \frac{1}{2} \right) = \frac{x^2}{n^2} \cdot \frac{1}{2} n^2 = \frac{x^2}{2}.$$

We observe that in all cases $\lim_n Y_n = \frac{1}{2}x^2$. Theorem VIII.4.2 applies ($L = 1$), so $y(x) = \frac{1}{2}x^2$, $x > 0$. $\square$

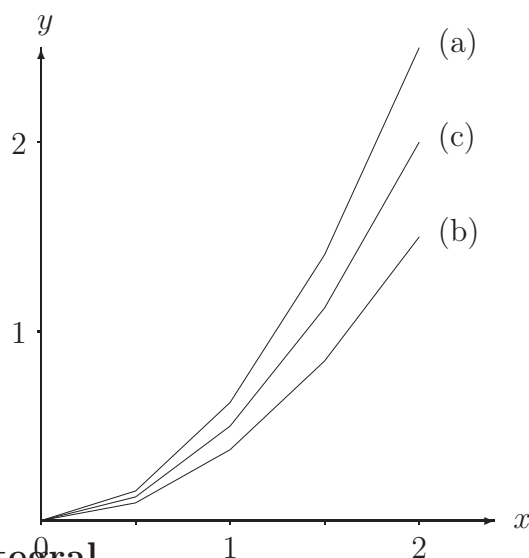
In the example, for the approximation $Y_n \approx y(x)$, Theorem VIII.4.2 gives the error estimate $|y(x) - Y_n| \leq x^2/n$ ($L = 1$, $b - a = x$, $h_n = x/n$). The observed errors are a) $y(x) - Y_n = -\frac{x^2}{2n}$, b) $y(x) - Y_n = \frac{x^2}{2n}$, $y(x) - Y_n = 0$, so these are consistent with the estimate.

If the solution $y(x)$ of (P) is to be computed at several points, algorithm A1–A3 can be started separately for each computation point, as in the example. When computing numerically, however, it is worthwhile to put the algorithm to more efficient use by noting that, according to formula (A3), the number Y_n comes to be computed in practice recursively in the form

$$Y_0 = 0, \quad Y_k = Y_{k-1} + f(\xi_k)(x_k - x_{k-1}), \quad k = 1 \dots n.$$

The numbers $Y_k = \sum_{i=1}^k f(\xi_i)(x_i - x_{i-1})$, $k = 1 \dots n - 1$ are then obtained as intermediate results of the summation. These are the approximations, according to algorithm A1–A3, for the numbers $y(x_k)$, when the partition points are x_i , $i = 0 \dots k$ on the interval $[a, x_k]$. Under the same assumptions as above, the error of these too is $\mathcal{O}(h_n)$, more precisely $|y(x_k) - Y_k| \leq L(x_k - a)h_n$, $k = 1 \dots n - 1$. It is therefore worthwhile to take all the numbers Y_k into account, whereby one obtains a picture of the function $y(x)$ over the whole interval under consideration without additional work.

EXAMPLE 1 (continued) In the figure we have chosen $x = 2$, $n = 4$ and joined the points (x_k, Y_k) computed with the choices of the example into a polygonal line. This represents, approximately, the graph of the function $y(x)$ on the interval $[0, 2]$. The result is most accurate in case (c), but in the other cases too the accuracy increases for fixed x as $n \rightarrow \infty$.



The definite integral

The number $y(b) = \lim_n Y_n$ computed according to algorithm A1–A3, where $y(x)$ is the solution of problem (P), has its own name and notation in mathematics: the number is called the *definite integral of the function f over the interval $[a, b]$* , denoted

$$y(b) = \int_a^b f(x) dx$$

and read 'the integral from a to b of $f(x) dx$ '. One further says that a is the *lower limit* of the integral, b is the *upper limit*, and $[a, b]$ is the *interval of integration*. The variable x , i.e. the *variable of integration*, is a 'dummy' (like a summation index), i.e. the variable can be changed without changing the meaning of the integral.

The definite integral is thus a real number depending on the function f to be integrated and on the interval of integration, which in practice can be computed as the limit of a sequence according to algorithm (A1)–(A3). When in the computational formula (A3) for Y_n one uses the abbreviated notation Δx_k for the differences $x_k - x_{k-1}$, the computation of the definite integral in the manner

presented above can be condensed into the *sum formula*

$$\boxed{\int_a^b f(x) dx = \lim_n \sum_{k=1}^n f(\xi_k) \Delta x_k.} \quad (1)$$

Here the limit notation ' $\lim_n$ ' incorporates, first, the restrictions imposed in algorithm A1–A3 concerning the choice of the points x_k and ξ_k ; second, the condition ($\star$); and third, the requirement that all sequences $\{Y_n\}$ according to algorithm (A1)–(A3) converge to the same limit (= the value of the definite integral). By what was presented above, the last-mentioned requirement is satisfied (and the sum formula is thus valid) at least under the assumptions that problem (P) is solvable (the basic assumption for now!) and that f is Lipschitz continuous on the interval $[a, b]$.

Under the above assumptions, a more general computational formula for the solution of problem (P) was also derived above, which can now be written as a definite integral:

$$y(x) = \int_a^x f(t) dt, \quad a < x \leq b. \quad (2)$$

Since the solution of (P) is (an) antiderivative of f on the interval (a, b) , formula (2) together with the sum formula (1) gives a method for determining the antiderivative that is independent of the integration technique developed so far (and also of the differentiation rules!).*

On the other hand, if on the interval (a, b) an antiderivative F of f is known (e.g. as an elementary function or as a series) and F is continuous on the interval $[a, b]$ (a necessary additional assumption!), then computing the number $\int_a^b f(x) dx$ can be done directly by means of F . Namely, under these assumptions the solution of problem (P) is $y(x) = F(x) - F(a)$, so that

$$\int_a^b f(x) dx = y(b) = F(b) - F(a).$$

This computational formula for the definite integral, based on a known antiderivative F , is usually presented as the *substitution formula*

$$\boxed{\int_a^b f(x) dx = \left|_{x=a}^{x=b} F(t) = \left|_a^b F(x).} \quad (3)$$

*Behind the antiderivative notation ' $\int f(x) dx$ ' lies the sum formula of the definite integral. The notation was introduced by **G.W. Leibniz** with the sum formula's 'passages to the limit' $\sum \hookrightarrow \int$ and $\Delta x_k \hookrightarrow dx$ as $n \rightarrow \infty$ in mind. According to Leibniz's own diary entries, the exact day the integral notation was invented was 29 October 1675.

Here the right-hand side is read 'substitution from a to b of $F(x)$ '. Formula (3) is made valid also in the case $a > b$ when one agrees that the *interchange rule* holds for the definite integral

$$\boxed{\int_a^b f(x) dx = - \int_b^a f(x) dx.} \quad (4)$$

For convenience, this is assumed to hold also when $a = b$, so that it becomes agreed that

$$\int_a^a f(x) dx = 0.$$

Let us remark about the concept of the definite integral that, for now, it is nothing more than a special notation for the number $y(b)$ in the case where problem (P) has a solution $y(x)$. The same idea is echoed by the substitution formula (3) and the interchange rule (4). In the next section we shall see that the definite integral becomes a concept independent of the solvability of (P) (and, more generally, of the existence of an antiderivative) when its definition is taken directly to be the sum formula (1), which was arrived at above as a numerical computational formula for the number $y(b)$. In later sections the concept of the definite integral gains still more 'life' from various applications. As will be seen, in applications one arrives at the definite integral directly through the sum formula. In that case the substitution formula (3) appears as a shortcut to computing the value of the integral 'exactly', when a continuous antiderivative F can be found for the function f (by the methods of the previous sections) on the interval $[a, b]$.

In the following we present three central properties of the definite integral. These remain valid when the sum formula (1) is henceforth taken as the basis of the definition, but for now the properties are justified by appealing to the assumed solvability of problem (P).

Additivity. Linearity. The comparison principle

If problem (P) is solvable, then for every $c \in (a, b)$ one can write $y(b) = y(c) + [y(b) - y(c)]$, i.e. expressed by means of the definite integral,

$$\boxed{\int_a^b f(x) dx = \int_a^c f(x) dx + \int_c^b f(x) dx.} \quad (5)$$

On the basis of this rule one says that the definite integral is *additive with respect to the interval of integration*. When the interchange rule (4) is also taken into

account, one finds that the additivity rule (5) is valid regardless of the order of magnitude of the numbers a, b, c , provided that problem (P) is solvable when $\min\{a, b, c\}$ is in place of a and $\max\{a, b, c\}$ is in place of b .

Let $F(x)$ be the solution of (P), and in addition let $G(x)$ be the solution of (P) when g is in place of f . Then if $\alpha, \beta \in \mathbb{R}$, the function $y(x) = \alpha F(x) + \beta G(x)$ is the solution of (P) when $\alpha f + \beta g$ is in place of f . In particular, then, $y(b) = \alpha F(b) + \beta G(b)$, i.e. the *linearity rule* holds for the definite integral (cf. the corresponding rule VIII.1: (I-1) for the indefinite integral)

$$\boxed{\int_a^b [\alpha f(x) + \beta g(x)] dx = \alpha \int_a^b f(x) dx + \beta \int_a^b g(x) dx, \quad \alpha, \beta \in \mathbb{R}.} \quad (6)$$

If, in addition to the assumptions mentioned, one assumes that $f(x) \leq g(x) \forall x \in [a, b]$, then $F'(x) - G'(x) = f(x) - g(x) \leq 0$, $x \in (a, b)$, so that $F(x) - G(x)$ is decreasing on the interval $[a, b]$ (Theorem V.6.3). In particular $F(b) - G(b) \leq F(a) - G(a) = 0$, i.e. $F(b) \leq G(b)$. Consequently the *comparison principle* holds for definite integrals

$$\boxed{f(x) \leq g(x) \quad \forall x \in [a, b] \Rightarrow \int_a^b f(x) dx \leq \int_a^b g(x) dx.} \quad (7)$$

Since $\pm f(x) \leq |f(x)|$ and $\int_a^b [\pm f(x)] dx = \pm \int_a^b f(x) dx$ (the linearity rule!), applying the comparison principle gives in particular

$$\boxed{\left| \int_a^b f(x) dx \right| \leq \int_a^b |f(x)| dx.} \quad (8)$$

This is called the *triangle inequality for integrals*.*

EXAMPLE 2 By the substitution formula (3),

$$\begin{aligned} \int_0^2 \sin x dx &= \left| -\cos x \right|_0^2 = \left| \cos x \right|_2^0 = 1 - \cos 2, \\ \int_x^{x^2} t dt &= \left| \frac{1}{2} t^2 \right|_x^{x^2} = \frac{1}{2} (x^4 - x^2), \quad x \in \mathbb{R}. \end{aligned}$$

EXAMPLE 3 Compute $\int_0^2 f(x) dx$ when $f(x) = \max\{\sqrt{x}, x^2\}$.

*The inequality (8) is akin to the triangle inequality of an ordered field, which holds for the finite sums of the sum formula (1).

Solution We first observe that $f(x) = \sqrt{x}$ on the interval $[0, 1]$ and $f(x) = x^2$ on the interval $[1, 2]$. Then, using first the additivity rule (5) and then the substitution formula (3), we obtain

$$\int_0^2 f(x) dx = \int_0^1 \sqrt{x} dx + \int_1^2 x^2 dx = \left[\frac{2}{3} x^{3/2} \right]_0^1 + \left[\frac{1}{3} x^3 \right]_1^2 = 3. \quad \square$$

EXAMPLE 4 Determine the antiderivative on $\mathbb{R}$ of the function $f(x) = \min\{3x, 4 - x^2\}$ by making use of the definite integral.

Solution The function $y(x) = \int_0^x f(t) dt$ is the sought antiderivative with the additional condition $y(0) = 0$. Since $f(x) = 3x$ when $x \in [-4, 1]$, and $f(x) = 4 - x^2$ when $x \geq 1$ or $x \leq -4$, we have

$$x \in [-4, 1] \Rightarrow y(x) = \int_0^x 3t dt = \left[\frac{3}{2} t^2 \right]_0^x = \frac{3}{2} x^2,$$

$$x \in [1, \infty) \Rightarrow y(x) = y(1) + [y(x) - y(1)]$$

$$\begin{aligned} &= \frac{3}{2} + \int_1^x (4 - t^2) dt \\ &= \frac{3}{2} + \left[4t - \frac{1}{3} t^3 \right]_1^x = -\frac{1}{3} x^3 + 4x - \frac{13}{6}, \end{aligned}$$

$$x \in (-\infty, -4] \Rightarrow y(x) = y(-4) + [y(x) - y(-4)]$$

$$\begin{aligned} &= 24 + \int_{-4}^x (4 - t^2) dt \\ &= 24 + \left[4t - \frac{1}{3} t^3 \right]_{-4}^x = -\frac{1}{3} x^3 + 4x + \frac{56}{3}. \end{aligned}$$

Here $y(1) = 3/2$ and $y(-4) = 24$ were obtained from the first expression. The general antiderivative is $F(x) = y(x) + C$, $C \in \mathbb{R}$, so that

$$\int f(x) dx = \begin{cases} -\frac{1}{3} x^3 + 4x + \frac{56}{3} + C, & \text{when } x < -4, \\ \frac{3}{2} x^2 + C, & \text{when } -4 \leq x \leq 1, \\ -\frac{1}{3} x^3 + 4x - \frac{13}{6} + C, & \text{when } x > 1. \end{cases} \quad \square$$

EXAMPLE 5 Simplify the expression $\frac{d}{dx} \int_{\sqrt{x}}^{1/\sqrt{x}} e^{-t^2} dt$, $x > 0$.

Solution The function e^{-t^2} has on $\mathbb{R}$ an antiderivative $F(t)$ in series form, but for the simplification it suffices to know that $F(t)$ exists:

$$\begin{aligned}\frac{d}{dx} \int_{\sqrt{x}}^{1/\sqrt{x}} e^{-t^2} dt &= \frac{d}{dx} [F(\frac{1}{\sqrt{x}}) - F(\sqrt{x})] \\ &= -\frac{1}{2x\sqrt{x}} f(\frac{1}{\sqrt{x}}) - \frac{1}{2\sqrt{x}} f(\sqrt{x}) \\ &= -\frac{1}{2\sqrt{x}} \left(\frac{e^{-1/x}}{x} + e^{-x} \right). \quad \square\end{aligned}$$

EXAMPLE 6 Estimate the error in the approximation $\int_{10}^{20} \frac{x^4}{x^5+1} dx \approx \ln 2$.

Solution On the interval of integration

$$\frac{x^4}{x^5+1} = \frac{1}{x} - \frac{1}{x(x^5+1)},$$

so that by the linearity rule (6) and the substitution formula (3)

$$\int_{10}^{20} \frac{x^4}{x^5+1} dx = \int_{10}^{20} \frac{1}{x} dx - \delta = \left| \ln x \right|_{10}^{20} - \delta = \ln 2 - \delta,$$

where further, by the comparison principle (7) and the substitution formula,

$$0 \leq \delta = \int_{10}^{20} \frac{1}{x(x^5+1)} dx \leq \int_{10}^{20} \frac{1}{x^6} dx = \left| -\frac{1}{5x^5} \right|_{10}^{20} < 2 \cdot 10^{-6}.$$

Thus the approximation is an upper estimate whose error is less than $2 \cdot 10^{-6}$.
□

EXERCISES

1. Solve problem (P), or show that it is unsolvable, when $[a, b] = [0, 1]$ and

$$\begin{aligned}\text{a) } f(x) &= \begin{cases} \ln x, & \text{when } x > 0, \\ 0, & \text{when } x = 0, \end{cases} & \text{b) } f(x) &= \begin{cases} \frac{1}{x}, & \text{when } x > 0, \\ 0, & \text{when } x = 0, \end{cases} \\ \text{c) } f(x) &= \begin{cases} \frac{1}{\sqrt{x}}, & \text{when } x > 0, \\ 1, & \text{when } x = 0, \end{cases} & \text{d) } f(x) &= \begin{cases} \frac{1}{\sqrt{|2x-1|}}, & \text{when } x \neq \frac{1}{2}, \\ 0, & \text{when } x = \frac{1}{2}. \end{cases}\end{aligned}$$

2. Solve the initial value problem $y'(x) = f(x)$, $x > 0$, $y(0) = 0$ using algorithm A1–A3 and uniform partitions of the interval $[0, x]$, when a) $f(x) = x^2$, b) $f(x) = x^3$, c) $f(x) = e^{-x}$, d) $f(x) = 2^x$.
Additional information: $\sum_{k=1}^n k^2 = \frac{1}{6} n(n+1)(2n+1)$, $\sum_{k=1}^n k^3 = \frac{1}{4} n^2(n+1)^2$.

3. Compute the following limits by interpreting them as definite integrals.

$$\begin{array}{ll} \text{a)} \quad \lim_{n \rightarrow \infty} \frac{1}{n^4} [1^3 + 2^3 + \dots + (4n-1)^3] & \text{b)} \quad \lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{1}{n+k} \\ \text{c)} \quad \lim_{n \rightarrow \infty} \frac{\pi}{n} \sum_{k=1}^{n-1} \sin \frac{k\pi}{n} & \text{d)} \quad \lim_{n \rightarrow \infty} \frac{1}{n^2} \sum_{k=0}^{n-1} \sqrt{n^2 - k^2} \\ \text{e)} \quad \lim_{n \rightarrow \infty} \frac{1}{n^2} \sum_{k=0}^n \sqrt{n^2 + k^2} & \text{f)} \quad \lim_{n \rightarrow \infty} \sum_{k=0}^n \frac{n}{n^2 + k^2} \end{array}$$

4. Compute, by means of rules (3)–(6), exactly (if possible) or as a numerical approximation:

$$\begin{array}{lll} \text{a)} \quad \int_{-2}^2 (x^2 + 3)^2 dx & \text{b)} \quad \int_4^9 \left(\sqrt{x} - \frac{1}{\sqrt{x}} \right) dx & \text{c)} \quad \int_1^{10} \frac{1}{x^3 + x} dx \\ \text{d)} \quad \int_{-1}^1 2^x dx & \text{e)} \quad \int_0^4 |\sin \theta| d\theta & \text{f)} \quad \int_0^\pi \max\{\cos x, \sin 2x\} dx \\ \text{g)} \quad \int_1^2 |x^3 + x^2 - 3| dx & \text{h)} \quad \int_0^\pi |x - \cos x| dx & \text{i)} \quad \int_0^4 \min\{4x, e^x\} dx \end{array}$$

5. Compute the derivatives of the following functions with respect to the free variable.

$$\begin{array}{lll} \text{a)} \quad \int_\pi^x \sin^3 t dt & \text{b)} \quad \int_x^{2\pi} (\sin^2 u - u^4 e^{-u}) du & \text{c)} \quad \int_x^{5x} \frac{e^s}{s^2 + 1} ds \\ \text{d)} \quad \int_0^{3 \sinh 2x} \sqrt{9 + t^2} dt & \text{e)} \quad \int_{-\pi}^t \frac{\cos y}{1 + y^2} dy & \text{f)} \quad \int_{\sin \theta}^{\cos \theta} \sqrt{1 - x^2} dx \end{array}$$

6. Determine the antiderivatives on $\mathbb{R}$ of the following functions by making use of the definite integral:

$$\begin{array}{ll} \text{a)} \quad f(x) = |x| - |x - 2| & \text{b)} \quad f(x) = |x^2 - 7x + 10| \\ \text{c)} \quad f(x) = \max\{x^2 + 2x + 3, 9 - 2x - x^2\} \end{array}$$

7. Determine the smallest values of the following functions and draw the graphs of the functions.

$$\text{a) } f(x) = \int_0^1 |x - t| dt \quad \text{b) } f(x) = \int_0^\pi (x \cos t - t \cos x)^2 dt$$

8. Prove:

$$\begin{aligned} \text{a) } 1 &\leq \int_0^1 \frac{1+x^{20}}{1+x^{21}} \leq \frac{22}{21} & \text{b) } \frac{2}{\sqrt[4]{e}} &\leq \int_0^2 e^{x^2-x} dx \leq 2e^2 \\ \text{c) } \int_3^5 \frac{x}{e \ln x} dx &> 2 & \text{d) } 0 < \int_{50}^{100} \frac{x^3}{x^6+8x+9} dx &< 1.5 \cdot 10^{-4} \\ \text{e) } \int_{100}^{300} \frac{x^5}{x^6+x-1} dx &= \ln 3 - \delta, \quad 0 < \delta < 2 \cdot 10^{-11} \end{aligned}$$

9. (*) Justify the approximate computational formula

$$\int_1^2 \frac{e^{-x}}{x} dx \approx \ln 2 + \sum_{k=1}^n \frac{(-1)^k}{k \cdot k!} (2^k - 1), \quad n \in \mathbb{N}, \quad n \gg 1$$

and estimate its error for $n = 10$.

$$10. (*) \text{ Let } F(x) = \int_0^{2x-x^2} \cos\left(\frac{1}{1+t^2}\right) dt, \quad G(x) = \int_4^{x^2} e^{t^2} dt.$$

- a) Investigate whether F attains its smallest or largest value at some x .
b) Compute the limit $\lim_{x \rightarrow 2} G(x)/(x^3 - 8)$.

11. (*) In problem (P), let $[a, b] = [-1, 1]$, $f(0) = 0$, and the definition of f elsewhere than at the origin be

$$\text{a) } f(x) = \frac{1}{x} \left(\sin \frac{1}{x} + x \cos \frac{1}{x} \right), \quad \text{b) } f(x) = \frac{1}{x} \left(\sin \frac{1}{x^2} + x^2 \cos \frac{1}{x^2} \right).$$

Show that problem (P) a) is not solvable, b) is solvable.

VIII.5 The Riemann integral

Consider a function f defined on a closed interval $[a, b]$ that is *bounded*, i.e. there exists $M \in \mathbb{R}_+$ such that

$$|f(x)| \leq M \quad \forall x \in [a, b].$$

A general *partition* of the interval $[a, b]$ will henceforth be denoted by the symbol X . As in the previous section, a partition means a finite ordered set $X = \{x_0, x_1, \dots, x_n\}$, where $n \in \mathbb{N}$ and

$$a = x_0 < x_1 < \dots < x_n = b.$$

The most essential parameter of the partition X is its *mesh parameter*, defined by

$$h_X = \max_{k=1 \dots n} (x_k - x_{k-1}).$$

Associated with the partition X we also introduce a *set of sample points* $\Xi_X = \{\xi_1, \dots, \xi_n\}$, where $\xi_k \in [x_{k-1}, x_k]$, $k = 1 \dots n$ (otherwise Ξ_X is freely selectable).

With the above notation, to each pair (X, Ξ_X) we associate a real number $\sigma(f, X, \Xi_X)$, defined by

$$\sigma(f, X, \Xi_X) = \sum_{k=1}^n f(\xi_k)(x_k - x_{k-1}).$$

As in the previous section, we use the limit notation $\text{Lim}_{h_X \rightarrow 0} \sigma(f, X, \Xi_X) = A$ to describe a situation in which the sum always approaches the same limit A ($A \in \mathbb{R}$) as $h_X \rightarrow 0$. Like the limit of a function, the limit ‘Lim’ can also be defined in two ways, either by means of sequences or as an ‘ (ε, δ) -definition’, cf. Definition V.2.1 and Theorem V.2.2. Let us again formulate the latter definition as a theorem. (The proof is omitted, cf. the proof of Theorem V.2.2.)

DEFINITION VIII.5.1 $\text{Lim}_{h_X \rightarrow 0} \sigma(f, X, \Xi_X) = A \in \mathbb{R}$, if for every sequence $\{\sigma(f, X_n, \Xi_{X_n})\}$ satisfying $h_{X_n} \rightarrow 0$, it holds that $\lim_{n \rightarrow \infty} \sigma(f, X_n, \Xi_{X_n}) = A$.

***THEOREM VIII.5.2 (Lim: (ε, δ) -criterion)** $\text{Lim}_{h_X \rightarrow 0} \sigma(f, X, \Xi_X) = A$ exactly when for every $\varepsilon > 0$ there exists $\delta > 0$ such that for every pair (X, Ξ_X) with $h_X < \delta$,

$$|\sigma(f, X, \Xi_X) - A| < \varepsilon.$$

According to Definition VIII.5.1, the limit ‘Lim’ means the common limit of the sequences $\{\sigma(f, X_n, \Xi_{X_n})\}$ picked from all possible sums $\sigma(f, X, \Xi)$ that satisfy

the condition $h_{X_n} \rightarrow 0$ (no other conditions!). Insofar as such a common limit exists, Definition VIII.5.1 also provides an algorithm for computing it: Choose any sequence of increasingly fine (e.g. equally spaced) partitions X_n , associate to each X_n some set of sample points Ξ_n (e.g. the midpoints of the subintervals or one of the endpoints) and compute $A = \lim_n \sigma(f, X_n, \Xi_n) = \lim_n A_n$.

In the previous section it was noted that if f has an antiderivative F on the interval (a, b) , and F is in addition continuous on $[a, b]$, then for every X there exists a set of sample points Ξ_X such that

$$\sigma(f, X, \Xi_X) = F(b) - F(a) = \left| F(x) \right|_a^b = \int_a^b f(x) dx.$$

It was further shown that insofar as the definite integral $\int_a^b f(x) dx$ is defined in this way (under the assumptions mentioned) and it is additionally assumed that f is Lipschitz continuous on $[a, b]$, then for both every partition X and every set of sample points Ξ_X (Theorem VIII.4.2)

$$\sigma(f, X, \Xi_X) = \int_a^b f(x) dx + \mathcal{O}(h_X).$$

On the basis of Definition VIII.5.1 (or Theorem VIII.5.2) it follows that under the (rather strong) assumptions mentioned,

$$\lim_{h_X \rightarrow 0} \sigma(f, X, \Xi_X) = \int_a^b f(x) dx.$$

More generally, if f possibly does not satisfy the conditions mentioned, this computational formula is turned into a definition of the definite integral that is independent of the antiderivative.

DEFINITION VIII.5.3 (The Riemann* integral) Let f be defined and bounded on $[a, b]$. If

$$\lim_{h_X \rightarrow 0} \sigma(f, X, \Xi_X) = A \quad (A \in \mathbb{R}),$$

*The German mathematician **Georg Friedrich Bernhard Riemann** (1826–1866) is one of the great names of nineteenth-century (and indeed of all-time) mathematics. Riemann was a ‘pure mathematician’ more clearly than his predecessors, of whom even the most notable (e.g. Euler, Lagrange, Cauchy, Gauss) studied physics or other applications of mathematics alongside mathematics. Besides integrals, Riemann’s work concerned, among other things, the theory of complex functions (*Riemann surfaces*), the distribution of prime numbers, and the mathematical foundations of geometry. Many of Riemann’s results were pioneering and pointed the way for later mathematical research, which from the nineteenth century onward diverged ever more clearly from physics.

then f is said to be *Riemann integrable* (or integrable in the sense of Riemann) on $[a, b]$. The number A is called the *Riemann integral* of f (the Riemann integral) on $[a, b]$ and is denoted

$$A = \int_a^b f(x) dx.$$

In connection with this definition, the sums $\sigma(f, X, \Xi_X)$ are called *Riemann sums*. Henceforth we simply say that f is *integrable* on $[a, b]$ if f is defined, bounded, and Riemann integrable on that interval.

EXAMPLE 1 Let $c \in \mathbb{R}$ and define

$$f(x) = \begin{cases} 0, & \text{when } x < 1/2 \\ c, & \text{when } x = 1/2 \\ 1, & \text{when } x > 1/2 \end{cases}$$

Investigate the integrability of f on $[0, 1]$.

Solution If $X = \{x_0, \dots, x_n\}$, $n \geq 2$, is a partition of the interval $[0, 1]$, then for some $j \in \mathbb{N}$ we have $x_{j-1} < 1/2 < x_{j+1}$. Then $f(\xi_k) = 0$ when $k \leq j-1$ and $f(\xi_k) = 1$ when $k \geq j+2$, so that

$$\begin{aligned} \sigma(f, X, \Xi_X) &= \sum_{k=j}^{j+1} f(\xi_k)(x_k - x_{k-1}) + \sum_{k=j+2}^n (x_k - x_{k-1}) \\ &= f(\xi_j)(x_j - x_{j-1}) + f(\xi_{j+1})(x_{j+1} - x_j) + (1 - x_{j+1}). \end{aligned}$$

As $h_X \rightarrow 0$, we have $1 - x_{j+1} \rightarrow 1/2$, since $0 < x_{j+1} - 1/2 < x_{j+1} - x_{j-1} < 2h_X$, and by the triangle inequality

$$\begin{aligned} |f(\xi_j)(x_j - x_{j-1}) + f(\xi_{j+1})(x_{j+1} - x_j)| &\leq \max\{|c|, 1\}(x_{j+1} - x_{j-1}) \\ &\leq 2 \max\{|c|, 1\}h_X \rightarrow 0. \end{aligned}$$

Thus f is integrable on $[0, 1]$ and $\int_0^1 f(x) dx = \frac{1}{2}$. $\square$

EXAMPLE 2 The Dirichlet function

$$f(x) = \begin{cases} 1, & \text{if } x \in \mathbb{Q} \\ 0, & \text{otherwise} \end{cases}$$

is not integrable on $[0, 1]$, for if one chooses $\Xi_X \subset \mathbb{Q}$, then $\sigma(f, X, \Xi_X) = 1$, and if one chooses $\Xi_X \cap \mathbb{Q} = \emptyset$, then $\sigma(f, X, \Xi_X) = 0$, whatever X may be. $\square$

The examples show that not all integrable functions are continuous, nor are all bounded functions integrable. In Example 1 the value of the integral does not depend on the value of the function at the point of discontinuity. — More generally too, it holds that if f is integrable on $[a, b]$, then the value of the integral does not change if f is redefined at finitely many points (or even on a convergent sequence of points, see Exercise 8). Thus the integral is ‘insensitive’ to individual point values of the function, in the same way as the limit of a function.

We now show that the central properties of the definite integral established in the previous section, namely additivity with respect to the interval of integration, linearity with respect to the integrand, and the comparison principle for integrals, remain valid also for integrals in the sense of Definition VIII.5.3.

THEOREM VIII.5.4 (Additivity of the integral) If f is integrable on the intervals $[a, b]$ and $[b, c]$ ($a < b < c$), then f is integrable on $[a, c]$, and

$$\int_a^c f(x) dx = \int_a^b f(x) dx + \int_b^c f(x) dx.$$

Proof Consider the Riemann sums $\sigma(f, X, \Xi_X)$ on the interval $[a, c]$. Assume first that $b \in X$, and split the sum into two parts:

$$\sigma(f, X, \Xi_X) = \sum_{k: x_k \leq b} f(\xi_k)(x_k - x_{k-1}) + \sum_{k: x_k > b} f(\xi_k)(x_k - x_{k-1}).$$

By Definitions VIII.5.3 and VIII.5.1,

$$\begin{aligned} \sum_{k: x_k \leq b} f(\xi_k)(x_k - x_{k-1}) &\rightarrow \int_a^b f(x) dx, \quad \text{as } h_X \rightarrow 0, \\ \sum_{k: x_k > b} f(\xi_k)(x_k - x_{k-1}) &\rightarrow \int_b^c f(x) dx, \quad \text{as } h_X \rightarrow 0, \end{aligned}$$

since f was integrable on the intervals $[a, b]$ and $[b, c]$. We thus conclude that, provided the restriction $b \in X$ holds,

$$h_X \rightarrow 0 \Rightarrow \sigma(f, X, \Xi_X) \rightarrow \int_a^b f(x) dx + \int_b^c f(x) dx.$$

Consider next a Riemann sum $\sigma(f, X, \Xi_X)$, where $X = \{x_k\}$ and $x_{m-1} < b < x_m$ for some m (so that $b \notin X$). Compare this with the sum $\sigma(f, X', \Xi_{X'})$, where X' is obtained from X by adding only the partition point b and $\Xi_{X'}$ is chosen so that the sample points $\xi_k \in \Xi_X$ and $\xi'_l \in \Xi_{X'}$ are the same on the subintervals common

to the two partitions, i.e. when $[x_{k-1}, x_k] = [x'_{l-1}, x'_l]$. Then $x'_{m-1} = x_{m-1}$, $x'_m = b$ and $x'_{m+1} = x_m$, so that

$$\begin{aligned} \sigma(f, X, \Xi_X) - \sigma(f, X', \Xi_{X'}) \\ = f(\xi_m)(x_m - x_{m-1}) - f(\xi'_m)(b - x_m) - f(\xi'_{m+1})(x_{m+1} - b). \end{aligned}$$

If $|f(x)| \leq M \forall x \in [a, c]$, then using the triangle inequality we conclude

$$\begin{aligned} |\sigma(f, X, \Xi_X) - \sigma(f, X', \Xi_{X'})| &\leq 2M(x_m - x_{m-1}) \\ &\leq 2Mh_X \rightarrow 0, \quad \text{as } h_X \rightarrow 0. \end{aligned}$$

Thus the sum $\sigma(f, X, \Xi_X)$ has the same limit regardless of whether $b \in X$ is true or not. $\square$

Using the rule for interchanging the limits of integration agreed in the previous section,

$$\int_b^a f(x) dx = - \int_a^b f(x) dx,$$

we see that the assertion of Theorem VIII.5.4 is true regardless of the ordering of the numbers a, b, c , as long as f is integrable on the intervals $[\min\{a, b\}, \max\{a, b\}]$ and $[\min\{b, c\}, \max\{b, c\}]$.

THEOREM VIII.5.5 (Linearity of the integral) If f and g are integrable on $[a, b]$, then $\alpha f + \beta g$ is integrable on $[a, b]$ for every $\alpha, \beta \in \mathbb{R}$ and

$$\int_a^b [\alpha f(x) + \beta g(x)] dx = \alpha \int_a^b f(x) dx + \beta \int_a^b g(x) dx.$$

Proof If $\{X_n\}$ and $\{\Xi_n\}$ are partitions and sets of sample points on $[a, b]$, then the corresponding Riemann sums satisfy

$$\sigma(\alpha f + \beta g, X_n, \Xi_n) = \alpha \sigma(f, X_n, \Xi_n) + \beta \sigma(g, X_n, \Xi_n), \quad \alpha, \beta \in \mathbb{R}.$$

The asserted linearity rule follows from this by Definition VIII.5.3 and the combination rules for limits of sequences (Theorem I.7.3), when one assumes $\lim_n h_{X_n} = 0$. $\square$

THEOREM VIII.5.6 (Comparison principle for integrals) If f and g are integrable on $[a, b]$, then

$$f(x) \leq g(x) \quad \forall x \in [a, b] \Rightarrow \int_a^b f(x) dx \leq \int_a^b g(x) dx.$$

Proof This assertion too reduces to the theory of sequences: The comparison principle evidently holds for the Riemann sums $\sigma(f, X_n, \Xi_n)$ and $\sigma(g, X_n, \Xi_n)$, so it also holds for the limits (Proposition I.7.4 [V1]). $\square$

*Riemann's upper and lower integrals

Definition VIII.5.3 of Riemann integrability is concrete in the sense that it also directly provides an algorithm suitable for the numerical computation of the integral. In what follows we approach the Riemann integral in another, somewhat more abstract way, which leads to an alternative criterion for Riemann integrability (Theorem VIII.5.8). This criterion will subsequently turn out to be convenient in various theoretical considerations (see also Exercises 6–10).

Let f be a function defined and bounded on $[a, b]$ and $X = \{x_k, k = 0 \dots n\}$ a partition of $[a, b]$. Since f is bounded, the sets $Y_k = \{f(x) \mid x \in [x_{k-1}, x_k]\}$ are bounded. Thus there exist numbers (cf. Section I.11)

$$\bar{f}_k = \sup Y_k, \quad \underline{f}_k = \inf Y_k, \quad k = 1 \dots n.$$

The sums

$$\begin{aligned} \bar{\sigma}(f, X) &= \sum_{k=1}^n \bar{f}_k (x_k - x_{k-1}) \quad (\text{upper sum}), \\ \underline{\sigma}(f, X) &= \sum_{k=1}^n \underline{f}_k (x_k - x_{k-1}) \quad (\text{lower sum}). \end{aligned}$$

are called the *Riemann upper and lower sums* of the function f associated with the partition X . Note that if f is continuous on $[a, b]$, then f attains its maximum and minimum values on every subinterval $[x_{k-1}, x_k] \subset [a, b]$ (Theorem V.1.12), whence $\bar{f}_k = f(\xi_k)$ and $\underline{f}_k = f(\eta_k)$ for some $\xi_k, \eta_k \in [x_{k-1}, x_k]$. In this case, then, the Riemann upper and lower sums are special cases of Riemann sums.

Let $\mathcal{X}$ now be the set of all possible partitions X of the interval $[a, b]$. Then the sets of real numbers $\{\bar{\sigma}(f, X) \mid X \in \mathcal{X}\}$ and $\{\underline{\sigma}(f, X) \mid X \in \mathcal{X}\}$ are bounded, for if $|f(x)| \leq M$, $x \in [a, b]$, then $|\bar{f}_k| \leq M$ and $|\underline{f}_k| \leq M \quad \forall k$, whence

$$|\bar{\sigma}(f, X)|, |\underline{\sigma}(f, X)| \leq M \sum_{k=1}^n (x_k - x_{k-1}) = M(b - a).$$

The sets of numbers mentioned therefore have both a supremum and an infimum.

DEFINITION VIII.5.7 If f is defined and bounded on $[a, b]$ and $\mathcal{X}$ is the set of all possible partitions X of $[a, b]$, then the *Riemann upper and lower integrals* of the function f on $[a, b]$ are defined by

$$\begin{aligned} \int_a^b f(x) dx &= \inf_{X \in \mathcal{X}} \bar{\sigma}(f, X) \quad (\text{upper integral}), \\ \int_a^b f(x) dx &= \sup_{X \in \mathcal{X}} \underline{\sigma}(f, X) \quad (\text{lower integral}). \end{aligned}$$

According to Definition VIII.5.7, then, every function defined and bounded on $[a, b]$ is both ‘upper integrable’ and ‘lower integrable’ on that interval.

EXAMPLE 3 If $f(x) = 1$ when $x \in \mathbb{Q}$, and $f(x) = 0$ otherwise, then

$$\overline{\int_0^1} f(x) dx = 1, \quad \underline{\int_0^1} f(x) dx = 0. \quad \square$$

The example leads to the following very elegant criterion for Riemann integrability. The proof is presented at the end of the section.

*THEOREM VIII.5.8 (**Riemann integrability**) A function f defined and bounded on $[a, b]$ is Riemann integrable on that interval exactly when the upper and lower integrals of f on that interval are equal, in which case the Riemann integral of f = the common value of the upper and lower integrals.

In what follows we present two assertions concerning the Riemann upper and lower sums, which are useful in theoretical considerations together with Theorem VIII.5.8. First we show that under *refinement* of a partition the upper sum decreases or stays the same and correspondingly the lower sum increases or stays the same.

THEOREM VIII.5.9 If X and X' are partitions of $[a, b]$ and $X' \supset X$ (i.e. X' is a refinement of X), then

$$\underline{\sigma}(f, X) \leq \underline{\sigma}(f, X') \leq \overline{\sigma}(f, X') \leq \overline{\sigma}(f, X).$$

Proof The refinement can be thought of as carried out by adding to the partition X one point at a time until one arrives at the partition X' . The assertion is true if it is true for every such partial refinement. So let $X = \{x_k, k = 0 \dots n\}$ and let $X' = X \cup \{x'_k\}$, where $x'_k \in (x_{k-1}, x_k)$. Then if $\overline{f}_k$ and $\underline{f}_k$ are defined as above (associated with the partition X) and we set

$$\alpha_k = \sup_{x \in [x_{k-1}, x'_k]} f(x), \quad \beta_k = \sup_{x \in [x'_k, x_k]} f(x),$$

then $\alpha_k \leq \overline{f}_k$ and $\beta_k \leq \overline{f}_k$ (cf. Exercise I.11:2c), so that

$$\alpha_k(x'_k - x_{k-1}) + \beta_k(x_k - x'_k) \leq \overline{f}_k(x_k - x_{k-1}).$$

Since the partitions are the same outside the interval $[x_{k-1}, x_k]$, we conclude that necessarily $\overline{\sigma}(f, X') \leq \overline{\sigma}(f, X)$. In a similar way one shows that $\underline{\sigma}(f, X) \leq$

$\underline{\sigma}(f, X')$. By the definition of the upper and lower sums we also have $\underline{\sigma}(f, X') \leq \overline{\sigma}(f, X')$, so the assertion follows. $\square$

The following assertion makes the upper and lower integrals concrete by means of sequences. The assertion relies essentially on the previous one.

*THEOREM VIII.5.10 Let f be defined and bounded on $[a, b]$. Then for every sequence $\{X_n\}$ of partitions of $[a, b]$ with $h_{X_n} \rightarrow 0$,

$$\lim_n \overline{\sigma}(f, X_n) = \overline{\int_a^b} f(x) dx, \quad \lim_n \underline{\sigma}(f, X_n) = \underline{\int_a^b} f(x) dx.$$

Proof Let $\{X_n\}$ be a sequence as assumed and let $\varepsilon > 0$. By the definition of the upper integral there exists a partition $X = \{x_k, k = 0 \dots K\}$ of $[a, b]$ such that

$$0 \leq \overline{\sigma}(f, X) - \overline{\int_a^b} f(x) dx < \frac{\varepsilon}{2}.$$

Set $X'_n = X_n \cup X$, so that X'_n is a refinement of the partitions X_n and X . In the refinement $X_n \hookrightarrow X'_n$, the subintervals generated by X_n that split are exactly those whose interior contains at least one point of the set X . There are at most $K - 1$ such subintervals, and each of these has length at most h_{X_n} . Since only these subintervals affect the difference $\overline{\sigma}(f, X_n) - \overline{\sigma}(f, X'_n)$, denoting the contributions of the subintervals mentioned in the sums $\overline{\sigma}(f, X_n)$ and $\overline{\sigma}(f, X'_n)$ by the symbols $\overline{\sigma}_0(f, X_n)$ and $\overline{\sigma}_0(f, X'_n)$ and assuming that $|f(x)| \leq M$, $x \in [a, b]$, we can estimate, by Theorem VIII.5.9 and the triangle inequality,

$$\begin{aligned} 0 \leq \overline{\sigma}(f, X_n) - \overline{\sigma}(f, X'_n) &= |\overline{\sigma}_0(f, X_n) - \overline{\sigma}_0(f, X'_n)| \\ &\leq |\overline{\sigma}_0(f, X_n)| + |\overline{\sigma}_0(f, X'_n)| \leq (K - 1) \cdot 2M \cdot h_{X_n}. \end{aligned}$$

On the other hand, since X'_n is also a refinement of X , we have (Theorem VIII.5.9)

$$\overline{\sigma}(f, X'_n) \leq \overline{\sigma}(f, X).$$

Combining the inequalities gives

$$\begin{aligned} \overline{\int_a^b} f(x) dx &\leq \overline{\sigma}(f, X_n) \leq \overline{\sigma}(f, X'_n) + 2M(K - 1) h_{X_n} \\ &\leq \overline{\sigma}(f, X) + 2M(K - 1) h_{X_n} \\ &< \overline{\int_a^b} f(x) dx + \frac{\varepsilon}{2} + 2M(K - 1) h_{X_n}. \end{aligned}$$

Since $h_{X_n} \rightarrow 0$ as $n \rightarrow \infty$ and since K is independent of n , we have here $2M(K-1)h_{X_n} < \varepsilon/2$ from some index N onward, whence

$$\int_a^b f(x) dx \leq \overline{\sigma}(f, X_n) < \int_a^b f(x) dx + \varepsilon, \quad n > N.$$

Here $\varepsilon > 0$ was arbitrary and $N \in \mathbb{N}$, so by the definition of convergence of a sequence the part of the assertion concerning the upper sums is proved. The subassertion concerning the lower sums is proved in a similar way. $\square$

As an example of applying Theorems VIII.5.8 and VIII.5.10, let us prove

THEOREM VIII.5.11 (Integrability on a subinterval) If a function f defined and bounded on $[a, b]$ is Riemann integrable on that interval, then f is Riemann integrable also on every subinterval $[c, d] \subset [a, b]$.

Proof Let $\{X_n\}$ be a sequence of partitions of the interval such that $h_{X_n} \rightarrow 0$ and $c, d \in X_n \forall n$. Then for every n , X_n contains a partition X'_n of the interval $[c, d]$ with $h_{X'_n} \leq h_{X_n} \rightarrow 0$. Writing $X_n = \{x_k, k = 0 \dots n\}$ and $X'_n = \{x_k, k = l-1 \dots m\}$, we have

$$\begin{aligned} \overline{\sigma}(f, X_n) - \underline{\sigma}(f, X_n) &= \sum_{k=1}^n (\overline{f}_k - \underline{f}_k)(x_k - x_{k-1}) \geq \sum_{k=l}^m (\overline{f}_k - \underline{f}_k)(x_k - x_{k-1}) \\ &= \overline{\sigma}(f, X'_n) - \underline{\sigma}(f, X'_n) \geq 0. \end{aligned}$$

Since here, by the assumptions and Theorems VIII.5.8 and VIII.5.10,

$$\overline{\sigma}(f, X_n) - \underline{\sigma}(f, X_n) \rightarrow \int_a^b f(x) dx - \int_a^b f(x) dx = 0,$$

we have $\overline{\sigma}(f, X'_n) - \underline{\sigma}(f, X'_n) \rightarrow 0$ (Proposition 1.7.4 [V2]). Since also $h_{X'_n} \rightarrow 0$, it follows by Theorem VIII.5.10 that the upper and lower integrals of f on $[c, d]$ are equal, i.e. f is Riemann integrable on that interval (Theorem VIII.5.8). $\square$

Proof of Theorem VIII.5.8. $\Rightarrow$ Assume that f is Riemann integrable on $[a, b]$ in the sense of Definition VIII.5.3. Let $\{X_n\}$ be a sequence of partitions of $[a, b]$ with $h_{X_n} \rightarrow 0$. Fix an index n and write $X_n = \{x_k, k = 0 \dots K\}$. Now for each $k = 1 \dots K$ we can choose $\xi_k, \xi'_k \in [x_{k-1}, x_k]$ such that

$$\begin{aligned} 0 \leq \overline{f}_k - f(\xi_k) &< 2^{-n}, & \overline{f}_k &= \sup_{x \in [x_{k-1}, x_k]} f(x), \\ 0 \leq f(\xi'_k) - \underline{f}_k &< 2^{-n}, & \underline{f}_k &= \inf_{x \in [x_{k-1}, x_k]} f(x). \end{aligned}$$

Then

$$\begin{aligned} 0 &\leq \bar{\sigma}(f, X_n) - \sigma(f, \Xi_n, X_n) < 2^{-n}(b-a), & \Xi_n &= \{\xi_k\}, \\ 0 &\leq \sigma(f, \Xi'_n, X_n) - \underline{\sigma}(f, X_n) < 2^{-n}(b-a), & \Xi'_n &= \{\xi'_k\}. \end{aligned}$$

Since here $\sigma(f, \Xi_n, X_n) \rightarrow A \in \mathbb{R}$ and $\sigma(f, \Xi'_n, X_n) \rightarrow A$ by assumption, it follows that also $\bar{\sigma}(f, X_n) \rightarrow A$ and $\underline{\sigma}(f, X_n) \rightarrow A$. By Theorem VIII.5.10, A is then the common value of the upper and lower integrals of f .

$\boxed{\Leftarrow}$ Assume that the upper and lower integrals of f on $[a, b]$ have a common value A . Then if $\{X_n\}$ is a sequence of partitions of $[a, b]$ and $h_{X_n} \rightarrow 0$, then by Theorem VIII.5.10 $\bar{\sigma}(f, X_n) \rightarrow A$ and $\underline{\sigma}(f, X_n) \rightarrow A$. Then also $\sigma(f, \Xi_{X_n}, X_n) \rightarrow A \forall \Xi_{X_n}$, since $\underline{\sigma}(f, X_n) \leq \sigma(f, \Xi_{X_n}, X_n) \leq \bar{\sigma}(f, X_n)$ (Proposition I.7.4 [V2]). Thus f is integrable in the sense of Definition VIII.5.3. $\square$

EXERCISES

1. a) Let $f(x) = x^2$ when $x \in \mathbb{Z}$, and $f(x) = 0$ when $x \in \mathbb{R}$, $x \notin \mathbb{Z}$. Show that f is integrable on every interval $[a, b] \subset \mathbb{R}$ and that $\int_a^b f(x) dx = 0$.
 b) Let $f(x) = 1$ when $x \in \{2^{-k}, k = 0, 1, \dots\}$ and $f(x) = 0$ otherwise. By computing the upper sums $\bar{\sigma}(f, X_n)$, where X_n is the equally spaced partition of $[0, 1]$ into 2^n subintervals, show that $\int_0^1 f(x) dx = 0$.
2. a) Prove: If f is integrable on $[-a, a]$, then

$$\begin{aligned} f \text{ even} &\Rightarrow \int_{-a}^a f(x) dx = 2 \int_0^a f(x) dx, \\ f \text{ odd} &\Rightarrow \int_{-a}^a f(x) dx = 0. \end{aligned}$$

- b) Prove: If f is L -periodic and integrable on $[0, L]$, then f is integrable on $[a, a+L]$ for every $a \in \mathbb{R}$ and

$$\int_a^{a+L} f(x) dx = \int_0^L f(x) dx.$$

3. Let $f(x) = \begin{cases} 1/x, & \text{when } x > 0 \\ 0, & \text{when } x = 0 \end{cases}$

- a) Verify that f is defined on $[0, 1]$ but not bounded.
- b) Show that f is not Riemann integrable on $[0, 1]$ by constructing a sequence of Riemann sums for which $h_{X_n} \rightarrow 0$ and $\sigma(X_n, \Xi_{X_n}) \rightarrow \infty$.

4. Show that if f is integrable on every interval $[a, b] \subset \mathbb{R}$ and the integral over every interval = 0, then $f(x) = 0$ at every point where f is either left- or right-continuous.
5. Using Proposition VIII.5.9, show that no Riemann lower sum can be greater than an upper sum, i.e. for all partitions X, X' of $[a, b]$ we have $\underline{\sigma}(f, X) \leq \overline{\sigma}(f, X')$.
6. (*) Show that Definition VIII.5.3 does not change if the partitions X_n of $[a, b]$ in the limit 'Lim' (Definition VIII.5.1) are restricted to be equally spaced. Can the choice of the sets of sample points Ξ_{X_n} also be restricted (e.g. $\Xi_{X_n} \subset X_n$) without changing the definition of Riemann integrability?
7. (*) Show that if f is defined on $[a, b]$ but is not bounded, then there exists a sequence of Riemann sums for which $h_{X_n} \rightarrow 0$ and $|\sigma(X_n, \Xi_{X_n})| \rightarrow \infty$ (cf. Exercise 3). — Which assumption of Definition VIII.5.3 is therefore unnecessary?
8. (*) Let X be a (finite or countable) set formed from the terms of a convergent sequence of real numbers, let f be defined and bounded on $[a, b]$ and let g be defined on $[a, b]$ so that $g(x) = f(x)$ when $x \notin X$. About the set $\{g(x) \mid x \in X \cap [a, b]\} \subset \mathbb{R}$ only that it is bounded is known. Show that if f is Riemann integrable on $[a, b]$, then so is g and $\int_a^b g(x) dx = \int_a^b f(x) dx$.
9. (*) a) Let f be defined and bounded on $[a, b]$ and $\mathcal{X}$ the set of partitions of $[a, b]$. To each $X = \{x_k, k = 0 \dots n\} \in \mathcal{X}$ associate the number

$$\delta(f, X) = \sum_{k=1}^n (\overline{f}_k - \underline{f}_k),$$

where $\overline{f}_k$ and $\underline{f}_k$ are the supremum and infimum of f on the subinterval $[x_{k-1}, x_k]$. Show that if the set $\{\delta(f, X) \mid X \in \mathcal{X}\}$ is bounded, then f is Riemann integrable on $[a, b]$.

b) Show that if f is defined and monotonic (increasing or decreasing) on $[a, b]$, then f is Riemann integrable on $[a, b]$.

10. (*) a) Show that if f is defined, bounded and Riemann integrable on $[a, b]$, then the following functions are also Riemann integrable on $[a, b]$:

$$f_+(x) = \max\{f(x), 0\}, \quad f_-(x) = \min\{f(x), 0\}.$$

b) Show that of the following assertions the first is true and the second false:

$$\begin{aligned} f \text{ Riemann integrable} &\Rightarrow |f| \text{ Riemann integrable,} \\ |f| \text{ Riemann integrable} &\Rightarrow f \text{ Riemann integrable.} \end{aligned}$$

c) Show that if f and g are defined, bounded and Riemann integrable on $[a, b]$, then so is the product fg . *Hint:* Write $f = f_+ + f_-$, $g = g_+ + g_-$ (see part a)).

VIII.6 The fundamental theorem of calculus

Let us return to the problem (P) considered in Section VIII.4, in which we sought an antiderivative $y(x)$ of a given function f on the interval (a, b) under the additional conditions that $y(x)$ is continuous on $[a, b]$ and $y(a) = 0$. The question of the solvability of the problem was at that time left open. We now show that a sufficient condition for solvability is that the function f is continuous on $[a, b]$. This is one of the most fundamental results of mathematical analysis, and it is indeed known as the *fundamental theorem of calculus**.

***THEOREM VIII.6.1 (The fundamental theorem of calculus)** If f is continuous on $[a, b]$, then

1. f is Riemann integrable on $[a, b]$.
2. The function $F(x) = \int_a^x f(t) dt$ is continuously differentiable on $[a, b]$ and $F'(x) = f(x)$ on that interval.

In what follows we prove the first (more demanding) subassertion of Theorem VIII.6.1 in a weakened form, by reinstating the Lipschitz assumption made in Section VIII.4

$$|f(x_1) - f(x_2)| \leq L|x_1 - x_2| \quad \forall x_1, x_2 \in [a, b].$$

THEOREM VIII.6.2 A function that is Lipschitz continuous on $[a, b]$ is Riemann integrable.

Proof Let $X_n = \{x_k, k = 0 \dots n\}$ be a partition of $[a, b]$. Since f is (at least) continuous on $[a, b]$, f attains its maximum and minimum values on the intervals $[x_{k-1}, x_k]$ (Theorem V.1.12), so that the Riemann upper and lower sums associated with the partition X_n (see the previous section) can be written

$$\overline{\sigma}(f, X_n) = \sum_{k=1}^n f(\xi_k)(x_k - x_{k-1}), \quad \underline{\sigma}(f, X_n) = \sum_{k=1}^n f(\eta_k)(x_k - x_{k-1}),$$

*The formulations of the theorem in the literature vary. — Often only the second subassertion of Theorem VIII.6.1 is presented as the fundamental theorem, in which case the theorem is fairly light in content, see Theorem VIII.6.3 below.

where $\xi_k, \eta_k \in [x_{k-1}, x_k]$. On the basis of this and the assumed Lipschitz continuity,

$$\begin{aligned} 0 \leq \overline{\sigma}(f, X_n) - \underline{\sigma}(f, X_n) &= \sum_{k=1}^n [f(\xi_k) - f(\eta_k)](x_k - x_{k-1}) \\ &\leq \sum_{k=1}^n L|\xi_k - \eta_k|(x_k - x_{k-1}) \\ &\leq \sum_{k=1}^n Lh_{X_n}(x_k - x_{k-1}) = L(b-a)h_{X_n}. \end{aligned}$$

We have deduced that if $\{X_n\}$ is a sequence of partitions with $h_{X_n} \rightarrow 0$, then

$$\overline{\sigma}(f, X_n) - \underline{\sigma}(f, X_n) = \mathcal{O}(h_{X_n}) \rightarrow 0.$$

The assertion follows from this and Theorems VIII.5.10 and VIII.5.8. $\square$

According to the first assertion of the fundamental theorem of calculus, Theorem VIII.6.2 is true also if f is merely continuous on $[a, b]$. The main idea of the proof is the same in this case too, but the proof must rely on a deeper logic of continuity, more precisely on uniform continuity (see Section V.9). Let us note here only that, on the basis of the mere continuity assumption, the final conclusion of the above proof construction becomes

$$\overline{\sigma}(f, X_n) - \underline{\sigma}(f, X_n) = o(1) \rightarrow 0, \quad \text{as } h_{X_n} \rightarrow 0.$$

The asserted integrability (the first assertion of the fundamental theorem of calculus) is thus also proved in this way.

In order to prove the second assertion of the fundamental theorem of calculus, we consider (with applications in mind, see the example below) a more general situation in which the function f is integrable on $[a, b]$ but not necessarily continuous on the whole interval $[a, b]$. In this case too the function $F(x) = \int_a^x f(t) dt$ is defined on the whole interval $[a, b]$ (Theorem VIII.5.11). The second assertion of the following theorem proves the second subassertion of Theorem VIII.6.1, so the fundamental theorem of calculus is at the same time completely proved.

THEOREM VIII.6.3 Let f be defined, bounded and Riemann integrable on $[a, b]$ and $F(x) = \int_a^x f(t) dt$, $x \in [a, b]$. Then:

1. If $|f(x)| \leq M \forall x \in [a, b]$, then F is Lipschitz continuous on $[a, b]$ with constant $L = M$.

2. If f has a right-hand limit $f(x^+)$ at a point $x \in [a, b)$, then F is right differentiable at x and $D_+F(x) = f(x^+)$. Correspondingly, if f has a left-hand limit $f(x^-)$ at a point $x \in (a, b]$, then F is left differentiable at x and $D_-F(x) = f(x^-)$.

Proof 1. If $-M \leq f(x) \leq M$, $x \in [a, b]$, then by the comparison principle for integrals (Theorem VIII.5.6) we have on every subinterval $[x_1, x_2] \subset [a, b]$ ($x_1 < x_2$)

$$-M(x_2 - x_1) = \int_{x_1}^{x_2} (-M) dt \leq \int_{x_1}^{x_2} f(t) dt \leq \int_{x_1}^{x_2} M dt = M(x_2 - x_1).$$

By the additivity of the integral (Theorem VIII.5.4) it follows that

$$|F(x_2) - F(x_1)| = \left| \int_{x_1}^{x_2} f(t) dt \right| \leq M|x_2 - x_1| \quad \forall x_1, x_2 \in [a, b] \Rightarrow \text{assertion 1.}$$

2. Let the right-hand limit $f(x^+)$ exist at a point $x \in [a, b)$ and let $\varepsilon > 0$. Then there exists $\delta \in (0, b-x]$ such that $|f(t) - f(x^+)| < \varepsilon$ when $t \in [x, x+\delta)$ (Theorem V.2.2). Then if $0 < \Delta x < \delta$, by the linearity, additivity and comparison principle of the integral,

$$\begin{aligned} \left| \frac{F(x + \Delta x) - F(x)}{\Delta x} - f(x^+) \right| &= \left| \frac{1}{\Delta x} \int_x^{x+\Delta x} [f(t) - f(x^+)] dt \right| \\ &\leq \frac{1}{\Delta x} \int_x^{x+\Delta x} |f(t) - f(x^+)| dt \leq \frac{1}{\Delta x} \int_x^{x+\Delta x} \varepsilon dt = \varepsilon. \end{aligned}$$

Since here $\varepsilon > 0$ was arbitrary and the inequality is valid for every $\Delta x \in (0, \delta)$, where $\delta > 0$, F is by definition right differentiable at x and $D_+F(x) = f(x^+)$. The second part of the assertion is proved similarly. $\square$

EXAMPLE 1 : Law of motion. If, in the rectilinear motion of a body (mass = m), a force $f(t)$ (t = time) acts in the direction of motion, then the law of conservation of momentum on the time interval $[0, \infty)$ is

$$mv(t) - mv(0) = \int_0^t f(t') dt',$$

where $v(t)$ = the velocity of the body at time t and the (Riemann) integral on the right-hand side is the impulse of the force f on the time interval $[0, t]$. If f is continuous at a point $t > 0$, then by Theorem VIII.6.3 the law of conservation of momentum yields (by differentiating both sides) the equation of motion

$$mv'(t) = f(t).$$

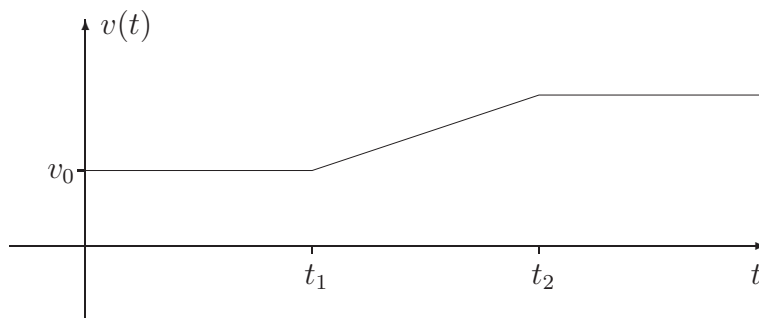
If f is discontinuous at a point t (physically possible!), then the equation of motion is (in general) not valid at that instant. The law of conservation of momentum, by contrast, retains its validity, requiring only that f be (Riemann) integrable on the time intervals $[0, t]$. The law of conservation of momentum can therefore be regarded as the ‘original’ physical law, and the equation of motion rather as a consequence of it under the additional assumption mentioned (f continuous at t). For example, if $v(0) = v_0$ and

$$f(t) = \begin{cases} F, & \text{when } t \in [t_1, t_2], \\ 0, & \text{otherwise,} \end{cases}$$

where $F \neq 0$ is a constant and $0 < t_1 < t_2$, then the (physically correct!) solution according to the law of conservation of momentum is

$$v(t) = \begin{cases} v_0, & \text{when } t \in [0, t_1], \\ v_0 + \frac{F}{m}(t - t_1), & \text{when } t \in [t_1, t_2], \\ v_0 + \frac{F}{m}(t_2 - t_1), & \text{when } t \in [t_2, \infty). \end{cases}$$

The equation of motion $mv'(t) = f(t)$ holds in this case on the intervals $(0, t_1)$, (t_1, t_2) and (t_2, ∞) . At the points t_1 and t_2 the velocity $v(t)$ is, by Theorem VIII.6.3, continuous (since f is bounded) and both left and right differentiable. Since f has a jump discontinuity at these points, the one-sided derivatives are unequal, and hence $v(t)$ is not differentiable at the points t_1 and t_2 . $\square$



The derivative of the function $\int_{g_1(x)}^{g_2(x)} f(t) dt$

By the additivity of the integral, the function in the heading can be expressed in the form

$F(g_2(x)) - F(g_1(x))$, where $F(x) = \int_a^x f(t) dt$. By the chain rule for differentiation and Theorem VIII.6.3, we arrive at the differentiation rule

$$\frac{d}{dx} \int_{g_1(x)}^{g_2(x)} f(t) dt = f(g_2(x))g_2'(x) - f(g_1(x))g_1'(x)$$

under the following assumptions:

- (i) $g_1(t)$ and $g_2(t)$ are differentiable at the point $t = x$.
- (ii) $f(t)$ is defined, bounded and Riemann integrable on some interval $[a, b]$, where $a < \min\{g_1(x), g_2(x)\}$ and $b > \max\{g_1(x), g_2(x)\}$.
- (iii) $f(t)$ is continuous at the points $g_1(x)$ and $g_2(x)$.

EXAMPLE 2 For every $x \neq 0$ (also when $x = 1$ (!)) we have

$$\frac{d}{dx} \int_x^{1/x} \frac{e^t}{t} dt = -\frac{1}{x^2} (x e^{1/x}) - \frac{e^x}{x} = -\frac{1}{x} (e^x + e^{1/x}),$$

since assumptions (i)–(iii) hold. For the same reason we have for every $x > 0$

$$\frac{d}{dx} \int_x^{x^2} \frac{e^t}{t} dt = 2x \cdot \frac{e^{x^2}}{x^2} - \frac{e^x}{x} = \frac{1}{x} (2e^{x^2} - e^x).$$

If, on the other hand, $x \leq 0$, then this computation is not valid, since then assumption (ii) (and in the case $x = 0$ also (iii)) does not hold. $\square$

Integration by parts and substitution in the definite integral

For the definite integral the following computational formulas hold under sufficient regularity assumptions, cf. the corresponding formulas for the indefinite integral in Section VIII.2.

1. Integration by parts formula

$$\int_a^b f'(x)g(x) dx = \left[f(x)g(x) - \int_a^b f(x)g'(x) dx \right]_a^b.$$

2. Substitution formula

$$\int_a^b f(x) dx = \int_\alpha^\beta f(u(t))u'(t) dt, \quad u(\alpha) = a, \quad u(\beta) = b.$$

The integration by parts formula is valid when f and g are continuously differentiable on $[a, b]$. Then $f'g$ and fg' are continuous on $[a, b]$, whence, by the fundamental theorem of calculus and the linearity of the integral,

$$\begin{aligned}\int_a^b f'(x)g(x) dx + \int_a^b f(x)g'(x) dx &= \int_a^b [f'(x)g(x) + f(x)g'(x)] dx \\ &= \int_a^b \frac{d}{dx} f(x)g(x) dx = \left| f(x)g(x) \right|_a^b.\end{aligned}$$

The substitution formula for the definite integral requires somewhat longer justification. Let, for example, $\alpha < \beta$ (the validity of the formula does not require this). Then one must assume

- (i) u is continuously differentiable on $[\alpha, \beta]$,
- (ii) f is continuous on $[A, B]$, for which $t \in [\alpha, \beta] \Rightarrow u(t) \in [A, B]$.

Setting

$$F(x) = \int_a^x f(s)ds, \quad x \in [A, B],$$

we have, by the fundamental theorem of calculus,

$$F'(x) = f(x), \quad x \in [A, B].$$

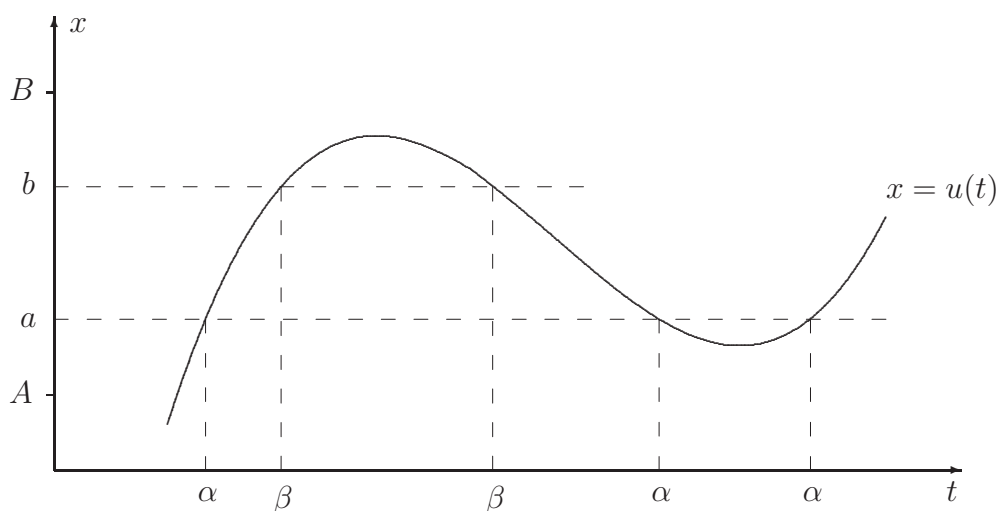
Then, by assumptions (i)-(ii),

$$\frac{d}{dt}F(u(t)) = f(u(t))u'(t), \quad t \in [\alpha, \beta],$$

so that

$$\begin{aligned}\int_\alpha^\beta f(u(t))u'(t) dt &= \left| F(u(t)) \right|_\alpha^\beta \\ &= F(u(\beta)) - F(u(\alpha)) \\ &= F(b) - F(a) = \int_a^b f(x) dx.\end{aligned}$$

In the case $\alpha > \beta$ one must speak of the interval $[\beta, \alpha]$, otherwise the justification is the same. Note that since injectivity of the function u was not required, α and β are not necessarily unique, cf. the figure.



EXAMPLE 3 When, in the integral

$$\int_0^2 \frac{e^{-x}}{1 + \sqrt{x}} dx$$

one makes the substitution

$$\sqrt{x} = t \geq 0 \Leftrightarrow x = u(t) = t^2,$$

then

$$dx = u'(t)dt = 2t dt$$

and

$$u(\alpha) = 0 \Rightarrow \alpha = 0, \quad u(\beta) = 2 \Rightarrow \beta = \sqrt{2},$$

so that the integral takes the form

$$\int_0^2 \frac{e^{-x}}{1 + \sqrt{x}} dx = \int_0^{\sqrt{2}} \frac{e^{-t^2}}{1 + t} 2t dt = 2 \int_0^{\sqrt{2}} \frac{te^{-t^2}}{1 + t} dt. \quad \square$$

In the transformed integral of the example (unlike in the original), the integrand is regular (smooth) on the whole interval of integration. Regularity of the integrand is generally an advantage when the value of the integral is computed by numerical methods, see Section [VIII.9](#) below.

The mean value theorem for integrals

The following theorem is the third and last in the series of mean value theorems, cf. Theorems [V.1.10](#) and [V.6.1](#). Compared with the earlier mean value theorems,

this theorem is not very independent, for it follows easily from the fundamental theorem of calculus and the mean value theorem of differential calculus (Exercise 8a).

THEOREM VIII.6.4 (Mean value theorem for integrals) If f is continuous on $[a, b]$, then for some $\xi \in (a, b)$

$$\int_a^b f(x) dx = f(\xi)(b - a).$$

The following more general (and also more independent) form of Theorem VIII.6.4 is called the *generalized* mean value theorem for integrals.

THEOREM VIII.6.5 If f is continuous on $[a, b]$, g is Riemann integrable on $[a, b]$ and either $g(x) \geq 0$ or $g(x) \leq 0$ on the whole interval $[a, b]$, then for some $\xi \in [a, b]$

$$\int_a^b f(x)g(x) dx = f(\xi) \int_a^b g(x) dx.$$

Proof If $g(x) \geq 0$ and $f_{\min}$ and $f_{\max}$ are the minimum and maximum values of f on $[a, b]$, then we conclude

$$\begin{aligned} f_{\min} g(x) &\leq f(x)g(x) \leq f_{\max} g(x), \quad x \in [a, b] \\ \Rightarrow f_{\min} \int_a^b g(x) dx &\leq \int_a^b f(x)g(x) dx \leq f_{\max} \int_a^b g(x) dx \\ \Rightarrow \int_a^b f(x)g(x) dx &= \eta \int_a^b g(x) dx, \quad \eta \in [f_{\min}, f_{\max}]. \end{aligned}$$

By the (first) mean value theorem (Theorem V.1.10), here $\eta = f(\xi)$ for some $\xi \in [a, b]$, so the assertion follows. In the case $g(t) \leq 0$ one can use the already proved assertion, with $-g$ in place of g . $\square$

The number

$$K(f) = \frac{1}{b-a} \int_a^b f(x) dx.$$

is called the *average value* of the function f on $[a, b]$. If $g(x) \geq 0 \forall x \in [a, b]$ and $\int_a^b g(x) dx > 0$, then the number

$$K_g(f) = \frac{\int_a^b f(x)g(x) dx}{\int_a^b g(x) dx}$$

is called the *weighted average* of f on $[a, b]$ and the function g is then called the *weight function*. By Theorem VIII.6.5, for a function continuous on $[a, b]$ we have $K_g(f) = f(\xi)$ for some $\xi \in [a, b]$.

The integral form of Taylor's theorem

In many applications (e.g. when estimating the error of numerical methods, see Exercise 11) the following alternative *integral form* of Taylor's theorem is very convenient — cf. Theorem VII.4.3, which is called the *mean value form* of Taylor's theorem.

THEOREM VIII.6.6 (Taylor's theorem – integral form) If f is $n + 1$ times continuously differentiable on $[x_0, x]$ ($x > x_0$) or $[x, x_0]$ ($x < x_0$) and $T_n(x, x_0)$ is the Taylor polynomial of f of degree n at the point x_0 , then

$$f(x) - T_n(x, x_0) = \int_{x_0}^x \frac{1}{n!} (x - t)^n f^{(n+1)}(t) dt.$$

Proof We start from the obvious identity

$$f(x) = f(x_0) + \int_{x_0}^x f'(t) dt,$$

which is the same as the assertion of the theorem in the case $n = 0$. If here f is twice continuously differentiable on the interval of integration (the assumption in the case $n = 1$), we can integrate by parts as follows:

$$\begin{aligned} f(x) &= f(x_0) - \int_{x_0}^x (x - t) f'(t) + \int_{x_0}^x (x - t) f''(t) dt \\ &= f(x_0) + f'(x_0)(x - x_0) + \int_0^x (x - t) f''(t) dt. \end{aligned}$$

This proves the assertion in the case $n = 1$. Continuing the integration by parts, one sees correspondingly that the assertion is true when $n = 2$, etc. (General proof: Exercise 8b.) $\square$

EXERCISES

1. If f is continuous on $[a, b]$, then what is the smallest value of the function

$$g(x) = \int_a^b [f(t) - x]^2 dt$$

and where is it attained?

2. Compute:

$$\begin{array}{lll} \text{a)} \frac{d}{dx} \int_2^x \frac{\sin t}{t} dt & \text{b)} \frac{d}{dt} \int_t^3 \frac{\sin x}{x} dx & \text{c)} \frac{d}{dx} \int_{x^2}^x \frac{\sin t}{t} dt \\ \text{d)} \frac{d}{dx} x^2 \int_0^{x^2} \frac{\sin u}{u} du & \text{e)} \frac{d}{dt} \int_{-t^{-1}}^t \frac{\cos y}{1+y^2} dy & \\ \text{f)} \frac{d}{d\theta} \int_{\sin \theta}^{\cos \theta} \frac{1}{1-x^2} dx & \text{g)} \frac{d}{dx} F(\sqrt{x}), \quad F(t) = \int_0^t \cos(x^2) dx & \\ \text{h)} H'(2), \quad H(x) = 3x \int_4^{x^2} e^{-\sqrt{t}} dt & & \end{array}$$

3. Solve for $y(x)$:

$$\text{a)} \quad y(x) = 1 - \int_0^x y(t) dt \quad \text{b)} \quad y(x) = \pi \left(1 + \int_1^x y(t) dt \right)$$

4. Compute by integrating by parts in parts a)–f) and by substitution in parts g)–l):

$$\begin{array}{lll} \text{a)} \int_0^1 x e^{-x} dx & \text{b)} \int_0^\pi x \cos x dx & \text{c)} \int_0^\pi x \sin x dx \\ \text{d)} \int_0^2 \sqrt{x} \ln x dx & \text{e)} \int_0^1 \operatorname{Arctan} x dx & \text{f)} \int_0^\pi e^{-x} \sin x dx \\ \text{g)} \int_{-2}^2 \frac{1}{2x^2 + 4x + 3} dx & \text{h)} \int_0^{\ln 2} \frac{e^x}{1+e^x} dx & \text{i)} \int_1^4 \frac{\sqrt{x}}{\sqrt[4]{x}+1} dx \\ \text{j)} \int_0^1 \sqrt{\frac{1-x}{1+x}} dx & \text{k)} \int_0^{\pi/4} \frac{1}{1+\tan x} dx & \text{l)} \int_0^\pi \frac{\sin x}{2-\cos x} dx \end{array}$$

5. Let $R > 0$ and $ab \neq 0$. Compute

$$\text{a)} \int_0^R x^2 \sqrt{R^2 - x^2} dx, \quad \text{b)} \int_0^{\pi/2} \frac{1}{a^2 \cos^2 x + b^2 \sin^2 x} dx.$$

6. Simplify the following function expressions using a substitution ($x > 0$).

$$\text{a)} \quad f(x) = \int_0^x \sqrt{t} \sqrt{x^2 - t^2} dx \quad \text{b)} \quad f(x) = \int_{-x/2}^{x/2} \frac{1}{(x^2 - t^2)^{3/2}} dt$$

7. Compute the a) average value, b) average value weighted by the function $g(x) = x$, of the function $f(x) = \sin^2 x$ on $[0, \pi]$.

8. a) Prove the mean value theorem for integrals by applying the mean value theorem of differential calculus to the function $F(x) = \int_a^x f(t) dt$. b) Prove Theorem VIII.6.6 by induction. c) Derive from Theorem VIII.6.6 the mean value form of Taylor's theorem (Theorem VII.4.3) by making use of Theorem VIII.6.5.
9. (*) Does the function $F(x) = \int_0^{2x-x^2} \cos\left(\frac{1}{1+t^2}\right) dt$ have a largest or smallest value? Justify your answer!
10. (*) Verify the *summation by parts* formula

$$\sum_{k=1}^n f_k g_k = \left| \begin{array}{l} k=n \\ k=1 \end{array} \right. F_k g_k - \sum_{k=1}^{n-1} F_k (g_{k+1} - g_k), \quad F_k = \sum_{i=1}^k f_i.$$

What is the corresponding formula for the definite integral?

11. (*) Let f be three times continuously differentiable on $[a-h, a+h]$. Using Theorem VIII.6.6, derive the error formula for the central difference approximation

$$f'(a) - \frac{f(a+h) - f(a-h)}{2h} = -\frac{1}{h} \int_{a-h}^{a+h} k(t) f'''(t) dt,$$

$$\text{where } k(t) = \begin{cases} \frac{1}{2}(a-h-t)^2, & \text{when } t \in [a-h, a], \\ \frac{1}{2}(a+h-t)^2, & \text{when } t \in [a, a+h]. \end{cases}$$

Derive from this the mean value form of the error formula (cf. Proposition VII.5.3)

$$f'(a) - \frac{f(a+h) - f(a-h)}{2h} = -\frac{1}{6} h^2 f'''(\xi), \quad \xi \in [a-h, a+h].$$

VIII.7 Extensions of the Riemann integral

A basic assumption in the definition of the Riemann integral is that the integrand f is both defined and bounded on the whole interval of integration, which must be a (finite) closed interval. Without these restrictions the integral cannot in general be defined as a limit of Riemann sums (or of upper and lower sums), i.e. the restrictions are necessary for f to be Riemann integrable. For many applications, however, the restrictions mentioned are unnecessarily strong or even artificial. For example, since it is known that the value of the integral does not depend on the values of f at a single (or at finitely many) point, it seems pointless to require at all that f be defined at every point of the interval of integration. To remove such unnecessary restrictions it has become customary to construct various extensions of the definition of the Riemann integral, which are called *improper* (literally ‘unsuitable’ or ‘contrary to good manners’) Riemann integrals. The extensions do not lead to a very clear-cut general definition of the integral; rather, their purpose is mainly to ‘patch up’ the original definition so that the most common cases arising in applications are covered. In what follows we present the most common of these extensions.*

If f is bounded and Riemann integrable on $[a, b]$, then by the Lipschitz continuity of the function

$$F(x) = \int_a^x f(t) dt$$

(Theorem VIII.6.3),

$$\int_{a+\varepsilon}^b f(t) dt = F(b) - F(a + \varepsilon) = \int_a^b f(t) dt + \mathcal{O}(\varepsilon),$$

when $0 < \varepsilon < b - a$, so that

$$\int_a^b f(x) dx = \lim_{\varepsilon \rightarrow 0^+} \int_{a+\varepsilon}^b f(x) dx.$$

The limit on the right does not depend on the values of f at the point a — in fact the limit does not depend even on whether f is defined at this point. On this basis it is quite natural to define the integral as that limit (if it exists) even when f is not defined at the point a and/or f is not bounded as this point is approached.

*The problems of the Riemann integral were noticed already in the second half of the nineteenth century, when the theory of measures and integrals developed vigorously. The definitive solution to the problem came with the definition presented in 1906 by the Frenchman **Henri Lebesgue** (1875-1941), which has since been known as the *Lebesgue integral*. The Lebesgue integral removes the blemishes of the Riemann integral, but its definition requires deeper measure-theoretic considerations.

EXAMPLE 1 If $f(x) = 1/\sqrt{x-a}$, then f is not bounded on $[a, b]$ (nor defined at the point a), but the above limit exists:

$$\begin{aligned} \int_{a+\varepsilon}^b \frac{1}{\sqrt{x-a}} dx &= \left|_{x=a+\varepsilon}^{x=b} 2\sqrt{x-a} \right. = 2\sqrt{b-a} - 2\sqrt{\varepsilon} \\ &\rightarrow 2\sqrt{b-a}, \quad \text{as } \varepsilon \rightarrow 0^+. \quad \square \end{aligned}$$

When the above limit procedure is applied at both endpoints of the interval of integration, we arrive at the following definition.

DEFINITION VIII.7.1 (**First extension of the Riemann integral**) Let $a < c < b$. If f is bounded and Riemann integrable on the intervals $[a + \varepsilon, c]$ and $[c, b - \varepsilon]$ for every $\varepsilon > 0$ ($\varepsilon < \min\{c - a, b - c\}$), then one defines

$$\int_a^b f(x) dx = \lim_{\varepsilon \rightarrow 0^+} \int_{a+\varepsilon}^c f(x) dx + \lim_{\varepsilon \rightarrow 0^+} \int_c^{b-\varepsilon} f(x) dx,$$

insofar as the limits on the right-hand side exist. One then says that f is integrable on $[a, b]$.

The definition can be presented slightly more briefly in the form

$$\int_a^b f(x) dx = \lim_{\varepsilon_1, \varepsilon_2 \rightarrow 0^+} \int_{a+\varepsilon_1}^{b-\varepsilon_2} f(x) dx,$$

where the limit notation on the right means that $\varepsilon_1 \rightarrow 0^+$ and $\varepsilon_2 \rightarrow 0^+$ independently of each other. If it is assumed that f has an antiderivative F on (a, b) , then, following the abbreviated notation, by Definition VIII.7.1

$$\int_a^b f(x) dx = \lim_{\varepsilon_1, \varepsilon_2 \rightarrow 0^+} \left|_{a+\varepsilon_1}^{b-\varepsilon_2} F(x) \right. = F(a^+) - F(b^-).$$

Thus in this case f is integrable exactly when the limits $F(a^+)$ and $F(b^-)$ exist, i.e. when F is continuous on $[a, b]$.

EXAMPLE 1 (continued) The function $f(x) = 1/\sqrt{x-a}$ has a continuous antiderivative $F(x) = 2\sqrt{x-a}$ on $[a, b]$. When computing the value of the integral, it is in practice safe (and accepted) to skip the limit process of Definition VIII.7.1, i.e. to compute directly

$$\int_a^b \frac{1}{\sqrt{x-a}} dx = \left|_{x=a}^{x=b} 2\sqrt{x-a} \right. = 2\sqrt{b-a}. \quad \square$$

Since it is natural to regard additivity of the integral as a property also of the (improper) integrals extended in the above manner, the following extension is obtained immediately.

DEFINITION VIII.7.2 (Second extension of the Riemann integral) If $a = c_0 < c_1 < \dots < c_n = b$ and f is integrable on the intervals $[c_{k-1}, c_k]$, $k = 1 \dots n$, then f is integrable on $[a, b]$ and

$$\int_a^b f(x) dx = \sum_{k=1}^n \int_{c_{k-1}}^{c_k} f(x) dx.$$

According to this definition, for example, a function piecewise continuous on $[a, b]$ is integrable on that interval. According to the definition, the value of the integral is then independent of how f is defined (or whether it is defined at all) at the points of discontinuity.

EXAMPLE 2 If $f(x) = k$ when $k-1 < x < k$, $k = 1 \dots n$ ($f(k)$ either defined or not, for $k = 0 \dots n$), then by Definition VIII.7.2

$$\int_0^n f(x) dx = \sum_{k=1}^n \int_{k-1}^k f(x) dx = \sum_{k=1}^n k = \frac{1}{2}n(n+1). \quad \square$$

Finally, we extend the concept of the integral to cover the infinite intervals $[a, \infty)$, $(-\infty, b]$ and $(-\infty, \infty)$:

DEFINITION VIII.7.3 (Third extension of the Riemann integral) If f is integrable on the intervals $[a, M]$ for every $M > a$ and the limit

$$A = \lim_{M \rightarrow \infty} \int_a^M f(x) dx,$$

exists, then f is said to be integrable on $[a, \infty)$, the number A is called the integral of f over that interval, and one writes

$$A = \int_a^\infty f(x) dx.$$

Correspondingly one defines

$$\begin{aligned} \int_{-\infty}^b f(x) dx &= \lim_{M \rightarrow \infty} \int_{-M}^b f(x) dx, \\ \int_{-\infty}^\infty f(x) dx &= \int_{-\infty}^c f(x) dx + \int_c^\infty f(x) dx, \quad c \in \mathbb{R}. \end{aligned}$$

EXAMPLE 3 The function $f(x) = 1/(1 + x^2)$ is integrable on $(-\infty, \infty)$, for by Definition VIII.7.3

$$\begin{aligned} \int_{-\infty}^{\infty} \frac{1}{1+x^2} &= \int_{-\infty}^0 \frac{1}{1+x^2} dx + \int_0^{\infty} \frac{1}{1+x^2} dx \\ &= \lim_{M \rightarrow \infty} \int_{-M}^0 \frac{1}{1+x^2} dx + \lim_{M \rightarrow \infty} \int_0^M \frac{1}{1+x^2} dx \\ &= \lim_{M \rightarrow \infty} \left[\text{Arctan } x \right]_{-M}^0 + \lim_{M \rightarrow \infty} \left[\text{Arctan } x \right]_0^M = \frac{\pi}{2} + \frac{\pi}{2} = \pi. \quad \square \end{aligned}$$

If f is integrable on the interval under consideration (finite or infinite) either in the ordinary or in the extended sense, then one says that the integral of f over that interval *converges*. If f is not integrable, then one says that the integral *diverges*. As in the example, the convergence of an integral is easily determined by means of the antiderivative, insofar as one can be found on the interval of integration. Indeed, by Definition VIII.7.3, on the infinite intervals of integration $[a, \infty)$, $(-\infty, b]$ and $(-\infty, \infty)$ too it holds that: if the antiderivative of f on (a, b) is F , then the value of the integral $\int_a^b f(x) dx$ (if it exists) is

$$\int_a^b f(x) dx = \lim_{x \rightarrow b^+} F(x) - \lim_{x \rightarrow a^-} F(x),$$

i.e. the integral converges exactly when both limits on the right-hand side exist. If no antiderivative valid on the whole interval of integration can be found, the interval can always be split into a finite number of subintervals, the additivity of the integral used, and the convergence of the integral studied separately on each subinterval.

EXAMPLE 4 ‘Cutting corners’ somewhat in the limits, one can compute

$$\begin{aligned} \int_{-\infty}^{\infty} \frac{1}{1+x^2} dx &= \left[\text{Arctan } x \right]_{-\infty}^{\infty} = \frac{\pi}{2} - \left(-\frac{\pi}{2} \right) = \pi \quad (\text{converges}), \\ \int_0^1 \frac{1}{x} &= \left[\ln x \right]_0^1 = \infty \quad (\text{diverges}). \end{aligned}$$

The following computation, by contrast, is careless:

$$\int_{-\infty}^{\infty} \frac{1}{x^2} dx = \left[-\frac{1}{x} \right]_{-\infty}^{\infty} = 0,$$

for it went unnoticed that $F(x) = -1/x$ is not an antiderivative of the function $f(x) = 1/x^2$ on the whole interval $(-\infty, \infty)$. By splitting the integral into two parts over the intervals $(-\infty, 0]$ and $[0, \infty)$ one sees that it diverges. $\square$

Comparison of improper integrals

When the convergence of an improper integral cannot be studied directly by means of the antiderivative, or when doing so is difficult, convergence can usually be determined by comparing the integral with a simpler, computable integral. The comparison starts from the already familiar comparison principle (Theorem VIII.5.6)

$$f(x) \leq g(x) \quad \forall x \in [a, b] \Rightarrow \int_a^b f(x) dx \leq \int_a^b g(x) dx.$$

In applying this to improper integrals on a finite interval $[a, b]$ it suffices to assume that $f(x) \leq g(x)$ holds on the open interval (a, b) (Definition VIII.7.1) or on the open subintervals $(c_{k-1}, c_k) \subset (a, b)$, $k = 1 \dots n$ (Definition VIII.7.2), for since the comparison principle is valid for ordinary Riemann integrals, it remains valid also for the limits in the definitions. In comparing the integrals $\int_a^\infty f(x) dx$ and $\int_a^\infty g(x) dx$ (Definition VIII.7.3) it correspondingly suffices to assume that $f(x) \leq g(x)$ on the intervals (c_{k-1}, c_k) , $k = 1, 2, \dots$ where $a = c_0 < c_1 < \dots$ and $\lim_k c_k = \infty$.

Consider as an example the integral $\int_a^\infty f(x) dx$. (The integral $\int_{-\infty}^b f(x) dx$ reduces to this by the change of variable $t = -x$.) For functions that do not change sign on the interval of integration (except possibly at isolated points, which do not affect the value of the integral), one can use the following comparison principles in convergence studies of integrals, cf. the corresponding principles for positive-term series (Theorem I.12.1). For simplicity we assume here the functions defined and the inequalities valid on the whole interval $[a, \infty)$.

THEOREM VIII.7.4 (Majorant and minorant principles for integrals)

If f and g are defined on $[a, \infty)$, integrable on $[a, M]$ for every $M > a$ and $0 \leq f(x) \leq g(x) \quad \forall x \in [a, \infty)$, then

$$\begin{aligned} \int_a^\infty g(x) dx \text{ converges} &\Rightarrow \int_a^\infty f(x) dx \text{ converges,} \\ \int_a^\infty f(x) dx \text{ diverges} &\Rightarrow \int_a^\infty g(x) dx \text{ diverges.} \end{aligned}$$

Proof The assertions are logically equivalent, so it suffices to prove the first. Assume, then, that $\int_a^\infty g(x) dx$ converges. Since $f(x) \geq 0$ on $[a, \infty)$, the function $F(x) = \int_a^x f(t) dt$ is increasing on that interval:

$$F(x_2) - F(x_1) = \int_{x_1}^{x_2} f(t) dt \geq 0, \quad \text{when } a \leq x_1 < x_2.$$

For the same reason $G(x) = \int_a^x g(t) dt$ is increasing on $[a, \infty)$, and since $f(x) \leq g(x)$, $x \in [a, \infty)$ and the integral $\int_a^\infty g(t) dt$ converges, we can conclude:

$$F(x) \leq G(x) \leq \lim_{x \rightarrow \infty} G(x) = \int_a^\infty g(t) dt, \quad x \in [a, \infty).$$

Thus $F(x)$ is both increasing and bounded on $[0, \infty)$. By Theorem V.2.8 the limit $\lim_{x \rightarrow \infty} F(x) = \int_a^\infty f(x) dx$ then exists. $\square$

EXAMPLE 5 By Theorem VIII.7.4 the integral $\int_1^\infty e^{-x}/x dx$ converges, for on $[1, \infty)$ we have $e^{-x}/x \leq e^{-x}$ and the integral $\int_0^\infty e^{-x} dx$ converges:

$$\int_1^\infty e^{-x} dx = \left| -e^{-x} \right|_1^\infty = 1. \quad \square$$

The majorant and minorant principles of Theorem VIII.7.4 generalize under suitable assumptions to cover other improper integrals as well. For example, assuming that $0 \leq f(x) \leq g(x) \forall x \in (a, b)$, for the integrals in the sense of Definition VIII.7.1 we have

$$\int_a^b g(x) dx \text{ converges} \quad \Rightarrow \quad \int_a^b f(x) dx \text{ converges}.$$

EXAMPLE 6 The integral $\int_0^1 e^{-x}/x dx$ can be studied by the change of variable $t = 1/x$, whereupon Theorem VIII.7.4 applies. A more direct argument, however, suffices: The integral diverges, since $e^{-x}/x \geq e^{-1}x^{-1} > 0$, $x \in (0, 1]$ and $\int_0^1 x^{-1} dx$ diverges. $\square$

Cauchy's criterion for integrals

The following general convergence criterion works for all improper integrals of the form $\int_a^\infty f(x) dx$, cf. the corresponding Cauchy criterion for series (Theorem I.12.4).

THEOREM VIII.7.5 (Cauchy's criterion for integrals) Let f be integrable on $[a, M]$ for every $M > a$. Then the integral $\int_a^\infty f(x) dx$ converges exactly when for every $\varepsilon > 0$ there exists $M > a$ such that

$$\left| \int_{M_1}^{M_2} f(x) dx \right| < \varepsilon, \quad \text{when } M_1, M_2 > M.$$

Proof $\Rightarrow$ If the integral converges and $M_1, M_2 > a$, then

$$\begin{aligned} \int_{M_1}^{M_2} f(x) dx &= \int_a^{M_2} f(x) dx - \int_a^{M_1} f(x) dx \\ &\rightarrow \int_a^\infty f(x) dx - \int_a^\infty f(x) dx = 0, \quad \text{as } M_1, M_2 \rightarrow \infty, \end{aligned}$$

so Cauchy's criterion is satisfied.

$\Leftarrow$ Assume that Cauchy's criterion is satisfied and set

$$A_n = \int_a^n f(x) dx, \quad n \in \mathbb{N}, \quad n > a.$$

By assumption

$$A_n - A_m = \int_m^n f(x) dx \rightarrow 0, \quad \text{as } n, m \rightarrow \infty,$$

so $\{A_n\}$ is a Cauchy sequence, and hence (Theorem I.10.5)

$$A_n \rightarrow A \in \mathbb{R}, \quad \text{as } n \rightarrow \infty.$$

Using this result and appealing again to the assumption, we find that if $\varepsilon > 0$, then by choosing $M \in \mathbb{R}$ and $n \in \mathbb{N}$ large enough

$$\begin{aligned} \left| \int_a^M f(x) dx - A \right| &\leq \left| \int_a^M f(x) dx - A_n \right| + |A_n - A| \\ &= \left| \int_n^M f(x) dx \right| + |A_n - A| < \varepsilon. \end{aligned}$$

Thus f is integrable on $[a, \infty)$:

$$\lim_{M \rightarrow \infty} \int_a^M f(x) dx = A. \quad \square$$

EXAMPLE 7 Does the integral $\int_0^\infty \frac{\cos x}{\sqrt{x}} dx$ converge or diverge?

Solution Split the integral into two parts:

$$\int_0^\infty \frac{\cos x}{\sqrt{x}} dx = \int_0^{\pi/2} \frac{\cos x}{\sqrt{x}} dx + \int_{\pi/2}^\infty \frac{\cos x}{\sqrt{x}} dx.$$

Integrating by parts in the first part, we find

$$\int_0^{\pi/2} \frac{\cos x}{\sqrt{x}} dx = \left| 2\sqrt{x} \cos x + \int_0^{\pi/2} 2\sqrt{x} \sin x dx \right|_0^{\pi/2} = 2 \int_0^{\pi/2} \sqrt{x} \sin x dx.$$

The function $f(x) = \sqrt{x} \sin x$ is continuous on $[0, \pi/2]$, so this part of the integral converges. In the remaining part of the integral we integrate by parts the other way:

$$\int_{\pi/2}^M \frac{1}{\sqrt{x}} \cos x dx = \left| \frac{1}{\sqrt{x}} \sin x + \frac{1}{2} \int_{\pi/2}^M x^{-3/2} \sin x dx \right|_{\pi/2}^M.$$

Here

$$\left| \frac{1}{\sqrt{x}} \sin x \right|_{\pi/2}^M \rightarrow -\sqrt{\frac{2}{\pi}}, \quad \text{as } M \rightarrow \infty,$$

so the question of convergence shifts to the integral

$$\int_{\pi/2}^\infty x^{-3/2} \sin x dx.$$

Estimating by means of the triangle inequality (Section VIII.4, inequality (8)), we find that if $\pi/2 < M_1 < M_2$, then

$$\begin{aligned} \left| \int_{M_1}^{M_2} x^{-3/2} \sin x dx \right| &\leq \int_{M_1}^{M_2} x^{-3/2} |\sin x| dx \\ &\leq \int_{M_1}^{M_2} x^{-3/2} dx \\ &= -2 \left| x^{-1/2} \right|_{M_1}^{M_2} \rightarrow 0, \quad \text{as } M_1, M_2 \rightarrow \infty, \end{aligned}$$

so by Cauchy's criterion this part of the integral converges as well. Thus the answer is: It converges! $\square$

Integrals and series

We have already seen above that the convergence theories of integrals $\int_a^\infty f(x) dx$ and of series resemble each other. From a more computational standpoint too, the connection between integrals and series is significant, for it turns out that integrals can be used both to simplify convergence studies of series and to speed up the numerical computation of the sums of series.

Consider as an example the series $\sum_{k=1}^\infty f(k)$, where it is assumed that f is defined on $[1, \infty)$ and that in addition, for some $n \in \mathbb{N}$:

- (i) f is decreasing on $[n, \infty)$ and $\lim_{x \rightarrow \infty} f(x) = 0$.

From assumption (i) it follows that $f(x) \geq 0$ on $[n, \infty)$ and for every $k \geq n$ we have

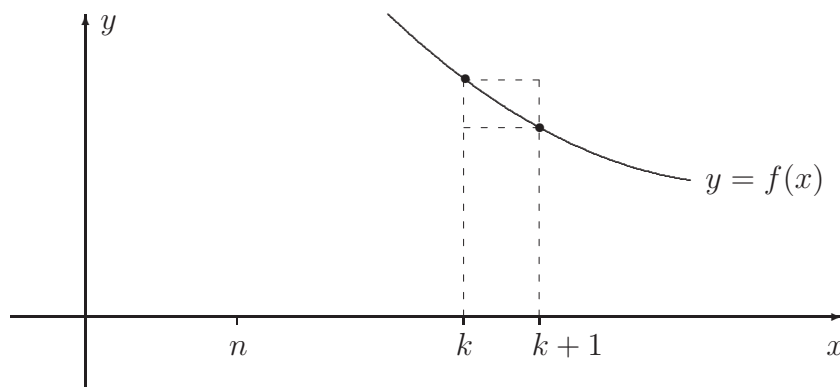
$$f(k+1) \leq f(x) \leq f(k), \quad \text{when } x \in [k, k+1].$$

Thus

$$\int_k^{k+1} f(k+1) dx \leq \int_k^{k+1} f(x) dx \leq \int_k^{k+1} f(k) dx,$$

i.e.

$$f(k+1) \leq \int_k^{k+1} f(x) dx \leq f(k).$$



Summing the above result over the indices $k = n \dots N-1$ ($N > n$) and using the additivity of the integral gives

$$\sum_{k=n+1}^N f(k) \leq \int_n^N f(x) dx \leq \sum_{k=n}^{N-1} f(k) = \sum_{k=n+1}^{N-1} f(k) + f(n).$$

Since here $\lim_n f(n) = 0$ by assumption (i), then, by the Cauchy criteria for series and integrals (Theorems I.12.4 and VIII.7.5), we conclude that the series $\sum_{k=1}^{\infty} f(k)$ converges exactly when the integral $\int_n^{\infty} f(x) dx$ converges. Moreover, in the limit $N \rightarrow \infty$ we obtain the estimates

$$\begin{aligned} \sum_{k=n+1}^{\infty} f(k) &\leq \int_n^{\infty} f(x) dx \leq \sum_{k=n+1}^{\infty} f(k) + f(n) \\ \Leftrightarrow \int_n^{\infty} f(x) dx - f(n) &\leq \sum_{k=n+1}^{\infty} f(k) \leq \int_n^{\infty} f(x) dx. \end{aligned}$$

From this follows a result that is interesting for the computation of the sum of the series:

$$\begin{aligned} \sum_{k=1}^{\infty} f(k) &= \sum_{k=1}^n f(k) + \int_n^{\infty} f(x) dx - \varepsilon_n \\ &= \sum_{k=1}^{n-1} f(k) + \int_n^{\infty} f(x) dx + \delta_n, \end{aligned}$$

where $0 \leq \varepsilon_n \leq f(n)$ and $0 \leq \delta_n \leq f(n)$ ($\delta_n = f(n) - \varepsilon_n$). This means that when the ‘tail’ of the series (starting from index n or $n+1$) is computed as an integral, then of the approximations obtained in this way

$$\sum_{k=1}^{\infty} f(k) \approx \sum_{k=1}^{n-1} f(k) + \int_n^{\infty} f(x) dx \quad (\text{a})$$

$$\approx \sum_{k=1}^n f(k) + \int_n^{\infty} f(x) dx \quad (\text{b})$$

(a) gives a lower approximation and (b) an upper approximation for the sum of the series, and the error of each is at most $f(n)$ = the n th term of the series (= the difference of the approximations). The result is thus valid provided that the above assumption (i) holds.

EXAMPLE 8 For which values of α ($\alpha \in \mathbb{R}$) does the series $\sum_{k=1}^{\infty} 1/k^\alpha$ converge?

Solution The series undoubtedly diverges if $\alpha \leq 0$. If $\alpha > 0$, then assumption (i) holds with $n = 1$. The integral $\int_1^{\infty} x^{-\alpha} dx$ converges exactly when $\alpha > 1$, so the series converges under exactly the same condition. $\square$

The question in the example was already solved earlier (Theorem I.12.2) but more laboriously. Besides simplifying the convergence study, comparison with an integral often also facilitates the numerical computation of the sum of the series.

EXAMPLE 8 (continued) For the sum of the convergent (super-harmonic) series in the example, by approximation (b) we have

$$\sum_{k=1}^{\infty} \frac{1}{k^{\alpha}} \approx \sum_{k=1}^n \frac{1}{k^{\alpha}} + \frac{1}{\alpha-1} n^{1-\alpha}, \quad \alpha > 1.$$

This is an upper approximation, which can be compared with mere truncation of the series, i.e. the lower approximation

$$\sum_{k=1}^{\infty} \frac{1}{k^{\alpha}} \approx \sum_{k=1}^n \frac{1}{k^{\alpha}}.$$

The error of the former is at most $n^{-\alpha}$, that of the latter $n^{1-\alpha}/(\alpha-1) + \mathcal{O}(n^{-\alpha})$. E.g. if $\alpha = 5/4$, the difference in accuracy is in practice dramatic, see Exercise 7. $\square$

Let us note that in the example an even more efficient algorithmic alternative is the average of approximations (a) and (b), i.e.

$$\sum_{k=1}^{\infty} \frac{1}{k^{\alpha}} \approx \sum_{k=1}^n \frac{1}{k^{\alpha}} + \frac{1}{\alpha-1} n^{1-\alpha} - \frac{1}{2} n^{-\alpha}.$$

Its error is of order of magnitude $\mathcal{O}(n^{-1-\alpha})$ (see Exercise VIII.9:11).

EXERCISES

1. Compute or show to be divergent in parts a)–o). In parts p)–ö), determine the values of k ($k \in \mathbb{R}$) for which the integral converges.

$$\begin{array}{lll}
 \text{a)} \int_0^6 \frac{2x}{x^2-4} dx & \text{b)} \int_0^6 \frac{2x}{|x^2-4|^{2/3}} dx & \text{c)} \int_0^1 \frac{x^5}{(1-x^2)^{3/2}} dx \\
 \text{d)} \int_0^1 x^{-4/5} \ln x dx & \text{e)} \int_0^{\pi/2} \tan x dx & \text{f)} \int_0^\pi \frac{x}{\cos^2 x} dx \\
 \text{g)} \int_0^2 \frac{x^2}{\sqrt{2x-x^2}} dx & \text{h)} \int_{-\infty}^\infty \frac{1}{x^2+7x+12} dx & \text{i)} \int_0^\infty \frac{1}{x^3+1} dx \\
 \text{j)} \int_{-\infty}^\infty \frac{1}{x^2+7x+13} dx & \text{k)} \int_0^\infty \frac{1}{x^3-1} dx & \text{l)} \int_1^\infty \frac{1+\sqrt{x}}{x^2+x} dx \\
 \text{m)} \int_1^\infty \frac{1+\sqrt{x}}{x\sqrt{x}+2x} dx & \text{n)} \int_0^\infty \frac{1}{\sqrt{e^x-1}} dx & \text{o)} \int_{-\infty}^\infty e^{-|x|} \cos x dx \\
 \text{p)} \int_0^1 \frac{1+x+x^2}{x^k} dx & \text{q)} \int_1^\infty \frac{1+x+x^2}{x^k} dx & \text{r)} \int_0^\infty \frac{1}{kx^2+1} dx \\
 \text{s)} \int_0^1 \frac{x^{k-1}+x^{-k}}{1+x} dx & \text{t)} \int_0^{\pi/2} \sin^k x dx & \text{u)} \int_0^\infty x^k e^{-x} dx \\
 \text{v)} \int_0^\infty x^k \sin x dx & \text{x)} \int_0^\infty \frac{|k|^x}{\sqrt{e^x+1}} dx & \text{y)} \int_0^\infty x^k \ln x dx \\
 \text{z)} \int_0^1 \frac{e^x}{(1-x^2)^k} dx & \text{ä)} \int_0^\infty |x^2-1|^k e^{-x} dx & \text{b)} \int_{-\infty}^\infty \frac{\ln|x|}{|x^3+1|^k} dx \\
 \text{ö)} \int_{-\infty}^\infty \left[\sqrt[4]{|\sin x \cos x|} e^{|x|} (\ln|x|)^8 \right]^k dx
 \end{array}$$

2. Derive a reduction formula for the following integrals, where $n \in \mathbb{N} \cup \{0\}$.

$$\text{a)} \int_0^\infty x^n e^{-x} dx \quad \text{b)} \int_0^\infty \frac{1}{(x^2+1)^{n+1}} dx \quad \text{c)} \int_0^1 (\ln x)^n dx$$

3. Prove convergence using the substitution $t = 1/x$:

$$\text{a)} \int_0^1 e^{-1/x} dx \quad \text{b)} \int_0^1 \cos \frac{1}{x} dx \quad \text{c)} \int_0^\infty \frac{1}{x} \sin \frac{1}{x} dx$$

4. Simplify and sketch the graph:

$$f(x) = \int_0^\pi \frac{x \sin t}{\sqrt{1+2x \cos t + x^2}} dx.$$

5. The *Gamma function* (Γ -function) is defined by

$$\Gamma(x) = \int_0^\infty t^{x-1} e^{-t} dt.$$

Verify the following assertions and, on their basis, sketch the behavior of the Γ -function. (In part e), $\Gamma(1/2) = \sqrt{\pi} \approx 1.77$, see Proposition [XIII.4.1.](#))

- a) Γ is defined on $(0, \infty)$.
 - b) $\Gamma(x+1) = x\Gamma(x) \quad \forall x > 0$.
 - c) $\Gamma(n) = (n-1)! \quad \forall n \in \mathbb{N}$.
 - d) $\lim_{x \rightarrow 0^+} \Gamma(x) = \lim_{x \rightarrow \infty} \Gamma(x) = \infty$.
 - e) $\Gamma(1/2) = 2\Gamma(3/2) = \int_{-\infty}^\infty e^{-x^2} dx$.
6. a) One wishes to determine for which values of α ($\alpha \in \mathbb{R}$) the series

$$\sum_{k=2}^\infty \frac{1}{k(\ln k)^\alpha}$$

converges. Solve the question by comparison with an integral.

- b) Let $n \in \mathbb{N}$ be a large number. Using an integral, justify the estimates

$$\sum_{k=1}^n k^\alpha = \begin{cases} \frac{1}{\alpha+1} n^{\alpha+1} + \mathcal{O}(n^\alpha), & \text{when } \alpha > -1, \\ \ln n + \mathcal{O}(n^{-1}), & \text{when } \alpha = -1. \end{cases}$$

What is the remainder term $\mathcal{O}(n^\alpha)$ exactly in the cases $\alpha = 0$ and $\alpha = 1$?

7. A computer is available that computes the sum $s_n = \sum_{k=1}^n k^{-5/4}$ in time $10^{-10}n$ seconds. Using the sums s_n , one wishes to compute the limit $s = \lim_n s_n$ to six decimal places (error $< 5 \cdot 10^{-7}$). Estimate the computation time
- a) in billions of years using the approximation $s \approx s_n$,
 - b) in microseconds using the approximation $s \approx s_n + 4/\sqrt[4]{n}$.
8. (*) *Euler's constant* $\gamma = 0.5772156649\dots$ is defined by

$$\gamma = \lim_{n \rightarrow \infty} \left(\sum_{k=1}^n \frac{1}{k} - \ln n \right) = \sum_{k=1}^\infty a_k, \quad a_k = \frac{1}{k} - \ln \left(1 + \frac{1}{k} \right).$$

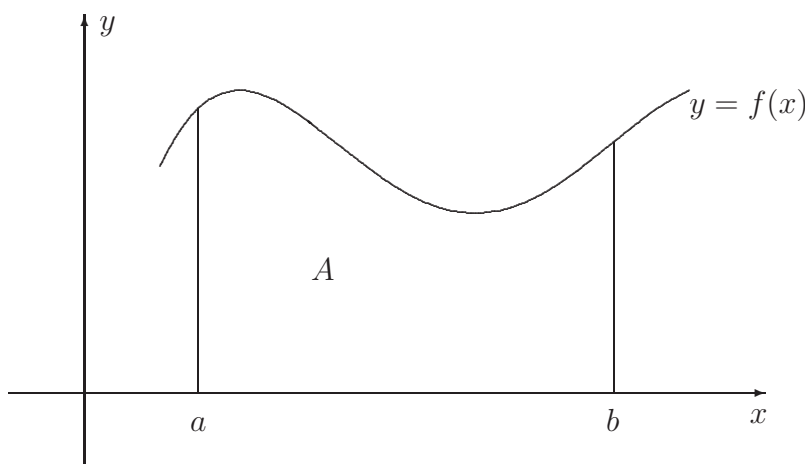
- a) Justify the latter formula and the convergence of the series in question.
 - b) Show that for large $n \in \mathbb{N}$ we have $\sum_{k=1}^n a_k + \frac{1}{2n} = \gamma + \mathcal{O}(n^{-2})$.
9. (*) Estimate the number $n!$ both from above and from below by comparing the number $\ln(n!) = \ln 1 + \ln 2 + \dots + \ln n$ with an integral. How many digits does $1000!$ have (in base ten)?

VIII.8 Area and arc length

In this section we consider the use of the definite integral in two problems of plane geometry: determining the area of a *plane region* (a subset of $\mathbb{R}^2$ or E^2) bounded by a curved line, and computing the arc length of a simple curve arc.

Let $f(x) \geq 0$ on $[a, b]$. Consider the subset of $\mathbb{R}^2$ bounded by the curve $y = f(x)$ and the lines $y = 0$, $x = a$ and $x = b$,

$$A = \{ (x, y) \in \mathbb{R}^2 \mid x \in [a, b] \text{ \& } 0 \leq y \leq f(x) \}.$$



We wish to determine the area of A . This is a non-negative real number, denoted henceforth by the symbol $\mu(A)$. — This is in fact a function of the type $\mu : \mathcal{M} \rightarrow [0, \infty)$, where $\mathcal{M}$ consists of subsets of $\mathbb{R}^2$, more precisely of the so-called *measurable* sets, whose area is definable. The number $\mu(A)$ is accordingly called also the (area) *measure* of A *. Considering for the time being only plane regions of the assumed form, we impose on the area measure μ the following two axioms:

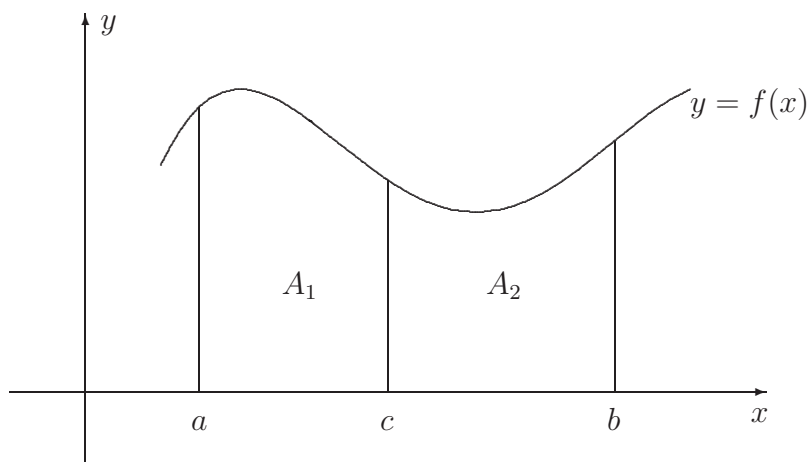
A1. *Comparison principle*: If $0 \leq m \leq f(x) \leq M \forall x \in [a, b]$, then

$$m(b - a) \leq \mu(A) \leq M(b - a).$$

*The concepts ‘measure’ and ‘measurability’ refer to a branch of mathematics called *measure theory*. The basic ideas of the more general measure theory are not considered in this connection; the matter will be taken up later in the context of multivariable analysis.

A2. *Additivity*: If $A_1 = \{(x, y) \in A \mid x \in [a, c]\}$ and $A_2 = \{(x, y) \in A \mid x \in [c, b]\}$, where $a < c < b$, then A is measurable exactly when A_1 and A_2 are both measurable, and

$$\mu(A) = \mu(A_1) + \mu(A_2).$$



Assume now that A is such that the area measure $\mu(A)$ according to axioms A1–A2 is definable, and let us investigate what follows from this. First, by applying the additivity axiom A2 repeatedly, we find that if $X = \{x_k, k = 0 \dots n\}$ is a partition of $[a, b]$ (i.e. $a = x_0 < x_1 < \dots < x_n = b$), and we set

$$A_k = \{(x, y) \in A \mid x_{k-1} \leq x \leq x_k\}, \quad k = 1 \dots n,$$

then each A_k is measurable and

$$\mu(A) = \sum_{k=1}^n \mu(A_k).$$

By the comparison principle A1 we have here

$$\begin{aligned} m_k &\leq f(x) \leq M_k \quad \forall x \in [x_{k-1}, x_k] \\ \Rightarrow \quad m_k(x_k - x_{k-1}) &\leq \mu(A_k) \leq M_k(x_k - x_{k-1}) \\ \Rightarrow \quad \mu(A_k) &= \eta_k(x_k - x_{k-1}), \quad \eta_k \in [m_k, M_k]. \end{aligned}$$

If f is continuous on $[a, b]$, then by Weierstrass's theorem (Theorem V.1.12) we can here choose

$$m_k = \min_{x \in [x_{k-1}, x_k]} \{f(x)\}, \quad M_k = \max_{x \in [x_{k-1}, x_k]} \{f(x)\},$$

whereupon, by the first mean value theorem (Theorem V.1.10), there exists $\xi_k \in [x_{k-1}, x_k]$ such that $f(\xi_k) = \eta_k$.

Combining the above conclusions, we find that if f is continuous on $[a, b]$, then, associated with each partition X of $[a, b]$, $\mu(A)$ is expressible as a Riemann sum associated with that partition:

$$\mu(A) = \sum_{k=1}^n f(\xi_k)(x_k - x_{k-1}), \quad \xi_k \in [x_{k-1}, x_k], \quad k = 1 \dots n.$$

Since, by the fundamental theorem of calculus and the definition of the Riemann integral, the sums on the right-hand side of this formula approach the definite integral $\int_a^b f(x) dx$ as the partition becomes finer, regardless of the choice of the points ξ_k , it follows that $\mu(A)$ must be defined by the formula

$$\boxed{\mu(A) = \int_a^b f(x) dx.} \quad (1)$$

For the measure so defined, axioms A1–A2 also hold, by the corresponding properties of the integral (Theorems VIII.5.6 and VIII.5.4). It has thus been concluded that if f is continuous on $[a, b]$, then A is measurable in the sense of axioms A1–A2 if (and only if!) the measure is defined by the formula (1).

EXAMPLE 1 The area of the *parabolic segment*

$$A = \{ (x, y) \in \mathbb{R}^2 \mid x \in [0, 2] \ \& \ 0 \leq y \leq 2x - x^2 \}$$

is, by formula (1),*

$$\mu(A) = \int_0^2 (2x - x^2) dx = \left| x^2 - \frac{1}{3}x^3 \right|_0^2 = \underline{\underline{\frac{4}{3}}}. \quad \square$$

The computational formula (1) follows also from merely assuming that f is bounded and Riemann integrable on $[a, b]$. Indeed, if X is a partition of $[a, b]$, then, by the above conclusions, $\mu(A)$ (if it exists) can be estimated by the Riemann upper and lower sums (see Section VIII.5):

$$\underline{\sigma}(f, X) \leq \mu(A) \leq \overline{\sigma}(f, X).$$

It then follows from Theorem VIII.5.8 that the measure must be defined by the formula (1), whereupon axioms A1–A2 also hold. We have arrived at the following result.

*The area of a parabolic segment was first computed by the most notable mathematician of antiquity, **Archimedes** (287–212 BC). Archimedes's method was based on polygons approximating the segment, see Exercise VIII.9:7.

THEOREM VIII.8.1 If f is defined, bounded and Riemann integrable on $[a, b]$ and $f(x) \geq 0$, $x \in [a, b]$, then the area

$$A = \{ (x, y) \in \mathbb{R}^2 \mid x \in [a, b] \ \& \ 0 \leq y \leq f(x) \}$$

$\mu(A)$ of the plane region is determined uniquely from axioms A1–A2 by the formula (1).

EXAMPLE 2 Compute the area of the quarter disk

$$A = \{ (x, y) \in \mathbb{R}^2 \mid x, y \geq 0 \ \& \ x^2 + y^2 \leq R^2 \}.$$

Solution Computing by formula (1) gives

$$\begin{aligned} \mu(A) &= \int_0^R \sqrt{R^2 - x^2} \, dx \\ &\quad [\text{substitution } x = R \sin t, \, dx = R \cos t \, dt] \\ &= \int_0^{\pi/2} R^2 \cos^2 t \, dt \\ &= \int_0^{\pi/2} \frac{1}{2} R^2 (t + \sin t \cos t) = \underline{\underline{\frac{1}{4} \pi R^2}}. \quad \square \end{aligned}$$

EXAMPLE 3 Determine $\mu(A)$ when

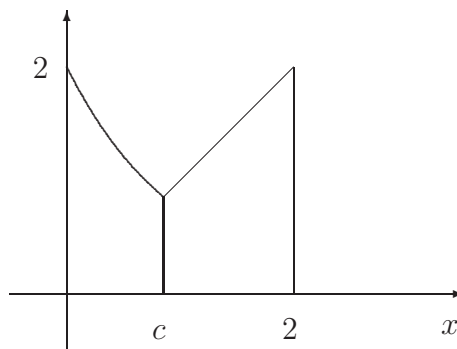
$$A = \{ (x, y) \in \mathbb{R}^2 \mid x \in [0, 2] \ \& \ 0 \leq y \leq \max\{x, 2e^{-x}\} \}.$$

Solution Here one must first solve

$$x = 2e^{-x} \Rightarrow x = c \approx 0.852606,$$

whereupon (cf. the figure)

$$f(x) = \begin{cases} 2e^{-x}, & \text{when } x \in [0, c], \\ x, & \text{when } x \in [c, 2]. \end{cases}$$

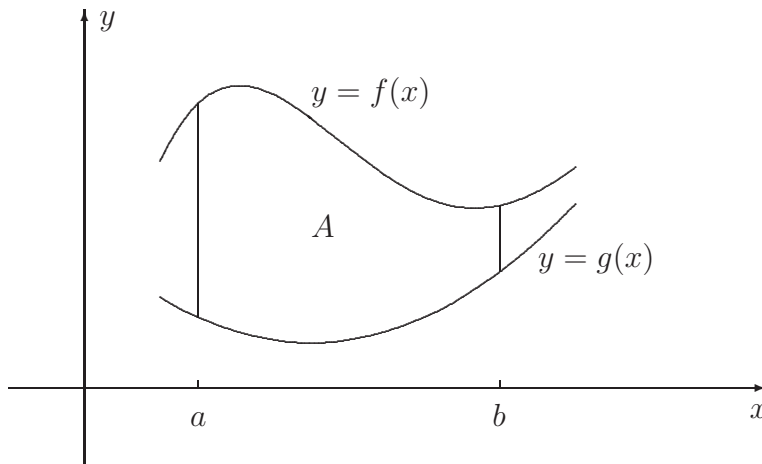


By formula (1) and the additivity of the integral we obtain

$$\begin{aligned}
 \mu(A) &= \int_0^2 f(x) dx = \int_0^c 2e^{-x} dx + \int_c^2 x dx \\
 &= \left[-2e^{-x} \right]_0^c + \left[\frac{1}{2}x^2 \right]_c^2 \\
 &= 4 - 2e^{-c} - \frac{1}{2}c^2 \approx \underline{\underline{2.78393}}. \quad \square
 \end{aligned}$$

By agreeing on further properties for the area measure, one can also compute, by means of integrals, the areas of more general plane regions. Let us consider here only, as an example, the area of the region between two curves. Let $0 \leq g(x) \leq f(x) \forall x \in [a, b]$. We wish to determine $\mu(A)$ when

$$A = \{(x, y) \in \mathbb{R}^2 \mid x \in [a, b] \ \& \ g(x) \leq y \leq f(x)\}.$$



Setting

$$\begin{aligned}
 A_1 &= \{(x, y) \in \mathbb{R}^2 \mid x \in [a, b] \ \& \ 0 \leq y \leq g(x)\}, \\
 A_2 &= \{(x, y) \in \mathbb{R}^2 \mid x \in [a, b] \ \& \ 0 \leq y \leq f(x)\},
 \end{aligned}$$

then, assuming the additivity law $\mu(A_2) = \mu(A_1) + \mu(A)$ (a new axiom for the area measure!), we obtain the computational formula

$$\mu(A) = \mu(A_2) - \mu(A_1) = \int_a^b [f(x) - g(x)] dx.$$

This can be considered valid whenever f and g are integrable on $[a, b]$ and $f(x) \geq g(x) \forall x \in [a, b]$. If the latter assumption does not hold, the formula is applied by putting $\max\{f, g\}$ in place of f and $\min\{f, g\}$ in place of g , whereupon the general computational formula for the area of the region between two curves on $[a, b]$ becomes

$$\mu(A) = \int_a^b |f(x) - g(x)| dx. \quad (2)$$

EXAMPLE 4 Compute the area of the region between the curves $y = x$ and $y = 2e^{-x}$ on $[0, 2]$.

Solution By formula (2) and the previous example,

$$\begin{aligned} \mu(A) &= \int_0^2 |2e^{-x} - x| dx = \int_0^c (2e^{-x} - x) dx + \int_c^2 (x - 2e^{-x}) dx \\ &= \left[-2e^{-x} - \frac{1}{2}x^2 \right]_0^c + \left[\frac{1}{2}x^2 + 2e^{-x} \right]_c^2 \\ &= 4 + 2e^{-2} - 4e^{-c} - c^2 \approx \underline{\underline{1.83852}}. \quad \square \end{aligned}$$

Arc length

According to the definition presented in Section V.4, the *arc length measure* $\mu(S)$ of the arc

$$S = \{ (x, y) \in \mathbb{R}^2 \mid x \in [a, b] \ \& \ y = f(x) \}$$

can be defined as the least upper bound of the sums associated with the partitions X of $[a, b]$

$$\begin{aligned} \sigma(f, X) &= \sum_{k=1}^n \sqrt{(x_k - x_{k-1})^2 + [f(x_k) - f(x_{k-1})]^2} \\ \mu(S) &= \sup_X \sigma(f, X), \end{aligned}$$

insofar as the set of sums is bounded. If it is assumed that f is continuous on $[a, b]$ and differentiable on (a, b) , then by the mean value theorem of differential calculus there exist $\xi_k \in (x_{k-1}, x_k)$, $k = 1 \dots n$ such that

$$f(x_k) - f(x_{k-1}) = f'(\xi_k)(x_k - x_{k-1}), \quad k = 1 \dots n,$$

whereupon

$$\sigma(f, X) = \sum_{k=1}^n \sqrt{1 + [f'(\xi_k)]^2} (x_k - x_{k-1}).$$

Under the assumptions mentioned, the sum $\sigma(f, X)$ is thus a Riemann sum associated with the partition X of $[a, b]$ and the function $g(x) = \sqrt{1 + [f'(x)]^2}$. We conclude that insofar as g is Riemann integrable on $[a, b]$ (by the fundamental theorem of calculus it suffices that f be continuously differentiable on $[a, b]$), the arc length measure is definable and the integral formula

$$\boxed{\mu(S) = \int_a^b \sqrt{1 + [f'(x)]^2} dx.} \quad (3)$$

holds.

EXAMPLE 5 Compute the arc length of the semicircular arc

$$S = \{ (x, y) \in \mathbb{R}^2 \mid -R \leq x \leq R \text{ \& } y = \sqrt{R^2 - x^2} \}.$$

Solution Here $f(x) = \sqrt{R^2 - x^2}$, so by formula (3)

$$\mu(S) = \int_{-R}^R \sqrt{1 + \frac{x^2}{R^2 - x^2}} dx = \int_{-R}^R \frac{R}{\sqrt{R^2 - x^2}} dx.$$

This is a convergent (improper) integral. With the substitution $x = R \cos t$, $t \in [0, \pi]$ we obtain

$$\mu(S) = \int_{\pi}^0 (-R) dt = \int_0^{\pi} R dt = \underline{\underline{\pi R}}. \quad \square$$

EXAMPLE 6 Compute the arc length of the curve $y = x^2$ on $[0, 1]$.

Solution By formula (3) and Example 2 of Section VIII.2 (substitution $2x = t$),

$$\begin{aligned} \mu(S) &= \int_0^1 \sqrt{1 + 4x^2} dx = \int_{t=0}^{t=2} \frac{1}{4} \left[t\sqrt{1 + t^2} + \ln(t + \sqrt{1 + t^2}) \right] \\ &= \frac{1}{2}\sqrt{5} + \frac{1}{4} \ln(2 + \sqrt{5}) \approx \underline{\underline{1.47894}}. \quad \square \end{aligned}$$

EXAMPLE 7 The arc length of the curve $y = \sin x$ on $[0, \pi]$ is

$$\mu(S) = \int_0^{\pi} \sqrt{1 + \cos^2 x} dx.$$

This is a so-called *elliptic integral*, which cannot be evaluated in closed form.

□

As the examples show, the arc length integral can, even in fairly simple cases, be difficult or even impossible from the standpoint of integration in closed form. The alternative in such a case is to compute the integral directly by numerical methods (see the next section).

EXERCISES

1. Compute the areas of the following plane regions (exactly or approximately).

- a) $0 \leq y \leq |\cos 3x|$, $x \in [0, \pi]$
- b) $0 \leq \min\{x, \cos x\}$, $x \in [0, \pi/2]$
- c) $\min\{\cos x, \sin 2x\} \leq y \leq \max\{\cos x, \sin 2x\}$, $x \in [0, \pi]$
- d) $|x| \leq y \leq \sqrt{x^2 + 1}$, $|x| \leq \sqrt{a^2 - 1}$, $a > 1$

2. Compute the area of the region between the following curves on the given interval (exactly or approximately).

- a) $y = \sinh x$, $y = \cosh x$, $[-\ln 2, \ln 2]$
- b) $y = \ln x$, $y = e^x$, $[1/2, 2]$
- c) $y = \cos x$, $y = x$, $[0, \pi]$
- d) $y = \sin x$, $y = x^2$, $[0, 1]$

3. Compute an approximate value for the area of the bounded plane region between the curves $y = e^x$ and $y = 3 - x^2$.

4. Show that the area of the region enclosed by the ellipse

$$S : \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \quad (a, b > 0)$$

$$\text{is } \mu(A) = \pi ab.$$

5. Compute the arc lengths of the following curves on the given interval.

- a) $y = 2\sqrt{x}$, $[0, 4]$
- b) $y = \ln x$, $[1, e]$
- c) $y = \ln \cos x$, $[0, \pi/4]$
- d) $y = (1 - x/3)\sqrt{x}$, $[0, 3]$

6. Show that the length of the half-arc of the ellipse $S : (x/a)^2 + (y/b)^2 = 1$ & $y \geq 0$ is given by the (elliptic) integral

$$\mu(S) = \int_{-\pi/2}^{\pi/2} \sqrt{a^2 \cos^2 t + b^2 \sin^2 t} dt.$$

7. (*) Compute the area of the plane region between the curves $y = e^{-|x|} \cos x$ and $y = e^{-|x|} \sin x$.

8. (*) Compute the areas of the following regions ($a > 0$):
- The region between the cycloid $x = a(t - \sin t)$, $y = a(1 - \cos t)$ and the x -axis on the interval $x \in [0, 2\pi a]$.
 - The region enclosed by the astroid $|x|^{2/3} + |y|^{2/3} = a^{2/3}$.
9. (*) Compute the arc lengths of the following arcs ($a > 0$):
- The cycloid arc $S: x = a(t - \sin t)$, $y = a(1 - \cos t)$, $t \in [0, 2\pi]$.
 - The astroid arc $S: |x|^{2/3} + |y|^{2/3} = a^{2/3}$, $x, y \geq 0$.
10. (*) Under gravity acting in the direction $-\vec{j}$, a chain hanging freely in the xy -plane follows a *catenary*

$$y = a \cosh \left(\frac{x - b}{a} \right) + c,$$

where $a > 0$ and $b, c \in \mathbb{R}$.

- Compute the constants a, b, c and sketch the behavior of the catenary, when the ends of the chain are at the points $(-2, 0)$ and $(2, 0)$ and the length of the chain $= 8$.
- If the ends of the chain are at the points $(-2, 4)$ and $(4, 4)$ and the chain passes through the origin, then what is the length of the chain?

VIII.9 Numerical integration

Numerical integration means the (approximate) computation of a definite integral, i.e. of the real number

$$I(f, a, b) = \int_a^b f(x) dx$$

by numerical means. In what follows, fairly strong regularity requirements are imposed on the integrand f . The minimum requirement is that f be continuous on $[a, b]$, in which case f is also Riemann integrable on that interval (Theorem VIII.6.1).

The methods of numerical integration are usually so-called *numerical quadratures* (or quadrature rules)*, in which the integral is approximated by a finite sum of the form

$$\int_a^b f(x) dx \approx \sum_{i=1}^N w_i f(x_i). \quad (1)$$

Here the points x_i , $i = 1 \dots N$ (usually $x_i \in [a, b]$) are called *quadrature points* and the numbers w_i *quadrature weights*. In choosing the quadrature points and weights, the interval of integration is often first divided into subintervals $[x_{k-1}, x_k]$, $k = 1 \dots n$ (as in the original definition of the definite integral). Then, by the additivity of the integral,

$$I(f, a, b) = \int_a^b f(x) dx = \sum_{k=1}^n \int_{x_{k-1}}^{x_k} f(x) dx = \sum_{k=1}^n I_k. \quad (2)$$

If the integrand is regular and the subintervals $[x_{k-1}, x_k]$ are short enough, the partial integrals

$$I_k = \int_{x_{k-1}}^{x_k} f(x) dx$$

can be computed approximately by using a relatively simple quadrature of the form (1), where $a = x_{k-1}$ and $b = x_k$. The basis for the simplification of the quadrature on a short interval is Taylor's theorem, according to which a regular function is approximately a (low-degree) polynomial on a short interval (see Section VII.4). The starting idea of the error analysis of numerical integration is indeed precisely comparison with polynomials.

*Quadrature means literally 'squaring'. The term refers to the connection between the integral and area.

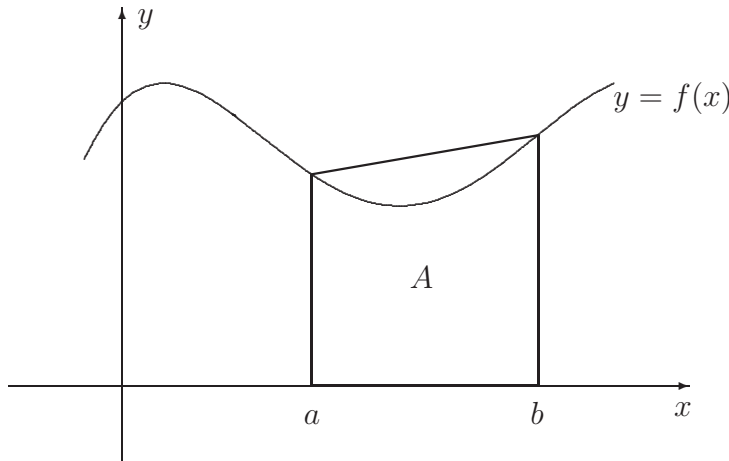
In what follows we regard the interval $[a, b]$ as a short subinterval of the original interval of integration and f as (sufficiently) regular on that interval. Under such starting assumptions, the simplest numerical quadratures are:

$$\text{Midpoint rule:} \quad \int_a^b f(x) dx \approx (b-a)f\left(\frac{a+b}{2}\right).$$

$$\text{Trapezoidal rule:} \quad \int_a^b f(x) dx \approx \frac{1}{2}(b-a)[f(a) + f(b)].$$

$$\text{Simpson's rule:} \quad \int_a^b f(x) dx \approx \frac{1}{6}(b-a) \left[f(a) + 4f\left(\frac{a+b}{2}\right) + f(b) \right].$$

Of these, Simpson's rule in particular (used on subintervals) is the method of numerical integration in general use by calculators and computers.* The trapezoidal rule is often also referred to by the name *Trapezoid*. As its name indicates, the rule gives as the value of the integral the area of a *trapezoid*:



When the above rules are used on subintervals in the decomposition (2), the result is an approximation of the original integral of the form (1).

One then says that this is a *composite* quadrature. For example, the *composite midpoint rule* is a quadrature of the form (1), where one chooses

$$x_i = \frac{1}{2}(x_{i-1} + x_i), \quad w_i = x_i - x_{i-1}, \quad i = 1 \dots n = N.$$

*Simpson's rule is a classic of numerical integration, which in its day was much used in hand computation as well. The rule was devised by the English mathematician **Thomas Simpson** (1710-1761).

The composite midpoint rule is thus one of the Riemann sums approximating the integral. Of the approximations based directly on Riemann sums, this is generally the most accurate (see the error estimates below).

If the partition in the decomposition (2) is equally spaced and in the corresponding composite quadrature (1) one sets $h = x_i - x_{i-1}$ (= the spacing of consecutive quadrature points) and $f_i = f(x_i)$, then the *composite trapezoid* takes the form

$$\int_a^b f(x) dx \approx h \left[\frac{1}{2}f_0 + f_1 + \cdots + f_{n-1} + \frac{1}{2}f_n \right]$$

and the *composite Simpson's rule* the form

$$\int_a^b f(x) dx \approx \frac{h}{3} [f_0 + 4f_1 + 2f_2 + \cdots + 2f_{2n-2} + 4f_{2n-1} + f_{2n}].$$

In the composite Simpson's rule, every other quadrature point is a partition point of the subinterval division and every other (equipped with the largest weight coefficient) is the midpoint of a subinterval.

EXAMPLE 1 When the integral

$$\int_0^1 x e^{-x^2} / (x+1) dx$$

is approximated using the equally spaced composite trapezoid and Simpson's rule, the following results are obtained:

points	Trapezoid	Simpson
3	0.17578506065833	0.20372346078015
21	0.20230210857337	0.20256800959472
201	0.20256528212086	0.20256794027440
2001	0.20256791368608	0.20256794026753

Here Simpson is clearly faster than the Trapezoid. This is to be expected, since the integrand is smooth, see the error estimates below. $\square$

EXAMPLE 2 The same comparison setup as in the previous example. We compute

$$\int_0^1 e^{-x} / (\sqrt{x} + 1) dx.$$

The results in this case are:

points	Trapezoid	Simpson
3	0.47363365	0.43418824
21	0.40741212	0.40604377
201	0.40520890	0.40516458
2001	0.40513820	0.40513679

The convergence is now clearly slower than in the previous example, and there is no significant difference between the two methods here. The reason is evidently that the integrand is less regular than in the previous example (continuous, but not continuously differentiable on the interval of integration). $\square$

In the above examples it is noteworthy that when the substitution $\sqrt{x} = t$ is made in the integral of Example 2, the result is the integral of Example 1 (with a factor of 2). The substitution is therefore worth making here, even though it does not otherwise (from the standpoint of integration in closed form) make the problem easier. By integration by parts, series expansions, etc., one can also often manipulate the problem so that the integrand becomes more regular, whereupon numerical quadratures work better.

On error estimates

The error of a composite quadrature based on a subinterval division is first estimated separately on each subinterval. In what follows we consider the three examples presented above: Midpoint rule, Trapezoid and Simpson.

In estimating the error on a single subinterval, the first task is to investigate for how high-degree polynomials the quadrature is exact, i.e. to determine the index m such that the quadrature is exact for a polynomial of degree m but not of degree $m + 1$. In the example cases mentioned, the result is as follows:

Rule	Exact for polynomials of degree
Midpoint rule	$m = 1$
Trapezoid	$m = 1$
Simpson	$m = 3$

Let us begin with the midpoint rule. Assume that the function f is twice continuously differentiable on the (sub)interval $[a, b]$ under consideration. Set $c = (a + b)/2$ and use the symbol $E(f)$ for the integration error:

$$E(f) = \int_a^b f(x) dx - (b - a)f(c).$$

The estimation of the integration error is based on two basic observations: First, one sees that

$$E(f + g) = E(f) + E(g). \quad (\text{a})$$

(More generally, $E(f)$ is linear with respect to f , i.e. $E(\alpha f + \beta g) = \alpha E(f) + \beta E(g)$, $\alpha, \beta \in \mathbb{R}$.) The second basic observation is based on the above table:

$$f(x) = A + Bx \quad (A, B \in \mathbb{R}) \quad \Rightarrow \quad E(f) = 0. \quad (\text{b})$$

Let $p(x)$ = the Taylor polynomial of f of degree $n = 1$ at the point c , i.e.

$$p(x) = f(c) + f'(c)(x - c).$$

Then, by observations (a)–(b), and since $(f - p)(c) = 0$,

$$E(f) = E(f - p + p) = E(f - p) + E(p) = E(f - p) = \int_a^b (f - p)(x) dx.$$

By Taylor's theorem (Theorem VII.4.3),

$$f(x) - p(x) = \frac{1}{2} f''(\xi)(x - c)^2 = f''(\xi) \omega(x), \quad x \in [a, b],$$

where $\xi = \xi(x) \in (a, b)$ and $\omega(x) = \frac{1}{2}(x - c)^2 \geq 0$. Thus if

$$M_1 = \min_{x \in [a, b]} f''(x), \quad M_2 = \max_{x \in [a, b]} f''(x),$$

then we can conclude

$$\begin{aligned} M_1 \omega(x) &\leq (f - p)(x) \leq M_2 \omega(x), \quad x \in [a, b] \\ \Rightarrow \quad M_1 \int_a^b \omega(x) dx &\leq \int_a^b (f - p)(x) dx \leq M_2 \int_a^b \omega(x) dx \\ \Leftrightarrow \quad \frac{1}{24} M_1 (b - a)^3 &\leq \int_a^b (f - p)(x) dx \leq \frac{1}{24} M_2 (b - a)^3. \end{aligned}$$

Thus for some $\eta \in [M_1, M_2]$

$$E(f) = \frac{\eta}{24} (b - a)^3.$$

Since f'' is continuous on $[a, b]$, we have $M_1 = f''(x_1)$ and $M_2 = f''(x_2)$ for some $x_1, x_2 \in [a, b]$ (Theorem V.1.12), whence $\eta = f''(\xi)$ for some $\xi \in [a, b]$ (Theorem V.1.10). For the midpoint rule the error formula

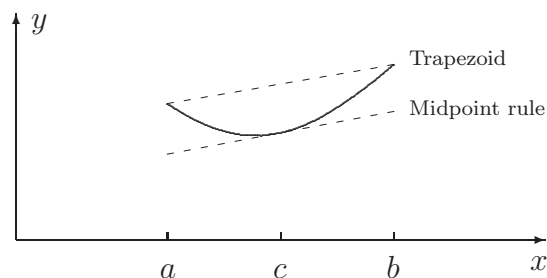
$$E(f) = \frac{1}{24} (b - a)^3 f''(\xi), \quad \xi \in [a, b] \quad (\text{Midpoint rule}).$$

has thus been derived.

For the Trapezoid an error formula can be derived in a similar manner. In this case, however, it is advisable to choose as the comparison polynomial p , instead of the Taylor polynomial, the first-degree interpolation polynomial, for which $p(a) = f(a)$ and $p(b) = f(b)$ (see Section VII.6). One obtains the error formula (Exercise 4)

$$E(f) = -\frac{1}{12} (b-a)^3 f''(\xi), \quad \xi \in [a, b] \quad (\text{Trapezoid}).$$

From the error formulas of the midpoint rule and the trapezoid one sees that if f is concave up ($f''(x) > 0$ on $[a, b]$), then the midpoint rule gives too small a value for the integral and the Trapezoid too large a value. In connection with the above error analysis, the phenomenon can be explained by the fact that the midpoint rule integrates correctly the Taylor polynomial $T_1(x, c)$ of f , whereas the Trapezoid integrates correctly the interpolation polynomial of f , cf. the figure.



For Simpson's rule too an error formula can be derived by the same type of reasoning as above. Assuming that f is four times continuously differentiable on $[a, b]$, one obtains the error formula (Exercise 8)

$$E(f) = -\frac{1}{90} h^5 f^{(4)}(\xi), \quad h = (b-a)/2, \quad \xi \in [a, b] \quad (\text{Simpson}).$$

From the above error formulas one can further derive error estimates for the composite quadratures. Consider again the midpoint rule as an example, restricting to an equally spaced partition, i.e. assuming the interval $[a, b]$ divided into subintervals $[x_{k-1}, x_k]$, $k = 1 \dots n$, where $x_k - x_{k-1} = h = (b-a)/n \forall k$. Let the function f be twice continuously differentiable on $[a, b]$, and let M_1 and M_2 be the minimum and maximum values of f'' on $[a, b]$. Then if $E_k(f)$ is the integration error of the midpoint rule on the subinterval $[x_{k-1}, x_k]$, then, by the above error formula,

$$\frac{1}{24} M_1 h^3 \leq E_k(f) \leq \frac{1}{24} M_2 h^3.$$

Summing over k and noting that $nh = b - a$, we obtain for the total error $E(f) = \sum_{k=1}^n E_k(f)$ the estimates

$$\frac{1}{24}(b-a)M_1h^2 \leq E(f) \leq \frac{1}{24}(b-a)M_2h^2.$$

Here $M_1 = f(x_1)$, $M_2 = f(x_2)$ for some $x_1, x_2 \in [a, b]$, so we arrive at a mean value form of the error formula as above. In the case of the equally spaced trapezoid and Simpson one proceeds correspondingly, whereupon the following error formulas result:

Equally spaced, composite midpoint rule:	$E(f) = \frac{1}{24}(b-a)f''(\xi)h^2$
Equally spaced, composite trapezoid:	$E(f) = -\frac{1}{12}(b-a)f''(\xi)h^2$
Equally spaced, composite Simpson:	$E(f) = -\frac{1}{180}(b-a)f^{(4)}(\xi)h^4$

Here $\xi \in [a, b]$ in all cases, and h is the spacing of the nearest quadrature points, i.e. either h = the length of the subinterval (Midpoint rule, Trapezoid), or h = half the length of the subinterval (Simpson).

In the case of a non-equally-spaced partition, the error of the above composite quadratures can be estimated in the form $|E(f)| \leq \dots$, where on the right-hand side, in place of $\pm f^{(k)}(\xi)$, one has the maximum value of $|f^{(k)}(x)|$ on $[a, b]$ and h is the largest spacing of consecutive quadrature points. In general, if the error of a composite numerical quadrature based on a subinterval division is of order of magnitude $\mathcal{O}(h^r)$ (with the integrand f sufficiently regular), but not of order $\mathcal{O}(h^{r+1})$ (however regular f may be), then one says that the *order of accuracy* of the method in question is r . As one can conclude from the above error analysis, in general $r = m + 1$ if the numerical quadrature integrates exactly, on the subintervals, a polynomial of degree m but not a polynomial of degree $m + 1$. The composite midpoint rule and trapezoid are thus second-order methods, and the order of the composite Simpson is $r = 4$. Examples of first-order methods are the Riemann sum formulas in which the quadrature points (i.e. the sample points ξ_k , cf. Section VIII.4) are not the midpoints of the subintervals (see Exercise 6). Of the approximations based on Riemann sums, the composite midpoint rule is thus in a class of its own for accuracy.

Application example: Adaptive Simpson

In calculators and numerical-symbolic computer software, behind the numerical integration commands (e.g. Mathematica: `NIntegrate`) there is often the com-

posite Simpson, and possibly also higher-order composite quadratures. In these programs the division into subintervals is not usually carried out with equal spacing, but *adaptively*, adjusting to the integrand. The idea of the adaptive Simpson method is to determine, by ‘sniffing’, how large the fourth derivative $f^{(4)}(x)$ of the integrand is on the subinterval under consideration $[x_{k-1}, x_k]$. The basic idea is simple: If the subinterval is short (as, in the course of the computation, it before long is), then one can assume that $f^{(4)}(x)$ is approximately constant $= M_k$ on that interval. Then the constant M_k can be found by interpreting the results given by the algorithm itself *a posteriori* (i.e. after the computation) as follows: First apply Simpson’s rule on $[x_{k-1}, x_k]$ – result I_1 . Then halve the interval and apply Simpson’s rule separately on each subinterval – result I_2 . If now the exact value of the integral $= I$ on $[x_{k-1}, x_k]$, then by Simpson’s error formula

$$\begin{cases} I - I_1 = -\frac{1}{90}M_k h^5, \\ I - I_2 = -\frac{1}{90}M_k \cdot 2 \cdot \left(\frac{h}{2}\right)^5 \end{cases} \Rightarrow M_k = 96h^{-5}(I_1 - I_2), \quad h = \frac{1}{2}(x_k - x_{k-1}).$$

In reality $f^{(4)}$ is not quite constant even on a short interval (if it were, the exact value of the integral I would also be found!), but the estimate obtained is generally sufficient for adaptive purposes: By means of it, the algorithm can be steered to refine the partition where the number M_k given by the computation is large in absolute value. The algorithm strives more precisely to refine the partition so that the error density, i.e. the integration error on a subinterval divided by the length of the subinterval, is approximately the same on every subinterval. (Such a partition can be shown to be optimal in terms of computational work.)

The adaptive algorithm works in practice surprisingly well also in most situations in which the function is not at all as regular as is assumed in the above computation (i.e. four times continuously differentiable). This happens, for instance, in the integration problem of Example 2, in which the integrand is not even continuously differentiable ($f(x) \approx 1 - \sqrt{x}$ near the point $x = 0$). By computing the numbers M_k in the manner described, the algorithm ends up refining the partition strongly near the origin and, thanks to this ‘machine intelligence’, is able to compute the value of the integral to the required accuracy almost as fast as in the situation of Example 1 (see Exercise 10c).

EXERCISES

1. The integral $\int_0^1 f(x) dx$ is computed by the equally spaced composite Trapezoid, dividing the interval of integration into n subintervals. Compute the error of the approximation thus obtained exactly, when a) $f(x) = x(1-x)$, b) $f(x) = e^x$.

2. For the following integrals, compute approximate values using the 11-point equally spaced composite Trapezoid (10 subintervals) and Simpson's rule (5 subintervals) and compare with the exact value.

$$\text{a) } \int_0^1 \frac{1}{1+x^2} dx \quad \text{b) } \int_0^1 \frac{4x^3}{1+x} dx \quad \text{c) } \int_0^1 \frac{1}{1+\sqrt[4]{x}} dx$$

3. The so-called *corrected trapezoidal formula* of numerical integration is

$$\int_a^b f(x) dx \approx \frac{1}{2}(b-a)[f(a) + f(b)] - \frac{1}{12}(b-a)^2[f'(b) - f'(a)].$$

Show that the formula is exact for polynomials of degree $m = 3$. What is the corresponding composite quadrature if the division into subintervals is equally spaced?

4. Let f be twice continuously differentiable on $[a, b]$ and $g = f - p$, where p is the first-degree (Lagrange) interpolation polynomial for which $p(a) = f(a)$ and $p(b) = f(b)$. Show by integration by parts that

$$-\frac{1}{2} \int_a^b (x-a)(b-x)g''(x) dx = \int_a^b g(x) dx.$$

Derive the Trapezoid error formula from this result.

5. a) Let p be the second-degree interpolation polynomial of the function f , defined by the conditions $p(a-h) = f(a-h)$, $p(a) = f(a)$ and $p(a+h) = f(a+h)$ on $[a-h, a+h]$. Show that Simpson's rule on that interval is the same as the approximation

$$\int_{a-h}^{a+h} f(x) dx \approx \int_{a-h}^{a+h} p(x) dx.$$

- b) Prove Simpson's error formula in the case where $f^{(4)}$ is constant on $[a-h, a+h]$ (i.e. f is a polynomial of degree 4).
6. Approximating the integral by a Riemann sum in which the sample points on the subintervals $[x_{k-1}, x_k]$ are chosen as $\xi_k = x_{k-1}$ corresponds to the approximation $\int_a^b f(x) dx \approx (b-a)f(a)$ on the subintervals. Derive for it the error formula

$$E(f) = \frac{1}{2}(b-a)^2 f'(\xi), \quad \xi \in [a, b].$$

What is the error formula on the whole interval if the division into subintervals is equally spaced?

7. (*) (Archimedes's algorithm) A line intersects the parabola $K : y = x^2$ at the points (a, a^2) and (b, b^2) , whereupon the line cuts off from the parabola a segment A on $[a, b]$ ($a < b$). Let A_n be the approximation of the area $\mu(A)$ of the segment obtained by dividing $[a, b]$ into 2^n subintervals of equal length ($n \in \mathbb{N}$) and using the composite trapezoid. Show, by algebraic means alone, that the sequence $\{A_n\}$ is a geometric series. Compute $\mu(A)$ ($= \lim_n A_n$) on this basis.
8. (*) a) (Cf. Problem 5a.) Let p be the generalized third-degree interpolation polynomial of the function f , defined by the conditions $p(a-h) = f(a-h)$, $p(a+h) = f(a+h)$, $p(a) = f(a)$ and $p'(a) = f'(a)$ on $[a-h, a+h]$. Show that Simpson's rule on that interval is the same as the approximation

$$\int_{a-h}^{a+h} f(x) dx \approx \int_{a-h}^{a+h} p(x) dx.$$

b) Derive Simpson's error formula. *Hint:* Theorem VII.6.4.

9. (*) Let f be four times continuously differentiable on $[a, b]$. Derive for the corrected trapezoidal formula (Exercise 3) the error formula

$$E(f) = \frac{1}{720} (b-a)^5 f^{(4)}(\xi), \quad \xi \in [a, b].$$

Hint: Hermite interpolation and Theorem VII.6.4!

10. (*) Consider the error of numerical integration when computing the integral $\int_0^1 \sqrt{x} dx$ by composite quadratures in which the interval $[0, 1]$ is divided in various ways into n subintervals.
- a) Show that if the partition is equally spaced, then the error of both the composite trapezoid and the composite Simpson is $\sim n^{-3/2}$ for large values of n .
- b) Choose as the partition points of $[0, 1]$ the points $x_k = (k/n)^{4/3}$, $k = 0 \dots n$ and use the composite trapezoid. Show that the error $= \mathcal{O}(n^{-2} \ln n)$.
- c) Choose as the partition points of $[0, 1]$ the points $x_k = (k/n)^{8/3}$, $k = 0 \dots n$ and use the composite Simpson. Show that the error $= \mathcal{O}(n^{-4} \ln n)$. (The adaptive Simpson algorithm ends up approximately at this partition for large values of n .)
11. (*) (Acceleration of a slow series) A convergent integral $\int_n^\infty f(x) dx$, where $n \in \mathbb{N}$ is a large number, can be computed approximately using either a) the composite trapezoid or b) the composite, corrected trapezoid (Exercise 3) in the division of the interval $[n, \infty)$ into subintervals $[k, k+1]$,

$k \in \mathbb{N}$, $k \geq n$. Starting from these ideas and the error estimates of the quadratures mentioned (see Exercise 9), derive the following results, when $f(x) = 1/x^\alpha$, $\alpha > 1$:

$$\text{a) } \sum_{k=1}^{\infty} \frac{1}{k^\alpha} = \sum_{k=1}^n \frac{1}{k^\alpha} + \frac{1}{\alpha-1} n^{1-\alpha} - \frac{1}{2} n^{-\alpha} + \mathcal{O}(n^{-\alpha-1}).$$

$$\text{b) } \sum_{k=1}^{\infty} \frac{1}{k^\alpha} = \sum_{k=1}^n \frac{1}{k^\alpha} + \frac{1}{\alpha-1} n^{1-\alpha} - \frac{1}{2} n^{-\alpha} + \frac{\alpha}{12} n^{-\alpha-1} + \mathcal{O}(n^{-\alpha-3}).$$

Hint: Note that

$$\text{a) } \sum_{k=n+1}^{\infty} f(k) = \sum_{k=n}^{\infty} \frac{1}{2} [f(k) + f(k+1)] - \frac{1}{2} f(n),$$

$$\begin{aligned} \text{b) } \sum_{k=n+1}^{\infty} f(k) &= \sum_{k=n}^{\infty} \left(\frac{1}{2} [f(k) + f(k+1)] - \frac{1}{12} [f'(k+1) - f'(k)] \right) \\ &\quad - \frac{1}{2} f(n) - \frac{1}{12} f'(n). \end{aligned}$$

Chapter IX

Differential equations

”The laws of nature are written in the language of mathematics”, declared the physicist and astronomer **Galileo Galilei** (1564-1642). Galileo’s still-timely dictum brings to mind especially *differential equations*, for nearly all laws of nature are of this kind. Apart from the laws of nature, differential equations are nowadays encountered in the most varied (human-made) mathematical models of physics, biology, economics, and so on. — The ‘matriarch’ of all such models may be taken to be the model of exponential growth or decay examined earlier in Sections [VI.2](#) and [VI.4](#).

In this chapter we first go through the basic concepts related to differential equations (Section [IX.1](#)), after which we examine the classical solution methods for differential equations and application examples related to them (Sections [IX.2](#)–[IX.6](#)). The traditional technique for solving differential equations is largely based on (and comparable to) the technique of integration covered in Sections [VIII.1](#)–[VIII.3](#), a branch of mathematical craftsmanship. For certain special types of differential equations the solutions can be found by an ‘educated guess’ alone; in other cases one tries to reduce the solving to ‘integrations’, that is, to *quadratures*.

Within the mathematical abundance of differential equations, those solvable by traditional methods may be regarded as rarities, but in applications such special cases are nonetheless fairly common. Sections [IX.2](#)–[IX.6](#) cover the most common of these special cases. Section [IX.7](#) examines *numerical* methods, which are more generally applicable than the traditional solution methods and with which the solutions of differential equations, and also of *systems of differential equations* formed by several equations, can be computed approximately. The main theorem of the theory of differential equations, the *Picard–Lindelöf theorem*, also rests on the ideas of approximate solving; it is presented in Section [IX.8](#).

IX.1 Basic concepts of differential equations

The general form of a *differential equation* (henceforth abbreviated DE), or more precisely of an *ordinary differential equation* (ODE), is

$$F(x, y, y', \dots, y^{(n)}) = 0,$$

where x is a real variable, $y = y(x)$ is an unknown real-valued function, and F is some (known) expression in $n + 2$ real variables (this is a function of type $F : \mathbb{R}^{n+2} \rightarrow \mathbb{R}$, cf. Section IV.3). When the equation is written more precisely in the form

$$F(x, y(x), y'(x), \dots, y^{(n)}(x)) = 0,$$

one sees more clearly that the equation has only one free variable (x). The name 'ordinary' DE comes precisely from this property. ('Non-ordinary' differential equations are not discussed for the time being.)

In the above differential equation the index $n \in \mathbb{N}$ is the *order* of the equation. If the equation is written in the form

$$y^{(n)} = f(x, y, \dots, y^{(n-1)}),$$

then this form is called the *normal form* of the differential equation. The normal form of a first-order differential equation is thus

$$y' = f(x, y).$$

The normal form is advantageous for the mathematical theory in that, on its basis, the solvability conditions of a differential equation can be stated much more easily than in the more general form.

Particular solution. General solution

If the order of a differential equation is n , then its solution, more precisely a *particular solution*, is any function $y(x)$ that on some open interval (a, b) (one may have $a = -\infty$ and/or $b = \infty$) is n times differentiable and satisfies the equation on that interval. By the *general solution* of a differential equation one means a set of functions that contains all (or 'almost all', see the remarks below) solutions. If the order of the equation is n , then the general solution is as a rule of the form

$$y = Y(x, C_1, \dots, C_n),$$

where $C_1, \dots, C_n$ are *constants* that can be chosen freely (or, e.g., freely from some intervals), and Y is some function of $n+1$ variables. Insofar as the constants can be chosen entirely freely, the general solution is thus the set of functions

$$\mathcal{Y} = \{ y(x) = Y(x, C_1, \dots, C_n) \mid C_1, \dots, C_n \in \mathbb{R} \}.$$

By 'solving' a differential equation one usually means determining the general solution.

EXAMPLE 1 The general solutions in $\mathbb{R}$ (on the interval $(-\infty, \infty)$) of the differential equations

$$\text{a) } y' = e^{2x}, \quad \text{b) } y' = 2y$$

are

$$\text{a) } y(x) = \frac{1}{2} e^{2x} + C \quad (C \in \mathbb{R}), \quad \text{b) } y(x) = C e^{2x} \quad (C \in \mathbb{R}). \quad \square$$

EXAMPLE 2 Solve: a) $y''' = 0$, b) $y''' + 2y'' = 0$.

Solution a) Integrating gives

$$\begin{aligned} y'''(x) = 0 &\Rightarrow y''(x) = C_1 \\ &\Rightarrow y' = C_1 x + C_2 \\ &\Rightarrow y(x) = \frac{1}{2} C_1 x^2 + C_2 x + C_3. \end{aligned}$$

Here one may just as well write $2C_1$ in place of C_1 , whereby the general solution takes the more natural form

$$y(x) = C_1 x^2 + C_2 x + C_3 \quad (C_1, C_2, C_3 \in \mathbb{R}).$$

b) The solution can here be found by first writing $y''(x) = u(x)$, whereby the equation simplifies to the form $u' + 2u = 0$. Its general solution is $u(x) = C_1 e^{-2x}$. Since $u = y''$, integrating yields

$$\begin{aligned} y''(x) = C_1 e^{-2x} &\Rightarrow y'(x) = \int C_1 e^{-2x} dx = -\frac{1}{2} C_1 e^{-2x} + C_2 \\ &\Rightarrow y(x) = \int y'(x) dx = \frac{1}{4} C_1 e^{-2x} + C_2 x + C_3. \end{aligned}$$

Writing $4C_1$ in place of C_1 gives the general solution the form

$$y(x) = C_1 e^{-2x} + C_2 x + C_3 \quad (C_1, C_2, C_3 \in \mathbb{R}). \quad \square$$

Singular solutions

If a differential equation has other solutions besides those contained in the general-solution expression $y = Y(x, C_1, \dots, C_n)$, then such 'surprise solutions' are called *singular solutions*. Whether a solution is 'singular' or not may depend on the form in which the general solution is presented.

EXAMPLE 3 The general solutions of the differential equations

$$\text{a) } y' = y^2, \quad \text{b) } (y')^2 = y$$

are

$$\text{a) } y(x) = \frac{1}{C - x}, \quad \text{b) } y(x) = \frac{1}{4}(x - C)^2,$$

as one may easily verify ($C \in \mathbb{R}$). The equations also have an obvious solution that is not obtained from the general-solution expression, namely

$$y(x) = 0.$$

This must therefore be regarded as a singular solution. In case a), however, the general solution can also be presented in the form

$$\text{a) } y(x) = \frac{C}{1 - Cx}$$

(in the earlier solution $1/C$ has been written in place of C), whereby $y(x) = 0$ is contained in it ($C = 0$). In case b) an even 'more singular' solution is

$$y(x) = \begin{cases} 0, & \text{if } x \leq C, \\ \frac{1}{4}(x - C)^2, & \text{if } x > C. \end{cases} \quad \square$$

The differential equation of a family of curves

The general solution of a differential equation can be interpreted geometrically as a *family of curves* (i.e. a set of curves). If a family of curves is known, then its differential equation can be derived by differentiation. Namely, if the functions of the family are of the form $y = Y(x, C_1, \dots, C_n)$, then differentiating n times gives a system of $n + 1$ equations, from which the constants $C_1, \dots, C_n$ can (at least in principle) be eliminated. This yields for that family of curves a differential equation of order n .

EXAMPLE 4 Of which differential equation is the general solution

$$\text{a) } y = (C + x)e^x, \quad \text{b) } y = \frac{C_1}{x + C_2} ?$$

Solution a) Differentiating once gives the system of equations

$$\begin{cases} y = Ce^x + xe^x \\ y' = Ce^x + xe^x + e^x \end{cases}$$

Subtraction gives the differential equation

$$y' - y = e^x.$$

b) Differentiate twice:

$$\begin{cases} y = C_1(x + C_2)^{-1} \\ y' = -C_1(x + C_2)^{-2} \\ y'' = 2C_1(x + C_2)^{-3} \end{cases}$$

Eliminating C_1 and C_2 gives the differential equation

$$yy'' = 2(y')^2. \quad \square$$

Orthogonal trajectories

If the differential equation of a one-parameter family of curves $y = Y(x, C)$ is

$$y' = f(x, y),$$

then by solving the differential equation

$$y' = -\frac{1}{f(x, y)}$$

one finds the *orthogonal trajectories* of the family of curves (cf. Example VII.2:2).

EXAMPLE 5 For the family of circles $x^2 + y^2 = C^2$, implicit differentiation gives the differential equation

$$x + yy' = 0.$$

The general solution of the differential equation of the orthogonal trajectories

$$x - y/y' = 0$$

is $y = Cx$, as one could (geometrically) have guessed. $\square$

Initial and boundary conditions

Since the solution of a differential equation contains undetermined constants, application situations require additional conditions in order for the solution to be unique. The additional conditions must be inferred from the application situation, i.e. they belong to the mathematical model just as the differential equation itself does.

If the order of a differential equation is n , then the solution generally has n undetermined constants, so n additional conditions are needed. The simplest way to impose the additional conditions is to fix, at some point x_0 , the values of the derivatives $y^{(k)}(x_0)$ for $k = 0 \dots n - 1$. This gives an *initial value problem*

$$\begin{cases} F(x, y', \dots, y^{(n)}) = 0, & x \in (a, b), \\ y(x_0) = A_0, \\ \vdots \\ y^{(n-1)}(x_0) = A_{n-1}. \end{cases}$$

Here one may have $x_0 \in (a, b)$ or also $x_0 = a$ or $x_0 = b$. If x_0 is an endpoint of the interval, then the initial conditions must be interpreted as one-sided limits of the derivatives $y^{(k)}(x)$ (or as one-sided derivatives) as $x \rightarrow a^+$ ($x_0 = a$) or $x \rightarrow b^-$ ($x_0 = b$). The requirement is then that these exist for the solutions of the differential equation.

EXAMPLE 6 If a body (mass = m) is at rest at the instant $t = 0$ at the point x_0 , and a force $f(t)$ acts on the body when $t > 0$, then the body's position $x(t)$ at time t is found by solving the initial value problem

$$\begin{cases} mx''(t) = f(t), & t > 0, \\ x(0) = x_0, \\ x'(0) = 0. \end{cases}$$

The initial conditions can be stated more precisely in the form $x(0^+) = x_0$, $D_+x(0) = 0$. $\square$

Additional conditions can also be imposed at several points. Besides the initial value problem, the most typical is the *two-point boundary value problem*, in which the conditions are imposed at the endpoints of the interval under consideration. If $n = 2$, then the normal form of the two-point boundary value problem is

$$\begin{cases} y'' = f(x, y, y'), & x \in (a, b), \\ y(a) = A, \quad y(b) = B. \end{cases}$$

In imposing the boundary conditions here, one must assume that the solution is right-continuous at a and left-continuous at b . Since the solution is in any case continuous on (a, b) (being even twice differentiable), these additional assumptions mean the same as continuity on $[a, b]$.

EXAMPLE 7 Solve the two-point boundary value problem

$$\begin{cases} y''' + 2y'' = 0, & x \in (0, 1), \\ y(0) = 1, \ y'(0) = -1, \ y(1) = 0. \end{cases}$$

Solution The general solution is (see Example 2)

$$y(x) = C_1 e^{-2x} + C_2 x + C_3,$$

so we obtain the system of equations

$$\begin{cases} y(0) = C_1 + C_3 = 1 \\ y'(0) = -2C_1 + C_2 = -1 \\ y(1) = e^{-2}C_1 + C_2 + C_3 = 0 \end{cases} \Rightarrow \begin{cases} C_1 = 0 \\ C_2 = -1 \\ C_3 = 1 \end{cases}$$

The solution is thus $y(x) = -x + 1$. $\square$

Systems of differential equations

By a *system of differential equations* one means a system of equations formed by several differential equations. A *normal-form* ordinary system of differential equations is, for some $n \in \mathbb{N}$, $n \geq 2$, of the form

$$\begin{cases} y_1' = f_1(x, y_1, \dots, y_n), \\ y_2' = f_2(x, y_1, \dots, y_n), \\ \vdots \\ y_n' = f_n(x, y_1, \dots, y_n). \end{cases}$$

Here x is the independent variable and the functions $y_i(x)$, $i = 1 \dots n$, are unknown. The solution (particular or general) is the (ordered) set formed by the functions y_i that satisfy the system on some open interval,

$$\mathbf{y}(x) = (y_1(x), \dots, y_n(x)).$$

This is in fact a vector formed by the functions, i.e. a vector-valued function. In the cases $n = 2, 3$ the solution can, if desired, be interpreted as a parametric

curve in the plane or in space (with x as parameter here, cf. Section IV.4). In an *initial value problem* it is required that the solution satisfy, in addition to the differential equations, n additional conditions of the form

$$y_i(x_0) = A_i, \quad i = 1 \dots n.$$

A normal-form higher-order differential equation

$$y^{(n)} = f(x, y', \dots, y^{(n-1)})$$

can always be written as a normal-form system of differential equations. Namely, when one writes

$$y_1 = y, \quad y_2 = y', \quad \dots, \quad y_n = y^{(n-1)},$$

these equations together with the differential equation form the system

$$\begin{cases} y_1' &= y_2, \\ &\vdots \\ y_{n-1}' &= y_n, \\ y_n' &= f(x, y_1, \dots, y_n). \end{cases}$$

This is of the above normal form.

EXAMPLE 8 The system form of the differential equation $y''' = x^2y^3 + x(y')^2 + y''$ is

$$\begin{cases} y_1' = y_2, \\ y_2' = y_3, \\ y_3' = x^2y_1^3 + xy_2^2 + y_3. \end{cases} \quad \square$$

Writing a higher-order differential equation in system form helps both in theoretical considerations and in numerical solution methods (see Sections IX.5, IX.7, IX.8). Sometimes the system form is advantageous also in the classical solution methods (see Section IX.3).

Regularity of solutions

An initial value problem can be regarded as an implicit definition of the function $y(x)$, whereby, with the help of implicit differentiation, it is possible to compute higher derivatives of the solution function at the initial point x_0 . In this way one can often also determine how regular the solution is on an interval on which it exists (being n times differentiable).

EXAMPLE 9 Let $a > 0$ and consider the initial value problem

$$\begin{cases} y' = x + y^2, & x \in (-a, a), \\ y(0) = 0. \end{cases}$$

If the problem is assumed to be solvable (the question is examined later in Section IX.8), then from the differential equation one can infer that the solution is, on that interval, arbitrarily many times differentiable (smooth). Namely, first, since the solution $y(x)$ is differentiable on $(-a, a)$ (the basic assumption), the differential equation shows that it is also twice differentiable:

$$y''(x) = \frac{d}{dx}[x + (y(x))^2] = 1 + 2y(x)y'(x) = 1 + 2y(x)[x + (y(x))^2], \quad x \in (-a, a).$$

By this, y'' is in turn differentiable on $(-a, a)$, i.e. y is three times differentiable, and so on. We conclude that $y(x)$ is arbitrarily many times differentiable on $(-a, a)$. The derivatives of the solution function at the initial point $x = 0$ are also directly computable, for since $y(0) = 0$ (the initial condition), the differential equation forces $y'(0) = 0$, whereby the above expression for $y''(x)$ gives $y''(0) = 1$. Continuing the implicit differentiation, one sees that all derivatives of the solution function at the point $x = 0$ are determined uniquely (!). $\square$

Solving by quadratures

In what follows we consider mainly those special cases of differential equations for which it is possible to compute an 'exact' solution by reducing the problem to finding antiderivatives of known functions. In this case one says that the differential equation is solvable by *quadratures*, i.e. by 'integrations'. (Quadrature means literally 'squaring', cf. the footnote in Section VIII.9.) Solvability by quadratures does not require that the antiderivatives be elementary functions.

EXAMPLE 10 The initial value problem

$$\begin{cases} y''' = \sin x/x = 1 - \frac{x^2}{6} + \frac{x^4}{120} - \dots, & x > 0, \\ y(0) = 1, \quad y'(0) = y''(0) = 0 \end{cases}$$

is solved by three successive quadratures:

$$\begin{aligned}
 y''(x) &= y''(x) - y''(0) = \int_0^x y'''(t) dt = \int_0^x (\sin t/t) dt \\
 &= x - \frac{x^3}{18} + \frac{x^5}{600} - \dots \\
 y'(x) &= y'(x) - y'(0) = \int_0^x y''(t) dt \\
 &= \frac{x^2}{2} - \frac{x^4}{72} + \frac{x^6}{3600} - \dots \\
 y(x) &= y(0) + [y(x) - y(0)] = y(0) + \int_0^x y'(t) dt \\
 &= 1 + \frac{x^3}{6} - \frac{x^5}{360} + \frac{x^7}{25200} - \dots \quad \square
 \end{aligned}$$

EXAMPLE 9 (continued) The differential equation $y' = x + y^2$ of the earlier example is not solvable by quadratures. (The equation is of *Riccati* type, see Exercise IX.4:10.) $\square$

The last example does not mean that there are no solutions, only that the solving does not reduce to 'integrations'. The solution must then be determined by other means. For example, if the initial value $y(x_0)$ is known, one can use numerical methods for initial value problems (see Section IX.7). Taylor polynomials also give approximate information about the behavior of the solution (cf. Exercise 10).

EXERCISES

1. Verify that $y = 2x + Ce^x$ is the general solution of the differential equation $y' = y + 2(1 - x)$. Draw the solution curves passing through the points $(0, 1)$ and $(0, -1)$.
2. Solve by quadratures (general solution or solution of the initial value problem):

$$\text{a) } y'' = \sin x \quad \text{b) } y''' = 24x + \cos x \quad \text{c) } y'' = \ln x$$

$$\text{d) } y'' = \frac{1}{x^2 + 1}, \quad x \in \mathbb{R}, \quad y(0) = 1, \quad y'(0) = 0$$

$$\text{e) } y'' = \frac{1}{x^2 - 2x}, \quad x \in (0, 2), \quad y(1) = y'(1) = 0$$

3. Determine the general solutions of the following differential equations by reducing the equations to first order (substitution $u(x) = y^{(k)}(x)$).

a) $y'' - y' = 0$ b) $2y''' + 3y'' = 0$ c) $y^{(5)} - 5y^{(4)} = 0$

4. Determine the differential equations of the following families of curves

a) $y = \sin(x + C)$ b) $y = C_1 + C_2 \ln |x|$ c) $y = C_1 + \frac{1}{x + C_2}$

d) $y = (1 + C_1) \ln |x + C_2| - C_1 x + C_2$ e) $x = C_1 e^y + C_2 e^{-y} + 3$

f) circles tangent to the x -axis g) circles tangent to the line $y = x$

5. As the curve $y = F(x)$ slides along the y -axis, a family of curves is formed. Determine the orthogonal trajectories of this family when a) $F(x) = e^x$, b) $F(x) = \ln |x|$.
6. The curve $y = u(x)$ intersects the solution curves of the differential equation $y' = x + y^2$ orthogonally. Of which differential equation is u a solution?
7. Express as a normal-form system of differential equations:

a) $yy'' + xy' = 0$ b) $y'' = (x + y')^2 + y'''$ c) $y^{(4)} = \frac{y'y''}{1 + x + y''}$

d) $\begin{cases} u' = (u + v)^2, \\ v'' = x + uv' \end{cases}$ e) $\begin{cases} u'' = uv, \\ v'' = -xuv \end{cases}$ f) $\begin{cases} u^{(4)} = u''v'' + v, \\ v'' = x + u''' + 2v' \end{cases}$

8. The initial value problem

$$\begin{cases} y' = xy + \sin y, & x \in \mathbb{R}, \\ y(x_0) = y_0 \end{cases}$$

has a unique solution for every $(x_0, y_0) \in \mathbb{R}^2$. a) Deduce that the solution is arbitrarily many times differentiable (smooth) on $\mathbb{R}$. b) Show that the solution curve either touches the x -axis tangentially or does not touch it at all.

9. The initial value problem $xy' = x + y$, $y(1) = 1$ defines a curve S passing through the point $P = (1, 1)$. Determine the center of curvature of S at the point P .

10. Determine the Taylor polynomial $T_n(x, 0)$ of the solution function $y(x)$ of the initial value problem for a general or the given value of n :
- a) $y'' = y$, $y(0) = y'(0) = 1$
 - b) $y'' = -y$, $y(0) = 1$, $y'(0) = 0$
 - c) $y' = x + y^2$, $y(0) = 1$; $n = 3$
 - d) $y' = xy + \sin y$, $y(0) = 1$; $n = 3$
 - e) $yy'' + y' + y = 0$, $y(0) = 1$, $y'(0) = 0$; $n = 4$
 - f) $y'' = yy' - x^2$, $y(0) = y'(0) = 1$; $n = 4$
11. (*) (Tractrix) A truck's trailer is attached to the tractor by a kingpin located at the origin. The tractor's nose points in the direction of the positive y -axis. The trailer's center line lies on the x -axis, and the midpoint of the trailer's axle (i.e. the one carrying the wheels) is at the point $(a, 0)$, $a > 0$. As the tractor moves along the positive y -axis, the midpoint of the trailer's axle traces a curve $y = y(x)$ on the interval $0 < x \leq a$. Derive for this "tractrix" the differential equation

$$y' = -\frac{\sqrt{a^2 - x^2}}{x},$$

and determine the curve $y = y(x)$ as its solution. Take the initial condition into account as well.

12. (*) The following families of curves are given either in parametric form or by means of polar coordinates. Derive the differential equations of the families of curves and of their orthogonal trajectories in the normal form $y' = f(x, y)$.

$$\begin{array}{ll} \text{a) } \begin{cases} x = t + \cos t + C, \\ y = 1 + \sin t \end{cases} & \text{b) } \begin{cases} x = e^t + t + C, \\ y = 2e^t - t + 2C \end{cases} \\ \text{c) } r = C \cos \varphi & \text{d) } r = C\varphi \quad \text{e) } r = Ce^\varphi \end{array}$$

13. (*) A curve $S : y = y_0(x)$ is called the *envelope* of the one-parameter family of curves $y = Y(x, C)$ if S is tangent to every curve of the family (that is, S and each curve of the family share a common point and a common tangent there).
- a) Show that if the differential equation of the family of curves is $F(x, y, y') = 0$, then $y = y_0(x)$ is also a (singular) solution of this differential equation.
 - b) The family of lines $y = Cx + 2C^2$, $C \in \mathbb{R}$ has as its envelope a certain second-degree polynomial curve (a parabola). Determine it and verify by drawing a figure!

14. (*) (Military exercise 3) The trajectory of a cannonball is a parametric curve $\vec{r} = \vec{r}(t)$ ($t = \text{time}$). At the point $(x, y, z) \hat{=} \vec{r}$ of the trajectory the forces acting on the ball are

$$\vec{G} = -mg\vec{k},$$

$$\vec{T} = T_1(x, y, z, t)\vec{i} + T_2(x, y, z, t)\vec{j},$$

$$\vec{F} = -k|\vec{v}|\vec{v},$$

where $\vec{G}$ is gravity ($m = \text{mass}$, $g = \text{acceleration due to Earth's gravity}$), $\vec{T}$ represents the wind, and $\vec{F}$ an air-resistance force proportional to the square of the speed ($\vec{v} = \vec{r}'$, $k = \text{constant}$). Express the projectile's equation of motion $m\vec{r}'' = \vec{G} + \vec{T} + \vec{F}$ as a normal-form system of differential equations of size $n = 6$ by writing

$$(x, y, z, x', y', z') = (y_1, \dots, y_6).$$

IX.2 Separable differential equation

A first-order differential equation in normal form is *separable* if it is of the form

$$y' = \frac{f(x)}{g(y)},$$

that is, if the variables x and y separate on the right-hand side. By writing $1/g(y)$ in place of $g(y)$ and/or $1/f(x)$ in place of $f(x)$, one obtains other manifestations of separability. The assumed form is convenient mainly for formulating the solvability assumptions. In what follows we assume of the functions f and g that for some $x_1 < x_2$ and $y_1 < y_2$ the following hold:

- (i) f is continuous on $[x_1, x_2]$ and g is continuous on $[y_1, y_2]$,
- (ii) $g(y) \neq 0 \quad \forall y \in [y_1, y_2]$.

Now choose $x_0 \in (x_1, x_2)$, $y_0 \in (y_1, y_2)$ (open intervals!) and consider the initial value problem

$$\begin{cases} y' = f(x)/g(y), & x \in (x_0 - \delta, x_0 + \delta) \\ y(x_0) = y_0 \end{cases} \quad (\star)$$

Here δ is first chosen so that

$$0 < \delta \leq \delta_1 = \min\{x_0 - x_1, x_2 - x_0\},$$

so that $[x_0 - \delta, x_0 + \delta] \subset [x_1, x_2]$. In what follows we also make a second assumption, concerning the solvability of the initial value problem $(\star)$. This condition too restricts the choice of δ , but for now we leave open how. The assumption is formulated as follows:

- (iii) The initial value problem $(\star)$ has a differentiable solution $y(x)$ on the interval $(x_0 - \delta, x_0 + \delta)$, and $y(x) \in [y_1, y_2] \quad \forall x \in (x_0 - \delta, x_0 + \delta)$.

By the assumptions (i)–(iii), the functions $f(x)$, $y(x)$, and $g(y(x))$ are continuous on $(x_0 - \delta, x_0 + \delta)$ and $g(y(x)) \neq 0$ on that interval. Appealing to this and to the fundamental theorem of calculus, one can deduce

$$\begin{aligned} y'(x) = \frac{f(x)}{g(y(x))} &\Leftrightarrow g(y(x))y'(x) = f(x) \\ &\Leftrightarrow \frac{d}{dx} \int_{y_0}^{y(x)} g(t) dt = \frac{d}{dx} \int_{x_0}^x f(t) dt, \quad x \in (x_0 - \delta, x_0 + \delta) \end{aligned}$$

that is,

$$y'(x) = \frac{f(x)}{g(y(x))} \Leftrightarrow \frac{d}{dx}[G(y(x)) - F(x)] = 0, \quad x \in (x_0 - \delta, x_0 + \delta),$$

$$G(y) = \int_{y_0}^y g(t)dt, \quad F(x) = \int_{x_0}^x f(t)dt.$$

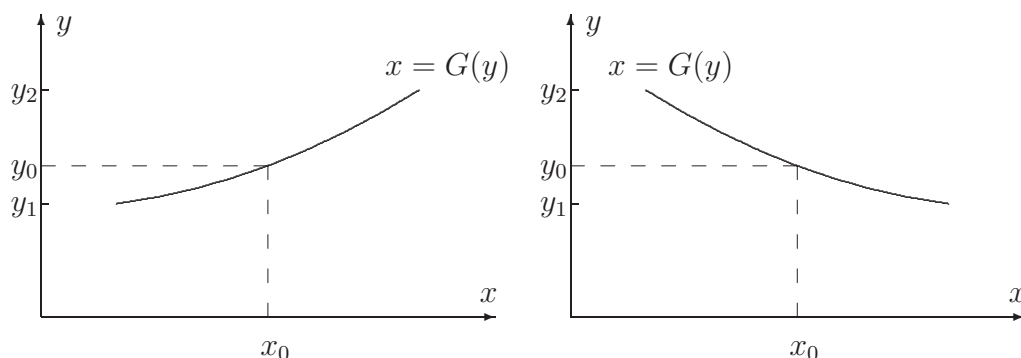
By this, $G(y(x)) - F(x) = C = \text{constant}$ on $(x_0 - \delta, x_0 + \delta)$ (Theorem V.6.7). On the other hand, since $G(y(x_0)) = G(y_0) = 0$ and $F(x_0) = 0$, it follows that $C = 0$, so we have deduced:

$$G(y(x)) = F(x), \quad x \in (x_0 - \delta, x_0 + \delta).$$

Let us now verify that this equation implicitly defines the function $y(x)$. First, from the assumptions (i)–(ii) on the function g and from Theorem V.1.12 it follows that

$$\text{either: } g(y) \geq c > 0 \quad \forall y \in [y_1, y_2], \quad \text{or: } g(y) \leq -c < 0 \quad \forall y \in [y_1, y_2].$$

(Here the largest value of $c = \text{the minimum value of } |g(y)| \text{ on } [y_1, y_2]$.) On this basis the function $G(y)$ is either strictly increasing or strictly decreasing on $[y_1, y_2]$, so G is one-to-one on that interval.



Thus, since G is one-to-one on $[y_1, y_2]$ and since by assumption (iii) $y(x) \in [y_1, y_2]$ when $x \in (x_0 - \delta, x_0 + \delta)$, we may write

$$G(y(x)) = F(x) \Leftrightarrow y(x) = G^{-1}(F(x)), \quad x \in (x_0 - \delta, x_0 + \delta).$$

With assumptions (i)–(iii) in force, we have thus deduced:

$$y'(x) = \frac{f(x)}{g(y(x))} \Leftrightarrow y(x) = G^{-1}(F(x)), \quad x \in (x_0 - \delta, x_0 + \delta).$$

Starting from the solution formula obtained, one sees from the above chain of reasoning that the $y(x)$ given by the formula is the unique (differentiable) solution of the initial value problem $(\star)$, insofar as the condition $y(x) \in [y_1, y_2] \quad \forall x \in (x_0 - \delta, x_0 + \delta)$ contained in assumption (iii) holds. It thus remains only to verify that this condition too is satisfied for the obtained (only possible) solution $y(x) = G^{-1}(F(x))$.

We first note that from the above estimates for the function $g(y)$ and from the comparison principle for integrals (Theorem VIII.5.6) it follows that

$$\left| \int_{y_0}^{y(x)} g(t) dt \right| \geq c |y(x) - y_0| \quad \forall x \in (x_0 - \delta, x_0 + \delta).$$

On the other hand, by assumption (i) and Theorems V.1.12 and VIII.5.6,

$$\left| \int_{x_0}^x f(t) dt \right| \leq M |x - x_0|, \quad x \in [x_1, x_2],$$

where $M = \max_{x \in [x_1, x_2]} |f(x)|$. Combining the estimates yields

$$\begin{aligned} c |y(x) - y_0| &\leq |G(y(x))| = |F(x)| \leq M |x - x_0| \\ \Rightarrow |y(x) - y_0| &\leq \frac{M}{c} |x - x_0| < \frac{M\delta}{c} \quad \forall x \in (x_0 - \delta, x_0 + \delta). \end{aligned}$$

Consequently, when δ is chosen so that

$$0 < \delta \leq \min\{\delta_1, \delta_2\}, \quad \delta_2 = \frac{c}{M} \min\{y_0 - y_1, y_2 - y_0\},$$

the fulfillment of assumption (iii) is ensured, and thus we have proved

THEOREM IX.2.1 If $x_0 \in (x_1, x_2)$ and $y_0 \in (y_1, y_2)$, then, assumptions (i), (ii) being in force, there exists $\delta > 0$ such that the initial value problem $(\star)$ has a unique solution $y(x)$ on the interval $(x_0 - \delta, x_0 + \delta) \subset [x_1, x_2]$. The solution formula on that interval is

$$\int_{y_0}^{y(x)} g(t) dt = \int_{x_0}^x f(t) dt.$$

An easily remembered, formal procedure for finding the general solution of a separable differential equation is the scheme

$$\begin{aligned} \frac{dy}{dx} = \frac{f(x)}{g(y)} &\Rightarrow g(y) dy = f(x) dx \\ &\Rightarrow \int g(y) dy = \int f(x) dx. \end{aligned}$$

Here the first step is 'separation' and the second 'integration'. The procedure is valid under the conditions of Theorem IX.2.1, and the solution satisfying the initial condition $y(x_0) = y_0$ is thus obtained by changing the indefinite integrals into definite ones.

EXAMPLE 1 In the initial value problem solved earlier in Section VI.2

$$\begin{cases} y' = y, & x \in \mathbb{R}, \\ y(0) = y_0 \end{cases}$$

the differential equation is of the separable type mentioned above ($f(x) = 1$, $g(y) = 1/y$), so by Theorem IX.2.1 the solution formula is

$$\int_{y_0}^{y(x)} \frac{1}{t} dt = \int_0^x dt.$$

The formula is valid if either $y_0, y(x) \in (0, \infty)$ or $y_0, y(x) \in (-\infty, 0)$, in which case the solution is

$$\ln(y/y_0) = x \Leftrightarrow y(x) = y_0 e^x, \quad x \in \mathbb{R}.$$

The obtained solution is valid also in the case $y_0 = 0$, to which the separation procedure does not apply. $\square$

EXAMPLE 2 Determine the general solution of the differential equation $1 + y^2 + xyy' = 0$.

Solution Separating and integrating gives

$$\begin{aligned} \frac{y dy}{1 + y^2} &= -\frac{dx}{x} \Rightarrow \int \frac{y dy}{1 + y^2} = -\int \frac{dx}{x} \\ &\Rightarrow \frac{1}{2} \ln(1 + y^2) = -\ln|x| + C. \end{aligned}$$

Writing $\ln C$, $C > 0$ in place of C , simplification gives

$$x^2(1 + y^2) = C^2.$$

Solving this for $y = y(x)$ gives a two-branch solution

$$y(x) = \pm \frac{1}{x} \sqrt{C^2 - x^2}, \quad C > 0.$$

This is valid on the intervals $(-C, 0)$ and $(0, C)$. $\square$

In the example the change $C \mapsto \ln C$ was based on the fact that $\ln C$ takes all possible real values as $C \in \mathbb{R}_+$. By such manipulations the solution can often be brought into a simpler form.

EXAMPLE 3 Solve the differential equation $y' = \sin y$.

$$\begin{aligned}
 \text{Solution} \quad \frac{dy}{\sin y} = dx &\Rightarrow \int \frac{dy}{\sin y} = \int dx \\
 &\Rightarrow \ln \left| \tan \frac{1}{2}y \right| = x + \ln C, \quad C > 0 \\
 &\Leftrightarrow \left| \tan \frac{1}{2}y \right| = Ce^x, \quad C > 0 \\
 &\Leftrightarrow \tan \frac{1}{2}y = Ce^x, \quad C \neq 0 \quad (\pm C \hookrightarrow C) \\
 &\Leftrightarrow y(x) = 2 \arctan(Ce^x), \quad C \neq 0 \\
 &\Leftrightarrow y(x) = 2 \operatorname{Arctan}(Ce^x) + 2n\pi, \quad C \neq 0, \quad n \in \mathbb{Z}.
 \end{aligned}$$

The obtained solution is valid also when $C = 0$, since $y(x) = 2n\pi$ is a solution of the original differential equation for every $n \in \mathbb{Z}$. These solutions may be counted as belonging to the general solution, even though the separation procedure does not reach them directly. Additional singular solutions not belonging to the general solution are

$$y(x) = (2n + 1)\pi, \quad n \in \mathbb{Z}.$$

These are the limits of the above solution functions as $C \rightarrow \pm\infty$. $\square$

Autonomous differential equation

A separable differential equation of the form

$$y' = f(y),$$

where f does not depend on the free variable x , is called *autonomous*. As in Example 3 above, each zero y_i of f corresponds to a constant solution

$$y(x) = y_i.$$

This is a singular solution, or possibly one that can be included in the general-solution expression, cf. the examples above. The separation procedure does not yield such solutions directly, because the function $g(y) = 1/f(y)$ is not continuous in a neighborhood of the zeros of f (cf. Theorem IX.2.1).

EXAMPLE 4 Solve the differential equation $y' = y(1 - y)$.

Solution Separating and integrating gives

$$\begin{aligned} dx = \frac{dy}{y(1-y)} &\Rightarrow \int dx = x = \int \frac{1}{y(1-y)} dy = \int \left(\frac{1}{y} + \frac{1}{1-y} \right) dy \\ &\Rightarrow x = \ln \left| \frac{y}{1-y} \right| + \ln C, \quad C > 0. \end{aligned}$$

By this,

$$\begin{aligned} \frac{y}{1-y} = \pm C^{-1} e^x &\Leftrightarrow y(x) = \frac{e^x}{e^x \pm C}, \quad C > 0 \\ &\Leftrightarrow y(x) = \frac{e^x}{e^x + C}, \quad C \neq 0. \end{aligned}$$

The restriction $C \neq 0$ can be removed from the obtained general-solution expression, since $y(x) = 1$ is also a solution. This is the common limit of all solutions (the asymptotic solution) as $x \rightarrow \infty$. A further solution is $y(x) = 0$, obtained as the limit of the general-solution expression as $C \rightarrow \infty$. $\square$

EXAMPLE 5 : Logistic growth model. If the growth of a physical quantity (e.g. a population) as a function of time t is described by the autonomous differential equation

$$y' = ay - by^2,$$

where a and b are (dimensionless) positive constants, then the growth model (and also the differential equation) is called *logistic*. By the substitution $y(t) = (a/b)u(t)$ the logistic DE transforms into the form

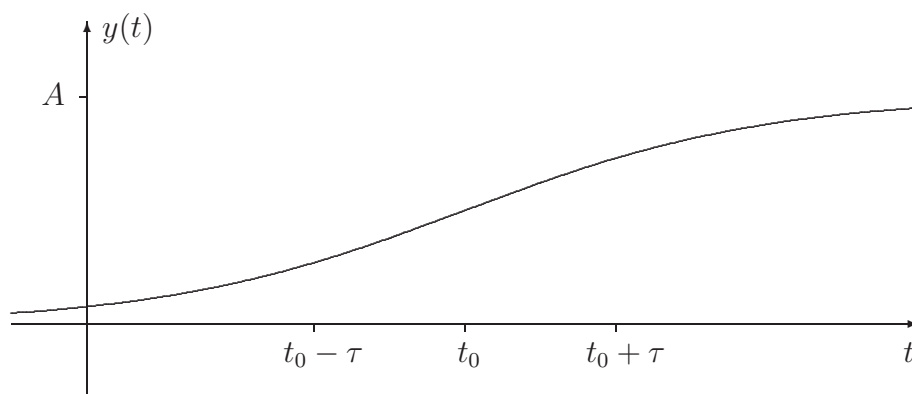
$$u' = a(u - u^2).$$

Solution by separation (cf. the previous example):

$$a dt = \frac{du}{u(1-u)} \Rightarrow u(t) = \frac{e^{at}}{e^{at} + C}, \quad C \in \mathbb{R}.$$

The solution is physically meaningful (increasing) only when $C > 0$. In this case one can write $C = e^{at_0}$, $t_0 \in \mathbb{R}$. Denoting further $\tau = 1/a$ (the time constant), the solutions take the form

$$y(t) = AY \left(\frac{t - t_0}{\tau} \right), \quad A = a/b, \quad \tau = 1/a, \quad Y(t) = \frac{e^t}{e^t + 1} \quad (t_0 \in \mathbb{R}). \quad \square$$



The differential equation $y' = f(y/x)$

A *homogeneous* differential equation $y' = f(y/x)$ transforms, by the substitution

$$u(x) = y(x)/x \Rightarrow y(x) = xu(x) \Rightarrow y' = xu' + u$$

into the equation

$$xu' = f(u) - u.$$

This is separable, so the homogeneous equation is reducible to a separable one.

If $f(u_0) - u_0 = 0$, then a singular solution is

$$u(x) = u_0 \Rightarrow y(x) = u_0 x.$$

The general solution is found by separation.

EXAMPLE 6 Solve the differential equation

$$y' = \frac{x+y}{x-y}.$$

Solution This is homogeneous, so we make the substitution $u = y/x$:

$$\Rightarrow x \frac{du}{dx} = \frac{1+u}{1-u} - u = \frac{1+u^2}{1-u}.$$

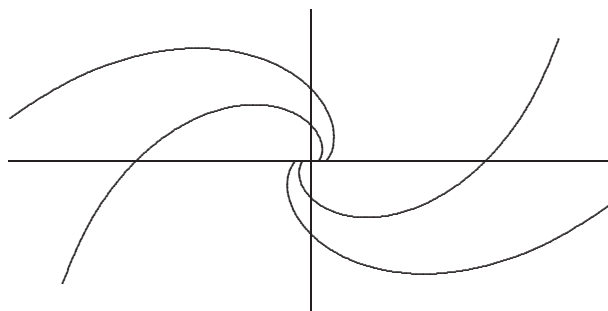
Separating and integrating yields

$$\begin{aligned}\int \frac{dx}{x} &= \ln |x| = \int \frac{1-u}{1+u^2} du \\ &= \operatorname{Arctan} u - \frac{1}{2} \ln(1+u^2) + \ln C \\ \Leftrightarrow \operatorname{Arctan} u &= \ln \left(\frac{|x|\sqrt{1+u^2}}{C} \right) \\ \Leftrightarrow \operatorname{Arctan} \frac{y}{x} &= \ln \left(\frac{\sqrt{x^2+y^2}}{C} \right) \quad (C > 0).\end{aligned}$$

Passing to polar coordinates gives a simpler form for the solution

$$\varphi = \ln \frac{r}{C} \Leftrightarrow r = Ce^\varphi \quad (C > 0).$$

The solutions are so-called *logarithmic spirals*. $\square$



The differential equation $y' = f(ax + by + c)$

The differential equation $y' = f(ax + by + c)$, where $a, b, c \in \mathbb{R}$ and $b \neq 0$, is also reducible to separable form. Here the natural substitution is

$$u = ax + by + c \Rightarrow y = \frac{1}{b}(u - ax - c),$$

whereby

$$\frac{dy}{dx} = \frac{1}{b} \left(\frac{du}{dx} - a \right),$$

and consequently one obtains for u a separable (in fact autonomous) DE

$$u' = bf(u) + a.$$

EXAMPLE 7 Solve the differential equation

$$y' = (x - y)^2 + 1.$$

Solution By the substitution

$$x - y = u \Leftrightarrow y = x - u \Rightarrow y' = 1 - u'$$

one obtains the differential equation

$$u' = -u^2.$$

Separation gives the general solution $u(x) = 1/(x + C)$, so the general solution of the original equation is

$$y(x) = x - \frac{1}{x + C}.$$

In addition, there is the singular solution corresponding to the solution $u(x) = 0$ of the transformed equation

$$y(x) = x. \quad \square$$

EXERCISES

1. Determine the general solutions of the following separable differential equations (or ones reducible to separable), or solve the initial value problem. In the case of an initial value problem, also determine on which interval the solution is valid.

a) $x^2y' = y^2$ b) $y' = -2xy$ c) $y' = (1 - y)^2$ d) $y' = y^2 - 1$
 e) $(1 + x)y' = 1 + y$ f) $1 + y^2 - xy' = 0$ g) $(1 + x^3)y' = x^2y$
 h) $(1 - x^2)y' = 1 - y^2$ i) $y' = \tan y$ j) $xy' = \cot y$
 k) $(x - 3)y' = -y$, $y(-1) = 1$ l) $y' = y^2$, $y(0) = 4$
 m) $y' + 5x^4y^2 = 0$, $y(0) = 1$ n) $y' - 5x^4y^2 = 0$, $y(0) = 1$
 o) $y' \operatorname{Arctan} y = 1$, $y(1) = 1$ p) $y' \sin x = y \ln y$, $y(\pi/2) = 1$
 q) $(1 + e^x)yy' = e^x$, $y(1) = 1$ r) $\cos^2 x \cos(\ln y)y' = y$, $y(\pi/4) = 2$
 s) $y' = \frac{y}{x - y}$ t) $(3x + y)y' = y$ u) $xyy' = x^2 + y^2$
 v) $(2x^2 + y^2)y' = 2xy$ x) $xy' = y + \tan \frac{y - x}{x}$ y) $y' = e^{x-2y}$
 z) $xy' = xe^{y/x} + y$, $y(1) = 0$ ä) $xy' = y \ln \frac{y}{x}$, $y(1) = 1$
 ä) $(x + y)y' = 1 - x - y$, $y(0) = 1$ ö) $y' = (2x + y + 3)^2$, $y(0) = 0$

2. Solve the initial value problem $e^y y' = x + e^y - 1$, $y(0) = y_0$ by the substitution $u = x + e^y$. For which values of y_0 is the solution valid on all of $\mathbb{R}$?
3. Determine the solution curves of the differential equation $y' = 2x|y - 1|$ that are tangent to the x -axis.
4. Determine the orthogonal trajectories of the following families of curves.

a) $y = Cx^2$ b) $y = e^x + C$ c) $x^2 + 2y^2 = C^2$ d) $e^{x+y} = (x + C)^2$

5. Determine the curves having the property: the distance from a point P of the curve to the point where the curve's normal at P meets the x -axis $= a = \text{constant}$.

6. Determine the function $y(x)$ continuous on $[0, \infty)$ that satisfies the *integral equation*

$$1 + \int_0^x \frac{y(t)}{t^2 + 1} dt = y(x), \quad x \geq 0.$$

7. (Hypetia, Utopia, and Apatia) In the realms of H, U, and A the standard of living was $= 1$ in the year 2000 and $= 1.01$ in 2001. Predict the standards of living of these realms in the year 2100, given that the rate of growth of the standard of living is, in H, directly proportional to the square of the standard of living; in U, directly proportional to the standard of living; and in A, inversely proportional to the standard of living.
8. In 1960 the world's population was 3.0 billion and the growth rate at that time was 50 million/yr. If it is assumed that the population $y(t) \rightarrow 12$ billion as $t \rightarrow \infty$, then what should the population have been in the year 2000 according to the logistic growth model? (The actual population was 6.1 billion.)
9. After the parachute opens, a parachutist's rate of descent obeys the law of motion $mv' = mg - av - bv^2$, where $m = 80$ kg, $g = 9.8$ m/s², $a = 120$ kg/s and $b = 4.0$ kg/m. Determine $v(t)$, $t \geq 0$ (time unit = s), given that at the instant the parachute opens, $t = 0$, the rate of descent is 20 m/s. How is the terminal velocity $v_\infty = \lim_{t \rightarrow \infty} v(t)$ most easily found? Sketch $v(t)$ on the time intervals $[0, 1]$ and $[0, 60]$.
10. Light rays propagating in space in the direction $-\vec{k}$ are reflected from the surface of revolution $S : z = u(r)$, $r = \sqrt{x^2 + y^2}$ in such a way that every reflected ray passes through the origin. The function $u(r)$ must then satisfy the homogeneous differential equation $ru' = \sqrt{r^2 + u^2} + u$ (cf. Exercise VII.2:12). Solve it!
11. (*) The differential equation

$$y' = f\left(\frac{2x - y + 1}{x - 2y + 1}\right)$$

reduces to a homogeneous one by the substitutions $x = x_0 + t$, $y = y_0 + u$, when x_0 and y_0 are chosen suitably. Solve the equation in this way, when $f(x) = x$.

12. (*) The following differential equations reduce to homogeneous ones by the substitutions $x = t^\alpha$, $y = z^\beta$, when α and β are chosen suitably. Solve!
- a) $42(x^2 - xy^2)y' + y^3 = 0$ b) $(x^2y^2 - 1)y' + 2xy^3 = 0$

13. (*) A cylindrical vessel contains 200 liters of water. An opening of area $= 20 \text{ cm}^2$ is made in the bottom of the vessel. Water discharges from the opening at speed $v = \sqrt{2gh}$, where $g = 9.8 \text{ m/s}^2$ and h = the height of the water in the vessel at the given instant. (The discharge speed is assumed the same at all points of the opening.) Compute in what time the vessel empties, when the height of the water is initially $h = 1 \text{ m}$.
14. (*) The curve $S : y = y(x)$, $x \geq 0$, where $y(x)$ is continuously differentiable on $[0, \infty)$, has the property: the arc length of the curve on $[0, x]$ equals the area of the region between the curve and the x -axis on that interval. Determine all curves that satisfy this condition and, in addition, the initial condition $y(0) = 2$.

IX.3 Reducible second-order DEs

The general second-order differential equation in normal form $y'' = f(x, y, y')$ can be solved by quadratures in certain special cases. Of these we consider here only two (fairly common in applications), in which f does not depend on either y or x , so that the differential equation is of the form either $y'' = f(x, y')$ or $y'' = f(y, y')$. In both cases we are dealing with a differential equation reducible to first order. Insofar (and only insofar) as the first-order DE in question is separable, reducible to separable form, or otherwise solvable by quadratures, the original DE can be solved by quadratures.

The differential equation $y'' = f(x, y')$ reduces immediately to first order, for by the substitution $y' = u$ it can be written as the system

$$\begin{cases} u' = f(x, u), \\ y' = u. \end{cases}$$

If here the first equation can be solved into the form $u = U(x, C_1)$, then the latter equation is solved by a single quadrature into the form $y(x) = Y(x, C_1, C_2)$. Solvable by quadratures is, for example, a DE of the form $y'' = f(y')$; see the application example at the end of the section. We now turn to the second special case mentioned.

The differential equation $y'' = f(y, y')$

In this case too, we first transform the DE to be solved into a system by the substitution $y' = u$:

$$\begin{cases} u' = f(y, u), \\ y' = u. \end{cases}$$

Here we express u as a composite function in the form $u = u(x) = U(y(x)) = U(y)$. (If the solution is $y = Y(x)$, then $U(y) = (u \circ Y^{-1})(y)$ on intervals where Y is invertible.) With the said notation and on the basis of the above differential equations, it follows that

$$\begin{aligned} f(y, U(y)) &= \frac{du}{dx} = \frac{d}{dx}U(y(x)) = \frac{dU}{dy} \cdot \frac{dy}{dx} = \frac{dU}{dy} \cdot u(x) = \frac{dU}{dy} \cdot U(y) \\ &\Rightarrow \frac{dU}{dy} = \frac{f(y, U)}{U}. \end{aligned}$$

Insofar as the obtained first-order DE can be solved into the form $U(y) = V(y, C_1)$, one can further solve y from the separable (autonomous) equation

$$y' = U(y) = V(y, C_1).$$

The original second-order DE has thus been broken down into two successive first-order DEs, the latter of which is separable.

The above solution method becomes easier to remember in practice if we simply write $u(x) = u(y)$ (i.e. think of y as the variable instead of x), and carry out a formal termwise division in the system of differential equations:

$$\left. \begin{array}{l} \frac{du}{dx} = f(y, u) \\ \frac{dy}{dx} = u \end{array} \right\} \Rightarrow \frac{du}{dy} = \frac{f(y, u)}{u}.$$

EXAMPLE 1 Solve the differential equation $y'' = yy'$.

Solution Here $f(y, u) = yu$, so the above solution method gives

$$\begin{aligned} \frac{du}{dy} = y &\Rightarrow u = \int y dy = \frac{1}{2}y^2 + \frac{C_1}{2} \\ \Leftrightarrow \frac{dy}{dx} = \frac{1}{2}y^2 + \frac{C_1}{2} &\Rightarrow x = \int \frac{2}{y^2 + C_1} dy, \quad C_1 \in \mathbb{R}. \end{aligned}$$

From here on we must distinguish the cases $C_1 > 0$, $C_1 = 0$, and $C_1 < 0$. First, if $C_1 > 0$, we write C_1^2 ($C_1 > 0$) in place of C_1 , obtaining

$$\begin{aligned} x &= \frac{2}{C_1^2} \int \frac{1}{(y/C_1)^2 + 1} dy = \frac{2}{C_1} \operatorname{Arctan} \frac{y}{C_1} - \frac{2C_2}{C_1} \\ \Rightarrow \underline{\underline{y(x) = C_1 \tan \left(\frac{C_1 x}{2} + C_2 \right), \quad C_1 > 0, \quad C_2 \in \mathbb{R}.}} \end{aligned}$$

In the case $C_1 < 0$ we obtain, correspondingly, by writing $-C_1^2$ in place of C_1 ,

$$\begin{aligned} x &= \frac{2}{C_1^2} \int \frac{1}{(y/C_1)^2 - 1} dy = -\frac{2}{C_1} \operatorname{artanh} \frac{y}{C_1} - \frac{2C_2}{C_1} \\ \Rightarrow \underline{\underline{y(x) = -C_1 \tanh \left(\frac{C_1 x}{2} + C_2 \right), \quad C_1 > 0, \quad C_2 \in \mathbb{R}.}} \end{aligned}$$

Finally, in the case $C_1 = 0$ the solutions are

$$x = \int \frac{2}{y^2} dy = -\frac{2}{y} - C \Rightarrow \underline{\underline{y(x) = -\frac{2}{x+C}, \quad C \in \mathbb{R}.}}$$

In addition to these, we must also include the singular solutions of the autonomous differential equations $y' = \frac{1}{2}(y^2 + C_1)$ when $C_1 = -C^2 \leq 0$:

$$\underline{\underline{y(x) = C, \quad C \in \mathbb{R}. \quad \square}}$$

The differential equation $y'' = f(y)$

The special case $y'' = f(y)$ of the DE type examined above is also common in applications. According to the above solution scheme, this reduces to separable first-order equations, so the equation can always be solved by quadratures.

EXAMPLE 2 : Harmonic oscillation. A body of mass $= m$ is attached to a spring whose spring constant $= k$. At time t the body's distance from the equilibrium position is $y(t)$, so the equation of motion is

$$my'' = -ky.$$

Solve $y(t)$ with the initial conditions $y(0) = 0$, $y'(0) = v_0$.

Solution Here $f(y) = -(k/m)y$, so we get

$$\begin{aligned} \frac{du}{dy} &= \frac{f(y)}{u} = -\frac{ky}{mu} \\ \Rightarrow mu \frac{du}{dy} + ky &= 0 \\ \Leftrightarrow \frac{d}{dy} \left(\frac{1}{2} mu^2 + \frac{1}{2} ky^2 \right) &= 0 \\ \Leftrightarrow \frac{1}{2} mu^2 + \frac{1}{2} ky^2 &= C. \end{aligned}$$

In this intermediate result $u = y' =$ the body's velocity, $E_k = \frac{1}{2}mu^2 =$ the kinetic energy, and $E_p = \frac{1}{2}ky^2 =$ the spring's potential energy. The result is known as the energy principle: Total energy = constant. Since at the initial instant $y = 0$, $u = v_0$, we must have

$$C = \frac{1}{2}mv_0^2.$$

Continuing from the obtained law of conservation of energy with this value of C , writing

$$T = \sqrt{\frac{m}{k}}, \quad L = v_0 T,$$

assuming that $|y| < L$, and using the initial condition $y(0) = 0$, we get

$$T \frac{dy}{dt} = \sqrt{L^2 - y^2} \Rightarrow \int_0^{y(t)} \frac{1}{\sqrt{L^2 - y^2}} dy = \int_0^t \frac{1}{T} dt = \frac{t}{T}.$$

On the interval $t/T \in (-\pi/2, \pi/2)$ the solution can be written in the form

$$\text{Arcsin}(y/L) = t/T \Leftrightarrow y(t) = L \sin(t/T), \quad t/T \in (-\pi/2, \pi/2).$$

From the final result one sees that the restriction $t/T \in (-\pi/2, \pi/2)$ can be removed. We thus obtained a solution $y(t) = v_0 T \sin(t/T)$, valid for all values of t , where $T = \sqrt{m/k}$. $\square$

EXAMPLE 3 : Explosion. A fast chemical reaction is described by the DE system

$$u' = 3y^2, \quad y' = u,$$

where $u = u(t)$ and $y = y(t)$ ($t = \text{time}$). Assume the initial conditions $y(0) = 1$ and $u(0) = 0$. Compute the time T taken by the reaction, i.e. determine the largest T such that the initial value problem has a solution on the interval $(0, T)$.

Solution Differentiating the latter equation and eliminating u , one sees that y satisfies the differential equation $y'' = 3y^2$. In the solution, however, we start from the original system form, since it is already suited to the solution scheme:

$$\begin{aligned} \frac{du}{dy} = \frac{3y^2}{u} &\Rightarrow u \frac{du}{dy} - 3y^2 = 0 \\ &\Leftrightarrow \frac{d}{dy} \left(\frac{1}{2} u^2 - y^3 \right) = 0 \\ &\Leftrightarrow \frac{1}{2} u^2 - y^3 = C. \end{aligned}$$

Since $y = 1$ and $u = 0$ when $t = 0$, we have $C = -1$, so

$$u = y' = \sqrt{2} \sqrt{y^3 - 1} \quad (y \geq 1).$$

Separating and integrating, and using the initial condition $y(0) = 1$, we get

$$\frac{1}{\sqrt{2}} \int_1^{y(t)} \frac{dx}{\sqrt{x^3 - 1}} = \int_0^t dx = t \Leftrightarrow y(t) = \Phi^{-1}(t),$$

where

$$\Phi(s) = \frac{1}{\sqrt{2}} \int_1^s \frac{dx}{\sqrt{x^3 - 1}}, \quad s \geq 1.$$

The function Φ is not an elementary function, but since $x^3 - 1 = (x - 1)(x^2 + x + 1) > 3(x - 1)$ when $x > 1$,

$$\frac{1}{\sqrt{x^3 - 1}} < \frac{1}{\sqrt{3}} \frac{1}{\sqrt{x - 1}}, \quad \text{when } x > 1,$$

so (by the majorant principle, see Section VIII.7) $\Phi(s)$ is defined as a convergent integral for every $s > 1$. Further, since

$$\frac{1}{\sqrt{x^3 - 1}} \sim \frac{1}{x^{3/2}}, \quad \text{when } x \gg 1,$$

Φ has a limit as $s \rightarrow \infty$, and this too can be computed as a convergent integral:

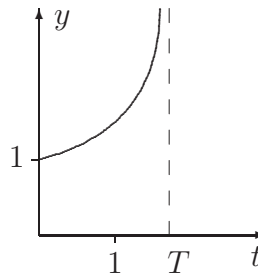
$$\lim_{s \rightarrow \infty} \Phi(s) = \int_1^\infty \frac{dx}{\sqrt{x^3 - 1}} = \Phi_\infty.$$

Since $\Phi'(s) = 1/\sqrt{s^3 - 1} > 0$ when $s > 1$, Φ is strictly increasing on $[1, \infty)$. Thus $\Phi : [1, \infty) \mapsto [0, \Phi_\infty)$ is a bijection, and consequently $y(t) = \Phi^{-1}(t)$ is also a bijection:

$$y : [0, T) \longrightarrow [1, \infty),$$

where

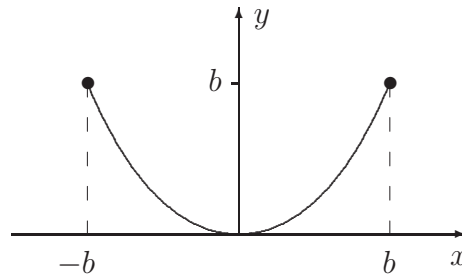
$$\begin{aligned} T = \Phi_\infty &= \frac{1}{\sqrt{2}} \int_1^\infty \frac{dx}{\sqrt{x^3 - 1}} \\ &\approx \underline{\underline{1.71732}}. \quad \square \end{aligned}$$



Application example: The catenary

Problem description

A chain is hung so that its ends are at the points (b, b) and $(-b, b)$, $b > 0$, and the chain passes through the origin. The direction of gravity $= -\vec{j}$ and the weight of the chain per unit length $= \rho$.

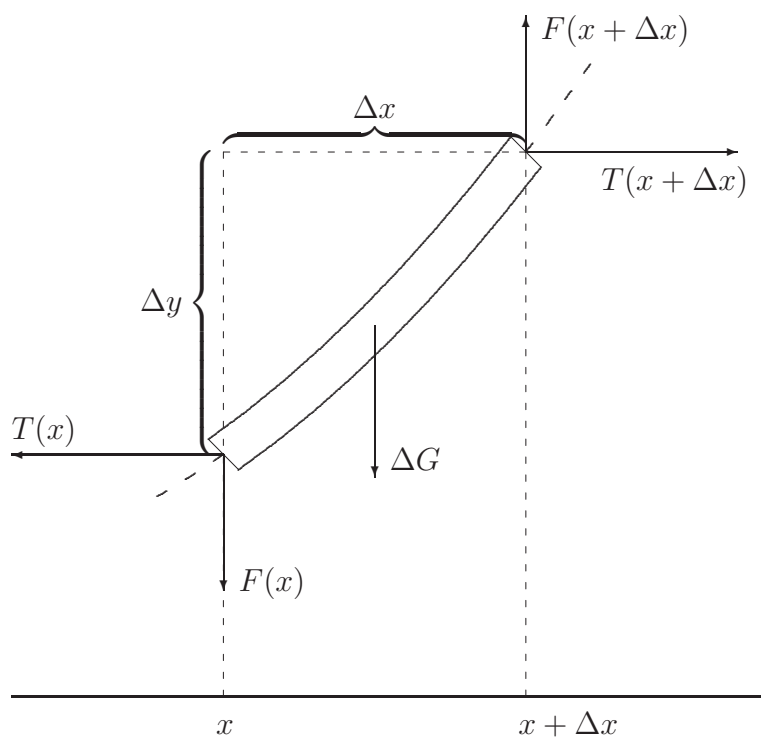


Problem Determine the shape of the chain, i.e. the *catenary*

$$y = y(x), \quad x \in [-b, b].$$

Solution

Since the problem description is physical, we first need a mathematical model derived from physical principles. To derive it, consider a small piece of chain on the interval $[x, x + \Delta x]$. Marked in the figure are the horizontal binding forces (T), the vertical binding forces (F), and the gravitational force (ΔG) acting on the piece of chain. Since it is a chain, there are no binding moments (bending moments).



The equilibrium equations are

$$\begin{aligned} T(x + \Delta x) - T(x) &= 0, & (\text{horizontal direction}) \\ F(x + \Delta x) - F(x) &= \Delta G, & (\text{vertical direction}) \\ F(x)\Delta x - T(x)\Delta y &= \mathcal{O}(\Delta G\Delta x). & (\text{moment balance}) \end{aligned}$$

The length of the piece of chain is approximately (cf. Section VIII.8)

$$\begin{aligned}\Delta s &\cong \sqrt{(\Delta x)^2 + (\Delta y)^2} = \sqrt{1 + \left(\frac{\Delta y}{\Delta x}\right)^2} \Delta x \\ &\approx \sqrt{1 + [y'(x)]^2} \Delta x,\end{aligned}$$

so the gravitational force ΔG is approximately

$$\Delta G = \rho g \Delta s \approx \rho g \sqrt{1 + [y'(x)]^2} \Delta x.$$

Dividing by Δx gives, in the limit $\Delta x \rightarrow 0$, the following mathematical model:

$$\begin{cases} T' = 0, \\ F' = \rho g \sqrt{1 + (y')^2}, \\ F - Ty' = 0. \end{cases}$$

From the first and third equations

$$T(x) = T_0 = \text{const}, \quad F(x) = T_0 y'(x).$$

Substituting into the middle equation gives the differential equation of the catenary:

$$ay'' = \sqrt{1 + (y')^2}, \quad a = \frac{T_0}{\rho g}.$$

This becomes separable by the substitution $u(x) = y'(x)$:

$$\begin{aligned}au' = \sqrt{1 + u^2} &\Rightarrow \int \frac{du}{\sqrt{1 + u^2}} = \int \frac{dx}{a} \\ &\Rightarrow \operatorname{arsinh} u = \frac{x}{a} - \frac{c}{a} \quad (C = -\frac{c}{a}) \\ &\Leftrightarrow u(x) = \sinh \left(\frac{x - c}{a} \right).\end{aligned}$$

From this and the condition $y(0) = 0$ it follows that

$$y(x) = \int_0^x u(t) dt = a \cosh \left(\frac{x - c}{a} \right) - a \cosh \frac{c}{a}.$$

Since $y(b) = y(-b)$, we must have $c = 0$, i.e.

$$y(x) = a \left(\cosh \frac{x}{a} - 1 \right).$$

The condition $y(b) = b$ (which is still unused) leads to the transcendental equation

$$\cosh \alpha = \alpha + 1, \quad \alpha = b/a.$$

The solution (the catenary) is thus

$$y(x) = \frac{b}{\alpha} \left(\cosh \frac{\alpha x}{b} - 1 \right), \quad \cosh \alpha = \alpha + 1, \quad \alpha > 0.$$

The numerical value of the constant α is

$$\alpha = 1.6161375137743..$$

From the above solution one sees that the general catenary (with gravity acting in the direction $-\vec{j}$) is of the form

$$y(x) = a \cosh \left(\frac{x - c}{a} \right) + C,$$

where $a \in \mathbb{R}_+$ and $c, C \in \mathbb{R}$ are undetermined constants. The constants are determined, for example, by giving (as in the example) three points through which the chain passes. Alternatively, one may know the horizontal supporting force T_0 acting at the suspension point and/or the length of the chain, which on the interval $[x_1, x_2]$ is (cf. Section VIII.8)

$$\begin{aligned} L &= \int_{x_1}^{x_2} \sqrt{1 + [y'(x)]^2} dx = \int_{x_1}^{x_2} \sqrt{1 + \sinh^2 \left(\frac{x - c}{a} \right)} dx \\ &= \int_{x_1}^{x_2} \cosh \left(\frac{x - c}{a} \right) dx = a \left| \sinh \left(\frac{x - c}{a} \right) \right|_{x_1}^{x_2}. \quad \square \end{aligned}$$

EXERCISES

1. Solve by quadratures (general solution or solution of the initial value problem). In the initial value problems, also determine the largest interval on

which the solution is valid.

- a) $y'' = (y')^2$ b) $2yy'' = 1 + (y')^2$ c) $y^2y'' = y'$ d) $y'' = (y')^3e^y$
 e) $y'y''' = 2(y'')^2$ f) $x^2y'' = (y')^2$ g) $y'' = 2x(y')^2$ h) $x^3y'' = 2(y')^2$
 i) $(1 + y^2)y'' = y' + (y')^3$, $y(0) = 1$, $y'(0) = -1$
 j) $y'' = 1 - (y')^2$, $y(0) = 0$, $y'(0) = -1$
 k) $y^3y'' = -1$, $y(0) = 1$, $y'(0) = -1$
 l) $\sqrt{y}y'' = 1$, $y(0) = 1$, $y'(0) = -2$
 m) $3y^2y'y'' = -1$, $y(0) = y'(0) = 1$
 n) $2yy'' - 3(y')^2 = 4y^2$, $y(0) = 2$, $y'(0) = 4$
 o) $2x^2y''y''' = -1$, $y(1) = 0$, $y'(1) = 1$, $y''(1) = 2$
 p) $2y^{(4)} + 3(y'')^2 = 0$, $y(0) = y'(0) = 0$, $y''(0) = y'''(0) = -1$

2. Determine the solution curve of the equation $yy'' + 2(y')^2 = 0$ that is tangent to the line $y = x$ at the point $(1, 1)$.
3. For which curves $y = y(x)$ does the following hold: the arc length of the curve on $[0, x]$ is proportional to $y'(x)$?
4. Determine the curves $S : y = y(x)$ whose (signed) curvature at the point $P = (x, y) \in S$ satisfies the formula

$$\kappa = \frac{y''}{[1 + (y')^2]^{3/2}} = y^{-1} \sin \alpha \cos \alpha,$$

where $\alpha \in [0, \pi)$ is the direction angle of the curve's tangent at the point P .

5. A body of mass m moving along the x -axis is acted on by a force $\vec{F} = f(x)\vec{i}$ directed along the x -axis, where $x = x(t)$ is the body's position at time t . From the equation of motion $mx'' = f(x)$ derive the energy principle

$$\frac{1}{2}mv_2^2 - \frac{1}{2}mv_1^2 = \int_{x_1}^{x_2} f(x) dx,$$

where v_i = the body's speed at the point x_i ($i = 1, 2$).

6. (*) In a mathematical pendulum a point mass swings on a weightless rod (length R), so the equation of motion is $R\theta'' = -g \sin \theta$, where $\theta = \theta(t)$ is the swing angle measured from the equilibrium position.
 - a) By multiplying the equation of motion by θ' , derive an energy principle

of the form $F(\theta, \theta') = \text{constant}$.

b) Using the result of part a), determine $\theta(t)$, $t \geq 0$ with the initial conditions

$$\theta(0) = 0, \theta'(0) = \frac{1}{T}, \quad T = \frac{1}{2} \sqrt{\frac{R}{g}}.$$

(In this special case $\theta(t)$ is an elementary function!)

7. (*) (The hare and the eagle-owl) At instant H a hare sets off from the origin running along the positive y -axis at speed v (constant). An eagle-owl at the point $(4, 0)$ notices the hare at the same instant and starts flying low so that its flight direction is at all times toward the hare. Derive for the eagle-owl's flight curve $y = y(x)$ a differential equation of order 2, assuming that the owl's flight speed is a) v , b) $2v$. Taking the initial conditions into account as well, determine the flight curve. At what point (if any) does the owl catch the hare?
8. (*) (Shrovetide, 2nd run) A snowboarder descends from the origin to the point $P = (a, -b)$ along the curve $y = -y(x)$ ($a > 0$, $b \geq 0$, direction of gravity $-\vec{j}$). If the rider's initial speed = 0 and there is no friction, then by the energy principle the rider's speed at the point $(x, -y(x))$ is determined by the equation $\frac{1}{2}[v(x)]^2 = gy(x)$, where g is the acceleration due to Earth's gravity. a) Show that the descent time to the point P is

$$t = \int_0^a \sqrt{\frac{1 + [y'(x)]^2}{2gy(x)}} dx.$$

b) From this expression it can be shown that the descent is the fastest possible if $y(x)$ satisfies the differential equation $2yy'' + 1 + (y')^2 = 0$. Show that this curve of fastest descent (the *brachistochrone*) is an arc of a cycloid.

IX.4 First-order linear DE

Besides the separable differential equation (or one reducible to it), another type of differential equation very common in applications is the *linear* DE. In this section the object of study is the first-order linear differential equation, whose general form is

$$y' + P(x)y = R(x). \quad (1)$$

Here P is the so-called *coefficient function*. Both P and the *right-hand side* of the equation, i.e. R , are assumed continuous on the interval (a, b) on which the solution is sought. As usual, the solution y is required to be differentiable on (a, b) . Since differentiability implies continuity, the differential equation shows that y' too is continuous on (a, b) .

Introducing the *operator* notation

$$Ly = \frac{dy}{dx} + P(x)y^*,$$

equation (1) can be written briefly as

$$Ly = R(x).$$

The equation is called linear because

$$L(c_1y_1 + c_2y_2) = c_1Ly_1 + c_2Ly_2, \quad c_1, c_2 \in \mathbb{R},$$

that is, L is a *linear* operator.[†]

If even a single particular solution $y = y_0(x)$ of the linear equation (1) is known, then finding the general solution of the equation becomes considerably easier. Indeed, if $y = y(x)$ is any other solution, then from the equations

$$\begin{cases} Ly = R(x), \\ Ly_0 = R(x) \end{cases}$$

it follows, by the linearity of L , that

$$0 = Ly - Ly_0 = L(y - y_0),$$

*In the case $P = 0$ we have $L = D$, cf. Section V.3. More generally one can write $L = D + P(x)$, in which case it is the sum of the operators $f \mapsto Df$ and $f \mapsto P(x)f$ as 'functions of a function'.

[†]If an operator is not linear, then it is *nonlinear*. Nonlinear is, for instance, the operator $L(y) = y' + y^2$.

that is, the difference $u(x) = y(x) - y_0(x)$ satisfies the simpler equation

$$y' + P(x)y = 0. \quad (2)$$

This is called the (first-order linear) *homogeneous* equation. If $R \neq 0$, then equation (1) is correspondingly called *inhomogeneous* or, especially in the context of solving, the *complete* equation. We have deduced:

The general solution of the linear equation (1)

$$\begin{aligned} &= \text{a particular solution of the complete equation (1)} \\ &+ \text{the general solution of the homogeneous equation (2)}. \end{aligned}$$

Insofar as a particular solution $y_0(x)$ is known, what remains is thus to find the general solution of the homogeneous equation (2). This is easy, since the equation is separable: by the methods of Section IX.2 the solution is obtained as

$$y(x) = Ce^{-\int P(x)dx}.$$

(Here there is no need to include another arbitrary constant C_1 in the antiderivative $\int P(x) dx$, since its effect is already contained in the constant C as the factor e^{-C_1} .) Equation (1) is thus solved by a single quadrature:

$$y(x) = y_0(x) + Ce^{-\int P(x)dx}.$$

If no particular solution is known, the equation is solved by two quadratures. Indeed, a particular solution can always be found using the procedure known as *variation of the constant* (which contains one quadrature). Variation of the constant means making, in the general solution of the homogeneous equation, the change $C \mapsto C(x)$ and testing the resulting function

$$y(x) = C(x)e^{-\int P(x)dx}$$

as a possible solution of the complete equation. Substituting into the equation gives

$$Ly = C'(x)e^{-\int P(x)dx} + C(x)L[e^{-\int P(x)dx}] = R(x).$$

Here $L[e^{-\int P(x)dx}] = 0$, so we get

$$\begin{aligned} C'(x)e^{-\int P(x)dx} = R(x) &\Leftrightarrow C'(x) = e^{\int P(x)dx} R(x) \\ &\Leftrightarrow C(x) = \int e^{\int P(x)dx} R(x) dx + C. \end{aligned}$$

By the fundamental theorem of calculus, this solution is valid on the interval (a, b) on which P and R are continuous. Leaving the constant C unfixed in the particular solution, the general solution of equation (1) can be written directly as

$$y(x) = e^{-\int P(x)dx} C(x) = e^{-\int P(x)dx} \left(C + \int e^{\int P(x)dx} R(x) dx \right).$$

The solution satisfying the initial condition $y(x_0) = Y_0$ ($x_0 \in (a, b)$) is unique and can be written directly by means of definite integrals:

$$y(x) = e^{-\int_{x_0}^x P(t)dt} \left(Y_0 + \int_{x_0}^x e^{\int_{x_0}^s P(t)dt} R(s) ds \right). \quad (3)$$

If here only x_0 is fixed and Y_0 is allowed to vary ($Y_0 \hookrightarrow C$), one again obtains the general solution.

EXAMPLE 1 Solve the differential equation

$$y' + \frac{2y}{1-x^2} = \frac{1+x}{(1-x)^3}.$$

Solution First solve the homogeneous equation:

$$\begin{aligned} \frac{dy}{dx} + \frac{2y}{1-x^2} = 0 &\Rightarrow \int \frac{dy}{y} = \int \frac{2dx}{x^2-1} \\ &\Rightarrow \ln|y| = \ln \left| \frac{x-1}{x+1} \right| + \ln|C| \\ &\Rightarrow y = C \frac{x-1}{x+1}. \end{aligned}$$

General solution by variation of the constant:

$$\begin{aligned} y = C(x) \frac{x-1}{x+1} &\Rightarrow C'(x) \frac{x-1}{x+1} = \frac{1+x}{(1-x)^3} \\ \Rightarrow C'(x) &= -\frac{(x+1)^2}{(x-1)^4} = -\frac{[(x-1)+2]^2}{(x-1)^4} \\ &= -\frac{1}{(x-1)^2} - \frac{4}{(x-1)^3} - \frac{4}{(x-1)^4} \\ \Rightarrow C(x) &= \frac{1}{x-1} + \frac{2}{(x-1)^2} + \frac{4}{3(x-1)^3} + C \\ &= \frac{3x^2+1}{3(x-1)^3} + C \\ \Rightarrow y(x) &= C(x) \frac{x-1}{x+1} = \frac{3x^2+1}{3(x-1)^2(x+1)} + C \frac{x-1}{x+1}. \end{aligned}$$

The solution is valid on the intervals $(-\infty, -1)$, $(-1, 1)$, and $(1, \infty)$. $\square$

The integrating-factor method

A more straightforward way than variation of the constant to solve equation (1) is to first multiply the equation by the so-called *integrating factor*, defined as

$$H(x) = e^{\int P(x)dx}.$$

Since $H' = P(x)H$,

$$H(x)[y' + P(x)y] = \frac{d}{dx}[H(x)y].$$

Consequently, and since $H(x) > 0 \forall x$, we obtain the solution scheme

$$\begin{aligned} y' + P(x)y = R(x) &\Leftrightarrow H(x)[y' + P(x)y] = H(x)R(x) \\ &\Leftrightarrow [H(x)y]' = H(x)R(x) \\ &\Leftrightarrow H(x)y(x) = \int H(x)R(x) dx \\ &\Leftrightarrow y(x) = H(x)^{-1} \int H(x)R(x) dx, \quad x \in (a, b). \end{aligned}$$

The reasoning again rests on the fundamental theorem of calculus and the continuity assumptions made. The result is the same as above, and the solution of the initial value problem is again obtained by changing the indefinite integrals into definite ones.

EXAMPLE 2 Solve the differential equation $y' + y \cot x = e^{\cos x}$.

Solution Since $\int \cot x dx = \ln |\sin x| + C$, on, e.g., the interval $(0, \pi)$ (on which $\cot x$ is continuous) one can choose as the integrating factor

$$H(x) = e^{\ln \sin x} = \sin x.$$

Following the above solution scheme, we deduce

$$\begin{aligned} y' + y \cot x = e^{\cos x} &\Leftrightarrow y' \sin x + y \cos x = \sin x e^{\cos x} \\ &\Leftrightarrow (y \sin x)' = \sin x e^{\cos x} \\ &\Leftrightarrow y \sin x = \int \sin x e^{\cos x} dx = -e^{\cos x} + C \\ &\Leftrightarrow y(x) = \frac{C - e^{\cos x}}{\sin x}. \end{aligned}$$

The solution is valid on the intervals $(n\pi, (n+1)\pi)$, $n \in \mathbb{Z}$. $\square$

The constant-coefficient equation

A special case of equation (1) very common in applications is the *constant-coefficient* first-order linear DE, which is of the form

$$y' + ay = f(x) \quad (a \in \mathbb{R}, a \neq 0). \quad (4)$$

(The case $a = 0$ is passed over as already familiar.) Its solution with the initial condition $y(x_0) = Y_0$ is, according to the solution formula (3),

$$\begin{aligned} y(x) &= e^{-a(x-x_0)} \left(Y_0 + \int_{x_0}^x e^{a(t-x_0)} f(t) dt \right) \\ &= Y_0 e^{-a(x-x_0)} + \int_{x_0}^x e^{a(t-x)} f(t) dt. \end{aligned}$$

If one does not wish to use this formula directly (in the general solution $Y_0 \leftrightarrow C$), a particular solution can often be found easily also by an 'educated guess'. The prerequisite is that $f(x)$ be of a sufficiently simple form.

EXAMPLE 3 Find a particular solution $y_0(x)$ of the equation $y' + y = f(x)$ when

$$\text{a) } f(x) = x^2, \quad \text{b) } f(x) = e^x, \quad \text{c) } f(x) = e^{-x}, \quad \text{d) } f(x) = \cos 2x.$$

Solution a) Let us try whether a particular solution might be a second-degree polynomial of the form $y_0(x) = Ax^2 + Bx + C$. Substituting into the equation gives

$$Ax^2 + (2A + B)x + (B + C) = x^2.$$

This holds for every x when

$$\begin{cases} A &= 1 \\ 2A + B &= 0 \\ B + C &= 0 \end{cases} \Leftrightarrow \begin{cases} A = 1 \\ B = -2 \\ C = 2 \end{cases}$$

So a particular solution is $y_0(x) = x^2 - 2x + 2$.

Solution b) Let us try a particular solution of the form $y_0(x) = Ae^x$:

$$2Ae^x = e^x \quad \forall x \Leftrightarrow A = \frac{1}{2}.$$

So $y_0(x) = \frac{1}{2}e^x$ works.

Solution c) Here the attempt $y_0(x) = Ae^{-x}$ fails, since this happens to be a solution of the homogeneous equation. Let us try

$$y_0(x) = Axe^{-x} : \quad Ae^{-x} - Axe^{-x} + Axe^{-x} = e^{-x} \Rightarrow A = 1.$$

So $y_0(x) = xe^{-x}$ works.

Solution d) Attempt: $y_0(x) = A \cos 2x + B \sin 2x$

$$\begin{aligned} \Rightarrow (A + 2B) \cos 2x + (-2A + B) \sin 2x &= \cos 2x \\ \Rightarrow \begin{cases} A + 2B = 1 \\ -2A + B = 0 \end{cases} &\Leftrightarrow \begin{cases} A = 1/5 \\ B = 2/5 \end{cases} \end{aligned}$$

So $y_0(x) = \frac{1}{5}(\cos 2x + 2 \sin 2x)$ is a particular solution. $\square$

When looking for a particular solution, it is worth noting also the linearity rule:

$$\begin{aligned} Ly_1 = f_1(x) \quad \& \quad Ly_2 = f_2(x) \\ \Rightarrow L(c_1y_1 + c_2y_2) &= c_1f_1(x) + c_2f_2(x). \end{aligned}$$

EXAMPLE 3 (continued) Find a particular solution when

$$\text{e) } f(x) = 4 \sin^2 x - 5x^2 - 2, \quad \text{f) } f(x) = 6 \sinh x - 10 \cos 2x.$$

Solution Since

$$\text{e) } f(x) = -2 \cos 2x - 5x^2, \quad \text{f) } f(x) = 3e^x - 3e^{-x} - 10 \cos 2x,$$

using the linearity rule and the results of parts a)-d), we get

$$\begin{aligned} \text{e) } y_0(x) &= -2 \cdot \frac{1}{5}(\cos 2x + 2 \sin 2x) - 5(x^2 - 2x + 2) \\ &= -\frac{2}{5} \cos 2x - \frac{4}{5} \sin 2x - 5x^2 + 10x - 10, \\ \text{f) } y_0(x) &= 3 \cdot \frac{1}{2} e^x - 3xe^{-x} - 10 \cdot \frac{1}{5}(\cos 2x + 2 \sin 2x) \\ &= \frac{3}{2} e^x - 3xe^{-x} - 2 \cos 2x - 4 \sin 2x. \quad \square \end{aligned}$$

The rules in the following table are generalizations of the above example. In the table, the differential equation (4) is assumed, where $a \neq 0$. In addition, $n \in \mathbb{N} \cup \{0\}$ and $\omega \neq 0$.

$f(x)$	$y_0(x)$ of the form
x^n	$A_n x^n + \cdots + A_0$
$x^n e^{bx}, \quad b \neq -a$	$(A_n x^n + \cdots + A_0) e^{bx}$
$x^n e^{-ax}$	$A x^{n+1} e^{-ax}$
$x^n (A \cos \omega x + B \sin \omega x)$	$(A_n x^n + \cdots + A_0) \cos \omega x + (B_n x^n + \cdots + B_0) \sin \omega x$

Periodic solutions

A special case of the constant-coefficient equation (4) common in applications is one in which the right-hand side f of the equation is periodic. In this case a particular solution y_p can be found that is likewise periodic, with the period of y_p = the period of f . (In the case $a = 0$ there is in general no periodic solution, see Exercise 7.) Let f be a function defined on all of $\mathbb{R}$, continuous and L -periodic. We seek an L -periodic solution by determining it on an interval of one period's length, e.g. on $[0, L]$. On this interval it suffices to impose the periodicity condition $y(0) = y(L)$, so we must find a function y_p that is continuous on $[0, L]$, differentiable on $(0, L)$, and satisfies

$$\begin{cases} y' + ay = f(x), & x \in (0, L), \\ y(0) = y(L). \end{cases}$$

The solution is of the form $y(x) = Ce^{-ax} + y_0(x)$, where y_0 is some particular solution of the differential equation, e.g. (cf. the solution formula above)

$$y_0(x) = \int_0^x e^{a(t-x)} f(t) dt.$$

The periodicity condition determines the constant C , giving the solution (assuming $a \neq 0$)

$$y_p(x) = \frac{y_0(L) - y_0(0)}{1 - e^{-aL}} e^{-ax} + y_0(x), \quad x \in [0, L].$$

When y_p is extended to all of $\mathbb{R}$ by the periodicity condition $y_p(x \pm L) = y_p(x)$, it can be deduced (Exercise 7a) that the y_p so defined is a solution of the differential equation on $\mathbb{R}$. Since y_p is thus a particular solution, the general solution on $\mathbb{R}$ is

$$y(x) = Ce^{-ax} + y_p(x).$$

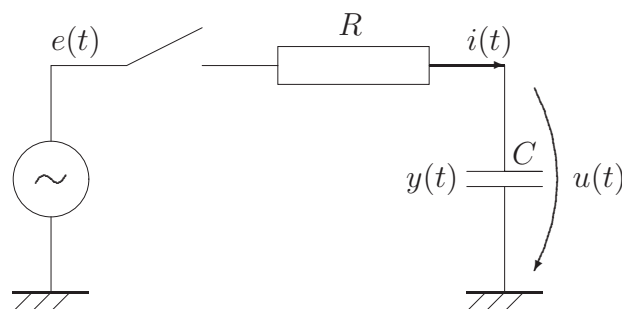
The initial condition $y(x_0) = Y_0$ determines the value of C as $C = (Y_0 - y_p(x_0))e^{ax_0}$.

In applications one usually has $a > 0$ and, in place of the variable x , time (t), so the above solution has two parts: the periodic part $y_p(t)$ and the so-called transient, which 'dies out' as $t \rightarrow \infty$. Below is an example from electrical engineering.

Application example: RC circuit

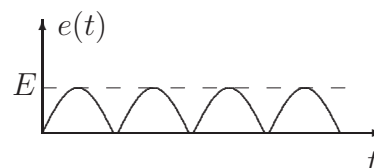
Problem description

In the electric circuit shown, a resistor (R) and a capacitor of capacitance $= C$ are connected in series. The current in the circuit at time t is $i(t)$, the charge of the capacitor $= y(t)$, and the voltage across the capacitor $= u(t)$.



The circuit is fed with a full-wave rectified voltage

$$e(t) = E|\sin \omega t|.$$



Mathematical model (Assume the switch closed)

$$\begin{aligned} Ri(t) + u(t) &= e(t), \\ y(t) &= Cu(t), \\ y'(t) &= i(t). \end{aligned}$$

Initial assumptions

At the instants $t \leq t_0$ we have $i(t) = q(t) = 0$. The switch is closed at the instant $t = t_0$ and kept closed thereafter.

Problem

Determine the charge $y(t)$ of the capacitor for $t \geq t_0$.

Solution

Denote $Q = EC$ and $\tau = RC$ (τ = the time constant). Eliminating $i(t)$ and $u(t)$, we obtain the initial value problem to be solved

$$\begin{cases} \tau y' + y = Q|\sin \omega t|, & t > t_0, \\ y(t_0) = 0. \end{cases}$$

Since the right-hand side of the differential equation is periodic, with period

$$T = \pi/\omega,$$

a particular solution y_p can be found for the equation that is likewise T -periodic. It is continuous on $[0, T]$, differentiable on $(0, T)$, and satisfies

$$\begin{cases} \tau y_p' + y_p = Q \sin \omega t, & t \in (0, T), \\ y_p(0) = y_p(T). \end{cases}$$

First we find a particular solution y_0 of the differential equation on $(0, T)$:

$$\begin{aligned} y_0(t) &= A \sin \omega t + B \cos \omega t \\ \Rightarrow (A - \tau \omega B) \sin \omega t + (\tau \omega A + B) \cos \omega t &= Q \sin \omega t, \quad 0 < t < T \\ \Leftrightarrow \begin{cases} A - \tau \omega B = Q \\ \tau \omega A + B = 0 \end{cases} \\ \Leftrightarrow A = \frac{Q}{k^2 + 1}, \quad B = -\frac{kQ}{k^2 + 1}, \quad k = \tau \omega. \end{aligned}$$

(Here the new parameter k is dimensionless.) The sought T -periodic solution must thus be, on $[0, T]$, of the form

$$y_p(t) = Ce^{-t/\tau} + \frac{Q}{k^2 + 1}(\sin \omega t - k \cos \omega t).$$

The constant C is determined by the periodicity condition:

$$\begin{aligned} y_p(0) = y_p(T) &\Leftrightarrow C - \frac{kQ}{k^2 + 1} = Ce^{-T/\tau} + \frac{kQ}{k^2 + 1} \\ &\Leftrightarrow C = \frac{2kQ}{(k^2 + 1)(1 - e^{-T/\tau})}. \end{aligned}$$

So on $[0, T]$ we have

$$y_p(t) = \frac{Q}{k^2 + 1} \left(\frac{2k}{1 - e^{-T/\tau}} e^{-t/\tau} + \sin \omega t - k \cos \omega t \right),$$

where

$$Q = EC, \quad \tau = RC, \quad T = \frac{\pi}{\omega}, \quad k = \tau\omega = \pi \frac{\tau}{T}.$$

The solution satisfying the initial condition $y(t_0) = 0$ is

$$y(t) = -y_p(t_0)e^{-(t-t_0)/\tau} + y_p(t), \quad t \geq t_0.$$

If it happens that $y_p(t_0) = 0$, the transient does not 'wake up'. $\square$

EXERCISES

1. Solve (general solution or solution of the initial value problem):

- a) $y' + 2xy = 2xe^{-x^2}$ b) $(1 + x^2)y' - 2xy = (1 + x^2)^2$
 c) $\cos xy' - \sin xy = xe^x$ d) $y' - y = \cosh x$
 e) $xy' + 2y = x^3$, $y(1) = 1$ f) $y' + y \cos x = \sin x \cos x$, $y(0) = 1$
 g) $y' + y \tan x = \sin^3 x$, $y(0) = 1$ h) $y' + |x - |x||y = x$, $y(0) = 0$
 i) $y' + 2y = x^3 - x$, $y(0) = 1$ j) $y' - y = e^x - \sin x$, $y(0) = 0$

2. a) The differential equation $y' + P(x)y = (x + 1)^2e^x$ has the solution $y = (x^2 - 1)e^x$. Determine the general solution.

b) Let the functions f and g be differentiable, with f' and g' continuous on (a, b) , and let $f(x) \neq 0 \forall x \in (a, b)$. Of which linear differential equation is $y(x) = Cf(x) + g(x)$, $C \in \mathbb{R}$ the general solution on (a, b) ?

3. Determine the function $y(x)$ continuous on $\mathbb{R}$ that satisfies the equation

$$2 \int_0^x ty(t) dt = x^2 + y(x), \quad x \in \mathbb{R}.$$

4. Show that the *Bernoulli* DE $y' = A(x)y + B(x)y^k$ ($k \in \mathbb{R}$, $k \neq 0$, $k \neq 1$) reduces to a linear one by the substitution $u = y^{1-k}$. Using this principle, solve

- a) $xy' + y = x^3y^2$ b) $y' + y = y^2(\cos x - \sin x)$
 c) $3y' + y = (1 - 2x)y^4$ d) $y' + 2y/(1 - x) = 4(x^2 - x)\sqrt{y}$

5. At the point P of the curve $y = y(x)$ a tangent is drawn, which meets the y -axis at the point Q . Determine all curves with the property: the area of the triangle OPQ (O = the origin) $= a^2 = \text{constant}$.

6. The function $R(x)$ is continuous on $\mathbb{R}$ and $-2x^2 \leq R(x) \leq x^2 \forall x \in \mathbb{R}$. The function $y(x)$ is the solution of the initial value problem $y' + 3x^2y = R(x)$, $y(0) = 0$. What values can $y(-1)$ take?
7. Let f and y_p be functions defined on all of $\mathbb{R}$, continuous and L -periodic, and let y_p additionally be a solution of the differential equation $y' + ay = f(x)$ ($a \in \mathbb{R}$) on the interval $(0, L)$.
- a) Show that $y'_p + ay_p = f(x) \forall x \in \mathbb{R}$.
- b) Let F and Y be the averages of f and y_p on $[0, L]$. Show that $F = aY$.
— Conclusion if $a = 0$?

8. Let

$$f(x) = \begin{cases} 0, & \text{if } x = k \in \mathbb{Z}, \\ 2x - 2k - 1, & \text{if } x \in (k, k+1), k \in \mathbb{Z}. \end{cases}$$

Determine the function y_p that is (1) periodic, (2) defined and continuous on all of $\mathbb{R}$, and (3) a solution of the differential equation $y' + y = f(x)$ on the intervals $(k, k+1)$, $k \in \mathbb{Z}$. Sketch y_p graphically on $[0, 2]$. Is the differential equation satisfied at the point $x = 1$?

9. (*) a) Show that the solution of the initial value problem $y' = x + |y|$, $y(-2) = 1$ has a minimum $y(\ln 2 - 1) = \ln 2 - 1$.
- b) Determine the general solution of the differential equation $y' = |y - x|$. Also compute $y_1(1)$ and $y_2(1)$ for the particular solutions satisfying the conditions $y_1(-1) = -1/2$ and $y_2(-1) = 1/2$.
10. (*) The *Riccati* DE is of the form $y' = A(x) + B(x)y + C(x)y^2$. If a particular solution y_0 is known for it, then the equation reduces to a Bernoulli DE (see Exercise 4) by the substitution $y = y_0 + u$. Using this principle, solve the following differential equations using the additional information given.
- a) $y' = 1 + x + x^2 - (2x + 1)y + y^2$, $y_0(x)$ = a polynomial of degree 1
- b) $y' = y^2 - x^2y - (x - 1)^2$, $y_0(x)$ = a polynomial of degree 2
- c) $x^2y' + (xy - 2)^2 = 0$, $y_0(x) = a/x$ ($a \in \mathbb{R}$)
11. (*) The function $y(x)$ is differentiable on $(0, \infty)$, right-continuous at $x = 0$, with $y(0^+) = 1$. It is further known that $y'(x) + y(x) \geq f(x)$ when $x > 0$, where f is a function continuous on $[0, \infty)$. Show that for every $x > 0$

$$y(x) \geq e^{-x} + \int_0^x e^{t-x} f(t) dt.$$

12. (*) Let $\tau > 0$. Solve the differential equation

$$\tau y' + y = |2k - t|, \quad t \in (0, \infty) \cap [2k - 1, 2k + 1], \quad k \in \mathbb{N} \cup \{0\}$$

in the form $y(t) = Ce^{-t/\tau} + y_p(t)$, where y_p is periodic with period $T = 2$. Sketch graphically the solution satisfying the initial condition $y(0) = 2$ for the values 10, 1, and 0.1 of τ .

IX.5 Linear, constant-coefficient DEs.

Euler's differential equation

In the previous section we already solved the first-order linear, constant-coefficient differential equation $y' + ay = f(x)$. Common in applications is also the corresponding second-order differential equation, whose general form is

$$y'' + ay' + by = f(x), \quad a, b \in \mathbb{R}. \quad (1)$$

To solve this, we first consider the corresponding homogeneous equation

$$y'' + ay' + by = 0. \quad (2)$$

When we seek its solution with the ansatz

$$y(x) = e^{rx},$$

then substituting into the equation gives, for determining the constant r , the *characteristic equation*

$$r^2 + ar + b = 0 \quad (\text{characteristic equation}).$$

The roots of the characteristic equation are

$$r = -\frac{a}{2} \pm \sqrt{\frac{a^2}{4} - b},$$

whence there are three possibilities:

1. $a^2 - 4b > 0$. In this case the characteristic equation has two distinct real roots r_1 and r_2 , and hence the homogeneous equation (2) has the solutions

$$y_1(x) = e^{r_1 x}, \quad y_2(x) = e^{r_2 x}.$$

A solution is also $y(x) = C_1 y_1(x) + C_2 y_2(x)$ ($C_1, C_2 \in \mathbb{R}$), for if equation (2) is written as $Ly = 0$, then L is a linear operator, so that $L(C_1 y_1 + C_2 y_2) = C_1 L y_1 + C_2 L y_2 = 0$. Thus every $y(x)$ of the form

$$y(x) = C_1 e^{r_1 x} + C_2 e^{r_2 x}, \quad C_1, C_2 \in \mathbb{R}$$

is a solution of the homogeneous equation (2). We now show that this is the general solution of equation (2), i.e. that any solution is of this form. With this in mind, we first write the characteristic equation in the form

$$r^2 + ar + b = (r - r_1)(r - r_2) = r^2 - (r_1 + r_2)r + r_1 r_2 = 0,$$

where thus $a = -r_1 - r_2$ and $b = r_1 r_2$. Let $y(x)$ be a solution of the homogeneous equation (2). Then, differentiating the function $u(x) = y'(x) - r_2 y(x)$, one finds that

$$u' - r_1 u = y'' - (r_1 + r_2)y' + r_1 r_2 y = y'' + ay' + by = 0.$$

Thus $u(x)$ and $y(x)$ can be solved from the system of differential equations

$$\begin{cases} u' - r_1 u = 0, \\ y' - r_2 y = u(x). \end{cases}$$

When the general solution $u(x) = C_1 e^{r_1 x}$ ($C_1 \in \mathbb{R}$) of the first equation here is substituted into the latter equation, one finds its particular solution to be $y_0(x) = (r_1 - r_2)^{-1} C_1 e^{r_1 x}$ (cf. the previous section), and hence the general solution to be

$$y(x) = \frac{C_1}{r_1 - r_2} e^{r_1 x} + C_2 e^{r_2 x}.$$

Here one may just as well write $(r_1 - r_2)C_1$ in place of C_1 , so one finds that every solution of equation (2) is of the form $y(x) = C_1 e^{r_1 x} + C_2 e^{r_2 x}$, $C_1, C_2 \in \mathbb{R}$. Every such function was also a solution, so this is the general solution.

2. $a^2 - 4b = 0$. In this case the characteristic equation has the double root $r = -a/2$, so in the above system $r_1 = r_2 = r$. When the general solution $u(x) = C_1 e^{rx}$ ($C_1 \in \mathbb{R}$) of the first equation is substituted into the latter, its particular solution is $y_0(x) = C_1 x e^{rx}$ (cf. the previous section), and hence the general solution is

$$y(x) = (C_1 x + C_2) e^{rx}, \quad r = -\frac{a}{2}.$$

This is also the general solution of equation (2).

3. $a^2 - 4b < 0$. In this case the roots of the characteristic equation form a conjugate pair

$$r_{1,2} = \alpha \pm i\beta, \quad \alpha = -\frac{a}{2}, \quad \beta = \sqrt{b - \frac{a^2}{4}}.$$

By the substitution $y(x) = e^{\alpha x} u(x)$ equation (2) transforms into the form

$$u'' + \beta^2 u = 0.$$

This can be solved as a DE reducible to first order (see Section IX.3), but one can also reason more directly that the solutions are $u_1(x) = \cos \beta x$ and $u_2(x) = \sin \beta x$, and hence the general solution is evidently $u(x) = C_1 \cos \beta x + C_2 \sin \beta x$, $C_1, C_2 \in \mathbb{R}$. In this way the general solution of equation (2) is obtained as

$$y(x) = e^{\alpha x} (C_1 \cos \beta x + C_2 \sin \beta x).$$

EXAMPLE 1 Solve the boundary value problem

$$\begin{cases} y'' - y' + 2y = 0, & x \in (0, 1), \\ y(0) = 1, \quad y(1) = 0. \end{cases}$$

Solution The roots of the characteristic equation $r^2 - r - 2$ are $r_1 = 2$, $r_2 = -1$, so the general solution of the differential equation is

$$y(x) = C_1 e^{2x} + C_2 e^{-x}.$$

The boundary conditions are satisfied when

$$\begin{cases} C_1 + C_2 = 1 \\ e^2 C_1 + e^{-1} C_2 = 0 \end{cases} \Leftrightarrow \begin{cases} C_1 = -1/(e^3 - 1), \\ C_2 = e^3/(e^3 - 1). \end{cases}$$

Thus the solution of the boundary value problem is

$$y(x) = \frac{1}{e^3 - 1}(-e^{2x} + e^{3-x}). \quad \square$$

The complete constant-coefficient equation

Since the differential equation (1) is linear, its solution obeys the same general principle as for the first-order linear DE: the general solution of the complete equation = the general solution of the homogeneous equation + a particular solution $y_0(x)$ of the complete equation. To determine the particular solution, we consider here only 'educated guess' methods, which work when $f(x)$ in equation (1) is of a sufficiently simple form. (A more general method is presented in the next section; see also Exercise 9.)

EXAMPLE 2 Solve $y'' + y = x^2 + 2e^{2x}$.

Solution We try a particular solution of the form

$$y(x) = Ax^2 + Bx + C + De^{2x}.$$

Substituting into the equation, one finds this to be a solution when $A = 1$, $B = 0$, $C = -2$ and $D = 2/5$. The general solution of the homogeneous equation is

$$y(x) = C_1 \cos x + C_2 \sin x,$$

so the general solution of the complete equation is

$$y(x) = C_1 \cos x + C_2 \sin x + x^2 - 2 + \frac{2}{5}e^{2x}. \quad \square$$

If more generally the right-hand side $R(x)$ of equation (1) is of the form

$$\text{a) } R(x) = x^n e^{\alpha x}, \quad \text{or} \quad \text{b) } R(x) = x^n (A \cos \omega x + B \sin \omega x),$$

where $n \in \{0, 1, 2, \dots\}$, $\alpha \in \mathbb{R}$, and $\omega \in \mathbb{R}$, $\omega \neq 0$, then a particular solution of equation (1) can be found correspondingly in the form

$$\text{a) } y(x) = p(x) e^{\alpha x},$$

$$\text{b) } y(x) = p(x) \cos \omega x + q(x) \sin \omega x,$$

where p and q are polynomials. As a rule p and q are of degree n . An exception arises in those cases where the ansatz given by the rule happens to be a solution of the homogeneous equation. In case a) this is possible when $n = 0$ or $n = 1$, in case b) when $n = 0$. Then the degree of the polynomial must be raised by one — in case a) possibly by two — in order to find a particular solution.

EXAMPLE 3 Solve the initial value problem

$$\begin{cases} y'' + 2y' + y = e^{-x}, & x \in \mathbb{R}, \\ y(1) = y'(1) = 0. \end{cases}$$

Solution The characteristic equation has the double root $r = -1$, so the general solution of the homogeneous equation is $y(x) = (C_1 + C_2 x)e^{-x}$. Since both $f(x) = e^{-x}$ and xe^{-x} are solutions of the homogeneous equation, a particular solution must be sought in the form

$$y(x) = (Ax^2 + Bx + C)e^{-x}.$$

This turns out to be a solution when we choose $A = 1/2$ and $B, C \in \mathbb{R}$, i.e. we obtained directly the general solution

$$y(x) = \left(\frac{1}{2}x^2 + Bx + C\right)e^{-x}, \quad B, C \in \mathbb{R}.$$

(The same result would have been reached if in the ansatz one had 'more wisely' chosen $B = C = 0$ and added the general solution of the homogeneous equation only afterwards.) The initial conditions are satisfied when $A = -1$ and $B = 1/2$, so the solution of the initial value problem is

$$y(x) = \frac{1}{2}(x-1)^2 e^{-x}. \quad \square$$

Complex-valued solutions

When solving constant-coefficient differential equations it is often convenient to carry out the computations using complex-valued functions, even when one is aiming at a real final result. The method is especially convenient when the roots of the characteristic equation are complex numbers, or when trigonometric functions appear on the right-hand side of the complete equation (1). When computing with complex functions, one admits in the solution ansatz of the homogeneous equation

$$y(x) = e^{rx}$$

directly also complex values of r . This is then a complex-valued function of a real variable, whose derivative is defined in the usual way, i.e. as the limit of the difference quotient. When, in the definition of the derivative, we write in the case of the function in question

$$z = rx, \quad \Delta z = r\Delta x,$$

then one sees that

$$\frac{y(x + \Delta x) - y(x)}{\Delta x} = r \frac{e^{z+\Delta z} - e^z}{\Delta z}.$$

Here $\Delta x \rightarrow 0 \Rightarrow \Delta z \rightarrow 0$, so by the differentiation rule for the complex function e^z (see Section VI.3) one can conclude that, as expected,

$$\boxed{\frac{d}{dx} e^{rx} = r e^{rx}, \quad r \in \mathbb{C}.}$$

EXAMPLE 4 Solve the differential equation (2) using complex functions, when $a^2 - 4b < 0$.

Solution The roots of the characteristic equation are $r_{1,2} = \alpha \pm i\beta$, so the general solution is obtained as before by solving the system

$$\begin{cases} u' - r_1 u = 0, \\ y' - r_2 y = u(x). \end{cases}$$

As before (cf. the case $r_1, r_2 \in \mathbb{R}$ above), the general solution is obtained as

$$\begin{aligned} y(x) &= C_1 e^{r_1 x} + C_2 e^{r_2 x} \\ &= e^{\alpha x} (C_1 e^{i\beta x} + C_2 e^{-i\beta x}), \quad C_1, C_2 \in \mathbb{C}. \end{aligned}$$

— Note that here there is no reason to restrict the coefficients C_1, C_2 to be real. Taking into account Euler's formula

$$e^{\pm i\beta x} = \cos \beta x \pm i \sin \beta x,$$

one sees that the solution can be expressed equivalently in the form

$$y(x) = e^{\alpha x}(D_1 \cos \beta x + D_2 \sin \beta x),$$

$$\text{where } \begin{cases} D_1 = C_1 + C_2, \\ D_2 = i(C_1 - C_2). \end{cases}$$

From this one sees that by choosing $C_2 = \overline{C_1}$ (the conjugate of C_1) one obtains the general real solution in the form presented earlier. $\square$

By the procedure of the example one can also find the general solution of the homogeneous equation (2) in the complex-coefficient case ($a, b \in \mathbb{C}$), should such a situation arise. Instead, let us examine a computational technique that is very common in applications, in which for the real-coefficient complete equation

$$Ly = y'' + ay' + by = \begin{cases} \sin \omega x \\ \cos \omega x \end{cases}$$

one seeks a particular solution using complex functions. The method rests on the following simple observation: If $f(x) = f_1(x) + if_2(x)$, where f_1 and f_2 are real-valued, then y is a particular solution of the equation $Ly = f$ precisely when $y = y_1 + iy_2$, where $Ly_1 = f_1$ and $Ly_2 = f_2$. (This is easily verified from the linearity of L and the fact that the coefficients a, b are real — in the complex-coefficient case the rule does not hold.)

EXAMPLE 5 Find a particular solution of the differential equation $y'' - y' + y = \sin \omega x$ ($\omega \neq 0$) using complex functions.

Solution Since $\sin \omega x = \operatorname{Im}(e^{i\omega x})$, the problem can be solved by finding a complex-valued particular solution of the differential equation

$$y'' - y' + y = e^{i\omega x},$$

and taking the imaginary part of the solution. The complex solution is easily found by substituting the ansatz $y(x) = Ae^{i\omega x}$ ($A \in \mathbb{C}$) into the equation:

$$\begin{aligned} (-\omega^2 - i\omega + 1)Ae^{i\omega x} &= e^{i\omega x} \quad \forall x \\ \Rightarrow A &= \frac{1}{1 - \omega^2 - i\omega} = \frac{1 - \omega^2 + i\omega}{\omega^4 - \omega^2 + 1} \\ \Rightarrow y(x) &= \frac{1 - \omega^2 + i\omega}{\omega^4 - \omega^2 + 1} e^{i\omega x} = \frac{1 - \omega^2 + i\omega}{\omega^4 - \omega^2 + 1} (\cos \omega x + i \sin \omega x). \end{aligned}$$

The required solution is the imaginary part of this, i.e.

$$y(x) = \frac{1}{\omega^4 - \omega^2 + 1} [\omega \cos \omega x + (1 - \omega^2) \sin \omega x]. \quad \square$$

In the example one could of course have managed with the real ansatz $y(x) = A \cos \omega x + B \sin \omega x$ ($A, B \in \mathbb{R}$) as well, but the computation would have been more unpleasant.

Euler's differential equation

A differential equation of the form

$$x^2 y'' + axy' + by = f(x)$$

is called the (second-order) *Euler* DE type. In the basic form

$$y'' + ax^{-1}y' + bx^{-2}y = x^{-2}f(x)$$

$x = 0$ is a point of discontinuity of the coefficients, so Euler's differential equation must be considered separately on the intervals $(-\infty, 0)$ and $(0, \infty)$ — in applications usually on the interval $(0, \infty)$. On each interval the equation reduces to a constant-coefficient one by the substitution

$$|x| = e^t \Leftrightarrow t = \ln |x|,$$

for when, e.g. on the interval $(0, \infty)$, we write

$$y(x) = y(e^t) = u(t) = u(\ln x),$$

then we obtain

$$\begin{aligned} y'(x) &= \frac{1}{x} u'(\ln x) = \frac{1}{x} u'(t), \\ y''(x) &= \frac{1}{x^2} u''(\ln x) - \frac{1}{x^2} u'(\ln x) = \frac{1}{x^2} [u''(t) - u'(t)], \end{aligned}$$

so the equation is brought to the constant-coefficient form

$$u''(t) + (a - 1)u'(t) + bu(t) = f(e^t).$$

On the interval $(-\infty, 0)$ the right-hand side becomes $f(-e^t)$; otherwise the result is the same.

In practice one need not transform the homogeneous Euler equation to constant-coefficient form, since a solution can be sought directly with the ansatz

$$y(x) = x^r.$$

Then, to determine r , one obtains a second-degree characteristic equation. Depending on the nature of the roots, the general solution is one of the following:

$$\text{a) } y(x) = C_1 x^{r_1} + C_2 x^{r_2},$$

$$\text{b) } y(x) = x^r (C_1 + C_2 \ln |x|),$$

$$\text{c) } y(x) = x^\alpha [C_1 \cos(\beta \ln |x|) + C_2 \sin(\beta \ln |x|)].$$

In case a) the roots are real and distinct, in case b) there is a real double root, and in case c) the roots form a conjugate pair $\alpha \pm i\beta$. The claimed forms of the general solution can be deduced by means of the above transformation.

The complete Euler equation is solved in favorable cases by an 'educated guess'. The form of the guess can be derived by thinking of the transformation to the constant-coefficient situation.

EXAMPLE 6 Solve the initial value problem

$$\begin{cases} x^2 y'' + 3xy' + y = 1/x, & x > 0, \\ y(1) = 1, & y'(1) = 0. \end{cases}$$

Solution In the homogeneous equation the substitution $y(x) = x^r$ leads to the characteristic equation

$$r(r-1) + 3r + 1 = r^2 + 2r + 1 = 0.$$

This has the double root $r = -1$, so the general solution of the homogeneous equation is

$$y(x) = \frac{1}{x} (C_1 + C_2 \ln x).$$

Since the right-hand side $f(x) = x^{-1}$ is a solution of the homogeneous equation, as is $x^{-1} \ln |x|$, a particular solution of the complete equation must be sought in the form

$$y_0(x) = Ax^{-1}(\ln x)^2.$$

When this, together with the derivatives,

$$\begin{aligned}y'_0(x) &= Ax^{-2}[-(\ln x)^2 + 2 \ln x], \\y''_0(x) &= Ax^{-3}[2(\ln x)^2 - 6 \ln x + 2],\end{aligned}$$

is substituted into the equation, one obtains a solution when $A = 1/2$. Thus the general solution is

$$y(x) = \frac{1}{x} \left(C_1 + C_2 \ln x + \frac{1}{2}(\ln x)^2 \right).$$

The initial conditions are satisfied when we choose $C_1 = C_2 = 1$. $\square$

Higher-order constant-coefficient DEs

The general linear, constant-coefficient, homogeneous, n th-order differential equation

$$y^{(n)} + a_{n-1}y^{(n-1)} + \cdots + a_0y = 0$$

can be solved with the ansatz $y(x) = e^{rx}$, as in the cases $n = 1, 2$. One obtains the characteristic equation

$$r^n + a_{n-1}r^{n-1} + \cdots + a_0 = 0,$$

to each root of which there corresponds a number of linearly independent solutions equal to the multiplicity of the root. If the root is a real root of multiplicity m , then these solutions are

$$x^k e^{rx}, \quad k = 0 \dots m-1.$$

If it is a conjugate pair $\alpha \pm i\beta$ of multiplicity m , then the corresponding solutions are

$$x^k e^{\alpha x} \cos \beta x, \quad x^k e^{\alpha x} \sin \beta x, \quad k = 0 \dots m-1.$$

The general solution of the homogeneous equation is obtained as a linear combination of all these solutions (n in total).

EXAMPLE 7 Solve $y''' - y = x^2$, $x \in \mathbb{R}$.

Solution A particular solution is $y(x) = -x^2$. In solving the homogeneous equation, the characteristic equation becomes $r^3 - 1 = 0$. The roots are

$$r_1 = 1, \quad r_{2,3} = -\frac{1}{2} \pm i\frac{\sqrt{3}}{2},$$

so the general solution is

$$y(x) = C_1 e^x + e^{-x/2} \left(C_2 \cos \frac{\sqrt{3}x}{2} + C_3 \sin \frac{\sqrt{3}x}{2} \right) - x^2. \quad \square$$

As in the example, the solution of the complete equation can often be found by trial. There is hardly any practical need for more general methods.

Let us note finally that the general n th-order homogeneous Euler DE

$$x^n y^{(n)} + a_{n-1} x^{n-1} y^{(n-1)} + \dots + a_0 y = 0$$

is solved with the ansatz $y(x) = x^r$ in the same way as above.

EXERCISES

1. Solve (the general solution in parts a)–n), the general solution for all pairs $(a, b) \in \mathbb{R}^2$ in parts o)–t), the solution of the initial or boundary value problem in parts u)–ö):

- | | |
|---|---|
| a) $y'' + y' - 30y = 0$ | b) $y'' + 4y' + y = 0$ |
| c) $y'' + 6y' + 10y = 0$ | d) $y'' + 4y' + 6y = 0$ |
| e) $y'' + 4y' + 5y = 3x - 2$ | f) $y'' - y' - 2y = x^2$ |
| g) $y'' - 7y' + 6y = \sin x$ | h) $y'' + 4y = \sin 3x$ |
| i) $y'' + 2y' + 5y = e^{-x}$ | j) $y'' + y' - 2y = x + e^x$ |
| k) $y'' - 4y = xe^{2x}$ | l) $y'' + 4y' + 4y = (x - 1)^2 e^{-2x}$ |
| m) $y'' - 6y' + 9y = (xe^x)^3$ | n) $y'' + 4y' + 4y = e^{-2x} + \sin x$ |
| o) $y'' + ay' = e^{bx}$ | p) $y'' - a^2 y = e^{bx}$ |
| q) $y'' + a^2 y = e^{bx}$ | r) $y'' + a^2 y = \sin bx$ |
| s) $y'' + a^2 y = x \sin bx$ | t) $y'' + 2y' + (1 + a^2)y = e^{bx}$ |
| u) $y'' + 2y' + y = 0, \quad y(0) = 0, \quad y'(0) = 1$ | |
| v) $y'' + 2y' + 2y = 0, \quad y(0) = 1, \quad y'(0) = 0$ | |
| x) $y'' - 4y' - 5y = 0, \quad y(0) = 1, \quad \lim_{x \rightarrow \infty} y(x) = 0$ | |
| y) $y'' + 3y' + 2y = 0, \quad y(0) = y(1) = 1$ | |
| x) $y'' - 4y' + 5y = \sin x, \quad y(0) = 0, \quad y'(0) = -1$ | |
| å) $y'' - 3y' + x^2 - 1 = 0, \quad y(0) = 1, \quad y'(0) = 0$ | |
| ä) $y'' - 5y' + 6y = e^x, \quad y(0) = y(1) = 0$ | |
| ö) $4y'' + 8y' + 5y = x^2, \quad y(0) = y(\pi) = 1$ | |

2. For the following differential equations, first find a complex-valued particular solution when the right-hand side of the equation is $f(x) = e^{i\omega x}$ ($\omega \in \mathbb{R}$, $\omega \neq 0$), and use it to determine a real particular solution.
- a) $y'' - 2y' + 2y = \sin \omega x$ b) $y'' - 3y' + 2y = \sin \omega x$
 c) $y'' + 4y' + 4y = \cos \omega x$ d) $y'' + y' + y = \sin \omega x - 2 \cos \omega x$
3. (Resonance) A body of mass $= m$ and position $= y(t)$ at time t is subject to a spring force and, in addition, an external load $f(t)$, so that $y(t)$ satisfies the equation of motion $my'' = -ky + f(t)$. Determine $y(t)$, $t \geq 0$ with the initial conditions $y(0) = y'(0) = 0$, when $f(t) = F \sin \omega t$ ($F = \text{constant}$). Consider separately the cases a) $\omega \neq \omega_0$ and b) $\omega = \omega_0$, where $\omega_0 = \sqrt{k/m}$ (the so-called resonance frequency).
4. (Bump) The front wheel of a car hits a bump, so that the equation of motion of the wheel in the vertical direction is $my'' = F_1 + F_2$, where $m =$ the mass of the wheel, $F_1 = -ky$ is the spring force and $F_2 = -cy'$ is the resisting force of the shock absorber. Sketch the solution $y(t)$ (time unit s) with the initial conditions $y(0) = 0$ m, $y'(0) = 60$ m/s, when $m = 15$ kg, $k = 6 \cdot 10^5$ kg/s² and the shock absorber is a) new: $c = 7500$ kg/s, b) due to be replaced soon: $c = 6000$ kg/s, c) due to be replaced immediately: $c = 4500$ kg/s.
5. Solve (in part i) for all $(a, b) \in \mathbb{R}^2$:
- a) $x^2y'' - xy' + y = 0$ b) $x^2y'' - 6xy' + 7y = 0$
 c) $x^2y'' - 2xy' + 2y = x$ d) $x^2y'' + xy' - y = (x+1)^2/x^2$
 e) $x^2y'' + xy' + y = x^3 - 2x$ f) $x^2y'' - 3xy' + 4y = x^2 \ln |x|$
 g) $(x^2 + 2x + 1)y'' + (x+1)y' + y = x^2$
 h) $(3x+2)^2y'' + (9x+6)y' - 36y = 81x + 18$
 i) $x^2y'' + (2a+1)xy' + by = 0$
6. a) Determine all functions differentiable on the interval $(0, \infty)$ that satisfy the equation $x^2y'(x) = 2 \int_0^x y(t) dt$, $x > 0$.
 b) Solve the differential equation $x^2y''' + 2(x^2 - x)y'' + (x^2 - 2x + 2)y' = x^3$ by making the substitution $u = y'/x$.
7. Solve (the general solution or the solution of the initial value problem):
- a) $y^{(5)} + 2y''' + y' = 0$ b) $y^{(7)} + 3y^{(6)} + 3y^{(5)} + y^{(4)} = 0$
 c) $y''' + 3y'' - 2y = \sin x$ d) $y^{(4)} + 3y''' + 3y'' + y' = 2e^{-2x} - 2x$
 e) $y^{(4)} + a^4y = x^2$, $a \geq 0$
 f) $y''' - (a+2)y'' + (2a+1)y' - ay = x + 1$, $a \in \mathbb{R}$
 g) $x^3y''' + 2ax^2y'' - 4a(xy' - y) = 0$, $a \in \mathbb{R}$
 h) $y''' - 5y'' + 17y' - 13y = 0$, $y(0) = y'(0) = 0$, $y''(0) = 24$

- i) $y''' - y'' - y' + y = e^x$, $y(0) = y'(0) = y''(0) = 0$
- j) $y''' - ay' = 0$, $y(0) = 0$, $y'(0) = 1$, $y''(0) = 2$, $a \in \mathbb{R}$
- k) $x^3 y''' + xy' - y = \sqrt{x}$, $y(1) = 1$, $y'(1) = y''(1) = 0$

8. (*) (Vertical firing) A bullet is fired straight upward with initial velocity 800 m/s. During the flight the bullet is subject to a resisting force proportional to the velocity, so that the height $y(t)$ satisfies the differential equation

$$y'' + ay' + g = 0,$$

where $g = 9.8 \text{ m/s}^2$, $a = 0.10 \text{ s}^{-1}$ during the bullet's ascent and $a = 0.20 \text{ s}^{-1}$ during the fall. Determine the peak height of the bullet's flight, the time of flight, and the return velocity of the bullet as it falls to the ground.

9. (*) In the constant-coefficient differential equation $y'' + ay' + by = f(x)$, let the roots of the characteristic equation be $r_1, r_2 \in \mathbb{C}$. By writing the differential equation as a system for the functions y and $u = y' - r_2 y$, show that the particular solution of the equation with initial conditions $y(x_0) = y'(x_0) = 0$ is

$$y_0(x) = \int_{x_0}^x e^{r_1(x-t)} \left[\int_{x_0}^t e^{r_2(t-s)} f(s) ds \right] dt.$$

Verify that the formula works when $a = -2$, $b = 1$, $x_0 = 0$ and $f(x) = x^2 - 4x + 2$.

IX.6 *The general 2nd-order linear DE

The general second-order linear differential equation is of the form

$$y'' + P(x)y' + Q(x)y = R(x). \quad (1)$$

The basic assumption is that both the *coefficient functions* P and Q and the *right-hand side* R of the equation are continuous on the interval (a, b) under consideration. A solution is then taken to be any function $y(x)$ twice differentiable on that interval that satisfies the equation. — One can then read off from the equation that y'' too is continuous on the interval (a, b) . In what follows we first consider the homogeneous special case of equation (1), in which $R(x) = 0$.

The homogeneous equation: solution space

When the homogeneous differential equation is written

$$Ly = y'' + P(x)y' + Q(x)y = 0, \quad (2)$$

then the operator L is again linear:

$$L(c_1y_1 + c_2y_2) = c_1Ly_1 + c_2Ly_2, \quad c_1, c_2 \in \mathbb{R}.$$

Thus if y_1 and y_2 are solutions of equation (2), then a solution is also any linear combination of them

$$y(x) = C_1y_1(x) + C_2y_2(x), \quad C_1, C_2 \in \mathbb{R}.$$

The central result of solvability theory is that this is in fact the general solution of equation (2), provided only that the functions y_1 and y_2 are *linearly independent* on the interval (a, b) . By this is meant that (cf. Section IV.5)

$$c_1y_1(x) + c_2y_2(x) = 0 \quad \forall x \in (a, b) \Rightarrow c_1 = c_2 = 0.$$

If this does not hold, then y_1 and y_2 are said to be *linearly dependent* on the interval (a, b) . In this case either $y_2 = cy_1$ or $y_1 = cy_2$ on that interval for some $c \in \mathbb{R}$.

***THEOREM IX.6.1** If in the differential equation (2) the coefficient functions P and Q are continuous on the interval (a, b) , then the equation has on that interval two linearly independent solutions. Furthermore, if y_1 and y_2 are any two linearly independent solutions of equation (2) on the interval (a, b) , then the general solution on that interval is

$$y(x) = C_1y_1(x) + C_2y_2(x), \quad C_1, C_2 \in \mathbb{R}.$$

EXAMPLE 1 In the basic form (2) of the Euler differential equation

$$x^2 y'' - 3xy' - 3y = 0$$

the coefficient functions are $P(x) = -3/x$ and $Q(x) = -3/x^2$. These are continuous on the intervals $(-\infty, 0)$ and $(0, \infty)$, and on these intervals the general solution is (see the previous section)

$$y(x) = C_1 x + C_2 x^3, \quad C_1, C_2 \in \mathbb{R}.$$

The result is in agreement with Theorem IX.6.1, for $y_1(x) = x$ and $y_2(x) = x^3$ are linearly independent on every interval (a, b) :

$$c_1 x + c_2 x^3 = 0 \quad \forall x \in (a, b) \quad \Leftrightarrow \quad c_1 = c_2 = 0. \quad \square$$

EXAMPLE 1 (continued) If the example's differential equation $x^2 y'' - 3xy' - 3y = 0$ is considered on the interval $(-\infty, \infty)$, then one observes that a solution is also

$$y_3(x) = |x|^3 = \begin{cases} x^3, & \text{when } x \geq 0, \\ -x^3, & \text{when } x < 0, \end{cases}$$

for this is twice differentiable and satisfies the equation on all of $\mathbb{R}$. This function cannot, however, be expressed in the form $C_1 y_1(x) + C_2 y_2(x)$, since $y_1(x) = x$ and $y_2(x) = x^3$ (and likewise their linear combinations) are odd, whereas y_3 is even. — The result does not contradict Theorem IX.6.1, since the theorem's assumption of the continuity of the coefficient functions P and Q does not hold on the interval $(-\infty, \infty)$. Note also that on the intervals $(0, \infty)$ and $(-\infty, 0)$ one has $y_3(x) = \pm y_2(x)$, so on these intervals y_3 brings nothing new to the general solution. $\square$

The result of Theorem IX.6.1 can be interpreted as saying that, under the assumptions made, the set $\mathcal{Y}$ of solutions of equation (2) is a 2-dimensional vector space (a function space, cf. Section IV.5). A basis of this space is any linearly independent pair of solutions $\{y_1, y_2\}$, so that the whole solution space can be represented in the form

$$\mathcal{Y} = \{y = c_1 y_1 + c_2 y_2 \mid c_1, c_2 \in \mathbb{R}\}.$$

A (partial) proof of Theorem IX.6.1 is given below. Before that, we consider consequences of the theorem with a view to solving equation (2) by quadratures.

The homogeneous equation: solution by quadratures

First of all, one must acknowledge that equation (2) does not always solve by quadratures. Not reducible to quadratures is, for example, even such a simple equation as

$$y'' = xy.$$

After this bad news, let it be noted that if for equation (2) even one nontrivial (i.e. nonzero) solution y_1 can be found, then a second solution linearly independent of y_1 — and hence the general solution — can be constructed by quadratures. In what follows we restrict ourselves to this case, i.e. we assume one solution $y_1 \neq 0$ to be known.

If a solution $y_1 \neq 0$ of equation (2) is known, then a second solution can be sought in the familiar way by variation of the constant, i.e. in the form

$$y(x) = C(x)y_1(x).$$

Then

$$y' = y_1 C' + y_1' C, \quad y'' = y_1 C'' + 2y_1' C' + y_1'' C.$$

Since $Ly_1 = 0$, there follows for C the differential equation

$$y_1(x)C'' + [2y_1'(x) + P(x)y_1(x)]C' = 0.$$

By the substitution $u = C'$ this becomes separable:

$$u' + K(x)u = 0, \quad K(x) = 2 \frac{y_1'(x)}{y_1(x)} + P(x) = 2 \frac{d}{dx} \ln |y_1(x)| + P(x).$$

By separating and integrating we obtain

$$\begin{aligned} \int \frac{du}{u} &= - \int K(x) dx \\ \Leftrightarrow \ln |u| &= -2 \ln |y_1(x)| - \int P(x) dx + A \quad (A \in \mathbb{R}). \end{aligned}$$

Choosing $A = 0$ (one solution $u \neq 0$ suffices!) we obtain

$$u(x) = \frac{e^{-\int P(x) dx}}{[y_1(x)]^2} = C'(x).$$

Consequently one solution is

$$\begin{aligned} y_2(x) &= y_1(x) \int u(x) dx \\ &= y_1(x) \int \frac{e^{-\int P(x) dx}}{[y_1(x)]^2} dx. \end{aligned}$$

The solution is valid at least on the open intervals between the zeros of y_1 . On such intervals the ratio $u = y_2/y_1$ is clearly not constant (since it is an antiderivative of a positive function), so the solution found is linearly independent of y_1 . The general solution of the homogeneous equation (2) can then (at least on the intervals mentioned) be written

$$y(x) = C_1 y_1(x) + C_2 y_2(x).$$

EXAMPLE 2 The differential equation

$$y'' + xy' - y = 0$$

has the obvious solution $y_1(x) = x$, so we seek a second one in the form

$$y(x) = xC(x) \quad \Rightarrow \quad y' = xC' + C, \quad y'' = xC'' + 2C'.$$

Substitution into the equation gives

$$xu' + (x^2 + 2)u = 0, \quad u = C'.$$

Solution by separation (assuming $x > 0$):

$$\begin{aligned} \int \frac{du}{u} &= \ln u = - \int \left(x + \frac{2}{x} \right) dx = \ln \left(x^{-2} e^{-x^2/2} \right) + \ln A \\ \Rightarrow \quad u &= C' = Ax^{-2} e^{-x^2/2}. \end{aligned}$$

Choosing $A = 1$, imposing the condition $\lim_{x \rightarrow \infty} C(x) = 0$ and integrating by parts, we obtain

$$C(x) = - \int_x^\infty u(t) dt = -x^{-1} e^{-x^2/2} + \int_x^\infty e^{-t^2/2} dt.$$

The general solution is therefore

$$y(x) = C_1 x + C_2 x C(x) = C_1 x + C_2 \left(-e^{-x^2/2} + x \int_x^\infty e^{-t^2/2} dt \right), \quad C_1, C_2 \in \mathbb{R}.$$

Although $x > 0$ was assumed in the intermediate steps of the computation, the solution is valid on all of $\mathbb{R}$. $\square$

The homogeneous equation: solvability theory

Let us return to Theorem IX.6.1, which has not been proved. The deepest of the theorem's assertions concerns — as usual — the existence of the solutions

y_1 and y_2 . In what follows this result, like the other assertions of Theorem IX.6.1, is derived from the following basic theorem, which in turn is a special case of a much more general assertion concerning the solvability of the initial value problem. That assertion is not proved at this stage; instead we return to the matter later in Section IX.8. Applied to the special case in question, the solvability assertion is the following:

*THEOREM IX.6.2 If P and Q are continuous on the interval (a, b) and $x_0 \in (a, b)$, then the initial value problem

$$\begin{cases} y'' + P(x)y' + Q(x)y = 0, & x \in (a, b), \\ y(x_0) = A, \\ y'(x_0) = B \end{cases}$$

has a unique solution for every $A, B \in \mathbb{R}$.

In what follows, then, we prove the assertion: Theorem IX.6.2 $\Rightarrow$ Theorem IX.6.1. A central role in the proof is played by the following concept.

DEFINITION IX.6.3 The *Wronskian determinant* of functions y_1 and y_2 differentiable on the interval (a, b) is

$$W_{12}(x) = \begin{vmatrix} y_1(x) & y_2(x) \\ y_1'(x) & y_2'(x) \end{vmatrix} = (y_1 y_2' - y_2 y_1')(x), \quad x \in (a, b).$$

PROPOSITION IX.6.4 If y_1 and y_2 are differentiable on the interval (a, b) and $W_{12}(x_0) = (y_1 y_2' - y_2 y_1')(x_0) \neq 0$ for some $x_0 \in (a, b)$, then y_1 and y_2 are linearly independent on the interval (a, b) .

Proof If for some $c_1, c_2 \in \mathbb{R}$ one has

$$c_1 y_1(x) + c_2 y_2(x) = 0, \quad x \in (a, b),$$

then by differentiation it follows that also

$$c_1 y_1'(x) + c_2 y_2'(x) = 0, \quad x \in (a, b).$$

When these equations are written at the point x_0 , one obtains the system

$$\begin{cases} y_1(x_0)c_1 + y_2(x_0)c_2 = 0, \\ y_1'(x_0)c_1 + y_2'(x_0)c_2 = 0. \end{cases}$$

Since $W_{12}(x_0) \neq 0$, the only solution of the system is $c_1 = c_2 = 0$.* Thus from the assumption $c_1 y_1(x) + c_2 y_2(x) = 0$, $x \in (a, b)$ it follows that $c_1 = c_2 = 0$, so y_1 and y_2 are linearly independent on the interval (a, b) . $\square$

EXAMPLE 3 The functions $y_1(x) = x^3$ and $y_2(x) = |x|^3$ are (twice) differentiable and linearly independent on $\mathbb{R}$ (cf. Example 1 above). Computing the Wronskian determinant of these functions, one finds that $W_{12}(x) = 0 \forall x \in \mathbb{R}$. On the basis of this (counter)example one concludes:

$$y_1 \text{ and } y_2 \text{ linearly independent on the interval } (a, b) \\ \not\Rightarrow W_{12}(x_0) \neq 0 \text{ for some } x_0 \in (a, b). \quad \square$$

On the basis of the example, the (implication) assertion of Proposition IX.6.4 does not hold in reverse. If instead the assumptions are strengthened so that y_1 and y_2 are assumed to be solutions of the differential equation (2) on the interval (a, b) , then the converse assertion is also true. Under such an assumption one obtains the following result, considerably stronger than Proposition IX.6.4.

THEOREM IX.6.5 If y_1 and y_2 are solutions of the differential equation (2) on the interval (a, b) (P and Q continuous on the interval (a, b)) and $W_{12}(x)$ is the Wronskian determinant of the functions y_1, y_2 on the interval (a, b) , then

$$\begin{aligned} W_{12}(x_0) &= 0 \quad \text{for some } x_0 \in (a, b) \\ \Leftrightarrow W_{12}(x) &= 0 \quad \text{for every } x \in (a, b) \\ \Leftrightarrow y_1 \text{ and } y_2 &\text{ linearly dependent on the interval } (a, b). \end{aligned}$$

Proof By the assumptions

$$\begin{cases} y_1'' + P(x)y_1' + Q(x)y_1 = 0, & x \in (a, b), \\ y_2'' + P(x)y_2' + Q(x)y_2 = 0, & x \in (a, b). \end{cases}$$

Multiplying the first equation by y_2 and the second by y_1 and subtracting, we get

$$y_1 y_2'' - y_2 y_1'' + P(x)(y_1 y_2' - y_2 y_1') = 0, \quad x \in (a, b).$$

*If $a_{ij}, x_i, b_i \in \mathbb{R}$, $i, j = 1, 2$ and $W = a_{11}a_{22} - a_{21}a_{12} \neq 0$, then

$$\begin{cases} a_{11}x_1 + a_{12}x_2 = b_1 \\ a_{21}x_1 + a_{22}x_2 = b_2 \end{cases} \Leftrightarrow \begin{cases} x_1 = W^{-1}(a_{22}b_1 - a_{12}b_2), \\ x_2 = W^{-1}(a_{11}b_2 - a_{21}b_1). \end{cases}$$

But

$$y_1 y_2'' - y_2 y_1'' = \frac{d}{dx}(y_1 y_2' - y_2 y_1').$$

Thus it has been deduced that $y(x) = W_{12}(x)$ is, for every $x_0 \in (a, b)$, a solution of the initial value problem

$$\begin{cases} y' + P(x)y = 0, & x \in (a, b) \\ y(x_0) = W_{12}(x_0) \end{cases} \quad (3)$$

Since the solution is (cf. Section IX.4)

$$W_{12}(x) = W_{12}(x_0) e^{-\int_{x_0}^x P(t)dt}, \quad x \in (a, b)$$

and since $e^t \neq 0 \forall t \in \mathbb{R}$, the first partial assertion follows: $W_{12}(x_0) = 0 \Leftrightarrow W_{12}(x) = 0 \forall x \in (a, b)$.

In the case $y_1(x) = 0 \forall x \in (a, b)$ the second partial assertion is also true, so assume that $y_1(x_0) \neq 0$ for some $x_0 \in (a, b)$. Then, by the continuity of y_1 , $y_1(x) \neq 0$ on some interval $(x_0 - \delta, x_0 + \delta) \subset (a, b)$, $\delta > 0$. On this interval y_2/y_1 is differentiable and

$$\frac{d}{dx} \left[\frac{y_2(x)}{y_1(x)} \right] = \frac{W_{12}(x)}{[y_1(x)]^2}.$$

Consequently, if $W_{12}(x) = 0$ on the interval (a, b) , then for some $C \in \mathbb{R}$

$$\frac{y_2(x)}{y_1(x)} = C \Leftrightarrow -C y_1(x) + y_2(x) = 0 \quad \text{on the interval } (x_0 - \delta, x_0 + \delta).$$

Writing $y(x) = -C y_1(x) + y_2(x)$, $x \in (a, b)$, we thus have $y(x) = 0$ on the said subinterval, so in particular $y(x_0) = y'(x_0) = 0$. Since y is also a solution of the differential equation (2) on the interval (a, b) , y therefore solves the initial value problem of Theorem IX.6.2 with $A = B = 0$. Since $y = 0$ is also an obvious solution, and since the solution is unique by that theorem, we must have $y(x) = -C y_1(x) + y_2(x) = 0$, $x \in (a, b)$. Then y_1 and y_2 are linearly dependent on the interval (a, b) . The reasoning rested on the assumption that $W_{12}(x) = 0$ on the interval (a, b) , so the part $\Rightarrow$ of the second partial assertion is proved. The remaining part $\Leftarrow$ has already been proved, since this is the logically equivalent form of the assertion of Proposition IX.6.4. $\square$

Now we are ready to present (relying on Theorem IX.6.2) the

proof of Theorem IX.6.1 Let y_1 and y_2 be the solutions of the initial value problem of Theorem IX.6.2 when a) $A = 1$, $B = 0$, b) $A = 0$, $B = 1$. Then the Wronskian determinant of the functions y_1 and y_2 at x_0 is

$$W_{12}(x_0) = \begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix} = 1.$$

By Proposition IX.6.4, y_1 and y_2 are linearly independent on the interval (a, b) , so the first partial assertion is proved.

To prove the second partial assertion, assume that y_1 and y_2 are any two linearly independent solutions of the differential equation (2) on the interval (a, b) , and that u is a third solution of the same DE. Then, if the coefficients c_1 and c_2 can be determined so that the function $y(x) = c_1y_1(x) + c_2y_2(x)$ satisfies $y(x_0) = u(x_0)$, $y'(x_0) = u'(x_0)$, then u and y are solutions of the same initial value problem, whence by Theorem IX.6.2 one must have $u(x) = y(x)$, $x \in (a, b)$. The coefficients are determined by the system

$$\begin{cases} y_1(x_0)c_1 + y_2(x_0)c_2 = u(x_0), \\ y_1'(x_0)c_1 + y_2'(x_0)c_2 = u'(x_0). \end{cases}$$

This has a unique solution under the condition $y_1(x_0)y_2'(x_0) - y_1'(x_0)y_2(x_0) \neq 0$, i.e. the condition $W_{12}(x_0) \neq 0$, where W_{12} is the Wronskian determinant of the functions y_1 and y_2 . But by the assumption and Theorem IX.6.5, $W_{12}(x_0) \neq 0$ for every $x_0 \in (a, b)$. Thus u can be expressed as a linear combination of the functions y_1 and y_2 . $\square$

The complete equation: variation of constants

If the general solution of the homogeneous equation (2) is known, then to solve the complete equation (1) it again suffices to find a particular solution for it. The general method is based on variation of constants: If the general solution of the homogeneous equation is $y(x) = C_1y_1(x) + C_2y_2(x)$, then a particular solution of the complete equation is found in the form

$$y(x) = C_1(x)y_1(x) + C_2(x)y_2(x).$$

Then

$$y' = C_1(x)y_1'(x) + C_2(x)y_2'(x) + C_1'(x)y_1(x) + C_2'(x)y_2(x).$$

To simplify this expression, we impose the additional condition

$$C_1'(x)y_1(x) + C_2'(x)y_2(x) = 0$$

(which turns out to be possible!), so that

$$\begin{aligned} y' &= C_1(x)y_1'(x) + C_2(x)y_2'(x), \\ y'' &= C_1(x)y_1''(x) + C_2(x)y_2''(x) + C_1'(x)y_1'(x) + C_2'(x)y_2'(x) \end{aligned}$$

and consequently

$$\begin{aligned} Ly &= y'' + P(x)y' + Q(x)y \\ &= C_1(x)Ly_1 + C_2(x)Ly_2 + C_1'(x)y_1'(x) + C_2'(x)y_2'(x). \end{aligned}$$

Here $Ly_1 = Ly_2 = 0$, so, taking into account the additional condition assumed, we obtain the system

$$\begin{cases} y_1(x)C_1' + y_2(x)C_2' = 0, \\ y_1'(x)C_1' + y_2'(x)C_2' = R(x). \end{cases}$$

Since here (Theorem IX.6.5)

$$(y_1y_2' - y_2y_1')(x) = W_{12}(x) \neq 0,$$

the system is solvable:

$$C_1' = -\frac{y_2(x)R(x)}{W_{12}(x)}, \quad C_2' = \frac{y_1(x)R(x)}{W_{12}(x)}.$$

By integrating $C_1(x)$ and $C_2(x)$ from these equations, a particular solution of the complete equation (1) has been found. The general solution of equation (1) is thus

$$y(x) = C_1y_1(x) + C_2y_2(x) - y_1(x) \int \frac{y_2(x)R(x)}{W_{12}(x)} dx + y_2(x) \int \frac{y_1(x)R(x)}{W_{12}(x)} dx.$$

EXAMPLE 4 Solve the Euler differential equation

$$x^2y'' + 4xy' + 2y = f(x), \quad x > 0.$$

Solution The general solution of the homogeneous equation is (see the previous section)

$$y(x) = \frac{C_1}{x} + \frac{C_2}{x^2} = C_1y_1(x) + C_2y_2(x).$$

In the above formulas we have

$$R(x) = \frac{f(x)}{x^2}, \quad W_{12}(x) = (y_1y_2' - y_1'y_2)(x) = -\frac{1}{x^4},$$

so variation of constants gives

$$\begin{aligned} C_1' &= f(x) \quad \Rightarrow \quad C_1(x) = \int f(x) dx, \\ C_2' &= -xf(x) \quad \Rightarrow \quad C_2(x) = -\int xf(x) dx. \end{aligned}$$

The particular solution is therefore

$$y(x) = \frac{1}{x} \int f(x) dx - \frac{1}{x^2} \int x f(x) dx.$$

This is also the general solution when indefinite integration constants are included in the integrals on the right-hand side. $\square$

Some well-known differential equations

The second-order linear, homogeneous differential equation is a common type of differential equation in applications. Below are a few well-known differential equations, named after the mathematicians who studied them.

Legendre: $(x^2 - 1)y'' + 2xy' - n(n + 1)y = 0$

Chebyshev: $(1 - x^2)y'' - xy' + n^2y = 0$

Hermite: $y'' - 2xy' + 2ny = 0$

Laguerre: $xy'' + (1 - x)y' + ny = 0$

Bessel: $x^2y'' + xy' + (x^2 - p^2)y = 0$

Of these, all except Bessel's differential equation have as a solution a polynomial of degree n when $n \in \mathbb{N} \cup \{0\}$ (the most interesting case for applications). These polynomials are named in the same way as the differential equations. In Bessel's differential equation $p \geq 0$ is a real parameter

Series solutions

If a second-order linear, homogeneous differential equation has no 'recognizable' solution at all, then one may attempt to solve it by the *series method*. Here a solution is sought either as a power series or as an expression involving power series. The basic idea of the method is the same as in the series method for integration (see Section VIII.3).

As an example of the application of the series method, take Bessel's differential equation, whose basic form is

$$y'' + \frac{1}{x}y' + \left(1 - \frac{p^2}{x^2}\right)y = 0.$$

The general solution (in applications usually on the interval $(0, \infty)$) is written in the form

$$y(x) = C_1 J_p(x) + C_2 Y_p(x),$$

where J_p and Y_p are *Bessel functions*, more precisely Bessel functions of the *first kind* (J_p) and of the *second kind* (Y_p). (In the case $p = 1/2$ these are exceptionally elementary functions, see Exercise 7.) The function J_p is of the form

$$J_p(x) = \sum_{k=0}^{\infty} a_k x^{2k+p} = x^p F(x),$$

where the power series $F(x) = \sum_{k=0}^{\infty} a_k x^{2k}$ converges everywhere. For non-integer values, correspondingly,

$$Y_p(x) = \sum_{k=0}^{\infty} a_k x^{2k-p} = x^{-p} G(x), \quad p \notin \{0, 1, 2, \dots\}$$

where the series for G again converges everywhere. If p is an integer, then Y_p contains also a logarithmic term of the form $x^{-p} H(x) \ln |x|$, where $H(x)$ can be expressed as a power series.

Let $p = 0$, and try a series solution $y(x) = \sum_{k=0}^{\infty} a_k x^k$. Substitution into the equation gives

$$\sum_{k=2}^{\infty} k(k-1) a_k x^{k-2} + \sum_{k=1}^{\infty} k a_k x^{k-2} + \sum_{k=0}^{\infty} a_k x^k = 0.$$

When in the first two sums the summation index k is replaced by $k+2$, we obtain

$$\sum_{k=0}^{\infty} (k+2)(k+1) a_{k+2} x^k + [a_1 x^{-1} + \sum_{k=0}^{\infty} (k+2) a_{k+2} x^k] + \sum_{k=0}^{\infty} a_k x^k = 0 \quad \forall x$$

i.e., combining the sums,

$$a_1 x^{-1} + \sum_{k=0}^{\infty} [(k+2)^2 a_{k+2} + a_k] x^k = 0 \quad \forall x.$$

One sees that this holds when we choose

$$\begin{aligned} a_1 &= a_3 = a_5 = \dots = 0 \\ a_0 &= 1, \quad a_{k+2} = -\frac{1}{(k+2)^2} a_k, \quad k = 0, 2, 4, \dots \end{aligned}$$

For the coefficients a_{2k} one obtains (after a little reflection) the expression

$$a_{2k} = (-1)^k \frac{1}{(k!)^2 \cdot 4^k}, \quad k = 0, 1, \dots$$

The solution, called J_0 , is thus

$$J_0(x) = \sum_{k=0}^{\infty} (-1)^k \frac{1}{(k!)^2} \left(\frac{x}{2}\right)^{2k}.$$

This series converges (and can be differentiated term by term) everywhere, so a solution valid on all of $\mathbb{R}$ has been found for the differential equation $xy'' + y' + xy = 0$. The second-kind solution $Y_0(x)$ is valid only separately on the intervals $(-\infty, 0)$ and $(0, \infty)$. This too can be expressed by means of power series (see Exercise 10).

EXERCISES

- Of what second-order linear and homogeneous differential equation are (on some open interval) solutions a) $y_1(x) = x + 1$ and $y_2(x) = x^2$, b) $y_1(x) = x$ and $y_2(x) = e^x$, c) $y_1(x) = x^2$ and $y_2(x) = \sin x$?
- Solve using the given additional information about a particular solution:
 - $xy'' - (x + 3)y' + y = 0$; polynomial
 - $x^2(\ln x - 1)y'' - xy' + y = 0$; polynomial
 - $(x^2 - 2x - 1)y'' - (2x - 1)y' + 2y = 0$; polynomial
 - $y'' + (\tan x - 2 \cot x)y' + 2(\cot^2 x)y = 0$; $\sin x$
 - $y'' + 2(1 - \tan^2 x)y = 0$; $\cos^2 x$
 - $(x - 2)y'' - (4x - 7)y' + (4x - 6)y = 0$; e^{ax}
- The differential equation $y'' + P(x)y' + Q(x)y = 0$ has an easily guessed nontrivial solution, if $1 + P(x) + Q(x) = 0$. Solve on the basis of this (and, if necessary, another) guess:
 - $(x - 1)y'' - xy' + y = 0$
 - $(2x - x^2)y'' + (x^2 - 2)y' + (2 - 2x)y = 0$
 - $y'' - (2 + 2x + x^{-1})y' + (1 + 2x + x^{-1})y = 0$
 - $(x + 1)y'' - xy' - y = (x + 1)^2$
- Assume that for the coefficient functions of the differential equation $y'' + P(x)y' + Q(x)y = 0$ the following holds: P is differentiable and P' and

Q are continuous on the interval (a, b) under consideration. Then, by the substitution $y(x) = K(x)u(x)$, the DE can be transformed into the form

$$u'' + I(x)u = 0.$$

How must $K(x)$ be chosen, and what is the expression for $I(x)$?

5. Solve by variation of constants:

- | | |
|----------------------------------|------------------------------------|
| a) $y'' + y = \tan x$ | b) $y'' - y = 1/(e^x + 1)$ |
| c) $y'' + 2y' + y = 1/(x^2 + 1)$ | d) $x^2 y'' + 2xy' - 2y = x^2 e^x$ |
| e) $y'' + 5y' + 4y = f(x)$ | f) $y'' + 4y' + 4y = f(x)$ |
| g) $y'' + 4y' + 5y = f(x)$ | h) $x^2 y'' + 5xy' + 3y = f(x)$ |
| i) $x^2 y'' + 5xy' + 4y = f(x)$ | j) $x^2 y'' - 2y = f(x)$ |

6. a) The *Legendre polynomial* of degree n is defined by

$$P_n(x) = \frac{1}{2^n n!} \frac{d^n}{dx^n} (x^2 - 1)^n.$$

Verify that P_n is a polynomial of degree n and show that it is a solution of Legendre's differential equation for $n = 0, 1, 2, 3$.

b) Determine the Hermite and Laguerre polynomials of degree $n \leq 2$, i.e. the polynomial solutions of the respective differential equations for $n = 0, 1, 2$.

Determine c) the general solution of Legendre's, d) Chebyshev's, e) Hermite's, f) Laguerre's DE for $n = 1$.

7. In the case $p = 1/2$ Bessel's differential equation can be solved with elementary functions. Determine the solution by the substitution $y(x) = u(x)/\sqrt{x}$.

8. The general solution of the following differential equations can be represented near the point $x = 0$ in the form $y(x) = C_1 \sum_{k=0}^{\infty} a_k x^k + C_2 \sum_{k=0}^{\infty} b_k x^k$, where $a_0 = b_1 = 1$ and $a_1 = b_0 = 0$. Determine the first four nonzero terms of both power series.

- | | | |
|---------------------------------|--|----------------------------|
| a) $y'' + 2xy' + 5y = 0$ | b) $y'' + 2xy + 6y = 0$ | c) $y'' + xy' - x^2 y = 0$ |
| d) $y'' + (x+1)y' + (x-1)y = 0$ | e) $(1-x^2)y'' - xy' + ay = 0, a \in \mathbb{R}$ | |

9. (*) The *Chebyshev polynomial* of degree n is defined on the interval $[-1, 1]$ by the formula

$$T_n(x) = \cos(n \operatorname{Arccos} x), \quad x \in [-1, 1].$$

- a) Show that T_n is a polynomial of degree n (substitute $x = \cos \theta$). b) Show that T_n is a solution of Chebyshev's differential equation on the interval $(-1, 1)$. c) Compute the coefficients of the polynomial T_5 and sketch $T_5(x)$ graphically on the interval $[-2, 2]$. What are the zeros of T_5 , and at what points is $T_5(x) = \pm 1$?
10. (*) Show that if $p = 0$, then Bessel's differential equation has on the interval $(0, \infty)$ a (second-kind) solution of the form $Y_0(x) = J_0(x) \ln x + F(x)$, where F can be expressed as an everywhere-convergent power series.
11. (*) Show that the general solution of the differential equation $y'' = xy$ can be represented as a power series in the form

$$y(x) = C_1 \sum_{k=0}^{\infty} a_k x^{3k} + C_2 \sum_{k=0}^{\infty} b_k x^{3k+1}, \quad a_0 = b_0 = 1.$$

Define the coefficient sequences $\{a_k\}$ and $\{b_k\}$ as recursive sequences and deduce that the series converges everywhere. What, more generally, is the general series solution of the differential equation $y'' = x^m y$ when $m \in \mathbb{N}$?

IX.7 Numerical solution methods for differential equations

In the previous sections we have developed the art of solving differential equations mainly from the point of view of so-called 'exact' solution. This body of solution technique is very traditional, and it has its own enduring value. — Especially when the solution can be expressed by means of elementary functions as a relatively simple expression, the traditional approach is hard to pass over. In applications, however, one very often encounters differential equations (and especially systems of them) on which none of the recipes offered so far has any effect. Then one must be content to seek the solution approximately, *numerically*, in practice with the aid of a computer or a powerful calculator.* Numerical solving thus differs considerably from the traditional methods, by which a solution is usually obtained (if at all) 'by hand'. There is nonetheless no difference in principle between numerical and 'exact' solving, for by numerical means too one obtains the exact solution, if the sequence of approximate solutions produced by the algorithm is thought of as computed 'all the way to the end'. — In fact this is precisely how we proceeded in Section VIII.4, when we sought a universally valid method of solution for the differential equation $y' = f(x)$.

In what follows we briefly consider the basics of the so-called *step methods* used in the numerical solution of initial value problems. We again take as our object of study the first-order initial value problem in normal form

$$\begin{cases} y' = f(x, y), & x > x_0, \\ y(x_0) = y_0. \end{cases} \quad (1)$$

In all step methods the idea is to start from the initial point x_0 and proceed in short steps, first to the point x_1 , then to the point x_2 , and so on. At each step $x_k \rightarrow x_{k+1}$ a number y_{k+1} is computed, which aims to be an approximation to the value of the exact solution $y(x)$ at the point x_{k+1} :

$$y_{k+1} \approx y(x_{k+1}), \quad k = 0, 1, \dots$$

When computing the number y_{k+1} , the numbers $y_0, y_1, \dots, y_k$ are already known (from the initial condition and from earlier steps), so y_{k+1} can be computed

*Differential equations were solved numerically even before the age of computers, but such solving was confined only to matters regarded as very important, such as determining the trajectories of artillery shells. In the computer era, the numerical solution of differential equations has become an everyday matter in the most diverse fields of science and engineering. At the same time the solution methods have grown into a broad methodological science, about which thick books have been written.

recursively. If only the preceding number y_k appears in the computation formula, it is a *one-step method*, otherwise a *multistep method*. Thus what ultimately distinguishes the various step methods from one another is only what kind of recursion formula or more general algorithm is used in computing the number y_{k+1} . The accuracy can be controlled both by the choice of method and by the *step lengths* $x_{k+1} - x_k$ (cf. the numerical integration methods in Section VIII.9). In the following we first consider the very simplest — and also the oldest — of the step methods.

Euler's method

Euler's method^{*} is a one-step method in which the number y_{k+1} is computed from the recursion formula

$$y_{k+1} = y_k + hf(x_k, y_k), \quad h = x_{k+1} - x_k, \quad k = 0, 1, \dots$$

The formula can be derived, e.g., as follows: Integrate the differential equation $y' = f(x, y)$ over the interval $[x_k, x_{k+1}]$ on both sides, which gives

$$y(x_{k+1}) = y(x_k) + \int_{x_k}^{x_{k+1}} f(x, y(x)) dx.$$

Assume that $f(x, y(x)) = y'(x)$ is approximately constant on the interval $[x_k, x_{k+1}]$ and, on this basis, make the approximation

$$\int_{x_k}^{x_{k+1}} f(x, y(x)) dx \approx hf(x_k, y(x_k)), \quad h = x_{k+1} - x_k.$$

(This is a simple numerical integration, cf. Section VIII.9.) One arrives at Euler's method when the result

$$y_{k+1} \approx y_k + hf(x_k, y_k), \quad y_k = y(x_k), \quad y_{k+1} = y(x_{k+1})$$

is 'mangled' into the form

$$y_{k+1} = y_k + hf(x_k, y_k), \quad y_k \approx y(x_k), \quad y_{k+1} \approx y(x_{k+1}).$$

^{*}The Swiss mathematician **Leonhard Euler** (1707-1783) is already familiar to us from the complex exponential function (Euler's formula) and from the differential equation named after him. *Euler's equations* (in an entirely different sense) are central to a branch of mathematics called the *calculus of variations* and (again in another sense) to fluid mechanics. Euler studied the mathematics and physics of his time very broadly, including astronomy, music, navigation, etc. Euler was one of the most significant mathematicians of the 18th century and one of the most versatile of all time.

If the step length h is assumed known (e.g. constant), then a step of Euler's method consists of one function evaluation, one multiplication and two additions:

$$x_k = x_{k-1} + h, \quad t_k = f(x_k, y_k), \quad y_{k+1} = y_k + ht_k.$$

EXAMPLE 1 Compute an approximation y_n to Napier's number e by solving the initial value problem $y' = y$, $y(0) = 1$ by Euler's method with constant step length $h = 1/n$, $n \in \mathbb{N}$. Estimate the error.

Solution Computing by Euler's method, we obtain

$$\begin{aligned} y_0 &= 1, \quad y_k = y_{k-1} + hy_{k-1} = (1+h)y_{k-1}, \quad k = 1, 2, \dots \\ \Rightarrow y_n &= (1+h)^n = \underline{\underline{\left(1 + \frac{1}{n}\right)^n}}. \end{aligned}$$

To estimate the error, consider the expression

$$\ln \frac{y_n}{e} = n \ln \left(1 + \frac{1}{n}\right) - 1.$$

By Taylor's theorem (see Section VII.4), $\ln(1+x) = x - \frac{1}{2}x^2 + \mathcal{O}(|x|^3)$, so it follows that

$$\begin{aligned} \ln \frac{y_n}{e} &= -\frac{1}{2n} + \mathcal{O}(n^{-2}) \quad \Rightarrow \quad \frac{y_n}{e} = e^{-\frac{1}{2n} + \mathcal{O}(n^{-2})} = 1 - \frac{1}{2n} + \mathcal{O}(n^{-2}) \\ &\Rightarrow \quad e - y_n = \underline{\underline{\frac{e}{2n} + \mathcal{O}(n^{-2})}}. \quad \square \end{aligned}$$

The error of Euler's method

The error estimate $y_n - y(x_n) = \mathcal{O}(h)$ obtained in the example above holds for Euler's method under fairly general conditions. To derive a more general error estimate, let the step length be $h = \text{constant}$, and write $Y_k = y(x_k) = y(x_0 + kh)$, where $y(x)$ is the solution of the initial value problem (1). Let us first estimate the so-called *consistency error* of Euler's method, i.e. the number δ_k in the equation

$$Y_{k+1} = Y_k + hf(x_k, Y_k) + \delta_k.$$

Writing $u(x) = f(x, y(x)) = y'(x)$, integration by parts gives (see also the derivation of Euler's method above)

$$\begin{aligned}\delta_k &= \int_{x_k}^{x_{k+1}} u(x) dx - hu(x_k) \\ &= \left| \int_{x_k}^{x_{k+1}} (x - x_{k+1})u(x) - \int_{x_k}^{x_{k+1}} (x - x_{k+1})u'(x) dx - hu(x_k) \right| \\ &= \int_{x_k}^{x_{k+1}} (x_{k+1} - x)y''(x) dx.\end{aligned}$$

Assume now (Assumption #1) that y is twice continuously differentiable and that the estimate $|y''(x)| \leq M$ holds (for the values of x in question). Then the consistency error is at most

$$|\delta_k| \leq \int_{x_k}^{x_{k+1}} (x_{k+1} - x) M dx = \frac{1}{2} Mh^2.$$

Using this estimate and the triangle inequality, one can now conclude:

$$\begin{aligned}\begin{cases} Y_{k+1} = Y_k + hf(x_k, Y_k) + \delta_k, \\ y_{k+1} = y_k + hf(x_k, y_k) \end{cases} \\ \Rightarrow |Y_{k+1} - y_{k+1}| \leq |Y_k - y_k| + h|f(x_k, Y_k) - f(x_k, y_k)| + \frac{1}{2} Mh^2.\end{aligned}$$

Assume (Assumption #2) that on the right-hand side one may use the estimate

$$|f(x, y_1) - f(x, y_2)| \leq L|y_1 - y_2|$$

($x = x_k$, $y_1 = Y_k$, $y_2 = y_k$), where L is a constant. Then it follows that

$$|Y_{k+1} - y_{k+1}| \leq (1 + hL) |Y_k - y_k| + \frac{1}{2} Mh^2.$$

Applying this inequality recursively for $k = 0, 1, \dots$ it follows (induction!) that

$$|Y_n - y_n| \leq \frac{1}{2} Mh^2 \sum_{k=0}^{n-1} (1 + hL)^k = \frac{M}{2L} [(1 + hL)^n - 1] h, \quad n = 1, 2, \dots$$

Using here also the estimate $(1 + hL)^n \leq (e^{hL})^n = e^{Ln h} = e^{L(x_n - x_0)}$, one obtains, under the assumptions made, as the error estimate for Euler's method

$$\boxed{|y(x_n) - y_n| \leq \frac{M}{2L} (e^{L(x_n - x_0)} - 1) h, \quad x_n = x_0 + nh.} \quad (2)$$

EXAMPLE 2 The initial value problem

$$\begin{cases} y' = e^{-x} + x \sin y, & x > 0, \\ y(0) = 1 \end{cases}$$

cannot be solved by traditional methods, so one must be content with a numerical solution. From the differential equation one sees that the solution satisfies $|y'(x)| \leq 1 + x$ for $x \geq 0$. Differentiating implicitly and using this estimate, it follows that

$$y''(x) = -e^{-x} + \sin y + xy' \cos y \Rightarrow |y''(x)| \leq 2 + x + x^2, \quad x \geq 0.$$

Since, further,

$$|f(x, y_1) - f(x, y_2)| = x|\sin y_1 - \sin y_2| \leq x|y_1 - y_2|, \quad x \geq 0,$$

one sees that the error estimate (2) for Euler's method is valid when in the estimate one sets

$$L = x_n = nh, \quad M = 2 + x_n + x_n^2 \quad (n \geq 1).$$

For example, if a solution is wanted on the interval $[0, 2]$, then one can choose $L = 2$, $M = 8$, whence the error is at most

$$|y(x_n) - y_n| < 2e^4 h < 110h \quad (x_n \leq 2).$$

Probably the error is considerably smaller than this estimate. $\square$

Systems of differential equations

A strong point of Euler's method, as of the other step methods (see the examples below), from the standpoint of applications is that the methods apply as such also to the solution of a differential-equation system in normal form, insofar as it is an initial value problem. Also initial value problems for higher-order differential equations are solved in this way, by first writing the problem in system form, cf. Section IX.1. Let an example illuminate the matter.

EXAMPLE 3 Present an algorithm based on Euler's method by which the initial value problem

$$\begin{cases} y''' = x + yy'y'', & x > 0, \\ y(0) = 1, \quad y'(0) = y''(0) = 0. \end{cases}$$

can be solved numerically.

Solution From the initial conditions and the system form of the differential equation to be solved

$$\begin{cases} y' = u, \\ u' = v, \\ v' = x + yuv \end{cases}$$

one obtains as the algorithm

$$y_0 = 1, \quad u_0 = v_0 = 0, \quad \begin{cases} y_{k+1} = y_k + hu_k, \\ u_{k+1} = u_k + hv_k, \\ v_{k+1} = v_k + h(x_k + y_k u_k v_k), \end{cases} \quad k = 0, 1, \dots \quad \square$$

*Other step methods

The 'list of possible methods' usable in initial value problems for ordinary differential equations is nowadays extraordinarily long. Below are just a few examples of frequently used or otherwise well-known step methods.

Implicit Euler

$$y_{k+1} = y_k + hf(x_{k+1}, y_{k+1})$$

Trapezoidal rule

$$y_{k+1} = y_k + \frac{h}{2} [f(x_k, y_k) + f(x_{k+1}, y_{k+1})]$$

Midpoint rule

$$y_{k+1} = y_k + hf\left(x_k + \frac{1}{2}h, \frac{1}{2}y_k + \frac{1}{2}y_{k+1}\right)$$

BDF-2

$$y_{k+1} = \frac{4}{3}y_k - \frac{1}{3}y_{k-1} + \frac{2}{3}hf(x_{k+1}, y_{k+1})$$

BDF-3

$$y_{k+1} = \frac{18}{11}y_k - \frac{9}{11}y_{k-1} + \frac{2}{11}y_{k-2} + \frac{6}{11}hf(x_{k+1}, y_{k+1})$$

BDF-4

$$y_{k+1} = \frac{48}{25}y_k - \frac{36}{25}y_{k-1} + \frac{16}{25}y_{k-2} - \frac{3}{25}y_{k-3} + \frac{12}{25}hf(x_{k+1}, y_{k+1})$$

Classical Runge-Kutta

$$\begin{aligned}
y_{k+1} &= y_k + \frac{h}{6}(V_1 + 2V_2 + 2V_3 + V_4), \\
V_1 &= f(x_k, y_k), \quad V_2 = f\left(x_k + \frac{1}{2}h, y_k + \frac{1}{2}hV_1\right) \\
V_3 &= f\left(x_k + \frac{1}{2}h, y_k + \frac{1}{2}hV_2\right), \quad V_4 = f(x_k + h, y_k + hV_3)
\end{aligned}$$

Of these, the implicit (as well as the ordinary) Euler method is a *first-order* method, i.e. the error is, under favorable conditions, of order $\mathcal{O}(h)$. The trapezoidal rule, the midpoint rule and BDF-2 are *second-order* methods (error $\mathcal{O}(h^2)$ for sufficiently regular solutions), BDF-3 is of order 3, and BDF-4 and the classical Runge-Kutta are both of order 4. Among the examples, BDF-2, BDF-3 and BDF-4 represent multistep methods (the step length assumed constant). To start these, one must first compute $y_1 \dots y_{m-1}$ by a separate algorithm (BDF- m). — The name BDF is an abbreviation of the words Backward Differentiation Formula (see Exercise 5).

In all the examples except the last, we are dealing with an *implicit* method. By this is meant that the number y_{k+1} to be computed also appears 'under' the function f in the computation formula. Then the number cannot be found by function evaluations alone; instead one must solve an equation (in the case of a system, a system of equations) of the form $y_{k+1} = F(y_{k+1})$. In practice this is solved approximately, for example by a few (1–3) Newton iterations or, more simply, by fixed-point iteration:

$$t_{j+1} = F(t_j), \quad j = 0 \dots m; \quad y_{k+1} = t_{m+1}.$$

(Cf. Section V.7 — iterative solution methods for systems of equations will be discussed later.) As the initial value for the iteration one may take, e.g., the value $t_0 = y_k + hf(x_k, y_k)$ given by Euler's method.

Implicit methods such as implicit Euler and BDF- m are very popular as 'black box' solvers of differential equations, for the reason that they are relatively reliable in varying situations. — Let it only be noted here that this reliability shows up especially in so-called *stiff* problems, in which the solution contains rapidly (exponentially) decaying transients. In such a problem, for example, implicit Euler is a considerably more reliable alternative than the ordinary Euler method, even though there is no difference in the formal accuracy of the methods (see Exercise 6). By contrast, in problems where special difficulties of the mentioned type do not occur, the simpler *explicit* methods (such as Euler and the classical Runge-Kutta) are still noteworthy alternatives and also in use.

EXERCISES

1. In the following initial value problems, compute by Euler's method approximations to the numbers $y(x_k)$, $k = 1 \dots 5$, where $x_k = k/5$ (step length $h = 0.2$). Compare with the exact solution!

$$\text{a) } \begin{cases} y' = -y^2, \\ y(0) = 1 \end{cases} \quad \text{b) } \begin{cases} y' = y^2, \\ y(0) = 1 \end{cases} \quad \text{c) } \begin{cases} y' = -2xy^2, \\ y(0) = 1 \end{cases} \quad \text{d) } \begin{cases} y' = 2xy^2, \\ y(0) = 1 \end{cases}$$

2. The initial value problem

$$y(0) = 1, \quad y' = \frac{x}{\sqrt{y^2 + 1}} + \frac{2x^2}{x^2 + 1}$$

is solved numerically on the interval $[0, 10]$ using Euler's method and constant step length h . Estimate the constants L and M in the error estimate for Euler's method on that interval. For what step lengths can one guarantee that the relative error of the numerical solution does not exceed the value 10^{-4} at any computation point?

3. The initial value problem $y' = -1/y$, $y(0) = 3$ is solved numerically on the interval $[0, 1]$ using step length $h = 0.2$. Compute by the following methods an approximation y_5 to the number $y(1)$: Euler, Trapezoidal rule, BDF-2, BDF-3, Classical Runge-Kutta. Compare with the exact value!
4. In the initial value problem $y' = f(x)$, $y(0) = 0$, the classical Runge-Kutta method reduces to a well-known numerical integration rule. Which rule is it?
5. The solution method BDF- m for the differential equation $y' = f(x, y)$ is derived by writing the equation at the point $x = x_{k+1}$, making the approximation

$$y'(x_{k+1}) \approx \frac{1}{h} \sum_{j=0}^m \alpha_j y_{k+1-j}, \quad y_i = y(x_i)$$

and requiring that this be exact for polynomials of degree $\leq m$. Verify, by computing the coefficients α_j , that the result is indeed the method BDF-2 when $m = 2$. What kind of method is BDF-1?

6. Let $\tau = 0.02$. We wish to solve numerically the initial value problem

$$\begin{cases} \tau y' + y = \sin x, & x > 0, \\ y(0) = 0.1. \end{cases}$$

Compare the solutions given by the ordinary and the implicit Euler method by computing with both methods $y_k \approx y(kh)$, $k = 1 \dots 5$ with step lengths $h = 0.01$, $h = 0.04$ and $h = 0.08$. Sketch all the solutions graphically and compare also with the exact solution! (See the application example in Section IX.4.)

7. A spacecraft moving in the plane is subject to the gravitational attraction of the Earth located at the origin, $\vec{F} = -(k/r^2) \vec{e}_r$, where k is a constant (polar coordinate system). At time t the spacecraft is at the position $(x(t), y(t))$ and its velocity is $u(t)\vec{i} + v(t)\vec{j}$. Write the equation of motion $m\vec{r}'' = \vec{F}$ of the spacecraft in Cartesian coordinates as a differential-equation system in normal form, and present an algorithm based on Euler's method for solving the equations numerically. Assume the initial values x_0, y_0, u_0, v_0 at time $t = 0$ to be known.
8. (*) When solving the initial value problem

$$\begin{cases} y' = \frac{y^2}{x+y+1} + 4x + 2, & x > 0, \\ y(0) = A \end{cases}$$

by Euler's method with step length $h = 0.1$, one observes, surprisingly, that the exact solution is obtained, i.e. $y_k = y(x_k) \forall k$ ($x_k = kh$). What is the value of A ?

IX.8 *The Picard–Lindelöf theorem

Let us return to studying the first-order initial value problem

$$\begin{cases} y' = f(x, y), \\ y(x_0) = y_0 \end{cases} \quad (1)$$

this time from a more theoretical standpoint. We pose the question: Under what conditions, imposed on the function f and the numbers x_0, y_0 , can one guarantee that the initial value problem has a unique solution in a neighborhood of the point x_0 ? The answer is given by the following famous theorem.

***THEOREM IX.8.1 (Picard–Lindelöf*)** Let $T = [a, b] \times [c, d] \subset \mathbb{R}^2$ and assume that $f : T \rightarrow \mathbb{R}$ satisfies the conditions

- (i) The function $x \mapsto f(x, y)$ is continuous on the interval $[a, b]$ for every $y \in [c, d]$.
- (ii) The function $y \mapsto f(x, y)$ is, for every $x \in [a, b]$, Lipschitz continuous on the interval $[c, d]$ with a constant that does not depend on x , i.e. $\exists L \in \mathbb{R}_+$ such that

$$|f(x, y_1) - f(x, y_2)| \leq L|y_1 - y_2| \quad \forall (x, y_1) \in T, (x, y_2) \in T.$$

Then, if $(x_0, y_0) \in (a, b) \times (c, d)$, there exists $\delta > 0$ such that the initial value problem

$$\begin{cases} y' = f(x, y), & x \in (x_0 - \delta, x_0 + \delta), \\ y(x_0) = y_0 \end{cases}$$

has a unique solution. Furthermore, if the assumptions hold when $\mathbb{R}$ takes the place of the interval $[c, d]$ ($T = [a, b] \times \mathbb{R}$), then the solution exists and is unique on the whole open interval (a, b) , when $(x_0, y_0) \in (a, b) \times \mathbb{R}$.

In the following we briefly go through the ideas and main stages of the proof of Theorem IX.8.1. (For carrying the proof through rigorously, our conceptual provisions are a little inadequate.) The starting point is the idea that the solution of the initial value problem is constructed by so-called *Picard iteration*. The idea

*The original version of the Picard–Lindelöf theorem was presented and proved, in a more general form than the one given here, by the French mathematician **Emile Picard** (1853–1941) in 1893. The Finnish mathematician **Ernst Lindelöf** (1870–1946) refined the result somewhat later. The latter part of the theorem’s assertion (concerning solvability on the whole interval under consideration) is Lindelöf’s work.

here, in turn, is the following: When one is near the point x_0 , one may assume that $y(x)$ is near y_0 . Accordingly, the first approximation to the solution is a constant:

$$y(x) \approx y_0(x) = y_0.$$

We now improve this step by step by solving recursively the initial value problems

$$\begin{cases} y'_k(x) = f(x, y_{k-1}(x)), \\ y_k(x_0) = y_0, \quad k = 1, 2, \dots \end{cases}$$

Since here y_{k-1} is known from earlier iteration rounds (or from the initial condition), each y_k is determined recursively as the integral of a known function:

$$y_0(x) = y_0, \quad y_k(x) = y_0 + \int_{x_0}^x f(t, y_{k-1}(t)) dt, \quad k = 1, 2, \dots \quad (2)$$

Thus Picard iteration has been defined. — Note that this is in fact a fixed-point iteration, in which instead of a 'point' one seeks a function (cf. Section V.7). The iteration is applied to the integrated form of the initial value problem

$$y(x) = y_0 + \int_{x_0}^x f(t, y(t)) dt,$$

i.e. Picard iteration is a matter of solving this *integral equation* iteratively. — By truncating the iteration one obtains an approximate solution, as in fixed-point iterations generally.

Picard iteration can thus be regarded as a method for approximately solving the initial value problem. Unlike the numerical methods considered in the previous section, however, Picard iteration is mainly a 'method of thought', since in practice the integrals involved in the iteration usually become difficult quickly.

EXAMPLE 1 Compute the first two Picard iterates ($y_k(x)$, $k = 1, 2$) for the initial value problem

$$\begin{cases} y' = x + y^2, \\ y(0) = 1. \end{cases}$$

Solution Here $f(x, y) = x + y^2$, $x_0 = 0$, $y_0 = 1$, so $y_0(x) = 1$ and

$$\begin{aligned} y_1(x) &= 1 + \int_0^x (t + 1) dt = 1 + x + \frac{1}{2}x^2, \\ y_2(x) &= 1 + \int_0^x \left[t + \left(1 + t + \frac{1}{2}t^2 \right)^2 \right] dt \\ &= 1 + x + \frac{3}{2}x^2 + \frac{2}{3}x^3 + \frac{1}{4}x^4 + \frac{1}{20}x^5. \end{aligned}$$

The next iterate $y_3(x)$ would already be a polynomial of degree 11. $\square$

Let us return to the proof of Theorem IX.8.1, in which the idea, then, is to construct the solution of the initial value problem by Picard iteration (2). The aim is then to show that the iteration converges to a function y continuous on the interval $[x_0 - \delta, x_0 + \delta]$ that satisfies

$$y(x) = y_0 + \int_{x_0}^x f(t, y(t)) dt, \quad x \in [x_0 - \delta, x_0 + \delta]. \quad (3)$$

By the choice of the parameter δ (δ sufficiently small) it is guaranteed that $[x_0 - \delta, x_0 + \delta] \subset [a, b]$ and that $y(x)$ does not 'escape' from the interval $[c, d]$, i.e. $(x, y(x)) \in T \quad \forall x \in [x_0 - \delta, x_0 + \delta]$ (cf. the corresponding idea in the proof of Theorem IX.2.1). Then, by assumption (i), the composite function $x \mapsto f(x, y(x))$ is continuous on the interval $[x_0 - \delta, x_0 + \delta]$, whence from equation (3) and the fundamental theorem of calculus it follows that y is continuously differentiable on the interval $[x_0 - \delta, x_0 + \delta]$. Then y is also a solution of the initial value problem (1) on the interval $(x_0 - \delta, x_0 + \delta)$.

To establish the convergence of the iteration (2), we introduce the *maximum norm* that measures a continuous function defined on the interval $[x_0 - \delta, x_0 + \delta]$

$$\|y\| = \max \{ |y(x)| \mid x_0 - \delta \leq x \leq x_0 + \delta \}.$$

For it (as for norms in general, cf. Section II.4) the triangle inequality holds

$$\|y_1 + y_2\| \leq \|y_1\| + \|y_2\|.$$

Let us now start from assumption (ii) of Theorem IX.8.1 and estimate

$$\begin{aligned} |y_{k+1}(x) - y_k(x)| &= \left| \int_{x_0}^x [f(t, y_k(t)) - f(t, y_{k-1}(t))] dt \right| \\ &\leq \left| \int_{x_0}^x L |y_k(t) - y_{k-1}(t)| dt \right| \\ &\leq L|x - x_0| \|y_k - y_{k-1}\| \\ &\leq L\delta \|y_k - y_{k-1}\|, \quad x \in [x_0 - \delta, x_0 + \delta], \quad k = 1, 2, \dots \end{aligned}$$

The result can be written

$$\|y_{k+1} - y_k\| \leq L\delta \|y_k - y_{k-1}\|, \quad k = 1, 2, \dots$$

Write now $q = L\delta$, and assume in what follows that $q < 1$ (which can be arranged by choosing $\delta < 1/L$). Using the obtained estimate recursively, it follows that

$$\|y_{k+1} - y_k\| \leq q^k \|y_1 - y_0\|, \quad k = 1, 2, \dots$$

By assumption (i), the function $g(x) = f(x, y_0)$ is continuous on the interval $[a, b]$, so by the iteration formula (2)

$$\|y_1 - y_0\| = \max_{x \in [x_0 - \delta, x_0 + \delta]} \left| \int_{x_0}^x f(t, y_0) dt \right| \leq C\delta,$$

where C is the maximum value of $|g|$ on the interval $[a, b]$. Combining the obtained estimates and using the triangle inequality, it follows that

$$\begin{aligned} \|y_n - y_k\| &= \|(y_n - y_{n-1}) + (y_{n-1} - y_{n-2}) + \dots + (y_{k+1} - y_k)\| \\ &\leq \|y_n - y_{n-1}\| + \|y_{n-1} - y_{n-2}\| + \dots + \|y_{k+1} - y_k\| \\ &\leq (q^{n-1} + q^{n-2} + \dots + q^k) C\delta \\ &< C\delta q^k \sum_{i=0}^{\infty} q^i = \frac{C\delta}{1-q} q^k, \quad 0 \leq k < n. \end{aligned}$$

From this one can conclude, first, that if δ is sufficiently small, then $\|y_n - y_0\| \leq C\delta/(1 - \delta) \leq \min\{y_0 - c, d - y_0\} \forall n$, whereby the escape of $y_n(x)$ from the interval $[c, d]$ is prevented. Second, one concludes that the sequence $\{y_k(x)\}$ is Cauchy for every $x \in [x_0 - \delta, x_0 + \delta]$ and hence converges:

$$y_k(x) \rightarrow y(x) \in [c, d], \quad x \in [x_0 - \delta, x_0 + \delta].$$

It can further be concluded that this convergence is in fact *uniform* on the interval $[x_0 - \delta, x_0 + \delta]$, i.e.

$$\|y_k - y\| = \max_{x \in [x_0 - \delta, x_0 + \delta]} |y_k(x) - y(x)| \rightarrow 0, \quad \text{as } k \rightarrow \infty,$$

and that the $y(x)$ constructed as the limit is continuous on the interval $[x_0 - \delta, x_0 + \delta]$. (This is the conceptually most demanding point of the proof!) Finally, by taking the limit on both sides in the iteration formula (2) as $k \rightarrow \infty$, one concludes that $y(x)$ solves the problem (3), whereby the part of the first assertion of Theorem IX.8.1 concerning the existence of the solution has been proved. The uniqueness of the solution can be established more straightforwardly, see Exercise 6.

In proving the second assertion of Theorem IX.8.1, there is no escape problem for the sequence $\{y_k(x)\}$, so the interval $[x_0 - \delta, x_0 + \delta]$ can be replaced by the interval $[a, b]$. By refining the above computation, it turns out that one can also estimate

$$|y_{k+1}(x) - y_k(x)| \leq C \frac{L^k |x - x_0|^k}{k!}, \quad x \in [a, b].$$

In studying the convergence of the sequence $\{y_k(x)\}$, the point of comparison now becomes, instead of a geometric series, the series of the exponential function

$$e^{L|x-x_0|} = \sum_{k=0}^{\infty} \frac{L^k |x-x_0|^k}{k!}.$$

This converges everywhere, so one can conclude in the same way as above that the function $y(x) = \lim_k y_k(x)$ is a solution of the integral equation (3), this time on the whole interval $[a, b]$. The initial value problem (1) has thus been solved uniquely on the interval (a, b) .

EXAMPLE 2 The initial value problem

$$\begin{cases} y' = \sin y, \\ y(x_0) = y_0 \end{cases}$$

satisfies the conditions of Theorem IX.8.1 with $T = \mathbb{R}^2$, since $f(y) = \sin y$ is Lipschitz continuous on all intervals with constant $L = 1$. The initial value problem therefore has a unique solution on the interval $(-\infty, \infty)$ for every $(x_0, y_0) \in \mathbb{R}^2$. If one chooses, e.g., $x_0 = 0$, then one sees that the solutions corresponding to the different values of y_0 are all found among the solutions computed in Example IX.2.3. This confirms that in the said example all solutions of the differential equation $y' = \sin y$ were found. $\square$

EXAMPLE 3 For the initial value problem

$$\begin{cases} y' = \sqrt{|y|}, \\ y(0) = 0 \end{cases}$$

the conditions of Theorem IX.8.1 are not satisfied for any choice of the set $T = (a, b) \times (c, d)$, since one must have $0 \in (a, b)$, whereupon $f(y) = \sqrt{|y|}$ is not Lipschitz continuous on the interval $[a, b]$. Indeed the solution is not unique (see Example IX.1:3b). $\square$

EXAMPLE 4 If in the linear initial value problem

$$\begin{cases} y' + P(x)y = R(x), \\ y(x_0) = y_0 \end{cases}$$

P and Q are continuous on the interval $[a, b]$, then the conditions of Theorem IX.8.1 hold when we choose $T = [a, b] \times \mathbb{R}$. Namely

- (i) The function $x \mapsto f(x, y) = -P(x)y + R(x)$ is continuous on the interval $[a, b]$ for every $y \in \mathbb{R}$.
- (ii) $|f(x, y_1) - f(x, y_2)| = |P(x)||y_1 - y_2| \leq L|y_1 - y_2|$, $(x, y_1), (x, y_2) \in T$, where $L = \max_{x \in [a, b]} |P(x)|$.

The initial value problem thus has a unique solution on the interval (a, b) when $x_0 \in (a, b)$. (The same was concluded in Section IX.4 by other means.) $\square$

Solvability of a system of differential equations

The real elegance of the Picard–Lindelöf theorem lies in the fact that the theorem’s assertion can be generalized very straightforwardly to concern a general differential-equation system in normal form (cf. Section IX.1)

$$\begin{cases} y_1'(x) = f_1(x, y_1, \dots, y_n), \\ y_2'(x) = f_2(x, y_1, \dots, y_n), \\ \vdots \\ y_n'(x) = f_n(x, y_1, \dots, y_n). \end{cases}$$

As initial conditions we assume

$$y_i(x_0) = A_i, \quad i = 1 \dots n,$$

where $x_0 \in (a, b)$. As an example, consider an initial value problem of size $n = 2$:

$$\begin{cases} y' = f_1(x, y, u), \\ u' = f_2(x, y, u), & x \in (a, b), \\ y(x_0) = y_0, \quad u(x_0) = u_0. \end{cases} \quad (4)$$

In the formulation of the Picard–Lindelöf theorem for this problem one sets

$$T = [a, b] \times [c_1, d_1] \times [c_2, d_2] \subset \mathbb{R}^3 \quad \text{or} \quad T = [a, b] \times \mathbb{R}^2 \subset \mathbb{R}^3$$

and assumes

- (i) The functions $x \mapsto f_1(x, y, u)$ and $x \mapsto f_2(x, y, u)$ are continuous on the interval $[a, b]$ for every $(y, u) \in [c_1, d_1] \times [c_2, d_2]$ or for every $(y, u) \in \mathbb{R}^2$.

- (ii) f_1 and f_2 satisfy the Lipschitz condition with respect to the variables y, u :
For some $L \in \mathbb{R}_+$

$$|f_i(x, y_1, u_1) - f_i(x, y_2, u_2)| \leq L \max\{|y_1 - y_2|, |u_1 - u_2|\}, \quad i = 1, 2$$

$$\forall (x, y_1, u_1) \in T, (x, y_2, u_2) \in T.$$

Under these assumptions one arrives at an assertion analogous to the one above, i.e. the initial value problem (4) has a unique solution, either near x_0 on the interval $(x_0 - \delta, x_0 + \delta)$, or, under stronger assumptions, on the whole interval (a, b) . The proof, too, is in its basic idea the same as above: the starting point is the construction of the solution by Picard iteration

$$\begin{cases} y_k(x) = y_0 + \int_{x_0}^x f_1(t, y_{k-1}(t), u_{k-1}(t)) dt, \\ u_k(x) = u_0 + \int_{x_0}^x f_2(t, y_{k-1}(t), u_{k-1}(t)) dt, \quad k = 1, 2, \dots \end{cases}$$

Let us finally return to the second-order linear and homogeneous initial value problem considered in Section IX.6

$$\begin{cases} y'' + P(x)y' + Q(x)y = 0, \\ y(0) = A, \quad y'(0) = B. \end{cases} \quad (5)$$

Writing here $u = y'$, one arrives at the system form (4), where

$$f_1(x, y, u) = u, \quad f_2(x, y, u) = -Q(x)y - P(x)u.$$

If P and Q are continuous on the interval (a, b) (as was assumed in Section IX.6), then $x \mapsto f_1(x, y, u)$ and $x \mapsto f_2(x, y, u)$ are continuous on the interval $[a_1, b_1]$ for every $(y, u) \in \mathbb{R}^2$, when $[a_1, b_1] \subset (a, b)$. Since

$$\begin{aligned} f_1(x, y_1, u_1) - f_1(x, y_2, u_2) &= u_1 - u_2, \\ f_2(x, y_1, u_1) - f_2(x, y_2, u_2) &= -Q(x)(y_1 - y_2) - P(x)(u_1 - u_2), \end{aligned}$$

one sees that condition (ii) too is satisfied when one sets

$$L = \max\{1, L_1\}, \quad L_1 = \max_{x \in [a_1, b_1]} \{|P(x)| + |Q(x)|\}.$$

Consequently, by the (stronger assertion of the) Picard–Lindelöf theorem, the initial value problem (5) has a unique solution on the interval (a_1, b_1) . Since this is true whenever $[a_1, b_1] \subset (a, b)$, the problem has a unique solution on the whole open interval (a, b) . Theorem IX.6.2 has thus been proved as a consequence of the Picard–Lindelöf theorem.

EXERCISES

1. What can be said, in light of the Picard–Lindelöf theorem, about the solvability of the following initial value problems and the uniqueness of their solutions on $\mathbb{R}$ or on subintervals of $\mathbb{R}$?

$$\begin{aligned} \text{a)} \quad y' &= \cos y + \frac{x}{y^2 + 1}, \quad y(0) = A \in \mathbb{R} \\ \text{b)} \quad y' &= \frac{y}{\cosh x} + e^{x-y^2}, \quad y(0) = A \in \mathbb{R} \\ \text{c)} \quad y' &= \frac{y}{\cos x} + e^{x-y^2}, \quad y(0) = A \in \mathbb{R} \\ \text{d)} \quad y' &= 8xy + \frac{\sin(xy)}{x^2 + 2x - 35}, \quad y(0) = A \in \mathbb{R} \\ \text{e)} \quad y'' &= \cos(xy), \quad y(0) = A \in \mathbb{R}, \quad y'(0) = B \in \mathbb{R} \end{aligned}$$

2. For the following initial value problems, determine all solutions valid in a neighborhood of the initial point. Compare with the assertion of the Picard–Lindelöf theorem.

$$\begin{array}{ll} \text{a)} \quad y' = y^2, \quad y(0) = 1 & \text{b)} \quad y' = y^2, \quad y(0) = 1000 \\ \text{c)} \quad y' = y\sqrt{|y|}, \quad y(0) = 1 & \text{d)} \quad y' = \sqrt[4]{|y|}, \quad y(0) = 1 \\ \text{e)} \quad y' = \sqrt[4]{|y|}, \quad y(0) = 0.0001 & \text{f)} \quad y' = \sqrt[4]{|y|}, \quad y(0) = 0 \end{array}$$

3. We wish to compute Napier's number e starting from the fact that $e = y(1)$, where $y(x)$ solves the initial value problem $y' = y$, $y(0) = 1$. What kind of formula for e is obtained when the problem is solved by Picard iteration?
4. In the following initial value problems, compute the Picard iterates $y_1(x)$ and $y_2(x)$ and compare with the exact solution. (Cf. Exercise [IX.7:1](#).)

$$\begin{array}{llll} \text{a)} \quad \begin{cases} y' = -y^2, \\ y(0) = 1 \end{cases} & \text{b)} \quad \begin{cases} y' = y^2, \\ y(0) = 1 \end{cases} & \text{c)} \quad \begin{cases} y' = -2xy^2, \\ y(0) = 1 \end{cases} & \text{d)} \quad \begin{cases} y' = 2xy^2, \\ y(0) = 1 \end{cases} \end{array}$$

5. For the initial value problem

$$\begin{array}{ll} \text{a)} \quad \begin{cases} y'' = -y, \\ y(0) = 1, \quad y'(0) = 0 \end{cases} & \text{b)} \quad \begin{cases} y'' = -y, \\ y(0) = 0, \quad y'(0) = 1 \end{cases} \end{array}$$

compute an approximate solution by iterating $2n$ times ($n \in \mathbb{N}$) with Picard iteration from the system form of the problem.

6. (*) Show that, under the assumptions of Theorem [IX.8.1](#), the solution of the initial value problem [\(1\)](#) (if it exists) is unique on the interval $(x_0 - \delta, x_0 + \delta)$ when δ is sufficiently small. *Hint:* Assume $y_1(x)$ and $y_2(x)$ to be solutions, start from the integral form [\(3\)](#) of the problem, use assumption (ii) and choose $\delta < 1/L$.

Chapter X

Matrices

”If a system of equations has more than 50 unknowns, it is advisable to use a computer to solve it”.*

In many fields of engineering there has been a long tradition of solving *linear systems of equations* already before the age of computers. In the mathematics of linear systems of equations too, in *linear algebra*, the traditions run long. They reach back to the 18th century, when a theory of the solvability of systems of equations based on the concept of the *determinant*, together with solution formulas, was developed. Determinants still find use today in theoretical considerations, in various computational formulas (cf. Section II.6), and more generally in hand computation when the size of the system is small. Otherwise the theory of determinants must be regarded as outdated, for the reason that the computational formulas of this theory are not suited to large-scale numerical use. In the age of computers, linear algebra has indeed largely broken away from the determinant tradition and from the old methods of hand computation.

In this chapter the object of study is the key concept of linear algebra, the *matrix*. Computation with matrices, that is, *matrix algebra*, is examined first in Section X.1. After this, in Sections X.2–X.4, we study linear systems of equations, their solution by *Gaussian elimination*, and the concept of the *inverse matrix*, which is central to that solution. In Section X.5 we briefly go through the old theory of determinants together with its computational formulas. In Sections X.6–X.7 the objects of study are the *linear maps* and *affine maps* based on matrix algebra, together with their geometric applications, the *geometric maps*. In the final subsection we present some application examples of linear systems of equations from traditional engineering mathematics.

*A remark by an engineer versed in the strength analysis of aircraft wings, in a professional publication of the field in the 1950s.

X.1 Matrix algebra

A general *linear system of equations* (or *-system*) of m equations in n unknowns is of the form

$$\begin{cases} a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n &= b_1, \\ a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n &= b_2, \\ \vdots &\vdots \\ a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mn}x_n &= b_m. \end{cases} \quad (1)$$

Here $m, n \in \mathbb{N}$, the a_{ij} are the *coefficients* of the system, and the numbers b_i , $i = 1 \dots m$, which form the so-called ‘right-hand side’ (or ‘constant term’) of the system, are also assumed to be known. The unknowns are thus x_i , $i = 1 \dots n$, and the problem is generally to find these, that is, to *solve* the system. In what follows we assume as a rule that $a_{ij}, x_i, b_i \in \mathbb{R}$. More generally one could assume that $a_{ij}, x_i, b_i \in \mathbb{K}$, where $\mathbb{K}$ is any field, since to write down and solve a system one needs only the arithmetic operations of the field. For instance one could have $\mathbb{K} = \mathbb{C}$ or (as often in simple examples) $\mathbb{K} = \mathbb{Q}$.

The system (1) is said to be of *size* $m \times n$ (*‘m by n’*). A system need not have a solution at all, or the solution may be non-unique. If a system has a unique solution no matter how the numbers b_i on the right-hand side are chosen, then the system is said to be *regular*. Otherwise, that is, if there is no solution for some (b_i) or the solution is non-unique, the system is irregular, or *singular*. The regularity or singularity of a system thus does not depend on the numbers b_i but is a property of the coefficient array (a_{ij}) alone. We shall see that the condition $m = n$ is a necessary condition for regularity, i.e. in a regular system the number of equations (m) and the number of unknowns (n) must match. In the case $m > n$ the system (1) is called *overdetermined*, in the case $m < n$ *underdetermined*. These are thus always singular systems.

EXAMPLE 1 Singular systems of type $m = n = 2$ are, for example,

$$\begin{cases} x_1 + x_2 = b_1, \\ x_1 + x_2 = b_2 \end{cases} \quad \text{and} \quad \begin{cases} x_1 + 2x_2 = b_1, \\ 2x_1 + 4x_2 = b_2. \end{cases} \quad \square$$

The starting point of the algebraic considerations to follow is the general linear system of equations (1). First we write the system in array form (1st stage of abstraction)

$$\begin{bmatrix} a_{11}x_1 + & \dots & + a_{1n}x_n \\ \vdots & & \vdots \\ a_{m1}x_1 + & \dots & + a_{mn}x_n \end{bmatrix} = \begin{bmatrix} b_1 \\ \vdots \\ b_m \end{bmatrix},$$

that is,

$$\mathbf{y} = \mathbf{b}, \quad (2)$$

where

$$\mathbf{b} = \begin{bmatrix} b_1 \\ \vdots \\ b_m \end{bmatrix}, \quad \mathbf{y} = \begin{bmatrix} y_1 \\ \vdots \\ y_m \end{bmatrix}, \quad y_i = \sum_{j=1}^n a_{ij}x_j.$$

Here $\mathbf{b}, \mathbf{y}$ are so-called *column vectors*. In what follows we use bold lowercase letters $\mathbf{a}, \mathbf{b}, \mathbf{x}, \mathbf{y}$, etc., as symbols for column vectors.

In the above column vectors $\mathbf{y}, \mathbf{b}$, the number m is the *size* (or type) of the vector, which can also be expressed by the term *m-vector*. The numbers b_i, y_i are the *entries* (or components) of those vectors. One can also use the notations

$$[\mathbf{b}]_i = b_i, \quad \mathbf{b} = (b_i), \quad \mathbf{b} = (b_i)_{i=1}^m.$$

The term ‘vector’ refers to the fact that for column vectors one can define the basic vector operations, that is, addition of vectors and multiplication by a scalar. The definitions are

$$\mathbf{x} + \mathbf{y} = \begin{bmatrix} x_1 + y_1 \\ \vdots \\ x_n + y_n \end{bmatrix}, \quad \lambda \mathbf{x} = \begin{bmatrix} \lambda x_1 \\ \vdots \\ \lambda x_n \end{bmatrix} \quad (\lambda \in \mathbb{R}).$$

In addition the vectors $\mathbf{x}, \mathbf{y}$ must be of the same size, otherwise addition cannot be defined. In these operations one thus adds and multiplies entrywise, as was done earlier in Sections II.2 and II.6 when the number pairs of $\mathbb{R}^2$ and the number triples of $\mathbb{R}^3$ were treated as vectors. Column vectors of size n may indeed be regarded as a generalization of these already familiar ‘algebra vectors’. (Treating number pairs or triples as column vectors does not affect the vector operations in any way.) In keeping with the earlier notation, we agree that the column vectors of size n form a vector space called $\mathbb{R}^n$. The *zero vector* of the space $\mathbb{R}^n$ is, as before, the vector all of whose entries are zero. This is denoted by the symbol $\mathbf{0}$. The natural *basis* of the space $\mathbb{R}^n$ is obtained by writing

$$\mathbf{x} = x_1 \mathbf{e}_1 + \dots + x_n \mathbf{e}_n = \sum_{i=1}^n x_i \mathbf{e}_i,$$

whereupon

$$\mathbf{e}_1 = \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}, \quad \mathbf{e}_2 = \begin{bmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{bmatrix}, \quad \dots \quad \mathbf{e}_n = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{bmatrix},$$

that is,

$$[\mathbf{e}_i]_j = \delta_{ij} = \begin{cases} 1, & \text{if } i = j, \\ 0, & \text{if } i \neq j. \end{cases}$$

Here the symbol δ_{ij} is the so-called *Kronecker delta* (-symbol). The vectors $\mathbf{e}_i$ are called the *unit vectors* of $\mathbb{R}^n$. Every $\mathbf{x} \in \mathbb{R}^n$ can thus be written (evidently uniquely) as a *linear combination* of the vectors $\mathbf{e}_i$, that is, $\{\mathbf{e}_1, \dots, \mathbf{e}_n\}$ is a basis of $\mathbb{R}^n$. Since the basis contains n vectors, one says that the *dimension* of $\mathbb{R}^n$ is n :

$$\dim \mathbb{R}^n = n.$$

For the time being it thus makes no difference whether the vectors of $\mathbb{R}^n$ stand ‘upright’ or ‘on their side’, as long as in equation (2) the equality sign is interpreted in the normal way, i.e.

$$\mathbf{y} = \mathbf{b} \quad \Leftrightarrow \quad y_i = b_i, \quad i = 1 \dots n.$$

In the next (second) stage of abstraction, however, a difference arises, when in equation (2) the coefficients a_{ij} are separated out into a distinct object by writing

$$\mathbf{y} = \begin{bmatrix} y_1 \\ \vdots \\ y_m \end{bmatrix} = \begin{bmatrix} a_{11} & \dots & a_{1n} \\ \vdots & & \vdots \\ a_{m1} & \dots & a_{mn} \end{bmatrix} \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} = \mathbf{A}\mathbf{x},$$

whereupon the original system of equations (1) becomes written in the form

$$\mathbf{A}\mathbf{x} = \mathbf{b}. \tag{3}$$

The object $\mathbf{A}$, which thus looks like a two-dimensional array of numbers, is called a *matrix*. The indices m, n determine that the matrix is of *type* or *size* $m \times n$ (‘ m by n ’). Here m is the number of (horizontal) *rows* and n the number of *columns* (or vertical rows) of the matrix. The numbers a_{ij} are called the *entries* or *elements* of the matrix. For these one can also use the notation $[\mathbf{A}]_{ij}$, and one can also write

$$\mathbf{A} = (a_{ij}) \quad \text{or} \quad \mathbf{A} = (a_{ij}, \quad i = 1 \dots m, \quad j = 1 \dots n).$$

By definition, a column vector of size m (an m -vector) is a matrix of type $m \times 1$. In what follows we nonetheless distinguish matrices more generally from vectors by using bold uppercase letters $\mathbf{A}$, $\mathbf{B}$, etc., as symbols for matrices.

Like a vector, a matrix too ‘comes to life’ only through the arithmetic operations defined for it. In equation (3) there already appears one of the most central operations: the multiplication of a matrix by a vector. Comparing equations (1)–(3) one sees at once what the definition of this product is: $\mathbf{Ax}$ is a column vector of size m whose entries $[\mathbf{Ax}]_i$ are defined by

$$\begin{aligned} [\mathbf{Ax}]_i &= \sum_{j=1}^n a_{ij}x_j, \quad i = 1 \dots m, \\ \mathbf{x} &= (x_i, i = 1 \dots n), \\ \mathbf{A} &= (a_{ij}, i = 1 \dots m, j = 1 \dots n). \end{aligned}$$

By definition, the number of columns of $\mathbf{A}$ must match the size of the vector $\mathbf{x}$ in order for $\mathbf{Ax}$ to be defined. Compatibility conditions of this type restrict all arithmetic operations between matrices.

A second way of expressing the product of a matrix and a vector is obtained by letting $\mathbf{a}_j$ = the j th column of $\mathbf{A}$, i.e.

$$\mathbf{a}_j = \begin{bmatrix} a_{1j} \\ \vdots \\ a_{mj} \end{bmatrix}.$$

When $\mathbf{A}$ is expressed by means of its columns, then from the above definition one sees that

$$\mathbf{A} = [\mathbf{a}_1 \dots \mathbf{a}_n] \quad \Rightarrow \quad \mathbf{Ax} = \sum_{j=1}^n x_j \mathbf{a}_j.$$

Thus the product of a matrix $\mathbf{A}$ and a vector $\mathbf{x} = (x_i)$ is a linear combination of the columns $\mathbf{a}_j$ of $\mathbf{A}$, with coefficients x_j . In particular,

$$\mathbf{A} = [\mathbf{a}_1 \dots \mathbf{a}_n] \quad \Rightarrow \quad \mathbf{Ae}_k = \mathbf{a}_k, \quad k = 1 \dots n.$$

From this, in turn, it is easy to deduce the following result, which will be useful in what follows.

PROPOSITION X.1.1 If $\mathbf{A}$ and $\mathbf{B}$ are matrices of size $m \times n$, then

$$\mathbf{Ax} = \mathbf{Bx} \quad \forall \mathbf{x} \in \mathbb{R}^n \quad \Rightarrow \quad \mathbf{A} = \mathbf{B}.$$

EXAMPLE 2 Compute $\mathbf{Ax}$ and $\mathbf{Bx}$, when

$$\mathbf{A} = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} 1 & 3 \\ 2 & -1 \\ 1 & 1 \end{bmatrix}, \quad \mathbf{x} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}.$$

Solution

$$\mathbf{Ax} = \begin{bmatrix} 1 \cdot 1 + 2 \cdot (-1) \\ 0 \cdot 1 + 1 \cdot (-1) \end{bmatrix} = (1) \begin{bmatrix} 1 \\ 0 \end{bmatrix} + (-1) \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} -1 \\ -1 \end{bmatrix},$$

$$\mathbf{Bx} = \begin{bmatrix} 1 \cdot 1 + 3 \cdot (-1) \\ 2 \cdot 1 + (-1) \cdot (-1) \\ 1 \cdot 1 + 1 \cdot (-1) \end{bmatrix} = (1) \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} + (-1) \begin{bmatrix} 3 \\ -1 \\ 1 \end{bmatrix} = \begin{bmatrix} -2 \\ 3 \\ 0 \end{bmatrix}. \quad \square$$

From the above definition one sees that multiplication by a matrix is a linear operation with respect to the vector being multiplied:

$$\mathbf{A}(\lambda \mathbf{x} + \mu \mathbf{y}) = \lambda \mathbf{Ax} + \mu \mathbf{Ay} \quad \forall \lambda, \mu \in \mathbb{R}.$$

(Here it is of course required that $\mathbf{x}$ and $\mathbf{y}$ be of the same size and compatible with $\mathbf{A}$.)

For matrices we define multiplication by a scalar and addition of matrices of the same size in the same way as for vectors, that is, the operations are carried out entrywise. Thus if $\mathbf{A} = (a_{ij})$ and $\mathbf{B} = (b_{ij})$ are of the same size, then

$$[\lambda \mathbf{A}]_{ij} = \lambda a_{ij}, \quad [\mathbf{A} + \mathbf{B}]_{ij} = a_{ij} + b_{ij}.$$

The vector operations in $\mathbb{R}^n$ can be interpreted as special cases of these operations ($\mathbf{A}$ and $\mathbf{B}$ of size $n \times 1$). Like vectors of the same size, matrices of the same size, equipped with these operations, also form a vector space. The zero element of this space is the so-called *zero matrix*, all of whose entries are zero (symbol $\mathbf{0}$).

The matrix product

The product of two matrices $\mathbf{A}$, $\mathbf{B}$, that is, the *matrix product*, can be defined provided that the matrices are compatible in size. Let $\mathbf{x}$ be a column vector of size n and let $\mathbf{A} = (a_{ij})$ be of size $m \times p$ and $\mathbf{B} = (b_{ij})$ of size $p \times n$ ($m, n, p \in \mathbb{N}$). Then the composite matrix-vector product $\mathbf{A}(\mathbf{Bx})$ is defined when $\mathbf{x} \in \mathbb{R}^n$. We wish to write this in the form

$$\mathbf{A}(\mathbf{Bx}) = (\mathbf{AB})\mathbf{x}, \quad \mathbf{x} \in \mathbb{R}^n \tag{4}$$

and to take this as the definition of the matrix product $\mathbf{AB}$. Then one must have

$$\begin{aligned} [(\mathbf{AB})\mathbf{x}]_i &= [\mathbf{A}(\mathbf{Bx})]_i = \sum_{k=1}^p a_{ik}[\mathbf{Bx}]_k = \sum_{k=1}^p a_{ik} \sum_{j=1}^n b_{kj}x_j \\ &= \sum_{j=1}^n \left(\sum_{k=1}^p a_{ik}b_{kj} \right) x_j = \sum_{j=1}^n [\mathbf{AB}]_{ij}x_j. \end{aligned}$$

Thus one sees that (4) holds when one defines

$$\begin{aligned} [\mathbf{AB}]_{ij} &= \sum_{k=1}^p a_{ik} b_{kj}, \quad i = 1 \dots m, \quad j = 1 \dots n, \\ \mathbf{A} &= (a_{ij}, \quad i = 1 \dots m, \quad j = 1 \dots p), \\ \mathbf{B} &= (b_{ij}, \quad i = 1 \dots p, \quad j = 1 \dots n). \end{aligned}$$

The matrix product is thus defined uniquely, since by Proposition X.1.1 (4) can hold for only one matrix $\mathbf{AB}$. From the definition one sees that the j th column of $(\mathbf{AB}) = \mathbf{A}\mathbf{b}_j$, where $\mathbf{b}_j = (b_{kj}, k = 1 \dots p)$ is the j th column of $\mathbf{B}$. Accordingly, if $\mathbf{B}$ is expressed by means of its columns, then the matrix product $\mathbf{AB}$ also has the representation

$$\mathbf{B} = [\mathbf{b}_1 \quad \dots \quad \mathbf{b}_n] \quad \Rightarrow \quad \mathbf{AB} = [\mathbf{A}\mathbf{b}_1 \quad \dots \quad \mathbf{A}\mathbf{b}_n].$$

The matrix product thus reduces to products of a matrix and a vector: one multiplies each column vector $\mathbf{b}_j$ of $\mathbf{B}$ by $\mathbf{A}$ and places the results as the columns of $(\mathbf{AB})$.

From the convention (4) underlying the matrix product one can deduce that the matrix product is associative. Indeed, if $\mathbf{A}$, $\mathbf{B}$, $\mathbf{C}$, $\mathbf{x}$ are compatible in size, then by the convention (4) one has for every $\mathbf{x} \in \mathbb{R}^n$

$$[\mathbf{A}(\mathbf{BC})]\mathbf{x} = \mathbf{A}[(\mathbf{BC})\mathbf{x}] = \mathbf{A}[\mathbf{B}(\mathbf{C}\mathbf{x})] = (\mathbf{AB})(\mathbf{C}\mathbf{x}) = [(\mathbf{AB})\mathbf{C}]\mathbf{x},$$

whereupon one must have (Proposition X.1.1)

$$\mathbf{A}(\mathbf{BC}) = (\mathbf{AB})\mathbf{C}.$$

The matrix product, on the other hand, is not commutative. First of all, the compatibility rules associated with the product are quite restrictive: the products $\mathbf{AB}$ and $\mathbf{BA}$ are not necessarily both defined, and even if they are, they need not be of the same type. If $\mathbf{A}$ and $\mathbf{B}$ are both of size $n \times n$, that is, *square matrices* of the same size, then in these respects all is well, but even then in general $\mathbf{AB} \neq \mathbf{BA}$.

Square matrices are a remarkably important special class of matrices for the algebra of linear systems of equations, of which more will be said in the next section. Let it be noted in this connection, however, that if in the case of square matrices of the same size it happens that $\mathbf{AB} = \mathbf{BA}$, then one says that the matrices $\mathbf{A}$ and $\mathbf{B}$ *commute*:

$$\mathbf{AB} = \mathbf{BA} \quad \Leftrightarrow \quad \mathbf{A} \text{ and } \mathbf{B} \text{ commute.}$$

Every square matrix commutes trivially at least with itself.

EXAMPLE 3 Of the nine possible mutual products of the matrices

$$\mathbf{A} = \begin{bmatrix} 1 & -1 \\ -1 & 2 \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \quad \mathbf{C} = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 0 & 1 \end{bmatrix}$$

the products $\mathbf{AC}$, $\mathbf{BC}$ and $\mathbf{CC}$ are not defined; the others are:

$$\begin{aligned} \mathbf{AB} &= \left[\mathbf{A} \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad \mathbf{A} \begin{bmatrix} 1 \\ 0 \end{bmatrix} \right] = \begin{bmatrix} 0 & 1 \\ 0 & -1 \end{bmatrix}, & \mathbf{BA} &= \begin{bmatrix} -1 & 2 \\ 0 & 0 \end{bmatrix}, \\ \mathbf{CA} &= \left[\mathbf{C} \begin{bmatrix} 1 \\ -1 \end{bmatrix} \quad \mathbf{C} \begin{bmatrix} -1 \\ 2 \end{bmatrix} \right] = \begin{bmatrix} 1 & -1 \\ 0 & 1 \\ -1 & 2 \end{bmatrix}, & \mathbf{CB} &= \begin{bmatrix} 0 & 1 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}, \\ \mathbf{AA} &= \begin{bmatrix} 2 & -3 \\ -3 & 5 \end{bmatrix}, & \mathbf{BB} &= \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}. \quad \square \end{aligned}$$

In the example the square matrices $\mathbf{A}$ and $\mathbf{B}$ thus do not commute. The product $\mathbf{AA}$ defined for a square matrix can be written $\mathbf{A}^2$, and more generally one can define raising a square matrix to a power:

$$\mathbf{A}^k = \underbrace{\mathbf{AA} \cdots \mathbf{A}}_{k \text{ factors}}, \quad k \in \mathbb{N} \quad (\mathbf{A} \text{ a square matrix}).$$

The example shows that in the case of a square matrix one can have $\mathbf{B}^2 = \mathbf{0}$ (= the zero matrix), even though $\mathbf{B} \neq \mathbf{0}$.

The transpose of a matrix

One central operation of matrix algebra still remains to be defined, *transposition*. By this we mean simply interchanging the rows and columns of a matrix $\mathbf{A}$ with each other. The result is denoted $\mathbf{A}^T$ and called the *transpose* of $\mathbf{A}$:

$$\boxed{[\mathbf{A}^T]_{ij} = [\mathbf{A}]_{ji}.$$

EXAMPLE 3 (continued) For the matrices of the example

$$\mathbf{A}^T = \mathbf{A}, \quad \mathbf{B}^T = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, \quad \mathbf{C}^T = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}. \quad \square$$

As in the example, in the case of a square matrix (and no other) it is possible that $\mathbf{A}^T = \mathbf{A}$, in which case one says that $\mathbf{A}$ is *symmetric*:

$$\mathbf{A}^T = \mathbf{A} \quad \Leftrightarrow \quad \mathbf{A} \text{ symmetric.}$$

The most important of the rules for computing transposes are

$$(\mathbf{A}^T)^T = \mathbf{A}, \quad (\mathbf{AB})^T = \mathbf{B}^T \mathbf{A}^T.$$

The first of these is evident, and the second (the rule for the transpose of a product) can also be derived straightforwardly from the definition of the matrix product (Exercise 11a). When the rule for the transpose of a product is applied to a multiple matrix product, one obtains the more general rule

$$(\mathbf{A}_1 \mathbf{A}_2 \dots \mathbf{A}_n)^T = \mathbf{A}_n^T \mathbf{A}_{n-1}^T \dots \mathbf{A}_1^T.$$

Thus the transpose of a product = the product of the transposes in reverse order.

Among the other rules of matrix algebra let us still mention the following, evident from the definitions (matrix compatibility is assumed):

$$\begin{aligned} \mathbf{A} + \mathbf{B} &= \mathbf{B} + \mathbf{A}, & (\mathbf{A} + \mathbf{B}) + \mathbf{C} &= \mathbf{A} + (\mathbf{B} + \mathbf{C}), \\ \mathbf{A}(\lambda \mathbf{B}) &= (\lambda \mathbf{A})\mathbf{B} = \lambda(\mathbf{AB}), & \lambda &\in \mathbb{R}, \\ \mathbf{A}(\mathbf{B} + \mathbf{C}) &= \mathbf{AB} + \mathbf{AC}, & (\mathbf{A} + \mathbf{B})\mathbf{C} &= \mathbf{AC} + \mathbf{BC}, \\ (\mathbf{A} + \mathbf{B})^T &= \mathbf{A}^T + \mathbf{B}^T, & (\lambda \mathbf{A})^T &= \lambda \mathbf{A}^T, \quad \lambda \in \mathbb{R}. \end{aligned}$$

Row vectors — the Euclidean inner product and norm

When a column vector $\mathbf{x} = (x_i)_{i=1}^n$ is interpreted as a matrix of size $n \times 1$, then this matrix has a transpose, which is denoted $\mathbf{x}^T$ and called a *row vector*. From the standpoint of vector algebra there is no difference between column and row vectors, so the vector space $\mathbb{R}^n$ may equally well be thought of as consisting of row vectors. From the standpoint of matrix algebra, on the other hand, column and row vectors are different objects. Especially interesting is the matrix product $\mathbf{x}^T \mathbf{y}$, where $\mathbf{y} = (y_i)_{i=1}^n$ is a column vector of the same size as $\mathbf{x}$. By the definition of the matrix product, $\mathbf{x}^T \mathbf{y}$ is a matrix of size 1×1 , that is, a scalar:

$$\mathbf{x}^T \mathbf{y} = \sum_{i=1}^n x_i y_i \in \mathbb{R}.$$

Here we have in fact defined, by means of matrix algebra, the *Euclidean inner product* of $\mathbb{R}^n$

$$\langle \mathbf{x}, \mathbf{y} \rangle = \mathbf{x}^T \mathbf{y} = \mathbf{y}^T \mathbf{x} = \sum_{i=1}^n x_i y_i, \quad \mathbf{x}, \mathbf{y} \in \mathbb{R}^n.$$

This is a generalization of the inner products of $\mathbb{R}^2$ and $\mathbb{R}^3$ defined earlier in Sections II.2 and II.6. The Cauchy–Schwarz inequality for the Euclidean inner product of $\mathbb{R}^n$ is (see Theorem II.4.2)

$$|\langle \mathbf{x}, \mathbf{y} \rangle| \leq |\mathbf{x}| |\mathbf{y}|, \quad \mathbf{x}, \mathbf{y} \in \mathbb{R}^n,$$

where

$$|\mathbf{x}| = \langle \mathbf{x}, \mathbf{x} \rangle^{1/2} = (\mathbf{x}^T \mathbf{x})^{1/2} = \left(\sum_{i=1}^n x_i^2 \right)^{1/2}$$

is the *Euclidean norm* of $\mathbb{R}^n$. Referring to the inner product and norm so defined, the space $\mathbb{R}^n$ is generally called the *Euclidean space* $\mathbb{R}^n$. The natural basis $\{\mathbf{e}_1, \dots, \mathbf{e}_n\}$ of $\mathbb{R}^n$ defined above is, as a basis of the Euclidean space, *orthonormal*:

$$\langle \mathbf{e}_i, \mathbf{e}_j \rangle = \delta_{ij} \quad (\text{Kronecker } \delta).$$

If $\mathbf{x} \in \mathbb{R}^n$, $\mathbf{y} \in \mathbb{R}^m$ and $\mathbf{A}$ is a matrix of size $m \times n$, then the inner product $\langle \mathbf{Ax}, \mathbf{y} \rangle = (\mathbf{Ax})^T \mathbf{y}$ is defined. Here, by the rule for the transpose of a product, $(\mathbf{Ax})^T = \mathbf{x}^T \mathbf{A}^T$, so that

$$\langle \mathbf{Ax}, \mathbf{y} \rangle = \langle \mathbf{x}, \mathbf{A}^T \mathbf{y} \rangle.$$

Let us further remark that if $\mathbf{A}$ is of size $m \times p$, $\mathbf{B}$ is of size $p \times n$, and we let $\mathbf{A}^T = [\mathbf{a}_1 \dots \mathbf{a}_m]$ (i.e. the i th row of $\mathbf{A} = \mathbf{a}_i^T$) and $\mathbf{B} = [\mathbf{b}_1 \dots \mathbf{b}_n]$, then by the definition of the matrix product

$$[\mathbf{AB}]_{ij} = \mathbf{a}_i^T \mathbf{b}_j = \langle \mathbf{a}_i, \mathbf{b}_j \rangle.$$

The product $\mathbf{AB}$ is thus an array formed from the $(\mathbb{R}^p)$ inner products between the rows of $\mathbf{A}$ and the columns of $\mathbf{B}$.

Complex vectors and matrices

Since matrix algebra is ultimately only a matter of the basic arithmetic operations among matrix entries and their combination, the matrix entries may be thought of as belonging, instead of to $\mathbb{R}$, to any field $\mathbb{K}$. In simple computational examples (such as the examples above) one often has $\mathbb{K} = \mathbb{Q}$. In more general mathematical considerations, as well as in many applications (e.g. in electrical engineering), one must allow the matrix entries to be complex-valued, in which case $\mathbb{K} = \mathbb{C}$. In this case the column vectors of size n form the Euclidean space $\mathbb{C}^n$, whose inner product is

$$\langle \mathbf{x}, \mathbf{y} \rangle = \sum_{i=1}^n x_i \bar{y}_i = \mathbf{x}^T \bar{\mathbf{y}}.$$

Here $\bar{\mathbf{y}}$, more generally $\bar{\mathbf{A}}$, denotes complex conjugation entrywise.

EXAMPLE 4 If

$$\mathbf{x} = [i, 1+i, 2-i]^T \in \mathbb{C}^3, \quad \mathbf{y} = [1-2i, 2i, -1-i]^T \in \mathbb{C}^3,$$

then

$$\begin{aligned} \langle \mathbf{x}, \mathbf{x} \rangle &= |\mathbf{x}|^2 = 1 + (1+1) + (4+1) = 8, & \langle \mathbf{y}, \mathbf{y} \rangle &= (1+4) + 4 + (1+1) = 11, \\ \langle \mathbf{x}, \mathbf{y} \rangle &= i(1+2i) + (1+i)(-2i) + (2-i)(-1+i) = -1+3i. \quad \square \end{aligned}$$

The matrix $\mathbf{A}^* = \bar{\mathbf{A}}^T$ is called the *adjoint* of $\mathbf{A}$. If $\mathbf{x} \in \mathbb{C}^n$, $\mathbf{y} \in \mathbb{C}^m$ and $\mathbf{A}$ is a complex matrix of size $m \times n$, then from the definitions of the adjoint and of the inner product of $\mathbb{C}^m$ one easily obtains (cf. the real case above)

$$\boxed{\langle \mathbf{A}\mathbf{x}, \mathbf{y} \rangle = \langle \mathbf{x}, \mathbf{A}^*\mathbf{y} \rangle.}$$

If $\mathbf{A} = \{a_{ij}\}_{i,j=1}^n$ is a square matrix and $\mathbf{A}^* = \mathbf{A}$, i.e. $a_{ji} = \bar{a}_{ij} \forall i, j$, then one says that $\mathbf{A}$ is *Hermitian*:

$$\mathbf{A}^* = \mathbf{A} \quad \Leftrightarrow \quad \mathbf{A} \text{ Hermitian.}$$

Every real and symmetric matrix is, by definition, also Hermitian.

EXAMPLE 5 A general Hermitian matrix of size 2×2 is of the form

$$\mathbf{A} = \begin{bmatrix} a & c \\ \bar{c} & b \end{bmatrix},$$

where $a, b \in \mathbb{R}$ and $c \in \mathbb{C}$. $\square$

EXERCISES

1. Compute $2\mathbf{A} + 3\mathbf{B}$, $\mathbf{A} - \mathbf{B}^T$, $\mathbf{AB}$ and $\mathbf{BA}$, when

$$\mathbf{A} = \begin{bmatrix} 1 & -1 & 1 \\ -3 & 2 & -1 \\ -2 & 1 & 0 \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 1 & 2 & 3 \end{bmatrix}.$$

2. Let

$$\mathbf{A} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 1 & 0 & -1 \end{bmatrix}, \quad \mathbf{C} = \begin{bmatrix} 0 & 2 \\ 1 & 1 \\ -1 & -1 \end{bmatrix}.$$

Compute $\mathbf{AB}$, $\mathbf{AC}$, $\mathbf{BC}$, $\mathbf{CA}$ and $\mathbf{BA}^T$.

3. Under what assumptions on the type of the matrices are the products $\mathbf{AB}$ and $\mathbf{BA}$

a) both defined, b) of the same type?

4. Let

$$\mathbf{a} = [1, 3, 5, 2], \quad \mathbf{b} = [-1, 3, 2, 4]^T, \quad \mathbf{A} = ((-1)^{i+j}, i = 1, 2, j = 1, 2, 3, 4).$$

Compute all of the mutual (pairwise) products of the vectors/matrices $\mathbf{a}$, $\mathbf{b}$, $\mathbf{A}$ that are defined.

5. The matrices $\mathbf{A}$ and $\mathbf{B}$ are of size 10×10 and their entries are $a_{ij} = i + j$ and $b_{ij} = i - j$. Compute the entry c_{ij} of the product matrix $\mathbf{C} = \mathbf{AB}$.

6. Let

$$\mathbf{A} = a \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} b & -c & -c \\ -c & d & d \\ -c & d & d \end{bmatrix}.$$

Determine the numbers a, b, c, d such that $\mathbf{A} \neq \mathbf{0}$, $\mathbf{B} \neq \mathbf{0}$, $\mathbf{A}^2 = \mathbf{A}$, $\mathbf{B}^2 = \mathbf{B}$ and $\mathbf{AB} = \mathbf{0}$.

7. Determine the matrix $\mathbf{A}$, given that

$$\mathbf{A} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 3 \\ 4 \\ 2 \end{bmatrix}, \quad \mathbf{A} \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix}, \quad \mathbf{A} \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 4 \\ 3 \\ 2 \\ 1 \end{bmatrix}.$$

8. Find all matrices $\mathbf{B}$ that commute with the matrix

$$\text{a) } \mathbf{A} = \begin{bmatrix} 0 & 1 \\ 0 & -1 \end{bmatrix}, \quad \text{b) } \mathbf{A} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}.$$

9. Let $\mathbf{A}$ and $\mathbf{B}$ be square matrices of the same size. Prove:

a) $(\mathbf{A} + \mathbf{B})^2 = \mathbf{A}^2 + 2\mathbf{AB} + \mathbf{B}^2 \Leftrightarrow \mathbf{A}$ and $\mathbf{B}$ commute

b) $\mathbf{A}$ and $\mathbf{B}$ symmetric $\nRightarrow \mathbf{AB}$ symmetric

c) $\mathbf{A}$ and $\mathbf{B}$ symmetric and commuting $\Rightarrow \mathbf{AB}$ symmetric

10. Compute a) $\mathbf{A}^k \forall k \in \mathbb{N}$, b) $\mathbf{B} = (\mathbf{I} + \mathbf{A})^{100}$, when

$$\mathbf{A} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \quad \mathbf{I} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

11. a) Prove the rule for the transpose of a matrix product $(\mathbf{AB})^T = \mathbf{B}^T \mathbf{A}^T$.
 b) Show that every square matrix can be represented uniquely in the form $\mathbf{A} = \mathbf{B} + \mathbf{C}$, where $\mathbf{B}$ is symmetric and $\mathbf{C}$ is *skew-symmetric*:
 $\mathbf{C} + \mathbf{C}^T = \mathbf{0}$.

12. Let

$$\mathbf{A} = \begin{bmatrix} 5 & -2 & 0 \\ -2 & 6 & 2 \\ 0 & 2 & 7 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix}, \quad \mathbf{x} = \begin{bmatrix} x \\ y \\ z \end{bmatrix}.$$

Simplify the expression $f(x, y, z) = \mathbf{x}^T \mathbf{A} \mathbf{x} + \mathbf{b}^T \mathbf{x}$ to a second-degree polynomial.

13. Compute $\mathbf{Ax}$, $\mathbf{Bx}$, $\mathbf{AB}$, $(\mathbf{AB})^*$ and $\mathbf{B}^* \mathbf{A}^*$, when

$$\mathbf{A} = \begin{bmatrix} i & 1-i & 2+i \\ 1-i & -2i & 3+2i \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} 2-i & i & -i \\ -i & 2 & 1+i \\ 3i & 2+2i & 1-i \end{bmatrix}, \quad \mathbf{x} = \begin{bmatrix} 3 \\ 3-i \\ 2+i \end{bmatrix}.$$

14. (*) If a real matrix $\mathbf{A} = (a_{ij})$ of size $m \times n$ is interpreted as an element of the space $\mathbb{R}^{mn}$, then the *matrix norm*

$$\|\mathbf{A}\| = \left(\sum_{i=1}^m \sum_{j=1}^n a_{ij}^2 \right)^{1/2}$$

corresponds to the Euclidean norm of that space. Show that $|\mathbf{Ax}| \leq \|\mathbf{A}\| |\mathbf{x}| \forall \mathbf{x} \in \mathbb{R}^n$.

15. (*) Let $x \in \mathbb{R}$ and define

$$\mathbf{A} = \begin{bmatrix} x & 1 & 0 & 0 \\ 0 & x & 1 & 0 \\ 0 & 0 & x & 1 \\ 0 & 0 & 0 & x \end{bmatrix}, \quad \mathbf{I} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad \exp(\mathbf{A}) = \mathbf{I} + \sum_{k=1}^{\infty} \frac{1}{k!} \mathbf{A}^k.$$

Compute the entries of the matrix $\exp(\mathbf{A})$ as functions of x .

16. (*) (Cattle in the pasture) In a herd of cattle there are 83 entirely brown animals, 77 horned ones, 36 that are bulls by sex, 22 brown horned ones, 15 brown bulls, 25 horned bulls and 7 brown horned bulls. Every animal in the herd belongs to one of the mentioned groups. How many animals are there in the herd? *Hint:* Divide the animals into mutually disjoint groups, e.g. brown horned bulls x_1 in number $\dots$ non-brown hornless cows x_8 in number. Write a linear system of equations and solve it!

X.2 Square matrices. The inverse matrix

In this and the next section the object of study is the linear system of equations

$$\mathbf{A}\mathbf{x} = \mathbf{b},$$

where there are equally many equations and unknowns, that is, $\mathbf{A}$ is a square matrix of size $n \times n$ and $\mathbf{b} \in \mathbb{R}^n$ ($n \in \mathbb{N}$). (For convenience we think of $\mathbf{A}$ and $\mathbf{b}$ as real — the results that follow hold as such also in the complex domain.)

An important special case of square matrices, important both for the matrix product and for the matrix-vector product, is the *unit matrix*, or *identity matrix*, whose symbol is $\mathbf{I}$ and whose definition is

$$[\mathbf{I}]_{ij} = (\delta_{ij}) = \begin{bmatrix} 1 & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 1 \end{bmatrix} = [\mathbf{e}_1 \dots \mathbf{e}_n].$$

If $\mathbf{A}$ and $\mathbf{I}$ are of size $n \times n$ and $\mathbf{x}$ is a column vector of size n , then

$$(a) \quad \mathbf{AI} = \mathbf{IA} = \mathbf{A}, \quad (b) \quad \mathbf{Ix} = \mathbf{x}.$$

By property (a), $\mathbf{I}$ is the unit element of matrix multiplication in operations between square matrices of the same size — hence the name ‘unit matrix’. By property (b), the map $\mathbf{x} \mapsto \mathbf{Ix}$ is the *identity map* of $\mathbb{R}^n$ — hence the name ‘identity matrix’.

If matrix multiplication has a ‘unit matrix’ associated with it, is there also an inverse matrix associated with it? — Here we come to the most central question of the theory of square matrices.

DEFINITION X.2.1 A square matrix $\mathbf{A}$ is *regular*, or *non-singular*, or *invertible*, if there exists a matrix $\mathbf{B}$ such that

$$\mathbf{AB} = \mathbf{BA} = \mathbf{I}.$$

One says that $\mathbf{B}$ is the *inverse matrix* of $\mathbf{A}$ and writes

$$\mathbf{B} = \mathbf{A}^{-1}.$$

If $\mathbf{A}$ is not regular, it is irregular, or *singular*.

The symbol $\mathbf{A}^{-1}$ for the inverse matrix is usually read ‘A minus one’. If $\mathbf{A}^{-1}$ exists, then it is unique. Indeed, if $\mathbf{B}$ and $\mathbf{C}$ are both inverse matrices of $\mathbf{A}$, then by the following claim one must have $\mathbf{B} = \mathbf{C}$.

PROPOSITION X.2.2 If there exist matrices $\mathbf{B}$ and $\mathbf{C}$ of size $n \times n$ such that $\mathbf{AB} = \mathbf{I}$ and $\mathbf{CA} = \mathbf{I}$, then $\mathbf{B} = \mathbf{C}$.

Proof By the assumptions and the associativity of the matrix product,

$$\mathbf{B} = \mathbf{IB} = (\mathbf{CA})\mathbf{B} = \mathbf{C}(\mathbf{AB}) = \mathbf{CI} = \mathbf{C}. \quad \square$$

If $\mathbf{A}$ is regular, then from Definition X.2.1 one sees immediately that $\mathbf{B} = \mathbf{A}^{-1}$ is also regular and $\mathbf{B}^{-1} = \mathbf{A}$. Carrying out transposition in the definition, one also sees that

$$\mathbf{AB} = \mathbf{BA} = \mathbf{I} \Leftrightarrow \mathbf{B}^T \mathbf{A}^T = \mathbf{A}^T \mathbf{B}^T = \mathbf{I}^T = \mathbf{I}.$$

Thus: if $\mathbf{A}$ is regular, then both $\mathbf{A}^{-1}$ and $\mathbf{A}^T$ are regular, and

$$\boxed{(\mathbf{A}^{-1})^{-1} = \mathbf{A}, \quad (\mathbf{A}^T)^{-1} = (\mathbf{A}^{-1})^T.}$$

If $\mathbf{A}$ and $\mathbf{B}$ are regular and of the same size, then the product $\mathbf{AB}$ is also regular. Indeed, by the rules of the matrix product and the definition of the inverse matrix,

$$\begin{aligned} (\mathbf{AB})(\mathbf{B}^{-1}\mathbf{A}^{-1}) &= \mathbf{A}(\mathbf{BB}^{-1})\mathbf{A}^{-1} = \mathbf{AIA}^{-1} = \mathbf{AA}^{-1} = \mathbf{I}, \\ (\mathbf{B}^{-1}\mathbf{A}^{-1})(\mathbf{AB}) &= \mathbf{B}^{-1}(\mathbf{A}^{-1}\mathbf{A})\mathbf{B} = \mathbf{B}^{-1}\mathbf{IB} = \mathbf{B}^{-1}\mathbf{B} = \mathbf{I}, \end{aligned}$$

so that

$$\boxed{(\mathbf{AB})^{-1} = \mathbf{B}^{-1}\mathbf{A}^{-1}.}$$

The result is easily generalized as follows (cf. the corresponding rule for transposes in the previous section):

PROPOSITION X.2.3 If $\mathbf{A}_k$, $k = 1 \dots m$ are regular square matrices of the same size, then the product $\mathbf{A} = \mathbf{A}_1\mathbf{A}_2 \dots \mathbf{A}_m$ is also regular and $\mathbf{A}^{-1} = \mathbf{A}_m^{-1}\mathbf{A}_{m-1}^{-1} \dots \mathbf{A}_1^{-1}$.

In solving a linear system of equations, the regularity of the coefficient matrix ‘settles’ the problem from a fundamental standpoint as follows:

PROPOSITION X.2.4 If in the system $\mathbf{Ax} = \mathbf{b}$ the coefficient matrix $\mathbf{A}$ is a regular square matrix of size $n \times n$, then the system has, for every $\mathbf{b} \in \mathbb{R}^n$, a unique solution

$$\mathbf{x} = \mathbf{A}^{-1}\mathbf{b}.$$

Proof Let $\mathbf{B} = \mathbf{A}^{-1}$ and reason:

$$\text{a) } \mathbf{AB} = \mathbf{I} \quad \Rightarrow \quad \mathbf{A}(\mathbf{Bb}) = (\mathbf{AB})\mathbf{b} = \mathbf{Ib} = \mathbf{b}.$$

$$\text{b) } \mathbf{BA} = \mathbf{I} \quad \Rightarrow \quad \mathbf{x} = \mathbf{Ix} = (\mathbf{BA})\mathbf{x} = \mathbf{B}(\mathbf{Ax}).$$

On this basis we note: a) $\mathbf{x} = \mathbf{Bb}$ is a solution of the system $\mathbf{Ax} = \mathbf{b}$.

b) If $\mathbf{Ax} = \mathbf{b}$, then $\mathbf{x} = \mathbf{Bb}$, i.e. this is the only possible solution. $\square$

By Proposition X.2.4, a possible singularity of the coefficient matrix of the system $\mathbf{Ax} = \mathbf{b}$ is revealed when the system is solved:

COROLLARY X.2.5 If the system $\mathbf{Ax} = \mathbf{b}$ ($\mathbf{A}$ of size $n \times n$) either has no solution for some $\mathbf{b} \in \mathbb{R}^n$ or the solution is not unique, then $\mathbf{A}$ is singular.

Regularity criteria for a square matrix

By Proposition X.2.4, the regularity of the coefficient matrix $\mathbf{A}$ guarantees the regularity, that is, the unique solvability, of the system $\mathbf{Ax} = \mathbf{b}$ for every $\mathbf{b} \in \mathbb{R}^n$. This claim also holds conversely, and in fact a much stronger result holds: if the system $\mathbf{Ax} = \mathbf{b}$ either has a unique solution when $\mathbf{b} = \mathbf{0}$ (that is, the only solution is $\mathbf{x} = \mathbf{0}$) or is solvable for every $\mathbf{b} \in \mathbb{R}^n$, then in either case $\mathbf{A}$ is a regular matrix. Let us formulate the result as follows:

*THEOREM X.2.6 (**Regularity of a square matrix**) If $\mathbf{A}$ is a square matrix of size $n \times n$, then the following claims are mutually equivalent.

E0. $\mathbf{A}$ is a regular matrix.

E1. The only solution of the system $\mathbf{Ax} = \mathbf{0}$ is $\mathbf{x} = \mathbf{0}$.

E2. The system $\mathbf{Ax} = \mathbf{b}$ has a solution for every $\mathbf{b} \in \mathbb{R}^n$.

Theorem X.2.6 contains six implication claims, of which the independent ones are, e.g.: $E0 \Rightarrow E1$, $E0 \Rightarrow E2$, $E1 \Rightarrow E0$ and $E2 \Rightarrow E0$. Of these, the first two are contained in Proposition X.2.4 and hence already clear. The other two, on the other hand, are deeper. These are contained as special cases in a more general claim about linear systems of equations, known as the *Fundamental Theorem of Linear Algebra*. The more general theorem (which concerns also systems of size $m \times n$, $m \neq n$) is not formulated here. Instead, let us note an interesting consequence of Theorem X.2.6:

COROLLARY X.2.7 For square matrices of the same size,

$$\mathbf{AB} = \mathbf{I} \quad \Rightarrow \quad \mathbf{BA} = \mathbf{I}.$$

Proof If $\mathbf{AB} = \mathbf{I}$, then the regularity criterion E2 is satisfied (see the proof of Proposition X.2.4, part a)), so that by Theorem X.2.6 $\mathbf{A}$ is regular. Then if $\mathbf{C} = \mathbf{A}^{-1}$, we have $\mathbf{AB} = \mathbf{CA} = \mathbf{I}$, whereupon one must have $\mathbf{C} = \mathbf{B}$ (Proposition X.2.2). Thus $\mathbf{BA} = \mathbf{I}$. $\square$

By Corollary X.2.7, then, already the condition $\mathbf{AB} = \mathbf{I}$ suffices to guarantee that $\mathbf{B} = \mathbf{A}^{-1}$ and $\mathbf{A} = \mathbf{B}^{-1}$, i.e. one of the conditions of Definition X.2.1 is superfluous (!).

In what follows we do not attempt to prove Theorem X.2.6 right away; instead, in the next two sections we first examine the solution of a linear system of equations by algorithmic means. It turns out that, as a by-product of the solution algorithm, Theorem X.2.6 also gets proved.*

In the remainder of this section we turn our attention to two special classes of square matrices, *orthogonal* and *triangular* matrices, and to a subclass of the former, *permutation* matrices. Permutation and triangular matrices will play a central role in the following in the solution algorithm for a linear system of equations. As we shall see, for these matrices the question of regularity can be settled without relying on Theorem X.2.6. In the case of a triangular matrix the basis is the second of the following two claims (which relies on the first). This resembles the claim $\text{E2} \Rightarrow \text{E0}$ of Theorem X.2.6 but rests on stronger assumptions. First let us formulate the original assumption-claim E2 in another way:

PROPOSITION X.2.8 We have: $\text{E2} \Leftrightarrow \mathbf{AB} = \mathbf{I}$ for some $\mathbf{B}$.

Proof $\Rightarrow$ If E2 is true, then there exist vectors $\mathbf{b}_k \in \mathbb{R}^n$ such that $\mathbf{Ab}_k = \mathbf{e}_k$, $k = 1 \dots n$. Then if $\mathbf{B} = [\mathbf{b}_1 \dots \mathbf{b}_n]$, we have $\mathbf{AB} = [\mathbf{Ab}_1 \dots \mathbf{Ab}_n] = [\mathbf{e}_1 \dots \mathbf{e}_n] = \mathbf{I}$. $\Leftarrow$ See the proof of Proposition X.2.4, part a). $\square$

PROPOSITION X.2.9 If $\mathbf{A}$ is of size $n \times n$ and the systems $\mathbf{Ax} = \mathbf{b}$ and $\mathbf{A}^T \mathbf{x} = \mathbf{b}$ are both solvable for every $\mathbf{b} \in \mathbb{R}^n$, then $\mathbf{A}$ is a regular matrix.

Proof By the assumptions and Proposition X.2.8 there exist matrices $\mathbf{B}$ and $\mathbf{C}$ such that $\mathbf{AB} = \mathbf{I}$ and $\mathbf{A}^T \mathbf{C} = \mathbf{I}$. Then also $(\mathbf{A}^T \mathbf{C})^T = \mathbf{C}^T \mathbf{A} = \mathbf{I}^T = \mathbf{I}$. Thus $\mathbf{AB} = \mathbf{C}^T \mathbf{A} = \mathbf{I}$, from which it follows (Proposition X.2.2) that $\mathbf{B} = \mathbf{C}^T (= \mathbf{A}^{-1})$, i.e. $\mathbf{A}$ is regular. $\square$

*In the literature dealing with linear algebra, the Fundamental Theorem of Linear Algebra (and Theorem X.2.6 contained in it) is usually proved non-algorithmically, by the means of abstract linear algebra.

Orthogonal matrices

A special class of regular matrices is formed by the *orthogonal* matrices. If a square matrix $\mathbf{A}$ is expressed by means of its columns in the form

$$\mathbf{A} = [\mathbf{a}_1 \dots \mathbf{a}_n],$$

then one says that $\mathbf{A}$ is orthogonal if $\{\mathbf{a}_1, \dots, \mathbf{a}_n\}$ is an orthonormal system in $\mathbb{R}^n$, i.e.

$$\langle \mathbf{a}_i, \mathbf{a}_j \rangle = \mathbf{a}_i^T \cdot \mathbf{a}_j = \delta_{ij}, \quad i, j = 1 \dots n.$$

By the definition of the matrix product (see the previous section) this is the same as the condition

$$\mathbf{A}^T \mathbf{A} = \mathbf{I}.$$

Then by Corollary X.2.7 we also have $\mathbf{A} \mathbf{A}^T = \mathbf{I}$, that is, in an orthogonal matrix the rows too form an orthonormal system (!) and $\mathbf{A}^{-1} = \mathbf{A}^T$. On the other hand, if $\mathbf{A}^T \mathbf{A} = \mathbf{I}$, then

$$\delta_{ij} = [\mathbf{A}^T \mathbf{A}]_{ij} = \mathbf{a}_i^T \mathbf{a}_j,$$

i.e. $\mathbf{A}$ is orthogonal. We have thus arrived (appealing in part at Corollary X.2.7 and hence at Theorem X.2.6) at the result

$\mathbf{A} \text{ orthogonal} \Leftrightarrow \mathbf{A}^{-1} = \mathbf{A}^T.$

If $\mathbf{A}$ is both orthogonal and symmetric, then $\mathbf{A} \mathbf{A} = \mathbf{I}$. One can thus note (this time without appealing to Theorem X.2.6) that:

PROPOSITION X.2.10 If $\mathbf{A}$ is an orthogonal and symmetric matrix, then $\mathbf{A}$ is regular and $\mathbf{A}^{-1} = \mathbf{A}$.

The permutation matrix $\mathbf{I}_p$

A class of orthogonal matrices particularly interesting for what follows consists of the *permutation matrices*, which have the same columns as the unit matrix $\mathbf{I}$ but in a changed (permuted) order. A permutation matrix is denoted

$$\mathbf{I}_p = [\mathbf{e}_{i_1} \ \mathbf{e}_{i_2} \ \dots \ \mathbf{e}_{i_n}],$$

where $p = (i_1, i_2, \dots, i_n)$ is a *permutation* of the ordered set of numbers $(1, 2, \dots, n)$ (that is, the same numbers in a different order). A well-known (easy) result of the theory of permutations is that the number of distinct permutations of the set $(1, 2, \dots, n)$ — and hence also the number of distinct permutation matrices

$\mathbf{I}_p$ of size $n \times n$ — is $n!$. Every permutation p can be attained by performing a ($(p$ -dependent)) number m of successive *transpositions*, in which two numbers are interchanged. The number m is not unique, since repeating the same transposition restores the original order. It is, however, possible to show (though not entirely easily) that the parity (evenness/oddness) of the number m is characteristic of the permutation, i.e. every permutation is either *even* or *odd*.

Every permutation matrix $\mathbf{I}_p$ is of course orthogonal, i.e. $\mathbf{I}_p^T \mathbf{I}_p = \mathbf{I}$. It is easily seen that the rows of $\mathbf{I}_p$ too are obtained by permuting the rows of $\mathbf{I}$, that is, $\mathbf{I}_p^T = \mathbf{I}_q$, where q is another permutation of the numbers $(1, 2, \dots, n)$ ($q = p$, if $\mathbf{I}_p$ is symmetric). Since $\mathbf{I} = \mathbf{I}_q^T \mathbf{I}_q = \mathbf{I}_p \mathbf{I}_p^T$, we have $\mathbf{I}_p^T \mathbf{I}_p = \mathbf{I}_p \mathbf{I}_p^T = \mathbf{I}$. Thus we have deduced (without appealing to Theorem X.2.6) that $\mathbf{I}_p^{-1} = \mathbf{I}_p^T$.

EXAMPLE 1 In the case $n = 3$ there are $3! = 6$ distinct permutation matrices, corresponding to the permutations $p = (1, 2, 3), (2, 1, 3), (1, 3, 2), (3, 2, 1), (2, 3, 1), (3, 1, 2)$. Of these the first and the last two are even (0 or 2 transpositions), the others odd (1 transposition). From the diagram (with $\bullet = 1$ and $\cdot = 0$ marked) one sees that if $p = (2, 3, 1)$, then $\mathbf{I}_p^T = \mathbf{I}_q$, where $q = (3, 1, 2)$.

$$\begin{bmatrix} \bullet & \cdot & \cdot \\ \cdot & \bullet & \cdot \\ \cdot & \cdot & \bullet \end{bmatrix} \quad \begin{bmatrix} \cdot & \bullet & \cdot \\ \bullet & \cdot & \cdot \\ \cdot & \cdot & \bullet \end{bmatrix} \quad \begin{bmatrix} \bullet & \cdot & \cdot \\ \cdot & \cdot & \bullet \\ \cdot & \bullet & \cdot \end{bmatrix} \quad \begin{bmatrix} \cdot & \cdot & \bullet \\ \cdot & \bullet & \cdot \\ \bullet & \cdot & \cdot \end{bmatrix} \quad \begin{bmatrix} \cdot & \cdot & \bullet \\ \bullet & \cdot & \cdot \\ \cdot & \bullet & \cdot \end{bmatrix} \quad \begin{bmatrix} \cdot & \bullet & \cdot \\ \cdot & \cdot & \bullet \\ \bullet & \cdot & \cdot \end{bmatrix} \quad \square$$

If $\mathbf{A} = [\mathbf{a}_1 \dots \mathbf{a}_n]$ is a matrix of size $n \times n$ and $\mathbf{I}_p = [\mathbf{e}_{i_1} \dots \mathbf{e}_{i_n}]$, then

$$\mathbf{A} \mathbf{I}_p = [\mathbf{A} \mathbf{e}_{i_1} \dots \mathbf{A} \mathbf{e}_{i_n}] = [\mathbf{a}_{i_1} \dots \mathbf{a}_{i_n}].$$

The operation $\mathbf{A} \mapsto \mathbf{A} \mathbf{I}_p$ thus permutes the columns of $\mathbf{A}$ into the order p dictated by the columns of $\mathbf{I}_p$. Correspondingly, since $(\mathbf{I}_p \mathbf{A})^T = \mathbf{A}^T \mathbf{I}_p^T$, one deduces that the operation $\mathbf{A} \mapsto \mathbf{I}_p \mathbf{A}$ permutes the columns of $\mathbf{A}^T$ (that is, the rows of $\mathbf{A}$) into the order dictated by the columns of $\mathbf{I}_p^T$ (that is, by the rows of $\mathbf{I}_p$). Summary:

$\mathbf{A} \mapsto \mathbf{A} \mathbf{I}_p :$	permutation of the columns of $\mathbf{A}$ according to the columns of $\mathbf{I}_p$.
$\mathbf{A} \mapsto \mathbf{I}_p \mathbf{A} :$	permutation of the rows of $\mathbf{A}$ according to the rows of $\mathbf{I}_p$.

EXAMPLE 1 (continued)

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 3 & 1 & 2 \\ 6 & 4 & 5 \\ 9 & 7 & 8 \end{bmatrix}, \quad \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} = \begin{bmatrix} 4 & 5 & 6 \\ 7 & 8 & 9 \\ 1 & 2 & 3 \end{bmatrix}. \quad \square$$

From the rules of matrix algebra of the previous section one can deduce that permutation of the columns of a matrix by the operation $\mathbf{A} \mapsto \mathbf{A} \mathbf{I}_p$ works equally

well for matrices of size $m \times n$, $m \neq n$, and correspondingly permutation of the rows for matrices of size $n \times m$. In the case of a square matrix one also sees that, under permutation of columns or rows, a regular matrix remains regular (since the product of regular matrices is regular). Likewise a singular matrix remains singular (Exercise 1d).

A special case of a permutation matrix is the *transposition matrix*, in which only two columns of $\mathbf{I}$ have been interchanged. If the interchanged columns are $\mathbf{e}_k$ and $\mathbf{e}_m$ ($k < m$), then the transposition matrix is $\mathbf{V} = [\mathbf{e}_1 \dots \mathbf{e}_{k-1} \mathbf{e}_m \mathbf{e}_{k+1} \dots \mathbf{e}_{m-1} \mathbf{e}_k \mathbf{e}_{m+1} \dots \mathbf{e}_n]$. This is not only orthogonal but also symmetric:

$$\mathbf{V} = \begin{bmatrix} 1 & & & & & & & & & & \\ & \ddots & & & & & & & & & \\ & & 0 & \dots & \dots & \dots & 1 & & & & \\ & & 1 & & & & & & & & \\ & & & \ddots & & & & & & & \\ & & & & 1 & & & & & & \\ & & 1 & \dots & \dots & \dots & 0 & & & & \\ & & & & & & & 1 & & & \\ & & & & & & & & 1 & & \\ & & & & & & & & & 1 & \\ & & & & & & & & & & 1 \end{bmatrix}$$

Since the permutation p is reached by successive transpositions, correspondingly $\mathbf{I}_p$ is obtained by multiplying $\mathbf{I}$ on the right by the corresponding transposition matrices $\mathbf{V}_i$, $i = 1 \dots m$, that is,

$$\mathbf{I}_p = \mathbf{I}\mathbf{V}_1 \dots \mathbf{V}_m = \mathbf{V}_1 \dots \mathbf{V}_m.$$

Since each $\mathbf{V}_i$ is symmetric, by the rule for the transpose of a product

$$\mathbf{I}_p^{-1} = \mathbf{I}_p^T = \mathbf{V}_m \dots \mathbf{V}_1.$$

Triangular matrices

DEFINITION X.2.11 A square matrix $\mathbf{A} = (a_{ij})$ is

- *upper triangular*, if $a_{ij} = 0$ when $i > j$,
- *lower triangular*, if $a_{ij} = 0$ when $i < j$,
- *diagonal*, or a *diagonal matrix*, if $a_{ij} = 0$ when $i \neq j$.

$$\begin{bmatrix} \# & \# & \# & \# \\ 0 & \# & \# & \# \\ 0 & 0 & \# & \# \\ 0 & 0 & 0 & \# \end{bmatrix} \quad \begin{bmatrix} \# & 0 & 0 & 0 \\ \# & \# & 0 & 0 \\ \# & \# & \# & 0 \\ \# & \# & \# & \# \end{bmatrix} \quad \begin{bmatrix} \# & 0 & 0 & 0 \\ 0 & \# & 0 & 0 \\ 0 & 0 & \# & 0 \\ 0 & 0 & 0 & \# \end{bmatrix}$$

upper triangular

lower triangular

diagonal

By definition, a diagonal matrix is a special case of both kinds of triangular matrix. For the matrix types of the definition, one often uses the special symbols **U** (upper triangular), **L** (lower triangular) and **D** (diagonal matrix). The entries a_{ii} of a square matrix **A** are generally called the *diagonal entries* of **A**, and the (ordered) set $(a_{ii}, i = 1 \dots n)$ the *diagonal* of **A**. The set $(a_{i,i+k}, i = 1 \dots n-k)$ is correspondingly the k th *superdiagonal* and $(a_{i-k,i}, i = k+1 \dots n)$ the k th *subdiagonal*. A diagonal matrix is often expressed by means of its diagonal entries using the notation

$$\mathbf{D} = \text{diag}(d_i), \quad d_i = [\mathbf{D}]_{ii}.$$

The question of the regularity of a triangular matrix is settled by

THEOREM X.2.12 (Regularity of a triangular matrix) If $\mathbf{A} = (a_{ij})$ is a triangular matrix of size $n \times n$, then **A** is regular exactly when the diagonal entries a_{ii} are all nonzero, and in that case

- (i) **A** diagonal/upper triangular/lower triangular
 $\Rightarrow \mathbf{A}^{-1}$ diagonal/upper triangular/lower triangular
- (ii) $[\mathbf{A}^{-1}]_{ii} = 1/a_{ii}, \quad i = 1 \dots n.$

Proof In the case of a diagonal matrix one sees that if $a_{kk} = 0$ for some $k \in \{1, \dots, n\}$, then the system $\mathbf{Ax} = \mathbf{0}$ has a non-unique solution $\mathbf{x} = x_k \mathbf{e}_k$, $x_k \in \mathbb{R}$, so that by Corollary X.2.5 **A** is singular. If $a_{ii} \neq 0$, $i = 1 \dots n$, then $\mathbf{AA}^{-1} = \mathbf{A}^{-1}\mathbf{A} = \mathbf{I}$ is satisfied with the choice given in the claim, that is,

$$\mathbf{A}^{-1} = \text{diag}(1/a_{ii}).$$

Next assume that **A** is a lower triangular matrix, in which case the system $\mathbf{Ax} = \mathbf{b}$ written out is

$$\begin{cases} a_{11}x_1 & = b_1, \\ a_{21}x_1 + a_{22}x_2 & = b_2, \\ \vdots & \vdots \\ a_{n1}x_1 + a_{n2}x_2 + \dots + a_{nn}x_n & = b_n. \end{cases}$$

If $a_{ii} \neq 0 \forall i$, then the system is solved by unwinding it recursively from the top:

$$\begin{aligned} x_1 &= a_{11}^{-1}b_1 = b_{11}b_1, \\ x_2 &= -a_{22}^{-1}a_{21}x_1 + a_{22}^{-1}b_2 = -a_{22}^{-1}a_{21}b_{11}b_1 + a_{22}^{-1}b_2 = b_{21}b_1 + b_{22}b_2, \end{aligned}$$

and in general (by induction!)

$$x_i = \sum_{j=1}^i b_{ij}b_j, \quad i = 1 \dots n,$$

where $b_{ii} = a_{ii}^{-1}$. Setting $b_{ij} = 0$, $j > i$, and $\mathbf{B} = \{b_{ij}\}$, then $\mathbf{B}$ is thus a lower triangular matrix, with diagonal entries $b_{ii} = a_{ii}^{-1}$, and the solution of the system is

$$\mathbf{x} = \mathbf{Bb}.$$

This was arrived at assuming that $a_{ii} \neq 0 \forall i$. If this assumption does not hold, then for some $k \in \{1, \dots, n\}$ we have: $a_{kk} = 0$ and $a_{ii} \neq 0$, $i = 1 \dots k-1$. Then one sees from the above algorithm that if one chooses $b_i = 0$, $i = 1 \dots k-1$, then one must have $x_i = 0$, $i = 1 \dots k-1$, whereupon the k th equation takes the form $0 = b_k$. Accordingly the system $\mathbf{Ax} = \mathbf{b}$ is not solvable in general, so that by Corollary X.2.5 $\mathbf{A}$ is singular.

If $\mathbf{A}$ is an upper triangular matrix, then the system $\mathbf{Ax} = \mathbf{b}$ unwinds from the bottom: one first solves the n th equation $a_{nn}x_n = b_n$, then the $(n-1)$ th equation, etc. In this case too one deduces that $\mathbf{A}$ is singular if $a_{kk} = 0$ for some k ; otherwise there is a matrix $\mathbf{B}$ for which $\mathbf{AB} = \mathbf{I}$. The matrix $\mathbf{B}$ is again of the same type as $\mathbf{A}$ (upper triangular), and the diagonal entries are $b_{ii} = a_{ii}^{-1}$.

Now let $\mathbf{A}$ more generally be a triangular matrix (upper or lower) with nonzero diagonal entries. Then $\mathbf{A}^T$ is also a triangular matrix whose diagonal entries are the same as those of $\mathbf{A}$. By the above reasoning the systems $\mathbf{Ax} = \mathbf{b}$ and $\mathbf{A}^T\mathbf{x} = \mathbf{b}$ are both solvable for every $\mathbf{b} \in \mathbb{R}^n$, so that $\mathbf{A}$ is a regular matrix (Proposition X.2.9). The question of the regularity of a triangular matrix is thus settled independently of Theorem X.2.6. $\square$

As a by-product of the above proof, one also obtained an algorithm for computing the inverse matrix of a triangular matrix $\mathbf{A}$: one solves (in the manner presented in the proof) the linear system of equations $\mathbf{Ax} = \mathbf{b}$ for a general $\mathbf{b} \in \mathbb{R}^n$. When the solution is expressed in the form

$$x_i = \sum_{j=1}^n b_{ij}b_j, \quad i = 1 \dots n \quad \Leftrightarrow \quad \mathbf{x} = \mathbf{Bb},$$

then $\mathbf{B} = \mathbf{A}^{-1}$, i.e. the coefficients b_{ij} are the entries of the inverse matrix. These are computed recursively, row by row, in the course of the algorithm. Only in the case of a diagonal (or otherwise special, see Exercise 7b) matrix is it possible to compute simple expressions for the entries of the inverse matrix $\mathbf{A}^{-1}$ by means of the entries of $\mathbf{A}$.

EXERCISES

1. Let $\mathbf{A}$, $\mathbf{B}$ and $\mathbf{C}$ be square matrices of the same size. Prove:

- a) $\mathbf{A}$ regular and symmetric $\Rightarrow \mathbf{A}^{-1}$ symmetric.
- b) $\mathbf{CA} = \mathbf{CB}$ and $\mathbf{C}$ regular $\Rightarrow \mathbf{A} = \mathbf{B}$.
- c) $\mathbf{AB}$ singular $\Rightarrow \mathbf{A}$ or $\mathbf{B}$ singular.
- d) $\mathbf{A}$ singular and $\mathbf{B}$ regular $\Rightarrow \mathbf{AB}$ and $\mathbf{BA}$ singular.
- e) $\mathbf{A}$ and $\mathbf{B}$ orthogonal $\Rightarrow \mathbf{AB}$ orthogonal.

2. Verify the rule for computing the inverse of a 2×2 matrix

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \frac{1}{D} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}, \quad D = ad - bc \neq 0.$$

3. a) Show that every orthogonal 2×2 matrix can be written, for some $\theta \in \mathbb{R}$, in one or the other of the following forms:

$$\mathbf{A} = \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix} \quad \text{or} \quad \mathbf{A} = \begin{bmatrix} \cos \theta & \sin \theta \\ \sin \theta & -\cos \theta \end{bmatrix}.$$

b) Let $\mathbf{A}$ be a 3×3 orthogonal matrix, of whose entries it is known that: $a_{11} = \frac{3}{7}$, $a_{12} = -\frac{2}{7}$, $a_{22} = \frac{6}{7}$, $a_{21} < 0$, $a_{31} > 0$, $a_{13} < 0$. Compute $\mathbf{A}$ and $\mathbf{A}^{-1}$.

c) Verify that in the system

$$\begin{cases} \sqrt{\frac{2}{3}}x_1 + \frac{1}{2}x_2 - \frac{1}{2\sqrt{3}}x_3 &= 1 \\ \frac{1}{\sqrt{3}}x_1 - \frac{1}{\sqrt{2}}x_2 + \frac{1}{\sqrt{6}}x_3 &= 2 \\ -\frac{1}{2}x_2 - \frac{\sqrt{3}}{2}x_3 &= 3 \end{cases}$$

the coefficient matrix is orthogonal. Solve it using this information!

d) The matrix

$$\mathbf{A} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 \\ -1 & 1 & 1 & -1 \end{bmatrix}$$

has the property: $\lambda \mathbf{A}$ is orthogonal for a certain $\lambda \in \mathbb{R}$. Using this information, determine the row vector $\mathbf{x}^T$ such that $\mathbf{x}^T \mathbf{A} = [7, 13, -3, -9]$.

4. a) Let $\mathbf{H} = \mathbf{I} - \mathbf{x}\mathbf{x}^T$, where $\mathbf{x} \in \mathbb{R}^n$, $\mathbf{x} \neq \mathbf{0}$ and $|\mathbf{x}| \neq 1$. Show that for a certain $\lambda \in \mathbb{R}$ we have $\mathbf{H}^{-1} = \mathbf{I} + \lambda \mathbf{x}\mathbf{x}^T$.
 b) Let $\mathbf{x} \in \mathbb{R}^n$, $\mathbf{x} \neq \mathbf{0}$ and $r = 2|\mathbf{x}|^{-2}$. Show that $\mathbf{H} = \mathbf{I} - r \mathbf{x}\mathbf{x}^T$ is a symmetric and orthogonal matrix.
 c) Show that if a matrix $\mathbf{A}$ satisfies $\mathbf{A} + \mathbf{A}^T = \mathbf{0}$ and $\mathbf{I} + \mathbf{A}$ is a regular matrix, then $\mathbf{B} = (\mathbf{I} + \mathbf{A})^{-1}(\mathbf{I} - \mathbf{A})$ is orthogonal.
5. Is the permutation $(4, 6, 2, 7, 9, 1, 3, 10, 5, 8)$ of the set $(1, 2, \dots, 10)$ even or odd?
6. Of the matrices $\mathbf{V}_1$ and $\mathbf{V}_2$ of size 3×3 it is known that, multiplying the matrix $\mathbf{A}$ on the left, $\mathbf{V}_1$ interchanges the order of rows 1 and 2 of $\mathbf{A}$ and $\mathbf{V}_2$ that of rows 2 and 3. What row permutation is then produced by $\mathbf{V}_1 \mathbf{V}_2$ and $\mathbf{V}_2 \mathbf{V}_1$? Determine also $\mathbf{V}_1$ and $\mathbf{V}_2$ as well as the mentioned products.
7. a) Let $\mathbf{A} = (a_{ij})$ be of size $n \times n$ and $\mathbf{D} = \text{diag}\{d_i, i = 1 \dots n\}$. Determine the entries of the matrices $\mathbf{AD}$ and $\mathbf{DA}$.
 b) Let $n, k \in \mathbb{N}$, $n \geq 2$, $k \leq n$, $\mathbf{A} = (a_{ij})$ of size $n \times n$, $a_{ii} = a \neq 0$ when $i \neq k$, $a_{kk} = b \neq 0$, $a_{ik} = c \neq 0$ when $i > k$ and $a_{ij} = 0$ otherwise. Determine the entries of the inverse matrix $\mathbf{A}^{-1}$.
8. Determine the inverse matrices of the following triangular matrices by solving the general linear system of equations whose coefficient matrix is the matrix in question.

$$\begin{array}{lll}
 \text{a) } \begin{bmatrix} 3 & 0 \\ -5 & 2 \end{bmatrix} & \text{b) } \begin{bmatrix} 4 & 0 & 0 \\ -2 & 2 & 0 \\ 3 & 1 & -1 \end{bmatrix} & \text{c) } \begin{bmatrix} 3 & 0 & -1 & 1 \\ 0 & 2 & 1 & 1 \\ 0 & 0 & -1 & 2 \\ 0 & 0 & 0 & 2 \end{bmatrix} \\
 \text{d) } \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 1 \end{bmatrix} & \text{e) } \begin{bmatrix} 1 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 2 \end{bmatrix} & \text{f) } \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 1 & 1 \end{bmatrix}
 \end{array}$$

9. (*) Appealing to Theorem X.2.6, show that for square matrices we have:
 - a) $\mathbf{AB}$ regular $\Rightarrow \mathbf{A}$ and $\mathbf{B}$ regular.
 - b) $\mathbf{A}^2 + 3\mathbf{A} + 2\mathbf{I} = \mathbf{0} \Rightarrow \mathbf{A}$ regular.

10. (*) Show that there exist permutation matrices $\mathbf{U}$ and $\mathbf{V}$ such that

$$\mathbf{A} = \begin{bmatrix} 0 & 2 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 2 & 2 & 2 & 1 \\ 1 & 2 & 2 & 0 \end{bmatrix} = \mathbf{U} \begin{bmatrix} 1 & 2 & 2 & 2 \\ 0 & 1 & 2 & 2 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 1 \end{bmatrix} \mathbf{V}.$$

Compute the inverse matrix $\mathbf{A}^{-1}$ using this information. Verify that $\mathbf{A}\mathbf{A}^{-1} = \mathbf{I}$.

11. (*) Let $a, b \in \mathbb{R}$ and $b \neq 0$. Show by means of matrix algebra that: the differential equation $y'' + ay' + by = x^{100}$ has as a particular solution a polynomial of degree 100.

X.3 Gaussian elimination

Linear systems of equations that are not utterly enormous in size (e.g. $n < 10^4$) are in practice often solved by *Gaussian elimination*. The algorithm has many variations, and these many names, but the main idea is the same in all of them.

Let us set out to solve the system

$$\begin{cases} a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n = b_1, \\ a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n = b_2, \\ \vdots \quad \quad \quad \vdots \quad \quad \quad \vdots \quad \quad \quad \vdots \\ a_{i1}x_1 + a_{i2}x_2 + \dots + a_{in}x_n = b_i, \\ \vdots \quad \quad \quad \vdots \quad \quad \quad \vdots \quad \quad \quad \vdots \\ a_{n1}x_1 + a_{n2}x_2 + \dots + a_{nn}x_n = b_n. \end{cases}$$

In Gaussian elimination one assumes that the system has a solution (x_i) and proceeds as if the solution had been substituted in place of the unknowns. The solution need not be unique — this becomes clear in the course of the algorithm. Likewise it becomes clear whether a solution exists at all.

In Gaussian elimination the system is transformed step by step into a simpler form. The transformed systems are equivalent to the original (that is, the transformations can be made in both directions), so that no information is lost in the course of the computation process.

Assuming that $a_{11} \neq 0$ (the case $a_{11} = 0$ is returned to later), the first so-called *elimination step* of Gaussian elimination is: solve from the 1st equation

$$x_1 = - \sum_{j=2}^n \frac{a_{1j}}{a_{11}} x_j + \frac{b_1}{a_{11}}$$

and substitute this expression in place of x_1 into the other equations. After simplification, the result is a system equivalent to the original

$$\begin{cases} a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n = b_1, \\ a_{22}^{(1)}x_2 + \dots + a_{2n}^{(1)}x_n = b_2^{(1)}, \\ \vdots \quad \quad \quad \vdots \quad \quad \quad \vdots \\ a_{i2}^{(1)}x_2 + \dots + a_{in}^{(1)}x_n = b_i^{(1)}, \\ \vdots \quad \quad \quad \vdots \quad \quad \quad \vdots \\ a_{n2}^{(1)}x_2 + \dots + a_{nn}^{(1)}x_n = b_n^{(1)}. \end{cases}$$

Here the first equation has been left in its original form, and we have written

$$\begin{aligned} a_{ij}^{(1)} &= a_{ij} - a_{i1}a_{1j}/a_{11}, \quad i, j = 2 \dots n, \\ b_i^{(1)} &= b_i - a_{i1}b_1/a_{11}, \quad i = 2 \dots n. \end{aligned}$$

Computing in array form, this transformation is carried out by multiplying the 1st equation on both sides by the number $-a_{i1}/a_{11}$, adding the result term by term to the i th equation, and replacing the previous i th equation by the result, $i = 2 \dots n$.

The system after the first elimination step is, in matrix form,

$$\begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ 0 & a_{22}^{(1)} & \dots & a_{2n}^{(1)} \\ \vdots & \vdots & & \vdots \\ 0 & a_{n2}^{(1)} & \dots & a_{nn}^{(1)} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2^{(1)} \\ \vdots \\ b_n^{(1)} \end{bmatrix}.$$

Since in this system x_1 appears only in the first equation, this equation can henceforth be left in peace and one can consider the reduced system

$$\begin{bmatrix} a_{22}^{(1)} & a_{23}^{(1)} & \dots & a_{2n}^{(1)} \\ a_{33}^{(1)} & a_{33}^{(1)} & \dots & a_{3n}^{(1)} \\ \vdots & \vdots & & \vdots \\ a_{n2}^{(1)} & a_{n3}^{(1)} & \dots & a_{nn}^{(1)} \end{bmatrix} \begin{bmatrix} x_2 \\ x_3 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} b_2^{(1)} \\ b_3^{(1)} \\ \vdots \\ b_n^{(1)} \end{bmatrix}.$$

If $a_{22}^{(1)} \neq 0$, one performs on this system the same kind of elimination transformation as above, whereupon it is transformed into the form

$$\begin{bmatrix} a_{22}^{(1)} & a_{23}^{(1)} & \dots & a_{2n}^{(1)} \\ 0 & a_{33}^{(2)} & \dots & a_{3n}^{(2)} \\ \vdots & \vdots & & \vdots \\ 0 & a_{n3}^{(2)} & \dots & a_{nn}^{(2)} \end{bmatrix} \begin{bmatrix} x_2 \\ x_3 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} b_2^{(1)} \\ b_3^{(2)} \\ \vdots \\ b_n^{(2)} \end{bmatrix}.$$

Here x_2 appears only in the first equation, so the system is again reduced. In general, after $k - 1$ elimination steps the whole system has been brought to the form

$$\begin{bmatrix} a_{11} & a_{12} & \dots & a_{1k} & \dots & a_{1n} \\ 0 & a_{22}^{(1)} & \dots & a_{2k}^{(1)} & \dots & a_{2n}^{(1)} \\ \vdots & \vdots & \ddots & \vdots & & \vdots \\ 0 & 0 & \dots & a_{kk}^{(k-1)} & \dots & a_{kn}^{(k-1)} \\ \vdots & \vdots & & \vdots & & \vdots \\ 0 & 0 & \dots & a_{nk}^{(k-1)} & \dots & a_{nn}^{(k-1)} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_k \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2^{(1)} \\ \vdots \\ b_k^{(k-1)} \\ \vdots \\ b_n^{(k-1)} \end{bmatrix}.$$

The precondition for reaching this form is thus that

$$a_{ll}^{(l-1)} \neq 0, \quad l = 1 \dots k-1,$$

where $a_{11}^{(0)} = a_{11}$. The entry $a_{kk}^{(k-1)}$ is called the *pivot element* of the next, i.e. the k th, elimination step. Provided that all pivot elements are nonzero and also $a_{nn}^{(n-1)} \neq 0$, one arrives after $n-1$ steps at a system whose coefficient matrix is a regular triangular matrix:

$$\begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ 0 & a_{22}^{(1)} & \dots & a_{2n}^{(1)} \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & a_{nn}^{(n-1)} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2^{(1)} \\ \vdots \\ b_n^{(k-1)} \end{bmatrix}.$$

As we shall soon see, the successful completion of the algorithm in this way shows that the coefficient matrix $\mathbf{A}$ of the original system is regular. In any case the transformed system is easily solved by unwinding it from the bottom, cf. the proof of Theorem X.2.12 above. This phase of the algorithm is called *back-substitution*.

EXAMPLE 1 (See Section X.8, Example 2) Solve by Gaussian elimination the system

$$\begin{bmatrix} 3 & -1 & 0 & 0 & -1 & 1 \\ 1 & -3 & 0 & 2 & -1 & 1 \\ 0 & 0 & 3 & 1 & -1 & -1 \\ 0 & -2 & 1 & 3 & -1 & -1 \\ 1 & -1 & 1 & 1 & -4 & 0 \\ -1 & 1 & 1 & 1 & 0 & -4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \\ x_6 \end{bmatrix} = F \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} + G \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \\ 0 \\ 0 \end{bmatrix},$$

where $F, G \in \mathbb{R}$.

Solution The solution is $\mathbf{x} = F\mathbf{x}_1 + G\mathbf{x}_2$, where $\mathbf{Ax}_1 = \mathbf{b}_1 = [0 \ 1 \ 0 \ 0 \ 0 \ 0]^T$ and $\mathbf{Ax}_2 = \mathbf{b}_2 = [0 \ 0 \ 0 \ 1 \ 0 \ 0]^T$, so two linear systems of equations must be solved. By Gaussian elimination both systems can be handled simultaneously: write the systems in array form

$$\left[\begin{array}{cccccc|cc} 3 & -1 & 0 & 0 & -1 & 1 & 0 & 0 \\ 1 & -3 & 0 & 2 & -1 & 1 & 1 & 0 \\ 0 & 0 & 3 & 1 & -1 & -1 & 0 & 0 \\ 0 & -2 & 1 & 3 & -1 & -1 & 0 & 1 \\ 1 & -1 & 1 & 1 & -4 & 0 & 0 & 0 \\ -1 & 1 & 1 & 1 & 0 & -4 & 0 & 0 \end{array} \right],$$

and extend the elimination to the whole array. Below is the transformed array after each elimination step.

$$(k = 1) \quad \left[\begin{array}{cccccc|cc} 3 & -1 & 0 & 0 & -1 & 1 & 0 & 0 \\ 0 & -\frac{8}{3} & 0 & 2 & -\frac{2}{3} & \frac{2}{3} & 1 & 0 \\ 0 & 0 & 3 & 1 & -1 & -1 & 0 & 0 \\ 0 & -2 & 1 & 3 & -1 & -11 & 0 & 1 \\ 0 & -\frac{2}{3} & 1 & 1 & -\frac{11}{3} & -\frac{1}{3} & 0 & 0 \\ 0 & \frac{2}{3} & 1 & 1 & -\frac{1}{3} & -\frac{11}{3} & 0 & 0 \end{array} \right]$$

$$(k = 2) \quad \left[\begin{array}{cccccc|cc} 3 & -1 & 0 & 0 & -1 & 1 & 0 & 0 \\ 0 & -\frac{8}{3} & 0 & 2 & -\frac{2}{3} & \frac{2}{3} & 1 & 0 \\ 0 & 0 & 3 & 1 & -1 & -1 & 0 & 0 \\ 0 & 0 & 1 & \frac{3}{2} & -\frac{1}{2} & -\frac{3}{2} & -\frac{3}{4} & 1 \\ 0 & 0 & 1 & \frac{1}{2} & -\frac{7}{2} & -\frac{1}{2} & -\frac{1}{4} & 0 \\ 0 & 0 & 1 & \frac{3}{2} & -\frac{1}{2} & -\frac{7}{2} & \frac{1}{4} & 0 \end{array} \right]$$

$$(k = 3) \quad \left[\begin{array}{cccccc|cc} 3 & -1 & 0 & 0 & -1 & 1 & 0 & 0 \\ 0 & -\frac{8}{3} & 0 & 2 & -\frac{2}{3} & \frac{2}{3} & 1 & 0 \\ 0 & 0 & 3 & 1 & -1 & -1 & 0 & 0 \\ 0 & 0 & 0 & \frac{7}{6} & -\frac{1}{6} & -\frac{7}{6} & -\frac{3}{4} & 1 \\ 0 & 0 & 0 & \frac{1}{6} & -\frac{19}{6} & -\frac{1}{6} & -\frac{1}{4} & 0 \\ 0 & 0 & 0 & \frac{7}{6} & -\frac{1}{6} & -\frac{19}{6} & \frac{1}{4} & 0 \end{array} \right]$$

$$(k = 4) \quad \left[\begin{array}{cccccc|cc} 3 & -1 & 0 & 0 & -1 & 1 & 0 & 0 \\ 0 & -\frac{8}{3} & 0 & 2 & -\frac{2}{3} & \frac{2}{3} & 1 & 0 \\ 0 & 0 & 3 & 1 & -1 & -1 & 0 & 0 \\ 0 & 0 & 0 & \frac{7}{6} & -\frac{1}{6} & -\frac{7}{6} & -\frac{3}{4} & 1 \\ 0 & 0 & 0 & 0 & -\frac{22}{7} & 0 & -\frac{1}{7} & -\frac{1}{7} \\ 0 & 0 & 0 & 0 & 0 & -2 & 1 & -1 \end{array} \right]$$

Step $k = 5$ is not needed in this case, since $a_{65}^{(4)} = 0$.

The solution is obtained from this by back-substitution. — Here, however, we use another possibility, equivalent in terms of computational labor, in which the elimination algorithm is used again ‘backwards’. Since the coefficient matrix is upper triangular, when eliminating backwards there changes in the matrix at each elimination step only the column at the position of the pivot element, or more precisely the part of it above the diagonal. The elimination phases backwards

are the following.

$$(k = 5) \quad \left[\begin{array}{cccccc|cc} 3 & -1 & 0 & 0 & -1 & 0 & \frac{1}{2} & -\frac{1}{2} \\ 0 & -\frac{8}{3} & 0 & 2 & -\frac{2}{3} & 0 & \frac{4}{3} & -\frac{1}{3} \\ 0 & 0 & 3 & 1 & -1 & 0 & -\frac{1}{2} & \frac{1}{2} \\ 0 & 0 & 0 & \frac{7}{6} & -\frac{1}{6} & 0 & -\frac{4}{3} & \frac{19}{12} \\ 0 & 0 & 0 & 0 & -\frac{22}{7} & 0 & -\frac{1}{7} & -\frac{1}{7} \\ 0 & 0 & 0 & 0 & 0 & -2 & 1 & -1 \end{array} \right]$$

$$(k = 4) \quad \left[\begin{array}{cccccc|cc} 3 & -1 & 0 & 0 & 0 & 0 & \frac{6}{11} & -\frac{5}{11} \\ 0 & -\frac{8}{3} & 0 & 2 & 0 & 0 & \frac{15}{11} & -\frac{10}{33} \\ 0 & 0 & 3 & 1 & 0 & 0 & -\frac{5}{11} & \frac{6}{11} \\ 0 & 0 & 0 & \frac{7}{6} & 0 & 0 & -\frac{175}{132} & \frac{35}{22} \\ 0 & 0 & 0 & 0 & -\frac{22}{7} & 0 & -\frac{1}{7} & -\frac{1}{7} \\ 0 & 0 & 0 & 0 & 0 & -2 & 1 & -1 \end{array} \right]$$

$$(k = 3) \quad \left[\begin{array}{cccccc|cc} 3 & -1 & 0 & 0 & 0 & 0 & \frac{6}{11} & -\frac{5}{11} \\ 0 & -\frac{8}{3} & 0 & 0 & 0 & 0 & \frac{40}{11} & -\frac{100}{33} \\ 0 & 0 & 3 & 0 & 0 & 0 & \frac{15}{11} & -\frac{9}{33} \\ 0 & 0 & 0 & \frac{7}{6} & 0 & 0 & -\frac{175}{132} & \frac{35}{22} \\ 0 & 0 & 0 & 0 & -\frac{22}{7} & 0 & -\frac{1}{7} & -\frac{1}{7} \\ 0 & 0 & 0 & 0 & 0 & -2 & 1 & -1 \end{array} \right]$$

$$(k = 2) \quad \left[\begin{array}{cccccc|cc} 3 & 0 & 0 & 0 & 0 & 0 & -\frac{9}{11} & \frac{15}{22} \\ 0 & -\frac{8}{3} & 0 & 0 & 0 & 0 & \frac{40}{11} & -\frac{100}{33} \\ 0 & 0 & 3 & 0 & 0 & 0 & \frac{15}{11} & -\frac{9}{33} \\ 0 & 0 & 0 & \frac{7}{6} & 0 & 0 & -\frac{175}{132} & \frac{35}{22} \\ 0 & 0 & 0 & 0 & -\frac{22}{7} & 0 & -\frac{1}{7} & -\frac{1}{7} \\ 0 & 0 & 0 & 0 & 0 & -2 & 1 & -1 \end{array} \right]$$

Finally the solution of the diagonal system:

$$\left[\begin{array}{cccccc|cc} 1 & 0 & 0 & 0 & 0 & 0 & -\frac{3}{11} & \frac{5}{22} \\ 0 & 1 & 0 & 0 & 0 & 0 & -\frac{15}{11} & \frac{25}{22} \\ 0 & 0 & 1 & 0 & 0 & 0 & -\frac{5}{22} & -\frac{3}{11} \\ 0 & 0 & 0 & 1 & 0 & 0 & -\frac{25}{22} & \frac{30}{22} \\ 0 & 0 & 0 & 0 & 1 & 0 & \frac{1}{22} & \frac{1}{22} \\ 0 & 0 & 0 & 0 & 0 & 1 & -\frac{1}{2} & \frac{1}{2} \end{array} \right].$$

The requested solution of the system is accordingly

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \\ x_6 \end{bmatrix} = \frac{F}{22} \begin{bmatrix} -6 \\ -30 \\ 5 \\ -25 \\ 1 \\ -11 \end{bmatrix} + \frac{G}{22} \begin{bmatrix} 5 \\ 25 \\ -6 \\ 30 \\ 1 \\ 11 \end{bmatrix}. \quad \square$$

The workload of Gaussian elimination

Gaussian elimination thus consists of two phases, the elimination phase and back-substitution. In English the terms ‘forward sweep’ and ‘backward sweep’ are also used. Of these the first ‘sweep’ is, in terms of workload (computation time), considerably heavier. This is seen from the general algorithmic form of the elimination phase, which is

$$\begin{aligned} a_{ij}^{(k)} &= a_{ij}^{(k-1)} - a_{ik}^{(k-1)} a_{kj}^{(k-1)} / a_{kk}^{(k-1)}, \quad i, j = k+1 \dots n, \\ b_i^{(k)} &= b_i^{(k-1)} - a_{ik}^{(k-1)} b_k^{(k-1)} / a_{kk}^{(k-1)}, \quad i = k+1 \dots n. \end{aligned}$$

Here it is advisable to compute the divisions $a_{ik}^{(k-1)} / a_{kk}^{(k-1)}$, $i = k+1 \dots n$ first, whereupon the algorithm is, in programming-language form,

$$\begin{aligned} l_{ik} &\leftarrow a_{ik} / a_{kk}, & i &= k+1 \dots n, \\ a_{ij} &\leftarrow a_{ij} - l_{ik} a_{kj}, & i, j &= k+1 \dots n, \\ b_i &\leftarrow b_i - l_{ik} b_k, & i &= k+1 \dots n. \end{aligned}$$

In estimating the workload, let us take as the unit of work an arithmetic operation consisting of one multiplication (or division) and one addition or subtraction. From the algorithm one sees that the most laborious part of the algorithm consists of the transformations $a_{ij}^{(k)} \leftarrow a_{ij}^{(k-1)}$. The workload these require in the whole elimination phase is, in the mentioned operations,

$$W = (n-1)^2 + (n-2)^2 + \dots + 1 = \frac{1}{6}(n-1)n(2n-1).$$

The other parts of the algorithm, such as the transformations $b_i^{(k)} \leftarrow b_i^{(k-1)}$ and the back-substitutions, are of order of magnitude $\mathcal{O}(n^2)$ in workload, so that the workload of Gaussian elimination is

$$W = \frac{1}{3}n^3 + \mathcal{O}(n^2).$$

From this one sees that, for example, solving a system of a thousand equations ($W \approx 0.3 \cdot 10^9$) on a computer is no problem in terms of computation time. On the other hand, a system of size $n = 10^6$ ($W \approx 0.3 \cdot 10^{18}$) may already be an overwhelming task. — Even systems this large are solved nowadays, but then one generally exploits special properties of the coefficient matrix. Another, traditionally popular alternative is to use *iterative* (approximate) solution methods suited to the task. In relation to iterative methods, a solution based on Gaussian elimination is called a *direct* solution method. For moderate values of n , the direct solution of a system by Gaussian elimination is still a competitive method, and as the performance of computers has grown it has also won ground from iterative methods. Especially when the algorithm goes through in the above straightforward way, it is also very easy to program.

The LU -decomposition of a square matrix

By Gaussian elimination, insofar as it succeeds in the manner described above, one obtains for the system $\mathbf{Ax} = \mathbf{b}$ a solution for every $\mathbf{b} \in \mathbb{R}^n$, so that by Theorem X.2.6 the coefficient matrix must be regular. There is, however, no need to appeal to that theorem (which has not yet been proved), since the regularity of the coefficient matrix is proved directly by Gaussian elimination. The reasoning is as follows: first one notes that the elimination phase of Gaussian elimination transforms the original system

$$\mathbf{Ax} = \mathbf{b}$$

into the form

$$\mathbf{Ux} = \mathbf{c} = \mathbf{Rb},$$

where $\mathbf{U}$ is the transformed coefficient matrix, that is, the upper triangular matrix whose diagonal entries are $a_{kk}^{(k-1)} \neq 0$, $k = 1 \dots n$. The matrix $\mathbf{R}$ is not directly visible after the elimination phase, but it can be found by considering the system

$$\mathbf{Ly} = \mathbf{b},$$

where $\mathbf{L}$ is the lower triangular matrix defined by

$$[\mathbf{L}]_{ik} = \begin{cases} 0, & \text{if } i < k, \\ 1, & \text{if } i = k, \\ l_{ik} = a_{ik}^{(k-1)} / a_{kk}^{(k-1)}, & \text{if } i > k. \end{cases}$$

Here $a_{ik}^{(k-1)}$, $i = k \dots n$, are the entries of the transformed coefficient matrix after $k - 1$ elimination steps:

$$\begin{array}{ccc} a_{k-1,k-1}^{(k-2)} & & \\ 0 & a_{kk}^{(k-1)} & \\ \vdots & \vdots & \\ 0 & a_{ik}^{(k-1)} & \rightarrow l_{ik} = a_{ik}^{(k-1)} / a_{kk}^{(k-1)} \\ \vdots & \vdots & \\ 0 & a_{nk}^{(k-1)} & \end{array}$$

From the form of the matrix $\mathbf{L}$ one sees that if Gaussian elimination is applied to the above system, then already the elimination phase gives the solution $\mathbf{y} = \mathbf{L}^{-1}\mathbf{b}$. On the other hand, comparing Gaussian elimination applied to the systems $\mathbf{L}\mathbf{y} = \mathbf{b}$ and $\mathbf{A}\mathbf{x} = \mathbf{b}$, one sees that the elimination-phase transformations $b_i^{(k)} \leftarrow b_i^{(k-1)}$ are in both cases exactly the same. Thus we have deduced:

$$\mathbf{y} = \mathbf{L}^{-1}\mathbf{b} = \mathbf{R}\mathbf{b}.$$

Since this holds $\forall \mathbf{b} \in \mathbb{R}^n$, we have $\mathbf{R} = \mathbf{L}^{-1}$ (Proposition X.1.1), and accordingly

$$\mathbf{A}\mathbf{x} = \mathbf{b} \Leftrightarrow \mathbf{L}\mathbf{U}\mathbf{x} = \mathbf{b}$$

Thus $\mathbf{A}\mathbf{x} = \mathbf{L}\mathbf{U}\mathbf{x} \forall \mathbf{x} \in \mathbb{R}^n$, so that

$$\mathbf{A} = \mathbf{L}\mathbf{U}.$$

This is called the *LU-decomposition* of the matrix. To compute it, nothing more is needed than the elimination phase of Gaussian elimination, provided one remembers to store the matrix $\mathbf{L}$, that is, the numbers

$$[\mathbf{L}]_{ik} = l_{ik} = a_{ik}^{(k-1)} / a_{kk}^{(k-1)}, \quad i = k + 1 \dots n, \quad k = 1 \dots n - 1.$$

These get computed in any case in the course of the elimination, but without being stored $\mathbf{L}$ does not remain visible (since the system's solution algorithm computes ready the vector $\mathbf{L}^{-1}\mathbf{b} = \mathbf{R}\mathbf{b}$).

EXAMPLE 1 (continued) By following the elimination phases ($k = 1 \dots 4$) one sees that in the example system the *LU-decomposition* of the coefficient matrix is

$$\mathbf{A} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ \frac{1}{3} & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & \frac{3}{4} & \frac{1}{3} & 1 & 0 & 0 \\ \frac{1}{3} & \frac{1}{4} & \frac{1}{3} & \frac{1}{7} & 1 & 0 \\ -\frac{1}{3} & -\frac{1}{4} & \frac{1}{3} & 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} 3 & -1 & 0 & 0 & -1 & 1 \\ 0 & -\frac{8}{3} & 0 & 2 & -\frac{2}{3} & \frac{2}{3} \\ 0 & 0 & 3 & 1 & -1 & -1 \\ 0 & 0 & 0 & \frac{7}{6} & -\frac{1}{6} & -\frac{7}{6} \\ 0 & 0 & 0 & 0 & -\frac{22}{7} & 0 \\ 0 & 0 & 0 & 0 & 0 & -2 \end{bmatrix} = \mathbf{L}\mathbf{U}. \quad \square$$

The LU -decomposition of a matrix thus succeeds by Gaussian elimination if $a_{kk}^{(k-1)} \neq 0$, $k = 1 \dots n$. Then $\mathbf{L}$ and $\mathbf{U}$ are regular (Theorem X.2.12), so that $\mathbf{A}$ is regular and

$$\mathbf{A}^{-1} = \mathbf{U}^{-1}\mathbf{L}^{-1}.$$

The mentioned condition on the pivot elements is thus a sufficient condition for the regularity of the matrix. In the next section we shall see that when Gaussian elimination is modified a little, it settles the question of the regularity of a square matrix in the general case too.

Computing the inverse matrix

Insofar as Gaussian elimination for the system $\mathbf{Ax} = \mathbf{b}$ succeeds in the manner described above (pivot elements $\neq 0$), the coefficient matrix is thus regular, whereupon the solution of the system can be written $\mathbf{x} = \mathbf{A}^{-1}\mathbf{b}$. If one also wants to compute the inverse matrix $\mathbf{A}^{-1}$ and not just the solution $\mathbf{x}$, then one (especially natural for hand computation) procedure is to apply Gaussian elimination symbolically, in such a way that the entries b_i of $\mathbf{b}$ are not given numerical values. Using the algorithm, the solution is then obtained in the form $\mathbf{x} = \mathbf{Bb}$, whereupon $\mathbf{A}^{-1} = \mathbf{B}$.

EXAMPLE 2 Compute the inverse matrix of the matrix

$$\mathbf{A} = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 1 & -1 & -1 \end{bmatrix}.$$

Solution By elimination and back-substitution one obtains

$$\begin{aligned} & \begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 1 & -1 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} \\ \mapsto & \begin{bmatrix} 1 & 1 & 0 \\ 0 & -1 & 1 \\ 0 & 0 & -3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} b_1 \\ -b_1 + b_2 \\ 3b_1 - 2b_2 + b_3 \end{bmatrix} \\ \mapsto & \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \frac{1}{3} \begin{bmatrix} b_1 + b_2 + b_3 \\ 2b_1 - b_2 - b_3 \\ -b_1 + 2b_2 - b_3 \end{bmatrix}. \end{aligned}$$

From the result the inverse matrix can be read off:

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 1 & 1 & 1 \\ 2 & -1 & -1 \\ -1 & 2 & -1 \end{bmatrix} \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} = \mathbf{A}^{-1}\mathbf{b}. \quad \square$$

A more straightforward way than the previous one to compute the inverse matrix is to make use of the matrix-algebra identity (cf. Section X.1)

$$\mathbf{A}^{-1} = \mathbf{A}^{-1}\mathbf{I} = \mathbf{A}^{-1}[\mathbf{e}_1 \dots \mathbf{e}_n] = [\mathbf{A}^{-1}\mathbf{e}_1 \dots \mathbf{A}^{-1}\mathbf{e}_n].$$

Accordingly the columns of $\mathbf{A}^{-1}$ can be computed by solving the linear system of equations $\mathbf{A}\mathbf{x} = \mathbf{b}$ with $\mathbf{b} = \mathbf{e}_i$, $i = 1 \dots n$. Since in Gaussian elimination one can handle several different vectors $\mathbf{b}$ at once (cf. Example 1), the columns of $\mathbf{A}^{-1}$ can also all be determined at one go. The computation of the inverse matrix can then be described as a two-phase operation (elimination and back-substitution)

$$[\mathbf{A} \mid \mathbf{I}] \mapsto [\mathbf{U} \mid \mathbf{L}^{-1}] \mapsto [\mathbf{I} \mid \mathbf{A}^{-1}].$$

In fact one proceeds in a corresponding way in the above symbolic computation, for when one writes $\mathbf{b} = \sum_{i=1}^n b_i \mathbf{e}_i$, then (cf. Section X.1)

$$\mathbf{A}^{-1}\mathbf{b} = \sum_{i=1}^n b_i \mathbf{A}^{-1}\mathbf{e}_i = [\mathbf{A}^{-1}\mathbf{e}_1 \dots \mathbf{A}^{-1}\mathbf{e}_n] \mathbf{b}.$$

EXAMPLE 2 (continued) The computation of the example organized differently:

$$\begin{aligned} [\mathbf{A} \mid \mathbf{I}] &= \left[\begin{array}{ccc|ccc} 1 & 1 & 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 & 1 & 0 \\ 1 & -1 & -1 & 0 & 0 & 1 \end{array} \right] \mapsto \left[\begin{array}{ccc|ccc} 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & -1 & 1 & -1 & 1 & 0 \\ 0 & 0 & -3 & 1 & -2 & 1 \end{array} \right] \\ &\mapsto \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \\ 0 & 1 & 0 & \frac{2}{3} & -\frac{1}{3} & -\frac{1}{3} \\ 0 & 0 & 1 & -\frac{1}{3} & \frac{2}{3} & -\frac{1}{3} \end{array} \right] = [\mathbf{I} \mid \mathbf{A}^{-1}]. \quad \square \end{aligned}$$

The inverse matrix can also be computed using the formula based on the LU -decomposition, $\mathbf{A}^{-1} = \mathbf{U}^{-1}\mathbf{L}^{-1}$, that is, by first computing separately, in the above way, the inverse matrices $\mathbf{L}^{-1}$ and $\mathbf{U}^{-1}$ and then their product. Both in this way and by computing according to the above algorithm, the workload comes to $W = n^3 + \mathcal{O}(n^2)$ (Exercise 6) — that is, about three times the workload compared with computing the LU -decomposition or solving a single system. It turns out, indeed, that when solving systems numerically it simply does not pay

to compute the inverse matrix. Indeed, even if one were to solve several systems with the coefficient matrix remaining the same (e.g. a varying load on a truss structure, cf. Section X.8), the repetition of the elimination phase of Gaussian elimination can be avoided by means of the LU -decomposition alone: when the decomposition matrices $\mathbf{L}$ and $\mathbf{U}$ are stored, the system $\mathbf{Ax} = \mathbf{b}$ can be broken into two successive systems:

$$\mathbf{LUx} = \mathbf{b} \quad \Leftrightarrow \quad \mathbf{Ly} = \mathbf{b}, \quad \mathbf{Ux} = \mathbf{y}.$$

These are solved by forward- and back-substitution, whereupon the workload of solving the system comes to $W \sim n^2$ (see Exercise 5b). If $\mathbf{A}^{-1}$ had been stored, the workload would be the same, so nothing is gained by computing the inverse matrix (!).

EXERCISES

- Write the following systems in array form and solve them by Gaussian elimination. Compute also the LU -decompositions of the coefficient matrices.

$$\text{a) } \begin{cases} x + y = 33 \\ 2x + 4y = 100 \end{cases} \qquad \text{b) } \begin{cases} x + ay = 1 \\ ax + y = 2 \end{cases} \quad (a \neq \pm 1)$$

$$\text{c) } \begin{cases} x_1 + 3x_2 - x_3 = 4 \\ -x_1 + x_2 + 2x_3 = 7 \\ 2x_1 + 6x_2 + 2x_3 = 20 \end{cases} \qquad \text{d) } \begin{cases} x_1 + 2x_2 + 3x_3 = 1 \\ x_1 + x_2 - x_3 = 2 \\ x_1 - 6x_3 = 4 \end{cases}$$

$$\text{e) } \begin{cases} x_1 + x_2 + 2x_3 = -1 \\ -x_1 + x_2 = 1 \\ 3x_1 - x_2 + 3x_3 = -3 \end{cases} \qquad \text{f) } \begin{cases} x_1 - x_2 + 4x_3 = 8 \\ x_1 - 2x_2 + 3x_3 = 0 \\ -2x_1 + 4x_2 - 5x_3 = 8 \end{cases}$$

$$\text{g) } \begin{cases} 2x_1 - x_2 = 1 \\ -x_1 + 2x_2 - x_3 = 2 \\ -x_2 + 2x_3 - x_4 = -1 \\ -x_3 + 2x_4 = 1 \end{cases} \qquad \text{h) } \begin{cases} 2x_1 - x_2 - x_4 = -4 \\ -x_1 + 2x_2 - x_3 = 1 \\ -x_2 + 2x_3 - x_4 = 4 \\ -x_1 - x_3 + 3x_4 = 10 \end{cases}$$

- Let

$$\mathbf{A} = \begin{bmatrix} 1 & 2 & 3 & 1 \\ 2 & 1 & 1 & 1 \\ 1 & 2 & 1 & 0 \\ 0 & 1 & 1 & 2 \end{bmatrix}, \quad \mathbf{b}_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}, \quad \mathbf{b}_2 = \begin{bmatrix} 5 \\ 3 \\ 4 \\ 0 \end{bmatrix}$$

Solve by Gaussian elimination the systems $\mathbf{Ax} = \mathbf{b}_1$ and $\mathbf{Ax} = \mathbf{b}_2$ and compute the LU -decomposition of the matrix $\mathbf{A}$.

3. Let

$$\mathbf{A} = \begin{bmatrix} 1 & 0 & 0 & 1 \\ 1 & 1 & 0 & 2 \\ 1 & 0 & 1 & 0 \\ 1 & 1 & 2 & 1 \end{bmatrix}.$$

Compute the inverse matrix $\mathbf{A}^{-1}$ by Gaussian elimination

a) by solving the system $\mathbf{Ax} = \mathbf{b}$ for a general $\mathbf{b} \in \mathbb{R}^4$ (symbolic computation),

b) by solving in array form the systems $\mathbf{Ax} = \mathbf{e}_i$, $i = 1 \dots 4$

4. Show that if the coefficient matrix $\mathbf{A}$ of a linear system of equations is symmetric, then Gaussian elimination keeps symmetric the part of the matrix to which the elimination has not yet advanced, i.e. after $(k-1)$ elimination steps

$$a_{ji}^{(k-1)} = a_{ij}^{(k-1)} \quad \forall i, j \geq k.$$

Deduce that the computational work required by Gaussian elimination is, in the case of a symmetric coefficient matrix, $W = \frac{1}{6}n^3 + \mathcal{O}(n^2)$.

5. Let $\mathbf{A}, \mathbf{B}$ be matrices of size $n \times n$, $\mathbf{L}, \mathbf{U}$ lower and upper triangular matrices of the same size and $\mathbf{b} \in \mathbb{R}^n$. Verify the following estimates for the workload of arithmetic operations (work unit = multiplication + addition).

a)	$\mathbf{A}, \mathbf{b} \mapsto \mathbf{Ab}$	$n^2 + \mathcal{O}(n)$
b)	$\mathbf{L}, \mathbf{b} \mapsto \mathbf{L}^{-1}\mathbf{b}$	$\frac{1}{2}n^2 + \mathcal{O}(n)$
c)	$\mathbf{A}, \mathbf{B} \mapsto \mathbf{AB}$	$n^3 + \mathcal{O}(n^2)$
d)	$\mathbf{L}, \mathbf{U} \mapsto \mathbf{LU}$	$\frac{1}{3}n^3 + \mathcal{O}(n^2)$

6. (*) Show that the workload of the operation $\mathbf{A} \mapsto \mathbf{A}^{-1}$ in the different variants of Gaussian elimination is ($\mathbf{A}$ of size $n \times n$, work unit = multiplication + addition)

a)	$[\mathbf{A} \mid \mathbf{I}] \mapsto [\mathbf{I} \mid \mathbf{A}^{-1}]$	$n^3 + \mathcal{O}(n^2)$
b)	$\mathbf{A} \mapsto \mathbf{A}^{-1} = \mathbf{U}^{-1}\mathbf{L}^{-1}$	$n^3 + \mathcal{O}(n^2)$

X.4 Gaussian elimination with pivoting. Singular systems

If in Gaussian elimination one arrives, after $k - 1$ elimination steps, at a situation in which the pivot element $a_{kk}^{(k-1)} = 0$, then one can proceed by using so-called *pivoting*. In pivoting one simply interchanges the order of the rows and/or columns of the matrix (corresponding to interchanging the order of the equations or unknowns in the system). The aim is to find some order in which the pivot element becomes nonzero, so that Gaussian elimination can be continued. Let us first consider so-called *partial*, or *row pivoting*, in which the procedure is as follows: one goes through the entries below the diagonal in the k th column

$$a_{ik}^{(k-1)}, \quad i = k + 1 \dots n.$$

One chooses (if possible) an index $i = l$ such that $a_{lk}^{(k-1)} \neq 0$, and performs in the system (that is, in the matrix $\mathbf{A}^{(k-1)}$ and the vector $\mathbf{b}^{(k-1)}$) the row interchange

$$k \text{ th row} \rightleftharpoons l \text{ th row}.$$

This corresponds to interchanging the order of the equations in the system. From Gaussian elimination one sees that, insofar as the rearrangement of rows is carried out on rows $i = k \dots n$, the interchange of order does not disturb the algorithm itself. In other words, the end result is the same if the rows are interchanged already in the original system and $k - 1$ elimination steps are then performed (Exercise 1a). After the pivoting operation the new pivot element is nonzero, whereupon Gaussian elimination can be continued.*

The above pivoting operation fails only in the case that

$$a_{ik}^{(k-1)} = 0, \quad i = k \dots n.$$

Gaussian elimination has then shown the original matrix $\mathbf{A}$ to be singular. Indeed, since then the unknown x_k does not appear at all in rows $i = k \dots n$ of the transformed system, a non-unique solution of the system $\mathbf{Ax} = \mathbf{0}$ is obtained by giving x_k an arbitrary value, setting $x_{k+1} = \dots = x_n = 0$, and solving x_i , $i = 1 \dots k - 1$ from the transformed system by back-substitution (in terms of x_k). Thus $\mathbf{A}$ is singular (Corollary X.2.5).

*In *automatic* row pivoting one performs the interchange of rows before every elimination step so that the pivot element becomes the largest in absolute value among the entries $a_{ik}^{(k-1)}$, $i = k \dots n$. This reduces the accumulation of rounding errors in the course of the solution algorithm (in floating-point computation). In many ‘black box’ solvers of computer software packages, automatic row pivoting is used (e.g. Mathematica: `LinearSolve`).

After the partial pivoting operation has failed (and thus the matrix has turned out to be singular), Gaussian elimination can still be continued by using so-called *full pivoting*, in which one goes through all the matrix entries beginning from row k and column k , that is, the entries

$$a_{ij}^{(k-1)}, \quad i, j = k \dots n.$$

Assume that for some (l, m) one has $a_{lm}^{(k-1)} \neq 0$. (Possibly one chooses the entry largest in absolute value, as in automatic row pivoting, cf. the footnote above.)

$$\left[\begin{array}{cccc} 0 & \dots & a_{km}^{(k-1)} & \dots & a_{kn}^{(k-1)} \\ \vdots & & \vdots & & \\ a_{lk}^{(k-1)} & \dots & a_{lm}^{(k-1)} & & \\ \vdots & & & & \\ a_{nk}^{(k-1)} & & & & \end{array} \right]$$

One then performs the interchange of rows and columns

$$k \text{ th row} \rightleftharpoons l \text{ th row}, \quad k \text{ th column} \rightleftharpoons m \text{ th column}.$$

The interchange of columns corresponds in the system to a re-indexing of the unknowns. Applied to indices $j \geq k$, this pivoting operation too does not disturb the algorithm itself, i.e. the operation can be thought of as carried out already in the original form of the system (Exercise 1a). Insofar as the pivoting operation succeeds in the assumed way, after the operation the new pivot element is $a_{lm}^{(k-1)} \neq 0$, whereupon Gaussian elimination can be continued. Otherwise one has arrived at a situation in which $a_{ij}^{(k-1)} = 0$ when $i = k \dots n$ and $j = 1 \dots n$. Gaussian elimination has then terminated in a *singular system in standard form*, which is thus, for some $k \in \{0, \dots, n-1\}$, of the form

$$\left[\begin{array}{cccc} u_{11} & \dots & & u_{1n} \\ 0 & \ddots & & \\ \vdots & & u_{kk} & \dots & u_{kn} \\ 0 & 0 & \dots & 0 \\ \vdots & \vdots & & \vdots \\ 0 & 0 & \dots & 0 \end{array} \right] \left[\begin{array}{c} y_1 \\ \vdots \\ y_k \\ \vdots \\ y_n \end{array} \right] = \left[\begin{array}{c} c_1 \\ \vdots \\ c_k \\ \vdots \\ c_n \end{array} \right].$$

Here $\mathbf{U} = \{u_{ij}\}$ is an upper triangular matrix whose last $n-k$ rows are zero rows and $u_{ii} \neq 0$ when $i = 1 \dots k$ (if $k = 0$, then $\mathbf{U} = \mathbf{A} = \mathbf{0}$). The coefficients u_{ij} and c_i have been obtained from the original system $\mathbf{Ax} = \mathbf{b}$ by Gaussian elimination

with pivoting, that is, by combining eliminations and pivoting operations (interchanges of rows and columns). The vector $\mathbf{y}$ contains the original unknowns x_i in the new order dictated by the column interchanges. If in the course of the algorithm only row pivoting has been used, then $\mathbf{y} = \mathbf{x}$.

From the above standard-form system one sees that it has a solution exactly when

$$c_{k+1} = \dots = c_n = 0.$$

In that case the solution is non-unique: y_i , $i = k + 1 \dots n$, can be chosen arbitrarily, after which $y_1, \dots, y_k$ are determined by back-substitution.

EXAMPLE 1 In the following, Gaussian elimination with full pivoting is applied to a singular system.

$$\begin{bmatrix} 0 & 0 & 3 \\ 1 & 2 & 0 \\ 2 & 4 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$$

$$\begin{bmatrix} 4 & 2 & 0 \\ 2 & 1 & 0 \\ 0 & 0 & 3 \end{bmatrix} \begin{bmatrix} x_2 \\ x_1 \\ x_3 \end{bmatrix} = \begin{bmatrix} b_3 \\ b_2 \\ b_1 \end{bmatrix} \quad (\text{interchange of rows } 1 \leftrightarrow 3, \text{ interchange of columns } 1 \leftrightarrow 2)$$

$$\begin{bmatrix} 4 & 2 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 3 \end{bmatrix} \begin{bmatrix} x_2 \\ x_1 \\ x_3 \end{bmatrix} = \begin{bmatrix} b_3 \\ b_2 - b_3/2 \\ b_1 \end{bmatrix} \quad (\text{elimination})$$

$$\begin{bmatrix} 4 & 0 & 2 \\ 0 & 3 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_2 \\ x_3 \\ x_1 \end{bmatrix} = \begin{bmatrix} b_3 \\ b_1 \\ b_2 - b_3/2 \end{bmatrix} \quad (\text{interchange of rows and columns } 2 \leftrightarrow 3)$$

One sees that the resulting standard-form system has a solution exactly when

$$b_2 - b_3/2 = 0 \Leftrightarrow b_3 = 2b_2,$$

and the general solution is then

$$\begin{cases} x_1 = t, \\ x_2 = -t/2 + b_3/4 = (b_2 - t)/2, \\ x_3 = b_1/3 \end{cases} \quad (t \in \mathbb{R}). \quad \square$$

The result of the example could have been seen more directly. The advantage of the general algorithm, however, is that it always works, even when the solver does not ‘see’ it.

The general product decomposition of a square matrix

By means of Gaussian elimination with pivoting too, one can determine the inverse matrix of a matrix $\mathbf{A}$, insofar as $\mathbf{A}$ is regular. When the pivoting operations are thought of as done before applying Gaussian elimination, then, first of all, the row pivotings correspond to a transformation of the system

$$\mathbf{Ax} = \mathbf{b} \quad \Leftrightarrow \quad \mathbf{VAx} = \mathbf{Vb},$$

where $\mathbf{V} = \mathbf{I}_p$ is the permutation matrix corresponding to the final order p of the rows. — Let us recall from Section X.2 that if the row interchanges correspond to transposition matrices $\mathbf{V}_i$, $i = 1 \dots m$, then

$$\mathbf{V} = \mathbf{V}_m \cdots \mathbf{V}_2 \mathbf{V}_1$$

(reverse order!) and $\mathbf{V}^{-1} = \mathbf{V}^T = \mathbf{V}_1 \mathbf{V}_2 \cdots \mathbf{V}_m$. The end result of the column interchanges, in turn, corresponds to a transformation $\mathbf{y} = \mathbf{W}^T \mathbf{x} \Leftrightarrow \mathbf{x} = \mathbf{Wy}$, where the permutation matrix $\mathbf{W} = \mathbf{I}_q$ corresponds to the final order q of the columns. If the transposition matrices corresponding to the column interchanges are $\mathbf{W}_i$, $i = 1 \dots l$, then

$$\mathbf{W} = \mathbf{W}_1 \mathbf{W}_2 \cdots \mathbf{W}_l$$

(same order as in the interchanges!) and $\mathbf{W}^{-1} = \mathbf{W}^T = \mathbf{W}_l \mathbf{W}_{l-1} \cdots \mathbf{W}_1$. When all the pivotings are taken into account, the system is transformed into the form

$$\mathbf{Ax} = \mathbf{b} \quad \Leftrightarrow \quad \mathbf{VAx} = \mathbf{Vb} \quad \Leftrightarrow \quad (\mathbf{VAW})(\mathbf{W}^T \mathbf{x}) = \mathbf{Vb} \quad \Leftrightarrow \quad \mathbf{By} = \mathbf{c},$$

where $\mathbf{c} = \mathbf{Vb}$ (the equations in the changed order p), $\mathbf{y} = \mathbf{W}^T \mathbf{x}$ (the unknowns in the changed order q) and

$$\mathbf{B} = \mathbf{VAW} \Leftrightarrow \mathbf{A} = \mathbf{V}^T \mathbf{BW}^T.$$

When Gaussian elimination is applied to the transformed system $\mathbf{By} = \mathbf{c}$, pivoting is no longer needed. As the end result of the elimination, $\mathbf{B}$ is transformed into an upper triangular matrix $\mathbf{U}$, whereupon there are two alternatives:

- a) $[\mathbf{U}]_{kk} \neq 0$, $k = 1 \dots n$. Then the elimination phase goes through in its entirety, $\mathbf{U}$ is regular, one obtains for $\mathbf{B}$ the LU -decomposition

$$\mathbf{B} = \mathbf{LU},$$

and for $\mathbf{A}$ correspondingly the product decomposition

$$\boxed{\mathbf{A} = \mathbf{V}^T \mathbf{LUW}^T.} \quad (\star)$$

Then $\mathbf{A}$ is regular and

$$\mathbf{A}^{-1} = \mathbf{WU}^{-1} \mathbf{L}^{-1} \mathbf{V} = \mathbf{WB}^{-1} \mathbf{V}.$$

- b) $[\mathbf{U}]_{ii} \neq 0$, $i = 1 \dots k < n$, $[\mathbf{U}]_{i,i} = 0$, $i = k \dots n$ (assuming $\mathbf{A} \neq \mathbf{0}$). Then $\mathbf{U}$ is of the above singular standard form, that is, $[\mathbf{U}]_{ij} = 0$ when $i > k$, $j = 1 \dots n$, and the elimination algorithm stops at row k . The stopping corresponds to setting, in the matrix $\mathbf{L}$,

$$[\mathbf{L}]_{ij} = \delta_{ij}, \quad \text{when } i \in \{k+1, \dots, n\}, j \in \{1, \dots, n\}.$$

When $\mathbf{L}$ is determined otherwise in the same way as in the case of a regular matrix, one sees that the product decomposition $(\star)$ is still valid.

The question of the regularity or singularity of a square matrix has thus been settled by Gaussian elimination as follows: every square matrix can be represented in the product form $(\star)$, where $\mathbf{U}$ is an upper triangular matrix and $\mathbf{L}, \mathbf{V}, \mathbf{W}$ are regular matrices. The matrix $\mathbf{U}$ is either regular or singular, which settles also the character of $\mathbf{A}$.

EXAMPLE 1 (continued) Here

$$\begin{aligned} \mathbf{A} &= \begin{bmatrix} 0 & 0 & 3 \\ 1 & 2 & 0 \\ 2 & 4 & 0 \end{bmatrix}, \quad \mathbf{V} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}, \\ \mathbf{W} &= \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}, \quad \mathbf{B} = \mathbf{VAW} = \begin{bmatrix} 4 & 0 & 2 \\ 0 & 3 & 0 \\ 2 & 0 & 1 \end{bmatrix}, \\ \mathbf{y} = \mathbf{W}^T \mathbf{x} &= \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} x_2 \\ x_3 \\ x_1 \end{bmatrix}, \quad \mathbf{c} = \mathbf{Vb} = \begin{bmatrix} b_3 \\ b_1 \\ b_2 \end{bmatrix}. \end{aligned}$$

From the transformed system

$$\begin{bmatrix} 4 & 0 & 2 \\ 0 & 3 & 0 \\ 2 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_2 \\ x_3 \\ x_1 \end{bmatrix} = \begin{bmatrix} b_3 \\ b_1 \\ b_2 \end{bmatrix}$$

one reaches upper triangular form in a single elimination step. Since the pivoting operations have been carried out already beforehand, they are no longer needed in connection with the elimination. $\square$

Rank

If $\mathbf{A}$ is singular, then the termination index k of the above Gaussian elimination with pivoting is called the *rank* of $\mathbf{A}$, denoted $k = r(\mathbf{A})$. The number $n - r(\mathbf{A})$ thus tells how many zero rows there are in the singular standard-form upper triangular matrix $\mathbf{U}$ of the product decomposition of $\mathbf{A}$. If $\mathbf{A}$ is regular, then $r(\mathbf{A}) = n$, whereupon also $r(\mathbf{A}^T) = n$. More generally,

$$r(\mathbf{A}^T) = r(\mathbf{A}).$$

This too can be justified by means of the product decomposition given by Gaussian elimination (we omit the justification).

A system of size $m \times n$

Gaussian elimination is suited also to a linear system of equations $\mathbf{A}\mathbf{x} = \mathbf{b}$ in which the numbers of equations and unknowns are not the same, i.e. $\mathbf{A}$ is of size $m \times n$, $m \neq n$ ($\mathbf{b} \in \mathbb{R}^m$). When Gaussian elimination with pivoting is applied to such a system, the end result is again a transformed standard-form system $\mathbf{U}\mathbf{y} = \mathbf{c}$, where $\mathbf{U} = \{u_{ij}, i = 1 \dots m, j = 1 \dots n\}$ is an upper triangular matrix (i.e. $u_{ij} = 0$ when $i > j$), and for some k , $0 \leq k \leq \min\{m, n\}$,

$$u_{ii} \neq 0, \text{ when } i = 1 \dots k, \quad u_{ij} = 0, \text{ when } i = k + 1 \dots m, j = 1 \dots n.$$

The index k is again called the *rank* of $\mathbf{A}$. If k is the largest possible, that is, if $k = \min\{m, n\}$, then one says that $\mathbf{A}$ has *full rank*. In this case (fully pivoted) Gaussian elimination can be continued until the algorithm hits the ‘edge’ ($m > n$) or the ‘bottom’ ($m < n$) of the matrix $\mathbf{A}$. The transformed system $\mathbf{U}\mathbf{y} = \mathbf{c}$ can be expressed in *block matrix* form

$$\begin{bmatrix} \mathbf{U}_{11} & \mathbf{U}_{12} \\ \mathbf{U}_{21} & \mathbf{U}_{22} \end{bmatrix} \begin{bmatrix} \mathbf{y}_1 \\ \mathbf{y}_2 \end{bmatrix} = \begin{bmatrix} \mathbf{c}_1 \\ \mathbf{c}_2 \end{bmatrix},$$

where $\mathbf{U}_{11} = \{u_{ij}, i, j = 1 \dots k\}$ is a regular upper triangular matrix, $\mathbf{U}_{12}$ is of size $k \times (n - k)$, $\mathbf{U}_{21}$ is a zero matrix of size $(m - k) \times k$, $\mathbf{U}_{22}$ is a zero matrix of size $(m - k) \times (n - k)$, and $\mathbf{c}_1$ is a k -vector. Since $\mathbf{U}_{21}$ and $\mathbf{U}_{22}$ are zero matrices, the system opens up into the form

$$\begin{cases} \mathbf{U}_{11} \mathbf{y}_1 + \mathbf{U}_{12} \mathbf{y}_2 = \mathbf{c}_1, \\ \mathbf{0} = \mathbf{c}_2. \end{cases}$$

From this one sees that, insofar as $k < m$ (this is always the case when $m > n$), the system has a both necessary and sufficient

$$\text{solvability condition: } \mathbf{0} = \mathbf{c}_2.$$

If $k < n$ (this is always the case when $m < n$), the solution is not unique even under this condition, since in the solution one can set $\mathbf{y}_1 = \mathbf{U}_{11}^{-1}(\mathbf{c}_1 - \mathbf{U}_{12}\mathbf{y}_2)$, whatever $\mathbf{y}_2 \in \mathbb{R}^{n-k}$ may be. One thus deduces in particular that a system $\mathbf{Ax} = \mathbf{b}$ of size $m \times n$, $m \neq n$, is always singular: either there is no solution for every $\mathbf{b} \in \mathbb{R}^m$, or the solution is not unique. The former situation prevails always when $m > n$, the latter always when $m < n$.

EXAMPLE 2 Determine the solvability conditions, general solution and rank of the coefficient matrix of the singular system

$$\begin{cases} x_1 + x_2 + x_3 = b_1 \\ 2x_1 + x_2 = b_2 \\ -x_1 + x_3 = b_3 \\ +x_2 + 2x_3 = b_4 \\ x_1 - x_2 - 3x_3 = b_5 \end{cases}$$

Solution Here $m = 5$ and $n = 3$. In Gaussian elimination one gets by without pivoting:

$$\begin{aligned} \begin{bmatrix} 1 & 1 & 1 \\ 2 & 1 & 0 \\ -1 & 0 & 1 \\ 0 & 1 & 2 \\ 1 & -1 & -3 \end{bmatrix} \begin{bmatrix} b_1 \\ b_2 \\ b_3 \\ b_4 \\ b_5 \end{bmatrix} &\longmapsto \begin{bmatrix} 1 & 1 & 1 \\ 0 & -1 & -2 \\ 0 & 1 & 2 \\ 0 & 1 & 2 \\ 0 & -2 & -4 \end{bmatrix} \begin{bmatrix} b_1 \\ -2b_1 + b_2 \\ b_1 + b_3 \\ b_4 \\ -b_1 + b_5 \end{bmatrix} \\ &\longmapsto \begin{bmatrix} 1 & 1 & 1 \\ 0 & -1 & -2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} b_1 \\ -2b_1 + b_2 \\ -b_1 + b_2 + b_3 \\ -2b_1 + b_2 + b_4 \\ 3b_1 - 2b_2 + b_5 \end{bmatrix} \end{aligned}$$

By the resulting singular standard form, the rank of the coefficient matrix is $k = 2$. The solvability conditions can be read off from the last three rows of the system:

$$\begin{cases} -b_1 + b_2 + b_3 = 0 \\ -2b_1 + b_2 + b_4 = 0 \\ 3b_1 - 2b_2 + b_5 = 0 \end{cases}$$

that is,

$$\begin{bmatrix} 1 & 0 & 0 & -2 & 3 \\ 0 & 1 & 0 & 1 & -2 \\ 0 & 0 & 1 & 1 & -1 \end{bmatrix} \begin{bmatrix} b_5 \\ b_4 \\ b_3 \\ b_2 \\ b_1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.$$

Since the matrix on the left here is a standard-form upper triangular matrix (if it were not, one would first apply Gaussian elimination), one sees that b_1 and b_2 can be chosen freely, after which b_3 , b_4 and b_5 are determined:

$$\begin{cases} b_3 = b_1 - b_2, \\ b_4 = 2b_1 - b_2, \\ b_5 = -3b_1 + 2b_2. \end{cases}$$

With these conditions in force, the general solution is obtained by back-substitution:

$$\begin{cases} x_1 = -b_1 + b_2 + t, \\ x_2 = 2b_1 - b_2 - 2t, \\ x_3 = t \end{cases} \quad (t \in \mathbb{R}).$$

The solution is not unique, because the rank of the coefficient matrix is not full: $r(\mathbf{A}) = 2 < \min\{m, n\} = 3$. $\square$

Proof of Theorem X.2.6

By the product decomposition ($\star$),

$$\mathbf{Ax} = \mathbf{b} \Leftrightarrow \mathbf{Uy} = \mathbf{Cb}, \quad \mathbf{y} = \mathbf{Wx}, \quad \mathbf{C} = \mathbf{L}^{-1}\mathbf{V}.$$

If $\mathbf{A}$ is singular, then at least the n th of the rows of $\mathbf{U}$ is a zero row, whereupon the n th equation of the system $\mathbf{Uy} = \mathbf{Cb}$ takes the form $0 = [\mathbf{Cb}]_n$. If $\mathbf{b} = \mathbf{0}$, then the system has a non-unique solution, where y_n is freely choosable. Each solution $\mathbf{y} \neq \mathbf{0}$ ($y_n \neq 0$) corresponds to a solution $\mathbf{x} = \mathbf{W}^T \mathbf{y} \neq \mathbf{0}$ of the system $\mathbf{Ax} = \mathbf{0}$. If, on the other hand, $\mathbf{b} \in \mathbb{R}^n$ is chosen so that $[\mathbf{Cb}]_n \neq 0$, then the mentioned equation is not satisfied, whereupon $\mathbf{Uy} \neq \mathbf{Cb} \forall \mathbf{y} \in \mathbb{R}^n$. Then also $\mathbf{Ax} \neq \mathbf{b}$, $\mathbf{x} = \mathbf{W}^T \mathbf{y}$, $\mathbf{y} \in \mathbb{R}^n$ (since $\mathbf{Uy} = \mathbf{Cb} \Leftrightarrow \mathbf{Ax} = \mathbf{b}$). Thus $\mathbf{Ax} \neq \mathbf{b} \forall \mathbf{x} \in \mathbb{R}^n$. We have deduced:

$$\mathbf{A} \text{ singular} \Rightarrow \begin{cases} \text{for some } \mathbf{x} \neq \mathbf{0} \text{ we have } \mathbf{Ax} = \mathbf{0}, \\ \text{for some } \mathbf{b} \in \mathbb{R}^n \text{ we have } \mathbf{Ax} \neq \mathbf{b} \forall \mathbf{x} \in \mathbb{R}^n. \end{cases}$$

These claims are equivalent to the sub-claims $E1 \Rightarrow E0$ and $E2 \Rightarrow E0$ of Theorem X.2.6, which are thus true. Since $E0 \Rightarrow E1$ and $E0 \Rightarrow E2$ were already true by Proposition X.2.4, the theorem is proved. $\square$

EXERCISES

- Show that the pivoting operations (interchanges of rows and columns) performed in the course of Gaussian elimination with pivoting can be performed before the eliminations without changing the end result.
 - Let $\mathbf{A}$ be a matrix of size $n \times n$ and let $\mathbf{A}_k$ be the matrix of size $k \times k$ consisting of the entries a_{ij} , $i, j = 1 \dots k$ of $\mathbf{A}$. Show that the system $\mathbf{Ax} = \mathbf{b}$ is solvable by Gaussian elimination without pivoting operations exactly when $\mathbf{A}_k$ is a regular matrix for every $k = 1 \dots n$.
- Transform to singular standard form by Gaussian elimination with pivoting, determine the solutions or establish unsolvability:

$$\text{a) } \begin{bmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 8 \\ 4 \\ 3 \end{bmatrix} \quad \text{b) } \begin{bmatrix} 0 & 1 & 1 \\ 1 & 2 & 1 \\ 2 & 4 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 5 \\ 6 \\ 12 \end{bmatrix}$$

$$\text{c) } \begin{bmatrix} 0 & 1 & -1 & 1 \\ 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & -1 \\ 1 & 2 & -1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} -1 \\ 1 \\ 1 \\ 0 \end{bmatrix}$$

$$\text{d) } \begin{bmatrix} 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 2 \\ 1 & 1 & 2 & 1 & 1 & 2 \\ 1 & 0 & 1 & 0 & 1 & 3 \\ 0 & 0 & 0 & 0 & 1 & 3 \\ 1 & 1 & 2 & 0 & 2 & 5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \\ x_6 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 2 \\ 0 \\ 0 \\ 1 \end{bmatrix}$$

- Determine the rank of $\mathbf{A}$, all solutions of the system $\mathbf{Ax} = \mathbf{0}$, and the product decomposition $\mathbf{A} = \mathbf{V}^T \mathbf{L} \mathbf{U} \mathbf{W}^T$, when $\mathbf{A} =$

$$\text{a) } \begin{bmatrix} 1 & 2 & -1 \\ -2 & -4 & 2 \\ -1 & -2 & 1 \end{bmatrix} \quad \text{b) } \begin{bmatrix} 1 & 1 & 1 \\ 2 & 2 & 1 \\ 3 & 3 & 2 \end{bmatrix} \quad \text{c) } \begin{bmatrix} 2 & 2 & 1 \\ 3 & 3 & 2 \\ 4 & 4 & 3 \end{bmatrix}$$

$$\text{d) } \begin{bmatrix} 2 & 1 & 1 & 1 \\ 4 & 2 & 2 & 3 \\ 0 & 0 & 0 & 1 \\ 1 & 2 & 2 & 0 \end{bmatrix} \quad \text{e) } \begin{bmatrix} 1 & 1 & 1 & 1 \\ 2 & 1 & 2 & 1 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 1 \end{bmatrix} \quad \text{f) } \begin{bmatrix} 1 & -2 & 1 & 0 \\ -1 & 2 & -1 & 1 \\ 2 & -4 & 2 & 4 \\ 1 & 1 & -1 & -1 \end{bmatrix}$$

4. Bring the following systems to singular standard form by Gaussian elimination with pivoting. Determine the solvability conditions and the general solution, and further the product decomposition of the coefficient matrix $\mathbf{A} = \mathbf{V}^T \mathbf{L} \mathbf{U} \mathbf{W}^T$.

$$\text{a) } \begin{bmatrix} 1 & -2 & 2 \\ -1 & 2 & 1 \\ 5 & -10 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} \quad \text{b) } \begin{bmatrix} 0 & 1 & -4 & -3 \\ 1 & 0 & 2 & 1 \\ 3 & 2 & -2 & -3 \\ 2 & 1 & 0 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \\ b_4 \end{bmatrix}$$

$$\text{c) } \begin{bmatrix} 1 & 2 & -1 & 2 & 2 \\ 2 & 4 & -2 & 0 & 4 \\ -1 & -2 & 1 & -2 & -2 \\ -1 & -2 & 0 & -3 & -3 \\ -2 & -4 & 4 & -2 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \\ b_4 \\ b_5 \end{bmatrix}$$

5. Transform to singular standard form and solve, if possible:

$$\text{a) } \begin{bmatrix} 1 & 0 & 4 \\ -1 & 3 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 3 \end{bmatrix} \quad \text{b) } \begin{bmatrix} 2 & 1 \\ 4 & 6 \\ 3 & 5 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 1 \\ 4 \\ -2 \end{bmatrix}$$

$$\text{c) } \begin{bmatrix} 1 & 2 & 2 \\ 2 & 4 & 6 \\ 3 & 6 & 8 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 2 \\ 2 \\ 4 \\ 1 \end{bmatrix} \quad \text{d) } \begin{bmatrix} 2 & 1 & 4 \\ 6 & 3 & 8 \\ 4 & 2 & 7 \\ 2 & 1 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 4 \\ 1 \\ 1 \end{bmatrix}$$

$$\text{e) } \begin{bmatrix} 1 & 4 & 7 & -3 \\ -2 & 3 & -6 & 1 \\ 0 & 11 & 8 & -5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 1 \\ 3 \\ 5 \end{bmatrix}$$

$$6. \text{ Let } \text{a) } \mathbf{A} = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \\ 3 & 4 & 1 & 2 \end{bmatrix}, \quad \text{b) } \mathbf{A} = \begin{bmatrix} 1 & 3 & 5 & -2 \\ 3 & -2 & 7 & 5 \\ 2 & -5 & 2 & 7 \end{bmatrix}.$$

Determine by Gaussian elimination the solvability conditions and general solution for the systems $\mathbf{A}\mathbf{x} = \mathbf{b}$ ($\mathbf{b} \in \mathbb{R}^3$) and $\mathbf{A}^T \mathbf{x} = \mathbf{b}$ ($\mathbf{b} \in \mathbb{R}^4$).

X.5 Determinant

Let $\mathbf{a}_1, \dots, \mathbf{a}_n \in \mathbb{R}^n$ be n column vectors of size n . We set

DEFINITION X.5.1 A real-valued function $V(\mathbf{a}_1, \dots, \mathbf{a}_n)$, $\mathbf{a}_i \in \mathbb{R}^n$, is a (normalized) *determinant function* if the following hold:

- (i) V is linear with respect to each vector $\mathbf{a}_i$.
- (ii) If the ordered index set $(i_1, \dots, i_n)$ is obtained from the set $(1, 2, \dots, n)$ by interchanging two elements, then

$$V(\mathbf{a}_{i_1}, \dots, \mathbf{a}_{i_n}) = -V(\mathbf{a}_1, \dots, \mathbf{a}_n).$$

- (iii) If $[\mathbf{e}_i]_j = \delta_{ij}$, then $V(\mathbf{e}_1, \dots, \mathbf{e}_n) = 1$.

Explanations:

- (i) Linearity means that for every $k \in \{1, \dots, n\}$ and $\forall \lambda, \mu \in \mathbb{R}$ it holds that

$$\begin{aligned} V(\mathbf{a}_1, \dots, \lambda \mathbf{a}_k + \mu \mathbf{b}_k, \dots, \mathbf{a}_n) \\ = \lambda V(\mathbf{a}_1, \dots, \mathbf{a}_k, \dots, \mathbf{a}_n) + \mu V(\mathbf{a}_1, \dots, \mathbf{b}_k, \dots, \mathbf{a}_n). \end{aligned}$$

- (ii) This *interchange rule* means that the value of the determinant changes to its opposite when two vectors are interchanged in the ordered set $(\mathbf{a}_1, \dots, \mathbf{a}_n)$ (the pairwise interchange familiar from Section X.2). If the two interchanged vectors are the same, the value of the determinant naturally does not change, so from the interchange rule there follows the *zero rule*

$$\mathbf{a}_i = \mathbf{a}_j, \quad i \neq j \quad \Rightarrow \quad V(\mathbf{a}_1, \dots, \mathbf{a}_n) = 0.$$

If more generally $p = (i_1, \dots, i_n)$ is a permutation of the index set $(1, \dots, n)$, then p is either even or odd, i.e. p is obtained by either an even or an odd number of pairwise interchanges (cf. Section X.2). The interchange rule thus generalizes to the rule: In an even permutation the value of the determinant is preserved, in an odd permutation it changes to its opposite.

- (iii) This is the normalization condition. Without this condition the determinant function could be multiplied by an arbitrary constant, in which case the following claim would not be true.

PROPOSITION X.5.2 The determinant function is a function, i.e. $V(\mathbf{a}_1, \dots, \mathbf{a}_n) \in \mathbb{R}$ is uniquely determined when $\mathbf{a}_1, \dots, \mathbf{a}_n \in \mathbb{R}^n$.

Proof We compute the value of the determinant function using the given rules (i)–(iii). Denote $[\mathbf{a}_j]_i = a_{ij}$, so that

$$\mathbf{a}_j = \sum_{i=1}^n a_{ij} \mathbf{e}_i$$

and thus

$$V(\mathbf{a}_1, \dots, \mathbf{a}_n) = V\left(\sum_{i=1}^n a_{i1} \mathbf{e}_i, \dots, \sum_{i=1}^n a_{in} \mathbf{e}_i\right).$$

By repeatedly using the linearity property (i) we see that this expands into the sum

$$V(\mathbf{a}_1, \dots, \mathbf{a}_n) = \sum_{i_1=1}^n \cdots \sum_{i_n=1}^n a_{i_1 1} \cdots a_{i_n n} V(\mathbf{e}_{i_1}, \dots, \mathbf{e}_{i_n}).$$

Here there are n^n terms, but by the zero rule all those terms in which the same index i_k appears twice can be dropped from the sum. Accordingly,

$$V(\mathbf{a}_1, \dots, \mathbf{a}_n) = \sum_p a_{i_1 1} \cdots a_{i_n n} V(\mathbf{e}_{i_1}, \dots, \mathbf{e}_{i_n}),$$

where the summation runs over all the distinct permutations $p = (i_1, \dots, i_n)$ of the set $(1, \dots, n)$ ($n!$ of them). From rules (ii) and (iii) it further follows that $V(\mathbf{e}_{i_1}, \dots, \mathbf{e}_{i_n}) = \sigma_p = \pm 1$ depending on whether the permutation p is even or odd. Thus

$$\boxed{V(\mathbf{a}_1, \dots, \mathbf{a}_n) = \sum_p \sigma_p a_{i_1 1} a_{i_2 2} \cdots a_{i_n n}} \quad (1)$$

and accordingly $V(\mathbf{a}_1, \dots, \mathbf{a}_n)$ is uniquely determined. $\square$

The properties (i)–(iii) of the determinant function can conversely be deduced from the computational formula (1), so this formula is also valid as a definition of the determinant function. A somewhat clearer form of the formula is obtained by writing (cf. Section X.2)

$$\mathbf{I}_p = [\mathbf{e}_{i_1} \cdots \mathbf{e}_{i_n}], \quad \Lambda_p = \{(i, j) \mid [\mathbf{I}_p]_{ij} = 1\}, \quad p = (i_1, \dots, i_n).$$

Then one sees that formula (1) is the same as

$$V(\mathbf{a}_1, \dots, \mathbf{a}_n) = \sum_p \sigma_p \prod_{(i,j) \in \Lambda_p} a_{ij}. \quad (2)$$

When actually computing the value of a determinant, formulas (1)–(2) are treacherous, since using them requires as many as $(n-1)n!$ multiplications alone. — The formulas are therefore generally used as algorithms only in the cases $n = 2$ and $n = 3$. Formulas (1)–(2) do, however, have more general theoretical use, as will be seen later.

The determinant of a matrix

The *determinant of a matrix* is defined simply by thinking of the vectors $\mathbf{a}_j$ in the determinant function $V(\mathbf{a}_1, \dots, \mathbf{a}_n)$ as the columns of the (square) matrix $\mathbf{A}$. The determinant is denoted $\det \mathbf{A}$ (sometimes $|\mathbf{A}|$), and so the definition is

$$\det \mathbf{A} = V(\mathbf{a}_1, \dots, \mathbf{a}_n), \quad \mathbf{A} = [\mathbf{a}_1 \dots \mathbf{a}_n].$$

Accordingly, in formula (2) $\sigma_p = \det \mathbf{I}_p$.

The determinant is thus defined only for square matrices. If $\mathbf{A}$ is given as an array, the determinant is denoted by vertical bars delimiting the array:

$$\det \mathbf{A} = \begin{vmatrix} a_{11} & \dots & a_{1n} \\ \vdots & & \vdots \\ a_{n1} & \dots & a_{nn} \end{vmatrix}, \quad \mathbf{A} = (a_{ij}).$$

By the normalization rule (iii), the determinant of the identity matrix is

$$\det \mathbf{I} = 1.$$

From formula (1) the rule

$$\det(\lambda \mathbf{A}) = \lambda^n \det \mathbf{A}, \quad \lambda \in \mathbb{R}.$$

also follows easily. The following two properties of the determinant can also be derived from the definition, but these are less obvious. — They are the most central claims of determinant theory.

THEOREM X.5.3 For the determinant of a matrix,

$$\det \mathbf{AB} = \det \mathbf{A} \det \mathbf{B}, \quad \det \mathbf{A}^T = \det \mathbf{A}.$$

Proof Let $\mathbf{A} = [\mathbf{a}_1 \dots \mathbf{a}_n]$ and $\mathbf{B} = [\mathbf{b}_1 \dots \mathbf{b}_n] = (b_{ij})$. Using the expansion formulas of Section X.1 we can write

$$\mathbf{AB} = [\mathbf{Ab}_1 \dots \mathbf{Ab}_n] = \left[\sum_{i_1=1}^n b_{i_1 1} \mathbf{a}_{i_1} \dots \sum_{i_n=1}^n b_{i_n n} \mathbf{a}_{i_n} \right],$$

whence, by the linearity rule (i) of the determinant, it follows that

$$\det \mathbf{AB} = \sum_{i_1=1}^n \dots \sum_{i_n=1}^n b_{i_1 1} \dots b_{i_n n} \det [\mathbf{a}_{i_1} \dots \mathbf{a}_{i_n}].$$

From this sum, by the zero rule, all terms in which the same index i_k appears twice can again be dropped. In the remaining terms $p = (i_1, \dots, i_n)$ is a permutation of the set $(1, \dots, n)$, so that by the interchange rule (ii)

$$\det [\mathbf{a}_{i_1} \dots \mathbf{a}_{i_n}] = \sigma_p \det [\mathbf{a}_1 \dots \mathbf{a}_n] = \sigma_p \det \mathbf{A}, \quad p = (i_1, \dots, i_n).$$

Here σ_p is defined as in formula (1), so

$$\det \mathbf{AB} = \det \mathbf{A} \sum_p \sigma_p b_{i_1 1} \dots b_{i_n n} = \det \mathbf{A} \det \mathbf{B}.$$

To prove the transpose rule for determinants, we first note that it holds for all permutation matrices $\mathbf{I}_p$. Indeed, since $\mathbf{I}_p^T \mathbf{I}_p = \mathbf{I}$, by the already proven product rule for determinants we have

$$\det \mathbf{I}_p^T \det \mathbf{I}_p = 1.$$

Since here $\det \mathbf{I}_p = \pm 1$ and $\det \mathbf{I}_p^T = \det \mathbf{I}_q = \pm 1$, we conclude that necessarily $\det \mathbf{I}_p^T = \det \mathbf{I}_p$. Taking this into account and writing $\mathbf{I}_p^T = \mathbf{I}_q$, by the computational formula (2) we have

$$\det \mathbf{A}^T = \sum_p \det \mathbf{I}_p \prod_{(i,j) \in \Lambda_p} a_{ji} = \sum_q \det \mathbf{I}_q \prod_{(i,j) \in \Lambda_q} a_{ij} = \det \mathbf{A}. \quad \square$$

From the rule $\det \mathbf{A}^T = \det \mathbf{A}$ it can be deduced that the interchange rule and the zero rule for determinants also hold in the form: $\det \mathbf{A}$ changes to its opposite if two rows of $\mathbf{A}$ are interchanged, and $\det \mathbf{A} = 0$ if two rows of $\mathbf{A}$ are the same.

If $\mathbf{A}$ is a nonsingular matrix, then applying the product rule for determinants to the product $\mathbf{A}^{-1} \mathbf{A} = \mathbf{I}$ shows that

$$\det \mathbf{A}^{-1} = (\det \mathbf{A})^{-1}.$$

We also see that

$$\mathbf{A} \text{ nonsingular} \Rightarrow \det \mathbf{A} \neq 0,$$

which is logically the same as

$$\det \mathbf{A} = 0 \Rightarrow \mathbf{A} \text{ singular}.$$

This implication holds in the converse direction as well, and herein lies the most important result of determinant theory from the standpoint of applications. The result itself is a direct consequence of the product decompositions of the previous section, the product rule for determinants, and the rule for the determinant of a triangular matrix presented below (Proposition X.5.6).

THEOREM X.5.4 (Determinant criterion) For a square matrix,

$$\mathbf{A} \text{ singular} \Leftrightarrow \det \mathbf{A} = 0.$$

Computing the determinant

The computational formula (1) is generally used only in the cases $n = 2, 3$:

$$\begin{aligned}
 n = 2 : \quad & \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} = a_{11}a_{22} - a_{21}a_{12} . \\
 n = 3 : \quad & \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} = a_{11}a_{22}a_{33} + a_{31}a_{12}a_{23} + a_{21}a_{32}a_{13} \\
 & \quad - a_{31}a_{22}a_{13} - a_{11}a_{32}a_{23} - a_{21}a_{12}a_{33} .
 \end{aligned}$$

These can also be easily derived directly from Definition X.5.1. For example:

$$\begin{aligned}
 \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} &= V \left(\begin{bmatrix} a_{11} \\ a_{21} \end{bmatrix}, \begin{bmatrix} a_{12} \\ a_{22} \end{bmatrix} \right) \\
 &= V \left(a_{11} \begin{bmatrix} 1 \\ 0 \end{bmatrix} + a_{21} \begin{bmatrix} 0 \\ 1 \end{bmatrix}, a_{12} \begin{bmatrix} 1 \\ 0 \end{bmatrix} + a_{22} \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right) \\
 &= a_{11}a_{12} \begin{vmatrix} 1 & 1 \\ 0 & 0 \end{vmatrix} + a_{11}a_{22} \begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix} + a_{21}a_{12} \begin{vmatrix} 0 & 1 \\ 1 & 0 \end{vmatrix} + a_{21}a_{22} \begin{vmatrix} 0 & 0 \\ 1 & 1 \end{vmatrix} \\
 &= a_{11}a_{12} \cdot 0 + a_{11}a_{22} \cdot 1 + a_{21}a_{12} \cdot (-1) + a_{21}a_{22} \cdot 0 = a_{11}a_{22} - a_{21}a_{12} .
 \end{aligned}$$

In the case $n = 3$ the computational formula can be remembered by grouping the summands according to formula (2) as follows (the so-called *Sarrus's rule*):

$$\begin{aligned}
 \text{sign } + \quad & \begin{vmatrix} \bullet & \cdot & \cdot \\ \cdot & \bullet & \cdot \\ \cdot & \cdot & \bullet \end{vmatrix} \quad \begin{vmatrix} \cdot & \bullet & \cdot \\ \cdot & \cdot & \bullet \\ \bullet & \cdot & \cdot \end{vmatrix} \quad \begin{vmatrix} \cdot & \cdot & \bullet \\ \bullet & \cdot & \cdot \\ \cdot & \bullet & \cdot \end{vmatrix} \\
 \text{sign } - \quad & \begin{vmatrix} \cdot & \cdot & \bullet \\ \cdot & \bullet & \cdot \\ \bullet & \cdot & \cdot \end{vmatrix} \quad \begin{vmatrix} \bullet & \cdot & \cdot \\ \cdot & \cdot & \bullet \\ \cdot & \bullet & \cdot \end{vmatrix} \quad \begin{vmatrix} \cdot & \bullet & \cdot \\ \bullet & \cdot & \cdot \\ \cdot & \cdot & \bullet \end{vmatrix}
 \end{aligned}$$

EXAMPLE 1

$$\begin{vmatrix} 1 & 1 & 3 \\ 2 & 0 & 2 \\ 3 & -1 & 1 \end{vmatrix} = 1 \cdot 0 \cdot 1 + 1 \cdot 2 \cdot 3 + 2 \cdot (-1) \cdot 3 - 3 \cdot 0 \cdot 3 - 1 \cdot (-1) \cdot 2 - 2 \cdot 1 \cdot 1 = 0.$$

As a matrix, this is thus singular. $\square$

Sarrus's rule is limited to determinants of size 3×3 . For expanding somewhat larger determinants by hand, the following so-called *minor expansion rule* is considerably more convenient than formulas (1)–(2). The rule is universally valid, and it is also usable as a definition of the determinant function (see Exercise 10).

THEOREM X.5.5 (Minor expansion rule) If $\mathbf{A}$ is a square matrix of size $n \times n$, then for every $k \in \{1, \dots, n\}$

$$\begin{aligned} \det \mathbf{A} &= \sum_{i=1}^n (-1)^{k+i} a_{ik} \det \mathbf{A}^{(i,k)} \\ &= \sum_{j=1}^n (-1)^{k+j} a_{kj} \det \mathbf{A}^{(k,j)}, \end{aligned}$$

where $\mathbf{A}^{(i,j)}$ is the $(n-1) \times (n-1)$ matrix obtained from $\mathbf{A}$ by deleting the i th row and the j th column.

Proof Consider the first of the claimed expansion rules, with $k = 1$. By the linearity rule (i) of the determinant we can write

$$\det \mathbf{A} = \sum_{i=1}^n a_{i1} \det \mathbf{A}_i,$$

where $\mathbf{A}_i$ is obtained from the matrix $\mathbf{A}$ by replacing the first column with the unit vector $\mathbf{e}_i$. In the matrix $\mathbf{A}_i$ we first interchange the first and the i th row, and then permute rows no. $2 \dots i$ so that the i th row moves to the second row without changing the order of the other rows. The latter operation requires $i-2$ pairwise row interchanges, so the interchanges total $i-1$. If the matrix obtained after the interchanges is denoted $\tilde{\mathbf{A}}_i$, then $\det \mathbf{A}_i = (-1)^{i-1} \det \tilde{\mathbf{A}}_i$. As a result of the interchanges, the first column of $\tilde{\mathbf{A}}_i$ is $\mathbf{e}_1$ and $\tilde{\mathbf{A}}_i^{(1,1)} = \mathbf{A}^{(i,1)}$. From the computational formula (1) for the determinant one then sees that $\det \tilde{\mathbf{A}}_i = \det \tilde{\mathbf{A}}_i^{(1,1)} = \det \mathbf{A}^{(i,1)}$. Combining the deductions, it follows that

$$\det \mathbf{A} = \sum_{i=1}^n (-1)^{i-1} a_{i1} \det \mathbf{A}^{(i,1)} = \sum_{i=1}^n (-1)^{i+1} a_{i1} \det \mathbf{A}^{(i,1)}.$$

The first of the claimed expansion rules is thus proven in the case $k = 1$. The other claimed rules reduce to this case by means of the interchange and transpose rules. $\square$

The determinants $\det \mathbf{A}^{(i,j)}$ are called the *minors* of the determinant $\det \mathbf{A}$ — hence the name of the expansion rules of Theorem X.5.5.

EXAMPLE 2 Choosing $k = 1$ and expanding the determinant along the first column, we obtain

$$\begin{vmatrix} 1 & 2 & -2 & 1 \\ 2 & -1 & 1 & 1 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 2 & 2 \end{vmatrix} = 1 \cdot \begin{vmatrix} -1 & 1 & 1 \\ 1 & 0 & 1 \\ 0 & 2 & 2 \end{vmatrix} - 2 \cdot \begin{vmatrix} 2 & -2 & 1 \\ 1 & 0 & 1 \\ 0 & 2 & 2 \end{vmatrix} - 1 \cdot \begin{vmatrix} 2 & -2 & 1 \\ -1 & 1 & 1 \\ 1 & 0 & 1 \end{vmatrix}.$$

The minors can be expanded by Sarrus's rule or by the minor expansion rule. We apply the latter, expanding the first two minors along the first column ($k = 1$) and the last one along the third row ($k = 3$):

$$\begin{vmatrix} -1 & 1 & 1 \\ 1 & 0 & 1 \\ 0 & 2 & 2 \end{vmatrix} = (-1) \cdot \begin{vmatrix} 0 & 1 \\ 2 & 2 \end{vmatrix} - 1 \cdot \begin{vmatrix} 1 & 1 \\ 2 & 2 \end{vmatrix} = (-1) \cdot (-2) - 1 \cdot 0 = 2,$$

$$\begin{vmatrix} 2 & -2 & 1 \\ 1 & 0 & 1 \\ 0 & 2 & 2 \end{vmatrix} = 2 \cdot \begin{vmatrix} 0 & 1 \\ 2 & 2 \end{vmatrix} - 1 \cdot \begin{vmatrix} -2 & 1 \\ 2 & 2 \end{vmatrix} = 2 \cdot (-2) - 1 \cdot (-6) = 2,$$

$$\begin{vmatrix} 2 & -2 & 1 \\ -1 & 1 & 1 \\ 1 & 0 & 1 \end{vmatrix} = 1 \cdot \begin{vmatrix} -2 & 1 \\ 1 & 1 \end{vmatrix} + 1 \cdot \begin{vmatrix} 2 & -2 \\ -1 & 1 \end{vmatrix} = 1 \cdot (-3) + 1 \cdot 0 = -3.$$

The value of the determinant comes out to $1 \cdot 2 - 2 \cdot 2 - 1 \cdot (-3) = 1$. $\square$

By the minor expansion rule, a general determinant of size $n \times n$ can ultimately be reduced to minors of size 2×2 . However, the required number of arithmetic operations is still of order $W \sim n!$ (see Exercise 9), so for large values of n this method too is useless — So how is it even possible to compute, say, a determinant of size $n = 100$? The answer is provided by the following result, which is easily proven using the minor expansion rule.

PROPOSITION X.5.6 The determinant of a triangular matrix $\mathbf{A} = \{a_{ij}\}$ is the product of the diagonal entries of the matrix:

$$\det \mathbf{A} = \prod_{i=1}^n a_{ii}.$$

Proof Choosing $k = 1$ in the minor expansion rule and expanding the determinant along either the first row (lower triangular) or the first column (upper triangular), a_{11} is the only nonzero entry of that row/column, so by Theorem X.5.5

$$\det \mathbf{A} = a_{11} \det \mathbf{A}^{(1,1)}.$$

Here $\mathbf{A}^{(1,1)}$ is again a triangular matrix, so the same rule applies over and over and leads to the claimed result. $\square$

Based on Proposition X.5.6, the value of the determinant can be computed — how else — by Gaussian elimination. Indeed, if the algorithm goes through without pivoting operations (row/column interchanges), it leads to the LU -decomposition

$$\mathbf{A} = \mathbf{L}\mathbf{U},$$

where $[\mathbf{L}]_{kk} = 1$ and $[\mathbf{U}]_{kk} = a_{kk}^{(k-1)}$, $k = 1 \dots n$ (cf. the previous section). Then, by the product rule for determinants and Proposition X.5.6,

$$\det \mathbf{A} = \det \mathbf{L} \det \mathbf{U} = \prod_{k=1}^n a_{kk}^{(k-1)}.$$

The value of the determinant is thus simply the product of the pivots in the elimination phase of Gaussian elimination (!). If pivoting operations are carried out in the elimination phase, the value of the determinant becomes

$$\det \mathbf{A} = (-1)^m \prod_{k=1}^n [\mathbf{U}]_{kk},$$

where m is the total number of row/column interchanges performed.

Using Gaussian elimination, the workload of computing the determinant thus becomes $W \sim \frac{1}{3}n^3$ (multiplications and additions) — that is, for large values of n this is a considerably more efficient method than any of the alternatives presented above. In purely numerical computations, and for large n , the determinant is in fact of little practical significance — after all, Gaussian elimination settles the question of a matrix's nonsingularity anyway. The concept of the determinant is, however, useful in various theoretical considerations in linear algebra, and the determinant also has uses in symbolic computation. A symbolic-computation problem might be, for example, one in which the matrix under consideration depends on some parameter (variable) s , i.e. $\mathbf{A} = (a_{ij}(s)) = \mathbf{A}(s)$, and one wishes, say, to determine for which values of s the matrix $\mathbf{A}(s)$ is singular. If $\mathbf{A}$ is not very large in size, then the determinant condition

$$\det \mathbf{A}(s) = 0$$

is a natural starting point for the solution and (especially in hand computation) often used.

Cramer's rule

Among the other uses of the determinant, the following two computational rules are worth mentioning (without proof, see Exercise 11).

THEOREM X.5.7 (Cramer's* rule) If $\mathbf{A}$ is a nonsingular square matrix of size $n \times n$, then the solution of the system $\mathbf{Ax} = \mathbf{b}$ is

$$\mathbf{x} = (x_i), \quad x_i = \frac{\det \mathbf{A}^{(i)}}{\det \mathbf{A}}, \quad i = 1 \dots n,$$

where $\mathbf{A}^{(i)}$ is obtained from $\mathbf{A}$ by replacing the i th column with $\mathbf{b}$.

THEOREM X.5.8 The inverse of a nonsingular matrix $\mathbf{A}$ is

$$\mathbf{A}^{-1} = \mathbf{B}^T, \quad [\mathbf{B}]_{ij} = (-1)^{i+j} \frac{\det \mathbf{A}^{(ij)}}{\det \mathbf{A}}.$$

Cramer's rule still has some use in hand computation for solving small systems of equations. Especially if the entries of the coefficient matrix are integers, the rule can be used to minimize the (laborious, in hand computation) divisions. In symbolic (hand) computation Cramer's rule is likewise useful in the same way as the determinant generally. — In numerical matrix computation, on the other hand, Cramer's rule has the dubious honor of being the 'world's worst' algorithm for solving a linear system of equations.

EXERCISES

1. Show that the determinant of an orthogonal matrix has only two possible values: Either $\det \mathbf{A} = 1$ or $\det \mathbf{A} = -1$.
2. Compute using Sarrus's rule:

$$\begin{array}{ccc} \text{a)} & \left| \begin{array}{ccc} 2 & 3 & -1 \\ -1 & 2 & 0 \\ 1 & 4 & 3 \end{array} \right| & \text{b)} & \left| \begin{array}{ccc} 0 & -2 & -3 \\ 1 & 0 & -2 \\ 3 & 4 & 0 \end{array} \right| & \text{c)} & \left| \begin{array}{ccc} -3 & 7 & 2 \\ -5 & 4 & 0 \\ 9 & -1 & -6 \end{array} \right| \end{array}$$

*The Swiss mathematician **Gabriel Cramer** (1704–1752) was one of the pioneers of determinant theory with his work "Introduction à l'analyse des lignes courbes algébriques" (1750).

3. Let $\alpha, \beta, \gamma \in \mathbb{R}$. Compute

$$\text{a) } \begin{vmatrix} 1 & \alpha & \alpha^2 \\ 1 & \beta & \beta^2 \\ 1 & \gamma & \gamma^2 \end{vmatrix} \quad \text{b) } \begin{vmatrix} \alpha + \beta & \alpha + 2\beta & \alpha + 3\beta \\ \alpha + 3\beta & \alpha + \beta & \alpha + 2\beta \\ \alpha + 2\beta & \alpha + 3\beta & \alpha + \beta \end{vmatrix} \quad \text{c) } \begin{vmatrix} 2 & \alpha & \alpha^2 \\ 1 & \alpha^2 & \alpha \\ \alpha & \alpha^3 & 1 \end{vmatrix}$$

4. Compute a) by minor expansion, b) and c) by Gaussian elimination, d) by both:

$$\text{a) } \begin{vmatrix} 1 & -1 & 0 & 2 \\ 2 & 1 & 0 & 0 \\ 1 & 1 & 2 & 2 \\ 0 & 0 & 1 & 1 \end{vmatrix} \quad \text{b) } \begin{vmatrix} 5 & 4 & 2 & 1 \\ 2 & 3 & 1 & -2 \\ -5 & -7 & -3 & 9 \\ 1 & -2 & -1 & 4 \end{vmatrix}$$

$$\text{c) } \begin{vmatrix} 2 & 1 & 0 & 0 & 4 \\ 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 1 \\ 3 & 0 & 0 & 0 & 5 \end{vmatrix} \quad \text{d) } \begin{vmatrix} 1 & 1 & 4 & 5 \\ 1 & 1 & 5 & 4 \\ 2 & 4 & 1 & 1 \\ 4 & 2 & 1 & 1 \end{vmatrix}$$

5. Compute $\det \mathbf{B}$, where $\mathbf{B} = (-5\mathbf{A}\mathbf{A}^T)^7$ and $\mathbf{A} = \begin{bmatrix} 13 & 8 & 6 \\ -13 & -8 & -4 \\ 8 & 5 & 5 \end{bmatrix}$.

6. Determine all real or complex values of λ for which the following matrices are singular.

$$\text{a) } \begin{bmatrix} 2 - \lambda & -4 \\ 2 & 6 - \lambda \end{bmatrix} \quad \text{b) } \begin{bmatrix} \lambda - 2 & 3 & 1 \\ \lambda - 4 & 3 & 2 \\ \lambda - 6 & \lambda & 3 \end{bmatrix}$$

$$\text{c) } \begin{bmatrix} 1 & 3 - \lambda & 4 \\ 4 - \lambda & 2 & -1 \\ 1 & \lambda - 6 & 2 \end{bmatrix} \quad \text{d) } \begin{bmatrix} 1 - \lambda & -1 & 2 \\ 1 & 2 - \lambda & -13 \\ -2 & 1 & 1 + \lambda \end{bmatrix}$$

7. Let $\mathbf{a}_1, \dots, \mathbf{a}_n \in \mathbb{R}^n$, and define

$$\mathbf{b}_k = \mathbf{a}_k, \quad \mathbf{b}_j = \mathbf{a}_j + \beta_j \mathbf{a}_k, \quad j = 1 \dots n, \quad j \neq k,$$

where $1 \leq k \leq n$ and $\beta_j \in \mathbb{R}$. Using the properties of the determinant function V , show that $V(\mathbf{b}_1, \dots, \mathbf{b}_n) = V(\mathbf{a}_1, \dots, \mathbf{a}_n)$. How does this result relate to Gaussian elimination applied to the matrix $\mathbf{A} = [\mathbf{a}_1 \dots \mathbf{a}_n]^T$?

8. Let the square matrix $\mathbf{A} = (a_{ij})$ be of size $n \times n$ and *tridiagonal*, i.e. $a_{ij} = 0$ when $|i - j| \geq 2$. Further let $a_{ii} = 1$, $i = 1 \dots n$, and $a_{ij} = \lambda$ when $|i - j| = 1$. a) Show that the determinant $D_n = \det \mathbf{A}$ satisfies the recurrence $D_n = D_{n-1} - \lambda^2 D_{n-2}$. b) Compute D_n , $n = 2 \dots 10$, when $\lambda = 2$. c) If $\lambda = 1$, for which values of n is $\mathbf{A}$ singular?
9. (*) Show that if a determinant of size $n \geq 3$ is computed by the minor expansion rule, the required number of multiplications is

$$W_n = n! \sum_{k=1}^{n-1} \frac{1}{k!}.$$

10. (*) Define the determinant function $V(\mathbf{a}_1, \dots, \mathbf{a}_n) = \det[\mathbf{a}_1 \dots \mathbf{a}_n]$ recursively using the first expansion rule ($k = 1$) of Theorem X.5.5 together with the rule $\det a = a$ for a determinant of size $n = 1$.
- a) Show by induction that the function defined in this way has the properties (i)–(iii) of the determinant function.
- b) Show that if $p = (i_1, \dots, i_n)$ is a permutation of the set $(1, \dots, n)$ and $\mathbf{I}_p = [\mathbf{e}_{i_1} \dots \mathbf{e}_{i_n}]$, then according to the definition in part a) either $\det \mathbf{I}_p = 1$ or $\det \mathbf{I}_p = -1$. Conclude that in the former case the permutation p is even (i.e. achievable by an even number of pairwise interchanges) and in the latter odd — so every permutation is of one type or the other.
11. (*) By choosing $\mathbf{b} = b_j \mathbf{e}_j$, $j = 1 \dots n$ derive Cramer's rule from the minor expansion rule. Further prove Theorem X.5.8 starting from Cramer's rule.

X.6 Linear maps

A *linear map* between the spaces $\mathbb{R}^n$ and $\mathbb{R}^m$ is a function of the type

$$\mathbf{f} : \mathbb{R}^n \rightarrow \mathbb{R}^m, \quad m, n \in \mathbb{N},$$

($\mathcal{D}_{\mathbf{f}} = \mathbb{R}^n$, $\mathcal{R}_{\mathbf{f}} \subset \mathbb{R}^m$) satisfying the condition

$$\mathbf{f}(\alpha \mathbf{x} + \beta \mathbf{y}) = \alpha \mathbf{f}(\mathbf{x}) + \beta \mathbf{f}(\mathbf{y}), \quad \mathbf{x}, \mathbf{y} \in \mathbb{R}^n, \quad \alpha, \beta \in \mathbb{R}. \quad (1)$$

When considering linear maps, it is customary to deviate slightly from the usual function notation: A linear map is denoted

$$\mathbf{f}(\mathbf{x}) = A\mathbf{x} \quad (\text{linear map}).$$

On the right, the parentheses around the variable are thus omitted, and this is also reflected in how it is read: one reads ' Ax ' rather than ' A at x '. In accordance with this reading, the mapping (i.e. the determination of the function value) is thought of as A *operating* on $\mathbf{x}$. The operation can be understood as an abstract multiplication (= the product of a matrix and a vector) that is *linear* with respect to both vector addition and multiplication by a scalar, i.e. obeys the usual distributive laws with respect to these operations:

$$A(\alpha \mathbf{x} + \beta \mathbf{y}) = \alpha A\mathbf{x} + \beta A\mathbf{y} \quad (\text{linear map}).$$

Capital letters A, B etc. are generally used as symbols for linear maps.

If $\mathbf{A}$ is a matrix of size $m \times n$, then by the rules of matrix algebra it is evident that the function $\mathbf{f}(\mathbf{x}) = \mathbf{A}\mathbf{x}$ is a linear map of the type $\mathbf{f} : \mathbb{R}^n \rightarrow \mathbb{R}^m$. It turns out that there are no other linear maps of this type:

PROPOSITION X.6.1 If $\mathbf{f} : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is a linear map, then there exists a unique matrix $\mathbf{A}$ of size $m \times n$ such that $\mathbf{f}(\mathbf{x}) = \mathbf{A}\mathbf{x}$, $\mathbf{x} \in \mathbb{R}^n$.

Proof When $\mathbf{x} \in \mathbb{R}^n$ is expressed in terms of the basis $\{\mathbf{e}_1, \dots, \mathbf{e}_n\}$ in the form

$$\mathbf{x} = \sum_{i=1}^n x_i \mathbf{e}_i,$$

then, by the linearity of $\mathbf{f}$ and the rules of matrix algebra (Section X.1),

$$\mathbf{f}(\mathbf{x}) = \sum_{i=1}^n x_i \mathbf{f}(\mathbf{e}_i) = \mathbf{A}\mathbf{x}, \quad \mathbf{A} = [\mathbf{f}(\mathbf{e}_1) \dots \mathbf{f}(\mathbf{e}_n)]. \quad \square$$

By Proposition X.6.1, there is a one-to-one correspondence between linear maps and matrices:

$$\text{linear map } A : \mathbb{R}^n \rightarrow \mathbb{R}^m \longleftrightarrow \text{matrix } \mathbf{A} \text{ of size } m \times n \quad (A\mathbf{x} = \mathbf{A}\mathbf{x}).$$

If the matrix $\mathbf{A}$ corresponds to the linear map A , then the correspondence is referred to by saying that $\mathbf{A}$ is the *matrix of the linear map* (or the representation matrix). Through the representation matrix, computation with linear maps can thus always be reduced to matrix algebra. Conversely, the multiplication of a matrix and a vector can, whenever desired, be regarded as a linear map, i.e. as an application of the computational rule of some linear map. In particular, the product of two matrices corresponds to the composition of two linear maps:

$$AB \leftrightarrow \mathbf{A}\mathbf{B}.$$

The restrictions on composition are the same as for the matrix product: one must assume that $B : \mathbb{R}^n \rightarrow \mathbb{R}^p$ and $A : \mathbb{R}^p \rightarrow \mathbb{R}^m$ for some $m, n, p \in \mathbb{N}$, in which case the product $\mathbf{A}\mathbf{B}$ is also defined.

The formal definition of the *inverse map* A^{-1} of a linear map $A : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is

$$A\mathbf{x} = \mathbf{y} \quad \Leftrightarrow \quad \mathbf{x} = A^{-1}\mathbf{y}.$$

The inverse map is definable in this way provided A is 1-1 ('one-to-one'), i.e. satisfies the condition

$$A\mathbf{x}_1 = A\mathbf{x}_2 \quad \Rightarrow \quad \mathbf{x}_1 = \mathbf{x}_2.$$

Since A is linear, this is the same as the condition

$$A\mathbf{x} = \mathbf{0} \quad \Rightarrow \quad \mathbf{x} = \mathbf{0}.$$

From earlier results (Section X.4) it can be deduced that this condition can hold only if $m \geq n$ (a necessary condition). If the condition holds, A is *invertible*, and the inverse map is then defined as a map of the type $A^{-1} : \mathbb{R}^m \rightarrow \mathbb{R}^n$. The domain, however, is not necessarily all of $\mathbb{R}^m$, since by the definition of the inverse map $A^{-1}\mathbf{y}$ is defined only for those vectors $\mathbf{y} \in \mathbb{R}^m$ for which $\mathbf{y} = \mathbf{A}\mathbf{x}$ for some $\mathbf{x} \in \mathbb{R}^n$, i.e. the system $\mathbf{A}\mathbf{x} = \mathbf{y}$ must have a solution. By earlier results, the solvability of the system $\mathbf{A}\mathbf{x} = \mathbf{y}$ for every $\mathbf{y} \in \mathbb{R}^m$ is possible only if $m \leq n$ (a necessary condition). We therefore conclude that the inverse map A^{-1} exists and is defined on all of $\mathbb{R}^m$ exactly when $m = n$ and the matrix $\mathbf{A}$ of A (a square matrix) is nonsingular. The inverse map is then a linear map of the type $A^{-1} : \mathbb{R}^n \rightarrow \mathbb{R}^n$ (of the same type as A), and the matrix of the inverse map is naturally $\mathbf{A}^{-1}$.

The linearity of the inverse map of a linear map $A : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is often exploited when solving linear systems of equations. A typical application is one in which the system $\mathbf{Ax} = \mathbf{b}$ is to be solved for several different values of $\mathbf{b}$ while $\mathbf{A}$ remains the same. If the solution is denoted as a map $\mathbf{b} \mapsto \mathbf{x}$ (the inverse map of the map $\mathbf{x} \mapsto \mathbf{b} = \mathbf{Ax}$) and it is assumed that the solutions $\mathbf{b}_1 \mapsto \mathbf{x}_1$ and $\mathbf{b}_2 \mapsto \mathbf{x}_2$ are known, then by linearity one knows (cf. Section X.3, Example 1)

$$\alpha \mathbf{b}_1 + \beta \mathbf{b}_2 \mapsto \alpha \mathbf{x}_1 + \beta \mathbf{x}_2 \quad (\alpha, \beta \in \mathbb{R}).$$

Change of basis in $\mathbb{R}^n$

A vector system (ordered set) $\{\mathbf{a}_1, \dots, \mathbf{a}_m\}$ in $\mathbb{R}^n$ is said to be *linearly independent* if

$$x_1 \mathbf{a}_1 + \dots + x_m \mathbf{a}_m = \mathbf{0} \quad \Rightarrow \quad x_i = 0, \quad i = 1 \dots m.$$

If the vectors $\mathbf{a}_i$ are collected as the columns of a matrix $\mathbf{A}$ ($\mathbf{A}$ of size $n \times m$), then one can write $x_1 \mathbf{a}_1 + \dots + x_m \mathbf{a}_m = \mathbf{Ax}$ (cf. Section X.1), so the condition of linear independence takes the form

$$\mathbf{x} \in \mathbb{R}^m \text{ \& } \mathbf{Ax} = \mathbf{0} \quad \Rightarrow \quad \mathbf{x} = \mathbf{0}, \quad \mathbf{A} = [\mathbf{a}_1, \dots, \mathbf{a}_m].$$

By the results of Section X.4, this condition can hold only when $m \leq n$. Thus in $\mathbb{R}^n$ every system formed by $n+1$ vectors (or more) is *linearly dependent*. On the other hand, if $m = n$, then a linearly independent system is formed at least by the natural basis $\{\mathbf{e}_1, \dots, \mathbf{e}_n\}$ of $\mathbb{R}^n$, which in the above condition corresponds to the matrix $\mathbf{A} = \mathbf{I}$. The number n is thus the critical number for linear independence, and this has already been named the *dimension* (= dimensionality) of the space $\mathbb{R}^n$. If $m = n$, then the above condition holds exactly when $\mathbf{A}$ is a nonsingular matrix (Theorem X.2.6). Consequently, and since $\mathbf{A}$ is nonsingular exactly when $\mathbf{A}^T$ is nonsingular (see Section X.2), the following new (geometric) interpretations are obtained for the nonsingularity of a square matrix:

$\begin{aligned} \mathbf{A} \text{ nonsingular} &\Leftrightarrow \mathbf{A} \text{ has linearly independent columns} \\ &\Leftrightarrow \mathbf{A} \text{ has linearly independent rows.} \end{aligned}$

If $\{\mathbf{a}_1, \dots, \mathbf{a}_n\}$ is a linearly independent vector system in $\mathbb{R}^n$, i.e. the matrix $\mathbf{A} = [\mathbf{a}_1, \dots, \mathbf{a}_n]$ is nonsingular, then an arbitrary vector $\mathbf{b} \in \mathbb{R}^n$ can be expressed uniquely as a linear combination of the vectors $\mathbf{a}_i$. Indeed, the coefficients x_i are uniquely determined by the system

$$\sum_{i=1}^n x_i \mathbf{a}_i = \mathbf{b} \quad \Leftrightarrow \quad \mathbf{Ax} = \mathbf{b}.$$

On this basis one can say that every linearly independent system of n vectors $\{\mathbf{a}_1, \dots, \mathbf{a}_n\}$ is a *basis* of $\mathbb{R}^n$. So if $\mathbf{A}$ is any nonsingular matrix of size $n \times n$, then the columns of $\mathbf{A}$ (as well as the rows) form a basis of $\mathbb{R}^n$. — Note that such a basis is *orthonormal*, i.e. $\langle \mathbf{a}_i, \mathbf{a}_j \rangle = \delta_{ij}$ (the Kronecker δ), exactly when $\mathbf{A}$ is orthogonal (cf. Section X.2). We therefore conclude that every nonsingular matrix of size $n \times n$ represents, through its columns (or rows), a basis of $\mathbb{R}^n$, and every orthogonal matrix represents an orthonormal basis.

If one wishes to carry out a *change of basis*, starting from the natural basis $\{\mathbf{e}_1, \dots, \mathbf{e}_n\}$ of $\mathbb{R}^n$, then the coordinates x'_i of a vector $\mathbf{x}$ in the basis $\{\mathbf{a}_1, \dots, \mathbf{a}_n\}$ can be found from the system

$$\mathbf{A}\mathbf{x}' = \mathbf{x} \quad \Rightarrow \quad \mathbf{x}' = \mathbf{A}^{-1}\mathbf{x}, \quad \mathbf{A} = [\mathbf{a}_1, \dots, \mathbf{a}_n].$$

A change of basis, i.e. the coordinate transformation $\mathbf{x} \mapsto \mathbf{x}'$, is thus a linear map. If the new basis is also orthonormal, then the matrix of the transformation is easy to compute: $\mathbf{A}^{-1} = \mathbf{A}^T$.

EXAMPLE 1 Determine the coordinates x', y', z' of the vector $x\vec{i} + y\vec{j} + z\vec{k}$ in the basis formed by the vectors $\vec{a}_1 = \vec{i} + \vec{j}$, $\vec{a}_2 = \vec{j} + \vec{k}$ and $\vec{a}_3 = \vec{i} + \vec{k}$.

Solution

$$\begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} x' \\ y' \\ z' \end{bmatrix} = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \quad \Rightarrow \quad \begin{bmatrix} x' \\ y' \\ z' \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 1 & 1 & -1 \\ -1 & 1 & 1 \\ 1 & -1 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix}. \quad \square$$

EXAMPLE 2 At a point $P = (r, \theta, \varphi)$ of the spherical coordinate system, the basis $\{\vec{e}_r, \vec{e}_\theta, \vec{e}_\varphi\}$ is used for spatial vectors, where (cf. Section II.8)

$$\begin{cases} \vec{e}_r &= \sin \theta \cos \varphi \vec{i} + \sin \theta \sin \varphi \vec{j} + \cos \theta \vec{k}, \\ \vec{e}_\theta &= \cos \theta \cos \varphi \vec{i} + \cos \theta \sin \varphi \vec{j} - \sin \theta \vec{k}, \\ \vec{e}_\varphi &= -\sin \varphi \vec{i} + \cos \varphi \vec{j}. \end{cases}$$

When one writes

$$\vec{F} = F_1 \vec{i} + F_2 \vec{j} + F_3 \vec{k} = F_r \vec{e}_r + F_\theta \vec{e}_\theta + F_\varphi \vec{e}_\varphi,$$

then the transformation formula $(F_r, F_\theta, F_\varphi) \mapsto (F_1, F_2, F_3)$ is in matrix form

$$\begin{bmatrix} \sin \theta \cos \varphi & \cos \theta \cos \varphi & -\sin \varphi \\ \sin \theta \sin \varphi & \cos \theta \sin \varphi & \cos \varphi \\ \cos \theta & -\sin \theta & 0 \end{bmatrix} \begin{bmatrix} F_r \\ F_\theta \\ F_\varphi \end{bmatrix} = \begin{bmatrix} F_1 \\ F_2 \\ F_3 \end{bmatrix}.$$

Since $\{\vec{e}_r, \vec{e}_\theta, \vec{e}_\varphi\}$ is an orthonormal system, the inverse transformation is obtained by transposing:

$$\begin{bmatrix} F_r \\ F_\theta \\ F_\varphi \end{bmatrix} = \begin{bmatrix} \sin \theta \cos \varphi & \sin \theta \sin \varphi & \cos \theta \\ \cos \theta \cos \varphi & \cos \theta \sin \varphi & -\sin \theta \\ -\sin \varphi & \cos \varphi & 0 \end{bmatrix} \begin{bmatrix} F_1 \\ F_2 \\ F_3 \end{bmatrix}. \quad \square$$

Rotation of the coordinate system

It was noted above that if $\mathbf{A}$ is an orthogonal matrix of size $n \times n$, then the columns of $\mathbf{A}$ (as well as the rows) define an orthonormal basis of $\mathbb{R}^n$. That basis is said to be *positively oriented* (or 'right-handed') if $\det \mathbf{A} = 1$, and *negatively oriented* ('left-handed') if $\det \mathbf{A} = -1$. ($\det \mathbf{A}$ has no other possible values, see Exercise X.5:1). If the direction of one basis vector is reversed, then, by determinant theory, the sign of $\det \mathbf{A}$ changes, i.e. the orientation of the basis changes.

EXAMPLE 3 If $\{\vec{a}, \vec{b}, \vec{c}\}$ is an orthonormal basis of spatial vectors, then the coordinate vectors of these vectors in the basis $\{\vec{i}, \vec{j}, \vec{k}\}$ form an orthonormal basis $\{\mathbf{a}, \mathbf{b}, \mathbf{c}\}$ of $\mathbb{R}^3$, i.e. the matrix $\mathbf{A} = [\mathbf{a} \ \mathbf{b} \ \mathbf{c}]$ is orthogonal (as in the previous example). Computed by geometric vector operations, the determinant of $\mathbf{A}$ is (cf. Section II.6)

$$\det \mathbf{A} = \det \mathbf{A}^T = (\vec{a} \times \vec{b}) \cdot \vec{c} = [\vec{a} \ \vec{b} \ \vec{c}].$$

Since $\{\vec{a}, \vec{b}, \vec{c}\}$ is an orthonormal system, either $\vec{c} = \vec{a} \times \vec{b}$, in which case $\det \mathbf{A} = |\vec{c}|^2 = 1$, or $\vec{c} = -(\vec{a} \times \vec{b})$, in which case $\det \mathbf{A} = -1$. In the former case the basis is thus positively oriented, in the latter negatively oriented. $\square$

EXAMPLE 4 The bases $\{\vec{e}_r, \vec{e}_\varphi\}$ of the polar coordinate system (cf. Section II.8) cover, for the various values of φ , all possible orthonormal, positively oriented systems of plane vectors. Consequently, all orthogonal matrices $\mathbf{A}$ of size 2×2 satisfying the condition $\det \mathbf{A} = 1$ are of the form

$$\mathbf{A} = \begin{bmatrix} \cos \varphi & \sin \varphi \\ -\sin \varphi & \cos \varphi \end{bmatrix}, \quad \varphi \in [0, 2\pi). \quad \square$$

If $\mathbf{A} = [\mathbf{a}_1 \ \dots \ \mathbf{a}_n]$ is an orthogonal matrix and $\det \mathbf{A} = 1$, then the change of basis $\{\mathbf{e}_1, \dots, \mathbf{e}_n\} \mapsto \{\mathbf{a}_1, \dots, \mathbf{a}_n\}$ is called a *rotation of the coordinate system*. Indeed, it can be shown that the change of basis can then be realized as a continuous transformation of the form

$$t \in [0, 1] \mapsto \{\mathbf{a}_1(t), \dots, \mathbf{a}_n(t)\}$$

such that the following conditions hold:

- (i) $\mathbf{a}_i(0) = \mathbf{e}_i$ and $\mathbf{a}_i(1) = \mathbf{a}_i$, $i = 1 \dots n$.
- (ii) The vector system $\{\mathbf{a}_1(t), \dots, \mathbf{a}_n(t)\}$ is orthonormal and positively oriented for every $t \in [0, 1]$.
- (iii) Continuity condition: The functions $t \in [0, 1] \mapsto \mathbf{a}_j(t) \in \mathbb{R}^n$ are continuous, i.e. the real functions $t \mapsto a_{ij}(t) = [\mathbf{a}_j(t)]_i$, $i, j = 1 \dots n$ are continuous on the interval $[0, 1]$.

The transformation satisfying the conditions can be imagined as rotating the vector system $\{\mathbf{e}_1, \dots, \mathbf{e}_n\}$ 'as a rigid body', so that orthogonality and orientation are preserved. Starting from the basic system $\{\mathbf{e}_1, \dots, \mathbf{e}_n\}$, it is thus possible, by such a continuous rotation, to reach any given orthonormal and positively oriented system $\{\mathbf{a}_1, \dots, \mathbf{a}_n\}$. If the system in question is negatively oriented, then at some point a 'jump' must occur (a reversal of the direction of one vector), so that the transformation is not continuous. According to this claim, then, all positively oriented orthonormal systems of $\mathbb{R}^n$ are mutually (likewise the negatively oriented ones among themselves) *rotationally equivalent*. In the case $n = 2$ the claim can be proven from the result of Example 4 (Exercise 5). The general case is more challenging — the detailed justification is omitted.

Subspaces of $\mathbb{R}^n$

Let m vectors $\mathbf{a}_j \in \mathbb{R}^n$, $j = 1 \dots m$ be given (one may have $m \leq n$ or $m > n$). Then the set of vectors defined by

$$W = \{\mathbf{v} \in \mathbb{R}^n \mid \mathbf{v} = \sum_{j=1}^m y_j \mathbf{a}_j \text{ for some } [y_1, \dots, y_m]^T \in \mathbb{R}^m\}$$

is a *subspace* of $\mathbb{R}^n$, since by its definition $W \subset \mathbb{R}^n$ and

$$\mathbf{v}_1, \mathbf{v}_2 \in W \quad \Rightarrow \quad \alpha_1 \mathbf{v}_1 + \alpha_2 \mathbf{v}_2 \in W \quad \forall \alpha_1, \alpha_2 \in \mathbb{R}.$$

We say that W is the space *spanned* by the vectors $\mathbf{a}_1, \dots, \mathbf{a}_m$. If the vectors $\mathbf{a}_i$ are linearly independent (in which case necessarily $m \leq n$), then $\dim W = m$ and $\{\mathbf{a}_1, \dots, \mathbf{a}_m\}$ is a basis of W . Whether the vectors $\mathbf{a}_i$ are linearly independent or not is determined by Gaussian elimination: form the matrix $\mathbf{A} = [\mathbf{a}_1 \dots \mathbf{a}_m]$ with the vectors $\mathbf{a}_j$ as columns and find all solutions $\mathbf{x} \in \mathbb{R}^m$ of the system $\mathbf{A}\mathbf{x} = \mathbf{0}$ (cf. Section X.4).

EXAMPLE 5 Are the vectors $\mathbf{a}_1 = [1, 2, 1, -1]^T$, $\mathbf{a}_2 = [2, -1, 3, -1]^T$ and $\mathbf{a}_3 = [0, 5, -1, -1]^T$ of $\mathbb{R}^4$ linearly independent?

Solution We transform the system $\mathbf{Ax} = \mathbf{0}$ by Gaussian elimination into the singular basic form $\mathbf{Ux} = \mathbf{0}$:

$$\begin{bmatrix} 1 & 2 & 0 \\ 2 & -1 & 5 \\ 1 & 3 & -1 \\ -1 & -1 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} \Leftrightarrow \begin{bmatrix} 1 & 2 & 0 \\ 0 & -5 & -5 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}.$$

On this basis

$$\mathbf{Ax} = \mathbf{0} \Leftrightarrow \mathbf{x} = t[-2, 1, 1]^T, \quad t \in \mathbb{R}.$$

Answer: No, since $-2\mathbf{a}_1 + \mathbf{a}_2 + \mathbf{a}_3 = \mathbf{0}$. $\square$

Gaussian elimination also reveals, more generally, the dimension of the subspace W spanned by given vectors $\mathbf{a}_j \in \mathbb{R}^n$, $j = 1 \dots m$, and which linearly independent subsystem of that vector system qualifies as a basis of W when the whole system is linearly dependent.

EXAMPLE 5 (continued) The columns of the matrix $\mathbf{A}$ of the example span a subspace W of $\mathbb{R}^4$ and the rows span a subspace U of $\mathbb{R}^3$. Determine the dimensions of these subspaces and a basis for each.

Solution In the algorithm of the example no row or column interchanges were made, so the result can be read off directly: $\dim W = \dim U = 2$, one basis of W = columns 1 and 2 of $\mathbf{A}$, and one basis of U = rows 1 and 2 of $\mathbf{A}$, i.e.

$$\begin{aligned} W &= \{\mathbf{v} = \alpha_1[1, 2, 1, -1]^T + \alpha_2[2, -1, 3, -1]^T, \quad \alpha_1, \alpha_2 \in \mathbb{R}\}, \\ U &= \{\mathbf{v} = \alpha_1[1, 2, 0] + \alpha_2[2, -1, 5], \quad \alpha_1, \alpha_2 \in \mathbb{R}\}. \quad \square \end{aligned}$$

EXAMPLE 6 The columns of the matrix

$$\mathbf{A} = \begin{bmatrix} 3 & 3 & 0 & 3 & 4 & 1 \\ 1 & -3 & 2 & -1 & -1 & 2 \\ -3 & 3 & -3 & 0 & 2 & -1 \\ 6 & -2 & 4 & 2 & 1 & 3 \end{bmatrix}$$

span a subspace W of $\mathbb{R}^4$ and the rows span a subspace U of $\mathbb{R}^6$. Determine the dimensions of these subspaces and a basis for each.

Solution The singular basic form of the system

$$\begin{bmatrix} 3 & 3 & 0 & 3 & 4 & 1 \\ 1 & -3 & 2 & -1 & -1 & 2 \\ -3 & 3 & -3 & 0 & 2 & -1 \\ 6 & -2 & 4 & 2 & 1 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \\ x_6 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

is obtained by pivoted Gaussian elimination as

$$\begin{bmatrix} 3 & 3 & 4 & 1 & 0 & 3 \\ 0 & -4 & -\frac{7}{3} & \frac{5}{3} & 2 & -2 \\ 0 & 0 & \frac{5}{2} & \frac{5}{2} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_5 \\ x_6 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}.$$

To reach this form no row interchanges were made. Columns, on the other hand, were interchanged so that the new order is 125634, as can be seen from the result. We therefore conclude: $\dim W = \dim U = 3$, one basis of U is formed by the first three rows of $\mathbf{A}$, and one basis of W by columns 1, 2 and 5 of $\mathbf{A}$. $\square$

In the above examples the dimensions of the spaces spanned by the columns and by the rows of the matrix are the same. This is no coincidence; the dimensions are always the same: the common value of the dimensions = the rank of the matrix (see Section X.4).

*Orthogonal projection in $\mathbb{R}^n$

Let $\{\mathbf{a}_1, \dots, \mathbf{a}_m\}$, $1 \leq m < n$, be a linearly independent vector system in $\mathbb{R}^n$, and let W be the (proper) subspace of $\mathbb{R}^n$ spanned by these vectors:

$$W = \{\mathbf{v} \in \mathbb{R}^n \mid \mathbf{v} = \sum_{i=1}^m x_i \mathbf{a}_i, \quad x_i \in \mathbb{R}\}.$$

Then if $\mathbf{u} \in \mathbb{R}^n$, the *orthogonal projection* of $\mathbf{u}$ onto the subspace W is defined as the vector $\mathbf{w}$ satisfying

$$\mathbf{w} \in W \quad \& \quad \langle \mathbf{u} - \mathbf{w}, \mathbf{v} \rangle = 0 \quad \forall \mathbf{v} \in W.$$

By writing here $\mathbf{v}$ as a linear combination of the vectors $\mathbf{a}_1, \dots, \mathbf{a}_m$ (possible for every $\mathbf{v} \in W$) and using the bilinearity of the inner product, one sees that it suffices to impose the orthogonality condition for the vectors $\mathbf{v} = \mathbf{a}_i$, $i = 1 \dots m$:

$$\langle \mathbf{u} - \mathbf{w}, \mathbf{v} \rangle = 0 \quad \forall \mathbf{v} \in W \quad \Leftrightarrow \quad \langle \mathbf{u} - \mathbf{w}, \mathbf{a}_i \rangle = 0, \quad i = 1 \dots m.$$

The search for the orthogonal projection is thus reduced to a system of equations. Writing further $\mathbf{w} = \sum_{j=1}^m x_j \mathbf{a}_j$ (possible, since $\mathbf{w} \in W$) and again using the bilinearity and symmetry of the inner product, one sees that this is the following linear system for the coefficient vector $\mathbf{x} = [x_1, \dots, x_m]^T \in \mathbb{R}^m$:

$$\mathbf{Ax} = \mathbf{b}, \quad [\mathbf{A}]_{ij} = \langle \mathbf{a}_i, \mathbf{a}_j \rangle, \quad [\mathbf{b}]_i = \langle \mathbf{u}, \mathbf{a}_i \rangle, \quad i, j = 1 \dots m.$$

This formulation is called the *normal form* of the projection problem.

In applications of linear algebra, the projection problem is often arrived at through a minimization problem: one wishes to find, in the sense of some norm $\|\cdot\|$, the vector in the subspace W closest to a given vector $\mathbf{u} \in \mathbb{R}^n$, i.e. to solve the minimization problem

$$\mathbf{v} \in W : \quad \|\mathbf{u} - \mathbf{v}\| = \min!$$

If the norm is associated with an inner product $\langle \cdot, \cdot \rangle$ so that $\|\mathbf{x}\|^2 = \langle \mathbf{x}, \mathbf{x} \rangle$ (see Section II.4), then the solution of the problem is obtained as the orthogonal projection $\mathbf{w} \in W$ of $\mathbf{u}$ with respect to that inner product. Indeed, if $\mathbf{v} \in W$ is arbitrary and we set $\mathbf{v}_0 = \mathbf{w} - \mathbf{v}$, then $\langle \mathbf{u} - \mathbf{w}, \mathbf{v}_0 \rangle = 0$ (because $\mathbf{v}_0 \in W$), whence

$$\begin{aligned} \|\mathbf{u} - \mathbf{v}\|^2 &= \|(\mathbf{u} - \mathbf{w}) + \mathbf{v}_0\|^2 = \langle (\mathbf{u} - \mathbf{w}) + \mathbf{v}_0, (\mathbf{u} - \mathbf{w}) + \mathbf{v}_0 \rangle \\ &= |\mathbf{u} - \mathbf{w}|^2 + 2\langle \mathbf{u} - \mathbf{w}, \mathbf{v}_0 \rangle + |\mathbf{v}_0|^2 \\ &= |\mathbf{u} - \mathbf{w}|^2 + |\mathbf{v}_0|^2. \end{aligned}$$

Thus the minimum value of $\|\mathbf{u} - \mathbf{v}\|$ is attained when $\mathbf{v}_0 = \mathbf{0} \Leftrightarrow \mathbf{v} = \mathbf{w}$.

The unique solvability of the projection problem is guaranteed by

THEOREM X.6.2 (Projection theorem in $\mathbb{R}^n$) If $W \subset \mathbb{R}^n$ is the subspace of $\mathbb{R}^n$ spanned by linearly independent vectors $\mathbf{a}_1, \dots, \mathbf{a}_m$ ($m < n$) and $\mathbf{u} \in \mathbb{R}^n$, then there exists a unique $\mathbf{w} \in W$ such that $\langle \mathbf{u} - \mathbf{w}, \mathbf{v} \rangle = 0 \quad \forall \mathbf{v} \in W$.

Proof The theorem claims that the matrix $\mathbf{A} = (\langle \mathbf{a}_i, \mathbf{a}_j \rangle)$ (a square matrix of size $m \times m$) is nonsingular. To show this, let $\mathbf{w} = \sum_{i=1}^m x_i \mathbf{a}_i$ and first compute

$$|\mathbf{w}|^2 = \langle \mathbf{w}, \mathbf{w} \rangle = \left\langle \sum_{i=1}^m x_i \mathbf{a}_i, \sum_{j=1}^m x_j \mathbf{a}_j \right\rangle.$$

By the bilinearity of the inner product and matrix algebra, this expands into the form

$$|\mathbf{w}|^2 = \sum_{i=1}^m \sum_{j=1}^m \langle \mathbf{a}_i, \mathbf{a}_j \rangle x_i x_j = \mathbf{x}^T \mathbf{A} \mathbf{x}.$$

On this basis we reason as follows:

$$\mathbf{A} \mathbf{x} = \mathbf{0} \quad \Rightarrow \quad \mathbf{x}^T \mathbf{A} \mathbf{x} = 0 \quad \Rightarrow \quad |\mathbf{w}|^2 = 0 \quad \Rightarrow \quad \mathbf{w} = \mathbf{0}.$$

Since $\{\mathbf{a}_1, \dots, \mathbf{a}_m\}$ was a linearly independent system, we further conclude: $\mathbf{w} = \mathbf{0} \Rightarrow \mathbf{x} = \mathbf{0}$. We have thus deduced: $\mathbf{A} \mathbf{x} = \mathbf{0} \Rightarrow \mathbf{x} = \mathbf{0}$, so $\mathbf{A}$ is nonsingular. $\square$

The solution $\mathbf{x} = \mathbf{A}^{-1}\mathbf{b}$ of the normal form of the projection problem thus gives the coordinates of the orthogonal projection $\mathbf{w}$ in the basis $\{\mathbf{a}_1, \dots, \mathbf{a}_m\}$ of W . For the solution one must first compute the matrix $\mathbf{A}$ and the vector $\mathbf{b}$. These depend on the vectors $\mathbf{u}$ and $\{\mathbf{a}_i\}$ and on the chosen inner product of $\mathbb{R}^n$, which thus need not be the Euclidean one. The dependence of the vector $\mathbf{b}$ on the vector $\mathbf{u}$ can be expressed in terms of matrix algebra:

$$\mathbf{u} = \sum_{j=1}^n u_j \mathbf{e}_j \quad \Rightarrow \quad [\mathbf{b}]_i = \langle \mathbf{u}, \mathbf{a}_i \rangle = \langle \mathbf{a}_i, \mathbf{u} \rangle = \sum_{j=1}^n \langle \mathbf{a}_i, \mathbf{e}_j \rangle u_j.$$

Thus $\mathbf{b} = \mathbf{B}\mathbf{u}$, where

$$[\mathbf{B}]_{ij} = \langle \mathbf{a}_i, \mathbf{e}_j \rangle, \quad i = 1 \dots m, \quad j = 1 \dots n.$$

(In the case of the Euclidean inner product, $[\mathbf{B}]_{ij} = [\mathbf{a}_i]_j$.) According to this interpretation, the orthogonal projection $\mathbf{u} \mapsto \mathbf{x}$ is a linear map of the type $P: \mathbb{R}^n \rightarrow \mathbb{R}^m$ and the matrix of P is $\mathbf{A}^{-1}\mathbf{B}$.

EXAMPLE 7 Compute the orthogonal projection of the vector $\mathbf{u} = [0, 1, 0, 0]^T$, with respect to the Euclidean inner product, onto the subspace $W \subset \mathbb{R}^4$ spanned by the vectors $\mathbf{a}_1 = [1, 1, 1, 1]^T$, $\mathbf{a}_2 = [1, 1, 1, 0]^T$ and $\mathbf{a}_3 = [0, 1, 1, 1]^T$. Determine also a unit vector $\mathbf{n} \in \mathbb{R}^4$ that is orthogonal to the subspace W .

Solution By the projection principle, the coordinates x_i of the vector $\mathbf{w} = x_1\mathbf{a}_1 + x_2\mathbf{a}_2 + x_3\mathbf{a}_3$ are determined by solving the system $\mathbf{A}\mathbf{x} = \mathbf{b}$, where

$$\mathbf{A} = (\langle \mathbf{a}_i, \mathbf{a}_j \rangle) = \begin{bmatrix} 4 & 3 & 3 \\ 3 & 3 & 2 \\ 3 & 2 & 3 \end{bmatrix}, \quad \mathbf{b} = \mathbf{B}\mathbf{u} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}.$$

The solution is $\mathbf{x} = \frac{1}{2}[-1, 1, 1]^T$, so the requested projection is

$$\mathbf{w} = \frac{1}{2}(-\mathbf{a}_1 + \mathbf{a}_2 + \mathbf{a}_3)^T = \frac{1}{2}[0, 1, 1, 0]^T.$$

The vector

$$\mathbf{n} = \lambda(\mathbf{u} - \mathbf{w}) = \frac{1}{2}\lambda[0, 1, -1, 0]^T$$

is orthogonal to W , so the requested unit vectors are

$$\mathbf{n} = \pm \frac{1}{\sqrt{2}}[0, 1, -1, 0]^T. \quad \square$$

EXERCISES

1. a) A linear map $A : \mathbb{R}^3 \mapsto \mathbb{R}^3$ satisfies: $A\mathbf{a} = (0, 2, 1)$, $A\mathbf{b} = (1, 2, 3)$ and $A\mathbf{c} = (1, -1, 0)$. Compute $A(3\mathbf{a} - 2\mathbf{b} + \mathbf{c})$.
 b) A linear map $A : \mathbb{R}^2 \mapsto \mathbb{R}^2$ satisfies: $A(1, 1) = (3, -1)$ and $A(2, -1) = (1, 2)$. Compute the matrix of A .
 c) A function $\mathbf{f} : \mathbb{R}^3 \mapsto \mathbb{R}^3$ satisfies: $\mathbf{f}(0, 1, 1) = (1, 0, 0)$, $\mathbf{f}(1, 0, 1) = (1, 1, 0)$ and $\mathbf{f}(1, -1, 0) = (1, 1, 1)$. Can $\mathbf{f}$ be a linear map?
2. Let $A : \mathbb{R}^n \mapsto \mathbb{R}^n$ be a linear map and $\mathbf{a}_1, \dots, \mathbf{a}_m \in \mathbb{R}^n$.
 a) Show that if $\{\mathbf{a}_1, \dots, \mathbf{a}_m\}$ is linearly dependent then so is $\{A\mathbf{a}_1, \dots, A\mathbf{a}_m\}$.
 b) Show that if $\{A\mathbf{a}_1, \dots, A\mathbf{a}_m\}$ is linearly independent, then so is $\{\mathbf{a}_1, \dots, \mathbf{a}_m\}$.
 c) Assume in addition that A is invertible. Show that then $\{\mathbf{a}_1, \dots, \mathbf{a}_m\}$ is linearly dependent/independent exactly when $\{A\mathbf{a}_1, \dots, A\mathbf{a}_m\}$ is linearly dependent/independent.
3. Compute the matrix-form coordinate transformation formulas $\mathbf{x} = \mathbf{A}\mathbf{x}'$ and $\mathbf{x}' = \mathbf{B}\mathbf{x}$ for the following changes of basis of $\mathbb{R}^n$ or of the corresponding geometric vector space.
 a) $\{\mathbf{e}_1, \mathbf{e}_2\} \mapsto \{\mathbf{e}_1 + 9\mathbf{e}_2, 6\mathbf{e}_1 + 8\mathbf{e}_2\}$
 b) $\{\vec{i}, \vec{j}\} \mapsto \{\vec{i} - 2\vec{j}, 3\vec{i} + \vec{j}\}$
 c) $\{\mathbf{e}_1, \mathbf{e}_2\} \mapsto \{\mathbf{a}, \mathbf{b}\}$, $\mathbf{a} = [5, -1]^T$, $\mathbf{b} = [1, 5]^T$
 d) $\{\mathbf{a}, \mathbf{b}\} \mapsto \{\mathbf{c}, \mathbf{d}\}$, $\mathbf{a} = [1, 1]^T$, $\mathbf{b} = [1, 2]^T$, $\mathbf{c} = [2, 1]^T$, $\mathbf{d} = [1, -1]^T$
 e) $\{\vec{i}, \vec{j}, \vec{k}\} \mapsto \{\vec{i} + 2\vec{j}, 3\vec{j} + 4\vec{k}, 6\vec{i} + 5\vec{k}\}$
 f) $\{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\} \mapsto \{\mathbf{e}_1 + \mathbf{e}_2 + \mathbf{e}_3, \mathbf{e}_1 - \mathbf{e}_2 + \mathbf{e}_3, -\mathbf{e}_1 + \mathbf{e}_2 + \mathbf{e}_3\}$
 g) $\{\mathbf{e}_1, \dots, \mathbf{e}_4\} \mapsto \{\mathbf{a}_1, \dots, \mathbf{a}_4\}$, $\mathbf{a}_i = \mathbf{e}_1 + \dots + \mathbf{e}_i$, $i = 1 \dots 4$
 h) $\{\mathbf{e}_1, \dots, \mathbf{e}_n\} \mapsto \{\mathbf{e}_1 + \mathbf{e}_2, \mathbf{e}_2 + \mathbf{e}_3, \dots, \mathbf{e}_{n-1} + \mathbf{e}_n, \mathbf{e}_1 + \mathbf{e}_n\}$, $n = 7$
4. Investigate whether the following vectors are linearly independent.
 a) $[0, 0, 4, 1]$, $[2, 1, -1, 1]$, $[1, -1, 2, 1]$, $[1, 2, 1, 1]$
 b) $[1, -1, 1, -1, 1]$, $[1, 1, 1, 1, 1]$, $[1, 1, -1, 1, 1]$, $[1, 1, 1, -1, 1]$
 c) $[1, 1, 1, 1, 1, 1, 1, 1, 1]$, $[0, 1, 1, 1, 1, 1, 1, 1, 1]$, $\dots$, $[0, 0, 0, 0, 0, 0, 0, 0, 1]$
5. Construct a continuous transformation $t \in [0, 1] \mapsto \{\mathbf{a}(t), \mathbf{b}(t)\}$ preserving the orthonormality and orientation of the basis, from the basis $\{\mathbf{e}_1, \mathbf{e}_2\}$ of $\mathbb{R}^2$ to a given orthonormal and positively oriented basis $\{\mathbf{a}, \mathbf{b}\}$.
Hint: Example 4.
6. Show that the following coordinate transformation formulas of $\mathbb{R}^3$ represent a rotation of the coordinate system. What is the inverse transformation

formula? The coordinates of which points remain unchanged under the rotation?

$$\text{a) } \begin{bmatrix} x' \\ y' \\ z' \end{bmatrix} = \frac{1}{7} \begin{bmatrix} 3 & -2 & 6 \\ 2 & -6 & -3 \\ 6 & 3 & -2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

$$\text{b) } \begin{bmatrix} x' \\ y' \\ z' \end{bmatrix} = \frac{1}{15} \begin{bmatrix} 10 & -5 & 10 \\ -11 & -2 & 10 \\ -2 & -14 & -5 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

7. The columns and rows of the following matrices span a subspace of $\mathbb{R}^n$ ($n = 3, 4, 5$ or 7). Determine, using pivoted Gaussian elimination, the dimension of that space and a basis.

$$\text{a) } \begin{bmatrix} 1 & 5 & 0 \\ 1 & 2 & 3 \\ 1 & 3 & 2 \end{bmatrix} \quad \text{b) } \begin{bmatrix} 3 & 4 & -2 & 1 \\ 2 & 1 & 3 & -2 \\ 1 & -2 & 8 & -5 \end{bmatrix}$$

$$\text{c) } \begin{bmatrix} 1 & 2 & -3 & 1 & 5 \\ 3 & 0 & 3 & -3 & 3 \\ 2 & -1 & 4 & -3 & 1 \\ 0 & 2 & -4 & 2 & 4 \end{bmatrix} \quad \text{d) } \begin{bmatrix} 1 & 0 & -1 & -1 & 3 & 3 & 5 \\ 1 & 1 & 1 & 2 & 2 & -1 & 3 \\ 1 & 2 & 3 & 5 & 1 & -5 & 1 \\ 3 & 2 & 1 & 3 & 7 & 2 & 11 \end{bmatrix}$$

8. (*) A general rotated (orthonormal and positively oriented) basis $\{\mathbf{a}, \mathbf{b}, \mathbf{c}\}$ of $\mathbb{R}^3$ arises by rotating the basis $\{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\}$ about the x_1 -axis by the angle α , the result about the x_2 -axis by the angle β , and finally the result of these rotations about the x_3 -axis by the angle γ . Determine the vectors $\mathbf{a}, \mathbf{b}, \mathbf{c}$ in terms of the angles α, β, γ . For which values of the angles does one obtain a) the spherical-coordinate basis $\{\mathbf{e}_r, \mathbf{e}_\theta, \mathbf{e}_\varphi\}$ corresponding to spherical surface coordinates θ, φ , b) the rotated basis of Exercise 6a?
9. (*) Let $W \subset \mathbb{R}^4$ be the subspace spanned by the vectors $\mathbf{a} = [1, 1, 1, 1]$ and $\mathbf{b} = [1, 1, -1, -1]$. Determine the orthogonal projection of the vector $\mathbf{u} = [1, 2, -1, 3]$ onto the subspace W a) in the sense of the Euclidean inner product, b) in the sense of the inner product $\langle \mathbf{x}, \mathbf{y} \rangle = x_1y_1 + x_2y_2 + 2x_3y_3 + 3x_4y_4$. What kind of minimization problem of the form

$$\mathbf{x} \in W : f(\mathbf{x}) = \min!$$

do these orthogonal projections solve?

X.7 Affine maps. Geometric maps

Affine maps are maps somewhat more general than linear maps, of the type

$$\mathbf{f} : \mathbb{R}^n \rightarrow \mathbb{R}^m, \quad \mathbf{f}(\mathbf{x}) = \mathbf{A}\mathbf{x} + \mathbf{b},$$

where $\mathbf{A}$ is a matrix of size $m \times n$ and $\mathbf{b} \in \mathbb{R}^m$. Affine maps form their own important class in the middle ground between linear maps ($\mathbf{b} = \mathbf{0}$) and the more general *nonlinear* (= non-linear) maps. In their properties, affine maps are close to linear maps. For example, if $\mathbf{f}_1(\mathbf{x}) = \mathbf{A}_1\mathbf{x} + \mathbf{b}_1$ is an affine map of the type $\mathbf{f}_1 : \mathbb{R}^n \rightarrow \mathbb{R}^p$ and $\mathbf{f}_2(\mathbf{x}) = \mathbf{A}_2\mathbf{x} + \mathbf{b}_2$ is an affine map of the type $\mathbf{f}_2 : \mathbb{R}^p \rightarrow \mathbb{R}^m$, then the composite map $\mathbf{f}_2 \circ \mathbf{f}_1$ is defined and also affine:

$$(\mathbf{f}_2 \circ \mathbf{f}_1)(\mathbf{x}) = \mathbf{A}_2(\mathbf{A}_1\mathbf{x} + \mathbf{b}_1) + \mathbf{b}_2 = \mathbf{A}_2\mathbf{A}_1\mathbf{x} + (\mathbf{A}_2\mathbf{b}_1 + \mathbf{b}_2), \quad \mathbf{x} \in \mathbb{R}^n.$$

Likewise one sees that if $\mathbf{f}(\mathbf{x}) = \mathbf{A}\mathbf{x} + \mathbf{b}$ is of the type $\mathbf{f} : \mathbb{R}^n \rightarrow \mathbb{R}^n$, then $\mathbf{f}$ is invertible exactly when $\mathbf{A}$ is a nonsingular matrix, in which case the inverse map $\mathbf{f}^{-1}$ is also affine.

In the previous section it was noted that the coordinate transformation corresponding to a change of basis of $\mathbb{R}^n$ is a linear map. If the change of basis is also combined with a *shift of the origin*, then the coordinate transformation becomes an affine map.

EXAMPLE 1 Express as affine maps the coordinate transformation formulas between the coordinate systems $(O, \vec{i}, \vec{j}, \vec{k})$ and $(O', \vec{a}, \vec{b}, \vec{c})$ of E^3 , when in the former coordinate system $O' = (1, -2, 2)$ and

$$\vec{a} = \vec{i} + \vec{j} - \vec{k}, \quad \vec{b} = \vec{j} + 2\vec{k}, \quad \vec{c} = \vec{i} + \vec{k}.$$

Solution If the coordinates of the point P are (x, y, z) and (x', y', z') , then it must hold that

$$x\vec{i} + y\vec{j} + z\vec{k} = \overrightarrow{OO'} + x'\vec{a} + y'\vec{b} + z'\vec{c}.$$

This can be interpreted as a vector-form transformation formula $(x', y', z') \mapsto (x, y, z)$, whose matrix form is an affine map:

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ -2 \\ 2 \end{bmatrix} + \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ -1 & 2 & 1 \end{bmatrix} \begin{bmatrix} x' \\ y' \\ z' \end{bmatrix}.$$

Solving this for $[x', y', z']^T$ gives the inverse transformation

$$\begin{bmatrix} x' \\ y' \\ z' \end{bmatrix} = \frac{1}{4} \begin{bmatrix} 1 & 2 & -1 \\ -1 & 2 & 1 \\ 3 & -2 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} + \frac{1}{4} \begin{bmatrix} 5 \\ 3 \\ -9 \end{bmatrix}. \quad \square$$

Geometric maps

By *geometric maps* one generally means maps of the Euclidean point space E^2 or E^3 that have some (relatively simple) geometric meaning. By reducing to coordinate space, geometric maps can always be interpreted as maps of $\mathbb{R}^2$ or $\mathbb{R}^3$ as well. In what follows we restrict ourselves to the simplest geometric maps, which are affine. These include, among others, the following maps $P \mapsto Q$ of the type $f: E^2 \rightarrow E^2$ or $f: E^3 \rightarrow E^3$, or compositions of these.

- *shift* i.e. *translation*: $\overrightarrow{OQ} = \overrightarrow{OP} + \vec{b}$ ($\vec{b}$ given)
- *rotation* in the plane about the origin: $\overrightarrow{OQ} = \overrightarrow{OP}$ rotated by the angle φ counterclockwise ($\varphi \in \mathbb{R}$ given)
- *parallel projection* onto a spatial plane T (onto a line S in the plane) in the direction of the line L , or of the vector $\vec{v}$ = the direction vector of L : $Q \in T$ or $Q \in S$ and $\overrightarrow{PQ} = \lambda \vec{v}$, $\lambda \in \mathbb{R}$ ($\vec{v}$ not parallel to T/S)
- *parallel projection onto a spatial line* S in the direction of the plane T , or of the vectors $\vec{v}_1, \vec{v}_2$ = the direction vectors of T : $Q \in S$ and $\overrightarrow{PQ} = \lambda \vec{a}_1 + \mu \vec{a}_2$, $(\lambda, \mu) \in \mathbb{R}^2$ ($\{\vec{v}, \vec{a}_1, \vec{a}_2\}$ linearly independent, $\vec{v}$ = the direction vector of S)
- *reflection* with respect to a given plane (in space) or line (in the plane)
- *scaling* i.e. *homothety* with respect to the origin: $\overrightarrow{OQ} = \lambda \overrightarrow{OP}$ ($\lambda \in \mathbb{R}$ given)

The simplest of the geometric maps are the translation and the homothety, whose affine forms are $\mathbf{f}(\mathbf{x}) = \mathbf{I}\mathbf{x} + \mathbf{b}$ and $\mathbf{f}(\mathbf{x}) = \lambda \mathbf{I}\mathbf{x}$ ($\mathbf{I}$ the identity matrix). A general affine map $\mathbf{f}(\mathbf{x}) = \mathbf{A}\mathbf{x} + \mathbf{b}$ is a composition of a linear map and a translation:

$$\mathbf{x} \mapsto \mathbf{A}\mathbf{x} \mapsto \mathbf{A}\mathbf{x} + \mathbf{b}.$$

If in the parallel projection the plane T /line S , or in the projection onto a line the line S , passes through the origin, then $\mathbf{f}(\mathbf{0}) = \mathbf{0}$, so the projection is a linear map. Special cases of these projections are the orthogonal projections, in which one projects perpendicularly onto the plane/line.

EXAMPLE 2 The plane T passes through the origin and its direction vectors are $\vec{v}_1 = \vec{i} + \vec{j}$ and $\vec{v}_2 = \vec{j} + \vec{k}$. The point $P \hat{=} x_1\vec{i} + x_2\vec{j} + x_3\vec{k} \hat{=} [x_1, x_2, x_3]^T = \mathbf{x}$ is projected perpendicularly to the point of the plane T

$$Q \hat{=} y_1\vec{i} + y_2\vec{j} + y_3\vec{k} = z_1\vec{v}_1 + z_2\vec{v}_2.$$

Determine the matrices $\mathbf{A}, \mathbf{B}$ of the corresponding linear maps

$$A: \mathbf{x} \mapsto \mathbf{y} = [y_1, y_2, y_3]^T, \quad B: \mathbf{x} \mapsto \mathbf{z} = [z_1, z_2]^T$$

.

Solution The projection condition is

$$(x_1\vec{i} + x_2\vec{j} + x_3\vec{k} - z_1\vec{v}_1 - z_2\vec{v}_2) \cdot \vec{v}_i = 0, \quad i = 1, 2$$

$$\Leftrightarrow \begin{cases} x_1 + x_2 - 2z_1 - z_2 = 0 \\ x_2 + x_3 - z_1 - 2z_2 = 0 \end{cases}$$

$$\Leftrightarrow \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} z_1 \\ z_2 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

$$\Leftrightarrow \mathbf{C}\mathbf{z} = \mathbf{D}\mathbf{x}.$$

Thus $\mathbf{z} = \mathbf{B}\mathbf{x}$, where

$$\mathbf{B} = \mathbf{C}^{-1}\mathbf{D} = \frac{1}{3} \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 2 & 1 & -1 \\ -1 & 1 & 2 \end{bmatrix}.$$

Since

$$y_1\vec{i} + y_2\vec{j} + y_3\vec{k} = z_1\vec{v}_1 + z_2\vec{v}_2 = z_1\vec{i} + (z_1 + z_2)\vec{j} + z_2\vec{k},$$

we have

$$\begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} z_1 \\ z_2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 0 & 1 \end{bmatrix} \mathbf{B} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}.$$

Thus $\mathbf{y} = \mathbf{A}\mathbf{x}$, where

$$\mathbf{A} = \frac{1}{3} \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 2 & 1 & -1 \\ -1 & 1 & 2 \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 2 & 1 & -1 \\ 1 & 2 & 1 \\ -1 & 1 & 2 \end{bmatrix}.$$

Check: Applying Gaussian elimination to the system $\mathbf{A}\mathbf{x} = \mathbf{0}$ gives the solution $\mathbf{x} = t[1, -1, 1]^T$, $t \in \mathbb{R}$. This is consistent with the fact that the normal vector of the plane T is $\vec{n} = \vec{i} - \vec{j} + \vec{k}$. $\square$

By means of the orthogonal projection, the reflection map can also be defined easily: If P_0 is the projection point of P , then $\overrightarrow{OQ} = \overrightarrow{OP} + 2\overrightarrow{PP}_0$ determines the mirror image Q .

EXAMPLE 2 (continued) The plane T passes through the point $(1, -1, 2)$ and its direction vectors are $\vec{v}_1 = \vec{i} + \vec{j}$ and $\vec{v}_2 = \vec{j} + \vec{k}$. Determine the reflection map $(x_1, x_2, x_3) \triangleq P \mapsto Q \triangleq (y_1, y_2, y_3)$ as an affine map $\mathbf{y} = \mathbf{A}\mathbf{x} + \mathbf{b}$.

Solution Denote $\mathbf{a} = [1, -1, 2]^T$ and let the projection point be $P_0 \triangleq (z_1, z_2, z_3)$. Then $\mathbf{z} - \mathbf{a} = A(\mathbf{x} - \mathbf{a})$, where A is the linear map computed in the previous example. Denoting its matrix now by the symbol $\mathbf{A}_0$, we thus have

$$\mathbf{z} - \mathbf{a} = \mathbf{A}_0(\mathbf{x} - \mathbf{a}), \quad \mathbf{A}_0 = \frac{1}{3} \begin{bmatrix} 2 & 1 & -1 \\ 1 & 2 & 1 \\ -1 & 1 & 2 \end{bmatrix}.$$

Consequently

$$\begin{aligned} \mathbf{y} &= \mathbf{x} + 2(\mathbf{z} - \mathbf{x}) = (2\mathbf{A}_0 - \mathbf{I})\mathbf{x} + 2(\mathbf{I} - \mathbf{A}_0)\mathbf{a} \\ &= \frac{1}{3} \begin{bmatrix} 1 & 2 & -2 \\ 2 & 1 & 2 \\ -2 & 2 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \frac{8}{3} \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} \\ &= \mathbf{A}\mathbf{x} + \mathbf{b}. \quad \square \end{aligned}$$

EXAMPLE 3 Interpret the rotation in the plane as a linear map $\mathbf{x} \mapsto \mathbf{y} = \mathbf{A}\mathbf{x}$.

Solution The rotation of a vector in the plane by a given angle φ can be interpreted either as a linear map of the type $A: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ or as a map of the complex plane $z = x_1 + ix_2 \mapsto w = y_1 + iy_2$. By the latter interpretation,

$$\begin{aligned} y_1 + iy_2 &= (\cos \varphi + i \sin \varphi)(x_1 + ix_2) \\ &= (x_1 \cos \varphi - x_2 \sin \varphi) + i(x_1 \sin \varphi + x_2 \cos \varphi), \end{aligned}$$

so the matrix of A is

$$\mathbf{A} = \begin{bmatrix} \cos \varphi & -\sin \varphi \\ \sin \varphi & \cos \varphi \end{bmatrix}. \quad \square$$

The *rotation of a point about a spatial line* can also be interpreted as an affine map: If P_0 is the perpendicular projection of P onto the line S (a special case of the parallel projection onto a spatial line), then $\overrightarrow{OQ} = \overrightarrow{OP_0} + \overrightarrow{P_0Q}$, where $\overrightarrow{P_0Q} =$ the vector $\overrightarrow{P_0P}$ rotated in the plane passing through the point P_0 whose normal is S .

EXAMPLE 3 (continued) When the vector $\vec{r} = x_1\vec{i} + x_2\vec{j} + x_3\vec{k}$ is rotated about the x_1 -axis, the component x_1 remains unchanged and $[x_2, x_3]^T$ transforms as

in a plane rotation. Consequently, the rotation corresponds to the linear map $\mathbf{f}(\mathbf{x}) = \mathbf{A}\mathbf{x}$, where

$$\mathbf{A} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \varphi & -\sin \varphi \\ 0 & \sin \varphi & \cos \varphi \end{bmatrix}. \quad \square$$

Compositions of translation, plane or spatial rotation, reflection, and scaling in E^n ($n = 2$ or $n = 3$) are called *Euclidean maps**. Two point sets $K_1 \subset E^2$ and $K_2 \subset E^2$ of the Euclidean plane (these may be geometric figures such as triangles) are said to be *similar* if there exists a Euclidean map $\mathbf{f} : E^2 \rightarrow E^2$ such that $K_2 = \mathbf{f}(K_1)$. If $\mathbf{f}$ consists only of translations, rotations, and reflections, then K_1 and $\mathbf{f}(K_2)$ are said to be *congruent*.

*Geometry of $\mathbb{R}^n$

Geometric thinking can be extended generally to the space $\mathbb{R}^n$, in which case the elements of $\mathbb{R}^n$ are the counterparts of the points of the (imagined) Euclidean point space E^n . The expression 'at the point $\mathbf{x}$ ' is then to be interpreted so that the point $P \in E^n$ and its counterpart $\mathbf{x} \in \mathbb{R}^n$ (= the coordinates of P in a chosen orthonormal basis of $\mathbb{R}^n$) are identified, or the latter is interpreted as the 'name' of the former. Once this way of thinking is adopted, the points, line segments, triangles, etc. of geometry can be naturally carried directly into the space $\mathbb{R}^n$.

In the following we list some of the most common concepts of geometry as carried into and generalized to the space $\mathbb{R}^n$:

- A *point* is an element $\mathbf{a} \in \mathbb{R}^n$.
- A *line* is the point set $S = \{\mathbf{x} \in \mathbb{R}^n \mid \mathbf{x} = \mathbf{x}_0 + t\mathbf{v}, t \in \mathbb{R}\}$, where $\mathbf{x}_0, \mathbf{v} \in \mathbb{R}^n$ and $\mathbf{v} \neq \mathbf{0}$.
- A *hyperplane* (a *plane* in the case $n = 3$) is the point set $T = \{\mathbf{x} \in \mathbb{R}^n \mid \mathbf{x} = \mathbf{x}_0 + \sum_{i=1}^{n-1} t_i \mathbf{v}_i, \mathbf{t} = (t_i) \in \mathbb{R}^{n-1}\}$, where $\mathbf{x}_0, \mathbf{v}_i \in \mathbb{R}^n$ and $\{\mathbf{v}_1, \dots, \mathbf{v}_{n-1}\}$ is linearly independent.
- An *m-plane* (an *m-dimensional plane*) is the point set $T = \{\mathbf{x} \in \mathbb{R}^n \mid \mathbf{x} = \mathbf{x}_0 + \sum_{i=1}^m t_i \mathbf{v}_i, \mathbf{t} = (t_i) \in \mathbb{R}^m\}$, where $2 \leq m \leq n-1$, $\mathbf{x}_0, \mathbf{v}_i \in \mathbb{R}^n$ and $\{\mathbf{v}_1, \dots, \mathbf{v}_m\}$ is linearly independent.

*A map $K \mapsto \mathbf{f}(K)$, where $\mathbf{f}$ is a composition of translations and rotations only, is called a *Euclidean motion*, and in mechanics (when $K \subset E^3$ represents a solid body) a rigid-body motion.

- A *line segment* is the point set $S = \{\mathbf{x} \in \mathbb{R}^n \mid \mathbf{x} = \lambda \mathbf{a}_1 + (1 - \lambda) \mathbf{a}_2, \lambda \in [0, 1]\}$,
where $\mathbf{a}_1, \mathbf{a}_2 \in \mathbb{R}^n$ and $\mathbf{a}_1 \neq \mathbf{a}_2$.
- A *triangle* is the point set
 $K = \{\mathbf{x} \in \mathbb{R}^n \mid \mathbf{x} = \sum_{i=1}^3 \lambda_i \mathbf{a}_i, \lambda_i \in [0, 1], \sum_{i=1}^3 \lambda_i = 1\}$, where
 $\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3 \in \mathbb{R}^n$ and $\{\mathbf{a}_2 - \mathbf{a}_1, \mathbf{a}_3 - \mathbf{a}_1\}$ is linearly independent.
- An *m-simplex* ($2 \leq m \leq n + 1$) is the point set
 $K = \{\mathbf{x} \in \mathbb{R}^n \mid \mathbf{x} = \sum_{i=1}^m \lambda_i \mathbf{a}_i, \lambda_i \in [0, 1], \sum_{i=1}^m \lambda_i = 1\}$, where
 $\mathbf{a}_i \in \mathbb{R}^n$ and $\{\mathbf{a}_2 - \mathbf{a}_1, \dots, \mathbf{a}_m - \mathbf{a}_1\}$ is linearly independent.
- An *m-parallelepiped* ($2 \leq m \leq n$) is the point set
 $K = \{\mathbf{x} \in \mathbb{R}^n \mid \mathbf{x} = \mathbf{x}_0 + \sum_{i=1}^m \lambda_i \mathbf{a}_i, \lambda_i \in [0, 1]\}$, where $\mathbf{x}_0, \mathbf{a}_i \in \mathbb{R}^n$ and
 $\{\mathbf{a}_1, \dots, \mathbf{a}_m\}$ is linearly independent
- A *ball* is the point set $B(\mathbf{x}_0, R) = \{\mathbf{x} \in \mathbb{R}^n \mid |\mathbf{x} - \mathbf{x}_0| \leq R\}$, where
 $\mathbf{x}_0 \in \mathbb{R}^n$ and $R > 0$.
- A *sphere* is the point set $K = \{\mathbf{x} \in \mathbb{R}^n \mid |\mathbf{x} - \mathbf{x}_0| = R\}$, where
 $\mathbf{x}_0 \in \mathbb{R}^n$ and $R > 0$.
- An *n-rectangle* (oriented along the coordinate axes) is the point set
 $K = \{\mathbf{x} = (x_i) \in \mathbb{R}^n \mid x_i \in [a_i, b_i], i = 1 \dots n\}$, where
 $a_i, b_i \in \mathbb{R}, a_i < b_i, i = 1 \dots n$.

By these definitions, a hyperplane = an $(n - 1)$ -dimensional plane, a 2-simplex = a line segment, a 3-simplex = a triangle, and a 4-simplex of $\mathbb{R}^3$ = a tetrahedron. In the definitions of the hyperplane and the m -plane, the vectors $\mathbf{v}_i$ are called the *direction vectors* of that plane. In the definition of a simplex, the points $\mathbf{a}_i$ are the *vertices* of the simplex. If in an n -rectangle $b_i - a_i = a, i = 1 \dots n$, then it is an *n-cube* (edge length a).

In the above definition, the hyperplane and the m -plane are presented in *parametric* form by means of their direction vectors $\mathbf{v}_i$. For a hyperplane one can also write the *equation of the hyperplane* in the form

$$T: \langle \mathbf{x} - \mathbf{x}_0, \mathbf{n} \rangle = \mathbf{n}^T (\mathbf{x} - \mathbf{x}_0) = 0,$$

where the vector $\mathbf{n}$ is the *normal* of T , i.e. perpendicular to the vectors $\mathbf{v}_i$. Such an $\mathbf{n}$ can be found, e.g. by means of the orthogonal projection, cf. Example 7 in the previous section. More generally, an m -plane can be defined by the system

$$T: \langle \mathbf{x} - \mathbf{x}_0, \mathbf{n}_i \rangle = 0, \quad i = 1 \dots n - m,$$

where the vectors $\mathbf{n}_i$ are mutually linearly independent normal vectors of T . Geometric problems involving hyperplanes and m -planes can typically be solved by means of matrix algebra.

EXAMPLE 4 Under what condition do the hyperplanes

$$T_i : \langle \mathbf{x} - \mathbf{a}_i, \mathbf{n}_i \rangle = 0, \quad i = 1, \dots, n$$

intersect in exactly one point?

Solution By matrix algebra, the equations of the planes T_i form a linear system $\mathbf{Ax} = \mathbf{b}$, where the i th row of $\mathbf{A} = \mathbf{n}_i^T$ and $[\mathbf{b}]_i = \langle \mathbf{a}_i, \mathbf{n}_i \rangle$. So the answer is: exactly on the condition that $\mathbf{A}$ is nonsingular, i.e. that the normal vectors $\mathbf{n}_i$ form a linearly independent system (a basis of $\mathbb{R}^n$). $\square$

*Linear optimization in $\mathbb{R}^n$

The general form of a real-valued linear map $L : \mathbb{R}^n \rightarrow \mathbb{R}$ of $\mathbb{R}^n$ is

$$L\mathbf{x} = \mathbf{a}^T \mathbf{x},$$

where $\mathbf{a} \in \mathbb{R}^n$ (assume henceforth: $\mathbf{a} \neq \mathbf{0}$). Finding the maximum or minimum value of such a function on a given set $K \subset \mathbb{R}^n$ is called a *linear optimization* problem. If T is a hyperplane whose equation is

$$T : \langle \mathbf{a}, \mathbf{x} - \mathbf{x}_0 \rangle = \mathbf{a}^T (\mathbf{x} - \mathbf{x}_0) = 0,$$

then $L\mathbf{x} = \mathbf{a}^T \mathbf{x}_0 = \text{constant}$ when $\mathbf{x} \in T$, i.e. such hyperplanes are the *level surfaces* of L . On the other hand, if one considers the values of L along the line passing through the point $\mathbf{c}$ and perpendicular to the level surfaces

$$S : \mathbf{x} = \mathbf{c} + t\mathbf{a}, \quad t \in \mathbb{R},$$

then one sees that on this line

$$L\mathbf{x} = L(\mathbf{c} + t\mathbf{a}) = \alpha t + \beta, \quad \alpha = |\mathbf{a}|^2 > 0, \quad \beta = \mathbf{a}^T \mathbf{c}.$$

If now, in the equation of the level surfaces of L , one chooses $\mathbf{x}_0 = \mathbf{c} + t\mathbf{a} \in S$, and denotes this level surface (hyperplane) by T_t , then by varying the parameter t the whole space $\mathbb{R}^n$ becomes 'swept' by the hyperplanes T_t . One then sees immediately that if L has a maximum or minimum value on the point set K ,

then it must be, respectively, the smallest / largest number in the set of real numbers

$$A = \{t \in \mathbb{R} \mid T_t \cap K \neq \emptyset\}.$$

The optimization problem is thus 'solved' geometrically. — Applying the idea in practice does, however, require that the set A can be determined. In the following example K is geometrically simple and n moderate, so the problem can be solved 'by hand'.

EXAMPLE 5 Determine the smallest and largest value of the function $f(x_1, x_2, x_3, x_4) = 2x_1 - 4x_2 + 5x_3 - 6x_4$ in the ball of $\mathbb{R}^4$

$$K = \{\mathbf{x} \in \mathbb{R}^4 \mid |\mathbf{x} - \mathbf{x}_0| \leq R\}, \quad \text{where } \mathbf{x}_0 = (1, -1, 3, 2) \text{ and } R = 3.$$

Solution Here $f(\mathbf{x}) = L\mathbf{x} = \mathbf{a}^T \mathbf{x}$, $\mathbf{a} = [2, -4, 5, -6]^T$, $\mathbf{x} = [x_1, x_2, x_3, x_4]^T$. If T_t is the hyperplane of $\mathbb{R}^4$ passing through the point $P_t \hat{=} \mathbf{x}_0 + t\mathbf{a}$, then

$$T_t \cap K \neq \emptyset \quad \Leftrightarrow \quad |t\mathbf{a}| = |t||\mathbf{a}| \leq 3 \quad \Leftrightarrow \quad |t| \leq \frac{3}{\sqrt{2^2 + 4^2 + 5^2 + 6^2}} = \frac{1}{3}.$$

Thus $\{t \in \mathbb{R} \mid T_t \cap K \neq \emptyset\} = [-1/3, 1/3]$. Since on the hyperplane T_t

$$f(\mathbf{x}) = \mathbf{a}^T(\mathbf{x}_0 + t\mathbf{a}) = \mathbf{a}^T \mathbf{x}_0 + |\mathbf{a}|^2 t = 9 + 81t, \quad \mathbf{x} \in T_t,$$

then

$$f_{\min} = 9 + 81 \cdot \left(-\frac{1}{3}\right) = \underline{\underline{-18}}, \quad f_{\max} = 9 + 81 \cdot \frac{1}{3} = \underline{\underline{36}}. \quad \square$$

Very common in applications is the linear optimization problem in which K is defined by means of *linear inequalities* in the form

$$K = \{\mathbf{x} \in \mathbb{R}^n \mid \mathbf{a}_i^T \mathbf{x} \leq c_i, \quad i = 1 \dots m\},$$

where $\mathbf{a}_i \in \mathbb{R}^n$, $\mathbf{a}_i \neq \mathbf{0}$, and $c_i \in \mathbb{R}$. The set K can be characterized geometrically as a *polytope* bounded by the hyperplanes $T_i : \mathbf{a}_i^T \mathbf{x} = c_i$ (usually $m > n$). Such a problem has traditionally been called a *linear programming* problem. Algorithms have been developed both for the direct (in principle exact) and for the approximate iterative solution of the problem.

EXERCISES

1. What does a) the rectangle with vertices at the points $(1, 1)$, $(3, 1)$, $(3, 2)$ and $(1, 2)$, b) the unit circle $S : x^2 + y^2 = 1$ become under the affine map

$$\mathbf{f}(x, y) = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} 5 \\ 6 \end{bmatrix} ?$$

Draw a figure! Determine also the equation of the curve $S' = \mathbf{f}(S)$.

2. Express as an affine map: $\mathbf{f} : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ or $\mathbf{f} : \mathbb{R}^3 \rightarrow \mathbb{R}^3$:
- The perpendicular projection in the plane onto the line $x + 2y + 1 = 0$
 - The projection onto the line $x - 3y - 2 = 0$ in the direction of the line $y = 2x$
 - The rotation in the plane about the point $(2, 3)$ by 60° counterclockwise
 - The projection onto the plane $T : x + 2z - 3y + 3 = 0$ in the direction of the vector $\vec{i} + \vec{j} - \vec{k}$
 - The reflection with respect to the plane $T : 3x - 2y + z - 5 = 0$
 - The reflection with respect to the spatial line $S : \vec{r} = (1 + t)\vec{i} + (t - 2)\vec{j} + (2t - 3)\vec{k}$
3. The plane T passes through the origin and its direction vectors are $\vec{a}_1 = \vec{i} + \vec{j} + \vec{k}$ and $\vec{a}_2 = \vec{j} - 2\vec{k}$. The line S passes through the point $(0, 1, 2)$ and its direction vector is $\vec{a}_3 = -\vec{i} + \vec{j} + \vec{k}$.
- Let $\mathbf{y} = [y_1, y_2, y_3]^T$ be the coordinates of the vector $x_1\vec{i} + x_2\vec{j} + x_3\vec{k}$ in the basis $\{\vec{a}_1, \vec{a}_2, \vec{a}_3\}$. Determine the linear map $\mathbf{x} \mapsto \mathbf{y}$, i.e. the matrix $\mathbf{C}$ such that $\mathbf{y} = \mathbf{C}\mathbf{x}$.
 - Using the result of part a) and matrix algebra, determine the affine map $\mathbf{f}(\mathbf{x}) = \mathbf{A}\mathbf{x} + \mathbf{b}$ that carries out the parallel projection of the point $P = (x_1, x_2, x_3)$ onto the line S in the direction of the plane T ($\mathbf{f} : \mathbb{R}^3 \rightarrow \mathbb{R}^3$).
 - Define an affine map of the type $\mathbf{f} : \mathbb{R}^3 \rightarrow \mathbb{R}^2$ that carries out the projection of the point $P = (x_1, x_2, x_3)$ in the direction of the line S to the point Q of the plane T , expressed in coordinates in the basis $\{\mathbf{a}_1, \mathbf{a}_2\}$.
4. The vector $\vec{b}$ is obtained from the vector $\vec{a} = 2\vec{i} + \vec{j} + 2\vec{k}$ by first rotating $\vec{a}$ about the x -axis by the angle 30° and then rotating the resulting vector further about the y -axis by 60° (the rotations counterclockwise as seen from the direction of the positive coordinate axes). Determine the matrices of both rotation operations (linear maps), from these the matrix $\mathbf{A}$ of the entire operation, and from $\mathbf{A}$ the vector $\vec{b}$. Which vector, when rotated in the stated ways, yields as the final result $\vec{b} = \vec{k}$?

5. Show that the linear map

$$\mathbf{f}(x, y) = \frac{1}{25} \begin{bmatrix} 7 & -24 \\ -24 & -7 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

is a reflection with respect to some line, and determine the equation of the line.

6. The vertices of triangle A are at the points $(0, 0)$, $(1, 0)$ and $(0, 1)$, and the vertices of triangle B are at the points $(-1, 1)$, $(-2, 2)$ and $(-3, 0)$. Determine all triangles C for which $C = \mathbf{f}(B)$ and $B = \mathbf{f}(A)$, where $\mathbf{f}$ is an affine map of the plane.
7. The following linear maps define the parallel projection onto the plane T passing through the origin, in the direction of the line S . Determine the equation of T and the direction vector of S .

$$\text{a) } \begin{bmatrix} x' \\ y' \\ z' \end{bmatrix} = \frac{1}{5} \begin{bmatrix} 4 & 1 & -3 \\ 1 & 4 & 3 \\ -1 & 1 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

$$\text{b) } \begin{bmatrix} x' \\ y' \\ z' \end{bmatrix} = \frac{1}{10} \begin{bmatrix} 7 & -6 & -9 \\ -2 & 6 & -6 \\ -1 & -2 & 7 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

8. Show that if an affine transformation $\mathbf{f} : \mathbb{R}^n \rightarrow \mathbb{R}^n$, where $n = 2$ or $n = 3$, is invertible, then $\mathbf{f}$
- a) maps a line to a line ($n = 2, 3$) and a plane to a plane ($n = 3$)
 - b) maps a line segment to a line segment and a triangle to a triangle
 - c) maps a tetrahedron to a tetrahedron ($n = 3$),
 - d) preserves the parallelism of lines and planes,
 - e) preserves the ratio of the lengths of two parallel line segments.
 - f) How does property c) change if $\mathbf{f}$ is not invertible?
9. At which point of the set $A \subset \mathbb{R}^4$ does the function

$$f(\mathbf{x}) = x_1 - 2x_2 - x_3 + 2x_4$$

attain its largest value, when A is

- a) the unit cube: $A = \{\mathbf{x} \in \mathbb{R}^4 \mid |x_i| \leq 1, i = 1 \dots 4\}$,
- b) the unit ball: $A = \{\mathbf{x} \in \mathbb{R}^4 \mid x_1^2 + \dots + x_4^2 \leq 1\}$?

10. (*) Determine the affine map $\mathbf{f} : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ that carries out the rotation about the line $S : x = 1 + 6t, y = 1 - 3t, z = -2t$, by the angle φ counterclockwise as seen from the direction $\vec{v} = 6\vec{i} - 3\vec{j} - 2\vec{k}$. *Hint:* Determine $\mathbf{f}$ first in a coordinate system in which S is a coordinate axis — see Exercise X.6: 6a!
11. (*) Solve the linear optimization problem $f(x, y, z) = x + 2y + 3z = \max!$ subject to the conditions

$$\begin{cases} x \geq 0, y \geq 0, z \geq 0, \\ 2x + y - 2z \leq 6, \\ -x + 5y + z \leq 8, \\ 4x + 2y + 7z \leq 23. \end{cases}$$

Hint: It suffices to examine the vertices of the polytope!

12. (*) (Home for the evening) From an airplane, a businessman sees his home house in the direction $-2\vec{i} - 3\vec{j} - \vec{k}$. The base of the house is a rectangular box of height = 1 and whose corners on the ground are at the points $(0, 0)$, $(3, 0)$, $(3, 2)$ and $(0, 2)$ of the xy -plane. The house has a symmetric gable roof, ridge height = 2 and direction = $\vec{i}$. The eaves on the sides and ends of the house extend to a distance 0.2 from the walls. Define an affine map $\mathbf{f} : \mathbb{R}^3 \mapsto \mathbb{R}^2$ that transforms the three-dimensional reality into the projection image seen by the businessman (assume a natural head position). Draw a picture of the house as the businessman sees it.

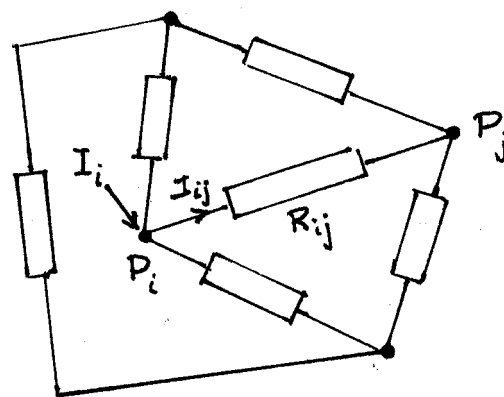
X.8 *Traditional application examples of linear systems

For large-scale scientific-technical computation performed by computers — for example numerical weather forecasting — it is typical that a considerable portion of the computing time used is spent solving large linear systems of equations. Systems of equations arise when the natural laws under consideration (in weather forecasting, the fundamental equations of fluid mechanics) are *discretized*, i.e. converted into a form suitable for numerical computation. Reducing the computation specifically to linear systems of equations is common in models also because a computer, in the final analysis, essentially only 'knows how' to solve numerically basic problems of this type.

Alongside modern scientific-technical computational tasks, the problem of a large linear system is also central in many more traditional — often much older than the computer — mathematical models that occur especially in the engineering sciences. In the following, two such classical application examples are presented, one from electrical engineering and one from structural mechanics.

Electric circuit: A resistor network

Consider a classical electric circuit that is a network formed of resistors only. Let the nodes of the network be P_i , $i = 1 \dots n$, and let the nodes P_i, P_j be connected by a resistor R_{ij} (if there is no direct connection, one sets $1/R_{ij} = 0$ in the system (3) below). The adjacent figure shows a general four-node network.



If we denote I_{ij} = the current through the resistor R_{ij} , positive in the direction away from P_i (cf. the figure), then by Kirchhoff's law (**G.R. Kirchhoff**, 1845)

$$\sum_{j=1}^n I_{ij} = I_i, \quad i = 1 \dots n, \quad (1)$$

where I_i = the current fed into node P_i from outside. On the other hand, by Ohm's law (**G.S. Ohm**, 1826)

$$R_{ij}I_{ij} = V_i - V_j, \quad i, j = 1 \dots n, \quad (2)$$

where V_i = the voltage at node P_i . Using Ohm's law (2), the currents I_{ij} in the system (1) can be expressed in terms of the voltages, giving the system

$$\sum_{j=1}^n a_{ij}V_j = I_i, \quad i = 1 \dots n, \quad (3)$$

where

$$a_{ii} = \sum_{j=1}^n (1/R_{ij}), \quad a_{ij} = -1/R_{ij}, \quad i \neq j.$$

This is a linear system of size $n \times n$, where $x_i = V_i$, $b_i = I_i$.

EXAMPLE 1 If in a four-node network (cf. the figure above) $R_{ij} = R$, $i, j = 1 \dots 4$, $i \neq j$, then for the voltages V_i one obtains the system

$$\begin{cases} 3V_1 - V_2 - V_3 - V_4 = RI_1 \\ -V_1 + 3V_2 - V_3 - V_4 = RI_2 \\ -V_1 - V_2 + 3V_3 - V_4 = RI_3 \\ -V_1 - V_2 - V_3 + 3V_4 = RI_4 \end{cases}$$

This is a singular system: If, e.g. $I_1 = I \neq 0$ and $I_2 = I_3 = I_4 = 0$, then there is no solution. If also $I_1 = 0$, then $V_i = V$, $i = 1 \dots 4$ is a solution for any $V \in \mathbb{R}$, i.e. there are infinitely many solutions. $\square$

The singularity problem of the example is a property of the system (3) more generally. Indeed, from the expressions for the coefficients a_{ij} above one sees that

$$\sum_{j=1}^n a_{kj} = \sum_{i=1}^n a_{ik} = 0, \quad k = 1 \dots n.$$

There are two consequences of this: First, if the equations of the system (3) are added together side by side, one obtains

$$\left(\sum_{i=1}^n a_{i1}\right)V_1 + \dots + \left(\sum_{i=1}^n a_{in}\right)V_n = \sum_{i=1}^n I_i,$$

i.e.

$$0 = \sum_{i=1}^n I_i. \quad (4)$$

Second, one sees that if $\{V_i\}$ is a solution of the system (4), then also $\{V_i + V\}$ is a solution for any $V \in \mathbb{R}$, since

$$\sum_{j=1}^n a_{ij}(V_j + V) = \sum_{j=1}^n a_{ij}V_j + \left(\sum_{j=1}^n a_{ij}\right)V = I_i + 0 = I_i, \quad i = 1 \dots n.$$

We have thus deduced: (a) If the system (3) has a solution, then condition (4) holds, i.e. this is a necessary condition for solvability. (b) If the system (3) is solvable, then there are infinitely many solutions. The system (3) is thus generally singular.

The singularity problem is a sign that the mathematical model has not been thought through (from the physical standpoint) quite to the end. — There is indeed a simple 'electrician's solution' to the problem: Ground one node P_k of the network. In the mathematical model this means the conditions to be imposed at node P_k

$$V_k = V_{ref}, \quad I_k = -\sum_{i \neq k} I_i. \quad (5)$$

The first condition assigns the voltage V_k some arbitrary reference value V_{ref} , e.g. $V_{ref} = 0$. This is permissible, because only voltage differences are physically significant. In the second condition, I_k can be interpreted as the grounding current, which is determined physically so that the balance condition (4) of the supply currents (also a physical condition!) holds. The condition $V_k = V_{ref}$ removes V_k from the set of unknowns, so the system (3) can be written in the form

$$\sum_{j \neq k} a_{ij}V_j = I_i - a_{ik}V_{ref} = J_i, \quad i = 1 \dots n.$$

From this, the k th equation can further be removed as unnecessary. Indeed, since

$$\sum_{i=1}^n a_{ij} = 0 \quad \Rightarrow \quad a_{kj} = -\sum_{i \neq k} a_{ij}, \quad j = 1 \dots n,$$

then, assuming the other equations ($i \neq k$) of the system (3) and the latter part of condition (5) to hold, one deduces

$$\begin{aligned} \sum_{j=1}^n a_{kj}V_j &= -\sum_{j=1}^n \left(\sum_{i \neq k} a_{ij}\right)V_j \\ &= -\sum_{i \neq k} \left(\sum_{j=1}^n a_{ij}V_j\right) = -\sum_{i \neq k} I_i = I_k. \end{aligned}$$

(The interchange of summation order made here is based on the commutative law of addition.) By this result, the k th equation is a consequence of the other equations and the additional condition (5), hence unnecessary.

After the above operations, the mathematical model of the electric circuit has reduced to a linear system of size $(n-1) \times (n-1)$. In the case $V_{ref} = 0$, the reduced model is obtained simply by removing from the array form of the system the k th column of the array (corresponding to the condition $V_k = 0$) and the k th row (removal of the k th equation). The model then takes the form

$$\sum_{j \neq k} a_{ij} V_j = I_i, \quad i = 1 \dots n, \quad i \neq k. \quad (6)$$

It turns out (the detailed justification is omitted) that the system (6) is always nonsingular.

EXAMPLE 1 (continued) If the grounding condition $V_4 = 0$ is imposed at node P_4 , then in the case $I_1 = I$, $I_2 = I_3 = 0$ the reduced system has the

$$\begin{cases} 3V_1 - V_2 - V_3 = RI \\ -V_1 + 3V_2 - V_3 = 0 \\ -V_1 - V_2 + 3V_3 = 0 \end{cases}$$

(unique) solution

$$V_1 = \frac{1}{2} RI, \quad V_2 = \frac{1}{4} RI, \quad V_3 = \frac{1}{4} RI.$$

From the solution, the effective resistance (total resistance) between nodes P_1 and P_4 is

$$R_{eff} = \frac{V_1 - V_4}{I} = \frac{1}{2} R. \quad \square$$

Truss structure

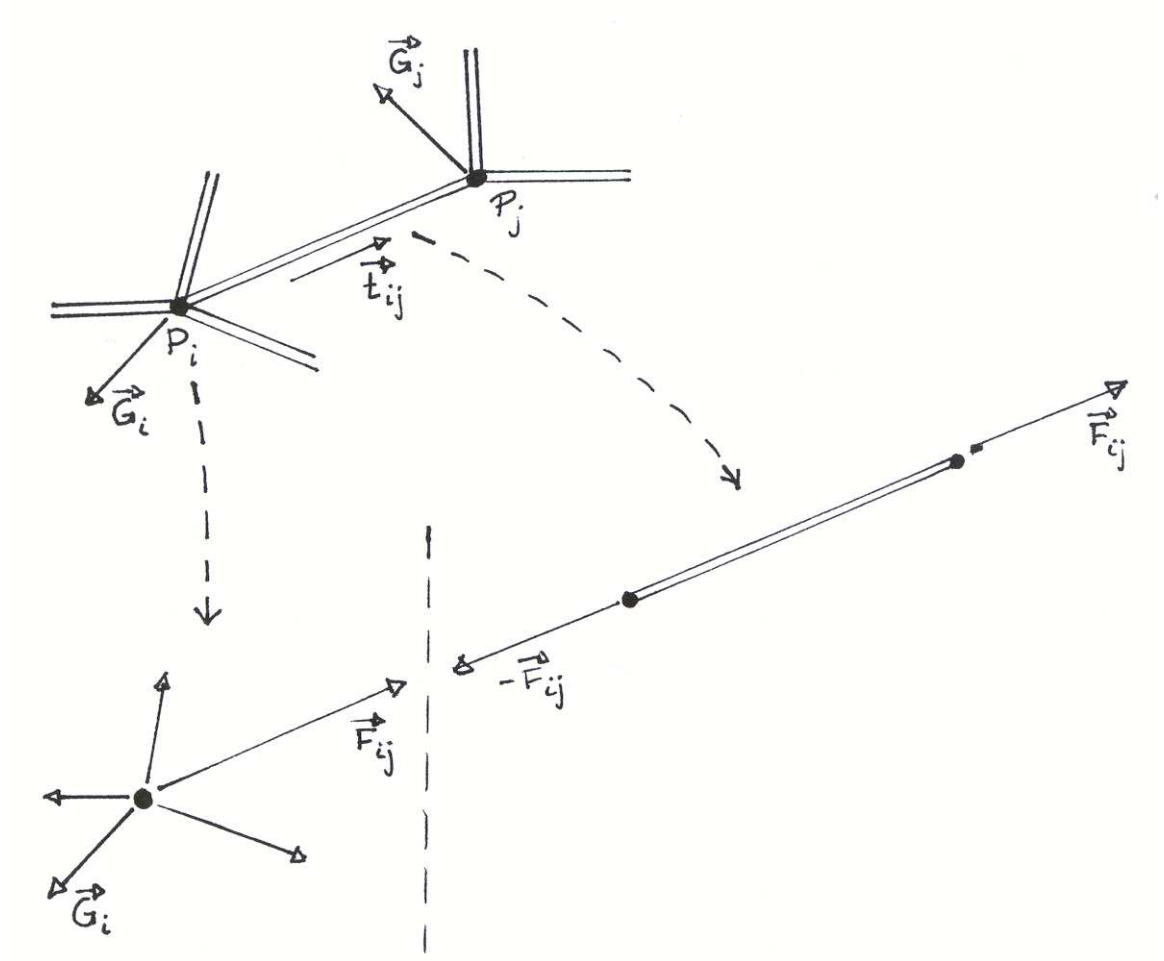
Consider a truss structure formed of elastic bars, in which the bars are joined to one another at the nodes P_i , $i = 1 \dots p$. Assume that the joints transmit from one bar to another only tension/compression, not torsion (e.g. a loose bolted joint). In the bar whose endpoints are P_i , P_j , let the force stretching the bar be

$$\vec{F}_{ij} = F_{ij} \vec{t}_{ij},$$

where $\vec{t}_{ij}$ is the unit vector in the direction of the vector $\overrightarrow{P_i P_j}$ (it is compression if $F_{ij} < 0$). Then force balance at the point P_i requires that

$$\sum_{j \in \Lambda_i} \vec{F}_{ij} + \vec{G}_i = \vec{0}, \quad i = 1 \dots p, \quad (7)$$

where $\{P_j \mid j \in \Lambda_i\}$ is the set of nodes to which P_i is connected by a bar, and $\vec{G}_i$ = the external load at node P_i .

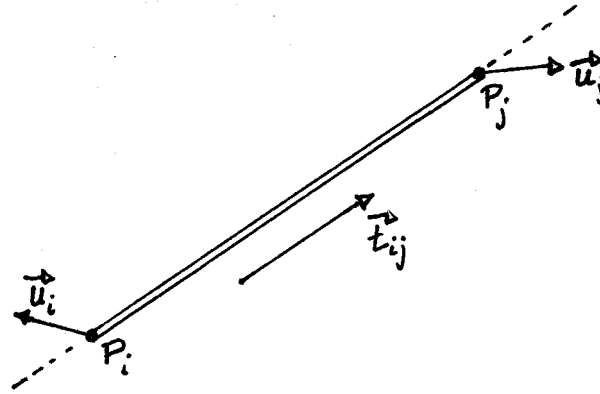


In the figure, the loading state has been decomposed into two so-called free-body diagrams, in each of which force balance prevails. — Note that the joint equilibrium condition (7) is very similar to Kirchhoff's law (1) for electric circuits, which is also an equilibrium condition (concerning currents).

When the vector equations in the system (7) are decomposed into coordinate form in a Cartesian coordinate system, the total number of scalar equations becomes $2p$ or $3p$ depending on whether it is a plane or a space truss structure. The number of unknowns F_{ij} is naturally the same as the number of bars in the structure. The system (7) may be directly uniquely solvable, in which case the

structure is said to be statically determinate. The more usual situation, however, is that the system (7) is underdetermined, i.e. statically indeterminate. Then in addition to the equilibrium laws a second law of physics is needed, Hooke's law (**R. Hooke**, 1678). By Hooke's law, each bar behaves like a spring when the truss is loaded, i.e. the force F_{ij} is directly proportional to the elongation of the bar.

Assume that when the truss is loaded, node P_i (position vector $\vec{r}_i$) moves to the location Q_i , and denote the displacement $\vec{u}_i = \overrightarrow{P_i Q_i}$. Assuming the displacements are small compared to the length of the bar, the elongation of each bar (a negative elongation means compression) can be computed approximately from the formula (cf. the figure)



$$|(\vec{r}_i + \vec{u}_i) - (\vec{r}_j + \vec{u}_j)| - |\vec{r}_i - \vec{r}_j| \approx (\vec{u}_j - \vec{u}_i) \cdot \vec{t}_{ij}.$$

(In the approximation it is assumed that the components of the displacements $\vec{u}_i$ and $\vec{u}_j$ perpendicular to the bar do not affect the length of the bar.) By Hooke's law, the elongation is directly proportional to the force, i.e.

$$F_{ij} = K_{ij}(\vec{u}_j - \vec{u}_i) \cdot \vec{t}_{ij}, \quad (8)$$

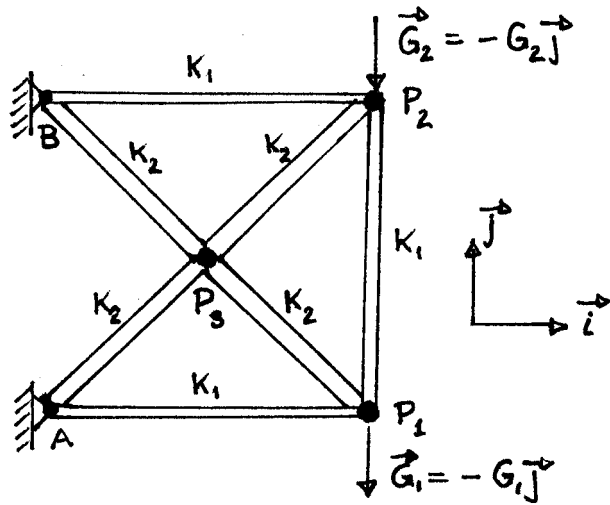
where K_{ij} is a coefficient characteristic of the bar (the spring constant). — Note again the analogy to electric circuits: the displacements $\vec{u}_i$ correspond to voltages, the forces F_{ij} to currents, and Hooke's law (8) compares to Ohm's law for resistors. If the bars are of uniform thickness and made of a homogeneous material, the spring constant K_{ij} is directly proportional to the cross-sectional area of the bar and inversely proportional to its length. The spring constant also depends on the quality of the material through the so-called modulus of elasticity (a material constant).

Using Hooke's law (8), the system (7) becomes linear, with the coordinates of the displacement vectors $\vec{u}_i$ as unknowns. The number of unknowns is thus either $2p$ (plane truss) or $3p$ (space truss). In order to make the system nonsingular, the truss must still be 'grounded', i.e. supported sufficiently. The support conditions may be, e.g. of the form $\vec{u}_k = \vec{0}$, whereby the displacement of node P_k

under loading is prevented. At such a node the force-balance conditions need not be written, since the balance is taken care of by the (unknown) support forces, which correspond to the grounding currents of the electric circuit. The support conditions must in general prevent those displacement modes that cause no loads in the structure ($F_{ij} = 0 \forall i, j$). In particular, the movement of the whole structure as a rigid body must be prevented in the mathematical model. Once such sufficient (physically often obvious) additional conditions have been imposed, the system (7)–(8) becomes nonsingular. The displacements $\vec{u}_i$ can then be solved first, and from these the (physically perhaps more interesting) loads F_{ij} from Hooke's laws (8).

EXAMPLE 2

In the adjacent (plane) truss there are two types of bars, spring constants K_1 and K_2 . There are 7 bars and the system (7) has $3 \times 2 = 6$ equations (at the supported nodes A, B the force balance is taken care of by the support forces), so the structure is statically indeterminate. Denoting the displacements at the nodes P_i by



$$\vec{u}_i = u_i \vec{i} + v_i \vec{j}, \quad i = 1, 2, 3,$$

then, with the shorthand notation

$$a = 1 + 2K_2/K_1, \quad b = 2K_2/K_1, \quad g_1 = G_1/K_2, \quad g_2 = -G_2/K_2$$

the system (7)–(8) takes the form

$$\left\{ \begin{array}{ccccccc} au_1 & - & v_1 & & & - & u_3 & + & v_3 & = & 0 \\ u_1 & - & av_1 & & + & bv_2 & - & u_3 & + & v_3 & = & g_1 \\ & & & au_2 & + & v_2 & - & u_3 & - & v_3 & = & 0 \\ & & - & bv_1 & + & u_2 & + & av_2 & - & u_3 & - & v_3 & = & g_2 \\ u_1 & - & v_1 & + & u_2 & + & v_2 & - & 4u_3 & & & = & 0 \\ -u_1 & + & v_1 & + & u_2 & + & v_2 & & & - & 4v_3 & = & 0 \end{array} \right.$$

This is a nonsingular system, from which the displacements u_i, v_i can be solved. The forces stretching or compressing the bars can then be computed from Hooke's

laws (8). For example, the bar P_1P_3 is stretched (or, when negative, compressed) by the force

$$F_{13} = K_2 [(u_3 - u_1)\vec{i} + (v_3 - v_1)\vec{j}] \cdot \frac{1}{\sqrt{2}}(-\vec{i} + \vec{j}) = \frac{K_2}{\sqrt{2}}(u_1 - v_1 - u_3 + v_3). \quad \square$$

EXERCISES

1. The nodes of a resistor circuit are the vertices of an equilateral triangle ABC and the center D of the triangle. On the sides AB , BC and CA there are resistors $R_1 = 1$, $R_2 = 2$ and $R_3 = 3$. The vertices A , B and C of the triangle are connected to the center by resistors $R_4 = R_5 = 2$ and $R_6 = 3$. Assuming a current supply I into node A and grounding at node C , form the linear system for the voltages of nodes A , B , D and compute the resistance between nodes A and C .
2. The vertices of a cube P_i , $i = 0 \dots 7$, are numbered so that P_0 and P_7 are opposite corners on one face of the cube. The vertices are nodes in a circuit whose wires run along the edges of the cube, with resistance $= R$ on each edge. Of the node voltages (x_i) it is known that $x_0 = E$ (voltage source) and $x_7 = 0$. Form the linear system for the voltages x_i , $i = 1 \dots 6$. Solve it and compute the resistance between nodes P_0 and P_7 .
3. (*) (Bridge) The nodes $P_1, \dots, P_7$ of a plane truss structure are at the points $(0, 0)$, $(1, 1)$, $(2, 0)$, $(3, 1)$, $(4, 0)$, $(5, 1)$ and $(6, 0)$. The nodes are connected by bars so that five adjacent isosceles right triangles are formed. Node P_1 is rigidly supported and the displacement of node P_7 in the direction $\vec{j}$ is prevented; there are no other supports. The loads acting on nodes P_3 and P_5 are $\vec{F}_3 = -aG\vec{j}$ and $\vec{F}_5 = -bG\vec{j}$, where $a, b \geq 0$ are dimensionless constants. There are no other loads. Denote the forces stretching (or, when negative, compressing) the bars by the symbols $T_i = x_iG$ (11 bars!) and use these to write the equilibrium equations of the nodes. Solve the system (possible, because the structure is statically determinate), and, based on the solution, answer the question: If each bar withstands compression up to G and tension without limit, then what kinds of loads are possible without the structure collapsing? Draw a picture in the ab -plane!

Chapter XI

Multivariable differential calculus

In this chapter the objects of study are *vector-variable* functions, i.e. functions of several real variables of type $f : \mathbb{R}^n \rightarrow \mathbb{R}$, and *vector-valued* functions of a vector variable of type $\mathbf{f} : \mathbb{R}^n \rightarrow \mathbb{R}^m$. Special cases of the former, functions of two and three real variables, are already familiar from Chapter IV.3. Vector-valued functions have also been encountered before, since both the parametric curves and parametric surfaces introduced in Chapter IV.4 and the linear and affine maps treated in Chapters X.6–X.7 are special cases of these. Vector-valued functions also include the *vector fields* of the plane and of space, in which a vector is understood as a geometric–physical ‘arrow vector’.

In Section XI.1 the notions of continuity and limit familiar from Chapters V.1–V.2 are generalized to functions of several variables. After that, the central content of the chapter consists of various extensions of the concept of the derivative to multivariable settings, together with the computational technique (differential calculus) and applications associated with these extensions. As more general notions of the derivative, this chapter introduces *partial derivatives* and the *gradient* formed from them (Sections XI.2–XI.3), the *divergence* and *curl* associated with vector fields (Sections XI.4–XI.5), and finally, in connection with more general vector-valued functions, i.e. *nonlinear maps*, the *Jacobian matrix* (Section XI.6). As applications we consider, among others, geometric problems such as determining the *tangent plane of a surface* (Section XI.3), the *partial differential equations* of physics (Sections XI.4–XI.5), *nonlinear systems of equations* and their numerical solution (Section XI.6), and *optimization* problems (Section XI.8). In Section XI.7 we present three powerful theorems of *nonlinear analysis* related to nonlinear systems of equations. In the final subsection (Section XI.9) we define the *Taylor polynomial* for functions of several variables.

XI.1 Functions of several variables: Continuity and limits

The algebra of functions of two and three real variables was considered earlier in Chapter IV.3. In this and the following sections the objects of study are, in addition to functions of two and three variables, the more general real-valued functions of n real variables of the form

$$y = f(x_1, \dots, x_n).$$

One may also use the notation of matrix algebra

$$y = f(\mathbf{x}), \quad \mathbf{x} \in \mathbb{R}^n$$

($\mathbf{x}$ a column or row vector) and speak of a function of a *vector variable*. It is also possible (and even customary in mathematical texts) to use, instead of matrix-algebra notation, the simple symbol x for the vector variable, i.e. to write

$$y = f(x), \quad x = (x_1, \dots, x_n) \in \mathbb{R}^n.$$

In what follows the various notations are used side by side, where in matrix-algebra notation $\mathbf{x}$ denotes a column vector. In the cases of two and three variables the vector variable is denoted in the old way by (x, y) or (x, y, z) .

The notions of the continuity and limit of a function, which so far have been attached only to functions of one variable (see Chapters V.1–V.2), can be generalized fairly straightforwardly to functions of several variables. We begin with functions of two variables.

The function $f(x, y)$

As a preliminary to the definitions of continuity and limit for a function of two variables, we need

DEFINITION XI.1.1 A sequence $\{(x_n, y_n)\}$, where $(x_n, y_n) \in \mathbb{R}^2$, $n \in \mathbb{N}$, *converges to* (approaches) the *ordered pair* (point) (x, y) exactly when $x_n \rightarrow x$ and $y_n \rightarrow y$.

When the convergence of a sequence of ordered pairs is denoted, as before, by ' $\rightarrow$ ', it has thus been agreed that:

$$(x_n, y_n) \rightarrow (x, y) \quad \Leftrightarrow \quad x_n \rightarrow x \quad \& \quad y_n \rightarrow y.$$

An equivalent convention is (see Exercise I.6: 3d)

$$(x_n, y_n) \rightarrow (x, y) \Leftrightarrow (x_n - x)^2 + (y_n - y)^2 \rightarrow 0.$$

According to this, $(x_n, y_n) \rightarrow (x, y)$ simply means that the (geometric) distance of the point (x_n, y_n) from the point (x, y) approaches (as a numerical sequence) 0, as $n \rightarrow \infty$. If in particular $(x, y) = (0, 0)$, then by the polar transformation $x_n = r_n \cos \varphi_n$, $y_n = r_n \sin \varphi_n$ we have

$$(x_n, y_n) \rightarrow (0, 0) \Leftrightarrow r_n \rightarrow 0 \quad (\text{polar coordinates}).$$

Since this imposes no restrictions on the sequence $\{\varphi_n\}$, the approach may take place e.g. along a ray on which $\varphi_n = \varphi = \text{constant}$, or it may be 'erratic' in direction, e.g. spiral-like. In any case there are infinitely many possible directions of approach (corresponding to rays). — This is an essential difference compared with the one-variable situation, where a point can be approached from only two different directions.

Once the two-dimensional 'approach' $(x_n, y_n) \rightarrow (x, y)$ has been defined in this way, generalizing Definitions V.1.1 (continuity) and V.2.1 (limit of a function) to the two-variable situation is straightforward:

DEFINITION XI.1.2 A function $f : \mathcal{D}_f \subset \mathbb{R}^2 \rightarrow \mathbb{R}$ is *continuous* at a point $(x, y) \in \mathcal{D}_f$ exactly when for every sequence of real ordered pairs $\{(x_n, y_n)\}$

$$(x_n, y_n) \in \mathcal{D}_f \quad \forall n \quad \& \quad (x_n, y_n) \rightarrow (x, y) \Rightarrow f(x_n, y_n) \rightarrow f(x, y).$$

DEFINITION XI.1.3 A function $f(x, y)$ has at a point (a, b) the *limit* $A \in \mathbb{R}$ if for every sequence of real ordered pairs $\{(x_n, y_n)\}$

$$(a, b) \neq (x_n, y_n) \in \mathcal{D}_f \quad \forall n \quad \& \quad (x_n, y_n) \rightarrow (a, b) \Rightarrow f(x_n, y_n) \rightarrow A,$$

and the hypothesis holds for some sequence $\{(x_n, y_n)\}$. The limit is denoted

$$\lim_{(x, y) \rightarrow (a, b)} f(x, y) = A.$$

As in the one-variable case, Definition XI.1.3 precludes defining the limit at an (isolated) point (a, b) toward which the approach $(x_n, y_n) \rightarrow (a, b)$ from the set $\mathcal{D}_f$ is not possible under the additional condition $(x_n, y_n) \neq (a, b) \quad \forall n$. In Definition XI.1.2 this additional condition is absent, so if $(a, b) \in \mathcal{D}_f$ is an isolated point of $\mathcal{D}_f$, then f is continuous at (a, b) (cf. Exercise V.1: 4).

EXAMPLE 1 Define the functions

$$f_1(x, y) = \frac{x^2 y}{x^2 + y^2}, \quad f_2(x, y) = \frac{xy}{x^2 + y^2}, \quad (x, y) \in \mathbb{R}^2, (x, y) \neq (0, 0).$$

Investigate whether the functions are continuous at the origin when we set $f_1(0, 0) = f_2(0, 0) = 0$.

Solution By the polar transformations

$$\begin{aligned} g_1(r, \varphi) &= f_1(r \cos \varphi, r \sin \varphi) = r \cos^2 \varphi \sin \varphi, \\ g_2(r, \varphi) &= f_2(r \cos \varphi, r \sin \varphi) = \cos \varphi \sin \varphi \end{aligned}$$

we conclude: If $(x_n, y_n) \hat{=} (r_n, \varphi_n)$, then

$$r_n \rightarrow 0 \quad \Rightarrow \quad |f_1(x_n, y_n)| = |g_1(r_n, \varphi_n)| \leq r_n \rightarrow 0 = f(0, 0),$$

so f_1 is continuous at $(0, 0)$. On the other hand, f_2 is discontinuous at the origin, since $r_n \rightarrow 0 \not\Rightarrow g_2(r_n, \varphi_n) \rightarrow 0$. $\square$

Limits and continuity obey the same combination rules as in the one-variable case, since the logic and algebra underlying these results do not essentially depend on the number of variables. The combination results for continuity are the following:

THEOREM XI.1.4 If $f : D_f \rightarrow \mathbb{R}$, $D_f \subset \mathbb{R}^2$, and $g : D_g \rightarrow \mathbb{R}$, $D_g \subset \mathbb{R}^2$, are continuous at a point $(x, y) \in D_f \cap D_g$, then λf ($\lambda \in \mathbb{R}$), $f + g$ and fg are also continuous at (x, y) . If in addition $g(x, y) \neq 0$, then f/g too is continuous at (x, y) .

THEOREM XI.1.5 If $f : D_f \rightarrow \mathbb{R}$, $D_f \subset \mathbb{R}^2$, is continuous at a point $(x, y) \in D_f$ and $g : D_g \rightarrow \mathbb{R}$, $D_g \subset \mathbb{R}$, is continuous at $f(x, y) \in D_g$, then the composite function $g \circ f$ is continuous at (x, y) .

Continuity at a single point in the sense of Definition [XI.1.2](#) extends in a natural way to continuity on a set: f is continuous on a set $A \subset \mathcal{D}_f$ if f is continuous at every point of A . As in the one-variable case, the usual 'single-expression functions' are continuous throughout their domain.

EXAMPLE 1 (continued) If in the example $f_1(0, 0)$ and $f_2(0, 0)$ are left undefined, then by Theorem [XI.1.4](#) the functions of the example are continuous throughout their common domain

$$\mathcal{D}_{f_1} = \mathcal{D}_{f_2} = \{(x, y) \in \mathbb{R}^2 \mid (x, y) \neq (0, 0)\}.$$

The function f_1 has at the origin the limit

$$\lim_{(x,y) \rightarrow (0,0)} f_1(x, y) = 0$$

(Definition XI.1.3), so f_1 can be made continuous at the origin (and hence on all of $\mathbb{R}^2$) by setting $f_1(0, 0) = 0$ (extension of a function!). The function f_2 does not have this limit, so extending f_2 to the origin is not possible. $\square$

EXAMPLE 2 The domain of a rational function $f(x, y) = p(x, y)/q(x, y)$ (p and q polynomials) is $\mathcal{D}_f = \{(x, y) \in \mathbb{R}^2 \mid q(x, y) \neq 0\}$. By Theorem XI.1.4, p and q are continuous throughout their domain ($= \mathbb{R}^2$), and so is f . $\square$

EXAMPLE 3 At which points is the function $f(x, y) = \sqrt{y - x^2}/(x - y^2)$ a) defined, b) continuous?

Solution a) The domain is $D_f = \{(x, y) \in \mathbb{R}^2 \mid y \geq x^2 \text{ \& } x \neq y^2\}$.
b) This is a composite and combined function $f = (g \circ f_1)/f_2$, where

$$\begin{aligned} f_1(x, y) &= y - x^2, \quad D_{f_1} = \mathbb{R}^2, \quad f_2(x, y) = x - y^2, \quad D_{f_2} = \mathbb{R}^2, \\ g(t) &= \sqrt{t}, \quad D_g = [0, \infty) \subset \mathbb{R}. \end{aligned}$$

By Theorems XI.1.4 and XI.1.5, f is continuous throughout its domain. $\square$

As in the one-variable case, the continuity and limit of a function of two variables can also be defined alternatively without appealing to sequences of numbers (or pairs). In the alternative ' (ε, δ) -definition' one needs a *neighborhood* corresponding to the open interval $(x - \delta, x + \delta)$, more precisely the *open δ -neighborhood of a point*, denoted $U_\delta(x, y)$ ($\delta > 0$). As in one dimension, this means a set that surrounds the point by an amount δ , or a distance proportional to δ , in every direction. Typically $U_\delta(x, y)$ is thought of as either square-shaped or disk-shaped, i.e.

$$\begin{aligned} \text{either : } & U_\delta(x, y) = (x - \delta, x + \delta) \times (y - \delta, y + \delta), \\ \text{or : } & U_\delta(x, y) = \{(x', y') \in \mathbb{R}^2 \mid (x' - x)^2 + (y' - y)^2 < \delta^2\}. \end{aligned}$$

The alternative definition of continuity is (cf. Definition V.1.2)

DEFINITION XI.1.6 A function $f : \mathcal{D}_f \subset \mathbb{R}^2 \rightarrow \mathbb{R}$ is continuous at a point $(x, y) \in \mathcal{D}_f$ exactly when for every $\varepsilon > 0$ there exists $\delta > 0$ such that

$$|f(x', y') - f(x, y)| < \varepsilon \quad \forall (x', y') \in U_\delta(x, y) \cap \mathcal{D}_f.$$

Definitions [XI.1.2](#) and [XI.1.6](#) can be shown to be equivalent in the same way as in the one-variable case (cf. Theorem [V.1.3](#)).

The neighborhood $U_\delta(x, y)$ is, as its name suggests, an example of an *open* set, whose more general definition is (cf. Definition [V.8.1](#) for sets $A \in \mathbb{C}$)

DEFINITION XI.1.7 A set $A \subset \mathbb{R}^2$ is *open* if for every $(x, y) \in A$ there exists $\delta > 0$ such that $U_\delta(x, y) \subset A$.

The function $f(x, y, z)$

Definitions [XI.1.2](#) and [XI.1.3](#) have obvious counterparts for functions of three variables $f(x, y, z)$. For example, the continuity condition at a point $(x, y, z) \in \mathcal{D}_f$ is

$$(x_n, y_n, z_n) \in \mathcal{D}_f \quad \forall n \quad \& \quad (x_n, y_n, z_n) \rightarrow (x, y, z) \quad \Rightarrow \quad f(x_n, y_n, z_n) \rightarrow f(x, y, z),$$

where (cf. Exercise [I.6:3e](#))

$$\begin{aligned} (x_n, y_n, z_n) \rightarrow (x, y, z) &\Leftrightarrow x_n \rightarrow x \quad \& \quad y_n \rightarrow y \quad \& \quad z_n \rightarrow z \\ &\Leftrightarrow (x_n - x)^2 + (y_n - y)^2 + (z_n - z)^2 \rightarrow 0. \end{aligned}$$

In particular, if $(x, y, z) = (0, 0, 0)$, then by the spherical-coordinate transformation $(x_n, y_n, z_n) \hat{=} (r_n, \theta_n, \varphi_n)$ we have

$$(x_n, y_n, z_n) \rightarrow (0, 0, 0) \quad \Leftrightarrow \quad r_n \rightarrow 0 \quad (\text{spherical coordinates}).$$

EXAMPLE 4 Under exactly what conditions on the numbers $\alpha, \beta, \gamma \in \mathbb{R}$ do we have

$$\lim_{(x,y,z) \rightarrow (0,0,0)} f(x, y, z) = 0, \quad \text{when} \quad f(x, y, z) = \frac{|x|^\alpha |y|^\beta |z|^\gamma}{x^2 + y^2 + z^2}?$$

Solution We make the spherical-coordinate transformation:

$$f(x, y, z) = g(r, \theta, \varphi) = r^{\alpha+\beta+\gamma-2} |\sin \theta|^{\alpha+\beta} |\cos \theta|^\gamma |\cos \varphi|^\alpha |\sin \varphi|^\beta.$$

We conclude that $r_n \rightarrow 0 \Rightarrow g(r_n, \theta_n, \varphi_n) \rightarrow 0$ is true exactly under the conditions

$$\alpha + \beta + \gamma > 2, \quad \alpha \geq 0, \quad \beta \geq 0, \quad \gamma \geq 0. \quad \square$$

The function $f(\mathbf{x})$, $\mathbf{x} \in \mathbb{R}^n$

For a general real-valued function f of n variables, the continuity condition at a point $\mathbf{x} \in \mathcal{D}_f$ is

$$\mathbf{x}_k \in \mathcal{D}_f \quad \forall k \quad \mathbf{x}_k \rightarrow \mathbf{x} \Rightarrow f(\mathbf{x}_k) \rightarrow f(\mathbf{x}),$$

where $\{\mathbf{x}_k\}$ is a vector sequence in $\mathbb{R}^n$ and *convergence* $\mathbf{x}_k \rightarrow \mathbf{x}$ to the *limit* $\mathbf{x} \in \mathbb{R}^n$ means:

$$\mathbf{x}_k \rightarrow \mathbf{x} \Leftrightarrow (\mathbf{x}_k)_i \rightarrow (\mathbf{x})_i = x_i, \quad i = 1 \dots n \Leftrightarrow |\mathbf{x}_k - \mathbf{x}| \rightarrow 0,$$

where $|\cdot|$ is the Euclidean norm of $\mathbb{R}^n$. In the n -dimensional counterparts of Definitions [XI.1.6](#) and [XI.1.7](#), the neighborhood $U_\delta(\mathbf{x})$ is interpreted either as an n -dimensional cube with edge length $= 2\delta$, or as an n -dimensional ball with radius $= \delta$.

*Continuity on a compact set

We set (cf. Definitions [V.9.1](#) and [V.9.5](#))

DEFINITION XI.1.8 A set $K \subset \mathbb{R}^n$ is

- *closed* if for every vector sequence $\{\mathbf{x}_k\}$ in $\mathbb{R}^n$:
 $\mathbf{x}_k \in K \quad \forall k \quad \& \quad \mathbf{x}_k \rightarrow \mathbf{x} \in \mathbb{R}^n \Rightarrow \mathbf{x} \in K,$
- *bounded* if $\exists C \in \mathbb{R}_+$ such that $|\mathbf{x}| \leq C \quad \forall \mathbf{x} \in K,$
- *compact* if K is closed and bounded.

An alternative definition of a closed set is (cf. Theorem [V.9.3](#))

THEOREM XI.1.9 A set $A \subset \mathbb{R}^n$ is closed exactly when the *complement* of A , $\mathcal{C}(A) = \{\mathbf{x} \in \mathbb{R}^n \mid \mathbf{x} \notin A\}$, is open.

Proof Exercise [4](#).

EXAMPLE 5 Compact sets $K \subset \mathbb{R}^n$ include, for example, the (closed) n -dimensional rectangular box

$$K = [a_1, b_1] \times [a_2, b_2] \times \dots \times [a_n, b_n]$$

and the (closed) n -dimensional ball

$$K = \{\mathbf{x} \in \mathbb{R}^n \mid |\mathbf{x} - \mathbf{x}_0| \leq R\}.$$

Every finite subset of $\mathbb{R}^n$ is also compact (Exercise [6](#)). $\square$

The notion of a compact set is central in the following generalization of Weierstrass's theorem V.1.12, whose case $n = 1$ was proved earlier as Theorem V.9.7. The more general proof is not given; let us note only that the logic of the proof is very similar to the one-variable case, see Section V.9. We first set (cf. Definition V.9.5)

DEFINITION XI.1.10 A function $f : \mathcal{D}_f \subset \mathbb{R}^n \rightarrow \mathbb{R}$ is *continuous on a compact set* $K \subset \mathcal{D}_f$ if for every vector sequence $\{\mathbf{x}_k\}$ in $\mathbb{R}^n$

$$\mathbf{x}_k \in K \ \forall k \quad \& \quad \mathbf{x}_k \rightarrow \mathbf{x} \Rightarrow f(\mathbf{x}_k) \rightarrow f(\mathbf{x}).$$

*THEOREM XI.1.11 If a function $f : \mathcal{D}_f \subset \mathbb{R}^n \rightarrow \mathbb{R}$ is continuous on a compact set $K \subset \mathcal{D}_f$, then f attains its smallest and largest values on K .

EXERCISES

- In parts a)–h) determine the limit or conclude that no limit exists. In parts i)–j) determine all values of the coefficients a, b, c for which the limit exists. Passing to polar or spherical coordinates may help.

$$\begin{array}{lll} \text{a)} \quad \lim_{(x,y) \rightarrow (2,-1)} (xy + y^2) & \text{b)} \quad \lim_{(x,y) \rightarrow (0,0)} \sqrt{x^2 + 2y^2} & \text{c)} \quad \lim_{(x,y) \rightarrow (0,0)} \frac{x^2 + y^2}{y} \\ \text{d)} \quad \lim_{(x,y) \rightarrow (0,0)} \frac{x}{x^2 + y^2} & \text{e)} \quad \lim_{(x,y) \rightarrow (0,0)} \frac{y^3}{x^2 + y^2} & \text{f)} \quad \lim_{(x,y) \rightarrow (0,0)} \frac{x^2 y^2}{x^2 + y^4} \\ \text{g)} \quad \lim_{(x,y) \rightarrow (0,0)} \frac{\sin(xy)}{x^2 + y^2} & \text{h)} \quad \lim_{(x,y,z) \rightarrow (0,0,0)} \frac{\sin(xyz)}{x^2 + y^2 + z^2} & \\ \text{i)} \quad \lim_{(x,y) \rightarrow (0,0)} \frac{xy}{ax^2 + bxy + cy^2} & \text{j)} \quad \lim_{(x,y,z) \rightarrow (0,0,0)} \frac{x^2 + y^2 - z^2}{ax^2 + by^2 + cz^2} & \end{array}$$

- In parts a)–i) extend the definition of the function f so that the function becomes continuous on the largest possible subset of $\mathbb{R}^2$ or $\mathbb{R}^3$. In parts

j)–m) determine all points at which f is discontinuous.

- a) $f(x, y) = \frac{x\sqrt[4]{|y|}}{|x| + |y|}$ b) $f(x, y) = \frac{\sqrt{(1+x^2)(1+y^2)} - 1}{x^2 + y^2}$
- c) $f(x, y) = \frac{x^3 - 8y^3}{x - 2y}$ d) $f(x, y) = \frac{x + y}{x^3 + y^3}$ e) $f(x, y) = \frac{x^2 + y}{x^4 - y^2}$
- f) $f(x, y) = \frac{\sin(x + y)}{x^2 - y^2}$ g) $f(x, y) = \frac{\sin(x^3 + y^3)}{x + y}$
- h) $f(x, y, z) = \frac{1 - \cos(xyz)}{x^2 y^2 z^2}$ i) $f(x, y, z) = \frac{x^2 - y^2 + z^2 + 2xz}{x + y + z}$
- j) $f(x, y) = \begin{cases} 4x^2 + y^2 - 47, & \text{when } x^2 + y^2 < 25 \\ x^2 - 2y^2 + 28, & \text{when } x^2 + y^2 \geq 25 \end{cases}$
- k) $f(x, y) = \begin{cases} x^2 + xy - 2y^2, & \text{when } 0 < |y| < x \\ 0, & \text{otherwise} \end{cases}$
- l) $f(x, y) = \begin{cases} y \sin\left(\frac{1}{x} + \frac{1}{y}\right), & \text{when } x \neq 0 \text{ and } y \neq 0 \\ 0, & \text{when } x = 0 \text{ or } y = 0 \end{cases}$
- m) $f(x, y, z) = \begin{cases} \frac{\sin(xyz)}{\sin(xy)}, & \text{when } xy \neq 0 \\ z, & \text{when } xy = 0 \end{cases}$

3. (*) Define $f : \mathbb{R}^2 \mapsto \mathbb{R}$ as follows:

$$f(x, y) = \begin{cases} 1, & \text{when } x^2 < y < 2x^2 \\ 0, & \text{otherwise} \end{cases}$$

Show that the function has the same limit at the origin when the origin is approached along any line $S : ax + by = 0$ ($a, b \in \mathbb{R}$), but that nevertheless the limit $\lim_{(x,y) \rightarrow (0,0)} f(x, y)$ does not exist.

4. (*) Prove Theorem [XI.1.9](#).

5. (*) Let $f(x, y) = p(x, y)/q(x, y)$, where p and q are defined and continuous on all of $\mathbb{R}^2$ (e.g. polynomials).

a) Show directly, using Theorem [XI.1.6](#), that $\mathcal{D}_f \subset \mathbb{R}^2$ is open.

b) Conclude that $\mathcal{D}_f$ is open by first showing, relying on Definition [XI.1.8](#), that $\{(x, y) \in \mathbb{R}^2 \mid q(x, y) = 0\}$ is a closed set and appealing to Theorem [XI.1.9](#).

6. (*) Let $\mathbf{a}_i \in \mathbb{R}^n$, $i = 1 \dots m$ ($m \in \mathbb{N}$), $\mathbf{a}_i \neq \mathbf{a}_j$ when $i \neq j$, and $K = \{\mathbf{a}_1, \dots, \mathbf{a}_m\}$.
- a) Show that $K \subset \mathbb{R}^n$ is a compact set.
 - b) Show that if $f : \mathcal{D}_f \subset \mathbb{R}^n \rightarrow \mathbb{R}$ is any function for which $\mathbf{a}_i \in \mathcal{D}_f$, $i = 1 \dots m$, then f is continuous on K in the sense of Definition [XI.1.10](#).
 - c) Verify that the assertion of Theorem [XI.1.11](#) is consistent with the result of part b).

XI.2 Partial derivatives

The *partial derivative* of a function of two or more variables means simply the derivative of the function with respect to one of the variables, the other variables being held constant in the limit process. The symbol for the partial derivative is ∂ , which is read the same as the ordinary derivative, i.e. 'dee' (or 'partial'). For example, the partial derivatives of a function of two variables $f(x, y)$ are denoted

$$\begin{aligned}\frac{\partial f}{\partial x}(x, y) &= \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x, y) - f(x, y)}{\Delta x}, \\ \frac{\partial f}{\partial y}(x, y) &= \lim_{\Delta y \rightarrow 0} \frac{f(x, y + \Delta y) - f(x, y)}{\Delta y}.\end{aligned}$$

Less space-consuming (and computationally faster) notations are

$$\partial_x f, \partial_y f, \quad \text{or} \quad f_x, f_y.$$

Of these, the first notation refers to the fact that, like the ordinary derivative, the partial derivative too is a 'function of a function', i.e. an operator. The subscripted notation is common especially when the variables are physical space and time variables.

The partial derivative behaves in many respects like the ordinary derivative. For instance, it is *linear* in the function being differentiated:

$$\partial_x [\lambda f(x, y) + \mu g(x, y)] = \lambda \partial_x f(x, y) + \mu \partial_x g(x, y), \quad \lambda, \mu \in \mathbb{R}.$$

Partial differentiation also satisfies, among others, the product rule

$$\frac{\partial}{\partial x}(fg) = \frac{\partial f}{\partial x}g + f\frac{\partial g}{\partial x},$$

since for this rule too it is a matter of ordinary differentiation with respect to one chosen variable.

EXAMPLE 1 The partial derivatives of the function

$$f(x, y) = \begin{cases} 0, & \text{when } (x, y) = (0, 0), \\ xy/(x^2 + y^2), & \text{otherwise} \end{cases}$$

away from the origin are

$$f_x(x, y) = \frac{y^3 - x^2y}{(x^2 + y^2)^2}, \quad f_y(x, y) = \frac{x^3 - xy^2}{(x^2 + y^2)^2}, \quad (x, y) \neq (0, 0).$$

Here the usual (one-variable) rules for differentiating a rational function have been used. Since $f(x, 0) = f(0, y) = 0$, $x, y \in \mathbb{R}$, the function f has partial derivatives at the origin as well:

$$f_x(0, 0) = f_y(0, 0) = 0. \quad \square$$

The function of the example is discontinuous at the origin (see the previous section, Example 1). — Thus a function may be discontinuous at a single point even though the partial derivatives exist everywhere. This difference from one-variable functions (for which the existence of the derivative at a point guarantees continuity at that point) is explained by the fact that in the limits defining the partial derivatives the point is approached only in the directions of the coordinate axes, whereas in the definition of continuity of a function of several variables (see the previous section) there are infinitely many possible directions of approach. The property of a multivariable function being differentiable, i.e. its so-called *differentiability*, must therefore be defined as a coordinate-independent 'differentiability in every direction', in order for it to correspond to the one-variable notion of differentiability. The definition is given in the next section.

The partial derivatives of a function can (as functions) be differentiated further with respect to the various variables, yielding *second-order* partial derivatives; by differentiating these, *third-order* partial derivatives, and so on. Higher-order partial derivatives are denoted

$$\begin{aligned} \frac{\partial}{\partial x} \left(\frac{\partial f}{\partial x} \right) &= \frac{\partial^2 f}{\partial x^2} = \partial_x(\partial_x f) = (f_x)_x = f_{xx}, \\ \frac{\partial}{\partial x} \left(\frac{\partial f}{\partial y} \right) &= \frac{\partial^2 f}{\partial x \partial y} = \partial_x(\partial_y f) = (f_y)_x = f_{yx}, \\ \frac{\partial}{\partial x} \left(\frac{\partial^2 f}{\partial x \partial y} \right) &= \frac{\partial^3 f}{\partial x^2 \partial y} = \partial_x^2(\partial_y f) = (f_y)_{xx} = f_{yxx}, \\ \frac{\partial^2}{\partial y^2} \left(\frac{\partial^2 f}{\partial x^2} \right) &= \frac{\partial^4 f}{\partial y^2 \partial x^2} = \partial_y^2(\partial_x^2 f) = (f_{xx})_{yy} = f_{xxyy}, \quad \text{jne.} \end{aligned}$$

EXAMPLE 2 The partial derivatives up to second order of the function $f(x, y) = \ln(x^2 + y^2)$ at points $(x, y) \neq (0, 0)$ (i.e. in the domain of f) are

$$\begin{aligned} f_x &= \frac{2x}{x^2 + y^2}, \quad f_y = \frac{2y}{x^2 + y^2} \\ f_{xx} &= \frac{2(y^2 - x^2)}{(x^2 + y^2)^2}, \quad f_{yy} = -f_{xx}, \quad f_{xy} = f_{yx} = -\frac{4xy}{(x^2 + y^2)^2}. \quad \square \end{aligned}$$

The result $f_{xy} = f_{yx}$ of the example is an instance of the more general *interchange rule* of partial differentiation

$$\boxed{f_{xy} = f_{yx} \quad (\text{interchange rule}).} \quad (1)$$

The rule is valid under mild regularity conditions that can usually be assumed. These conditions and the justification of the rule are given at the end of the section (Theorem [XI.2.1](#)). By the interchange rule, the order of differentiation is free in higher-order partial derivatives as well; e.g. $f_{xxy} = f_{xyx} = f_{yxx}$.

Partial derivatives of many variables

The interchange rule (1) holds also in the case of more than two variables, since when differentiating with respect to two variables the function can be regarded as depending only on those variables. Thus if the partial derivatives $\partial f / \partial x_k$ and $\partial^2 f / (\partial x_k \partial x_l)$ of a function $f(x_1, \dots, x_n)$ exist and are continuous for $k, l = 1 \dots n$, $k \neq l$, then

$$\frac{\partial^2 f}{\partial x_k \partial x_l} = \frac{\partial^2 f}{\partial x_l \partial x_k}, \quad k, l \in \{1, \dots, n\}, \quad k \neq l.$$

Since the order of differentiation is thereby free (under mild conditions that are usually assumed), for higher partial derivatives it suffices only to know how many times one differentiates with respect to each variable. A general partial derivative is then denoted using a so-called *multi-index*, i.e. an ordered set of indices (index vector) of the form

$$\alpha = (\alpha_1, \dots, \alpha_n), \quad \alpha_i \in \mathbb{N} \cup \{0\}.$$

Then $\partial^\alpha f$ denotes the partial derivative

$$\partial^\alpha f = \frac{\partial^{|\alpha|} f}{\partial x_1^{\alpha_1} \dots \partial x_n^{\alpha_n}} = \partial_1^{\alpha_1} \dots \partial_n^{\alpha_n} f,$$

where $\partial_i = \partial / \partial x_i$, and the *order* of the partial derivative has been denoted

$$|\alpha| = \sum_{i=1}^n \alpha_i.$$

If in the multi-index α we have $\alpha_k = 0$, then no differentiation is performed with respect to that variable, i.e. $\partial_k^0 f = f$.

EXAMPLE 3 If $x_1 = x$ and $x_2 = y$, then, taking the interchange rule into account,

$$\begin{aligned}\partial^{(1,1)} f(x, y) &= \partial_x \partial_y f(x, y) = f_{xy}(x, y), \\ \partial^{(0,2)} f(x, y) &= (\partial_y)^2 f(x, y) = f_{yy}(x, y), \\ \partial^{(2,1)} f(x, y) &= (\partial_x)^2 \partial_y f(x, y) = f_{xxy}(x, y). \quad \square\end{aligned}$$

EXAMPLE 4 Applying the product rule and the linearity rule,

$$\begin{aligned}\partial^{(1,1)}(fg)(x, y) &= \partial_x \partial_y (fg)(x, y) \\ &= \partial_x (f_y g + f g_y)(x, y) \\ &= (f_{xy} g + f_x g_y + f_y g_x + f g_{xy})(x, y). \quad \square\end{aligned}$$

EXAMPLE 5 **Quadratic form.** A general so-called *quadratic form* on $\mathbb{R}^n$ is a function of the form

$$f(x_1, \dots, x_n) = \frac{1}{2} \sum_{i,j=1}^n a_{ij} x_i x_j, \quad a_{ij} \in \mathbb{R},$$

or in matrix-algebra notation

$$f(\mathbf{x}) = \frac{1}{2} \mathbf{x}^T \mathbf{A} \mathbf{x}, \quad \mathbf{x} = (x_i), \quad \mathbf{A} = (a_{ij}).$$

Here one may assume that $a_{ij} = a_{ji}$ ($\mathbf{A}$ symmetric), since f depends only on the sums $a_{ij} + a_{ji}$ ($a_{ij} x_i x_j + a_{ji} x_j x_i = (a_{ij} + a_{ji}) x_i x_j$). When this assumption is made, the terms depending on a given variable x_k separate out in the form

$$f(x_1, \dots, x_n) = \frac{1}{2} a_{kk} x_k^2 + \sum_{j \neq k} a_{kj} x_k x_j + [\dots],$$

where $[\dots]$ does not contain the variable x_k . Hence

$$\frac{\partial f}{\partial x_k} = \sum_{j=1}^n a_{kj} x_j = [\mathbf{A} \mathbf{x}]_k, \quad k = 1 \dots n \quad \Rightarrow \quad \left(\frac{\partial f}{\partial x_i} \right) = \mathbf{A} \mathbf{x}.$$

The second-order partial derivatives are

$$\left(\frac{\partial^2 f}{\partial x_i \partial x_j} \right) = (a_{ij}) = \mathbf{A}.$$

The higher-order partial derivatives vanish. $\square$

Partial differential operators

When computing with partial derivatives, it is often convenient to think of the symbols ∂_x , ∂_y , etc. as operators detached from the individual objects of differentiation. For example, if ∂^α and ∂^β are differential operators of n variables, then for sufficiently regular functions f the interchange rule gives

$$\partial^\alpha [\partial^\beta f(x)] = \partial^\beta [\partial^\alpha f(x)] = \partial^{\alpha+\beta} f(x).$$

(Here $\alpha + \beta$ is the sum of the multi-indices as vectors.) This can be expressed more briefly as a computational rule concerning the operators alone,

$$\partial^\alpha \partial^\beta = \partial^\beta \partial^\alpha = \partial^{\alpha+\beta}.$$

Here $\partial^\alpha \partial^\beta$ denotes the *operator product*, i.e. the two-stage (composite) operation $f \mapsto \partial^\beta f \mapsto \partial^\alpha \partial^\beta f$. By the interchange rule, the differential operators ∂^α and ∂^β therefore *commute*, i.e. the commutative law $\partial^\alpha \partial^\beta = \partial^\beta \partial^\alpha$ holds. — Note that the operator ∂^α is by its very definition itself an $(|\alpha| - 1)$ -fold operator product whose factors are the individual differential operators $\partial_k = \partial/\partial x_k$.

Besides the operator product (i.e. the composite operator), another common way of combining differential operators is *linear combination*. Here the principle is the same as for functions in general:

$$(\lambda \partial^\alpha + \mu \partial^\beta) f = \lambda \partial^\alpha f + \mu \partial^\beta f, \quad \lambda, \mu \in \mathbb{R}.$$

EXAMPLE 6 Defining, for functions of two variables, the differential operator $A = \partial_x + \partial_y$, we have

$$\begin{aligned} A^2 &= (\partial_x + \partial_y)(\partial_x + \partial_y) \\ &= \partial_x(\partial_x + \partial_y) + \partial_y(\partial_x + \partial_y) \\ &= (\partial_x)^2 + \partial_x \partial_y + \partial_y \partial_x + (\partial_y)^2 \\ &= (\partial_x)^2 + 2\partial_x \partial_y + (\partial_y)^2. \end{aligned}$$

Accordingly, $A^2 f = A(Af) = f_{xx} + 2f_{xy} + f_{yy}$. More generally, A^k , $k \in \mathbb{N}$ expands by the binomial formula, e.g.

$$A^3 f = f_{xxx} + 3f_{xxy} + 3f_{xyy} + f_{yyy}. \quad \square$$

In the linear-combination formula the coefficients λ, μ may also be variable, i.e. functions. When forming products of such *variable-coefficient* operators, the differentiation acts on the coefficients as well, so that the result differs from the constant-coefficient case. When expanding operator products, one then also needs the product rule.

EXAMPLE 7 Define, for functions of two variables, the differential operators $A = x\partial_x + y\partial_y$ and $B = y^2\partial_x - x^2\partial_y$. Then

$$\begin{aligned} AB &= (x\partial_x + y\partial_y)(y^2\partial_x - x^2\partial_y) \\ &= x\partial_x(y^2\partial_x) - x\partial_x(x^2\partial_y) + y\partial_y(y^2\partial_x) - y\partial_y(x^2\partial_y) \\ &= xy^2(\partial_x)^2 - 2x^2\partial_y - x^3\partial_x\partial_y + 2y^2\partial_x + y^3\partial_y\partial_x - x^2y(\partial_y)^2 \\ &= xy^2(\partial_x)^2 - x^2y(\partial_y)^2 + (y^3 - x^3)\partial_x\partial_y + 2y^2\partial_x - 2x^2\partial_y, \\ BA &= AB - y^2\partial_x + x^2\partial_y. \quad \square \end{aligned}$$

In the example the differential operators do not commute, i.e. $AB \neq BA$. In the variable-coefficient case this is the rule. Constant-coefficient operators, by contrast, always commute (under the conditions of the interchange rule).

Chain rules

If in a function $f(x_1, \dots, x_n)$ the variables x_i are made into functions of a variable t , i.e. $x_i = x_i(t)$, we obtain the composite function $F(t) = f(x_1(t), \dots, x_n(t))$. For it the following generalization of the chain rule for a composite function holds, called the *chain rule*.

$$\begin{aligned} F(t) &= f(x_1(t), \dots, x_n(t)) \\ \Rightarrow F'(t) &= \sum_{k=1}^n \frac{\partial f}{\partial x_i} x'_i(t) \quad (\text{chain rule}). \end{aligned} \tag{2a}$$

If in place of t there is a vector variable $(u_1, \dots, u_m)$, then the more general form of the chain rule is

$$\begin{aligned} F(u_1, \dots, u_m) &= f(x_1(u_1, \dots, u_m), \dots, x_n(u_1, \dots, u_m)) \\ \Rightarrow \frac{\partial F}{\partial u_k} &= \sum_{i=1}^n \frac{\partial f}{\partial x_i} \frac{\partial x_i}{\partial u_k} \quad (\text{general chain rule}). \end{aligned} \tag{2b}$$

In Theorem [XI.2.2](#) below we present sufficient regularity conditions for the validity of the chain rule (2a) in the case $n = 2$. The corresponding conditions and the proof for the general rule (2b) ($n, m \in \mathbb{N}$) are easily generalized from this case.

EXAMPLE 8 The function $y(x)$ is the solution of the initial value problem

$$\begin{cases} y' = f(x, y), \\ y(0) = 1 \end{cases}$$

Determine the Taylor polynomial $T_2(x, 0)$ of y , given that $f(0, 1) = 2$, $f_x(0, 1) = -2$ and $f_y(0, 1) = 4$.

Solution By the differential equation and the initial condition, $y'(0) = f(0, y(0)) = f(0, 1) = 2$, so by the differential equation and the chain rule (2a) we have further

$$y''(x) = \frac{d}{dx}f(x, y(x)) = f_x(x, y(x)) + f_y(x, y(x))y'(x),$$

so $y''(0) = -2 + 4 \cdot 2 = 6$. Thus the requested Taylor polynomial is

$$T_2(x, 0) = y(0) + y'(0)x + \frac{1}{2}y''(0)x^2 = 1 + 2x + 3x^2. \quad \square$$

Implicit (partial) differentiation

One application of the chain rules (2a)–(2b) is the general principle of implicit differentiation or partial differentiation. Let, for example, F be a function of two variables, and suppose that the equation

$$F(x, y) = 0$$

defines a differentiable function $y = y(x)$. Then $F(x, y(x)) = 0$, so by the rule (2a) (assuming sufficient regularity of F)

$$0 = \frac{d}{dx}F(x, y(x)) = F_x(x, y(x)) + F_y(x, y(x))y'(x).$$

As long as $F_y(x, y(x)) \neq 0$, one can solve this for $y'(x)$:

$$y'(x) = -\frac{F_x(x, y)}{F_y(x, y)}, \quad y = y(x).$$

EXAMPLE 9 The equation

$$F(x, y) = xy^3 - e^{-xy^2} + y + \sin(y \cos x) = 0$$

defines a function $y(x)$ in a neighborhood of the origin. If one wishes to compute $y'(0)$, one must first compute $y(0) = y_0$ by solving (numerically)

$$y + \sin y = 1 \Rightarrow y = y_0.$$

After this, the chain rule (2a) gives

$$F_x(0, y_0) + F_y(0, y_0)y'(0) = 0,$$

where

$$\begin{aligned} F_x &= y^3 + y^2 e^{-xy^2} - y \sin x \cos(y \cos x) \\ F_y &= 3xy^2 + 2xy e^{-xy^2} + 1 + \cos x \cos(y \cos x) \\ \Rightarrow y'(0) &= -\frac{y_0^3 + y_0^2}{1 + \cos y_0}. \quad \square \end{aligned}$$

When the implicit function has several variables, the chain rule (2b) can be used for *implicit partial differentiation*. For example, if the equation

$$F(x, y, z) = 0$$

defines a function $z = z(x, y)$, then by the rule (2b)

$$\begin{aligned} 0 &= \partial_x F(x, y, z(x, y)) = \frac{\partial F}{\partial x} + \frac{\partial F}{\partial z} \frac{\partial z}{\partial x}, \\ 0 &= \partial_y F(x, y, z(x, y)) = \frac{\partial F}{\partial y} + \frac{\partial F}{\partial z} \frac{\partial z}{\partial y}. \end{aligned}$$

As long as $F_z(x, y, z) \neq 0$, one can solve this for $\partial z/\partial x$ and $\partial z/\partial y$.

EXAMPLE 10 In a neighborhood of the point $(x, y, z) = (1, 2, 0)$ define

$$z = f(x, y) \Leftrightarrow 4x^2 - y^2 + z^2 + e^z = 1.$$

Differentiating implicitly gives

$$8x + (2z + e^z) \frac{\partial z}{\partial x} = 0, \quad -2y + (2z + e^z) \frac{\partial z}{\partial y} = 0.$$

Here $\partial z/\partial x = f_x(x, y)$ and $\partial z/\partial y = f_y(x, y)$, so substituting $x = 1$, $y = 2$, $z = 0$ gives

$$f_x(1, 2) = -8, \quad f_y(1, 2) = 4.$$

By differentiating implicitly further, one can compute higher-order partial derivatives of f . For example,

$$\begin{aligned} 0 &= \partial_y \left(8x + (2z + e^z) \frac{\partial z}{\partial x} \right) = (2 + e^z) \frac{\partial z}{\partial x} \frac{\partial z}{\partial y} + (2z + e^z) \frac{\partial^2 z}{\partial x \partial y} \\ \Rightarrow 0 &= 3 \cdot (-8) \cdot 4 + 1 \cdot f_{xy}(1, 2) \Rightarrow f_{xy}(1, 2) = 96. \quad \square \end{aligned}$$

Differentiation under the integral sign

We wish to derive a differentiation rule for a function F defined by means of an integral in the form

$$F(x) = \int_a^b f(x, t) dt.$$

Since by the linearity of the integral

$$\frac{F(x + \Delta x) - F(x)}{\Delta x} = \int_a^b \frac{f(x + \Delta x, t) - f(x, t)}{\Delta x} dt,$$

it would appear that differentiation can be carried under the integral, turned into partial differentiation:

$$F(x) = \int_a^b f(x, t) dt \Rightarrow F'(x) = \int_a^b \partial_x f(x, t) dt. \quad (3a)$$

In the multivariable case this corresponds to the rule

$$\frac{\partial}{\partial x_k} \int_a^b f(x_1, \dots, x_n, t) dt = \int_a^b \frac{\partial}{\partial x_k} f(x_1, \dots, x_n, t) dt. \quad (3b)$$

The rules (3a)–(3b) really do hold quite generally, so they may be regarded (like the interchange rule (1)) as practically applicable general rules, with only rare exceptions (see Exercise 13a). — In unclear cases the justification of the rules (3a)–(3b) must ultimately rest directly on the definition of the derivative (see Exercise 21). In the case of an ordinary (Riemann) integral over the interval $[a, b]$, sufficient guarantees for the validity of the rule (3a) are provided by Theorem XI.2.3 below.

EXAMPLE 11 $F(x) = \int_1^2 \frac{e^{-xt}}{t} dt, \quad F'(0) = ?$

Solution The rule (3a) is valid (see Theorem XI.2.3), so

$$F'(0) = \left[\int_1^2 \partial_x \left(\frac{e^{-xt}}{t} \right) dt \right]_{x=0} = \left[\int_1^2 (-e^{-xt}) dt \right]_{x=0} = \int_1^2 (-1) dt = -1. \quad \square$$

Justifications of the rules of partial differentiation

In the following we justify the interchange rule of partial differentiation (1), the chain rule (2a), and the rule for interchanging differentiation and integration (3a),

restricting ourselves in the last case to the ordinary Riemann integral over the interval $[a, b]$. The continuity assumptions refer to the definitions of the previous section.

THEOREM XI.2.1 (Interchange rule of partial differentiation) If a function f of two variables is continuous and the partial derivatives f_x and f_y exist and are continuous in a neighborhood $U_\delta(x, y) \subset \mathcal{D}_f$ of the point (x, y) , and in addition the partial derivatives f_{xy} and f_{yx} exist in that neighborhood and are continuous at the point (x, y) , then $f_{xy}(x, y) = f_{yx}(x, y)$.

Proof Consider a sequence $\{(\Delta x_n, \Delta y_n)\}$ for which $(\Delta x_n, \Delta y_n) \rightarrow (0, 0)$ and $|\Delta x_n|^2 + |\Delta y_n|^2 < \delta^2 \forall n$, so that $(x + \Delta x_n, y + \Delta y_n) \in U_\delta(x, y) \forall n$. We examine the expression

$$\delta_n[f] = f(x + \Delta x_n, y + \Delta y_n) - f(x, y + \Delta y_n) - f(x + \Delta x_n, y) + f(x, y).$$

By grouping the terms of this expression in different ways, setting

$$F_n(t) = f(t, y + \Delta y_n) - f(t, y), \quad G_n(t) = f(x + \Delta x_n, t) - f(x, t)$$

and using the Mean Value Theorem of differential calculus (see Exercise 15), we obtain

$$\begin{aligned} \delta_n[f] &= F_n(x + \Delta x_n) - F_n(x) \\ &= F'_n(\xi_n) \Delta x_n \\ &= \partial_x [f(t, y + \Delta y_n) - f(t, y)]_{t=\xi_n} \Delta x_n \\ &= [f_x(\xi_n, y + \Delta y_n) - f_x(\xi_n, y)] \Delta x_n \\ &= [\partial_y [f_x(\xi_n, y)]_{y=\eta_n} \Delta y_n] \Delta x_n \\ &= f_{xy}(\xi_n, \eta_n) \Delta x_n \Delta y_n, \\ \delta_n[f] &= G_n(y + \Delta y_n) - G_n(y) \\ &= G'_n(\eta'_n) \Delta y_n \\ &= \partial_y [f(x + \Delta x_n, t) - f(x, t)]_{t=\eta'_n} \Delta y_n \\ &= [f_y(x + \Delta x_n, \eta'_n) - f_y(x, \eta'_n)] \Delta y_n \\ &= [\partial_x [f_y(x, \eta'_n)]_{x=\xi'_n} \Delta x_n] \Delta y_n \\ &= f_{yx}(\xi'_n, \eta'_n) \Delta x_n \Delta y_n. \end{aligned}$$

Here $|\xi_n - x|, |\xi'_n - x| \leq |\Delta x_n|$ and $|\eta_n - y|, |\eta'_n - y| \leq |\Delta y_n|$ (Exercise 15), so $(\xi_n, \eta_n) \rightarrow (x, y)$ and $(\xi'_n, \eta'_n) \rightarrow (x, y)$, because $(\Delta x_n, \Delta y_n) \rightarrow (0, 0)$. Consequently, and because f_{xy} and f_{yx} are by hypothesis continuous at the point (x, y) , it follows that

$$f_{xy}(\xi_n, \eta_n) \rightarrow f_{xy}(x, y) \quad \text{and} \quad f_{yx}(\xi'_n, \eta'_n) \rightarrow f_{yx}(x, y), \quad \text{as } n \rightarrow \infty.$$

But by the above results $f_{xy}(\xi_n, \eta_n) = f_{yx}(\xi'_n, \eta'_n) \forall n$, so we must also have $f_{xy}(x, y) = f_{yx}(x, y)$. $\square$

THEOREM XI.2.2 (Chain rule) If in the composite function $F(t) = f(x(t), y(t))$ the function $f(x, y)$ is continuous in a neighborhood $U_\delta(x(t), y(t))$ of the point $(x(t), y(t))$, the partial derivatives f_x and f_y exist in that neighborhood and are continuous at the point $(x(t), y(t))$, and the derivatives $x'(t)$ and $y'(t)$ exist, then the differentiation rule

$$F'(t) = f_x(x(t), y(t))x'(t) + f_y(x(t), y(t))y'(t)$$

holds.

Proof Let $\varepsilon > 0$ be sufficiently small (a more precise condition shortly) and consider a sequence $\{\Delta t_n\}$ for which $0 < |\Delta t_n| < \varepsilon \forall n$ and $\Delta t_n \rightarrow 0$ ($n \rightarrow \infty$). Write $x(t) = x$, $y(t) = y$ and $x(t + \Delta t_n) = x + \Delta x_n$, $y(t + \Delta t_n) = y + \Delta y_n$, $n = 1, 2, \dots$. Since $x'(t)$ and $y'(t)$ exist,

$$\Delta x_n = x'(t)\Delta t_n + o(|\Delta t_n|), \quad \Delta y_n = y'(t)\Delta t_n + o(|\Delta t_n|).$$

Consequently $\varepsilon > 0$ can be chosen so that $(\Delta x_n)^2 + (\Delta y_n)^2 < \delta^2 \forall n$ holds when $|\Delta t_n| < \varepsilon \forall n$, whereupon $(x + \Delta x_n, y + \Delta y_n) \in U_\delta(x, y) \forall n$. Assuming this, by the Mean Value Theorem of differential calculus

$$\begin{aligned} F(t + \Delta t_n) - F(t) &= f(x + \Delta x_n, y + \Delta y_n) - f(x, y) \\ &= [f(x + \Delta x_n, y + \Delta y_n) - f(x, y + \Delta y_n)] + [f(x, y + \Delta y_n) - f(x, y)] \\ &= f_x(\xi_n, y + \Delta y_n)\Delta x_n + f_y(x, \eta_n)\Delta y_n, \end{aligned}$$

where $\xi_n \in [x, x + \Delta x_n]$ or $\xi_n \in [x + \Delta x_n, x]$ and correspondingly $\eta_n \in [y, y + \Delta y_n]$ or $\eta_n \in [y + \Delta y_n, y]$. (If $\Delta x_n = 0$, set $\xi_n = x$, resp. $\eta_n = y$ if $\Delta y_n = 0$.) Dividing through by Δt_n , it follows that $\Delta x_n/\Delta t_n \rightarrow x'(t)$ and $\Delta y_n/\Delta t_n \rightarrow y'(t)$ as $n \rightarrow \infty$, and by the continuity assumptions $f_x(\xi_n, y + \Delta y_n) \rightarrow f_x(x, y)$ and $f_y(x, \eta_n) \rightarrow f_y(x, y)$. Consequently the limit $\lim_{\Delta t \rightarrow 0} [F(t + \Delta t) - F(t)]/\Delta t = F'(t)$ exists and the asserted differentiation rule holds. $\square$

The continuity assumptions of the following theorem refer to Definition [XI.1.10](#). The proof rests on the deeper logic of continuity (uniform continuity, cf. Section [V.9](#)). We present only a simplified version of the proof, based on slightly strengthened assumptions.

***THEOREM XI.2.3 (Differentiation under the integral sign)** If $f(x, t)$ is continuous and $\partial_x f = f_x(x, t)$ exists and is continuous in the rectangle $K_\delta = [c - \delta, c + \delta] \times [a, b]$ for some $\delta > 0$, then the differentiation rule [\(3a\)](#) is valid when $x = c$.

Proof (simplified) Assume in addition that f_x satisfies the Lipschitz condition

$$|f_x(x_1, t) - f_x(x_2, t)| \leq L|x_1 - x_2| \quad \forall (x_1, t), (x_2, t) \in K_\delta. \quad (\star)$$

If $0 < |\Delta x| \leq \delta$, then by the assumptions (f continuous, f_x existing) and the Mean Value Theorem of differential calculus

$$f(c + \Delta x, t) - f(c, t) - f_x(c, t)\Delta x = [f_x(\xi, t) - f_x(c, t)]\Delta x, \\ \text{where } \xi \in (c - |\Delta x|, c + |\Delta x|) \subset (c - \delta, c + \delta).$$

By this and the Lipschitz condition ($\star$) we can estimate

$$\left| \frac{f(c + \Delta x, t) - f(c, t)}{\Delta x} - f_x(c, t) \right| \leq L|\xi - c| \leq L|\Delta x|, \quad t \in [a, b].$$

Consequently, and since $f(c + \Delta x, t)$, $f(c, t)$ and $f_x(c, t)$ are (with respect to t) Riemann-integrable over $[a, b]$ by the assumptions and the Fundamental Theorem of Calculus, we conclude

$$\int_a^b \frac{f(c + \Delta x, t) - f(c, t)}{\Delta x} dt = \int_a^b f_x(c, t) dt + \mathcal{O}(|\Delta x|) \\ \Rightarrow F'(c) = \lim_{\Delta x \rightarrow 0} \int_a^b \frac{f(c + \Delta x, t) - f(c, t)}{\Delta x} dt = \int_a^b f_x(c, t) dt. \quad \square$$

EXERCISES

1. Compute the partial derivatives with respect to all variables:

$$\begin{array}{lll} \text{a) } xy^3 + x \sin(xy) & \text{b) } \ln \sin(x - 2y) & \text{c) } \operatorname{Arcsin} \frac{y}{x} \\ \text{d) } x^y & \text{e) } \log_y x & \text{f) } z^{xy} \quad \text{g) } z^{xy} \quad \text{h) } e^{xy(s^2 + st)} \end{array}$$

2. At which points do the following functions have both partial derivatives?

$$\text{a) } \sqrt{x^2 + y^2} \quad \text{b) } (x + y)|x + y| \quad \text{c) } \sqrt{|x^2 - y^2|}$$

3. Compute

$$\begin{array}{l} \text{a) } (3\partial_x - 2\partial_y)^2(x^2 + xy), \quad (x, y) = (0, 0) \\ \text{b) } (\partial_x + \partial_y)(\partial_y + \partial_z)xy^2z^3, \quad (x, y, z) = (1, 1, 1) \\ \text{c) } \partial_x xy \partial_y xy^2, \quad (x, y) = (2, 1) \\ \text{d) } (x\partial_y - y\partial_x)^2(xy - y^2), \quad (x, y) = (1, 1) \\ \text{e) } (\partial_1 + \dots + \partial_n)(x_1^2 + \dots + x_n^2), \quad (x_1, \dots, x_n) = (1, 2, \dots, n) \\ \text{f) } \partial_1 \dots \partial_n \ln(x_1^2 + \dots + x_n^2), \quad (x_1, \dots, x_n) = (1, 1, \dots, 1) \end{array}$$

4. Compute the partial derivatives of the function f at the given point P . If m is given, also compute the higher-order partial derivatives up to order m . In parts j)–m) use implicit partial differentiation.
 - a) $f(x, y) = x - y + 2$, $P = (3, 2)$, $m = 2$
 - b) $f(x, y) = xy + x^2$, $P = (2, 0)$, $m = 2$
 - c) $f(x, y) = \operatorname{Arctan}(y/x)$, $P = (-1, 1)$
 - d) $f(x, y) = \sin(x\sqrt{y})$, $P = (\pi/3, 4)$
 - e) $f(x, y) = 1/\sqrt{x^2 + y^2}$, $P = (-3, 4)$
 - f) $f(x, y, z) = xyz$, $P = (1, -1, 2)$, $m = 3$
 - g) $f(x, y, z) = \ln(1 + e^{xyz})$, $P = (2, 0, -1)$
 - h) $f(x, y, z) = x^{y \ln z}$, $P = (e, 2, e)$
 - i) $f(x_1, x_2, x_3, x_4) = (x_1 - x_2^2)/(x_3 + x_4^2)$, $P = (3, 1, -1, -2)$
 - j) $z = f(x, y) : x^2 - 2y^2 - 3yz^3 - z = 0$, $P : x = 2, y = z = -1$
 - k) $z = f(x, y) : \cos xyz = \sin(x - y + 2z)$, $P : x = y = 1, z = \pi/2$
 - l) $z = f(x, y) : 2xz^3 - xyz = 1$, $P : x = y = z = 1, m = 2$
 - m) $u = f(x, y, z) : xyz u + xy u^2 + xu^3 = 1$, $P : x = u = 1, z = y = -1$
5. Using the partial derivatives of the function $f(x, y)$, compute the expressions $A^2 f$, $B^2 f$ and $(AB - BA)f$, when
 - a) $A = \partial_x + \partial_y$, $B = \partial_x - \partial_y$
 - b) $A = 2\partial_x + \partial_y$, $B = \partial_x + 3\partial_y$
 - c) $A = y\partial_x$, $B = x\partial_y$
 - d) $A = x\partial_x + y\partial_y$, $B = y\partial_x - x\partial_y$
6. Show that:
 - a) $[(\partial_x)^2 + (\partial_y)^2] \ln(x^2 + y^2) = 0$, $(x, y) \neq (0, 0)$
 - b) $[(\partial_x)^2 + (\partial_y)^2 + (\partial_z)^2] \frac{1}{\sqrt{x^2 + y^2 + z^2}} = 0$, $(x, y, z) \neq (0, 0, 0)$
7. Compute both directly and using the chain rule:
 - a) $\partial u / \partial t$, when $u = \sqrt{x^2 + y^2}$, $x = e^{st}$, $y = 1 + s^2 \cos t$
 - b) $\partial z / \partial x$, when $z = \operatorname{Arctan}(u/v)$, $u = 2x + y$, $v = 2x - y$
 - c) dz/dt , when $z = txy^2$, $x = t + \ln(y + t^2)$, $y = e^t$
8. Suppose the expressions $f_x(x, y)$, $f_y(x, y)$, $f_{xx}(x, y)$, etc. for the partial derivatives of the function $f(x, y)$ are known. Using these, compute:
 - a) $\frac{\partial}{\partial x} f(2x, 3y)$
 - b) $\frac{\partial}{\partial x} f(2y, 3x)$
 - c) $\frac{\partial}{\partial y} f(yf(x, t), f(y, t))$
 - d) $\frac{\partial^2}{\partial s \partial t} f(t \sin s, t \cos s)$
 - e) $\frac{\partial^3}{\partial t^2 \partial s} f(s^2 - t, s + t^2)$

9. It is known about the function $y(x)$ that y satisfies on the interval $(-a, a)$ the differential equation $y' = f(x, y)$ and that $y(0) = 0$. It is known about the function $f(x, y)$ that f is regular in a neighborhood of the origin and $f(0, 0) = 1$, $f_x(0, 0) = -1$, $f_y(0, 0) = 2$, $f_{xx}(0, 0) = 4$, $f_{xy}(0, 0) = -2$, $f_{yy}(0, 0) = -6$. Determine the third-degree Taylor polynomial $T_3(x, 0)$ of the function $y(x)$.
10. A differential equation $f(x, y) + g(x, y)y' = 0$ is said to be *exact* if $f(x, y) = F_x(x, y)$ and $g(x, y) = F_y(x, y)$ for some F . Conclude that the general solution of an exact differential equation is $F(x, y) = C$. Apply this solution idea:
- a) $e^{x^2}(y' + 2xy) = 0$ b) $3x^2 + 6xy^2 + (6x^2y + 4y^3)y' = 0$
 c) $e^{-y} + (1 - xe^{-y})y' = 0$ d) $2x(x + \cos^2 y) + (2y - x^2 \sin 2y)y' = 0$
11. Compute $f'(0)$:

a) $f(x) = \int_0^1 \frac{e^{xt}}{1+t^2} dt$ b) $f(x) = \int_1^2 \ln t e^{-xt^2} dt$
 c) $f(x) = \int_0^\pi \sin xt \cos t dt$ d) $f(x) = \int_0^1 \ln(1 + xt + t^2)e^{-xt} dt$

12. The function f is defined in a neighborhood of the origin by the formula

$$f(x, y) = \int_0^1 \frac{\sin xt \cos yt}{1 + x + y + t} dt.$$

Compute the partial derivatives of f up to second order at the origin.

13. a) Let $F(x) = \int_0^\infty x^3 e^{-x^2 t} dt$, $x \in \mathbb{R}$. Show that F is differentiable everywhere, but $F'(0)$ cannot be computed by differentiating under the integral sign.
 b) Verify that by the differentiation rule (3a)

$$\frac{d}{dx} \int_1^\infty \frac{e^{-xt}}{t} dt = -\frac{e^{-x}}{x}, \quad x > 0.$$

Confirm the result by changing the integration variable to $s = xt$.

14. Compute the following integrals by differentiating the given function $F(x)$, $x >$

0.

$$\begin{aligned}
\text{a) } I_n &= \int_0^\infty t^n e^{-t} dt, \quad n \in \mathbb{N}, \quad F(x) = \int_0^\infty e^{-xt} dt \\
\text{b) } I_n &= \int_0^\infty \frac{1}{(t^2 + 1)^n} dt, \quad n = 2, 3, 4, \quad F(x) = \int_0^\infty \frac{1}{t^2 + x^2} dt \\
\text{c) } I_n(x) &= \int_0^1 t^{x-1} (\ln t)^n dt, \quad n \in \mathbb{N}, \quad F(x) = \int_0^1 t^{x-1} dt
\end{aligned}$$

15. To which (one-variable) functions is the Mean Value Theorem of differential calculus applied in the proof of Theorem [XI.2.1](#), and on which intervals? — Check the applicability of the mean value theorem based on the assumptions of the theorem, and also check that the reasoning works in the case $\Delta x_n = 0$ or $\Delta y_n = 0$.
16. (*) Investigate which of the assumptions of Theorem [XI.2.1](#) hold when $(x, y) = (0, 0)$ and

$$f(x, y) = \begin{cases} 0, & \text{when } (x, y) = (0, 0) \\ \frac{x^3 y - xy^3}{x^2 + y^2}, & \text{otherwise} \end{cases}$$

Does the interchange rule hold?

17. (*) Show that if the function $f(x, y)$ and the partial derivatives f_x and f_y are continuous in the open rectangle $A = (a, b) \times (c, d)$ and $f_{xx} = f_{xy} = f_{yx} = f_{yy} = 0$ in A , then $f(x, y) = a + bx + cy$, $(x, y) \in A$ for some $(a, b, c) \in \mathbb{R}^3$ (i.e. f is a first-degree polynomial).
18. (*) An airplane flies upward along the space curve

$$S: y = x^2, \quad z = \frac{1}{3}(2x + y^2),$$

where $z > 0$ is the altitude above the ground (units km). The air temperature near the flight path is (unit °C)

$$T(x, y, z) = -10(z^2 - z + 1) + (2x^2 + 3y)/(1 + z^2).$$

The outside air temperature is also measured aboard the plane — let the measured value be $T(t)$ at time t (min). At a certain instant the plane is at the point $(1, 1, 1)$ and its speed is 6 km/min. What is, at that instant, the instantaneous rate of change $T'(t)$ (unit °C/min) of the outside temperature computed from the reading $T(t)$?

19. (*) Let

$$F(x, y) = \int_0^\infty \frac{e^{-xt} - e^{-yt}}{t} dt, \quad x > 0, y > 0.$$

By computing $\partial F/\partial x$ and $\partial F/\partial y$, show that $F(x, y) = \ln(y/x)$.

20. (*) Compute $y(0)$ and $y'(0)$, where $y(x)$ is defined in a neighborhood of the point $x = 0$ by the formula

$$\int_0^1 \frac{e^{xyt}}{x + y + t} dt = \ln 2.$$

21. (*) Verify that by the rule (3a) the derivative of the Γ -function $\Gamma(x) = \int_0^\infty t^{x-1} e^{-t} dt$ is

$$\Gamma'(x) = \int_0^\infty t^{x-1} \ln t e^{-t} dt, \quad x > 0.$$

Confirm this differentiation rule directly from the definition of the derivative by showing that there exists a constant $C(x)$ depending only on x such that

$$\left| \frac{\Gamma(x + \Delta x) - \Gamma(x)}{\Delta x} - \int_0^\infty t^{x-1} \ln t e^{-t} dt \right| \leq C(x) |\Delta x|,$$

when $x > 0$ and $|\Delta x| \leq x/2$. *Hint:* First prove the auxiliary result:

$$|e^x - 1 - x| \leq \frac{1}{2} x^2 e^{|x|}, \quad x \in \mathbb{R}.$$

XI.3 Gradient

Let us recall from Section VII.1 that if a function f of one variable is differentiable at a point x , then

$$f(x + \Delta x) = f(x) + df(x, \Delta x) + o(|\Delta x|),$$

where $df(x, \Delta x) = f'(x)\Delta x$ is the differential of f . The corresponding result for a function of two variables $f(x, y)$ is an expansion of the form

$$f(x + \Delta x, y + \Delta y) = f(x, y) + df(x, y, \Delta x, \Delta y) + o(h), \quad h = \sqrt{(\Delta x)^2 + (\Delta y)^2}.$$

Insofar as this holds in some neighborhood of the point (x, y) , we again say that $df(x, y, \Delta x, \Delta y)$ is the *differential* of f at the point (x, y) . To derive the expression for the differential we assume, as in the chain rule above (Theorem XI.2.2), that f is continuous in a neighborhood $U_\delta(x, y)$ of the point (x, y) and that the partial derivatives f_x and f_y exist in that neighborhood and are continuous at the point (x, y) . Then (cf. the proof of Theorem XI.2.2)

$$\begin{aligned} f(x + \Delta x, y + \Delta y) - f(x, y) &= [f(x + \Delta x, y + \Delta y) - f(x, y + \Delta y)] + [f(x, y + \Delta y) - f(x, y)] \\ &= f_x(\xi, y + \Delta y)\Delta x + f_y(x, \eta)\Delta y \\ &= f_x(x, y)\Delta x + f_y(x, y)\Delta y + r(x, y, \Delta x, \Delta y), \end{aligned}$$

where

$$r(x, y, \Delta x, \Delta y) = [f_x(\xi, y + \Delta y) - f_x(x, y)]\Delta x + [f_y(x, \eta) - f_y(x, y)]\Delta y.$$

Since here $|\xi - x| \leq |\Delta x|$ and $|\eta - y| \leq |\Delta y|$ and f_x and f_y are continuous at the point (x, y) , we have $f_x(\xi, y + \Delta y) - f_x(x, y) = o(1)$ and $f_y(x, \eta) - f_y(x, y) = o(1)$ as $h \rightarrow 0$. Thus (cf. the rules of order-of-magnitude algebra in Section VII.5)

$$r(x, y, \Delta x, \Delta y) = o(1)\Delta x + o(1)\Delta y = o(h), \quad h = \sqrt{(\Delta x)^2 + (\Delta y)^2}.$$

We have concluded that the expansion

$$\boxed{f(x + \Delta x, y + \Delta y) = f(x, y) + f_x(x, y)\Delta x + f_y(x, y)\Delta y + o(h).} \quad (1)$$

holds. The sought expression for the differential (under the assumptions made) is thus

$$df(x, y, \Delta x, \Delta y) = f_x(x, y)\Delta x + f_y(x, y)\Delta y.$$

EXAMPLE 1 Estimate $f(1.01, 1.98)$ using the differential $df(1, 2, 0.01, -0.02)$, where $f(x, y) = xy - 2y^3$.

Solution Here $f_x(x, y) = y$ and $f_y(x, y) = x - 6y^2$, so

$$\begin{aligned} f(1.01, 1.98) &= f(1 + 0.01, 2 - 0.02) \\ &\approx f(1, 2) + f_x(1, 2) \cdot 0.01 + f_y(1, 2) \cdot (-0.02) \\ &= -14 + 2 \cdot 0.01 + (-23) \cdot (-0.02) = -13.52. \end{aligned}$$

The exact value is $f(1.01, 1.98) = -13.524984$. $\square$

When the vector notation

$$\Delta \vec{r} = \Delta x \vec{i} + \Delta y \vec{j},$$

is introduced in the differential expansion (1), and the vector notation $\vec{r} = x\vec{i} + y\vec{j}$ is also used in place of the variables x, y , then the expansion can be written in the form

$$f(\vec{r} + \Delta \vec{r}) = f(\vec{r}) + \nabla f(\vec{r}) \cdot \Delta \vec{r} + o(|\Delta \vec{r}|), \quad (2)$$

where ∇f is the *gradient* of f , defined by

$$\nabla f(x, y) = f_x(x, y)\vec{i} + f_y(x, y)\vec{j}.$$

For a function of three variables too, a differential expansion of the form (2) can be derived in the same way as above. In that case $\Delta \vec{r} = \Delta x \vec{i} + \Delta y \vec{j} + \Delta z \vec{k}$, and the gradient is defined by

$$\nabla f(x, y, z) = f_x(x, y, z)\vec{i} + f_y(x, y, z)\vec{j} + f_z(x, y, z)\vec{k}.$$

On the basis of the above results, the gradient may be regarded as a 'generalized derivative'. The one-variable derivative operator $D = d/dx$ thus corresponds, in the two- and three-variable cases, to the vector-valued differential operator

$$\nabla = \begin{cases} \vec{i} \partial_x + \vec{j} \partial_y & (d = 2), \\ \vec{i} \partial_x + \vec{j} \partial_y + \vec{k} \partial_z & (d = 3). \end{cases}$$

The symbol 'grad' is also used for the gradient. It is read 'gradient', 'grad', or 'nabla'.

EXAMPLE 2 The gradient of the function $f(x, y, z) = x^2 + 3y^2 + xyz$ is

$$\begin{aligned} \nabla f &= (\vec{i} \partial_x + \vec{j} \partial_y + \vec{k} \partial_z)(x^2 + 3y^2 + xyz) \\ &= (2x + yz)\vec{i} + (6y + xz)\vec{j} + xy\vec{k}. \quad \square \end{aligned}$$

The gradient of many variables

The differential expansions derived above are easily generalized to the more general functions of n variables. It is then most convenient to switch to matrix-algebra notation, i.e. index the variables,

$$x \hookrightarrow x_1, \quad y \hookrightarrow x_2, \quad \dots,$$

and write in place of $\vec{r}$ the column vector $\mathbf{x}$:

$$\vec{r} \hookrightarrow \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = \mathbf{x}, \quad \Delta \vec{r} \hookrightarrow \Delta \mathbf{x}.$$

With these notations the differential expansion (2) takes the form

$$f(\mathbf{x} + \Delta \mathbf{x}) = f(\mathbf{x}) + \sum_{k=1}^n \frac{\partial f}{\partial x_k}(\mathbf{x}) \Delta x_k + o(|\Delta \mathbf{x}|),$$

where $|\cdot|$ denotes the Euclidean norm of $\mathbb{R}^n$. The result can be written in the language of matrix algebra in the form

$$\boxed{\begin{aligned} f(\mathbf{x} + \Delta \mathbf{x}) &= f(\mathbf{x}) + \langle \nabla f(\mathbf{x}), \Delta \mathbf{x} \rangle + o(|\Delta \mathbf{x}|), \\ \nabla f(\mathbf{x}) &= \left[\frac{\partial f}{\partial x_1}(\mathbf{x}), \frac{\partial f}{\partial x_2}(\mathbf{x}), \dots, \frac{\partial f}{\partial x_n}(\mathbf{x}) \right]^T. \end{aligned}} \quad (3)$$

In the expansion (3) the term $df(\mathbf{x}, \Delta \mathbf{x}) = \langle \nabla f(\mathbf{x}), \Delta \mathbf{x} \rangle = (\nabla f(\mathbf{x}))^T \Delta \mathbf{x}$ is again the *differential*. Regardless of the number of variables, the differential can be used to estimate changes in the value of the function, in the same way as in Example 1 above.

EXAMPLE 3 The loading of a truss structure (see Section X.8) causes the displacements of the endpoints P and Q of an individual bar

$$\begin{aligned} \vec{u} &= \overrightarrow{PP'} = u_1 \vec{i} + u_2 \vec{j} + u_3 \vec{k}, \\ \vec{v} &= \overrightarrow{QQ'} = v_1 \vec{i} + v_2 \vec{j} + v_3 \vec{k}. \end{aligned}$$

Assuming that $|\vec{u}|, |\vec{v}| \ll |\vec{a}|$, $\vec{a} = \overrightarrow{QP}$, determine the elongation of the bar approximately by means of the differential.

Solution The elongation is a function of six variables

$$f(u_1, u_2, u_3, v_1, v_2, v_3) = |\vec{a} + \vec{u} - \vec{v}| - |\vec{a}|.$$

Here

$$\begin{aligned} |\vec{a} + \vec{u} - \vec{v}|^2 &= (\vec{a} + \vec{u} - \vec{v}) \cdot (\vec{a} + \vec{u} - \vec{v}) \\ &= |\vec{a}|^2 + 2\vec{a} \cdot (\vec{u} - \vec{v}) + |\vec{u} - \vec{v}|^2, \end{aligned}$$

so

$$f(u_1, \dots, v_3) = \left[|\vec{a}|^2 + \sum_{k=1}^3 2a_k(u_k - v_k) + \sum_{k=1}^3 (u_k - v_k)^2 \right]^{1/2} - |\vec{a}|.$$

Then $f(\mathbf{0}) = 0$ and

$$\frac{\partial f}{\partial u_k}(\mathbf{0}) = \frac{a_k}{|\vec{a}|}, \quad \frac{\partial f}{\partial v_k}(\mathbf{0}) = -\frac{a_k}{|\vec{a}|}, \quad k = 1, 2, 3,$$

so by the differential expansion (3)

$$f(u_1, \dots, v_3) \approx \sum_{k=1}^3 \frac{a_k}{|\vec{a}|} (u_k - v_k) = \underline{\underline{\vec{t} \cdot (\vec{u} - \vec{v})}},$$

where $\vec{t} = \vec{a}/|\vec{a}|$ is the unit vector in the direction of $\vec{a}$. $\square$

Differentiability

The expansion (3) is valid provided that f is continuous in a neighborhood of $\mathbf{x}$

$$U_\delta(\mathbf{x}) = \{\mathbf{x}' \in \mathbb{R}^n \mid |\mathbf{x}' - \mathbf{x}| < \delta\}, \quad \delta > 0$$

and in addition the partial derivatives $\partial f / \partial x_k$, $k = 1 \dots n$ exist in that neighborhood and are continuous at the point $\mathbf{x}$. Such background assumptions are dispensed with when we lay down the following more general definition of *differentiability* and, at the same time, of the gradient.

DEFINITION XI.3.1 A function $f : D_f \rightarrow \mathbb{R}$, $D_f \subset \mathbb{R}^n$, is *differentiable* at a point $\mathbf{x} \in D_f$ if f is defined in some neighborhood of the point $\mathbf{x}$ and there exists a column vector $\mathbf{h} \in \mathbb{R}^n$ such that

$$f(\mathbf{x} + \Delta\mathbf{x}) = f(\mathbf{x}) + \mathbf{h}^T \Delta\mathbf{x} + o(|\Delta\mathbf{x}|), \quad \text{when } |\Delta\mathbf{x}| \rightarrow 0.$$

Then $\mathbf{h}$ is the *gradient* of f at the point $\mathbf{x}$, denoted $\mathbf{h} = \nabla f(\mathbf{x})$.

If f is differentiable, then the gradient can be computed from the partial derivatives according to formula (3):

PROPOSITION XI.3.2 If f is differentiable at a point $\mathbf{x} \in \mathbb{R}^n$, then the partial derivatives $\partial f / \partial x_k$, $k = 1 \dots n$ exist at the point $\mathbf{x}$, and $[\nabla f(\mathbf{x})]_k = \partial f(\mathbf{x}) / \partial x_k$, $k = 1 \dots n$.

Proof Setting $\Delta \mathbf{x} = \Delta x_k \mathbf{e}_k$ in the expansion (3) ($\mathbf{e}_k$ = Euclidean unit vector), the expansion can be written in the form

$$f(x_1, \dots, x_{k-1}, x_k + \Delta x_k, x_{k+1}, \dots, x_n) - f(x_1, \dots, x_n) = [\nabla f(\mathbf{x})]_k \Delta x_k + o(|\Delta x_k|).$$

Dividing by Δx_k and letting $\Delta x_k \rightarrow 0$, it follows that $[\nabla f(\mathbf{x})]_k = \partial f(\mathbf{x}) / \partial x_k$. $\square$

As a summary of the above, we may note that continuity of the function in a neighborhood of the point $\mathbf{x}$, the existence of the partial derivatives in that neighborhood, and continuity at the point $\mathbf{x}$ together suffice to guarantee differentiability; on the other hand, differentiability at the point $\mathbf{x}$ guarantees only the existence of the partial derivatives at that point, not elsewhere. Comparing with the considerations of the previous section, we now see that differentiability is the most natural definition of what it means for a function of several variables to be differentiable. For, first, on the basis of the expansion (3) it is evident that differentiability immediately entails continuity at the point in question, as in the one-variable case (cf. Section V.3). Second, we see that in justifying the chain rules of partial differentiation, the continuity and other assumptions concerning the outer function can be replaced by differentiability. For if we consider the function $F(t) = f(x(t), y(t))$ of Theorem XI.2.2, write $x(t) = x$, $y(t) = y$, $x(t + \Delta t) = x + \Delta x$, $y(t + \Delta t) = y + \Delta y$ and assume that f is differentiable at the point (x, y) , then by the expansion (1)

$$\frac{F(t + \Delta t) - F(t)}{\Delta t} = f_x(x, y) \frac{\Delta x}{\Delta t} + f_y(x, y) \frac{\Delta y}{\Delta t} + (\Delta t)^{-1} o(h).$$

If furthermore the derivatives $x'(t)$ and $y'(t)$ exist, then here

$$h = \sqrt{(\Delta x)^2 + (\Delta y)^2} = \sqrt{[x'(t)]^2 + [y'(t)]^2} \Delta t + o(|\Delta t|),$$

so

$$(\Delta t)^{-1} o(h) = (\Delta t)^{-1} o(|\Delta t|) \rightarrow 0, \quad \text{when } \Delta t \rightarrow 0.$$

Thus the chain rule $F'(t) = f_x(x, y)x'(t) + f_y(x, y)y'(t)$ follows.

The correspondence between one-variable differentiability and multivariable differentiability is also reflected in the following generalization of the Mean Value Theorem of differential calculus. The proof is left as an exercise (Exercise 7c).

THEOREM XI.3.3 (Mean value theorem in $\mathbb{R}^n$) Let $\mathbf{a}, \mathbf{b} \in \mathbb{R}^n$, $\mathbf{a} \neq \mathbf{b}$ and let $f(\mathbf{x})$ be continuous at the point $\mathbf{x}(t) = t\mathbf{a} + (1-t)\mathbf{b}$ for every $t \in [0, 1]$ and differentiable at the point $\mathbf{x}(t)$ for every $t \in (0, 1)$. Then for some $\xi \in (0, 1)$

$$f(\mathbf{b}) - f(\mathbf{a}) = \langle \nabla f(\mathbf{x}(\xi)), \mathbf{b} - \mathbf{a} \rangle.$$

Directional derivatives

The gradient is used in studying functions of several variables in the same way as the derivative in the one-variable case. In particular, the gradient can be used to compute how the values of the function change near a given point $\mathbf{x}$ when one moves from that point in the direction indicated by some unit vector of $\mathbb{R}^n$. Namely, if $\mathbf{e}$ is such a vector, i.e.

$$\mathbf{e} = (e_i), \quad |\mathbf{e}|^2 = \sum_{i=1}^n e_i^2 = 1,$$

then setting $\Delta \mathbf{x} = \Delta t \mathbf{e}$ in the expansion (3) yields

$$f(\mathbf{x} + \Delta t \mathbf{e}) = f(\mathbf{x}) + \partial_{\mathbf{e}} f(\mathbf{x}) \Delta t + o(|\Delta t|), \quad (4)$$

where

$$\partial_{\mathbf{e}} f(\mathbf{x}) = \sum_{k=1}^n e_k \frac{\partial f}{\partial x_k}(\mathbf{x}) = \langle \nabla f(\mathbf{x}), \mathbf{e} \rangle.$$

The $\partial_{\mathbf{e}} f$ so defined can be called the *directional derivative* of f in the direction $\mathbf{e}$, since by the definition (4)

$$\partial_{\mathbf{e}} f(\mathbf{x}) = \lim_{\Delta t \rightarrow 0} \frac{f(\mathbf{x} + \Delta t \mathbf{e}) - f(\mathbf{x})}{\Delta t} = g'_{\mathbf{e}}(0), \quad g_{\mathbf{e}}(t) = f(\mathbf{x} + t\mathbf{e}).$$

In particular, if we choose $\mathbf{e} = \mathbf{e}_k$ (Euclidean unit vector), then $\partial_{\mathbf{e}} f = \partial f / \partial x_k$. Thus the partial derivatives are special cases of directional derivatives.

In the two- or three-variable case the expansion (4) can, if desired, be written in the form

$$f(\vec{r} + \Delta t \vec{e}) = f(\vec{r}) + \vec{e} \cdot \nabla f(\vec{r}) \Delta t + o(|\Delta t|),$$

so that the directional derivative is the differential operator

$$\partial_{\mathbf{e}} = \vec{e} \cdot \nabla = \begin{cases} e_x \partial_x + e_y \partial_y & (d = 2), \\ e_x \partial_x + e_y \partial_y + e_z \partial_z & (d = 3). \end{cases}$$

If in the multivariable case $\nabla f(\mathbf{x}) \neq \mathbf{0}$, then for the direction $\mathbf{e}$ one may choose $\mathbf{e} = \pm[\nabla f(\mathbf{x})]/|\nabla f(\mathbf{x})|$, whereupon correspondingly $\partial_{\mathbf{e}}f(\mathbf{x}) = \pm|\nabla f(\mathbf{x})|$. Since, on the other hand, by the Cauchy–Schwarz inequality

$$|\partial_{\mathbf{e}}f(\mathbf{x})| = \left| \sum_{k=1}^n e_k \frac{\partial f}{\partial x_k} \right| \leq |\mathbf{e}| |\nabla f(\mathbf{x})| = |\nabla f(\mathbf{x})|,$$

we have concluded that the largest value of the directional derivative $= |\nabla f(\mathbf{x})|$ in the direction of the gradient, and the smallest value $= -|\nabla f(\mathbf{x})|$ in the opposite direction. Thus:

The function increases most strongly in the direction of the gradient, decreases most strongly in the direction of the negative gradient, and changes least in the directions $\mathbf{e}$ for which $\langle \nabla f, \mathbf{e} \rangle = 0$.

EXAMPLE 4 Compute the derivative of the function $f(x, y, z) = x^3 - 2x^2yz + 4yz^2 - 2x - y + z$ in the direction $\vec{e} = (\vec{i} - 2\vec{j} - \vec{k})/\sqrt{6}$ at the point $(1, -1, 2)$, and determine the direction $\vec{n}$ in which the function increases fastest at that point.

Solution The gradient is

$$\begin{aligned} \nabla f(x, y, z) &= (3x^2 - 4xyz - 2)\vec{i} + (-2x^2z + 4z^2 - 1)\vec{j} + (-2x^2y + 8yz + 1)\vec{k} \\ \Rightarrow \nabla f(1, -1, 2) &= 9\vec{i} + 11\vec{j} - 13\vec{k}. \end{aligned}$$

The derivative in the direction $\vec{e}$ is

$$\vec{e} \cdot \nabla f(1, -1, 2) = \frac{1}{\sqrt{6}} [1 \cdot 9 - 2 \cdot 11 - 1 \cdot (-13)] = \underline{\underline{0}}.$$

The direction of fastest increase is

$$\begin{aligned} \vec{n} &= \frac{\nabla f(1, -1, 2)}{|\nabla f(1, -1, 2)|} = \frac{1}{\sqrt{371}} (9\vec{i} + 11\vec{j} - 13\vec{k}) \\ &\approx \underline{\underline{0.47\vec{i} + 0.57\vec{j} - 0.67\vec{k}}}. \quad \square \end{aligned}$$

EXAMPLE 5 The vertices of a triangle are at the points $P_1 = (1, 0)$, $P_2 = (0, 2)$ and $P_3 = (2, 3)$. Each vertex of the triangle is moved the same (small) distance Δs so that P_1 moves in the direction $\vec{j}$, P_2 in the direction $-\vec{i}$ and P_3 in the direction $\vec{e}$. How should the direction $\vec{e}$ be chosen so that the area of the triangle a) changes as little as possible, b) increases as much as possible, c) decreases as much as possible?

Solution If the vertices of the triangle after the change are $P_1 = (x_1, y_1)$, $P_2 = (x_2, y_2)$ and $P_3 = (x_3, y_3)$, then the area is (cf. Section II.6)

$$\begin{aligned} f(\mathbf{x}) = f(x_1, y_1, x_2, y_2, x_3, y_3) &= \frac{1}{2} \begin{vmatrix} x_3 - x_1 & y_3 - y_1 \\ x_2 - x_1 & y_2 - y_1 \end{vmatrix} \\ &= \frac{1}{2} (x_3 - x_1)(y_2 - y_1) - \frac{1}{2} (x_2 - x_1)(y_3 - y_1). \end{aligned}$$

If $\vec{e} = \cos \varphi \vec{i} + \sin \varphi \vec{j}$, then the displacement of the vertices is described by the vector

$$\Delta \mathbf{x} = \Delta s [0, 1, -1, 0, \cos \varphi, \sin \varphi]^T.$$

The gradient of the function f is

$$\begin{aligned} \nabla f(\mathbf{x}) &= \frac{1}{2} [y_3 - y_2, x_2 - x_3, y_1 - y_3, x_3 - x_1, y_2 - y_1, x_1 - x_2]^T \\ \Rightarrow \nabla f(1, 0, 0, 2, 2, 3) &= \frac{1}{2} [1, -2, -3, 1, 2, 1]^T. \end{aligned}$$

The change in area estimated by means of the differential is thus

$$\Delta f \approx \langle \nabla f(1, 0, 0, 2, 2, 3), \Delta \mathbf{x} \rangle = \frac{\Delta s}{2} (1 + 2 \cos \varphi + \sin \varphi).$$

The area changes least when $1 + 2 \cos \varphi + \sin \varphi = 0$, i.e. when either $\vec{e} = -\vec{j}$ ($\varphi = 3\pi/2$) or $\vec{e} = (-4\vec{i} + 3\vec{j})/5$. The area changes most when

$$D_\varphi(1 + 2 \cos \varphi + \sin \varphi) = -2 \sin \varphi + \cos \varphi = 0,$$

i.e. when $\vec{e} = \pm(2\vec{i} + \vec{j})/\sqrt{5}$. In these directions, correspondingly, $\Delta f \approx \frac{\Delta s}{2}(1 \pm \sqrt{5})$.

Answer:

$$\text{a) } \vec{e} = -\vec{j} \text{ or } \vec{e} = \frac{1}{5}(-4\vec{i} + 3\vec{j}), \text{ b) } \vec{e} = \frac{1}{\sqrt{5}}(2\vec{i} + \vec{j}), \text{ c) } \vec{e} = -\frac{1}{\sqrt{5}}(2\vec{i} + \vec{j}). \quad \square$$

Level surfaces

In the same way as the tangent and normal of a curve can be determined by means of the derivative (cf. Section VII.2), the partial derivatives can be used to determine the *tangent plane of a surface* and the normal of this plane, i.e. the *normal of the surface* (a line or vector). If the equation of a surface $S \subset E^3$ is

$$S : F(x, y, z) = 0,$$

then the normal of the surface (and, via the normal, the tangent plane) at a given point can be computed most quickly by regarding the surface as a level surface of the function F . It is evident that a unit vector $\vec{t}$ is a tangent vector of a level surface of F at the point (x_0, y_0, z_0) exactly when the directional derivative of F in the direction of $\vec{t}$ vanishes at that point, i.e. when $\vec{t} \cdot \nabla F(x_0, y_0, z_0) = 0$. On this basis $\vec{n} = \nabla F(x_0, y_0, z_0)$ is a normal vector of the surface at that point. The same result is reached if the surface under consideration is the level surface of F passing through the point (x_0, y_0, z_0) , whose equation is $F(x, y, z) = c = F(x_0, y_0, z_0)$. It has thus been concluded that

$\vec{n} = \nabla F(x_0, y_0, z_0)$ is the normal vector of a level surface of the function F at the point (x_0, y_0, z_0) .

EXAMPLE 6 Determine the equation of the tangent plane of the surface

$$S: F(x, y, z) = x^2 + y^2 + z^2 - xy - yz - xz = 1.$$

at the point $(1, 1, 2)$.

Solution The gradient of the function F is

$$\nabla F(x, y, z) = (2x - y - z)\vec{i} + (2y - x - z)\vec{j} + (2z - y - x)\vec{k},$$

so the normal vector of the surface at the point $(1, 1, 2)$ is

$$\vec{n} = \nabla F(1, 1, 2) = -\vec{i} - \vec{j} + 2\vec{k}.$$

The equation of the tangent plane at the point $(1, 1, 2) \hat{=} \vec{r}_0$ is

$$T: (\vec{r} - \vec{r}_0) \cdot \vec{n} = 0 \Leftrightarrow \underline{\underline{x + y - 2z + 2 = 0}}. \quad \square$$

EXAMPLE 7 If the equation of a surface is given in the form $S: z = f(x, y)$, then by writing the equation in the form $F(x, y, z) = z - f(x, y) = 0$ we obtain the normal vector at a point $(x, y, z) \in S$:

$$\vec{n} = \nabla F(x, y, z) = -f_x(x, y)\vec{i} - f_y(x, y)\vec{j} + \vec{k}. \quad \square$$

EXAMPLE 8 Determine the tangent vector of the curve

$$\begin{cases} x^2 - y^2 + z^2 = 1, \\ x^2 + y^2 - 2z^2 = 0 \end{cases}$$

at the point $(1, 1, 1)$.

Solution This is the intersection curve of two surfaces. The surfaces are level surfaces of the functions $F_1(x, y, z) = x^2 - y^2 + z^2$ and $F_2(x, y, z) = x^2 + y^2 - 2z^2$, whose normal vectors at the common point $(x, y, z) = (1, 1, 1)$ of the surfaces are

$$\begin{aligned}\vec{n}_1 &= \nabla F_1(x, y, z) = 2x\vec{i} - 2y\vec{j} + 2z\vec{k} = 2(\vec{i} - \vec{j} + \vec{k}), \\ \vec{n}_2 &= \nabla F_2(x, y, z) = 2x\vec{i} + 2y\vec{j} - 4z\vec{k} = 2(\vec{i} + \vec{j} - 2\vec{k}).\end{aligned}$$

The requested tangent vector $\vec{t}$ is perpendicular to these, so for example

$$\vec{t} = \frac{1}{4} \vec{n}_1 \times \vec{n}_2 = \underline{\underline{\vec{i} + 3\vec{j} + 2\vec{k}}}. \quad \square$$

The normal of a parametric surface

Consider a parametrized surface in $\mathbb{R}^3$ (cf. Section IV.4)

$$S : \vec{r} = \vec{r}(u, v) \Leftrightarrow \begin{cases} x = x(u, v), \\ y = y(u, v), \\ z = z(u, v). \end{cases}$$

If we assume here that the functions x , y and z are differentiable at the point (u, v) , then it is evident that the vectors

$$\begin{aligned}\vec{t}_u &= \lim_{\Delta u \rightarrow 0} \frac{\vec{r}(u + \Delta u, v) - \vec{r}(u, v)}{\Delta u} = \frac{\partial \vec{r}}{\partial u}(u, v), \\ \vec{t}_v &= \lim_{\Delta v \rightarrow 0} \frac{\vec{r}(u, v + \Delta v) - \vec{r}(u, v)}{\Delta v} = \frac{\partial \vec{r}}{\partial v}(u, v)\end{aligned}$$

are parallel to the tangent plane of the surface, i.e. *tangent vectors of the surface* at the point $P \hat{=} \vec{r}(u, v)$. It is reasonable to expect of the parametrization that these vectors be linearly independent, in which case the normal vector of the surface at the point P is

$$\vec{n} = \frac{\partial \vec{r}}{\partial u} \times \frac{\partial \vec{r}}{\partial v} \quad (\text{normal of a parametric surface}).$$

EXAMPLE 9 (Cf. Example 7.) If the equation of a surface is $S : z = f(x, y)$, then from the parametrized representation

$$S : \vec{r}(x, y) = x\vec{i} + y\vec{j} + f(x, y)\vec{k}$$

we obtain the tangent and normal vectors of S at the point $(x, y, f(x, y))$:

$$\begin{aligned}\vec{t}_x &= \partial_x \vec{r}(x, y) = \vec{i} + f_x(x, y)\vec{k}, \\ \vec{t}_y &= \partial_y \vec{r}(x, y) = \vec{j} + f_y(x, y)\vec{k}, \\ \vec{n} &= \vec{t}_x \times \vec{t}_y = -f_x(x, y)\vec{i} - f_y(x, y)\vec{j} + \vec{k}. \quad \square\end{aligned}$$

EXAMPLE 10 (Cf. Example IV.4: 14.) Determine the normal of the parametric surface

$$S : \begin{cases} x = (2 - 2v) \cos u - v \sin u, \\ y = v \cos u + (2 - 2v) \sin u, \\ z = 3v \end{cases}$$

at the point $(x, y, z) = (2, 0, 0)$.

Solution Since $(2, 0, 0) \hat{=} \vec{r} = 2\vec{i} = \vec{r}(0, 0)$, we compute

$$\begin{aligned} \vec{t}_u(0, 0) &= \frac{\partial \vec{r}}{\partial u}(0, 0) = \frac{\partial x}{\partial u}(0, 0)\vec{i} + \frac{\partial y}{\partial u}(0, 0)\vec{j} + \frac{\partial z}{\partial u}(0, 0)\vec{k} = 2\vec{j}, \\ \vec{t}_v(0, 0) &= \frac{\partial \vec{r}}{\partial v}(0, 0) = -2\vec{i} + \vec{j} + 3\vec{k} \\ \Rightarrow \vec{n} &= \vec{t}_u \times \vec{t}_v = 6\vec{i} + 4\vec{k} \uparrow\uparrow \underline{\underline{3\vec{i} + 2\vec{k}}}. \quad \square \end{aligned}$$

EXERCISES

1. Compute an approximate value for the function at the given point using the differential.

a) x^2y^3 , $(3.1, 0.9)$	b) $\text{Arctan}(y/x)$, $(3.01, 2.98)$
c) $24(x^2 + xy + y^2)^{-1}$, $(2.1, 1.9)$	d) $\sin(\pi xy + \ln y)$, $(0.01, 1.05)$
e) $\sqrt{x + 2y + 3z}$, $(1.9, 1.8, 1.1)$	f) xe^{y+z^2} , $(2.05, -3.92, 1.97)$
2. The edges of a rectangular box have been measured to within 1%. Estimate how large an effect (%) the measurement errors have at most on the box's
 - a) volume,
 - b) base area,
 - c) diagonal, computed from the measurements.
3. The distance of a point A from an observation point B has been sought by triangulation, choosing a second observation point C and measuring the angles ABC and ACB . The measurement results are: $\angle ACB = 45 \pm 0.1^\circ$, $\angle ABC = 105 \pm 0.1^\circ$, and the distance between the points B and C has been measured as 265 ± 0.5 m. Compute the sought distance d and estimate the error Δd by means of the differential (maximum error).

4. The height of a tower is measured by angle measurement from two points A and B , which are in the same direction as seen from the tower. The measured angles are $50 \pm 1^\circ$ and $35 \pm 1^\circ$. In addition, the distance between the points A and B has been measured as 100 ± 1 m. What is the computed value of the tower's height, and how large can the measurement error be at most, estimated by means of the differential?
5. An orienteer, who is at the origin of the map, is heading to a control that, according to the map, is in the direction $\varphi = 30^\circ$ (polar coordinate). Due to rain and sweat the map paper has shrunk by 1% in the x -direction and 0.6% in the y -direction. Estimate by how many % the orienteer should correct the direction of the control and the distance measured from the map in order to reach the correct values.
6. The area of a quadrilateral is a function $f(\mathbf{x}) = f(x_1, y_1, x_2, y_2, x_3, y_3, x_4, y_4)$, where (x_i, y_i) , $i = 1, \dots, 4$ are the vertices of the quadrilateral. Compute ∇f at the point $\mathbf{x} = (0, 0, 1, 0, 1, 1, 0, 1)$, and by means of the differential the approximate area of the quadrilateral whose vertices are at the points $(0.03, -0.05)$, $(1.02, -0.02)$, $(0.96, 1.01)$, $(-0.01, 0.98)$.
7. a) Let $f(0, 0) = 0$ and $f(x, y) = xy \sin(1/\sqrt{x^2 + y^2})$ otherwise. Show that f is differentiable at the origin but f_x and f_y are not continuous at the origin.
 b) Let $f(x, y) = 0$ when $x = 0$ or $y = 0$ and $f(x, y) = |xy|^\alpha (x^2 + y^2)^\beta$ otherwise ($\alpha, \beta \in \mathbb{R}$). Determine $A \subset \mathbb{R}^2$ and $B \subset \mathbb{R}^2$ such that f is differentiable at the origin exactly when $(\alpha, \beta) \in A$ and differentiable at every point $(x, y) \in \mathbb{R}^2$ exactly when $(\alpha, \beta) \in B$.
 c) Prove Theorem [XI.3.3](#) by considering the function $F(t) = f(t\mathbf{a} + (1-t)\mathbf{b})$.
8. Compute the directional derivative at the given point, both in the given direction and in the direction of fastest increase of the function:
 - a) e^{xy+y} , $(0, 0)$, $\vec{i} + \vec{j}$
 - b) $\sin(\pi xy) \cos(\pi y^2)$, $(1, 2)$, $\vec{i} - 2\vec{j}$
 - c) xy^2z^3 , $(-3, 2, 1)$, $6\vec{i} - 2\vec{j} + 3\vec{k}$
 - d) $xy^2 + yz^3$, $(3, -2, -1)$, $2\vec{i} + 2\vec{j} - 3\vec{k}$
9. Below, a function $f(x, y)$ and a point (x_0, y_0) are given. In each case determine (i) the gradient $\nabla f(x_0, y_0)$, (ii) the equation of the tangent of the curve $f(x, y) = f(x_0, y_0)$ at the point (x_0, y_0) , (iii) the equation of the tangent plane of the surface $z = f(x, y)$ at the point (x_0, y_0, z_0) , $z_0 = f(x_0, y_0)$.

- a) $f(x, y) = x^2 - y^2$, $(2, -1)$ b) $f(x, y) = x^2/(x + y)$, $(1, 1)$
 c) $f(x, y) = e^{x^2y+x^2}$, $(2, 0)$ d) $f(x, y) = \ln(x^2 + y^2)$, $(1, -2)$
10. Determine the tangent plane of the following surfaces at the given point P , as well as those points of the surfaces at which the tangent plane is parallel to the xy -plane.
- a) $x^2 - 2xy - y^2 + z^2 - 2x - 2y - 3z = 0$, $P = (0, 0, 0)$
 b) $xy + yz - 2xz + z^3 = 1$, $P = (1, 1, 1)$
 c) $e^{x+2y-z} = x + y^2 + 2z + 1$, $P = (0, 0, 0)$
11. Determine the normal of the level surface of the following functions and the equation of the tangent plane of that surface at the given point.
- a) $f(x, y, z) = x^2y + y^2z + z^2x$, $(1, -1, 1)$
 b) $f(x, y, z) = \cos(x + 2y + 3z)$, $(\pi/2, \pi, \pi)$
 c) $f(x, y, z) = ye^{-x^2}$, $(0, 1, 1)$
 d) $f(x, y, z) = \sin x \sin 2y \sin 3z \sin(3x - 2y + z)$, $(\pi/6, \pi/6, \pi/6)$
12. Determine the tangent vector of the space curve at the given point P :
- a) $\begin{cases} x^2 - y^2 + 2z^2 = 13, \\ 3x + 2y - z = 3, \end{cases} \quad P = (-2, 3, -3)$
 b) $\begin{cases} x^3 - xyz^3 + yz^5 = 1, \\ xy^2 + yz + 2xz^3 = 0, \end{cases} \quad P = (1, 2, -1)$
13. (*) The ends of a stick are at the points $A = (1, 1, -2)$ and $B = (-1, -2, 4)$. The stick is moved so that its ends slide along the curves

$$S_1 : \begin{cases} x^3 + y^2 + z = 0, \\ x + y^2 + z^3 = -5 \end{cases} \quad \text{and} \quad S_2 : \begin{cases} x^3 + y^2 + z = 7, \\ x + y^2 + z^3 = 67 \end{cases}$$

(the length of the stick is preserved). Let the ends of the stick, after a small displacement, be at the points $A' \in S_1$ and $B' \in S_2$. Compute the limit of the ratio $|AA'|/|BB'|$ as $|AA'| \rightarrow 0$.

14. (*) A line, one of whose points is A , moves so that the line is at all times parallel to the xy -plane, the point A moves at constant speed upward along the z -axis, and at the same time the line rotates at constant angular velocity about the z -axis counterclockwise, as seen from the direction of the positive z -axis. The line is parallel to the x -axis exactly when A is at one of the

points $(0, 0, 3n)$, $n \in \mathbb{Z}$. Determine a parametric representation of the (ruled) surface traced out by the moving line, as well as the normal and tangent plane of that surface at the intersection point of the surface and the line $x = 1$, $y = 2$ that is closest to the xy -plane.

15. (*) (Jyrki's descent) A fell rises from a plain. In the sector $A : 0 \leq y \leq 2x$ the height of the fell above the plain is (units km)

$$h(x, y) = e^{(-x^2 - 2xy + y^2)/100}, \quad (x, y) \in A.$$

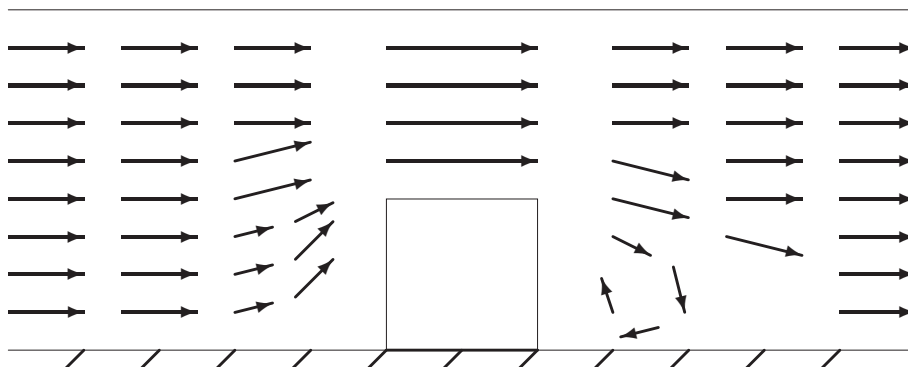
The skier Jyrki starts from the point $(3, 6)$ and at each instant points his skis so that the descent is as steep as possible. Compute Jyrki's descent route!

XI.4 Vector fields: Divergence and curl

In this section we consider functions that depend on two or three physical space variables (x, y or x, y, z), possibly also on a physical time variable t . The functions may be either scalar functions, i.e. real-valued (or scalar-valued, literally 'scaling functions'), or vector-valued, i.e. *vector fields*. In what follows we consider only physical vector fields, which have either two (a plane field) or three (a space field) components. The scalar functions, which are also called *scalar fields*, may thus be of type $f(x, y)$, $f(x, y, z)$, $f(x, y, t)$ or $f(x, y, z, t)$ (sometimes one wishes to write the time variable first, e.g. $f(t, x, y, z)$), and the vector fields e.g. of type

$$\vec{F}(x, y, z, t) = F_1(x, y, z, t) \vec{i} + F_2(x, y, z, t) \vec{j} + F_3(x, y, z, t) \vec{k}.$$

Physical vector fields include e.g. the flow field formed by the velocity vectors of a flowing liquid, gas, etc., a force field (e.g. one associated with gravitation), and the electric field and magnetic field associated with electromagnetic phenomena, and so on. Such physical fields are either static or dynamic, depending on whether or not time (t) is among the variables. A vector field is often depicted graphically as an 'arrow field' (in the figure, a flow field).



When acting on a vector field, the differential operators $\partial_x, \partial_y, \partial_z, \partial_t$ are understood in the same way as in the multiplication of a scalar and a vector, i.e. the operators act on each component of the vector field. E.g.

$$\partial_x \vec{F}(x, y, z, t) = \frac{\partial F_1}{\partial x} \vec{i} + \frac{\partial F_2}{\partial x} \vec{j} + \frac{\partial F_3}{\partial x} \vec{k}.$$

Gradient fields

If f is a scalar function (scalar field) of the type mentioned above, then the gradient of the function is understood as acting only on the space variables x, y or x, y, z . The gradient then acts as a differential operator that transforms a scalar function into a vector-valued function, i.e. a vector field. Many physical vector fields are such *gradient fields*. If $\vec{F}$ is a gradient field and

$$\vec{F} = -\nabla u$$

(the minus sign for physical convenience), then u is said to be the (*scalar*) *potential* of the field $\vec{F}$.

EXAMPLE 1 Physical examples of potentials and gradient fields include, for instance :

<u>Potential</u>	<u>Gradient field</u>
$u = \text{gravitational potential}$	$\vec{G} = -\nabla u = \text{gravitational field}$
$u = \text{electric potential}$	$\vec{E} = -\nabla u = \text{electric field}$
$u = \text{temperature}$	$\vec{J} = -\lambda \nabla u = \text{heat flux density}$

The relation $\vec{J} = -\lambda \nabla u$ is called, in the theory of heat, Fourier's law. According to it, then, heat energy flows in the direction of the negative gradient of temperature $-\nabla u$, and the flux density (unit W/m²) is proportional to the absolute value of the gradient. The constant of proportionality is called the thermal conductivity of the material. $\square$

Divergence and curl. The Laplacian

The differential operator $\nabla = \partial_x \vec{i} + \partial_y \vec{j} + \partial_z \vec{k}$ can also be understood as an operator whose objects are vector-valued functions, i.e. vector fields. In that case the operator ∇ operates on the target function $\vec{F}$ via either the dot product (scalar product) or the cross product (vector product). Taking the target function to be

$$\vec{F}(x, y, z) = F_1(x, y, z)\vec{i} + F_2(x, y, z)\vec{j} + F_3(x, y, z)\vec{k},$$

we thus obtain the definitions of the *divergence* of the field $\vec{F}$

$$\nabla \cdot \vec{F} = \partial_x F_1 + \partial_y F_2 + \partial_z F_3 \quad (\text{divergence})$$

and the *curl*

$$\nabla \times \vec{F} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \partial_x & \partial_y & \partial_z \\ F_1 & F_2 & F_3 \end{vmatrix} \quad (\text{curl}).$$

The expanded form of the formula for the curl is

$$\nabla \times \vec{F} = (\partial_y F_3 - \partial_z F_2)\vec{i} - (\partial_x F_3 - \partial_z F_1)\vec{j} + (\partial_x F_2 - \partial_y F_1)\vec{k}.$$

The divergence and curl are also denoted by the symbols div and rot (or $\overrightarrow{\text{rot}}$). Common ways of reading them are 'nabla dot' and 'nabla cross', corresponding to the reading 'nabla f' of the gradient ∇f .

EXAMPLE 2 The divergence and curl of the function $\vec{F}(x, y, z) = x^2\vec{i} + y^2z\vec{j} - (2xz + yz^2)\vec{k}$ are

$$\begin{aligned} \nabla \cdot \vec{F} &= 2x + 2yz - 2x - 2yz = 0, \\ \nabla \times \vec{F} &= (\partial_y F_3 - \partial_z F_2)\vec{i} + (\partial_z F_1 - \partial_x F_3)\vec{j} + (\partial_x F_2 - \partial_y F_1)\vec{k} \\ &= -(y^2 + z^2)\vec{i} + 2z\vec{j}. \quad \square \end{aligned}$$

Via the dot and cross products one also obtains definitions of, among others, the operator products

$$(1) \quad \nabla \cdot \nabla \quad (2) \quad \nabla(\nabla \cdot) \quad (3) \quad \nabla \times (\nabla \times) \quad (4) \quad \nabla \times \nabla$$

The last of these is the zero operator:

$$\nabla \times \nabla = \vec{0}.$$

The zero operator is also $\nabla \cdot (\nabla \times)$, i.e. (cf. Section II.6)

$$\nabla \cdot \nabla \times \vec{F} = (\nabla \times \nabla) \cdot \vec{F} = 0.$$

Between the operators (1), (2) and (3) there is the relation (Exercise 3f — cf. Section II.6)

$$\nabla \times (\nabla \times \vec{F}) = \nabla(\nabla \cdot \vec{F}) - (\nabla \cdot \nabla)\vec{F}.$$

The operator $\nabla \cdot \nabla$, which can operate on both a scalar and a vector, is called the *Laplacian*. This is a very common differential operator in various laws of physics (see the remarks below), which is why it has an established separate symbol Δ . The notation ∇^2 is also used. It is read 'Laplace' or 'nabla squared'. The definition is thus

$$\begin{aligned} \nabla \cdot \nabla &= (\vec{i}\partial_x + \vec{j}\partial_y + \vec{k}\partial_z) \cdot (\vec{i}\partial_x + \vec{j}\partial_y + \vec{k}\partial_z) \\ &= \partial_x^2 + \partial_y^2 + \partial_z^2 = \Delta. \end{aligned}$$

The operators $\nabla \star$ and $\vec{u} \cdot \nabla$

General 'nabla rules' can often be simplified by using the operator notation $\nabla \star$, where ' $\star$ ' may be 'dot' ($\nabla \cdot$), 'cross' ($\nabla \times$) or 'empty' (∇). In the last case the target is always interpreted as a scalar field, i.e. $\nabla \vec{F} = \nabla F$. The operator notation $\nabla \star$ is convenient, for example, in the generalization of the product rule: If $f = f(x, y, z)$ is a scalar field and $\vec{F}$ a three-dimensional vector field, then

$$\nabla \star (f \vec{F}) = \nabla f \star \vec{F} + f \nabla \star \vec{F}.$$

(In the case $\star =$ 'empty' this is interpreted as: $\nabla(fF) = F\nabla f + f\nabla F$.) The justification is left as an exercise (Exercise 3abc).

Among *variable-coefficient* vector differential operators, one worth mentioning, common in applications, is the operator

$$\vec{u} \cdot \nabla = u_1 \partial_x + u_2 \partial_y + u_3 \partial_z,$$

where $\vec{u} = \vec{u}(x, y, z)$ (or $\vec{u} = \vec{u}(x, y, z, t)$) is a vector field. This is a scalar operator that can act on a scalar or vector field. In the former (but not the latter) case $\vec{u} \cdot \nabla f$ can also be computed in two stages:

$$f \mapsto \nabla f \mapsto \vec{u} \cdot \nabla f.$$

When combining variable-coefficient operators, one must note that the differentiation then acts on the coefficients as well (cf. Section XI.2).

The source of a vector field. The Poisson and Laplace equations

When $\vec{F}$ is some physical field, such as an electric, magnetic, flow, etc. field, the divergence

$$\nabla \cdot \vec{F} = \rho$$

is called the *source* of the field $\vec{F}$. The relation between the field and the source ρ is, in such applications, a law of nature in which the source is interpreted as the cause of the field $\vec{F}$ — hence the name 'source'. A typical starting point is indeed that the source is known and the field is to be determined. For example, in determining the gravitational field ρ is proportional to the mass density, and in the case of the electric field to the charge density.

If the source ρ of a vector field $\vec{F}$ is known and it is furthermore known that $\vec{F}$ is a gradient field, i.e. there exists a potential u such that $\vec{F} = -\nabla u$ (this is the

case e.g. for the gravitational and electric fields), then the relation between the field and the source $\nabla \vec{F} = \rho$ can be written in the form

$$-\Delta u = \rho.$$

This is called *Poisson's equation*. At points where the field $\vec{F}$ is *source-free*, i.e. $\nabla \cdot \vec{F} = 0$, the potential of the field satisfies *Laplace's equation**

$$\Delta u = 0.$$

A function that satisfies Laplace's equation is called *harmonic*.

In applications, Poisson's or Laplace's equation typically arises in a *boundary value problem*, in which on the boundary ∂A of the (open) set A under consideration — or, as is more often said in such contexts, the *region* A ($A \subset \mathbb{R}^2$ or $A \subset \mathbb{R}^3$) — some *boundary condition* is imposed. In an application the boundary condition is determined on physical grounds. For example, if the so-called (homogeneous) *Dirichlet condition* is imposed on the boundary, we obtain the basic form of the boundary value problem:

$$\begin{cases} -\Delta u = \rho & \text{at the interior points of } A, \\ u = 0 & \text{on the boundary } \partial A. \end{cases}$$

This can be shown, under certain regularity conditions (concerning the function ρ and the boundary curve/surface $S = \partial A$), to be uniquely solvable. The solving itself must generally be done by numerical (approximate) means, apart from a few exceptional cases.

EXAMPLE 3 If A is a disk of radius R ($A \subset \mathbb{R}^2$) or a ball ($A \subset \mathbb{R}^3$) and $\rho = Q =$ constant, then the solution of the boundary value problem

$$\begin{cases} -\Delta u = Q & \text{at the interior points of } A, \\ u = 0 & \text{on the boundary } \partial A \end{cases}$$

is in both cases a second-degree polynomial:

$$\begin{aligned} A \subset \mathbb{R}^2 : \quad u(x, y) &= \frac{Q}{4}(R^2 - x^2 - y^2). \\ A \subset \mathbb{R}^3 : \quad u(x, y, z) &= \frac{Q}{6}(R^2 - x^2 - y^2 - z^2). \quad \square \end{aligned}$$

*The names of Laplace's and Poisson's equations refer to the French mathematician-physicists **P.S. de Laplace** (1749-1827) and **S.D. Poisson** (1781-1840), who studied the theory of equations of this type. The foundation for the practical solution of the equations was laid at around the same time by **J.B.J. Fourier** (1768-1830). Fourier too was a mathematician-physicist, who studied heat conduction phenomena in particular. The *Fourier series* and *Fourier transforms* that originated from these studies have made Fourier's name immortal.

Poisson's and Laplace's equations are examples of equations that go by the general name *partial differential equation* (abbrev. PDE). Besides in Poisson's and Laplace's equations, the Laplacian appears in many other partial differential equations of physics. Let us mention here only the *diffusion equation* and the *wave equation*, which describe time-dependent phenomena, and whose basic forms are (in applications various variations of these occur):

$$\text{Diffusion equation : } u_t = \Delta u.$$

$$\text{Wave equation : } u_{tt} = \Delta u.$$

Of the diffusion-equation type is, for example, the (time-dependent) heat equation.

The vortex field. Maxwell's equations

The curl of a physical vector field

$$\nabla \times \vec{F} = \vec{\omega}$$

is called the *vortex field* of the field $\vec{F}$. Vortex fields have physical significance for example in fluid mechanics, but they occupy an especially central place in the theory of electromagnetic fields. This is seen already from the basic equations of electromagnetic fields, *Maxwell's equations* (**J.C. Maxwell**, 1864):

$$\left\{ \begin{array}{l} \nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}, \\ \nabla \times \vec{H} = \frac{\partial \vec{D}}{\partial t} + \vec{J}, \\ \nabla \cdot \vec{D} = \rho, \\ \nabla \cdot \vec{B} = 0. \end{array} \right.$$

The names of the fields appearing here are:

$$\begin{array}{ll} \vec{E} : & \text{electric field} \\ \vec{H} : & \text{magnetic field} \\ \rho : & \text{charge density} \end{array} \quad \begin{array}{ll} \vec{D} : & \text{electric flux density} \\ \vec{B} : & \text{magnetic flux density} \\ \vec{J} : & \text{current density} \end{array}$$

When solving Maxwell's equations, one must take into account the mutual dependence of the fields $\vec{E}, \vec{D}$ and $\vec{H}, \vec{B}$, which is characteristic of the medium. In

the simplest case the dependences can be expressed by means of two material constants ϵ, μ in the form

$$\vec{D} = \epsilon \vec{E}, \quad \vec{B} = \mu \vec{H}.$$

A vector field $\vec{F}$ is called *irrotational* in a region where $\nabla \times \vec{F} = \vec{0}$. All gradient fields, i.e. fields expressible by means of a scalar potential, are irrotational (because $\nabla \times \nabla = \vec{0}$). By Maxwell's equations, the electric field $\vec{E}$ is irrotational in a stationary situation, in which the fields are time-independent. Both source-free and irrotational is, for example, the gravitational field in empty (massless) space, and more generally a gradient field in a source-free region.

Curl in the plane

If a plane vector field is interpreted as a space vector field (independent of the z -coordinate and parallel to the xy -plane), then the curl of the field is

$$\nabla \times [F_1(x, y) \vec{i} + F_2(x, y) \vec{j}] = (\partial_x F_2 - \partial_y F_1) \vec{k}.$$

Here it is customary to write

$$\text{rot } \vec{F} = \frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y} \quad (\text{plane vector field}),$$

whereby the curl in the plane becomes defined as a scalar operator (like div).

EXAMPLE 4 By Maxwell's equations, in a stationary situation the vortex field of the magnetic field $\vec{H}$ is the electric current density $\vec{J}$. If we assume a typical application in which the current flows in a straight conductor parallel to the z -axis, then $\vec{J}$ is of the form

$$\vec{J}(x, y, z) = \begin{cases} I(x, y) \vec{k}, & \text{when } (x, y) \in A, \\ \vec{0}, & \text{otherwise.} \end{cases}$$

Here $A \subset \mathbb{R}^2$ represents the cross-section of the conductor and I is the (scalar) current density in the conductor. The magnetic field is parallel to the xy -plane and independent of the z -coordinate, i.e. of the form $\vec{H} = H_1(x, y) \vec{i} + H_2(x, y) \vec{j}$, so that Maxwell's equation $\nabla \times \vec{H} = \vec{J}$ reduces, on the cross-section of the conductor, to the scalar equation

$$\text{rot } \vec{H} = \partial_x H_2 - \partial_y H_1 = I(x, y).$$

Outside the conductor the magnetic field is irrotational: $\text{rot } \vec{H} = 0$. $\square$

*The Navier–Stokes flow equations

The basic equations governing the flow of a liquid or gas (e.g. water or air) are so-called *conservation laws*, which express mathematically (as partial differential equations) the conservation of mass, momentum and energy in the flow. The vector and scalar fields appearing in the conservation laws of mass and momentum are

$$\begin{aligned}\vec{u} &: \text{flow velocity} \\ p &: \text{pressure} \\ \rho &: \text{(mass) density}\end{aligned}$$

If the mass density of the flowing substance can be assumed constant (e.g. water, and in many flow situations also air) and moreover the flow is free of external forces, then the conservation laws of momentum and mass can be written as the *Navier–Stokes equations* (**C.-L. Navier, G.G. Stokes, 1822**)

$$\begin{cases} \vec{u}_t - \nu \Delta \vec{u} + (\vec{u} \cdot \nabla) \vec{u} + \rho^{-1} \nabla p = \vec{0}, \\ \nabla \cdot \vec{u} = 0. \end{cases}$$

Here ν is the (kinematic) viscosity of the flowing substance.

EXERCISES

1. Compute the divergence and curl:

$$\begin{aligned}\text{a) } & x\vec{i} + y\vec{j} + z\vec{k} & \text{b) } & yz\vec{i} + xz\vec{j} + xy\vec{k} & \text{c) } & xy^2\vec{i} - yz^2\vec{j} + zx^2\vec{k} \\ \text{d) } & f(x)\vec{i} + g(y)\vec{j} + h(z)\vec{k} & \text{e) } & f(y, z)\vec{i} + g(x, z)\vec{j} + h(x, y)\vec{k}\end{aligned}$$

2. Let $\vec{F}(x, y, z) = (xy - z^2)\vec{i} + xyz\vec{j} + (x - y^2 - z^2)\vec{k}$.

a) Compute $\nabla \cdot \vec{F}$, $\nabla(\nabla \cdot \vec{F})$, $\nabla \times \vec{F}$, $\nabla \times (\nabla \times \vec{F})$ and $\Delta \vec{F}$ and verify that $\nabla \times (\nabla \times \vec{F}) = \nabla(\nabla \cdot \vec{F}) - \Delta \vec{F}$.

b) Compute $(\vec{F} \cdot \nabla)\vec{F}$ and $(\vec{F} \times \nabla) \cdot \vec{F}$.

3. Let $\vec{F}$ and $\vec{G}$ be vector fields of the plane or of space (possibly time-dependent) and f and g scalar fields. Justify the following rules (assuming sufficient differentiability):

$$\text{a) } \nabla(fg) = g\nabla f + f\nabla g \qquad \text{b) } \nabla \cdot (f\vec{F}) = \vec{F} \cdot \nabla f + f \nabla \cdot \vec{F}$$

- c) $\nabla \times (f\vec{F}) = \nabla f \times \vec{F} + f \nabla \times \vec{F}$ d) $\partial_t(\nabla \star \vec{F}) = \nabla \star \vec{F}_t$
e) $\nabla \cdot (\vec{F} \times \vec{G}) = (\nabla \times \vec{F}) \cdot \vec{G} - (\nabla \times \vec{G}) \cdot \vec{F}$
f) $\nabla \times (\nabla \times \vec{F}) = \nabla(\nabla \cdot \vec{F}) - \Delta \vec{F}$
g) $\square(\vec{F} \circ \vec{G}) = (\square \vec{F}) \circ \vec{G} + \vec{F} \circ \square \vec{G}$ and $\square(f\vec{F}) = (\square f)\vec{F} + f\square \vec{F}$,
when $\square \in \{\partial_x, \partial_y, \partial_z, \partial_t\}$ and $\circ \in \{\cdot, \times\}$
h) $(\vec{u} \square) \circ f\vec{F} = \square f \vec{u} \circ \vec{F} + f\vec{u} \circ \square \vec{F}$, when $\square \in \{\partial_x, \partial_y, \partial_z, \partial_t\}$ and $\circ \in \{\cdot, \times\}$

4. a) Show that the following functions are harmonic in their domain:

$$xy, \quad x^2 - y^2, \quad x^3 - 3xy^2, \quad x^3y - xy^3, \quad \ln(x^2 + y^2), \quad \operatorname{Arctan} \frac{y}{x}$$

$$e^{ax}(A \sin ay + B \cos ay) \quad (a, A, B \in \mathbb{R})$$

b) The Laplacian of many variables is defined by $\Delta = \partial_1^2 + \dots + \partial_n^2$. Investigate for which real functions f the following holds: $f(|\mathbf{x}|) = f(\sqrt{x_1^2 + \dots + x_n^2})$ is harmonic away from the origin.

5. The following partial differential equations are claimed to have solutions of the given form. Verify the claims (the functions f and g are assumed twice differentiable).

- a) $u_{tt} = c^2 u_{xx}$, $u(x, t) = f(x + ct) + g(x - ct)$
b) $u_{xx} + u_{yy} = 2u_{xy}$, $u(x, y) = xf(x + y) + yg(x + y)$
c) $u_t = u_{xx}$, $u(x, t) = e^{-\omega^2 t}(A \sin \omega x + B \sin \omega x)$ ($\omega, A, B \in \mathbb{R}$)
d) $u_t = u_{xx}$, $u(x, t) = t^{-1/2} e^{-x^2/4t}$
e) $u_t = u_{xx} + u_{yy}$, $u(x, y, t) = t^{-1} e^{-(x^2+y^2)/4t}$

6. Show that the general solution of the partial differential equation $u_{xy} = 0$ in $\mathbb{R}^2$ is $u(x, y) = f(x) + g(y)$, where $f : \mathbb{R} \rightarrow \mathbb{R}$ and $g : \mathbb{R} \rightarrow \mathbb{R}$ are any functions differentiable on $\mathbb{R}$.

7. (*) Prove (assuming sufficient differentiability):

$$\nabla \times (\vec{F} \times \vec{G}) = (\nabla \cdot \vec{G})\vec{F} + (\vec{G} \cdot \nabla)\vec{F} - (\nabla \cdot \vec{F})\vec{G} - (\vec{F} \cdot \nabla)\vec{G}.$$

8. (*) Let $A \subset \mathbb{R}^2$ be an equilateral triangle with vertices $(0, 0)$, $(a, 0)$ and $(a/2, \sqrt{3}a/2)$ ($a > 0$). Show that the boundary value problem

$$\begin{cases} -\Delta u = Q & \text{at the interior points of } A, \\ u = 0 & \text{on the boundary } \partial A \end{cases}$$

($Q = \text{constant}$) has a polynomial solution.

9. (*) Consider Maxwell's equations in a homogeneous medium, where $\rho = 0$, $\vec{J} = \vec{0}$ and the fields $\vec{D}, \vec{E}$ and $\vec{B}, \vec{H}$ are related by $\vec{D} = \epsilon \vec{E}$ and $\vec{B} = \mu \vec{H}$ (ϵ and μ material constants). Show that all the fields mentioned, insofar as they are sufficiently regular, satisfy the vector-valued wave equation

$$\vec{F}_{tt} = c^2 \Delta \vec{F},$$

where $c = (\epsilon\mu)^{-1/2}$ (= the speed of light in the medium).

XI.5 The operators grad, div, rot, and Δ in curvilinear coordinate systems

Because of physical symmetries, a vector field/scalar field is often to be expressed in a curvilinear polar, cylindrical, or spherical coordinate system. In that case the vector differential operators must also be transformed from the Cartesian system into the curvilinear system in question. As an example, consider the transformation into the plane polar coordinate system.

First we must investigate how the operators ∂_x and ∂_y are expressed in terms of the corresponding operators ∂_r and ∂_φ of the polar coordinate system. The starting point is the chain rule (see Section XI.2), according to which

$$\begin{aligned} u(x, y) &= v(r, \varphi) = v(r(x, y), \varphi(x, y)) \\ \Rightarrow \begin{cases} \partial_x u = \frac{\partial r}{\partial x} \partial_r v + \frac{\partial \varphi}{\partial x} \partial_\varphi v, \\ \partial_y u = \frac{\partial r}{\partial y} \partial_r v + \frac{\partial \varphi}{\partial y} \partial_\varphi v. \end{cases} \end{aligned}$$

Here $r = r(x, y) = \sqrt{x^2 + y^2}$, so

$$\frac{\partial r}{\partial x} = \frac{x}{r} = \cos \varphi, \quad \frac{\partial r}{\partial y} = \frac{y}{r} = \sin \varphi.$$

Differentiating implicitly with respect to x (implicit partial differentiation!) then gives

$$\begin{aligned} x = r \cos \varphi &\Rightarrow 1 = \frac{\partial r}{\partial x} \cos \varphi - r \sin \varphi \frac{\partial \varphi}{\partial x} \\ &\Rightarrow 1 = \cos^2 \varphi - r \sin \varphi \frac{\partial \varphi}{\partial x} \\ &\Rightarrow \frac{\partial \varphi}{\partial x} = -\frac{1}{r} \sin \varphi. \end{aligned}$$

Similarly, differentiating with respect to y gives

$$\begin{aligned} 0 &= \frac{\partial r}{\partial y} \cos \varphi - r \sin \varphi \frac{\partial \varphi}{\partial y} = \sin \varphi \cos \varphi - r \sin \varphi \frac{\partial \varphi}{\partial y} \\ &\Rightarrow \frac{\partial \varphi}{\partial y} = \frac{1}{r} \cos \varphi. \end{aligned}$$

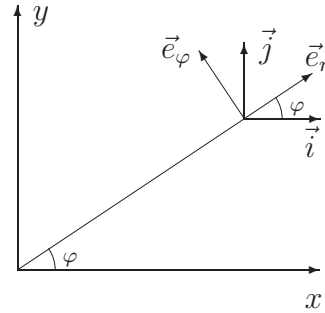
This yields the transformation formulas

$$\begin{aligned} \partial_x &= \cos \varphi \partial_r - \frac{1}{r} \sin \varphi \partial_\varphi, \\ \partial_y &= \sin \varphi \partial_r + \frac{1}{r} \cos \varphi \partial_\varphi. \end{aligned}$$

When, in addition, we take into account the transformation formulas between the basis vectors $\vec{i}, \vec{j}$ and $\vec{e}_r, \vec{e}_\varphi$

$$\begin{cases} \vec{i} = \cos \varphi \vec{e}_r - \sin \varphi \vec{e}_\varphi, \\ \vec{j} = \sin \varphi \vec{e}_r + \cos \varphi \vec{e}_\varphi, \end{cases}$$

then for the operator ∇ we first obtain the expression



$$\begin{aligned} \nabla &= \vec{i} \partial_x + \vec{j} \partial_y = (\cos \varphi \vec{e}_r - \sin \varphi \vec{e}_\varphi) \left(\cos \varphi \partial_r - \frac{1}{r} \sin \varphi \partial_\varphi \right) \\ &\quad + (\sin \varphi \vec{e}_r + \cos \varphi \vec{e}_\varphi) \left(\sin \varphi \partial_r + \frac{1}{r} \cos \varphi \partial_\varphi \right) \end{aligned}$$

and after simplifications

$$\nabla = \vec{e}_r \partial_r + \frac{1}{r} \vec{e}_\varphi \partial_\varphi \quad (\text{polar coordinate system}).$$

EXAMPLE 1 Compute the gradient of the function $u(x, y) = x/(x^2 + y^2)$ in the polar coordinate system.

Solution Since

$$u(x, y) = \frac{x}{x^2 + y^2} = \frac{r \cos \varphi}{r^2} = \frac{\cos \varphi}{r} = v(r, \varphi),$$

then by the transformation formula above

$$\nabla u = \frac{\partial v}{\partial r} \vec{e}_r + \frac{1}{r} \frac{\partial v}{\partial \varphi} \vec{e}_\varphi = \underline{\underline{-\frac{1}{r^2} (\cos \varphi \vec{e}_r + \sin \varphi \vec{e}_\varphi)}}. \quad \square$$

To compute the divergence of a given vector field $\vec{F}$, assume that $\vec{F}$ is known in the form

$$\vec{F} = F_r(r, \varphi, z) \vec{e}_r + F_\varphi(r, \varphi, z) \vec{e}_\varphi.$$

Then

$$\begin{aligned} \nabla \cdot \vec{F} &= \left(\vec{e}_r \partial_r + \frac{1}{r} \vec{e}_\varphi \partial_\varphi \right) \cdot (F_r \vec{e}_r + F_\varphi \vec{e}_\varphi) \\ &= (\vec{e}_r \partial_r) \cdot (F_r \vec{e}_r) + (\vec{e}_r \partial_r) \cdot (F_\varphi \vec{e}_\varphi) \\ &\quad + \frac{1}{r} (\vec{e}_\varphi \partial_\varphi) \cdot (F_r \vec{e}_r) + \frac{1}{r} (\vec{e}_\varphi \partial_\varphi) \cdot (F_\varphi \vec{e}_\varphi). \end{aligned}$$

Here it must be noted that the vectors $\vec{e}_r$ and $\vec{e}_\varphi$ are not constant vectors but depend on the coordinate φ . Hence we also need the derivatives $\partial_\varphi \vec{e}_r$ and $\partial_\varphi \vec{e}_\varphi$:

$$\begin{aligned}\partial_\varphi \vec{e}_r &= \partial_\varphi (\cos \varphi \vec{i} + \sin \varphi \vec{j}) = -\sin \varphi \vec{i} + \cos \varphi \vec{j}, \\ \partial_\varphi \vec{e}_\varphi &= \partial_\varphi (-\sin \varphi \vec{i} + \cos \varphi \vec{j}) = -\cos \varphi \vec{i} - \sin \varphi \vec{j},\end{aligned}$$

i.e.

$$\partial_\varphi \vec{e}_r = \vec{e}_\varphi, \quad \partial_\varphi \vec{e}_\varphi = -\vec{e}_r.$$

Using these formulas and the product rules of differentiation (see Exercise [XI.4: 3h](#)) the above expression can be expanded term by term:

$$\begin{aligned}(\vec{e}_r \partial_r) \cdot (F_r \vec{e}_r) &= \partial_r F_r \vec{e}_r \cdot \vec{e}_r + F_r \vec{e}_r \cdot \partial_r \vec{e}_r = \partial_r F_r + F_r \vec{e}_r \cdot \vec{e}_\varphi = \partial_r F_r, \\ (\vec{e}_r \partial_r) \cdot (F_\varphi \vec{e}_\varphi) &= \partial_r F_\varphi \vec{e}_r \cdot \vec{e}_\varphi + F_\varphi \vec{e}_r \cdot \partial_r \vec{e}_\varphi = 0, \\ \frac{1}{r} (\vec{e}_\varphi \partial_\varphi) \cdot (F_r \vec{e}_r) &= \frac{1}{r} [\partial_\varphi F_r \vec{e}_\varphi \cdot \vec{e}_r + F_r \vec{e}_\varphi \cdot \partial_\varphi \vec{e}_r] = \frac{1}{r} F_r \vec{e}_\varphi \cdot \vec{e}_\varphi = \frac{1}{r} F_r, \\ \frac{1}{r} (\vec{e}_\varphi \partial_\varphi) \cdot (F_\varphi \vec{e}_\varphi) &= \frac{1}{r} [\partial_\varphi F_\varphi \vec{e}_\varphi \cdot \vec{e}_\varphi + F_\varphi \vec{e}_\varphi \cdot \partial_\varphi \vec{e}_\varphi] \\ &= \frac{1}{r} [\partial_\varphi F_\varphi - F_\varphi \vec{e}_\varphi \cdot \vec{e}_r] = \frac{1}{r} \partial_\varphi F_\varphi.\end{aligned}$$

The final result:

$$\nabla \cdot \vec{F} = \partial_r F_r + \frac{1}{r} F_r + \frac{1}{r} \partial_\varphi F_\varphi \quad (\text{polar coordinate system}).$$

By combining the first two terms, we obtain for this an alternative so-called *conservation form*

$$\nabla \cdot \vec{F} = \frac{1}{r} \partial_r (r F_r) + \frac{1}{r} \partial_\varphi F_\varphi \quad (\text{conservation form}).$$

EXAMPLE 2 Away from the origin, the divergence of the vector field $\vec{F} = r^{-1} \vec{e}_r$ is

$$\nabla \cdot \vec{F} = \frac{1}{r} \partial_r \left(r \cdot \frac{1}{r} \right) = 0,$$

so the field is source-free in the region $A = \{(x, y) \in \mathbb{R}^2 \mid (x, y) \neq (0, 0)\}$. $\square$

The following table collects the computation rules for the operators ∇ , $\nabla \cdot$, $\nabla \times$, and Δ (in conservation form) in cylindrical and spherical coordinate systems.

Cylindrical coordinate system	
∇	$= \vec{e}_r \partial_r + \frac{1}{r} \vec{e}_\varphi \partial_\varphi + \vec{e}_z \partial_z$
$\nabla \cdot \vec{F}$	$= \frac{1}{r} \partial_r(r F_r) + \frac{1}{r} \partial_\varphi F_\varphi + \partial_z F_z$
$\nabla \times \vec{F}$	$= \frac{1}{r} \begin{vmatrix} \vec{e}_r & r\vec{e}_\varphi & \vec{e}_z \\ \partial_r & \partial_\varphi & \partial_z \\ F_r & rF_\varphi & F_z \end{vmatrix}$
Δ	$= \frac{1}{r} \partial_r(r \partial_r) + \frac{1}{r^2} \partial_\varphi^2 + \partial_z^2$
Spherical coordinate system	
∇	$= \vec{e}_r \partial_r + \frac{1}{r} \vec{e}_\theta \partial_\theta + \frac{1}{r \sin \theta} \vec{e}_\varphi \partial_\varphi$
$\nabla \cdot \vec{F}$	$= \frac{1}{r^2} \partial_r(r^2 F_r) + \frac{1}{r \sin \theta} \partial_\theta(\sin \theta F_\theta) + \frac{1}{r \sin \theta} \partial_\varphi F_\varphi$
$\nabla \times \vec{F}$	$= \frac{1}{r^2 \sin \theta} \begin{vmatrix} \vec{e}_r & r\vec{e}_\theta & r \sin \theta \vec{e}_\varphi \\ \partial_r & \partial_\theta & \partial_\varphi \\ F_r & rF_\theta & r \sin \theta F_\varphi \end{vmatrix}$
Δ	$= \frac{1}{r^2} \partial_r(r^2 \partial_r) + \frac{1}{r^2 \sin \theta} \partial_\theta(\sin \theta \partial_\theta) + \frac{1}{r^2 \sin^2 \theta} \partial_\varphi^2$

According to the table, the transformation of the Laplacian from the Cartesian coordinate system of the plane to the polar coordinate system is

$$\Delta = \partial_x^2 + \partial_y^2 = \frac{1}{r} \partial_r(r \partial_r) + \frac{1}{r^2} \partial_\varphi^2 = \partial_r^2 + \frac{1}{r} \partial_r + \frac{1}{r^2} \partial_\varphi^2.$$

This is reached by simplifying, in the manner described above, the operator expression

$$\Delta = \nabla \cdot \nabla = \left(\vec{e}_r \partial_r + \frac{1}{r} \vec{e}_\varphi \partial_\varphi \right) \cdot \left(\vec{e}_r \partial_r + \frac{1}{r} \vec{e}_\varphi \partial_\varphi \right)$$

(Exercise 4).

EXAMPLE 3 Let the field $\vec{F}$ be spherically symmetric, i.e. of the form $\vec{F} = F(r)\vec{e}_r$ in the spherical coordinate system. What is the function $F(r)$ if the field is source-free outside the origin?

Solution According to the table, the divergence of the field is $\nabla \cdot \vec{F} = r^{-2} \partial_r(r^2 F(r))$, so outside the origin ($r > 0$) we must have

$$\partial_r(r^2 F(r)) = 0 \Rightarrow r^2 F(r) = A \Rightarrow F(r) = \frac{A}{r^2}, \quad A \in \mathbb{R}. \quad \square$$

In a polar- or spherically symmetric situation, where the source ρ and the potential u of the vector field depend only on the radial coordinate r , Poisson's equation can be written as an ordinary differential equation (cf. the table above):

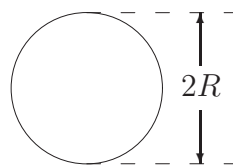
$$\text{Polar symmetry : } \frac{1}{r} \frac{d}{dr} \left(r \frac{du}{dr} \right) = u'' + \frac{1}{r} u' = -\rho(r).$$

$$\text{Spherical symmetry : } \frac{1}{r^2} \frac{d}{dr} \left(r^2 \frac{du}{dr} \right) = u'' + \frac{2}{r} u' = -\rho(r).$$

These are both of Euler type (cf. Section IX.5), so they can be solved by quadratures. The quickest way to solve them is directly from the conservation form.

EXAMPLE 4 The temperature on the cross-section of a round resistance wire satisfies the equation

$$-\Delta u = Q = \text{constant}.$$



Determine u , given that on the outer surface $u(R) = 0$.

Solution Start from the conservation form (polar symmetry):

$$\begin{aligned} -\frac{1}{r} \frac{d}{dr}(ru') &= Q \Rightarrow \frac{d}{dr}(ru') = -Qr \\ \Rightarrow ru' &= -\frac{1}{2}Qr^2 + A \Rightarrow u' = -\frac{1}{2}Qr + \frac{A}{r} \\ \Rightarrow u(r) &= -\frac{1}{4}Qr^2 + A \ln r + B \quad (A, B \in \mathbb{R}) \\ \Rightarrow A &= 0, \quad B = \frac{1}{4}QR^2 \\ \Rightarrow u(r) &= \underline{\underline{\frac{1}{4}Q(R^2 - r^2)}}. \end{aligned}$$

Expressed in the Cartesian coordinate system, the solution is

$$u(x, y) = \frac{Q}{4} (R^2 - x^2 - y^2).$$

The logarithmic term $A \ln r = \frac{1}{2} A \ln(x^2 + y^2)$ had to be dropped from the solution, because otherwise the equation $-\Delta u = Q$ would not hold at the origin (elsewhere it would). $\square$

EXERCISES

1. Compute the divergence $\nabla \cdot \vec{F}$ of the vector field

$$\vec{F} = (x^2 + y^2 + z^2)(x\vec{i} + y\vec{j} + z\vec{k})$$

a) in the Cartesian coordinate system, b) by first transforming $\vec{F}$ into the spherical coordinate system.

2. Show that the following functions, defined in the polar or spherical coordinate system, are harmonic in their domain.

$$\text{a) } u(r, \varphi) = \frac{\cos \varphi}{r} \quad \text{b) } u(r, \varphi) = \frac{\sin 2\varphi}{r^2} \quad \text{c) } u(r, \theta, \varphi) = \frac{\sin \theta \cos \varphi}{r^2}$$

3. Concerning the vector field $\vec{F} = F_1\vec{i} + F_2\vec{j} + F_3\vec{k}$ it is known that the functions F_i are differentiable in $\mathbb{R}^3$ and that the field is irrotational in $\mathbb{R}^3$. Determine the Cartesian components F_i of the field, given in addition that in the spherical coordinate system the field is of the form

$$\vec{F} = r^3 \sin^2 \theta \vec{e}_r + F_\theta(r, \theta, \varphi) \vec{e}_\theta.$$

4. Simplify the expression of the Laplacian in the polar coordinate system, i.e. show that

$$\left(\vec{e}_r \partial_r + \frac{1}{r} \vec{e}_\varphi \partial_\varphi \right) \cdot \left(\vec{e}_r \partial_r + \frac{1}{r} \vec{e}_\varphi \partial_\varphi \right) = \frac{1}{r} \partial_r(r \partial_r) + \frac{1}{r^2} \partial_\varphi^2.$$

5. A current flowing in an electric conductor, with density $\vec{J} = J(r)\vec{e}_z$ (cylindrical coordinate system), produces a magnetic field of the form $\vec{H} = H(r)\vec{e}_\varphi$, where $H(0) = 0$. Determine $H(r)$ in terms of $J(r)$ from Maxwell's equation $\nabla \times \vec{H} = \vec{J}$.

6. On the cross-section

$$A = \{(x, y) \in \mathbb{R}^2 \mid R_1^2 \leq x^2 + y^2 \leq R_2^2\}$$

of a body shaped like a cylinder or cylindrical shell made of homogeneous material, the temperature u satisfies at the interior points of A Poisson's equation $-u_{xx} - u_{yy} = q$, where q depends only on the polar coordinate r , together with the boundary conditions prescribed on the boundary ∂A . Determine u in the following cases:

- a) $R_1 = 0$, $R_2 = R$, $q(r) = Q = \text{constant}$, $u(R) = u_0$
- b) $R_1 = 0$, $R_2 = R$, $q(r) = Q = \text{constant}$, $u'(R) = -ku(R)$
- c) $R_1 = 0$, $R_2 = R$, $q(r) = Q(1 - r^2/R^2)$, $u'(R) = 0$
- d) $R_1 = R$, $R_2 = 2R$, $q(r) = 0$, $u(R_1) = u_0$, $u'(R_2) = 0$
- e) $R_1 = R$, $R_2 = 2R$, $q(r) = Q = \text{constant}$, $u(R_1) = u(R_2) = 0$

7. In a body shaped like a ball or spherical shell made of homogeneous material,

$$A = \{(x, y, z) \in \mathbb{R}^3 \mid R_1^2 \leq x^2 + y^2 + z^2 \leq R_2^2\}$$

the temperature u satisfies at the interior points of A Poisson's equation $-\Delta u = q$, where q depends only on the spherical coordinate r , together with the boundary conditions prescribed on the boundary ∂A . Determine u in the cases of the previous exercise.

8. (*) Derive the differentiation formulas for the basis vectors of the spherical coordinate system

$$\begin{aligned} \partial_\theta \vec{e}_r &= \vec{e}_\theta, & \partial_\theta \vec{e}_\theta &= -\vec{e}_r, & \partial_\theta \vec{e}_\varphi &= \vec{0}, \\ \partial_\varphi \vec{e}_r &= \sin \theta \vec{e}_\varphi, & \partial_\varphi \vec{e}_\theta &= \cos \theta \vec{e}_\varphi, & \partial_\varphi \vec{e}_\varphi &= -\sin \theta \vec{e}_r - \cos \theta \vec{e}_\theta \end{aligned}$$

and by means of these the expressions of the gradient and the Laplacian in the spherical coordinate system.

9. (*) (Kepler's law) A spacecraft of mass $= m$ moves in a planar central-force field $\vec{F} = \nabla(k/r)$ ($k = \text{constant}$, $r = \text{distance from the origin}$) so that the spacecraft's polar coordinates at time t are $r(t)$ and $\varphi(t)$. Write the equation of motion $m\vec{r}'' = \vec{F}$ of the spacecraft as a system of differential equations for the functions $r(t)$ and $\varphi(t)$, and show that Kepler's second law holds

$$[r(t)]^2 \varphi'(t) = \text{constant}.$$

XI.6 Nonlinear systems of equations: the Jacobian matrix and Newton's method

Consider a general system of m equations and n unknowns of the form

$$\begin{cases} f_1(x_1, \dots, x_n) = 0 \\ \vdots \\ f_m(x_1, \dots, x_n) = 0 \end{cases}$$

where the functions f_i , $i = 1 \dots m$, are real-valued functions of n real variables, i.e. functions of the type $f_i : \mathcal{D}_{f_i} \rightarrow \mathbb{R}$, $\mathcal{D}_{f_i} \subset \mathbb{R}^n$. The system is *nonlinear* if it is not linear, i.e. if at least one of the functions f_i is a non-affine map (cf. Section X.7). When we define

$$\mathbf{f} = (f_1, \dots, f_m), \quad \mathcal{D}_{\mathbf{f}} = \mathcal{D}_{f_1} \cap \mathcal{D}_{f_2} \cap \dots \cap \mathcal{D}_{f_m},$$

then the above system can be conveniently written in vector form:

$$\mathbf{f}(\mathbf{x}) = \mathbf{0}.$$

Here, then, $\mathbf{f}$ is a function (map) of the type $\mathbf{f} : \mathcal{D}_{\mathbf{f}} \rightarrow \mathbb{R}^m$, $\mathcal{D}_{\mathbf{f}} \subset \mathbb{R}^n$. This is a *vector-valued function of a vector variable*, which can be interpreted as a generalization of the linear and affine maps defined earlier in Sections X.6–X.7. The term *vector field* (cf. Section XI.4) can also be used. When systems of equations are handled by the methods of matrix algebra, $\mathbf{f}$ is usually interpreted as a column vector. As will be seen below, matrix algebra is needed, among other things, when one wants to determine algorithmically the *solutions* of the system, i.e. the set

$$X = \{\mathbf{x} \in \mathcal{D}_{\mathbf{f}} \mid \mathbf{f}(\mathbf{x}) = \mathbf{0}\}.$$

When systems of equations are solved in practice (numerically), most commonly $m = n$.

EXAMPLE 1 The vector-form representations of the nonlinear systems of equations

$$\text{a) } \begin{cases} 3x + y + z + 9 = 0 \\ 7x + y + 3z + 27 = 0 \\ x^2 + y^2 + z^2 - 81 = 0 \end{cases} \quad \text{b) } \begin{cases} x + y + z - 16 = 0 \\ x^2 + y^2 + z^2 - 81 = 0 \end{cases}$$

are

$$\begin{aligned} \text{a) } \mathbf{f}(x, y, z) &= (3x + y + z + 9, 7x + y + 3z + 27, x^2 + y^2 + z^2 - 81) \\ &= (0, 0, 0), \\ \text{b) } \mathbf{f}(x, y, z) &= (x + y + z - 16, x^2 + y^2 + z^2 - 81) \\ &= (0, 0), \end{aligned}$$

or, by means of column vectors,

$$\begin{aligned} \text{a) } \mathbf{f}(x, y, z) &= \begin{bmatrix} 3x + y + z + 9 \\ 7x + y + 3z + 27 \\ x^2 + y^2 + z^2 - 81 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}, \\ \text{b) } \mathbf{f}(x, y, z) &= \begin{bmatrix} x + y + z - 16 \\ x^2 + y^2 + z^2 - 81 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}. \end{aligned}$$

By algebraic means (or, in case b), by geometric means) the solutions can be found to be

$$\text{a) } X = \{ (0, 0, -9), (6, -6, 3) \}, \quad \text{b) } X = \emptyset. \quad \square$$

A vector-valued function $\mathbf{f}$ is said to be *differentiable* at a point $\mathbf{x} \in \mathcal{D}_{\mathbf{f}}$ if each component function f_i is differentiable at that point (in the sense of Definition XI.3.1). If $\mathbf{f}$ is differentiable, then combining the differential expansions of the component functions yields the result

$$\mathbf{f}(\mathbf{x} + \Delta\mathbf{x}) = \mathbf{f}(\mathbf{x}) + \mathbf{J} \mathbf{f}(\mathbf{x}) \Delta\mathbf{x} + \mathbf{r}(\mathbf{x}),$$

where $\mathbf{J}$ is the *Jacobian* matrix* of $\mathbf{f}$

$$\mathbf{J} \mathbf{f}(\mathbf{x}) = \begin{bmatrix} \frac{\partial f_1}{\partial x_1}(\mathbf{x}) & \cdots & \frac{\partial f_1}{\partial x_n}(\mathbf{x}) \\ \vdots & & \\ \frac{\partial f_m}{\partial x_1}(\mathbf{x}) & \cdots & \frac{\partial f_m}{\partial x_n}(\mathbf{x}) \end{bmatrix} \quad (\text{Jacobian matrix})$$

and for the remainder function $\mathbf{r}$ the estimate

$$|\mathbf{r}(\mathbf{x})| = o(|\Delta\mathbf{x}|), \quad \text{as } |\Delta\mathbf{x}| \rightarrow 0$$

*The German mathematician **K.G.J. Jacobi** lived in the years 1804–1851.

holds. In the case $m = 1$ the Jacobian matrix is the same as the transposed gradient, i.e. $\mathbf{J}f = (\nabla f)^T$. In the case $n = 1$, on the other hand, the Jacobian matrix reduces to the (ordinary) derivative of a vector-valued function of one variable. In the cases $m = 2, 3$ this is already familiar as the derivative of a parametric curve (Section VII.2). Thus the Jacobian matrix is again a generalization of the derivative concept, in two different ways. More generally, the definition shows that the Jacobian matrix can be represented either by means of the gradients ∇f_i or the partial derivatives $\partial_j \mathbf{f}$, in the forms

$$\mathbf{J}\mathbf{f} = [\nabla f_1 \dots \nabla f_m]^T = [\partial_1 \mathbf{f} \dots \partial_n \mathbf{f}].$$

EXAMPLE 2 When the system of equations

$$\begin{cases} x^3 - 2xy^5 - x - 2 = 0 \\ -x^3y + 2xy^2 + y^5 + 3 = 0 \end{cases}$$

is written in the form $\mathbf{f}(x, y) = \mathbf{0}$, $\mathbf{f} = [f_1, f_2]^T$, then the Jacobian matrix is

$$\mathbf{J}\mathbf{f}(x, y) = \begin{bmatrix} \partial_x f_1 & \partial_y f_1 \\ \partial_x f_2 & \partial_y f_2 \end{bmatrix} = \begin{bmatrix} 3x^2 - 2y^5 - 1 & -10xy^4 \\ -3x^2 + 2y^2 & -x^3 + 4xy + 5y^4 \end{bmatrix}.$$

One solution of the system is $(x, y) = (2, 1)$. At this point,

$$\mathbf{J}\mathbf{f}(2, 1) = \begin{bmatrix} 9 & -20 \\ -10 & 5 \end{bmatrix}. \quad \square$$

If $\mathbf{f}$ is differentiable at the point $\mathbf{a}$, then the approximation

$$\mathbf{f}(\mathbf{x}) \approx \mathbf{f}(\mathbf{a}) + \mathbf{J}\mathbf{f}(\mathbf{a})(\mathbf{x} - \mathbf{a})$$

is called, as before (cf. Section V.3), the *linearization* of $\mathbf{f}$ at the point $\mathbf{a}$. (More correctly, it is a matter of 'affinization'.) Since the error of the approximation is of order $o(|\mathbf{x} - \mathbf{a}|)$ (in accordance with the definition of differentiability), the conclusion is:

Differentiable map $\approx$ affine map locally.

EXAMPLE 2 (continued) In the neighborhood of the point $(-1, 1) \hat{=} \mathbf{a} = [-1, 1]^T$, approximately

$$\mathbf{f}(x, y) \approx \mathbf{f}(-1, 1) + \mathbf{J}(-1, 1)(\mathbf{x} - \mathbf{a}) = \begin{bmatrix} 0 \\ 3 \end{bmatrix} + \begin{bmatrix} 0 & 10 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} x + 1 \\ y - 1 \end{bmatrix}. \quad \square$$

Newton's method for systems of equations

Consider the system of equations $\mathbf{f}(\mathbf{x}) = \mathbf{0}$, where $\mathbf{f}$ is of the type $\mathbf{f} : \mathbb{R}^n \rightarrow \mathbb{R}^n$. In Newton's method one makes a linearizing approximation in the system at the point $\mathbf{x}_k$, i.e. one solves

$$\mathbf{f}(\mathbf{x}_k) + \mathbf{J}(\mathbf{x}_k)(\mathbf{x} - \mathbf{x}_k) = \mathbf{0}.$$

Denoting the solution by $\mathbf{x} = \mathbf{x}_{k+1}$, the result is the iteration formula

$$\mathbf{J}(\mathbf{x}_k)(\mathbf{x}_{k+1} - \mathbf{x}_k) = -\mathbf{f}(\mathbf{x}_k)$$

i.e.

$$\mathbf{x}_{k+1} = \mathbf{x}_k - \mathbf{J}(\mathbf{x}_k)^{-1} \mathbf{f}(\mathbf{x}_k) \quad (\text{Newton}).$$

The basic idea here is exactly the same as in the single-equation ($n = 1$) case (cf. Section V.7): it is a *fixed-point iteration* of the form

$$\mathbf{x}_{k+1} = \mathbf{F}(\mathbf{x}_k), \quad k = 0, 1, \dots,$$

where $\mathbf{F}(\mathbf{x}) = \mathbf{x} - \mathbf{J}(\mathbf{x})^{-1} \mathbf{f}(\mathbf{x})$.

Newton's method converges in the solution of systems of equations too, under assumptions similar to those in the single-equation case. — Let us recall from Section XI.1 that the convergence of a vector sequence $\{\mathbf{x}_k\}$ toward a vector $\mathbf{c} \in \mathbb{R}^n$ means:

$$\mathbf{x}_k \rightarrow \mathbf{c} \quad \Leftrightarrow \quad |\mathbf{x}_k - \mathbf{c}| \rightarrow 0 \quad (k \rightarrow \infty),$$

where $|\cdot|$ is the Euclidean norm of $\mathbb{R}^n$. As in the case $n = 1$, the convergence of Newton's method is also (under sufficient assumptions) *quadratic*, in a manner defined more precisely in the following convergence theorem. The theorem, together with its proof, is a fairly straightforward generalization of Theorem V.7.4, but the details of the proof are more technical. (The proof relies on the multi-variable Taylor's theorem, see Section XI.9 below, cf. also Exercise VII.4:16.) We omit the proof.

THEOREM XI.6.1 Let $\mathbf{f} : \mathcal{D}_{\mathbf{f}} \rightarrow \mathbb{R}^n$, $\mathcal{D}_{\mathbf{f}} \subset \mathbb{R}^n$, $\mathbf{f} = (f_i)$, and let $\mathbf{f}(\mathbf{c}) = \mathbf{0}$, $\mathbf{c} \in \mathbb{R}^n$. Let, further, the partial derivatives $\partial^\alpha f_i(\mathbf{x})$ exist and be continuous in a neighborhood of the point $\mathbf{c}$ for $i = 1 \dots n$ and $|\alpha| \leq 2$, and let the Jacobian matrix $\mathbf{J} \mathbf{f}(\mathbf{c})$ be nonsingular. Then Newton's iteration $\mathbf{x}_{k+1} = \mathbf{x}_k - \mathbf{J}(\mathbf{x}_k)^{-1} \mathbf{f}(\mathbf{x}_k)$ converges toward $\mathbf{c}$, provided that $|\mathbf{x}_0 - \mathbf{c}|$ is sufficiently small, and the convergence is quadratic:

$$|\mathbf{x}_{k+1} - \mathbf{c}| \leq C |\mathbf{x}_k - \mathbf{c}|^2 \quad (C = \text{constant}).$$

EXAMPLE 3 Using Newton's method, find a solution to the system of equations

$$\begin{cases} x^3 - 2xy^5 - x = 2.04 \\ -x^3y + 2xy^2 + y^5 = -0.05 \end{cases}$$

near the point $(-1, 1)$.

Solution Here

$$\mathbf{f}(x, y) = \begin{bmatrix} x^3 - 2xy^5 - x - 2.04 \\ -x^3y + 2xy^2 + y^5 + 0.05 \end{bmatrix}$$

and the Jacobian matrix $\mathbf{J}(x, y)$ is the same as in Example 2. Computing according to the iteration formula

$$[x_{k+1}, y_{k+1}]^T = [x_k, y_k]^T - \mathbf{J}(x_k, y_k)^{-1} \mathbf{f}(x_k, y_k)$$

gives

$$\begin{aligned} (x_0, y_0) &= (-1, 1) \\ (x_1, y_1) &= (-0.942, 1.004) \\ (x_2, y_2) &= (-0.918940, 1.006156) \\ (x_3, y_3) &= (-0.914247, 1.006622) \\ (x_4, y_4) &= (-0.914214, 1.006633) \\ (x_5, y_5) &= (-0.914216, 1.006633) \\ &\vdots \quad \square \end{aligned}$$

In the example, Newton's iteration converged rapidly because the initial guess was close to the solution. If a good initial guess is not available, one can first try to locate the solutions by some simpler method. For example, if the system is not large, one can investigate by trial where the function

$$F(\mathbf{x}) = \sum_{i=1}^n [f_i(\mathbf{x})]^2$$

takes small values. Even more straightforward is the 'trial and error' method: one iterates with varying initial guesses until, by chance, one hits the so-called *basin of attraction*, i.e. sufficiently close to some solution. If the iteration does not converge after, say, ten rounds, one changes the initial guess.

EXAMPLE 4 Solve the system of equations

$$\begin{cases} x^2 + y^2 = 1 \\ e^x + \sin y = 1 \end{cases}$$

Solution Here

$$\mathbf{f}(x, y) = \begin{bmatrix} x^2 + y^2 - 1 \\ e^x + \sin y - 1 \end{bmatrix}, \quad \mathbf{J}\mathbf{f}(x, y) = \begin{bmatrix} 2x & 2y \\ e^x & \cos y \end{bmatrix},$$

so Newton's iteration takes the form

$$\begin{bmatrix} x_{k+1} \\ y_{k+1} \end{bmatrix} = \begin{bmatrix} x_k \\ y_k \end{bmatrix} - \begin{bmatrix} 2x_k & 2y_k \\ e^{x_k} & \cos y_k \end{bmatrix}^{-1} \begin{bmatrix} x_k^2 + y_k^2 - 1 \\ e^{x_k} + \sin y_k - 1 \end{bmatrix}.$$

There are two solutions:

$$A: (x, y) = (0.5538\ldots, -0.8327\ldots)$$

$$B: (x, y) = (-0.8089\ldots, 0.5879\ldots)$$

Below are results depending on the initial guess. The mark * denotes a failure (a singular Jacobian matrix or divergence of the iteration).

$$\begin{array}{lll} (1, 0) \rightarrow A & (0, 0) \rightarrow * & (-5, 0) \rightarrow B \\ (1, 1) \rightarrow B & (0.1, 0) \rightarrow A & (-10, 0) \rightarrow B \\ (-1, 1) \rightarrow B & (-0.1, 0) \rightarrow * & (3, 3) \rightarrow B \\ (-1, -1) \rightarrow B & (5, 0) \rightarrow * & (3, -3) \rightarrow A \\ (0, -1) \rightarrow A & (0, 5) \rightarrow B & (3, -1.5) \rightarrow A \\ (1, -1) \rightarrow A & (0, -5) \rightarrow A & (3, -1.2) \rightarrow B \end{array} \quad \square$$

Large systems of equations. The continuation method

Newton's method is also commonly used to solve very large nonlinear systems of equations. Such problems often arise as part of a larger computational process, in which case it is also common for a good initial guess to be available from the earlier stages of the computation. This is the case, for example, when solving systems of differential equations by numerical step methods (see Section IX.7).

If one merely wants to solve a single large system of equations without prior information about the location of the solution(s), finding a working initial guess for Newton's iteration can be very difficult. In such a situation a rather commonly used approach is the so-called *continuation method*, in which the problem to be solved is parametrized into the form

$$P(t), \quad 0 \leq t < 1.$$

Assume that $P(1)$ is the problem one actually wants to solve, and that $P(0)$ is easy to solve. Then the idea of the continuation method is to solve successively

$$P(t_0), P(t_1), \dots, P(t_N), \quad t_N = 1,$$

where (for example) $t_k = k\Delta t$. If Δt is small and the parametrization is done well, the solution of the problem $P(t_{k-1})$ is a good initial guess for Newton's iteration when solving the problem $P(t_k)$. In this way one can advance step by step to the final state $t_N = 1$.^{*} If no better idea is available, a general system of equations can be parametrized, for instance, by a transformation of the form

$$\mathbf{f}(\mathbf{x}) = \mathbf{0} \hookrightarrow \mathbf{g}(\mathbf{x}) + t[\mathbf{f}(\mathbf{x}) - \mathbf{g}(\mathbf{x})] = \mathbf{0},$$

where $\mathbf{g}(\mathbf{x}) = \mathbf{0}$ is some easily solvable (e.g. linear) system of equations.

EXAMPLE 5 For the system of equations

$$\begin{cases} x^3 - 2y^3 + x + 2y = 10 \\ x^3 + y^3 - 2xy^2 - 3x - y = 10 \end{cases}$$

Newton's algorithm with the initial guesses $(x_0, y_0) = (1, 1)$ and $(x_0, y_0) = (-1, -1)$ gives the solution $(x, y) = (-2.25872, -2.42834)$. Let us try whether some other solution can be found by the continuation method: we parametrize the system into the form

$$\mathbf{f}(t, \mathbf{x}) = \begin{bmatrix} t(x^3 - 2y^3) + x + 2y \\ t(x^3 + y^3 - 2xy^2) - 3x - y \end{bmatrix} = \begin{bmatrix} 10 \\ 10 \end{bmatrix}$$

and solve this for the parameter values $t_k = 0.2k$, $k = 0 \dots 5$. When $k = 0$, the solution is $(x, y) = (-6, 8)$. From here on ($k = 1 \dots 5$) the solution obtained with the previous value of k is used as the initial guess for Newton's iteration. Result:

t	Solution
0.2	(10.5223, 8.55714)
0.4	(7.60605, 6.14808)
0.6	(6.30833, 5.07754)
0.8	(5.53177, 4.43769)
1.0	(5.00000, 4.00000)

The solution $(x, y) = (5, 4)$ can also be found by varying the initial guesses. For example, $(x_0, y_0) = (3, 3)$ or even $(x_0, y_0) = (0, 0)$ lead to it. $\square$

^{*}In designing an algorithm based on the continuation method, the physical background of the problem is usually of help. For example, in a structural mechanics problem where one wants to simulate large deformations of a structure under load, it is natural to proceed as in an experimental setup: one parametrizes the magnitude of the load, i.e. one starts from the unloaded state and increases the load in small steps until the desired final state is reached. The parameter t can then be imagined as a time variable associated with the experimental setup. Introducing a time variable (time parametrization) is natural more generally as well, whenever a time-independent (equilibrium) solution is sought for a problem originating in physics.

Simplified Newton iteration

When solving a nonlinear system of equations by Newton's method, the following computational operations must be performed on each iteration round:

- Function evaluation: $\mathbf{x}_k \mapsto \mathbf{f}(\mathbf{x}_k) = \mathbf{y}_k$.
- *Updating* of the Jacobian matrix $\mathbf{x}_k \mapsto \mathbf{Jf}(\mathbf{x}_k) = \mathbf{A}_k$.
- Solution of a linear system of equations: $\mathbf{y}_k \mapsto \mathbf{u}_k = \mathbf{A}_k^{-1}\mathbf{y}_k$ ($\mathbf{A}_k\mathbf{u}_k = \mathbf{y}_k$).
- Subtraction: $\mathbf{x}_k, \mathbf{u}_k \mapsto \mathbf{x}_k - \mathbf{u}_k = \mathbf{x}_{k+1}$.

Both updating the Jacobian matrix and, especially, solving the linear system of equations are relatively heavy operations if the size of the system is large. When solving very large systems, one may therefore 'slack off' a little in the updating, i.e. the update is not performed on every iteration round. This also makes the (direct) solution of the linear systems easier, because the coefficient matrix of the systems stays the same between updates. It then suffices to compute the *LU* decomposition of the Jacobian matrix after each update and keep it stored until the next update (cf. Section X.3). Depending on how often the update is performed, various variations of Newton's method are obtained, from which the most efficient can be chosen for the problem at hand.

In the simplest of these variations, the Jacobian matrix is computed only once, at the initial-guess point. Denoting $\mathbf{A} = \mathbf{J}(\mathbf{x}_0)$, the iteration formula becomes

$$\mathbf{x}_{k+1} = \mathbf{x}_k - \mathbf{A}^{-1}\mathbf{f}(\mathbf{x}_k), \quad k = 0, 1, \dots$$

This extreme simplification of Newton's method is not, in practice, the equal of the genuine Newton's method, since it converges (insofar as it converges) only linearly (cf. Exercise V.7:15, for $n = 1$). — On the other hand, this method is well suited as a 'method of thinking' for theoretical considerations, see the next section.

EXAMPLE 6 In the situation of Example 3,

$$\mathbf{A} = \mathbf{Jf}(-1, 1) = \begin{bmatrix} 0 & 10 \\ -1 & 2 \end{bmatrix}, \quad \mathbf{A}^{-1} = \frac{1}{10} \begin{bmatrix} 2 & -10 \\ 1 & 0 \end{bmatrix},$$

so the Newton iteration formula, simplified in the above manner, becomes

$$\begin{bmatrix} x_{k+1} \\ y_{k+1} \end{bmatrix} = \begin{bmatrix} x_k \\ y_k \end{bmatrix} - \frac{1}{10} \begin{bmatrix} 2 & -10 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} x_k^3 - 2x_k y_k^5 - x_k - 2.04 \\ -x_k^3 y_k + 2x_k y_k^2 + y_k^5 - 0.05 \end{bmatrix}.$$

The iteration converges, but clearly more slowly than the genuine Newton iteration (cf. Example 3):

$$\begin{aligned}(x_0, y_0) &= (-1, 1) \\(x_1, y_1) &= (-0.942, 1.004) \\(x_2, y_2) &= (-0.929318, 1.005191) \\(x_3, y_3) &= (-0.923142, 1.005780) \\(x_4, y_4) &= (-0.919696, 1.006109) \\(x_5, y_5) &= (-0.917651, 1.006304) \\&\vdots \quad \square\end{aligned}$$

EXERCISES

1. In the following cases, compute the Jacobian matrix of $\mathbf{f}$ at the first given point, approximate $\mathbf{f}$ by an affine map, and use this to compute the value of $\mathbf{f}$ approximately at the second given point. Compare with the exact value.

- a) $\mathbf{f}(x, y) = (x^3 + y^3, x^3 - y^3)$, $(1, 1)$, $(0.9, 1.1)$
- b) $\mathbf{f}(x, y) = (e^x \sin \pi y, e^{-x} \cos \pi y)$, $(0, 1/3)$, $(-0.04, 0.30)$
- c) $\mathbf{f}(x, y, z) = (x^2 y, x^2 z, y^2 - z^2)$, $(1, 3, 3)$ $(0.99, 3.02, 2.97)$
- d) $\mathbf{f}(x, y, z) = (x^2 + yz, y^2 - x \ln z)$, $(2, 2, 1)$, $(1.98, 2.01, 1.03)$
- e) $\mathbf{f}(\mathbf{x}) = (x_1 x_2, x_1^2 - x_3^2, x_2^2 + x_3 x_4, x_2 x_4 + x_4^2)$, $(1, 2, -2, -1)$, $(1.01, 2.02, -2.02, -1.01)$

2. The system of equations

$$[\mathbf{f}(x, y, z)]^T = [x^3 - x^2 y + x z^3, x y z + z^2, x^3 + y^3 + z^3] = \mathbf{b}^T$$

has an easy solution when $\mathbf{b}^T = [1, 2, 3]$. By approximating $\mathbf{f}$ by an affine map near that solution point, compute the solution of the system approximately when $\mathbf{b} = [1.04, 1.98, 3.02]^T$.

3. Compute the Jacobian matrices $\mathbf{J}(r, \varphi, z)$ and $\mathbf{J}(r, \theta, \varphi)$ of the coordinate transformations $(r, \varphi, z) \mapsto (x, y, z)$ (cylindrical $\hookrightarrow$ Cartesian) and $(r, \theta, \varphi) \mapsto (x, y, z)$ (spherical $\hookrightarrow$ Cartesian). At which points is $\mathbf{J}$ a singular matrix?
4. Let $\mathbf{f}$ be of the type $\mathbf{f} : \mathbb{R}^n \rightarrow \mathbb{R}^p$ and $\mathbf{g}$ of the type $\mathbf{g} : \mathbb{R}^p \rightarrow \mathbb{R}^m$. Derive the rule for computing the Jacobian matrix of the composite function $\mathbf{F}(\mathbf{x}) = (\mathbf{g} \circ \mathbf{f})(\mathbf{x}) = \mathbf{g}[\mathbf{f}(\mathbf{x})]$, $\mathbf{JF}(\mathbf{x}) = \mathbf{Jg}[\mathbf{f}(\mathbf{x})]\mathbf{Jf}(\mathbf{x})$. *Hint:* The chain rule!

5. Find a solution to the following systems of equations by Newton's method, starting from the initial guess $(x_0, y_0) = (1, 1)$.

$$\text{a) } \begin{cases} x^4 + y^4 = 2xy^5 \\ x + x^2 + y^4 = 4 \end{cases} \quad \text{b) } \begin{cases} x^3 + 3x = y^4 + 4y \\ \cos \pi x + xye^{-y} = 0 \end{cases}$$

6. Find the solutions of the following systems of equations to four significant figures using the two-dimensional Newton's method. In parts a) and b), compute the solution also by an alternative approach, in which one of the unknowns is first eliminated and the one-dimensional Newton's method is used.

- a) $y - e^x = 0, \quad x - \sin y = 0$
 b) $x^2 + y^2 = 1, \quad y = e^x$ (two solutions)
 c) $y - \sin x = 0, \quad x^2 + (y + 1)^2 = 2$ (two solutions)
 d) $x^2 - xy + 2y^2 = 10, \quad x^3y^2 = 2$ (four solutions)
 e) $\sin x + \sin y = 1, \quad x^3 = y^2 < 30$ (four solutions)

7. Using the (two-dimensional) Newton's method, determine at which points the angle between the tangent of the folium of Descartes $x^3 + y^3 = 3xy$ and some coordinate axis is 45° .

8. Determine approximately the solutions of the following systems of equations by first eliminating one of the unknowns and then using Newton's method.

$$\text{a) } \begin{cases} x^2 + y^2 + z^2 = 1 \\ z = xy \\ 6xz = 1 \end{cases} \quad \text{b) } \begin{cases} x^2 + y^2 + z^2 = 1 \\ y = \sin z \\ z + z^2 + z^3 = x + y \end{cases} \quad \text{c) } \begin{cases} x^2 + y^4 = 1 \\ z = x^3y \\ e^x = 2y - z \end{cases}$$

9. The system of equations

$$x + yze^x = x^3 + y^3z = y + e^{xyz} = 2.1$$

has a solution near the point $(0, 1, 2)$. Iterate three times from this initial guess using the simplified version of Newton's method, in which the Jacobian matrix is not updated. Compare with the results given by the correct Newton iteration.

10. (*) Using the continuation method and Newton's method, find some solution to the system of equations

$$\begin{cases} 3x^3 - y^3 + z^3 + x - 2z + 40 = 0 \\ 4x^3 - 3y^3 - z^3 - 2x - 3y - 40 = 0 \\ x^3 + y^3 + z^3 - 5x - 5y + 5z - 40 = 0 \end{cases}$$

11. (*) Determine the line that is tangent to the curve

$$K : x^2y^2 - 2xy^3 + y^4 - 4x^2y + 6xy^2 - 2y^3 + 4x^2 - 3y^2 - 12x + y = 20$$

at two different points.

12. (*) (Space parking) In interstellar space, at the point $P = (x, y, z)$ (unit of length = light-year), there is a space station. The space station is acted on by the gravitational attraction of the nearest stars $\vec{G} = -k\nabla u$, where k is a constant and

$$u(x, y, z) = \sum_{i=1}^n \frac{m_i}{|P - P_i|},$$

where m_i is the mass of the star T_i and P_i its position. For space parking, determine all points P where $\vec{G} = \vec{0}$, when $m_i = iM$, $i = 1 \dots n$ ($M = \text{constant}$) and

- a) $n = 3$, $P_1 = (0, 0, 0)$, $P_2 = (1, 0, 0)$, $P_3 = (0, 1, 0)$,
 b) $n = 4$, $P_1 = (0, 0, 0)$, $P_2 = (1, 0, 0)$, $P_3 = (0, 1, 0)$, $P_4 = (0, 0, 1)$.

XI.7 Inverse function theorem. Implicit function theorem.

Contraction mapping theorem

In this section we consider the basic question of the solvability of a nonlinear system of equations when the number of equations equals the number of unknowns. Let $\mathbf{f}(\mathbf{x}) = (f_1(\mathbf{x}), \dots, f_n(\mathbf{x}))$, $\mathbf{x} \in \mathcal{D}_{\mathbf{f}} \subset \mathbb{R}^n$, and consider the system of equations

$$\mathbf{f}(\mathbf{x}) = \mathbf{y}, \quad (\star)$$

where $\mathbf{y} \in \mathbb{R}^n$. Insofar as the system has a unique solution $\mathbf{x} \in \mathcal{D}_{\mathbf{f}}$ for some $\mathbf{y}$, the solution can be formally written

$$\mathbf{x} = \mathbf{f}^{-1}(\mathbf{y}),$$

where $\mathbf{f}^{-1}(\mathbf{y}) = (g_1(\mathbf{y}), \dots, g_n(\mathbf{y}))$ is the *inverse function* of $\mathbf{f}$. If the system has a unique solution for every $\mathbf{y} \in B$ ($B \subset \mathbb{R}^n$) and $A = \mathbf{f}^{-1}(B) = \{\mathbf{x} \in \mathcal{D}_{\mathbf{f}} \mid \mathbf{f}(\mathbf{x}) \in B\}$, then $\mathbf{f} : A \rightarrow B$ is a bijection.

EXAMPLE 1 Let $\mathbf{f}(x, y) = (x + y, x^2 - y^2)$, $\mathbf{x} = (x, y) \in \mathbb{R}^2$. Denoting $\mathbf{y} = (u, v)$, the system $(\star)$ is

$$\begin{cases} x + y = u, \\ x^2 - y^2 = v. \end{cases}$$

From the form $u(x - y) = v$ of the latter equation one sees that if $u \neq 0$, then the system has a unique solution $x = \frac{1}{2}(u^2 + v)/u$, $y = \frac{1}{2}(u^2 - v)/u$. On the line $S : x + y = 0$ we have $\mathbf{f}(x, y) = (0, 0)$, so on this line $\mathbf{f}$ is not one-to-one. We conclude that if $A = \{(x, y) \in \mathbb{R}^2 \mid x + y \neq 0\}$ and $B = \{(u, v) \in \mathbb{R}^2 \mid u \neq 0\}$, then $\mathbf{f} : A \rightarrow B$ is a bijection and

$$\mathbf{f}^{-1}(\mathbf{y}) = \left(\frac{u^2 + v}{2u}, \frac{u^2 - v}{2u} \right), \quad \mathbf{y} = (u, v) \in B. \quad \square$$

Local inverse function. Inverse function theorem

In Example 1 the system $(\star)$ was solved completely. More commonly, solving a nonlinear system completely is difficult if not impossible, nor is there any more general solvability theory comparable to the theory of linear systems. Instead, one can, under relatively general conditions, resolve the question of the local solvability of the system according to the following (typical in applications, cf. the previous section) problem setting: If $\mathbf{a} \in \mathcal{D}_{\mathbf{f}}$ and $\mathbf{b} = \mathbf{f}(\mathbf{a})$, does the system

($\star$) have (and under what conditions) a unique solution near $\mathbf{a}$, when $\mathbf{y}$ is likewise near $\mathbf{b}$?

DEFINITION XI.7.1 A function $\mathbf{f} : \mathcal{D}_{\mathbf{f}} \rightarrow \mathbb{R}^n$, $\mathcal{D}_{\mathbf{f}} \subset \mathbb{R}^n$, is *locally invertible* at a point $\mathbf{a} \in \mathcal{D}_{\mathbf{f}}$ if $\exists \delta > 0$ such that $\mathbf{f}$ is defined in a neighborhood $U_{\delta}(\mathbf{a}) = \{\mathbf{x} \in \mathbb{R}^n \mid |\mathbf{x} - \mathbf{a}| < \delta\}$ and, restricted to this neighborhood, is a one-to-one map. The inverse function

$$\mathbf{x} = \mathbf{f}^{-1}(\mathbf{y}) \Leftrightarrow \mathbf{y} = \mathbf{f}(\mathbf{x}) \quad \& \quad \mathbf{x} \in U_{\delta}(\mathbf{a})$$

is called the *local inverse function* of $\mathbf{f}$.

EXAMPLE 1 (continued) By solving the system $\mathbf{f}(\mathbf{x}) = \mathbf{y}$, the function $\mathbf{f}$ of the example was found to be invertible when restricted to the set $A = \{(x, y) \in \mathbb{R}^2 \mid x + y \neq 0\}$. Also in accordance with Definition XI.7.1, $\mathbf{f}$ is locally invertible at every point $\mathbf{a} = (a, b) \in A$, since A is an open set, so that $\forall (a, b) \in A \exists \delta > 0$ such that $U_{\delta}(a, b) \subset A$, whereby $\mathbf{f}$ restricted to this neighborhood is one-to-one. (More precisely, here $U_{\delta}(a, b) \subset A$ exactly when $\delta \leq d$, where $d = |a + b|/\sqrt{2}$ is the distance of the point $P = (a, b)$ from the line $S : x + y = 0$.) $\square$

In the case of a function of one real variable, sufficient conditions for local invertibility can easily be given in terms of the derivative. Assume:

- (i) f is differentiable on the interval $(a - \delta, a + \delta)$ for some $\delta > 0$.
- (ii) $f'(x)$ is continuous at the point $x = a$.
- (iii) $f'(a) \neq 0$.

On the basis of these assumptions, either $f'(x) > 0$ or $f'(x) < 0$ on the interval $[a - \rho, a + \rho]$ for some $\rho \in (0, \delta)$ (Proposition V.1.4), whereby f is strictly monotonic on that interval (Theorem V.6.3) and hence one-to-one. Thus conditions (i)–(iii) are sufficient for the local invertibility of f at the point a . Since under these conditions f is both strictly monotonic and continuous on the interval $[a - \rho, a + \rho]$, we also conclude (Theorem V.1.10): the equation $f(x) = y$ has a unique solution on the interval $[a - \rho, a + \rho]$ whenever $|y - f(a)| \leq \varepsilon$, where $\varepsilon = \min\{|f(a) - f(a - \rho)|, |f(a) - f(a + \rho)|\} > 0$. Furthermore, under conditions (i)–(iii) it also holds (Theorem V.3.3, $\delta = \rho$): the local inverse function f^{-1} of f is differentiable at the point $b = f(a)$ and $Df^{-1}(b) = 1/f'(a)$.

EXAMPLE 2 The function $f(x) = x^2$ satisfies assumptions (i)–(iii) at points $a \neq 0$, so at these points f is locally invertible. — This is clear in any case, since f is known to be invertible on the intervals $(-\infty, 0]$ and $[0, \infty)$. On the interval

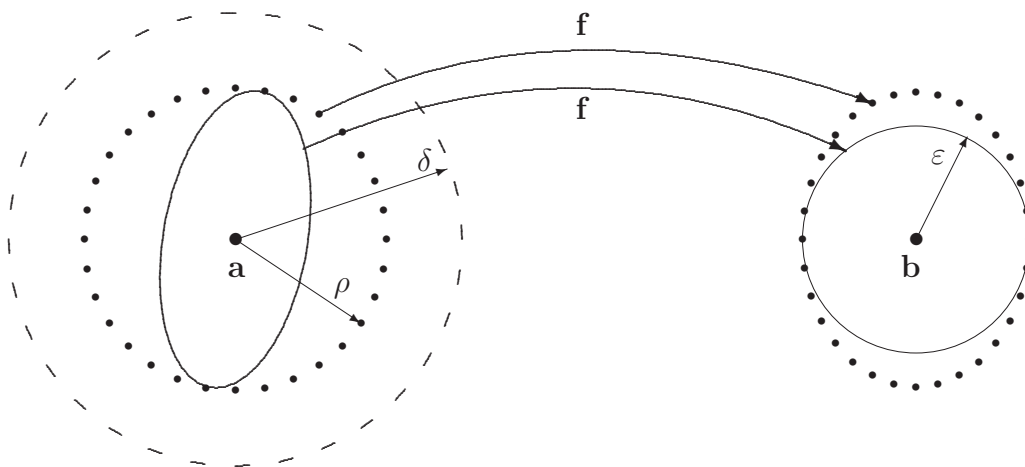
$(-\delta, \delta)$, f is not one-to-one for any $\delta > 0$, so f is not locally invertible at the point $a = 0$. The function $f(x) = x^3$, on the other hand, is one-to-one on the interval $(-\infty, \infty)$, so f is locally invertible at every point, including the point $a = 0$, where $f'(a) = 0$. $\square$

The example shows that the case $f'(a) = 0$ is, from the standpoint of the local invertibility of f , an open case, i.e. f may or may not be locally invertible at the point a .

The statements mentioned above, based on assumptions (i)–(iii), also have counterparts concerning functions of several variables, which are collected below into a single theorem. The proof of this *inverse function theorem* rests essentially on another notable theorem, the *contraction mapping theorem*, which is stated and proved below. The proof of the inverse function theorem is presented only at the end of the section.

***THEOREM XI.7.2 (Inverse function theorem)** Assume that the function $\mathbf{f} : \mathcal{D}_{\mathbf{f}} \rightarrow \mathbb{R}^n$, $\mathcal{D}_{\mathbf{f}} \subset \mathbb{R}^n$ satisfies the conditions: (i) $\exists \delta > 0$ such that f is differentiable in a neighborhood $U_{\delta}(\mathbf{a}) \subset \mathcal{D}_{\mathbf{f}}$ of the point $\mathbf{a}$. (ii) The partial derivatives $\partial f_i(\mathbf{x})/\partial x_j$, $i, j = 1 \dots n$ are continuous at the point $\mathbf{x} = \mathbf{a}$. (iii) The Jacobian matrix $\mathbf{J}\mathbf{f}(\mathbf{x}) = (\partial f_i(\mathbf{x})/\partial x_j)$ is nonsingular at the point $\mathbf{x} = \mathbf{a}$. Then there exist $\rho \in (0, \delta)$ and $\varepsilon > 0$ such that $\mathbf{f}$ is one-to-one on the set $K = \{\mathbf{x} \in \mathbb{R}^n \mid |\mathbf{x} - \mathbf{a}| \leq \rho\}$ and the system $\mathbf{f}(\mathbf{x}) = \mathbf{y}$ has a solution $\mathbf{x} \in K$ whenever $|\mathbf{y} - \mathbf{b}| \leq \varepsilon$, where $\mathbf{b} = \mathbf{f}(\mathbf{a})$. Moreover, the local inverse function $\mathbf{f}^{-1}$ of $\mathbf{f}$ is differentiable at the point $\mathbf{b}$ and $\mathbf{J}\mathbf{f}^{-1}(\mathbf{b}) = [\mathbf{J}\mathbf{f}(\mathbf{a})]^{-1}$.

In the figure, the sets $B = \{\mathbf{y} \in \mathbb{R}^n \mid |\mathbf{y} - \mathbf{b}| \leq \varepsilon\}$ and $\mathbf{f}^{-1}(B)$ are bounded by a solid line, and the sets K and $\mathbf{f}(K)$ by a dotted line. According to the statement of the theorem, the local inverse function $\mathbf{f}^{-1}$ is defined on B and $\mathbf{f}^{-1}(B) \subset K$.



EXAMPLE 1 (continued) In the example, the local invertibility of $\mathbf{f}$ can also be determined without solving the system $(\star)$: By Theorem XI.7.2, $\mathbf{f}$ is locally invertible at the point (a, b) whenever $a + b \neq 0$, since

$$\mathbf{J}\mathbf{f}(a, b) = \begin{bmatrix} 1 & 1 \\ 2a & -2b \end{bmatrix} \Rightarrow \det \mathbf{J}(a, b) = -2(a + b) \neq 0, \text{ when } a + b \neq 0. \quad \square$$

EXAMPLE 3 (Cf. Example 2 in the previous section.) By the inverse function theorem, the system

$$\mathbf{f}(x, y) = \mathbf{y} \Leftrightarrow \begin{cases} x^3 - 2xy^5 - x = u \\ -x^3y + 2xy^2 + y^5 = v \end{cases}$$

has, in a small neighborhood of the point $\mathbf{a} = (2, 1)$, a unique solution when $|\mathbf{y} - \mathbf{b}| \leq \varepsilon$, where $\mathbf{b} = (2, 3)$ and $\varepsilon > 0$ is sufficiently small. In a small neighborhood of the point $\mathbf{x} = \mathbf{a}$, approximately

$$\mathbf{f}(\mathbf{x}) \approx \begin{bmatrix} 2 \\ 3 \end{bmatrix} + \begin{bmatrix} 9 & -20 \\ -10 & 5 \end{bmatrix} \begin{bmatrix} x - 2 \\ y - 1 \end{bmatrix},$$

so the conditions $\mathbf{f}(\mathbf{x}) = \mathbf{y}$ and $\mathbf{y} \in B : |\mathbf{y} - \mathbf{b}| \leq \varepsilon \Leftrightarrow (u - 2)^2 + (v - 3)^2 \leq \varepsilon^2$ correspond, near the point $\mathbf{x} = \mathbf{a}$, approximately to the condition

$$\begin{aligned} & [9(x - 2) - 20(y - 1)]^2 + [-10(x - 2) + 5(y - 1)]^2 \leq \varepsilon^2 \\ \Leftrightarrow & 181(x - 2)^2 - 460(x - 2)(y - 1) + 425(y - 1)^2 \leq \varepsilon^2. \end{aligned}$$

According to this, the set $\mathbf{f}^{-1}(B)$ is bounded approximately by a second-degree curve — it is a (slanted) *ellipse* with center $\mathbf{a} = (2, 1)$, cf. the figure above. $\square$

Implicit function theorem

A slightly more general form of the inverse function theorem is the *implicit function theorem*, in which one considers a system of the form

$$\mathbf{F}(\mathbf{x}, \mathbf{y}) = \mathbf{0},$$

where $\mathbf{x} \in \mathbb{R}^n$, $\mathbf{y} \in \mathbb{R}^p$ and $\mathbf{F} : D_{\mathbf{F}} \rightarrow \mathbb{R}^p$, $D_{\mathbf{F}} \subset \mathbb{R}^{n+p}$. Let a point $(\mathbf{x}, \mathbf{y}) = (\mathbf{a}, \mathbf{b})$ be given, at which $\mathbf{F}(\mathbf{a}, \mathbf{b}) = \mathbf{0}$. We ask: Can $\mathbf{y}$ be solved from the system in the form $\mathbf{y} = \mathbf{g}(\mathbf{x})$ in a neighborhood of the point $(\mathbf{a}, \mathbf{b})$? — A sufficient condition for solvability is again obtained by examining the Jacobian matrix of $\mathbf{F}$

$$\mathbf{J} \mathbf{F}(\mathbf{x}, \mathbf{y}) = \begin{bmatrix} \frac{\partial F_1}{\partial x_1} & \cdots & \frac{\partial F_1}{\partial x_n} & \frac{\partial F_1}{\partial y_1} & \cdots & \frac{\partial F_1}{\partial y_p} \\ \vdots & & & & & \\ \frac{\partial F_p}{\partial x_1} & \cdots & \frac{\partial F_p}{\partial x_n} & \frac{\partial F_p}{\partial y_1} & \cdots & \frac{\partial F_p}{\partial y_p} \end{bmatrix}.$$

When this is written in the form

$$\mathbf{J} \mathbf{F}(\mathbf{x}, \mathbf{y}) = [\mathbf{J}_{\mathbf{x}} \mathbf{F}(\mathbf{x}, \mathbf{y}), \mathbf{J}_{\mathbf{y}} \mathbf{F}(\mathbf{x}, \mathbf{y})],$$

where $\mathbf{J}_{\mathbf{x}} \mathbf{F}$ contains the first n columns of $\mathbf{J} \mathbf{F}$, then the critical matrix for solvability is $\mathbf{J}_{\mathbf{y}} \mathbf{F}(\mathbf{a}, \mathbf{b})$ (of size $p \times p$). If this is nonsingular, then the solution generally succeeds. Stated more precisely, this claim is an adaptation of Theorem [XI.7.2](#), and the proof (which is not given) also follows the line of thought of the proof of that theorem, see the end of the section

***THEOREM XI.7.3 (Implicit function theorem)** Assume that the function $\mathbf{F}(\mathbf{x}, \mathbf{y})$,

where $\mathbf{x} \in \mathbb{R}^n$, $\mathbf{y} \in \mathbb{R}^p$ and $\mathbf{F} : D_{\mathbf{F}} \rightarrow \mathbb{R}^p$, $D_{\mathbf{F}} \subset \mathbb{R}^{n+p}$, satisfies the conditions: (i) $\exists \delta > 0$ such that $\mathbf{F}$ is differentiable in a neighborhood $U_{\delta}(\mathbf{a}, \mathbf{b}) \subset D_{\mathbf{F}}$ of the point $(\mathbf{a}, \mathbf{b})$.

(ii) The partial derivatives $[\mathbf{J}_{\mathbf{x}} \mathbf{F}(\mathbf{x}, \mathbf{y})]_{ij} = \partial F_i(\mathbf{x}, \mathbf{y}) / \partial x_j$, $i = 1 \dots p$, $j = 1 \dots n$ and $[\mathbf{J}_{\mathbf{y}} \mathbf{F}(\mathbf{x}, \mathbf{y})]_{ij} = \partial F_i(\mathbf{x}, \mathbf{y}) / \partial y_j$, $i, j = 1 \dots p$ are continuous at the point $(\mathbf{x}, \mathbf{y}) = (\mathbf{a}, \mathbf{b})$. (iii) The matrix $\mathbf{J}_{\mathbf{y}} \mathbf{F}(\mathbf{a}, \mathbf{b})$ is nonsingular. Then there exist $\rho \in (0, \delta)$ and $\varepsilon > 0$ such that the system $\mathbf{F}(\mathbf{x}, \mathbf{y}) = \mathbf{0}$ is solvable under the condition $|\mathbf{y} - \mathbf{b}| \leq \rho$ uniquely into the form $\mathbf{y} = \mathbf{g}(\mathbf{x})$ whenever $|\mathbf{x} - \mathbf{a}| \leq \varepsilon$. Moreover, $\mathbf{g}$ is differentiable at the point $\mathbf{a}$ and $\mathbf{J} \mathbf{g}(\mathbf{a}) = -[\mathbf{J}_{\mathbf{y}} \mathbf{F}(\mathbf{a}, \mathbf{b})]^{-1} \mathbf{J}_{\mathbf{x}} \mathbf{F}(\mathbf{a}, \mathbf{b})$.

A special case of Theorem [XI.7.3](#) is Theorem [XI.7.2](#), when $\mathbf{F}(\mathbf{x}, \mathbf{y}) = \mathbf{x} - \mathbf{f}(\mathbf{y})$ ($p = n$). If the differentiability of $\mathbf{g}$ is assumed, then the rule for computing the Jacobian matrix $\mathbf{J} \mathbf{g}(\mathbf{a})$ can be derived by implicit partial differentiation (Exercise [7](#)).

EXAMPLE 4 At which points of the plane curve $S : F(x, y) = x^2 - 2xy - y^2 - 1 = 0$ can one locally solve from the curve's equation a) y in terms of x , b) x in terms of y ?

Solution a) The solvability condition is $F_y(x, y) = -2x - 2y \neq 0 \Leftrightarrow x + y \neq 0$. Since $x + y = 0 \Rightarrow F(x, y) = 2x^2 - 1$, the only exceptional points are $(1/\sqrt{2}, -1/\sqrt{2})$ and $(-1/\sqrt{2}, 1/\sqrt{2})$. At the other points the solution succeeds.

b) The solvability condition is $F_x(x, y) = 2x - 2y \neq 0 \Leftrightarrow y \neq x$. This condition holds at all points of the curve (since $y = x \Rightarrow F(x, y) = -2x^2 - 1 < 0$), so the solution succeeds at all points. $\square$

EXAMPLE 5 The system of equations

$$\begin{cases} 2x^2 + xy - yz = 0 \\ 2xz - 2y^2 + z^2 = 0 \end{cases}$$

is solved in a neighborhood of the point $(x, y, z) = (1, 2, 2)$ into the form $y = y(x)$, $z = z(x)$. Compute $y'(1)$ and $z'(1)$.

Solution At the point $(x, y, z) = (1, 2, 2)$,

$$\mathbf{J}_{y,z}\mathbf{F}(x, y, z) = \begin{bmatrix} \partial_y F_1 & \partial_z F_1 \\ \partial_y F_2 & \partial_z F_2 \end{bmatrix} = \begin{bmatrix} x - z & -y \\ -4y & 2x + 2z \end{bmatrix} = \begin{bmatrix} -1 & -2 \\ -8 & 6 \end{bmatrix}.$$

This is a nonsingular matrix, so the solution succeeds in the claimed manner. Since $\partial_x F_1 = 4x + y = 6$ and $\partial_x F_2 = 2z = 4$ when $(x, y, z) = (1, 2, 2)$, we obtain

$$\mathbf{J}\mathbf{g}(1) = \begin{bmatrix} y'(1) \\ z'(1) \end{bmatrix} = - \begin{bmatrix} -1 & -2 \\ -8 & 6 \end{bmatrix}^{-1} \begin{bmatrix} 6 \\ 4 \end{bmatrix} = \begin{bmatrix} 2 \\ 2 \end{bmatrix}.$$

The same result is reached by differentiating the equations

$$2x^2 + xy(x) - y(x)z(x) = 0, \quad 2xz(x) - 2[y(x)]^2 + [z(x)]^2 = 0,$$

implicitly, setting $(x, y, z) = (1, 2, 2)$, and solving the derivatives $y'(1)$ and $z'(1)$ from the resulting (linear) system of equations. $\square$

*Contraction mapping theorem

In the local solvability theory of the system $\mathbf{f}(\mathbf{x}) = \mathbf{y}$, the starting point is the same as when seeking the solution in practice, i.e. the system is written in the

form $\mathbf{x} = \mathbf{F}(\mathbf{x})$, so that the question is whether $\mathbf{F}$ has a unique fixed point near $\mathbf{a}$. As one might guess, $\mathbf{F}$ is chosen so that the fixed point is found by fixed-point iteration. Guarantees for the convergence of the iteration, and at the same time for the uniqueness of the fixed point in a small neighborhood of the given point $\mathbf{a}$, are provided by

***THEOREM XI.7.4 (Contraction mapping theorem*)** Let $\mathbf{F} : \mathcal{D}_{\mathbf{F}} \rightarrow \mathbb{R}^n$, $\mathcal{D}_{\mathbf{F}} \subset \mathbb{R}^n$ and $K = \{\mathbf{x} \in \mathbb{R}^n \mid |\mathbf{x} - \mathbf{a}| \leq \rho\} \subset \mathcal{D}_{\mathbf{F}}$ for some $\mathbf{a} \in \mathbb{R}^n$ and $\rho > 0$. Then if

(i) $\mathbf{F}$, restricted to the set K , is a *contraction (mapping)*, i.e.

$$|\mathbf{F}(\mathbf{x}) - \mathbf{F}(\mathbf{y})| \leq L|\mathbf{x} - \mathbf{y}|, \quad \mathbf{x}, \mathbf{y} \in K, \quad \text{where } L < 1,$$

(ii) $\mathbf{F}(K) \subset K$,

then

(1) $\mathbf{F}$ has exactly one fixed point $\mathbf{c}$ in K .

(2) The fixed-point iteration $\mathbf{x}_{k+1} = \mathbf{F}(\mathbf{x}_k)$, $k = 0, 1, \dots$ converges toward $\mathbf{c}$ for every $\mathbf{x}_0 \in K$.

Proof First, assumption (ii) implies that the fixed-point iteration $\mathbf{x}_{k+1} = \mathbf{F}(\mathbf{x}_k)$ satisfies

$$\mathbf{x}_0 \in K \Rightarrow \mathbf{x}_k \in K \quad \forall k,$$

so, by assumption (i),

$$\begin{aligned} |\mathbf{x}_{k+1} - \mathbf{x}_k| &= |\mathbf{F}(\mathbf{x}_k) - \mathbf{F}(\mathbf{x}_{k-1})| \\ &\leq L|\mathbf{x}_k - \mathbf{x}_{k-1}| \\ &\vdots \\ &\leq L^k|\mathbf{x}_1 - \mathbf{x}_0|. \end{aligned}$$

*The contraction mapping theorem is also known as the *Banach fixed-point theorem*, because the theorem is valid in *Banach spaces*, which are more general than the Euclidean spaces $\mathbb{R}^n$. Very many statements concerning the solvability of mathematical equations rely on the contraction mapping theorem in its more general forms. Such is, for example, the Picard–Lindelöf theorem concerning the solvability of differential equations (Section IX.8).

From this it follows that if $l > k$, then

$$\begin{aligned} |\mathbf{x}_l - \mathbf{x}_k| &\leq |\mathbf{x}_l - \mathbf{x}_{l-1}| + \dots + |\mathbf{x}_{k+1} - \mathbf{x}_k| \\ &\leq \sum_{i=k}^{l-1} L^i |\mathbf{x}_1 - \mathbf{x}_0| \\ &\leq |\mathbf{x}_1 - \mathbf{x}_0| \sum_{i=k}^{\infty} L^i \\ &= L^k (1 - L)^{-1} |\mathbf{x}_1 - \mathbf{x}_0|. \end{aligned}$$

Thus, more generally,

$$|\mathbf{x}_k - \mathbf{x}_l| < L^{\min\{k, l\}} (1 - L)^{-1} |\mathbf{x}_1 - \mathbf{x}_0|.$$

According to this, $\{\mathbf{x}_k\}$ is a *Cauchy sequence* in $\mathbb{R}^n$, i.e.

$$|\mathbf{x}_k - \mathbf{x}_l| \rightarrow 0, \quad \text{when } \min\{k, l\} \rightarrow \infty.$$

Denoting here $\mathbf{x}_k = ((\mathbf{x}_k)_i, i = 1 \dots n)$, it follows that

$$|(\mathbf{x}_k)_i - (\mathbf{x}_l)_i| \leq |\mathbf{x}_k - \mathbf{x}_l| \rightarrow 0, \quad \text{when } \min\{k, l\} \rightarrow \infty, \quad i = 1 \dots n.$$

Thus the real-number sequence $\{(\mathbf{x}_k)_i, k = 0, 1, \dots\}$ is a Cauchy sequence for every i . By the Cauchy criterion (Theorem [I.10.5](#)) there then exist real numbers $c_i, i = 1 \dots n$ such that $\lim_k (\mathbf{x}_k)_i = c_i, i = 1 \dots n$, whereby also $\lim_k |\mathbf{c}_k - \mathbf{c}| = 0, \mathbf{c} = (c_i)$ (cf. Section [XI.1](#)). We have thus concluded that the sequence $\{\mathbf{x}_k\}$ (and, more generally, every Cauchy sequence in $\mathbb{R}^n$) converges:

$$\mathbf{x}_k \rightarrow \mathbf{c} \in \mathbb{R}^n.$$

But since K is, by its definition, closed (in fact compact, see Definition [XI.1.8](#)), we have $\mathbf{c} \in K$. Then, since $\mathbf{F}$ is continuous on K by assumption (i), it follows that $\mathbf{F}(\mathbf{x}_k) \rightarrow \mathbf{F}(\mathbf{c})$, whereby, by the iteration formula $\mathbf{x}_{k+1} = \mathbf{F}(\mathbf{x}_k)$, we have $\mathbf{c} = \mathbf{F}(\mathbf{c})$, i.e. $\mathbf{c}$ is a fixed point. It has thus been shown that at least one fixed point $\mathbf{c} \in K$ exists, and that the fixed-point iteration converges, for every $\mathbf{x}_0 \in K$, toward some fixed point lying in K . Finally, we show that there is only one fixed point in K : Let $\mathbf{c}_1$ and $\mathbf{c}_2$ be fixed points of $\mathbf{F}$ in K , i.e. let $\mathbf{c}_1 = \mathbf{F}(\mathbf{c}_1)$, $\mathbf{c}_2 = \mathbf{F}(\mathbf{c}_2)$ and $\mathbf{c}_1, \mathbf{c}_2 \in K$. Assumption (i) then implies

$$|\mathbf{c}_1 - \mathbf{c}_2| = |\mathbf{F}(\mathbf{c}_1) - \mathbf{F}(\mathbf{c}_2)| \leq L |\mathbf{c}_1 - \mathbf{c}_2| \quad \Leftrightarrow \quad (1 - L) |\mathbf{c}_1 - \mathbf{c}_2| \leq 0.$$

Since here $1 - L > 0$ (assumption (i)), we must have $|\mathbf{c}_1 - \mathbf{c}_2| = 0 \Leftrightarrow \mathbf{c}_1 = \mathbf{c}_2$. The contraction mapping theorem is thus proved. $\square$

*Proof of the inverse function theorem

We start from the differential expansion

$$\mathbf{f}(\mathbf{x}) = \mathbf{b} + \mathbf{Jf}(\mathbf{a})(\mathbf{x} - \mathbf{a}) + \mathbf{r}(\mathbf{x}), \quad \mathbf{b} = \mathbf{f}(\mathbf{a}), \quad \mathbf{x} \in K, \quad (\text{a})$$

where $K = \{\mathbf{x} \in \mathbb{R}^n \mid |\mathbf{x} - \mathbf{a}| \leq \rho\}$, $\rho \in (0, \delta)$. On this basis,

$$\mathbf{f}(\mathbf{x}_1) - \mathbf{f}(\mathbf{x}_2) = \mathbf{Jf}(\mathbf{a})(\mathbf{x}_1 - \mathbf{x}_2) + \mathbf{r}(\mathbf{x}_1) - \mathbf{r}(\mathbf{x}_2), \quad \mathbf{x}_1, \mathbf{x}_2 \in K. \quad (\text{b})$$

When the $\mathbb{R}^n$ mean value theorem (Theorem [XI.3.3](#)) is applied here on the left-hand side to the functions $f_i = (\mathbf{f})_i$, it follows that

$$f_i(\mathbf{x}_1) - f_i(\mathbf{x}_2) = [\nabla f_i(\boldsymbol{\xi}_i)]^T(\mathbf{x}_1 - \mathbf{x}_2), \quad i = 1 \dots n, \quad (\text{c})$$

where $\boldsymbol{\xi}_i \in K$ (more precisely $\boldsymbol{\xi}_i = t_i(\mathbf{x}_1 - \mathbf{x}_2)$, $t_i \in (0, 1)$). By the continuity assumptions, here further

$$\nabla f_i(\boldsymbol{\xi}_i) = \nabla f_i(\mathbf{a}) + \mathbf{d}_i, \quad |\mathbf{d}_i| \rightarrow 0 \text{ when } \rho \rightarrow 0, \quad i = 1 \dots n. \quad (\text{d})$$

(Here some technical details are omitted.) Combining (b), (c), and (d), one sees that

$$\mathbf{r}(\mathbf{x}_1) - \mathbf{r}(\mathbf{x}_2) = [\mathbf{d}_1 \dots \mathbf{d}_n]^T(\mathbf{x}_1 - \mathbf{x}_2),$$

so, by the Cauchy–Schwarz inequality,

$$|\mathbf{r}(\mathbf{x}_1) - \mathbf{r}(\mathbf{x}_2)| = \left(\sum_{i=1}^n \langle \mathbf{d}_i, \mathbf{x}_1 - \mathbf{x}_2 \rangle^2 \right)^{1/2} \leq L|\mathbf{x}_1 - \mathbf{x}_2|, \quad (\text{e})$$

where

$$L = \left(\sum_{i=1}^n |\mathbf{d}_i|^2 \right)^{1/2} \rightarrow 0, \quad \text{when } \rho \rightarrow 0.$$

Since $\mathbf{Jf}(\mathbf{a})$ is nonsingular, equation (b), rewritten, is

$$\mathbf{x}_1 - \mathbf{x}_2 = [\mathbf{Jf}(\mathbf{a})]^{-1}[\mathbf{f}(\mathbf{x}_1) - \mathbf{f}(\mathbf{x}_2)] - [\mathbf{Jf}(\mathbf{a})]^{-1}[\mathbf{r}(\mathbf{x}_1) - \mathbf{r}(\mathbf{x}_2)]. \quad (\text{f})$$

When the triangle inequality and the estimate (e) are applied here, it follows that

$$\begin{aligned} |\mathbf{x}_1 - \mathbf{x}_2| &\leq |[\mathbf{Jf}(\mathbf{a})]^{-1}[\mathbf{f}(\mathbf{x}_1) - \mathbf{f}(\mathbf{x}_2)]| + |[\mathbf{Jf}(\mathbf{a})]^{-1}[\mathbf{r}(\mathbf{x}_1) - \mathbf{r}(\mathbf{x}_2)]| \\ &\leq C|\mathbf{f}(\mathbf{x}_1) - \mathbf{f}(\mathbf{x}_2)| + CL|\mathbf{x}_1 - \mathbf{x}_2|, \quad \mathbf{x}_1, \mathbf{x}_2 \in K, \end{aligned} \quad (\text{g})$$

where $C = \|[\mathbf{Jf}(\mathbf{a})]^{-1}\|$ (see Exercise [X.1:14](#)). Since C is a constant depending only on the matrix $[\mathbf{Jf}(\mathbf{a})]^{-1}$ and $L \rightarrow 0$ as $\rho \rightarrow 0$, one can choose $\rho \in (0, \delta)$ such that

$$CL \leq \frac{1}{2}, \quad (\text{h})$$

whereby from inequalities (g) and (h) it follows that

$$|\mathbf{x}_1 - \mathbf{x}_2| \leq 2C|\mathbf{f}(\mathbf{x}_1) - \mathbf{f}(\mathbf{x}_2)| \quad \forall \mathbf{x}_1, \mathbf{x}_2 \in K. \quad (\text{i})$$

According to this, for all $\mathbf{x}_1, \mathbf{x}_2 \in K$ we have $\mathbf{f}(\mathbf{x}_1) = \mathbf{f}(\mathbf{x}_2) \Rightarrow \mathbf{x}_1 = \mathbf{x}_2$, so $\mathbf{f}$ is one-to-one on K when $\rho \in (0, \delta)$ has been chosen so that condition (h) holds. The first part of the inverse function theorem is thus proved.

To prove the second part of the inverse function theorem, we show that when ρ has been chosen in the above manner, $\exists \varepsilon > 0$ such that the conditions of the contraction mapping theorem hold when $|\mathbf{y} - \mathbf{b}| \leq \varepsilon$ and the system to be solved is written in the form

$$\mathbf{f}(\mathbf{x}) = \mathbf{y} \Leftrightarrow \mathbf{x} = \mathbf{F}(\mathbf{x}), \quad \mathbf{F}(\mathbf{x}) = \mathbf{x} - [\mathbf{Jf}(\mathbf{a})]^{-1}[\mathbf{f}(\mathbf{x}) - \mathbf{y}], \quad \mathbf{x} \in K.$$

— Note that the fixed-point iteration $\mathbf{x}_{k+1} = \mathbf{F}(\mathbf{x}_k)$ is then the same as the simplified Newton iteration considered in the previous section, applied to the system $\mathbf{f}(\mathbf{x}) - \mathbf{y} = \mathbf{0}$ (with initial guess $\mathbf{x}_0 = \mathbf{a}$).

By the differential expansion (a), the expression for $\mathbf{F}$ is obtained in the form

$$\mathbf{F}(\mathbf{x}) = \mathbf{a} + [\mathbf{Jf}(\mathbf{a})]^{-1}(\mathbf{y} - \mathbf{b}) - [\mathbf{Jf}(\mathbf{a})]^{-1}\mathbf{r}(\mathbf{x}), \quad \mathbf{x} \in K.$$

Using the estimate (e) in this expression, it follows that

$$\begin{aligned} |\mathbf{F}(\mathbf{x}_1) - \mathbf{F}(\mathbf{x}_2)| &= |[\mathbf{Jf}(\mathbf{a})]^{-1}[\mathbf{r}(\mathbf{x}_1) - \mathbf{r}(\mathbf{x}_2)]| \\ &\leq CL|\mathbf{x}_1 - \mathbf{x}_2|, \quad \mathbf{x}_1, \mathbf{x}_2 \in K \end{aligned} \quad (\text{j})$$

and since $\mathbf{r}(\mathbf{a}) = \mathbf{0}$, it also follows that

$$\begin{aligned} |\mathbf{F}(\mathbf{x}) - \mathbf{a}| &= |[\mathbf{Jf}(\mathbf{a})]^{-1}(\mathbf{y} - \mathbf{b}) - [\mathbf{Jf}(\mathbf{a})]^{-1}[\mathbf{r}(\mathbf{x}) - \mathbf{r}(\mathbf{a})]| \\ &\leq C|\mathbf{y} - \mathbf{b}| + CL|\mathbf{x} - \mathbf{a}|, \quad \mathbf{x} \in K \end{aligned} \quad (\text{k})$$

(C as above). Now choose ε such that

$$0 < \varepsilon \leq \frac{\rho}{2C}. \quad (\text{l})$$

Then if $|\mathbf{y} - \mathbf{b}| \leq \varepsilon$, from inequalities (j), (k), (h), and (l) it follows that

$$\begin{cases} |\mathbf{F}(\mathbf{x}_1) - \mathbf{F}(\mathbf{x}_2)| \leq \frac{1}{2}|\mathbf{x}_1 - \mathbf{x}_2|, & \mathbf{x}_1, \mathbf{x}_2 \in K, \\ |\mathbf{F}(\mathbf{x}) - \mathbf{a}| \leq \frac{\rho}{2} + \frac{\rho}{2} = \rho, & \mathbf{x} \in K. \end{cases}$$

By the first of these conditions $\mathbf{F}$ is a contraction mapping on K , and by the second $\mathbf{F}(K) \subset K$, so the conditions of the contraction mapping theorem are

satisfied. By that theorem, the system $\mathbf{f}(\mathbf{x}) = \mathbf{y} \Leftrightarrow \mathbf{x} = \mathbf{F}(\mathbf{x})$ has a unique solution in K when ρ and ε have been chosen in the above manner and $|\mathbf{y} - \mathbf{b}| \leq \varepsilon$. The second part of the inverse function theorem is thus proved.

To prove the last part, we choose in equation (f) $\mathbf{x}_1 = \mathbf{x} = \mathbf{f}^{-1}(\mathbf{y})$, where $0 < |\mathbf{y} - \mathbf{b}| \leq \varepsilon$ (so that $\mathbf{x} \neq \mathbf{a}$) and $\mathbf{x}_2 = \mathbf{a}$, whereby the equation takes the form

$$\mathbf{f}^{-1}(\mathbf{y}) - \mathbf{a} = [\mathbf{Jf}(\mathbf{a})]^{-1}(\mathbf{y} - \mathbf{b}) - [\mathbf{Jf}(\mathbf{a})]^{-1}\mathbf{r}(\mathbf{x}). \quad (\text{m})$$

By inequality (i) we have (for the same $\mathbf{x}$ and $\mathbf{y}$)

$$|\mathbf{x} - \mathbf{a}| \leq 2C|\mathbf{y} - \mathbf{b}|.$$

On the basis of this and the estimate $|\mathbf{Jf}(\mathbf{a})|^{-1}\mathbf{r}(\mathbf{x})| \leq C|\mathbf{r}(\mathbf{x})|$,

$$\frac{|\mathbf{Jf}(\mathbf{a})|^{-1}\mathbf{r}(\mathbf{x})|}{|\mathbf{y} - \mathbf{b}|} \leq \frac{C|\mathbf{r}(\mathbf{x})|}{|\mathbf{x} - \mathbf{a}|} \cdot \frac{|\mathbf{x} - \mathbf{a}|}{|\mathbf{y} - \mathbf{b}|} \leq 2C^2 \frac{|\mathbf{r}(\mathbf{x})|}{|\mathbf{x} - \mathbf{a}|}.$$

Since here $|\mathbf{y} - \mathbf{b}| \rightarrow 0 \Rightarrow |\mathbf{x} - \mathbf{a}| \rightarrow 0$ (by the previous inequality) and since $|\mathbf{r}(\mathbf{x})|/|\mathbf{x} - \mathbf{a}| \rightarrow 0$ as $|\mathbf{x} - \mathbf{a}| \rightarrow 0$, we have $|\mathbf{Jf}(\mathbf{a})|^{-1}\mathbf{r}(\mathbf{x})| = o(|\mathbf{y} - \mathbf{b}|)$. Equation (m) can thus be written

$$\mathbf{f}^{-1}(\mathbf{y}) = \mathbf{f}^{-1}(\mathbf{b}) + [\mathbf{Jf}(\mathbf{a})]^{-1}(\mathbf{y} - \mathbf{b}) + \mathbf{r}(\mathbf{y}), \quad |\mathbf{r}(\mathbf{y})| = o(|\mathbf{y} - \mathbf{b}|).$$

Consequently, $\mathbf{f}^{-1}$ is differentiable at the point $\mathbf{b}$ and $\mathbf{Jf}^{-1}(\mathbf{b}) = [\mathbf{Jf}(\mathbf{a})]^{-1}$. $\square$

EXERCISES

- Consider the function $\mathbf{f}(x, y) = (2x^2 - y^2, -x^2 + 2y^2)$. a) Determine the range $\mathbf{f}(\mathbb{R}^2)$ of $\mathbf{f}$. b) For which sets $A \subset \mathbb{R}^2$ is $f : A \rightarrow f(A)$ a bijection? c) Show that $\mathbf{f}$ is locally invertible at the point (a, b) if $ab \neq 0$. Is this condition also necessary for local invertibility? d) Determine the expressions for the local inverse functions of $\mathbf{f}$ at the points $(1, 1)$, $(-1, 1)$, $(1, -1)$, and $(-1, -1)$.
- Let $\mathbf{f}(x, y) = (x - y + e^{x+2y}, x + y + e^{x-2y})$ and let $\mathbf{f}^{-1}$ be the local inverse function of $\mathbf{f}$ in a neighborhood of the origin. Further, let
 - $K_a(x_0, y_0) = \{(x, y) \in \mathbb{R}^2 \mid |x - x_0| \leq a \text{ \& \& } |y - y_0| \leq a\}$,
 - $K_a(x_0, y_0) = \{(x, y) \in \mathbb{R}^2 \mid (x - x_0)^2 + (y - y_0)^2 \leq a^2\}$,
 where $a > 0$ is a small number. By approximating $\mathbf{f}$ by an affine map, determine approximately the sets $\mathbf{f}(K_a(0, 0))$ and $\mathbf{f}^{-1}(K_a(1, 1))$. Draw the figures!

3. Show that the following functions are bijections as maps $\mathbf{f} : \mathbb{R}^2 \rightarrow \mathbb{R}^2$. Determine the expression for the inverse function, if possible. Also determine the points at which $\mathbf{Jf}$ is singular.
 - a) $\mathbf{f}(x, y) = (x^3 + y^3, x^3 - y^3)$
 - b) $\mathbf{f}(x, y) = ((x + y)^3, (x - y)^3)$
 - c) $\mathbf{f}(x, y) = (x - y, x + y^3)$
 - d) $\mathbf{f}(x, y) = (e^x - y, x + x^3 + y + y^3)$
4. Let $\mathbf{f} = (e^x \cos y, e^x \sin y)$.
 - a) Show that $\mathbf{f}$ is everywhere locally invertible.
 - b) Determine $\mathbf{f}(A)$, $A = \mathbb{R} \times [0, 2\pi]$. Is $\mathbf{f} : A \rightarrow \mathbf{f}(A)$ one-to-one?
 - c) Let $A = \{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 < R^2\}$. What is the largest value of R for which $\mathbf{f} : A \rightarrow \mathbf{f}(A)$ is a bijection?
5. At which points (x, y) are the assumptions of the inverse function theorem satisfied for the function $\mathbf{f}(x, y) = (x^2 + 2y, y^2 - 2x)$? Determine the Jacobian matrix $\mathbf{Jf}^{-1}(u, v)$, $(u, v) = \mathbf{f}(x, y)$ at these points.
6. The arm of a robot moving in the plane consists of an upper arm (a line segment) OP attached at the origin and a forearm PQ attached at the point P . The length of each arm = 1. The hand (Q) is controlled by adjusting the polar angle φ between the upper arm and the x -axis and the angle θ between the forearm and the upper arm. The angles can be varied over the ranges $\varphi \in [0, \pi]$ and $\theta \in [-5\pi/6, 5\pi/6]$ ($\theta > 0$ counterclockwise).
 - a) Determine the position of the robot's hand as a function $\mathbf{f}(\varphi, \theta)$.
 - b) Which points of the plane can the hand reach, i.e. what is the range of $\mathbf{f}$? Figure!
 - c) Verify that $\mathbf{f}$ is not one-to-one. At which points $(\varphi, \theta) \in (0, \pi) \times (-5\pi/6, 5\pi/6)$ are the conditions of the inverse function theorem satisfied?
 - d) The robot's hand is to be moved from the point $(1.4, 0.2)$ to the point $(1.42, 0.22)$. Using the differential, estimate the angle changes $\Delta\varphi$ and $\Delta\theta$ with which this motion is achieved, when the arm is in a position where $\theta < 0$.
7. Derive the rule for computing the Jacobian matrix of the implicit function $\mathbf{y} = \mathbf{g}(\mathbf{x})$ (Theorem [XI.7.3](#)) by implicit partial differentiation from the system

$$\mathbf{F}(\mathbf{x}, \mathbf{g}(\mathbf{x})) = \mathbf{0} \quad \Leftrightarrow \quad F_i(\mathbf{x}, g_1(\mathbf{x}), \dots, g_p(\mathbf{x})) = 0, \quad i = 1 \dots p.$$

8. Assume that the system $F(x, y, u, v) = 0$, $G(x, y, u, v) = 0$ holds when $(x, y, u, v) = (a, b, c, d)$, and that near that point the system is solvable both into the form $(u, v) = \mathbf{g}(x, y)$ and into the form $(x, y) = \mathbf{h}(u, v)$. From the last part of the implicit function theorem (under the assumptions

of that theorem), derive the result $\mathbf{Jg}(a, b)\mathbf{Jh}(c, d) = \mathbf{I}$. In what other way can this be deduced?

9. Let f be a regular function of the type $f: \mathbb{R}^2 \rightarrow \mathbb{R}$. If the system

$$\begin{cases} f(f(x, y), z) = 0 \\ f(x, f(y, z)) = 0 \end{cases}$$

holds when $(x, y, z) = (a, b, c)$, then under what condition on the partial derivatives f_x and f_y of f can it be guaranteed that near the point (a, b, c) the system is solvable into the form $x = x(z)$, $y = y(z)$?

10. Let $a, b, c > 0$. Show that the system

$$\begin{cases} \frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1 \\ \frac{(2x - a)^2}{a^2} + \frac{4y^2}{b^2} = 1 \end{cases}$$

defines, in a neighborhood of the point $P = (a, b, c)$, a parametric curve $S: x = f(z)$ & $y = g(z)$. What is the tangent vector of S at the point P ?

11. Show that the equation or system defines an implicit function of the given form in a neighborhood of the given point P , and solve the additional task:

a) $x^6 + 2y^4 - 3x^2y = 0$, $P = (1, 1)$, $y = f(x)$. Equation of the tangent of the curve $y = f(x)$ at the point P ?

b) $\sin(yz) + y^2e^z = x$, $P = (e\pi^2, \pi, 1)$, $z = f(x, y)$. Estimate $f(27, 3)$ using the differential.

c) $ye^{xz} + \sin(x - y + z) = 0$, $P = (0, 0, 0)$, $z = f(x, y)$. Equation of the tangent plane of the surface $z = f(x, y)$ at the origin?

d) $x = z + y \sin z$, $P = (0, 0, 0)$, $z = f(x, y)$. Compute $f_{xy}(0, 0)$.

e) $x - y - z + 1 = 0$ & $x^2 + y^2 - 2z = 0$, $P = (1, 1, 1)$, $(x, y) = \mathbf{f}(z)$. Geometric interpretation?

f) $2x^2 + 3y^2 + z^2 = 47$ & $x^2 + 2y^2 - z = 0$, $P = (-2, 1, 6)$, $(y, z) = (f(x), g(x))$. Compute $f'(-2)$ and $g'(-2)$.

g) $y^5 + xy + z^2 = 4$ & $e^{xz} = y^2$, $(3, 1, 0)$, $(x, y) = (f(z), g(z))$. Compute $f'(0)$ and $g'(0)$.

h) $2x^2 + 3y^2 + z^2 = 47$ & $x^2 + 2y^2 - z = 0$, $P = (-2, 1, 6)$, $(y, z) = (f(x), g(x))$. Compute $f'(-2)$ and $g'(-2)$.

12. Show that the following systems define an implicit function $\mathbf{g}$ of the given form near the given point, and compute the Jacobian matrix $\mathbf{Jg}$ at that point.

$$\text{a) } \begin{cases} x - u^3 - v^3 = 0 \\ y - uv + v^2 = 0, \end{cases} \quad (u, v) = \mathbf{g}(x, y), \quad (x, y, u, v) = (2, 0, 1, 1)$$

$$\text{b) } \begin{cases} xe^y - u^3z + \sin v = 0 \\ x^3u^2 + yv - uz \cos v = 0, \end{cases} \quad (u, v) = \mathbf{g}(x, y, z), \\ (x, y, z, u, v) = (1, 0, 1, 1, 0)$$

$$\text{c) } \begin{cases} xy^2 + zu + v^2 = 3 \\ x^3z + 2y - uv = 2 \\ xu + yv - xyz = 1, \end{cases} \quad (x, y, z) = \mathbf{g}(u, v), \\ (x, y, z, u, v) = (1, 1, 1, 1, 1)$$

13. (*) Show that the function $\mathbf{f}(x, y) = (2x - xy, 4y - xy)$ is locally invertible at the point (a, b) exactly when $a + 2b \neq 4$.
14. (*) Assume that the equation $F(x, y, z) = 0$ defines (under the assumptions of the implicit function theorem) the implicit functions $x = x(y, z)$, $y = y(z, x)$, and $z = z(x, y)$. Show that:

$$\frac{\partial x}{\partial y} \cdot \frac{\partial y}{\partial z} \cdot \frac{\partial z}{\partial x} = -1.$$

Verify the validity of the rule when $F(x, y, z) = x^3y^2z - 1$. Formulate and prove the corresponding more general statement concerning the implicit functions defined by the equation $F(x_1, \dots, x_n) = 0$ (n of them, $n \geq 2$).

XI.8 Multivariable extremum problems

We begin with a definition.

DEFINITION XI.8.1 A function $f : \mathcal{D}_f \rightarrow \mathbb{R}$, $\mathcal{D}_f \subset \mathbb{R}^n$, has at a point $\mathbf{c} \in \mathcal{D}_f$ a *local maximum* if for some $\delta > 0$

$$0 < |\mathbf{x} - \mathbf{c}| < \delta \Rightarrow \mathbf{x} \in \mathcal{D}_f \text{ and } f(\mathbf{x}) < f(\mathbf{c}),$$

and a *local minimum* if for some $\delta > 0$

$$0 < |\mathbf{x} - \mathbf{c}| < \delta \Rightarrow \mathbf{x} \in \mathcal{D}_f \text{ and } f(\mathbf{x}) > f(\mathbf{c}).$$

If f has at the point $\mathbf{c}$ a local maximum or minimum, then f has a *local extremum* at the point $\mathbf{c}$. If $\mathbf{c} \in \mathcal{D}_f$ is not a local extremum point of the function f , but for some $\delta > 0$

$$\begin{aligned} \text{either : } & 0 < |\mathbf{x} - \mathbf{c}| < \delta \Rightarrow \mathbf{x} \in \mathcal{D}_f \text{ and } f(\mathbf{x}) \leq f(\mathbf{c}), \\ \text{or : } & 0 < |\mathbf{x} - \mathbf{c}| < \delta \Rightarrow \mathbf{x} \in \mathcal{D}_f \text{ and } f(\mathbf{x}) \geq f(\mathbf{c}), \end{aligned}$$

then $\mathbf{c}$ is said to be a *flat point* of f .

EXAMPLE 1 The function $f(x, y) = ax^2 + by^2$ has a local extremum point at the origin if $ab > 0$. It is a local minimum if $a > 0$ and $b > 0$, and a local maximum if $a < 0$ and $b < 0$. If $ab = 0$, then the origin is a flat point. In the case $ab < 0$ the origin is neither an extremum point nor a flat point. $\square$

A local extremum of a function is also called *relative*, in which case the maximum/minimum of the function over the whole set under consideration (e.g. the whole domain) is called *absolute*. The term *global* extremum is also used, especially when it concerns the whole domain of the function.

EXAMPLE 1 (continued) If the set $A \subset \mathbb{R}^2$ contains the origin, then in the case $ab > 0$ the value $f(0, 0) = 0$ is the absolute minimum/maximum of the function on A . $\square$

With the aid of the gradient, the local minima and maxima of a real-valued function can easily be located, if the function is differentiable. Indeed, on the basis of the considerations of Section XI.3, the following result is quite evident:

PROPOSITION XI.8.2 If $f : \mathcal{D}_f \rightarrow \mathbb{R}$, $\mathcal{D}_f \subset \mathbb{R}^n$, is differentiable at the point $\mathbf{c} \in \mathcal{D}_f$ and $\mathbf{c}$ is a local extremum or flat point of f , then $\nabla f(\mathbf{c}) = \mathbf{0}$.

The zeros of the gradient are more generally called the *critical points* of the function.

EXAMPLE 2 Find the critical points of the function $f(x, y) = x^2 + 2y^2 + 2xy + 4x + 14y$. (Cf. Section IV.3, Example 1.)

Solution

$$\nabla f(x, y) = \vec{0} \quad \Leftrightarrow \quad \begin{cases} \partial_x f(x, y) = 2x + 2y + 4 = 0, \\ \partial_y f(x, y) = 2x + 4y + 14 = 0 \end{cases} \quad \Leftrightarrow \quad \begin{cases} x = 3, \\ y = -5. \end{cases}$$

Thus the only critical point is $(3, -5)$. $\square$

In the example, the critical point is the absolute minimum of the function, as was determined in Section IV.3 by purely algebraic means. — A critical point need not, more generally, be a local extremum point or a flat point. For example, the critical point (the origin) of the function

$$f(x, y) = x^2 - y^2$$

is a so-called *saddle point* — the term hardly needs closer definition. Classifying the critical points of a function as extremum points, flat points, or saddle points is a problem whose (algebraic) solution is of more general interest. We return to this a little later. In this context we restrict ourselves to extremum problems in which the classification of critical points is not of central importance.

Constrained optimization

Multivariable extremum problems arising in applications can usually be formulated as *constrained optimization* problems. In a typical constrained optimization problem, one wants to find the minimum or maximum of a real-valued function $f(\mathbf{x})$, $\mathcal{D}_f \subset \mathbb{R}^n$ on a set A , whose definition is given in the form

$$\mathbf{x} \in A \quad \Leftrightarrow \quad g_i(\mathbf{x}) \geq 0, \quad i = 1 \dots m.$$

In applications, this is usually about maximizing or minimizing something of practical importance (such as economic gain or loss), i.e. *optimization*, subject to the *constraints* that are unavoidable in the problem setting. — From the standpoint of designing (or using) optimization algorithms, it is worth noting that inequality constraints can also include equality constraints, in the form

$$g(\mathbf{x}) = 0 \quad \Leftrightarrow \quad \begin{cases} g(\mathbf{x}) & \geq 0, \\ -g(\mathbf{x}) & \geq 0. \end{cases}$$

Constraints are called *linear* if the functions g_i are first-degree polynomials (affine maps). If f is also of this type, then it is a *linear programming* problem, cf. Section X.7.

As an example, consider the two-variable situation in which the optimization problem is posed as

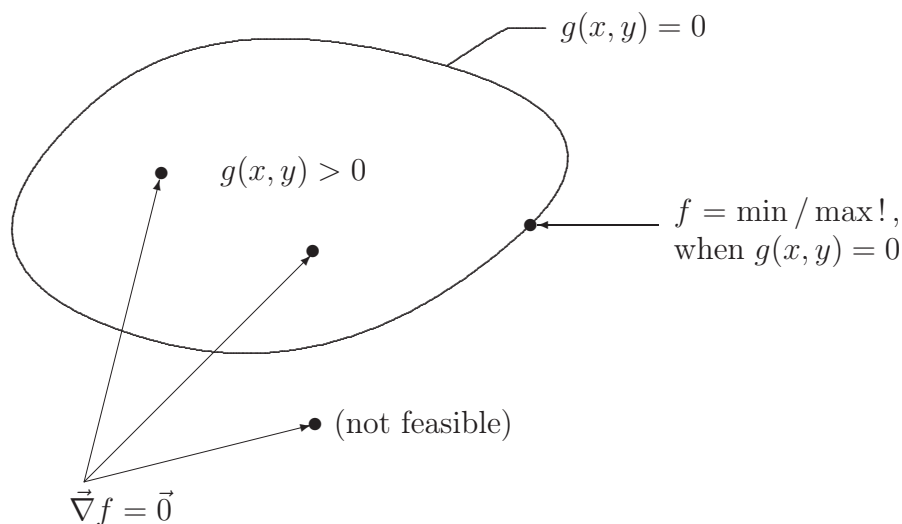
$$f(x, y) = \min / \max! \quad \text{subject to} \quad g_i(x, y) \geq 0, \quad i = 1 \dots m.$$

In what follows, assume that both f and the constraint functions g_i are defined and differentiable in $\mathbb{R}^2$. The problem can be split into two parts: First solve the *unconstrained* problem by going through all the critical points of f . Among the critical points, the *feasible* ones are those at which the prescribed constraints hold. Assume that these are finite in number, so that a selection can be made: choose from among the points the one at which f attains its smallest/largest value.

The optimization problem is not yet solved; the more difficult phase is still ahead: one must separately determine the minimum/maximum of the function on the *boundary* ∂A of the set A under consideration, which is defined

$$(x, y) \in \partial A \quad \Leftrightarrow \quad g_i(x, y) = 0 \quad \text{for some } i.$$

The figure shows the whole optimization problem in the case of one constraint ($m = 1$).



An optimization problem subject to one or more equality constraints is called a *constrained* extremum problem. In the two-variable case, the problem is naturally solved if one assumes that each constraint defines a parametric curve of the form

$$g_i(x, y) = 0 \Leftrightarrow x = x_i(t), \quad y = y_i(t), \quad t \in \mathbb{R}, \quad i = 1 \dots m,$$

and assumes further that the boundary ∂A can be split into curve arcs

$$\partial A_i = \{(x, y) \in \mathbb{R}^2 \mid x = x_i(t), \quad y = y_i(t), \quad t \in [a_i, b_i]\},$$

which together cover ∂A and whose common points are at most the endpoints $(x_i(a_i), y_i(a_i))$ and $(x_i(b_i), y_i(b_i))$. Under these assumptions, the remaining part of the optimization problem is reduced to one-variable constrained optimization problems:

$$F_i(t) = f(x_i(t), y_i(t)) = \min/\max ! \quad \text{when} \quad a_i \leq t \leq b_i, \quad i = 1 \dots m.$$

These problems are solved in the normal way, i.e. by finding the critical points of the functions F_i (= zeros of the derivative) on the open intervals (a_i, b_i) and examining the endpoints a_i, b_i separately. Insofar as the functions F_i have a finite number of feasible critical points, a finite set of points has been found, among which the extrema of the function to be optimized are necessarily located.

EXAMPLE 3 Solve the constrained optimization problems

$$\begin{aligned} f(x, y) = 6x^2 - 4xy + 9y^2 + 16x - 22y = \min ! \quad \text{and} \quad f(x, y) = \max ! \\ \text{when} \quad x \geq 0, \quad x^2 + (y - 1)^2 \leq 1. \end{aligned}$$

Solution There is one critical point:

$$\nabla f(x, y) = \vec{0} \Leftrightarrow \begin{cases} 12x - 4y + 16 = 0, \\ -4x + 18y - 22 = 0 \end{cases} \Leftrightarrow \begin{cases} x = -1, \\ y = 1. \end{cases}$$

This is not feasible, so we proceed to examine the function on the boundary ∂A of the set A under consideration:

$$(x, y) \in \partial A \Leftrightarrow x = 0 \quad \text{or} \quad x^2 + (y - 1)^2 = 1.$$

This splits into two parts:

$$\begin{aligned} \partial A_1 : \quad x = 0 \quad \& \quad x^2 + (y - 1)^2 \leq 1 \quad \Leftrightarrow \quad x = 0 \quad \& \quad y \in [0, 2]. \\ \partial A_2 : \quad x^2 + (y - 1)^2 = 1 \quad \& \quad x \geq 0. \end{aligned}$$

The boundary part ∂A_1 is a line segment, on which

$$F_1(y) = f(0, y) = 9y^2 - 22y.$$

The critical point $y = 11/9$ of the function F_1 is feasible, so we record the value of f at this point and at the endpoints of the segment ∂A_1 :

$$f(0, 11/9) = -121/9 = -13.4444.., \quad f(0, 0) = 0, \quad f(0, 2) = -8.$$

We move on to the boundary part ∂A_2 . This is a semicircular arc, which is parametrized in the form

$$x = \sin t, \quad y = 1 - \cos t, \quad t \in [0, \pi].$$

The function to be examined is therefore

$$\begin{aligned} F_2(t) &= f(\sin t, 1 - \cos t) \\ &= 6 \sin^2 t + 4 \sin t \cos t + 9 \cos^2 t + 12 \sin t + 4 \cos t - 13, \quad t \in [0, \pi]. \end{aligned}$$

The critical points of the function F_2 are found by solving the equation

$$F_2'(t) = \dots = -3 \sin 2t + 4 \cos 2t + 12 \cos t - 4 \sin t = 0.$$

By numerical means, the only feasible solution obtained is

$$t = 0.927295.., \quad (x(t), y(t)) = (0.800000.., 0.400000..) = (4/5, 2/5).$$

At this point $F_2(0.927295..) = 8$. Since the endpoints of the boundary parts ∂A_1 and ∂A_2 are the same, it has been concluded that the smallest and largest values of f are found in the set $\{-121/9, 0, -8, 8\}$. The solutions of the optimization problems are therefore

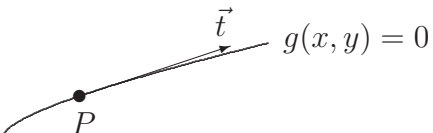
$$\begin{aligned} f_{\min} &= -121/9 = -13.4444.. \quad \text{at the point } (x, y) = (0, 11/9) = (0, 1.1111..), \\ f_{\max} &= 8 \quad \text{at the point } (x, y) = (4/5, 2/5) = (0.8, 0.4). \quad \square \end{aligned}$$

The method of Lagrange multipliers

The solution method presented above can, in principle, be applied to a constrained optimization problem in more variables as well. The difficulty, however, arises from the constrained extremum problems that must be solved when optimizing the objective function on the boundary of the set under consideration. In the case of more than two variables, solving such problems by parametrization, i.e. by reducing the number of free variables, is often awkward. In constrained

extremum problems of three or more variables, one therefore generally uses another, much more elegant and more generally applicable method, called, after its inventor **J. L. Lagrange** (1736–1813), the *method of Lagrange multipliers*.

Let us again consider first the two-variable situation. Suppose we seek the local extremum points of the function $f(x, y)$ on the curve S defined by the constraint $g(x, y) = 0$. Let $P = (x_0, y_0) \in S$ be one such extremum point and let $\vec{t}$ be the tangent vector of S at the point P . Then we must have

$$\vec{t} \cdot \nabla f(x_0, y_0) = 0.$$


On the other hand, since the curve $g(x, y) = 0$ is a level curve of the function g , it also holds (cf. Section [XI.3](#)) that

$$\vec{t} \cdot \nabla g(x_0, y_0) = 0.$$

Consequently, insofar as $\nabla g(x_0, y_0) \neq \vec{0}$ (which is assumed in what follows — the points at which $g(x_0, y_0) = 0$ and $\nabla g(x_0, y_0) = \vec{0}$ must be examined separately), we must have, for some $\lambda \in \mathbb{R}$,

$$\nabla f(x_0, y_0) + \lambda \nabla g(x_0, y_0) = \vec{0}.$$

The sought local extremum points can thus be found by solving the system of equations

$$\begin{cases} f_x(x, y) + \lambda g_x(x, y) = 0, \\ f_y(x, y) + \lambda g_y(x, y) = 0, \\ g(x, y) = 0. \end{cases}$$

Here there are three unknowns and three equations, so at least the formal side is in order.

The above system is arrived at in a more easily memorable way by defining the function

$$F(x, y, \lambda) = f(x, y) + \lambda g(x, y).$$

Here the function g appearing in the constraint has been brought in weighted by the coefficient λ — the *Lagrange multiplier*. The above system is now arrived at simply by seeking the critical points of F from the conditions

$$\begin{cases} \partial_x F(x, y, \lambda) = 0, \\ \partial_y F(x, y, \lambda) = 0, \\ \partial_\lambda F(x, y, \lambda) = 0. \end{cases}$$

The above geometric reasoning also works in the three-variable case. Namely, if $P = (x_0, y_0, z_0)$ is a local extremum point of the function $f(x, y, z)$ on the surface $S : g(x, y, z) = 0$, then for every tangent vector $\vec{t}$ of the surface at the point P it must hold that

$$\vec{t} \cdot \nabla f(x_0, y_0, z_0) = 0.$$

Then $\nabla f(x_0, y_0, z_0)$ is a normal of the surface S , so for some $\lambda \in \mathbb{R}$

$$\nabla f(x_0, y_0, z_0) + \lambda \nabla g(x_0, y_0, z_0) = \vec{0}$$

(assuming that $\nabla g(x_0, y_0, z_0) \neq \vec{0}$). The extremum points are then found among the critical points of the function

$$F(x, y, z, \lambda) = f(x, y, z) + \lambda g(x, y, z).$$

EXAMPLE 4 Find the extrema of the function $f(x, y, z) = xy + yz - xz$ on the surface of the unit sphere.

Solution The constraint is $x^2 + y^2 + z^2 = 1$, so we seek the critical points of the function

$$F(x, y, z, \lambda) = xy + yz - xz + \lambda(x^2 + y^2 + z^2 - 1) :$$

$$\begin{cases} \partial_x F = y - z + 2\lambda x = 0, \\ \partial_y F = x + z + 2\lambda y = 0, \\ \partial_z F = y - x + 2\lambda z = 0, \\ \partial_\lambda F = x^2 + y^2 + z^2 - 1 = 0. \end{cases}$$

The first three equations, in matrix form, are

$$\begin{bmatrix} 2\lambda & 1 & -1 \\ 1 & 2\lambda & 1 \\ -1 & 1 & 2\lambda \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \mathbf{0}.$$

We seek a nontrivial solution for this, so the coefficient matrix must be singular. The determinant of the coefficient matrix is

$$\det(\mathbf{A}) = 8\lambda^3 - 6\lambda - 2 = 2(\lambda - 1)(2\lambda + 1)^2,$$

so we must have either $\lambda = 1$ or $\lambda = -1/2$. When $\lambda = 1$, the system becomes

$$\begin{bmatrix} 2 & 1 & -1 \\ 1 & 2 & 1 \\ -1 & 1 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \mathbf{0} \quad \stackrel{\text{(Gauss)}}{\Leftrightarrow} \quad \begin{bmatrix} 2 & 1 & -1 \\ 0 & \frac{3}{2} & \frac{3}{2} \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \mathbf{0}.$$

Its solution, together with the constraint $x^2 + y^2 + z^2 = 1$, is

$$(x, y, z) = t(1, -1, 1), \quad t = \pm 1/\sqrt{3},$$

and the value of the function f at these points is

$$f\left(\pm \frac{1}{\sqrt{3}}, \mp \frac{1}{\sqrt{3}}, \pm \frac{1}{\sqrt{3}}\right) = \underline{\underline{-1}}.$$

If $\lambda = -1/2$, then the system becomes

$$\begin{bmatrix} -1 & 1 & -1 \\ 1 & -1 & 1 \\ -1 & 1 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \mathbf{0} \Leftrightarrow \begin{bmatrix} -1 & 1 & -1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \mathbf{0}.$$

The solutions satisfying the constraint are $(x, y, z) = (t - s, t, s)$, where $t, s \in \mathbb{R}$ satisfy

$$(t - s)^2 + t^2 + s^2 = 1 \Leftrightarrow 2(t^2 + s^2 - ts) = 1.$$

The value of the function f at these points becomes

$$\begin{aligned} f(x, y, z) &= (t - s)t + ts - (t - s)s \\ &= t^2 + s^2 - ts = \underline{\underline{1/2}}. \end{aligned}$$

Thus

$$\begin{aligned} f_{\min} &= -1 \quad \text{at the points} \quad \pm \frac{1}{\sqrt{3}}(1, -1, 1), \\ f_{\max} &= 1/2 \quad \text{on the curve } S : (x, y, z) = (t - s, t, s), \quad (t, s) \in \mathbb{R}^2, \quad t^2 + s^2 - ts = 1/2. \end{aligned}$$

The maximum points form a great circle of the unit sphere in the plane $T : x - y + z = 0$. $\square$

The method of Lagrange multipliers works, apart from exceptional situations, in the case of several constraints as well. If one must find the extrema of the function $f(x_1, \dots, x_n)$ subject to the constraints

$$g_i(x_1, \dots, x_n) = 0, \quad i = 1 \dots m < n,$$

then these are found among the critical points of the function

$$F(\mathbf{x}, \boldsymbol{\lambda}) = f(\mathbf{x}) + \sum_{i=1}^m \lambda_i g_i(\mathbf{x}).$$

For example, in the case $n = 3$, $m = 2$ one must examine the function

$$F(x, y, z, \lambda, \mu) = f(x, y, z) + \lambda g_1(x, y, z) + \mu g_2(x, y, z).$$

The idea of the method can, in this case too, be justified geometrically: The constraints can be assumed to define a space curve K , which is the intersection of the surfaces

$$S_1 : g_1(x, y, z) = 0, \quad S_2 : g_2(x, y, z) = 0.$$

If the normals of the surfaces at the point (x_0, y_0, z_0) of the curve are

$$\vec{n}_i = \nabla g_i(x_0, y_0, z_0), \quad i = 1, 2,$$

then the tangent $\vec{t}$ of the curve K is perpendicular to both of these. The extremum condition is of the same form as in the case of one constraint, i.e. we must have

$$\vec{t} \cdot \nabla f(x_0, y_0, z_0) = 0.$$

Since here $\vec{t}$ is thus perpendicular to the vectors $\vec{n}_1$ and $\vec{n}_2$, then, insofar as $\vec{n}_1$ and $\vec{n}_2$ are linearly independent, we must have $\nabla f(x_0, y_0, z_0) = \lambda \vec{n}_1 + \mu \vec{n}_2$ for some $(\lambda, \mu) \in \mathbb{R}^2$, i.e.

$$\nabla f(x_0, y_0, z_0) = \lambda \nabla g_1(x_0, y_0, z_0) + \mu \nabla g_2(x_0, y_0, z_0).$$

EXAMPLE 5 Minimize $f(x, y, z) = x + 2y - z$ subject to $x^2 + y^2 + z^2 = 12$ and $x + y + z = 1$.

Solution We find the critical points of the function

$$F(x, y, z, \lambda, \mu) = x + 2y - z + \lambda(x^2 + y^2 + z^2 - 12) + \mu(x + y + z - 1) :$$

$$\begin{cases} 2\lambda x + \mu = -1 \\ 2\lambda y + \mu = -2 \\ 2\lambda z + \mu = 1 \\ x^2 + y^2 + z^2 = 12 \\ x + y + z = 1 \end{cases} \Rightarrow \begin{cases} 2\lambda(x + y + z) + 3\mu = -2 \\ x + y + z = 1 \end{cases} \Rightarrow 2\lambda = -3\mu - 2.$$

From the first three and the fourth equation of the system it also follows that

$$4\lambda^2(x^2 + y^2 + z^2) = (\mu + 1)^2 + (\mu + 2)^2 + (\mu - 1)^2 = 48\lambda^2.$$

Here, by the previous result, $48\lambda^2 = 12(3\mu + 2)^2$, so it follows that

$$105\mu^2 + 140\mu + 42 = 0 \Rightarrow (\mu, \lambda) = \begin{cases} (-0.877485, & 0.316228), \\ (-0.455848, & -0.316228) \end{cases}$$

$$\Rightarrow (x, y, z) = \begin{cases} (-0.19371, & -1.77485, & 2.96856), \\ (0.86038, & 2.44152, & -2.30190). \end{cases}$$

Of these, the first point gives the minimum value $f_{\min} = \underline{\underline{-6.71197}}$. $\square$

When the method of Lagrange multipliers is used, solving the resulting nonlinear system does not, of course, always succeed by hand calculation. The solutions are then found by numerical means, most commonly by Newton's method. Combined with numerical solution, the method of Lagrange multipliers in fact shows its best side, since the solution algorithm becomes in this way very straightforward and easily implemented (by machine).

EXAMPLE 6 In Example 3, the extremum problem

$$\begin{aligned} f(x, y) &= 6x^2 - 4xy + 9y^2 + 16x - 22y = \max ! \\ \text{subject to } g(x, y) &= x^2 + (y - 1)^2 - 1 = 0 \end{aligned}$$

was solved by parametrizing the constraint. When the Lagrange multiplier is used, we seek the critical points of the function

$$F(x, y, \lambda) = 6x^2 - 4xy + 9y^2 + 16x - 22y + \lambda[x^2 + (y - 1)^2 - 1],$$

i.e. we solve the system

$$\begin{cases} 6x - 2y + x\lambda - 8 = 0, \\ -2x + 9y + (y - 1)\lambda - 11 = 0, \\ x^2 + (y - 1)^2 - 1 = 0. \end{cases}$$

When Newton's method is used in the solution, the iteration formula becomes

$$\begin{bmatrix} x_{k+1} \\ y_{k+1} \\ \lambda_{k+1} \end{bmatrix} = \begin{bmatrix} x_k \\ y_k \\ \lambda_k \end{bmatrix} - \begin{bmatrix} \lambda_k + 6 & -2 & x_k \\ -2 & \lambda_k + 9 & y_k - 1 \\ 2x_k & 2y_k - 2 & 0 \end{bmatrix}^{-1} \begin{bmatrix} 6x_k - 2y_k + x_k\lambda_k - 8 \\ -2x_k + 9y_k + (y_k - 1)\lambda_k - 11 \\ x_k^2 + (y_k - 1)^2 - 1 \end{bmatrix}.$$

With the initial guess $(x_0, y_0, \lambda_0) = (1, 0.5, 0)$ one obtains

k	x_k	y_k	λ_k
0	1.000000	0.500000	0.0000
1	1.000000	0.750000	-12.5000
2	0.815287	0.136146	-14.9283
3	0.818006	0.376563	-14.8590
4	0.800355	0.399765	-14.9999
5	0.800000	0.400000	-15.0000

The convergence was somewhat lucky, since the initial guess turned out to be poor as regards the Lagrange multiplier. A better initial guess could have been computed, e.g. as follows (a more general procedure: see Exercise 12):

$$\vec{a} = \nabla f(x_0, y_0) = 26\vec{i} - 17\vec{j}, \quad \vec{b} = \nabla g(x_0, y_0) = 2\vec{i} - \vec{j},$$

$$\vec{a} + \lambda \vec{b} \approx \vec{0} \Rightarrow \lambda \approx -\frac{\vec{a} \cdot \vec{b}}{|\vec{b}|^2} = -13.8 = \lambda_0. \quad \square$$

EXERCISES

1. Determine the largest and smallest values of the function under the given constraints. Use parametrization in the constraints (if there are any).
 - a) $x - x^2 + y^2$, $0 \leq x \leq 2$, $0 \leq y \leq 1$
 - b) $xy - 2x$, $-1 \leq x \leq 1$, $0 \leq y \leq 1$
 - c) $xy - x^3y^2$, $0 \leq x, y \leq 1$
 - d) $xy(a - 2x - y)$, $|x| \leq a$, $|y| \leq a$
 - e) $xy(1 - x - 2y)$, $x, y \geq 0$, $x + y \leq 1$
 - f) $x^2 - 2y^2 - xy - x$, $x, y \geq -1$, $x + y \leq 1$
 - g) $xy - x^2$, $x^2 + y^2 \leq 1$
 - h) $3 + x - x^2 - y^2$, $2x^2 + y^2 \leq 1$
 - i) $y^2 - 2x^2$, $x^2 - y \leq 1$, $x + y \leq 1$
 - j) $xy + x - y$, $x^2 + y^2 \leq 1$, $x + y + 1 \geq 0$, $x - y - 1 \leq 0$
 - k) $x^3 + 3y^3$, $0 \leq xy \leq 1$, $x^2 + y^2 \leq 4$
 - l) $\sin x + \sin y - \sin(x + y)$, $0 \leq x, y \leq \pi$
 - m) $\sin x \cos y$, $x, y \geq 0$, $x + y \leq 2\pi$
 - n) $\sin x \sin y \sin(x + y)$, $x, y \geq 0$, $x + y \leq \pi$
 - o) $(x - 3y)e^y$, $|x| \leq 1$, $|y| \leq 1$
 - p) $xe^{-x^2-y^2}$, $x^2 + y^2 \leq R^2$
 - q) $(x + 2y)e^{-x^2-y^2}$, $(x, y) \in \mathbb{R}^2$
 - r) xy^2e^{-xy} , $x, y \geq 0$
 - s) $(2x - y)(1 + x^2 + y^2)^{-1}$, $(x, y) \in \mathbb{R}^2$
 - t) $(1 + 2x + 3y)(1 + x^2 + y^2)^{-1}$, $(x, y) \in \mathbb{R}^2$
 - u) $xy^2 + yz^2$, $x^2 + y^2 + z^2 \leq 1$
 - v) $xy + yz$, $x^2 + y^2 + z^2 \leq 1$
 - x) xy^2z^3 , $x^2 + y^2 + z^2 \leq 1$
 - y) $xyz e^{-x^2-2y^2-3z^2}$, $(x, y, z) \in \mathbb{R}^3$
 - z) $(x + y + z)e^{-x^2-2y^2-3z^2}$, $(x, y, z) \in \mathbb{R}^3$
 - å) $(x + 2y + 3z + 4u)e^{-x^2y^2z^2u^2}$, $(x, y, z, u) \in \mathbb{R}^4$

2. We want to determine the minimum and maximum values of the function $f(x, y, z) = xy + z^2$ on the spherical surface $S : x^2 + y^2 + z^2 = 1$. Solve the problem a) by parametrizing S with spherical surface coordinates, b) using Lagrange multipliers. c) What are the minimum and maximum values of f subject to $x^2 + y^2 + z^2 \leq 1$?
3. Using Lagrange multipliers, determine the smallest and largest values of the function

$$\text{a) } f(x, y, z) = x^2 + y^2 + z^2 \quad \text{b) } f(x, y, z) = xy$$

on the intersection curve of the sphere $S_1 : x^2 + y^2 + z^2 + 4x + 4y = 0$ and the cylinder $S_2 : x^2 + y^2 + 2x + 2y = 6$.

4. Solve the following constrained extremum problems using Lagrange multipliers.
- a) $x^2 + y^2 = \min!/\max!$ on the curve $S : x^2 + 8xy + 7y^2 = 45$
 - b) $x^2 + y^2 = \min!/\max!$ on the curve $S : x^4 + x^2y^2 + 5y^4 = 95$
 - c) $x^3y^5 = \max!$ subject to $x + y = 8$
 - d) $x + 2y - 3x = \min!/\max!$ on the spherical surface $x^2 + y^2 + z^2 = 1$
 - e) $x^2 + y^2 + z^2 = \min!$ subject to $xyz^2 = 2$
 - f) $xyz = \min!/\max!$ on the spherical surface $x^2 + y^2 + z^2 = 12$
 - g) $x = \min!/\max!$ on the curve $S : z = x + y$ & $x^2 + 2y^2 + 2z^2 = 8$
 - h) $x^2 + y^2 + z^2 = \min!/\max!$ on the curve $S : z^2 = x^2 + y^2$ & $x - 2z = 3$
 - i) $z = \min!/\max!$ on the curve $S : x^2 + y^2 = 8$ & $x + y + z = 1$
 - j) $x^3 + y^3 + z^3 = \min!/\max!$ on the curve $S : x + y + z = 3$ & $x^2 + y^2 + z^2 = 27$
 - k) $\mathbf{a}^T \mathbf{x} = \min!/\max!$ subject to $|\mathbf{x}| = 1$ ($\mathbf{a}, \mathbf{x} \in \mathbb{R}^n$)
 - l) $x_1^{2m} + \dots + x_n^{2m} = \min!$ subject to $x_1 + \dots + x_n = a \neq 0$ ($m \in \mathbb{N}$)
5. Solve using the Lagrange multiplier:
- a) What is the shortest distance from the origin to the curve $S : x^2 + 8xy + 7y^2 = 45$?
 - b) Which point Q of the surface $S : xy + 2yz + 3xz = 0$ is closest to the point $P = (-2, 2, -2)$?
6. One vertex of a tetrahedron is at the origin and the other three vertices are in the directions $\vec{i}$, $\vec{i} - \vec{j} + \vec{k}$, and $-\vec{i} + \vec{j} + \vec{k}$ from the origin. In addition, the distance from the origin to the opposite face of the tetrahedron $= 3$. What is the smallest volume of the tetrahedron under these conditions?
7. a) Show that among rectangular boxes (rectangular parallelepipeds) having the same total surface area of the faces, the cube has the largest volume.

b) A lidless box in the shape of a rectangular parallelepiped is made from two different materials so that the bottom material is (per unit area) five times more expensive than the material of the walls of the box. How should the dimensions of the box be chosen when one wants to minimize the material costs subject to the box's volume $= a^3$?

8. a) An ellipsoid-shaped gemstone

$$A: \frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} \leq 1$$

is to be cut into the shape of a rectangular parallelepiped. How should the dimensions of the parallelepiped be chosen so that as little of the stone as possible is wasted?

b) Show that if α, β and γ are the angles of a triangle, then

$$\sin \frac{\alpha}{2} \sin \frac{\beta}{2} \sin \frac{\gamma}{2} \leq \frac{1}{8}.$$

9. Compute (if necessary by numerical means) the smallest and largest values of the function $f(x, y) = (x - y^2)e^{xy}$ on the set a) $A: x^2 + y^2 \leq 4$, b) $A: x^4 + y^4 \leq 16$.

10. (*) Prove the statement: The *geometric mean* of the numbers $a_k \geq 0$, $k = 1 \dots n$ is at most equal to the *arithmetic mean*, i.e.

$$\sqrt[n]{a_1 \cdots a_n} \leq \frac{1}{n}(a_1 + \dots + a_n).$$

Hint: Set up a suitable constraint!

11. (*) Find the extremum points of the function $f(x, y, z) = (x + 2y)e^{xz} + xyz + y^3$ on the spherical surface $S: (x - 3)^2 + (y + 3)^2 + (z + 4)^2 = 16$. Method: Lagrange multipliers and Newton!

12. (*) In the general method of Lagrange multipliers, one seeks a critical point $(\mathbf{c}, \boldsymbol{\mu})$ of the function $F(\mathbf{x}, \boldsymbol{\lambda}) = f(\mathbf{x}) + \sum_{i=1}^m \lambda_i g_i(\mathbf{x})$, $\mathbf{x} \in \mathbb{R}^n$, $m < n$. Assume that in some neighborhood $U_\delta(\mathbf{c})$ of $\mathbf{c}$ the vectors $\nabla g_i(\mathbf{x})$, $i = 1 \dots m$ are linearly independent and that an initial guess $\mathbf{x}_0 \in U_\delta(\mathbf{c})$ is available. Show that if the initial guess $\boldsymbol{\lambda}_0$ is determined by means of $\mathbf{x}_0$ by computing

$$|\nabla f(\mathbf{x}_0) - \sum_{i=1}^m \lambda_i \nabla g_i(\mathbf{x}_0)|^2 = \min! \Rightarrow (\lambda_i) = \boldsymbol{\lambda}_0,$$

($|\cdot|$ = the Euclidean norm of $\mathbb{R}^n$), then: $\mathbf{x}_0 = \mathbf{c} \Rightarrow \boldsymbol{\lambda}_0 = \boldsymbol{\mu}$. To what formula for $\boldsymbol{\lambda}_0$ does this procedure lead in the case $m = 1$? Apply the procedure to the constrained extremum problem

$$\begin{aligned} f(x, y, z) &= x^2 + y^2 + z^2 = \min! \\ \text{subject to } &x^3y + xy^3 + 2 = 0, \quad x^3y + y^3z + xz^3 + 1 = 0, \end{aligned}$$

when the initial guess is $(x_0, y_0, z_0) = (1, -1, 1)$. Determine the critical point of F approximately with one step of Newton's iteration.

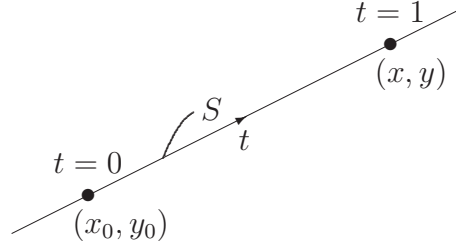
XI.9 Multivariable Taylor polynomials

Consider a two-variable function $f = f(x, y)$ on the line segment S joining the points (x_0, y_0) and (x, y) . On this segment f can be interpreted as a one-variable function

$$g(t) = f(x_0 + h_x t, y_0 + h_y t), \quad t \in [0, 1],$$

where we have set

$$h_x = x - x_0, \quad h_y = y - y_0.$$



Assume that the partial derivatives ∂^α of the function f exist up to order $|\alpha| = n + 1$ and are continuous on the segment S . Then it can be deduced from the chain rules of partial differentiation that the function g is $n + 1$ times continuously differentiable on the interval $[0, 1]$, so that by Taylor's theorem, for every $t \in (0, 1]$,

$$g(t) = \sum_{k=0}^n \frac{1}{k!} g^{(k)}(0) t^k + \frac{1}{(n+1)!} g^{(n+1)}(\xi) t^{n+1},$$

where $\xi \in (0, t)$. Choosing in particular $t = 1$, the result is

$$f(x, y) = g(1) = \sum_{k=0}^n \frac{1}{k!} g^{(k)}(0) + \frac{1}{(n+1)!} g^{(n+1)}(\xi). \quad (\star)$$

Here the derivative $g'(t)$ can be computed by the chain rule in the form

$$\begin{aligned} g'(t) &= h_x f_x(x_0 + h_x t, y_0 + h_y t) + h_y f_y(x_0 + h_x t, y_0 + h_y t) \\ &= [(h_x \partial_x + h_y \partial_y) f](x_0 + h_x t, y_0 + h_y t). \end{aligned}$$

By applying this rule again, one sees that

$$g^{(k)}(t) = [(h_x \partial_x + h_y \partial_y)^k f](x_0 + h_x t, y_0 + h_y t).$$

Here h_x and h_y are, during differentiation, treated as constants (since the differentiation is with respect to the variable t). By this result, the expression $(\star)$ for $f(x, y)$ can be interpreted as the *two-variable Taylor formula*

$$\begin{aligned} f(x, y) &= T_n(x, y, x_0, y_0) + R_n(x, y), \quad \text{where} \\ T_n(x, y, x_0, y_0) &= \sum_{k=0}^n \frac{1}{k!} [(h_x \partial_x + h_y \partial_y)^k f](x_0, y_0), \\ R_n(x, y) &= \frac{1}{(n+1)!} [(h_x \partial_x + h_y \partial_y)^{n+1} f](x(\xi), y(\xi)), \\ h_x &= x - x_0, \quad h_y = y - y_0, \\ x(\xi) &= x_0 + h_x \xi, \quad y(\xi) = y_0 + h_y \xi, \quad \xi \in (0, 1). \end{aligned}$$

Here $T_n(x, y, x_0, y_0)$ is the *Taylor polynomial* of f of degree n at the point (x_0, y_0) and R_n is the *remainder term*. Noting that (by the binomial formula, cf. Section XI.2)

$$(h_x \partial_x + h_y \partial_y)^k f = \sum_{l=0}^k \binom{k}{l} h_x^{k-l} h_y^l \partial_x^{k-l} \partial_y^l f, \quad \binom{k}{l} = \frac{k!}{(k-l)! l!},$$

one obtains for the polynomial $T_n(x, y, x_0, y_0)$ a more concrete expression

$$T_n(x, y, x_0, y_0) = \sum_{k=0}^n \sum_{l=0}^k \frac{1}{(k-l)! l!} \frac{\partial^k f}{\partial x^{k-l} \partial y^l}(x_0, y_0) (x - x_0)^{k-l} (y - y_0)^l.$$

From this and the easily verified differentiation formula

$$\left[\frac{\partial^{i+j}}{\partial x^i \partial y^j} (x - x_0)^m (y - y_0)^n \right] \bigg|_{\substack{x=x_0 \\ y=y_0}} = \begin{cases} m! n!, & \text{if } i = m \text{ and } j = n, \\ 0, & \text{otherwise} \end{cases}$$

one sees that the polynomial $p(x, y) = T_n(x, y, x_0, y_0)$ satisfies the conditions

$$\partial^\alpha p(x_0, y_0) = \partial^\alpha f(x_0, y_0), \quad |\alpha| \leq n.$$

The Taylor polynomial is also uniquely determined by these conditions, as in the one-variable case (cf. Section VII.4).

EXAMPLE 1 Determine the Taylor polynomial $T_2(x, y, 0, 0)$ of the function

$$f(x, y) = e^{-x} + xy + y^2 + \cos x \cos y.$$

Solution By differentiation one finds

$$\begin{aligned} f(0,0) &= 2, & f_x(0,0) &= -1, & f_y(0,0) &= 0, \\ f_{xx}(0,0) &= 0, & f_{xy}(0,0) &= f_{yx}(0,0) = f_{yy}(0,0) &= 1, \end{aligned}$$

so

$$\begin{aligned} T_2(x, y, 0, 0) &= 2 + (-1) \cdot x + 0 \cdot y + \frac{1}{2!0!} \cdot 0 \cdot x^2 + 2 \cdot \frac{1}{1!1!} \cdot 1 \cdot xy + \frac{1}{0!2!} \cdot 1 \cdot y^2 \\ &= 2 - x + xy + \frac{1}{2}y^2. \quad \square \end{aligned}$$

The Taylor formula thus holds provided that the partial derivatives of f are continuous up to order $n+1$ on the connecting segment between the point of consideration (x, y) and the point (x_0, y_0) . The Taylor formula is thus, even in the two-variable case, one-dimensional in its underlying idea. If one wants the formula to hold as the point of consideration moves within some set A in a neighborhood of the point (x_0, y_0) , then one must assume, in addition to sufficient regularity of f , that A is so-called *star-shaped* with respect to the point (x_0, y_0) . This means that from the point (x_0, y_0) there is a 'line of sight' to every point $(x, y) \in A$, i.e. the connecting segment between the points lies entirely in the set A . With this in mind, from the Taylor formula one can derive the following general polynomial approximation result for a two-variable function.

THEOREM XI.9.1 If the partial derivatives of the function $f = f(x, y)$ are continuous up to order $n+1$ on a compact set A that is star-shaped with respect to the point (x_0, y_0) , then for every $(x, y) \in A$

$$f(x, y) = T_n(x, y, x_0, y_0) + \mathcal{O}(h^{n+1}), \quad h = \sqrt{(x - x_0)^2 + (y - y_0)^2}.$$

If the partial derivatives of f are continuous on A up to order n , then

$$f(x, y) = T_n(x, y, x_0, y_0) + o(h^n), \quad (x, y) \in A.$$

Proof If the partial derivatives of f are continuous up to order $n+1$ on the compact set A , then they are also bounded (see Theorem [XI.1.11](#)):

$$|\partial^\alpha f(x, y)| \leq C_\alpha \quad \forall (x, y) \in A, \quad |\alpha| \leq n+1.$$

Choosing $C = \max_{|\alpha|=n+1} C_\alpha$, the following estimate for the remainder term of

the Taylor formula follows

$$\begin{aligned} |R_n(x, y)| &\leq \frac{C}{(n+1)!} \sum_{l=0}^{n+1} \binom{n+1}{l} |x - x_0|^{n+1-l} |y - y_0|^l \\ &= \frac{C}{(n+1)!} (|x - x_0| + |y - y_0|)^{n+1} \\ &\leq \frac{C(\sqrt{2})^{n+1}}{(n+1)!} h^{n+1} = \mathcal{O}(h^{n+1}), \quad (x, y) \in A. \end{aligned}$$

When regularity assumptions one degree weaker hold, one concludes correspondingly (cf. Theorem VII.4.4)

$$f(x, y) = T_{n-1}(x, y, x_0, y_0) + R_{n-1}(x, y) = T_n(x, y, x_0, y_0) + o(h^n). \quad \square$$

In the case $n = 1$, the latter statement of Theorem XI.9.1 is: If f , f_x , f_y are continuous in a (star-shaped) neighborhood of the point (x_0, y_0) , then

$$\begin{aligned} f(x, y) &= T_1(x, y, x_0, y_0) + o(h) \\ &= f(x_0, y_0) + f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0) + o(h). \end{aligned}$$

The result is already familiar, since under these assumptions f is differentiable at the point (x_0, y_0) (cf. Section XI.3). The latter statement of Theorem XI.9.1 can also be exploited when determining the Taylor polynomial of f : According to the statement, it suffices to find any polynomial $p(x, y)$ for which

$$f(x, y) - p(x, y) = o(h^n), \quad h = \sqrt{(x - x_0)^2 + (y - y_0)^2}.$$

Such a polynomial is necessarily unique — thus $p(x, y) = T_n(x, y, x_0, y_0)$ (cf. Proposition VII.4.5).

EXAMPLE 1 (continued) For the function of the example

$$\begin{aligned} f(x, y) &= 1 - x + \frac{1}{2}x^2 + \mathcal{O}(|x|^3) + xy + y^2 + (1 - \frac{1}{2}x^2)(1 - \frac{1}{2}y^2) + \mathcal{O}(x^4 + y^4) \\ &= 2 - x + xy + \frac{1}{2}y^2 + \mathcal{O}(|x|^3 + |y|^3) \\ &= 2 - x + xy + \frac{1}{2}y^2 + \mathcal{O}(h^3), \end{aligned}$$

so on this basis as well

$$T_2(x, y, 0, 0) = 2 - x + xy + \frac{1}{2}y^2. \quad \square$$

Taylor polynomials of several variables

The above considerations generalize easily to functions of three or more variables. The Taylor polynomial of a three-variable function $f = f(x, y, z)$ is defined as

$$T_n(x, y, z, x_0, y_0, z_0) = \sum_{k=0}^n \frac{1}{k!} (h_x \partial_x + h_y \partial_y + h_z \partial_z)^k f(x_0, y_0, z_0),$$

$$h_x = x - x_0, \quad h_y = y - y_0, \quad h_z = z - z_0.$$

More generally, if $f = f(\mathbf{x}) = f(x_1, \dots, x_d)$, then the Taylor polynomial of f of degree n at the point $\mathbf{a} = (a_1, \dots, a_d) \in \mathbb{R}^d$ is

$$T_n(\mathbf{x}, \mathbf{a}) = \sum_{k=0}^n \frac{1}{k!} \left[\left(\sum_{i=1}^d h_i \partial_i \right)^k f \right](\mathbf{a}), \quad h_i = x_i - a_i, \quad i = 1 \dots d.$$

Let it be mentioned without further justification that this expression can be expanded into the following form, which in appearance hardly differs from the one-variable situation (cf. also the case $d = 2$ above):

$$T_n(\mathbf{x}, \mathbf{a}) = \sum_{|\alpha| \leq n} \frac{1}{\alpha!} \partial^\alpha f(\mathbf{a}) (\mathbf{x} - \mathbf{a})^\alpha,$$

$$\alpha! = \prod_{i=1}^d \alpha_i!, \quad \mathbf{x}^\alpha = \prod_{i=1}^d x_i^{\alpha_i},$$

$$\alpha = (\alpha_1, \dots, \alpha_d), \quad \mathbf{x} = (x_1, \dots, x_d).$$

In the formula, the summation runs over all distinct multi-indices (i.e. ordered index sets) for which $|\alpha| = \alpha_1 + \dots + \alpha_d \leq n$.

The Taylor polynomial $p(\mathbf{x}) = T_n(\mathbf{x}, \mathbf{a})$ is also always uniquely determined as a polynomial of degree n satisfying the conditions

$$\partial^\alpha p(\mathbf{a}) = \partial^\alpha f(\mathbf{a}), \quad |\alpha| \leq n.$$

Insofar as f and the partial derivatives of f up to order n are continuous in a neighborhood of the point $\mathbf{a}$, it also suffices to find (by any means) a polynomial p of degree n for which

$$f(\mathbf{x}) = p(\mathbf{x}) + o(|\mathbf{x} - \mathbf{a}|^n).$$

Then $p(\mathbf{x}) = T_n(\mathbf{x}, \mathbf{a})$.

The polynomial $T_2(\mathbf{x}, \mathbf{a})$ – the Hessian matrix

Taylor polynomials of several variables are seldom computed beyond degree $n = 2$. By the above formula, the second-degree Taylor polynomial of f is

$$T_2(\mathbf{x}, \mathbf{a}) = f(\mathbf{a}) + \sum_{i=1}^d \partial_i f(\mathbf{a})(x_i - a_i) + \frac{1}{2} \sum_{i=1}^d \sum_{j=1}^d \partial_i \partial_j f(\mathbf{a})(x_i - a_i)(x_j - a_j).$$

This can be expressed in matrix form

$$T_2(\mathbf{x}, \mathbf{a}) = f(\mathbf{a}) + [\nabla f(\mathbf{a})]^T (\mathbf{x} - \mathbf{a}) + \frac{1}{2} (\mathbf{x} - \mathbf{a})^T \mathbf{H}(\mathbf{a}) (\mathbf{x} - \mathbf{a}),$$

where $\nabla f(\mathbf{a})$ is the gradient and $\mathbf{H}(\mathbf{a})$ is the so-called *Hessian matrix* of f , consisting of the second-order partial derivatives of f :

$$[\mathbf{H}(\mathbf{a})]_{ij} = \frac{\partial^2 f}{\partial x_i \partial x_j}(\mathbf{a}), \quad i, j = 1 \dots d.$$

EXAMPLE 2 Determine the Taylor polynomial $T_2(\mathbf{x}, \mathbf{0})$ of the function

$$f(x_1, \dots, x_d) = \prod_{i=1}^d e^{c_i x_i}.$$

Solution Since

$$f(\mathbf{0}) = 1, \quad \frac{\partial f}{\partial x_i}(\mathbf{0}) = c_i, \quad i = 1 \dots d, \quad \frac{\partial^2 f}{\partial x_i \partial x_j} = c_i c_j, \quad i, j = 1 \dots d,$$

so

$$\begin{aligned} T_2(\mathbf{x}, \mathbf{0}) &= 1 + \left(\sum_{i=1}^d x_i \frac{\partial}{\partial x_i} \right) f(\mathbf{0}) + \frac{1}{2} \left(\sum_{i=1}^d x_i \frac{\partial}{\partial x_i} \right)^2 f(\mathbf{0}) \\ &= 1 + \sum_{i=1}^d c_i x_i + \frac{1}{2} \sum_{i,j=1}^d c_i c_j x_i x_j. \end{aligned}$$

Denoting $\mathbf{c} = [c_1, \dots, c_d]^T$ and $\mathbf{H} = (c_i c_j) = \mathbf{c} \mathbf{c}^T$ (= the Hessian matrix of f at the origin), the result takes the form

$$\begin{aligned} T_2(\mathbf{x}, \mathbf{0}) &= 1 + \mathbf{c}^T \mathbf{x} + \frac{1}{2} \mathbf{x}^T \mathbf{H} \mathbf{x} \\ &= 1 + \mathbf{c}^T \mathbf{x} + \frac{1}{2} (\mathbf{c}^T \mathbf{x})^2. \end{aligned}$$

A shorter route to the same result is obtained by remembering to use the properties of the real function $\exp(x)$:

$$\begin{aligned} f(\mathbf{x}) &= \exp(\mathbf{c}^T \mathbf{x}) = 1 + \mathbf{c}^T \mathbf{x} + \frac{1}{2}(\mathbf{c}^T \mathbf{x})^2 + \mathcal{O}(|\mathbf{c}^T \mathbf{x}|^3) \\ &= 1 + \mathbf{c}^T \mathbf{x} + \frac{1}{2}(\mathbf{c}^T \mathbf{x})^2 + \mathcal{O}(|\mathbf{x}|^3). \quad \square \end{aligned}$$

EXERCISES

- Compute the Taylor polynomials of the following functions of the given degree n at the given point. When the opportunity arises, use one-variable Taylor polynomials as an aid.
 - $f(x, y) = 2x^4 - 5y^3 + 2xy^2$, $(0, 0)$, $n = 3$
 - $f(x, y) = \sqrt{x^3 y}$, $(1, 4)$, $n = 4$
 - $f(x, y) = \frac{1}{2 + x - 2y}$, $(2, 1)$, $n = 3$
 - $f(x, y) = y^2 \ln x$, $(1, 0)$, $n = 4$
 - $f(x, y) = \sin(xy)$, $(0, 0)$, $n \in \mathbb{N}$
 - $f(x, y) = e^{xy} \sin y$, $(0, 0)$, $n = 3$
 - $f(x, y) = \cos(x + \sin y)$, $(0, 0)$, $n = 4$
 - $f(x, y) = \int_0^{x+y^2-1} e^{-t^2} dt + \int_0^{\pi/2} \cos(xt^2 + yt) dt$, $(0, 1)$, $n = 2$
 - $x^2 + 2y^2 + 3z^2 = 6 \Leftrightarrow z = f(x, y)$, $(1, 1, 1)$, $n = 2$
 - $x + xy^2 + 2z - \sin z = 0 \Leftrightarrow z = f(x, y)$, $(0, 0, 0)$, $n = 2$
 - $f(x, y, z) = e^{xyz}$, $(0, 0, 0)$, $n \in \mathbb{N}$
 - $f(x, y, z) = \frac{1}{x + y + z}$, $(1, 0, 0)$, $n = 3$
 - $f(\mathbf{x}) = \cos(x_1 + \dots + x_n)$, $\mathbf{0}$, $n = 2$
 - $f(\mathbf{x}) = \frac{1}{x_1 + \dots + x_n}$, $(1, \dots, 1)$, $n = 2$
- Compute the partial derivative $f_{xxxxxyy}(0, 0)$ of the function $f(x, y) = \ln(1 + x^2 + y^2)$ by means of Taylor polynomials.

3. Estimate how large the error of the following approximations is at most in the disk $A : x^2 + y^2 \leq a^2$ ($a > 0$).

$$\begin{array}{ll} \text{a)} & e^{x+2y} \approx 1 + x + 2y \\ \text{b)} & e^{x+2y} \approx 1 + x + 2y + \frac{1}{2}x^2 + 2xy + 2y^2 \\ \text{b)} & \cos(xy) \approx 1 - \frac{1}{2}x^2y^2 \\ \text{d)} & 2xy + 2\cos(x-y) \approx 2 + x^2 + y^2 \end{array}$$

4. Determine the second-degree Taylor polynomial of the function and, by means of it, the Hessian matrix at the origin:

$$\begin{array}{lll} \text{a)} & x \sin(x+y) & \text{b)} \quad xy \cos(x-y) & \text{c)} \quad e^{x+y} + e^{x-y} \\ \text{d)} & \frac{1-x+x^2}{1+y-y^2} & \text{e)} \quad \cos(x+y-2z) & \text{f)} \quad \frac{e^{xy} + e^{yz} + e^{xz}}{1+x+y+z} \end{array}$$

5. The equation $x^y = y^x \Leftrightarrow y \ln x = x \ln y$ defines a curve, possibly several, in a neighborhood of the point (a, a) ($a > 0$). Investigate the matter by approximating the function $f(x, y) = y \ln x - x \ln y$ by its first-, second-, and third-degree Taylor polynomials at the point (a, a) .

Chapter XII

Eigenvalues and quadratic forms of a matrix

Matrices and matrix algebra have earlier (Chapter X) been studied mainly from the point of view of the theory and solution of systems of linear equations. In this chapter we open up an entirely new perspective on matrix computation and its applications, as the objects of study become certain special numbers and vectors associated with a matrix, called *eigenvalues* and *eigenvectors*.

In Section XII.1 we define the eigenvalues and eigenvectors of a matrix, that is, taken together, the *eigenpairs*, and we present the central claims of eigenvalue theory. It turns out that the eigenvalues and eigenvectors of a general real matrix can be complex. The eigenvalue problem for a symmetric matrix, by contrast, is solved entirely in $\mathbb{R}^n$, and it turns out that in this case the eigenvectors can also be used to form an orthonormal basis of $\mathbb{R}^n$. The matrix *diagonalization* based on this occupies a central place in Section XII.2, where the eigenvalue theory of a symmetric matrix is applied to the quadratic forms of $\mathbb{R}^n$. In this way, first of all, a problem left open in Section XI.8 is (at least partly) solved: Is a critical point of a function $f(\mathbf{x})$ a local minimum, a maximum, or neither? In connection with quadratic forms, Section XII.2 also solves the problems of the geometric classification of second-degree curves and second-degree surfaces by means of the eigenvalue theory of matrices.

In Section XII.3 it is shown that symmetric matrices and their eigenvalue theory are also keys to the concept of the *curvature of a surface*. In the final subsection we briefly introduce the concept of a *tensor*, as which, for example, the curvature of a surface can be understood. The tensor and its various manifestations lead us to reconsider also the concepts of vector and scalar.

XII.1 Eigenvalues of a matrix

Let us begin with a definition.

DEFINITION XII.1.1 If $\mathbf{A}$ is a square matrix of size $n \times n$, then a complex number $\lambda \in \mathbb{C}$ is an *eigenvalue* (Swedish *egenvärde*, German *Eigenwert*) of $\mathbf{A}$, and a vector $\mathbf{x} \in \mathbb{C}^n$, $\mathbf{x} \neq \mathbf{0}$, is an *eigenvector* associated with the eigenvalue λ , if

$$\mathbf{A}\mathbf{x} = \lambda\mathbf{x}.$$

In the definition, $\mathbf{A}$ may be either a real or a complex matrix. Thus the definition does not simplify to the form $\lambda \in \mathbb{R}$, $\mathbf{x} \in \mathbb{R}^n$ even in the case where $\mathbf{A}$ is real.

It is seen at once from the definition that eigenvectors are not unique: if $\mathbf{x}$ is an eigenvector, then so is $\alpha\mathbf{x}$ for every $\alpha \in \mathbb{C}$, $\alpha \neq 0$. The eigenvectors can therefore be *normalized* in an arbitrary way, for example by the condition $|\mathbf{x}| = 1$.

The equation $\mathbf{A}\mathbf{x} = \lambda\mathbf{x}$ defining the eigenvalues and eigenvectors of a matrix, that is, the *eigenpairs*, is called the *eigenvalue equation*. The computation of the eigenpairs begins by first determining from this equation all the eigenvalues. The set of eigenvalues, which is called the *spectrum* of $\mathbf{A}$, will henceforth be denoted by the symbol $\sigma(\mathbf{A})$:

$$\sigma(\mathbf{A}) = \{\text{eigenvalues of } \mathbf{A}\} \subset \mathbb{C}.$$

By means of the spectrum, one obtains yet another characterization of the regularity of a matrix:

$\mathbf{A} \text{ regular} \quad \Leftrightarrow \quad 0 \notin \sigma(\mathbf{A}).$

When the eigenvalue equation is written in the form

$$(\mathbf{A} - \lambda\mathbf{I})\mathbf{x} = \mathbf{0}$$

($\mathbf{I}$ = identity matrix), then by matrix algebra and the theory of determinants this has a nontrivial solution, that is, $\lambda \in \sigma(\mathbf{A})$, precisely when

$$\det(\mathbf{A} - \lambda\mathbf{I}) = 0.$$

This, then, is the *eigenvalue condition*. Since λ appears here only on the diagonal of $(\mathbf{A} - \lambda\mathbf{I})$, the minor-determinant rule (Theorem X.5.5) shows that the eigenvalue condition reduces to an equation of the form

$$p(\lambda) = 0,$$

where $p(\lambda)$ is a polynomial of degree n , more precisely

$$p(\lambda) = (-1)^n \lambda^n + \{\text{lower-degree terms}\}.$$

The eigenvalue condition $p(\lambda) = 0$ is called the *characteristic equation* of $\mathbf{A}$, and the polynomial $p(\lambda) = \det(\mathbf{A} - \lambda \mathbf{I})$ the *characteristic polynomial* of $\mathbf{A}$. The eigenvalues of $\mathbf{A}$ are thus the roots of the characteristic polynomial of $\mathbf{A}$, and consequently the spectrum $\sigma(\mathbf{A})$ of $\mathbf{A}$ consists of at least one and at most n complex numbers (the fundamental theorem of algebra). If $\mathbf{A}$ is a real matrix, then the characteristic polynomial $p(\lambda)$ has real coefficients, in which case the roots are either real or form conjugate pairs. If $\lambda \in \sigma(\mathbf{A})$ is a root of $p(\lambda)$ of multiplicity m , then the number $m_a(\lambda) = m$ is called the *algebraic multiplicity* of that eigenvalue. By the fundamental theorem of algebra,

$$\sum_{\lambda \in \sigma(\mathbf{A})} m_a(\lambda) = n.$$

If $m_a(\lambda) = 1$, one has a *simple* eigenvalue.

Once an eigenvalue $\lambda \in \sigma(\mathbf{A})$ has been found (possibly by numerical means), the corresponding eigenvectors are obtained from the eigenvalue equation $(\mathbf{A} - \lambda \mathbf{I})\mathbf{x} = \mathbf{0}$, which by the definition of an eigenvalue is a singular system of equations in $\mathbb{C}^n$. When this system is reduced by Gaussian elimination to its singular basic form, all the solutions are obtained. In general, if $\mathbf{x}_1 \in \mathbb{C}^n$ and $\mathbf{x}_2 \in \mathbb{C}^n$ are solutions of the eigenvalue equation, then so is $\alpha_1 \mathbf{x}_1 + \alpha_2 \mathbf{x}_2 \forall \alpha_1, \alpha_2 \in \mathbb{C}$, so the eigenvectors associated with a given eigenvalue form a subspace of $\mathbb{C}^n$. This subspace is called the *eigenspace* associated with that eigenvalue. If the dimension of the eigenspace is ν , that is, if to the same eigenvalue λ one can associate exactly ν linearly independent eigenvectors, then one says that the number ν (always $\nu \geq 1$) is the *geometric multiplicity* of that eigenvalue, and one writes

$$m_g(\lambda) = \nu.$$

To each eigenvalue $\lambda \in \sigma(\mathbf{A})$ there are thus associated two multiplicities, the algebraic multiplicity $m_a(\lambda)$ and the geometric multiplicity $m_g(\lambda)$, both of which are natural numbers. It can also be shown that one always has

$$m_g(\lambda) \leq m_a(\lambda), \quad \lambda \in \sigma(\mathbf{A}).$$

EXAMPLE 1 Determine the eigenvalues and eigenvectors of the matrix

$$\mathbf{A} = \begin{bmatrix} 2 & 0 & 4 \\ 0 & 6 & 0 \\ 4 & 0 & 2 \end{bmatrix}.$$

Solution The determinant condition for the eigenvalues is

$$\begin{aligned}\det(\mathbf{A} - \lambda \mathbf{I}) &= \begin{vmatrix} 2 - \lambda & 0 & 4 \\ 0 & 6 - \lambda & 0 \\ 4 & 0 & 2 - \lambda \end{vmatrix} \\ &= (6 - \lambda)(\lambda^2 - 4\lambda - 12) = -(6 - \lambda)^2(\lambda + 2) = 0\end{aligned}$$

(the determinant expanded along the second row or column), so the eigenvalues and their algebraic multiplicities are

$$\lambda_1 = 6, \quad m_a(\lambda_1) = 2, \quad \lambda_2 = -2, \quad m_a(\lambda_2) = 1.$$

The eigenvectors are obtained by solving the eigenvalue equation separately for each eigenvalue:

$$(\mathbf{A} - \lambda_1 \mathbf{I})\mathbf{x} = \mathbf{0} \Leftrightarrow \begin{bmatrix} -4 & 0 & 4 \\ 0 & 0 & 0 \\ 4 & 0 & -4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \mathbf{0}$$

$$\Leftrightarrow \begin{bmatrix} -4 & 0 & 4 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \mathbf{0}.$$

$$(\mathbf{A} - \lambda_2 \mathbf{I})\mathbf{x} = \mathbf{0} \Leftrightarrow \begin{bmatrix} 4 & 0 & 4 \\ 0 & 8 & 0 \\ 4 & 0 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \mathbf{0}$$

$$\Leftrightarrow \begin{bmatrix} 4 & 0 & 4 \\ 0 & 8 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \mathbf{0}.$$

One sees that to the eigenvalue λ_1 there are associated two linearly independent eigenvectors, for example

$$\mathbf{x}_1 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \quad \mathbf{x}_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix},$$

and to the eigenvalue λ_2 there is associated one linearly independent eigenvector, for example

$$\mathbf{x}_3 = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}.$$

The geometric multiplicities of the eigenvalues are therefore

$$m_g(\lambda_1) = 2, \quad m_g(\lambda_2) = 1. \quad \square$$

EXAMPLE 2 Compute the eigenvalues and eigenvectors of the matrix

$$\mathbf{A} = \begin{bmatrix} 1 & 0 & -2 \\ 0 & 1 & -2 \\ 0 & 1 & -1 \end{bmatrix}.$$

Solution The characteristic polynomial is

$$p(\lambda) = (1 - \lambda)(\lambda^2 + 1),$$

so the eigenvalues are

$$\lambda_1 = 1, \quad \lambda_2 = i, \quad \lambda_3 = -i.$$

Computing in the same way as in the previous example, the eigenvectors are found to be

$$\mathbf{x}_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \quad \mathbf{x}_2 = \begin{bmatrix} 1+i \\ 1+i \\ 1 \end{bmatrix}, \quad \mathbf{x}_3 = \begin{bmatrix} 1-i \\ 1-i \\ 1 \end{bmatrix}.$$

The multiplicities of the eigenvalues are

$$m_a(\lambda_i) = m_g(\lambda_i) = 1, \quad i = 1, 2, 3. \quad \square$$

EXAMPLE 3 Compute the eigenvalues and eigenvectors of the diagonal matrix $\mathbf{A} = \text{diag}(d_i, i = 1 \dots n)$.

Solution The characteristic equation is

$$\det(\mathbf{A} - \lambda \mathbf{I}) = \prod_{i=1}^n (d_i - \lambda) = 0.$$

If $d_i \neq d_j$ when $i \neq j$, then the eigenvalues and their algebraic multiplicities are

$$\lambda_i = d_i, \quad m_a(\lambda_i) = 1, \quad i = 1 \dots n.$$

If $d_i = d \forall i$ (in which case $\mathbf{A} = d\mathbf{I}$), then $\mathbf{A}$ has one eigenvalue of multiplicity n :

$$\lambda = d, \quad m_a(\lambda) = n.$$

If, more generally, the number d appears on the diagonal of $\mathbf{A}$ exactly m times, then the algebraic multiplicity of the eigenvalue $\lambda = d$ is $m_a(\lambda) = m$. The

geometric multiplicity is also $m_g(\lambda) = m$, since the eigenvalue equation shows that the standard basis vectors $\mathbf{e}_i$ of $\mathbb{R}^n$ are eigenvectors for every i for which $d_i = d$. In particular, if $d_i = d \ \forall i$, then every vector $\mathbf{x} \in \mathbb{R}^n$, $\mathbf{x} \neq \mathbf{0}$, is an eigenvector. $\square$

In the examples so far, the algebraic and geometric multiplicities of the eigenvalues have been equal. The following example shows that this is not always so.

EXAMPLE 4 Determine the eigenvalues and eigenvectors of $\mathbf{A}$ when

$$\mathbf{A} = \begin{bmatrix} a & 1 & & \\ & a & 1 & \\ & & a & \ddots \\ & & & \ddots & 1 \\ & & & & a \end{bmatrix}$$

that is

$$A = (a_{ij})_{i,j=1}^n, \quad a_{ij} = \begin{cases} a, & \text{when } i = j, \\ 1, & \text{when } j = i + 1, \\ 0, & \text{otherwise.} \end{cases}$$

Solution This is a triangular matrix, whose characteristic equation is

$$\det(\mathbf{A} - \lambda \mathbf{I}) = (a - \lambda)^n = 0.$$

Accordingly $\lambda = a$ is an eigenvalue of multiplicity n ($m_a(\lambda) = n$). When $\lambda = a$, the eigenvalue equations are

$$(\mathbf{A} - \lambda \mathbf{I})\mathbf{x} = \mathbf{0} \Leftrightarrow \begin{cases} x_2 = 0 \\ x_3 = 0 \\ \vdots \\ x_n = 0 \end{cases}$$

so $\mathbf{A}$ has only one linearly independent eigenvector $\mathbf{x} = [1, 0, \dots, 0]^T$. Thus the geometric multiplicity of the sole eigenvalue is $m_g(\lambda) = 1$. $\square$

Let us at this point also record a general result restricting the linear dependence of the eigenvectors of a matrix.

THEOREM XII.1.2 The eigenvectors of a matrix associated with distinct eigenvalues are mutually linearly independent.

Proof Let $\mathbf{A}\mathbf{a}_i = \lambda_i\mathbf{a}_i$, $i = 1 \dots m$, where $\mathbf{a}_i \neq \mathbf{0}$ and $\lambda_i \neq \lambda_j$ when $i \neq j$. Setting

$$\mathbf{B} = (\mathbf{A} - \lambda_1\mathbf{I})(\mathbf{A} - \lambda_2\mathbf{I}) \cdots (\mathbf{A} - \lambda_{m-1}\mathbf{I}) = \prod_{k=1}^{m-1} (\mathbf{A} - \lambda_k\mathbf{I}),$$

we have, by the assumptions,

$$\mathbf{B}\mathbf{a}_j = \prod_{k=1}^{m-1} (\lambda_j - \lambda_k) \mathbf{a}_j = \begin{cases} \mathbf{0}, & j = 1 \dots m-1, \\ \prod_{k=1}^{m-1} (\lambda_m - \lambda_k) \mathbf{a}_m, & j = m. \end{cases}$$

Suppose now that

$$\sum_{j=1}^m x_j \mathbf{a}_j = \mathbf{0}.$$

Then it follows that

$$\mathbf{0} = \mathbf{B} \sum_{j=1}^m x_j \mathbf{a}_j = \sum_{j=1}^m x_j \mathbf{B}\mathbf{a}_j = \prod_{k=1}^{m-1} (\lambda_m - \lambda_k) x_m \mathbf{a}_m,$$

so we must have $x_m = 0$. Thus

$$\sum_{j=1}^{m-1} x_j \mathbf{a}_j = \mathbf{0}.$$

If $m = 2$, this equation holds only when $x_1 = 0$, in which case we have concluded that $x_1 = x_2 = 0$. If $m > 2$, then by multiplying the equation on both sides by the matrix

$$\mathbf{B} = \prod_{k=1}^{m-2} (\mathbf{A} - \lambda_k\mathbf{I})$$

we conclude in the same way as above that $x_{m-1} = 0$, and so on. Thus the vectors $\mathbf{a}_1, \dots, \mathbf{a}_m$ are linearly independent:

$$\sum_{j=1}^m x_j \mathbf{a}_j = \mathbf{0} \quad \Rightarrow \quad x_m = x_{m-1} = \dots = x_1 = 0. \quad \square$$

Theorem [XII.1.2](#) has the following obvious corollary:

COROLLARY XII.1.3 If $\mathbf{A}$ is of size $n \times n$ and all the eigenvalues of $\mathbf{A}$ are simple, then the eigenvectors of $\mathbf{A}$ can be used to form a basis of $\mathbb{C}^n$.

Symmetric matrix

The eigenvalue theory of a real symmetric matrix can be distilled into the following beautiful theorem. The complete proof of the theorem is better suited to a more extensive presentation of linear algebra, so here we confine ourselves to proving only two of the (easier) partial claims contained in the theorem.

THEOREM XII.1.4 Let $\mathbf{A}$ be a real symmetric matrix of size $n \times n$. Then the eigenvalues of $\mathbf{A}$ are real, the algebraic and geometric multiplicities of the eigenvalues are equal, and the eigenvectors of $\mathbf{A}$ can be used to form an orthonormal basis of $\mathbb{R}^n$.

Proof (partial) If

$$\mathbf{Ax} = \lambda \mathbf{x}, \quad \lambda \in \mathbb{C}, \quad \mathbf{x} \in \mathbb{C}^n, \quad \mathbf{x} \neq \mathbf{0},$$

then

$$\langle \mathbf{Ax}, \mathbf{x} \rangle = \lambda \langle \mathbf{x}, \mathbf{x} \rangle = \lambda |\mathbf{x}|^2,$$

where $\langle \cdot, \cdot \rangle$ is the Euclidean inner product of $\mathbb{C}^n$ and $|\cdot|$ is the Euclidean norm:

$$\langle \mathbf{x}, \mathbf{y} \rangle = \sum_{i=1}^n x_i \bar{y}_i, \quad |\mathbf{x}| = \sqrt{\langle \mathbf{x}, \mathbf{x} \rangle}.$$

Taking the complex conjugate of both sides of the resulting equation, we obtain

$$\overline{\langle \mathbf{Ax}, \mathbf{x} \rangle} = \bar{\lambda} |\mathbf{x}|^2 = \bar{\lambda} |\mathbf{x}|^2.$$

But by the assumed reality and symmetry of $\mathbf{A}$ we have (cf. Chapter X.1)

$$\overline{\langle \mathbf{Ax}, \mathbf{x} \rangle} = \langle \mathbf{x}, \mathbf{Ax} \rangle = \langle \mathbf{Ax}, \mathbf{x} \rangle,$$

so $\lambda |\mathbf{x}|^2 = \bar{\lambda} |\mathbf{x}|^2$. Here $|\mathbf{x}|^2 \in \mathbb{R}_+$, so $\lambda = \bar{\lambda}$, that is, $\lambda \in \mathbb{R}$.

Let λ and μ be two distinct eigenvalues of $\mathbf{A}$ and $\mathbf{x}$ and $\mathbf{y}$ the corresponding eigenvectors:

$$\mathbf{Ax} = \lambda \mathbf{x} \quad \& \quad \mathbf{Ay} = \mu \mathbf{y}.$$

By what has already been shown, $\lambda, \mu \in \mathbb{R}$, so it follows that

$$\begin{aligned} \langle \mathbf{Ax}, \mathbf{y} \rangle &= \langle \lambda \mathbf{x}, \mathbf{y} \rangle = \lambda \langle \mathbf{x}, \mathbf{y} \rangle, \\ \langle \mathbf{x}, \mathbf{Ay} \rangle &= \langle \mathbf{x}, \mu \mathbf{y} \rangle = \mu \langle \mathbf{x}, \mathbf{y} \rangle. \end{aligned}$$

But since $\mathbf{A}$ is real and symmetric,

$$\langle \mathbf{Ax}, \mathbf{y} \rangle = \langle \mathbf{x}, \mathbf{Ay} \rangle,$$

so it follows that

$$(\lambda - \mu)\langle \mathbf{x}, \mathbf{y} \rangle = 0 \Rightarrow \langle \mathbf{x}, \mathbf{y} \rangle = 0.$$

Combining the conclusions so far, we have proved that the eigenvalues of $\mathbf{A}$ are real, and that the eigenvectors associated with distinct eigenvalues are orthogonal. The remainder of the proof is omitted. $\square$

From the result of Theorem XII.1.4 it follows that in the case of a real symmetric matrix, complex numbers are not needed in the eigenvectors either. Indeed, if $\mathbf{x} \in \mathbb{C}^n$ is an eigenvector and $\mathbf{x} = \mathbf{u} + i\mathbf{v}$, $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$, then by the reality of $\mathbf{A}$, λ , $\mathbf{u}$, and $\mathbf{v}$

$$\begin{aligned} \mathbf{A}\mathbf{x} = \lambda\mathbf{x} &\Leftrightarrow \mathbf{A}\mathbf{u} + i\mathbf{A}\mathbf{v} = \lambda\mathbf{u} + i\lambda\mathbf{v} \\ &\Leftrightarrow \mathbf{A}\mathbf{u} = \lambda\mathbf{u} \quad \& \quad \mathbf{A}\mathbf{v} = \lambda\mathbf{v}. \end{aligned}$$

The eigenvalue equation $\mathbf{A}\mathbf{x} = \lambda\mathbf{x}$ of a real symmetric matrix can therefore be solved entirely in $\mathbb{R}^n$.

EXAMPLE 5 The orthonormal eigenvectors of the symmetric matrix

$$\mathbf{A} = \begin{bmatrix} 2 & 0 & 4 \\ 0 & 6 & 0 \\ 4 & 0 & 2 \end{bmatrix}$$

are (cf. Example 1)

$$\mathbf{x}_1 = \frac{1}{\sqrt{2}}[1, 0, 1]^T, \quad \mathbf{x}_2 = [0, 1, 0]^T, \quad \mathbf{x}_3 = \frac{1}{\sqrt{2}}[-1, 0, 1]^T. \quad \square$$

Diagonalizable matrix

Let $\mathbf{A}$ be a real symmetric matrix of size $n \times n$ and let $\mathbf{C}$ be a real matrix whose columns are the orthonormal eigenvectors of $\mathbf{A}$:

$$\mathbf{C} = [\mathbf{c}_1 \dots \mathbf{c}_n], \quad \mathbf{A}\mathbf{c}_i = \lambda_i\mathbf{c}_i, \quad i = 1 \dots n, \quad \langle \mathbf{c}_i, \mathbf{c}_j \rangle = \delta_{ij}.$$

Here the eigenvalue equations $\mathbf{A}\mathbf{c}_i = \lambda_i\mathbf{c}_i$ can be collected into the matrix equation

$$[\mathbf{A}\mathbf{c}_1 \dots \mathbf{A}\mathbf{c}_n] = [\lambda_1\mathbf{c}_1 \dots \lambda_n\mathbf{c}_n] = [\mathbf{C}(\lambda_1\mathbf{e}_1) \dots \mathbf{C}(\lambda_n\mathbf{e}_n)]$$

that is (cf. Chapter X.1)

$$\mathbf{A}\mathbf{C} = \mathbf{C}\mathbf{\Lambda},$$

where $\mathbf{\Lambda} = [\lambda_1 \mathbf{e}_1 \dots \lambda_n \mathbf{e}_n] = \text{diag}(\lambda_i)$. Since $\mathbf{C}$ is orthogonal, $\mathbf{C}^{-1} = \mathbf{C}^T$, so it follows that

$$\boxed{\mathbf{C}^T \mathbf{A} \mathbf{C} = \mathbf{\Lambda} \quad \Leftrightarrow \quad \mathbf{A} = \mathbf{C} \mathbf{\Lambda} \mathbf{C}^T.}$$

In this result, then, the columns of $\mathbf{C}$ are the orthonormal eigenvectors of $\mathbf{A}$ in some order and $\mathbf{\Lambda}$ is a diagonal matrix whose diagonal contains the eigenvalues of $\mathbf{A}$ in the order determined by $\mathbf{C}$, such that $\mathbf{A} \mathbf{c}_i = \lambda_i \mathbf{c}_i$, $\lambda_i = [\mathbf{\Lambda}]_{ii}$. Let us give a somewhat more general definition at this point.

DEFINITION XII.1.5 (Similarity transformation) If $\mathbf{A}$ and $\mathbf{B}$ are square matrices of the same size, and there exists a regular matrix $\mathbf{C}$ such that $\mathbf{B} = \mathbf{C}^{-1} \mathbf{A} \mathbf{C}$, then one says that $\mathbf{A}$ and $\mathbf{B}$ are *similar*, and the map $\mathbf{A} \mapsto \mathbf{B}$ is called a *similarity transformation*. The similarity transformation is *orthogonal* if $\mathbf{C}$ is an orthogonal matrix. If $\mathbf{A}$ is similar to a diagonal matrix, then one says that $\mathbf{A}$ is *diagonalizable*.

A real symmetric matrix is thus, by definition, diagonalizable, and in this case the matrix $\mathbf{C}$ can also be chosen so that the diagonalizing similarity transformation is orthogonal. More generally, one can choose any n linearly independent eigenvectors as the columns of $\mathbf{C}$. The existence of such a system of eigenvectors is also a necessary condition for a matrix to be diagonalizable:

THEOREM XII.1.6 A matrix $\mathbf{A}$ of size $n \times n$ is diagonalizable precisely when $\mathbf{A}$ has n linearly independent eigenvectors. Then, if the columns of the matrix $\mathbf{C}$ are any linearly independent eigenvectors of $\mathbf{A}$ and the diagonal entries of the diagonal matrix $\mathbf{\Lambda}$ are the corresponding eigenvalues, then $\mathbf{A} = \mathbf{C} \mathbf{\Lambda} \mathbf{C}^{-1}$.

Proof If $\mathbf{A}$ has linearly independent eigenvectors $\mathbf{c}_i$, $i = 1 \dots n$, then forming the matrix $\mathbf{C} = [\mathbf{c}_1 \dots \mathbf{c}_n]$ with these as columns, one verifies as above that $\mathbf{A} \mathbf{C} = \mathbf{C} \mathbf{\Lambda} \Leftrightarrow \mathbf{A} = \mathbf{C} \mathbf{\Lambda} \mathbf{C}^{-1}$, where $\mathbf{\Lambda} = \text{diag}(\lambda_i)$ and λ_i = the eigenvalue associated with the eigenvector $\mathbf{c}_i$. This proves the part $\boxed{\Leftarrow}$ of the claim. To prove the part $\boxed{\Rightarrow}$, let $\mathbf{A} = \mathbf{C}^{-1} \mathbf{D} \mathbf{C}$, where $\mathbf{C}$ is regular and $\mathbf{D} = \text{diag}(d_i)$ is a diagonal matrix. Then

$$\mathbf{D} \mathbf{e}_i = d_i \mathbf{e}_i \quad \Leftrightarrow \quad (\mathbf{C}^{-1} \mathbf{D} \mathbf{C})(\mathbf{C}^{-1} \mathbf{e}_i) = d_i \mathbf{C}^{-1} \mathbf{e}_i, \quad i = 1 \dots n,$$

so the matrix $\mathbf{A} = \mathbf{C}^{-1} \mathbf{D} \mathbf{C}$ has eigenvectors $\mathbf{c}_i = \mathbf{C}^{-1} \mathbf{e}_i$ and corresponding eigenvalues d_i . The vectors $\mathbf{c}_i$ are linearly independent, since

$$\sum_{i=1}^n x_i \mathbf{c}_i = \mathbf{0} \quad \Leftrightarrow \quad \mathbf{C} \sum_{i=1}^n x_i \mathbf{c}_i = \sum_{i=1}^n x_i \mathbf{e}_i = \mathbf{0} \quad \Leftrightarrow \quad x_i = 0, \quad i = 1 \dots n.$$

Thus $\{d_i, i = 1 \dots n\} \subset \sigma(\mathbf{A})$ and $\mathbf{A}$ has n linearly independent eigenvectors corresponding to these eigenvalues. Then we must also have $\{d_i, i = 1 \dots n\} = \sigma(\mathbf{A})$, since otherwise this would lead to a contradiction with Theorem XII.1.2. $\square$

EXAMPLE 6 The matrix

$$\mathbf{A} = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$$

has only one eigenvector (Example 4), so $\mathbf{A}$ is not diagonalizable (Theorem XII.1.6). $\square$

EXAMPLE 7 Define a diagonalizing transformation $\mathbf{A} \mapsto \mathbf{D} = \mathbf{C}^{-1}\mathbf{A}\mathbf{C}$ for the matrix

$$\mathbf{A} = \begin{bmatrix} 2 & 0 & 4 \\ 0 & 6 & 0 \\ 4 & 0 & 2 \end{bmatrix}.$$

Solution is obtained directly from Examples 1 and 5 of the previous section:

$$\begin{bmatrix} 6 & 0 & 0 \\ 0 & 6 & 0 \\ 0 & 0 & -2 \end{bmatrix} = \begin{bmatrix} \frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \\ 0 & 1 & 0 \\ -\frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} 2 & 0 & 4 \\ 0 & 6 & 0 \\ 4 & 0 & 2 \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{2}} & 0 & -\frac{1}{\sqrt{2}} \\ 0 & 1 & 0 \\ \frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \end{bmatrix}.$$

Here an orthogonal transformation has been chosen. It is also possible to choose as the columns of $\mathbf{C}$, for example, the vectors $\mathbf{c}_1 = [1, 1, 1]^T$, $\mathbf{c}_2 = [1, 0, -1]^T$ and $\mathbf{c}_3 = [0, 1, 0]^T$, since these too are linearly independent eigenvectors of $\mathbf{A}$ (corresponding to the eigenvalues $\lambda_1 = 6$, $\lambda_2 = -2$ and $\lambda_3 = 6$, see Example 1 of the previous section). With this choice, the required transformation is

$$\begin{bmatrix} 6 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & 6 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 1 & -1 & 1 \end{bmatrix}^{-1} \begin{bmatrix} 2 & 0 & 4 \\ 0 & 6 & 0 \\ 4 & 0 & 2 \end{bmatrix} \begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 1 & -1 & 1 \end{bmatrix} \quad \square$$

EXERCISES

1. Prove:

- If $\lambda \in \sigma(\mathbf{A})$, then $\lambda^p \in \sigma(\mathbf{A}^p) \forall p \in \mathbb{N}$.
- If $\mathbf{A}$ is regular and $\lambda \in \sigma(\mathbf{A})$, then $\lambda^{-1} \in \sigma(\mathbf{A}^{-1})$.
- If $\mathbf{A}$ is real, then $\sigma(\mathbf{A}^T) = \sigma(\mathbf{A})$.
- If $1 \in \sigma(\mathbf{A})$ or $2 \in \sigma(\mathbf{A})$, then $\mathbf{B} = 2\mathbf{I} - 3\mathbf{A} + \mathbf{A}^2$ is singular.

2. Determine the eigenvalues and eigenvectors of the following matrices, as well as the algebraic and geometric multiplicities of the eigenvalues.

$$\begin{array}{lll} \text{a)} \begin{bmatrix} 1 & -2 \\ -2 & 1 \end{bmatrix} & \text{b)} \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} & \text{c)} \begin{bmatrix} 4 & 4 \\ -1 & 0 \end{bmatrix} \\ \text{d)} \begin{bmatrix} 3 & 1 & 1 \\ 1 & 3 & 1 \\ 1 & 1 & 3 \end{bmatrix} & \text{e)} \begin{bmatrix} -2 & 1 & 1 \\ 0 & 0 & 4 \\ 0 & -1 & -4 \end{bmatrix} & \text{f)} \begin{bmatrix} 1 & 2 & 1 \\ 0 & 2 & -2 \\ 1 & 0 & 3 \end{bmatrix} \\ \text{g)} \begin{bmatrix} 2 & 2 & 3 \\ 4 & 1 & 0 \\ 1 & -2 & 0 \end{bmatrix} & \text{h)} \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 2 & 0 & 0 \\ 9 & 0 & 3 & 0 \\ 1 & 0 & 0 & 4 \end{bmatrix} & \text{i)} \begin{bmatrix} 1 & 0 & 0 & 4 \\ 0 & 1 & 3 & 0 \\ 0 & 2 & 1 & 0 \\ 1 & 0 & 0 & 1 \end{bmatrix} \end{array}$$

3. Let

$$\mathbf{A} = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 2 & 0 \\ 2 & 1 & -1 \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} 1 & t & t \\ t & 1 & t \\ t & t & 1 \end{bmatrix}, \quad \mathbf{C} = \begin{bmatrix} \alpha & 1 & 1 & \beta \\ 4 & \alpha & \beta & 2 \\ 4 & \beta & \alpha & 2 \\ \beta & 3 & 3 & \alpha \end{bmatrix}.$$

- a) Eigenvectors of the matrix $\mathbf{A}$ are $[1, 1, 1]^T$, $[1, 0, 1]^T$ and $[0, 0, 1]^T$. Using this information, compute $\mathbf{A}^{11}\mathbf{x}$ when $\mathbf{x} = [2, 1, 1]^T$.
- b) The eigenvalues of the matrix $\mathbf{B}$ are real functions $t \mapsto \lambda_i(t)$. Sketch their graphs!
- c) The matrix $\mathbf{C}$ has an eigenvalue λ and a corresponding eigenvector $[0, 1, -1, 0]^T$. What condition does this imply for the numbers α, β and λ ?
4. a) Show that if a square matrix satisfies $\mathbf{A}^2 = \mathbf{A}$, then $\sigma(\mathbf{A}) \subset \{0, 1\}$. Show by an example that the alternatives $\sigma(\mathbf{A}) = \{0\}$, $\sigma(\mathbf{A}) = \{1\}$ and $\sigma(\mathbf{A}) = \{0, 1\}$ are all possible.
- b) The linear map $\mathbf{x} \mapsto \mathbf{A}\mathbf{x}$, where $\mathbf{A}$ is of size 3×3 , represents the projection onto the plane $T : x + 3y - 2z = 0$ in the direction of the line $S : x = y = z$. Show that $\sigma(\mathbf{A}) = \{0, 1\}$ and that $\mathbf{A}$ has three linearly independent eigenvectors. What are the multiplicities of the eigenvalues? Does $\mathbf{A}^2 = \mathbf{A}$ also hold?
- c) Let $\mathbf{A} = \mathbf{a}\mathbf{a}^T$, where $\mathbf{a} \in \mathbb{R}^n$, $n \geq 2$ and $|\mathbf{a}| = 1$. Show that $\sigma(\mathbf{A}) = \{0, 1\}$. What are the geometric multiplicities of the eigenvalues?
5. A matrix of size 3×3 has the property: the eigenvalues of the matrix are $\lambda_1 = -1$, $\lambda_2 = 1$, $\lambda_3 = 2$ and the corresponding eigenvectors are

$[1, 1, 1]^T$, $[1, 1, -1]^T$ and $[0, 1, 2]^T$. How many distinct matrices satisfying these conditions exist? Determine one!

6. Investigate whether the following matrices $\mathbf{A}$ are diagonalizable. In the affirmative case, compute for the matrix some (real or complex) product decomposition of the form $\mathbf{A} = \mathbf{C}\mathbf{D}\mathbf{C}^{-1}$, where $\mathbf{D}$ is diagonal.

$$\begin{array}{llll} \text{a)} \begin{bmatrix} 1 & 0 \\ 4 & -3 \end{bmatrix} & \text{b)} \begin{bmatrix} 3 & 1 \\ 5 & -1 \end{bmatrix} & \text{c)} \begin{bmatrix} -3 & 2 \\ -2 & 1 \end{bmatrix} & \text{d)} \begin{bmatrix} 1 & 2 \\ -4 & -3 \end{bmatrix} \\ \text{e)} \begin{bmatrix} 1 & 1 & 0 \\ 0 & 2 & 0 \\ -2 & 5 & -1 \end{bmatrix} & \text{f)} \begin{bmatrix} -2 & -1 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 0 \end{bmatrix} & \text{g)} \begin{bmatrix} 0 & 2 & 0 \\ -2 & 2 & 2 \\ 1 & 1 & 0 \end{bmatrix} & \\ \text{h)} \begin{bmatrix} -1 & -1 & 0 \\ 1 & -2 & 3 \\ 1 & -2 & -6 \end{bmatrix} & \text{i)} \begin{bmatrix} 1 & 0 & 0 & 9 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 \end{bmatrix} & \text{j)} \begin{bmatrix} 1 & 0 & 0 & 3 \\ 0 & 1 & 3 & 0 \\ 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 \end{bmatrix} & \end{array}$$

7. a) Show that similarity is an equivalence relation among square matrices of the same size. What is the equivalence class that contains the identity matrix $\mathbf{I}$?
 b) Let $\mathbf{A}$ and $\mathbf{B}$ be the matrices of a linear map $A : \mathbb{R}^n \rightarrow \mathbb{R}^n$ in two different bases of the space $\mathbb{R}^n$. Show that $\mathbf{A}$ and $\mathbf{B}$ are similar.
8. Determine the eigenvalues of the following matrices and form from the eigenvectors an orthonormal, positively oriented basis of $\mathbb{R}^2$, $\mathbb{R}^3$, or $\mathbb{R}^4$.

$$\begin{array}{llll} \text{a)} \begin{bmatrix} 0 & 2 \\ 2 & 0 \end{bmatrix} & \text{b)} \begin{bmatrix} 1 & 1 \\ 2 & 2 \end{bmatrix} & \text{c)} \begin{bmatrix} 5 & 2 \\ 2 & 8 \end{bmatrix} & \text{d)} \begin{bmatrix} 0 & -2 \\ -2 & 1 \end{bmatrix} \\ \text{e)} \begin{bmatrix} 0 & -1 & -1 \\ -1 & 0 & 1 \\ -1 & 1 & 0 \end{bmatrix} & \text{f)} \begin{bmatrix} 4 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{bmatrix} & \text{g)} \begin{bmatrix} 1 & 2 & 0 \\ 2 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix} & \\ \text{h)} \begin{bmatrix} 3 & 1 & 1 \\ 1 & 3 & 1 \\ 1 & 1 & 3 \end{bmatrix} & \text{i)} \begin{bmatrix} 2 & 0 & 0 & 2 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 2 & 0 & 0 & 1 \end{bmatrix} & \text{j)} \begin{bmatrix} 1 & 0 & 0 & 2 \\ 0 & 1 & 2 & 0 \\ 0 & 2 & 1 & 0 \\ 2 & 0 & 0 & 1 \end{bmatrix} & \end{array}$$

9. (*) By the *Cayley–Hamilton theorem*, a square matrix satisfies its own characteristic equation, i.e. if $p(\lambda)$ is the characteristic polynomial of the matrix,

then $p(\mathbf{A}) = \mathbf{0}$. a) Prove the theorem in the case where $\mathbf{A}$ is diagonalizable. b) Show that the theorem is also true for the non-diagonalizable matrix of Example [4](#)

XII.2 Classification of quadratic forms

A general *quadratic form* of $\mathbb{R}^n$ is a function $f(\mathbf{x}) = \mathbf{x}^T \mathbf{A} \mathbf{x}$, $\mathbf{x} \in \mathbb{R}^n$, where $\mathbf{A}$ is a real symmetric matrix of size $n \times n$ (see Example XI.2: 5). In this section we consider the classification of a quadratic form f according to the signs of the values that f takes away from the origin. In the general case, the solution of the classification problem rests on the eigenvalue theory of a symmetric matrix presented in the previous section and on the matrix diagonalization based on it (Theorems XII.1.4 and XII.1.6). This solution also yields the *principal-axis coordinate system* of the quadratic form, in which the quadratic form (i.e. the matrix $\mathbf{A}$) is diagonalized.

In what follows we first classify quadratic forms and solve the classification problem, restricting to the nontrivial cases $\mathbf{A} \neq \mathbf{0}$ and $n \geq 2$. After this we consider two applications of the theory of quadratic forms: the classification of the critical point of a function of n variables as a possible local extremum (Application #1) and the geometric classification of second-degree curves and surfaces (Application #2).

If $\mathbf{x} \in \mathbb{R}^n$, $\mathbf{x} \neq \mathbf{0}$ and we write $\mathbf{e} = \mathbf{x}/|\mathbf{x}|$, then for the quadratic form $f(\mathbf{x}) = \mathbf{x}^T \mathbf{A} \mathbf{x}$ we have

$$f(\mathbf{x}) = f(|\mathbf{x}|\mathbf{e}) = |\mathbf{x}|^2 \mathbf{e}^T \mathbf{A} \mathbf{e} = |\mathbf{x}|^2 f(\mathbf{e}). \quad (\star)$$

Accordingly, when investigating the signs of the values of the quadratic form it suffices to determine the range of f on the unit sphere of $\mathbb{R}^n$

$$B = \{\mathbf{x} \in \mathbb{R}^n \mid |\mathbf{x}| = 1\}.$$

According to the range $f(B)$, the quadratic form is classified as follows.

DEFINITION XII.2.1 The quadratic form $f(\mathbf{x}) = \mathbf{x}^T \mathbf{A} \mathbf{x}$ is

- *positive definite*, if $f(\mathbf{x}) > 0 \quad \forall \mathbf{x} \in B$,
- *negative definite*, if $f(\mathbf{x}) < 0 \quad \forall \mathbf{x} \in B$,
- *positive semidefinite* (semidefinite), if $f(\mathbf{x}) \geq 0 \quad \forall \mathbf{x} \in B$ and $f(\mathbf{x}) = 0$ for some $\mathbf{x} \in B$,
- *negative semidefinite*, if $f(\mathbf{x}) \leq 0 \quad \forall \mathbf{x} \in B$ and $f(\mathbf{x}) = 0$ for some $\mathbf{x} \in B$,
- *indefinite* (non-definite), if $f(\mathbf{x}) > 0$ and $f(\mathbf{y}) < 0$ for some $\mathbf{x}, \mathbf{y} \in B$.

In the simplest case, that of two variables, the definiteness problem of a quadratic form is easily solved. Writing $\mathbf{x} = (x, y)$ and $\mathbf{A} = (a_{ij})$, the general quadratic form of two variables is a function

$$f(x, y) = a_{11}x^2 + 2a_{12}xy + a_{22}y^2, \quad a_{ij} \in \mathbb{R}.$$

The type of definiteness can be determined, for example, by directly computing the range of f on the unit circle $B = \{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 = 1\}$. This is done by parametrizing B in the form $x = \cos t$, $y = \sin t$, $t \in [0, 2\pi]$, whereupon the object of study becomes the function

$$f(\cos t, \sin t) = F(t) = a_{11} \cos^2 t + 2a_{12} \cos t \sin t + a_{22} \sin^2 t.$$

The range $[F_{\min}, F_{\max}]$ of F on the interval $[0, 2\pi]$ determines the type of definiteness of f .

A second method is to determine the zeros of f along the lines through the origin $S_t : y = tx$, $t \in \mathbb{R}$ and (separately) along the line $S_y : x = 0$:

$$\begin{aligned} y = tx & : f(x, tx) = (a_{11} + 2a_{12}t + a_{22}t^2)x^2 = F(t)x^2 \\ x = 0 & : f(0, y) = a_{22}y^2. \end{aligned}$$

According to whether $f(x, y) = 0$ only at the origin, along one line, or along two lines intersecting at the origin, one can conclude that f is respectively definite, semidefinite, or indefinite.

EXAMPLE 1 Classify the quadratic form $f(x, y) = x^2 + axy + 4y^2$ when
a) $a = 3$, b) $a = 4$, c) $a = 5$.

Solution

- a) $f(\cos t, \sin t) = \cos^2 t + 3 \cos t \sin t + \sin^2 t$
 $= (\cos t + \frac{3}{2} \sin t)^2 + \frac{1}{4} \sin^2 t > 0 \quad \forall t \in [0, 2\pi]$
 $\Rightarrow f$ is positive definite.
- b) $f(x, y) = (x + 2y)^2$
 $\Rightarrow f$ is positive semidefinite.
- c) $f(0, y) = 4y^2 > 0$, when $y \neq 0$,
 $f(x, tx) = (1 + 5t + 4t^2)x^2 = 0$, when $t = -1$ or $t = -1/4$
 $\Rightarrow f$ is indefinite. $\square$

Diagonalization of a quadratic form

In the case of more than two variables, the classification problem of a quadratic form is most naturally solved by means of the eigenvalue theory of a symmetric matrix: one carries out the diagonalization of the matrix $\mathbf{A}$

$$\mathbf{D} = \mathbf{C}^T \mathbf{A} \mathbf{C} \Leftrightarrow \mathbf{A} = \mathbf{C} \mathbf{D} \mathbf{C}^T,$$

where $\mathbf{D} = \text{diag}(\lambda_i)$ consists of the eigenvalues of $\mathbf{A}$ and $\mathbf{C}$ of the orthonormal eigenvectors. Then $f(\mathbf{x})$ takes the form

$$f(\mathbf{x}) = \mathbf{x}^T \mathbf{C} \mathbf{D} \mathbf{C}^T \mathbf{x} = (\mathbf{C}^T \mathbf{x})^T \mathbf{D} (\mathbf{C}^T \mathbf{x}).$$

Thus if one makes in $\mathbb{R}^n$ the change of variable

$$\mathbf{y} = \mathbf{C}^T \mathbf{x} \Leftrightarrow \mathbf{x} = \mathbf{C} \mathbf{y},$$

then f takes the form

$$f(\mathbf{x}) = g(\mathbf{y}) = \mathbf{y}^T \mathbf{D} \mathbf{y} = \sum_{i=1}^n \lambda_i y_i^2.$$

The quadratic form $g(\mathbf{y})$ is called — with very good reason — the *diagonalized form* of f . Note that, $\mathbf{C}$ being orthogonal, the transformation $\mathbf{x} \mapsto \mathbf{y} = \mathbf{C}^T \mathbf{x}$ preserves the length of the vector:

$$|\mathbf{y}|^2 = \mathbf{y}^T \mathbf{y} = \mathbf{x}^T \mathbf{C} \mathbf{C}^T \mathbf{x} = \mathbf{x}^T \mathbf{x} = |\mathbf{x}|^2.$$

By Definition [XII.2.1](#), the classification problem of a quadratic form is thus solved by means of the eigenvalues of $\mathbf{A}$ as follows:

$$f(\mathbf{x}) = \mathbf{x}^T \mathbf{A} \mathbf{x} :$$

$$f \text{ positive definite} \quad \Leftrightarrow \quad \lambda > 0 \quad \forall \lambda \in \sigma(\mathbf{A}).$$

$$f \text{ negative definite} \quad \Leftrightarrow \quad \lambda < 0 \quad \forall \lambda \in \sigma(\mathbf{A}).$$

$$f \text{ positive semidefinite} \quad \Leftrightarrow \quad \lambda \geq 0 \quad \forall \lambda \in \sigma(\mathbf{A}) \text{ and } 0 \in \sigma(\mathbf{A}).$$

$$f \text{ negative semidefinite} \quad \Leftrightarrow \quad \lambda \leq 0 \quad \forall \lambda \in \sigma(\mathbf{A}) \text{ and } 0 \in \sigma(\mathbf{A}).$$

$$f \text{ indefinite} \quad \Leftrightarrow \quad \lambda, \mu \in \sigma(\mathbf{A}) \text{ for some } \lambda > 0 \text{ and } \mu < 0.$$

Taking into account also the scaling result ($\star$), then by the above classifications one has in particular

- f positive definite $\Rightarrow f(\mathbf{x}) \geq \lambda|\mathbf{x}|^2 \forall \mathbf{x} \in \mathbb{R}^n$,
- f negative definite $\Rightarrow f(\mathbf{x}) \leq -\lambda|\mathbf{x}|^2 \forall \mathbf{x} \in \mathbb{R}^n$,

where $\lambda > 0$ is the smallest of the numbers $|\lambda_i|$, $\lambda_i \in \sigma(\mathbf{A})$.

EXAMPLE 1 (continued) In the example we have

$$f(x, y) = \mathbf{x}^T \mathbf{A} \mathbf{x} = \begin{bmatrix} x & y \end{bmatrix} \begin{bmatrix} 1 & \frac{a}{2} \\ \frac{a}{2} & 4 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}.$$

The classifications of the example in the cases a) $a = 3$, b) $a = 4$, c) $a = 5$ are also reached by computing

$$\text{a) } \sigma(\mathbf{A}) = \left\{ \frac{1}{2}(5 \pm 3\sqrt{2}) \right\}, \quad \text{b) } \sigma(\mathbf{A}) = \{0, 5\}, \quad \text{c) } \sigma(\mathbf{A}) = \left\{ \frac{1}{2}(5 \pm \sqrt{34}) \right\}. \quad \square$$

EXAMPLE 2 Classify the quadratic form $f(x, y, z) = x^2 + 3y^2 + z^2 + 4xz$ of $\mathbb{R}^3$.

Solution

$$\begin{aligned} f(x, y, z) &= \begin{bmatrix} x & y & z \end{bmatrix} \begin{bmatrix} 1 & 0 & 2 \\ 0 & 3 & 0 \\ 2 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} \\ &= \mathbf{x}^T \mathbf{A} \mathbf{x}, \quad \mathbf{x} = \begin{bmatrix} x & y & z \end{bmatrix}^T. \end{aligned}$$

Here $\sigma(2\mathbf{A}) = \{-2, 6\}$ (Example 1 of the previous section), so $\sigma(\mathbf{A}) = \{-1, 3\}$ and f is therefore indefinite. $\square$

Principal-axis coordinate system of a quadratic form

The type of definiteness of a quadratic form $\mathbf{x}^T \mathbf{A} \mathbf{x}$ can be inferred solely from the eigenvalues of $\mathbf{A}$, but of interest is also the information about the kind of coordinate system in which the quadratic form is diagonalized. This information is conveyed by the orthonormal eigenvectors of $\mathbf{A}$, which determine the diagonalizing coordinate transformation $\mathbf{y} = \mathbf{C}^T \mathbf{x}$. Accordingly, the entries of the vector $\mathbf{y}$ are the coordinates of the vector $\mathbf{x}$ in the basis $\{\mathbf{c}_1, \dots, \mathbf{c}_n\}$ formed by the columns of $\mathbf{C}$. In what follows it is assumed that the basis $\{\mathbf{c}_1, \dots, \mathbf{c}_n\}$ is also positively oriented (reversing, if necessary, the direction of one vector), in

which case $\det \mathbf{C} = 1$ and the change of basis $\{\mathbf{e}_i\} \hookrightarrow \{\mathbf{c}_i\}$ is a rotation of the coordinate system (see Section X.6).

The rotated coordinate system in which the quadratic form $f(\mathbf{x}) = \mathbf{x}^T \mathbf{A} \mathbf{x}$ is diagonalized is called the *principal-axis coordinate system* of that quadratic form. The basis of this coordinate system is thus formed by the orthonormal and positively oriented eigenvectors of $\mathbf{A}$.

EXAMPLE 2 (continued) In the example, the (orthonormal and positively oriented) basis vectors of the principal-axis coordinate system are

$$\mathbf{c}_1 = \frac{1}{\sqrt{2}} [1 \ 0 \ 1]^T, \quad \mathbf{c}_2 = [0 \ 1 \ 0]^T, \quad \mathbf{c}_3 = \frac{1}{\sqrt{2}} [-1 \ 0 \ 1]^T$$

corresponding to the eigenvalues $\lambda_1 = \lambda_2 = 3$ and $\lambda_3 = -1$ of $\mathbf{A}$ (cf. the previous section). Denoting the transformed coordinates by $\mathbf{y} = (\xi, \eta, \zeta)$, the rotation of the coordinate system is

$$\mathbf{y} = \mathbf{C}^T \mathbf{x} \quad \Leftrightarrow \quad \begin{bmatrix} \xi \\ \eta \\ \zeta \end{bmatrix} = \begin{bmatrix} \frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \\ 0 & 1 & 0 \\ -\frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix},$$

and in this principal-axis coordinate system we thus have

$$f(x, y, z) = g(\xi, \eta, \zeta) = 3\xi^2 + 3\eta^2 - \zeta^2. \quad \square$$

Application 1: Classification of the critical point of a function

The classification of a critical point of a function $f(\mathbf{x})$ as a possible local extremum (cf. Proposition XI.8.2) is a question that has not so far been examined. We now return to this question in the light of the theory of quadratic forms. Let $\mathbf{c}$ be a critical point of the function f , i.e. $\nabla f(\mathbf{c}) = \mathbf{0}$. Then if f is sufficiently regular in a neighborhood of $\mathbf{c}$, then by the multivariable Taylor formula (see Section XI.9) one has

$$f(\mathbf{x}) = f(\mathbf{c}) + \frac{1}{2}(\mathbf{x} - \mathbf{c})^T \mathbf{H}(\mathbf{c})(\mathbf{x} - \mathbf{c}) + o(|\mathbf{x} - \mathbf{c}|^2),$$

where $\mathbf{H}(\mathbf{c})$ is the Hessian matrix of the function at the point $\mathbf{c}$:

$$[\mathbf{H}(\mathbf{c})]_{ij} = \frac{\partial^2 f}{\partial x_i \partial x_j}(\mathbf{c}), \quad i, j = 1 \dots n.$$

To settle whether $\mathbf{c}$ is a local extremum of f or not, one must therefore evidently first investigate what kinds of values the function

$$g(\mathbf{x}) = \frac{1}{2}(\mathbf{x} - \mathbf{c})^T \mathbf{H}(\mathbf{c})(\mathbf{x} - \mathbf{c})$$

takes in the immediate neighborhood of the point $\mathbf{c}$. Shifting the origin to the point $\mathbf{c}$ by the coordinate transformation $\mathbf{x}' = \mathbf{x} - \mathbf{c}$, the object of study becomes the quadratic form $h(\mathbf{x}') = g(\mathbf{x}' + \mathbf{c}) = (\mathbf{x}')^T \mathbf{A} \mathbf{x}'$, $\mathbf{A} = \frac{1}{2} \mathbf{H}(\mathbf{c})$ of $\mathbb{R}^n$.

Let the quadratic form $h(\mathbf{x}')$ be positive definite. Then, under the stated assumptions, one has for some $\delta > 0$

$$f(\mathbf{x}) \geq f(\mathbf{c}) + \lambda |\mathbf{x} - \mathbf{c}|^2 - c_\delta |\mathbf{x} - \mathbf{c}|^2, \quad \text{when } 0 < |\mathbf{x} - \mathbf{c}| < \delta,$$

where $\lambda > 0$ and $c_\delta \rightarrow 0$ as $\delta \rightarrow 0$. Thus if $\delta > 0$ is chosen so that $c_\delta \leq \lambda/2$, then for that value of δ one has

$$f(\mathbf{x}) \geq f(\mathbf{c}) + \frac{1}{2} \lambda |\mathbf{x} - \mathbf{c}|^2, \quad 0 < |\mathbf{x} - \mathbf{c}| < \delta.$$

Hence f has a local minimum at $\mathbf{c}$. In a corresponding way one concludes that $\mathbf{c}$ is a local maximum point of f if $h(\mathbf{x}')$ is negative definite, and that $\mathbf{c}$ is neither a local extremum nor a flat point of f if $h(\mathbf{x}')$ is indefinite. Combined, the conclusions are thus the following.

PROPOSITION XII.2.2 If for a function $f : A \rightarrow \mathbb{R}$, $A \subset \mathbb{R}^n$, the approximation

$$f(\mathbf{x}) = f(\mathbf{c}) + g(\mathbf{x} - \mathbf{c}) + o(|\mathbf{x} - \mathbf{c}|^2), \quad \mathbf{x} \in U_\delta(\mathbf{a}) \subset A,$$

holds in some neighborhood of the point $\mathbf{c} \in A$, where g is a quadratic form, then

1. g positive definite $\Rightarrow \mathbf{c}$ is a local minimum point of f .
2. g negative definite $\Rightarrow \mathbf{c}$ is a local maximum point of f .
3. g indefinite $\Rightarrow \mathbf{c}$ is neither a local extremum nor a flat point of f .

Note that indefiniteness of a quadratic form is possible only in the situation of two or more variables, since if $n = 1$, then $\mathbf{x}^T \mathbf{A} \mathbf{x} = Ax^2$, so the quadratic form is either positive definite ($A > 0$), negative definite ($A < 0$), or $= 0$ ($A = 0$). The condition $f''(c) > 0$ is familiar from single-variable analysis as a (sufficient) condition for a local minimum. Its counterpart in the situation of n variables is thus the condition

$$\mathbf{x}^T \mathbf{A} \mathbf{x} > 0 \quad \forall \mathbf{x} \in \mathbb{R}^n, \mathbf{x} \neq \mathbf{0}; \quad \mathbf{A} = \left(\frac{\partial^2 f}{\partial x_i \partial x_j}(\mathbf{c}) \right),$$

that is, the counterpart of the second derivative $f''(c)$ in the extremum analysis of n variables is the Hessian matrix formed by all the second-order partial derivatives.

EXAMPLE 3 Investigate whether the following functions have a local extremum at the origin:

$$\text{a) } f(x, y) = \cos x + 2y - 2e^y, \quad \text{b) } f(x, y) = e^{-x^2}e^{-y^2} + 3 \sin xy.$$

Solution a) $f(x, y) = -1 + (-\frac{1}{2}x^2 - y^2) + \dots$ The quadratic-form part is negative definite. Thus f has a local maximum at the origin.

b) $f(x, y) = 1 + (-x^2 - y^2 + 3xy) + \dots$ The quadratic-form part is indefinite. Thus the origin is neither a local extremum nor a flat point of f . $\square$

EXAMPLE 4 Determine the nature of the critical point $(3, -5)$ of the function $f(x, y) = x^2 + 2y^2 + 2xy + 4x + 14y$ (cf. Example 2 of Section XI.8).

Solution Let us first shift the critical point to the origin:

$$\begin{aligned} x &= 3 + x', \quad y = -5 + y' \\ g(x', y') &= f(3 + x', -5 + y') = (x')^2 + 2x'y' + 2(y')^2 - 29. \end{aligned}$$

Since the quadratic-form part of g is positive definite, f has a local minimum at the point $(3, -5)$. From the form of the function one sees that this is also the absolute minimum. $\square$

Proposition XII.2.2 leaves open the case in which the quadratic form $g(\mathbf{x}')$ is semidefinite. As in the case $n = 1$ ($f''(c) = 0$), the nature of the possible extremum must be settled case by case.

EXAMPLE 5 Determine the nature of the critical point $(0, 0)$ when

$$f(x, y) = x^2 + 2xy + y^2 + a(x + y)^3 + bx^n, \quad a, b \in \mathbb{R}, \quad n \in \{3, 4, \dots\}.$$

Solution Since the quadratic-form part of f

$$g(x, y) = x^2 + 2xy + y^2 = (x + y)^2$$

is positive semidefinite, the investigation must be continued at least along the line $y = -x$:

$$f(x, -x) = bx^n.$$

We conclude: If $b = 0$, the origin is a flat point. If $b > 0$ and n is even, the origin is a local minimum point of f . Otherwise, that is, if $b < 0$ and n is even, or if $b \neq 0$ and n is odd, f takes both positive and negative values in every neighborhood of the origin, so the origin is neither a local extremum nor a flat point. $\square$

Application 2: Second-degree curves and surfaces

Let us recall from Section II.7 that the equation of a general second degree curve in the plane is of the form

$$Ax^2 + By^2 + 2Cxy + Dx + Ey + F = 0,$$

where $A, \dots, F \in \mathbb{R}$ and at least one of the numbers A, B, C is nonzero, and the equation of a general second degree surface in space is of the form

$$Ax^2 + By^2 + Cz^2 + 2Dxy + 2Eyz + 2Fxz + Gx + Hy + Iz + J = 0,$$

where $A, \dots, J \in \mathbb{R}$ and at least one of the numbers $A, \dots, F$ is nonzero. Writing

$$\text{Plane:} \quad \mathbf{A} = \begin{bmatrix} A & C \\ C & B \end{bmatrix}, \quad \mathbf{x} = \begin{bmatrix} x \\ y \end{bmatrix}, \quad \mathbf{a} = \begin{bmatrix} D \\ E \end{bmatrix}, \quad c = F,$$

$$\text{Space:} \quad \mathbf{A} = \begin{bmatrix} A & D & F \\ D & B & E \\ F & E & C \end{bmatrix}, \quad \mathbf{x} = \begin{bmatrix} x \\ y \\ z \end{bmatrix}, \quad \mathbf{a} = \begin{bmatrix} G \\ H \\ I \end{bmatrix}, \quad c = J,$$

the equations of the second-degree curve and surface take the form

$$\mathbf{x}^T \mathbf{A} \mathbf{x} + \mathbf{a}^T \mathbf{x} + c = 0.$$

Let us carry out here the coordinate transformation

$$\mathbf{x} = \mathbf{C} \mathbf{y} + \mathbf{b},$$

where $\mathbf{y} = [\xi, \eta]^T$ (plane) or $\mathbf{y} = [\xi, \eta, \zeta]^T$ (space), and $\mathbf{C}$ and $\mathbf{b}$ are chosen so that the equation takes the simplest possible form in the transformed coordinates. First, $\mathbf{C}$ is chosen so that the quadratic form $\mathbf{x}^T \mathbf{A} \mathbf{x}$ is diagonalized, that is, the orthonormal and positively oriented eigenvectors of $\mathbf{A}$ are chosen as the columns of $\mathbf{C}$. Then the equation takes the form

$$\mathbf{y}^T \mathbf{D} \mathbf{y} + \mathbf{d}^T \mathbf{y} + d = 0$$

where $\mathbf{D}$ is a diagonal matrix (with the eigenvalues of $\mathbf{A}$ on the diagonal) and

$$\mathbf{d} = \mathbf{C}^T (2\mathbf{A} \mathbf{b} + \mathbf{a}), \quad d = \mathbf{b}^T (\mathbf{A} \mathbf{b} + \mathbf{a}) + c.$$

If $\mathbf{A}$ is a regular matrix, then $\mathbf{b}$ is determined so that $\mathbf{d} = \mathbf{0}$; otherwise the choice of $\mathbf{b}$ is used to reduce the term $\mathbf{d}^T \mathbf{y} + d$ to the simplest possible form. (If it is a surface in space and $\lambda = 0$ is a double eigenvalue of $\mathbf{A}$, the simplification of

the equation may also be affected by the choice of $\mathbf{C}$, see Exercise 10.) Once the equation has been reduced in this way, the curve or surface it defines can be classified geometrically.

If it is a plane curve, then the above transformations lead to one of the following basic forms:

1. $\frac{\xi^2}{a^2} + \frac{\eta^2}{b^2} = q, \quad q \in \{1, 0, -1\}, \quad a, b > 0.$
2. $\frac{\xi^2}{a^2} - \frac{\eta^2}{b^2} = q, \quad q \in \{1, 0, -1\}, \quad a, b > 0.$
3. $\eta = a\xi^2 \quad \text{or} \quad \xi = a\eta^2, \quad a \in \mathbb{R}, \quad a \neq 0.$
4. $\xi^2 = a \quad \text{or} \quad \eta^2 = a, \quad a \in \mathbb{R}.$

In case 1 the curve is an *ellipse* if $q = 1$; otherwise it is a point ($q = 0$) or the empty set ($q = -1$). In case 2 the curve is a *hyperbola* if $q = \pm 1$; otherwise it is two lines intersecting each other ($q = 0$). In case 3 the curve is a *parabola*, and in case 4 it is either two parallel lines ($a > 0$), one line ($a = 0$), or the empty set ($a < 0$). Case 1 is reached when $\mathbf{A}$ is definite (eigenvalues of the same sign), case 2 when $\mathbf{A}$ is indefinite (eigenvalues of opposite sign), and cases 3 and 4 when $\mathbf{A}$ is semidefinite (one eigenvalue = 0).

EXAMPLE 6 Find a coordinate system in which the level curves of the function

$$f(x, y) = 13x^2 - 8xy + 7y^2 - 42x + 36y$$

take the basic form of a second-degree curve. Make a figure!

Solution Let us first write the equation of the level curve $S : f(x, y) = c$ in the form

$$\begin{bmatrix} x & y \end{bmatrix} \begin{bmatrix} 13 & -4 \\ -4 & 7 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} -42 & 36 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} - c = 0.$$

The eigenvalues and orthonormal, positively oriented eigenvectors of the matrix $\mathbf{A} = \begin{bmatrix} 13 & -4 \\ -4 & 7 \end{bmatrix}$ are

$$\lambda_1 = 15, \quad \mathbf{c}_1 = \frac{1}{\sqrt{5}} \begin{bmatrix} 2 \\ -1 \end{bmatrix}, \quad \lambda_2 = 5, \quad \mathbf{c}_2 = \frac{1}{\sqrt{5}} \begin{bmatrix} 1 \\ 2 \end{bmatrix}.$$

Let us carry out the coordinate transformation in two stages. First a rotation into the principal-axis coordinate system (x', y') of the quadratic form (with $\mathbf{b} = \mathbf{0}$ above):

$$\begin{bmatrix} x \\ y \end{bmatrix} = \frac{1}{\sqrt{5}} \begin{bmatrix} 2 & 1 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} x' \\ y' \end{bmatrix} \Leftrightarrow \begin{bmatrix} x' \\ y' \end{bmatrix} = \frac{1}{\sqrt{5}} \begin{bmatrix} 2 & -1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}.$$

Result:

$$\begin{aligned} f(x, y) = c &\Leftrightarrow 15(x')^2 + 5(y')^2 - \frac{120}{\sqrt{5}}x' + \frac{30}{\sqrt{5}}y' - c = 0 \\ &\Leftrightarrow 15\left(x' - \frac{4}{\sqrt{5}}\right)^2 + 5\left(y' + \frac{3}{\sqrt{5}}\right)^2 - 57 - c = 0. \end{aligned}$$

The second stage is a shift of the origin, to the coordinates

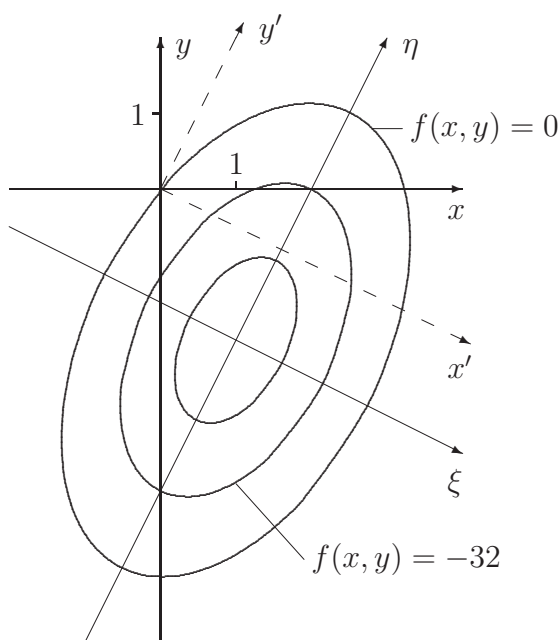
$$\xi = x' - \frac{4}{\sqrt{5}}, \quad \eta = y' + \frac{3}{\sqrt{5}}.$$

Final result:
$$f(x, y) = c \Leftrightarrow 3\xi^2 + \eta^2 = \frac{1}{5}(57 + c).$$

Accordingly, the level curve is either an ellipse ($c > -57$), a point ($c = -57$), or the empty set ($c < -57$). The common principal axes of the ellipses are the lines $\xi = 0$ and $\eta = 0$, and at their common center

$$(\xi, \eta) = (0, 0) \Leftrightarrow (x', y') = (4/\sqrt{5}, -3/\sqrt{5}) \Leftrightarrow (x, y) = (1, -2)$$

f attains its absolute minimum value $f_{\min} = -57$. $\square$



Second-degree surfaces can, by means of the basic form given by the above coordinate transformation, be classified into the following nine main types (in the equations the order of the coordinates may be interchanged).

1. *Ellipsoid*: $\frac{\xi^2}{a^2} + \frac{\eta^2}{b^2} + \frac{\zeta^2}{c^2} = 1.$
2. *Hyperboloid of one sheet*: $\frac{\xi^2}{a^2} + \frac{\eta^2}{b^2} - \frac{\zeta^2}{c^2} = 1.$
3. *Hyperboloid of two sheets*: $\frac{\xi^2}{a^2} + \frac{\eta^2}{b^2} - \frac{\zeta^2}{c^2} = -1.$
4. *Cone*: $\frac{\xi^2}{a^2} + \frac{\eta^2}{b^2} - \frac{\zeta^2}{c^2} = 0.$
5. *Elliptic paraboloid*: $\zeta = \frac{\xi^2}{a^2} + \frac{\eta^2}{b^2}.$
6. *Hyperbolic paraboloid*: $\zeta = \frac{\xi^2}{a^2} - \frac{\eta^2}{b^2}.$
7. *Elliptic cylinder*: $\frac{\xi^2}{a^2} + \frac{\eta^2}{b^2} = 1.$
8. *Hyperbolic cylinder*: $\frac{\xi^2}{a^2} - \frac{\eta^2}{b^2} = 1.$

9. *Parabolic cylinder:* $\eta = a\xi^2$.

The hyperbolic paraboloid is also called a *saddle surface*. In cases 1–4 $\mathbf{A}$ is a regular matrix, and in case 1 the eigenvalues are moreover of the same sign. In cases 5–8 $\mathbf{A}$ has a simple, and in case 9 a double, eigenvalue $\lambda = 0$. In addition to these main types, the equation of a second-degree surface can represent two planes intersecting each other, two parallel planes, a single plane, a line in space, a point, or nothing.

EXAMPLE 7 Classify, for the various values of c , the second-degree surface

$$5x^2 + 2y^2 + 2z^2 + 2xy - 4yz - 2xz + 10x + 2y - 2z + c = 0.$$

Solution The characteristic equation of the matrix

$$\mathbf{A} = \begin{bmatrix} 5 & 1 & -1 \\ 1 & 2 & -2 \\ -1 & -2 & 2 \end{bmatrix}$$

is

$$\begin{vmatrix} 5 - \lambda & 1 & -1 \\ 1 & 2 - \lambda & -2 \\ -1 & -2 & 2 - \lambda \end{vmatrix} = -\lambda^3 + 9\lambda^2 - 18\lambda = 0.$$

The eigenvalues are $\lambda_1 = 0$, $\lambda_2 = 3$ and $\lambda_3 = 6$. The corresponding orthonormal and positively oriented eigenvectors, collected as the columns of the matrix $\mathbf{C}$, are:

$$\mathbf{C} = \begin{bmatrix} 0 & \frac{1}{\sqrt{3}} & \frac{2}{\sqrt{6}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{3}} & \frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{3}} & -\frac{1}{\sqrt{6}} \end{bmatrix}.$$

Let us first perform the rotation of the coordinate system $\mathbf{C}\mathbf{x}' = \mathbf{x} \Leftrightarrow \mathbf{x}' = \mathbf{C}^T\mathbf{x}$, where $\mathbf{x} = [x, y, z]^T$, $\mathbf{x}' = [x', y', z']^T$, whereupon the equation of the surface takes the form

$$\begin{aligned} 0 &= 3(y')^2 + 6(z')^2 + \frac{5}{\sqrt{3}}y' + \frac{22}{\sqrt{6}}z' + c \\ &= 3\left(y' + \frac{1}{\sqrt{3}}\right)^2 + 6\left(z' + \frac{2}{\sqrt{6}}\right)^2 + c - 5. \end{aligned}$$

From here one reaches the basic form by a shift of the origin:

$$\xi = x', \quad \eta = y' + \frac{1}{\sqrt{3}}, \quad \zeta = z' + \frac{2}{\sqrt{6}} \quad \Rightarrow \quad 3\eta^2 + 6\zeta^2 = 5 - c.$$

From this one sees that if $c < 5$, then the surface is an elliptic cylinder. The symmetry axis of the cylinder is the line $S : \eta = \zeta = 0$. The line S passes through the point $(\xi, \eta, \zeta) = (0, 0, 0)$, that is, the point $(x, y, z) = (-1, 0, 0)$, and its direction vector in the original coordinate system is $\mathbf{t} = [0, 1, 1]^T \hat{=} \vec{j} + \vec{k}$ (the eigenvector of $\mathbf{A}$ corresponding to the eigenvalue $\lambda_1 = 0$, which determines the direction of the ξ -axis). In the case $c = 5$ the equation defines only the line S , and in the case $c > 5$ nothing. — The line S is in fact the line of intersection of the planes $T_1 : x - y + z + 1 = 0$ and $T_2 : 2x + y - z + 2 = 0$, since the original equation is the same as

$$(x - y + z + 1)^2 + (2x + y - z + 2)^2 = 5 - c. \quad \square$$

EXERCISES

- Classify the following quadratic forms by determining, in part a), the range on the unit circle, and in parts b) and c), by the substitution $y = tx$.
 a) $3x^2 - 16xy + 16y^2$ b) $4x^2 - 16xy + 16y^2$ c) $5x^2 - 16xy + 16y^2$
- Determine the principal-axis coordinate system and the diagonalized form of the quadratic form:
 a) $f(x, y) = xy$ b) $f(x, y) = 4x^2 - 16xy + 16y^2$
 c) $f(x, y) = -3x^2 - 3y^2 + 4xy$ d) $f(x, y, z) = 2x^2 + 3y^2 + 2z^2 - 6xz$
 e) $f(x, y, z, u) = x^2 + y^2 + z^2 + u^2 + 2yz + 2xu$
- Determine the quadratic form $f(x, y, z)$ whose principal-axis coordinate system has basis vectors

$$\mathbf{c}_1 = \frac{1}{7} \begin{bmatrix} 3 \\ -2 \\ 6 \end{bmatrix}, \quad \mathbf{c}_2 = \frac{1}{7} \begin{bmatrix} -2 \\ 6 \\ 3 \end{bmatrix}, \quad \mathbf{c}_3 = \frac{1}{7} \begin{bmatrix} -6 \\ -3 \\ 2 \end{bmatrix}$$

and whose diagonalized form in the principal-axis coordinate system is

$$g(\xi, \eta, \zeta) = 49\xi^2 - 98\eta^2 + 147\zeta^2.$$

4. Investigate whether the origin is a local extremum of the function:

a) $f(x, y) = \cos(x - 2y) - \sin xy$ b) $f(x, y) = e^x y - x e^y + \sin(x - y)$

c) $f(x, y) = 1 - x + 2y + e^{\frac{1}{2}xy} + \sin(x - 2y) + 2 \cos(x + y)$

d) $f(x, y) = 1 + x + 2y + (1 + x + 2y)^{-1} + 2 \sin(xy) - 12 \cos y$

e) $f(x, y) = e^{x^2+y^2} \sqrt[3]{7 + 42xy}$

f) $f(x, y, z) = x + y + z + e^{-x-y-z} + (x + y) \sin(x + y) + \cos(y + z)$

g) $f(x, y, z) = x + y + z - e^{x+y+z} - (x + y) \sin(x + y) + \cos(y + z)$

5. Classify the origin as a critical point of the function for various values of the parameter a :

a) $f(x, y) = \frac{1 + x^2 + 2y^2}{1 + axy}$ b) $f(x, y) = \sin xy + a \cos(x + y)$

6. Find all the critical points of the following functions and classify them.

a) $x^2 + 2y^2 - 4x + 4y$ b) $xy - x + y$ c) $x^3 + y^3 - 3xy$

d) $x^4 + y^4 - 4xy$ e) $\cos(x + y)$ f) $\frac{x}{y} + \frac{8}{x} - y$

g) $\frac{1}{x - y + x^2 + y^2}$ h) $\frac{xy}{1 + x + y}$ i) $(x - 3y)e^{x^2+y^2}$

7. Find a Cartesian coordinate system in which the level curves of the following functions are in the simplest basic form of a second-degree curve. Sketch the course of the level curves.

a) $16x^2 + 9y^2 + 24xy$ b) $2x^2 + 3y^2 + 2xy$ c) $xy + y^2$

d) $xy + x + 2y + 1$ e) $x^2 + 4xy + 4y^2 + 6x + 8y$

f) $2x^2 + 3y^2 + 2xy - 4x - 4y$ g) $3x^2 + 4xy + y^2 + 2x$

8. Find, for the following second-degree surfaces, a coordinate system in which the equation of the surface reduces to its basic form. Classify the surfaces.

a) $xy + yz + xz = 0$ b) $xy - yz + xz = 0$ c) $xy + yz + x + y + z = 0$

d) $xy + yx - xz + 2x - 3y + z + 1 = 0$

e) $x^2 + y^2 + z^2 + 8xy + 6yz - 24y + 8z = 0$

f) $x^2 + y^2 + 3z^2 + 2xy + 4yz + 4xz + x + y + z - 18 = 0$

9. (*) Show that for some $m \in \mathbb{N}$ one has: If the origin is traversed along the curves

$$y = kx^n, \quad k \in \mathbb{R}, \quad n \in \mathbb{N}, \quad n \neq m \quad \text{or} \quad x = ky^n, \quad k \in \mathbb{R}, \quad n \in \mathbb{N},$$

then the function $f(x, y) = y^2 + 3x^{50}y + 2x^{100}$ has, along each curve, a local minimum at the origin. Show that the origin is nevertheless neither a local minimum point nor even a flat point of f .

10. The equation $\mathbf{x}^T \mathbf{A} \mathbf{x} + \mathbf{b}^T \mathbf{x} + c = 0$, $\mathbf{x}^T = [x, y, z]$, of a second-degree surface has been brought, by a rotation of the coordinate system, to the form

$$\xi^2 + G\xi + H\eta + I\zeta + J = 0$$

($\mathbf{A}$ having a double eigenvalue $\lambda = 0$). Show that by a new rotation of the coordinate system, followed by a shift of the origin, the equation can be brought to the basic form $\eta' = a(\xi')^2$ or $(\xi')^2 = a$ ($a \in \mathbb{R}$). Apply the procedure:

- a) $x^2 + x + 2y - 2z - 4 = 0$ b) $(x + y + 2z)^2 + x + y - z = 0$
 c) $x^2 + y^2 + 4z^2 - 2xy + 4xz - 4yz - 1 = 0$
 d) $x^2 + 4y^2 + z^2 + 4xy + 2xz + 4yz - 4x - 3y - 4z + 4 = 0$
 e) $9x^2 + 4y^2 + z^2 - 12xy - 6xz + 4yz + 6x - 4y + 2z + 2 = 0$

11. (*) Determine the principal-axis coordinate system and classify the surface:

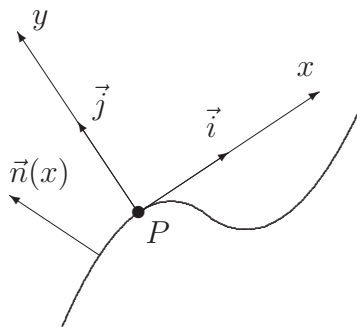
- a) $17x^2 + 76y^2 + 54z^2 - 60xy + 12xz + 48yz + 4x + 16y + 36z + 6 = 0$
 b) $x^2 - 68y^2 + 18z^2 + 36xy + 60xz - 96yz - 22x - 53y + 26z + 121 = 0$
 c) $13x^2 + 40y^2 + 45z^2 - 36xy + 24xz + 12yz - 34x - 17y + 44z + 124 = 0$
 d) $32x^2 - 20y^2 + 135z^2 - 24xy + 156xz - 132yz + 52x - 44y + 90z + 15 = 0$
 e) $48x^2 + 117y^2 + 31z^2 - 36xy - 60xz + 96yz - 216x - 66y + 86z + 300 = 0$

XII.3 Curvature of a surface

Let us recall from the theory of curves that the curvature of a curve tells how rapidly the unit tangent vector of the curve, or alternatively the principal normal vector pointing in the direction of curving, changes as one moves along the curve, see Section VII.3, formulas (3)–(4).

If one wishes to determine the curvature of a plane curve at a given point P of the curve, then the most convenient coordinate system for the computational formulas is the one whose origin is at P and whose x -axis is parallel to the tangent of the curve (figure). Let the equation of the curve in this coordinate system be $y = f(x)$, where f is twice continuously differentiable near the point $x = 0$. Then the unit normal vector of the curve near P is

$$\vec{n}(x) = \frac{1}{\sqrt{1 + [f'(x)]^2}} [-f'(x)\vec{i} + \vec{j}].$$

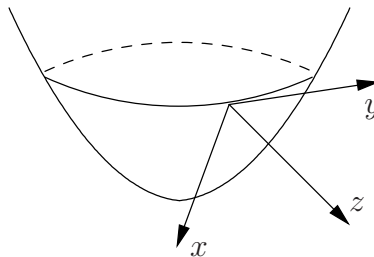


In the situation of the figure the curve is, in the coordinate system (x, y) , curving downward near P ($f''(0) < 0$), so in this case the principal normal vector of the curve near $P = -\vec{n}(x)$. Taking into account that $f'(0) = 0$, differentiation gives

$$\frac{d\vec{n}}{dx}(0) = -f''(0)\vec{i}.$$

In the chosen coordinate system, then, the (signed) curvature of the curve at P is $\kappa = f''(0)$ (cf. formulas (4) and (6) of Section VII.3 for the principal normal vector $\vec{n}$).

Let us now consider a surface whose equation is $z = f(x, y)$. Correspondingly as above, we assume here that the coordinate system is chosen so that the xy -plane is the tangent plane of the surface at the origin (figure). We also assume that the partial derivatives of f up to the second order are continuous near the point $(x, y) = (0, 0)$.



The assumed situation can be thought of as having been reached by choosing some point P of a given surface, shifting the origin of the coordinate system to that point, and then choosing at this point a (rotated) coordinate system in which the xy -plane is the tangent plane of the surface at P .

The normal of the surface at the point $(x, y, f(x, y))$ is (cf. Section XI.3)

$$\vec{n}(x, y) = \frac{-f_x \vec{i} - f_y \vec{j} + \vec{k}}{\sqrt{1 + f_x^2 + f_y^2}}.$$

By the choice of coordinate system, $f_x(0, 0) = f_y(0, 0) = 0$, so it follows that

$$\begin{aligned} \frac{\partial \vec{n}}{\partial x}(0, 0) &= -a_{11} \vec{i} - a_{12} \vec{j}, \\ \frac{\partial \vec{n}}{\partial y}(0, 0) &= -a_{12} \vec{i} - a_{22} \vec{j}, \end{aligned}$$

where

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} \\ a_{12} & a_{22} \end{bmatrix} = \begin{bmatrix} f_{xx}(0, 0) & f_{xy}(0, 0) \\ f_{xy}(0, 0) & f_{yy}(0, 0) \end{bmatrix}.$$

The matrix $\mathbf{A}$ is called (under the assumptions made) the *curvature matrix* of the surface at the point considered and in the coordinates (x, y) chosen in the stated way in the tangent plane of the surface. The numbers a_{11} and a_{22} express the curvature of the surface in the directions of the coordinate axes: near the origin the curves $S_1 : y = 0, z = f(x, 0)$ and $S_2 : x = 0, z = f(0, y)$ are approximately circular arcs whose centers are at the points $(0, 0, 1/a_{11})$ and $(0, 0, 1/a_{22})$. The number a_{12} is the *twist* of the surface at the origin. The twist tells how rapidly (and in which direction) the normal of the surface 'tips to the side' as one moves in the direction of the coordinate lines (cf. the torsion of a space curve: Exercise VII.3:10).

The curvature matrix naturally depends on the point of the surface under consideration, but it also depends on the chosen coordinate system (x, y) in the tangent plane of the surface. It is worth investigating how the curvature matrix changes when the coordinate axes x, y are rotated about the z -axis, that is, when one makes the coordinate transformation

$$\mathbf{C} \begin{bmatrix} \xi \\ \eta \end{bmatrix} = \begin{bmatrix} x \\ y \end{bmatrix} \Leftrightarrow \begin{bmatrix} \xi \\ \eta \end{bmatrix} = \mathbf{C}^T \begin{bmatrix} x \\ y \end{bmatrix},$$

where $\mathbf{C}$ is an orthogonal matrix (of size 2×2). The answer to this question is obtained by thinking of the function $f(x, y)$ as approximated near the origin according to the Taylor formula:

$$f(x, y) = \frac{1}{2} a_{11} x^2 + \frac{1}{2} a_{22} y^2 + a_{12} xy + \dots$$

Only the terms shown in the expansion matter for the curvature, since the second-order partial derivatives of the remainder term vanish at the origin. We conclude, then, that the curvature matrix transforms under a rotation of the coordinate system in the same way as a quadratic form. In other words, in the rotated (ξ, η) -coordinate system the curvature matrix is

$$\mathbf{B} = \mathbf{C}^T \mathbf{A} \mathbf{C}.$$

If here in particular one chooses as the columns of $\mathbf{C}$ the orthonormal and positively oriented eigenvectors of $\mathbf{A}$, then $\mathbf{B}$ is diagonal:

$$\mathbf{B} = \begin{bmatrix} \kappa_1 & 0 \\ 0 & \kappa_2 \end{bmatrix}.$$

One says that the diagonal entries κ_1, κ_2 of $\mathbf{B}$ are the *principal curvatures* of the surface at the point considered, and that the coordinate system (ξ, η) is at that point the *principal-curvature coordinate system*. Choosing $\zeta = z$, then in the coordinate system (ξ, η, ζ) the equation of the surface is, in a neighborhood of the origin, of the form

$$\zeta = \frac{1}{2} \kappa_1 \xi^2 + \frac{1}{2} \kappa_2 \eta^2 + \dots$$

A sufficiently regular surface is thus, in the immediate neighborhood of each of its points P , approximately either (a) an elliptic paraboloid ($\kappa_1 \kappa_2 > 0$), (b) a parabolic cylinder or plane ($\kappa_1 \kappa_2 = 0$; a plane if $\kappa_1 = \kappa_2 = 0$), or (c) a hyperbolic paraboloid ($\kappa_1 \kappa_2 < 0$), cf. the classification of second-degree surfaces above. According to this division, one says that P is (a) an *elliptic*, (b) a *parabolic*, or (c) a *hyperbolic* point of the surface.

If the point P of the surface is the origin of the principal-curvature coordinate system (ξ, η, ζ) in the above manner and the surface is viewed from the direction of the positive ζ -axis (i.e. the normal of the surface), then in the elliptic case the surface curves in every direction either toward the observer (if $\kappa_1 > 0$ and $\kappa_2 > 0$) or away from the observer (if $\kappa_1 < 0$ and $\kappa_2 < 0$). In the hyperbolic case the surface curves toward the observer in one principal-curvature direction and away from the observer in the other, that is, P is a so-called *saddle point*. In the parabolic case the surface curves essentially only in one principal-curvature direction. In all cases the curving takes place in the principal-curvature directions in such a way that the line of intersection of the $\xi\zeta$ -plane and the surface is near the origin approximately a circular arc whose center is $Q_1 = (0, 0, 1/\kappa_1)$, and correspondingly the line of intersection of the $\eta\zeta$ -plane and the surface is approximately a circular arc whose center is $Q_2 = (0, 0, 1/\kappa_2)$. The numbers $R_i = 1/|\kappa_i|$ (which may also be $R_i = \infty$) are accordingly called the *principal radii of curvature* and the points Q_i the *centers of curvature*. The centers of

curvature can be understood geometrically in that the normal of the surface rotates approximately about the center of curvature as the point of intersection of the normal and the surface moves from the point P under consideration in the corresponding principal-curvature direction. Accordingly, the point P is elliptic precisely when the centers of curvature are on the same side of the surface and at a finite distance, and hyperbolic precisely when the centers of curvature are on opposite sides of the surface and at a finite distance.

EXAMPLE 1 When moving on the spherical surface

$$S : (x - x_0)^2 + (y - y_0)^2 + (z - z_0)^2 = R^2$$

in any direction, the normal vector $\vec{n}$ pointing away from the center $Q = (x_0, y_0, z_0)$ of the sphere rotates about the center. Both centers of curvature are therefore at the point Q . We conclude that if the origin of a Cartesian coordinate system (ξ, η, ζ) is on the spherical surface and the $\xi\eta$ -plane is the tangent plane of the spherical surface, then every such coordinate system is a principal-curvature coordinate system. If the normal vector $\vec{n}$ is chosen in the stated way, then the curvature matrix is

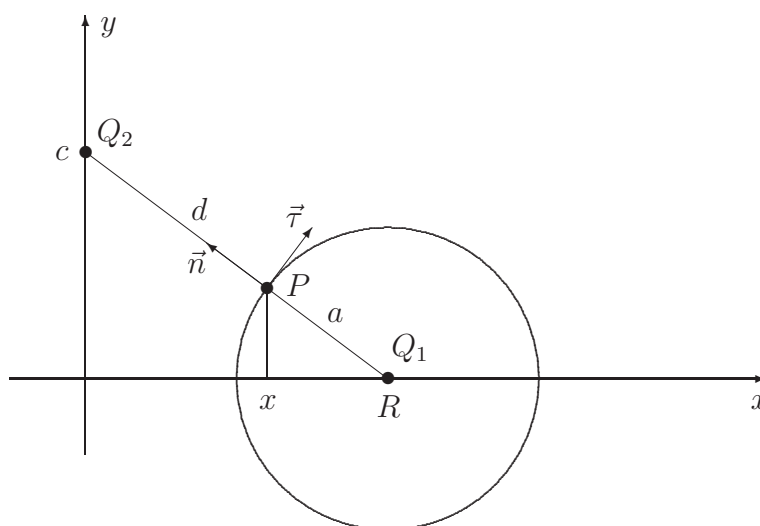
$$\mathbf{A} = \begin{bmatrix} -\frac{1}{R} & 0 \\ 0 & -\frac{1}{R} \end{bmatrix}.$$

Every point of the surface is elliptic. $\square$

EXAMPLE 2 **Torus.** The circular arc in the xy -plane

$$K = \{(x, y) \in \mathbb{R}^2 \mid (x - R)^2 + y^2 = a^2\}, \quad \text{where } 0 < a < R,$$

rotates in space about the y -axis. Compute the principal curvatures of the surface thus created, the *torus*, at its point $P = (x, y, 0)$ in the xy -plane.



Solution Geometrically one can conclude that the principal-curvature directions at the point P are $\vec{\tau}$ and $\vec{k}$, where $\vec{\tau}$ is the unit tangent vector of S parallel to the xy -plane and $\vec{k}$ is the normal of the xy -plane. Indeed, when one moves from the point P in the direction $\vec{\tau}$, the normal of the surface rotates about the point $Q_1 = (R, 0, 0)$. When moving in the direction $\vec{k}$, the normal rotates about the point $Q_2 = (0, c, 0)$ on the y -axis (see figure), since the y -axis is the symmetry axis of the surface. The points Q_1, Q_2 are thus the centers of curvature. When the normal $\vec{n}$ of the surface is chosen to point away from the point Q_1 , then in the principal-curvature coordinate system (ξ, η, ζ) chosen in this way, $Q_1 = (0, 0, -a)$ and (see figure)

$$Q_2 = \begin{cases} (0, 0, d), & \text{if } x < R, \\ (0, 0, -d), & \text{if } x > R, \end{cases}$$

where d is the distance of the point P from the y -axis in the direction $\pm\vec{n}$. In the case $x < R$ one concludes, by means of similar triangles, that (see figure)

$$\frac{a+d}{R} = \frac{a}{R-x} \quad \Rightarrow \quad d = \frac{ax}{R-x},$$

so the principal curvatures are in this case

$$\underline{\underline{\kappa_1 = -\frac{1}{a}, \quad \kappa_2 = \frac{1}{a} \left(\frac{R}{x} - 1 \right)}}.$$

These are found to be valid also in the case $x \geq R$. The point $P = (x, y, 0)$ is accordingly elliptic when $x > R$, parabolic when $x = R$, and hyperbolic when $x < R$. $\square$

EXAMPLE 3 Determine the principal curvatures and principal-curvature coordinate system of the surface $S : z = \sin xy$ at the origin.

Solution The tangent plane of the surface at the origin is the xy -plane, and the curvature matrix at the origin is

$$\mathbf{A} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}.$$

Its eigenvalues are $\lambda_1 = +1$ and $\lambda_2 = -1$, and the corresponding eigenvectors $[1, 1]^T$ and $[-1, 1]^T$. Thus the principal curvatures are $\kappa_1 = +1$ and $\kappa_2 = -1$, and the basis vectors of the principal-curvature coordinate system are

$$\vec{e}_1 = (\vec{i} + \vec{j})/\sqrt{2}, \quad \vec{e}_2 = (-\vec{i} + \vec{j})/\sqrt{2}.$$

In this coordinate system the equation of the surface is, in series form,

$$\begin{aligned} z &= xy - \frac{(xy)^3}{3!} + \dots \\ &= \frac{1}{2}(\xi^2 - \eta^2) - \frac{1}{48}(\xi^2 - \eta^2)^3 + \dots \end{aligned}$$

The surface is thus, near the origin, approximately a hyperbolic paraboloid (saddle surface), that is, the origin is a hyperbolic point of the surface. $\square$

EXAMPLE 4 Determine the principal curvatures and principal directions of the surface $S: z = xy - y^2$ at the point $P = (1, 1, 0)$.

Solution If $z(x, y) = xy - y^2$, then $z_x(1, 1) = 1$ and $z_y(1, 1) = -1$, so the unit normal vector of the surface at the point $(1, 1, 0)$ is

$$\vec{n} = \frac{1}{\sqrt{3}}(-\vec{i} + \vec{j} + \vec{k}).$$

Unit tangent vectors are

$$\vec{t}_1 = \frac{1}{\sqrt{2}}(\vec{i} + \vec{j}), \quad \vec{t}_2 = \vec{n} \times \vec{t}_1 = \frac{1}{\sqrt{6}}(-\vec{i} + \vec{j} - 2\vec{k}),$$

and $\{\vec{t}_1, \vec{t}_2, \vec{n}\}$ is an orthonormal, positively oriented system. Let us carry out the coordinate transformation $(x, y, z) \mapsto (\xi, \eta, \zeta)$ into the coordinate system $\{P, \vec{t}_1, \vec{t}_2, \vec{n}\}$:

$$\begin{bmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{6}} & -\frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{3}} \\ 0 & -\frac{2}{\sqrt{6}} & \frac{1}{\sqrt{3}} \end{bmatrix} \begin{bmatrix} \xi \\ \eta \\ \zeta \end{bmatrix} = \begin{bmatrix} x - 1 \\ y - 1 \\ z \end{bmatrix} \Leftrightarrow \begin{cases} x = \frac{\xi}{\sqrt{2}} - \frac{\eta}{\sqrt{6}} - \frac{\zeta}{\sqrt{3}} + 1, \\ y = \frac{\xi}{\sqrt{2}} + \frac{\eta}{\sqrt{6}} + \frac{\zeta}{\sqrt{3}} + 1, \\ z = -\frac{2\eta}{\sqrt{6}} + \frac{\zeta}{\sqrt{3}}. \end{cases}$$

Substituting into the equation $z = y(x - y)$ gives

$$\begin{aligned} -\frac{2}{\sqrt{6}}\eta + \frac{1}{\sqrt{3}}\zeta &= \left(\frac{1}{\sqrt{2}}\xi + \frac{1}{\sqrt{6}}\eta + \frac{1}{\sqrt{3}}\zeta + 1\right) \left(-\frac{2}{\sqrt{6}}\eta - \frac{2}{\sqrt{3}}\zeta\right) \\ \Leftrightarrow \quad 3\sqrt{3}\zeta + \sqrt{6}\xi\zeta + 2\sqrt{2}\eta\zeta + \zeta^2 + \sqrt{3}\xi\eta + \eta^2 &= 0. \end{aligned}$$

This implicitly defines a function $\zeta = \zeta(\xi, \eta)$, for which $\zeta(0, 0) = \zeta_\xi(0, 0) = \zeta_\eta(0, 0) = 0$. By implicit differentiation one can compute the second-order partial derivatives (cf. Exercise 3):

$$\zeta_{\xi\xi}(0, 0) = 0, \quad \zeta_{\xi\eta}(0, 0) = -\frac{1}{3}, \quad \zeta_{\eta\eta}(0, 0) = -\frac{2}{3\sqrt{3}}.$$

The curvature matrix in the chosen coordinate system is thus

$$\mathbf{A} = \begin{bmatrix} 0 & -\frac{1}{3} \\ -\frac{1}{3} & -\frac{2}{3\sqrt{3}} \end{bmatrix}.$$

The eigenvalues of $\mathbf{A}$ are

$$\lambda_1 = \frac{1}{3\sqrt{3}}, \quad \lambda_2 = -\frac{1}{\sqrt{3}},$$

and the eigenvectors lie on the lines

$$\lambda_1 : \xi + \sqrt{3}\eta = 0, \quad \lambda_2 : \sqrt{3}\xi - \eta = 0.$$

Direction vectors of these lines are

$$\begin{aligned} \vec{v}_1 &= \sqrt{3}\vec{t}_1 - \vec{t}_2 = \frac{2}{\sqrt{6}}(2\vec{i} + \vec{j} + \vec{k}), \\ \vec{v}_2 &= \vec{t}_1 + \sqrt{3}\vec{t}_2 = \sqrt{2}(\vec{j} - \vec{k}). \end{aligned}$$

Thus the principal curvatures and principal directions in the original coordinate system are

$$\begin{aligned} \kappa_1 &= \frac{1}{\underline{\underline{3\sqrt{3}}}}, \quad \text{direction } \underline{\underline{2\vec{i} + \vec{j} + \vec{k}}}, \\ \kappa_2 &= -\frac{1}{\underline{\underline{\sqrt{3}}}}, \quad \text{direction } \underline{\underline{\vec{j} - \vec{k}}}. \end{aligned}$$

The point P is a hyperbolic point of the surface. $\square$

EXERCISES

- Determine the principal curvatures and principal-curvature coordinate system of the surface at the given point:
 - $z = (x + y)(x - 2y)$, $(0, 0, 0)$
 - $z = 2 \sin xy + \cos(x + y)$, $(0, 0, 1)$
 - $z = 1 - x + 2y + e^{x-2y}$, $(0, 0, 2)$
 - $x^2 + 2y^2 - 2xy + z^2 = 1$, $(0, 0, -1)$
 - $\cos(x + y) + \sin(x + y + z) - \cos z = 0$, $(0, 0, 0)$
- The curve $K : y = 5 \sin x$ rotates in space about the x -axis. Determine the principal radii of curvature and principal directions of the resulting surface of revolution at the point
 - $(\pi/6, 5/2, 0)$,
 - $(\pi/2, 4, -3)$,
 - $(3\pi/4, -5/2, -5/2)$.

3. Consider a surface $S : F(x, y, z) = 0$, where the partial derivatives of F are continuous up to the second order. Show that if the plane $T : z = z_0$ is the tangent plane of S at the point $P = (x_0, y_0, z_0) \in S$ and $F_z(P) \neq 0$, then the curvature matrix of S at the point P is

$$\mathbf{A} = -\frac{1}{F_z(P)} \begin{bmatrix} F_{xx}(P) & F_{xy}(P) \\ F_{xy}(P) & F_{zz}(P) \end{bmatrix}.$$

4. (*) Determine the principal curvatures and principal-curvature coordinate system of the surface at the given point:
- a) $9x^2 + 4y^2 + z^2 = 14$, $(1, -1, 1)$ b) $2x^2 - 2y^2 - z^2 = 1$, $(1, 0, -1)$
c) $xy + 2yz - 3xz = 0$, $(1, 1, 1)$ d) $xy^2z^3 = 1$, $(1, 1, 1)$
5. (*) A line S_t moves in space according to a parameter $t \in \mathbb{R}$ so that the line passes through the point $(0, 0, 2t)$ and the direction vector of the line is $\vec{a}(t) = \cos t \vec{i} + \sin t \vec{j}$. Determine the principal curvatures and principal-curvature directions, at the origin, of the ruled surface generated by the line.

XII.4 *Tensors, vectors, and scalars

As noted above, the quadratic form $\mathbf{x}^T \mathbf{A} \mathbf{x}$ has the property that when the coordinate system is rotated the matrix of the quadratic form transforms by similarity:

$$\mathbf{x} \mapsto \mathbf{y} = \mathbf{C}^T \mathbf{x} \Rightarrow \mathbf{A} \mapsto \mathbf{B} = \mathbf{C}^T \mathbf{A} \mathbf{C}.$$

It has also been seen that the curvature matrix of a surface has this same property, when the rotation of the coordinate system takes place about the normal of the surface. In general, an object that has the form of a matrix in every coordinate system, and that transforms according to the aforementioned rule when the coordinate system is rotated, is called a *tensor*.* A quadratic form, seen through its matrix, is thus a tensor, in fact a *symmetric tensor*. The curvature of a surface is also a symmetric tensor, and instead of the curvature matrix one usually speaks of the *curvature tensor*. Speaking in this way, the curvature comes to be more clearly understood as a local geometric property of the surface, which merely appears different in different coordinate systems.

If $f = f(x, y, z)$ is twice continuously differentiable, then the Hessian matrix computed at a given point

$$\mathbf{A} = \begin{bmatrix} f_{xx} & f_{xy} & f_{xz} \\ f_{xy} & f_{yy} & f_{yz} \\ f_{xz} & f_{yz} & f_{zz} \end{bmatrix}$$

transforms, when the coordinate system is rotated, according to the aforementioned similarity rule. — This is seen by approximating f with the second-degree Taylor polynomial. One can then think that the 'second derivative' of f is not a matrix of second partial derivatives tied to any single coordinate system, but rather a tensor, of which these matrices are manifestations (!).

Also a linear map $A : \mathbb{R}^n \rightarrow \mathbb{R}^n$ can be understood as a tensor, for if

$$A : \mathbf{x} \mapsto \mathbf{y} \Leftrightarrow \mathbf{y} = \mathbf{A} \mathbf{x},$$

then one sees that under the coordinate transformation

$$\mathbf{x} = \mathbf{C} \mathbf{x}', \quad \mathbf{y} = \mathbf{C} \mathbf{y}' \quad (\mathbf{C}^{-1} = \mathbf{C}^T)$$

*The tensor is in mathematics a more general concept than the one presented here. Tensors are associated not only with rotations of a Cartesian coordinate system, but with transformations between more general curvilinear coordinate systems. The differential calculus associated with such transformations, so-called *tensor calculus*, is a branch of mathematics of its own. Tensors and tensor calculus have much use in physics.

the linear map takes the form

$$\mathbf{y}' = \mathbf{B}\mathbf{x}', \quad \mathbf{B} = \mathbf{C}^T \mathbf{A} \mathbf{C}.$$

EXAMPLE 1 Magnetic permeability. In electromagnetics, the simplest material law relating the magnetic flux density ($\vec{B}$) and the magnetic field strength ($\vec{H}$) is

$$\vec{B} = \mu \vec{H},$$

where the material constant μ is the magnetic permeability of the medium. This law is the same in all coordinate systems. More generally, if the material is anisotropic, the dependence of $\vec{B}$ on $\vec{H}$ may be a more general linear map of the form

$$\mathbf{B} = \boldsymbol{\mu} \mathbf{H} \quad \Leftrightarrow \quad \begin{bmatrix} B_1 \\ B_2 \\ B_3 \end{bmatrix} = \begin{bmatrix} \mu_{11} & \mu_{12} & \mu_{13} \\ \mu_{21} & \mu_{22} & \mu_{23} \\ \mu_{31} & \mu_{32} & \mu_{33} \end{bmatrix} \begin{bmatrix} H_1 \\ H_2 \\ H_3 \end{bmatrix}.$$

Then $\boldsymbol{\mu}$ is a tensor, that is, the matrix (μ_{ij}) transforms by similarity when the coordinate system is rotated. $\square$

EXAMPLE 2 Of a twice continuously differentiable function $f = f(x, y)$ it is known that $f_{xx}(0, 0) = 24$, $f_{yy}(0, 0) = -24$ and $f_{xy}(0, 0) = -7$. Compute $f_{\xi\xi}(0, 0)$, $f_{\xi\eta}(0, 0)$ and $f_{\eta\eta}(0, 0)$ in the coordinate system (ξ, η) whose basis vectors are

$$\vec{e}_\xi = \frac{1}{5}(4\vec{i} + 3\vec{j}), \quad \vec{e}_\eta = \frac{1}{5}(-3\vec{i} + 4\vec{j}).$$

Solution The coordinate transformation $(x, y) \mapsto (\xi, \eta)$ is determined by the condition

$$\begin{bmatrix} x \\ y \end{bmatrix} = \frac{1}{5} \begin{bmatrix} 4 & -3 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} \xi \\ \eta \end{bmatrix} = \mathbf{C} \begin{bmatrix} \xi \\ \eta \end{bmatrix}.$$

Since the second-order partial derivatives of f form a tensor,

$$\begin{aligned} \begin{bmatrix} f_{\xi\xi}(0, 0) & f_{\xi\eta}(0, 0) \\ f_{\xi\eta}(0, 0) & f_{\eta\eta}(0, 0) \end{bmatrix} &= \mathbf{C}^T \begin{bmatrix} f_{xx}(0, 0) & f_{xy}(0, 0) \\ f_{xy}(0, 0) & f_{yy}(0, 0) \end{bmatrix} \mathbf{C} \\ &= \frac{1}{25} \begin{bmatrix} 4 & 3 \\ -3 & 4 \end{bmatrix} \begin{bmatrix} 24 & -7 \\ -7 & -24 \end{bmatrix} \begin{bmatrix} 4 & -3 \\ 3 & 4 \end{bmatrix} \\ &= \begin{bmatrix} 0 & -25 \\ -25 & 0 \end{bmatrix}. \end{aligned}$$

Thus $f_{\xi\xi}(0, 0) = f_{\eta\eta}(0, 0) = 0$, $f_{\xi\eta} = -25$. — One function satisfying the assumptions is

$$f(x, y) = (4x + 3y)(3x - 4y) = -25\xi\eta. \quad \square$$

Vectors

If a tensor is an object that 'looks like a matrix' in every (Cartesian) coordinate system, then a vector of the plane or space can correspondingly be interpreted as an object whose manifestation is a pair or triple of numbers in every coordinate system. Like a tensor, this pair or triple transforms in a systematic way when the coordinate system is rotated. For example, if $\vec{v} = x\vec{i} + y\vec{j} + z\vec{k}$ is a space vector, then the transformation rule $\mathbf{x} \hat{=} (x, y, z) \mapsto (x', y', z') \hat{=} \mathbf{x}'$ on passing to a rotated coordinate system whose basis vectors have coordinates, in the basis $\{\vec{i}, \vec{j}, \vec{k}\}$, equal to the columns of the orthogonal matrix $\mathbf{C}$, is (cf. Section X.6)

$$\mathbf{C}\mathbf{x}' = \mathbf{x} \quad \Leftrightarrow \quad \mathbf{x}' = \mathbf{C}^T \mathbf{x}.$$

Thus not only the tensor's, but also the vector's 'innermost essence' is revealed only when the coordinate system is rotated (!). — One also observes that a symmetric tensor and a vector share the common feature that there exists a coordinate system, or coordinate systems, in which the manifestation of the tensor/vector as a matrix/element of $\mathbb{R}^n$ takes the simplest possible form. In the case of a symmetric tensor this simplest form is a diagonal matrix. The simplest form of a vector of the plane or space is a pair of numbers $(a, 0)$ or a triple $(a, 0, 0)$, where $a > 0$ is the length of the vector. On passing to such a coordinate system one in fact returns to the original geometric idea of a vector — for a vector was originally a directed line segment. The constancy of the direction of the vector as the coordinate system is rotated is ensured by the aforementioned transformation rule. According to it, then, the manifestation of the vector as an element of $\mathbb{R}^n$ is not constant, but precisely changes as the coordinate system is rotated, in the systematic way expressed by the transformation rule.

EXAMPLE 3 One can imagine (especially in the cases $n = 2$ and $n = 3$) that the Euclidean space $\mathbb{R}^n$ is the coordinate space of the point space E^n in some chosen (Cartesian) coordinate system. If now a chosen point $P \in E^n$ is represented by a vector $\mathbf{x}$ (position vector), then when the coordinate system is rotated the coordinates of P transform according to the formula $\mathbf{x}' = \mathbf{C}^T \mathbf{x}$. The transformation formula ensures that the point 'stays in place', that is, the position vector of a point is a vector precisely because of this requirement. Correspondingly, if one considers the transformation of a function $f(\mathbf{x})$ under a rotation of the coordinate system, then by the transformation rule $g(\mathbf{x}') = f(\mathbf{C}\mathbf{x}')$ one means that the 'function stays in place', i.e. the value of the function at a given (fixed) point does not change. $\square$

EXAMPLE 4 Let $f = f(x, y, z)$ be a differentiable function. Let us rotate the coordinate system by the transformation $\mathbf{C}\mathbf{x}' = \mathbf{x} \Leftrightarrow \mathbf{x}' = \mathbf{C}^T \mathbf{x}$ ($\mathbf{x} = [x, y, z]^T$, $\mathbf{x}' =$

$[x', y', z']^T$) and write $g(\mathbf{x}') = f(\mathbf{x}) = f(\mathbf{C}\mathbf{x}')$. Let $\nabla f(0, 0, 0) = a_1\vec{i} + a_2\vec{j} + a_3\vec{k}$. Then if $\mathbf{a} = [a_1, a_2, a_3]^T$, by the definition of differentiability we have

$$f(x, y, z) = f(0, 0, 0) + \mathbf{a}^T \mathbf{x} + o(|\mathbf{x}|).$$

By the chain rules, g too is differentiable at the origin, so for some $\mathbf{b} = [b_1, b_2, b_3]^T$ we have

$$g(x', y', z') = g(0, 0, 0) + \mathbf{b}^T \mathbf{x}' + o(|\mathbf{x}'|) = f(0, 0, 0) + \mathbf{b}^T \mathbf{x}' + o(|\mathbf{x}'|).$$

Assuming in these expansions that $\mathbf{C}\mathbf{x}' = \mathbf{x}$, then $|\mathbf{x}'| = |\mathbf{x}|$ and $g(x', y', z') = f(x, y, z)$, so we must have

$$\mathbf{a}^T \mathbf{x} = \mathbf{b}^T \mathbf{x}' = \mathbf{b}^T \mathbf{C}^T \mathbf{x} = (\mathbf{C}\mathbf{b})^T \mathbf{x} \Rightarrow \mathbf{C}\mathbf{b} = \mathbf{a}.$$

The result means that the gradient of a function (interpreted as a column vector) transforms like a vector when the coordinate system is rotated. The same result is also reached by computing the gradient of g directly by means of the chain rules from the formula $g(\mathbf{x}') = f(\mathbf{C}\mathbf{x}')$. $\square$

The gradient of a function has already earlier been interpreted as a vector, but then the interpretation was based only on the 'appearance' of the gradient in a fixed coordinate system. Now we can therefore confirm that the interpretation also withstands the rotation of the coordinate system:

The gradient *is* a vector.

Scalars

If the true nature of a tensor and of a vector is revealed only under a rotation of the coordinate system, then so it is with a scalar. A scalar is an object whose manifestation in every coordinate system is a number (here a real number), and indeed always the same number. For example, if $f = f(\mathbf{x})$, $\mathbf{x} \in \mathbb{R}^n$, is a real-valued function, then the value of f at a given (fixed) point is a scalar. The scalar associated with a vector is the length of the vector, since the length is a real number that does not change under a rotation of the coordinate system. The scalar associated with two vectors $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$ is (as its name says) the inner product, for if $\mathbf{x}' = \mathbf{C}^T \mathbf{x}$ and $\mathbf{y}' = \mathbf{C}^T \mathbf{y}$, then

$$(\mathbf{x}')^T \mathbf{y}' = (\mathbf{C}^T \mathbf{x})^T \mathbf{C}^T \mathbf{y} = \mathbf{x}^T \mathbf{C} \mathbf{C}^T \mathbf{y} = \mathbf{x}^T \mathbf{I} \mathbf{y} = \mathbf{x}^T \mathbf{y}.$$

An interesting scalar associated with tensors is the *trace* of the tensor:

DEFINITION XII.4.1 If T is a tensor of $\mathbb{R}^n$ whose matrix in a given orthonormal coordinate system is $\mathbf{A} = (a_{ij})$, then the *trace* of T is

$$\text{tr } T = \sum_{i=1}^n a_{ii}.$$

THEOREM XII.4.2 The trace of a tensor is a scalar.

Proof Let $\mathbf{C}$ be an orthogonal matrix of size $n \times n$ and $\mathbf{B} = \mathbf{C}^T \mathbf{A} \mathbf{C}$. The theorem claims that

$$\sum_{i=1}^n [\mathbf{B}]_{ii} = \sum_{i=1}^n [\mathbf{A}]_{ii}.$$

Let us start from the definition of matrix product and interchange the order of summation:

$$\begin{aligned} \sum_{i=1}^n [\mathbf{B}]_{ii} &= \sum_{i=1}^n \sum_{j=1}^n [\mathbf{C}^T]_{ij} [\mathbf{A} \mathbf{C}]_{ji} \\ &= \sum_{i=1}^n \sum_{j=1}^n [\mathbf{C}^T]_{ij} \sum_{k=1}^n [\mathbf{A}]_{jk} [\mathbf{C}]_{ki} \\ &= \sum_{j=1}^n \sum_{k=1}^n [\mathbf{A}]_{jk} \left(\sum_{i=1}^n [\mathbf{C}]_{ki} [\mathbf{C}^T]_{ij} \right). \end{aligned}$$

Here, by the orthogonality of $\mathbf{C}$,

$$\sum_{i=1}^n [\mathbf{C}]_{ki} [\mathbf{C}^T]_{ij} = [\mathbf{C} \mathbf{C}^T]_{kj} = [\mathbf{I}]_{kj} = \delta_{kj},$$

so

$$\sum_{i=1}^n [\mathbf{B}]_{ii} = \sum_{j=1}^n \sum_{k=1}^n [\mathbf{A}]_{jk} \delta_{kj} = \sum_{j=1}^n [\mathbf{A}]_{jj}. \quad \square$$

EXAMPLE 5 Let us consider the 'second derivative' of a function $u = u(x, y)$, that is, the tensor T whose matrix is

$$\mathbf{A} = \begin{bmatrix} u_{xx}(x, y) & u_{xy}(x, y) \\ u_{xy}(x, y) & u_{yy}(x, y) \end{bmatrix}.$$

When the coordinate system is rotated by the transformation $\mathbf{C}[\xi, \eta]^T = [x, y]^T$ and we write $v(\xi, \eta) = u(x(\xi, \eta), y(\xi, \eta))$, then the matrix of T in the rotated coordinate system is

$$\begin{bmatrix} v_{\xi\xi}(\xi, \eta) & v_{\xi\eta}(\xi, \eta) \\ v_{\xi\eta}(\xi, \eta) & v_{\eta\eta}(\xi, \eta) \end{bmatrix} = \mathbf{C}^T \begin{bmatrix} u_{xx}(x, y) & u_{xy}(x, y) \\ u_{xy}(x, y) & u_{yy}(x, y) \end{bmatrix} \mathbf{C}.$$

By Theorem XII.4.2 one has

$$(\partial_\xi^2 + \partial_\eta^2)v(\xi, \eta) = (\partial_x^2 + \partial_y^2)u(x, y). \quad \square$$

From the example one can draw the significant conclusion (valid more generally in $\mathbb{R}^n$ as well) that the Laplace operator Δ 'looks the same' in all Cartesian coordinate systems. Since Δ also operates on both scalars and vectors like a scalar, one can say, in two different senses:

Δ is a scalar operator.

EXERCISES

1. Of a twice continuously differentiable function $f(x, y)$ it is known that: $f_{xx}(0, 0) = 0$, $f_{yy}(0, 0) = -2$ and $f_{xy}(0, 0) = 2$. Compute $f_{\xi\xi}(0, 0)$ and $f_{\eta\eta}(0, 0)$ in a rotated coordinate system in which $f_{\xi\eta}(0, 0) = 0$.
2. In a certain material there is between the magnetic flux density and the magnetic field strength the dependence $\mathbf{B} = \boldsymbol{\mu}\mathbf{H}$, where $\boldsymbol{\mu}$ is a tensor whose matrix representation is

$$\boldsymbol{\mu} = \begin{bmatrix} \mu & 0 & 0 \\ 0 & 2\mu & 0 \\ 0 & 0 & 3\mu \end{bmatrix}$$

in the coordinate system whose basis vectors are

$$\vec{e}_1 = \frac{1}{5}(3\vec{i} + 4\vec{j}), \quad \vec{e}_2 = \frac{1}{25}(12\vec{i} - 9\vec{j} + 20\vec{k}), \quad \vec{e}_3 = \frac{1}{25}(16\vec{i} - 12\vec{j} - 15\vec{k}).$$

What is the representation of $\boldsymbol{\mu}$ in the basis $\{\vec{i}, \vec{j}, \vec{k}\}$ of the standard coordinate system?

3. Show that if $\vec{a}$ and $\vec{F}$ are vector fields (i.e. vector-valued functions), then
a) $\nabla \cdot \vec{F}$ is a scalar field, b) $(\vec{a} \cdot \nabla)\vec{F}$ is a vector field.
4. (*) a) Show that the differential operator

$$A = a\partial_x^2 + b\partial_y^2 + c\partial_x\partial_y,$$

where a , b and c are scalars ($a, b, c \in \mathbb{R}$), remains invariant under a rotation of the coordinate system of $\mathbb{R}^2$ precisely when $a = b$ and $c = 0$, that is, when $A = a\Delta$.

b) What does the partial differential equation $u_{xx} - u_{yy} = 0$ become in a rotated coordinate system whose basis vectors are

$$\vec{e}_1 = \frac{1}{\sqrt{2}}(\vec{i} + \vec{j}), \quad \vec{e}_2 = \frac{1}{\sqrt{2}}(-\vec{i} + \vec{j})?$$

c) If $\vec{a} = \vec{i} + \vec{j}$, then what form does the differential operator

$$(\vec{a} \cdot \nabla)^2 = (\partial_x)^2 + (\partial_y)^2 + 2\partial_x\partial_y$$

take in the rotated coordinate system of part b)?

Chapter XIII

Multivariable integral calculus

The content of this chapter consists of various extensions of the concept of the definite integral to more dimensions and to various geometric situations. The types of integrals to be considered are the *plane* and *volume integrals* associated with functions of two or more variables, the *line* or *curve integrals*, and the *surface integrals*. As applications we treat, among others, the computation of areas, volumes, arc lengths, *centroids*, *moments of inertia*, and so on.

In Section [XIII.1](#) we first examine the *measure-theoretic* foundations of plane integrals. It turns out that a plane integral, like the other types of integrals as well, is a matter of integration with respect to some *measure*. The measure associated with a plane integral is the *area measure* assigned to subsets of $\mathbb{R}^2$, which in Section [XIII.1](#) is defined more precisely as the *Jordan* measure. Besides the properties of this measure and its relation to the integral, Section [XIII.1](#) also considers the numerical computation of a plane integral 'directly from the definition'.

A central part of multivariable integral calculus is inevitably the classical computational technique for integrals, whose foundation is *Fubini's theorem* (Section [XIII.2](#)). The theorem reduces plane and volume integrals to successive, i.e. *iterated*, one-dimensional integrals, which under favorable conditions can be evaluated in closed form. The use of Fubini's theorem is examined in Sections [XIII.2–XIII.3](#).

In the classical computational technique for integrals, a *change of variable* often helps as well, for example passing to curvilinear coordinate systems in the plane or in space. The computational technique of change of variable is examined in Section [XIII.4](#).

The diverse applications of plane and volume integrals are surveyed in Section [XIII.5](#). Finally, in Sections [XIII.6–XIII.7](#) we examine the line and surface in-

tegrals associated with plane and space curves and with curved surfaces in $\mathbb{R}^3$, together with their applications.

XIII.1 Area measure and plane integrals

The determination of the area of a plane region by means of the definite integral was examined earlier in Section VIII.8. In this section the objects of study are the more general *plane integrals* of the form

$$\int_A f d\mu = I(f, A, \mu), \quad (\star)$$

where $A \subset \mathbb{R}^2$, f is a (two-variable) function defined on A , and μ is the *area measure* of $\mathbb{R}^2$. The integral $I(f, A, \mu)$ is more precisely a real number that depends on the function f , the set A , and the measure μ , as is apparent from the way it is read: the integral of f over A with respect to the measure μ . — This is a generalization of the already familiar definite integral; namely, as will be observed below, the notation $(\star)$ is valid also when f is an (integrable) function defined on the interval $A = [a, b]$, $I(f, A, \mu) = \int_a^b f(x) dx$ and μ is the *length measure* of $\mathbb{R}$.

Jordan area measure

Since the plane integral is defined relative to the area measure, the latter must first be defined. A *measure* is, more generally, a *set function* that associates to a *measurable* set (here a subset of $\mathbb{R}^2$) a real number:

$$\mu : \mathcal{M} \longrightarrow \mathbb{R}, \quad \mathcal{M} = \{\text{measurable sets } A \subset \mathbb{R}^2\}.$$

In what follows, the area measure is taken to be the *Jordan measure*^{*}, which will subsequently be defined so that the following three axioms hold. In the first of the axioms one posits the measure of a so-called *basic rectangle* oriented along the coordinate axes. The other two are axioms (or consequences of axioms) of more general *positive* measures as well.

A1. *Measure of a basic rectangle*: $T = \{(x, y) \in \mathbb{R}^2 \mid x \in [a_1, b_1] \ \& \ y \in [a_2, b_2]\}$:

$$\mu(T) = (b_1 - a_1)(b_2 - a_2).$$

A2. *Additivity*: $A, B \in \mathcal{M} \ \& \ A \cap B = \emptyset$

$$\Rightarrow A \cup B \in \mathcal{M} \ \& \ \mu(A \cup B) = \mu(A) + \mu(B).$$

^{*}The Jordan measure is also called the *Peano–Jordan* measure. The terms refer to the mathematicians **Camille Jordan** (French, 1838–1922) and **Giuseppe Peano** (Italian, 1858–1932).

A3. *Positivity*: $\mu(A) \geq 0 \quad \forall A \in \mathcal{M}$.

Note that, in accordance with axiom A1, the measure μ depends in principle on the chosen (Cartesian) coordinate system. For the branch of mathematics dealing with measures, *measure theory*, such a starting point is typical, and the independence of the measure from the coordinate system (if true) must then always be shown separately.

The empty set $\emptyset$ is always regarded as measurable, so that by axiom A2 one must have $\mu(\emptyset) = 0$. More generally, if $A \in \mathcal{M}$ and $\mu(A) = 0$, one says that A has *measure zero*. Sets of measure zero occupy an important place in measure theory.*

EXAMPLE 1 The line segments $A = \{a\} \times [b, c]$ and $B = [a, b] \times \{c\}$ parallel to the coordinate axes can be interpreted as special cases of a rectangle, so that by axiom A1 $\mu(A) = \mu(B) = 0$. $\square$

EXAMPLE 2 If $A = [a_1, b_1] \times [a_2, b_2]$ and $B = [b_1, c_1] \times [a_2, b_2]$, then $A \cup B = [a_1, c_1] \times [a_2, b_2]$, so that by axiom A1 we have $\mu(A \cup B) = \mu(A) + \mu(B)$. $\square$

In the latter example the additivity axiom A2 holds even though $A \cap B \neq \emptyset$. This is because (by the previous example) $A \cap B$ has measure zero. — More generally, in axiom A2 the condition $A \cap B = \emptyset$ may be weakened to the condition $\mu(A \cap B) = 0$ (see the definition of the Jordan measure below and Exercise 3c).

Integral over a bounded set

Let $A \subset \mathbb{R}^2$ be a bounded set, so that it is contained in some basic rectangle:

$$A \subset T = [a_1, b_1] \times [a_2, b_2]$$

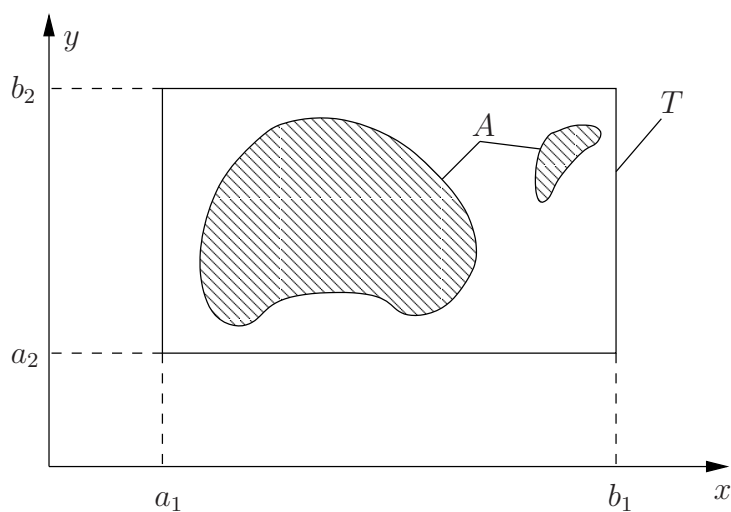
(see the figure below). Let f be a bounded function defined on A , and let $f_0 =$ the *zero extension* of f outside A :

$$f_0(x, y) = \begin{cases} f(x, y), & \text{if } (x, y) \in A, \\ 0, & \text{otherwise.} \end{cases}$$

Then one first agrees that

$$\boxed{\int_A f \, d\mu = \int_T f_0 \, d\mu, \quad A \subset T.}$$

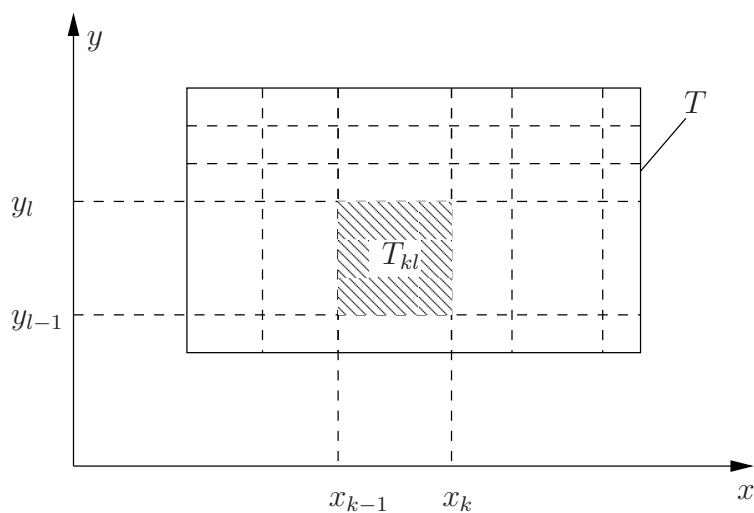
* Associated with measure zero is the peculiar mathematical term *almost everywhere*, which means: everywhere except on a (sub)set of measure zero.



As with the definite integral, the starting point for defining the integral $\int_T f_0 d\mu$ is a *partition* (subdivision) of T , in this case into subsets shaped like basic rectangles:

$$\mathcal{T}_h = \{T_{kl}, k = 1, \dots, m, l = 1, \dots, n\},$$

$$T_{kl} = [x_{k-1}, x_k] \times [y_{l-1}, y_l], \quad \begin{cases} a_1 = x_0 < x_1 < \dots < x_m = b_1, \\ a_2 = y_0 < y_1 < \dots < y_n = b_2. \end{cases}$$



The parameter h used in indexing the partition (i.e. the set of rectangles $\mathcal{T}_h$) is

again the *mesh parameter*. Its definition in two dimensions is

$$h = \max\{h_x, h_y\}, \quad h_x = \max_k \{x_k - x_{k-1}\}, \quad h_y = \max_l \{y_l - y_{l-1}\}.$$

The integral $\int_T f_0 d\mu$ is now defined as with the definite integral, i.e. by means of *Riemann sums* through a limit process (cf. Definitions VIII.5.3 and VIII.5.1)

$$\int_T f_0 d\mu = \lim_{h \rightarrow 0} \sum_{k=1}^m \sum_{l=1}^n f_0(\xi_{kl}, \eta_{kl}) \mu(T_{kl}),$$

where $(\xi_{kl}, \eta_{kl}) \in T_{kl}$. An alternative criterion for integrability, convenient especially in theoretical considerations, is obtained by defining, in connection with the partition $\mathcal{T}_h$,

$$\begin{aligned} - \text{upper sum : } \quad \bar{\sigma}_h(f_0, \mathcal{T}_h) &= \sum_{k=1}^m \sum_{l=1}^n M_{kl} \mu(T_{kl}), \quad M_{kl} = \sup_{(x,y) \in T_{kl}} f_0(x, y), \\ - \text{lower sum : } \quad \underline{\sigma}_h(f_0, \mathcal{T}_h) &= \sum_{k=1}^m \sum_{l=1}^n m_{kl} \mu(T_{kl}), \quad m_{kl} = \inf_{(x,y) \in T_{kl}} f_0(x, y). \end{aligned}$$

Then f is integrable over A exactly when (cf. Theorem VIII.5.8)

$$\sup_{\mathcal{T}_h} \underline{\sigma}_h(f_0, \mathcal{T}_h) = \underline{I}(f, \mu, A) = \bar{I}(f, \mu, A) = \inf_{\mathcal{T}_h} \bar{\sigma}_h(f_0, \mathcal{T}_h),$$

and the value of the integral = the common value $I(f, A, \mu)$ of these *lower* and *upper integrals*.

From both ways of defining it, one sees that $\int_T f_0 d\mu$ does not depend on the choice of the rectangle T as long as $T \supset A$ (Exercise 1). — This result was already anticipated above when we agreed: $\int_A f d\mu = \int_T f_0 d\mu$, $T \supset A$.

Linearity. Additivity. Comparison principle

If the set of functions integrable over the bounded set A is denoted by the symbol $\mathcal{R}_A$, then the following general properties are common to plane integrals and to one-dimensional definite integrals (see Theorems VIII.5.4–VIII.5.6). These also

hold for the other types of integrals to be defined later.

Linearity : $f, g \in \mathcal{R}_A \Rightarrow \alpha f + \beta g \in \mathcal{R}_A \quad \forall \alpha, \beta \in \mathbb{R} \quad \text{and}$

$$\int_A (\alpha f + \beta g) d\mu = \alpha \int_A f d\mu + \beta \int_A g d\mu.$$

Additivity : $f \in \mathcal{R}_A \quad \& \quad f \in \mathcal{R}_B \quad \& \quad A \cap B = \emptyset$

$$\Rightarrow \int_{A \cup B} f d\mu = \int_A f d\mu + \int_B f d\mu.$$

Comparison principle : $f, g \in \mathcal{R}_A \quad \& \quad f \leq g \quad \text{on } A \Rightarrow \int_A f d\mu \leq \int_A g d\mu.$

Linearity and the comparison principle follow directly from the definition, as in the one-dimensional case (see Section VIII.5). Additivity is most simply derived from linearity: When the zero extensions of f with respect to the sets A , B and $A \cup B$ are denoted respectively by the symbols f_A , f_B and f_0 , then

$$A \cap B = \emptyset \quad \Rightarrow \quad f_0 = f_A + f_B.$$

Consequently, if $T = [a_1, b_1] \times [a_2, b_2] \supset A \cup B$, then by linearity

$$\int_T f_0 d\mu = \int_T f_A d\mu + \int_T f_B d\mu \quad \Leftrightarrow \quad \int_{A \cup B} f d\mu = \int_A f d\mu + \int_B f d\mu.$$

The measure of a bounded set

In the aforementioned definitions the integral's dependence on the measure is quite simple, since the definitions require only the measures of the rectangles T_{kl} in accordance with axiom A1. But once the integral has been defined using the simplest properties of the measure, one can in fact define the measure of a more general set A by using the integral (!). The procedure is as follows: Consider the function

$$f(x, y) = 1, \quad (x, y) \in A.$$

The zero extension of this is called the *characteristic function* of A and is denoted

$$\chi_A(x, y) = \begin{cases} 1, & \text{if } (x, y) \in A, \\ 0, & \text{if } (x, y) \notin A. \end{cases}$$

As long as χ_A is Riemann integrable over the rectangle $T \supset A$, by the aforementioned definitions we have

$$\int_A d\mu = \int_T \chi_A d\mu, \quad T = [a_1, b_1] \times [a_2, b_2] \supset A.$$

DEFINITION XIII.1.1 (**Jordan measure**) A bounded set $A \subset \mathbb{R}^2$ is *Jordan measurable* exactly when the characteristic function χ_A of A is Riemann integrable over the rectangles $T = [a_1, b_1] \times [a_2, b_2] \supset A$, and the *Jordan measure* of A is then

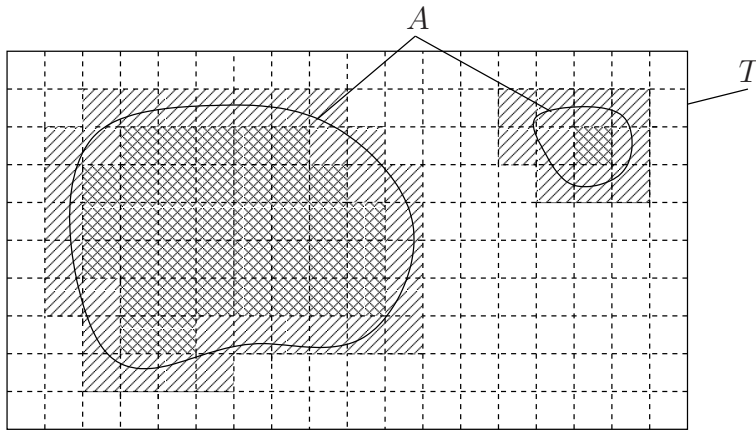
$$\mu(A) = \int_A d\mu.$$

The fulfillment of the measure axioms A1–A3 is evident from this definition and the aforementioned general properties of the integral.

Not all bounded sets are measurable (see Exercise 2a), but for every bounded set $A \subset \mathbb{R}^2$ one can define

$$\begin{aligned} \text{outer measure: } \bar{\mu}(A) &= \inf_{\mathcal{T}_h} \bar{\sigma}_h(\chi_A, \mathcal{T}_h), \\ \text{inner measure: } \underline{\mu}(A) &= \sup_{\mathcal{T}_h} \underline{\sigma}_h(\chi_A, \mathcal{T}_h), \end{aligned}$$

where inf and sup are computed over all partitions of the basic rectangle $T \supset A$. The set A is measurable exactly when $\bar{\mu}(A) = \underline{\mu}(A)$.



In the figure,

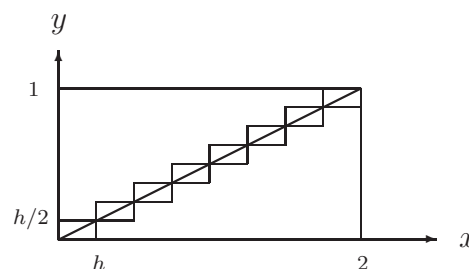
$$\begin{aligned} \mu(\text{cross-hatch}) &= \underline{\sigma}_h(\chi_A, \mathcal{T}_h) = \sum_{T_{kl} \in \mathcal{T}_h: T_{kl} \subset A} \mu(T_{kl}), \\ \mu(\text{diagonal lines}) &= \bar{\sigma}_h(\chi_A, \mathcal{T}_h) - \underline{\sigma}_h(\chi_A, \mathcal{T}_h). \end{aligned}$$

EXAMPLE 3 A = the line segment with endpoints $(0,0)$ and $(2,1)$. Show: $\mu(A) = 0$.

Solution Partition $T = [0, 2] \times [0, 1] \supset A$ into rectangles of size $h \times h/2$, where $2/h = n \in \mathbb{N}$. Then (cf. the figure)

$$\bar{\sigma}_h(\chi_A, \mathcal{T}_h) = \frac{2}{h} \cdot \frac{h^2}{2} = h.$$

Consequently $\bar{\mu}(A) \leq h = 2/n \forall n \in \mathbb{N}$, so $\bar{\mu}(A) = 0$ and hence $\mu(A) = 0$. $\square$



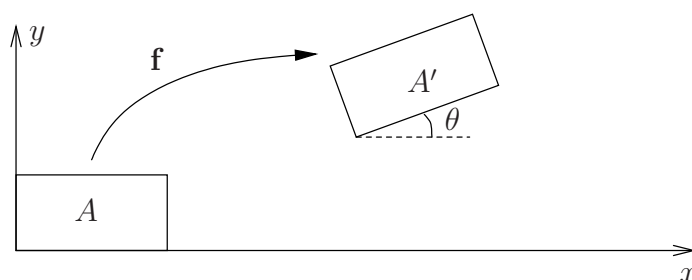
Let us mention one further theorem, whose partial proof is left as an exercise (Exercise 11). The theorem's continuity assumption (which is strengthened in the exercise) refers to Definition XI.1.10.

*THEOREM XIII.1.2 If f is continuous on the rectangle $T = [a, b] \times [c, d]$ and $A \subset T$ is measurable, then f is Riemann integrable over A .

Translation, reflection, and rotation invariance

In order for the Jordan measure to correspond to the ordinary area known from geometry, the measure must be *translation*-, *reflection*-, and *rotation-invariant*. By this is meant that if A is measurable and A' is (geometrically) *congruent* to A , i.e. $A' = \mathbf{f}(A)$, where for some $a, b, \theta \in \mathbb{R}$

$$\mathbf{f}(x, y) = (a + x \cos \theta \mp y \sin \theta, b + x \sin \theta \pm y \cos \theta),$$



then $\mu(A') = \mu(A)$. The map $\mathbf{f} : A \rightarrow A'$ is a special case of an affine transformation consisting of a *translation* (translation vector $\vec{r}_0 = a\vec{i} + b\vec{j}$), a *rotation* (angle of rotation θ), and a possible *reflection* (variation of sign). A combined translation and rotation (figure) is called a *Euclidean motion*.

Translation and reflection invariance are quite evident from the definition of the measure μ . With respect to rotation, on the other hand, one might suspect that the coordinate dependence of the definition would cause problems. In reality there are none, and the following holds (see Exercises 6,13)

THEOREM XIII.1.3 The Jordan area measure is translation-, reflection-, and rotation-invariant; that is, geometrically congruent sets have equal measures.

Midpoint rule

Like single-variable integrals, plane integrals too can be computed numerically directly from the definition, i.e. by approximating the integral with a Riemann sum:

$$\int_A f d\mu = \int_T f_0 d\mu \approx \sum_{k=1}^m \sum_{l=1}^n f_0(\xi_{kl}, \eta_{kl}) \mu(T_{kl}), \quad T \supset A.$$

If here one chooses (ξ_{kl}, η_{kl}) = the midpoint of T_{kl} , then the approximation is called the *composite midpoint rule*. As in one dimension (cf. Section VIII.9), this is generally the most accurate of the approximations based on Riemann sums.

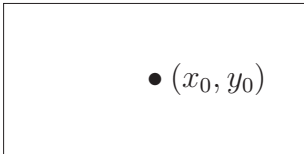
EXAMPLE 4 If $A = [a, b] \times [c, d]$, then the integral $\int_A x d\mu$ can be computed by partitioning A into $m \times n$ equal-sized rectangles and using the composite midpoint rule:

$$\begin{aligned} \int_A x d\mu &\approx \sum_{k=1}^m \sum_{l=1}^n \left[a + \frac{(k - \frac{1}{2})(b - a)}{m} \right] \frac{b - a}{m} \cdot \frac{c - d}{n} \\ &= (b - a)(c - d) \left[a + \frac{b - a}{m^2} \sum_{k=1}^m (k - \frac{1}{2}) \right] \\ &= (b - a)(c - d) \left[a + \frac{1}{2}(b - a) \right] = \frac{1}{2}(a + b)\mu(A). \end{aligned}$$

Since the result does not depend on the parameters m, n , one must have $\int_A x d\mu = \frac{1}{2}(a + b)\mu(A)$. Computing in the same way gives $\int_A y d\mu = \frac{1}{2}(c + d)\mu(A)$. $\square$

In the example an exact result was already obtained with a single rectangle ($m = n = 1$, $T_{11} = A$). More generally, the midpoint rule integrates a first-degree polynomial exactly over a rectangle:

$$\begin{aligned} f(x, y) &= a_0 + a_1x + a_2y \quad (a_0, a_1, a_2 \in \mathbb{R}) \\ \Rightarrow \int_T f d\mu &= f(x_0, y_0)\mu(T). \end{aligned}$$



On the basis of this property and the two-variable Taylor formula (Section XI.9), an error estimate for the midpoint rule can be derived in the same way as in one dimension (Exercise 15).

EXAMPLE 5 Let $A = [0, 1] \times [0, 1]$ and consider the integral $\int_A x^2 y^2 d\mu$. When A is partitioned into squares of size $h \times h$ and in each square the evaluation point is chosen as (a) the lower-left corner, (b) the midpoint, the following approximate values are obtained for the integral (exact value = $1/9$; the number of summed terms $N = h^{-2}$ is also given in the table):

h	N	Approx. (a)	Approx. (b)
0.1	10^2	0.081225000..	0.110556250..
0.01	10^4	0.107813722..	0.111105555..
0.001	10^6	0.110778138..	0.111111055..
0.0001	10^8	0.111077781..	0.111111110..

□

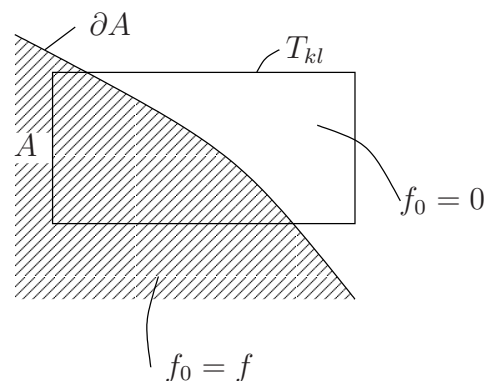
If the set A is a more general set than a rectangle, then the numerical approximation based on Riemann sums

$$\int_A f d\mu = \int_T f_0 d\mu \approx \sum_{k=1}^m \sum_{l=1}^n f_0(\xi_{kl}, \eta_{kl}) \mu(T_{kl})$$

is not necessarily very good, because the zero extension f_0 of f is (generally) discontinuous on those rectangles T_{kl} that intersect the boundary curve ∂A (figure). In that case the estimate

$$\int_{T_{kl}} f_0 d\mu \approx f_0(\xi_{kl}, \eta_{kl}) \mu(T_{kl})$$

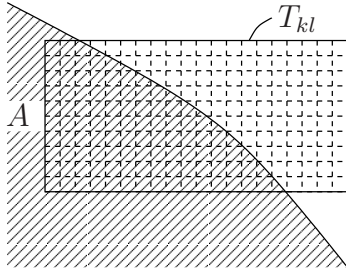
is as a rule poor, no matter how $(\xi_{kl}, \eta_{kl}) \in T_{kl}$ is chosen.



If one wishes to improve the algorithm, it is advisable to use more accurate approximations for the integrals

$$\int_{T_{kl}} f_0 d\mu = \int_{T_{kl} \cap A} f d\mu$$

when $T_{kl} \cap \partial A \neq \emptyset$. For example, the partition can be locally refined and the same algorithm applied again (with T_{kl} in place of T , and $A \cap T_{kl}$ in place of A).



The length measure of $\mathbb{R}$

In defining the (Jordan) *length measure* of $\mathbb{R}$, one writes in place of axiom A1

A1. *Measure of a closed interval:* $A = [a, b] : \mu(A) = b - a.$

The other axioms remain unchanged. If $A \subset \mathbb{R}$ is a bounded set, one defines

$$\int_A f d\mu = \int_T f_0 d\mu, \quad T = [a, b] \supset A,$$

where f_0 is the zero extension of f , and here further

$$\int_T f_0 d\mu = \lim_{h \rightarrow 0} \sum_{k=1}^n f_0(\xi_k) \mu(T_k),$$

where $T_k = [x_{k-1}, x_k]$, $\xi_k \in T_k$ and $a = x_0 < x_1 < \dots < x_n = b$. Since $\mu(T_k) = x_k - x_{k-1}$, here $\int_T f_0 d\mu = \int_a^b f_0(x) dx$ (Definition VIII.5.3). The definite (Riemann) integral is thus a matter of integration with respect to the length measure of $\mathbb{R}$. The Jordan measure of a bounded set $A \subset \mathbb{R}$ is defined as above, i.e. by means of the integral:

$$\mu(A) = \int_A d\mu.$$

EXAMPLE 6 If $A = [0, 1] \cup [2, 4]$, then by the additivity of the integral (or directly from the definition)

$$\int_A f d\mu = \int_{[0,1]} f d\mu + \int_{[2,4]} f d\mu = \int_0^1 f(x) dx + \int_2^4 f(x) dx.$$

In particular, $\mu(A) = \int_A d\mu = 1 + 2 = 3.$ $\square$

EXERCISES

1. Show that if T_1 and T_2 are basic rectangles containing the set $A \subset \mathbb{R}^2$, then so is $T_1 \cap T_2$. Conclude from this that the definition $\int_A f d\mu = \int_T f_0 d\mu$ gives the same result regardless of the choice of the basic rectangle $T \supset A$.
2. a) Show that the set $A = ([0, 1] \cap \mathbb{Q}) \times [0, 1]$ is not Jordan measurable.
b) Give an example of a set A for which $A \subset [0, 1] \times [0, 1]$, $\overline{\mu}(A) = 1$ and $\underline{\mu}(A) = 0.99$.
3. Prove the following claims concerning bounded sets $A, B \subset \mathbb{R}^2$:
a) If $A \subset B$, then $\underline{\mu}(A) \leq \underline{\mu}(B)$ and $\overline{\mu}(A) \leq \overline{\mu}(B)$.
b) If $\mu(B) = 0$, then $\underline{\mu}(A \cup B) = \underline{\mu}(A)$ and $\overline{\mu}(A \cup B) = \overline{\mu}(A)$.
c) If $\mu(A \cap B) = 0$, then $\underline{\mu}(A \cup B) = \underline{\mu}(A) + \underline{\mu}(B)$ and $\overline{\mu}(A \cup B) = \overline{\mu}(A) + \overline{\mu}(B)$.
4. Compute approximations to the Jordan outer and inner measures of the set $A = \{(x, y) \in \mathbb{R}^2 \mid x \in [0, 1] \text{ \& } 0 \leq y \leq x^2\}$ by partitioning $T = [0, 1] \times [0, 1]$ into rectangles of size $h \times h^2$ ($h^{-1} \in \mathbb{N}$), and on this basis the measure $\mu(A)$.
5. Let $A \subset \mathbb{R}^2$ be a bounded set and f and g defined and bounded on A . Show that if $f(x, y) \leq g(x, y) \forall (x, y) \in A$, then $\overline{I}(f, \mu, A) \leq \underline{I}(g, \mu, A)$.
6. a) Let A = the right triangle with vertices at the points $(0, 0)$, $(a, 0)$ and $(0, b)$. Show from the definition of the Jordan measure that $\mu(A) = \frac{1}{2}|a||b|$.

b) Appealing to the result of part (a) and the additivity of the measure, conclude that the measure of a rectangle is rotation-invariant.
7. Using the composite midpoint rule, compute the integral $\int_A (1 + x - 2y) d\mu$, where $A \subset \mathbb{R}^2$ is a T -shaped region whose corner points are $(2, 0)$, $(3, 0)$, $(3, 3)$, $(5, 3)$, $(5, 4)$, $(0, 4)$, $(0, 3)$ and $(2, 3)$.
8. Let $A = [-a/2, a/2] \times [-b/2, b/2]$. By partitioning A into equal-sized rectangles and using the composite midpoint rule, verify:

$$\int_A x^2 d\mu = \frac{1}{12} a^3 b, \quad \int_A y^2 d\mu = \frac{1}{12} a b^3, \quad \int_A |xy| d\mu = \frac{1}{16} a^2 b^2.$$

9. Let $A = [0, 1] \times [0, 1]$. Compute the following integrals by partitioning A into squares of size $h \times h$ ($h^{-1} \in \mathbb{N}$), using the composite midpoint rule, and computing the limit of the result as $h \rightarrow 0$.

$$\text{a) } \int_A xy^2 d\mu \quad \text{b) } \int_A x^2 y^2 d\mu \quad \text{c) } \int_A e^{x+y} d\mu$$

10. (*) In Section VIII.8 it was claimed that if

$$A = \{ (x, y) \in \mathbb{R}^2 \mid x \in [a, b] \ \& \ 0 \leq y \leq f(x) \},$$

where f is bounded, non-negative, and Riemann integrable on the interval $[a, b]$, then $\mu(A) = \int_a^b f(x) dx$. Prove the claim anew by showing that

$$\underline{\mu}(A) = \int_a^b f(x) dx, \quad \overline{\mu}(A) = \int_a^b f(x) dx$$

and by appealing to Theorem VIII.5.8.

11. (*) Prove the claim of Theorem XIII.1.2 when, instead of continuity, it is assumed that f satisfies the Lipschitz continuity condition

$$|f(x_1, y_1) - f(x_2, y_2)| \leq L(|x_1 - x_2| + |y_1 - y_2|), \quad (x_1, y_1), (x_2, y_2) \in T.$$

Hint: Compare the upper and lower sums $\overline{\sigma}(f_0, \mathcal{T}_h)$ and $\underline{\sigma}(f_0, \mathcal{T}_h)$ ($\mathcal{T}_h$ a partition of T).

12. (*) Let $A \subset \mathbb{R}^2$ be a bounded set and f defined on A . Define

$$f_+(x, y) = \begin{cases} f(x, y), & \text{if } f(x, y) > 0, \\ 0, & \text{otherwise.} \end{cases}$$

a) Show: $\overline{I}(f_+, A) - \underline{I}(f_+, A) \leq \overline{I}(f, A) - \underline{I}(f, A)$.

b) Conclude: If f is Riemann integrable over A , then so are f_+ , $f_- = f - f_+$ and $|f|$.

13. (*) Starting from the result of Exercise 6, prove: If $A \subset \mathbb{R}^2$ is bounded, then $\underline{\mu}(A)$ and $\overline{\mu}(A)$ are rotation-invariant. Conclude that if A is Jordan measurable, then $\mu(A)$ is rotation-invariant.
14. (*) For the integral $\int_A f d\mu$, where A is the triangle with vertices $(0, 0)$, $(1, 0)$ and $(0, 1)$, and $f(x, y) = x + 2y$, an approximate value is computed by partitioning $T = [0, 1] \times [0, 1] \supset A$ into squares of size $h \times h$ and computing $\int_T f_0 d\mu$ by the composite midpoint rule. a) Show that the result is an overestimate whose error $= \frac{3}{2}h + \mathcal{O}(h^2)$. b) What is the exact value of the integral?

15. (*) Let the partial derivatives of f up to second order be continuous on the rectangle $A = [-a/2, a/2] \times [-b/2, b/2]$. Using Taylor's theorem, the comparison principle for integrals, and the results of Exercise 8, verify the error estimate for the midpoint rule

$$\left| \int_A f \, d\mu - f(0,0) ab \right| \leq \frac{ab}{24} (M_{11}a^2 + M_{22}b^2) + \frac{1}{16} M_{12} a^2 b^2,$$

where M_{11} , M_{22} and M_{12} are the maximum values on A of the absolute values of the partial derivatives f_{xx} , f_{yy} and f_{xy} . Show further that if A is partitioned into rectangles of size $h_1 \times h_2$ ($a/h_1 \in \mathbb{N}$, $b/h_2 \in \mathbb{N}$), then the composite midpoint rule satisfies the error estimate

$$|E(f)| \leq \frac{ab}{24} \left(M_{11}h_1^2 + M_{22}h_2^2 + \frac{3}{2} M_{12} h_1 h_2 \right).$$

What is $E(f)$ exactly, if f is a second-degree polynomial and the (constant) values F_{11} , F_{22} and F_{12} of the partial derivatives f_{xx} , f_{yy} and f_{xy} are known?

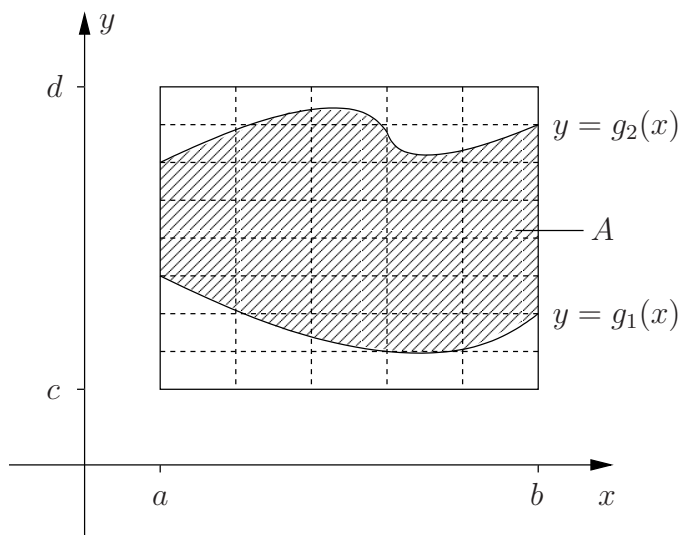
XIII.2 Computational technique for plane integrals

A plane integral can be reduced to single-variable integration problems, whereby under favorable conditions the integral can be evaluated using antiderivatives, i.e. in closed form. In this section we examine such a computational technique.

Consider a set $A \subset \mathbb{R}^2$ that is so-called *x-projectable*, i.e. of the form

$$A = \{(x, y) \in \mathbb{R}^2 \mid x \in [a, b], \ g_1(x) \leq y \leq g_2(x)\},$$

where it is assumed that g_1 and g_2 are bounded on the interval $[a, b]$. Let $T = [a, b] \times [c, d]$ be a rectangle containing A , and let $\mathcal{T}_h$ be some partition of T .



Let f be defined on T and bounded, and consider the Riemann sum defining the integral $\int_A f \, d\mu$. Let $\mathcal{T}_h = \{T_{kl}, \ k = 1 \dots m, \ l = 1 \dots n\}$ and choose

$$(\xi_{kl}, \eta_{kl}) = (\xi_k, \eta_l) \in T_{kl}, \quad k = 1 \dots m, \ l = 1 \dots n.$$

Then the sum takes the form

$$\sum_{T_{kl} \in \mathcal{T}_h} f_0(\xi_{kl}, \eta_{kl}) \mu(T_{kl}) = \sum_{k=1}^m \sum_{l=1}^n f_0(\xi_k, \eta_l) (x_k - x_{k-1})(y_l - y_{l-1}).$$

As long as the function $y \mapsto f_0(\xi_k, y)$ is Riemann integrable on the interval $[c, d]$, the inner sum has the limit

$$\sum_{l=1}^n f_0(\xi_k, \eta_l) (y_l - y_{l-1}) \xrightarrow{h \rightarrow 0} \int_c^d f_0(\xi_k, y) \, dy = \int_{g_1(\xi_k)}^{g_2(\xi_k)} f(\xi_k, y) \, dy.$$

When we denote

$$F(x) = \int_{g_1(x)}^{g_2(x)} f(x, y) dy, \quad x \in [a, b],$$

then one concludes that, as long as F is in turn Riemann integrable on the interval $[a, b]$, when the partition $\mathcal{T}_h$ is sufficiently fine we have approximately

$$\sum_{T_{kl} \in \mathcal{T}_h} f_0(\xi_k, \eta_l) \mu(T_{kl}) \approx \sum_{l=1}^n F(\xi_k)(x_k - x_{k-1}) \approx \int_a^b F(x) dx.$$

It thus seems evident that, when the functions g_1 , g_2 and f are sufficiently regular,

$$h \rightarrow 0 \Rightarrow \sum_{T_{kl} \in \mathcal{T}_h} f_0(\xi_k, \eta_l) \mu(T_{kl}) \rightarrow \int_a^b F(x) dx.$$

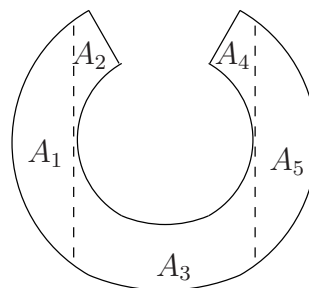
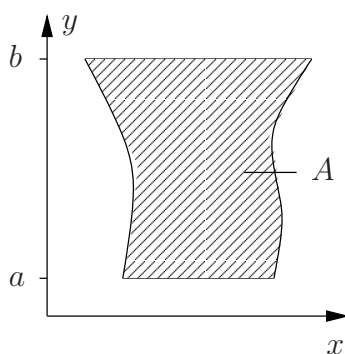
Thus we have derived (or at least made plausible) the computational formula

$$\begin{aligned} A &= \{(x, y) \in \mathbb{R}^2 \mid x \in [a, b], g_1(x) \leq y \leq g_2(x)\} \\ \Rightarrow \int_A f d\mu &= \int_a^b \left[\int_{g_1(x)}^{g_2(x)} f(x, y) dy \right] dx. \end{aligned} \tag{1}$$

According to this, then, a plane integral over an x -projectable set can be computed as successive one-dimensional, so-called *iterated integrals*. This is one form of the statement known as *Fubini's theorem*. The theorem is formulated more precisely and proved at the end of the section.

If $A \subset \mathbb{R}^2$ is y -projectable, i.e. (cf. the figure on the left)

$$A = \{(x, y) \in \mathbb{R}^2 \mid y \in [a, b], g_1(y) \leq x \leq g_2(y)\},$$



then in the integration formula (1) one must interchange x and y , whereby the formula takes the form

$$\int_A f \, d\mu = \int_a^b \left[\int_{g_1(y)}^{g_2(y)} f(x, y) \, dx \right] dy. \quad (2)$$

If A is of neither projectable type, then formulas (1) and (2) can generally be used once A is first divided into suitable parts (cf. the figure above) and the additivity of the integral is used:

$$\int_A f \, d\mu = \sum_{i=1}^m \int_{A_i} f \, d\mu, \quad A = \left(\bigcup_{i=1}^m A_i \right) \cup B, \quad \mu(A_i \cap A_j) = \mu(B) = 0, \quad i \neq j.$$

On the basis of the iteration formulas (1)–(2), a plane integral too can, under favorable conditions, be computed using antiderivatives. For example, let A be x -projectable and suppose that an antiderivative of the function $f(x, y)$ with respect to the variable y can be found, i.e. a function $F(x, y)$ for which

$$\frac{\partial}{\partial y} F(x, y) = f(x, y), \quad (x, y) \in A.$$

Then by formula (1)

$$\int_A f(x, y) \, d\mu = \int_a^b \left[\int_{y=g_1(x)}^{y=g_2(x)} F(x, y) \right] dx = \int_a^b G(x) \, dx,$$

where

$$G(x) = F(x, g_2(x)) - F(x, g_1(x)).$$

If an antiderivative H of the function G can further be found, then the integral can be computed in closed form:

$$\int f \, d\mu = \left|_a^b H(x) = H(b) - H(a).\right.$$

The success of the computation in this way thus depends not only on the function f , but also on the functions g_1 and g_2 , i.e. on the shape of the boundary curve of A . Success is guaranteed, for example, when f , g_1 and g_2 are polynomials, since then F , G and H are also polynomials.

With the iteration formulas (1)–(2) in mind, the following notations are used for plane integrals

$$\int_A f \, dx dy \quad \text{or} \quad \iint_A f \, dx dy.$$

In these one thus writes $d\mu = dxdy$, a notation that refers to the area of the 'differential rectangle' $[x, x + dx] \times [y, y + dy]$. This notation is used henceforth.

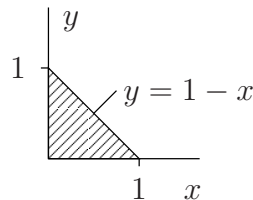
EXAMPLE 1 Let $A = [0, 1] \times [0, 1]$. Compute $\int_A \sqrt{|x - y|} dxdy$.

Solution By formula (1)

$$\begin{aligned}
 \int_A \sqrt{|x - y|} dxdy &= \int_0^1 \left[\int_0^1 \sqrt{|x - y|} dy \right] dx \\
 &= \int_0^1 \left[\int_0^x \sqrt{x - y} dy + \int_x^1 \sqrt{y - x} dy \right] dx \\
 &= \int_0^1 \left[-\frac{2}{3} \Big|_{y=0}^{y=x} (x - y)^{3/2} + \frac{2}{3} \Big|_{y=x}^{y=1} (y - x)^{3/2} \right] dx \\
 &= \int_0^1 \frac{2}{3} [x^{3/2} + (1 - x)^{3/2}] dx \\
 &= \Big|_0^1 \frac{4}{15} [x^{5/2} - (1 - x)^{5/2}] = \frac{8}{15}. \quad \square
 \end{aligned}$$

EXAMPLE 2 A = the triangle with vertices $(0, 0)$, $(1, 0)$ and $(0, 1)$. Compute

$$I = \int_A xy^2 dxdy.$$



Solution

$$\begin{aligned}
I &= \int_0^1 \left[\int_0^{1-x} xy^2 dy \right] dx \\
&= \int_0^1 \left[\int_{y=0}^{y=1-x} \frac{1}{3} xy^3 \right] dx \\
&= \int_0^1 \frac{1}{3} x(1-x)^3 dx \\
&= \left| -\frac{1}{12} x(1-x)^4 + \int_0^1 \frac{1}{12} (1-x)^4 dx \right|_0^1 \quad (\text{integration by parts}) \\
&= \int_0^1 \frac{1}{12} (1-x)^4 dx = \left| -\frac{1}{60} (1-x)^5 \right|_0^1 = \underline{\underline{\frac{1}{60}}}. \quad \square
\end{aligned}$$

EXAMPLE 3 Compute $\mu(A)$, where A is a disk of radius R .

Solution By formula (1)

$$\begin{aligned}
\mu(A) &= \int_A d\mu = \int_{-R}^R \left[\int_{-\sqrt{R^2-x^2}}^{\sqrt{R^2-x^2}} dy \right] dx \\
&= \int_{-R}^R \left[\int_{-\sqrt{R^2-x^2}}^{\sqrt{R^2-x^2}} y \right] dx = 2 \int_{-R}^R \sqrt{R^2-x^2} dx.
\end{aligned}$$

The substitution $x = R \sin t$, $dx = R \cos t dt$, $t \in [-\frac{\pi}{2}, \frac{\pi}{2}]$ gives

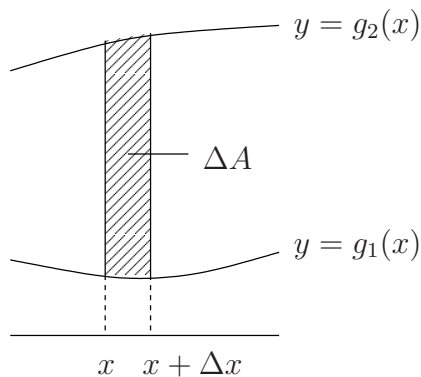
$$\begin{aligned}
2 \int_{-R}^R \sqrt{R^2-x^2} dx &= 2R^2 \int_{-\pi/2}^{\pi/2} \cos^2 t dt \\
&= R^2 \left| t + \sin t \cos t \right|_{-\pi/2}^{\pi/2} = \pi R^2. \quad \square
\end{aligned}$$

As a generalization of Example 3, one sees from formula (1) that the area measure of an x -projectable set can be computed by the formula

$$\mu(A) = \int_a^b [g_2(x) - g_1(x)] dx,$$

which is already familiar from Section VIII.8. Using the formula corresponds to computing the area by 'slicing' A into parts of the form $\Delta A = \{(x, y) \in$

$A \mid x \in [x, x + \Delta x]\}$, whereby $\mu(\Delta A) \approx [g_2(x) - g_1(x)]\Delta x$ for small values of Δx and assuming that g_1 and g_2 are continuous at the point x . (The continuity assumption can be weakened, cf. Section VIII.8.)



Improper plane integrals

If in the plane integral $\int_A f \, dx dy$ either A is not bounded or f is not bounded on A , the definition of the plane integral can be extended in a manner analogous to the case of the one-dimensional Riemann integral, cf. Section VIII.7. If the extension of the definition yields a unique (real) value for the integral, one again says that the *improper* integral so defined *converges*. Let us illuminate the matter first with examples.

EXAMPLE 4 Let $A = [0, 1] \times [0, 1]$ and $B = [0, \infty) \times [0, \infty)$. Compute

$$\text{a) } \int_A \frac{y^2}{\sqrt{x}} \, dx dy, \quad \text{b) } \int_B \frac{x}{1 + x^2 + y^2} \, dx dy.$$

Solution a) Since f is not bounded on A (not even defined when $x = 0$), this is an improper integral. It can (at least be attempted to) be computed as a limit

$$\int_A f \, dx dy = \lim_{\varepsilon \rightarrow 0^+} \int_{A_\varepsilon} f \, dx dy,$$

where $A_\varepsilon = [\varepsilon, 1] \times [0, 1]$. Here $\int_{A_\varepsilon} f \, dx dy$ is an ordinary Riemann integral, so the iteration formula (1) applies:

$$\begin{aligned} \int_{A_\varepsilon} f \, dx dy &= \int_\varepsilon^1 \left[\int_0^1 \frac{y^2}{\sqrt{x}} \, dy \right] dx = \int_\varepsilon^1 \frac{1}{\sqrt{x}} \left[\int_0^1 y^2 \, dy \right] dx \\ &= \int_\varepsilon^1 \frac{1}{\sqrt{x}} \left[\frac{y^3}{3} \right]_{y=0}^{y=1} dx = \int_\varepsilon^1 \frac{1}{3\sqrt{x}} \, dx = \left[\frac{2}{3} \sqrt{x} \right]_\varepsilon^1 \rightarrow \frac{2}{3}, \quad \text{when } \varepsilon \rightarrow 0^+. \end{aligned}$$

The integral thus converges. The correct value of the integral would also have been obtained by relying directly on formula (1), i.e. by bypassing the limit process:

$$\int_A f \, dx dy = \int_0^1 \left[\int_0^1 \frac{y^2}{\sqrt{x}} \, dy \right] dx = \int_0^1 \frac{1}{\sqrt{x}} \, dx \cdot \int_0^1 y^2 \, dy = 2 \cdot \frac{1}{3} = \frac{2}{3}.$$

b) This integral too converges, and the computation based directly on formula (2)

$$\begin{aligned} \int_A \frac{x}{(1+x^2+y^2)^2} \, dx dy &= \int_0^\infty \left[\int_0^\infty \frac{x}{(1+x^2+y^2)^2} \, dx \right] dy \\ &= \int_0^\infty \left[\left. \frac{1}{2(1+x^2+y^2)} \right|_{x=0}^{x=\infty} \right] dy \\ &= \int_0^\infty \frac{1}{2(1+y^2)} \, dy = \left. \frac{1}{2} \operatorname{Arctan} y \right|_{y=0}^{y=\infty} = \frac{\pi}{4} \end{aligned}$$

gives the correct result. One can verify this by first computing

$$\int_{T_{ab}} \frac{x}{(1+x^2+y^2)^2} \, dx dy = \int_0^b \left[\int_0^a \frac{x}{(1+x^2+y^2)^2} \, dx \right] dy,$$

where $T_{ab} = [0, a] \times [0, b]$, and observing that the value of this integral $\rightarrow \pi/4$ whenever $a \rightarrow \infty$ and $b \rightarrow \infty$. $\square$

On the basis of the example, formulas (1)–(2) appear to apply directly to improper integrals as well. Most often this is indeed the case in practice, but it is only a general rule, as the following example shows.

EXAMPLE 5 Using rules (1) and (2), compute the integral

$$I = \int_A f(x, y) \, dx dy, \quad A = [0, \infty) \times [0, 1], \quad f(x, y) = xy(2 - xy)e^{-xy}.$$

Solution

$$\text{Rule (1):} \quad I = \int_0^\infty \left[\left. xy^2 e^{-xy} \right|_{y=0}^{y=1} \right] dx = \int_0^\infty x e^{-x} \, dx = 1.$$

$$\text{Rule (2):} \quad I = \int_0^1 \left[\lim_{a \rightarrow \infty} \left. x^2 y e^{-xy} \right|_{x=0}^{x=a} \right] dy = \int_0^1 \left[\lim_{a \rightarrow \infty} a^2 y e^{-ay} \right] dy = 0. \quad \square$$

The contradiction in the example can be reduced to the following chain of reasoning, in which the inference marked with a question mark (and only that one!) is wrong:

$$\begin{aligned}
 1 &= \int_0^\infty \left[\int_0^1 f(x, y) dy \right] dx \\
 &= \lim_{a \rightarrow \infty} \int_0^a \left[\int_0^1 f(x, y) dy \right] dx \\
 &= \lim_{a \rightarrow \infty} \int_0^1 \left[\int_0^a f(x, y) dx \right] dy \\
 &= \int_0^1 \lim_{a \rightarrow \infty} \left[\int_0^a f(x, y) dx \right] dy \quad (?) \\
 &= \int_0^1 \left[\int_0^\infty f(x, y) dx \right] dy = 0.
 \end{aligned}$$

In the example the problem turns out to be the sign variation of the integrand: If this is allowed, the iteration formulas (1)–(2) do not necessarily give consistent results. The problem is resolved by removing the possibility of sign variation from the definition of the improper integral as follows:

Define the *positive part* f_+ and *negative part* f_- of the function $f(x, y)$ by setting, on the domain D_f of f ,

$$\begin{aligned}
 f_+(x, y) &= \begin{cases} f(x, y), & \text{if } f(x, y) > 0, \\ 0, & \text{otherwise,} \end{cases} \\
 f_-(x, y) &= \begin{cases} -f(x, y), & \text{if } f(x, y) < 0, \\ 0, & \text{otherwise.} \end{cases}
 \end{aligned}$$

Then if $A \subset D_f$ is a bounded set and f is Riemann integrable over A , then so are f_+ and f_- (Exercise XIII.1:12). Since also $f = f_+ - f_-$, it follows that

$$\int_A f d\mu = \int_A f_+ d\mu - \int_A f_- d\mu.$$

If A is not bounded or f is not bounded on A , then this is taken as the definition of the integral, i.e. one imposes the requirement that f_+ and f_- are both integrable over A . The possibility of sign variation of the function is thus removed from the extended definition of the integral.

EXAMPLE 5 (continued). Here

$$f_+(x, y) = \begin{cases} xy(2 - xy)e^{-xy}, & \text{if } 0 \leq xy \leq 2, \\ 0, & \text{otherwise,} \end{cases}$$

so that when $A = [0, \infty) \times [0, 1]$, we obtain

$$\begin{aligned} \int_A f_+ dx dy &= \int_0^1 \left[\int_0^\infty f_+(x, y) dx \right] dy \\ &= \int_0^1 \left[\int_0^{2/y} xy(2 - xy)e^{-xy} dx \right] dy \\ &= \int_0^1 \left[\int_{x=0}^{x=2/y} x^2 y e^{-xy} \right] dy = \int_0^1 4e^{-2} y^{-1} dy. \end{aligned}$$

The resulting integral does not converge, so f_+ is not integrable over A , and hence, by the definition, neither is f . $\square$

Let us note finally that even under the aforementioned definition, the iteration formulas (1)–(2) do not apply to improper integrals with complete generality, but the exceptions to these (general) rules are rather exotic, see Exercise 10.

Fubini's theorem

The aim is to justify the iteration formula (1) for the plane integral. Here it suffices to restrict to the special case where A is a rectangle, i.e. to the computational formula

$$\int_T f d\mu = \int_a^b \left[\int_c^d f(x, y) dy \right] dx, \quad T = [a, b] \times [c, d].$$

For if here

$$T \supset A = \{(x, y) \in \mathbb{R}^2 \mid x \in [a, b] \text{ \& } g_1(x) \leq y \leq g_2(x)\},$$

then this is formula (1), when one chooses $f = f_0 =$ the zero extension of f outside A .

THEOREM XIII.2.1 (Iteration formula for the plane integral – Fubini's* theorem) Let f be defined, bounded, and Riemann integrable on the rectangle $T = [a, b] \times [c, d]$. Suppose further that $f(x, y)$ is Riemann integrable with respect to the variable y on the interval $[c, d]$ for every $x \in [a, b]$, and with respect to the variable x on the interval $[a, b]$ for every $y \in [c, d]$. Then the functions

$$F(x) = \int_c^d f(x, y) dy, \quad G(y) = \int_a^b f(x, y) dx$$

*The Italian mathematician **Guido Fubini** lived from 1897 to 1943.

are Riemann integrable on the intervals $[a, b]$ and $[c, d]$, and

$$\int_T f(x, y) dx dy = \int_a^b F(x) dx = \int_c^d G(y) dy.$$

Proof Let $\mathcal{T}_h = \{T_{kl}, k = 1 \dots m, l = 1 \dots n\}$, $T_{kl} = [x_{k-1}, x_k] \times [y_{l-1}, y_l]$, be a partition of the rectangle T , and let

$$m_{kl} = \inf_{(x,y) \in T_{kl}} f(x, y), \quad M_{kl} = \sup_{(x,y) \in T_{kl}} f(x, y).$$

Then $m_{kl} \leq f(x, y) \leq M_{kl} \forall (x, y) \in T_{kl}$, so that by the comparison principle for integrals

$$m_{kl}(y_l - y_{l-1}) \leq \int_{y_{l-1}}^{y_l} f(x, y) dy \leq M_{kl}(y_l - y_{l-1}) \quad \forall x \in [x_{k-1}, x_k].$$

Summing this over the index l yields

$$m_k = \sum_{l=1}^n m_{kl}(y_l - y_{l-1}) \leq F(x) \leq \sum_{l=1}^n M_{kl}(y_l - y_{l-1}) = M_k \quad \forall x \in [x_{k-1}, x_k],$$

from which in turn follows

$$\begin{aligned} \sum_{k=1}^m \sum_{l=1}^n m_{kl}(x_k - x_{k-1})(y_l - y_{l-1}) &= \sum_{k=1}^m m_k(x_k - x_{k-1}) \\ &\leq \int_a^b F(x) dx \leq \overline{\int_a^b F(x) dx} \\ &\leq \sum_{k=1}^m M_k(x_k - x_{k-1}) \\ &= \sum_{k=1}^m \sum_{l=1}^n M_{kl}(x_k - x_{k-1})(y_l - y_{l-1}), \end{aligned}$$

i.e.

$$\underline{\sigma}(f, \mathcal{T}_h) \leq \int_a^b F(x) dx \leq \overline{\int_a^b F(x) dx} \leq \overline{\sigma}(f, \mathcal{T}_h),$$

where $\underline{\sigma}(f, \mathcal{T}_h)$ and $\overline{\sigma}(f, \mathcal{T}_h)$ are the Riemann lower and upper sums associated with the partition $\mathcal{T}_h$. Taking here the supremum on the left and the infimum on the right over all partitions $\mathcal{T}_h$, it follows by the definition of the plane integral (cf. the previous section)

$$\int_T f d\mu \leq \underline{\int_a^b F(x) dx} \leq \overline{\int_a^b F(x) dx} \leq \int_T f d\mu.$$

According to this, the upper and lower integrals of F on the interval $[a, b]$ are equal, so F is Riemann integrable on $[a, b]$ (Theorem VIII.5.8), and from the chain of inequalities the claimed (first) computational formula also follows. The second computational formula is justified similarly.

Let us mention that in Fubini's theorem the assumptions about the integrability of the function f separately with respect to the variables x and y are not necessarily needed; it suffices that f is integrable over T . In that case the functions F and G in the statement must be defined as either upper or lower integrals (any of the four options); otherwise the statement of the theorem retains its validity, and the proof too remains essentially the same (!). Thus if one assumes only that f is Riemann integrable over T , then, for example, the computational formula holds

$$\int_T f \, dx dy = \int_a^b \left[\int_c^d f(x, y) \, dy \right] dx.$$

Let us finally remark that, by Theorems VIII.6.1 and XIII.1.2, the conditions of Fubini's theorem are satisfied in the assumed form if f is continuous on the rectangle $T = [a, b] \times [c, d]$.

EXERCISES

- Let $f(x, y) = e^{x^4 - 2xy - y^2}$ and $A = [-1, 1] \times [0, 2]$. Compute $\int_A f_{xy}(x, y) \, dx dy$.
- Compute the integral of the given function $f(x, y)$ over the given set A :
 - $xy + y^2$, $A: 0 \leq x \leq 1 \text{ \& } 0 \leq y \leq 1 - x$
 - $(x + y)e^{x+y}$, $A: x, y \geq 0 \text{ \& } x + y \leq 1$
 - $\cos y$, $A: 0 \leq x \leq \pi \text{ \& } |y| \leq x$
 - $y^2 e^{xy}$, $A: 0 \leq x \leq 2 \text{ \& } x \leq y \leq 2$
 - xy^2 , $A: 0 \leq x \leq 1 \text{ \& } x^2 \leq y \leq \sqrt{x}$
 - $x \cos y$, $A: 0 \leq x \leq 1 \text{ \& } 0 \leq y \leq 1 - x^2$
 - $\sqrt{a^2 - y^2}$, $A =$ triangle with vertices $(0, 0)$, $(0, a)$ and (a, a) ($a > 0$)
 - $xy(1 + x^4)^{-1}$, $A =$ triangle with vertices $(0, 0)$, $(1, 0)$ and $(1, 1)$
- Determine the area $\mu(A)$, where A is defined by the given conditions ($a > 0$).
 - $ax \geq y^2$, $x^3 \leq ay^2$
 - $ax \geq y^2$, $x^2 + y^2 \leq a^2$
 - $a^2 y \leq x^3$, $y \geq 3x - 2a$
 - $x^2 - y^2 \leq a^2$, $|y| \leq x/2$
 - $x, y \geq 0$, $ax \geq y^2$, $ay \geq x^2$, $8xy^2 \leq a^3$

4. Determine and classify the critical points of the function

$$f(x, y) = \int_A (10 - xu - yv^2)^2 dudv, \quad A = [0, 1] \times [0, 1]$$

5. Does the integral $\int_A (x - y)^{-1} dx$ converge or diverge, where
a) $A = [0, 1] \times [0, 1]$, b) $A = [0, 1] \times [-1, 0]$?

6. Compute the following integrals for all values $\alpha \in \mathbb{R}$ for which the integral converges. a) $\int_A x^\alpha y^\alpha dx dy$, $A = [1, \infty) \times [1, \infty)$

b) $\int_A |x - 2y|^\alpha dx dy$, $A = [1, 2] \times [0, 1]$

c) $\int_A |x - 2y|^\alpha dx dy$, $A = [1, 2] \times [1, 2]$

d) $\int_A x^\alpha y e^{-xy} dx dy$, $A = [0, 1] \times [0, \infty)$

7. The following integrals are of the form $\int_A f(x, y) dx dy$. First determine A and then compute the value of the integral by choosing a suitable order of integration!

$$\begin{array}{ll} \text{a) } \int_0^1 \left[\int_x^1 e^{-y^2} dy \right] dx & \text{b) } \int_0^{\pi/2} \left[\int_y^{\pi/2} \frac{\sin x}{x} dx \right] dy \\ \text{c) } \int_0^1 \left[\int_{x^2}^x xy^{-1} e^y dy \right] dx & \text{d) } \int_0^1 \left[\int_x^1 \frac{y^\alpha}{x^2 + y^2} dy \right] dx \quad (\alpha > 0) \end{array}$$

8. It is known that $\int_{\mathbb{R}} e^{-x^2} dx = \sqrt{\pi}$.^{*} On this basis compute

$$\begin{array}{ll} \text{a) } \int_{\mathbb{R}^2} |x| e^{-x^2 - y^2} dx dy & \text{b) } \int_{\mathbb{R}^2} (x^2 + y^2) e^{-x^2 - y^2} dx dy \\ \text{c) } \int_{\mathbb{R}^2} |x + y| e^{-x^2 - y^2} dx dy & \text{d) } \int_{\mathbb{R}^2} |x - y^2| e^{-|x| - y^2} dx dy \end{array}$$

9. (*) a) Show by comparison with an integral that the 2-dimensional series

$$\sum_{i=1}^{\infty} \sum_{j=1}^{\infty} \frac{1}{(i+j)^\alpha}$$

converges exactly when $\alpha > 2$. b) Show that the error of the approximation

$$\sum_{i=1}^{\infty} \sum_{j=1}^{\infty} \frac{1}{(i+j)^\alpha} \approx \sum_{i=1}^N \sum_{j=1}^N \frac{1}{(i+j)^\alpha} \quad (\alpha > 2, N \in \mathbb{N})$$

^{*}See Proposition [XIII.4.1](#) below.

is of order $\mathcal{O}(N^{2-\alpha})$. c) Show that more precisely

$$\sum_{i=1}^{\infty} \sum_{j=1}^{\infty} \frac{1}{(i+j)^{\alpha}} = \sum_{i=1}^N \sum_{j=1}^N \frac{1}{(i+j)^{\alpha}} + \frac{2-2^{2-\alpha}}{(\alpha-2)(\alpha-1)} N^{2-\alpha} + \mathcal{O}(N^{1-\alpha}).$$

10. (*) (Riemann versus Fubini) Arrange the rational numbers in the interval $[0, 1]$ into a sequence $\{x_n, n = 0, 1, 2, \dots\}$ and define, on the set $A = [0, 1] \times [0, \infty)$, a function f as follows:

$$f(x, y) = \begin{cases} e^{n-y}, & \text{if } x = x_n \text{ and } y \geq n, \\ 0, & \text{otherwise.} \end{cases}$$

- a) Show that f is Riemann integrable over A (extended definition). b) Show that of the formulas (1)–(2), the latter works and the former does not, i.e.

$$\int_A f \, dx dy = \int_0^{\infty} \left[\int_0^1 f(x, y) \, dx \right] dy \neq \int_0^1 \left[\int_0^{\infty} f(x, y) \, dy \right] dx.$$

XIII.3 Volume integrals

An integral of the form

$$\int_A f d\mu, \quad A \subset \mathbb{R}^3, \quad f = f(x, y, z),$$

where μ is the *volume measure* of $\mathbb{R}^3$, is called a *volume integral*. The volume measure and integral are defined on the same principle as in the plane, with the sole difference that in place of the basic rectangle there is a rectangular basic box, whose measure is taken to be (the axiom of the volume measure)

$$T = [a_1, b_1] \times [a_2, b_2] \times [a_3, b_3] : \quad \mu(T) = (b_1 - a_1)(b_2 - a_2)(b_3 - a_3).$$

The volume measure defined as a Jordan measure (by means of outer and inner measures, cf. Section XIII.1) is translation-, reflection-, and rotation-invariant; that is, it is independent of the chosen (Cartesian) coordinate system.

Like plane integrals, volume integrals too can be computed directly, numerically, from the definition of the integral. If $\mathcal{T}_h$ is a partition of the box $T \supset A$ into sub-boxes whose edges are at most h in length, then computing the Riemann sum associated with the partition generally requires $N \sim h^{-3}$ arithmetic operations, whereas the corresponding number in the plane is $N \sim h^{-2}$ and in one dimension $N \sim h^{-1}$. In all cases the accuracy depends on the mesh parameter h in essentially the same way (for example, error $\sim h^2$), so that, in aiming for the same accuracy, the amount of work grows sharply as the dimension of the integral increases. (The same phenomenon afflicts all methods of numerical integration — and numerical computational methods more generally.) Of the methods based on Riemann sums, the most accurate is again the (composite) *midpoint rule*, which integrates a first-degree polynomial exactly in each sub-box.

EXAMPLE 1 Let $A = [0, 1] \times [0, 1] \times [0, 1]$ and consider computing $\int_A 3x^2y^2z^2 d\mu$. When A is partitioned into rectangular boxes of size $h \times h \times h$ and one integrates over each box by the midpoint rule, the sum to be computed has $N = h^{-3}$ terms. In this way the following approximate values are obtained for the integral (exact value = $1/9$, cf. Example XIII.1:5):

h	N	Approx.
0.1	10^3	0.110279859..
0.01	10^6	0.111102777..
0.001	10^9	0.111111027..
0.0001	10^{12}	0.111111110..

□

A volume integral too can be reduced to successive one-dimensional integrals, whereby under favorable conditions the integral can be computed in closed form. First, if $T = [a_1, b_1] \times [a_2, b_2] \times [a_3, b_3]$ is a rectangular box, and we denote $Q = [a_1, b_1] \times [a_2, b_2]$, then following the idea used in the proof of Fubini's theorem we obtain the iteration formula

$$\int_T f \, d\mu = \int_Q F(x, y) \, dx dy, \quad F(x, y) = \int_{a_3}^{b_3} f(x, y, z) \, dz.$$

When here we further use the computational formula of Fubini's theorem, the result is a triple iteration formula

$$\int_T f \, d\mu = \int_{a_1}^{b_1} \left\{ \int_{a_2}^{b_2} \left[\int_{a_3}^{b_3} f(x, y, z) \, dz \right] dy \right\} dx.$$

EXAMPLE 1 (continued) The integral of the example resolves into a product of three integrals:

$$\begin{aligned} \int_A f \, d\mu &= \int_0^1 \left\{ \int_0^1 \left[\int_0^1 3x^2 y^2 z^2 \, dz \right] dy \right\} dx \\ &= 3 \int_0^1 x^2 \left\{ \int_0^1 y^2 \left[\int_0^1 z^2 \, dz \right] dy \right\} dx \\ &= 3 \int_0^1 x^2 \, dx \int_0^1 y^2 \, dy \int_0^1 z^2 \, dz = 3 \cdot \frac{1}{3} \cdot \frac{1}{3} \cdot \frac{1}{3} = \frac{1}{9}. \quad \square \end{aligned}$$

If $A \subset \mathbb{R}^3$ is a more general set than a rectangular box, then one defines, as in the plane,

$$\int_A f \, d\mu = \int_T f_0 \, d\mu,$$

where f_0 is the zero extension of f to the rectangular box $T \supset A$. By applying the above iteration formula here, one obtains various formulas for resolving the integral, adapted to the shape of A . For example, if A is represented in the form

$$A = \{(x, y, z) \in \mathbb{R}^3 \mid x \in B \subset \mathbb{R} \text{ \& } (y, z) \in C(x) \subset \mathbb{R}^2\},$$

then one obtains the decomposition formula

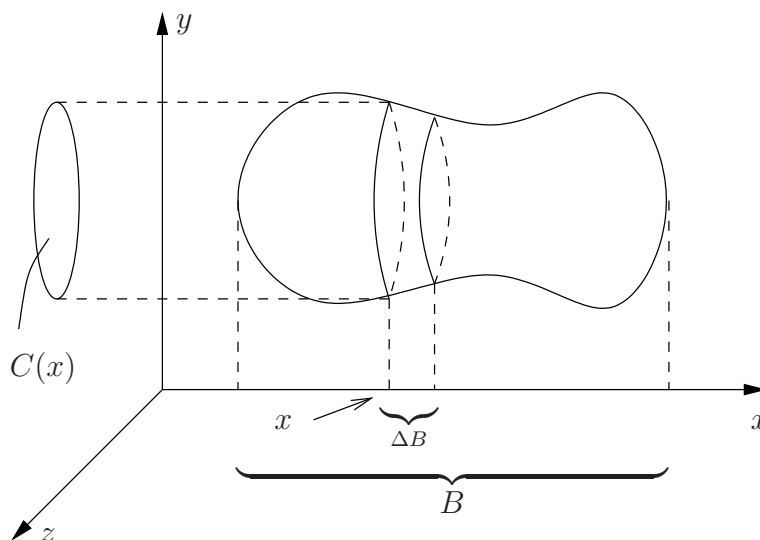
$$\int_A f \, d\mu = \int_B \left[\int_{C(x)} f(x, y, z) \, dy dz \right] dx, \quad (\star)$$

where the inner plane integral can be further decomposed by the methods of the previous section. The decomposition rule $(\star)$ can be illustrated geometrically as a 'slicing principle', in which A is divided into parts

$$\Delta A = \{(x, y, z) \in A \mid x \in \Delta B\},$$

whereby approximately (cf. the figure)

$$\int_{\Delta A} f d\mu \approx \left[\int_{C(x)} f(x, y, z) dydz \right] \mu(\Delta B) \quad (\mu = \text{length measure}).$$



Bearing in mind that a volume integral ultimately reduces to successive one-dimensional integrals, the following notations are often used for volume integrals

$$\int_A f dx dy dz \quad \text{or} \quad \iiint f dx dy dz,$$

where one has thus written $d\mu = dx dy dz$.*

EXAMPLE 2 A = the tetrahedron with vertices $(0, 0, 0)$, $(1, 0, 0)$, $(0, 1, 0)$ and $(0, 0, 1)$. Compute $\int_A f dx dy dz$, where $f(x, y, z) = xyz$.

Solution By the decomposition formula ($\star$), first of all

$$\int_A f dx dy dz = \int_0^1 \left[\int_{C(x)} xyz dy dz \right] dx,$$

where the set $C(x)$ in the yz -plane is determined by the conditions $y \geq 0$, $z \geq 0$ and

$$x + y + z \leq 1 \Leftrightarrow y + z \leq 1 - x,$$

*In applications (especially in physics) the measure notation for the volume integral is often dV .

i.e. $C(x)$ is the triangle with vertices at the points $(0, 0, 0)$, $(0, 1 - x, 0)$ and $(0, 0, 1 - x)$. Applying the decomposition rules of the previous section to the inner integral, we obtain

$$\begin{aligned}
 \int_A f \, dx dy dz &= \int_0^1 \left\{ \int_0^{1-x} \left[\int_0^{1-x-y} xyz \, dz \right] dy \right\} dx \\
 &= \int_0^1 \left\{ \int_0^{1-x} \left[\int_{z=0}^{z=1-x-y} \frac{1}{2} xyz^2 \, dz \right] dy \right\} dx \\
 &= \int_0^1 \left[\int_0^{1-x} \frac{1}{2} xy(1-x-y)^2 \, dy \right] dx \\
 (\text{int. by parts}) \quad &= \int_0^1 \left[\int_{y=0}^{y=1-x} -\frac{1}{6} xy(1-x-y)^3 + \int_0^{1-x} \frac{1}{6} x(1-x-y)^3 \, dy \right] dx \\
 &= \int_0^1 \left[\int_{y=0}^{y=1-x} -\frac{1}{24} x(1-x-y)^4 \right] dx \\
 &= \int_0^1 \frac{1}{24} x(1-x)^4 \, dx \\
 (\text{int. by parts}) \quad &= \left| -\frac{1}{120} x(1-x)^5 + \int_0^1 \frac{1}{120} (1-x)^5 \, dx \right|_0^1 \\
 &= \left| -\frac{1}{720} (1-x)^6 \right|_0^1 = \underline{\underline{\frac{1}{720}}}. \quad \square
 \end{aligned}$$

The decomposition rule $(\star)$ is especially convenient when f does not depend on the variables y, z . In that case the integral reduces directly to a one-dimensional one, provided that the area measure of the sets $C(x)$ is easily determined:

$$\int_A f \, dx dy dz = \int_B f(x) \mu(C(x)) \, dx, \quad f = f(x) \quad (\mu = \text{area measure}).$$

EXAMPLE 3 Compute the volume of a ball of radius R .

Solution We use the decomposition formula $(\star)$, whereby

$$B = [-R, R], \quad C(x) = \{(y, z) \in \mathbb{R}^2 \mid y^2 + z^2 \leq R^2 - x^2\}.$$

Since $C(x)$ is a disk of radius $\sqrt{R^2 - x^2}$, we have $\mu(C(x)) = \pi(R^2 - x^2)$. Thus

$$\mu(A) = \int_A d\mu = \int_{-R}^R \pi(R^2 - x^2) \, dx = \int_{-R}^R \pi \left(R^2 x - \frac{1}{3} x^3 \right) = \frac{4}{3} \pi R^3. \quad \square$$

EXAMPLE 4 Compute $\int_A x \, dx \, dy \, dz$, where $A = \{(x, y, z) \mid x^2 + y^2 + z^2 \leq R^2, x \geq 0\}$ (hemisphere).

Solution In the decomposition formula ($\star$), here $B = [0, R]$ and $C(x)$ is the same as in the previous example, so we obtain

$$\begin{aligned} \int_A f \, dx \, dy \, dz &= \int_B f(x) \mu(C(x)) \, dx \\ &= \int_0^R x \cdot \pi(R^2 - x^2) \, dx \\ &= \pi \left| \left(\frac{1}{2} R^2 x^2 - \frac{1}{4} x^4 \right) \right|_0^R = \underline{\underline{\frac{1}{4} \pi R^4}}. \quad \square \end{aligned}$$

EXAMPLE 5 The plane region bounded by the lines $x = 0$, $y = 0$ and the curve $y = e^{-x}$ is rotated about the x -axis, generating a set A in the space $\mathbb{R}^3$. Compute the volume $\mu(A)$.

Solution A is not bounded, so this is an improper integral. In formula ($\star$), $B = [0, \infty)$ and $C(x)$ is a disk of radius e^{-x} , so

$$\mu(A) = \int_A dx \, dy \, dz = \int_0^\infty \pi(e^{-x})^2 \, dx = \int_0^\infty \pi e^{-2x} \, dx = \left| -\frac{\pi}{2} e^{-2x} \right|_0^\infty = \underline{\underline{\frac{\pi}{2}}}. \quad \square$$

EXAMPLE 6 Two cylinders of radius R intersect each other perpendicularly. What is the volume measure of the set A lying inside both cylinders?

Solution Let the axis of one cylinder be the z -axis and that of the other the y -axis, whereby

$$A = \{(x, y, z) \in \mathbb{R}^3 \mid x^2 + y^2 \leq R^2 \text{ \& } x^2 + z^2 \leq R^2\}.$$

When this is cut by planes parallel to the yz -plane, one finds that

$$\begin{aligned} A &= \{(x, y, z) \in \mathbb{R}^3 \mid x \in [-R, R] \text{ \& } (y, z) \in C(x)\}, \quad \text{where} \\ C(x) &= \{(y, z) \in \mathbb{R}^2 \mid x^2 + y^2 \leq R^2 \text{ \& } x^2 + z^2 \leq R^2\} \\ &= \{(y, z) \in \mathbb{R}^2 \mid y^2 \leq R^2 - x^2 \text{ \& } z^2 \leq R^2 - x^2\}. \end{aligned}$$

The cross-section $C(x)$ is accordingly a square with side length $= 2\sqrt{R^2 - x^2}$. By formula ($\star$), then,

$$\mu(A) = \int_{-R}^R \mu(C(x)) \, dx = \int_{-R}^R 4(R^2 - x^2) \, dx = 4 \left| R^2 x - \frac{1}{3} x^3 \right|_{-R}^R = \underline{\underline{\frac{16}{3} R^3}}. \quad \square$$

Volume integrals in $\mathbb{R}^n$

A general n -dimensional volume integral is of the form $I(f, A, \mu) = \int_A f d\mu$, where $A \subset \mathbb{R}^n$, $f = f(\mathbf{x}) = f(x_1, \dots, x_n)$, and μ is the n -dimensional volume measure (Jordan measure). Among its properties is that the measure of an n -dimensional rectangular box is (axiom)

$$\mu(T) = \prod_{i=1}^n (b_i - a_i), \quad T = [a_1, b_1] \times [a_2, b_2] \times \cdots \times [a_n, b_n].$$

The alternative notations for the integral

$$\int_A f dx_1 \cdots dx_n, \quad \iiint \cdots \int f dx_1 \cdots dx_n.$$

again refer to the fact that the integral can be reduced to successive one-dimensional integrals. The basic form of the iteration formula in the case $A = T$ is

$$\int_T f d\mu = \int_{a_1}^{b_1} \left[\int_{a_2}^{b_2} \cdots \left[\int_{a_n}^{b_n} f(x_1, \dots, x_n) dx_n \right] \cdots \right] dx_1.$$

EXAMPLE 7 Compute the volume $\mu(A)$ of the four-dimensional ball of radius R .

Solution When we write

$$A = \{(x_1, x_2, x_3, x_4) \in \mathbb{R}^4 \mid x_1 \in [-R, R] \text{ \& } (x_2, x_3, x_4) \in C(x_1)\},$$

then (cf. the decomposition formula $(\star)$ above)

$$\int_A d\mu = \int_{-R}^R \mu(C(x_1)) dx_1,$$

where μ is the volume measure of $\mathbb{R}^3$. Here $C(x_1) \subset \mathbb{R}^3$ is a ball of radius $\sqrt{R^2 - x_1^2}$, so

$$\begin{aligned} \mu(A) &= \int_{-R}^R \frac{4}{3} \pi (R^2 - x_1^2)^{3/2} dx_1 \quad (\text{subst. } x_1 = R \sin t) \\ &= \frac{4\pi}{3} R^4 \int_{-\pi/2}^{\pi/2} \cos^4 t dt = \frac{4\pi}{3} R^4 \cdot \frac{3\pi}{8} = \underline{\underline{\frac{1}{2} \pi^2 R^4}}. \quad \square \end{aligned}$$

As for the theoretical properties of the n -dimensional Jordan measure, let us note only (without proof) that these properties are analogous to those in the plane. For example, the measure is translation- and rotation-invariant and can be defined by means of upper and lower integrals, cf. Section XIII.1. The following generalization of Theorem XIII.1.2 also holds:

*THEOREM XIII.3.1 If f is continuous on the box $T = [a_1, b_1] \times [a_2, b_2] \times \cdots \times [a_n, b_n]$ and $A \subset T$ is measurable, then f is Riemann integrable over A .

*The volume of a parallelepiped $K \subset \mathbb{R}^n$

Let us recall from Section X.7 that the n -dimensional parallelepiped in the space $\mathbb{R}^n$ with vertex at the point $\mathbf{x}_0$ and edge vectors $\mathbf{a}_i$ emanating from that vertex is the set

$$K = \{\mathbf{x} \in \mathbb{R}^n \mid \mathbf{x} = \mathbf{x}_0 + \sum_{i=1}^n t_i \mathbf{a}_i, \ t_i \in [0, 1], \ i = 1 \dots n\}.$$

If one writes $\mathbf{a}_i = \sum_{j=1}^n a_{ij} \mathbf{e}_j$ and forms the matrix $\mathbf{A}$ from the coefficients a_{ij} , then in the cases $n = 2$ and $n = 3$ it is known that the area ($n = 2$) or volume ($n = 3$) of K can be computed by the formula $\mu(K) = |\det \mathbf{A}|$ (see Section II.6). We show that this holds in general.

PROPOSITION XIII.3.2 The volume measure of the n -dimensional parallelepiped K in the space $\mathbb{R}^n$ is $\mu(K) = |\det \mathbf{A}|$, where the rows of the matrix $\mathbf{A}$ are the coefficients a_{ij} of the edge vectors $\mathbf{a}_i$ of K in the basis $\{\mathbf{e}_1, \dots, \mathbf{e}_n\}$.

Proof We take as known that the volume measure of $\mathbb{R}^n$ is rotation-invariant. Then the basis vector $\mathbf{e}_n$ can be chosen perpendicular to the hyperplane (subspace) spanned by the vectors $\mathbf{a}_i$, $i = 1 \dots n-1$, whereby $a_{in} = 0$ for $i = 1 \dots n-1$. The rotation also does not affect the value of the determinant $\det \mathbf{A}$, since the rotation corresponds to the transformation $\mathbf{A}^T = [\mathbf{a}_1, \dots, \mathbf{a}_n] \hookrightarrow [\mathbf{C}\mathbf{a}_1, \dots, \mathbf{C}\mathbf{a}_n] = \mathbf{C}\mathbf{A}^T$, i.e. $\mathbf{A} \hookrightarrow \mathbf{A}\mathbf{C}^T$, where $\det \mathbf{C} = 1$. In the assumed coordinate system the parallelepiped can be represented in the form

$$K = \{\mathbf{x} \in \mathbb{R}^n \mid \mathbf{x} = \mathbf{x}_0 + t_n a_{nn} \mathbf{e}_n + \mathbf{y}, \ t_n \in [0, 1] \ \& \ \mathbf{y} \in A(t_n)\},$$

where $A(t_n) \subset \mathbb{R}^{n-1}$ is the $(n-1)$ -dimensional parallelepiped whose edge vectors are $\mathbf{b}_i = \sum_{j=1}^{n-1} a_{ij} \mathbf{e}_j$, $i = 1 \dots n-1$, and whose vertex is at the point $\mathbf{y}_0(t_n) = t_n \sum_{j=1}^{n-1} a_{nj} \mathbf{e}_j$. Since the volume measure of $\mathbb{R}^{n-1}$ is translation-invariant, the measure of the set $A(t_n)$ is independent of t_n ; we denote $\mu(A(t_n)) = V_{n-1}$. Corresponding to the decomposition formula ($\star$), the measure of K is then

$$\mu(K) = \int_0^{|a_{nn}|} \left[\int_{A(t_n)} dx_1 \dots dx_{n-1} \right] dx_n = |a_{nn}| V_{n-1}.$$

If we denote $\mu(K) = V_n$, then we have obtained the recurrence formula

$$V_n = |a_{nn}| V_{n-1}.$$

On the other hand, if we denote $D_n = |\det \mathbf{A}|$ and note that $a_{in} = 0$ for $i \neq n$, then by the cofactor rule (Theorem X.5.5) we obtain the corresponding recurrence formula

$$D_n = |a_{nn}| D_{n-1}.$$

Since the recurrence formulas are of the same form and since $V_n = D_n$ for $n = 2, 3$, we conclude that $V_n = D_n \forall n$. $\square$

EXERCISES

1. Compute the volume $\mu(A)$, where $A \subset \mathbb{R}^3$ is defined by the given conditions.

- a) $0 \leq x \leq 1 \ \& \ 0 \leq y \leq x \ \& \ 0 \leq z \leq 1 - x^2$
- b) $0 \leq y \leq 1 \ \& \ 0 \leq x \leq y \ \& \ 0 \leq z \leq 1 - x^2$
- c) $x, y \geq 0 \ \& \ x + y \leq 1 \ \& \ 0 \leq z \leq 1 - x^2 - y^2$
- d) $1 \leq x \leq 2 \ \& \ 0 \leq y \leq x \ \& \ 0 \leq z \leq 1/(x + y)$
- e) $0 \leq z \leq 1 - x^2 - 2y^2$
- f) $0 \leq x \leq \sqrt[4]{\pi} \ \& \ 0 \leq y \leq x \ \& \ 0 \leq z \leq x^2 \sin(y^4)$
- g) $x^2 \leq z \leq 1 - y^2$
- h) $x^2 + y^2 \leq 8 \ \& \ y - 4 \leq z \leq 8 - x$

2. a)–e) In cases a)–e) of the previous exercise, compute the integrals $\int_A x \, d\mu$, $\int_A y \, d\mu$ and $\int_A z \, d\mu$.

3. Compute $\int_A f(x, y, z) \, dx dy dz$ for the given f and A :

- a) $f: xyz$, $A: 0 \leq x \leq 1 \ \& \ -2 \leq y \leq 0 \ \& \ 1 \leq z \leq 4$
- b) $f: x^2 + y^2 + z^2$, $A: 0 \leq x, y, z \leq a \ (a > 0)$
- c) $f: (1 - x - y)^5$, $A: x, y, z \geq 0 \ \& \ x + y + z \leq 1$
- d) $f: xyz^4$, $A: x, y, z \geq 0 \ \& \ x + y + z \leq 1$
- e) $f: 3 + 2xy$, $A: z \geq 0 \ \& \ x^2 + y^2 + z^2 \leq 4$
- f) $f: x$, $A: x, y, z \geq 0 \ \& \ (x/a) + (y/b) + (z/c) \leq 1 \ (a, b, c > 0)$
- g) $f: xy + z^2$, $A: 0 \leq z \leq 1 - |x| - |y|$
- h) $f: yz^2 e^{-xyz}$, $A: 0 \leq x, y, z \leq 1$
- i) $f: y$, $A: 0 \leq x, y, z \leq 1 \ \& \ 1 - y \leq z \leq 2 - x - y$
- j) $f: (x + y + z)^{-3}$, $A: 1 \leq z \leq 2 \ \& \ 0 \leq y \leq z \ \& \ 0 \leq x \leq y + z$
- k) $f: \cos x \cos y \cos z$, $A: x, y, z \geq 0 \ \& \ x + y + z \leq \pi$

4. Based on the representation of the set $A \subset \mathbb{R}^3$

$$A = \{(x, y, z) \in \mathbb{R}^3 \mid (x, y) \in B \subset \mathbb{R}^2 \ \& \ z \in C(x, y) \subset \mathbb{R}\}$$

derive the decomposition formula

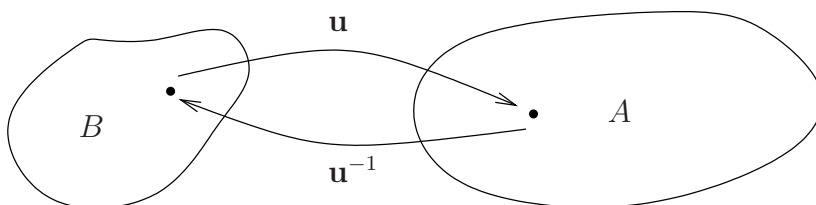
$$\int_A f \, d\mu = \int_B \left[\int_{C(x, y)} f(x, y, z) \, dz \right] dx dy.$$

Apply the formula when $A = \{(x, y, z) \in \mathbb{R}^3 \mid x, y, z \geq 0 \ \& \ x^2 + y + z \leq 1\}$ and $f(x, y, z) = xy^2$.

5. The tetrahedron $K \subset \mathbb{R}^3$ is bounded by the planes $z = 0$, $x = 2y$, $x = -y$ and $y + z = a$ ($a > 0$). Compute $\int_K z^2 dx dy dz$.
6. The base of a tent is a disk of radius $R = 2$ m lying in the xy -plane, and every cross-section of the tent parallel to the xz -plane is an isosceles triangle of height $h = 2$ m. Compute the volume of the tent.
7. Show that the volume of the set $A \subset \mathbb{R}^3$ lying inside the ellipsoid $S : x^2/a^2 + y^2/b^2 + z^2/c^2 = 1$ is $\mu(A) = 4\pi abc/3$ (cf. Exercise VIII.8:4).
8. Let $a, b > 0$, $a \neq b$. Compute the volume $\mu(A)$ of the set $A \subset \mathbb{R}^3$ lying inside the cylinders $S_1 : x^2 + y^2 = a^2$ and $S_2 : x^2 + z^2 = b^2$ as a one-dimensional integral. Can the integral be computed in closed form?
9. The set $\{(x, y, z) \in \mathbb{R}^3 \mid \sqrt{x^2 + y^2} \leq z \leq a\}$ represents the wine in a full wine glass. A ball of radius R is carefully lowered into the glass, causing wine to overflow, until the ball is fully submerged or comes to rest in the glass. For what value of the ratio R/a does the most wine overflow?
10. Let $A = [0, 1] \times [0, 1] \times \dots \times [0, 1] \subset \mathbb{R}^n$. Compute $\int_A f(\mathbf{x}) dx_1 \dots dx_n$, where
 - a) $f(\mathbf{x}) = \sum_{i=1}^n x_i$, b) $f(\mathbf{x}) = \prod_{i=1}^n x_i$, c) $f(\mathbf{x}) = e^{-(x_1 + x_2 + \dots + x_n)}$.
11. Let $K \subset \mathbb{R}^n$ be the n -simplex whose vertices are the origin and the points $(a, 0, \dots, 0)$, $(0, a, 0, \dots, 0)$, $\dots$, $(0, \dots, 0, a)$ ($a > 0$). Show that the n -dimensional volume of K is $\mu(K) = a^n/n!$.
12. The edge vectors of the parallelepiped $K \subset \mathbb{R}^4$ are $[1, 1, 1, 1]^T$, $[0, 1, 1, 0]^T$, $[0, 0, 1, 1]^T$ and $[1, 1, 0, 1]^T$. Compute the four-dimensional volume $\mu(K)$ of K .
13. (*) In the following integrals one integrates over some set $A \subset \mathbb{R}^3$. First determine A and compute the value of the integral by choosing a suitable order of integration.
 - a) $\int_0^1 \left[\int_z^1 \left(\int_0^x e^{x^3} dy \right) dx \right] dz$ b) $\int_0^1 \left[\int_0^{1-x} \left(\int_y^1 \frac{\sin(\pi z)}{z(2-z)} dz \right) dy \right] dx$
14. (*) The axes of three cylinders of radius R intersect one another perpendicularly at the same point. Show that the volume measure of the set A lying inside all three cylinders is $\mu(A) = 8(2 - \sqrt{2})R^3$.
15. (*) a) Compute the n -dimensional volume of the ball of radius R in $\mathbb{R}^n$ for $n = 5$ and $n = 6$. b) For what value of n is the volume of the unit ball ($R = 1$) the largest? Justify!

XIII.4 Transforming plane and space integrals

Let $A, B \subset \mathbb{R}^n$, $n \in \mathbb{N}$ and let $\mathbf{u} : B \rightarrow A$ be a bijection (or an 'almost bijection', see the remarks below).



We wish to compute the integral $\int_A f d\mu = \int_A f dx_1 \dots dx_n$ in the transformed form

$$\int_A f d\mu = \int_B g d\mu', \quad g(\mathbf{t}) = f(\mathbf{u}(\mathbf{t})), \quad \mathbf{t} \in B.$$

Here g is the counterpart of f on B mediated by the map $\mathbf{u}$, and μ' is another, as yet unknown measure on $\mathbb{R}^n$. This is thus a change of variables, i.e. a substitution

$$\mathbf{x} = \mathbf{u}(\mathbf{t}), \quad \mathbf{t} \in B.$$

In the transformation, one must first determine the function g . Insofar as the starting point is (as is usually the case) a known map $\mathbf{u}$, this is a straightforward operation. The formal definition of the measure μ' is also easy. Namely, if $\mathbf{u}(\Delta B) = \Delta A \subset A$ and we choose $f(\mathbf{x}) = 1$, $\mathbf{x} \in \Delta A$ (so that $g(\mathbf{t}) = 1$, $\mathbf{t} \in \Delta B$), then by the connection between measure and integral (cf. Section XIII.1) and the assumed connection between the integrals,

$$\mu'(\Delta B) = \int_{\Delta B} d\mu' = \int_{\Delta A} d\mu = \mu(\Delta A).$$

Thus $\mu'(\Delta B) = \mu(\Delta A)$, i.e. the measure of ΔB is determined as the Jordan measure of the corresponding set $\Delta A = \mathbf{u}(\Delta B)$. — Here it must be noted that the map $\mathbf{u}$ must be (at least almost) injective in order for μ' to really be a measure. Namely, since a measure is additive, for the measurable subsets of B we must have

$$\Delta B_1 \cap \Delta B_2 = \emptyset \quad \Rightarrow \quad \mu'(\Delta B_1 \cup \Delta B_2) = \mu'(\Delta B_1) + \mu'(\Delta B_2).$$

If $\mathbf{u}$ were not injective, then for some $\mathbf{x}_1, \mathbf{x}_2 \in B$, $\mathbf{x}_1 \neq \mathbf{x}_2$ we would have $\mathbf{u}(\mathbf{x}_1) = \mathbf{u}(\mathbf{x}_2)$, or more generally $\mathbf{u}(\Delta B_1) = \mathbf{u}(\Delta B_2) = \Delta A$, where $\Delta B_1, \Delta B_2 \subset B$ and

$\Delta B_1 \cap \Delta B_2 = \emptyset$. Then $\mu'(\Delta B_1 \cup \Delta B_2) = \mu(\Delta A)$ and $\mu'(\Delta B_1) + \mu'(\Delta B_2) = 2\mu(\Delta A) \neq \mu(\Delta A)$, unless $\mu(\Delta A) = 0$. We thus conclude that non-injectivity of $\mathbf{u}$ can be permitted at most in a set of measure zero (with respect to the measure μ').

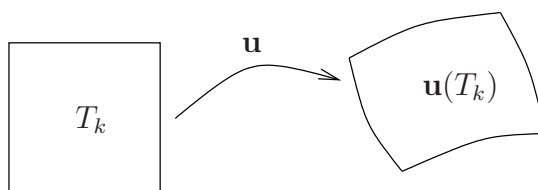
Insofar as $\mathbf{u}$ is injective, or 'almost' so in the manner mentioned, the question of transforming the integral is thus in principle solved. So far, however, it is not clear how one can compute in practice by means of the measure μ' (other than by returning to the original variables). This computational question is indeed the most central one, insofar as one wants to derive any benefit from the transformation.

The transformation ratio

Let us examine the transformed integral $\int_B g d\mu'$. Assume that B is bounded and $g(\mathbf{t})$ is bounded on B , so that (cf. Sections XIII.1 and XIII.3) $\int_B g d\mu' = \int_T g_0 d\mu'$, where $T \supset B$ is an n -dimensional rectangular basic box and g_0 = the extension of g by zero outside B . Assume in what follows that $\mathbf{u} : T \rightarrow \mathbf{u}(T)$ is (almost) injective and that $\mathbf{u}(T')$ is Jordan measurable whenever $T' \subset T$ is Jordan measurable. Then if $\mathcal{T}_h = \{T_k\}$ is a partition of T into subboxes whose edge lengths are at most h , then, analogously to integration with respect to the Jordan measure, the sum formula holds

$$\int_T g_0 d\mu' = \lim_{h \rightarrow 0} \sum_k g_0(\mathbf{t}_k) \mu'(T_k) = \lim_{h \rightarrow 0} \sum_k g_0(\mathbf{t}_k) \mu(\mathbf{u}(T_k)),$$

where $\mathbf{t}_k \in T_k$.



Assume now that for the map $\mathbf{u}$ one can define on T , as a continuous function $J(\mathbf{t})$, the so-called *transformation ratio* (measure ratio, magnification ratio) such

that for every $T_k \in \mathcal{T}_h$ and every $\mathbf{t}_k \in T_k$ we have*

$$\left| \frac{\mu(\mathbf{u}(T_k))}{\mu(T_k)} - J(\mathbf{t}_k) \right| \leq \varepsilon(h) \rightarrow 0 \quad \text{as } h \rightarrow 0,$$

where $\varepsilon(h)$ depends only on the density parameter h (not on k , nor more generally on $\mathcal{T}_h$).

On this basis, in the aforementioned sum formula one can write approximately $\mu(\mathbf{u}(T_k)) \approx J(\mathbf{t}_k)\mu(T_k)$. Assuming that $|g_0(\mathbf{t})| \leq C$, $\mathbf{t} \in T$ (g was bounded), for this approximation one obtains, by the assumptions made, the error estimate

$$\left| \sum_k g_0(\mathbf{t}_k)\mu(\mathbf{u}(T_k)) - \sum_k g_0(\mathbf{t}_k)J(\mathbf{t}_k)\mu(T_k) \right| \leq C\varepsilon(h) \sum_k \mu(T_k) = C\varepsilon(h)\mu(T).$$

Based on the assumption $\lim_{h \rightarrow 0} \varepsilon(h) = 0$ and applying the sum formula for the integral again, it follows that

$$\lim_{h \rightarrow 0} \sum_k g_0(\mathbf{t}_k)\mu(\mathbf{u}(T_k)) = \lim_{h \rightarrow 0} \sum_k g_0(\mathbf{t}_k)J(\mathbf{t}_k)\mu(T_k) = \int_T g_0(\mathbf{t})J(\mathbf{t}) d\mu.$$

Here furthermore $\int_T g_0(\mathbf{t})J(\mathbf{t}) d\mu = \int_B g(\mathbf{t})J(\mathbf{t}) d\mu$ (since $g_0J = (gJ)_0$), so we have concluded:

$$\int_A f d\mu = \int_B g d\mu' = \int_B g(\mathbf{t})J(\mathbf{t}) d\mu.$$

Thus, when transforming the integral $\int_A f d\mu$, in addition to the transformations $f(\mathbf{x}) = f(\mathbf{u}(\mathbf{t})) = g(\mathbf{t})$ and $A \hookrightarrow B$, $\mathbf{u}(B) = A$, one must also carry out the measure transformation $d\mu' = J(\mathbf{t}) d\mu$ based on the local transformation ratio $J(\mathbf{t})$; that is, the transformation formula is

$$\boxed{\int_A f(\mathbf{x}) dx_1 \dots dx_n = \int_B g(\mathbf{t})J(\mathbf{t}) dt_1 \dots dt_n.} \quad (\star)$$

EXAMPLE 1 Let $A = \{(x, y) \in \mathbb{R}^2 \mid x \in [1, 2] \text{ \& } x \leq y \leq 2x\}$. Compute $\int_A xy dx dy$ using the substitution $(x, y) = \mathbf{u}(\mathbf{t}) = \mathbf{u}(t, s) = (t, st)$.

Solution First we note that $A = \mathbf{u}(B)$, where $B = [1, 2] \times [1, 2]$, and that $\mathbf{u} : B \rightarrow A$ is a bijection. To compute the transformation ratio, consider the

*The definition of the transformation ratio can more generally be restricted to boxes for which $T_k \cap U = \emptyset$, where $\mu'(U) = 0$. Then the transformation ratio need not be continuous on all of T , or even defined there.

rectangle $\Delta T = [t, t + \Delta t] \times [s, s + \Delta s]$, where $t, s \geq 1$ and $0 < \Delta t, \Delta s \leq h$. Then

$$\begin{aligned}\Delta A &= \mathbf{u}(\Delta T) = \{(x, y) \in \mathbb{R}^2 \mid t \leq x \leq t + \Delta t \ \& \ sx \leq y \leq (s + \Delta s)x\} \\ \Rightarrow \mu(\Delta A) &= \int_t^{t+\Delta t} \left(\int_{sx}^{(s+\Delta s)x} dy \right) dx = \int_t^{t+\Delta t} (\Delta s)x \, dx \\ &= \frac{1}{2} \Delta s [(t + \Delta t)^2 - t^2] = t \Delta t \Delta s + \frac{1}{2} (\Delta t)^2 \Delta s.\end{aligned}$$

Since $\mu(\Delta A)/\mu(\Delta T) = t + \frac{1}{2}\Delta t = t + \mathcal{O}(h)$, then by the definition of the transformation ratio $J(t, s) = t$. Thus by formula $(\star)$

$$\int_A f(x, y) \, dx dy = \int_B f(t, st) t \, dt ds = \int_1^2 t^3 \, dt \cdot \int_1^2 s \, ds = \frac{15}{4} \cdot \frac{3}{2} = \frac{45}{8}.$$

Check (without a change of variables):

$$\int_A xy \, dx dy = \int_1^2 \left[\int_x^{2x} xy \, dy \right] dx = \int_1^2 \left[\frac{1}{2} xy^2 \right]_{y=x}^{y=2x} dx = \int_1^2 \frac{3}{2} x^3 \, dx = \frac{45}{8}. \quad \square$$

As in the example, the most central computational problem in transforming an integral is determining the transformation ratio. In what follows we solve this problem by means of differential calculus. Let us begin with the one-dimensional integral.

The transformation ratio in $\mathbb{R}$

Let $A = [a, b]$ and $B = [c, d]$ be closed intervals and $u : B \rightarrow A$ a bijection. Assume that u is continuously differentiable on the interval B . Then if $T_k = [t_{k-1}, t_k] \subset B$, where $t_k - t_{k-1} \leq h$, then for some $\xi_k \in T_k$ we have (Theorem V.6.1).

$$\mu(u(T_k)) = |u(t_k) - u(t_{k-1})| = |u'(\xi_k)|(t_k - t_{k-1}) = |u'(\xi_k)|\mu(T_k).$$

Here u' (and consequently also $|u'|$) is continuous, so $|u'(\xi_k)| - |u'(t)| = o(1) \forall t \in T_k$, as $h \rightarrow 0$.^{*} Consequently, for every $t \in T_k$

$$\mu(u(T_k)) = |u'(t)|\mu(T_k) + o(1)\mu(T_k), \quad \text{as } h \rightarrow 0.$$

^{*}The reasoning here relies on the deeper logic of continuity: under the assumptions made, $|u'|$ is uniformly continuous on the interval $[a, b]$ (Theorem V.9.13, see also Theorem V.9.12).

Thus by the definition of the transformation ratio $J(t) = |u'(t)|$, so that the transformation formula (★) takes the form

$$\int_a^b f(x) dx = \int_B f(u(t)) |u'(t)| dt, \quad u(B) = [a, b]. \quad (\text{a})$$

Since it was assumed that $u : B \rightarrow [a, b]$ is a continuous bijection, u is on the interval B either strictly increasing or strictly decreasing. Then if $u(\alpha) = a$ and $u(\beta) = b$,

$$\begin{aligned} u \text{ increasing } (u'(x) \geq 0) &\Rightarrow B = [\alpha, \beta], \\ u \text{ decreasing } (u'(x) \leq 0) &\Rightarrow B = [\beta, \alpha]. \end{aligned}$$

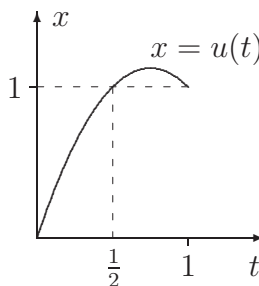
In both cases formula (a) can be written in the form

$$\int_a^b f(x) dx = \int_B f(u(t)) |u'(t)| dt = \int_\alpha^\beta f(u(t)) u'(t) dt. \quad (\text{b})$$

This computational formula (based on the interchange rule for the definite integral) concerns substitution in a definite integral, cf. Section VIII.6. Formula (b) is valid (under certain conditions, see the section mentioned) even if u is not a bijection. Formula (a), on the other hand, is not valid if u' changes sign on the interval B .

EXAMPLE 2 If in the integral $\int_0^1 x dx$ we make the substitution $x = u(t) = 3t - 2t^2$, then $u'(t) = 3 - 4t$, $u(0) = 0$, $u(1) = 1$, and the substitution formula (b) for the definite integral is valid:

$$\begin{aligned} \int_0^1 x dx &= \int_0^1 f(u(t)) u'(t) dt \\ &= \int_0^1 (9t - 18t^2 + 8t^3) dt \\ &= \left| \frac{9}{2}t^2 - 6t^3 + 2t^4 \right|_0^1 = \frac{1}{2}. \end{aligned}$$



Formula (a), on the other hand, gives a wrong result:

$$\int_0^1 f(u(t)) |u'(t)| dt = \int_0^{3/4} (9t - 18t^2 + 8t^3) dt - \int_{3/4}^1 (9t - 18t^2 + 8t^3) dt = \frac{49}{64}.$$

Both ways give the correct result if the transformed interval of integration is chosen to be $B = [0, \frac{1}{2}]$, so that $u : B \rightarrow [0, 1]$ is a bijection (cf. the figure). $\square$

The example shows that the transformation formula $(\star)$ for the definite integral generally gives a wrong result if the transformation map is not injective. The problem is more troublesome in several dimensions, where non-injectivity is not always easy to detect. In several dimensions there is also no simple shortcut, comparable to the substitution formula for the definite integral, for eliminating the problem.

The transformation ratio in $\mathbb{R}^n$

Let us first consider a two-dimensional transformation

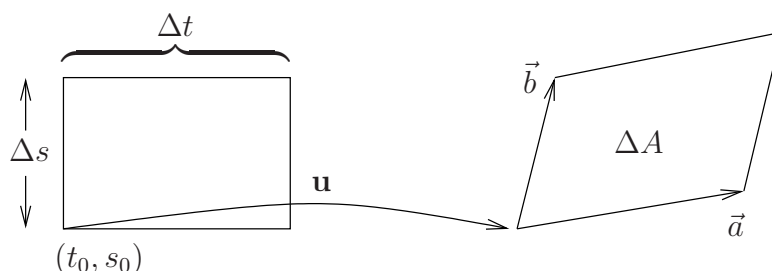
$$\begin{cases} x = u(t, s), \\ y = v(t, s), \end{cases} \quad \mathbf{t} = (t, s) \in T, \quad \mathbf{u} = (u, v),$$

where T is a basic rectangle. Assume that u and v are continuously differentiable on T (the partial derivatives exist and are continuous on T , Definition XI.1.10). Then the way the rectangle $\Delta T = [t_0 + \Delta t] \times [s_0 + \Delta s] \subset T$ is mapped can be studied approximately with the linearizing approximation (cf. Section XI.6)

$$\begin{cases} x(t, s) \approx u(t_0, s_0) + u_t(t_0, s_0)(t - t_0) + u_s(t_0, s_0)(s - s_0), \\ y(t, s) \approx v(t_0, s_0) + v_t(t_0, s_0)(t - t_0) + v_s(t_0, s_0)(s - s_0). \end{cases}$$

This is an affine map, according to which the rectangle is mapped to a parallelogram. The parallelogram is spanned by the vectors

$$\begin{aligned} \vec{a} &= \Delta t [u_t(t_0, s_0)\vec{i} + v_t(t_0, s_0)\vec{j}], \\ \vec{b} &= \Delta s [u_s(t_0, s_0)\vec{i} + v_s(t_0, s_0)\vec{j}]. \end{aligned}$$



When we define the Jacobian matrix of the map $\mathbf{u}$ (cf. Section XI.6)

$$\mathbf{J}\mathbf{u}(t, s) = \begin{bmatrix} u_t & u_s \\ v_t & v_s \end{bmatrix},$$

then we see that the area of the parallelogram is

$$\mu(\Delta A) = |\vec{a} \times \vec{b}| = |\det[\mathbf{J}\mathbf{u}(t_0, s_0)]| |\Delta t| |\Delta s|.$$

Since $\mu(\Delta T) = |\Delta t| |\Delta s|$, the transformation ratio is therefore $J(t, s) = |\det[\mathbf{J}\mathbf{u}(t, s)]|$. Under the assumed regularity conditions this turns out to be the correct result (we omit the more detailed justification), so the transformation ratio for the plane integral is obtained by means of the *Jacobian determinant* $\det[\mathbf{J}\mathbf{u}(t, s)]$:

Transformation ratio = absolute value of the Jacobian determinant of the map $\mathbf{u}$.

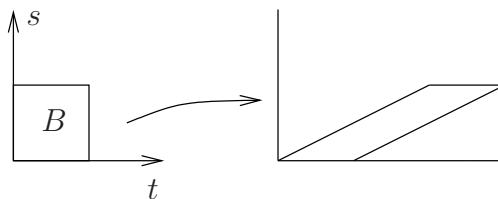
The above result is valid also more generally for the n -dimensional space integral. Namely, if $\mathbf{u} : T \rightarrow \mathbf{u}(T)$ is a nonlinear map ($T \subset \mathbb{R}^n$ a basic box), then its linearization at the point $\mathbf{t}$ is an affine map of $\mathbb{R}^n$ that maps a rectangular box to a parallelepiped. The determinant formed from the edge vectors of this parallelepiped = the determinant of the Jacobian (matrix) of $\mathbf{u}$ at the point $\mathbf{t}$. Since the absolute value of the determinant is the volume of the parallelepiped (Proposition XIII.3.2), the above rule is valid. Thus, when transforming a general n -dimensional integral, the transformation formula for the measure is

$$d\mu = |\det(\mathbf{J}\mathbf{u})| dt_1 \cdots dt_n.$$

EXAMPLE 3 A = parallelogram with vertices $(0, 0)$, $(1, 0)$, $(2, 1)$ and $(3, 1)$. Compute $\int_A xy \, dx dy$.

Solution We make the change of variables

$$\begin{cases} x = u(t, s) = t + 2s, \\ y = v(t, s) = s, \end{cases}$$



so that $\mathbf{u} : B \rightarrow A$, $B = [0, 1] \times [0, 1]$ is a bijection and

$$\begin{aligned} \mathbf{J} &= \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} \Rightarrow J = |\det \mathbf{J}| = 1 \\ \Rightarrow \int_A xy \, dx dy &= \int_0^1 \left[\int_0^1 (t + 2s)s \, ds \right] dt \\ &= \int_0^1 \left(\frac{1}{2}t + \frac{2}{3} \right) dt = \underline{\underline{\frac{11}{12}}}. \quad \square \end{aligned}$$

EXAMPLE 4 Compute the integral

$$\int_A e^{-x-y-z} \, dx dy dz, \quad A = \{(x, y, z) \in \mathbb{R}^3 \mid x+y \geq 0 \ \& \ y+z \geq 0 \ \& \ x+z \geq 0\}.$$

Solution We make a change of variables suited to the shape of A

$$\begin{cases} u = x + y \\ v = y + z \\ w = x + z \end{cases} \Leftrightarrow \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 1 & -1 & 1 \\ 1 & 1 & -1 \\ -1 & 1 & 1 \end{bmatrix} \begin{bmatrix} u \\ v \\ w \end{bmatrix},$$

so that the transformation ratio is

$$J = \left(\frac{1}{2} \right)^3 \left| \begin{bmatrix} 1 & -1 & 1 \\ 1 & 1 & -1 \\ -1 & 1 & 1 \end{bmatrix} \right| = \frac{1}{2}.$$

The map $(u, v, w) \in B \mapsto (x, y, z) \in A$ is an obvious bijection when $B = [0, \infty) \times [0, \infty) \times [0, \infty)$, so, applying the transformation formula (★) and Fubini's rule, the result is

$$\begin{aligned} \int_A e^{-x-y-z} \, dx dy dz &= \int_B e^{-\frac{1}{2}(u+v+w)} \frac{1}{2} \, du dv dw \\ &= \frac{1}{2} \int_0^\infty \int_0^\infty \int_0^\infty e^{-\frac{1}{2}u} e^{-\frac{1}{2}v} e^{-\frac{1}{2}w} \, du dv dw = \frac{1}{2} \left[\int_0^\infty e^{-\frac{1}{2}u} \, du \right]^3 = \underline{\underline{4}}. \quad \square \end{aligned}$$

Integrals in curvilinear coordinate systems

In physical applications the transformation formulas for integrals come into use most often when one wishes to pass from a Cartesian coordinate system to a curvilinear polar, cylindrical, or spherical coordinate system. If in the plane one passes to polar coordinates, then the Jacobian matrix of the transformation

$$x = r \cos \varphi = u(r, \varphi), \quad y = r \sin \varphi = v(r, \varphi)$$

is

$$\mathbf{J} = \begin{bmatrix} \cos \varphi & -r \sin \varphi \\ \sin \varphi & r \cos \varphi \end{bmatrix}.$$

The absolute value of this determinant is $J = r$, so the transformation formula becomes

$$\boxed{\int_A f(x, y) \, dx dy = \int_B g(r, \varphi) r \, dr d\varphi.}$$

For example, if A = a disk of radius R centered at the origin, then the formula is valid when $B = [0, R] \times [0, 2\pi]$. The slight non-injectivity of the transformation causes no harm here, since it is confined to a null-measure subset ($r = 0$, $\varphi = 0$ or $\varphi = 2\pi$).

EXAMPLE 5 Compute $\int_{\mathbb{R}^2} e^{-x^2-y^2} \, dx dy$.

Solution We pass to polar coordinates. The coordinate transformation $\mathbf{u} : B \rightarrow \mathbb{R}^2$ is (almost) a bijection when we choose $B = \{(r, \varphi) \in \mathbb{R}^2 \mid r \geq 0, 0 \leq \varphi \leq 2\pi\}$, so

$$\begin{aligned} \int_{\mathbb{R}^2} e^{-x^2-y^2} \, dx dy &= \int_B e^{-r^2} r \, dr d\varphi \\ &= \int_0^{2\pi} \left[\int_0^\infty e^{-r^2} r \, dr \right] d\varphi = 2\pi \int_0^\infty \left(-\frac{1}{2} e^{-r^2} \right) = \underline{\underline{\pi}}. \quad \square \end{aligned}$$

The integral of the example, computed in Cartesian coordinates, is

$$\begin{aligned} \int_{\mathbb{R}^2} e^{-x^2-y^2} \, dx dy &= \int_{\mathbb{R}^2} e^{-x^2} \cdot e^{-y^2} \, dx dy \\ &= \left[\int_{-\infty}^\infty e^{-x^2} \, dx \right] \left[\int_{-\infty}^\infty e^{-y^2} \, dy \right] = \left[\int_{-\infty}^\infty e^{-x^2} \, dx \right]^2. \end{aligned}$$

We thus obtained an amusing result:

PROPOSITION XIII.4.1 $\int_{-\infty}^\infty e^{-x^2} \, dx = \sqrt{\pi}$.

When passing to cylindrical coordinates in three dimensions, the transformation ratio is the same as when passing to polar coordinates in the plane, i.e. $J(r, \varphi, z) = r$. When passing to spherical coordinates, the transformation ratio is obtained as

$$J = |\det \mathbf{J}| = \begin{vmatrix} \sin \theta \cos \varphi & r \cos \theta \cos \varphi & -r \sin \theta \sin \varphi \\ \sin \theta \sin \varphi & r \cos \theta \sin \varphi & r \sin \theta \cos \varphi \\ \cos \theta & -r \sin \theta & 0 \end{vmatrix} = r^2 \sin \theta.$$

The transformation formulas for the integral to cylindrical and spherical coordinates are therefore

Cylindrical coord.:	$\int_A f(x, y, z) \, dx dy dz = \int_B g(r, \varphi, z) \, r \, dr d\varphi dz$
Spherical coord.:	$\int_A f(x, y, z) \, dx dy dz = \int_B g(r, \theta, \varphi) \, r^2 \sin \theta \, dr d\theta d\varphi$

EXAMPLE 6 Compute $I = \int_A (x^2 + y^2) \, dx dy dz$, where A is a *spherical shell*, i.e. the region between two spherical surfaces:

$$A = \{ (x, y, z) \in \mathbb{R}^3 \mid R_1^2 \leq x^2 + y^2 + z^2 \leq R_2^2 \}.$$

Solution The computation is easiest by means of spherical coordinates:

$$\begin{aligned} I &= \int_0^{2\pi} \left\{ \int_0^\pi \left[\int_{R_1}^{R_2} r^2 \sin^2 \theta \cdot r^2 \sin \theta \, dr \right] d\theta \right\} d\varphi \\ &= \int_0^{2\pi} d\varphi \cdot \int_0^\pi \sin^3 \theta \, d\theta \cdot \int_{R_1}^{R_2} r^4 \, dr = 2\pi \cdot \frac{4}{3} \cdot \frac{1}{5} (R_2^5 - R_1^5) = \underline{\underline{\frac{8\pi}{15} (R_2^5 - R_1^5)}}. \quad \square \end{aligned}$$

EXERCISES

1. Transform the integral $\int_A f(x, y) \, dx dy$ using the given transformation $\mathbf{u}$ by first computing exactly the ratio $\mu(\Delta A)/\mu(\Delta T)$, where $\Delta T = [t, t + \Delta t, s, s + \Delta s]$ and $\Delta A = \mathbf{u}(\Delta T)$, and the transformation ratio as the limit of this as $\max\{\Delta t, \Delta s\} \rightarrow 0$:
 - a) $A : x \in [1, 2] \ \& \ -x^2 \leq y \leq 3x^2, \ \mathbf{u}(t, s) = (t, st^2)$
 - b) $A : x \in [1, \infty) \ \& \ 1/x^2 \leq y \leq 2/x^2, \ \mathbf{u}(t, s) = (t, s/t^2)$
2. Compute using a suitable transformation:
 - a) $\int_A (x + 2y)^4 (x - 2y)^6 \, dx dy, \ A : |x + 2y| \leq 1 \ \& \ |x - 2y| \leq 2$
 - b) $\int_A (2x + 3y)^2 (x - 5y)^2 \, dx dy, \ A : |2x + 3y| \leq 4 \ \& \ |x - 5y| \leq 3$
3. Compute the following integrals using a substitution of the form $x = as \cos t, \ y = bs \sin t$.

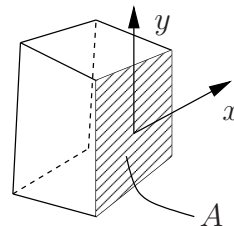
$$\begin{aligned} \text{a) } & \int_{\mathbb{R}^2} e^{-3x^2 - 4y^2} \, dx dy & \text{b) } & \int_{\mathbb{R}^2} \frac{1}{(1 + x^2 + 4y^2)^2} \, dx dy \\ \text{c) } & \int_A \ln \left(1 + \frac{x^2}{4} + \frac{y^2}{9} \right) \, dx dy, & A : & x, y \geq 0 \ \& \ 9x^2 + 4y^2 \leq 36 \end{aligned}$$

4. a) The set $A \subset \mathbb{R}^2$ lies in the coordinate quadrant $x, y > 0$ and is bounded by the curves $y^3 = ax^2$, $y^3 = bx^2$, $x^4 = cy^3$ and $x^4 = dy^3$, where $0 < a < b$ and $0 < c < d$. Compute the area $\mu(A)$ using the substitution $y^3 = ux^2$, $x^4 = vy^3$.
 b) Solve Exercise XIII.3:7 using the change of variables $(x, y, z) = (au, bv, cw)$.
5. Compute the integral $\int_A f dx dy dz$ by passing to cylindrical or spherical coordinates:
- $A: x^2 + y^2 \leq z \leq \sqrt{2 - x^2 - y^2}$, $f(x, y, z) = z$
 - $A: x^2 + y^2 + z^2 \leq 12$, $z \geq x^2 + y^2$, $f(x, y, z) = 1$
 - $A: x^2 + y^2 + z^2 \leq R^2$, $x, y, z \geq 0$, $f(x, y, z) = x$
 - $A: x^2 + y^2 + z^2 \leq R^2$, $x, y, z \geq 0$, $f(x, y, z) = xyz$
 - $A: 1 \leq x^2 + y^2 \leq 4$, $0 \leq y \leq x$, $0 \leq z \leq 1$, $f(x, y, z) = x^2 + y^2$
 - $A: 4 \leq x^2 + y^2 + z^2 \leq 9$, $z \geq 0$, $f(x, y, z) = z$
 - $A: x^2 + y^2 \geq R^2$, $x^2 + y^2 + z^2 \leq R^2$, $f(x, y, z) = x^2 + y^2$
6. The half-cone $z = \sqrt{x^2 + y^2}$ divides spheres centered at the origin into two parts. Compute the ratio of the volumes of these parts by means of spherical coordinates.
7. (*) Let $f(x, y) = e^{(y-x)/(y+x)}$ and A the triangle with vertices $(0, 0)$, $(1, 0)$ and $(0, 1)$. Compute the integral $\int_A f dx dy$ a) by means of polar coordinates,
 b) using the substitution $u = y - x$, $v = y + x$.
8. (*) Compute the plane integral $\int_0^\infty \left[\int_0^x \frac{e^{-x-2y}}{x+2y} dy \right] dx$.
9. (*) Compute the volume $\mu(A)$, where $A \subset \mathbb{R}^3$ is defined by the conditions
- $$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} \leq 1 \quad \& \quad z + y \geq b \quad (a, b, c > 0).$$
10. (*) Let K be a parallelepiped with one vertex at the origin, the other end-points of the edges emanating from the origin being $(2, 1, 1)$, $(-1, 2, 2)$ and $(0, -2, 1)$. Compute $\int_K xye^z dx dy dz$ using a suitable change of variables.

XIII.5 Applications of integrals: Density and total amount

In physical contexts, plane integrals are often called surface integrals and the space integral over $\mathbb{R}^3$ a volume integral.

A plane integral becomes a 'surface integral' when the plane is thought of as a spatial plane in $\mathbb{R}^3$ (or E^3) and the area measure as the Jordan measure defined on that plane. Physically, A may be, for example, the side surface of a plate-like body or the planar part of the outer surface of a more general three-



Understood in the above way, the plane integral also becomes, as a mathematical concept, a special case of the more general *surface integral*, in which the surface may be curved. Surface integrals in this more general sense are treated later in Section XIII.7. The one-dimensional integral, too, can be interpreted 'spatially' by thinking of it as a length measure defined on a spatial line. Understood this way, the one-dimensional integral is a special case of the *line integral*, i.e. the *curve integral*, associated with a curve in space (or in the plane). These are treated in the next section.

In physical applications, the set over which one integrates is generally called a region (in the case of a volume integral, sometimes a 'volume'). The term *region* (domain or region) is also used in mathematics for sets of a certain type. In what follows, however, let 'region' be understood as a 'physical set'.

Density and total amount

Let $A \subset \mathbb{R}^d$ be a set or, more physically, a region, where $d \in \{1, 2, 3\}$. In a typical situation where integrals are applied, one knows the *density* (density function, distribution) of a physical quantity in the region A , and the task is to compute the *total amount* of the quantity from the formula

Total amount in A = integral of the density over A .
--

If the density function = f and the total amount is denoted by the symbol F , then the computational formula is

$$F = \int_A f \, d\mu,$$

where μ is the Jordan measure of $\mathbb{R}^d$. The density function and the total amount can also be vector-valued: for example, if $A \subset \mathbb{R}^3$ and the density function is a vector field $\vec{f}(x, y, z) = f_1(x, y, z)\vec{i} + f_2(x, y, z)\vec{j} + f_3(x, y, z)\vec{k}$ defined on A , then the total amount is

$$\vec{F} = \int_A \vec{f} \, d\mu = \vec{i} \int_A f_1 \, dx dy dz + \vec{j} \int_A f_2 \, dx dy dz + \vec{k} \int_A f_3 \, dx dy dz.$$

Examples of applications of the above way of thinking are, say, the following:

density	total amount
$\rho = \text{mass density [kg/m}^d]$	$m = \text{total mass [kg]}$
$\sigma = \text{charge density [C/m}^d]$	$Q = \text{total charge [C]}$
$\vec{f} = \text{force density [N/m}^d]$	$\vec{F} = \text{total force [N]}$

If a physical quantity is distributed over a planar (or a more general, see Section XIII.7) surface, then the density is called *surface density* (e.g. surface charge density, unit C/m²). For a quantity distributed over a line (or a more general curve, see Section XIII.6), one correspondingly speaks of *line density* (e.g. the mass line density in a wire, unit kg/m).

EXAMPLE 1 Let the density of air at height x above the ground be $\rho(x) = \rho_0 e^{-x/a}$, where $\rho_0 = 1 \text{ kg/m}^3$ and $a = 10 \text{ km}$. How much air (unit = kg) is there in an imagined, vertical, 100 km tall tube whose cross-sectional area = 1 m^2 ?

Solution The mass density (line density) of air in the imagined tube is

$$f(x) = 1 \text{ m}^2 \cdot \rho(x) = e^{-x/a} \frac{\text{kg}}{\text{m}},$$

so the total mass is

$$\begin{aligned} m &= 1 \frac{\text{kg}}{\text{m}} \cdot \int_0^{100 \text{ km}} e^{-x/a} \, dx \\ &\approx 1 \frac{\text{kg}}{\text{m}} \cdot \int_0^\infty e^{-x/a} \, dx = 1 \frac{\text{kg}}{\text{m}} \cdot a = 1 \frac{\text{kg}}{\text{m}} \cdot 10000 \text{ m} = \underline{\underline{10000 \text{ kg}}}. \quad \square \end{aligned}$$

EXAMPLE 2 On a square forest clearing $A = [0, a] \times [0, a]$, snow is distributed unevenly so that the surface mass density is

$$\rho(x, y) = 36\rho_0 \left[1 + \left(\frac{x+y}{a} \right)^2 \right], \quad \rho_0 = 1 \text{ kg/m}^2.$$

How large is the cumulative snowfall [cm] of the snow, if for evenly distributed snow $1 \text{ kg/m}^2 \hat{=} 1 \text{ cm}$ holds?

Solution The total mass of the snow is

$$\begin{aligned} m &= \int_A \rho \, dx dy = 36\rho_0 \int_0^a \int_0^a \left[1 + \left(\frac{x+y}{a} \right)^2 \right] dx dy \\ &= 36\rho_0 \int_0^a \left\{ \int_{y=0}^{y=a} \left[y + \frac{a}{3} \left(\frac{x+y}{a} \right)^3 \right] dy \right\} dx \\ &= 36\rho_0 \int_0^a \left[a + \frac{1}{3a^2}(x+a)^3 - \frac{1}{3a^2}x^3 \right] dx \\ &= 36\rho_0 \left[ax + \frac{1}{12a^2}(x+a)^4 - \frac{1}{12a^2}x^4 \right]_0^a \\ &= 78\rho_0 a^2 \hat{=} \underline{\underline{78 \text{ cm}}} \quad (\rho_0 a^2 \hat{=} 1 \text{ cm}). \quad \square \end{aligned}$$

EXAMPLE 3 In a hemispherical sand pile $A : x^2 + y^2 + z^2 \leq R^2, z \geq 0$ the mass density of the sand is

$$\rho(x, y, z) = \rho_0 + \rho_1(x, y, z) = \rho_0 + 0.036\rho_0(1 - z/R),$$

where ρ_0 (= constant) is the density of dry sand and ρ_1 is the density of water bound in the sand. What percentage of the whole mass of the sand pile is water?

Solution The mass of the dry sand is

$$m_0 = \int_A \rho_0 \, dx dy dz = \rho_0 \mu(A) = \frac{2}{3}\pi\rho_0 R^3 \approx 0.667\pi\rho_0 R^3$$

and that of the water (cf. Example XIII.3:4)

$$m_1 = \int_A \rho_1 \, dx dy dz = 0.036\rho_0 \left(\frac{2}{3}\pi R^3 - \frac{1}{4}\pi R^3 \right) = 0.015\pi\rho_0 R^3.$$

The relative proportion of water is thus $0.015/0.682 \approx 0.022$. Answer: 2.2%.

□

The axioms of total amount

In the above examples, the integral formula connecting density and total amount was taken as 'given', i.e. as a physical starting assumption. It is possible, however, to derive the integral formula from much simpler assumptions, which can be regarded as the common basic axioms of the mathematical models used above. Let the density function be f and denote the total amount in A ($A \subset \mathbb{R}^d$) by the symbol $F(A)$. Assume:

- A1. Additivity: If $\mu(A \cap B) = 0$, then $F(A \cup B) = F(A) + F(B)$.
- A2. Comparison principle: If for every $\mathbf{x} \in A$ we have $m \leq f(\mathbf{x}) \leq M$, then $m\mu(A) \leq F(A) \leq M\mu(A)$.

If now $A \subset \mathbb{R}^d$ and one wishes to determine the total amount $F(A)$, then the density function f is interpreted as defined outside A by the extension by zero f_0 . Then if $T \supset A$ is a closed interval / basic rectangle / rectangular basic box and $\mathcal{T}_h$ is a partition of T , then from the assumptions A1–A2 (with $f = f_0$) it follows that

$$\underline{\sigma}_h(f_0, \mathcal{T}_h) \leq F(A) \leq \overline{\sigma}_h(f_0, \mathcal{T}_h),$$

where $\underline{\sigma}_h(f_0, \mathcal{T}_h)$ and $\overline{\sigma}_h(f_0, \mathcal{T}_h)$ are the lower and upper sums associated with the partition $\mathcal{T}_h$ (cf. Sections VIII.5 and XIII.1). Since this estimate is valid for every $\mathcal{T}_h$, then insofar as f is integrable over A , one must have $F(A) = \int_A f d\mu$. The integral formula thus follows from the assumed axioms A1–A2 (which in applications are usually easy to accept).

EXAMPLE 4 Let $A = [a, b]$ and $F(A)$ the area of the planar region between the curve $y = f(x)$ and the x -axis. Then if $f(x) \geq 0 \forall x \in A$ and f is integrable on the interval A , then from the axioms A1–A2 the integral formula $F(A) = \int_A f dx$ follows. — Cf. the discussion of Section VIII.8.

Moment

If $\vec{f} = f_1\vec{i} + f_2\vec{j} + f_3\vec{k}$ is a force density in $\mathbb{R}^3$, then the moment density of these forces with respect to the point in space $P_0 \hat{=} \vec{r}_0 = x_0\vec{i} + y_0\vec{j} + z_0\vec{k}$, at the point (x, y, z) , is $\vec{m} = (\vec{r} - \vec{r}_0) \times \vec{f}$, where $\vec{r} = x\vec{i} + y\vec{j} + z\vec{k}$. Accordingly, the total moment of the forces distributed over the region $A \subset \mathbb{R}^3$ with respect to the point P_0 is

$$\vec{M} = \int_A \vec{m} d\mu = \int_A (\vec{r} - \vec{r}_0) \times \vec{f} d\mu,$$

where μ is the volume measure of $\mathbb{R}^3$. The formula is valid also if $\vec{f}$ is a surface density on a spatial plane T ($A \subset T$) or a line density on a spatial line S ($A \subset S$), in which case μ is correspondingly the 2-dimensional or 1-dimensional Jordan measure. For example, if $\vec{f}$ is a surface density in a region A of the xy -plane and $P_0 =$ the origin, then the computational formula for the total moment is

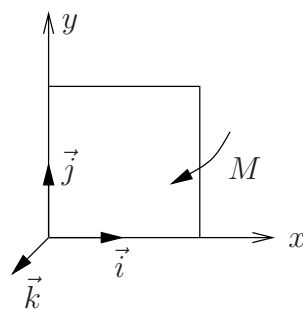
$$\begin{aligned}\vec{M} &= \int_A (x\vec{i} + y\vec{j}) \times (f_1\vec{i} + f_2\vec{j} + f_3\vec{k}) \, dxdy \\ &= \vec{i} \int_A y f_3(x, y) \, dxdy - \vec{j} \int_A x f_3(x, y) \, dxdy + \vec{k} \int_A [x f_2(x, y) - y f_1(x, y)] \, dxdy.\end{aligned}$$

EXAMPLE 5 Lock gate. A canal lock gate is square, with side length 4 meters. The gate opens like a door. On one side of the gate there is water over the entire height of the gate (4 m); on the other side there is no water. Determine the total force acting on the gate and the moment M that twists the gate open.

Solution The gate is subjected to a pressure in the direction of the normal (see the figure)

$$\begin{aligned}\vec{f}(x, y) &= \rho_0 g(a - y)\vec{k}, \quad (x, y) \in A, \\ A &= [0, a] \times [0, a], \quad a = 4 \text{ m}, \\ \rho_0 g &= 1000 \text{ G/m}^3, \quad G \approx 9.8 \text{ N}.\end{aligned}$$

The total force acting on the gate is the integral of the pressure (surface density):



$$\begin{aligned}\vec{F} &= \int_A \vec{f} \, dxdy = 1000 \text{ G m}^{-3} \vec{k} \int_0^a \left[\int_0^a (a - y) \, dy \right] dx \\ &= 500 \text{ G m}^{-3} a^3 \vec{k} = \underline{\underline{32000 \text{ G} \vec{k}}}.\end{aligned}$$

The $\vec{j}$ -component of the moment (positive when twisting the gate shut) is

$$\begin{aligned}M_y &= - \int_A x f_3(x, y) \, dxdy = -1000 \text{ G m}^{-3} \int_0^a \left[\int_0^a x(a - y) \, dy \right] dx \\ &= -1000 \text{ G m}^{-3} \cdot \frac{1}{4} a^4 = -64000 \text{ G m}.\end{aligned}$$

Thus the moment twisting the hatch open is

$$M = -M_y \approx \underline{\underline{6.3 \cdot 10^5 \text{ Nm}}}. \quad \square$$

Center of gravity. Centroid

The most common example of a force distributed over space is gravitation, whose density is

$$\vec{f}(x, y, z) = \rho(x, y, z)g\vec{e}, \quad (x, y, z) \in A.$$

Here A may represent, for example, a solid body, ρ = mass density [kg/m³], g is the acceleration of gravity (at the earth's surface $g \approx 9.8$ m/s²), and $\vec{e}$ = the direction vector (unit vector) of the gravitational force. In addition to the total force

$$\vec{G} = \int_A \vec{f} \, dxdydz = mg\vec{e}, \quad m = \int_A \rho \, dxdydz$$

the total moment of the gravitational forces is of interest, which with respect to the point $P_0 \hat{=} \vec{r}_0 = x_0\vec{i} + y_0\vec{j} + z_0\vec{k}$ is

$$\vec{M} = \int_A (\vec{r} - \vec{r}_0) \times \vec{e} \rho g \, dxdydz = g \left(\int_A (\vec{r} - \vec{r}_0) \rho \, dxdydz \right) \times \vec{e}.$$

The point P_0 is called the center of gravity of the body if the moment of the gravitational forces with respect to $P_0 = \vec{0}$ regardless of the vector $\vec{e}$ (i.e. regardless of the orientation of the body relative to the gravitational field). This is exactly the case when

$$\vec{0} = \int_A (\vec{r} - \vec{r}_0) \rho \, dxdydz = \int_A \rho \vec{r} \, dxdydz - \vec{r}_0 \int_A \rho \, dxdydz,$$

so the position vector of the center of gravity is

$$\boxed{\vec{r}_0 = \frac{1}{m(A)} \int_A \rho \vec{r} \, dxdydz, \quad m(A) = \int_A \rho \, dxdydz.}$$

Here $m(A)$ = the mass of the body. The coordinates of the center of gravity are thus

$$x_0 = \frac{1}{m(A)} \int_A x \rho(x, y, z) \, dxdydz, \quad \text{etc.}$$

If $\rho = \rho_0 = \text{constant}$, then the position vector of the center of gravity is

$$\vec{r}_0 = \frac{1}{\mu(A)} \int_A \vec{r} \, dxdydz \quad (\rho = \text{constant}).$$

The mathematical generalization of this is the *centroid* of a set $A \subset \mathbb{R}^n$, which is defined by

$$\mathbf{x}_0 = \frac{1}{\mu(A)} \int_A \mathbf{x} \, d\mu = \frac{1}{\mu(A)} \sum_{i=1}^n \left(\int_A x_i \, d\mu \right) \mathbf{e}_i,$$

where μ is the n -dimensional Jordan measure. Since the functions $f_i(\mathbf{x}) = x_i$ are undoubtedly integrable over every measurable set (Theorem [XIII.3.1](#)), the centroid of a set $A \subset \mathbb{R}^n$ is defined whenever A is measurable and $\mu(A) \neq 0$.

EXAMPLE 6 Determine the centroids of the following sets.

- a) $A = \{(x, y, z) \in \mathbb{R}^3 \mid x^2 + y^2 \leq R^2 \text{ \& } x \geq 0 \text{ \& } z \in [-H/2, H/2]\}$
 b) $A = \{(x, y, z) \in \mathbb{R}^3 \mid x^2 + y^2 + z^2 \leq R^2 \text{ \& } x \geq 0\}$

Solution a) Here $\mu(A) = \frac{1}{2}\pi R^2 H$ (as is easily found by integration), and by symmetry $\int_A y \, dx dy dz = \int_A z \, dx dy dz = 0$, so $y_0 = z_0 = 0$. Computed in cylindrical coordinates, the x -coordinate of the centroid is

$$\begin{aligned} x_0 &= \frac{1}{\mu(A)} \int_A x \, dx dy dz = \frac{2}{\pi R^2 H} \int_0^R \left\{ \int_{-\pi/2}^{\pi/2} \left[\int_{-H/2}^{H/2} r \cos \varphi \cdot r \, dz \right] d\varphi \right\} dr \\ &= \frac{2}{\pi R^2 H} \cdot \int_0^R r^2 \, dr \cdot \int_{-\pi/2}^{\pi/2} \cos \varphi \, d\varphi \cdot \int_{-H/2}^{H/2} dz \\ &= \frac{2}{\pi R^2 H} \cdot \frac{1}{3} R^3 \cdot 2 \cdot H = \underline{\underline{\frac{4}{3\pi} R}}. \end{aligned}$$

b) Here too $y_0 = z_0 = 0$, and (see Examples [XIII.3:3–4](#))

$$x_0 = \frac{1}{\mu(A)} \int_A x \, dx dy dz = \frac{3}{2\pi R^3} \cdot \frac{\pi R^4}{4} = \underline{\underline{\frac{3}{8} R}}. \quad \square$$

Pappus's rule

Let A be a region of the xy -plane for which $y \geq 0 \, \forall (x, y) \in A$. We wish to compute the volume of the solid of revolution V generated when A rotates about the x -axis. Let $B = \{x \in \mathbb{R} \mid (x, y) \in A \text{ for some } y \in \mathbb{R}\}$. Then V can be represented in the form

$$\begin{aligned} V &= \{(x, y, z) \in \mathbb{R}^3 \mid x \in B \text{ \& } (y, z) \in C(x)\}, \\ \text{where } C(x) &= \{(y, z) \in \mathbb{R}^2 \mid \sqrt{y^2 + z^2} \in D(x)\}, \\ \text{where } D(x) &= \{y \in \mathbb{R} \mid (x, y) \in A\} \subset [0, \infty). \end{aligned}$$

On this basis we have first (cf. Section [XIII.3](#))

$$\mu(V) = \int_B \mu(C(x)) \, dx.$$

Since here $C(x)$ is circularly symmetric about the origin of the yz -plane, then with the polar coordinate transformation $y = r \cos \varphi$, $z = r \sin \varphi$ the area $\mu(C(x))$ can be computed in the form

$$\mu(C(x)) = \int_{r \in D(x)} \int_{\varphi=0}^{2\pi} r \, dr d\varphi = 2\pi \int_{D(x)} r \, dr = 2\pi \int_{D(x)} y \, dy.$$

Consequently

$$\mu(V) = 2\pi \int_B \left[\int_{D(x)} y \, dy \right] dx = 2\pi \int_A y \, dx dy.$$

Here one can further write $\int_A y \, dx dy = y_0 \mu(A)$, where $\mu(A)$ = the area of A and y_0 = the y -coordinate of the centroid of A . In this way one obtains the computational formula known as *Pappus's* (Pappos's) *rule**

$$\mu(V) = \mu(A) \cdot s, \quad s = 2\pi y_0 \quad (\text{Pappus's rule}).$$

Here s = the distance traveled by the centroid of A in the rotation.

EXAMPLE 7 When the circle $K : x^2 + (y - R)^2 = a^2$, where $R \geq a$, rotates about the x -axis, then the volume of the region enclosed by the resulting surface of revolution (the torus) is, by Pappus's rule, $\mu(V) = \pi a^2 \cdot 2\pi R = 2\pi^2 a^2 R$. $\square$

Mass measure. Generalizations of the Jordan measure

If $A \subset \mathbb{R}^3$ represents a three-dimensional body whose mass density ρ is not constant, then the center of gravity of the body is defined by

$$\vec{r}_0 = \frac{1}{m(A)} \int_A \vec{r} \rho \, dx dy dz, \quad m(A) = \int_A \rho \, dx dy dz.$$

When one writes here $dm = \rho \, dx dy dz$, the expression for the center of gravity can be presented a little more elegantly in the form

$$\vec{r}_0 = \frac{1}{m(A)} \int_A \vec{r} \, dm, \quad m(A) = \int_A dm.$$

This defines the mass measure m associated with the mass density ρ . It is indeed a measure (additive and, taken as dimensionless, positive) which thus measures, instead of the volume of A , the total mass contained in A .

*Pappus's rule has also been called *Guldin's rule*, after the Swiss mathematician **Paul Guldin** (1577-1643). The rule was, however, discovered by the Greek **Pappos** as early as the 4th century.

More generally, if $\rho(\mathbf{x})$ is a non-negative function defined on $\mathbb{R}^n$ that is integrable over the set $A \subset \mathbb{R}^n$, then a measure can be defined for A

$$\mu_\rho(A) = \int_A \rho d\mu = \int_A d\mu_\rho.$$

The function ρ is then called the *density function* of the *measure* μ_ρ . A bounded set $A \subset \mathbb{R}^n$ is considered μ_ρ -measurable when ρ is integrable (in the ordinary or extended sense) over A . — The Jordan length, area, and volume measures can thus be seen as special cases of more general measures: the 'mass content' of a set according to the Jordan measure is obtained by setting the density function of the measure to $\rho = 1$.

Moment of inertia

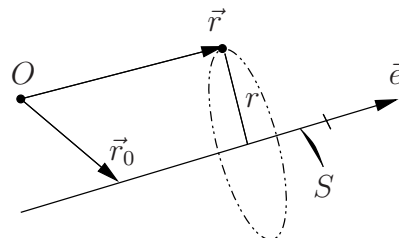
If $A \subset \mathbb{R}^3$ represents a solid body and S is a spatial line, then the moment of inertia of the body with respect to the line S is defined by

$$I_S = \int_A r^2 \rho dx dy dz = \int_A r^2 dm,$$

where m is the mass measure defined above (ρ = mass density) and $r = r(x, y, z)$ is the distance of the point (x, y, z) from the line S . The inertia density in the body is thus $f = r^2 \rho$.*

If the line passes through the point $P_0 \hat{=} \vec{r}_0$ and its direction vector is the unit vector $\vec{e}$, then (cf. the figure)

$$r(x, y, z) = |\vec{e} \times (\vec{r} - \vec{r}_0)|.$$



Let the origin above be the center of gravity of the body and denote R = the distance of the origin from the line S , so that

$$\vec{e} \times (\vec{r} - \vec{r}_0) = \vec{e} \times \vec{r} - R\vec{n}, \quad |\vec{n}| = 1$$

*The moment of inertia physically defines the rotational inertia of the body with respect to the axis of rotation S . If $\theta(t)$ is the angle of rotation at time t , then the (Newton) equation of rotational motion is $I_S \theta'' = M_S$, where M_S is the moment acting on the body in the direction of S .

($\vec{n}$ = a constant vector), and consequently

$$r^2 = (\vec{e} \times \vec{r} - R\vec{n}) \cdot (\vec{e} \times \vec{r} - R\vec{n}) = |\vec{e} \times \vec{r}|^2 - 2R\vec{n} \cdot \vec{e} \times \vec{r} + R^2.$$

Substituting this into the integral formula above, one obtains

$$I_S = \int_A |\vec{e} \times \vec{r}|^2 dm - 2R\vec{n} \cdot \vec{e} \times \int_A \vec{r} dm + R^2 \int_A dm.$$

Here the first term = the moment of inertia of the body with respect to the line S' passing through the origin and parallel to the line S , the second term = 0, since the origin is the center of gravity of the body, and the third term = mR^2 , where $m = m(A)$ is the mass of the body. For the moment of inertia one has thus obtained the reduction formula known as *Steiner's rule*

$$I_S = I_{S'} + mR^2 \quad (\text{Steiner's rule}).$$

Here, then, S and S' are parallel lines, S' passes through the center of gravity of the body, R = the distance between the lines, and m = the mass of the body.

EXAMPLE 8 A homogeneous steel ball of radius R has mass = m . Compute the moment of inertia of the ball with respect to a) a line S passing through the center of the ball, b) a line S' tangent to the ball.

Solution a) By symmetry the moment of inertia is independent of the direction vector $\vec{e}$ of S . Choosing $\vec{e} = \vec{k}$ and noting that $m = \rho_0 \cdot \frac{4}{3}\pi R^3$, where ρ_0 is the mass density of the ball, then (see Example 6 of the previous section)

$$I_S = \int_A (x^2 + z^2) \rho_0 dx dy dz = \rho_0 \cdot \frac{8}{15} \pi R^5 = \underline{\underline{\frac{2}{5} mR^2}}.$$

b) The origin is the center of gravity of the body, so by part a) and Steiner's rule

$$I_{S'} = \frac{2}{5} mR^2 + mR^2 = \underline{\underline{\frac{7}{5} mR^2}}. \quad \square$$

*The inertia tensor

Let us further consider computing the moment of inertia I_S of a body by means of matrix algebra. If the direction vector (unit vector) of the line S is $\vec{e} = e_1\vec{i} + e_2\vec{j} + e_3\vec{k}$, then

$$\begin{aligned} \vec{e} \times \vec{r} &= \vec{e} \times (x\vec{i} + y\vec{j} + z\vec{k}) \\ &= (e_2z - e_3y)\vec{i} + (e_3x - e_1z)\vec{j} + (e_1y - e_2x)\vec{k}, \end{aligned}$$

so

$$\begin{aligned} I_S &= \int_A |\vec{e} \times \vec{r}|^2 dm = \int_A [(e_2z - e_3y)^2 + (e_3x - e_1z)^2 + (e_1y - e_2x)^2] dm \\ &= \sum_{i=1}^3 \sum_{j=1}^3 I_{ij} e_i e_j = \mathbf{e}^T \mathbf{I} \mathbf{e}, \end{aligned}$$

where $\mathbf{e}^T = [e_1, e_2, e_3]$ and $\mathbf{I} = (I_{ij})$ is the symmetric so-called inertia matrix, whose entries are

$$\begin{aligned} I_{11} &= \int_A (y^2 + z^2) dm, & I_{22} &= \int_A (x^2 + z^2) dm, & I_{33} &= \int_A (x^2 + y^2) dm, \\ I_{12} &= I_{21} = - \int_A xy dm, & I_{13} &= I_{31} = - \int_A xz dm, & I_{23} &= I_{32} = - \int_A yz dm. \end{aligned}$$

By the definition of the moment of inertia, the diagonal entries I_{ii} are moments of inertia with respect to the coordinate axes. When in these entries one writes $y^2 + z^2 = -x^2 + (x^2 + y^2 + z^2)$ etc., then one sees that for the moment of inertia I_S the computational formula also holds

$$\boxed{I_S = J_0 - \mathbf{e}^T \mathbf{J} \mathbf{e}}, \quad (\star)$$

where

$$J_0 = \int_A (x^2 + y^2 + z^2) dm, \quad \mathbf{J} = \int_A \begin{bmatrix} x^2 & xy & xz \\ xy & y^2 & yz \\ xz & yz & z^2 \end{bmatrix} dm.$$

The entries of the matrix $\mathbf{J}$ are called products of inertia. Thus, if one wishes to compute the moment of inertia of a body with respect to a given line S , one must first compute, in some coordinate system, the product-of-inertia matrix $\mathbf{J}$, i.e. the integrals

$$J_{11} = \int_A x^2 dm, \quad J_{12} = \int_A xy dm, \quad J_{13} = \int_A xz dm, \quad \text{etc.}$$

(6 different ones). After this the moment of inertia is determined by formula $(\star)$, where $\mathbf{e}$ is the unit vector in the direction of the line S and $J_0 = J_{11} + J_{22} + J_{33}$.

If the unit vector $\vec{e}$ above is expressed in a rotated coordinate system $\{\vec{i}', \vec{j}', \vec{k}'\}$ as the coordinate vector $\mathbf{e}'$, then $\mathbf{e} = \mathbf{C} \mathbf{e}'$, where $\mathbf{C}$ is an orthogonal matrix (cf. Section X.6). In formula $(\star)$ this transformation means that $\mathbf{e}'$ replaces $\mathbf{e}$ and $\mathbf{C}^T \mathbf{J} \mathbf{C}$ replaces $\mathbf{J}$ (J_0 does not change). The product-of-inertia matrix, and likewise the inertia matrix, thus transforms under a rotation of the coordinate

system like a (symmetric) tensor, cf. Section XII.4. One therefore says that it is an *inertia tensor*, whereby inertia (rotational inertia, cf. the footnote above) becomes understood as a property of the body independent of the rotation of the coordinate system, relative to the chosen origin of the coordinate system (the center of rotation). One concludes further that by a rotation of the coordinate system one can always find a *principal-inertia coordinate system* in which the inertia matrix is diagonal (!). Namely, this is found by solving the eigenvalue problem of the inertia matrix (cf. Section XII.2). The eigenvalues, i.e. the diagonal entries of the inertia matrix in the principal-inertia coordinate system, are called the *principal moments of inertia*.

EXAMPLE 9 Determine the moment of inertia of a homogeneous cube (side length a , mass density ρ_0 , mass $m = \rho_0 a^3$) with respect to a diagonal.

Solution Let $A = [-a/2, a/2] \times [-a/2, a/2] \times [-a/2, a/2]$, so that by symmetry $I_{ij} = 0$ when $i \neq j$ (a principal-inertia coordinate system). The principal moments of inertia are also, by symmetry, equal, i.e. for every $i = 1, 2, 3$

$$I_{ii} = \rho_0 \int_A (y^2 + z^2) dx dy dz = 2\rho_0 \int_A z^2 dx dy dz = 2\rho_0 a^2 \int_{-a/2}^{a/2} z^2 dz = \frac{1}{6} ma^2.$$

Since the inertia matrix is diagonal and the diagonal entries are equal, the inertia matrix is the same in all coordinate systems whose origin is at the center of the cube (the center of gravity). Thus the answer: $I_S = \underline{\underline{\frac{1}{6}ma^2}}$. $\square$

*Probability measure

In the branch of mathematics called *probability theory* one considers so-called *random variables* $\mathbf{x} \in \mathbb{R}^n$. To each random variable is associated a *probability measure* P characteristic of that variable, whose density function f ($f(\mathbf{x}) \geq 0 \forall \mathbf{x}$) is called the *probability density* or *distribution*. A probability measure is required to have all of $\mathbb{R}^n$ be P -measurable and

$$P(\mathbb{R}^n) = \int_{\mathbb{R}^n} dP = \int_{\mathbb{R}^n} f d\mu = 1.$$

The probability measure of a set $A \subset \mathbb{R}^n$ is called the *probability* of A . The center of gravity of $\mathbb{R}^n$ defined by means of the probability measure

$$\mathbf{x}_0 = \int_{\mathbb{R}^n} \mathbf{x} dP = \int_{\mathbb{R}^n} \mathbf{x} f(\mathbf{x}) d\mu$$

is called the *expected value* (of the random variable under consideration).

EXAMPLE 10 Let $A \subset \mathbb{R}^n$ be measurable and $\mu(A) > 0$ (μ = the volume measure of $\mathbb{R}^n$) and let the probability density of the random variable $\mathbf{x}$ be the *uniform distribution*

$$f(\mathbf{x}) = \begin{cases} 1/\mu(A), & \text{if } \mathbf{x} \in A, \\ 0, & \text{otherwise.} \end{cases}$$

Then

$$P(B) = \int_B dP = \int_B f(\mathbf{x}) d\mu = \frac{\mu(A \cap B)}{\mu(A)}, \quad B \subset \mathbb{R}^n.$$

The expected value = the centroid of A . $\square$

EXERCISES

1. In a braided sweet-bread loaf, which lies on the x -axis on the interval $[0, a]$, $a = 44$ cm, the raisins are distributed so that the line density of the raisins is

$$\rho(x) = 1.5\rho_0 \left[1 + \frac{x}{a} \left(1 - \frac{x}{a} \right) \right], \quad x \in [0, a], \quad \rho_0 = \frac{1}{\text{cm}}.$$

How many (equal-sized) raisins have been put in the loaf?

2. The mean distance of a point $P_0 \hat{=} \vec{r}_0$ from the points of a set $A \subset \mathbb{R}^2$ can be defined as the integral $d = \frac{1}{\mu(A)} \int_A |\vec{r} - \vec{r}_0| d\mu$. Compute the mean distance of the point $(2, 0)$ from the square $A = [0, 1] \times [0, 1]$.
3. On the surface of a triangular plate $A : x \geq 0 \ \& \ x + |y| \leq a$ there acts a pressure $p = p_0(1 - x^2/a^2)$ in the direction of the surface normal, $p_0 =$ constant. Compute the total force due to the pressure and the moment with respect to the origin.
4. In the corner of a building there is a tetrahedral sand pile

$$A = \{(x, y, z) \in \mathbb{R}^3 \mid x, y, z \geq 0 \ \& \ x + y + z \leq a\}.$$

In the pile the mass density is

$$\rho(x, y, z) = \rho_0 + \frac{\rho_0}{30} \left(1 - \frac{x + y + z}{a} \right)^2,$$

where ρ_0 (= constant) is the mass density of dry sand and the rest of the density represents water bound in the sand. If the whole mass of the pile is 100 kg, then how many kilograms of water are in the pile?

5. In parts a)–k), determine the centroid of the set A ; in the other parts, the center of gravity of the body $A \subset \mathbb{R}^3$ when the mass density ρ is given ($a > 0$, $\rho_0 = \text{constant}$).
- $A \subset \mathbb{R}^2: x, y \geq 0 \ \& \ x + y \leq 1$
 - $A \subset \mathbb{R}^2: 0 \leq x \leq 1 \ \& \ -x^2 \leq y \leq x^3$
 - $A \subset \mathbb{R}^2: 0 \leq x \leq 1 \ \& \ x^m \leq y \leq \sqrt[m]{x} \ (m \in \mathbb{N}, m \geq 2)$
 - $A \subset \mathbb{R}^2: x = t - \sin t \ \& \ 0 \leq y \leq 1 - \cos t, \ t \in [0, 2\pi]$
 - $A \subset \mathbb{R}^2: x^2 + y^2 \leq a^2 \ \& \ 0 \leq y \leq x$
 - $A \subset \mathbb{R}^3: x, y, z \geq 0 \ \& \ x + 2x + 3y \leq 6$
 - $A \subset \mathbb{R}^3: 0 \leq x \leq 1 \ \& \ 0 \leq y \leq x^2 \ \& \ 0 \leq z \leq xy^2$
 - $A \subset \mathbb{R}^3: 0 \leq x, y, z \leq 1 \ \& \ x + y + z \leq 2$
 - $A \subset \mathbb{R}^3: 0 \leq x, y, z \leq 4 \ \& \ (x-1)^2 + (y-2)^2 + (z-3)^2 \geq 1$
 - $A \subset \mathbb{R}^3: x, y, z \geq 0 \ \& \ x^2 + y^2 + z^2 \leq a^2$
 - $A \subset \mathbb{R}^3: 0 \leq y \leq x \ \& \ z \geq 0 \ \& \ x^2 + y^2 + z^2 \leq a^2$
 - $A: x, y, z \geq 0 \ \& \ x + y + z \leq a, \ \rho = \rho_0 z/a$
 - $A: 0 \leq x \leq a \ \& \ a|y| \leq x^2 \ \& \ |z| \leq a, \ \rho = \rho_0 x/a$
 - $A: 0 \leq x, y, z \leq a, \ \rho = \rho_0(x^2 + y^2 + z^2)/a^2$
 - $A: 0 \leq x \leq a \ \& \ y^2 + z^2 \leq 4x^2, \ \rho = \rho_0 x/a$
 - $A: 0 \leq x \leq a\pi/2 \ \& \ y^2 + z^2 \leq a^2 \cos^2(x/a), \ \rho = \rho_0(x/a)^2$
 - $A: x^2 + y^2 \leq a^2 \ \& \ 0 \leq z \leq 2a, \ \rho = \rho_0 z$
 - $A: x, y, z \geq 0 \ \& \ x^2 + y^2 + z^2 \leq a^2, \ \rho = \rho_0(x^2 + y^2)/a^2$
 - $A: x \geq 0 \ \& \ y^2 + z^2 \leq a^6/(a^2 + x^2)^2, \ \rho = \rho_0 x/a$
 - $A: x, y \geq 0 \ \& \ 0 \leq z \leq e^{-(x+y)/a}, \ \rho = \rho_0(x^2 + y^2)/a^2$
6. On a triangular dam gate with vertices at the points $A = (0, 0, 0)$, $B = (0, 0, -L)$ and $C = (0, L, 0)$, there acts the water pressure $\vec{p}(y, z) = \rho g z \vec{i}$. The gate is just being opened (in a controlled manner), so that, in addition to the pressure load, there act on the gate a moment $\vec{M} = M \vec{k}$ restraining the opening and the support forces of the hinges $\vec{F}_A = F_A \vec{i}$ and $\vec{F}_B = F_B \vec{i}$. Compute the total force of the pressure load and its moment with respect to the hinge A , and by means of these further (from force and moment balance) the unknown quantities M , F_A and F_B .
7. In the following a planar region A is given. Using Pappus's rule, determine in part a) the centroid of A and in the other parts the volume $\mu(V)$, where V is the solid of revolution generated when A rotates about the x -axis.
- $x^2 + y^2 \leq R^2, \ y \geq 0$.
 - Triangle with vertices $(2, 1)$, $(1, 2)$ and $(4, 2)$.
 - Parallelogram with vertices $(3, 2)$, $(1, 3)$, $(4, 6)$ and $(6, 5)$.
 - The region between the curves $y = x^2$ and $y = \sqrt{x}$ on the interval $x \in [0, 1]$.

- e) The region between the curves $x = \sin y$ and $x = -\sin y$ on the interval $y \in [0, \pi]$.
- f) The region between the positive y -axis and the curve $x = e^{-y}$.
8. The moment of inertia, with respect to the z -axis, of the body bounded by the coordinate planes and the planes $z = a$ and $x + y + z = 3a$ ($a > 0$) is $I_z = kma^2$, where m = the mass of the body. Compute the coefficient k when the mass density of the body is a) $\rho = \rho_0$, b) $\rho = \rho_0(x + y + z)/a$ ($\rho_0 = \text{constant}$).
9. a) A homogeneous body is bounded by the plane $z = a$ and the paraboloid $x^2 + y^2 = az$, $a > 0$. Using cylindrical coordinates, compute the coefficients k_x , k_y and k_z in the formulas $I_x = k_x ma^2$, $I_y = k_y ma^2$ and $I_z = k_z ma^2$, where I_x, I_y, I_z are the moments of inertia of the body with respect to the coordinate axes and m = the mass of the body.
- b) For which value of α ($\alpha \in \mathbb{R}$) are the moments of inertia of the homogeneous body $A : 0 \leq z/a \leq (r/a)^\alpha$ (cylindrical coordinates, $a > 0$) with respect to all three Cartesian coordinate axes equal?
- c) The mass density of a ball of radius R in spherical coordinates is $\rho(r) = \rho_0(r/R)^\alpha$ ($\rho_0 = \text{constant}$) and its mass = m . For which value of α is the moment of inertia of the ball, with respect to a line through the center of the ball, $I_S = \frac{1}{2}mR^2$?
10. A body made of birch wood (homogeneous, density = 500 kg/m³) has the shape of a rectangular box whose edge lengths are 20, 30 and 60 cm. Compute the moments of inertia of the body with respect to a) the edges, b) the diagonals.
11. A homogeneous ball of radius R has center = $(3a, -2a, 6a)$ and mass = m . Determine some principal-inertia coordinate system and the principal moments of inertia when the center of rotation = the origin. *Hint:* See Exercise X.6: 6a.
12. If x and y are independent random numbers, then the probability that (x, y) falls in the set $A \subset \mathbb{R}^2$ is $P(A) = \mu(A \cap B)$, where μ is the area measure of $\mathbb{R}^2$ and $B = [0, 1] \times [0, 1]$. With what probability does one have a) $x^2 + y^2 < 1$, b) $x + y \geq 3/2$, c) $x + y = 1$?
13. (*) a) The coordinate planes and the plane $x + y + z = a > 0$ bound a homogeneous body of mass = m . Determine the principal-inertia coordinate system and the principal moments of inertia of the body when the center of rotation = the origin.

- b) A body has mass $= m$ and its center of gravity is at the origin. Show that if the product-of-inertia matrix of the body with respect to the origin $= \mathbf{J}$, then the product-of-inertia matrix with respect to the point $P = (x_0, y_0, z_0)$ is $\mathbf{J} + \mathbf{a}\mathbf{a}^T$, where $\mathbf{a}^T = [x_0, y_0, z_0]$.
- c) Solve the problem of part a) when the center of rotation is the center of gravity.
14. (*) (Monument) The memorial of an unknown mathematician is made of solid iron following the instructions below, in a coordinate system in which the positive x -axis points east and the positive y -axis points north. The ground plane is the xy -plane and the unit of length is the meter.
1. Viewed from the east, the profile of the monument is the region bounded by the lines $z = 0$, $y = 2$ and the curve $z = y^2$.
 2. Viewed from above, the profile of the monument is a rhombus with vertices at the points $A = (-\frac{3}{2}, 0)$, $B = (1, 0)$, $C = (0, 2)$ and $D = (\frac{5}{2}, 2)$.
 3. Every vertical, east–west cross-section of the monument is a triangle whose apex is on the ground on the segment BC and whose side opposite this apex is horizontal.
- a) How much does the monument weigh? ($\rho = 7800 \text{ kg/m}^3$)
- b) Determine the center of gravity of the monument.
- c) The monument is supported by fastening the vertex A to a concrete weight sunk into the ground. How large must the mass of this counterweight be, at minimum, so that the monument does not topple?

XIII.6 Line integrals

In this section we consider integrals of the form

$$I(f, S, \mu) = \int_S f d\mu,$$

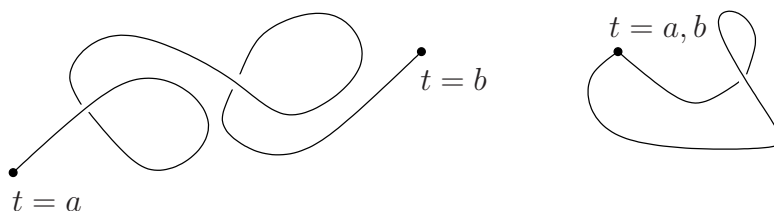
where $S \subset \mathbb{R}^2$ or $S \subset \mathbb{R}^3$ is a *curve* (in the plane or in space) and μ is the *arc-length measure* associated with the curve. Such integrals are called *line integrals* (or *curve integrals*). The arc-length measure along a line is the ordinary Jordan length measure. In the case of a plane curve (arc) $S : y = f(x)$ & $x \in [a, b]$, the arc-length measure has been defined earlier (Section V.4) and also interpreted by means of an integral (Section VIII.8). In the following we generalize this concept and study how line integrals — and the arc-length measures themselves — are computed in practice.

We take the same starting point as in Section IV.4, i.e. we assume the curve S to be parametrized by means of a real variable t on the interval $[a, b]$. Then, for example, a space curve is defined as a set of points of $\mathbb{R}^3$ (or E^3)

$$S = \{(x, y, z) \mid x = x(t), y = y(t), z = z(t), t \in [a, b]\}.$$

The functions $x(t)$, $y(t)$, $z(t)$ (in the case of a plane curve $x(t)$, $y(t)$) are assumed in what follows to be at least continuous on the interval $[a, b]$, so that the curve has no 'breaks'. Regularity assumptions on the functions will be added later as needed.*

If the parametrization of the curve $t \in [a, b] \mapsto \vec{r}(t)$ is injective on the half-open interval $[a, b)$, then it is a *simple curve*, i.e. a *Jordan curve*. If in addition $\vec{r}(a) = \vec{r}(b)$, then the curve is *closed*.



*Note that if the functions $x(t)$, $y(t)$ and $z(t)$ are merely continuous, one cannot draw far-reaching conclusions about the 'appearance' of the curve S . For example, one can construct on the interval $[0, 1]$ continuous functions $x(t)$ and $y(t)$ such that $S = \{(x(t), y(t)) \mid t \in [0, 1]\} = [0, 1] \times [0, 1]$ (!). The first to construct such a 'space-filling curve' was **G. Peano** in 1890.

In the computational technique presented below, only sufficient regularity is required of the parametrization of the curve. The curve thus need not be a Jordan curve; it may be self-intersecting or even wind on top of itself.

EXAMPLE 1 The unit circle or an arc of it can be parametrized in the form

$$S = \{ (x, y) \in \mathbb{R}^2 \mid x = \cos t, y = \sin t, t \in [a, b] \}.$$

This is a Jordan curve (a circular arc) if $b - a < 2\pi$, and a closed Jordan curve if $b - a = 2\pi$. More generally it is a parametric curve, which can be interpreted, e.g. as representing uniform circular motion (time parametrization). $\square$

EXAMPLE 2 A helix is a (non-closed) Jordan curve in space of the form

$$S = \{ (x, y, z) \in \mathbb{R}^3 \mid x = R \cos t, y = R \sin t, z = ct, t \in [a, b] \},$$

where $R > 0$ and $c \neq 0$. $\square$

The arc-length measure

Let us first generalize the concept of the arc-length measure from Section V.4 to a parametric space curve $S = \{(x(t), y(t), z(t)) \mid t \in [a, b]\}$. (The plane curve is covered as the special case $z(t) = 0$.) Let $\{t_k, k = 0, \dots, n\}$ be a partition of the interval $[a, b]$ ($a = t_0 < t_1 < \dots < t_n = b, n \in \mathbb{N}$), let $\mathcal{P} = \{P_k = (x(t_k), y(t_k), z(t_k)), k = 0, \dots, n\}$, and let, further, $S_{\mathcal{P}}$ be the polygonal line consisting of the segments $P_{k-1}P_k, k = 1 \dots n$. Then if for the arc-length measure one assumes the basic axioms

- axiom 1: segment AB : measure = $|\overrightarrow{AB}|$,
- axiom 2: additivity of the measure on a polygonal line,

then the length measure of $S_{\mathcal{P}}$ is

$$\mu(S_{\mathcal{P}}) = \sum_{k=1}^n |\vec{r}(t_k) - \vec{r}(t_{k-1})|.$$

As earlier in Section V.4, the measure of the curve S is then defined by

$$\mu(S) = \sup_{\mathcal{P}} \mu(S_{\mathcal{P}}).$$

The measure is defined exactly when the set of real numbers $\{\mu(S_{\mathcal{P}})\}$ is bounded, i.e. S is measurable, i.e. *rectifiable*, exactly under this condition.

From the above definition one sees that the arc-length measure resembles the Jordan measures in that the basic axioms require only the measure of a simple basic set (here a segment) and the additivity principle. The approximation principle used in defining arc length, according to which the curve is approximated by finite combinations of basic sets, also resembles the construction of the Jordan measures.

Arc-length parametrization

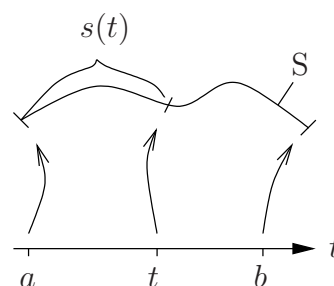
Let S be a rectifiable Jordan curve. Then if $S(t)$, $t \in [a, b]$ is the part of the curve S corresponding to parameter values on the interval $[a, t]$, then from the definition of the arc-length measure and the assumed injectivity of the map $t \mapsto \vec{r}(t)$ it follows that

$$t_1, t_2 \in [a, b] \ \& \ t_1 < t_2 \Rightarrow \mu(S(t_2)) - \mu(S(t_1)) \geq |\vec{r}(t_2) - \vec{r}(t_1)| > 0.$$

Consequently, if one denotes

$$s(t) = \mu(S(t)),$$

then $s(t)$ is strictly increasing on the interval $[a, b]$.



Thus to every $s \in [0, \mu(S)]$ there corresponds a unique $t \in [a, b]$ such that $\mu(S(t)) = s$. On the basis of this invertible one-to-one correspondence, the curve S can equally well be parametrized by means of the arc length s , i.e. one can write

$$S = \{P \in E^d \mid P \hat{=} \vec{r} = \vec{r}(s), \ 0 \leq s \leq \mu(S)\}.$$

This is called the *arc-length parametrization*. Computing it in practice is a problem in itself, and it is not in fact useful when computing line integrals in practice. As a 'conceptual parametrization' for curve-theoretic considerations, however, it is natural, for the reason that by definition

$$|\vec{r}'(s)| = 1.$$

Because of this property, the arc-length parametrization is the 'easiest' parametrization when one wishes to study the geometric properties of a curve (independent

of the parametrization), cf. the discussions of Sections VII.2 and VII.3. The arc-length parametrization is also what one has in mind in the general notation for the arc-length measure

$$d\mu = ds \quad (\text{arc-length measure}).$$

This notation is adopted in what follows.

The computational technique for line integrals

When computing line integrals in practice, one starts from a parametrization $t \mapsto \vec{r}(t)$ of the curve, assumed known (which is rarely the arc-length parametrization), and tries by means of it to 'straighten' the integral into an ordinary definite integral over the interval $[a, b]$. This means the transformation

$$\int_S f ds = \int_{[a,b]} g d\mu',$$

where $g(t) = f(x(t), y(t), z(t))$ and μ' is the counterpart of the arc-length measure on the interval $[a, b]$. To determine the measure μ' one proceeds in the same way as in a change of variables (cf. Section XIII.4). If $\Delta S(t)$ is the part of the curve S corresponding to the parameter interval $[t, t + \Delta t] \subset [a, b]$, then insofar as one can determine the *transformation ratio*

$$J(t) = \lim_{\Delta t \rightarrow 0^+} \frac{\mu(\Delta S(t))}{\Delta t},$$

and moreover $J(t)$ is sufficiently regular (e.g. continuous or piecewise continuous), then

$$\int_S f ds = \int_a^b g(t) J(t) dt.$$

The line integral has thus been reduced to an ordinary definite integral with respect to the Jordan measure.

To determine the transformation ratio $J(t)$, assume that $x(t)$, $y(t)$ and $z(t)$ are continuously differentiable on the interval $[a, b]$. (The assumption can be relaxed somewhat, see the comments below.) Let ΔS be the part of the curve on the interval $[t, t + \Delta t] \subset [a, b]$ ($\Delta t > 0$) and let $\{t_j, j = 0 \dots n\}$ be a partition of the interval $[t, t + \Delta t]$, i.e. $t = t_0 < t_1 < \dots < t_n = t + \Delta t$. Then by the mean value theorem of differential calculus

$$x(t_j) - x(t_{j-1}) = x'(\xi_j)(t_j - t_{j-1}), \quad \xi_j \in (t_{j-1}, t_j).$$

Since x' is continuous on the interval $[a, b]$ and $\xi_j \in [t, t + \Delta t]$, then here

$$x'(\xi_j) = x'(t) + o(1), \quad \text{as } \Delta t \rightarrow 0.$$

[Here one relies on the deeper logic of continuity (uniform continuity), cf. the corresponding reasoning in Section XIII.4.] Estimating in the same way $y(t_j) - y(t_{j-1})$ and $z(t_j) - z(t_{j-1})$, one concludes that

$$\begin{aligned} & \sqrt{[x(t_j) - x(t_{j-1})]^2 + [y(t_j) - y(t_{j-1})]^2 + [z(t_j) - z(t_{j-1})]^2} \\ &= \sqrt{[x'(t)]^2 + [y'(t)]^2 + [z'(t)]^2} (t_j - t_{j-1}) + o(1)(t_j - t_{j-1}), \quad \text{as } \Delta t \rightarrow 0. \end{aligned}$$

Summing over j and using the definition of arc length, it follows that

$$\mu(\Delta S) = |\vec{r}'(t)| \Delta t + o(\Delta t), \quad \text{as } \Delta t \rightarrow 0.$$

By the definition of the transformation ratio, $J(t) = |\vec{r}'(t)|$. Thus the transformation formula for the arc-length measure for a space curve is

$$ds = |\vec{r}'(t)| dt = \sqrt{[x'(t)]^2 + [y'(t)]^2 + [z'(t)]^2} dt,$$

and for the line integral one has obtained the computational formula

$$\boxed{\int_S f ds = \int_a^b g(t) |\vec{r}'(t)| dt, \quad g(t) = f(\vec{r}(t)).} \quad (\star)$$

The formula is valid under somewhat weaker conditions than those assumed. For example, the following assumptions are together sufficient:

- (i) $x(t)$, $y(t)$, $z(t)$ are continuous on the interval $[a, b]$.
- (ii) $x'(t)$, $y'(t)$, $z'(t)$ exist and are continuous on the open intervals (c_{i-1}, c_i) , $i = 1 \dots m$, where $a = c_0 < c_1 < \dots < c_m = b$ ($m \in \mathbb{N}$).
- (iii) $g(t) |\vec{r}'(t)|$ is Riemann integrable (in the extended sense) on the interval $[a, b]$.

In applications there is hardly any need to go beyond these relaxations.

If in formula $(\star)$ one sets $f = 1$ ($\Rightarrow g = 1$), the result is the formula for arc length

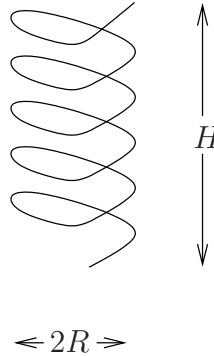
$$\boxed{\mu(S) = \int_a^b |\vec{r}'(t)| dt.}$$

Note that if one uses the arc-length parametrization ($t = s$), then $a = 0$, $b = \mu(S)$ and $|\vec{r}'(s)| = 1$, so that the formula takes the form

$$\mu(S) = \int_0^{\mu(S)} ds = \left|_0^{\mu(S)} s = \mu(S).\right.$$

This is of course only a tautology, of no practical use in computing arc length.

EXAMPLE 3 A helical spring made of a thin, uniformly thick, homogeneous wire is the space curve $x = R \cos t$, $y = R \sin t$, $z = Ht/(12\pi)$, $0 \leq t \leq 12\pi$. Compute a) the length of the spring, b) the moment of inertia with respect to the line $x = R$, $y = 0$, when the mass of the spring $= m$.



Solution a)

$$\begin{aligned} \mu(S) &= \int_0^{12\pi} \sqrt{[x'(t)]^2 + [y'(t)]^2 + [z'(t)]^2} dt \\ &= \int_0^{12\pi} \sqrt{R^2 + \left(\frac{H}{12\pi}\right)^2} dt \\ &= \underline{\underline{\sqrt{(12\pi R)^2 + H^2}}}. \end{aligned}$$

b) Let the mass density of the wire per unit length $= \rho_0$, so that by part a) $m = \rho_0 \sqrt{(12\pi R)^2 + H^2}$. The moment of inertia with respect to the line $S : x = R$, $y = 0$ is

$$\begin{aligned} I_S &= \int_S [(x - R)^2 + y^2] \rho_0 ds \\ &= \int_0^{12\pi} \rho_0 R^2 [(1 - \cos t)^2 + \sin^2 t] \sqrt{R^2 + \left(\frac{H}{12\pi}\right)^2} dt \\ &= \frac{mR^2}{12\pi} \int_0^{12\pi} (2 - 2 \cos t) dt \\ &= \underline{\underline{2mR^2}}. \end{aligned}$$

This result could have been reached without integration too: since the entire mass of the wire is at the constant distance R from the z -axis, the moment of inertia with respect to the z -axis is $I_z = mR^2$, so that from Steiner's rule (see the previous section) it follows that $I_S = mR^2 + mR^2 = 2mR^2$. $\square$

Plane curves

In the case of a plane curve, $|\vec{r}'(t)| = \sqrt{[x'(t)]^2 + [y'(t)]^2}$. If the curve is given in the form $y = f(x)$ and one chooses $t = x$, then one obtains the (already familiar, cf. Section VIII.8) result $|\vec{r}'(x)| = \sqrt{1 + [f'(x)]^2}$. If, on the other hand, the curve is given in polar form $r = f(\varphi)$, then it is natural to choose the parameter $t = \varphi$, so that

$$\begin{aligned} x'(\varphi) &= \frac{d}{d\varphi}[f(\varphi) \cos \varphi] = f'(\varphi) \cos \varphi - f(\varphi) \sin \varphi, \\ y'(\varphi) &= \frac{d}{d\varphi}[f(\varphi) \sin \varphi] = f'(\varphi) \sin \varphi + f(\varphi) \cos \varphi, \end{aligned}$$

and consequently

$$|\vec{r}'(\varphi)| = \sqrt{[x'(\varphi)]^2 + [y'(\varphi)]^2} = \sqrt{[f(\varphi)]^2 + [f'(\varphi)]^2}.$$

Thus, when computing line integrals over plane curves, one can, depending on the representation of the curve, use the measure transformations

$$ds = \begin{cases} \sqrt{[x'(t)]^2 + [y'(t)]^2} dt & (x = x(t), y = y(t)) \\ \sqrt{1 + [f'(x)]^2} dx & (y = f(x)) \\ \sqrt{[f(\varphi)]^2 + [f'(\varphi)]^2} d\varphi & (r = f(\varphi)) \end{cases}$$

EXAMPLE 4 Compute the length $\mu(S)$ of the semicircular arc $S : x^2 + y^2 = R^2, y \geq 0$ in three different ways.

Solution (a) $x = R \cos t, y = R \sin t, t \in [0, \pi]$

$$\Rightarrow \mu(S) = \int_0^\pi \sqrt{[x'(t)]^2 + [y'(t)]^2} dt = \int_0^\pi R dt = \underline{\underline{\pi R}}.$$

(b) $y = \sqrt{R^2 - x^2} = f(x), x \in [-R, R]$

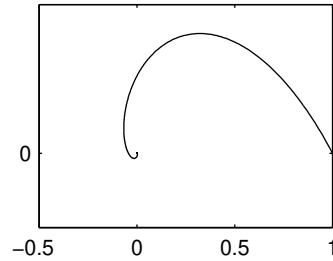
$$\begin{aligned} \Rightarrow \mu(S) &= \int_{-R}^R \sqrt{1 + [f'(x)]^2} dx = \int_{-R}^R \frac{R}{\sqrt{R^2 - x^2}} dx \\ &= \int_{-\pi/2}^{\pi/2} R dt = \underline{\underline{\pi R}}. \quad [\text{substitution } x = R \sin t] \end{aligned}$$

(c) $r = R = f(\varphi)$, $\varphi \in [0, \pi]$

$$\Rightarrow \mu(S) = \int_0^\pi \sqrt{[f(\varphi)]^2 + [f'(\varphi)]^2} d\varphi = \int_0^\pi R d\varphi = \underline{\underline{\pi R}}. \quad \square$$

EXAMPLE 5 The arc length of the logarithmic spiral $r = e^{-\varphi}$ on the interval $\varphi \in [0, \infty)$ is

$$\begin{aligned} \mu(S) &= \int_0^\infty \sqrt{[f(\varphi)]^2 + [f'(\varphi)]^2} d\varphi \\ &= \int_0^\infty \sqrt{2} e^{-\varphi} d\varphi = \underline{\underline{\sqrt{2}}}. \quad \square \end{aligned}$$



EXAMPLE 6 Compute the line integral $\int_S f ds$, when a) $f(x, y) = y$, b) $f(x, y) = x$ and $S = \{(x, y) \in \mathbb{R}^2 \mid y = x^3/3 \text{ \& } x \in [0, 1]\}$.

Solution

$$\begin{aligned} \text{a)} \quad \int_S y ds &= \int_0^1 \frac{1}{3} x^3 \sqrt{1 + x^4} dx \\ &= \left| \frac{1}{18} (1 + x^4)^{3/2} \right|_0^1 = \underline{\underline{\frac{1}{18} (2\sqrt{2} - 1)}}. \end{aligned}$$

$$\begin{aligned} \text{b)} \quad \int_S x ds &= \int_0^1 x \sqrt{1 + x^4} dx \quad [\text{substitution } x^2 = t, \ x dx = \tfrac{1}{2} dt] \\ &= \frac{1}{2} \int_0^1 \sqrt{1 + t^2} dt \quad [\text{substitution } t = \sinh u, \ dt = \cosh u du] \\ &= \frac{1}{2} \int_0^{\ln(\sqrt{2}+1)} \cosh^2 u du \\ &= \frac{1}{8} \int_0^{\ln(\sqrt{2}+1)} (e^{2u} + e^{-2u} + 2) du \\ &= \frac{1}{16} \left| e^{2u} - e^{-2u} + 4u \right|_0^{\ln(\sqrt{2}+1)} = \underline{\underline{\frac{1}{4} [\sqrt{2} + \ln(\sqrt{2} + 1)]}}. \quad \square \end{aligned}$$

EXERCISES

1. A wire of length $= 10\pi$ has been bent, starting from the point $(3, 0, 0)$, to follow a) the helix $S : x = 3 \cos t, y = 3 \sin t, z = 4t, t \geq 0$, b) the curve $S : x = 3 \cos^2 t, y = 4 \sin^2 t, z = 5 \sin t \cos t, t \geq 0$. Compute the other endpoint of the wire.
2. Compute the line integral $\int_S f ds$ for the given f and S :
 - a) $f(x, y) = xy, S : x = 2 \cos t, y = \sin t, t \in [0, \frac{\pi}{2}]$
 - b) $f(r, \varphi) = r^2 \varphi, S : r = \varphi, \varphi \in [0, 4\pi]$ (polar coord.)
 - c) $f(x, y, z) = x^2 z, S : r = 1, z = \varphi, \varphi \in [0, 4\pi]$ (cylindrical coord.)
 - d) $f(x, y, z) = x^2 z^3, S : r = 1, \theta = 2\pi/3, \varphi \in [0, 3\pi]$ (spherical coord.)
3. The (mathematical) centroid of a curve arc $S : t \in [a, b] \mapsto \vec{r}(t)$ is computed by the formula $\vec{r}_0 = \frac{1}{\mu(S)} \int_S \vec{r} ds$. Apply the formula:
 - a) $\vec{r} = R \cos t \vec{i} + R \sin t \vec{j}, t \in [0, \pi]$ (semicircle)
 - b) $\vec{r} = R(t - \sin t) \vec{i} + R(1 - \cos t) \vec{j}, t \in [0, 2\pi]$ (arc of a cycloid)
 - c) $\vec{r} = R \cos^3 t \vec{i} + R \sin^3 t \vec{j}, t \in [0, \pi/2]$ (arc of an astroid)
4. The endpoints of a straight wire are $(0, 0)$ and $(2a, 0)$ and the mass density of the wire per unit length is $\rho(x) = \rho_0(1 + x/a)$ ($\rho_0 = \text{constant}, a > 0$). The wire is bent, starting from the origin, to follow the circle $x^2 + y^2 = 2ay, x \geq 0$. Determine the center of gravity of the bent wire.
5. A jogger starts from the point $(1, 0)$ (unit of length = km) to run counter-clockwise around the running track $S : x^2 + y^2 = 1$. At which point is the jogger after running for one hour, if their running speed is $v(s) = v_0 e^{-0.05s}$, where $v_0 = 12$ km/h and s = the distance run in kilometers?
6. (*) Represent the following curves by means of the arc-length parametrization.
 - a) $x = R(t - \sin t), y = R(1 - \cos t), t \in [0, 2\pi]$
 - b) $y = x^2, x \geq 0$
7. (*) (Shrovetide, 3rd descent) Show that in Exercise IX.3:8 the shortest descent time of the snowboarder to the point $P = (a, 0)$ ($a > 0$) is $t_{\min} = \sqrt{2\pi a/g}$.
8. (*) The cylinders $L_1 : x^2 + y^2 = R^2, L_2 : (y - R)^2 + z^2 = R^2$ and the sphere $K : (x - R)^2 + y^2 + z^2 = R^2$ intersect one another along three space curves. Show that the lengths of these intersection curves are

- a) $\mu(L_1 \cap L_2) = \int_0^{\pi/2} \sqrt{f(\sin t)} dt, \quad f(x) = [1 - x^2(1 - x)^2]/(2x - x^2),$
b) $\mu(L_1 \cap K) = \mu(L_2 \cap K) = \int_{\pi/6}^{\pi/2} \sqrt{g(\sin t)} dt, \quad g(x) = (2x - x^2)/(2x - 1).$
9. (*) (A rope over the roof) On the xy -plane a long building has been erected, whose roof lies on the surface $z = 4 - x^2/2$ and whose eaves are along the lines $x = \pm 2, z = 2$. Over the building a (very flexible) rope has been thrown so that both ends of the rope reach the xy -plane. a) What is the minimum length of such a rope? b) What is the rope's minimum length with the additional condition that the rope passes over the eaves at the points $(2, 2, 2)$ and $(-2, -2, 2)$? c) As what kind of plane curve does the rope appear in part b) when viewed from far away in the direction of the positive z -axis?

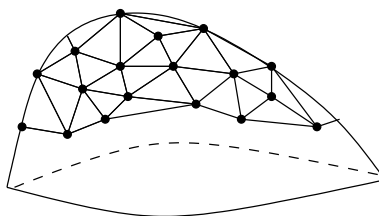
XIII.7 Surface integrals

By a *surface integral* one means an integral of the form

$$I(f, A, \mu) = \int_A f d\mu,$$

where A is a part of the surface $S \subset \mathbb{R}^3$ and μ is the *area measure on the surface* S . In what follows, in connection with the area measure of a curved surface, the notation $d\mu = dS$ is used. The relation of this measure to the (Jordan) area measure of the plane is of the same type as that of the arc-length measure to the length measure of $\mathbb{R}$. Thought of this way, the surface integral is a generalization of the plane integral, and especially in applications the term may be used even when speaking of plane integrals (cf. Section [XIII.5](#)).

Like the arc-length measure of a curve, the area measure of a curved surface too can be defined via approximations that 'flatten' the surface. A natural procedure is to approximate the surface by triangular planar surfaces. When the triangles are placed as a mesh whose nodes lie on the surface, then as the mesh is refined the total area of the triangles must approach the area measure $\mu(A)$ of A . If this is agreed on as the definition of measurability, then triangle approximations of the surface can be used for the numerical computation of area 'directly from the definition'. Such approximations have other uses as well, for example in displaying surfaces graphically (computer graphics) or even in constructing real surfaces (e.g. curved mirror surfaces).



As an alternative to direct 'number crunching', it is again possible to treat area and surface integrals by means of differential calculus, whereby one obtains for the surface integral a computational formula as a plane integral. By means of Fubini's theorem this reduces further to one-dimensional integrals, which under favorable (though in practice rather rare) circumstances can be computed in closed form. In deriving the computational formula for the surface integral, a natural starting point is the parametrization of the surface, i.e. its representation

as a *parametric surface* (cf. Section IV.4). If the parametrization is expressed in vector form

$$\vec{r} = \vec{r}(u, v) = x(u, v)\vec{i} + y(u, v)\vec{j} + z(u, v)\vec{k}$$

and one assumes that $(x(u, v), y(u, v), z(u, v)) \in A \Leftrightarrow (u, v) \in B \subset \mathbb{R}^2$, then the surface integral can be transformed into the form

$$\int_A f dS = \int_B g d\mu', \quad g(u, v) = f(x(u, v), y(u, v), z(u, v)),$$

where μ' is the counterpart, in the parameter plane, of the area measure of the surface S . The transformation to the parameter plane is again of the form

$$d\mu' = J du dv,$$

where $J = J(u, v)$ is the (generally computable) transformation ratio. When the transformation ratio is known, the computational formula is complete:

$$\int_A f dS = \int_B g J du dv.$$

To compute the transformation ratio, consider the way the rectangle

$$\Delta B = [u_0, u_0 + \Delta u] \times [v_0, v_0 + \Delta v] \subset B$$

is transformed. Assume the functions $x(u, v), y(u, v), z(u, v)$ to be sufficiently regular so that one can use the linearizing approximation

$$\vec{r}(u, v) \approx \vec{r}(u_0, v_0) + (u - u_0) \frac{\partial \vec{r}}{\partial u}(u_0, v_0) + (v - v_0) \frac{\partial \vec{r}}{\partial v}(u_0, v_0).$$

Accordingly, ΔB is mapped onto the tangent plane that touches the surface at the point $P_0 \hat{=} \vec{r}(u_0, v_0)$. The image is a parallelogram of this plane, spanned by the vectors

$$\vec{v}_1 = \Delta u \partial_u \vec{r}, \quad \vec{v}_2 = \Delta v \partial_v \vec{r}.$$

The area of the parallelogram is $|\vec{v}_1 \times \vec{v}_2| = |\partial_u \vec{r} \times \partial_v \vec{r}| |\Delta u| |\Delta v|$, so one concludes that the transformation ratio is*

$$J = \left| \frac{\partial \vec{r}}{\partial u} \times \frac{\partial \vec{r}}{\partial v} \right|.$$

*The computational formula for the transformation ratio can be justified more precisely by assuming that the partial derivatives of the functions $x(u, v), y(u, v), z(u, v)$ are continuous on a basic rectangle $T \supset B$,

If the surface is given in the form $z = f(x, y)$, i.e. $\vec{r} = \vec{r}(x, y) = x\vec{i} + y\vec{j} + f(x, y)\vec{k}$, then

$$\begin{aligned}\partial_x \vec{r} \times \partial_y \vec{r} &= \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 1 & 0 & f_x \\ 0 & 1 & f_y \end{vmatrix} = -f_x \vec{i} - f_y \vec{j} + \vec{k} \\ \Rightarrow J(x, y) &= \sqrt{1 + [f_x(x, y)]^2 + [f_y(x, y)]^2}.\end{aligned}$$

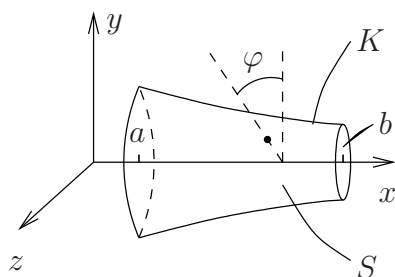
Since here $\vec{n} = (-f_x \vec{i} - f_y \vec{j} + \vec{k})/J = n_x \vec{i} + n_y \vec{j} + n_z \vec{k}$ is the unit normal vector of the surface, the result can also be expressed in the form $J(x, y) = 1/|n_z(x, y)|$.

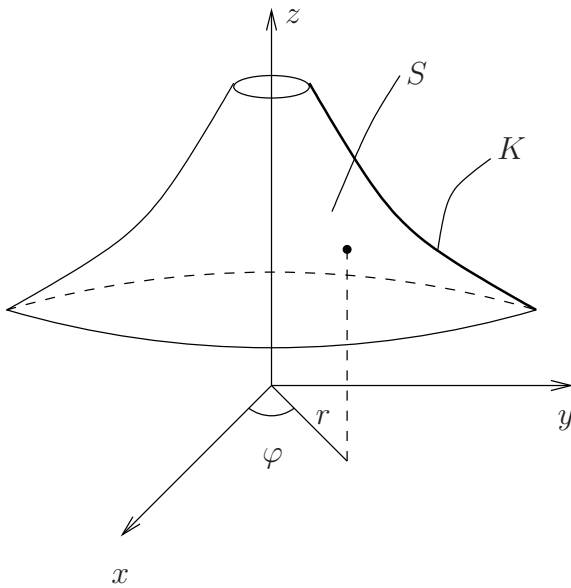
On a spherical surface of radius R

$$\begin{aligned}\vec{r} = \vec{r}(\theta, \varphi) &= R \sin \theta \cos \varphi \vec{i} + R \sin \theta \sin \varphi \vec{j} + R \cos \theta \vec{k} \\ \Rightarrow J(\theta, \varphi) &= |\partial_\theta \vec{r} \times \partial_\varphi \vec{r}| = R^2 \sin \theta.\end{aligned}$$

For a more general surface of revolution, generated when the curve $K : y = f(x)$, $x \in [a, b]$ rotates about the x -axis, or the curve $K : y = f(z)$, $z \in [a, b]$ about the z -axis (see the figures below), the transformation ratio can be computed in an analogous way (Exercise 1). The results collected:

Surface	Variables	Transformation ratio
$\vec{r} = \vec{r}(u, v)$	u, v	$ \partial_u \vec{r} \times \partial_v \vec{r} $
$z = f(x, y)$	x, y	$\sqrt{1 + [f_x(x, y)]^2 + [f_y(x, y)]^2}$
spherical surface $r = R$	θ, φ	$R^2 \sin \theta$
surface of revolution: $y = f(x) \hookrightarrow$ 	x, φ	$ f(x) \sqrt{1 + [f'(x)]^2}$
surface of revolution: $z = f(r) \hookrightarrow$ 	r, φ	$r \sqrt{1 + [f'(r)]^2}$





EXAMPLE 1 A mass m is uniformly distributed over a spherical surface of radius R . Compute the moment of inertia with respect to the z -axis.

Solution

$$\begin{aligned}
 I &= \int_A \rho_0 (x^2 + y^2) dS \\
 &= \int_B \rho_0 R^2 \sin^2 \theta \cdot R^2 \sin \theta d\theta d\varphi \quad (B = [0, \pi] \times [0, 2\pi]) \\
 &= \rho_0 R^4 \int_0^{2\pi} d\varphi \cdot \int_0^\pi \sin^3 \theta d\theta \\
 &= 2\pi \rho_0 R^4 \left[-\cos \theta + \frac{1}{3} \cos^3 \theta \right]_{\theta=0}^{\theta=\pi} \\
 &= \frac{8\pi}{3} \rho_0 R^4.
 \end{aligned}$$

Since

$$\begin{aligned}
 m &= \int_A \rho_0 dS = \rho_0 R^2 \int_0^{2\pi} d\varphi \cdot \int_0^\pi \sin \theta d\theta \\
 &= \rho_0 \cdot 4\pi R^2,
 \end{aligned}$$

we have $I = \underline{\underline{\frac{2}{3} m R^2}}$. $\square$

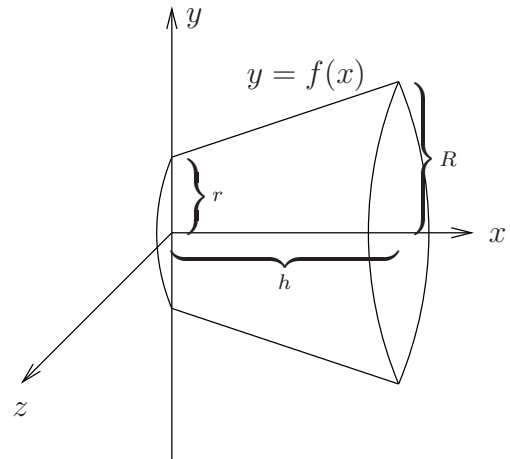
EXAMPLE 2 Compute $\int_A f \, dS$, when $f(x, y, z) = xy$ and

$$A = \left\{ (x, y, z) \in \mathbb{R}^3 \mid z = x^2 + \frac{1}{2}y^2 \text{ \& } (x, y) \in B \right\}, \quad B = [0, 1] \times [0, 1].$$

Solution

$$\begin{aligned} \int_A f \, dS &= \int_B xy \sqrt{1 + 4x^2 + y^2} \, dx dy \\ &= \int_0^1 x \left(\int_0^1 y \sqrt{1 + 4x^2 + y^2} \, dy \right) dx \\ &= \int_0^1 x \left[\frac{1}{3} (1 + 4x^2 + y^2)^{3/2} \right]_{y=0}^{y=1} dx \\ &= \int_0^1 \left[\frac{1}{3} x (2 + 4x^2)^{3/2} - \frac{1}{3} x (1 + 4x^2)^{3/2} \right] dx \\ &= \int_0^1 \left[\frac{1}{60} (2 + 4x^2)^{5/2} - \frac{1}{60} (1 + 4x^2)^{5/2} \right] dx \\ &= \frac{1}{60} (36\sqrt{6} - 25\sqrt{5} - 4\sqrt{2} + 1). \quad \square \end{aligned}$$

EXAMPLE 3 A truncated right circular cone has height $= h$, base radius $= R$ and, at the truncation, radius $= r$. Compute the lateral area $\mu(A)$.



Solution This is a surface of revolution, generated when the segment

$$K : y = f(x) = r + (R - r) \frac{x}{h}, \quad x \in [0, h]$$

rotates about the x -axis. Computing with the parameters $(x, \varphi) \in B = [0, 1] \times [0, 2\pi]$, one obtains

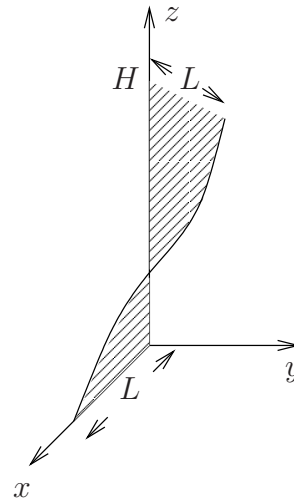
$$\begin{aligned}
 \mu(A) &= \int_B f(x) \sqrt{1 + [f'(x)]^2} dx d\varphi \\
 &= 2\pi \int_0^h \sqrt{1 + \left(\frac{R-r}{h}\right)^2} \left[r + (R-r)\frac{x}{h} \right] dx \\
 &= \pi(R+r)h \sqrt{1 + \left(\frac{R-r}{h}\right)^2} \\
 &= \underline{\underline{\pi(R+r)\sqrt{h^2 + (R-r)^2}}}. \quad \square
 \end{aligned}$$

EXAMPLE 4 Compute the area of the ruled surface

$$\vec{r}(u, v) = v[\cos(\omega u)\vec{i} + \sin(\omega u)\vec{j}] + u\vec{k}, \quad (u, v) \in [0, H] \times [0, L].$$

Solution

$$\begin{aligned}
 \partial_u \vec{r} \times \partial_v \vec{r} &= \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ -\omega v \sin(\omega u) & \omega v \cos(\omega u) & 1 \\ \cos(\omega u) & \sin(\omega u) & 0 \end{vmatrix} \\
 &= -\sin(\omega u)\vec{i} + \cos(\omega u)\vec{j} - \omega v\vec{k} \\
 \Rightarrow J &= |\partial_u \vec{r} \times \partial_v \vec{r}| = \sqrt{1 + \omega^2 v^2}.
 \end{aligned}$$



$$\begin{aligned}
\mu(A) &= \int_0^H \left(\int_0^L \sqrt{1 + \omega^2 v^2} dv \right) du \\
&= H \int_0^L \sqrt{1 + \omega^2 v^2} dv \\
&\quad [\text{substitution } \omega v = \sinh t, \quad dv = \omega^{-1} \cosh t dt, \quad \alpha = \operatorname{arsinh}(\omega L)] \\
&= H\omega^{-1} \int_0^\alpha \cosh^2 t dt \\
&= H\omega^{-1} \left[\frac{1}{2} \sinh t \cosh t + \frac{1}{2} t \right]_0^\alpha \\
&= \frac{1}{2} \left[\ln(\omega L + \sqrt{\omega^2 L^2 + 1}) + \omega L \sqrt{\omega^2 L^2 + 1} \right] H\omega^{-1}. \quad \square
\end{aligned}$$

Solid angle

Let $V \subset \mathbb{R}^3$ be a subset of space, in practice e.g. a solid body, a surface, or part of a surface. In spherical coordinates, define the set $B \subset [0, \pi] \times [0, 2\pi]$ by the condition

$$B = \{ (\theta, \varphi) \mid P = (r, \theta, \varphi) \in V \text{ for some } r > 0 \}.$$

Then V is seen from the origin in the direction $\vec{e}_r(\theta, \varphi)$ exactly when $(\theta, \varphi) \in B$. Let A be the part of the unit sphere S corresponding to the set B of the spherical-surface coordinates (θ, φ) . Then one says that V is seen from the origin in the *solid angle*

$$\Omega = \int_A dS = \int_B \sin \theta d\theta d\varphi.$$

Its largest value (when $A = S$) is 4π .

EXAMPLE 5 In what solid angle is the cylindrical surface

$$A = \{(x, y, z) \in \mathbb{R}^3 \mid x^2 + y^2 = R^2, 0 \leq z \leq H\}$$

seen from the origin?

Solution On this surface the spherical-surface coordinates take the values

$$(\theta, \varphi) \in B = [\theta_0, \pi/2] \times [0, 2\pi], \quad \cos \theta_0 = \frac{R}{\sqrt{R^2 + H^2}},$$

so

$$\Omega = \int_B \sin \theta \, d\theta d\varphi = \int_0^{2\pi} d\varphi \cdot \int_{\theta_0}^{\pi/2} \sin \theta \, d\theta = 2\pi \cos \theta_0 = 2\pi \frac{H}{\sqrt{R^2 + H^2}}. \quad \square$$

EXERCISES

- We wish to compute the area-measure transformation ratio J for the surface of revolution generated when a) the curve $K : y = f(x)$ of the xy -plane rotates about the x -axis, b) the curve $K : y = f(z)$ of the yz -plane rotates about the z -axis. Compute the transformation ratio for the parametrizations
 - $y = f(x) \cos \varphi, \quad z = f(x) \sin \varphi,$
 - $x = r \cos \varphi, \quad y = r \sin \varphi, \quad z = f(r).$
- Compute the area of the surface that the cylinder $L : x^2 + y^2 = a^2$ ($a > 0$) cuts
 - from the cone $z^2 = x^2 + y^2,$
 - from the saddle surface $az = xy,$
 - from the paraboloid $az = x^2 + y^2.$
- Compute the area $\mu(A)$, where A is
 - the boundary surface ∂V of the set $V = \{(x, y, z) \in \mathbb{R}^3 \mid x^2 + y^2 \leq z \leq \sqrt{x^2 + y^2}\},$
 - the parametric surface $x = 8u^2, \quad y = v^2, \quad z = 4uv, \quad (u, v) \in [0, 1] \times [0, 3].$
- Compute the centroid of a hemisphere (surface).
 - A mass m is uniformly distributed over the surface $S : x^2 + y^2 = R^2, \quad 0 \leq z \leq H.$ Compute, using cylindrical coordinates, the moments of inertia with respect to the coordinate axes.
- Let $f(x) \geq 0$ when $x \in [a, b]$ and S the surface generated when the curve $K : y = f(x) \quad \& \quad x \in [a, b]$ rotates about the x -axis. Prove *Guldin's rule*: the area of S = the length of K times the distance traveled by the centroid of K in the rotation.
- Compute in what solid angle the object is seen from the origin:
 - A disk of radius R on the plane $z = a > 0$, with center on the z -axis.
 - A ball of radius R whose center is at distance $= a \geq R$ from the origin.
 - The set $A : z \geq \sqrt{x^2 + y^2} + a, \quad a \geq 0$ bounded by a half-cone.
 - The paraboloid $z = x^2 + y^2 + a, \quad a \in \mathbb{R}.$

7. (*) The surface A that the cylinder $x^2 + y^2 = Rx$ cuts out of the sphere $x^2 + y^2 + z^2 = R^2$ is called *Viviani's window*. Compute the area of this surface.

8. (*) Let $A \subset \mathbb{R}^3$ be the set enclosed by the cylinders

$$S_1 : x^2 + y^2 = R^2, \quad S_2 : y^2 + z^2 = R^2, \quad S_3 : x^2 + z^2 = R^2.$$

Compute the area of the boundary surface ∂A of A .

9. (*) (The Nth nuclear power plant) The parametric representation of the outer surface of the shell of a power plant's cooling tower is (cf. Example IV.4: 14)

$$\begin{cases} x = a[(2 - 2v) \cos u - v \sin u], \\ y = a[v \cos u + (2 - 2v) \sin u], \\ z = 3av, \end{cases}$$

where $a=50$ m and $(u, v) \in B = [0, 2\pi] \times [0, 1]$. The shell is made of concrete and its thickness is 20 cm. Compute an approximate value for the amount of concrete needed using a surface integral.

10. (*) a) A plate

$$A = \left\{ (x, y, z) \mid \frac{x^2}{a^2} + \frac{y^2}{b^2} \leq 1, \quad z = c \right\} \quad (a, b, c > 0)$$

is seen from the origin in the solid angle Ω . Derive for Ω a computational formula of the form $\Omega = \int_0^{2\pi} f(\varphi) d\varphi$. b) Compute in what solid angle the spatial square

$$K = \{ (x, y, z) \mid 0 \leq x, y \leq 1, \quad z = 1 \}$$

is seen from the origin.

Chapter XIV

Gauss's and Stokes's theorems

Gauss's theorem and *Stokes's theorem* are central results of multivariable differential and integral calculus. The theorems are known more for their usefulness than for their mathematical grandeur, and in everyday contexts they are often referred to by the names 'Gauss's formula' and 'Stokes's formula'. Somewhat simplified, these formulas are about generalizing the integral formula

$$\int_a^b f'(x) dx = f(b) - f(a)$$

so that it applies to vector-valued functions of several (in practice usually two or three) variables, that is, to vector fields. Gauss's theorem (formula) is invoked very frequently when the various *conservation laws* of physics are to be written in the form of partial differential equations. Stokes's theorem too has this kind of use, especially in electromagnetism.

In Section [XIV.1](#) we first examine *path integrals*, which are akin to line integrals. These are useful in connection with Gauss's and Stokes's theorems and, more generally, in applications of physical vector fields. In Section [XIV.2](#) Gauss's theorem is derived in the plane and in space, and the generalized form of the theorem is presented. The starting point is provided by the *Green's formulas* for plane and volume integrals. In Section [XIV.3](#) we examine, in the light of examples, the use of Gauss's theorem when one wishes to derive the partial differential equations of physics or the continuity conditions of physical vector fields at material interfaces.

In Section [XIV.4](#) Stokes's theorem is derived, first restricted to the plane. When the result is generalized to surfaces in space, the concept of *orientability* of a surface is needed. In Section [XIV.5](#) we use Stokes's theorem to answer a question about vector fields that is central in physics: Under what conditions is an irrotational field a gradient field, that is, expressible by means of a scalar potential?

XIV.1 Vector fields and path integrals

In Section [XIII.6](#) we examined line integrals, in which one integrates over a curve in the plane or in space with respect to the arc-length measure. In this section the object of study is another class of line integrals, for which we will henceforth use the name *path integrals*. Path integrals are characterized by the fact that the integration along the curve proceeds in a definite direction; hence the name *path*, which can be interpreted as a directed curve. In addition to the direction, path integrals are typically such that the measure associated with the integration is not the arc-length measure but some other one-dimensional measure, which is case-dependent. The direction of the path affects the path integral in such a way that if only the direction changes, i.e. the path stays otherwise (as a curve) the same, then the value of the path integral changes to its negative. Through the parametrization of the path this corresponds to the interchange rule for the definite integral, see the examples below.

In what follows we restrict ourselves to those path integrals that, in applications, are usually associated with physical vector fields (such as force, electric, and magnetic fields). Path integrals associated with vector fields will also be useful in connection with Gauss's and Stokes's theorems.*

To distinguish it from a curve, a path will henceforth be denoted by the symbol p or, more precisely,

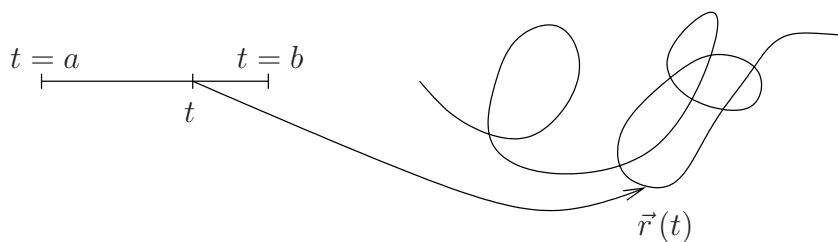
$$p : A \rightarrow B,$$

in which case the notation indicates both the endpoints of the path (curve) and its direction. Naturally the direction of the path is determined by the parametrization: the direction is either the direction of increasing parameter or the opposite direction. If the path is parametrized on the interval $t \in [a, b]$, then the path can be denoted more precisely, like a parametric curve:

$$p : t \in [a, b] \mapsto \vec{r}(t).$$

The parametrization need not be one-to-one, so a path (like a parametric curve) may intersect itself or wind over itself.

*In mathematical texts the names of the various line integrals are not entirely standardized. For instance, one may speak of 'line integrals of vector fields' when what is meant is path integrals in the sense of this text.



In what follows the objects of study are plane or space paths of the form $t \in [a, b] \mapsto (x(t), y(t))$ or $t \in [a, b] \mapsto (x(t), y(t), z(t))$. The functions $x(t)$, $y(t)$, $z(t)$ are assumed to be either continuously differentiable on the interval $[a, b]$ or to satisfy the corresponding, somewhat weaker regularity conditions, cf. Section XIII.6.

As before, the *length of the path* can be computed as an integral

$$\mu(p) = \int_a^b |\vec{r}'(t)| dt.$$

This, however, is a line integral of the already familiar kind, in which the measure is the arc-length measure and the direction of the path is immaterial.

Path integrals (also the most significant ones for what follows) are

$$\int_p f dx, \quad \int_p f dy, \quad \int_p f dz, \quad (\star)$$

where $f(x, y, z)$ (in the plane $f(x, y)$) is a known function. The measure associated with these integrals is the ordinary (1-dimensional) Jordan measure. If the parametrization of the path on an interval $t \in [a, b]$ is known, then, for example, $\int_p f dx$ is computed simply by writing $dx = x'(t)dt$ (as in a change of variable). If we further assume that $t = a$ corresponds to the initial point of the path and $t = b$ to the final point, then we obtain the computational formula

$$\int_p f dx = \int_a^b f(x(t), y(t), z(t)) x'(t) dt.$$

A change of parametrization corresponds in this formula to a (further) change of variable, so the parametrization does not affect the value of the integral.

EXAMPLE 1 A plane path p runs from the point $(0, 0)$ to the point $(1, 1)$ along the curve $y = x^2$, and the path $-p$ runs along the same curve from $(1, 1)$ to $(0, 0)$. Compute $\int_p xy dy$ and $\int_{-p} xy dy$.

Solution In both integrals we choose the parameter $t = x$:

$$\begin{aligned}\int_p xy \, dy &= \int_0^1 x \cdot x^2 \cdot 2x \, dx = \int_0^1 2x^4 \, dx = \underline{\underline{\frac{2}{5}}}, \\ \int_{-p} xy \, dy &= \int_1^0 2x^4 \, dx = - \int_0^1 2x^4 \, dx = \underline{\underline{-\frac{2}{5}}}. \quad \square\end{aligned}$$

In the latter integral of the example the interchange rule for the definite integral was used. — In fact, once the interchange rule is taken into account, the definite integral is itself interpretable as a path integral: $\int_a^b f(x) \, dx = \int_p f \, dx$, where p is the path in $\mathbb{R}$ whose initial point is a and whose final point is b (!).

The path integral $\int_p \vec{F} \cdot d\vec{r}$

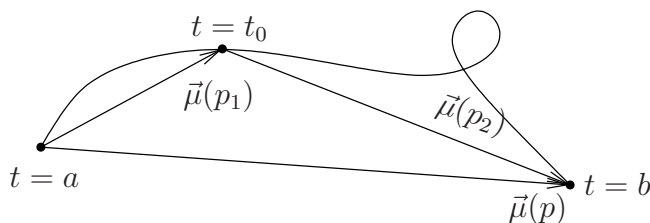
In applications in physics, path integrals are often associated with vector fields. A typical example is a (point-like) body moving in a force field $\vec{F}(x, y, z)$, whose trajectory is known over some time interval $t \in [a, b]$. In this case the work done by the force field on the body is computed as a path integral along the trajectory traversed by the body. The trajectory is thus interpreted as a path $p : t \mapsto \vec{r}(t)$, $t \in [a, b]$. If $\vec{F} = \text{constant}$, then, by the laws of physics, $\text{work} = |\vec{F}| \cdot s$, where s = the distance traveled along the path in the direction in which the force acts, i.e.

$$W = \vec{F} \cdot [\vec{r}(b) - \vec{r}(a)] \quad (\vec{F} \text{ constant}).$$

As far as the work done by a constant force field is concerned, the path is thus ‘measured’ by the vector

$$\vec{\mu}(p) = \vec{r}(b) - \vec{r}(a).$$

If this is interpreted as the (vector-valued) measure of the path, then one sees that this measure too is additive: If p_1 and p_2 are subpaths of p corresponding to parameter values on the intervals $[a, t_0]$ and $[t_0, b]$, $t_0 \in (a, b)$, then $\vec{\mu}(p) = \vec{\mu}(p_1) + \vec{\mu}(p_2)$, cf. the figure.



What if the force field $\vec{F}$ is not constant, but is continuous? In that case we proceed as with integrals in general: We introduce a partition $\{t_k, k = 0, \dots, n\}$, $a = t_0 < t_1 < \dots < t_n = b$, of the interval $[a, b]$, whereupon the path p splits into consecutive subpaths Δp_k , $k = 1 \dots n$ corresponding to parameter values on the intervals $[t_{k-1}, t_k]$. Since the function $t \mapsto \vec{r}(t)$ is continuous, the endpoints $\vec{r}(t_{k-1}) = \vec{r}_{k-1}$ and $\vec{r}(t_k) = \vec{r}_k$ of the subpaths Δp_k come ever closer to one another as the partition is refined. Then the force $\vec{F}$ is approximately constant on the subpath Δp_k (since $\vec{F}$ was continuous), so the work done by the force on this subpath is approximately

$$\Delta W_k \approx \vec{F}(\vec{r}_{k-1}) \cdot \vec{\mu}(\Delta p_k), \quad \vec{\mu}(\Delta p_k) = \vec{r}_k - \vec{r}_{k-1}.$$

As the fineness parameter of the partition $h = \max_k |\vec{r}_k - \vec{r}_{k-1}| \rightarrow 0$, we obtain for the total work done by the force field the integral expression

$$W = \lim_{h \rightarrow 0} \sum_{k=1}^n \Delta W_k = \int_p \vec{F} \cdot d\vec{\mu}.$$

This is thus to be interpreted as a path integral with respect to the vector measure $\vec{\mu}$ (!). The integral takes a somewhat more concrete form when the notation $d\vec{\mu} = d\vec{r}$ is used, so that the integral formula for the work is

$$W = \int_p \vec{F} \cdot d\vec{r} \quad (\text{work integral}).$$

By definition, approximate values for W are obtained by means of sums. For example,

$$W \approx \sum_{k=1}^n \vec{F}(\vec{r}_{k-1}) \cdot (\vec{r}_k - \vec{r}_{k-1}).$$

When computed in this way, no strong regularity assumptions on the path p are needed. For example, the work W can be computed even if p is not rectifiable (i.e. the length measure is not defined). This is because the work integral measures only the displacement in the direction in which the force acts, not the arc length.

If the parametrization $p : t \in [a, b] \mapsto \vec{r}(t)$ is assumed, then in the work integral one can write $d\vec{r} = \vec{r}'(t)dt$, which gives the computational formula

$$W = \int_a^b \vec{F}(\vec{r}(t)) \cdot \vec{r}'(t) dt.$$

If, on the other hand, the force field is expressed in coordinate form

$$\vec{F}(x, y, z) = F_1(x, y, z)\vec{i} + F_2(x, y, z)\vec{j} + F_3(x, y, z)\vec{k},$$

then, writing $d\vec{r} = dx \vec{i} + dy \vec{j} + dz \vec{k}$, the work integral decomposes into the sum of the path integrals ($\star$):

$$W = \int_p (F_1 dx + F_2 dy + F_3 dz).$$

On a sufficiently regular path the work integral can be expressed in a third way as well, since

$$d\vec{r} = \frac{\vec{r}'(t)}{|\vec{r}'(t)|} |\vec{r}'(t)| dt = \vec{t} ds,$$

where $\vec{t}$ is the unit tangent vector in the direction of the path. Accordingly, the work integral can, if desired, also be associated with the arc-length measure by the computational formula

$$W = \int_p \vec{F} \cdot \vec{t} ds.$$

As is apparent from the derivation of the formula, here $\vec{t} ds = \vec{r}'(t) dt$, so the dependence on the arc-length measure is only apparent.

EXAMPLE 2 Determine the work done by the force field

$$\vec{F} = (x^2 - y)\vec{i} - 2xy\vec{j}$$

as a body moves along the path

$$p: x(t) = 2 \cos t, \quad y(t) = \sin t, \quad t \in [0, 2\pi].$$

Solution $dx = -2 \sin t dt, \quad dy = \cos t dt, \quad t \in [0, 2\pi]$

$$\begin{aligned} \Rightarrow W &= \int_0^{2\pi} [(4 \cos^2 t - \sin t)(-2 \sin t) - (4 \cos t \sin t) \cos t] dt \\ &= \int_0^{2\pi} (-12 \cos^2 t \sin t + 2 \sin^2 t) dt \\ &= \left| 4 \cos^3 t - \cos t \sin t + t \right|_0^{2\pi} = \underline{\underline{2\pi}}. \quad \square \end{aligned}$$

Gradient field and work integral

If the force field $\vec{F}$ is a gradient field, i.e. expressible by means of a scalar potential u in the form

$$\vec{F} = -\nabla u,$$

then the work integral can be computed very simply. Indeed, in this case, by the chain rule of differentiation (Section XI.2),

$$\vec{F}(\vec{r}(t)) \cdot \vec{r}'(t) = -\nabla u(\vec{r}(t)) \cdot \vec{r}'(t) = -\frac{d}{dt}u(\vec{r}(t)),$$

so that

$$W = \int_a^b \vec{F}(\vec{r}(t)) \cdot \vec{r}'(t) dt = - \left|_a^b u(\vec{r}(t)) = u(\vec{r}(a)) - u(\vec{r}(b)).\right.$$

Thus the work integral of a gradient field is determined solely by the endpoints of the path:

The work done by a gradient field
= the potential difference between the initial and final points of the path.

EXAMPLE 2 (continued). In the example the initial and final points of the path are the same. Since $W \neq 0$, the field $\vec{F}$ of the example is not a gradient field. If, on the other hand, for example

$$\vec{F} = (x^2 - y^2)\vec{i} - 2xy\vec{j},$$

then it is clear without any further computation that $W = 0$, since

$$\vec{F} = -\nabla\left(-\frac{1}{3}x^3 + xy^2\right). \quad \square$$

Vector-valued path integrals

In applications in physics (for example, in electromagnetism) there also occur vector-valued path integrals of the form

$$\int_p f d\vec{r} \quad \text{or} \quad \int_p \vec{F} \times d\vec{r}.$$

These can be computed by means of the parametrization on the same principle as the work integral.

EXAMPLE 2 (continued) Compute $\int_p x d\vec{r}$ and $\int_p \vec{r} \times d\vec{r}$, when p is the path of the example.

Solution

$$\begin{aligned}
\int_p x \, d\vec{r} &= \int_0^{2\pi} x(t)[x'(t)\vec{i} + y'(t)\vec{j}] \, dt \\
&= \int_0^{2\pi} 2 \cos t (-2 \sin t \vec{i} + \cos t \vec{j}) \, dt \\
&= -\vec{i} \int_0^{2\pi} 4 \cos t \sin t \, dt + \vec{j} \int_0^{2\pi} 2 \cos^2 t \, dt \\
&= -\vec{i} \left[2 \sin^2 t \right]_0^{2\pi} + \vec{j} \left[t + \cos t \sin t \right]_0^{2\pi} = \underline{\underline{2\pi\vec{j}}}, \\
\int_p \vec{r} \times d\vec{r} &= \int_0^{2\pi} [x(t)\vec{i} + y(t)\vec{j}] \times [x'(t)\vec{i} + y'(t)\vec{j}] \, dt \\
&= \int_0^{2\pi} (2 \cos t \vec{i} + \sin t \vec{j}) \times (-2 \sin t \vec{i} + \cos t \vec{j}) \, dt \\
&= \int_0^{2\pi} (2 \cos^2 t + 2 \sin^2 t) \vec{k} \, dt = \vec{k} \int_0^{2\pi} 2 \, dt = \underline{\underline{4\pi\vec{k}}}. \quad \square
\end{aligned}$$

EXERCISES

1. Compute the path integral $\int_p (9x^2y \, dx - 11xy^2 \, dy)$, when the path p runs from the origin to the point $(1, 1)$ a) along the curve $x(t) = t^2$, $y(t) = t^3$, b) along the straight line, c) along the curve $\vec{r}(t) = t\vec{i} + t^\alpha\vec{j}$, $\alpha > 0$.
2. Compute the path integral $\int_p (x \, dy - y \, dx)$, when the path p runs from the point $(1, 0)$ along the logarithmic spiral $r = e^{-\varphi}$ to the origin.
3. Compute the path integral $\int_p [(y - x) \, dx + xy \, dy]$, when the path p is defined as follows:
 - a) From the point $(1, 0)$ to the point $(-1, 0)$ along the unit circle counter-clockwise
 - b) From the point $(1, 0)$ to the point $(-1, 0)$ along the unit circle clockwise
 - c) The polygonal line $ABCD$, where $A = (1, 0)$, $B = (1, 1)$, $C = (-1, 1)$ and $D = (-1, 0)$
 - d) From the point $(1, 0)$ along the unit circle back to the starting point, circling counterclockwise

- e) From the origin along the x -axis to the point $(\pi, 0)$ and back to the origin along the curve $y = \sin x$
- f) From the point $(a, 0)$ back to the starting point, circling counterclockwise around the ellipse $x^2/a^2 + y^2/b^2 = 1$ ($a, b > 0$).
4. Compute the following path integrals.
- a) $\int_p \vec{F} \cdot d\vec{r}$, when $\vec{F}(x, y, z) = \sqrt{y}\vec{i} + 2x\vec{j} + 3y\vec{k}$ and the path runs from the origin to the point $(3, 9, 27)$ along the curve $\vec{r}(t) = t\vec{i} + t^2\vec{j} + t^3\vec{k}$.
- b) $\int_p \vec{F} \cdot d\vec{r}$, when $\vec{F}(x, y, z) = x^3\vec{i} + y^2\vec{j} + z\vec{k}$ and the path runs from the origin to the point $(1, 1, 2)$ along the curve $S: x = y, z = x^2 + y^2$.
- c) $\int_p \vec{F} \times d\vec{r}$, when $\vec{F}(x, y, z) = xyz\vec{i} + y^2\vec{k}$ and the path p follows the curve of intersection of the plane $x = y$ and the surface $z = x^2$ from the origin to the point $(2, 2, 4)$.
5. The initial point of the path p is $(-1, 1, -1)$ and the final point is $(1, 2, 3)$. Using this information, compute the following path integrals (work integrals) by first writing the integrals in the form $\int_p \nabla u \cdot d\vec{r}$.
- a) $\int_p (yz \, dx + zx \, dy + xy \, dz)$ b) $\int_p (yz^2 \, dx + xz^2 \, dy + 2xyz \, dz)$
- c) $\int_p [e^x y \, dx + (e^x + z^2) \, dy + 2yz \, dz]$ d) $\int_p \sin \frac{\pi(x+y+z)}{6} (dx + dy + dz)$
6. Define $g(u, v) = \int_p (y \, dx + 2x \, dy)$, where p runs from the origin to the point (u, v) along a straight line. What is the maximum value of g on the unit circle?
7. (*) A body of mass $= m$ moves during the time interval $[t_1, t_2]$ along a path $p: t \mapsto \vec{r}(t)$ from a point P_1 to a point P_2 . During the motion the body is acted on by a force field $\vec{F}$. From the equation of motion $m\vec{r}'' = \vec{F}$ derive the energy principle

$$\frac{1}{2} m(v_2^2 - v_1^2) = \int_p \vec{F} \cdot d\vec{r}, \quad \text{where } v_i = |\vec{r}'(t_i)|, \quad i = 1, 2.$$

XIV.2 Gauss's theorem

Let us begin to examine the generalization of the integral formula

$$\int_a^b f'(x) dx = f(b) - f(a),$$

first in one dimension. Let $A \subset \mathbb{R}$ be a finite union of disjoint, closed intervals:

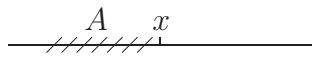
$$A = \bigcup_{i=1}^n A_i, \quad A_i = [a_i, b_i], \quad A_i \cap A_j = \emptyset, \quad i \neq j.$$

Then the above formula can be written in the form

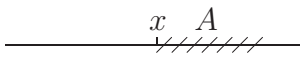
$$\begin{aligned} \int_A f' dx &= \sum_{i=1}^n [f(b_i) - f(a_i)] \\ &= \sum_{x \in \partial A} \omega(x) f(x), \end{aligned}$$

where ∂A is the boundary of A (consisting of the points $a_i, b_i, i = 1 \dots n$) and $\omega(x)$ takes the values ± 1 according to the following rule:

$$\omega(x) = \begin{cases} +1, & \text{if } (x - \delta, x) \subset A \text{ for some } \delta > 0, \\ -1, & \text{if } (x, x + \delta) \subset A \text{ for some } \delta > 0. \end{cases}$$



$\omega(x) = +1$



$\omega(x) = -1$

Let us now move to two dimensions. Let $f = f(x, y)$ be defined and continuously differentiable on a set $A \subset \mathbb{R}^2$ (about which some simplifying assumptions will be made right away) and consider the integral

$$\int_A \frac{\partial f}{\partial x} dx dy.$$

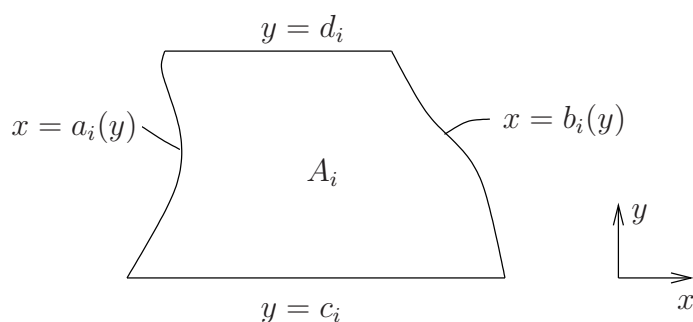
In what follows it is assumed that the set A is closed and can be divided into finitely many parts A_i of simple shape in such a way that the parts touch one another at most along their boundaries, i.e.

$$A = \bigcup_{i=1}^n A_i, \quad A_i \cap A_j = \partial A_i \cap \partial A_j, \quad i \neq j.$$

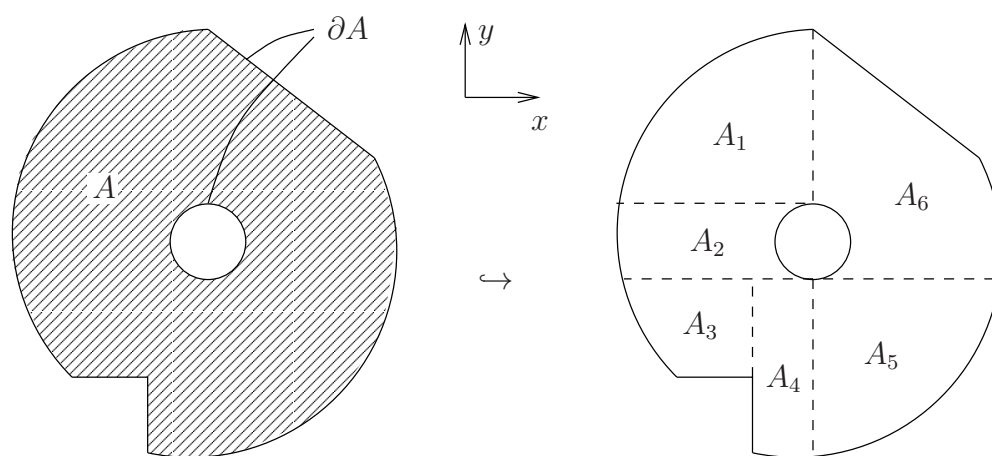
The parts A_i are further assumed to all be y -projectable. More precisely, it is assumed that each A_i is of the form

$$A_i = \{ (x, y) \in \mathbb{R}^2 \mid a_i(y) \leq x \leq b_i(y) \ \& \ y \in [c_i, d_i] \},$$

where the functions $a_i(y)$, $b_i(y)$ are continuous on the interval $[c_i, d_i]$.



The assumption on the set A is thus that a partition of A in the above manner is possible. Considering the ‘practical sets’ encountered in applications, the assumption is not very restrictive, cf. the figure.



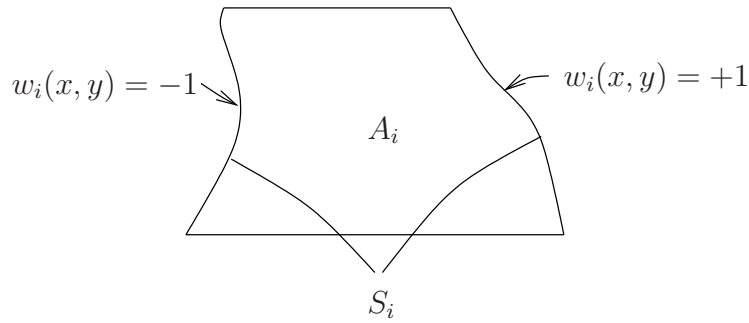
Once the partition of A has been made in the above manner, the integral under study can be broken into parts by the additivity principle (since $\mu(A_i \cap A_j) = 0$, $i \neq j$):

$$\int_A \frac{\partial f}{\partial x} dx dy = \sum_{i=1}^n \int_{A_i} \frac{\partial f}{\partial x} dx dy.$$

Here, by Fubini's theorem,

$$\begin{aligned} \int_{A_i} \frac{\partial f}{\partial x} dx dy &= \int_{c_i}^{d_i} \left(\int_{a_i(y)}^{b_i(y)} \frac{\partial f}{\partial x} dx \right) dy \\ &= \int_{c_i}^{d_i} [f(b_i(y), y) - f(a_i(y), y)] dy = \int_{S_i} \omega_i(x, y) f(x, y) dy, \end{aligned}$$

where $S_i \subset \partial A_i$ and the function w_i is defined as in the figure.



When the result obtained

$$\int_{A_i} \frac{\partial f}{\partial x} dx dy = \int_{S_i} \omega_i(x, y) f(x, y) dy$$

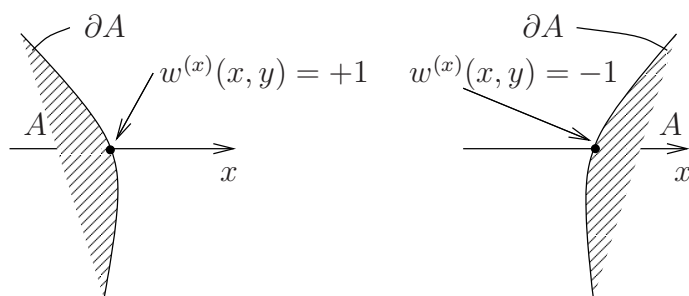
is summed over i and one notes that

$$i \neq j \ \& \ (x, y) \in S_i \cap S_j \Rightarrow \omega_i(x, y) + \omega_j(x, y) = 0,$$

then one sees that on the right-hand side the line integrals over the shared boundary lines of the subsets A_i (i.e. those lying in the interior of A) cancel. Using the additivity principle for the integral on both sides, the result is thus brought to the form

$$\int_A \frac{\partial f}{\partial x} dx dy = \int_{\partial A} \omega^{(x)} f dy,$$

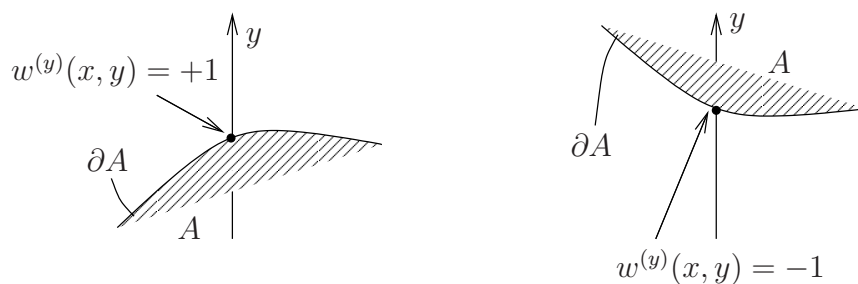
where $\omega^{(x)}(x, y) = 0$ on the parts of the boundary line ∂A parallel to the x -axis, and otherwise $\omega^{(x)}(x, y)$ takes the values ± 1 as shown in the figure.



Assuming that A can be divided in the same way into x -projectable parts, one obtains correspondingly the integral formula

$$\int_A \frac{\partial f}{\partial y} dx dy = \int_{\partial A} \omega^{(y)} f dx,$$

where $\omega^{(y)}(x, y) = 0$ on the parts of the boundary line ∂A parallel to the y -axis, and otherwise $\omega^{(y)}(x, y)$ takes the values ± 1 as shown in the figure.



Note that in the integral formulas obtained, the integrals computed over the boundary line ∂A are path integrals (cf. the previous section). These decompose into finite sums, in which each term is a definite integral over a closed interval, the measure being the ordinary length measure of $\mathbb{R}$. Thus the formulas do not even require the boundary line ∂A to be rectifiable. By assuming more regularity of the boundary line, however, the results are brought to a more easily memorable form.

The above partitions divide the boundary line ∂A into parts that are either of the form

$$S_i = \{ (x, y) \in \mathbb{R}^2 \mid x = f_i(y) \ \& \ y \in [a_i, b_i] \}$$

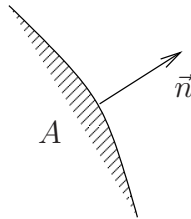
or of the form

$$S_i = \{ (x, y) \in \mathbb{R}^2 \mid y = g_i(x) \ \& \ x \in [c_i, d_i] \}.$$

Here it is assumed that the functions f_i are continuous on the closed intervals $[a_i, b_i]$ and that the derivatives f'_i are continuous on the open intervals (a_i, b_i) . Correspondingly, it is assumed that the functions g_i are continuous on the closed intervals $[c_i, d_i]$ and the derivatives g'_i continuous on the open intervals (c_i, d_i) . (These additional assumptions too are not, in practice, very restrictive, cf. the partition figure above.) In what follows, a set A that satisfies all the assumptions made will be called a *basic region*. A basic region is thus a set that can be divided into finitely many x -projectable parts, and likewise into finitely many y -projectable parts, in such a way that the boundary curves lying on the boundary line ∂A of these parts satisfy the regularity assumptions made. A basic region is a compact set (cf. Section XI.1).

If A is a basic region, then on the boundary ∂A , except at isolated points, the *outward normal* of the boundary is defined, i.e. the unit vector perpendicular to the boundary and pointing away from the set A , which will be denoted by

$$\vec{n}(x, y) = n_x(x, y)\vec{i} + n_y(x, y)\vec{j}, \quad |\vec{n}| = 1.$$

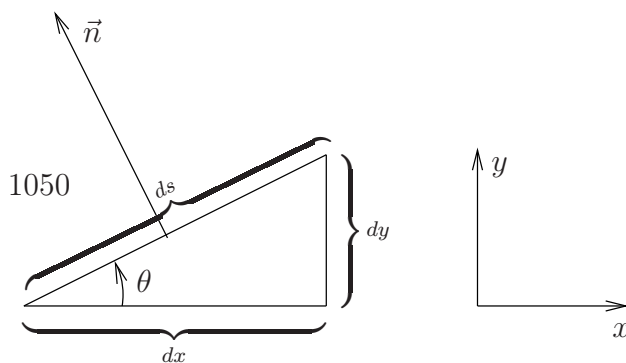


At those points $(x_i, y_i) \in \partial A$ where the outward normal $\vec{n}$ is defined, by the definition of the function $\omega^{(x)}$ we have

$$\omega^{(x)}(x, y) = \begin{cases} +1, & \text{if } n_x(x, y) > 0, \\ 0, & \text{if } n_x(x, y) = 0, \\ -1, & \text{if } n_x(x, y) < 0, \end{cases}$$

and $\omega^{(y)}(x, y)$ is determined correspondingly from $n_y(x, y)$. When these relations are taken into account, one can write (cf. the figure),

$$\omega^{(x)} dy = n_x ds, \quad \omega^{(y)} dx = n_y ds,$$



$$|n_x| = \sin \theta = \frac{dy}{ds}$$

$$|n_y| = \cos \theta = \frac{dx}{ds}$$

whereupon one obtains the easily memorable *Green's plane formulas*

$$\begin{aligned} \int_A \frac{\partial f}{\partial x} dxdy &= \int_{\partial A} n_x f ds, \\ \int_A \frac{\partial f}{\partial y} dxdy &= \int_{\partial A} n_y f ds. \end{aligned} \quad (\text{Green's plane formulas})$$

Let it be emphasized once more that in these formulas the arc-length measure has been introduced only for the sake of the mnemonic. In reality the integrals on the right-hand side of the formulas are path integrals.

Let us next consider a vector field defined on A

$$\vec{F}(x, y) = F_1(x, y)\vec{i} + F_2(x, y)\vec{j},$$

which is assumed to be continuously differentiable. Applying Green's plane formulas, we obtain

$$\begin{aligned} \int_A \nabla \cdot \vec{F} dxdy &= \int_A \frac{\partial F_1}{\partial x} dxdy + \int_A \frac{\partial F_2}{\partial y} dxdy \\ &= \int_{\partial A} (n_x F_1 + n_y F_2) ds = \int_{\partial A} \vec{n} \cdot \vec{F} ds. \end{aligned}$$

Thus we have derived

THEOREM XIV.2.1 (Gauss's theorem in the plane) If $A \subset \mathbb{R}^2$ is a basic region and $\vec{F}$ is a continuously differentiable vector field defined on A , then

$$\int_A \nabla \cdot \vec{F} dxdy = \int_{\partial A} \vec{n} \cdot \vec{F} ds.$$

The computational formula of Theorem [XIV.2.1](#), which will henceforth be called *Gauss's plane formula*, has a counterpart in three (and more) dimensions as well. Let $V \subset \mathbb{R}^3$ and assume that V can be divided into finitely many xy -projectable,

yz -projectable, and xz -projectable parts in the same manner as above. For example, the xy -projectable parts V_i are then of the form

$$V_i = \{ (x, y, z) \in \mathbb{R}^3 \mid (x, y) \in B_i \ \& \ a_i(x, y) \leq z \leq b_i(x, y) \}.$$

Here the sets $B_i \subset \mathbb{R}^2$ are again assumed to be basic regions of the plane, and it is further assumed that the functions a_i and b_i are continuous on B_i and continuously differentiable at the interior points of B_i (with continuous partial derivatives). A set V that satisfies these and the corresponding assumptions concerning the yz - and xz -projectable partitions is called a basic region of $\mathbb{R}^3$. If V satisfies these conditions, then in a manner corresponding to the plane case one sees that the integral formula

$$\int_V \frac{\partial f}{\partial z} dx dy dz = \int_{\partial V} \omega^{(z)} f dx dy$$

is correct, where $\omega^{(z)}$ takes the values $0, \pm 1$ by the corresponding principle as in the plane. Here one can write $dx dy = |n_z| dS$, where n_z is the z -component of the unit normal vector of ∂V and dS refers to the area measure of ∂V (cf. Section [XIII.7](#)). Taking also into account the sign variation of $\omega^{(z)}$, the integral formula takes the form

$$\int_V \frac{\partial f}{\partial z} dx dy dz = \int_{\partial V} n_z f dS,$$

where $\vec{n}$ is the outward normal (unit vector) of ∂V . This is one of the three *Green's space formulas* — the other two are obvious. Combining these formulas yields

THEOREM XIV.2.2 (Gauss's theorem in space) If $V \subset \mathbb{R}^3$ is a basic region and $\vec{F}$ is a continuously differentiable vector field defined on V , then

$$\boxed{\int_V \nabla \cdot \vec{F} dx dy dz = \int_{\partial V} \vec{n} \cdot \vec{F} dS.}$$

In this *Gauss's space formula*, too, the area measure of the surface ∂A serves only as a mnemonic. In reality the right-hand side of the formula consists of plane integrals, as is apparent from the derivation of the formula.

In Gauss's space formula the right-hand side is often written in the form

$$\int_{\partial A} \vec{n} \cdot \vec{F} dS = \int_{\partial A} \vec{F} \cdot d\vec{a},$$

where $d\vec{a} = \vec{n} dS$ is a 'vectorized' measure. Correspondingly, in the plane formula one can write

$$\vec{n} \cdot \vec{F} ds = \vec{F} \cdot d\vec{n}, \quad d\vec{n} = \vec{n} ds.$$

EXAMPLE 1 Let $V = \{(x, y, z) \in \mathbb{R}^3 \mid x^2 + y^2 + z^2 \leq R^2\}$ and $\vec{F}(x, y, z) = x\vec{i} + 2y\vec{j} + 3z\vec{k}$. Compute the integral $\int_{\partial V} \vec{n} \cdot \vec{F} dS$ in two different ways.

Solution a) On the surface ∂V we have $\vec{n} = (x\vec{i} + y\vec{j} + z\vec{k})/R$, so computing directly by means of spherical-surface coordinates we obtain

$$\begin{aligned} \int_{\partial V} \vec{n} \cdot \vec{F} dS &= \int_{\partial V} R^{-1}(x^2 + 2y^2 + 3z^2) dS \\ &= \int_0^\pi \int_0^{2\pi} R^{-1}(R^2 \sin^2 \theta \cos^2 \varphi + 2R^2 \sin^2 \theta \sin^2 \varphi + 3 \cos^2 \theta) R^2 \sin \theta d\theta d\varphi \\ &= R^3 \int_0^\pi \sin^3 \theta d\theta \int_0^{2\pi} \cos^2 \varphi d\varphi + 2R^3 \int_0^\pi \sin^3 \theta d\theta \int_0^{2\pi} \sin^2 \varphi d\varphi \\ &\quad + 3R^3 \int_0^\pi \cos^2 \theta \sin \theta d\theta \int_0^{2\pi} d\varphi \\ &= R^3 \left(\frac{4}{3} \cdot \pi + 2 \cdot \frac{4}{3} \cdot \pi + 3 \cdot \frac{2}{3} \cdot 2\pi \right) = \underline{\underline{8\pi R^3}}. \end{aligned}$$

b) By Gauss's (space) formula

$$\begin{aligned} \int_{\partial V} \vec{n} \cdot \vec{F} dS &= \int_V \nabla \cdot \vec{F} dxdydz = \int_V 6 dxdydz \\ &= 6\mu(V) = 6 \cdot \frac{4}{3} \pi R^3 = \underline{\underline{8\pi R^3}}. \quad \square \end{aligned}$$

Generalized Gauss's theorem

In the same way as above when Gauss's plane formula was derived from Green's formulas, one can conclude that Gauss's plane formula holds also if, on both sides of the formula, the dot product is replaced by the cross product, i.e. the formula remains valid under the substitutions

$$\nabla \cdot \vec{F} \hookrightarrow \nabla \times \vec{F}, \quad \vec{n} \cdot \vec{F} \hookrightarrow \vec{n} \times \vec{F}.$$

Furthermore, the formula also holds under the substitutions

$$\nabla \cdot \vec{F} \hookrightarrow \nabla f, \quad \vec{n} \cdot \vec{F} \hookrightarrow f\vec{n}.$$

Gauss's space formula (Theorem [XIV.2.2](#)) holds under the same substitutions. Using the notations $d\vec{n} = \vec{n} ds$ (plane), $d\vec{a} = \vec{n} dS$ (space) and the general product

symbol $*$, the results can be combined into the form

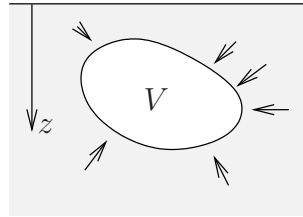
$$\boxed{\begin{aligned}\int_A \nabla * \vec{F} \, dx dy &= \int_{\partial A} d\vec{n} * \vec{F} \quad (\text{plane}), \\ \int_V \nabla * \vec{F} \, dx dy dz &= \int_{\partial V} d\vec{a} * \vec{F} \quad (\text{space}).\end{aligned}}$$

In the case $*$ = ‘empty’ (multiplication of a scalar and a vector) the vector field $\vec{F}$ is here to be interpreted as a scalar field: $\vec{F} \hookrightarrow f$. The results, which thus hold under the assumptions of Gauss's theorem, are known as the *generalized Gauss's theorem*.

EXAMPLE 2 On the boundary surface ∂V of a body $V \subset \mathbb{R}^3$ immersed in water there acts a pressure

$$\vec{f}(x, y, z) = -\rho_0 g z \vec{n}$$

(ρ_0 = density of water). What is the total force $\vec{F}$ exerted on the body?



Solution By the generalized Gauss's theorem ($*$ = ‘empty’)

$$\begin{aligned}\vec{F} &= -\rho_0 g \int_{\partial V} z \vec{n} \, dS = -\rho_0 g \int_V \nabla z \, dV \\ &= -\rho_0 g \vec{e}_z \int_V dV = \underline{\underline{-m(V)g \vec{e}_z}},\end{aligned}$$

where $m(V)$ = the mass of the volume of water displaced by the body. The result is known as

Archimedes' principle. $\square$

EXERCISES

1. a) Compute $\int_A (\partial f / \partial y) \, dx dy$, when A is the unit disk centered at the origin and $f(x, y) = 2x - 3y^2 + (x^2 + y^2 - 1) \sin(1 + xy)$.
- b) Compute $\int_{\partial A} \vec{F} \cdot d\vec{n}$, when A is the region enclosed by the ellipse $S : x^2/a^2 + y^2/b^2 = 1$ and $\vec{F} = (x + e^y)\vec{i} + (\sin x + 2y)\vec{j}$.

2. Let $V = \{(x, y, z) \in \mathbb{R}^3 \mid |x| \leq 1 \ \& \ |y| \leq 1 \ \& \ |z| \leq 1\}$ and $\vec{F}(x, y, z) = x\vec{i} + y^2\vec{j} - \vec{k}$. Compute the integrals $\int_{\partial V} d\vec{a} \cdot \vec{F}$ and $\int_{\partial V} d\vec{a} \times \vec{F}$ a) directly as surface integrals, b) as volume integrals using the generalized Gauss's theorem.
3. Let $V = \{(x, y, z) \in \mathbb{R}^3 \mid x^2 + y^2 + z^2 \leq 1 \ \& \ x \geq 0 \ \& \ y \geq 0\}$. Compute $\int_{\partial V} [(x+y)\vec{i} - 2xz\vec{j} + (y-z)\vec{k}] \times d\vec{a}$ a) directly, b) by converting to a volume integral.
4. Compute $\int_{\partial V} \vec{F} \cdot d\vec{a}$ and $\int_{\partial V} \vec{F} \times d\vec{a}$, when $\vec{F}(x, y, z) = 3xz^2\vec{i} - x\vec{j} - y\vec{k}$ and $V = \{(x, y, z) \in \mathbb{R}^3 \mid 0 \leq x \leq 1 \ \& \ y \geq 0 \ \& \ y^2 + z^2 \leq 1\}$.
5. Let $\vec{F}$ be a regular vector field and u a regular scalar field on a basic region $V \subset \mathbb{R}^3$. Show that $\int_{\partial V} \nabla \times \vec{F} \cdot d\vec{a} = 0$ and $\int_{\partial V} \nabla u \times d\vec{a} = \vec{0}$.
6. If $A \subset \mathbb{R}^2$ and $V \subset \mathbb{R}^3$ are basic regions, then what is the geometric meaning of the following integrals?
 a) $\frac{1}{2} \int_{\partial A} \vec{r} \cdot d\vec{n}$ b) $\frac{1}{3} \int_{\partial V} \vec{r} \cdot d\vec{a}$ c) $\frac{1}{2\mu(V)} \int_{\partial V} (x^2 + y^2 + z^2) d\vec{a}$
7. Let u , f and $\vec{F}$ be sufficiently regular scalar and vector fields on a basic region $V \subset \mathbb{R}^3$. Show that

$$\int_V \nabla u * \vec{F} \, dx dy dz = \int_{\partial V} u \, d\vec{a} * \vec{F} - \int_V u \nabla * \vec{F} \, dx dy dz,$$

where $\vec{F} = f$ if $*$ = 'empty'. What is the 2-dimensional counterpart of the formula? And the 1-dimensional counterpart, when $V = [a, b]$?

8. (*) (Zeppelin) In calm weather the air pressure $p(x, y, z)$ and density $\rho(x, y, z)$ satisfy the equilibrium law

$$\frac{\partial p}{\partial z} = -g\rho,$$

where z = the height above the ground and g = the acceleration of Earth's gravity. If $V \subset \mathbb{R}^3$ is an airship floating in the air, does Archimedes' principle hold?

XIV.3 Applications of Gauss's theorem

In physics, Gauss's space formula (Theorem XIV.2.2) is often presented in the form

$$\int_V \nabla \cdot \vec{F} dV = \int_{\partial V} \vec{F} \cdot d\vec{a}, \quad V \subset \mathbb{R}^3,$$

where we have written $dV = dx dy dz$. If $A \subset S$, where S is a surface in space, then the surface integral

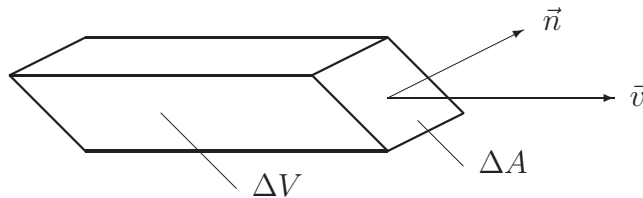
$$\phi = \int_A \vec{F} \cdot d\vec{a} = \int_A \vec{F} \cdot \vec{n} dS \quad (\star)$$

is the *flux* of the vector field $\vec{F}$ through A . In view of this connection, in physical applications the vector field itself is often called the flux density. In the following two application examples we use Gauss's formula to derive a conservation law describing a physical phenomenon in the form of a partial differential equation. The first example is from fluid mechanics and the second from the theory of heat.

Application example: The law of conservation of mass in a flow

Consider a flowing liquid or gas whose velocity at time t at the point (x, y, z) is $\vec{v} = \vec{v}(t, x, y, z)$ (a vector field, unit m/s) and whose density is $\rho = \rho(t, x, y, z)$ (a scalar field, unit kg/m³). Let $A \subset S$, where S is a regular (or at least piecewise regular) surface in space. Consider a small, nearly planar piece ΔA of A , whose area = ΔS . If we assume that ρ and $\vec{v}$ are nearly constant in the neighborhood of ΔA (continuity assumption!), then we can conclude that during the (short) time interval $[t, t + \Delta t]$ the particles that pass through the surface piece ΔA are those which at time t are in the set

$$\Delta V = \{ (x, y, z) \hat{=} \vec{r} \mid \vec{r} = \vec{r}_0 - \tau \vec{v}, \quad r_0 \in A, \quad \tau \in [0, \Delta t] \}.$$



Let $\vec{n}$ be the unit normal vector of S pointing in the direction of the flow, i.e. $\vec{n} \cdot \vec{v} \geq 0$. If we assume $\vec{v}$ constant and likewise $\vec{n}$ constant on ΔA (i.e. the

surface planar), then, under these assumptions, ΔV is a parallelepiped whose cross-sectional area in the plane perpendicular to the flow is $\Delta S \vec{v} \cdot \vec{n} / |\vec{v}|$. If, in addition, ρ too is assumed constant, then the mass of the liquid/gas in ΔV is, under the assumptions made,

$$\begin{aligned} \Delta m &= \rho \mu(\Delta V) = \rho [\Delta S \vec{v} \cdot \vec{n} / |\vec{v}|] |\vec{v}| \Delta t = \rho \vec{v} \cdot \vec{n} \Delta S \Delta t \\ \Rightarrow \quad \frac{\Delta m}{\Delta S \Delta t} &= \rho \vec{v} \cdot \vec{n}. \end{aligned}$$

It may be assumed that the approximations made hold at nearly all points of A , except for a possible subset of measure zero (with respect to the area measure). At points other than these exceptional points, the assumptions mentioned may also be assumed to hold exactly, as limits, when $\Delta t \rightarrow 0$ and ΔA shrinks to a point. Under these assumptions, the mass flux through the surface A (unit kg/s) can be computed by the integral formula ($\star$), where the mass-flux density (unit kg/m²/s) is defined as

$$\vec{F} = \rho \vec{v}.$$

Assume now that S is a closed surface enclosing a basic region $V \subset \mathbb{R}^3$ ($S = \partial V$). Then the total mass of the liquid/gas in V at time t is

$$m(t) = \int_V \rho(t, x, y, z) dV.$$

Assume that $m(t)$ can change only because of the flow through the boundary surface of V (i.e. in no other way does mass ‘vanish’ or ‘arise’). Then the balance law

$$m'(t) = - \int_{\partial V} \rho \vec{v} \cdot d\vec{a}$$

holds, where $d\vec{a} = \vec{n} dS$ points away from V , i.e. $\vec{n}$ = the outward normal of V . When Gauss's formula is used here on the right and on the left one writes (cf. Section XI.2)

$$m'(t) = \frac{d}{dt} \int_V \rho(t, x, y, z) dV = \int_V \frac{\partial \rho}{\partial t}(t, x, y, z) dV,$$

then the balance law takes the form

$$\int_V [\rho_t + \nabla \cdot (\rho \vec{v})] dV = 0. \quad (\text{a})$$

Now let $t \in \mathbb{R}$ be a fixed instant of time, $(x, y, z) \in \mathbb{R}^3$ a fixed point, and assume that ρ and $\vec{v}$ are continuously differentiable (with continuous partial derivatives) in a neighborhood of the point $(t, x, y, z) \in \mathbb{R}^4$. Then if in the balance law (a)

one chooses $V = B_\varepsilon$ = the ball (solid sphere) of radius ε centered at (x, y, z) , then

$$\begin{aligned} 0 &= \int_{B_\varepsilon} [\rho_t + \nabla \cdot (\rho \vec{v})] dV \\ &= [\rho_t + \nabla \cdot (\rho \vec{v})](t, x, y, z) \mu(B_\varepsilon) + o(1)\mu(B_\varepsilon), \quad \text{as } \varepsilon \rightarrow 0. \end{aligned}$$

Accordingly, at the point (t, x, y, z) under consideration, we must have

$$\boxed{\rho_t + \nabla \cdot (\rho \vec{v}) = 0.} \quad (\text{b})$$

This is known as the (differential) law of conservation of mass of fluid mechanics. — Note that since the continuity assumptions made in deriving it are not dictated by physics, the integral form of the conservation law (a) (which holds under weaker regularity conditions for every basic region V) is, as a law of nature, more fundamental than the differential form (b). Cf. the corresponding situation in Example VIII.6: 1.

Application example: Heat conduction

Consider the problem of heat conduction when space $\mathbb{R}^3$ is filled with a material of known physical properties. The basic quantities associated with the problem are the temperature $u = u(t, x, y, z)$, the heat-flux density $\vec{J} = \vec{J}(t, x, y, z)$ (unit W/m^2), and the heat-source density $\rho(t, x, y, z)$ (unit W/m^3). As material coefficients we further assume known the specific heat $c = c(x, y, z, u)$ of the material and its thermal conductivity $\lambda = \lambda(x, y, z, u)$. (Here it is assumed that the material coefficients may depend, in addition to the spatial variables, on the temperature.)

If $V \subset \mathbb{R}^3$, then the total amount of thermal energy contained in V at time t is (by the definition of specific heat)

$$E(t) = \int_V cu \, dV.$$

If it is assumed that the energy balance of V is affected only by the heat source in V and by heat conduction through the boundary surface of V , then the energy balance equation in V is

$$E'(t) = \int_V \rho \, dV - \int_{\partial V} \vec{J} \cdot d\vec{a}.$$

When here one writes

$$E'(t) = \frac{d}{dt} \int_V cu \, dV = \int_V cu_t \, dV, \quad \int_{\partial V} \vec{J} \cdot d\vec{a} = \int_V \nabla \cdot \vec{J} \, dV,$$

then the balance equation takes the form

$$\int_V (cu_t + \nabla \cdot \vec{J} - \rho) \, dV = 0.$$

By reasoning (and assumptions) similar to those in the previous example, there follows from this the differential law of conservation of energy

$$cu_t + \nabla \cdot \vec{J} = \rho.$$

If we further assume Fourier's law of heat conduction

$$\vec{J} = -\lambda \nabla u,$$

then the law of conservation of energy takes the form known as the heat conduction equation

$$cu_t - \nabla \cdot (\lambda \nabla u) = \rho.$$

If $\rho = 0$ and c and λ are constants (a homogeneous material, with c and λ having no temperature dependence), then the heat conduction equation simplifies to the form

$$u_t = k \Delta u, \quad k = \lambda/c.$$

*Discontinuities of vector fields

Let a vector field $\vec{F}$ be defined on $\mathbb{R}^3$ and continuously differentiable. Then the source $\rho = \nabla \cdot \vec{F}$ of the field is continuous on $\mathbb{R}^3$, and by Gauss's theorem

$$\int_V \rho \, dV = \int_{\partial V} \vec{F} \cdot d\vec{a} \quad (V \subset \mathbb{R}^3 \text{ basic region}). \quad (1)$$

On the other hand, if, in addition to the regularity assumptions mentioned, one assumes only (1), then by the earlier reasoning it follows that one must have $\nabla \cdot \vec{F} = \rho$. Thus if $\vec{F}$ is continuously differentiable on $\mathbb{R}^3$ and ρ is continuous, then

$$\text{Condition (1) holds} \quad \Leftrightarrow \quad \nabla \cdot \vec{F} = \rho \quad \text{in } \mathbb{R}^3.$$

But what if $\vec{F}$ is not continuously differentiable, or ρ is not continuous, but there is a relation between $\vec{F}$ and ρ of the form (1)? — In that case it is natural to agree

that condition (1) defines ρ as the source of the field $\vec{F}$. In physical applications one then thinks that condition (1) in fact expresses the law of nature in its original form (cf. the derivation of the law of conservation of mass above).

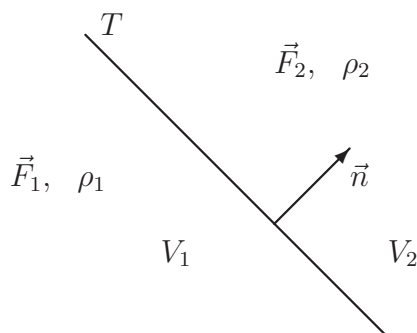
Expressing the dependence between a vector field and its source in the integral form of ‘Gauss’s law’ (1) is especially useful when one wishes to derive the continuity conditions of physical vector fields on material interfaces. As an example, consider the imposition of such conditions on a plane (on a planar material interface). Let T be a plane in space that divides $\mathbb{R}^3$ into two open parts V_1 and V_2 (i.e. $\mathbb{R}^3$ is divided into disjoint parts V_1 , V_2 and T), and let $\vec{F}$ be a piecewise continuously differentiable vector field of the form

$$\vec{F}(x, y, z) = \begin{cases} \vec{F}_1(x, y, z), & \text{when } (x, y, z) \in V_1, \\ \vec{F}_2(x, y, z), & \text{when } (x, y, z) \in V_2, \end{cases}$$

where $\vec{F}_1$ and $\vec{F}_2$ are both defined and continuously differentiable on all of $\mathbb{R}^3$. Correspondingly, let ρ be a piecewise continuous function of the form

$$\rho(x, y, z) = \begin{cases} \rho_1(x, y, z), & \text{when } (x, y, z) \in V_1, \\ \rho_2(x, y, z), & \text{when } (x, y, z) \in V_2, \end{cases}$$

where ρ_1 and ρ_2 are both defined and continuous on all of $\mathbb{R}^3$.

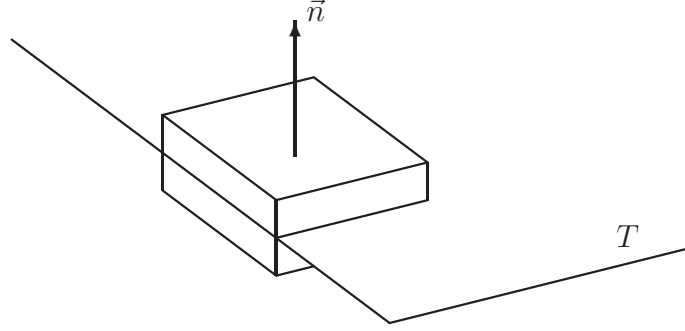


In addition to the regularity assumptions mentioned above, it is further assumed that the field $\vec{F}$ and the function ρ are bound by condition (1), i.e. ρ is the source of the field $\vec{F}$. Then if in that condition one chooses $V \subset V_1$ or $V \subset V_2$, it follows that one must have $\nabla \cdot \vec{F}_i = \rho_i$ in V_i , $i = 1, 2$, i.e.

$$\nabla \cdot \vec{F}(x, y, z) = \rho(x, y, z), \quad (x, y, z) \in V_1 \cup V_2. \quad (2)$$

Let us next investigate what follows from condition (1) when the set V is chosen so that $V \cap T \neq \emptyset$. With this in mind, consider a point $(x, y, z) \in T$ and define a

'pillbox' $B_{h,d}$ surrounding this point by the conditions: (i) $B_{h,d}$ is a rectangular box whose center $= (x, y, z)$ and two of whose faces are parallel to the plane T . (ii) The length of the edge of $B_{h,d}$ perpendicular to the plane T is $= d$. (iii) The rectangle $T_h = B_{h,d} \cap T$ is independent of d , and the length of the longest side of T_h is $= h$.



Choose in condition (1) $V = B_{h,d}$, where $T_h = B_{h,d} \cap T$ is fixed and $d \rightarrow 0$. Then there follows

$$\lim_{d \rightarrow 0} \int_{\partial B_{h,d}} \vec{F} \cdot d\vec{a} = \int_{T_h} \vec{n} \cdot (\vec{F}_2 - \vec{F}_1) d\mu = \lim_{d \rightarrow 0} \int_{B_{h,d}} \rho dV = 0,$$

where μ is the area measure of the plane T and $\vec{n}$ is the unit normal vector of T pointing in the direction of the subset V_2 (see the figures). When in this result one further lets h approach 0 and takes into account the continuity of the fields $\vec{F}_1$ and $\vec{F}_2$ at the point (x, y, z) , it follows that one must have $\vec{n} \cdot (\vec{F}_1 - \vec{F}_2)(x, y, z) = 0$. This holds at every point of T , so we have concluded that on the plane T the continuity condition

$$\vec{n} \cdot \vec{F}_1(x, y, z) = \vec{n} \cdot \vec{F}_2(x, y, z), \quad (x, y, z) \in T \quad (3)$$

holds. Both this and (2) are thus consequences of condition (1) (within the framework of the regularity assumptions made).

On the other hand, if one starts from the same regularity assumptions and further assumes (2) and (3), then from these together there follows (1). Indeed, if V is a basic region and $A_i = V \cap V_i \neq \emptyset$, $i = 1, 2$, then on the basis of assumption (2) and Gauss's theorem one can conclude

$$\begin{aligned} (2) \quad &\Rightarrow \sum_{i=1}^2 \int_{A_i} \nabla \cdot \vec{F}_i dV = \sum_{i=1}^2 \int_{A_i} \rho_i dV = \int_V \rho dV \\ &\Rightarrow \sum_{i=1}^2 \int_{\partial A_i} \vec{F}_i \cdot d\vec{a} = \int_V \rho dV. \end{aligned}$$

Since here

$$\partial A_1 \cup \partial A_2 = \partial V \cup (\partial A_1 \cap \partial A_2), \quad \partial A_1 \cap \partial A_2 \subset T,$$

by assumption (3) one further concludes

$$\sum_{i=1}^2 \int_{\partial A_i} \vec{F}_i \cdot d\vec{a} = \int_{\partial V} \vec{F} \cdot d\vec{a} + \int_{\partial A_1 \cap \partial A_2} \vec{n} \cdot (\vec{F}_1 - \vec{F}_2) d\mu = \int_{\partial V} \vec{F} \cdot d\vec{a}.$$

Thus condition (1) holds. We have thus concluded that, within the framework of the regularity assumptions made,

Condition (1) $\Leftrightarrow$ $\begin{cases} \nabla \cdot \vec{F} = \rho & \text{in the set } V_1 \cup V_2, \\ \vec{n} \cdot (\vec{F}_1 - \vec{F}_2) = 0 & \text{on the interface } T. \end{cases}$

The conclusion is the same also if the interface $\partial V_1 \cap \partial V_2$ is part of a curved, sufficiently regular surface S . Concerning the continuity conditions of a physical vector field on a material interface, we have thus concluded that if a vector field has a piecewise continuous source in the vicinity of a material interface, then the normal component of the field is continuous on the material interface. The tangential component of the field, on the other hand, need not be continuous.

In the same way as in condition (1), the relation between a vector field and its curl field can also be expressed as a definition in integral form. If $\vec{F}$ is continuously differentiable on $\mathbb{R}^3$, then the curl field of $\vec{F}$ is $\vec{\omega} = \nabla \times \vec{F}$ (cf. Section XI.4). Then, on the basis of the generalized Gauss's theorem,

$$\int_V \vec{\omega} dV = \int_{\partial V} d\vec{a} \times \vec{F} \quad (V \subset \mathbb{R}^3 \text{ basic region}). \quad (4)$$

If $\vec{F}$ is not continuously differentiable on all of $\mathbb{R}^3$, or $\vec{\omega}$ is not continuous, then condition (4) is taken as the definition of the curl field $\vec{\omega}$. If now $\vec{F}$ satisfies the same conditions as above and it is assumed that the curl field determined by definition (4) is of the form

$$\vec{\omega}(x, y, z) = \begin{cases} \vec{\omega}_1(x, y, z), & \text{when } (x, y, z) \in V_1, \\ \vec{\omega}_2(x, y, z), & \text{when } (x, y, z) \in V_2, \end{cases}$$

where $\vec{\omega}_1$ and $\vec{\omega}_2$ are both defined and continuous on all of $\mathbb{R}^3$, then in the same way as above one concludes that

Condition (4) $\Leftrightarrow$ $\begin{cases} \nabla \times \vec{F} = \vec{\omega} & \text{in the set } V_1 \cup V_2, \\ \vec{n} \times (\vec{F}_1 - \vec{F}_2) = \vec{0} & \text{on the interface } T. \end{cases}$

In this case the tangential component of the field is continuous on the material interface, since the tangential component is (cf. Section II.6)

$$\vec{F}_t = \vec{F} - (\vec{n} \cdot \vec{F}) \vec{n} = -\vec{n} \times (\vec{n} \times \vec{F}).$$

The normal component of the field need not be continuous.

EXAMPLE 1 A static electric field $\vec{E}$ and the electric flux density $\vec{D}$ satisfy Maxwell's equations (cf. Section XI.4)

$$\nabla \times \vec{E} = \vec{0}, \quad \nabla \cdot \vec{D} = \rho,$$

where ρ is the charge density. Assume that V_1 and V_2 above represent two different materials, in which the prevailing material laws are

$$\vec{D} = \epsilon_i \vec{E} \quad \text{in } V_i, \quad i = 1, 2,$$

where ϵ_i (= the electric permittivity of material i) is assumed constant in V_i . If ρ is assumed piecewise continuous as above, then the continuity conditions on the material interface are

$$\begin{cases} \vec{n} \times (\vec{E}_1 - \vec{E}_2) = \vec{0}, \\ \vec{n} \cdot (\vec{D}_1 - \vec{D}_2) = 0 \end{cases} \Leftrightarrow \begin{cases} \vec{n} \times \vec{E}_1 = \vec{n} \times \vec{E}_2, \\ \epsilon_1 \vec{n} \cdot \vec{E}_1 = \epsilon_2 \vec{n} \cdot \vec{E}_2. \end{cases}$$

Thus the tangential component of the electric field is continuous on the material interface. The normal component, on the other hand, is discontinuous unless either $\epsilon_1 = \epsilon_2$ or $\vec{n} \cdot \vec{E}_i = 0$. $\square$

EXAMPLE 1 (continued) In the example, let

$$V_1 = \{(x, y, z) \in \mathbb{R}^3 \mid x + y + z < 0\}, \quad V_2 = \{(x, y, z) \in \mathbb{R}^3 \mid x + y + z > 0\},$$

and $\epsilon_1/\epsilon_2 = 2$. Compute the electric field $\vec{E}_2(0, 0, 0)$, given that $\vec{E}_1(0, 0, 0) = E(\vec{i} + 2\vec{j} - \vec{k})$.

Solution The material interface is the plane $T: x + y + z = 0$. Its unit normal vector pointing toward material 2 is $\vec{n} = (\vec{i} + \vec{j} + \vec{k})/\sqrt{3}$, so

$$\vec{n} \cdot \vec{E}_1(0, 0, 0) = \frac{2E}{\sqrt{3}}, \quad \vec{n} \times \vec{E}_1(0, 0, 0) = \frac{E}{\sqrt{3}}(-3\vec{i} + 2\vec{j} + \vec{k}).$$

By the continuity conditions,

$$\vec{n} \cdot \vec{E}_2(0, 0, 0) = \frac{\epsilon_1}{\epsilon_2} \vec{n} \cdot \vec{E}_1(0, 0, 0), \quad \vec{n} \times \vec{E}_2(0, 0, 0) = \vec{n} \times \vec{E}_1(0, 0, 0),$$

so that

$$\begin{aligned}
 \vec{E}_2(0, 0, 0) &= [\vec{n} \cdot \vec{E}_2(0, 0, 0)] \vec{n} - \vec{n} \times [\vec{n} \times \vec{E}_2(0, 0, 0)] \\
 &= 2 [\vec{n} \cdot \vec{E}_1(0, 0, 0)] \vec{n} - \vec{n} \times [\vec{n} \times \vec{E}_1(0, 0, 0)] \\
 &= \frac{4E}{3} (\vec{i} + \vec{j} + \vec{k}) + \frac{E}{3} (\vec{i} + 4\vec{j} - 5\vec{k}) \\
 &= \underline{\underline{\frac{E}{3} (5\vec{i} + 8\vec{j} - \vec{k})}}. \quad \square
 \end{aligned}$$

EXERCISES

1. Compute the flux of the vector field $\vec{F}$ through the sphere of radius R centered at the origin:
 - a) $\vec{F} = x^2 y^4 z \vec{k}$
 - b) $\vec{F} = x \vec{i} - 2y \vec{j} + 4z \vec{k}$
 - c) $\vec{F} = y e^z \vec{i} + x^2 e^z \vec{j} + xy \vec{k}$
 - d) $\vec{F} = (x^2 + y^2) \vec{i} + (y^2 - z^2) \vec{j} + z \vec{k}$
 - e) $\vec{F} = x^2 \vec{i} + 3yz^2 \vec{j} + (3y^2 z + x^2) \vec{k}$
2. Compute the flux of the vector field $\vec{F} = x^2 \vec{i} + y^2 \vec{j} + z^2 \vec{k}$ through the boundary surface ∂V of the given region $V \subset \mathbb{R}^3$:
 - a) $V : (x - 2)^2 + y^2 + (z - 3)^2 \leq 9$
 - b) $V : x^2 + y^2 + 4(z - 1)^2 \leq 4$
 - c) $V : x, y, z \geq 0 \text{ \& } x + y + z \leq 3$
 - d) $V : x^2 + y^2 \leq 2y \text{ \& } 0 \leq z \leq 4$
3. A tetrahedral region $V \subset \mathbb{R}^3$ is bounded by the planes $z = 0$, $x = 2y$, $x = -y$ and $y + z = a$ ($a > 0$). Compute the flux of the vector field

$$\vec{F} = (3xyz + e^{yz}) \vec{i} + (x^2 + y^2 z) \vec{j} + (y^2 - 2yz^2) \vec{k}$$

through the boundary surface ∂V of V , from the inside outward.

4. Let $S : x^2 + y^2 + z^2 = 1$, $B \subset S$ and

$$V = \{(x, y, z) \in \mathbb{R}^3 \mid (x, y, z) = t(u, v, w), t \in [0, R] \text{ \& } (u, v, w) \in B\}.$$

Compute the volume $\mu(V)$ of V a) by means of spherical coordinates, b) by applying Gauss's theorem to the vector field $\vec{F} = x \vec{i} + y \vec{j} + z \vec{k}$.

5. In a rotating flow field the mass-flux density at the point $(x, y, z) \hat{=} \vec{r}$ is $\vec{F} = Q \vec{r} \times (2\vec{i} - \vec{j} + 3\vec{k})$, where $Q = 24 \text{ kg/m}^3/\text{s}$ and the unit of $\vec{r} = \text{m}$. Let V be a cube one of whose vertices is at the origin, whose edges emanating from the origin lie along the positive coordinate axes, and whose side length is $a = 2 \text{ m}$. Compute the mass flux (unit kg/s) through each face of V , counted as positive from the inside of the cube outward.

6. A vector field $\vec{F}$ and its source ρ are, in a) spherical coordinates, b) cylindrical coordinates, of the form $\vec{F} = F(r)\vec{e}_r$, $\rho = \rho(r)$. Compute $F(r)$ in terms of the source $\rho(r)$ by using Gauss's formula in a suitably chosen region V .
7. Let u and v be regular functions on a basic region $A \subset \mathbb{R}^2$ or $V \subset \mathbb{R}^3$. By applying Gauss's theorem to the vector field $\vec{F} = u\nabla v - v\nabla u$, show that

$$\int_A (u\Delta v - v\Delta u) \, dxdy = \int_{\partial V} \left(u \frac{\partial v}{\partial n} - v \frac{\partial u}{\partial n} \right) ds,$$

$$\int_V (u\Delta v - v\Delta u) \, dxdydz = \int_{\partial V} \left(u \frac{\partial v}{\partial n} - v \frac{\partial u}{\partial n} \right) dS,$$

where $\partial/\partial n$ denotes the directional derivative in the direction of the normal of ∂A or ∂V .

8. (Conservation of cars) In a one-dimensional traffic flow, $\rho(t, x)$ is the car density on the road (unit cars/km) and $v(t, x)$ the speed (unit km/h). a) Define the car flux on the road and derive the law of conservation of cars. b) At the point $x = 0$ road 1 narrows into road 2. On road 1 ($x < 0$) the traffic flow is steady: $\rho = 26$ cars/km and $v = 120$ km/h. On road 2 too the traffic flow is steady (ρ and v constant), such that the maximum capacity of the road, 3000 cars/h, is not exceeded. Can the point $x = 0$ be free of congestion in such a traffic situation? If not, then by at least how many cars does the congestion grow in an hour?
9. The basic equations of heat conduction are

$$\nabla \cdot \vec{J} = \rho, \quad \vec{J} = -\lambda \nabla u,$$

where $\vec{J}$ is the heat-flux density, ρ the heat-source density, u the temperature and λ the thermal conductivity of the material. Let $\lambda = \lambda_i = \text{constant}$ in V_i , $i = 1, 2$, where $\lambda_2 = 3\lambda_1$ and

$$V_{1,2} = \{ (x, y, z) \in \mathbb{R}^3 \mid x + 2y + 2z \leq 0 \}.$$

Determine the limit $\vec{J}_2(0, 0, 0)$ of the heat-flux vector $\vec{J}$ as the origin is approached from the side of material 2 (V_2), given that the corresponding limit from the side of material 1 is $\vec{J}_1(0, 0, 0) = J(-\vec{i} + \vec{j} + 3\vec{k})$ (J constant). The source ρ is constant in each material and the temperature u is continuous on the material interface.

10. (*) Let $A \subset \mathbb{R}^2$ or $V \subset \mathbb{R}^3$ be a basic region. Show that if u_1 and u_2 are solutions of the boundary value problem

$$\begin{cases} \Delta u = \rho & \text{in } A \text{ (in } V), \\ u = 0 & \text{on the boundary } \partial A \text{ (on the boundary } \partial V) \end{cases}$$

and moreover are sufficiently regular, then for the function $u = u_1 - u_2$

$$\int_A |\nabla u|^2 dx dy = 0 \quad \left(\int_V |\nabla u|^2 dx dy dz = 0 \right).$$

Conclude that the solution of the boundary value problem — insofar as it exists and is sufficiently regular — is unique.

XIV.4 Stokes's theorem

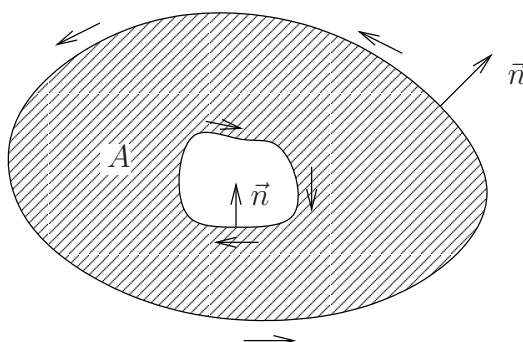
Let us first consider a basic region A of the plane (see Section XIV.2) and a vector field $\vec{F}(x, y) = F_1(x, y)\vec{i} + F_2(x, y)\vec{j}$ defined on A . When $\vec{F}$ is interpreted as a vector field in space, then the curl of $\vec{F}$ is (cf. Section XI.4)

$$\nabla \times \vec{F} = \left(\frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y} \right) \vec{k} = (\text{rot } \vec{F}) \vec{k},$$

so that, on the basis of Green's plane formulas (see Section XIV.2),

$$\begin{aligned} \int_A \text{rot } \vec{F} \, dxdy &= \int_{\partial A} (-n_y F_1 + n_x F_2) \, ds \\ &= \int_{\partial A} \vec{t} \cdot \vec{F} \, ds, \end{aligned}$$

where $\vec{t}$ is the tangent vector of the boundary oriented as shown in the figure, i.e. $\vec{t}$ points to the right as seen from the direction of the outward normal of the boundary. This is called the *positive orientation* of the boundary.



The integral over ∂A , computed in the above manner, is called a *circulation integral* and is denoted by the symbol $\oint_{\partial A}$. Noting that (cf. Section XIV.1)

$$\vec{t} \, ds = d\vec{r}$$

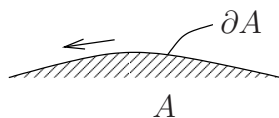
one sees that the circulation integral is a combination of path integrals (in fact work integrals), in which each separate part of the boundary ∂A is traversed in the manner determined by the orientation. The result, in the notation mentioned, is thus

$$\int_A \text{rot } \vec{F} \, dxdy = \oint_{\partial A} \vec{F} \cdot d\vec{r}. \quad (1)$$

Let next $\vec{F}$ be a vector field in space and A a basic region on a plane T in space. Choose Cartesian coordinates so that T is the xy -plane, and write in these coordinates $\vec{F} = F_1\vec{i} + F_2\vec{j} + F_3\vec{k}$ and $\text{rot}\vec{F} = \partial_x F_2 - \partial_y F_1$. Then one sees that the integral formula (1) retains its validity, because on the right-hand side of the formula $d\vec{r} \cdot \vec{k} = 0$, so that $\vec{F} \cdot d\vec{r} = (F_1\vec{i} + F_2\vec{j}) \cdot d\vec{r}$. Since on the left-hand side of the formula one can also write $\text{rot}\vec{F} = (\nabla \times \vec{F}) \cdot \vec{k}$, one can draw a more general conclusion: If $\vec{F}$ is a continuously differentiable vector field in space and A is a basic region on a plane in space whose unit normal vector is $\vec{n}$, then the integral formula

$$\int_A (\nabla \times \vec{F}) \cdot \vec{n} dS = \oint_{\partial A} \vec{F} \cdot d\vec{r} \quad (2)$$

holds. Here the direction of the normal $\vec{n}$ and the orientation of the boundary ∂A must be tied to each other so that when the boundary line is viewed from the side indicated by the normal $\vec{n}$, the situation is as shown in the figure.



Formula (2) is a special case of a still more general result known as *Stokes's* theorem*. By Stokes's theorem, formula (2) can, under certain conditions, be generalized to apply to curved surfaces in space as well. This more general result is the integral formula (Stokes's formula)

$$\boxed{\int_A (\nabla \times \vec{F}) \cdot d\vec{a} = \oint_{\partial A} \vec{F} \cdot d\vec{r}.} \quad (3)$$

Here $\vec{F}$ is a vector field in space, A is a (sufficiently regularly shaped) region on a (sufficiently regular) surface S in space and $d\vec{a} = \vec{n} dS$, where dS refers to the area measure of S and $\vec{n}$ = the unit normal vector of the surface.

In formula (3) it is required that the normal $\vec{n}$ of the surface and the orientation of the boundary line ∂A be tied to each other in the manner described above. Arranging such an orientation is not a problem in the case where A is on a plane in space. On curved surfaces in space, however, problems may arise, and in the general Stokes's theorem the surface A must indeed be restricted by a geometric condition: the surface must be *orientable* in the assumed manner. Let us examine the matter in the light of examples.

*Sir George Gabriel Stokes (1819-1903) was an English physicist-mathematician.

EXAMPLE 1 A hemisphere is orientable (cf. the figure), so Stokes's formula (3) is valid for it. The formula can also be proved using spherical coordinates: Let the sphere have radius R and

$$\vec{F} = F(r, \theta, \varphi) \vec{e}_r + F_\theta(r, \theta, \varphi) \vec{e}_\theta + F_\varphi(r, \theta, \varphi) \vec{e}_\varphi.$$

Then (cf. Section XI.4)

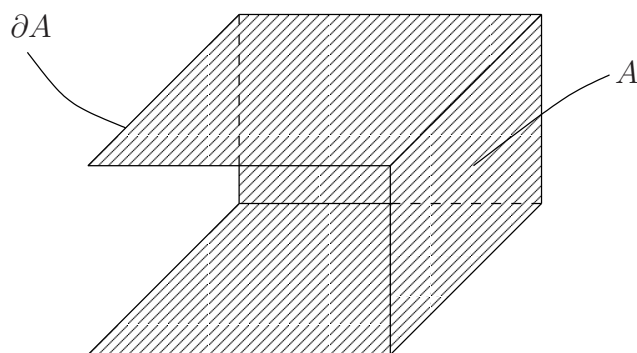
$$\nabla \times \vec{F} = \frac{1}{r \sin \theta} [\partial_\theta (\sin \theta F_\varphi) - \partial_\varphi F_\theta] \vec{e}_r + [\dots] \vec{e}_\theta + [\dots] \vec{e}_\varphi,$$

so that when one chooses $\vec{n} = \vec{e}_r$ (the other option would be $\vec{n} = -\vec{e}_r$), then

$$\begin{aligned} \int_A (\nabla \times \vec{F}) \cdot d\vec{a} &= \int_0^{\pi/2} \int_0^{2\pi} \frac{1}{R \sin \theta} [\partial_\theta (\sin \theta F_\varphi) - \partial_\varphi F_\theta] R^2 \sin \theta d\theta d\varphi \\ &= R \int_0^{2\pi} \left(\int_0^{\pi/2} \partial_\theta (\sin \theta F_\varphi) d\theta \right) d\varphi - R \int_0^{\pi/2} \left(\int_0^{2\pi} \partial_\varphi F_\theta d\varphi \right) d\theta \\ &= R \int_0^{2\pi} \left[\sin \theta F_\varphi(\theta, \varphi) \right]_{\theta=0}^{\theta=\pi/2} d\varphi - R \int_0^{\pi/2} \left[F_\theta(\theta, \varphi) \right]_{\varphi=0}^{\varphi=2\pi} d\theta \\ &= R \int_0^{2\pi} F_\varphi\left(\frac{\pi}{2}, \varphi\right) d\varphi. \end{aligned}$$

On the boundary line ∂A we have $\theta = \frac{\pi}{2}$ and $\vec{F} \cdot d\vec{r} = \vec{F} \cdot (R d\varphi \vec{e}_\varphi) = R F_\varphi d\varphi$, so formula (3) is confirmed to be valid for the hemisphere. $\square$

EXAMPLE 2 In a second example we consider a surface obtained by removing two adjacent faces from the outer surface of a rectangular box, cf. the figure.



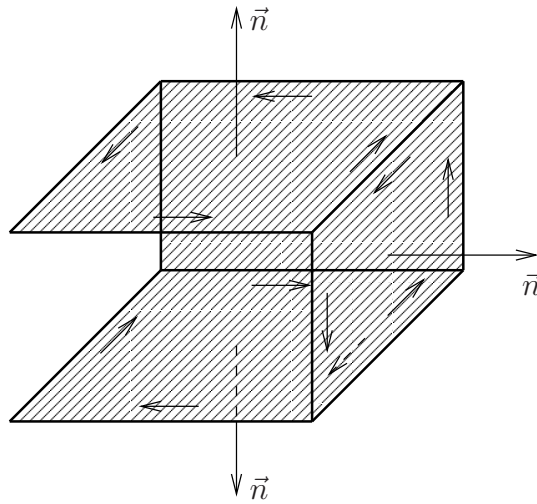
In this case the surface consists of planar surfaces A_i , for each of which Stokes's formula (3) holds, i.e.

$$\int_{A_i} (\nabla \times \vec{F}) \cdot d\vec{a} = \oint_{\partial A_i} \vec{F} \cdot d\vec{r}.$$

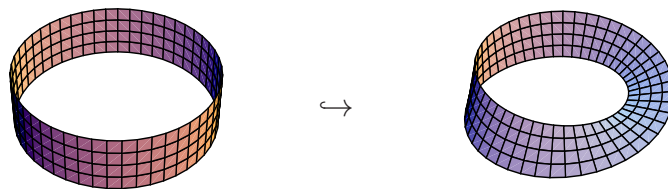
When the direction of the normal on each part is chosen to point outside the box, one sees that

$$\begin{aligned} \int_A (\nabla \times \vec{F}) \cdot d\vec{a} &= \sum_i \int_{A_i} (\nabla \times \vec{F}) \cdot d\vec{a} \\ &= \sum_i \oint_{\partial A_i} \vec{F} \cdot d\vec{r} \\ &= \int_{\partial A} \vec{F} \cdot d\vec{r}, \end{aligned}$$

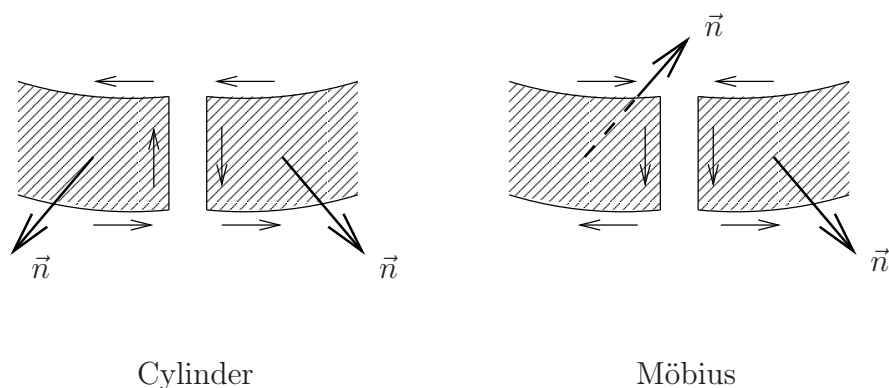
since the line integrals over the shared edges $\partial A_i \cap \partial A_j$ of the parts cancel in the summation, cf. the figure. Thus formula (3) is valid in this case as well. The same result would also be reached by removing whichever faces one likes from the box (1–5 of them). $\square$



EXAMPLE 3 As a third example we consider a surface obtained from a cylindrical shell by cutting the shell across, rotating one end by 180° , and joining the ends together again.



The resulting surface is the so-called *Möbius strip**. For the Möbius strip Stokes's formula (3) is not valid. It is, however, valid if the joining seam is left open and the line integrals on both sides of the seam are taken into account in formula (3). In the case of the Möbius strip these seam integrals are computed in the same direction, so that they do not cancel when the seam is joined together, unlike in the case of the cylindrical surface, cf. the figure.



The Möbius strip is thus an example of a non-orientable surface for which Stokes's formula (3) does not hold. The Möbius strip is in fact a *one-sided* surface on which the direction of the normal $\vec{n}$ cannot be chosen consistently so that it varies continuously as one moves along the surface. (If the surface is cut open, then at the gluing point the normal is discontinuous.) $\square$

In the case of a simple surface, Stokes's theorem can be proved using a parametrization of the surface, but the proof is then rather technical and requires relatively strong regularity assumptions. A more transparent and more generally valid proof is obtained by approximating the surface with triangular planar surfaces, in the same way as in the definition of the area measure (cf. Section XIII.7). Since Stokes's plane formula holds for every planar triangular surface, it follows by summation that Stokes's formula (3) also holds for the approximation of the surface A made up of triangles — provided that in the summation the line integrals over the shared sides of adjacent triangles cancel. The orientability condition lies precisely here. Insofar as the surface is orientable, by refining the triangular mesh one obtains the limit result (3), provided that the vector field $\vec{F}$, the surface A (the parametrization of the surface), and the boundary line ∂A are sufficiently regular. As far as the vector field is concerned, it suffices that it be continuously differentiable (in reality somewhat less suffices too), and the

*August Möbius (1790–1868) was a German mathematician.

assumptions concerning the surface A are, in practice, not very restrictive either, cf. the examples above.

Stokes's theorem, like Gauss's theorem, has its higher-dimensional counterparts. These belong to the branch of mathematics called *differential geometry*. It is also into such broader contexts that the more precise proof of Stokes's theorem most naturally fits, which is passed over here.

EXAMPLE 4 A body moves in the force field

$$\vec{F} = \frac{F}{a} (y\vec{i} - 2z\vec{j} + 3x\vec{k})$$

along a circular path that is the curve of intersection of the sphere $x^2 + y^2 + z^2 = a^2$ and the plane $x + y + z = 0$. Using Stokes's theorem, determine the work done by the force field during one revolution, when the direction of circulation is clockwise as seen from the point $(10, 10, 10)$.

Solution The trajectory of the body is ∂A , where A is the disk of radius a centered at the origin on the plane $x + y + z = 0$. By Stokes's theorem

$$W = \oint_{\partial A} \vec{F} \cdot d\vec{r} = \int_A (\nabla \times \vec{F}) \cdot \vec{n} dS.$$

Here

$$\nabla \times \vec{F} = \frac{F}{a} \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \partial_x & \partial_y & \partial_z \\ y & -2z & 3x \end{vmatrix} = \frac{F}{a} (2\vec{i} - 3\vec{j} - \vec{k}).$$

The assumed direction of circulation corresponds to $\vec{n} = -\frac{1}{\sqrt{3}}(\vec{i} + \vec{j} + \vec{k})$, so $(\nabla \times \vec{F}) \cdot \vec{n} = \frac{2F}{\sqrt{3}a}$. Thus

$$W = \int_A \frac{2F}{\sqrt{3}a} dS = \frac{2F}{\sqrt{3}a} \mu(A) = \frac{2F}{\sqrt{3}a} \pi a^2 = \underline{\underline{(2\pi/\sqrt{3}) Fa}}. \quad \square$$

Generalized Stokes's theorem

Like Gauss's theorem, Stokes's theorem too can be generalized so that in place of the dot product there may be a more general vector product, i.e. either the cross product or the multiplication of a scalar and a vector. The generalized Stokes's formula is

$$\boxed{\oint_{\partial A} d\vec{r} * \vec{F} = \int_A (d\vec{a} \times \nabla) * \vec{F}.}$$

Here again one must set $\vec{F} \hookrightarrow f$ when $*$ = ‘empty’. The generalized formula holds under the same conditions as the basic formula (3), and it can also be justified in the same way as above, i.e. by starting from the plane form of the formula.

EXERCISES

- Let $A = \{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 \leq 1 \text{ \& } x \geq 0\}$. Compute the following circulation integrals over ∂A (circling counterclockwise) both directly as line integrals and by first converting them into plane integrals using the (generalized) Stokes's formula.
 - $\oint_{\partial A} [y^2 dx + (x + y^2) dy]$
 - $\oint_{\partial A} (x + y^2) d\vec{r}$
 - $\oint_{\partial A} (xy^2 \vec{i} + |y| \vec{j}) \cdot d\vec{r}$
 - $\oint_{\partial A} (y^3 \vec{i} - x \vec{j}) \times d\vec{r}$
- Compute $\oint_{\partial A} \vec{F} \cdot d\vec{r}$ a) directly as a line integral, b) using Stokes's theorem, when $\vec{F} = x\vec{k}$ and $A \subset S$ corresponds to the parameter values $(u, v) \in [0, 1] \times [0, 3]$ on the surface $S : x = 8u^2, y = v^2, z = 4uv$.
- Using Stokes's theorem, show that if S is the curve of intersection of the sphere $K : x^2 + y^2 + z^2 = R^2$ and the plane $T : x + y + z = 0$, then $\oint_S (y dx + z dy + x dz) = \pm \sqrt{3} \pi R^2$.
- Compute the surface integral $\int_A \nabla \times (\vec{k} \times \vec{r}) \cdot d\vec{a}$, when $S, A \subset S$ and the direction of the unit normal vector $\vec{n} = n_x \vec{i} + n_y \vec{j} + n_z \vec{k}$ of S is given by the following conditions.
 - $S : z = 0, \quad A : x^2 + y^2 \leq 1, \quad n_z \geq 0$
 - $S : x^2 + y^2 + z^2 = 1, \quad A : z \geq 0, \quad n_z \geq 0$
 - $S : x^2 + y^2 + z^2 = 1, \quad A : z \leq 0, \quad n_z \leq 0$
- We wish to compute the circulation integral $\oint_S \vec{F} \cdot d\vec{r}$ over the closed curve $S : \vec{r} = \cos t \vec{i} + \sin t \vec{j} + \sin 2t \vec{k}, t \in [0, 2\pi]$, when $\vec{F} = (e^x - y^3) \vec{i} + (e^y + x^3) \vec{j} + e^z \vec{k}$. Compute the integral using Stokes's formula by noting that S lies on the surface $z = 2xy$.
- Let S be the curve of intersection of the cylinder $K : (x-1)^2 + 4y^2 = 16$ and the plane $T : 2x + y + z = 3$ and let $\vec{F} = [\sin(x^2) + y^2 + z^2] \vec{i} + (2xy + z) \vec{j} + (xz + 2yz) \vec{k}$. Compute $\oint_S \vec{F} \cdot d\vec{r}$, when the orientation of S is counterclockwise as seen from far along the positive z -axis.

7. Choose an orientation for the surface $A : z = 9 - x^2 + y^2$, $x^2 + y^2 \leq 9$ and compute $\int_A \nabla \times \vec{F} \cdot d\vec{a}$, when $\vec{F} = -y\vec{i} + x^2\vec{j} + z\vec{k}$.
8. Using Stokes's formula, compute the surface integral

$$\int_A \nabla \times [(x - z^2)\vec{i} + (x^3 + z)\vec{j} + xy\vec{k}] \cdot d\vec{a},$$

where A is the part of the conical surface $S : x^2 + y^2 = (z - 1)^2$ between the planes $z = 0$ and $z = 1$.

9. The electric current density $\vec{J}$ and the magnetic field strength $\vec{H}$ are, in cylindrical coordinates, of the form $\vec{J} = J(r)\vec{e}_z$, $\vec{H} = H(r)\vec{e}_\varphi$. Determine $H(r)$ in terms of the function $J(r)$ by using Maxwell's equation $\nabla \times \vec{H} = \vec{J}$ and Stokes's formula on a suitably chosen surface A .
10. Show that the generalized Stokes's formula is correct in the case where $\vec{F}$ is a vector field in space and $A \subset T$, where T is a plane in space.
11. Let S be a simple closed curve on a plane T in space whose unit normal vector is $\vec{n} = a\vec{i} + b\vec{j} + c\vec{k}$. Show that the area of the surface A enclosed by S is

$$\mu(A) = \frac{1}{2} \left| \oint_S [(bz - cy)dx + (cx - az)dy + (ay - bx)dz] \right|.$$

12. Define a surface A by the parametrization

$$A : \begin{cases} x = \cos 2u(a + bv \sin nu), \\ y = \sin 2u(a + bv \sin nu), \\ z = bv \cos nu, \end{cases} \quad (u, v) \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right) \times \left[-\frac{\pi}{2}, \frac{\pi}{2}\right],$$

where $0 < b < a$ and $n \in \mathbb{N}$. What does the boundary ∂A of A look like? For which values of n is A an orientable surface?

13. (*) Compute the circulation integral $\oint_S \vec{F} \cdot d\vec{r}$, when $\vec{F} = ye^x\vec{i} + (x^2 + e^x)\vec{j} + e^z\vec{k}$ and S is the closed parametric curve

$$\vec{r} = (2 \cos t - 3 \sin t)\vec{i} + (2 + 3 \cos t + 6 \sin t)\vec{j} + (1 + 6 \cos t - 2 \sin t)\vec{k}, \\ t \in [0, 2\pi].$$

XIV.5 Irrotational vector field

If $\vec{F}$ is a physical flow, electric, magnetic, or other field, the field $\vec{\omega} = \nabla \times \vec{F}$ is called the curl field of the field (cf. Section XI.4). Stokes's theorem expresses the connection between the field $\vec{F}$ and the curl field $\vec{\omega}$:

$$\oint_{\partial A} \vec{F} \cdot d\vec{r} = \int_A \vec{\omega} \cdot d\vec{a}.$$

This connection has much use, for example in electromagnetism, where curl fields play a prominent role already in the basic equations (Maxwell's equations, cf. Section XI.4). If $\vec{F}$ is an irrotational field, i.e. if $\vec{\omega} = \vec{0}$, then Stokes's theorem gives the result

$$\oint_{\partial A} \vec{F} \cdot d\vec{r} = 0. \quad (1)$$

In the terms of Section XIV.1 this can be read: The work done by an irrotational vector field along a closed path = 0. The field $\vec{F}$ is then said to be *conservative* (energy-preserving).

If $\vec{F}$ is a gradient field, i.e. $\vec{F} = -\nabla u$, then $\vec{F}$ is irrotational. The result (1) then holds — and was known even without Stokes's theorem, since the work done by a gradient field = 0 if the initial and final points of the path are the same, cf. Section XIV.1. — But what if the question is asked the other way around: What does a sufficiently regular, but otherwise as general as possible, vector field $\vec{F}$ look like which is either (a) conservative or (b) irrotational? The following focuses on answering these questions. Let us first consider vector fields in the plane.

Let $A \subset \mathbb{R}^2$ be an open set. Such a set is said to be *connected* if any two points of A can be joined by a continuous, rectifiable parametric curve whose endpoints are the two points mentioned and which lies entirely in A . In what follows, an open and connected set is called a *domain*^{*}.

THEOREM XIV.5.1 If a plane vector field $\vec{F}$ is continuous and conservative on a domain $A \subset \mathbb{R}^2$, then $\vec{F}$ is a gradient field, i.e. $\vec{F} = -\nabla u$, where u is continuously differentiable on A .

Proof Let $\vec{r}_0 \hat{=} (x_0, y_0) \in A$ be a fixed point and $\vec{r} \hat{=} (x, y) \in A$ a varying point of A . Let further p be a simple, rectifiable parametric curve (path) $\vec{r} = \vec{r}(t)$,

^{*}A domain is defined somewhat more generally as a set $B = A \cup S$, where A is open and connected and $S \subset \partial A$. If $S = \emptyset$, then $B = A$ is an *open domain*. If $S = \partial A$, then $B = \bar{A}$ is a *closed domain*.

$t \in [a, b]$, for which $\vec{r}(a) = \vec{r}_0$ and $\vec{r}_b = \vec{r} \hat{=} (x, y)$. Define

$$u(p, x, y) = - \int_p \vec{F} \cdot d\vec{r}.$$

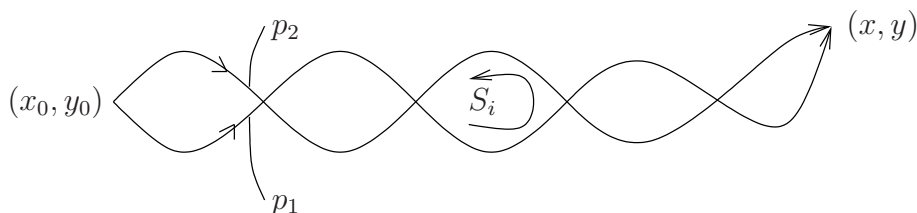
If u does not depend on the choice of the path p , i.e. $u(p, x, y) = u(x, y)$, then by continuing the path from the point $\vec{r}$ it follows from the continuity assumption that

$$u(\vec{r} + \Delta\vec{r}) = -\vec{F}(\vec{r}) \cdot \Delta\vec{r} + o(|\Delta\vec{r}|).$$

Accordingly u is differentiable and $\nabla u = -\vec{F}$. It thus suffices to show that $u(p, x, y)$ depends only on the endpoints of the path. Let p_1 and p_2 be two paths with the same initial point (x_0, y_0) and the same final point (x, y) . If the corresponding geometric curves are S_1 and S_2 , then

$$S_1 \cup S_2 = \bigcup_i S_i, \quad S_i \subset A,$$

where the S_i are closed curves (cf. the figure).



Then

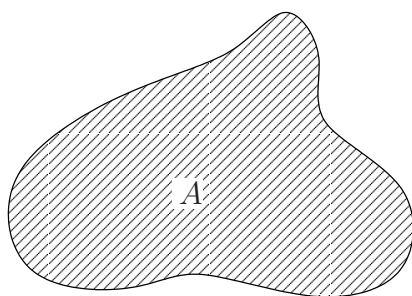
$$\int_{p_1} \vec{F} \cdot d\vec{r} - \int_{p_2} \vec{F} \cdot d\vec{r} = \sum_i \oint_{S_i} \vec{F} \cdot d\vec{r}.$$

Since $\vec{F}$ is conservative, $\oint_{S_i} \vec{F} \cdot d\vec{r} = 0$, so that

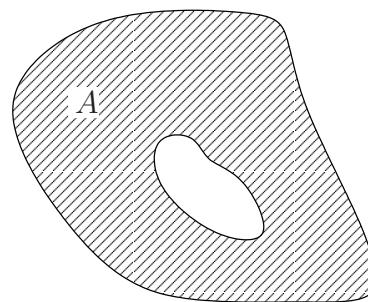
$$\int_{p_1} \vec{F} \cdot d\vec{r} = \int_{p_2} \vec{F} \cdot d\vec{r}.$$

Thus $u(p, x, y) = u(x, y)$ is independent of the choice of the path. $\square$

A domain $A \subset \mathbb{R}^2$ is said to be *simply connected* if $B \subset A$ holds whenever B is a domain for which $\partial B \subset A$. Writing $S = \partial B$, a somewhat more intuitive formulation of the condition is: $A \subset \mathbb{R}^2$ is simply connected if every closed curve $S \subset A$ can be shrunk to a point in such a way that the curve remains in A the whole time as it shrinks. A simply connected set is a ‘lake without islands’.



simply connected



not simply connected

The following corollary of Stokes's theorem is one of the most central results in the theory of vector fields. The result has much use in physics.

THEOREM XIV.5.2 If a plane vector field $\vec{F}$ is continuously differentiable and irrotational ($\text{rot } \vec{F} = 0$) on a simply connected domain A , then $\vec{F}$ is a gradient field.

Proof The potential of the field is determined as a path integral as above, whereupon it again suffices to show that the value of the integral depends only on the endpoints of the path. To prove this, choose again two paths having the same endpoints, whereupon (cf. the proof above)

$$\int_{p_1} \vec{F} \cdot d\vec{r} - \int_{p_2} \vec{F} \cdot d\vec{r} = \sum_i \oint_{S_i} \vec{F} \cdot d\vec{r}.$$

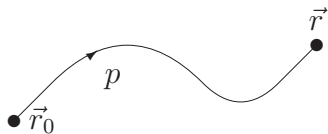
Here too the right-hand side $= 0$, since on the basis of the assumed simple connectedness the regions A_i enclosed by the curves S_i are contained in the set A , whereupon, by Stokes's plane formula (formula (1) of the previous section) and the irrotationality of $\vec{F}$,

$$\oint_{S_i} \vec{F} \cdot d\vec{r} = \int_{A_i} \text{rot } \vec{F} \, dxdy = 0. \quad \square$$

On the basis of the proofs of Theorems integral:

[XIV.5.1](#)–[XIV.5.2](#), the potential of a gradient field can be computed as a work

$$u(\vec{r}) = - \int_{p: \vec{r}_0 \rightarrow \vec{r}} \vec{F} \cdot d\vec{r}.$$



In the computational formula the path p can be chosen freely, as long as it remains in the domain A (in which $\vec{F}$ is defined). If the geometry of A imposes no restrictions, the path can be chosen, for example, to follow straight lines parallel to the coordinate axes, whereupon the computational formula for the potential becomes

$$\begin{aligned}\vec{F}(x, y) &= F_1(x, y)\vec{i} + F_2(x, y)\vec{j} : \\ u(x, y) &= - \int_{x_0}^x F_1(t, y_0) dt - \int_{y_0}^y F_2(x, t) dt.\end{aligned}$$

EXAMPLE 1 Is the vector field

$$\vec{F}(x, y) = (x^3 - y^3 + \cos x)\vec{i} - (3xy^2 + e^{-y})\vec{j}$$

a gradient field?

Solution Since

$$\text{rot } \vec{F} = -\partial_x(3xy^2 + e^{-y}) - \partial_y(x^3 - y^3 + \cos x) = 0,$$

$\vec{F}$ is a gradient field on $\mathbb{R}^2$ (Theorem XIV.5.2). With the choice $(x_0, y_0) = (0, 0)$ the potential is obtained as

$$\begin{aligned}u(x, y) &= - \int_0^x (t^3 + \cos t) dt + \int_0^y (3xt^2 + e^{-t}) dt \\ &= -\frac{1}{4}x^4 - \sin x + xy^3 - e^{-y} + 2.\end{aligned}$$

The general potential function is

$$u(x, y) = -\frac{1}{4}x^4 + xy^3 - \sin x - e^{-y} + C \quad (C = \text{constant}). \quad \square$$

EXAMPLE 2 In polar coordinates a vector field $\vec{F} = \frac{1}{r}\vec{e}_\varphi$ is defined. Investigate whether $\vec{F}$ is a gradient field on the domain A , when

- a) $A = \{(x, y) \in \mathbb{R}^2 \mid x > 0 \ \& \ y > 0\}$,
- b) $A = \{(x, y) \in \mathbb{R}^2 \mid (x, y) \neq (0, 0)\}$.

Solution In polar coordinates (cf. Section XI.4)

$$\vec{F} = F_r \vec{e}_r + F_\varphi \vec{e}_\varphi : \quad \text{rot } \vec{F} = \frac{1}{r} \partial_r(r F_\varphi) - \frac{1}{r} \partial_\varphi F_r.$$

Here $F_r = 0$ and $F_\varphi = 1/r$, so $\vec{F}$ is irrotational on every domain that does not contain the origin.

a) Here A is simply connected, so $\vec{F}$ is a gradient field. By choosing $r_0 > 0$ the potential $u(x, y) = v(r, \varphi)$ can be computed as a path integral

$$\begin{aligned} v(r, \varphi) &= - \int_{r_0}^r F_r(t, 0) dt - \int_0^\varphi F_\varphi(r, t) r dt = -\varphi \\ \Rightarrow \quad u(x, y) &= \underline{\underline{-\text{Arcsin} \left(\frac{y}{\sqrt{x^2 + y^2}} \right)}}. \end{aligned}$$

b) Here A is not simply connected, so we investigate separately whether $\vec{F}$ is conservative. Choose a path $p : \vec{r} = \vec{r}(\varphi)$, $0 \leq \varphi \leq 2\pi$ so that $\vec{r}(\varphi)$ circles the circle of radius r_0 :

$$\vec{r}(\varphi) = r_0 \vec{e}_r, \quad r_0 > 0, \quad 0 \leq \varphi \leq 2\pi.$$

Then the initial and final points of the path are the same, $\vec{r}(\varphi) \hat{=} (x, y) \in A$ for every φ and

$$\int_p \vec{F} \cdot d\vec{r} = \int_0^{2\pi} \frac{1}{r_0} \cdot r_0 d\varphi = 2\pi \neq 0,$$

Thus $\vec{F}$ is not conservative on A and hence not a gradient field on A either. $\square$

Irrotational vector field in space

By comparing the proof of Theorem XIV.5.1 with the three-dimensional version of Stokes's theorem, one sees that the proof (and hence also the assertion of Theorem XIV.5.1) is valid as it stands for a vector field in space as well. Insofar as the geometry of the domain imposes no restrictions, the potential of a three-dimensional, conservative vector field $\vec{F} = F_1(x, y, z)\vec{i} + F_2(x, y, z)\vec{j} + F_3(x, y, z)\vec{k}$ can be computed, for example, as a path integral

$$u(x, y, z) = - \int_{x_0}^x F_1(t, y_0, z_0) dt - \int_{y_0}^y F_2(x, t, z_0) dt - \int_{z_0}^z F_3(x, y, t) dt.$$

Theorem XIV.5.2 too generalizes to three dimensions, as long as simple connectedness is defined suitably. Consider simple, closed, rectifiable curves $S \subset A$ for

which a surface $B \subset \mathbb{R}^3$ satisfying the conditions of Stokes's theorem can be found such that $\partial B = S$. If for every such curve a surface B can further be determined such that $B \subset A$, then the domain A is said to be simply connected. For example,

$$A = \{ (x, y, z) \in \mathbb{R}^3 \mid a^2 < x^2 + y^2 + z^2 < R^2 \} \quad (0 \leq a < R)$$

is simply connected (!). On the other hand,

$$A = \{ (x, y, z) \in \mathbb{R}^3 \mid x^2 + y^2 + z^2 < R^2 \ \& \ x^2 + y^2 > a^2 \} \quad (0 \leq a < R)$$

is not simply connected, since if

$$S = \{ (x, y, z) \in \mathbb{R}^3 \mid x^2 + y^2 = b^2 \ \& \ z = 0 \},$$

then $S \subset A$ when $a < b < R$, but if $B \subset \mathbb{R}^3$ is a surface, then the conditions $S = \partial B$ and $B \subset A$ cannot hold simultaneously.

EXAMPLE 3 In spherical coordinates a vector field is defined by

$$\vec{F}(r, \theta, \varphi) = \frac{1}{r^3}(2 \cos \theta \vec{e}_r + \sin \theta \vec{e}_\theta), \quad r \neq 0.$$

Determine the potential of the field, if one exists.

Solution Here $\vec{F} = F_r \vec{e}_r + F_\theta \vec{e}_\theta + F_\varphi \vec{e}_\varphi$, where

$$F_r = 2r^{-3} \cos \theta, \quad F_\theta = r^{-3} \sin \theta, \quad F_\varphi = 0,$$

so that (see Section XI.5)

$$\nabla \times \vec{F} = 0 \vec{e}_r + 0 \vec{e}_\theta + \frac{1}{r}[\partial_r(rF_\theta) - \partial_\theta F_r] \vec{e}_\varphi = \vec{0} \quad (r \neq 0).$$

Since excluding the origin does not cause connectedness problems in three dimensions, $\vec{F}$ is a gradient field. The potential can be computed by taking as the starting point, for example, $\vec{r}_0 = (r_0, 0, 0)$, $r_0 > 0$, and proceeding along paths parallel to the coordinate lines:

$$\begin{aligned} u(r, \theta, \varphi) &= - \int_{r_0}^r F_r(t, 0, 0) dt - \int_0^\theta F_\theta(r, t, 0) r dt - \int_0^\varphi F_\varphi(r, \theta, t) r \sin \theta dt \\ &= - \int_{r_0}^r 2t^{-3} dt - \int_0^\theta r^{-2} \sin t dt \\ &= \left[t^{-2} + r^{-2} \right]_{r_0}^\theta \cos t \\ &= r^{-2} \cos \theta + \text{constant} (= -r_0^{-2}). \end{aligned}$$

A valid potential is thus $u(r, \theta, \varphi) = \underline{\underline{r^{-2} \cos \theta}}$. $\square$

EXERCISES

1. Verify that the vector field $\vec{F}(x, y) = (y + 2x)\vec{i} + x\vec{j}$ is irrotational. Using this observation, compute the path integral $\int_p \vec{F} \cdot d\vec{r}$, where p runs from the point $(1, 0)$ through the origin to the point $(1, 2)$ along an arc of a circle.
2. Determine a function f such that $f(0) = 1$ and the plane vector field

$$\vec{F}(x, y) = f(y)[2xy\vec{i} + x^2(y + 1)\vec{j}]$$

is irrotational. What is then the potential of the field?

3. Show that the following vector fields are irrotational in all of space and determine the potentials of the fields.
 - a) $z\vec{i} + z \cos y\vec{j} + (x + \sin y)\vec{k}$
 - b) $e^x \sin y\vec{j} + e^x \cos y\vec{j} + z\vec{k}$
 - c) $2xyzi + x^2z\vec{j} + x^2y\vec{k}$
 - d) $-2xe^{-y}\vec{i} + (x^2e^{-y} + \sin z)\vec{j} + y \cos z\vec{k}$
 - e) $(x^2 + y^2 + z^2)(x\vec{i} + y\vec{j} + z\vec{k}) = r^2\vec{r}$
4. It is known of a vector field $\vec{F}$ that it is irrotational in all of space and of the form $\vec{F}(x, y, z) = xy^2\vec{i} + (\cos z + x^2y)\vec{j} + yf(z)\vec{k}$. Determine the scalar potential of the field and, by means of it, the path integral $\int_p \vec{F} \cdot d\vec{r}$, where the initial point of p is $(0, 0, 0)$ and the final point $(2, -1, 3)$.
5. (*) (Disturbance in a field) An obstacle

$$A = \{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 \leq R^2\},$$

is introduced into a plane vector field $\vec{U}_0 = u_0\vec{i}$ ($u_0 = \text{constant}$), whereupon the field is disturbed into the form $\vec{U} = u(x, y)\vec{i} + v(x, y)\vec{j}$. It is known of the disturbed field $\vec{U}$ that it is both source-free and irrotational outside the obstacle, i.e. $\text{div } \vec{U} = \text{rot } \vec{U} = 0$ everywhere except in A . It is also known that $\vec{U} \cdot \vec{n} = 0$ on the boundary ∂A (i.e. when $x^2 + y^2 = R^2$, $\vec{n}$ = the normal of ∂A) and that $\vec{U}(x, y) \rightarrow \vec{U}_0$ when $x^2 + y^2 \rightarrow \infty$. The problem is to find a field $\vec{U}$ satisfying these conditions. Since the region outside A is not simply connected, $\vec{U}$ is not necessarily a gradient field (even though it is irrotational).

- a) Show that the problem has a solution whose polar-coordinate form is

$$\vec{U} = -\nabla f(r) \cos \varphi + g(r)\vec{e}_\varphi$$

i.e. determine the functions $f(r), g(r)$ in as general a form as possible so that the given conditions are satisfied. (The problem has no other solutions.)

b) Verify that the above solution becomes unique when an additional condition of the form $v(R, 0) = v_0$ (here Cartesian coordinates) is imposed.

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